FURTHER PROPERTIES OF THREE-DIMENSIONAL SU(2) GAUGE THEORY WITH MASSIVE FERMIONS

Abstract. We study further properties of three-dimensional fundamental SU(2) gauge theory with chamber fermions. At weak coupling and positive mass, the action-density and meson correlations on disjoint supports converge along the full cutoff sequence on each fixed physical torus. The laws of the block fields converge in total variation and define a compatible probability law. This law determines the coupling and the flavour number almost surely. A controlled expansion of the action-density two-point function identifies its next short-distance coefficient as $(8 - 2N_f)/(3\pi)$, uniformly in the torus side and in the specified infinite-volume limits. The proof combines comparison of adjacent cutoffs, estimates for derivatives of the source energy, and lattice coefficient calculations. We also prove moment and mass-comparison estimates, extend the scalar bilinears to diquarks, obtain charged-sector energy bounds and current correlations, and continue fixed-volume correlations in the mass. On each fixed torus, we identify the divergent subtraction for the scalar condensate and prove convergence of the subtracted quantity, with analytic dependence on coupling and mass. The construction paper supplies the model and the underlying renormalization estimates; the additional estimates and limit arguments are proved here.

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1. Introduction

We study the massive fundamental SU(2) theory constructed in [QCDM1]. Its regularization uses the chamber Dirac operator with Wilson parameter $r = 1/2$. Fix $N_f \geq 1$ and a physical mass

window $0 < m_- \le m_+ < 1/2$. At spacing $a = a_K = L^{-K}$ the bare parameters are $\beta = \mathsf{b}/a$ and $m_0 = a\mathsf{m}$. The gauge measure on a torus is

$$\nu_K(dU) = Z_K^{-1} e^{-\beta A(U)} |\det(D_+[U] + m_0)|^{2N_f}\, dU, \qquad A(U) = \sum_p (1 - \tfrac{1}{2}\operatorname{tr} U_p).$$

Here $D_+[U]$ is one internal block of the massless chamber operator for one flavour.

Theorem 0. For sufficiently weak gauge coupling and masses in a fixed compact positive interval, the following hold.

(i) On each fixed physical torus, the action-density and meson correlations with pairwise disjoint supports converge along the full cutoff sequence, uniformly in the permitted coupling and mass labels. At every fixed depth the block-field laws converge in total variation. Their limits define one probability law on compatible block fields. Within any finite family of flavour numbers with common construction constants, this law determines the coupling label b and N_f almost surely. Laws at different pairs (b, N_f) are mutually singular; at fixed b, N_f, changing the positive mass gives mutually absolutely continuous laws.

(ii) For nonnegative test functions f, h supported in separated balls of radius d, with the uniform size conditions specified below, the continuum action-density two-point function satisfies

$$S_2((0,f),(0,h)) = \frac{9}{16\pi^2}\mathsf{b}^{-2}\left[\mathcal{K}_6(f,h) + \frac{8-2N_f}{3\pi}\mathsf{b}^{-1}\mathcal{K}_5(f,h)\right] + o(d)\mathsf{b}^{-3}\|f\|_\infty\|h\|_\infty,$$

where $\mathcal{K}_q(f,h) = \iint f(x)h(y)|x-y|^{-q}\,dx\,dy$. The error is uniform in the side of the torus and in the admitted labels. The same law holds for the growing-torus and volume-first continuum limits. These correlations determine b and N_f.

(iii) Conditional fermion loops have cutoff-independent variance bounds, including at coinciding supports. The scalar-bilinear family extends to diquarks and has lower energy bounds in its charged sectors. The conserved-current two-point function has a short-distance law.

(iv) On a fixed torus, subsequences give correlations holomorphic in coupling and mass off the total diagonal. At fixed real coupling, analytic continuation in mass and the cutoff comparison give full convergence on disjoint supports throughout the stated real mass interval, including zero and some negative labels. Dilation transfers this conclusion to any real mass label on the corresponding admitted sides.

(v) For a mass window with $Lm_- < m_+$ and the additional field-scale hypotheses stated below, the scalar condensate on a fixed torus converges after subtracting

$$\mathsf{A}_0 a_K^{-2} + (\mathsf{A}_1\mathsf{m} + \mathsf{A}_2/\mathsf{b})a_K^{-1} + (w_1\mathsf{m}/\mathsf{b} + w_2/\mathsf{b}^2)K.$$

The five real coefficients are independent of the cutoff, torus and physical labels. The convergence holds along the full cutoff sequence, with all derivatives in coupling and mass on the admitted window.

■□

The cutoff and block-law statements are Thms. Q, U and W and Cor. 11.6. The short-distance expansion is Thm. X, with its test class and uniform error stated there. The further observable estimates are given below and in sections 5, 6 and 8; the mass continuation is Thms. M and T and Prop. 11.28. The condensate subtraction and its hypotheses are given in Thm. S.

The main comparison places two successive cutoffs on the same block lattice. After discarding histories with an early large-field region, we restart the two flows from that lattice. The estimates show that the flow forgets the differences contributed by the discarded scales. Their errors are summable in the cutoff. Comparing the densities before taking source derivatives gives convergence of the block laws; taking derivatives gives convergence of correlations on disjoint supports. These conclusions keep the physical torus fixed.

The next short-distance coefficient requires a further argument. We multiply every gauge fluctuation by one complex parameter and control the resulting source-energy derivatives on a disc of radius greater than one. Their terms of orders four and six give the free coefficient and its first correction. The remaining Taylor terms have a uniform bound. A finite-dimensional Laplace expansion identifies the retained coefficients with lattice Gaussian expressions; removing the block constraint and comparing the lattice kernels with their continuum limits gives the displayed coefficient. We first prove this estimate in a cutoff window. The summable cutoff comparison then passes it to the continuum. A two-volume chessboard estimate transfers it from the least torus to the other volumes and the infinite-volume states.

For the condensate, the adjacent-cutoff comparison must retain the contact terms. Integrating in the mass expresses them as differences of vacuum energies. We expand those energies in the same field-scale parameter and control the remainder. The already established bound on the second mass derivative leaves an affine dependence on the mass, up to a summable error. Exact lattice rescaling then determines the divergent powers of the cutoff and the logarithmic term displayed above.

The other results use additional fermion-source estimates, beginning with the following fluctuation bound.

At fixed K, an infinite-lattice state means any weak limit of these measures as the torus side tends to infinity through the sides permitted by [QCDM1, Prop. 10.20]. Such states exist by [QCDM1, Prop. 2.5]. For a bounded compactly supported function u put

$$\mathsf{G}_+ = a(D_+[U] + a\mathsf{m})^{-1}, \qquad \mathsf{L}(u) = \mathrm{Tr}(\mathsf{G}_+ \hat{u}).$$

The hat denotes multiplication by u at the site of each chamber. The trace is over chambers, spin and colour. It is finite at each cutoff, even on the infinite lattice.

Theorem A (Fluctuations and changes of mass)**.** Choose the odd integer $L \geq 13$ and the auxiliary constants as in [QCDM1, Prop. 10.20]. There is $\mathsf{b}_1 < \infty$ such that, for every fixed $\mathsf{b} \geq \mathsf{b}_1$ and every $R < \infty$, there is a constant $C_R < \infty$ with

$$\mathsf{E}_\nu |\mathsf{L}(u) - \mathsf{E}_\nu \mathsf{L}(u)|^2 \leq C_R \|u\|_\infty^2.$$

This holds for every $K \geq 1$, every $\mathsf{m} \in [m_-, m_+]$, every infinite-lattice torus-limit state ν, and every bounded real u whose support lies in a ball of radius R. The constant is independent of the location of the ball and the chosen state. On a fixed allowed physical torus the same assertion holds with a constant depending on its volume.

On a fixed physical torus, fix $0 < g \leq \gamma$, $\ell \in \mathfrak{L}(g)$ and an allowed coupling label b, as in section 2. Let f_K be Borel maps from the cutoff configuration spaces to a fixed Polish space. If their laws at two masses $\mathsf{m}, \mathsf{m}' \in [m_-, m_+]$ converge along a common cutoff sequence to P, Q, then P and Q are mutually absolutely continuous and

$$\mathrm{TV}(P, Q) \leq C_{g,\ell} |\mathsf{m}' - \mathsf{m}|.$$

The constant is independent of the cutoff, the two masses and b. The cutoff measure at each mass uses that mass in the determinant defining ν_K. ■□

The trace need not have a finite limit without subtraction. The theorem controls its fluctuation at coinciding supports. For its scalar one-point counterpart, Thm. S identifies the subtraction and proves convergence under the hypotheses there. The proof of the variance bound uses a determinant with a complex source in one internal block and the conjugate source in the other. The parts of the vacuum energy depending on just one source are holomorphic. Removing them leaves a mixed energy bounded uniformly in the cutoff. The first coefficient of the resulting holomorphic determinant gives the variance bound by Parseval's identity. Thm. B gives the explicit constants and proves the variance assertion of Thm. A. For a constant source the determinant changes the mass. The same circle bound controls the density for a short change of mass. Chaining these bounds across the mass interval gives the comparison of limiting gauge laws in Prop. 3.13.

This estimate also enters the analysis of parity-odd scalar bilinears. We treat those bilinears together with mesons, diquarks and antidiquarks. Their separated correlations have a common continuum subsequence and a compact symplectic flavour action by $G = \mathrm{USp}(2N_f)^2$. Coordinate reflection positivity gives an enlarged Hilbert space and a time-translation semigroup. Bounds obtained by counting fermion propagators give a lower bound $H \geq |q|\mathsf{m}$ on quark number q, and $H \geq 2\mathsf{m}$ on the complement of the G-invariant vectors. Both sectors $q = 2$ and $q = -2$ are nonzero. These claims and the distinction between coordinate reflection positivity and full rotation covariance are stated in sections 5 and 6.

We also prove a short-distance two-point law for the conserved currents. The current is a point-split lattice bilinear. Its comparison with a site bilinear requires a source that acts on spin, as well as on internal and flavour indices. The source construction extends to those matrices; an energy estimate for the propagator then controls the point splitting. The result concerns current two-point functions and lattice reflection-positive current states. It does not assume a joint reconstruction of all the currents with the scalar fields of [QCDM1]; see section 8.

For the original action-density and meson family on a fixed physical torus, we allow supports with empty total intersection. Individual pairs of supports may overlap. A polymer contributing to a mixed derivative is either large or far from at least one insertion; in both cases the local-energy estimate supplies decay. A common subsequence then gives correlations off the total diagonal, holomorphic in the coupling and mass near the permitted real parameter window. The neighbourhood depends on the physical volume. This is a separate fixed-volume conclusion from the volume-first construction used for the charged sectors.

At fixed real coupling we continue the separated correlations meromorphically in the mass. The same subsequence and discrete exceptional set work for all their orders and test functions. A quantitative lower bound for the normalized partition function gives a zero-free neighbourhood of a longer real mass interval, with width at least $c/[\ell^3(1 + \log \ell)]$. The real-mass cumulant estimates remain uniform in the volume. These statements, including the continued short-distance laws, are proved in section 10.

The sections on determinants, source energies and additional bilinears establish the fluctuation, mass-comparison and charged-sector results. Sections 9 and 10 develops the analytic parameter domains. The cutoff proof and its consequences occupy sections 11 to 13. Section 14 proves the next short-distance coefficient. The appendices supply the comparisons of meshes, the detailed history estimates, and the pure-gauge coefficient calculation. Throughout we use the chamber model and notation of [QCDM1].

2. Model, sources and parameter ranges

We recall the definitions needed for the observables and the estimates. The lattice construction is in [QCDM1, §2]. Spatial coordinates in operator formulas are in lattice units; supports and distances are in physical units unless stated otherwise.

2.1. The chamber operator.

Let W_3 be the group of signed permutation matrices. A place is $v = (x, q)$ with $x \in \mathbb{Z}^3$ and $q \in W_3$. Put

$$(N_0, N_1, N_2, N_3) = (e_1, e_2 - e_1, e_3 - e_2, -e_3), \qquad t_s = 3|N_s|,$$

and let w_s be reflection in $N_s^{\perp}$. The adjacent places and the corresponding unit vectors are

$$v \cdot s = (x + \delta_{s0} q e_1, q w_s), \qquad u_s(v) = q N_s/|N_s|.$$

Set $U(v, v \cdot s) = 1$ for $s = 1, 2, 3$ and $U(v, v \cdot 0) = U_{\langle x, x+qe_1\rangle}$. The spinor fibre is

$$\mathbb{C}^2_{\mathrm{s}} \otimes \mathbb{C}^2_{\tau} \otimes \mathbb{C}^2_{\mathrm{c}} \otimes \mathbb{C}^{N_f}.$$

The factors denote spin, internal index, colour and flavour. With the Pauli matrices σ_μ and τ_μ write

$$\gamma_\mu = \sigma_\mu \otimes \tau_3, \qquad \gamma_4 = 1 \otimes \tau_1, \qquad e_\pm = (1 \pm \tau_3)/2.$$

The massless operator is

$$(D[U]f)(v) = \frac{1}{2}\sum_{s=0}^{3} t_s f(v) - \sum_{s=0}^{3} t_s(\tfrac{1}{2} - \gamma(u_s(v)))U(v, v\cdot s)f(v\cdot s). \tag{2.i}$$

The two internal blocks satisfy $D_- = D_+^*$ and $\mathrm{Re}(D_\pm + m_0) \geq m_0$. In particular $\|\mathsf{G}_\pm\| \leq \mathsf{m}^{-1}$, where $\mathsf{G}_\pm = a(D_\pm + m_0)^{-1}$.

On $T_\ell = (\mathbb{R}/\ell\mathbb{Z})^3$ the links are periodic and spinors are antiperiodic in the three coordinate directions. The finite-volume functional is

$$\Omega_K(F) = \int dU \int d\bar\psi\, d\psi\, e^{-\beta A(U) - \bar\psi(D[U]+m_0)\psi} F, \qquad \langle F\rangle_K = \Omega_K(F)/\Omega_K(1).$$

Haar measures have total mass one. The Berezin integral has the convention giving $\det(D[U]+m_0)$ at $F = 1$. Conditional expectations at fixed U are the Gaussian Berezin functional with covariance $(D[U]+m_0)^{-1}$. On the infinite lattice we integrate this functional against an infinite-lattice torus-limit gauge state. All polynomial insertions have finite support at fixed a.

2.2. **Physical sources.** Choose Hermitian flavour matrices t_j with $t_1 = N_f^{-1/2}1$ and $\mathrm{tr}(t_j t_l) = \delta_{jl}$. Set

$$(c_0, c_1, c_2, c_3) = (1, i, i, 1), \quad \Gamma_{a,j} = 1_{\mathrm{s}} \otimes c_a\tau_a \otimes 1_{\mathrm{c}} \otimes t_j, \quad \mathfrak{B} = \{\Gamma_{a,j} : 0 \leq a \leq 3,\ 1 \leq j \leq N_f^2\}.$$

These matrices have operator norm at most one. The physical action and meson insertions are

$$O_K(w) = a^{-1}\sum_p w(x(p))(1 - \tfrac{1}{2}\,\mathrm{tr}\, U_p), \tag{2.ii}$$

$$\Phi_K(J) = a\sum_x \sum_{v: x_v = x} \bar\psi(v) J(ax)\psi(v), \tag{2.iii}$$

where J takes values in the complex span of $\mathfrak{B}$. Here $x(p)$ is the physical plaquette centre, and each positive coordinate orientation occurs once. The source partition function is

$$Z_K[w, J] = \Omega_K(e^{O_K(w)+\Phi_K(J)}) = \int e^{-\beta A(U) + O_K(w)} \det(D[U] + m_0 - a\hat J)\, dU.$$

For $\alpha = 0$ set $\Phi_{\alpha,K}(f) = O_K(f)$, and for $\alpha = \Gamma \in \mathfrak{B}$ set $\Phi_{\alpha,K}(f) = \Phi_K(f\Gamma)$. We use

$$\|f\|_0 = \|f\|_\sharp = \max\{\|f\|_\infty, 40M\,\mathrm{Lip}\, f\}, \qquad \|f\|_\Gamma = \|f\|_\infty.$$

The constants M, ϱ, ϱ_J and r_M belong to the source estimates of [QCDM1, §10].

Joint cumulants are derivatives of the logarithm of the normalized source partition function at zero. Their centered moments are the sums over set partitions with no singleton blocks. We use the same definition for the enlarged bilinear family introduced below.

2.3. **Permitted parameters and limits.** Fix the mass window and N_f. Choose the constants of [QCDM1, Prop. 10.20], and decrease its upper bound γ on g as specified in section 4. These choices depend on the fixed mass window and N_f, and not on K or the volume. Put

$$I_g = [g^{-2} + \bar B, L(g^{-2} + B_\sharp) - \bar B], \qquad 0 < g \leq \gamma, \quad \mathsf{b} \in I_g, \quad \mathsf{m} \in [m_-, m_+].$$

The allowed sides form the cofinal set $\mathfrak{L}(g) = \{2L^m : m \geq m_g\}$ of that proposition. More explicitly, with $M = L^{m'}$,

$$X_g = L(g^{-2} + B_\sharp), \qquad R(x) = \min\{L^q : q \geq 0,\ L^q \geq (\log x)^{r_c}\},$$
$$m_g = 1 + m' + \max\{\lceil \log_L R(X_g + \bar B)\rceil, \lceil \log_L C_{\mathrm{s}}\rceil\}.$$

Here $r_c \geq 2$ and $C_s \geq 2$ are the fixed local-chart constants of the construction. This condition supplies the room required by the local estimates. The intervals I_g cover all sufficiently large b: for example take $g^{-2} = \mathsf{b} - \bar{B}$ and increase the lower threshold on b so that the upper endpoint also exceeds b. This explains the parameter formulation in Thm. A.

We use two infinite-volume limits. The volume-first construction takes torus limits at fixed K and then lets $K \to \infty$. The growing-torus construction lets the physical side and the cutoff tend to infinity together. The coefficient theorem Thm. X applies to both constructions under its stated hypotheses. Constants may depend on g unless stronger uniformity is stated. The analytic results in section 9 keep the physical torus fixed throughout the cutoff limit.

3. Determinant bounds at coinciding supports

The estimates for disjoint supports do not bound the variance of a loop against itself. We obtain this bound from a different use of the fermion determinant. A source in one species gives a holomorphic determinant; its conjugate source in the other species gives the squared absolute value. After removing the two terms of the vacuum energy which depend on just one species, the remaining energy is bounded uniformly in the cutoff. Cauchy's formula then bounds the coefficients of the determinant, including the loop.

We retain the setting of section 2, with the further choices in section 4. Write ν_K for the gauge measure, and let $\mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$ be the infinite-lattice states obtained as limits of these torus measures. All the constants below are uniform in the admitted data except where dependence on g, a torus side, or a support radius is stated.

The two physical resolvents and the simplest loop are

$$\mathsf{G}_\pm = a(D_\pm[U] + m_0)^{-1}, \qquad \mathsf{G}_- = \mathsf{G}_+^*, \qquad \mathsf{L}(f) = \mathrm{Tr}(\mathsf{G}_+ \hat{f}).$$

Here $\hat{f}$ denotes multiplication by f, and the trace includes the position, spin and colour variables but no flavour variable. When a source acts on flavour, we extend $\mathsf{G}_\pm$ by the identity on that space. The resolvent bound is $\|\mathsf{G}_\pm\| \leq \mathsf{m}^{-1}$. On the infinite lattice all functions in a determinant or trace have finite support; consequently the relevant operator has finite rank. Set

$$\Xi^f_\nu(z) = \mathsf{E}_\nu |\det(1 + z\mathsf{G}_+ \hat{f})|^{2N_f}. \tag{3.i}$$

We say that the circle bound holds at (f, ρ, C) if there is a function h, holomorphic near $|z| \leq \rho$, such that $h(0) = 0$ and

$$\Xi^f_\nu(z) \leq C \exp(2N_f \operatorname{Re} h(z)), \qquad |z| = \rho. \tag{3.ii}$$

The function h absorbs the mean of the loop. Its size need not be uniform in K.

Theorem B (Circle bound)**.** There are constants C_{mix} and $C'_{\mathscr{C}}$, independent of the admitted data, for which the following holds. Put

$$\rho_* = 2\varrho_J \leq \tfrac{1}{2} m_-, \qquad \rho_* \leq \frac{r_M}{2L}, \qquad c_V(g) = C + 2 + C_P(1 + \log(L(g^{-2} + B_\sharp))).$$

The constants on the right are those of the partition bounds in [QCDM1, Props. 10.14 and 10.17].

(1) On a torus of side ℓ, every bounded real f satisfies the circle bound with

$$\rho = \rho_*/\|f\|_\infty, \qquad C = \exp((c_V(g) + C_{\mathrm{mix}})\ell^3).$$

The assertion for $f = 0$ is understood without the quotient. One can take

$$N_f h_K(z) = E_K(0, -e_+ z f 1) - E_K(0, 0), \tag{3.iii}$$

which is holomorphic on $|z|\|f\|_\infty < r_M/L$.

(2) For $R > 0$, let $i(R)$ be the least integer $i \geq 1$ with $L^i \geq 2R + 3$, and put

$$C(R) = \exp\left((c_V(g) + C_{\mathrm{mix}})L^{3i(R)} + 2C'_{\mathscr{C}}(1 + R^3)\right). \tag{3.iv}$$

If f is real, $\|f\|_\infty \leq 1$, and its support lies in a ball of radius R, the circle bound holds at $(f, \rho_*, C(R))$ on every admitted torus of side $\ell \geq 2L^{i(R)}$ and for every $\nu \in \mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$.

(3) For any such measure and any real u supported in that ball,

$$\mathrm{Var}_\nu\, \mathsf{L}(u) \leq \frac{\|u\|_\infty^2}{\rho_*^2} \min\left\{C(R)^{1/N_f}, \frac{C(R) - 1}{N_f^2}\right\}. \tag{3.v}$$

The same assertion on a fixed torus holds with $C(R)$ replaced by $\exp((c_V(g) + C_{\mathrm{mix}})\ell^3)$.

■□

The estimate concerns random variables of the gauge field. It does not identify the Grassmann bilinear with a random distribution. The relation between bilinear correlations and conditional loops is supplied by the conditional Gaussian identities.

3.1. **Coefficients of a determinant.**

Proposition 3.1. If the circle bound holds at (f, ρ, C), then

$$\mathsf{E}_\nu|\mathsf{L}(f) - h'(0)|^{2N_f} \leq \rho^{-2N_f} C, \tag{3.vi}$$

$$\mathsf{E}_\nu|\mathsf{L}(f) - h'(0)|^2 \leq \frac{C - 1}{N_f^2 \rho^2}. \tag{3.vii}$$

In particular,

$$\mathrm{Var}_\nu\, \mathsf{L}(f) \leq \rho^{-2} \min\{C^{1/N_f}, (C - 1)/N_f^2\}.$$

More generally, if Z_U is a random holomorphic polynomial with $Z_U(0) = 1$, and $\mathcal{F}$ is holomorphic near the closed disc with $\mathcal{F}(0) = 0$, write $e^{-\mathcal{F}(z)} Z_U(z) = \sum_{n \geq 0} d_n(U) z^n$. Then

$$\sum_{n \geq 0} \rho^{2n} \mathsf{E}_\nu |d_n|^2 = \frac{1}{2\pi} \int_0^{2\pi} e^{-2\,\mathrm{Re}\,\mathcal{F}(\rho e^{i\theta})} \mathsf{E}_\nu |Z_U(\rho e^{i\theta})|^2 \, d\theta. \tag{3.viii}$$

■

Proof. Parseval's identity holds at each U; Tonelli's theorem permits integration of its nonnegative terms. Apply it first to $Z_U(z) = \det(1 + z\mathsf{G}_+ \hat{f})^{N_f}$ and $\mathcal{F} = N_f h$. The first two coefficients are $d_0 = 1$ and $d_1 = N_f(\mathsf{L}(f) - h'(0))$. This gives eq. 3.vii.

For the higher moment, use the first coefficient of $e^{-h(z)} \det(1 + z\mathsf{G}_+ \hat{f})$ instead. Cauchy's formula and Jensen's inequality give

$$|\mathsf{L}(f) - h'(0)|^{2N_f} \leq \rho^{-2N_f} \frac{1}{2\pi} \int_0^{2\pi} |e^{-h(z)} \det(1 + z\mathsf{G}_+ \hat{f})|^{2N_f} \, d\theta, \qquad z = \rho e^{i\theta}.$$

Integrate over ν. Finally the mean minimizes the squared distance from a constant, and the L^{2N_f} bound implies the L^2 bound with constant C^{1/N_f}. □

The logarithm gives bounds for every moment. The useful fact is that its real part has mean zero on a circle after subtracting h. This controls each coefficient by the positive part of that real part, whose exponential is bounded by eq. 3.ii.

Lemma 3.2 (Moments of the logarithmic coefficients)**.** Under the hypotheses of Thm. B(ii), let $h(z) = \sum_{n \geq 1} h_n z^n$ give the circle bound for f, and put $\lambda_n = \mathrm{Tr}((\mathsf{G}_+ \hat{f})^n)$. Then, for every $n \geq 1$,

$$\mathsf{E}_\nu \exp\left[\frac{N_f \rho_*^n}{2} \left|\frac{(-1)^{n-1}}{n} \lambda_n - h_n\right|\right] \leq 1 + C(R). \tag{3.ix}$$

Consequently, for every real $q \geq 1$ and bounded real u supported in a ball of radius R,

$$\|\mathsf{L}(u) - \mathsf{E}_\nu \mathsf{L}(u)\|_{L^q(\nu)} \leq \frac{4q}{eN_f\rho_*}(1 + C(R))^{1/q}\|u\|_\infty. \tag{3.x}$$

On a fixed torus the same statements hold with $C(R)$ replaced by $\exp((c_V(g) + C_{\mathrm{mix}})\ell^3)$. ∎

Proof. Since $\rho_* \leq \mathsf{m}/2$ and $\|\mathsf{G}_+\| \leq \mathsf{m}^{-1}$, the finite-rank determinant has a logarithm near $|z| \leq \rho_*$, normalized to vanish at zero. At each configuration write

$$\varphi_U(z) = \log\det(1 + z\mathsf{G}_+\hat{f}) - h(z) = \sum_{n\geq 1} d_n z^n, \qquad d_n = \frac{(-1)^{n-1}}{n}\lambda_n - h_n.$$

For $\eta_U(\theta) = \operatorname{Re}\varphi_U(\rho_* e^{i\theta})$, the circle mean is zero and the nth Fourier coefficient is $\rho_*^n d_n/2$. Thus, writing averages with respect to $d\theta/(2\pi)$,

$$\rho_*^n|d_n| \leq 2 \operatorname*{avg}_\theta |\eta_U(\theta)| = 4 \operatorname*{avg}_\theta \eta_U(\theta)^+.$$

Jensen's inequality and $e^{x^+} \leq 1 + e^x$ give

$$e^{N_f\rho_*^n|d_n|/2} \leq \operatorname*{avg}_\theta (1 + e^{2N_f\eta_U(\theta)}).$$

The determinant coefficients are measurable functions of U. Tonelli's theorem and the circle bound therefore prove eq. 3.ix.

For $u \neq 0$, apply its $n = 1$ case to $f = u/\|u\|_\infty$. Set $Y = \mathsf{L}(f) - h_1$ and $\lambda = N_f\rho_*/2$. The elementary inequality $y^q \leq (q/(e\lambda))^q e^{\lambda y}$ for $y \geq 0$ yields

$$\|Y\|_{L^q(\nu)} \leq \frac{q}{e\lambda}(1 + C(R))^{1/q}.$$

Centering costs at most a factor two. Linearity in u proves eq. 3.x; the case $u = 0$ is immediate. □

For the passage to an infinite-lattice state it is preferable to avoid extracting a convergent family of the functions h. The next observation expresses their existence in terms of the determinant alone.

Lemma 3.3. Let $\rho\|f\|_\infty < \mathsf{m}$. The function $\log \Xi_\nu^f$ is subharmonic on $|z|\|f\|_\infty < \mathsf{m}$, and the circle bound holds at (f, ρ, C) if and only if

$$\frac{1}{2\pi}\int_0^{2\pi} \log \Xi_\nu^f(\rho e^{i\theta})\, d\theta \leq \log C. \tag{3.xi}$$

∎

Proof. Put $Z_U(z) = \det(1 + z\mathsf{G}_+\hat{f})^{N_f}$. The resolvent bound makes this determinant nonzero in the stated disc. Tilting ν by $|Z_U(z)|^2/\Xi_\nu^f(z)$ gives a probability measure ν_z, and differentiation gives

$$\partial_z\partial_{\bar{z}} \log \Xi_\nu^f(z) = \mathsf{E}_{\nu_z}|\partial_z \log Z_U(z)|^2 - |\mathsf{E}_{\nu_z}\partial_z \log Z_U(z)|^2 \geq 0.$$

The circle bound implies eq. 3.xi by the mean value property for $\operatorname{Re} h$. Conversely the boundary function $\log \Xi_\nu^f$ is real analytic. Let H be holomorphic near the closed disc, with real part equal to this boundary function and $H(0)$ equal to its real circle mean M. Such an H is obtained by its Fourier series. Then $h = (H - M)/(2N_f)$ gives the circle bound with $C = e^M$. □

3.2. **The two parts of the vacuum energy.** Write $e_\pm = (1 \pm \tau_3)/2$. A split source is a pair $J_1 = e_+ J_1 e_+$, $J_2 = e_- J_2 e_-$. For a function F of the source set

$$D^{\mathrm{mix}}F(J_1, J_2) = F(J_1 + J_2) - F(J_1) - F(J_2) + F(0). \tag{3.xii}$$

The vacuum energy $E_K(0, J)$ is the energy defined by the source construction in [QCDM1, Prop. 10.6]; put $E_K^{\mathrm{mix}} = D^{\mathrm{mix}}E_K(0, \cdot)$.

Proposition 3.4 (Mixed vacuum energy)**.** For split sources with $\|J_b\|_\infty \le \rho_*$,

$$|E_K^{\mathrm{mix}}(J_1, J_2)| \le C_{\mathrm{mix}} \ell^3. \tag{3.xiii}$$

More precisely, set

$$\lambda_j = L^{j-K}, \qquad n_j = \ell^3 \lambda_j^{-3}, \qquad \mathfrak{l}_k = \mathsf{t}_k^{-1/2} (\log \mathsf{t}_k)^{-p_0}, \qquad \mathsf{t}_k = L^{K-k} \mathsf{b},$$

and retain $\phi_k = \phi(x_k)$ from the source construction. The energy contribution at level j satisfies, for $\lambda_j \|J_b\|_\infty < r_M$,

$$\left| \sum_X D^{\mathrm{mix}} e_j(X; 0, \cdot)(J_1, J_2) \right| \le C_{\mathrm{gm3}} M^{-3} n_j \left(\phi_{j-1}^3 + \phi_{j-1}^2 e^{-c_{\mathsf{t}} \ell / (2\lambda_j)} + \mathfrak{l}_{j-1}^2 \right) \frac{\lambda_j^2 \|J_1\|_\infty \|J_2\|_\infty}{r_M^2}. \tag{3.xiv}$$

The bound is on the sum of the energy contributions at a level. ■

Its proof occupies section 4. The three terms have different origins. Independent colour averages remove the product of two linear gauge vertices. The remaining terms with fermion legs have three powers of ϕ, apart from a term caused by polymers crossing a torus seam. Two scalar vertices need not cancel; their contribution is controlled by $\mathfrak{l}^2$.

Proof of Thm. B. For $J = e_+ A + e_- B$, block diagonalization gives

$$\frac{\det(D + m_0 + a\hat{J})}{\det(D + m_0)} = \det(1 + \mathsf{G}_+ \hat{A}) \det(1 + \mathsf{G}_- \hat{B}). \tag{3.xv}$$

At $A = zf1$, $B = \bar{z} f1$, integration gives

$$\Xi_{\nu_K}^f(z) = \frac{Z[0, -e_+ z f1 - e_- \bar{z} f1]}{Z[0, 0]}.$$

The partition bounds in [QCDM1, Prop. 10.14] bound this quotient by the exponential of

$$\mathrm{Re}(E_K(0, -e_+ z f1 - e_- \bar{z} f1) - E_K(0, 0)) + c_V(g) \ell^3.$$

The two pure-source energies are complex conjugates. Their sum is $2N_f \,\mathrm{Re}\, h_K(z)$, with h_K as in eq. 3.iii; the mixed part is bounded by Prop. 3.4. This proves the fixed-torus assertion.

For the local assertion place the support in a cube of side $L^{i(R)}$, at distance at least one from its boundary. The chessboard estimate in [QCDM1, proof of Prop. 10.17] reduces the source quotient to the Nth root of the quotient for the reflected source on the whole board, where $\ell^3 = N L^{3i(R)}$. Reflection alternates the source z with its conjugate. Thus the upper species source has the form $z f_{\mathrm{even}} + \bar{z} f_{\mathrm{odd}}$, and the lower source is its conjugate. Its mixed energy is bounded by $C_{\mathrm{mix}} \ell^3$. The comparison of the energy of separated source copies with the sum of their energies, by [QCDM1, Prop. 10.16 and equation (53)] at separation one, costs at most $2NC'_{\mathscr{C}}(1 + R^3)$. The pure-source terms, after conjugating the odd copies, give twice the real part of a holomorphic function which vanishes at zero. Taking the Nth root gives eq. 3.iv.

At fixed K and fixed support the determinants are polynomials of a common finite degree. Torus convergence of their coefficients follows from the convergence of the local Gaussian functionals in [QCDM1, Prop. 2.5]; their uniform resolvent bounds also give uniform convergence on the circle. Since $\rho_* \le m_-/2$, the determinants are nonzero there and their squared absolute values have positive expectations. The logarithms of these expectations therefore converge uniformly. Pass to the limit in eq. 3.xi and apply Lem. 3.3. The variance assertion now follows from Prop. 3.1, after dividing u by its supremum norm. □

3.3. **Loops with two or more propagators.** Some longer loops require less information about the energy. A flavour cycle forces the first mixed source monomial to have degree at least four. The ordinary energy estimate is already summable at this degree.

Theorem C. Suppose $2 \le k \le N_f$, and let $f_1, \dots, f_k$ be real bounded functions on the torus. Set

$$J_+ = \sum_{i=1}^{k} f_i E_{i,i+1}, \qquad E_{k,k+1} = E_{k,1}, \qquad \lambda_{\mathsf{q}} = \operatorname{Tr}(\mathsf{G}_+ \hat{f}_1 \cdots \mathsf{G}_+ \hat{f}_k).$$

If $\rho \max_i \|f_i\|_\infty \le \rho_*$, there is a holomorphic $\mathcal{F}$, with $\mathcal{F}(0) = 0$, such that

$$\mathsf{E}_{\nu_K} |\det(1 + z\mathsf{G}_+ \hat{J}_+)|^2 \le e^{(c_V(g)+C_4)\ell^3} |e^{\mathcal{F}(z)}|^2, \qquad |z| \le \rho.$$

Consequently

$$\operatorname{Var}_{\nu_K} \lambda_{\mathsf{q}} \le \rho^{-2k} e^{(c_V(g)+C_4)\ell^3}.$$

If $N_f \ge 2$, f, h are real and bounded, and $\rho \|f\|_\infty \|h\|_\infty \le \rho_*^2$, the same conclusion holds for

$$\Upsilon(\zeta) = \mathsf{E}_{\nu_K} |\det(1 - \zeta \mathsf{G}_+ \hat{f} \mathsf{G}_- \hat{h})|^2, \quad |\zeta| \le \rho,$$

with a holomorphic factor, and

$$\operatorname{Var}_{\nu_K} \operatorname{Tr}(\mathsf{G}_+ \hat{f} \mathsf{G}_- \hat{h}) \le \rho^{-2} e^{(c_V(g)+C_4)\ell^3}.$$

One can take

$$C_4 = C_E C_{\mathfrak{C}} M^{-3} (2L)^{-4} \frac{L}{L-1} \left(1 - (2L)^{-2}\right)^{-2}.$$

■□

Proof. Give the k upper-species edges separate parameters z_i, and the lower-species edges separate parameters w_i. Diagonal flavour rotations leave the determinants unchanged and multiply the edge parameters by phases u_i with $\prod_i u_i = 1$. Average the energy bound over this phase group in each species. The surviving monomials are powers of $\zeta = \prod_i z_i$ and $\xi = \prod_i w_i$. The terms depending only on ζ or only on ξ form the holomorphic factor. Every remaining monomial has a positive power of each and hence at least $2k \ge 4$ source factors.

The energy coefficients in [QCDM1, Prop. 10.6] are holomorphic for $\lambda_j \|J\| < r_M$, with polymer bound $C_{\mathfrak{C}} e^{-\kappa d_j(X)}$, with $\sum_X e^{-\kappa d_j(X)} \le C_E n_j M^{-3}$. Cauchy's estimate, at source size $\rho_* \le r_M/(2L)$, bounds the mixed sum at level j by

$$C_E C_{\mathfrak{C}} M^{-3} n_j (2L)^{-4} \lambda_j^4 \left(1 - (2L)^{-2}\right)^{-2}.$$

Since $\sum_{j=1}^{K} n_j \lambda_j^4 \le \ell^3 L/(L-1)$, its sum is at most $C_4 \ell^3$. The geometric mean of the source bounds gives the asserted determinant bound. The flavour trace vanishes until the cycle closes, so

$$\det(1 + z\mathsf{G}_+ \hat{J}_+) = 1 + (-1)^{k-1} z^k \lambda_{\mathsf{q}} + O(z^{2k}).$$

Use the coefficient of z^k in eq. 3.viii; centering can only decrease its squared norm.

For the second assertion use the two-flavour source

$$(zf\tau_+ + z'h\tau_-) \otimes E_{11} + (wf\tau_+ + w'h\tau_-) \otimes E_{22}.$$

The Schur determinant identity gives factors $\det(1 - zz'\mathsf{G}_+ \hat{f} \mathsf{G}_- \hat{h})$ and its conjugate when $(w, w') = (\bar{z}, \bar{z}')$. Phase averaging leaves only powers of zz' and ww'; a mixed term again has degree at least four. For $\zeta = zz'$ choose $|z| \|f\|_\infty = |z'| \|h\|_\infty = \sqrt{|\zeta| \|f\|_\infty \|h\|_\infty}$, which stays in the source domain. The same energy sum applies. The coefficient of ζ is the negative of the stated trace, so eq. 3.viii gives its variance bound. □

3.4. **Four propagators.** When $N_f \geq 2$, the four distinct species $(-,1),(+,1),(-,2),(+,2)$ give a source whose determinant detects a fourth Schatten norm. Averaging its four phases removes all source monomials of degree less than four. The remaining vacuum-energy terms are summable even when all four supports coincide.

Let E_l°, $1 \leq l \leq 4$, carry the lth species in this list to the next, cyclically, and put

$$J_Q(z) = 1_Q \sum_{l=1}^{4} z_l E_l^\circ, \qquad T_Q = 1_Q \mathsf{G}_+ 1_Q, \qquad G_0 = \{u \in \mathbb{T}^4 : u_1u_2u_3u_4 = 1\}. \tag{3.xvi}$$

The four images are orthogonal, so $\|J_Q(z)\|_\infty \leq \max_l |z_l|$. The same is true after conjugation by τ_2. For $\lambda_j = L^{j-K}$ define

$$C_{\mathrm{sh}} = \sum_{t \geq 0} (2t+3)^3 e^{-\kappa_\sigma M t},$$
$$C_{4\mathrm{c}} = 3584 (2L)^{-4} (1-(2L)^{-4})^{-1} C_{\mathfrak{E}} C_{\mathrm{cnt}} C_{\mathrm{sh}}.$$

Here C_{cnt} bounds the sum of $e^{-\kappa d_j(X)/2}$ over polymers through a cube; $C_{\mathfrak{E}}$ and κ_σ are the source-energy constants of [QCDM1, Prop. 10.6].

Lemma 3.5 (Averaging the four phases)**.** Let $N_f \geq 2$, $1 \leq k \leq K$, $0 < \sigma \leq \lambda_k$, and let S have physical diameter at most $\sqrt{3}\sigma$. For either $E_l' = E_l^\circ$ for all l or $E_l' = \tau_2 E_l^\circ \tau_2$ for all l, and $|z_l| \leq r_J/(2\lambda_k)$,

$$\left| \int_{G_0} \left[E_k\left(0, 1_S \sum_l u_l z_l E_l'\right) - E_k(0,0) \right] du \right| \leq C_{4\mathrm{c}}.$$

■

Proof. Write the energy increment as $\sum_{j \leq k} \sum_X F_{j,X}(z)$. It is holomorphic for $\max_l |z_l| < R_j = r_M/\lambda_j$, vanishes at zero, and satisfies

$$|F_{j,X}(z)| \leq B_{j,X} := C_{\mathfrak{E}} e^{-\kappa d_j(X)} e^{-\kappa_\sigma \operatorname{dist}(X,S)/L^j},$$

where the distance in this formula is in lattice units. The average over G_0 retains precisely the monomials $(z_1z_2z_3z_4)^n$. Their coefficients have modulus at most $B_{j,X} R_j^{-4n}$. Since $r_J = r_M/L$,

$$\left| \int_{G_0} F_{j,X}(uz)\, du \right| \leq \frac{(2L)^{-4}}{1-(2L)^{-4}} \left(\frac{\lambda_j}{\lambda_k}\right)^4 B_{j,X}.$$

Assign each polymer to a cube nearest S. The polymer sum at that cube is at most C_{cnt}. The cubes at distance between tML^j and $(t+1)ML^j$ are at most $(2t+3)^3 \#_j^+(S)$ in number. Therefore

$$\sum_X B_{j,X} \leq C_{\mathfrak{E}} C_{\mathrm{cnt}} C_{\mathrm{sh}} \#_j^+(S), \qquad \#_j^+(S) \leq 256 \left(1 + 6(\sigma/\lambda_j)^3\right).$$

The last count is the cube-count estimate used in [QCDM1, equation (53)], applied to a set of the stated diameter. Both $\sum_{j \leq k} (\lambda_j/\lambda_k)^4$ and $\sum_{j \leq k} \lambda_j/\lambda_k$ are at most two. Since $\sigma \leq \lambda_k$, the sum of the cube-count factors times $(\lambda_j/\lambda_k)^4$ is at most $256(2+12) = 3584$. This proves the lemma. □

Set

$$\ell_g(\sigma) = 1 + \log(X_g/\sigma + \bar{B}), \qquad X_g = L(g^{-2} + B_\sharp),$$
$$p_4 = \frac{4}{3} + \frac{64 C_{\mathscr{C}}}{\log L}, \qquad C_{\mathrm{S4}} = \left(\frac{8L^2}{r_J}\right)^4 e^{C_{4\mathrm{c}} + C + 2 + C_P + 2C_{\mathscr{C}}^\flat + 384 C_{\mathscr{C}}}.$$

The constants $C_{\mathscr{C}}$ and $C_{\mathscr{C}}^\flat$ are those of the separated-source counts and the chessboard estimate at an intermediate level in [QCDM1, Props. 10.16 and 10.18]. Let $\sigma_0(g) \in (0,1]$ be the largest number

such that $4L^2\sigma_0(g)\ell_g(\sigma_0(g))^{1/3} \le 1$. This is well defined: $\sigma\ell_g(\sigma)^{1/3}$ tends to zero as $\sigma \downarrow 0$, and its derivative is positive, since $\sigma\ell_g'(\sigma) \ge -1$ and $\ell_g(\sigma) > 1$.

Theorem D (Fourth Schatten moment in a cube)**.** Suppose $N_f \ge 2$, with the source and chessboard estimates of [QCDM1, Props. 10.15 and 10.18]. At every admitted datum, if $a_K \le \sigma \le \sigma_0(g)$ and Q is contained in a cube of physical side σ, then

$$\mathsf{E}_{\nu_K}\|1_Q\mathsf{G}_+1_Q\|_4^4 \le C_{\mathrm{S4}}\sigma^4\ell_g(\sigma)^{p_4}.$$

The constants C_{S4} and p_4 are independent of $g, \ell, K, \mathsf{b}, \mathsf{m}, \sigma, Q$. ■□

Proof. Choose the first level i_0 with $\lambda_{i_0} \ge 4\sigma$. A chessboard cube of physical side λ_{i_0} can be centered at a nearest lattice site to the center of Q. Its distance from Q to its walls is at least

$$\tfrac{1}{2}(\lambda_{i_0} - \sigma - a_K) \ge \delta := \lambda_{i_0}/4 \ge \sigma.$$

Choose the least integer n with $L^{3n} \ge \ell_g(\sigma)$ and put $k = i_0 + n$. Then

$$\lambda_k < 4L^2\sigma\ell_g(\sigma)^{1/3} \le 1, \qquad |\square|/\lambda_k^3 = L^{-3n}.$$

In particular $k \le K-1$, so all the required intermediate levels exist. Set $A = r_J/(2\lambda_k)$, $z = (-A, A, A, A)$, and use the source $s_u = (0, J_Q(uz))$, $u \in G_0$. It lies in the source domain at level k.

Eliminating three species blocks by Schur complements gives

$$\det(1 - a\mathbb{G}\hat{J}_Q(y)) = \det(1 - y_1y_2y_3y_4(T_Q^*T_Q)^2).$$

Here one uses $1_Q^2 = 1_Q$, $\mathsf{G}_- = \mathsf{G}_+^*$, and $\det(1-XY) = \det(1-YX)$. For $y = uz$ the product is $-A^4$. Thus the positive number

$$\frac{Z[s_u]}{Z[0,0]} = \mathsf{E}_{\nu_K}\det(1 + A^4(T_Q^*T_Q)^2) \ge 1 + A^4\mathsf{E}_{\nu_K}\|T_Q\|_4^4$$

is independent of u.

The chessboard estimate at level k bounds this quotient by

$$\exp\left(\operatorname{Re}\lambda_k^{\mathfrak{b}}(s_u) + (C + 2 + C_P(1 + \log x_k))L^{-3n} + \mathscr{C}_k(Q;\delta)\right).$$

The averaged energy $\lambda_k^{\mathfrak{b}}$ is the mean of $E_k(\mathsf{r}_{\mathrm{lin}}s_u) - E_k(0,0)$ over the reflected copies. Each reflected source has matrices E_l° or $\tau_2E_l^\circ\tau_2$. Hence Lem. 3.5 bounds the phase average of its energy by $C_{4\mathrm{c}}$. Moreover $x_k \le X_g/\sigma + \bar{B}$, so the volume term is at most $C + 2 + C_P$.

It remains to bound the separation count

$$\mathscr{C}_k(Q;\delta) = C_{\mathscr{C}}\sum_{j\le k}\#_j^+(Q)e^{-\kappa_E\delta/\lambda_j}.$$

The levels $j < i_0$, where $\lambda_j \le \delta$, contribute at most $2C_{\mathscr{C}}^{\flat}$ by the intermediate-level count. For $i_0 \le j \le k$, $M\lambda_j \ge 4\sigma$ and $\#_j^+(Q) \le 192$. Since $n < 1 + \log\ell_g(\sigma)/(3\log L)$,

$$\mathscr{C}_k(Q;\delta) \le 2C_{\mathscr{C}}^{\flat} + 384C_{\mathscr{C}} + \frac{64C_{\mathscr{C}}}{\log L}\log\ell_g(\sigma).$$

Average the logarithm of the positive quotient bound over G_0 and combine these estimates. Finally,

$$A^{-4} < \left(\frac{8L^2}{r_J}\right)^4\sigma^4\ell_g(\sigma)^{4/3},$$

which yields the stated constant and exponent. □

3.5. **Joining loops and products of bilinears.** Throughout this subsection, test functions are bounded and have finite support at fixed lattice spacing. This includes the restrictions of bounded compactly supported continuum functions to the lattice.

For nonnegative functions f, h, define

$$\Pi_U(f,h) = \mathrm{Tr}(\mathsf{G}_+ \hat{f} \mathsf{G}_- \hat{h}) = \|\hat{h}^{1/2} \mathsf{G}_+ \hat{f}^{1/2}\|_2^2. \tag{3.xvii}$$

The subscript on the norm denotes the Schatten norm. The charge-conjugation identity for the two-colour propagator gives $\|\mathsf{G}_+(v,v')\|_{\mathrm{HS}} = \|\mathsf{G}_+(v',v)\|_{\mathrm{HS}}$; thus $\Pi_U(f,h) = \Pi_U(h,f)$. It is nonnegative and increases with either function. Put

$$\theta_{\mathrm{jl}} = 3N_f \varrho_J^2,$$
$$C_{\mathrm{jl}}(R,d) = \exp\left[c_V(g) L^{3i(R)} + C'_{\mathscr{C}}(1+R^3) + 2C'_{\mathscr{C}}\left(1 + \log_L^+ \frac{1}{d} + (R/d)^3\right)\right].$$

Here $\log_L^+ x = \max\{0, \log_L x\}$, and the remaining constants are those of Thm. B. The source ceiling includes $2\varrho_J \le m_-/2$.

Lemma 3.6 (Exponential moments of a joining loop)**.** Let ν be one of the measures in Thm. B(ii). Suppose $f, h \ge 0$ are bounded, with supports in one ball of radius R, at distance at least $d > 0$ from each other. If neither is zero, then

$$\mathsf{E}_\nu \exp\left[\theta_{\mathrm{jl}} \frac{\Pi_U(f,h)}{\|f\|_\infty \|h\|_\infty}\right] \le C_{\mathrm{jl}}(R,d).$$

For every $q > 0$, including when either function is zero,

$$\mathsf{E}_\nu \Pi_U(f,h)^q \le \left(\frac{q}{e\theta_{\mathrm{jl}}}\right)^q C_{\mathrm{jl}}(R,d) (\|f\|_\infty \|h\|_\infty)^q.$$

■

Proof. Write $\tau_\pm = (\tau_1 \pm i\tau_2)/2$ and $J(z) = z_1 f \tau_+ + z_2 h \tau_-$, with the identity on flavour. The two off-diagonal species blocks give the determinant identity

$$\det(1 - a\mathbb{G}\hat{J}(z)) = \det(1 - z_1 z_2 T)^{N_f}, \qquad T = \mathsf{G}_+ \hat{f} \mathsf{G}_- \hat{h}.$$

The nonzero eigenvalues of T are those of B^*B, where $B = \hat{h}^{1/2} \mathsf{G}_+ \hat{f}^{1/2}$. Set $\rho_1 = 2\varrho_J/\|f\|_\infty$, $\rho_2 = 2\varrho_J/\|h\|_\infty$, and $t = \rho_1\rho_2$. Then $t\|B\|^2 \le 4\varrho_J^2/\mathsf{m}^2 \le 1/4$. Since $\log(1+x) \ge 3x/4$ on $[0, 1/4]$,

$$\det(1+tT)^{N_f} \ge \exp\left(\frac{3}{4} N_f t \,\mathrm{Tr}\, T\right) = \exp\left(\theta_{\mathrm{jl}} \frac{\Pi_U(f,h)}{\|f\|_\infty \|h\|_\infty}\right). \tag{3.xviii}$$

First work on a torus. The source quotient $\mathcal{Q}(z) = Z[0, J(z)]/Z[0,0]$ depends only on $z_1 z_2$. The chessboard estimate used in the proof of Thm. B, with one occupied cube of side $L^{i(R)}$, gives

$$|\mathcal{Q}(z)| \le \exp\left(\mathrm{Re}\, \lambda^{\mathfrak{b}}(J(z)) + c_V(g) L^{3i(R)} + C'_{\mathscr{C}}(1+R^3)\right), \qquad |z_i| \le \rho_i.$$

Here $\lambda^{\mathfrak{b}}$ is the average of the vacuum-energy increments over the reflected copies of the occupied cube. The estimate applies since $\|z_1\tau_+ + z_2\tau_-\| = \max\{|z_1|, |z_2|\}$ and $\|J(z)\|_\infty \le 2\varrho_J$. The separated-source energy estimate [QCDM1, Prop. 10.16 and equation (53)] gives

$$\begin{aligned}\mathrm{Re}\, \lambda^{\mathfrak{b}}(J(z)) \le{}& \mathrm{Re}\, \lambda^{\mathfrak{b}}(z_1 f \tau_+) + \mathrm{Re}\, \lambda^{\mathfrak{b}}(z_2 h \tau_-) \\ &+ 2C'_{\mathscr{C}}\left(1 + \log_L^+ \frac{1}{d} + (R/d)^3\right).\end{aligned}$$

Each reflection preserves the distance between the two supports. Each of the two pure-source terms is holomorphic in its parameter on a disc containing $|z_i| \le \rho_i$, and vanishes at zero. Average over $(z_1, z_2) = (u\rho_1, -\bar{u}\rho_2)$, $|u| = 1$. Their real parts have mean zero, whereas

$$\mathcal{Q}(u\rho_1, -\bar{u}\rho_2) = \mathsf{E}_{\nu_K} \det(1+tT)^{N_f} \ge 1$$

is constant in u. Taking logarithms before this average proves $\mathcal{Q}(\rho_1,-\rho_2)\le C_{\mathrm{jl}}(R,d)$. Now use eq. 3.xviii.

At fixed spacing, the source exponential is a local Grassmann element. Its expectations converge along the tori defining $\nu\in\mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$, by [QCDM1, Prop. 2.5]. Thus the same determinant bound holds for ν. The pointwise inequality eq. 3.xviii proves the exponential bound there. The moment bound follows from $x^q\le (q/(e\vartheta))^q e^{\vartheta x}$, with $\vartheta=\theta_{\mathrm{jl}}/(\|f\|_\infty\|h\|_\infty)$. □

Write h for conditional fermionic expectation at U, and $\mathbb{G}_a=a\mathbb{G}$. For a cycle $\mathsf{c}=(i_1\cdots i_k)$ define

$$w_{\mathsf{c}}(U)=-\operatorname{Tr}\left(\mathbb{G}_a(\hat f_{i_1}\otimes\Gamma_{i_1})\cdots\mathbb{G}_a(\hat f_{i_k}\otimes\Gamma_{i_k})\right).$$

The matrices Γ_i act on species and flavour; the trace here includes those variables.

Lemma 3.7 (Conditional products)**.** Let $m_0>0$, f_i be real bounded functions of finite support, and $c_i\in\mathbb{C}$. Set $X_I=\prod_{i\in I}(\Phi_{\Gamma_i}(f_i)-c_i)$. Then

$$\mathsf{h}[X_{[n]}]=\sum_{\pi\in S_n}\prod_{i:\pi(i)=i}(w_{(i)}-c_i)\prod_{\substack{\mathsf{c}\in\mathrm{cyc}(\pi)\\ |\mathsf{c}|\ge 2}}w_{\mathsf{c}}.$$

If $\alpha_\Gamma=\operatorname{tr}((e_+\otimes 1)\Gamma)$ and $\beta_\Gamma=\operatorname{tr}((e_-\otimes 1)\Gamma)$, then

$$w_{(i)}=-\alpha_{\Gamma_i}\mathsf{L}(f_i)-\beta_{\Gamma_i}\overline{\mathsf{L}(f_i)},\qquad |\alpha_\Gamma|,|\beta_\Gamma|\le N_f\|\Gamma\|.$$

For a cycle of length $k\ge 2$, with $i_0=i_k$,

$$|w_{\mathsf{c}}|\le\prod_{r=1}^{k}\|\Gamma_{i_r}\|\big(2N_f\Pi_U(|f_{i_{r-1}}|,|f_{i_r}|)\big)^{1/2}. \tag{3.xix}$$

For a partition $[n]=I\sqcup I'$, the difference $\mathsf{h}[X_{[n]}]-\mathsf{h}[X_I]\mathsf{h}[X_{I'}]$ is the sum over π with $\pi(I)\ne I$. Every such permutation has at least two edges joining I to I'. ■

Proof. Wick's rule groups a fermionic contraction into permutation cycles, with one minus sign for each cycle. Expand the factors $-c_i$. For a fixed permutation all moved indices must be present; summing over the remaining indices replaces each fixed-point factor by $w_{(i)}-c_i$. Tracing the two species blocks gives the formula for $w_{(i)}$.

Factor $\hat f_i=|\hat f_i|^{1/2}\widehat{\operatorname{sgn} f_i}|\hat f_i|^{1/2}$ and use the cyclic trace. A cycle is the trace of a product B_rW_r, where

$$B_r=|\hat f_{i_{r-1}}|^{1/2}\mathbb{G}_a|\hat f_{i_r}|^{1/2},\qquad W_r=\widehat{\operatorname{sgn} f_{i_r}}\otimes\Gamma_{i_r}.$$

Schatten Hölder gives $|\operatorname{Tr}\prod_r B_rW_r|\le\prod_r\|B_rW_r\|_k\le\prod_r\|B_r\|_2\|\Gamma_{i_r}\|$. The two species blocks, adjoint to each other, and the kernel symmetry in eq. 3.xvii give $\|B_r\|_2^2=2N_f\Pi_U(|f_{i_{r-1}}|,|f_{i_r}|)$. This proves eq. 3.xix. Finally, permutations preserving I give exactly the product of the two separate expectations. A cycle meeting both sets must cross from one to the other and back. □

Put

$$C_{\mathrm{W}}(p)=\frac{4p}{e\varrho_J},\qquad C_{\mathrm{mb}}(R,d)=\max\{1+C(R),C_{\mathrm{jl}}(R,d),C_{\mathrm{jl}}(R+3d,3d/4)\}.$$

The third term will also control cycles joining two distant collections of supports. The source ceiling gives $C_{\mathrm{W}}(p)\ge 1$ for $p\ge 1$.

Proposition 3.8 (Moments of conditional products)**.** Let ν be as in Thm. B(ii), and let the real bounded functions $f_1,\dots,f_n$ have supports in one ball of radius R, pairwise at distance at least $d>0$. Set $c_i=\omega_\nu(\Phi_{\Gamma_i}(f_i))$. For $q\ge 1$,

$$\left\|\mathsf{h}\left[\prod_{i=1}^{n}(\Phi_{\Gamma_i}(f_i)-c_i)\right]\right\|_{L^q(\nu)}\le n!\,C_{\mathrm{W}}(nq)^n C_{\mathrm{mb}}(R,d)^{1/q}\prod_i\|\Gamma_i\|\|f_i\|_\infty.$$

■

Proof. Apply Lem. 3.7. A fixed-point factor is bounded by $2N_f\|\Gamma_i\|\|\mathsf{L}(f_i)-\mathsf{E}_\nu\mathsf{L}(f_i)|$. At $p=nq$, its L^p norm is at most $C_{\mathrm{W}}(p)(1+C(R))^{1/p}\|\Gamma_i\|\|f_i\|_\infty$ by Lem. 3.2. A moved-index factor is bounded by $\|\Gamma_i\|(2N_f\Pi_U(|f_{\pi^{-1}(i)}|,|f_i|))^{1/2}$. By Lem. 3.6, its L^p norm is at most

$$\frac{1}{\varrho_J}\sqrt{\frac{p}{3e}}\,C_{\mathrm{jl}}(R,d)^{1/p}\|\Gamma_i\|(\|f_{\pi^{-1}(i)}\|_\infty\|f_i\|_\infty)^{1/2}.$$

Here $\varrho_J^{-1}\sqrt{p/(3e)}\le C_{\mathrm{W}}(p)$. Apply Hölder to the n factors of each permutation. Around each cycle the square roots of the function norms multiply to their full product. The powers of C_{mb} sum to $1/q$. There are $n!$ permutations. □

Corollary 3.9 (Variance at one site)**.** For $\nu\in\mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$, let $z(x;U)=\operatorname{tr}G^+_U(x,x)$, where $G^+_U=(D_+[U]+m_0)^{-1}$ and the trace includes the places over x, spin and colour. Then

$$a_K^{-1}\operatorname{Var}_\nu z(0)\le\rho_*^{-2}\min\{C(1)^{1/N_f},(C(1)-1)/N_f^2\}.$$

■□

Proof. Let $Q=a_K\mathbb{Z}^3\cap[0,1)^3$, which contains a_K^{-3} sites. For independent signs $\epsilon_x\in\{-1,1\}$ put $u_\epsilon=\sum_{x\in Q}\epsilon_x 1_{\{x\}}$. The cube lies in a ball of radius one, and $\mathsf{L}(u_\epsilon)=a_K\sum_{x\in Q}\epsilon_x z(x)$. Average eq. 3.v over the signs. All off-diagonal covariance terms vanish. Translation invariance makes the remaining sum equal to $a_K^2|Q|\operatorname{Var}_\nu z(0)=a_K^{-1}\operatorname{Var}_\nu z(0)$. □

3.6. **Moments of the action density.** Write $\widetilde{O}_\nu(h)=O(h)-\mathsf{E}_\nu O(h)$. The action density is a function of the links. Its bound below has the explicit factor $a_K^{-3/2}$, which must be retained when it is used in a product. Set

$$\begin{aligned}C_{\mathrm{nc}}&=C_{\mathrm{cnt}}\sum_{t\ge0}(2t+3)^3e^{-\kappa_\sigma Mt/2},\\ C_\lambda(R)&=256C_{\mathfrak{E}}C_{\mathrm{nc}}(1+16R^3),\\ C_O(R)&=\tfrac12C_\lambda(R)+c_V(g)L^{3i(R)}+C'_{\mathscr{E}}(1+R^3),\\ C_{\mathrm{ad}}(p,R)&=\frac{p}{e\varrho}\big(4e^{2C_O(R)}\big)^{1/p}.\end{aligned}$$

Here ϱ is the gauge-source radius: sources $(w,0)$ with $\|w\|_\sharp<2\varrho$ belong to $\mathcal{P}$.

Lemma 3.10 (Energy of a source in a ball)**.** If $s\in\mathscr{O}_K$ has support in a ball of radius R, then $|E_K(s)-E_K(0,0)|\le C_\lambda(R)a_K^{-3}$. The same bound holds for its chessboard average $\lambda^{\mathfrak{b}}(s)$ when $s\in\mathcal{P}$ is supported in one cell. ■

Proof. The locality estimate for $e_j(X;s)$, compared with the zero source, gives

$$|e_j(X;s)|\le C_{\mathfrak{E}}e^{-\kappa d_j(X)}e^{-\kappa_\sigma\operatorname{dist}(X,S)/L^j},\qquad S=\operatorname{supp}s.$$

Assign each polymer to a cube nearest S as in Lem. 3.5. The sum is at most $C_{\mathfrak{E}}C_{\mathrm{nc}}\#_j^+(S)$, and $\#_j^+(S)\le256(1+8R^3L^{3(K-j)})$. Summing over the levels gives at most $256(1+16R^3)L^{3K}$, since $K+1\le L^{3K}$ and the geometric sum is at most two. This proves the first claim. Each reflected source has the same norms and support radius, so its average satisfies the same estimate. □

Proposition 3.11 (Centered action-density moments)**.** Let ν be as in Thm. B(ii), and put $a=a_K$. For a nonzero real Lipschitz function h supported in a ball of radius R,

$$\mathsf{E}_\nu\exp\left(\frac{\varrho a^{3/2}}{\|h\|_\sharp}|\widetilde{O}_\nu(h)|\right)\le4e^{2C_O(R)}.$$

Consequently, for $p\ge1$,

$$\|\widetilde{O}_\nu(h)\|_{L^p(\nu)}\le C_{\mathrm{ad}}(p,R)\|h\|_\sharp a^{-3/2}.$$

These constants are independent of the cutoff, torus side, labels, measure and location of the support. ■

Proof. First take $\nu = \nu_K$ and the chessboard cube of Thm. B, containing the ball at distance at least one from its walls. Put $r = 2\varrho/\|h\|_\sharp$ and $\varphi(\varepsilon) = \lambda^{\mathfrak{b}}(\varepsilon h, 0)$. On $|\varepsilon| < r$ this is holomorphic, vanishes at zero, is real for real ε, and has modulus at most $C_\lambda(R)a^{-3}$ by Lem. 3.10. For $c = \varphi'(0) \in \mathbb{R}$, Cauchy's estimate gives

$$|\varphi(\varepsilon) - c\varepsilon| \le 2C_\lambda(R)a^{-3}(|\varepsilon|/r)^2, \qquad |\varepsilon| \le r/2.$$

The one-cell source bound gives, for real ε,

$$\mathsf{E}_{\nu_K} e^{\varepsilon O(h)} \le \exp\bigl(\varphi(\varepsilon) + c_V(g)L^{3i(R)} + C'_{\mathscr{C}}(1 + R^3)\bigr).$$

At $\lambda_0 = ra^{3/2}/2$ the quadratic error is at most $C_\lambda(R)/2$. Adding the bounds at $\pm\lambda_0$ yields $\mathsf{E}_{\nu_K} e^{\lambda_0|O(h)-c|} \le 2e^{C_O(R)}$. Jensen's inequality bounds $\lambda_0|\mathsf{E}_{\nu_K}O(h) - c|$ by $\log(2e^{C_O(R)})$. Centering therefore gives the first claim with $4e^{2C_O(R)}$.

At fixed spacing $O(h)$ is a bounded continuous function of finitely many links. Its mean and the centered exponential converge along the tori defining ν. This proves the same bound for every such infinite-lattice state. Finally use $y^p \le (p/(e\lambda_0))^p e^{\lambda_0 y}$. □

Corollary 3.12 (Products including action densities)**.** Consider n_B centered bilinears whose real bounded test functions have supports in one ball of radius R, at pairwise distance at least $d > 0$, and n_0 centered action densities with real Lipschitz test functions each supported in a ball of radius R. Write their product as X. For $q \ge 1$ and $q' = (n_0 + 1)q$,

$$\|\mathsf{h}[X]\|_{L^q(\nu)} \le n_B! C_{\mathrm{W}}(n_B q')^{n_B} C_{\mathrm{mb}}(R,d)^{1/q'} C_{\mathrm{ad}}(q',R)^{n_0} a_K^{-3n_0/2} \times \prod_{i\in B} \|\Gamma_i\| \|f_i\|_\infty \prod_{i\in 0} \|f_i\|_\sharp.$$

The first three factors are interpreted as one if $n_B = 0$. For a finite combination $F_j = \sum_k c_k X_{k,j}$ in the slab, fix the tuples, test functions and coefficients as j varies; only their lattice restrictions and centerings depend on j. Let $n_0^* = \max_k n_0(X_{k,j})$. Then $\sup_j a_{K_j}^{3n_0^*} \|\mathsf{h}[F_j^-]\|_2^2 < \infty$. ■□

Proof. The action densities factor out of conditional fermionic expectation. Apply Hölder to these n_0 factors and the conditional product of bilinears, all at exponent q', then use Props. 3.8 and 3.11. The final claim follows by the triangle inequality and reflection invariance, using $a_K \le 1$. □

3.7. **Comparing two masses.** A constant source changes the mass at every site. The circle bound therefore compares the gauge measures at two masses. We first control the derivative of the determinant and the density for a short change of mass. Iterating this comparison gives bounds throughout the permitted mass interval, uniformly in the cutoff. If the same observables converge at both masses along one cutoff sequence, their limiting laws are mutually absolutely continuous; see Prop. 3.13.

Keep the chamber parameter $r = 1/2$ and the constants of sections 2 and 4. Fix $0 < g \le \gamma$, $\ell \in \mathfrak{L}(g)$ and $\mathsf{b} \in I_g$. The physical torus T_ℓ remains fixed. At cutoff K, write $a = L^{-K}$ and, for $s > 0$, put

$$F_s(U) = \det(D_+[U] + as), \qquad Z_K(s) = \int e^{-(\mathsf{b}/a)A(U)} |F_s(U)|^{2N_f}\, dU, \tag{3.xx}$$

$$\nu_{K,s}(dU) = Z_K(s)^{-1} e^{-(\mathsf{b}/a)A(U)} |F_s(U)|^{2N_f}\, dU, \tag{3.xxi}$$

$$\mathfrak{l}_s(U) = \partial_s \log|F_s(U)|^2 = 2a\,\mathrm{Re}\,\mathrm{Tr}(D_+[U] + as)^{-1}. \tag{3.xxii}$$

The determinant and trace include chambers, spin and colour, but no flavour variable. We suppress K in F_s and $\mathfrak{l}_s$, and write E_s for expectation under $\nu_{K,s}$. The resolvent bound makes F_s nonzero.

At fixed K these densities are strictly positive and continuous on the compact configuration space, and depend smoothly on $s > 0$.

With ρ_* and $c_V(g)$ from Thm. B, set

$$\begin{aligned} M_g &= \exp\bigl((c_V(g) + C_{\mathrm{mix}})\ell^3\bigr), \\ \mathfrak{X}_\star &= \frac{64 M_g}{e^2 N_f^2 \rho_*^2}, \qquad \mathfrak{X}_n^\sharp = 4^n (1+M_g)^3 \mathfrak{X}_\star \quad (n \geq 1). \end{aligned} \tag{3.xxiii}$$

These constants are independent of K, of $\mathsf{b} \in I_g$ and of the masses in $[m_-, m_+]$. Their dependence on the physical volume enters through M_g.

Theorem E (Moments and changes of mass)**.** For every cutoff $K \geq 1$ the following assertions hold.

(i) Let $t \in [m_-, m_+]$, $s \in \mathbb{R}$, $|s-t| \leq r < \rho_*$ and $p \geq 1/2$. There is a real constant c, independent of U and p, such that

$$\mathsf{E}_t |\mathfrak{l}_s - c|^{2p} \leq \left(\frac{2}{N_f}\right)^{2p} \left(\frac{4\rho_*}{(\rho_* - r)^2}\right)^{2p} \left(\frac{p}{e}\right)^{2p} M_g. \tag{3.xxiv}$$

In particular,

$$\mathrm{Var}_{\nu_{K,t}} \mathfrak{l}_t \leq \mathfrak{X}_\star. \tag{3.xxv}$$

(ii) If $t \in [m_-, m_+]$, $s \in \mathbb{R}$ and $|s-t| \leq \rho_*/9$, then the density $\rho_{s|t} = d\nu_{K,s}/d\nu_{K,t}$ satisfies

$$\mathsf{E}_t \rho_{s|t}^2 \leq (1+M_g)^3, \qquad \mathsf{E}_t \rho_{s|t}^{-2} \leq (1+M_g)^3. \tag{3.xxvi}$$

(iii) Fix $m_- \leq \mathsf{m} < \mathsf{m}' \leq m_+$ and put

$$n_{\mathrm{ch}} = \left\lceil \frac{9(\mathsf{m}' - \mathsf{m})}{\rho_*} \right\rceil. \tag{3.xxvii}$$

For every $s, t \in [\mathsf{m}, \mathsf{m}']$,

$$\mathrm{Var}_{\nu_{K,t}} \mathfrak{l}_s \leq \mathfrak{X}_{n_{\mathrm{ch}}}^\sharp, \tag{3.xxviii}$$

$$|\operatorname{Cov}_{\nu_{K,t}}(\mathfrak{l}_s, \mathfrak{l}_t)| \leq \mathfrak{X}_{n_{\mathrm{ch}}}^\sharp. \tag{3.xxix}$$

For every Borel set E of configurations,

$$\nu_{K,t}(E) \leq (1+M_g)^3 \nu_{K,s}(E)^{2^{-n_{\mathrm{ch}}}}. \tag{3.xxx}$$

■□

Proof. Fix K and $t \in [m_-, m_+]$. Apply Thm. B on T_ℓ with $f = 1$, and denote its holomorphic function by h_t. Since $\rho_* \leq t/2$ and $\rho_* \leq r_M/(2L)$,

$$\Omega_U(z) = N_f \Bigl[\operatorname{Tr}\log\bigl(1 + z a (D_+[U] + at)^{-1}\bigr) - h_t(z)\Bigr]$$

is holomorphic near $|z| \leq \rho_*$ and vanishes at zero. The matrix logarithm is the power series at the identity. The exponential of its trace is F_{t+z}/F_t. The circle bound says

$$\mathsf{E}_t e^{2\operatorname{Re}\Omega_U(z)} \leq M_g, \qquad |z| \leq \rho_*. \tag{3.xxxi}$$

For interior points this follows from the maximum principle applied to $\log \Xi^1_{\nu_{K,t}} - 2N_f \operatorname{Re} h_t$, which is subharmonic by Lem. 3.3. Differentiation at real z gives

$$\mathfrak{l}_{t+z} = \frac{2}{N_f} \operatorname{Re}\Omega_U'(z) + 2\operatorname{Re} h_t'(z). \tag{3.xxxii}$$

Write

$$Y_U^+ = \frac{1}{2\pi} \int_0^{2\pi} \bigl(\operatorname{Re}\Omega_U(\rho_* e^{i\theta})\bigr)^+ \, d\theta.$$

The real part of Ω_U has mean zero on the circle, so the mean of its absolute value is $2Y_U^+$. Schwarz's integral formula, with $w = \rho_* e^{i\theta}$, therefore gives

$$\Omega_U(z) = \frac{1}{2\pi}\int_0^{2\pi} \operatorname{Re}\Omega_U(w)\frac{2z}{w-z}\,d\theta,$$
$$\Omega_U'(z) = \frac{1}{2\pi}\int_0^{2\pi} \operatorname{Re}\Omega_U(w)\frac{2w}{(w-z)^2}\,d\theta.$$

Consequently, for $|z| \le r < \rho_*$,

$$|\Omega_U(z)| \le \frac{4r}{\rho_* - r}Y_U^+, \qquad |\Omega_U'(z)| \le \frac{4\rho_*}{(\rho_*-r)^2}Y_U^+. \tag{3.xxxiii}$$

For $p \ge 1/2$, Jensen's inequality and $(y^+)^{2p} \le (p/e)^{2p}e^{2y}$ give

$$(Y_U^+)^{2p} \le \left(\frac{p}{e}\right)^{2p}\frac{1}{2\pi}\int_0^{2\pi} e^{2\operatorname{Re}\Omega_U(\rho_* e^{i\theta})}\,d\theta.$$

Similarly $e^{2y^+} \le 1 + e^{2y}$ and Jensen's inequality imply

$$e^{2Y_U^+} \le 1 + \frac{1}{2\pi}\int_0^{2\pi} e^{2\operatorname{Re}\Omega_U(\rho_* e^{i\theta})}\,d\theta.$$

Integrating these nonnegative bounds and using eq. 3.xxxi, we obtain

$$\mathsf{E}_t(Y_U^+)^{2p} \le (p/e)^{2p}M_g, \qquad \mathsf{E}_t e^{2Y_U^+} \le 1 + M_g. \tag{3.xxxiv}$$

Take $c = 2\operatorname{Re}h_t'(s-t)$ in eq. 3.xxxii. Equations eq. 3.xxxiii and eq. 3.xxxiv prove eq. 3.xxiv. At $s = t$, take $r = 0$ and $p = 1$ and use the minimizing property of the mean to get eq. 3.xxv.

For (ii), put $z = s - t$. The factor $e^{2N_f \operatorname{Re}h_t(z)}$ is independent of U and cancels from the normalized density:

$$\rho_{s|t}(U) = \frac{|F_s(U)/F_t(U)|^{2N_f}}{\mathsf{E}_t|F_s/F_t|^{2N_f}} = \frac{e^{2\operatorname{Re}\Omega_U(z)}}{\mathsf{E}_t e^{2\operatorname{Re}\Omega_U(z)}}. \tag{3.xxxv}$$

If $|z| \le \rho_*/9$, then $4|z|/(\rho_* - |z|) \le 1/2$, and hence $4|\Omega_U(z)| \le 2Y_U^+$. In particular,

$$\mathsf{E}_t e^{4|\Omega_U(z)|} \le 1 + M_g, \qquad \mathsf{E}_t e^{2|\Omega_U(z)|} \le 1 + M_g.$$

Cauchy–Schwarz, applied to $e^{|\Omega_U(z)|}$ and its reciprocal, gives

$$(1+M_g)^{-1} \le \mathsf{E}_t e^{-2|\Omega_U(z)|} \le \mathsf{E}_t e^{2\operatorname{Re}\Omega_U(z)} \le M_g.$$

Substitution in eq. 3.xxxv yields

$$\mathsf{E}_t\rho_{s|t}^2 \le (1+M_g)^3, \qquad \mathsf{E}_t\rho_{s|t}^{-2} \le (1+M_g)M_g^2 \le (1+M_g)^3.$$

For (iii), first let $s \ne t$, set $n' = \lceil 9|t-s|/\rho_* \rceil \le n_{\mathrm{ch}}$ and divide the segment from s to t into n' equal parts, with endpoints $u_i = s + i(t-s)/n'$. All these masses lie in the admitted interval. For any nonnegative Borel function Y, eq. 3.xxvi gives

$$\mathsf{E}_{u_{i+1}}Y = \mathsf{E}_{u_i}(Y\rho_{u_{i+1}|u_i}) \le (1+M_g)^{3/2}(\mathsf{E}_{u_i}Y^2)^{1/2}.$$

Iterating, the exponents $1/2, 1/4, \dots, 2^{-n'}$ sum to $1 - 2^{-n'}$, so

$$\mathsf{E}_t Y \le (1+M_g)^{3(1-2^{-n'})}(\mathsf{E}_s Y^{2^{n'}})^{2^{-n'}}. \tag{3.xxxvi}$$

With $Y = 1_E$ this proves eq. 3.xxx. For the variance take $Y = (\mathfrak{l}_s - c_s)^2$, where c_s is the constant in (i) at the centre s, and use (i) with $r = 0$ and $p = 2^{n'}$. Then

$$\operatorname{Var}_{\nu_{K,t}}\mathfrak{l}_s \le \mathsf{E}_t Y$$
$$\le (1+M_g)^3\frac{64\,4^{n'}}{e^2N_f^2\rho_*^2}M_g^{2^{-n'}} \le \mathfrak{X}^\sharp_{n_{\mathrm{ch}}}.$$

The last inequality uses $M_g \geq 1$. Cauchy–Schwarz and eq. 3.xxv give eq. 3.xxix. If $s = t$, the variance estimate is already (i), and the event bound follows from $\nu_{K,s}(E) \leq \nu_{K,s}(E)^{2^{-n_{\mathrm{ch}}}}$. □

We now pass from the cutoff measures to any pair of limits formed with the same observables. For probability measures P, Q on one measurable space, write

$$\mathrm{TV}(P,Q) = \sup_E |P(E) - Q(E)|, \qquad \mathrm{Aff}(P,Q) = \int \sqrt{pq}\, dR,$$

where R dominates P, Q and $p = dP/dR$, $q = dQ/dR$. The affinity is independent of R. We use

$$\mathrm{Ent}(P \mid Q) = \int \log \frac{dP}{dQ}\, dP$$

when $P \ll Q$, and assign the value $+\infty$ otherwise.

Proposition 3.13 (Limit laws at two masses)**.** Retain the fixed g, ℓ, b above, and choose $m_- \leq \mathsf{m} < \mathsf{m}' \leq m_+$. Let Y be a Polish space and let f_K be Borel maps from the cutoff configuration spaces to Y. Suppose that, along one sequence $K_j \to \infty$,

$$(f_{K_j})_* \nu_{K_j,\mathsf{m}} \implies P, \qquad (f_{K_j})_* \nu_{K_j,\mathsf{m}'} \implies Q.$$

Put $\Delta = \mathsf{m}' - \mathsf{m}$, let n_{ch} be eq. 3.xxvii, and set

$$x = \frac{N_f}{2} \sqrt{\mathfrak{X}_\star}\, \Delta = \frac{4\sqrt{M_g}}{e\rho_*}\, \Delta, \qquad H = \frac{N_f^2}{2} \mathfrak{X}^\sharp_{n_{\mathrm{ch}}} \Delta^2.$$

Then P and Q are mutually absolutely continuous, and for every Borel set $E \subset Y$,

$$Q(E) \leq (1 + M_g)^3 P(E)^{2^{-n_{\mathrm{ch}}}}, \qquad P(E) \leq (1 + M_g)^3 Q(E)^{2^{-n_{\mathrm{ch}}}}. \tag{3.xxxvii}$$

Moreover,

$$\mathrm{TV}(P,Q) \leq x, \tag{3.xxxviii}$$

$$\mathrm{Ent}(P \mid Q),\ \mathrm{Ent}(Q \mid P) \leq H, \tag{3.xxxix}$$

$$\mathrm{Aff}(P,Q) \geq e^{-H/2}, \tag{3.xl}$$

and $\mathrm{Aff}(P,Q) \geq \cos x$ when $x \leq \pi/2$. ■

Proof. We first prove the estimates before taking the limit. At fixed K, let $p_t = d\nu_{K,t}/dU$. Differentiation of eq. 3.xxi gives its logarithmic derivative,

$$\partial_t \log p_t = N_f(\mathfrak{l}_t - \mathsf{E}_t \mathfrak{l}_t), \qquad \partial_t \mathsf{E}_t F = N_f \,\mathrm{Cov}_{\nu_{K,t}}(F, \mathfrak{l}_t) \tag{3.xli}$$

for a bounded F independent of t. Every differentiation here is justified on a compact positive mass interval by the resolvent bound and compactness of the configuration space.

The functions $p_t^{1/2}$ form a continuously differentiable curve on the unit sphere of $L^2(dU)$, with speed

$$\|\partial_t p_t^{1/2}\|_2 = \frac{N_f}{2} \sqrt{\mathrm{Var}_{\nu_{K,t}} \mathfrak{l}_t}.$$

The angle between its endpoints is at most its length. Their inner product is the affinity, so

$$\arccos \mathrm{Aff}(\nu_{K,\mathsf{m}}, \nu_{K,\mathsf{m}'}) \leq \frac{N_f}{2} \int_{\mathsf{m}}^{\mathsf{m}'} \sqrt{\mathrm{Var}_{\nu_{K,t}} \mathfrak{l}_t}\, dt \leq x. \tag{3.xlii}$$

For any two probability densities, Cauchy–Schwarz gives

$$\mathrm{TV}(P,Q) = \frac{1}{2} \int |\sqrt{p} - \sqrt{q}|(\sqrt{p} + \sqrt{q})\, dR \leq \sqrt{1 - \mathrm{Aff}(P,Q)^2} \leq \arccos \mathrm{Aff}(P,Q).$$

Thus the total variation at the cutoff is at most x as well.

For entropy, integration of the first identity in eq. 3.xli gives

$$\log p_{\mathsf{m}} - \log p_{\mathsf{m}'} = N_f \int_{\mathsf{m}}^{\mathsf{m}'} (\mathsf{E}_s \mathfrak{l}_s - \mathfrak{l}_s)\, ds.$$

Taking its expectation at m and applying the second identity in eq. 3.xli, with s held fixed, yields

$$\mathrm{Ent}(\nu_{K,\mathsf{m}} \mid \nu_{K,\mathsf{m}'}) = N_f^2 \int_{\mathsf{m}}^{\mathsf{m}'} ds \int_{\mathsf{m}}^{s} dt\ \mathrm{Cov}_{\nu_{K,t}}(\mathfrak{l}_s, \mathfrak{l}_t), \tag{3.xliii}$$

$$\mathrm{Ent}(\nu_{K,\mathsf{m}'} \mid \nu_{K,\mathsf{m}}) = N_f^2 \int_{\mathsf{m}}^{\mathsf{m}'} ds \int_{s}^{\mathsf{m}'} dt\ \mathrm{Cov}_{\nu_{K,t}}(\mathfrak{l}_s, \mathfrak{l}_t). \tag{3.xliv}$$

Each triangle has area $\Delta^2/2$. Equation eq. 3.xxix therefore bounds both entropies by H.

We recall the measure-theoretic facts needed to preserve these estimates. Pushforward by a Borel map decreases total variation and entropy and increases affinity. For total variation the supremum is over fewer sets. For affinity, take $R = (P+Q)/2$ and condition the densities p, q on the map: conditional Cauchy–Schwarz gives

$$\mathsf{E}_R(\sqrt{pq} \mid f) \le \sqrt{\mathsf{E}_R(p \mid f)\mathsf{E}_R(q \mid f)}.$$

For entropy use the variational formula

$$\mathrm{Ent}(P \mid Q) = \sup_{v} \left(\int v\, dP - \log \int e^v\, dQ \right). \tag{3.xlv}$$

On a Polish space the supremum may be taken either over bounded Borel functions or over bounded continuous functions; see [DM22, equation (1.1) and the following paragraph]. Using $v \circ f$ in the Borel version proves the pushforward inequality. The continuous version makes entropy lower semicontinuous under simultaneous weak convergence. The same is true for total variation, since

$$\mathrm{TV}(P,Q) = \frac{1}{2} \sup_{\substack{v \in C_b(Y) \\ \|v\|_\infty \le 1}} \left| \int v\, dP - \int v\, dQ \right|.$$

Affinity is upper semicontinuous under that convergence. Indeed,

$$\mathrm{Aff}(P,Q) = \inf_{\substack{v \in C_b(Y) \\ \inf v > 0}} \left(\int v\, dP \right)^{1/2} \left(\int v^{-1}\, dQ \right)^{1/2}. \tag{3.xlvi}$$

Cauchy–Schwarz proves one inequality. For the reverse, truncate $\sqrt{q/p}$ to $[\epsilon, \epsilon^{-1}]$, taking the value ϵ^{-1} where $p = 0$. If the result is v_ϵ, then

$$\int v_\epsilon\, dP \le \mathrm{Aff}(P,Q) + 2\epsilon, \qquad \int v_\epsilon^{-1}\, dQ \le \mathrm{Aff}(P,Q) + 2\epsilon.$$

Approximate v_ϵ in $L^1(P+Q)$ by bounded continuous functions with values in the same interval, using Lusin's theorem and extension from compact sets. Their reciprocals converge in $L^1(P+Q)$ too. Letting ϵ decrease to zero proves eq. 3.xlvi. Its right-hand side is an infimum of continuous functions of (P,Q), as required.

Apply these pushforward and semicontinuity statements to

$$P_j = (f_{K_j})_* \nu_{K_j,\mathsf{m}}, \qquad Q_j = (f_{K_j})_* \nu_{K_j,\mathsf{m}'}.$$

The cutoff estimates give eq. 3.xxxviii, eq. 3.xxxix, and the cosine lower bound. Finite entropy in both orders gives mutual absolute continuity. Jensen's inequality then gives

$$-\log \mathrm{Aff}(P,Q) = -\log \int \left(\frac{dQ}{dP} \right)^{1/2} dP \le \frac{1}{2} \mathrm{Ent}(P \mid Q) \le H/2,$$

which proves eq. 3.xl.

It remains to retain the event comparison. By eq. 3.xxx, with preimages under f_{K_j}, every Borel $E \subset Y$ satisfies

$$Q_j(E) \le (1+M_g)^3 P_j(E)^\alpha, \qquad \alpha = 2^{-n_{\mathrm{ch}}}.$$

For a nonempty closed set $F \subset Y$, choose a compatible metric and put $G_\delta = \{y : \mathrm{dist}(y,F) < \delta\}$ and $F_\delta = \{y : \mathrm{dist}(y,F) \le \delta\}$. The weak-convergence inequalities for open and closed sets give

$$\begin{aligned} Q(F) &\le Q(G_\delta) \le \liminf_j Q_j(G_\delta) \\ &\le (1+M_g)^3 \limsup_j P_j(F_\delta)^\alpha \le (1+M_g)^3 P(F_\delta)^\alpha. \end{aligned}$$

As $\delta \downarrow 0$, $P(F_\delta) \downarrow P(F)$. Inner regularity extends the inequality from closed sets to all Borel sets. Interchanging the masses proves the other inequality in eq. 3.xxxvii. □

4. The mixed source energy

The ordinary source estimate gives one small factor for each source. Applied to the mixed energy, these two factors leave a nonsummable power of the scale. We need additional smallness in the coefficient of the two sources. We prove it in two stages. First, at zero gauge source, the activities with fermion legs are quadratic in ϕ_k. Second, at the trivial gauge configuration, independent colour averages in the two species remove the mixed terms of weighted degree two which have fermion legs. The scalar term left at this degree is bounded by I_k^2.

We recall the scale quantities of [QCDM1, §4]:

$$\lambda_j = L^{j-K}, \qquad \mathsf{t}_j = \lambda_j^{-1}\mathsf{b}, \qquad x_0 = L^K \mathsf{b}, \quad x_{j+1} = L^{-1}x_j - b_{j+1}, \quad |b_{j+1}| \le B_\sharp,$$
$$\phi(t) = t^{-1/2}(1+\log t)^{p_0}, \qquad \phi_j = \phi(x_j), \qquad \delta_j = A_1 \mathsf{t}_j^{-1/2}(\log \mathsf{t}_j)^{p_0}.$$

The fixed exponent is $p_0 \ge 6r_c + 1$. The running and calibration coefficients satisfy $|x_j - \mathsf{t}_j| \le \bar{B}$, and $\phi_i/\phi_j \le (2L^{-1/2})^{j-i}$ for $i \le j$. A level-j polymer is a connected union of blocks of physical side $M\lambda_j$; $d_j(X)$ is the least number of edges in a block tree meeting every block of X. In the formulas below δ_k is the fluctuation radius, h_F is the fixed Grassmann norm parameter, and c_{cl} is the radius on which the cluster expansion converges.

We use the norms, local extraction and source coordinates of [QCDM1, Prop. 10.1 and its proof]. The level-j norm of a polymer family is the supremum, over a level-j cube, of the sum of the Grassmann norms of all its summands containing that cube, weighted by $e^{\kappa d_j(X)}$. Summands include the local quadratic forms introduced by extraction. A pointed family is obtained by retaining a specified fermion leg or specified scalar bond in the localization; its norm uses the same weights. The constant K_μ^{s} bounds the increase of this norm under local mass extraction.

A regular pair $(\mathbf{U}, \mathbf{J})$ consists of the complex connection and gauge-current coordinate in the analytic core of [QCDM1, §4]. This domain includes the bounds for the local gauge potentials and the reduced block towers, as well as those for curvature and current. We use radius α_0 at every level. The gauge-current coordinate $\mathbf{J}$ is distinct from the physical fermion source J below; the trivial pair is $(1,0)$. There is no gauge source. The doubled coordinate domain is

$$2\mathscr{D}_k^\circ = \{\mathfrak{c} = (0, \mathsf{n}, (\sigma_i)_{i\le k}) : L^{k+1}\|\mathsf{n}\| < 2r_M, \|\sigma_i\| < 2B_\sigma \phi_i\}. \tag{4.i}$$

Its inner half is $\mathscr{D}_k^\circ$. The bare coordinate is $\mathsf{n} = -aJ$. The matrices σ_i record the changes in the extracted local mass at the preceding levels. At zero gauge source the coordinate distance appearing below is

$$\|\mathfrak{c} - \mathfrak{c}'\|_{k,y} = \sup_x e^{-\kappa_\sigma |x-y|/L^{k+1}} \max\left\{ \frac{L^{k+1}|\mathsf{n}-\mathsf{n}'|(x)}{r_M}, \max_{i\le k} \frac{|\sigma_i - \sigma_i'|(x)}{B_\sigma \phi_i} \right\},$$

where x, y are in fine lattice units. The matrices σ_i remain free in the first-order domain; the stronger estimate proved below is a conclusion, not an assumption on these coordinates.

The role of these coordinates can be seen in one step of [QCDM1, Prop. 10.1]. At level k, the remaining fields have a quadratic fermion form and local interaction activities. Completing the quadratic form after insertion of the block-average kernel gives, on the variables being integrated,

$$\Psi_k = H_k \Psi_{k+1} + W, \qquad \bar{\Psi}_k = \bar{\Psi}_{k+1} \bar{H}_k + \bar{W}.$$

Here H_k extends the prescribed coarse field to its fermion background, $\bar{H}_k$ is its row twin, and $W, \bar{W}$ are integrated with covariance Γ_k^{-1}. The row twin need not be the adjoint. On a history, this notation suppresses the stored variables, which remain unchanged. Substituting these expressions into the old activities expresses their fermion legs in terms of the new background and the fluctuation. The gauge fluctuation is denoted by B. The background pair $(1, 0)$ fixes the background connection and current; setting $B = 0$ is a further evaluation, not part of the definition of that pair.

The connected cluster expansion for the joined gauge and Berezin integration produces activities $\widetilde{E}^{(k+1)}(X; \mathfrak{c})$ on level-$(k+1)$ polymers. Each is an even polynomial in the remaining fermion fields, with coefficients depending on the background pair and the coordinates $\mathfrak{c}$. Its terms of degree zero contribute to the scalar energy. More precisely, let $\mathfrak{E}_{k+1}(X; \mathfrak{c})$ be their value at the trivial pair, including the explicit normalizer and Gaussian terms. The source-dependent constant removed from the activity is

$$e_{k+1}(X; \mathfrak{c}) = \mathfrak{E}_{k+1}(X; \mathfrak{c}) - \mathfrak{E}_{k+1}(X; 0).$$

Its sum contributes to the source energy. Terms of degree zero may still depend on the gauge background after this constant has been removed.

The quadratic part has a matrix kernel. Its symmetric zeroth moment determines the local mass increment $\nu_{k+1}(\mathfrak{c})$, with $\mu_{k+1} = \nu_{k+1}(0)$ the zero-source increment. The extraction rewrites this local quadratic contribution as a mass term and retains the remaining kernel and its localized correction forms in the activities. This is why the activity norms below also include those forms. For a physical source J, the mass change relative to zero source at level k is $-\lambda_k J + \sigma_k(J)$. The corrections are determined recursively by

$$\sigma_0(J) = 0, \qquad \sigma_{k+1}(J) = L\sigma_k(J) + \nu_{k+1}(\mathfrak{c}_k(J)) - \mu_{k+1}, \qquad \mathfrak{c}_k(J) = (0, -aJ, (\sigma_i(J))_{i \le k}).$$

The factor L rescales the preceding mass correction. The two extractions therefore give the outputs needed here: a scalar source energy and a new local mass coordinate, together with the activities carried to the next step.

We first estimate the step with n and the σ_i independent in eq. 4.i. We then substitute $\mathfrak{c}_k(J)$, as in [QCDM1, Lem. 10.4]. For split sources, restrict an expression by zeroing one species' entries while holding the other entries fixed; for activities with fermion legs, zero that species' generators as well. The mixed part is the full expression minus its two one-species restrictions, plus its restriction with both species removed. Turning off one physical source instead recomputes the entire chain: the other species' corrections σ_i can change as well. The later comparison of source chains estimates this additional change. Thus a bound for the mixed coordinate expression is an input to the bound for the mixed physical-source energy.

4.1. **Quadratic bounds at zero gauge source.**

Proposition 4.1. There are constants C_2^{s}, $B_\nu^{(2)}$ and $B_\sigma^{(2)}$, chosen independently of the admitted data, such that the following holds after decreasing γ. For $1 \le j \le K$, the new activity with fermion legs, including its extracted summands and every pointed family, has norm at most $C_2^{\mathrm{s}} \phi_{j-1}^2$ throughout $2\mathscr{D}_{j-1}^{\circ}$. The increment satisfies

$$|\nu_{k+1}(y; \mathfrak{c}) - \nu_{k+1}(y; \mathfrak{c}')| \le B_\nu^{(2)} \phi_k^2 \|\mathfrak{c} - \mathfrak{c}'\|_{k,y}, \tag{4.ii}$$

for $\mathfrak{c} \in \mathscr{D}_k^\circ$ and $\mathfrak{c}' \in 2\mathscr{D}_k^\circ$. The norm on the right is the weighted coordinate norm of [QCDM1, Prop. 10.1]. Along the source chain,

$$\|\sigma_j(0,J)\| \le B_\sigma^{(2)} \phi_{j-1}^2 \frac{\lambda_j \|J\|}{r_M}, \qquad B_\sigma^{(2)} = \frac{B_\nu^{(2)}}{1-4/L}. \tag{4.iii}$$

■

Proof. We give the additional estimates needed in the proof of [QCDM1, Prop. 10.1]. Write the potential as the sum of the scalar term V_0, the gauge–fermion vertex V_F, the old activity V_{old} with fermion legs, and the quadratic source term V_s. On the field disc,

$$\|V_F\| \le C_V h_F^2 \mathsf{C}_{\mathcal{A}} \delta_k, \quad \|V_0\| \le K_0^{\mathrm{s}} \delta_k, \quad \|V_s\| \le c'_\psi c_{\mathsf{S}} h_F^2 (\bar{B}_M + 2B_\sigma)\phi_k, \quad \delta_k \le \sqrt{2} A_1 \phi_k.$$

At $k=0$ the independent coordinate σ_0 may be nonzero even though its value on the physical source chain is zero. The first-step mass-insertion bound of [QCDM1, Lem. 7.11], applied with $N = \sigma_0$, gives

$$V_s = \mathcal{V}_0^{\mathrm{s}}[\sigma_0], \qquad \|V_s\| \le \tfrac{1}{9} c_\psi c_{\mathsf{S}} h_F^2 \|\sigma_0\|_\infty.$$

Its coefficients are independent of the gauge fluctuation when the background is a formal leg. Hence this bound holds at every fluctuation and is below $(c_\psi + C_G\beta) c_{\mathsf{S}} h_F^2 \|\sigma_0\|_\infty$ for every $\beta \ge 0$, as well as below the bound with c'_ψ. This supplies the mass-origin bound at the first step in the coefficient estimates below.

Here one can take

$$K_0^{\mathrm{s}} = 7K_P + c_{\mathsf{s}} K_{\mathrm{old}}^{\mathrm{s}} + \frac{13}{10} \mathsf{C}_{\mathcal{A}} K_\Lambda^{\mathrm{s}} K_M c_{\mathrm{cl}} / \alpha_0,$$

using the scalar localization constants at radius α_0. The old-activity argument, including its cell sums, gives

$$\|V_{\mathrm{old}}\| \le c'_\psi \Sigma_k, \qquad \Sigma_k = (1 + c_{\mathrm{cs}}) \sum_{i \le k} \omega_{k+1-i} \epsilon_i + C_{0,\sharp}^{\mathrm{cs}} h_F^2 \phi_k^2,$$

where ϵ_i includes the extraction factor K_μ^{s} and the zero-source bound $2C_2\phi_{i-1}^2$. The additional term in $C_{0,\sharp}^{\mathrm{cs}} = C_0^{\mathrm{cs}} + 3L^{1/2} C_{\mathrm{pc}} S_{1/4} c_{\mathsf{S}} \bar{B}_M$ uses $|\mu_i| \le 3\bar{B}_M \phi_i^2$ in the cell calculation. Here μ_i is the zero-source mass increment. Its local mass forms cost $c_{\mathsf{S}} h_F^2 |\mu_i|$ before they are read at a later level. Their sum is bounded by

$$\sum_{i \le k} \omega_{k+1-i} c_{\mathsf{S}} h_F^2 |\mu_i| \le 3 c_{\mathsf{S}} \bar{B}_M h_F^2 \phi_k \sum_{i \le k} \omega_{k+1-i} \phi_i \le 3L^{1/2} C_{\mathrm{pc}} S_{1/4} c_{\mathsf{S}} \bar{B}_M h_F^2 \phi_k^2.$$

The change of a cell moment under the bare matrix mass has an additional factor $L^i \|\mathsf{n}\| / r_M$. Reading that cell at level k costs L^{k-i}, and

$$L^{k-i} \frac{L^i \|\mathsf{n}\|}{r_M} < \frac{2}{L} \quad \text{on } 2\mathscr{D}_k^\circ.$$

Thus its sum is controlled by $h_F^2 \sum_{i \le k} \phi_i^2$, which is at most $h_F^2 \phi_k^2 / (1 - 2L^{-1/2})$. This is the term included in C_0^{cs}. The other cell terms are linear in the old activity norms ϵ_i and give the factor $1 + c_{\mathrm{cs}}$. These are the reasons the old-activity estimate has the quadratic remainder $C_{0,\sharp}^{\mathrm{cs}} h_F^2 \phi_k^2$.

Introduce the four-variable potential

$$V(z) = z_3 V_0(B) + V_F(z_1 B) + z_2 V_{\mathrm{old}}(z_1 B) + z_4 V_s(z_1 B).$$

Take the radii

$$r_1 = \frac{\min\{c_{\mathrm{cl}}/(8C_V h_F^2), \beta_0/2\}}{\mathsf{C}_{\mathcal{A}} \delta_k},$$

$$r_2 = \frac{c_{\mathrm{cl}}}{8c'_\psi \Sigma_k}, \qquad r_3 = \frac{c_{\mathrm{cl}}}{8K_0^{\mathrm{s}} \delta_k}, \qquad r_4 = \frac{c_{\mathrm{cl}}}{8c'_\psi c_{\mathsf{S}} h_F^2 (\bar{B}_M + 2B_\sigma)\phi_k}.$$

Each of the four terms has norm at most $c_{\mathrm{cl}}/8$ on this polydisc. In particular, scaling B by z_1 keeps the fine field in its analytic domain; the extra bound $\beta_0/2$ in r_1 also keeps the changes of the background legs within their Cauchy radius. The cluster expansion of [QCDM1, Lem. 7.5] therefore gives a holomorphic family

$$\widetilde{E}_{\geq 2}(X;z) = \sum_{n\in\mathbb{N}^4} a_n(X) z^n, \qquad \|a_n(X)\| \leq K'_M c_{\mathrm{cl}} \prod_{i=1}^{4} r_i^{-n_i} e^{-\kappa_+ d_{k+1}(X)}.$$

The coefficients take values in a finite Grassmann algebra, so Cauchy's estimate applies in its norm. After summing polymers, K'_M is replaced by K_M.

Assign weights $1,2,1,1$ to z_1,z_2,z_3,z_4. We examine the coefficients before estimating their sum. When $z_1=z_2=z_4=0$ the potential is scalar, so every coefficient involving only z_3 has Grassmann degree zero. When $z_2=z_3=z_4=0$, the substitution $B\mapsto -B$ replaces z_1 by $-z_1$. The zero-source gauge Gaussian, its tests and its local quadratic term are even in B; the decoupled fermion covariance is independent of B. The cluster construction respects this symmetry polymer by polymer. Its coefficient of z_1 is therefore zero.

For the coefficient linear in z_4, put $z_1=z_2=0$. Let $\psi^{\mathrm{bg}} = H_{0,k+1}\Psi_{k+1}$ and $\mathsf{N} = s_k + \sigma_k$, where s_k is the zero-source mass coordinate. At $B=0$ the source vertex is

$$V_s(0) = -L^{-k}\left[\mathsf{S}_1(\mathsf{N}(\psi^{\mathrm{bg}} + H_{0,k}W)) - \mathsf{S}_1(\mathsf{N}\psi^{\mathrm{bg}})\right].$$

Here $\mathsf{S}_1(\mathsf{N}\psi)$ denotes the quadratic mass form with coefficient N. Expanding this difference gives two terms linear in the fluctuation generators and one term quadratic in those generators. The linear terms have zero Berezin mean; the quadratic term is contracted to a scalar. At any power of z_3, a coefficient linear in z_4 has exactly one such vertex, independent of B, beside scalar vertices. Its refined Gaussian integral thus has degree zero, and the subsequent cluster operations preserve that degree. This proves that every coefficient of weighted degree at most one has no part with fermion legs.

The coefficient of z_2 has weight two and must be kept separately: it is the first variation of the old family at $B=0$. For an even potential V with legs, the cluster activity at vV is holomorphic for $|v| < c_{\mathrm{cl}}/\|V\|$ and has bound $(17/16)K'_M c_{\mathrm{cl}} e^{-\kappa_+ d_{k+1}(X)}$. Cauchy's estimate at $v=0$ bounds its first variation by $(17/16)K'_M\|V\|e^{-\kappa_+ d_{k+1}(X)}$. At $V = V_{\mathrm{old}}(0)$ this yields $(17/16)K_M c'_\psi \Sigma_k$ in the polymer norm.

Choose C^{s}_r so that $r_1^{-1}, r_3^{-1}, r_4^{-1} \leq x = C^{\mathrm{s}}_r \phi_k$, and put $y = r_2^{-1} = 8c'_\psi \Sigma_k / c_{\mathrm{cl}}$. For $n_2 = 0$, all remaining coefficients have $n_1+n_3+n_4 \geq 2$. Their Cauchy factors sum to

$$(1-x)^{-3} - 1 - 3x \leq 22x^2, \qquad 0 \leq x \leq \tfrac{1}{2}.$$

The quotient by x^2 has nonnegative coefficients and takes the value 22 at $x = 1/2$. For $n_2 \geq 1$, remove the first-variation coefficient already treated. The remaining sum is at most

$$y\big((1-x)^{-3}(1-y)^{-1} - 1\big) \leq y(28x + 2y), \qquad 0 \leq x, y \leq \tfrac{1}{2},$$

using $(1-x)^{-3} - 1 \leq 14x$ and $(1-y)^{-1} \leq 2$. Together these give

$$\|\widetilde{E}^{(k+1)}_{\geq 2}\|_{k+1} \leq \tfrac{17}{16} K_M c'_\psi \Sigma_k + K_M c_{\mathrm{cl}}(22x^2 + 28xy + 2y^2). \tag{4.iv}$$

For a pointed family, the insertion $(V_\psi)_z$ multiplies each monomial by a number in $[0,1]$ and therefore preserves its degree and does not increase its norm. Replace the part with legs by $tV_\psi + \zeta(V_\psi)_z$, for $|t| < 9/8$ and $|\zeta| < 1$. The pointed activity is

$$\int_0^1 \partial_\zeta \widetilde{E}_{\geq 2}[z_3 V_0, tV_\psi + \zeta(V_\psi)_z]\Big|_{\zeta=0}\, dt.$$

The scalar part costs $c_{\mathrm{cl}}/8$, and the part with legs costs at most $(9/8+1)3c_{\mathrm{cl}}/8$; their sum is $59c_{\mathrm{cl}}/64$. Cauchy's estimate in ζ gives the same Taylor coefficient bound as above. The three

vanishing-coefficient arguments still hold because pointing retains the monomials and commutes with scaling B. Thus eq. 4.iv holds at every pointer, with the same constants.

For completeness, the constants can be chosen in the following order. Put

$$R_a = 8c'_\psi(1+c_{\mathrm{cs}})C_{\mathrm{pc}}S_{1/4}/c_{\mathrm{cl}}, \qquad R_b = 8c'_\psi C^{\mathrm{cs}}_{0,\sharp}h_F^2/c_{\mathrm{cl}},$$

$$C^{\mathrm{s}}_{(2)} = \theta^{\mathrm{s}}_E C^{\mathrm{cs}}_{0,\sharp}h_F^2 + K^{\mathrm{s}}_\mu K_M c_{\mathrm{cl}}(22(C^{\mathrm{s}}_r)^2 + 15R_b), \qquad C^{\mathrm{s}}_2 = \max\{2C^{\mathrm{s}}_{(2)}, 2C_2\}.$$

The source norm parameter h_F is fixed with $4\theta^{\mathrm{s}}_E C_{\mathrm{pc}}S_{1/4}(1+c_{\mathrm{cs}}) \le 1/2$, as permitted in [QCDM1, Prop. 10.20]. Decrease γ so that, with $\phi_* = \phi(\gamma^{-2}/2)$,

$$C^{\mathrm{s}}_r\phi_* \le \tfrac{1}{2}, \qquad (R_aC^{\mathrm{s}}_2 + R_b)\phi_* \le C^{\mathrm{s}}_r, \qquad 240K^{\mathrm{s}}_\mu K_M c_{\mathrm{cl}}C^{\mathrm{s}}_r R_a\phi_* \le 1.$$

To close the induction, suppose $\epsilon_i \le C^{\mathrm{s}}_2\phi_{i-1}^2$ for $i \le k$. The old-family sum then satisfies

$$\sum_{i\le k}\omega_{k+1-i}\epsilon_i \le C^{\mathrm{s}}_2\phi_k\sum_{i\le k}\omega_{k+1-i}\phi_{i-1} \le C_{\mathrm{pc}}S_{1/4}C^{\mathrm{s}}_2\phi_k^2.$$

Consequently $y \le (R_aC^{\mathrm{s}}_2 + R_b)\phi_k^2 \le x \le 1/2$. After extraction the linear old term is bounded by

$$\tfrac{17}{16}K^{\mathrm{s}}_\mu K_M c'_\psi\Sigma_k \le \tfrac{1}{8}C^{\mathrm{s}}_2\phi_k^2 + \theta^{\mathrm{s}}_E C^{\mathrm{cs}}_{0,\sharp}h_F^2\phi_k^2.$$

For the terms involving y, use $28x + 2y \le 30C^{\mathrm{s}}_r\phi_k$. The third smallness condition and $C^{\mathrm{s}}_r\phi_k \le 1/2$ give

$$K^{\mathrm{s}}_\mu K_M c_{\mathrm{cl}}y(28x+2y) \le \tfrac{1}{8}C^{\mathrm{s}}_2\phi_k^2 + 15K^{\mathrm{s}}_\mu K_M c_{\mathrm{cl}}R_b\phi_k^2.$$

Adding the $22x^2$ term leaves $(C^{\mathrm{s}}_2/4 + C^{\mathrm{s}}_{(2)})\phi_k^2 \le 3C^{\mathrm{s}}_2\phi_k^2/4$. The other term in the definition of ϵ_{k+1} is $2C_2\phi_k^2 \le C^{\mathrm{s}}_2\phi_k^2$. This closes the induction for the activity and every pointed family at once.

The increment is the local zeroth moment of the quadratic kernel. We describe how its coordinate estimate uses the improved norm. A term f in the moment at y comes from a polymer X and a walk ω. It is holomorphic on $2\mathscr{D}_k^\circ$ and bounded by M_0, the product of its polymer bound and the row-sum bound of the walk. Restrict the bare-coordinate difference to $X \cup \omega$ and each σ_i difference to X; call the resulting coordinate difference Δ. Let T be the maximum of the normalized absolute differences on these sets. The value of f depends only on these entries, so $f(\mathfrak{c}') = f(\mathfrak{c}+\Delta)$. Since $\mathfrak{c}$ is in the inner domain, the line $\mathfrak{c} + \zeta\Delta/T$, $|\zeta| < 1$, lies in the doubled domain. Schwarz's lemma gives

$$|f(\mathfrak{c}') - f(\mathfrak{c})| \le 2M_0T;$$

for $T > 1$ this follows directly from the two bounds by M_0. Summing these inequalities is exactly the kernel and walk sum in [QCDM1, Prop. 10.1]: the reserved factor $e^{-\kappa d_{k+1}(X)/4}$ absorbs the polynomial $1+d_{k+1}(X)$, and $e^{-|\omega|/4}$ absorbs the distance in the weighted coordinate norm. Replacing the polymer size there by $C^{\mathrm{s}}_2\phi_k^2$ proves eq. 4.ii with $B^{(2)}_\nu = 2c_{\mathrm{inc}}h_F^{-2}C^{\mathrm{s}}_2$.

To evaluate this estimate on the source chain, we first check its domain by induction using the first-order increment estimate of [QCDM1, Prop. 10.1 and Lem. 10.4]. The inductive bound is

$$\|\sigma_i(0,J)\| \le B_\nu S_\uparrow\phi_i\frac{\lambda_i\|J\|}{r_M}, \qquad B_\nu S_\uparrow \le LB_\sigma,$$

where $S_\uparrow$ bounds $\sum_{i\le j}\phi_i/\phi_j$. It holds at level zero, where $\sigma_0 = 0$. On $\lambda_j\|J\| < 2r_M$, every earlier entry has $\lambda_i\|J\| < 2r_M/L$ and is therefore inside the doubled coordinate domain. Moreover, the normalized coordinate distance at step j is at most $\lambda_j\|J\|/r_M$: this is its bare-coordinate entry, and for $i < j$,

$$\frac{\|\sigma_i(0,J)\|}{B_\sigma\phi_i} \le L\frac{\lambda_i\|J\|}{r_M} \le \frac{\lambda_j\|J\|}{r_M}.$$

The ordinary increment estimate is applicable with one point $0 \in \mathscr{D}^\circ_{j-1}$ and the other $\mathfrak{c}_{j-1}(0,J) \in 2\mathscr{D}^\circ_{j-1}$. It gives

$$\|\nu_j(\mathfrak{c}_{j-1}(0,J)) - \mu_j\| \le B_\nu \phi_j \frac{\lambda_j \|J\|}{r_M}.$$

Summing this in the recurrence preserves the inductive bound because $L^{j-i}\lambda_i = \lambda_j$ and $\sum_{i\le j}\phi_i \le S_\uparrow \phi_j$. The next increment is therefore defined, and holomorphy follows by composition. The stronger estimate is applied to a chain whose existence and domain have already been established. Since $\nu_j(0) = \mu_j$, it gives

$$\|\nu_j(\mathfrak{c}_{j-1}(0,J)) - \mu_j\| \le B^{(2)}_\nu \phi^2_{j-1} \frac{\lambda_j \|J\|}{r_M}.$$

Finally, iterate $\sigma_j = L\sigma_{j-1} + \nu_j - \mu_j$ from $\sigma_0 = 0$. For $1 \le i \le j$,

$$L^{j-i}\lambda_i \phi^2_{i-1} = \lambda_j \phi^2_{i-1} \le \lambda_j \phi^2_{j-1} (4/L)^{j-i}.$$

It follows that

$$\|\sigma_j(0,J)\| \le B^{(2)}_\nu \frac{\lambda_j\|J\|}{r_M} \sum_{i=1}^{j} \phi^2_{i-1} \le \frac{B^{(2)}_\nu}{1-4/L} \phi^2_{j-1} \frac{\lambda_j \|J\|}{r_M},$$

which is eq. 4.iii. □

4.2. **Separation of the species.** Restrict to split coordinates in eq. 4.i. Let R_b set both the coordinates and the Grassmann generators of species b to zero. Define

$$R^{(0)} = R_1 R_2, \quad R^{(1)} = (1-R_1)R_2, \quad R^{(2)} = R_1(1-R_2), \quad R^{\mathrm{mix}} = (1-R_1)(1-R_2).$$

Let P_{col} average independent constant SU(2) colour rotations in the two species at the trivial pair. These operations are contractions except for the explicit factors 2 in $R^{(b)}$ and 4 in R^{mix}.

Lemma 4.2. At split coordinates the covariances, background maps and their walk terms are block diagonal, with each block depending only on its own coordinates. The normalizers and Gaussian logarithms are sums of two separate block contributions. The source chain stays split. The new activities are neutral under the independent phase rotations of the two species. At the trivial pair P_{col} commutes with reading at later levels and is the identity on the quadratic part. ■

Proof. Each assertion about operators follows by taking the two blocks in their defining inverse, average or walk formula. A logarithmic trace also separates into the two blocks. The extraction map respects independent phase rotations; hence its quadratic kernel and its source increment are block diagonal. This proves the assertion about the chain by induction. At the trivial pair the background maps are scalar in colour. A quadratic monomial which is neutral in each species lies wholly in one block; its colour-invariant kernel is a multiple of the identity in colour, and independent colour averaging fixes it. □

Lemma 4.3. Throughout the regular pair domain, the mixed scalar fluctuation family and its pointed families satisfy

$$\|R^{\mathrm{mix}} F_0^{(k+1)}\|_{k+1} \le C_{\mathrm{gm}} \phi^2_k.$$

The activities with fermion legs have mixed norm at most $4C^{\mathrm{s}}_2 \phi^2_k$. ■

Proof. The latter assertion follows from Prop. 4.1. Write $V = R^{(0)}V + R^{(1)}V + R^{(2)}V + R^{\mathrm{mix}}V$. The preceding bounds give $\|R^{(b)}V\| \le V_1 \phi_k$ and

$$\|R^{\mathrm{mix}}V\| \le 4c'_\psi \bar\Sigma \phi^2_k + C_\Lambda C_{\mathrm{gm}} \phi^3_k, \qquad \bar\Sigma = (1+c_{\mathrm{cs}}) C_{\mathrm{pc}} S_{1/4} C^{\mathrm{s}}_2 + C^{\mathrm{cs}}_{0,\sharp} h^2_F.$$

Here $C_\Lambda = (13/12) K^{\mathrm{s}}_\Lambda \mathsf{C}_{\mathcal{A}} \sqrt{2} A_1/\alpha_0$ is the scalar localization constant in the proof of [QCDM1, Prop. 10.1], and

$$V_1 = \sqrt{2}A_1(C_V h^2_F \mathsf{C}_{\mathcal{A}} + 2K^{\mathrm{s}}_0) + c'_\psi c_{\mathsf{S}} h^2_F (\bar B_M + 2B_\sigma) + 2c'_\psi \bar\Sigma.$$

Use parameters t_1, t_2, u multiplying the last three parts, of radii $c_{\mathrm{cl}}/(8V_1\phi_k)$ and $c_{\mathrm{cl}}/(8\|R^{\mathrm{mix}}V\|)$, respectively. At $u = 0$ the mixed activity vanishes if either $t_b = 0$: the remaining Berezin covariance and all vertices then read only the other species. Schwarz's lemma in t_1, t_2 costs $O(\phi_k^2)$; the difference between $u = 1$ and $u = 0$ costs $O(\|R^{\mathrm{mix}}V\|)$.

Scalar subtraction compares the actual pair with the trivial pair and costs a factor two. The same argument applies to pointed terms by the auxiliary differentiation used in the previous proof, and to scalar bond pointers by the localization bound [QCDM1, Lem. 7.4]. In its notation one may take

$$C_{\mathrm{gm}} = 17K_M^{\bullet}(64V_1^2/c_{\mathrm{cl}} + 64c'_{\psi}\bar{\Sigma}).$$

Decreasing γ makes the last, cubic term at most half the proposed quadratic bound and all parameter radii at least two. The induction starts with no old mixed family. □

Lemma 4.4. At the trivial pair put $S_b = R^{(b)}V_0$. Then

$$|S_b(Y;B)| \le v_{\mathsf{S}}\mathsf{l}_k \left(M^{-3} \sum_{b'\subset Y} \mathsf{t}_k |B(b')|^2 \right) e^{-\lambda_V d_k(Y)}, \qquad v_{\mathsf{S}} = \frac{2K_0^{\mathrm{s}}M^3}{\mathsf{C}_{\mathcal{A}}^2 A_1}. \tag{4.v}$$

■

Proof. At $k = 0$ there is no old scalar part, so $S_b = 0$. For $k \ge 1$, scalar localization gives $S_b(Y;0) = 0$ and the bound $2K_0^{\mathrm{s}}\delta_k e^{-\lambda_V d_k(Y)}$ on $|B(b')| < \mathsf{C}_{\mathcal{A}}\delta_k$. Its derivative at zero is invariant under $B \mapsto \mathrm{Ad}(u)B$. The Haar average of the nontrivial adjoint representation of SU(2) is zero, so this derivative vanishes. Apply Schwarz's lemma of order two to $S_b(Y;\zeta B)$, using $\max_{b'\subset Y}|B(b')|^2 \le \sum_{b'\subset Y}|B(b')|^2$ and $\delta_k\mathsf{t}_k = A_1\mathsf{t}_k^{1/2}(\log \mathsf{t}_k)^{p_0}$. □

4.3. **A representative of the total mixed activity.** We must use cancellations in the sum of the activities at one level. They need not hold in each polymer of its chosen localization. We therefore propagate a representative of that sum. The representative is given only at the trivial pair; no holomorphic extension in the pair is claimed or needed. To express a level-j family at level k, we substitute the composed fermion background maps in its legs. At the next step the substitution is $\Psi_k = H_k\Psi_{k+1} + W$, where H_k is the fermion background extension and W is the Gaussian Berezin fluctuation. Its fluctuation difference is obtained by subtracting the same expression with $W = 0$. The sequence in [QCDM1, Lem. 7.9] is $\omega_n = C_\omega L^{n/4}$.

The bound below separates the quadratic terms that contain a difference of background fields from those that contain only their values at a cell centre. The former have the gain supplied by the regularity of the background maps. The latter retain the cell's zeroth moment and must be estimated separately. Equality of two total activities does not provide this separation: the polymer norm takes absolute values before summing over polymers. We therefore specify both the local kernels of a representative and the grouped cell moments which will enter its bound.

Lemma 4.5. Let $1 \le j \le k < K$ and let n satisfy $L^k\|\mathsf{n}\| < 2r_M/L$. Let F be an even family at the trivial pair, with no scalar part and with legs of levels at most j. Write $F(X) = \sum_\sigma F_\sigma(X)$, assigning each summand to a cell of X, and put

$$\epsilon_F = \sup_{\square} \sum_{X\supset\square} e^{\kappa d_j(X)} \sum_\sigma \|F_\sigma(X)\|_{h^{(j)}}.$$

Express its quadratic part in the level-j field Φ as

$$\sum_u \sum_{y,y'} \bar{\Phi}(y)\beta_u(y,y')\Phi(y').$$

Each kernel β_u is assigned to a cell $c(u)$, with centre $z_{c(u)}$. Assume, with a fixed constant C, that

$$\sum_{u:c(u)=c} \sum_{y,y'} \|\beta_u(y,y')\|(1 + |y - z_c|_j + |y' - z_c|_j) \le Ch_F^{-2}\epsilon_F \tag{4.vi}$$

for every cell c. Here $|\cdot|_j$ and dist_j use level-j lattice units. The Grassmann weights $h^{(j)}$ and the background-map constants A_κ, C_H, α_D are those of [QCDM1, Lem. 7.9]. The units are obtained by moving diagonal terms $\bar\Phi(y)a_X(y)\Phi(y)$ from the cell assigned to X to the cell containing y. Assume

$$\|a_X(y)\| \le Ch_F^{-2}(\|F(X)\| + \|F_1(X)\| + \|G(X)\|)e^{-\alpha_D \,\mathrm{dist}_j(y,X)/2}, \tag{4.vii}$$

where $F = F_1 + F_2$ and $\|F_1\|_j, \|G\|_j \le \epsilon_F$. For sums of summands, replace each norm in this inequality by the sum of the corresponding norms. Each $a_X(y)$ is also required to be a fixed linear combination, with sum of absolute coefficients at most 2, of row and column sums at y of the quadratic kernels of $F_1(X)$ and $G(X)$. Their legs are expressed in Φ by the background maps. The families F_1 and G have the same even local structure as F; G may use a second mass matrix, with the same background-map bounds. All these matrices are in the admitted mass domain of [QCDM1, Lem. 7.9].

Let m_c^{tor} be the sum of the zeroth moments of these units at c. On the torus the columns are transported to z_c along the lifted paths, including their antiperiodic signs. For the resulting decomposition $m_c^{\mathrm{tor}} = \sum_p m_{c,p}$ into terms supported on polymers Y_p, use

$$\|m_c^{\mathrm{tor}}\|_{\mathrm{nat}} := \sum_p \|m_{c,p}\| e^{\kappa L^{-(k-j)} d_j(Y_p)} (1 + d_j(Y_p))^3.$$

The summands here are the grouped coefficients obtained after the cell cancellations. Let n_{lag} be the least nonnegative integer such that $2(2A_\kappa C_H)^2 L^{-n_{\mathrm{lag}}} \le 1$. When F is expressed at level k by the composed background maps, at the trivial pair and with no shift, its fluctuation difference satisfies

$$\|V_F\| \le c_\psi \left(\omega_{k+1-j}\epsilon_F + C_{\mathrm{hl}} L^{k-j} h_F^2 \sup_c \|m_c^{\mathrm{tor}}\|_{\mathrm{nat}} \right) \tag{4.viii}$$

when $k - j \ge n_{\mathrm{lag}}$. For $k - j < n_{\mathrm{lag}}$ the first term alone suffices. The constants depend on the fixed RG parameters and the constant C in the two hypotheses. The total of V_F is $P(H_k\Psi_{k+1} + W) - P(H_k\Psi_{k+1})$, where P is the total of the read family. The first variation G_F of the cluster activities at potential vV_F has norm at most $(17/16)K_M\|V_F\|$, and its total is the Berezin expectation of the total of V_F. ∎

Proof. Put $n = k - j$. We use the leg and walk estimates in the proof of [QCDM1, Lem. 7.9], at the trivial pair and with no shift. The comparison there between the coefficient at a variable pair and its value at the trivial pair is identically zero here. The remaining estimates concern the Grassmann coefficients, the background maps and the cell units in the hypotheses.

Suppose first that $n \ge n_{\mathrm{lag}}$. Express each old leg in the level-j field Φ, and then substitute the composed background map from level k into Φ. Each leg contributes L^{-n} relative to its old weight. Its two expansions into local and walk terms cost at most $(2A_\kappa C_H)^2$. The definition of n_{lag} makes their product at most $1/2$. For Grassmann degree $2p \ge 4$, the factors beyond the fourth leg are therefore bounded by $2^{-(2p-4)}$; this also pays the polynomials in the number of legs in the walk count. The four remaining legs give L^{-4n}, while a level-k cube contains L^{3n} level-j cells. Thus the part of degree at least four is bounded by $C\epsilon_F L^{-n}$ per level-k cube, with the same localization estimates as in that lemma.

For the quadratic part, fix a unit u assigned to c and put $z = z_c$. Carry its fields and kernel to z along the prescribed lifted paths. In the next display the transported quantities are written without additional notation; the transports include the antiperiodic signs. Adding and subtracting

the reference fields gives the exact identity

$$\sum_{y,y'} \bar\Phi(y)\beta_u(y,y')\Phi(y') = \bar\Phi(z)m_u\Phi(z) + \sum_{y,y'}[\bar\Phi(y)-\bar\Phi(z)]\beta_u(y,y')\Phi(y') + \sum_{y,y'}\bar\Phi(z)\beta_u(y,y')[\Phi(y')-\Phi(z)], \qquad m_u = \sum_{y,y'}\beta_u(y,y').$$

After summing the units assigned to c, the first term is $\bar\Phi(z_c)m_c^{\mathrm{tor}}\Phi(z_c)$. The other two terms each contain a difference. This is the point at which the grouped zeroth moment enters: the norm of this zeroth-moment term is estimated using m_c^{tor} after the cancellations in the cell have been made.

The background estimate with Hölder exponent $3/4$ bounds a transported difference along a path of length s in level-j units by $Ch_F sL^{-7n/4}$. The other leg costs $Ch_F L^{-n}$. For a moved unit, the path lies in its assigned cell and its length is paid by eq. 4.vi. For a polymer unit, the paths run through its polymer and the walk polymers of its two legs. Their lengths are bounded by a fixed multiple of $M(1+d_j(X))$ plus the walk lengths, where X is the original polymer. The unused polymer decay and the walk weights pay these factors, as in the proof of [QCDM1, Lem. 7.9]. We obtain

$$\text{sum of norms of the difference terms at one cell} \le C\epsilon_F L^{-11n/4}.$$

Summing over the L^{3n} old cells gives $C\epsilon_F L^{n/4}$. The estimate uses the actual lifted paths; no wrapping path is replaced by a shorter distance on the torus.

We also keep track of the diagonal terms that were moved between cells. A term $\bar\Phi(y)a_X(y)\Phi(y)$ is subtracted from the unit assigned to X and added to the unit assigned to the cell of y. These two copies cancel in the total. For the local estimate, split $a_X(y)=\sum_p a_{X,p}(y)$ according to the local or walk term in each of the two legs of the kernels from which it was formed. Let Z_p be their supporting polymer. The row-and-column hypothesis ensures that this is the same decomposition used for the kernels of F_1 and G, with uniformly bounded coefficients, and that $a_{X,p}(y)=0$ unless $y\in Z_p$. The localized row-sum estimate gives

$$\sum_{p,y}\|a_{X,p}(y)\|e^{\kappa(d_j(Z_p)-d_j(X))} \le CA_\kappa^2 h_F^{-2}(\|F_1(X)\|+\|G(X)\|).$$

For families with several summands, the right-hand side is the sum of their norms. The same estimate holds for G at its second mass matrix, since its background maps have the same constants. This is the estimate for moved pieces in [QCDM1, Prop. 10.1]. It controls their actual supports as well as their coefficients.

Changing the cell assignment changes a level-k cube sum only through terms whose two assignments lie on opposite sides of its boundary. The count in [QCDM1, Lem. 7.9] bounds these terms by $C\epsilon_F$: the boundary contains $O(L^{2n})$ old cells, the two legs give L^{-2n}, and the exponential in eq. 4.vii pays the distance of a moved site from its polymer. The polymer and walk weights pay the remaining diameter factors. This contribution is included in $\omega_{k+1-j}\epsilon_F$.

To localize the difference terms, take the level-k hull of each supporting polymer and enlarge it by the walk polymers of its two final legs. The hull estimate

$$d_k(Z^{\mathrm{up}}) \le L^{-n}d_j(Z)$$

and the weighted walk sums give the local polymer norm. The spare rate $\kappa(1-L^{-n})d_j(Z)$ pays the polynomials left by the path and cell counts. Together with the preceding degree and boundary estimates, these are the terms bounded by $\omega_{k+1-j}\epsilon_F$.

It remains to localize the zeroth-moment terms. Use the decomposition $m_c^{\mathrm{tor}}=\sum_p m_{c,p}$ from the statement, after its cell cancellations. Assign

$$\bar\Phi(z_c)m_{c,p}\Phi(z_c)$$

to the level-k hull of Y_p, enlarged by the walk polymers of the two legs. Those legs give $Ch_F^2L^{-2n}$, and the count of old cells gives L^{3n}. The hull estimate accounts for the exponential $e^{\kappa L^{-n}d_j(Y_p)}$ in the norm of the moment; the factor $(1+d_j(Y_p))^3$ pays the number of cells from which its hull can reach a fixed output cube. The hull and walk count of [QCDM1, Prop. 10.1] therefore gives

$$C_{\mathrm{hl}}L^n h_F^2 \sup_c \|m_c^{\mathrm{tor}}\|_{\mathrm{nat}}.$$

This is the second term of eq. 4.viii. It retains the torus moment itself, including any contribution of wrapping pieces.

For $n < n_{\mathrm{lag}}$, use the original summands without expanding their legs about cell centres. The count is at most $L^{3n_{\mathrm{lag}}}$, a fixed factor already included in C_ω. This proves the asserted bound without a separate moment term at these lags.

Finally substitute $\Psi_k = H_k\Psi_{k+1} + W$ and subtract the value at $W = 0$. Every surviving monomial contains a fluctuation leg. The factor c_ψ follows from the fluctuation weights, rather than from counting these monomials individually. Let h be the original leg weight, h_a a bound for its background image, and h_b the weight of its fluctuation image. Choose a positive allowance h_b' with $h_a + h_b' \le h$ and $h_b \le h_b'$. For a homogeneous product of degree q,

$$(h_a+h_b)^q - h_a^q \le \frac{h_b}{h_b'}\big((h_a+h_b')^q - h_a^q\big).$$

The background image uses at most half the leg weight; take the other half for h_b'. The localized W bound then gives $h_b/h_b' \le c_\psi$, as in the old-potential estimate of [QCDM1, §7]. Applying this inequality to the localized terms above proves eq. 4.viii, including its short-lag assertion.

All decompositions used here preserve the sum: the moved diagonal terms were added and subtracted, the cell expansion is an identity, and the walk series converge absolutely. Thus the total of V_F is $P(H_k\Psi_{k+1}+W) - P(H_k\Psi_{k+1})$. For the first variation, the cluster activity at potential vV_F is holomorphic on $|v| < c_{\mathrm{cl}}/\|V_F\|$ and has bound

$$\frac{17}{16}K_M' c_{\mathrm{cl}} e^{-\kappa_+ d_{k+1}(X)}.$$

Cauchy's estimate, followed by the polymer sum, yields $\|G_F\|_{k+1} \le (17/16)K_M\|V_F\|$. Since V_F is independent of B, the joined integral factors into its gauge integral and $\mu_{\Gamma_k^{-1}}(e^{v\sum_X V_F(X)})$. Differentiating its logarithm at $v = 0$, and using $\mu_{\Gamma_k^{-1}}(1) = 1$, identifies the total of G_F with $\mu_{\Gamma_k^{-1}}(\sum_X V_F(X))$. □

Put

$$q_k = \phi_k^3 + \phi_k^2 e^{-c_{\mathrm{t}}\ell/(2\lambda_{k+1})}, \qquad \bar q_k = q_k + \mathfrak{l}_k^2.$$

The following estimate contains the cancellation needed for Prop. 3.4.

Lemma 4.6. There are constants $E_3, N_3, C_{\mathfrak{d}}$, independent of the admitted data, with these properties. At level j, the total of $P_{\mathrm{col}}R^{\mathrm{mix}}E_{\ge 2}^{(j)}$ at the trivial pair has an even representative F^j, neutral in each species and invariant under P_{col}, with no scalar part and

$$\epsilon_j^{(3)} \le E_3 q_{j-1}. \tag{4.ix}$$

Its cell sums satisfy

$$\|m_c^{\mathrm{tor}}\|_{\mathrm{nat}} \le C_{\mathrm{pw}}h_F^{-2}\epsilon_j^{(3)}\left[\eta_{\min}^{-2}\left(C_{\mathrm{bg}}^{\mathrm{w}}\frac{L^j\|\mathsf{n}\|}{r_M} + C_A e^{-4c_{\mathrm{t}}\ell/\lambda_j}\right) + e^{-c_{\mathrm{t}}\ell/\lambda_j}\right] + C_{\mathfrak{d}}h_F^{-2}\phi_{j-1}^2 e^{-c_{\mathrm{t}}\ell/\lambda_j}. \tag{4.x}$$

If $f_j(\mathfrak{c})$ is the total scalar output of step j at the trivial pair, including its normalizers and Gaussian terms, then

$$|R^{\mathrm{mix}}f_j(\mathfrak{c})| \le E_3 M^{-3} n_j \bar q_{j-1}. \tag{4.xi}$$

For split $\mathfrak{c} \in 2\mathscr{D}^{\circ}_{j-1}$, write $\nu_{j,1} = e_+\nu_j e_+$ and $\nu_{j,2} = e_-\nu_j e_-$. If $b' \neq b$, then

$$\sup_y \|\nu_{j,b}(\mathfrak{c})(y) - \nu_{j,b}(R_{b'}\mathfrak{c})(y)\| \leq N_3 q_{j-1}. \tag{4.xii}$$

Thus the coordinates of species b are held fixed in this comparison. No source factor is included in this bound. ∎

Proof. The induction starts with no old mixed activity. Fix a split coordinate $\mathfrak{c} \in 2\mathscr{D}^{\circ}_k$ and suppose the three assertions hold through level k. All quantities in this proof are evaluated at the trivial pair. We first identify the terms whose totals must be cancelled, then construct a local family representing the remaining total.

The deformed potential. Let $V_g = R^{(0)}V_0$ be the zero-source scalar potential. For species b, let $V_{F,b}$ and $V_{s,b}$ be the gauge–fermion vertex and mass vertex restricted to its generators; let $V_{\mathrm{old},b} = R^{(b)}V_{\mathrm{old},\geq 2}$; and let $S_b = R^{(b)}V_0$ as in Lem. 4.4. The old mixed family is split into its value at $B = 0$ and its change from that value. With ten parameters $p = (z_1, z_2, x_1, x_2, y_1, y_2, d_1, d_2, u, u')$, put

$$V_g + \sum_{b=1}^{2}\big(V_{F,b}(z_bB) + x_bV_{s,b}(z_bB) + y_bV_{\mathrm{old},b}(z_bB) + d_bS_b(B)\big) + uV_{\mathrm{mix},0} + u'V_{\mathrm{mix},1}(B).$$

Here the last two terms mean

$$V_{\mathrm{mix},0} = R^{\mathrm{mix}}V_{\mathrm{old},\geq 2}(0), \qquad V_{\mathrm{mix},1}(B) = R^{\mathrm{mix}}\big(V_{\mathrm{old},\geq 2}(B) - V_{\mathrm{old},\geq 2}(0)\big) + R^{\mathrm{mix}}V_0(B).$$

In a term evaluated at z_bB, all its background maps are evaluated at the fine connection reconstructed from z_bB. At $p = (1, \dots, 1)$ this is the original potential. The parameter u isolates the part which can be propagated by the old representative; u' groups terms which already have three small factors. Indeed, the bound $\|R^{\mathrm{mix}}V_{\mathrm{old},\geq 2}\| \leq 4c'_\psi\bar{\Sigma}\phi_k^2$, followed by Schwarz's lemma in the scaling of B, gives

$$\|R^{\mathrm{mix}}(V_{\mathrm{old},\geq 2}(B) - V_{\mathrm{old},\geq 2}(0))\| \leq 8c'_\psi\bar{\Sigma}\phi_k^2 r_z^{-1}.$$

The scalar localization estimate in Lem. 4.3 supplies the other term. Thus

$$\|V_{\mathrm{mix},1}\| \leq \left(8c'_\psi\bar{\Sigma}C'_r + C_\Lambda C_{\mathrm{gm}}\right)\phi_k^3.$$

Here $C'_r = \sqrt{2}A_1\mathsf{C}_{\mathcal{A}}\max\{16C_Vh_F^2/c_{\mathrm{cl}}, 2/\beta_0\}$. The first term has acquired an extra factor ϕ_k from $r_z^{-1} \leq C'_r\phi_k$; the second has that factor from scalar localization.

Choose parameter radii

$$r_z = \frac{\min\{c_{\mathrm{cl}}/(16C_Vh_F^2), \beta_0/2\}}{\mathsf{C}_{\mathcal{A}}\delta_k}, \qquad r_x = \frac{c_{\mathrm{cl}}}{16c'_\psi c_{\mathsf{S}}h_F^2(\bar{B}_M + 2B_\sigma)\phi_k},$$

$$r_y = \frac{c_{\mathrm{cl}}}{32c'_\psi\bar{\Sigma}\phi_k^2}, \qquad r_d = \frac{c_{\mathrm{cl}}}{16v_{\mathsf{S}}\mathfrak{l}_k},$$

$$r_u = \frac{c_{\mathrm{cl}}}{64c'_\psi\bar{\Sigma}\phi_k^2}, \qquad r_{u'} = \frac{c_{\mathrm{cl}}}{16(8c'_\psi\bar{\Sigma}C'_r + C_\Lambda C_{\mathrm{gm}})\phi_k^3}.$$

The eleven displayed parts of the potential have total norm at most $11c_{\mathrm{cl}}/16$; the cluster tail costs at most $c_{\mathrm{cl}}/16$. Assign weight one to z_b, x_b, d_b, weight two to y_b, u, and weight three to u'. Since $\mathfrak{l}_k \leq \sqrt{2}\phi_k$, there is a fixed $C_r^{\mathrm{b}} \geq 2$ such that the inverse radii are bounded by $(C_r^{\mathrm{b}}\phi_k)$ to their respective weights. Decrease γ until $C_r^{\mathrm{b}}\phi_* \leq 1/2$. For a multi-index $v \in \mathbb{N}^{10}$ denote its weighted degree by $|v|_w$. Cauchy's estimate gives

$$\widetilde{E}(X;p) = \sum_v a_v(X)p^v, \qquad \|a_v(X)\| \leq K'_M c_{\mathrm{cl}} r^{-v}e^{-\kappa_+ d_{k+1}(X)}, \qquad r^{-v} \leq (C_r^{\mathrm{b}}\phi_k)^{|v|_w}. \tag{4.xiii}$$

Every radius is at least two, so these series converge at $p = (1, \dots, 1)$.

To identify a coefficient of the total, let μ_B denote the normalized zero-source gauge integral of this step, including its scalar potential V_g, and let μ_1, μ_2 be the decoupled Berezin means of the two species. The remaining background generators are not integrated. The total of the cluster activities is the logarithm of the joined integral. Hence the coefficient of two distinct linear parameters is their connected expectation under

$$\langle A\rangle_* := \mu_B\mu_1\mu_2(A), \qquad \kappa_*(A,B) = \langle AB\rangle_* - \langle A\rangle_*\langle B\rangle_*.$$

This identity holds in the even algebra of the retained background generators. It describes the sum over X of a coefficient in eq. 4.xiii, not its separate polymer terms.

The coefficients of weight two. Call an index mixed if it contains a parameter of each species, or contains u or u'. Every other coefficient is killed by R^{mix} polymer by polymer. For example, with $u = u' = 0$ and all parameters of species two zero, the potential has no generator of that species and does not depend on its coordinates. Its Berezin integrals use only μ_1, the unused normalized mean contributing one. Thus its activities also have no dependence on species two.

There are precisely ten mixed indices of weight at most two: the index u, and the nine products formed by one of z_1, x_1, d_1 and one of z_2, x_2, d_2. The vertices in these products are

$$l_b(B) = \partial_{z_b} V_{F,b}(z_b B)\big|_{z_b=0}, \qquad M_b = V_{s,b}(0), \qquad S_b(B).$$

The first is linear in B and quadratic in the fields of species b; M_b is independent of B and has only that species' generators; S_b is scalar.

The five products containing a mass vertex have zero total. For example, the independence of μ_1 and μ_2 gives

$$\langle M_1 M_2\rangle_* = \mu_1(M_1)\mu_2(M_2) = \langle M_1\rangle_*\langle M_2\rangle_*.$$

Since M_2 is independent of B, the same factorization holds for $\langle l_1(B)M_2\rangle_*$ and $\langle S_1(B)M_2\rangle_*$. Exchanging the species proves the cancellations for $x_1x_2, z_1x_2, x_1z_2, x_1d_2, d_1x_2$.

For the other three products containing a gauge vertex, integrate the fermion fluctuations first. Put $L_b(B) = \mu_b(l_b(B))$. The corresponding totals are

$$\kappa_B(L_1, L_2), \qquad \kappa_B(L_1, S_2), \qquad \kappa_B(S_1, L_2),$$

where $\kappa_B(A,B) = \mu_B(AB) - \mu_B(A)\mu_B(B)$. These need not vanish before colour averaging. Let $\alpha_g^{(b)}$ rotate the background generators of species b by the constant colour matrix g. At the trivial pair the vertex and covariance satisfy

$$L_b(\mathrm{Ad}(g)B) \circ \alpha_g^{(b)} = L_b(B).$$

The covariance is scalar in colour, which permits commuting the rotation through μ_b. Since L_b is linear in B, its average in the background colour variables is

$$P^{(b)} L_b(B) = L_b\left(\int_{\mathrm{SU}(2)} \mathrm{Ad}(g^{-1})B\, dg\right) = 0.$$

The adjoint representation has no invariant vector. The averages of the two species are independent, so averaging the first species kills $\kappa_B(L_1, L_2)$ and $\kappa_B(L_1, S_2)$, and averaging the second kills $\kappa_B(S_1, L_2)$. In Grassmann degree zero the colour average is the identity: the scalar part of L_b is itself an invariant linear form in B and vanishes. Thus these cancellations hold in every degree after $P_{\mathrm{col}} R^{\mathrm{mix}}$.

The ninth product, d_1d_2, is different. Its total is the gauge covariance $\kappa_B(S_1, S_2)$, so it has degree zero. Both vertices vanish when their own species coordinate is zero; therefore R^{mix} fixes this coefficient. There is no cancellation asserted for it. Its Cauchy estimate is

$$\sum_X |a_{d_1d_2}(X)| \le K_M c_{\mathrm{cl}} r_d^{-2} n_{k+1} M^{-3} = \frac{256 K_M v_{\mathsf{S}}^2}{c_{\mathrm{cl}}} \mathfrak{l}_k^2 n_{k+1} M^{-3}. \tag{4.xiv}$$

Propagation of the old mixed total. The remaining coefficient of weight two is u. Because $V_{\mathrm{mix},0}$ is independent of B, its total is $\mu_1\mu_2$ of the total of that vertex. If P_i is the total of the level-i family after it has been read at level k, the fluctuation vertex has total

$$R^{\mathrm{mix}}\sum_{i\le k}\bigl(P_i(H_k\Psi_{k+1}+W)-P_i(H_k\Psi_{k+1})\bigr).$$

The projection P_{col} commutes with these background maps, with R^{mix} and with $\mu_1\mu_2$. Thus the induction hypothesis permits replacing each total by that of F^i. Apply Lem. 4.5 to each F^i and add the resulting families. If V_F denotes this sum, the coefficient of u, after colour averaging, has the total of its first-variation family G_F. This replacement uses equality of totals, not equality of the chosen localizations.

In the cell bound, terms containing $L^i\|\mathsf{n}\|/r_M$ acquire the factor L^{k-i} from reading at level k; their product is less than $2/L$. The exponential terms have uniformly bounded products with L^{k-i}. These factors are included in c_{cs}, exactly as in the ordinary old-family count. The last term of eq. 4.x must instead be summed separately. We obtain

$$\|V_F\|\le c_\psi\Bigg[(1+c_{\mathrm{cs}})\sum_{i\le k}\omega_{k+1-i}\epsilon_i^{(3)} + C_{\mathrm{hl}}C_{\mathfrak{d}}\sum_{i\le k}L^{k-i}\phi_{i-1}^2e^{-c_{\mathrm{t}}\ell/\lambda_i}\Bigg].$$

Both sums are bounded by a multiple of q_k. For the first, monotonicity of ϕ_i and of the torus exponential gives

$$q_{i-1}\le\frac{\phi_{i-1}}{\phi_k}q_k,\qquad \sum_{i\le k}\omega_{k+1-i}\phi_{i-1}\le C_{\mathrm{pc}}S_{1/4}\phi_k.$$

For the second, write $v=k-i$ and divide by the torus term $\phi_k^2e^{-c_{\mathrm{t}}\ell/(2\lambda_{k+1})}$ in q_k. The quotient is at most

$$\sum_{v\ge0}L^v\exp\left[-\frac{c_{\mathrm{t}}\ell}{\lambda_{k+1}}(L^{v+1}-\tfrac12)\right]\le\sum_{v\ge0}L^ve^{-2c_{\mathrm{t}}(L^{v+1}-1/2)},$$

because $\ell/\lambda_{k+1}\ge2$. Consequently

$$\|V_F\|\le c_\psi\left((1+c_{\mathrm{cs}})C_{\mathrm{pc}}S_{1/4}E_3+C_{\mathrm{hl}}C_{\mathfrak{d}}S_{\mathrm{tor}}\right)q_k,\qquad S_{\mathrm{tor}}=\sum_{v\ge0}L^ve^{-2c_{\mathrm{t}}(L^{v+1}-1/2)}<\infty.\tag{4.xv}$$

The factor from the first variation is $(17/16)K_M$.

The remaining coefficients. For a multi-index of weighted degree at least three the Cauchy product has at least three powers of ϕ_k. Summing these bounds uses

$$N_{10}=\sum_{m\ge3}\binom{m+10}{10}2^{-(m-3)}<\infty.$$

It bounds their total norm by $4K_Mc_{\mathrm{cl}}N_{10}(C_r^{\mathrm{b}})^3\phi_k^3$ after $P_{\mathrm{col}}R^{\mathrm{mix}}$. Adding their degree-at-least-two parts to the first variation gives a pre-extraction representative R with

$$\|R\|_{k+1}\le\left[\tfrac{17}{4}K_Mc_\psi((1+c_{\mathrm{cs}})C_{\mathrm{pc}}S_{1/4}E_3+C_{\mathrm{hl}}C_{\mathfrak{d}}S_{\mathrm{tor}})+4K_Mc_{\mathrm{cl}}N_{10}(C_r^{\mathrm{b}})^3\right]q_k.$$

For clarity, the pre-extraction representative in this display is the explicitly defined family

$$R(X)=\bigl(R^{\mathrm{mix}}G_F(X)\bigr)_{\ge2}+\sum_{\substack{v\ \text{mixed}\\ |v|_w\ge3}}P_{\mathrm{col}}R^{\mathrm{mix}}(a_v(X))_{\ge2}.$$

It has the total of the mixed activity with legs, because all omitted low-weight totals were just shown to vanish or to be scalar. Each retained term is neutral in both species. The first-variation family is formed from colour-invariant representatives, so it is also invariant under P_{col}.

The scalar estimate can already be closed. The normalizers and Gaussian logarithms have no mixed part. In degree zero the colour projection is the identity, and the coefficient bound for R^{mix} costs at most four. The scalar output is therefore bounded by the first-variation contribution, the sum of coefficients of weight at least three, and eq. 4.xiv. Their sum is

$$M^{-3}n_{k+1}\left[(E_3/2+A_3)q_k+\frac{256K_M v_{\mathfrak{S}}^2}{c_{\mathrm{cl}}}\mathfrak{l}_k^2\right]\le E_3M^{-3}n_{k+1}\bar{q}_k,$$

with the constants chosen below. This proves eq. 4.xi; it also explains why the scalar term $\mathfrak{l}_k^2$ need not enter the size of the representative with fermion legs.

Extraction and cell sums. In the following estimates put $j=k+1$. Write $K_X^R=\sum_p K_{X,p}^R$ for the quadratic kernel on X, decomposed by expanding the two background legs into their local and walk terms, and let Z_p be their supporting polymers. For such a kernel define

$$M_y[K]=\frac{1}{2}\sum_{y'}(K(y,y')+K(y',y)),$$

with columns transported to y inside the fundamental cell centered at y. This is a linear function of K. The moved coefficients are $a_{X,p}(y)=M_y[K_{X,p}^R]$. The row-sum bound from the proof of [QCDM1, Prop. 10.1] is

$$\sum_{p,y}\|a_{X,p}(y)\|e^{\kappa[d_j(Z_p)-d_j(X)]}\le CA_\kappa^2h_F^{-2}\|R(X)\|.$$

After summing over X, it gives, for every cell c, every cube $\square$, and $n\ge 1$,

$$\sum_{y\in c,X,p}\|a_{X,p}(y)\|e^{\kappa L^{-n}d_j(Z_p)}(1+d_j(Z_p))^3\le\tfrac{1}{2}C_{\mathrm{pw}}h_F^{-2}\|R\|_j, \tag{4.xvi}$$

$$\sum_{Z_p\supset\square,y}\|a_{X,p}(y)\|e^{\kappa d_j(Z_p)}\le\tfrac{1}{2}C_{\mathrm{pw}}h_F^{-2}\|R\|_j. \tag{4.xvii}$$

The factor $1/2$ appears because C_{pw} in that proof bounds the two families at the source and at zero together; here there is one family. At lag zero use eq. 4.xvii. Equality of the totals identifies the sum of these moments with the moment of the actual total. This does not identify the moments polymer by polymer. The source increment uses only polymers of diameter less than one quarter of the torus side and reads their moments along lifted paths. Denote the actual mixed quadratic kernels by $K_{X,p}$. By [QCDM1, Prop. 10.1, equation (44)],

$$-\mathfrak{s}_jR^{\mathrm{mix}}\nu_j(y)=\sum_{\operatorname{diam}X<\ell/4,p}\operatorname{ord}_0[K_{X,p}](y),$$

where $\mathfrak{s}_j\ge\eta_{\min}^2$ is the zero-source normalization of the extraction map. On a nonwrapping Z_p, the moment ord_0 read along the lifted paths agrees with M_y. Equality of the total kernels therefore gives the exact identity

$$-\mathfrak{s}_jR^{\mathrm{mix}}\nu_j(y)=\sum_{X,p}M_y[K_{X,p}^R]-d(y), \tag{4.xviii}$$

where

$$d(y)=\sum_{X,p}{}'M_y[K_{X,p}]-\sum_{X,p}{}''\operatorname{ord}_0[K_{X,p}](y).$$

The first sum contains actual pieces for which either $\operatorname{diam}X\ge\ell/4$ or Z_p wraps around the torus. The second contains pieces with $\operatorname{diam}X<\ell/4$ whose Z_p wraps around the torus. This formula

accounts for both the omitted large polymers and the difference between the two moment conventions on wrapping paths. Every supporting polymer in this difference satisfies

$$d_{k+1}(Z_p) \geq \frac{\ell}{4c_{\mathcal{D}} M \lambda_{k+1}} - 1.$$

Here $c_{\mathcal{D}}$ is the fixed diameter constant, $\operatorname{diam} Y \leq c_{\mathcal{D}} M \lambda_j (1 + d_j(Y))$. Indeed, either the original polymer has diameter at least one quarter of the torus side, or Z_p contains a path wrapping around the torus. The tail of eq. 4.xvi therefore gains $e^{-c_{\mathrm{t}}\ell/\lambda_{k+1}}$, by the choice of c_{t} in [QCDM1, Prop. 10.1]. Apply this estimate to the actual mixed quadratic activity, whose norm is at most $4C_2^{\mathrm{s}}\phi_k^2$ by Prop. 4.1. Consequently

$$\sum_{y \in c} |d(y)| \leq 4C_{\mathrm{pw}} h_F^{-2} C_2^{\mathrm{s}} \phi_k^2 e^{-c_{\mathrm{t}}\ell/\lambda_{k+1}}. \tag{4.xix}$$

We now perform the extraction on the representative. With $\psi^{\mathrm{bg}} = H_{0,j}\Psi$, define the quadratic forms

$$P^R_{z;X,p} = -\mathfrak{s}_j^{-1} L^{-j} \mathsf{S}_1(M_z[K^R_{X,p}] 1_z \psi^{\mathrm{bg}}),$$
$$P^d_z = \mathfrak{s}_j^{-1} L^{-j} \mathsf{S}_1(d(z) 1_z \psi^{\mathrm{bg}}).$$

Here 1_z restricts the coefficient to z and S_1 is the mass bilinear used in the source extraction. Assign both forms to the cell containing z; their supporting polymers are Z_p and the cube of z, respectively. By eq. 4.xviii,

$$\sum_{z;X,p} P^R_{z;X,p} + \sum_z P^d_z = L^{-j} \mathsf{S}_1(R^{\mathrm{mix}} \nu_j \psi^{\mathrm{bg}}).$$

This is precisely the local form that the actual extraction adds to the mixed activity. Thus

$$F_{\mathfrak{c}}^{(j)} = R + \sum_{z;X,p} P^R_{z;X,p} + \sum_z P^d_z$$

represents the required extracted total. Its forms are neutral in both species and invariant under P_{col}, since their coefficients are split and the reconstruction maps at the trivial pair act trivially on colour. The first forms add at most $\eta_{\min}^{-2} c_{\mathsf{S}} M^{-3} C_{\mathrm{pw}} \|R\|_{k+1}$ to the polymer norm by eq. 4.xvii. The correction forms add at most

$$4\eta_{\min}^{-2} c_{\mathsf{S}} M^{-3} C_{\mathrm{pw}} C_2^{\mathrm{s}} \phi_k^2 e^{-c_{\mathrm{t}}\ell/\lambda_{k+1}}.$$

Thus a common bound for the norm and the cell-unit hypotheses is

$$\epsilon^{(3)}_{k+1} = K^{\mathrm{s}}_{\mu} \|R\|_{k+1} + 4(1 + \eta_{\min}^{-2} c_{\mathsf{S}} M^{-3} C_{\mathrm{pw}}) C_2^{\mathrm{s}} \phi_k^2 e^{-c_{\mathrm{t}}\ell/\lambda_{k+1}}. \tag{4.xx}$$

For the unit hypotheses, split the quadratic kernel into $K^R_X - \delta_{yy'} M_y[K^R_X]$, assigned to the cell of X, and the moved diagonal pieces, assigned to the cell of y. Their variation and first moments obey eq. 4.vi by the kernel bound and eq. 4.xvi. The coefficients of the two types of extraction forms, summed over a cell, are bounded respectively by

$$\tfrac{1}{2}\eta_{\min}^{-2} C_{\mathrm{pw}} h_F^{-2} \|R\|_{k+1}, \qquad 4\eta_{\min}^{-2} C_{\mathrm{pw}} h_F^{-2} C_2^{\mathrm{s}} \phi_k^2 e^{-c_{\mathrm{t}}\ell/\lambda_{k+1}}.$$

Each is at most $\eta_{\min}^{-2} C_{\mathrm{pw}} h_F^{-2} \epsilon^{(3)}_{k+1}$. The moved coefficient is exactly one half of a row sum plus one half of a column sum of $K^R_{X,p}$, as required in Lem. 4.5; its second family G is zero.

For a nonwrapping Z_p, the zeroth moment of $K^R_{X,p} - \delta_{yy'} M_y[K^R_{X,p}]$ is zero. The two moment conventions agree on this support, so the subtracted diagonal has exactly the same total moment as the original kernel. A wrapping piece contributes at most twice the double sum of its kernel. Assigning it to the hull of Z_p and using the tail of eq. 4.xvi bounds these contributions in the cell norm by

$$C_{\mathrm{pw}} h_F^{-2} \|R\|_j e^{-c_{\mathrm{t}}\ell/\lambda_j}.$$

A moved piece and its extraction form have joint coefficient

$$\left(1-\mathfrak{s}_{k+1}^{-1}\sum_{y}A_{k+1}^{\mathsf{n}}(y,z)\right)M_z[K_{X,p}^{R}],$$

where A_j^{n} is the zeroth-moment map of the background mass form. Its source variation and its torus comparison from [QCDM1, proof of Prop. 10.1] give the bound

$$\eta_{\min}^{-2}\left(C_{\mathrm{bg}}^{\mathrm{w}}\frac{L^j\|\mathsf{n}\|}{r_M}+C_Ae^{-4c_{\mathrm{t}}\ell/\lambda_j}\right)\frac{1}{2}C_{\mathrm{pw}}h_F^{-2}\|R\|_j$$

for the grouped moved pieces and their forms. Keeping them grouped is what produces the small coefficient $1-\mathfrak{s}_j^{-1}\sum_y A_j^{\mathsf{n}}(y,z)$.

The correction form P_z^d contributes $\mathfrak{s}_j^{-1}\sum_y A_j^{\mathsf{n}}(y,z)d(z)$. At zero source,

$$\left|\sum_{y}A_j^{0}(y,z)-\mathfrak{s}_j\right|\le C_Ae^{-4c_{\mathrm{t}}\ell/\lambda_j}.$$

The change with n, expanded in the local and walk terms of the two background legs, is at most $C_{\mathrm{bg}}^{\mathrm{w}}L^j\|\mathsf{n}\|/r_M$ in the same weighted cell norm. Since $L^j\|\mathsf{n}\|/r_M<2$, eq. 4.xix bounds these correction forms by

$$C_{\mathfrak{d}}h_F^{-2}\phi_k^2e^{-c_{\mathrm{t}}\ell/\lambda_j},\qquad C_{\mathfrak{d}}=4(1+\eta_{\min}^{-2}(C_A+2C_{\mathrm{bg}}^{\mathrm{w}}))C_{\mathrm{pw}}C_2^{\mathrm{s}}.$$

Combining the three types of terms proves eq. 4.x.

Choose

$$\begin{aligned}C_{\mathfrak{d}}&=4(1+\eta_{\min}^{-2}(C_A+2C_{\mathrm{bg}}^{\mathrm{w}}))C_{\mathrm{pw}}C_2^{\mathrm{s}},\\ A_3&=K_\mu^{\mathrm{s}}\left(\tfrac{17}{4}K_Mc_\psi C_{\mathrm{hl}}C_{\mathfrak{d}}S_{\mathrm{tor}}+4K_Mc_{\mathrm{cl}}N_{10}(C_r^{\mathrm{b}})^3\right)\\&\quad+4(1+\eta_{\min}^{-2}c_{\mathsf{S}}M^{-3}C_{\mathrm{pw}})C_2^{\mathrm{s}},\\ E_3&=2A_3+256K_Mv_{\mathsf{S}}^2/c_{\mathrm{cl}}.\end{aligned}$$

After extraction the coefficient of E_3q_k is at most $4\theta_E^{\mathrm{s}}(1+c_{\mathrm{cs}})C_{\mathrm{pc}}S_{1/4}\le 1/2$. Thus the norm is at most $(E_3/2+A_3)q_k$. These choices close both eq. 4.ix and eq. 4.xi. Finally, eq. 4.xviii, the first moment bound and eq. 4.xix give

$$\begin{aligned}|R^{\mathrm{mix}}\nu_j(y)|&\le\eta_{\min}^{-2}C_{\mathrm{pw}}h_F^{-2}\left(\tfrac{1}{2}\|R\|_j+4C_2^{\mathrm{s}}\phi_k^2e^{-c_{\mathrm{t}}\ell/\lambda_j}\right)\\&\le N_3q_k,\qquad N_3=\eta_{\min}^{-2}C_{\mathrm{pw}}h_F^{-2}(2E_3+4C_2^{\mathrm{s}}).\end{aligned}$$

To read this as an estimate for one species, apply R^{mix} to its local quadratic term. For $b'\neq b$ the result is

$$R^{\mathrm{mix}}(\bar\psi_b\nu_{j,b}(\mathfrak{c})\psi_b)=\bar\psi_b(\nu_{j,b}(\mathfrak{c})-\nu_{j,b}(R_{b'}\mathfrak{c}))\psi_b.$$

The preceding kernel estimate is therefore eq. 4.xii. □

4.4. **The source chain and the sum over scales.** The preceding lemma concerns free split coordinates. The vacuum energy is evaluated along the actual source chain. We next control the change in one chain when the source in the other species is added.

Lemma 4.7. For split sources and $\lambda_i\|J_b\|<r_M$,

$$\|\sigma_{i,1}(J_1,J_2)-\sigma_{i,1}(J_1,0)\|\le C_{\mathrm{s}3}q_{i-1}\frac{\lambda_i\|J_2\|}{r_M},\qquad C_{\mathrm{s}3}=\frac{3N_3}{1-4/L},\tag{4.xxi}$$

and the analogous assertion holds for species two. If two inner coordinates have the same n, and

$$T=\max_{i\le j-1}\frac{\|\sigma_i-\sigma_i'\|}{B_\sigma\phi_i}\le 1,$$

then

$$|f_j(\mathfrak{c}) - f_j(\mathfrak{c}')| \le 8K_M M^{-3} n_j C_{\mathrm{Lip}} \phi_{j-1} T, \tag{4.xxii}$$

where $C_{\mathrm{Lip}} = c'_\psi c_{\mathfrak{S}} h_F^2 B_\sigma + 2c'_\psi \bar{\Sigma} + 2\sqrt{2} A_1 K_0^{\mathrm{s}}$. ■

Proof. The chains exist and are holomorphic on $\lambda_j \|J_b\| < 2r_M$ by Prop. 4.1. On the smaller domain in the statement their coordinates are split and lie in the inner coordinate domain. Write

$$\mathfrak{c}_i^+ = \mathfrak{c}_i(J_1, J_2), \qquad \mathfrak{c}_i^0 = \mathfrak{c}_i(J_1, 0), \qquad \Delta_l = \sigma_{l,1}(J_1, J_2) - \sigma_{l,1}(J_1, 0).$$

The bare first-species coordinate is $-aJ_1$ in both chains. It cancels from their difference, leaving

$$\Delta_l = \sum_{i=1}^{l} L^{l-i} D_i, \qquad D_i = \nu_{i,1}(\mathfrak{c}_{i-1}^+) - \nu_{i,1}(\mathfrak{c}_{i-1}^0). \tag{4.xxiii}$$

Insert the coordinates with their second block set to zero into each increment:

$$\begin{aligned} D_i = {} & [\nu_{i,1}(\mathfrak{c}_{i-1}^+) - \nu_{i,1}(R_2 \mathfrak{c}_{i-1}^+)] \\ & + [\nu_{i,1}(R_2 \mathfrak{c}_{i-1}^+) - \nu_{i,1}(R_2 \mathfrak{c}_{i-1}^0)] \\ & + [\nu_{i,1}(R_2 \mathfrak{c}_{i-1}^0) - \nu_{i,1}(\mathfrak{c}_{i-1}^0)]. \end{aligned}$$

The first and third brackets are each bounded by $N_3 q_{i-1}$, by eq. 4.xii. The third bracket must be retained: even when the bare source J_2 is zero, the chain can have nonzero second-species mass corrections. The two coordinates in the middle bracket have the same bare coordinate, and their earlier first-species entries differ by Δ_l, $l < i$. Hence eq. 4.ii gives

$$\|D_i\| \le 2N_3 q_{i-1} + B_\nu^{(2)} \phi_{i-1}^2 \max_{l<i} \frac{\|\Delta_l\|}{B_\sigma \phi_l}. \tag{4.xxiv}$$

The maximum is zero at $i = 1$.

Let d_i be the supremum of $\|D_i\|$ on $\lambda_i \|J_2\| < r_M$, with the admissible J_1 fixed. The increment difference is holomorphic in J_2 and vanishes at $J_2 = 0$. Schwarz's lemma on each complex line through the origin, followed by eq. 4.xxiii, gives

$$\|D_i\| \le d_i \frac{\lambda_i \|J_2\|}{r_M}, \qquad \|\Delta_l\| \le \frac{\lambda_l \|J_2\|}{r_M} \sum_{i=1}^{l} d_i. \tag{4.xxv}$$

Here the identity $L^{l-i}\lambda_i = \lambda_l$ accounts for the scale factor in the second inequality.

We prove $d_i \le 3N_3 q_{i-1}$ by induction. The scale comparison from [QCDM1, Prop. 10.1] gives $\phi_{i'-1} \le q_\phi^{l-i'} \phi_{l-1}$ with $q_\phi^2 \le 4/L$. The exponential in $q_{i'-1}$ is at most the corresponding exponential in q_{l-1}. It follows that

$$\sum_{i'=1}^{l} q_{i'-1} \le \frac{q_{l-1}}{1 - 4/L}. \tag{4.xxvi}$$

When $\lambda_i \|J_2\| < r_M$ and $l < i$, the earlier scale satisfies $\lambda_l \|J_2\| < r_M/L$. The induction hypothesis and eq. 4.xxv therefore imply

$$\|\Delta_l\| \le \frac{3N_3}{L(1 - 4/L)} q_{l-1}.$$

Moreover,

$$\frac{q_{l-1}}{\phi_l} \le \phi_{i-1} \left(\phi_{i-1} + e^{-c_{\mathrm{t}} \ell/(2\lambda_i)} \right), \qquad l < i.$$

Thus the last term of eq. 4.xxiv is at most

$$\frac{3B_\nu^{(2)} \phi_{i-1} N_3}{L(1 - 4/L) B_\sigma} q_{i-1} \le N_3 q_{i-1}.$$

This proves the induction. Substitution into eq. 4.xxv and eq. 4.xxvi proves eq. 4.xxi; the other species is treated by interchanging the two blocks. The smallness conditions used here and below are

$$3B_\nu^{(2)}\phi_* \le L(1-4/L)B_\sigma, \qquad 2C_{\mathrm{s}3}\phi_* \le B_\sigma.$$

For the scalar estimate put $\Delta V = V(\mathfrak{c}') - V(\mathfrak{c})$. The two coordinates have the same n, so their fermion covariances, reconstruction maps, Gaussian terms and normalizing factors coincide. The potential can change only through the mass member and the old activities. The mass difference is bounded by

$$c'_\psi c_{\mathsf{S}} h_F^2 \|\sigma'_{j-1} - \sigma_{j-1}\| \le c'_\psi c_{\mathsf{S}} h_F^2 B_\sigma \phi_{j-1} T.$$

If $T = 0$ there is nothing to prove. Otherwise, the line

$$\mathfrak{c} + \zeta(\mathfrak{c}' - \mathfrak{c})/T, \qquad |\zeta| < 1,$$

lies in the doubled coordinate domain. On this line the old members with fermion variables have norm at most $c'_\psi \bar{\Sigma}\phi_{j-1}^2$, while their scalar members have norm at most $K_0^{\mathrm{s}}\delta_{j-1}$. Schwarz's lemma applied to each difference from its value at $\zeta = 0$ supplies the factor $2T$. Consequently

$$\begin{aligned}\|\Delta V\| &\le \left(c'_\psi c_{\mathsf{S}} h_F^2 B_\sigma \phi_{j-1} + 2c'_\psi \bar{\Sigma}\phi_{j-1}^2 + 2K_0^{\mathrm{s}}\delta_{j-1}\right) T \\ &\le C_{\mathrm{Lip}}\phi_{j-1} T,\end{aligned}$$

where we used $\phi_{j-1} \le 1$ and $\delta_{j-1} \le \sqrt{2}A_1\phi_{j-1}$.

At the trivial pair the potential $V(\mathfrak{c})$ has norm at most $11c_{\mathrm{cl}}/16$. On the disc

$$|v| < \frac{c_{\mathrm{cl}}}{4\|\Delta V\|},$$

the potential $V(\mathfrak{c}) + v\Delta V$ has norm at most $15c_{\mathrm{cl}}/16$. The tail has norm at most $c_{\mathrm{cl}}/16$, so the cluster theorem applies throughout this disc. The scalar total is holomorphic there and bounded by $K_M c_{\mathrm{cl}} n_j M^{-3}$: the local cluster bound is summed over the $n_j M^{-3}$ cubes. Schwarz's lemma for its difference from the value at $v = 0$ gives

$$|f_j(\mathfrak{c}') - f_j(\mathfrak{c})| \le 2K_M c_{\mathrm{cl}} n_j M^{-3} \frac{4\|\Delta V\|}{c_{\mathrm{cl}}}.$$

If the displayed radius is at most one, the same estimate follows from the individual bounds at the two admissible endpoints. Substituting the bound for $\|\Delta V\|$ proves eq. 4.xxii. □

Proof of Prop. 3.4. Let $\mathfrak{c}_{j-1}(J_1, J_2)$ be the source-chain coordinate, and write

$$G_j(J_1, J_2) = D^{\mathrm{mix}}[f_j \circ \mathfrak{c}_{j-1}](J_1, J_2).$$

This is the sum over polymers of the mixed step energy. The bound for $R^{\mathrm{mix}} f_j$ does not apply to G_j directly: a single-source part of f_j can still depend on the other bare source through the mass corrections in its own species. We compare with a coordinate in which the two species depend on separate sources.

Define $\mathfrak{c}^\flat(J_1, J_2)$ by taking every first-species entry from $\mathfrak{c}_{j-1}(J_1, 0)$ and every second-species entry from $\mathfrak{c}_{j-1}(0, J_2)$. It is in the inner domain, and its bare coordinate agrees with that of $\mathfrak{c}_{j-1}(J_1, J_2)$. In the decomposition

$$f_j = f_j(0) + R^{(1)} f_j + R^{(2)} f_j + R^{\mathrm{mix}} f_j,$$

the composition of $R^{(b)} f_j$ with $\mathfrak{c}^\flat$ depends only on J_b. Further, $R^{\mathrm{mix}} f_j(\mathfrak{c}^\flat)$ vanishes on each source axis, because the corresponding block of $\mathfrak{c}^\flat$ is taken from the zero-source chain. Consequently

$$D^{\mathrm{mix}}[f_j(\mathfrak{c}^\flat)] = R^{\mathrm{mix}} f_j(\mathfrak{c}^\flat).$$

At each corner of the mixed difference the original and separated coordinates have the same bare coordinate. At (J_1, J_2) their two mass corrections differ by

$$\sigma_{i,1}(J_1, J_2) - \sigma_{i,1}(J_1, 0), \qquad \sigma_{i,2}(J_1, J_2) - \sigma_{i,2}(0, J_2).$$

At $(J_1,0)$ only the second block differs, by $\sigma_{i,2}(J_1,0)$; at $(0,J_2)$ only the first block differs, by $\sigma_{i,1}(0,J_2)$. At $(0,0)$ there is no difference. The axis differences are also bounded by eq. 4.xxi, with the other source set to zero. Thus at each of the three nonzero corners the normalized coordinate difference satisfies

$$T \le \frac{C_{\mathrm{s3}}}{B_\sigma} \max_{i\le j-1} \frac{q_{i-1}}{\phi_i}, \qquad \phi_{j-1} T \le \frac{C_{\mathrm{s3}}}{B_\sigma} q_{j-1}.$$

For the second inequality, use

$$\frac{q_{i-1}}{\phi_i} \le \phi_{i-1}\left(\phi_{i-1} + e^{-c_{\mathrm{t}}\ell/(2\lambda_i)}\right) \le \phi_{j-1}\left(\phi_{j-1} + e^{-c_{\mathrm{t}}\ell/(2\lambda_j)}\right).$$

The condition $2C_{\mathrm{s3}}\phi_* \le B_\sigma$ gives $T \le 1$. Apply eq. 4.xxii at the three corners and eq. 4.xi to $\mathfrak{c}^\flat$. Since $q_{j-1} \le \bar{q}_{j-1}$, we obtain

$$\begin{aligned} |G_j(J_1,J_2)| &\le E_3 M^{-3} n_j \bar{q}_{j-1} + 3\cdot 8 K_M M^{-3} n_j C_{\mathrm{Lip}} \frac{C_{\mathrm{s3}}}{B_\sigma} q_{j-1} \\ &\le C_{\mathrm{gm3}} M^{-3} n_j \bar{q}_{j-1}, \end{aligned}$$

where

$$C_{\mathrm{gm3}} = E_3 + 24 K_M C_{\mathrm{Lip}} C_{\mathrm{s3}}/B_\sigma.$$

This estimate holds throughout $\lambda_j \|J_b\| < r_M$. For nonzero J_1, J_2, set

$$G(\zeta_1,\zeta_2) = G_j(\zeta_1 J_1, \zeta_2 J_2), \qquad R_b = \frac{r_M}{\lambda_j \|J_b\|}.$$

The function G is holomorphic on $|\zeta_b| < R_b$ and zero on both coordinate axes. First apply Schwarz's lemma in ζ_1. The quotient G/ζ_1 has a holomorphic extension through $\zeta_1 = 0$, is bounded by $\sup|G|/R_1$, and is zero when $\zeta_2 = 0$. A second application in ζ_2 gives

$$|G_j(J_1,J_2)| \le C_{\mathrm{gm3}} M^{-3} n_j \bar{q}_{j-1} \frac{\lambda_j\|J_1\|}{r_M} \frac{\lambda_j\|J_2\|}{r_M}.$$

If either source is zero, this inequality is immediate. Since the energy contribution is $f_j(\mathfrak{c}_{j-1}(J)) - f_j(0)$, the inequality is eq. 3.xiv.

For the sum over scales, the two source factors supply λ_j^2, whereas the number of sites contributes $n_j = \ell^3 \lambda_j^{-3}$. Their product leaves one inverse length:

$$n_j \lambda_j^2 = \ell^3 \lambda_j^{-1} = \ell^3 L^{K-j}.$$

Since $\rho_*/r_M \le (2L)^{-1}$, it follows that

$$|E_K^{\mathrm{mix}}(J_1,J_2)| \le C_{\mathrm{gm3}} M^{-3} (2L)^{-2} \ell^3 \sum_{j=1}^{K} L^{K-j} \bar{q}_{j-1}. \tag{4.xxvii}$$

The three terms in $\bar{q}_{j-1}$ are summed separately. Put $n = K - j$. Then $\mathsf{t}_{j-1} = L^{n+1}\mathsf{b}$ and

$$x_{j-1} \ge \mathsf{t}_{j-1} - \bar{B} \ge \tfrac{1}{2} L^{n+1} \mathsf{b},$$

because $\mathsf{b} > \gamma^{-2} \ge L\bar{B}$. The function $\phi(t) = t^{-1/2}(1+\log t)^{p_0}$ is decreasing on $[e^{2p_0}, \infty)$, which contains $[\frac{1}{2}\gamma^{-2}, \infty)$ by the choice of γ. Consequently

$$\begin{aligned} \phi_{j-1} &\le \phi(\tfrac{1}{2} L^{n+1}\mathsf{b}) \\ &\le \sqrt{2} L^{-(n+1)/2} \phi(\mathsf{b}) (1 + (n+1)\log L)^{p_0}. \end{aligned}$$

For the second inequality use $1 + a + b \le (1+a)(1+b)$ for nonnegative a, b on the logarithmic factor. Since $\phi(\mathsf{b}) \le \phi_* \le 1$,

$$L^n \phi_{j-1}^3 \le 2^{3/2} L^{-3/2} L^{-n/2} (1 + (n+1)\log L)^{3p_0}.$$

The geometric factor $L^{-n/2}$ makes this series summable despite its logarithmic polynomial.

For the scalar covariance left at weight two, the definition of $\mathfrak{l}_{j-1}$ gives the exact identity

$$L^n \mathfrak{l}_{j-1}^2 = \frac{1}{L\mathsf{b}}(\log \mathsf{b} + (n+1)\log L)^{-2p_0} \le L^{-1}((n+1)\log L)^{-2p_0}.$$

The scale powers cancel. Since $2p_0 > 1$, the upper bounds sum to

$$\frac{\zeta(2p_0)}{L(\log L)^{2p_0}}.$$

Finally, $\ell \ge 2$ and $\phi_{j-1} \le 1$ imply

$$L^n \phi_{j-1}^2 e^{-c_{\mathrm{t}}\ell/(2\lambda_j)} = L^n \phi_{j-1}^2 e^{-c_{\mathrm{t}}\ell L^n/2} \le L^n e^{-c_{\mathrm{t}} L^n}.$$

This series is summable because its exponential contains L^n. Thus all three contributions in eq. 4.xxvii are bounded uniformly in K and ℓ. One may take

$$S_{\mathrm{mix}} = 2^{3/2} L^{-3/2} \sum_{n\ge 0} L^{-n/2}(1+(n+1)\log L)^{3p_0} + \frac{\zeta(2p_0)}{L(\log L)^{2p_0}} + \sum_{n\ge 0} L^n e^{-c_{\mathrm{t}} L^n},$$
$$C_{\mathrm{mix}} = C_{\mathrm{gm3}} M^{-3} (2L)^{-2} S_{\mathrm{mix}}.$$

This proves eq. 3.xiii. □

Order of the constants. The mass window remains fixed. The new constants are chosen after the mass-independent source norm and localization constants of [QCDM1, Prop. 10.20], and before the final choice of γ. First choose $K_0^{\mathrm{s}}, C_r^{\mathrm{s}}, C_2^{\mathrm{s}}, B_\nu^{(2)}$; then $\bar\Sigma, V_1, C_{\mathrm{gm}}, v_{\mathsf{S}}, C_r^{\mathrm{b}}$; then $C_{\mathfrak{d}}, A_3, E_3, N_3, C_{\mathrm{s}3}, C_{\mathrm{Lip}}$ and C_{mix}. All are functions of constants already chosen. In addition to the three smallness conditions in the proof of Prop. 4.1, impose

$$C_r^{\mathrm{b}}\phi_* \le \tfrac12, \qquad \tfrac{17}{2}\, 16 K_M^{\bullet} C_\Lambda \phi_* \le \tfrac12, \qquad 3B_\nu^{(2)}\phi_* \le L(1-4/L)B_\sigma, \qquad 2C_{\mathrm{s}3}\phi_* \le B_\sigma.$$

They hold after decreasing γ, since $\phi_* \to 0$. They do not depend on the cutoff, volume, source or scale.

5. Scalar mesons and diquarks

We retain the setting of section 2 and write $D_U = D[U] + m_0$ for the massive operator. Constants may depend on the fixed coupling interval and mass window, but are independent of the cutoff, volume and infinite-volume state. In this section and the next two, sites are labelled by $x \in a\mathbb{Z}^3$. Thus (x,q) corresponds to the place $(a^{-1}x, q)$ in eq. 2.i, and x_v denotes the physical site of v.

The mesons $\bar\psi X \psi$ do not form an invariant family under all internal symmetries of the model. Pseudoreality of the colour representation permits a symmetry to replace a quark by a charge-conjugate antiquark. We first include the resulting diquarks among the fields, and then prove estimates for arbitrary joint correlations of this larger family.

5.1. **The labels and the exact symmetries.** Put $\epsilon = i\sigma_2$ on colour and $C_0 = \sigma_2 \otimes \epsilon$ on spin and colour. The identities

$$C_0^{\mathsf{T}} = C_0, \qquad C_0^{-1} = -C_0, \qquad C_0^{\dagger} = -C_0, \qquad \sigma_\mu^{\mathsf{T}} C_0^{-1} = -C_0^{-1}\sigma_\mu$$

will be used below. With $\chi = C_0 \bar\psi^{\mathsf{T}}$, write $\widehat\Psi = (\psi, \chi)$. The internal label space of this column is

$$W = \mathbb{C}^2_\tau \otimes \mathbb{C}^2_{\mathrm{d}} \otimes \mathbb{C}^{N_f}.$$

The second factor distinguishes ψ from χ. Let $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ on this factor, let $\mathcal{J} = 1 \otimes J \otimes 1$, and let $P_\pm = (1 \pm \tau_3)/2$. The complex label space and its reflection-real form are

$$\mathscr{S} = \{M \in \operatorname{End} W : \mathcal{J} M^{\mathsf{T}} \mathcal{J}^{-1} = M\}, \qquad M^* = \tau_3 M^\dagger \tau_3, \qquad \mathscr{S}_{\mathbb{R}} = \{M \in \mathscr{S} : M^* = M\}. \quad (5.\mathrm{i})$$

Thus $\mathcal{J}M$ is antisymmetric and $\dim_{\mathbb{C}} \mathscr{S} = 2N_f(4N_f-1)$. For X on $\mathbb{C}^2_\tau \otimes \mathbb{C}^{N_f}$, let $\widehat{X}$ act as X on the ψ components and as X^{T} on the χ components. It belongs to $\mathscr{S}$. When this notation is applied to a meson label Γ, we suppress its identity factors on spin and colour.

For a chamber v over the physical site x_v, define

$$\phi[M](v) = \frac{1}{2}\widehat{\Psi}(v)^{\mathsf{T}}(\mathcal{J}M \otimes C_0^{-1})\widehat{\Psi}(v), \qquad \Phi[M](f) = a\sum_x f(x) \sum_{v: x_v = x} \phi[M](v).$$

In particular $\phi[\widehat{X}] = \bar{\psi} X \psi$. The blocks of $\mathcal{J}M$ with two ψ indices give the diquarks; those with two χ indices give the antidiquarks. The label 0 will continue to denote the action-density field $O(f)$ and is not the zero matrix in $\mathscr{S}$.

Let

$$G_{\mathbb{C}} = \mathrm{Sp}(2N_f, \mathbb{C})^2, \qquad G = \mathrm{USp}(2N_f)^2. \tag{5.ii}$$

Each factor acts on $P_\pm W$ with its symplectic form $J \otimes 1$. We regard (S_+, S_-) as the block diagonal matrix $S = P_+S_+ + P_-S_-$ on W, and define the algebra automorphism α_S by $\alpha_S \widehat{\Psi}(v) = S^{-1}\widehat{\Psi}(v)$. It acts trivially on functions of the links.

Proposition 5.1 (The doubled field and its symmetries)**.** At every cutoff the following identities hold on a torus and for every infinite-volume state used in [QCDM1]. For $S \in G_{\mathbb{C}}$,

$$\alpha_S \phi[M] = \phi[SMS^{-1}], \qquad \omega \circ \alpha_S = \omega.$$

At a lattice reflection with mirror P,

$$\Theta_P \Phi[M](f) = \Phi[M^*](\overline{f} \circ \theta_P).$$

Every α_S with $S \in G$ commutes with Θ_P. All these fields are gauge invariant and even.

On a finite torus, let $\widehat{D}_U = \sum_{h=+,-} P_h \otimes D_{U,h}$ be the massive operator on the doubled space and let M be a function of the chambers with values in $\mathscr{S}$. Its Berezin integral

$$Z_U[M] = \int d\bar{\psi}\, d\psi \, \exp\left(-\bar{\psi} D_U \psi - \sum_v \phi[M(v)](v)\right)$$

is a polynomial satisfying

$$Z_U[M]^2 = \det(\widehat{D}_U + M), \qquad Z_U[0] = \det D_U.$$

These identities also hold for complex link variables whenever the displayed expressions are defined. ∎

Proof. The relation $\epsilon u^{\mathsf{T}} \epsilon^{-1} = u^{-1}$ for $u \in \mathrm{SL}(2,\mathbb{C})$ and the Pauli identities give $C_0 D_{U,h}^{\mathsf{T}} C_0^{-1} = D_{U,h}$, with the transpose also interchanging the chamber arguments. Consequently $K_h = C_0^{-1} D_{U,h}$ is symmetric. Since odd columns satisfy $b^{\mathsf{T}} A a = -a^{\mathsf{T}} A^{\mathsf{T}} b$,

$$\bar{\psi}_h D_{U,h} \psi_h = \frac{1}{2}\widehat{\Psi}_h^{\mathsf{T}}(J \otimes K_h)\widehat{\Psi}_h.$$

The full quadratic coefficient is therefore $(\mathcal{J} \otimes C_0^{-1})(\widehat{D}_U + M)$, which is antisymmetric. Its Berezin integral is a fixed sign times its Pfaffian. Squaring gives the determinant identity; the normalization at $M = 0$ fixes the square of the sign. For $\widehat{X}$, the two off-diagonal blocks of $\mathcal{J}\widehat{X}$ are $-X^{\mathsf{T}}$ and X. Reversing the odd variables in the first term gives $\phi[\widehat{X}] = \chi^{\mathsf{T}} C_0^{-1} X \psi = \bar{\psi} X \psi$.

The identity $S^{\mathsf{T}} \mathcal{J} S = \mathcal{J}$ gives $S^{-\mathsf{T}} \mathcal{J} = \mathcal{J} S$. Hence substitution of $S^{-1}\widehat{\Psi}$ fixes the quadratic action, because S commutes with $P_\pm$, and carries the source coefficient $\mathcal{J}M$ into $\mathcal{J}SMS^{-1}$. Its Berezin Jacobian is one. This proves invariance on a torus. On the infinite lattice the doubled Gaussian functional has covariance

$$\sum_h (-P_h \mathcal{J}) \otimes D_{U,h}^{-1} C_0.$$

The same symplectic identity preserves this covariance and therefore every Wick contraction. Averaging in U proves the assertion for the state.

If $\hat{n}$ is the unit normal to P, direct substitution in the reflection of the generators gives

$$\Theta_P \widehat{\Psi}(v) = (\mathcal{J}\tau_3 \otimes C_0^{-1}\sigma(\hat{n}))\widehat{\Psi}(\theta_P v).$$

For an antilinear map reversing the order of odd factors, the quadratic coefficient A is changed to $B^{\mathsf{T}} A^{\dagger} B$, where B is the displayed matrix. Inserting $A = \mathcal{J}M \otimes C_0^{-1}$ gives $\mathcal{J}(\tau_3 M^{\dagger}\tau_3) \otimes C_0^{-1}$. The same computation on the generators gives $\Theta_P \alpha_S \Theta_P = \alpha_{(S^{\dagger})^{-1}}$, which equals α_S for $S \in G$.

Under a colour transformation, χ transforms by the same matrix as ψ, because $\epsilon u^{-\mathsf{T}} \epsilon^{-1} = u$. The form C_0^{-1} is invariant under this transformation, so every $\phi[M]$ is gauge invariant. Each is quadratic in the generators and hence even. □

Lemma 5.2 (A basis obtained from mesons)**.** The real space $\mathscr{S}_{\mathbb{R}}$ has a basis

$$\mathfrak{B}^{\mathrm{ext}} = \{S_b \widehat{\Gamma}_b S_b^{-1} : b = 1, \dots, 2N_f(4N_f - 1)\}, \qquad S_b \in G, \quad \Gamma_b \in \mathfrak{B}.$$

The matrices in this basis have operator norm at most one. There is a constant $c(N_f)$ such that the sum of the absolute values of the coefficients of any $M \in \mathscr{S}$ in this basis is at most $c(N_f)\|M\|$. ■

Proof. Let V be the real span of the displayed symplectic images of meson basis elements. Unitarity and $[S, \tau_3] = 0$ imply $(SMS^{-1})^* = SM^*S^{-1}$, so $V \subset \mathscr{S}_{\mathbb{R}}$. We show directly that $V + iV = \mathscr{S}$. Write $r = (h, f)$ for a species–flavour index. Under $M \mapsto \mathcal{J}M$, the meson labels give all blocks with one ψ index and one χ index. Their matrix units are complex linear combinations of the meson basis. The remaining blocks are antisymmetric forms in two ψ indices or two χ indices, hence are spanned by pairs $r \neq s$. For such a pair let $S \in G$ be J on the plane spanned by ψ_r, χ_r and the identity elsewhere. Since $S^{-1} = -J$ on that plane, $\alpha_S \psi_r = \chi_r$ and $\alpha_S \chi_r = -\psi_r$. Thus

$$\alpha_S(\bar{\psi}_r \psi_s) = -\psi_r^{\mathsf{T}} C_0^{-1} \psi_s.$$

Applying partial charge conjugation to the other index instead gives the corresponding antidiquark. These images span every remaining block of $\mathcal{J}M$, proving the claim.

For the real-form conclusion, write any $M \in \mathscr{S}_{\mathbb{R}}$ as $v + iw$, with $v, w \in V$. Since $v^* = v$, $w^* = w$ and $M^* = M$, we have $w = 0$. Thus $V = \mathscr{S}_{\mathbb{R}}$. Select a basis from the spanning set. Unitary conjugation preserves operator norm, and the coefficient estimate is equivalence of norms on this fixed finite-dimensional space. □

5.2. **Uniform estimates for the enlarged family.** Put $\mathcal{I}^{\mathrm{ext}} = \{0\} \cup \mathfrak{B}^{\mathrm{ext}}$, $\|f\|_0 = \|f\|_\sharp$ and $\|f\|_M = \|f\|_\infty$ for $M \neq 0$. An extended tuple is a list $\mathcal{X} = ((\lambda_i, f_i))_{i=1}^n$ with these labels; its truncated function is the cumulant of the indicated fields. We extend it complex multilinearly in the labels and functions. Write

$$p = 432 C_{\mathscr{C}} / \log L, \qquad \kappa = 5 + p/2.$$

Here $C_{\mathscr{C}}$ is the source-energy counting constant in [QCDM1, Thm. F].

Proposition 5.3 (All scalar bilinears at separated supports)**.** For every $n \geq 2$, $R, d > 0$ and extended tuple whose real Lipschitz functions have supports in balls of radius R, at pairwise distances at least d,

$$|S_n^T(\mathcal{X})| \leq [B_{\mathrm{ext}}(R)(1 + d^{-1})^{\kappa - 1}]^n (n!)^{\kappa} \prod_i \|f_i\|_{\lambda_i}. \tag{5.iii}$$

This holds on every torus of the construction and in every fixed-cutoff infinite-volume state obtained from these tori. The constant $B_{\mathrm{ext}}(R)$ is independent of cutoff and volume. For arbitrary labels in $\mathscr{S}$, the right side has the additional factor $\prod_{i:\lambda_i \neq 0} c(N_f)\|\lambda_i\|$. ■

Proof. We first explain why the one-cell estimate of [QCDM1, Prop. 10.17] applies to a symplectic image of a meson and to cells of side less than one. Let a torus be divided into cubes of physical side $b = L^{-q}$, with an even number of cubes in each direction and walls through links. For a source s supported in one cube at distance $\delta > 0$ from its walls, the same fold used in that proposition has $\mathsf{N} = \ell^3/b^3$ copies. The upper and lower partition estimates cost $c_V \ell^3/\mathsf{N} = c_V b^3$ after taking the Nth root. The separated-energy estimate gives the same counting term $\mathscr{C}_K(\operatorname{supp} s; \delta)$; its proof is local at the support and uses the distance to the walls. Thus, writing $\lambda^{\mathfrak{b}}$ for the average of the reflected single-source energies,

$$\|e^{O(w)+\Phi_\Gamma(c)}\|_{\mathrm{ch}} \le \exp\left(\operatorname{Re}\lambda^{\mathfrak{b}}(w, c\Gamma) + c_V b^3 + \mathscr{C}_K(\operatorname{supp}(w,c); \delta)\right). \tag{5.iv}$$

The required source domains are unchanged, since reflection preserves the source norms and the source vanishes near the walls. To specify the seminorm, let $\operatorname{Fold} G$ be the product of the reflected copies of an even element G from one cell into all N cells. For the normalized torus functional ω_ℓ, put

$$\|G\|_{\mathrm{ch}} = \omega_\ell(\operatorname{Fold} G)^{1/\mathsf{N}}.$$

Reflection positivity makes the quantity under this root nonnegative. The multiple-cell chessboard inequality is

$$\left|\omega_\ell\left(\prod_i G_i\right)\right| \le \prod_i \|G_i\|_{\mathrm{ch}},$$

when the G_i belong to distinct cells; the unused cells contain 1. This is the same inequality as in [QCDM1, Prop. 10.17 and its proof], with the finer board.

For $M = S\widehat{\Gamma}S^{-1}$ in the chosen basis,

$$e^{O(w)+\Phi[M](c)} = \alpha_S(e^{O(w)+\Phi_\Gamma(c)}).$$

The automorphism α_S commutes with every reflection in the fold and preserves the torus functional. Therefore the two exponentials have equal chessboard seminorms, and eq. 5.iv bounds the extended source too. To apply the multiple-cell inequality, first bound each individual seminorm in this way and then multiply the bounds. Thus each cell may use its own matrix S: each individual fold is compared with its meson fold by one global symmetry.

Choose $q \ge 0$ least such that $\sqrt{3}L^{-q} < d$, and set $\eta = L^{-q}/(16n)$. Then

$$\eta^{-1} \le 16\sqrt{3}Ln(1 + d^{-1}).$$

First suppose $L^{K-q} \ge 16n$. Split each f_i by the product hat partition at mesh η. There are at most $(2R/\eta + 3)^3$ pieces for each entry. Each is supported in a cube of side 2η, its supremum norm does not increase, and its sharp norm increases by at most $1 + 80M/\eta$. Fix one piece from each entry. In each coordinate a piece forbids at most $4\eta L^K + 1$ positions for a wall of a board whose cells have $\Delta = L^{K-q}$ sites on a side. The n pieces forbid at most $\Delta/4 + n < \Delta$ positions. Choose walls at the remaining half-integral positions. They are at distance at least η from all pieces. The diameter of a cell is less than d, so no cell contains pieces from two different entries.

For the chosen pieces, let s_i be the corresponding meson source before its symplectic transformation, and let $\Lambda(z) = \sum_i \lambda^{\mathfrak{b}}(z_i s_i)$. The extended generating function is entire, and the chessboard estimate gives, on each coordinate face indexed by $A \subset \{1, \dots, n\}$,

$$\left|e^{-\Lambda(z)}\frac{Z^{\mathrm{ext}}(z)}{Z}\right| \le \exp\left(|A|c_V + \sum_{i\in A}\mathscr{C}_K(\operatorname{supp} s_i; \eta)\right), \qquad |z_i| < \frac{2\varrho_{\lambda_i}}{\|f_i\chi_i\|_{\lambda_i}}.$$

Here $\varrho_0 = \varrho$, $\varrho_M = \varrho_J$ for $M \ne 0$, and $b^3 \le 1$. Since every term of Λ depends on one variable, its mixed derivative in all $n \ge 2$ variables vanishes. The logarithmic derivative formula, as in the proof

of [QCDM1, Thm. F], is

$$\partial_{[n]} \log Z^{\mathrm{ext}}(0) = \sum_{\pi} (-1)^{|\pi|-1} (|\pi|-1)! \prod_{A \in \pi} \partial_A \left(e^{-\Lambda} Z^{\mathrm{ext}} / Z \right)(0).$$

The sum of the absolute combinatorial coefficients is at most $2^n n!$. Writing $G = e^{-\Lambda} Z^{\mathrm{ext}}/Z$, Cauchy's estimate on smaller coordinate circles, followed by the limit as their radii increase to the displayed radii, gives

$$|\partial_A G(0)| \le \exp\left(|A| c_V + \sum_{i \in A} \mathscr{C}_K(\operatorname{supp} s_i; \eta) \right) \prod_{i \in A} \frac{\|f_i \chi_i\|_{\lambda_i}}{2\varrho_{\lambda_i}}.$$

In the product over a partition, each index occurs exactly once. Hence both the source counts and the inverse Cauchy radii have the same total cost for every partition. The refined source count of that theorem gives

$$\mathscr{C}_K(\operatorname{supp} s_i; \eta) \le 216 C_{\mathscr{C}} (1 + \log_L(1/\eta)) + C,$$

because the support has diameter bounded by a fixed multiple of η. Hence its exponential contributes at most $C^n \eta^{-np/2}$.

Summing over the pieces and restoring the original test functions therefore gives

$$|S_n^T(\mathcal{X})| \le n! \left[\frac{C e^{c_V}}{\varrho_{\min}} (2R/\eta + 3)^3 (1 + 80M/\eta) \eta^{-p/2} \right]^n \prod_i \|f_i\|_{\lambda_i}.$$

Here $\varrho_{\min} = \min(\varrho, \varrho_J)$. The three powers of η^{-1} count the pieces, one comes from their sharp norms, and $p/2$ comes from the source-energy count. Since $\eta^{-1} \le Cn(1 + d^{-1})$, the right-hand side is at most

$$\left[C'(1+R)^3 (1+d^{-1})^{4+p/2} \right]^n n!\, n^{(4+p/2)n} \prod_i \|f_i\|_{\lambda_i}.$$

Applying $n^n \le e^n n!$ gives the factorial exponent $1 + (4 + p/2) = 5 + p/2$ in eq. 5.iii.

If $L^{K-q} < 16n$, then $a^{-1} < 16\sqrt{3} Ln(1 + d^{-1})$. The Pfaffian identity above and $\|\widehat{D}_U^{-1}\| \le m_0^{-1}$ give

$$|Z_U[M]| \le \det D_U \, \exp\left(\frac{\|M\|_1}{2m_0} \right).$$

For the source of the extended tuple, $M(v) = -a \sum_i z_i f_i(x_v) \lambda_i$; the doubled fibre has $16N_f$ components and there are 48 chambers per site. Thus the fermion contribution is bounded by $384 N_f m_-^{-1} \sum_x \| \sum_i z_i f_i(x) \lambda_i \|$. The action contribution is bounded by $2a^{-1} \sum_p |w(p)|$. A radius-R support has at most $27a^{-3}(1+R)^3$ sites and $81a^{-3}(1+R)^3$ plaquettes. Since $a \le 1$, both costs are bounded by

$$B_a(R) = C(1 + N_f/m_-)(1+R)^3 a^{-4}$$

times the appropriate test-function norm. Set $r_i = (B_a(R) \|f_i\|_{\lambda_i})^{-1}$, omitting zero test functions. On the face A, the normalized generating function has modulus at most $e^{|A|}$ for $|z_i| \le r_i$. Cauchy's estimate and the cumulant partition formula give

$$|S_n^T(\mathcal{X})| \le 2^n n! e^n B_a(R)^n \prod_i \|f_i\|_{\lambda_i}.$$

Use $a^{-1} < Cn(1 + d^{-1})$ and $n^{4n} \le e^{4n} (n!)^4$ to obtain

$$|S_n^T(\mathcal{X})| \le [C(1 + N_f/m_-)(1+R)^3 (1+d^{-1})^4]^n (n!)^5 \prod_i \|f_i\|_{\lambda_i}.$$

This is bounded by the claimed estimate. Multilinearity and Lem. 5.2 give arbitrary labels.

Finally the expectation of any fixed local Grassmann polynomial converges along the tori defining a state, as proved in [QCDM1, Prop. 2.5]. A cumulant is a polynomial in finitely many such expectations. Passing to the limit proves the same estimate in the infinite-volume state. □

5.3. **Bilinears on disjoint supports.** The separation loss in Prop. 5.3 grows with the number of insertions. For scalar bilinears it can be reduced to one factor, and removed altogether at orders at least four. The reason is that the logarithm of the conditional Pfaffian has a summable remainder after degree three. We control the first three degrees by meson source energies, subtract them, and apply the chessboard estimate to the resulting function.

Theorem F (The bound on disjoint supports)**.** Let $n \geq 2$, and let $f_1, \dots, f_n$ be bounded real functions with pairwise disjoint supports, each contained in a ball of radius R. For $\lambda_i \in \mathscr{S}$,

$$|S_n^T((\lambda_i, f_i)_i)| \leq B_D(R)^n (n!)^{4+p/2} \prod_i \|\lambda_i\| \, \|f_i\|_\infty \qquad (n \geq 4).$$

For $n = 2, 3$, if two of the supports are at distance at least $d > 0$, the same bound holds with the additional factor $(1 + d^{-1})^3$. The constants are independent of the cutoff and volume. The bounds hold on the tori, in their fixed-cutoff infinite-volume limits, and in the extended continuum limits of Prop. 5.18. For the latter, take the Lipschitz test functions used to define those limits. ■□

We prove the estimate using the gauge marginal ν_K and write E for its expectation. Let H_1 be the one-species, one-flavour space of antiperiodic fields with spin and colour, and put

$$G = \mathsf{G}_+, \qquad \mathsf{G}_- = G^*, \qquad N_4 = \operatorname{Tr}(GG^*)^2, \qquad \widehat{G} = \sum_{h=\pm} P_h \otimes \mathsf{G}_h.$$

Multiplication by a site function acts on all the chambers over that site. The norm of a matrix-valued site function is its supremum norm. Define

$$R_U(\mathbb{M}) = Z_U[-a\mathbb{M}]/Z_U[0], \qquad W_U^{(q)}(\mathbb{M}) = -\frac{1}{2q} \operatorname{Tr}(\widehat{G}\mathbb{M})^q.$$

Lemma 5.4 (The Pfaffian at one configuration)**.** The operator G is accretive and

$$\|G\| \leq \mathsf{m}^{-1}, \qquad N_4 \leq 3\|G^2\|_2^2.$$

For $\|\mathbb{M}\| \leq \mathsf{m}/2$,

$$R_U(\mathbb{M})^2 = \det(1 - \widehat{G}\mathbb{M}), \qquad |R_U(\mathbb{M})| = \exp \operatorname{Re} \sum_{q \geq 1} W_U^{(q)}(\mathbb{M}), \qquad \left| \sum_{q \geq 4} W_U^{(q)}(\mathbb{M}) \right| \leq N_f \|\mathbb{M}\|^4 N_4.$$

The trilinear form $\operatorname{Tr}(\mathsf{G}_h u \mathsf{G}_h v \mathsf{G}_h w)$ is symmetric in the scalar site functions u, v, w. ■

Proof. The real part of $D_+[U] + m_0$ is at least m_0. Its inverse is accretive, since, with $v = (D_+[U] + m_0)^{-1}u$, $\operatorname{Re}\langle u, Gu\rangle = a \operatorname{Re}\langle (D_+[U] + m_0)v, v\rangle \geq 0$. Write $G = X + iY$, with $X \geq 0$ and X, Y Hermitian, and set $S = X^2 + Y^2$, $K = i[X, Y]$. Cyclicity gives

$$N_4 = \operatorname{Tr} S^2 + \operatorname{Tr} K^2, \qquad \|G^2\|_2^2 = \operatorname{Tr} S^2 - \operatorname{Tr} K^2.$$

Moreover

$$\operatorname{Tr} K^2 = 2 \operatorname{Tr} X^2Y^2 - 2\|X^{1/2}YX^{1/2}\|_2^2 \leq \tfrac{1}{2} \operatorname{Tr} S^2,$$

where the last inequality follows from $\operatorname{Tr}(X^2 - Y^2)^2 \geq 0$. This proves the factor three.

The Pfaffian identity is Prop. 5.1 with source $-a\mathbb{M}$. Since $\|\widehat{G}\mathbb{M}\| \leq 1/2$, its logarithm is the convergent trace series. For $q \geq 4$, the Schatten inequality gives

$$|\operatorname{Tr} A^q| \leq \|A\|^{q-4} \|A\|_4^4, \qquad \|\widehat{G}\|_4^4 = 4N_f N_4.$$

Summing $\sum_{q \geq 4} 2^{4-q}/(2q) \leq 1/4$ proves the remainder bound. Finally the transpose identity $\mathsf{G}_h^{\mathsf{T}} = C_0^{-1} \mathsf{G}_h C_0$ in the proof of Prop. 5.1, together with cyclicity, reverses the order of the three multiplication operators. Cyclic permutations and reversal generate all permutations. □

For scalar site functions a, b, c, d, use the first flavour and write

$$J[a,b,c,d] = \begin{pmatrix} a & b \\ c & d \end{pmatrix}_\tau \otimes E_{11}.$$

Then $\|J[a,b,c,d]\| \le \max(\|a\|,\|d\|) + \max(\|b\|,\|c\|)$, and the source conjugation is $J[a,b,c,d]^\star = J[\bar d, \bar b, \bar c, \bar a]$. Set

$$\mathscr{E}(J) = E_K(0,J) - E_K(0,0), \qquad \overline{\mathscr{E}}(J) = \int_0^{2\pi} \mathscr{E}(V_\varphi J V_\varphi^{-1}) \frac{d\varphi}{2\pi}, \qquad V_\varphi = e^{i\varphi} e_+ + e_-.$$

The bar in $\overline{\mathscr{E}}$ denotes this average. For a holomorphic function F, write $F^{(q)}(J)$ for the coefficient of z^q in $F(zJ)$.

Lemma 5.5 (The energy coefficients)**.** For $\|J\| \le \rho_* = 2\varrho_J$,

$$\left|\frac{Z[0,J]}{Z[0,0]}\right| \le e^{\operatorname{Re}\mathscr{E}(J) + c_V \ell^3}, \qquad \left|\frac{Z[0,J]}{Z[0,0]}\right| \le e^{\operatorname{Re}\overline{\mathscr{E}}(J) + c_V \ell^3},$$

and

$$\sum_{q \ge 4} |\mathscr{E}^{(q)}(J)| \le 2C_4 \ell^3 (\|J\|/\rho_*)^4.$$

The latter bound also holds for $\overline{\mathscr{E}}$. Both energies respect $\star$-conjugation. In the Taylor series of $\overline{\mathscr{E}}(J[a,b,c,d])$, the degrees in b and c are equal. The coefficient of zw in $\mathscr{E}(ze_+A + we_-D)$ has modulus at most $C_{\mathrm{mix}}\ell^3$ when $\|A\|, \|D\| \le \rho_*$. ■

Proof. The first bound is the partition estimate used in Thm. B. The determinant is unchanged by $J \mapsto V_\varphi J V_\varphi^{-1}$. Averaging its logarithmic upper bound gives the second inequality. Conjugation of the local source energies gives the reality assertion. Averaging the phase kills every monomial whose degrees in b, c differ.

At level j the source energy is holomorphic for $\lambda_j \|J\| < r_M$ and bounded by $C_{\mathfrak{E}} e^{-\kappa d_j(X)}$. The inequality $\rho_* \le r_M/(2L)$ and Cauchy's formula bound the sum of degrees at least four by

$$C_{\mathfrak{E}} (1 - (2L)^{-1})^{-1} (2L)^{-4} (\|J\|/\rho_*)^4 \sum_{j,X} \lambda_j^4 e^{-\kappa d_j(X)}.$$

Use $\sum_X e^{-\kappa d_j(X)} \le C_E n_j M^{-3}$ and $\sum_j n_j \lambda_j^4 \le L\ell^3/(L-1)$. The definition of C_4 in Thm. C gives the claim. The mixed coefficient is a coefficient of E_K^{mix}; apply Prop. 3.4 and Cauchy's formula on the unit bidisc. The phase average preserves each bound. □

Put

$$\theta_q = \varrho_J^4/192,$$
$$\rho_q = \min\{2\varrho_J, (\theta_q N_f)^{1/4}\},$$
$$\rho_D = \frac{\varrho_J}{400 N_f^4},$$
$$c_N = c_V + C_{\mathrm{mix}}/8 + 2C_4,$$
$$c_D = c_V + C_{\mathrm{mix}} + 2C_4.$$

In particular $\rho_q \ge \varrho_J/4$.

Lemma 5.6 (A fourth Schatten moment)**.** At every cutoff and torus side, $\mathsf{E} e^{\theta_q N_f N_4} \le e^{c_N \ell^3}$. ■

Proof. Set $t = \varrho_J^2/16$ and use the flavour-scalar sources

$$J_\varphi = \begin{pmatrix} \varrho_J e^{i\varphi} & \sqrt{t} \\ -\sqrt{t} & \varrho_J e^{-i\varphi} \end{pmatrix} \otimes 1_{N_f}, \qquad J' = \begin{pmatrix} 0 & \sqrt{t} \\ \sqrt{t} & 0 \end{pmatrix} \otimes 1_{N_f}.$$

They are $\star$-real and have norm at most $5\varrho_J/4 < \mathsf{m}/2$. Their determinant quotients are positive: they are real by $\star$-conjugation, nonzero along the segment from zero, and equal to one at zero. With $G_\varphi = G(1-\varrho_J e^{i\varphi}G)^{-1}$, Schur complementation gives

$$R_U(\widehat{J}_\varphi) = \left[|\det(1-\varrho_J e^{i\varphi}G)|^2 \det(1+tG_\varphi^* G_\varphi)\right]^{N_f}, \quad R_U(\widehat{J}') = \det(1-tG^*G)^{N_f}.$$

The circle mean of the first logarithmic determinant is zero. Since $\varrho_J\|G\| \le 1/4$,

$$\operatorname*{avg}_\varphi \|G_\varphi\|_2^2 = \sum_{k\ge 0} \varrho_J^{2k}\|G^{k+1}\|_2^2 \ge \|G\|_2^2 + \tfrac{1}{3}\varrho_J^2 N_4, \qquad \|G_\varphi\|_4^4 \le (4/3)^4 N_4.$$

Apply $\log(1+y) \ge y - y^2/2$ and $\log(1-y) \ge -y-y^2$ for $0 \le y \le 1/2$. The terms involving $\|G\|_2^2$ cancel, leaving

$$\tfrac{1}{2}\operatorname*{avg}_\varphi \log R_U(\widehat{J}_\varphi) + \tfrac{1}{2}\log R_U(\widehat{J}') \ge \frac{N_f t}{2}(\varrho_J^2/3 - 209t/81)N_4 \ge \theta_q N_f N_4.$$

Hölder's inequality for this probability average bounds the expectation of its exponential by the exponential of the same average of logarithms of expectations. Use Lem. 5.5. In the averaged energy, all terms of degrees at most three vanish except half the coefficient of $y_1 y_4$ times ϱ_J^2: phase averaging kills unequal diagonal degrees, and the $y_2 y_3$ terms at the two sources cancel. That remaining term is at most $C_{\mathrm{mix}}\ell^3/8$. Degrees at least four cost $2C_4\ell^3$. This gives $c_N\ell^3$. The version of Hölder used here follows from finite weighted Hölder by Riemann sums; all functions are continuous on the compact finite-lattice gauge space. □

Define the following coefficients of the averaged energy: $\mathscr{E}_h^{(q)}(u) = \mathscr{E}^{(q)}(e_h u E_{11})$, and let $\mathscr{E}_{+-}(v,w)$, $\mathscr{E}_{++-}(u,v,w)$, $\mathscr{E}_{--+}(u,v,w)$ be the coefficients of bc, abc, dbc, respectively, in $\overline{\mathscr{E}}(J[a,b,c,d])$ at the indicated arguments. Thus, for example,

$$\mathscr{E}_{++-}(u,v,w) = \overline{\mathscr{E}}^{(3)}(J[u,v,w,0]) - \overline{\mathscr{E}}^{(3)}(J[u,0,0,0]).$$

Set

$$Q_h^{(q)}(u) = -q^{-1}\operatorname{Tr}(\mathsf{G}_h u)^q - \mathscr{E}_h^{(q)}(u), \qquad Q^{+-}(v,w) = \operatorname{Tr}(GvG^*w) + \mathscr{E}_{+-}(v,w),$$
$$Q^{++-}(u;v,w) = \operatorname{Tr}(GuGvG^*w) + \mathscr{E}_{++-}(u,v,w).$$

Lemma 5.7 (The first three loop coefficients)**.** For $q = 1,2,3$ and complex scalar site functions,

$$\mathsf{E} e^{\operatorname{Re}(\xi Q_h^{(q)}(u))} \le e^{(c_V + C_{\mathrm{mix}})\ell^3} \quad \text{if } |\xi|^{1/q}\|u\| \le 2\varrho_J.$$

Moreover

$$\mathsf{E} e^{\operatorname{Re}(\xi Q^{+-}(v,w))} \le e^{c_D\ell^3} \quad \text{if } (8|\xi|)^{1/2}\max(\|v\|,\|w\|) \le \rho_q,$$

and

$$\mathsf{E} e^{\operatorname{Re}(\xi Q^{++-}(u;v,w))} \le e^{c_D\ell^3} \quad \text{if } (8|\xi|)^{1/3}[\|u\| + \max(\|v\|,\|w\|)] \le \rho_q.$$

■

Proof. For the first bound take $h = +$ and $\xi = 1$ by homogeneity. The function $F(u) = \log\det(1-Gu) - \mathscr{E}(e_+ u E_{11})$ obeys $\mathsf{E} e^{2\operatorname{Re}F(e^{i\varphi}u)} \le e^{(c_V+C_{\mathrm{mix}})\ell^3}$. Indeed the positive determinant squared is the source $J[u,0,0,\bar{u}]$, and Prop. 3.4 controls its mixed energy. Average $2\operatorname{Re}F(e^{i\varphi}u)$ against $(1+\cos q\varphi)d\varphi/(2\pi)$. Only its qth coefficient survives. Weighted Hölder gives the assertion. Complex conjugation gives $h = -$.

For real v, w, take $J = J[0,v,w,0]$. The odd traces vanish and the second trace is $-\operatorname{Tr}(GvG^*w)$. By Lems. 5.4 and 5.5,

$$\mathsf{E} e^{-Q^{+-}(v,w) - \|J\|^4 N_4} \le e^{(c_V+2C_4)\ell^3}.$$

The same is true with v replaced by $-v$. If $\|J\| \le \rho_q$, Cauchy–Schwarz and Lem. 5.6 give both signs of $Q^{+-}/2$ with bound $e^{c_D\ell^3}$. For complex v, w, the real part of $\xi Q^{+-}(v,w)$ is the sum of four

such real bilinear terms. Hölder with four factors and rescaling both arguments by $(8|\xi|)^{1/2}$ give the stated radius.

For the third bound first take v, w real and $\rho = \|u\| + \max(\|v\|, \|w\|) \le \rho_q$. Use $J_{\varphi,\sigma} = J[e^{i\varphi}u, \sigma v, w, e^{-i\varphi}\bar{u}]$ with probability weight $(1 + \sigma\cos\varphi)d\varphi/(4\pi)$ on the circle times $\{1, -1\}$. In the trace expansion through degree three, the surviving cyclic words have species $(+,+,-)$ and $(-,-,+)$. Their sum is $-\operatorname{Re}\operatorname{Tr}(GuGvG^*w)$. In the energy the surviving terms are $\operatorname{Re}\mathscr{E}_{++-}(u,v,w)$ and the mixed quadratic diagonal coefficient, bounded by $C_{\mathrm{mix}}\ell^3$. The fourth-order remainders give $\rho^4 N_4$ and $2C_4\ell^3$. Thus

$$\mathsf{E}e^{-\operatorname{Re}Q^{++-}(u;v,w)-\rho^4 N_4} \le e^{c_D\ell^3}.$$

Use Cauchy–Schwarz and Lem. 5.6, then replace u by any phase multiple. Splitting each of v, w into real and imaginary parts gives four terms. Hölder and the rescaling $(8|\xi|)^{1/3}$ prove the claim. □

Choose orthonormal bases of P_hW, and write $m_{ss'}$ for the matrix entries of $\mathbb{M}$, with $h(s) = h$ on P_hW. Each set of indices has size $2N_f$. Define a polynomial $\Lambda(\mathbb{M})$ of degree at most three by the following rule. Expand

$$\sum_{q=1}^{3} W_U^{(q)}(\mathbb{M}) = -\sum_{q=1}^{3}\frac{1}{2q}\sum_{s_1,\dots,s_q}\operatorname{Tr}(\mathsf{G}_{h(s_1)}m_{s_1s_2}\cdots\mathsf{G}_{h(s_q)}m_{s_qs_1}).$$

For words in one species use polarization,

$$\operatorname{Tr}(\mathsf{G}_h u\mathsf{G}_h v) = \tfrac{1}{4}\{\operatorname{Tr}(\mathsf{G}_h(u+v))^2 - \operatorname{Tr}(\mathsf{G}_h(u-v))^2\},$$
$$\operatorname{Tr}(\mathsf{G}_h u\mathsf{G}_h v\mathsf{G}_h w) = \tfrac{1}{24}\sum_{\eta,\zeta=\pm1}\eta\zeta\operatorname{Tr}(\mathsf{G}_h(u+\eta v+\zeta w))^3.$$

Then make the replacements

$$-q^{-1}\operatorname{Tr}(\mathsf{G}_h u)^q \longmapsto \mathscr{E}_h^{(q)}(u),$$
$$\operatorname{Tr}(GuG^*v) \longmapsto -\mathscr{E}_{+-}(u,v),$$
$$\operatorname{Tr}(GuGvG^*w) \longmapsto -\mathscr{E}_{++-}(u,v,w).$$

For the word with two minus species use

$$\operatorname{Tr}(G^*uG^*vGw) = \overline{\operatorname{Tr}(G\bar{u}G\bar{w}G^*\bar{v})}$$

and the conjugate energy. These rules specify every term and its coefficient; $\Lambda(0) = 0$.

Proposition 5.8 (The normalized Pfaffian)**.** For $\|\mathbb{M}\| \le \rho_D$,

$$\mathsf{E}|R_U(\mathbb{M})| \le e^{\operatorname{Re}\Lambda(\mathbb{M})+c_D\ell^3}.$$

Its quadratic and cubic counterterms have the form

$$\Lambda^{(q)}(\mathbb{M}) = \sum_\rho \kappa_\rho \mathscr{F}_\rho^{(q)}(A_\rho\mathbb{M}), \qquad q = 2, 3,$$

where $\mathscr{F}_\rho$ is $\mathscr{E}$ or $\overline{\mathscr{E}}$, A_ρ acts pointwise and linearly, $\|A_\rho M\| \le 3\|M\|$, and

$$\sum_{\rho:q_\rho=2}|\kappa_\rho| \le 4N_f^2, \qquad \sum_{\rho:q_\rho=3}|\kappa_\rho| \le \tfrac{52}{3}N_f^3.$$

Here q_ρ is the degree of the corresponding term. ■

Proof. The replacement rule writes $\operatorname{Re}(\sum_{q\le3}W_U^{(q)} - \Lambda)$ as a sum of $R = 2 + 20N_f^2 + 80N_f^3$ variables covered by Lem. 5.7. There are two linear terms, of coefficient $1/2$ and argument norm at most $2N_f\|\mathbb{M}\|$; $16N_f^2$ same-species quadratic terms, of coefficient $1/8$ and argument norm at most $2\|\mathbb{M}\|$; $4N_f^2$ mixed quadratic terms, of coefficient $1/2$; $64N_f^3$ same-species cubic terms, of coefficient $1/48$ and argument norm at most $3\|\mathbb{M}\|$; and $16N_f^3$ mixed cubic terms, of coefficient $1/2$ and individual

argument norms at most $\|\mathbb{M}\|$. Add the remainder $N_f\|\mathbb{M}\|^4 N_4$ and apply Hölder with $R+1 \le 103N_f^3$ factors. The radius ρ_D ensures all the required inequalities:

$$103N_f^4\rho_D \le 2\varrho_J, \qquad (4\cdot 103N_f^3)^{1/2}\rho_D \le \rho_q, \qquad 2(4\cdot 103N_f^3)^{1/3}\rho_D \le \rho_q, \qquad 103N_f^3\rho_D^4 \le \theta_q.$$

The same-species quadratic and cubic conditions are weaker. Each factor is bounded by $e^{c_D\ell^3}$, proving the claim.

For the energy representation, polarization gives coefficient sum $\frac{1}{8}(2)(8N_f^2) + \frac{1}{2}(4N_f^2) = 4N_f^2$ in degree two. In degree three the sum is $\frac{1}{48}(4)(16N_f^3) + 2\frac{1}{2}(2)(8N_f^3) = 52N_f^3/3$. The two extra factors in the mixed cubic terms come from the displayed difference defining $\mathscr{E}_{++-}$ and its minus-species counterpart. Each resulting meson source uses at most three matrix entries, giving the norm bound on A_ρ. □

Lemma 5.9 (A cell containing several bilinears)**.** Let $\mathbb{M}$ be supported in one chessboard cell $\square$, at distance at least δ from its walls, with $\|\mathbb{M}\| \le \rho_D$. There is a polynomial $\lambda_\square(\mathbb{M})$ of degree at most three such that

$$\|e^{\Phi[\mathbb{M}]}\|_{\mathrm{ch}} \le \exp\{\operatorname{Re}\lambda_\square(\mathbb{M}) + c_D|\square| + \mathscr{C}_K(\operatorname{supp}\mathbb{M};\delta)\}.$$

Here $\Phi[\mathbb{M}] = a\sum_{x,v:x_v=x}\phi[\mathbb{M}(x)](v)$. The polynomial is holomorphic in the coefficients of $\mathbb{M}$. ■

Proof. Reflection carries $\mathbb{M}$ to its reflected argument, with $M^* = \tau_3 M^\dagger \tau_3$ after an odd number of reflections. Let $\mathbb{M}_r$ be these images and $\mathbb{M}^{\mathrm{ch}} = \sum_r \mathbb{M}_r$. They have disjoint supports and preserve the supremum norm. The fold identity and positivity give

$$\|e^{\Phi[\mathbb{M}]}\|_{\mathrm{ch}}^{\mathsf{N}} = \mathsf{E}R_U(\mathbb{M}^{\mathrm{ch}}) \ge 0.$$

Define

$$\lambda_\square(\mathbb{M}) = \mathsf{N}^{-1}\left(\sum_{r\ \mathrm{even}}\Lambda(\mathbb{M}_r) + \sum_{r\ \mathrm{odd}}\overline{\Lambda(\mathbb{M}_r)}\right).$$

The second conjugation makes this holomorphic in the original coefficients, and preserves its degree.

The linear part of Λ is additive. For a term of degree $q = 2, 3$ in Prop. 5.8, put $T = 2\varrho_J/(3\rho_D)$ and apply the source-energy almost-additivity estimate of [QCDM1, Prop. 10.16] to the sources $tA_\rho\mathbb{M}_r$, $|t| \le T$. They lie in the admitted source ball and are at distances at least 2δ. The defect is bounded by $\mathsf{N}\mathscr{C}_K(\operatorname{supp}\mathbb{M};\delta)$; this is the same reflected-support count used in eq. 5.iv. Cauchy's estimate in t gives an additional T^{-q}. Consequently

$$\left|\Lambda(\mathbb{M}^{\mathrm{ch}}) - \sum_r \Lambda(\mathbb{M}_r)\right| \le (4N_f^2T^{-2} + \tfrac{52}{3}N_f^3T^{-3})\mathsf{N}\mathscr{C}_K(\operatorname{supp}\mathbb{M};\delta) \le \mathsf{N}\mathscr{C}_K(\operatorname{supp}\mathbb{M};\delta).$$

Combine this with Prop. 5.8 and take the Nth root, using $\ell^3/\mathsf{N} = |\square|$. □

Proof of Thm. F. First assume $L^K \ge 16n$ and put $\eta = (16n)^{-1}$. Use the hat partition from the proof of Prop. 5.3. Each f_i is a sum of at most $(32nR+3)^3$ pieces g_i, supported in cubes Q_i of side 2η, with $\|g_i\|_\infty \le \|f_i\|_\infty$. For each tuple of pieces choose a board of side-one cells whose walls stay at least η from the Q_i. The same forbidden-wall count used there applies. Several pieces may lie in one cell, but their actual supports are disjoint.

Put $\rho_i = \rho_D/(\|\lambda_i\|\|g_i\|_\infty)$ and $\mathbb{M}_{\square,z} = \sum_{Q_i\subset\square} z_i g_i\lambda_i$. On $|z_i| \le \rho_i$ its norm is at most ρ_D, since at most one g_i is nonzero at each site. Subtract the polynomial $\Lambda_0(z) = \sum_\square \lambda_\square(\mathbb{M}_{\square,z})$ from the logarithm of the generating function and write

$$F(z) = e^{-\Lambda_0(z)}\omega\left(e^{\sum_i z_i\Phi[\lambda_i](g_i)}\right), \qquad F(0) = 1.$$

On a coordinate face indexed by A, chessboard positivity and Lem. 5.9 give

$$|F(z)| \le \exp\{|A|c_D^+ + \sum_{i\in A}\mathscr{C}_K(Q_i;\eta)\}, \qquad c_D^+ = \max(c_D, 0).$$

The source count obeys $e^{n\sup_Q \mathscr{C}_K(Q;\eta)} \le C_{\mathrm{co}}^n (16n)^{np/2}$. The partition formula for $\partial_{[n]} \log F$ and Cauchy's estimate, exactly as in Prop. 5.3, imply

$$|\partial_{[n]} \log F(0)| \le 2^n n! C_{\mathrm{co}}^n e^{nc_D^+} (16n)^{np/2} \prod_i \rho_i^{-1}.$$

For $n \ge 4$, $\partial_{[n]}\Lambda_0 = 0$. For $n = 2, 3$ it vanishes unless all pieces lie in one cell. In that case use the energy representation in Prop. 5.8. The mixed source-energy estimate of [QCDM1, Prop. 10.16], with two supports at distance d, bounds each derivative by its Cauchy radii times the count $\mathscr{C}_K(Q_1; d)$. Reflection averaging preserves this bound. The coefficient sums give

$$|\partial_{[n]}\Lambda_0(0)| \le 70 N_f^3 C_{\mathrm{co}} (1 + d^{-1})^3 \prod_i \rho_i^{-1}.$$

Here the constants are those of the refined count in [QCDM1, proof of Thm. F]; in particular $\sup_Q \mathscr{C}_K(Q; d) \le C_{\mathrm{co}}(1 + d^{-1})^3$ for the cubes of side 2η. Sum over at most $(32nR + 3)^{3n}$ piece tuples. The inequalities $n^n \le e^n n!$ and $32nR + 3 \le 35n(1 + R)$ give the stated factorial exponent $1 + 3 + p/2$.

If $L^K < 16n$, use $|R_U(\mathbb{M})| \le \exp(\|\mathbb{M}\|_1/(2\mathsf{m}))$. The trace norm includes 48 chambers and $16N_f$ internal components, hence the full generating function is at most $\exp \sum_i |z_i| b_i$, with

$$b_i = 384 N_f \mathsf{m}^{-1} \|\lambda_i\| \sum_x |f_i(x)|.$$

A ball of radius R contains at most $27a^{-3}(1 + R)^3$ sites. Since $a^{-1} < 16n$,

$$b_i \le C N_f m_-^{-1} n^3 (1 + R)^3 \|\lambda_i\| \|f_i\|_\infty.$$

The same partition formula on $|z_i| \le b_i^{-1}$ gives $2^n n! e^n \prod_i b_i$, bounded by $B_c(R)^n (n!)^4 \prod_i \|\lambda_i\| \|f_i\|_\infty$. Enlarge $B_D(R)$ to cover both regimes.

At fixed cutoff the local Gaussian functionals converge along the defining torus sequence, so every cumulant does. The constants are independent of the side. They therefore pass to these states and to each extended continuum limit. The functions and labels are extended by multilinearity; their supports remain disjoint throughout the limit. □

5.4. **Action densities and bilinears on disjoint supports.** The preceding bound for bilinears subtracts a polynomial of degree three. Action-density sources require a different organization. We refine a partition only near four or more bilinear insertions. A cell with at most three such insertions can be handled by one symplectic change of variables, after expanding its tensor of labels in rotated meson tensors. The remaining cells have size comparable to the lattice spacing and admit a determinant bound. The resulting sum over scales is integrable because refinement requires four nearby bilinear entries.

Theorem G (Mixed tuples on a torus)**.** Suppose $N_f \ge 2$, with the source estimates at intermediate levels in [QCDM1, Props. 10.15 and 10.18]. Let $n \ge 2$ and $d > 0$. For real Lipschitz test functions with pairwise disjoint supports on T_ℓ, two of which are at distance at least d, and labels $\lambda_i \in \{0\} \cup \mathscr{S}$,

$$|S_n^T((\lambda_i, f_i)_i)| \le C_n^{\mathrm{W}}(d; g)(1 + \ell)^{3n} \prod_{i:\lambda_i \ne 0} \|\lambda_i\| \prod_i \|f_i\|_{\lambda_i}.$$

The constant is independent of $K, \ell, \mathsf{b}, \mathsf{m}$, the labels and the functions. The estimate passes to every fixed-torus continuum subsequence for this family. Here $\|f\|_0 = \|f\|_\sharp$ and $\|f\|_\lambda = \|f\|_\infty$ for $\lambda \ne 0$. One may replace $(1 + \ell)^{3n}$ by $4^n e^{nc_V \ell^3}$. ■□

The factor depending on ℓ is retained in this theorem. For limits on growing tori we use the estimate at pairwise separated supports in Prop. 5.3.

5.4.1. *At most three bilinear labels.*

Lemma 5.10 (Rotated meson tensors)**.** Suppose $N_f \geq 2$ and $1 \leq m \leq 3$. There are finitely many $S_r \in G$ and meson basis labels $\Gamma_{r,s}$, chosen once for each N_f, m, such that every tensor of labels has an expansion

$$M_1 \otimes \cdots \otimes M_m = \sum_r a_r \bigotimes_{s=1}^{m} S_r \widehat{\Gamma}_{r,s} S_r^{-1}, \qquad \sum_r |a_r| \leq c_{\mathrm{lab}}(N_f) \prod_s \|M_s\|.$$

The same matrix S_r occurs in all factors of each summand. ∎

Proof. Let $\Omega(u,v) = u^{\mathsf{T}} \mathcal{J} v$ on W. The map

$$\iota(u \wedge v) = v(\mathcal{J}u)^{\mathsf{T}} - u(\mathcal{J}v)^{\mathsf{T}}$$

identifies $\Lambda^2 W$ with $\mathscr{S}$ and intertwines the $G_{\mathbb{C}}$ actions. Indeed $\mathcal{J}\iota(u \wedge v)$ is antisymmetric, and

$$x^{\mathsf{T}} \mathcal{J} \iota(u \wedge v) y = \Omega(x,u)\Omega(v,y) - \Omega(x,v)\Omega(u,y).$$

Nondegeneracy of Ω makes this an isomorphism to the alternating forms. The subspace of meson labels corresponds to $W_\psi \wedge W_\chi$, where W_ψ and W_χ are the two coordinate Lagrangians. The associated forms vanish on each Lagrangian separately, and their mixed blocks are arbitrary.

Choose symplectic bases p_α, q_α in each species, with $\Omega(q_\alpha, p_\beta) = \delta_{\alpha\beta}$. There are N_f indices in each species. Write $P_\alpha = \mathrm{span}(p_\alpha, q_\alpha)$. Let V_m be the span of the $G_{\mathbb{C}}$ images of $(W_\psi \wedge W_\chi)^{\otimes m}$. The corresponding span for G is the same: it is a complex subspace invariant under the compact group, hence under its Lie algebra and its complexification, and therefore under the connected group $G_{\mathbb{C}}$. We prove $V_m = (\Lambda^2 W)^{\otimes m}$.

A tensor of basis wedges has a multigraph on the indices α, one edge for each wedge. A loop is $p_\alpha \wedge q_\alpha$ up to sign. A meson wedge has one p and one q. The symplectic flip $p_\alpha \mapsto q_\alpha$, $q_\alpha \mapsto -p_\alpha$ fixes a loop and exchanges the two types at every nonloop edge incident to α.

If the nonloop edges form a forest, choose flips from the root outwards to make every edge a meson wedge. This puts the tensor in V_m.

Suppose two edges join the same distinct indices α, β. Under $\mathrm{SL}(P_\alpha) \times \mathrm{SL}(P_\beta)$,

$$(P_\alpha \wedge P_\beta)^{\otimes 2} \simeq \bigoplus_{A,B \in \{\mathrm{Sym}^2, \Lambda^2\}} A(P_\alpha) \otimes B(P_\beta).$$

These four irreducible summands are pairwise nonisomorphic. The meson tensor $(q_\alpha \wedge p_\beta) \otimes (p_\alpha \wedge q_\beta)$ has nonzero projection on each: at either vertex its factor is $p \otimes q$ or $q \otimes p$, with nonzero symmetric and alternating parts. Its orbit therefore spans this whole space. If a third edge is a loop or disjoint from these vertices, it is fixed by this operation, and a flip at an outside endpoint makes it a meson wedge when necessary. If it has one endpoint at α or β and one at γ, expand its transformed endpoint in p, q and use the flip at γ separately in each term. This proves the claim unless all three edges are parallel.

For a triangle on α, β, γ, regroup the six factors by vertices. The space is the sum of the eight pairwise nonisomorphic tensor products of the symmetric and alternating squares at the vertices. The meson tensor

$$(q_\alpha \wedge p_\beta) \otimes (q_\beta \wedge p_\gamma) \otimes (q_\gamma \wedge p_\alpha)$$

has nonzero projection on every summand, so its orbit spans the space.

It remains to treat three parallel edges between α and β. First suppose that the species of β contains an index γ distinct from both. The symplectic transformation

$$q_\beta \mapsto q_\beta + p_\gamma, \qquad q_\gamma \mapsto q_\gamma + p_\beta$$

fixes all other basis vectors. Let $x_i \in P_\alpha$ be basis vectors. The tensors

$$u_y = (x_1 \wedge q_\gamma) \otimes (x_2 \wedge p_\beta) \otimes (x_3 \wedge y), \qquad y = p_\beta, q_\beta,$$

belong to V_3 by the double-edge case. Subtracting u_{p_β} from its transform gives the tensor with all three ends at β equal to p_β. For u_{q_β} the expansion has four terms. The original term and the two terms with an endpoint at γ are already in V_3 by the same case; the remaining term has ends $(p_\beta, p_\beta, q_\beta)$. Permuting the tensor factors and flipping at β give every choice of the three endpoints.

The only remaining case is $N_f = 2$ and α, β in the same species. On its four-dimensional space $W_h = P_\alpha \oplus P_\beta$, wedge product defines a nondegenerate symmetric form b on $\Lambda^2 W_h$. The invariant bivector $\omega = p_\alpha \wedge q_\alpha + p_\beta \wedge q_\beta$ has $b(\omega, \omega) = 2$. Thus

$$\Lambda^2 W_h = \mathbb{C}\omega \oplus Z, \qquad Z = \omega^\perp, \qquad \dim Z = 5.$$

The standard exterior-square isogeny identifies the image of $\mathrm{Sp}(4, \mathbb{C})$ on Z with $\mathrm{SO}(Z, b)$, its kernel being $\{1, -1\}$; see [FH91, §§16.2, 18.2, pp. 245–247, 278–279]. The meson wedges span $\mathbb{C}\omega \oplus U$, where

$$U = \mathrm{span}\{q_\alpha \wedge p_\beta, q_\beta \wedge p_\alpha, p_\alpha \wedge q_\alpha - p_\beta \wedge q_\beta\}.$$

The first two vectors pair nontrivially with one another and the third has square -2, so U is nondegenerate. Witt's extension theorem [EKM08, Thm. 8.3] carries it to every nondegenerate three-plane in Z. The isometry may be chosen with determinant one by composing with a reflection in the orthogonal complement. Consequently a triple in $\Lambda^2 W_h$ whose Z components have nonzero Gram determinant lies in a common translate of $(\mathbb{C}\omega \oplus U)^3$. Such triples form a nonempty Zariski-open set. A linear functional annihilating V_3 vanishes on their tensor products, and hence, by polynomial continuation, on every tensor product. This proves the remaining case.

These cases exhaust multigraphs with at most three edges. The hypothesis $N_f \geq 2$ supplies γ when the parallel edges join different species; the four-dimensional argument handles its only other exception. Finally select a basis of rotated meson tensors in each degree. Equivalence of norms in these finite-dimensional tensor spaces gives $c_{\mathrm{lab}}(N_f)$ and the stated coefficient bound. □

5.4.2. *Scales and cell estimates.* Let p_n be the least integer with $L^{p_n} \geq 16n$, let $t_n \geq 1$ be the least integer with $A_n := L^{t_n} \geq 2n$, and let $q \geq 0$ be the least integer with $\sqrt{3} L^{-q} < d$. Put

$$u_j = L^{-q} A_n^{-j}, \qquad v_j = L^{-p_n} u_j. \tag{5.v}$$

Then $u_0 \leq 1$ and $u_0^{-1} \leq \sqrt{3} L (1 + d^{-1})$. Call the cutoff fine when $u_1 \geq 2nLa$. In this case let j_* be the largest integer with $u_{j_*} \geq 2nLa$, so $2nLa \leq u_{j_*} < 2nLA_n a$.

At mesh v_j use the product hats $\chi_\alpha^{(j)}(x) = \prod_\mu (1 - |x_\mu / v_j - \alpha_\mu|)_+$. Their enlarged supporting cubes $Q_\alpha^{(j)}$ have side $2v_j(1 + 2/A_n)$. Their diameter is at most $4v_j \leq u_j/8$, their volume is at least $8v_j^3$, and at most 27 of them contain any point. The refinement coefficients $c_{\alpha\beta}^{(j)} = \chi_\alpha^{(j)}(v_{j+1}\beta)$ satisfy

$$\chi_\alpha^{(j)} = \sum_\beta c_{\alpha\beta}^{(j)} \chi_\beta^{(j+1)}, \qquad \sum_\alpha c_{\alpha\beta}^{(j)} = 1.$$

They are nonnegative, and a nonzero coefficient implies $Q_\beta^{(j+1)} \subset Q_\alpha^{(j)}$. The refinement identity is linear interpolation on each axis. For containment use $v_j + v_{j+1}(1 + 2/A_n) \leq v_j(1 + 2/A_n)$.

For a cell side $u = L^i a \leq 1$, $i \geq 1$, choose $k(u) = i + n'$ if $u \leq \sigma_0(g)$, where n' is least with $L^{3n'} \geq \ell_g(u)$, and choose $k(u) = K$ otherwise. The constants ℓ_g, σ_0 are those preceding Thm. D. The level choice in its proof gives

$$\begin{aligned} u \leq \lambda_{k(u)} &\leq \frac{L}{\sigma_0(g)} u \ell_g(u)^{1/3}, \\ (C + 2 + C_P(1 + \log x_{k(u)})) \frac{u^3}{\lambda_{k(u)}^3} &\leq c_V, \\ \log_L \frac{\lambda_{k(u)}}{u} &\leq 1 + \log_L \frac{1}{\sigma_0(g)} + \frac{\log \ell_g(u)}{3 \log L}. \end{aligned} \tag{5.vi}$$

For completeness, in the first case $\lambda_{k(u)} < Lu\ell_g(u)^{1/3} \le (4L)^{-1}$, so the level exists. Also $x_k \le X_g/u + \bar{B}$ and $1 + \log x_k \le \ell_g(u) \le L^{3n'}$. In the other case $\lambda_K = 1 < u/\sigma_0(g)$, $u \le 1$, and $x_K \le X_g$. These observations give all three inequalities.

Lemma 5.11 (Refining a chessboard)**.** Let every wall of a board $\mathbb{B}$ be a wall of $\mathbb{B}'$. For even elements $G_{\square'}$ in the finer cells inside a cell $\square$,

$$\Big\| \prod_{\square' \subset \square} G_{\square'} \Big\|_{\mathrm{ch},\mathbb{B}} \le \prod_{\square' \subset \square} \|G_{\square'}\|_{\mathrm{ch},\mathbb{B}'}.$$

■

Proof. On even elements, two parallel reflections compose to a translation by twice the displacement of their planes. Therefore reflection in a coarse wall is the copy map along the intervening fine cells: an odd row of fine reflections has the same action. The fold on $\mathbb{B}$ is consequently a product of fine-cell copies of the $G_{\square'}$. Apply the chessboard inequality on $\mathbb{B}'$. Every original fine cell occurs once in each of the N coarse copies, giving the product of its seminorms to power N. Taking the Nth root proves the claim. □

Lemma 5.12 (A cell at the chosen level)**.** Let $s \in \mathcal{P}^{\#}_{k(u)}$ be supported in a cell of side u, at distance at least $\delta \le u$ from its walls. For $S \in G$,

$$\|\alpha_S G_s\|_{\mathrm{ch}} \le e^{\operatorname{Re}\lambda^{\flat}_{k(u)}(s) + c_V + \mathscr{C}_{k(u)}(\operatorname{supp} s;\delta)},$$

where

$$\mathscr{C}_{k(u)}(\operatorname{supp} s;\delta) \le C^{\flat}_{\mathscr{C}}[1 + (u/\delta)^3] + 64 C_{\mathscr{C}}(1+\sqrt{3}u/\delta)^3 \left[2 + \log_L \frac{u}{\sigma_0(g)\delta} + \frac{\log \ell_g(u)}{3\log L}\right].$$

■

Proof. The symplectic substitution preserves the seminorm, as in Prop. 5.3. Apply the intermediate-level cell estimate of [QCDM1, Prop. 10.18(iii)] and eq. 5.vi. In the counting term, levels with $\lambda_j \le \delta$ contribute at most $C^{\flat}_{\mathscr{C}}[1 + (u/\delta)^3]$ by the count in part (ii) of that proposition. At the other levels the enlarged cube count is at most $64(1+\sqrt{3}u/\delta)^3$. There are at most $1 + \log_L(\lambda_{k(u)}/\delta)$ such levels; the last bound in eq. 5.vi gives the result. □

Lemma 5.13 (A cell near the lattice scale)**.** Let $\square$ have side $u = L^i a \le 1$, $i \ge 1$, and put $k = k(u)$, $t_k = r_J/(2\lambda_k)$. If $M(x) \in \mathscr{S}$ vanishes outside its sites, $\|M\|_\infty \le t_k$, and w vanishes near its walls, then

$$\|e^{O(w)+\Phi[M]}\|_{\mathrm{ch}} \le \exp\Big[\frac{2}{a}\sum_{p:x(p)\in\square} |w(x(p))| + 192 N_f \log 2\,(u/a)^3 + c_V + 27 C_{\mathfrak{E}} C_{\mathrm{cnt}} C_{\mathrm{sh}}(256k + 1536(u/a)^3)\Big].$$

■

Proof. Fold the two sources into all N cells. Reflection preserves $\|M\|_\infty$, so the folded source has the same bound. Since each plaquette action lies in $[0,2]$, the action-density factor costs the first term times N.

Let S' be the union of the reflected sets of sites, let P project onto their chambers, and put $T = P\mathsf{G}_+P$. The Pfaffian identity in Lem. 5.4 and the singular-value determinant inequality give

$$|R_U(M^{\mathrm{ch}})|^2 \le \prod_{\sigma}(1+t_k\sigma)^{4N_f}, \qquad |R_U(M^{\mathrm{ch}})| \le 2^{N_f d'} \det(1+t_k^2 T^* T)^{N_f}.$$

Here σ runs over the singular values of T, $d' \le 192\mathsf{N}(u/a)^3$, and the second inequality uses $(1+x)^2 \le 2(1+x^2)$. The determinant is the positive conditional quotient for the meson source $J' = t_k 1_{S'}(e_{+-} - e_{-+}) \otimes 1_{N_f}$. It belongs to $\mathcal{P}_k^{\#}$.

The partition bound at level k costs at most $\mathsf{N}c_V$ by eq. 5.vi. For its energy increment, the polymer filing in the proof of Lem. 3.5 gives

$$|E_k(0,J') - E_k(0,0)| \le C_{\mathfrak{E}} C_{\mathrm{cnt}} C_{\mathrm{sh}} \sum_{j \le k} \#_j^+(S').$$

The reflected support count is at most $27\mathsf{N}$ times the count for one cell. That count is at most $256[1+6(u/\lambda_j)^3]$, and $\sum_{j\ge 1}(u/\lambda_j)^3 \le (u/a)^3$. Combine these inequalities and take the Nth root. □

5.4.3. *A partition stopped at four nearby bilinears.* Write $I_B = \{i : \lambda_i \neq 0\}$, $I_0 = [n] \setminus I_B$, and $n_B = |I_B|$, $n_0 = |I_0|$. Refine the hats as follows. At stage $j < j_*$, an entry is refined precisely when its cube lies within distance $2u_j$ of the cubes of at least four bilinear entries, counting itself if it is bilinear. Entries that have already stopped keep their cubes. All remaining entries stop at j_*. A history records the successive nodes. Its weight is the product of the coefficients $c_{\alpha\beta}^{(j)}$ used in its refinements. A final configuration is $F = (j_i, \alpha_i)_{i\le n}$; write $Q_i = Q_{\alpha_i}^{(j_i)}$, $\chi_i = \chi_{\alpha_i}^{(j_i)}$ and $\mathcal{X}_F = ((\lambda_i, f_i\chi_i))_i$.

Lemma 5.14 (The stopped partition)**.** At a fine cutoff,

$$S_n^T(\mathcal{X}) = \sum_H W(H) S_n^T(\mathcal{X}_{F(H)}), \qquad \sum_{H:F(H)=F} W(H) \le 1.$$

The cubes in each history are nested. Discard final configurations with a zero entry. If $j_i \ge 1$, there are four bilinear entries l, with $j_l \ge j_i - 1$, such that

$$|y - y'| \le \tfrac{5}{2} A_n u_{j_i} \qquad (y \in Q_i, y' \in Q_l).$$

In particular, for $i \in I_0$,

$$\sup_{Q_i} |f_i| \le \tfrac{5}{2} A_n u_{j_i} \operatorname{Lip} f_i, \qquad \|f_i\chi_i\|_\sharp \le C_\sharp \|f_i\|_\sharp, \quad C_\sharp = 1 + 5A_nL^{p_n} + 80ML^{p_n}/u_0.$$

If $j_i < j_*$, at most three final bilinear cubes meet $Q_i^{[2u_{j_i}]}$, and no final cube with $j_l \ge j_i + 2$ meets $Q_i^{[\sqrt{3}u_{j_i}]}$. Here $Q^{[r]}$ is the closed r-neighbourhood of Q. ■

Proof. Multilinearity and the hat refinement identity give the expansion. The sum of the weights of all chains ending at one specified node is one, by induction from $\sum_\alpha c_{\alpha\beta}^{(j)} = 1$. The admissible histories are a subset of these chains, so their weight sum is at most one.

If i was refined from $j = j_i - 1$, at least four bilinear cubes met $Q_i(j)^{[2u_j]}$. None of these entries could have stopped at $j' < j$. Indeed that earlier cube would then have all four cubes within $2u_{j'}$, since $\operatorname{diam}(Q_i(j)^{[2u_j]}) \le (33/8)u_j \le 2u_{j'}$. It would have been refined, a contradiction. Thus all four cubes at stage j have diameter at most $u_j/8$. Their final subcubes are within $(2+1/4)u_j \le (5/2)A_n u_{j_i}$ of one another as asserted. For an action-density entry choose a point in the support of any of these nonzero bilinear pieces. At that point $f_i = 0$, since the original supports are disjoint. The Lipschitz estimate follows. Together with $\operatorname{Lip}\chi_i \le 2/v_{j_i}$ it bounds the derivative of $f_i\chi_i$. At $j_i = 0$ use its supremum norm instead; the two estimates give $C_\sharp$.

When i stops before j_*, the count of nearby bilinear cubes is at most three. Nestedness preserves that assertion for the final cubes. If a cube of level $j_l \ge j_i + 2$ met $Q_i^{[\sqrt{3}u_{j_i}]}$, its refinement at $j_l - 1$ would supply four bilinear cubes within

$$\sqrt{3}u_{j_i} + \frac{17}{8}u_{j_l-1} \le \left(\sqrt{3} + \frac{17}{8A_n}\right) u_{j_i} \le 2u_{j_i}$$

of Q_i. This contradicts its stopping rule. □

Lemma 5.15 (Boards for the stopped pieces)**.** For each final configuration there are nested boards $\mathbb{B}_j$, of cell side u_j, such that Q_i is at distance at least v_{j_i} from the walls of $\mathbb{B}_{j_i}$. Let $\mathcal{T}$ be the maximal cells among those containing the Q_i at their own levels. They are disjoint and cover all pieces. If $\square \in \mathcal{T}$ has level $j < j_*$, it contains at most three bilinear entries, and each entry in it has level j or $j+1$. A maximal cell of level j_* contains only entries of that level. Every piece in a maximal cell of level j is at distance at least v_{j+1} from its walls. Every maximal cell has diameter less than d. ■

Proof. Choose wall offsets from the finest board upwards. Once the offset at level $j+1$ is chosen, nestedness leaves A_n offsets at level j, spaced by u_{j+1}. A cube of level j excludes an interval of length at most $5v_j$ if its distance from the wall is to be at least v_j. Thus all n cubes exclude at most $5A_n/16 + n < A_n$ offsets. At level j_* the offsets are spaced by a, and the same count is $\frac{5}{16}u_{j_*}/a + n < u_{j_*}/a$, since $u_{j_*} \geq 2nLa$. Choose the three coordinates independently. The sides are odd powers of L in lattice units and divide the torus with an even number of cells, so these are admitted boards.

Cells of these nested boards are either nested or disjoint. A maximal cell of level j cannot contain an entry of smaller level, by maximality. It cannot contain an entry of level at least $j+2$, by the last assertion of Lem. 5.14. That assertion also limits its bilinear entries to three when $j < j_*$. Walls of its board remain walls of the finer boards, giving the distance statement. Finally its diameter is at most $\sqrt{3}u_0 < d$. □

5.4.4. *Summing the scales.* For $y \in T_\ell^n$, let $r_i(y)$ be the distance to the third nearest other bilinear point when $i \in I_B$, and to the fourth nearest bilinear point when $i \in I_0$. Set $r_i = \infty$ if there are too few points, and $m_i(y) = \min(u_0, r_i(y))$. By Lem. 5.14, every final configuration with nonzero pieces satisfies

$$m_i(y) \leq \tfrac{5}{2} A_n u_{j_i} \qquad (y \in \prod_l Q_l).$$

Lemma 5.16 (Products of distances)**.** For nonnegative exponents α_e on the unordered pairs of $[m]$, suppose that, for every $W \subset [m]$ with $|W| \geq 2$,

$$S(W) := \sum_{e \subset W} \alpha_e \leq 3(|W| - 1) - \theta, \qquad 0 < \theta \leq 1.$$

Put $\bar{\alpha} = \sum_e \alpha_e$. For $0 < \sigma \leq 1$,

$$\int_{T_\ell^m} \prod_e \min(\sigma, |y_u - y_v|)^{-\alpha_e}\, dy \leq I_m(\theta, \bar{\alpha}) \sigma^{-\bar{\alpha}} (1+\ell)^{3m},$$

where, if B_m is the number of partitions of $[m]$,

$$I_m(\theta, A) = 2^A (5m^3)^m m B_m^m \left(1 + \sum_{H \geq 1} H^m 2^{-\theta(H-1)}\right).$$

■

Proof. At dyadic scale $\sigma 2^{-h}$, connect points closer than this distance and let $\mathcal{P}_h$ be the partition into connected components. These partitions refine and are eventually singletons outside a null set. A block has diameter at most $(|W|-1)\sigma 2^{-h}$. If h_e is the last level where the endpoints of e share a block, their distance is at least $\sigma 2^{-h_e - 1}$. Set $h_e = -1$ if they already lie in different blocks of $\mathcal{P}_0$; then their distance is at least σ, and $d_e = \sigma$. Consequently the integrand is bounded by

$$2^{\bar{\alpha}} \sigma^{-\bar{\alpha}} 2^{\sum_{h \geq 1} \sum_{W \in \mathcal{P}_h} S(W)}.$$

Fix a sequence of partitions. Represent each block by its least element. When a block at level h splits into c_W children, its $c_W - 1$ new representatives lie in balls of radius $(m-1)\sigma 2^{-h}$ about the

old representative. Integrate the representatives of $\mathcal{P}_0$ over the torus and these new representatives over their balls. The volume of the corresponding configuration set is at most

$$(5m^3)^m(1+\ell)^{3m}2^{-3\sum_{h\ge1}\sum_{W\in\mathcal{P}_h}(|W|-1)}.$$

Here $\sigma\le1$ and the telescoping identity $\sum_h h(|\mathcal{P}_{h+1}|-|\mathcal{P}_h|)=\sum_{h\ge1}(m-|\mathcal{P}_h|)$ were used. Multiplication by the integrand leaves a factor at most $2^{-\theta N}$, where N counts the positive levels whose partition is not discrete.

A strictly refining chain has at most m distinct partitions, and there are at most mB_m^m such chains. If its last change is at level H, there are at most H^m choices of the earlier change levels and $N\ge H-1$. Summing proves the estimate. □

Lemma 5.17 (The sum over stopped configurations)**.** Let $\mathfrak{F}$ be the final configurations with nonzero pieces, $P\ge0$, and $b_B,b_0>0$. Then

$$\sum_{F\in\mathfrak{F}}\prod_{i\in I_B}b_Bu_{j_i}\ell_g(u_{j_i})^P\prod_{i\in I_0}b_0\ell_g(u_{j_i})^P\le b_B^{n_B}b_0^{n_0}J_n(P;g)u_0^{-4n}(1+\ell)^{3n},$$

where $J_n(P;g)$ is finite and independent of the cutoff and torus side. ■

Proof. For an entry put $g_i=1$ on I_B, $g_i=0$ on I_0, and $\psi_i(j)=b_iu_j^{g_i}\ell_g(u_j)^P$. Divide by $|Q_i|\ge8L^{-3p_n}u_{j_i}^3$ and integrate over the cubes of a configuration. Using the preceding bound on m_i and the overlap count 27, the whole sum is at most $\int\prod_i\Phi_i(y)\,dy$, where

$$\Phi_i(y)=\frac{27}{8}L^{3p_n}\sum_{j:u_j\ge x_i}\psi_i(j)u_j^{-3},\qquad x_i=\frac{2m_i(y)}{5A_n}.$$

Going to the next coarser scale decreases each summand by at least $A_n^{-2}\le1/2$; also ℓ_g decreases. Thus

$$\Phi_i(y)\le\frac{27}{4}L^{3p_n}b_ix_i^{g_i-3}\ell_g(x_i)^P\le b_iK_n(P;g)m_i(y)^{g_i-3-\epsilon},\qquad\epsilon=\frac{1}{8n}.$$

One may take

$$K_n(P;g)=\frac{27}{4}L^{3p_n}(5A_n/2)^3(1+\log(5A_n/2))^P(1+\log(X_g+\bar B))^PC_{P,n},$$

where $C_{P,n}=\sup_{t\ge0}(1+t)^Pe^{-t/(8n)}$. This follows from $\ell_g(u/c)\le(1+\log c)\ell_g(u)$ and $\ell_g(u)\le(1+\log(X_g+\bar B))(1+\log(1/u))$.

First integrate the action-density coordinates with the bilinear coordinates fixed. If $n_B<4$, all $m_i=u_0$ and the result follows directly. Otherwise bound $m_i^{-3-\epsilon}$ by the sum, over four-element sets $T\subset I_B$, of $\min(u_0,\max_{l\in T}|y_i-y_l|)^{-3-\epsilon}$. Choose a diameter pair $e_T=\{u,v\}$ in T, of distance D_T. The maximum is at least $\max(|y_i-y_u|,D_T/2)$. Radial integration gives

$$\int_{T_\ell}m_i(y)^{-3-\epsilon}\,dy_i\le321n^4\left[\ell^3u_0^{-3-\epsilon}+\sum_{e\subset I_B}d_e^{-\epsilon}\right],\qquad d_e=\min(u_0,|y_u-y_v|).$$

Indeed the radial bound for one set is $\ell^3u_0^{-3-\epsilon}+(8\pi/3+8\pi/\epsilon)d_{e_T}^{-\epsilon}$, and each pair belongs to at most n^2 such four-element sets. Expanding the product of the n_0 integrals gives at most $(2n^2)^{n_0}$ terms. Each has a factor $(\ell^3u_0^{-3-\epsilon})^a$ and n_0-a pair factors of exponent ϵ.

For a bilinear coordinate, its three nearest bilinear neighbours give

$$m_i^{-2-\epsilon}\le\sum_{N_i:\,|N_i|=3}\prod_{l\in N_i}d_{\{i,l\}}^{-(2+\epsilon)/3}.$$

There are at most n^3 choices at each coordinate. In any resulting distance product, a subset of $w \geq 2$ bilinear indices carries total exponent at most

$$\frac{2+\epsilon}{3} w \min(3, w-1) + n\epsilon.$$

For $w = 2$ this is at most $5/2$; for $w = 3$ at most $11/2$; and for $w \geq 4$ at most $2w+1/4 \leq 3(w-1)-1/2$. Thus Lem. 5.16 applies with $\theta = 1/2$. The total exponent is at most $3n$, and including the factors $u_0^{-(3+\epsilon)a}$ raises it to at most $4n$. Collecting the finite choices gives the assertion, for example with

$$J_n(P; g) = K_n(P; g)^n (642n^6)^n \max_{1 \leq m \leq n} I_m(1/2, 3n).$$

□

5.4.5. *Proof of the mixed estimate.*

Proof of Thm. G. If any entry is zero, the cumulant vanishes. If the cutoff is coarse, $a^{-1} < 2nLA_n/u_0$. At each configuration the singular-value determinant estimate gives

$$|R_U(\textstyle\sum_{i \in I_B} z_i f_i \lambda_i)| \leq \exp\left(384 N_f \mathsf{m}^{-1} \textstyle\sum_{i \in I_B} |z_i| \|\lambda_i\| \sum_x |f_i(x)|\right).$$

Combine this with $|e^{O(w)}| \leq \exp(2a^{-1} \sum_p |w(x(p))|)$. There are $(\ell/a)^3$ sites and $3(\ell/a)^3$ plaquettes. Cauchy's estimate on the polydisc where each exponent is at most one and the partition formula therefore bound the cumulant by

$$C_n^{\mathrm{c}}(d) \ell^{3n} \prod_{i \in I_B} \|\lambda_i\| \prod_i \|f_i\|_{\lambda_i},$$

where

$$C_n^{\mathrm{c}}(d) = 2^n n! \left[e(6 + 384 N_f/m_-)(2\sqrt{3} n L^2 A_n (1 + d^{-1}))^4\right]^n.$$

For a fine cutoff use Lems. 5.14 and 5.15. Call a maximal cell early if its level is less than j_*, and final otherwise. In each early cell apply Lem. 5.10 to its at most three bilinear labels. Multilinearity expands the cumulant into terms in which all these labels are mesons conjugated by one $S_\square$. The sum of the absolute coefficients is at most $c_{\mathrm{lab}}(N_f)^n$ times the product of the norms of the labels in early cells. Keep the original labels in final cells.

Fix one term in this expansion. For an entry in a cell $\square$, put

$$\rho_i = \begin{cases} r_J/(2\lambda_{k(u_\square)} \|f_i\|_\infty), & i \in I_B,\ \square \text{ early}, \\ r_J/(2\lambda_{k(u_{j_*})} \|\lambda_i\| \|f_i\|_\infty), & i \in I_B,\ \square \text{ final}, \\ 2\varrho/(C_\sharp \|f_i\|_\sharp), & i \in I_0. \end{cases}$$

For $|z_i| < \rho_i$, the source in an early cell belongs to $\mathcal{P}^{\#}_{k(u_\square)}$. Disjoint supports make the source supremum norm a maximum. The same is true of the sharp norm for a sum of Lipschitz functions with disjoint supports: on a segment joining distinct supports, their two distances to their boundaries add to at most the segment length. Thus the Lipschitz constant is at most the largest one. The bound in Lem. 5.14 then gives the required sharp norm bound 2ϱ.

Subtract from the logarithm of the generating function the sum $\Lambda(z)$ of the early-cell energies $\lambda^{\mathfrak{b}}_{k(u_\square)}$. It is holomorphic on this polydisc. Its derivative in all n variables vanishes, since every cell has diameter less than d and therefore cannot contain both distinguished separated entries. Let $F(z)$ be the normalized generating function multiplied by $e^{-\Lambda(z)}$, so $F(0) = 1$.

Apply the ordinary chessboard inequality on $\mathbb{B}_0$, then Lem. 5.11 whenever a cell is replaced by finer cells. This ends at the maximal cells. In an early cell apply Lem. 5.12 with $\delta = v_{j+1} = u_\square/(A_n L^{p_n})$. In a final cell the matrix source has norm at most $r_J/(2\lambda_{k(u_{j_*})})$. Its action-density source satisfies

$|w| \le \varrho u_{j_*}$: use the vanishing estimate in Lem. 5.14, $j_* = j_i \ge 1$, $40M \operatorname{Lip} f_i \le \|f_i\|_\sharp$, and $C_\sharp \ge 5A_n$. Hence Lem. 5.13 applies. It follows, on a coordinate face indexed by B, that

$$|F(z)| \le \prod_{\square:\square \text{ contains an entry of } B} e^{K_\square},$$

where

$$K_\square = c_V + c_A + p_A \log \ell_g(u_\square) \quad \text{in an early cell,} \qquad K_\square = c_V + c_L + p_L \log \ell_g(u_{j_*}) \quad \text{in a final cell.}$$

All these numbers are nonnegative. Their constants are given below. The final-cell bound uses $u_{j_*}/a < 2nLA_n$ and

$$k(u_{j_*}) \le 1 + \log_L(2nLA_n/\sigma_0(g)) + \frac{\log \ell_g(u_{j_*})}{3 \log L}.$$

Cauchy's estimate bounds $\partial_B F(0)$ by the same product times $\prod_{i\in B} \rho_i^{-1}$. In a partition of $[n]$, a cell containing $n_\square$ entries occurs in at most $n_\square$ blocks. Thus the partition formula and the vanishing derivative of Λ give

$$|S_n^T(\mathcal{X}_{F,r})| \le 2^n n! \prod_i e^{K_{\square(i)}} \rho_i^{-1}.$$

Now $u_{j_i} \le u_{\square(i)} \le A_n u_{j_i}$, and eq. 5.vi shows that each bilinear factor on the right is at most

$$b_B u_{j_i} \ell_g(u_{j_i})^P \|f_i\|_\infty, \quad b_B = \frac{2LA_n e^{c_V + c_*}}{r_J \sigma_0(g)},$$

with an additional $\|\lambda_i\|$ in a final cell. Each action-density factor is at most $b_0 \ell_g(u_{j_i})^P \|f_i\|_\sharp$, where $b_0 = C_\sharp e^{c_V+c_*}/(2\varrho)$, $c_* = \max(c_A, c_L)$ and $P = \max(p_A, p_L) + 1/3$.

Sum the tensor coefficients and the stopped configurations, using Lem. 5.17. This gives the theorem with the larger of $C_n^{\mathrm{c}}(d)$ and

$$C_n^{\mathrm{f}}(d; g) = 2^n n! c_{\mathrm{lab}}(N_f)^n \max(b_B, b_0, 1)^n J_n(P; g)(\sqrt{3}L(1 + d^{-1}))^{4n}.$$

The constants used above can be taken to be

$$\begin{aligned}
c_A &= C_{\mathscr{C}}^\flat[1 + (A_n L^{p_n})^3] + 64 C_{\mathscr{C}}(1 + \sqrt{3}A_n L^{p_n})^3[2 + \log_L(A_n L^{p_n}/\sigma_0(g))],\\
p_A &= 64 C_{\mathscr{C}}(1 + \sqrt{3}A_n L^{p_n})^3/(3 \log L),\\
c_L &= 6\varrho(2nLA_n)^4 + 192 N_f \log 2\,(2nLA_n)^3\\
&\quad + 27 C_{\mathfrak{E}} C_{\mathrm{cnt}} C_{\mathrm{sh}}[256(1 + \log_L(2nLA_n/\sigma_0(g))) + 1536(2nLA_n)^3],\\
p_L &= 256 \cdot 27 C_{\mathfrak{E}} C_{\mathrm{cnt}} C_{\mathrm{sh}}/(3 \log L).
\end{aligned}$$

They are formed after the existing source constants and impose no further restriction on them. The alternative volume factor follows from $(1 + \ell)^3 \le 4e^{\ell^3} \le 4e^{c_V \ell^3}$. At fixed ℓ, every bound is uniform in the cutoff and passes to a convergent subsequence. □

Proposition 5.18 (The enlarged limits)**.** Let S be a volume-first limit of [QCDM1], with a fixed defining sequence. There is a subsequence along which all extended truncated functions of orders $n \ge 2$ converge on pairwise disjoint C_c^1 supports. Their limits S^{ext} restrict to S on meson tuples and satisfy eq. 5.iii.

They are symmetric, translation invariant, invariant under simultaneous conjugation of the labels by $G_{\mathbb{C}}$. At fixed labels, these truncated functions determine distributions of locally finite order away from the diagonals. The centered functional σ^{ext}, obtained by the moment–cumulant formula with one-point functions zero, is reflection positive at every coordinate plane, with the involution $M \mapsto M^*$. It defines a separable reflection Hilbert space $\mathcal{H}^{\mathrm{ext}}$. Time translation gives a strongly continuous self-adjoint contraction semigroup e^{-tH}, $H \ge 0$; spatial translations give a strongly continuous unitary representation with generators P_1, P_2 commuting strongly with H. ■

Proof. Fix finitely many pairwise disjoint closed supports, each in a bounded ball. The uniform bound makes the truncated functions equicontinuous in the product of the indicated norms. Enumerate rational unions of closed cubes, a countable dense family of C_c^1 functions on these supports, and the finite label basis. A diagonal subsequence makes every corresponding truncated function converge. Equicontinuity extends the convergence to every tuple, and multilinearity extends the labels to $\mathscr{S}$. The original meson functions converge along the given defining sequence, so their limits remain S.

Symmetry and $G_{\mathbb{C}}$-invariance are exact at every cutoff. They pass to the limit. Lattice translations approximate every real translation; the estimate and translation continuity in C^1 give translation invariance. For a positive-side tuple, replace each field by its lattice field minus its expectation. The reflection sends this centered field to the centered field with label M^* and reflected conjugate test function. Every reflection form is therefore the finite sum of products of truncated functions specified by the centered moment–cumulant formula. Approximate a coordinate plane by bond planes. The supports stay a positive distance from the planes, and the reflected test functions converge in their insertion norms. Lattice reflection positivity and the bound pass the form inequality to the limit.

On a product of open sets with disjoint closures, the bound makes the limiting truncated functional continuous in the C^1 norms. The Schwartz kernel theorem gives a distribution there. Uniqueness makes these local distributions agree on intersections, so they define one distribution off the diagonals.

On finite linear combinations of positive-time tuples, define the positive semidefinite form by

$$\langle \mathcal{X}, \mathcal{Y} \rangle = \sigma^{\mathrm{ext}}(\theta_3 \mathcal{X}^* \cdot \mathcal{Y}).$$

We first check that time translation descends to its quotient. Write $T_t \mathcal{X} = \tau_{te_3} \mathcal{X}$, $t \geq 0$. Translation invariance and $\theta_3 \tau_{te_3} = \tau_{-te_3} \theta_3$ give

$$\langle T_s v, T_t v \rangle = \langle v, T_{s+t} v \rangle, \qquad \|T_t v\|^2 = \langle v, T_{2t} v \rangle.$$

For a fixed v, these correlations are bounded uniformly for $s, t \geq 0$. Indeed, each is a finite sum of products of truncated correlations; the support radii stay fixed, the separations within each tuple stay fixed, and the reflected tuple stays a positive distance from the translated tuple.

Fix $t > 0$ and put $a_k = \|T_{kt} v\|^2$. Cauchy–Schwarz for the semidefinite form gives

$$a_k^2 = |\langle T_{(k-1)t} v, T_{(k+1)t} v \rangle|^2 \leq a_{k-1} a_{k+1} \qquad (k \geq 1).$$

If $a_0 = 0$, this implies $a_1 = 0$. If $a_0 > 0$ and $a_1 > a_0$, all subsequent terms are positive and the ratios a_{k+1}/a_k are nondecreasing. They are then bounded below by $a_1/a_0 > 1$, contrary to boundedness. Thus $\|T_t v\| \leq \|v\|$. In particular null vectors remain null, so T_t defines a contraction on the quotient and its completion $\mathcal{H}^{\mathrm{ext}}$. The empty tuple gives the unit vector Ω.

Translation is symmetric for the form, so each bounded operator T_t is self-adjoint. The semigroup law is the translation law on tuples. In particular

$$\langle v, T_t v \rangle = \|T_{t/2} v\|^2 \geq 0.$$

This is also the reflection form at the intervening plane. For a vector represented by finitely many tuples,

$$\|T_t v - v\|^2 = \langle v, T_{2t} v \rangle - 2 \operatorname{Re} \langle v, T_t v \rangle + \|v\|^2 \longrightarrow 0 \quad (t \downarrow 0),$$

by continuity in the translated test functions. Contraction extends strong continuity to the completion. The spectral theorem for a strongly continuous self-adjoint contraction semigroup now gives $T_t = e^{-tH}$ with $H \geq 0$.

Spatial translations preserve the form and have inverses. They extend to strongly continuous unitary operators, and their commutation with the semigroup on tuples gives strong commutation of the generators. For separability, use the positive-time members of the countable family used for extraction, and take finite linear combinations with rational real and imaginary coefficients. Any positive-time tuple can be approximated by such tuples with supports in fixed closed sets that remain separated and stay away from the reflection plane. Its squared reflection-norm error is a

finite combination of moments, each a polynomial in continuous truncated functions. The error therefore tends to zero, proving density of this countable set. □

Proposition 5.19 (The internal group on the enlarged space)**.** The map $[X] \mapsto [X]$ embeds the mesonic space $\mathcal{H}(S)$ isometrically into $\mathcal{H}^{\mathrm{ext}}$ and intertwines time and spatial translations. Moreover

$$V(S)[\mathcal{X}] = [S\mathcal{X}S^{-1}], \qquad S \in G,$$

is a strongly continuous unitary representation. It fixes Ω, commutes with H, P_1, P_2, and is trivial on $(-1,-1)$. Let Σ_3 be 1 on the ψ components and -1 on the χ components. The generator Q defined by

$$V(e^{i\varphi\Sigma_3}, e^{i\varphi\Sigma_3}) = e^{-i\varphi Q}$$

has spectrum in $2\mathbb{Z}$. The mesonic space lies in $\ker Q$, and both eigenspaces $Q = 2$ and $Q = -2$ are nonzero. Charge conjugation $V(J, J)$ exchanges them. ■

Proof. The two centered functionals agree on meson tuples, so the displayed embedding preserves inner products. Both translations act on the functions and are therefore intertwined. For $S \in G$, $(SMS^{-1})^* = SM^*S^{-1}$. Invariance of σ^{ext} proves that $V(S)$ preserves the form and its null space. Its inverse is $V(S^{-1})$. For a fixed tuple the orbit lies in a finite-dimensional span of label tuples and depends continuously on S; this proves strong continuity. The stated commutations hold on tuples and hence on the completion.

The central element $(-1,-1)$ is the identity on all quadratic labels. Setting $\varphi = \pi$ therefore gives $e^{-i\pi Q} = 1$, so the spectral theorem places the spectrum of Q in $2\mathbb{Z}$. Every meson is neutral, and the mesonic space lies in its kernel.

To see that the charge-two space is nonzero, choose the meson $N_f^{-1/2} \sum_f \bar{\psi}_{+,f}\psi_{-,f}$. Its matrix is $\tau_+ \otimes t_1$, $\tau_+ = (\tau_1 + i\tau_2)/2$, a linear combination of $\Gamma_{1,1}$ and $\Gamma_{2,1}$. The leading two-point formula of [QCDM1, Prop. 11.1] has a positive definite leading coefficient on this two-dimensional span. At a pair of reflected small positive test functions its relative error tends to zero as their scale tends to zero. Thus the associated one-field vector has positive norm. The element of G equal to $J \otimes 1$ on the plus half and to the identity on the minus half sends this meson to a nonzero multiple of $\sum_f \psi_{+,f}^{\mathsf{T}} C_0^{-1} \psi_{-,f}$. Its image under $V(S)$ has the same positive norm and charge two. Finally $J\Sigma_3 J^{-1} = -\Sigma_3$, so $V(J,J)QV(J,J)^{-1} = -Q$. This gives the nonzero charge-minus-two space as well. □

6. Energy in the charged sectors

The estimate needed in the continuum has constant one. A direct propagator estimate initially gives a constant depending on the cutoff. Reflection positivity removes that constant before the cutoff is taken to zero.

6.1. **A sharp decay rate on the chamber lattice.**

In this subsection lattice spacing is one. Let

$$L_\rho = \frac{1}{2}\sum_{s=1}^{3} t_s(1 - \widehat{R}_s) \quad \text{on } \ell^2(W_3), \qquad E = \{g : ge_1 = \pm e_3\},$$

where $\widehat{R}_s$ is right multiplication by the within-site chamber reflection w_s. The kernel of L_ρ consists of the constants; let $\lambda_\rho > 0$ be its first positive eigenvalue. These are the within-site operator and its gap used in [QCDM1, Section 5]. There are 16 elements in E and 48 in W_3. Define

$$\begin{aligned} &h(k) = -\lambda_{\min}(L_\rho - k1_E), \qquad k(\alpha) = \tfrac{3}{2}(\cosh\alpha - 1) + 3\sinh\alpha, \\ &e(\alpha) = h(k(\alpha)), \quad 0 \le \alpha \le \operatorname{artanh}(1/2), \qquad \mu(m_0) = \sup\{\alpha : e(\alpha) < m_0\}. \end{aligned} \tag{6.i}$$

Lemma 6.1 (Directional decay)**.** If $\varphi(v) = \phi((x_v)_3)$ is real and bounded, with increments of modulus at most $\alpha \le \operatorname{artanh}(1/2)$, then

$$\operatorname{Re}(e^{\varphi} D_U e^{-\varphi}) \ge m_0 - e(\alpha).$$

If two sets of sites A, B are at separation at least d in their third coordinates, then, for $0 \le \alpha < \mu(m_0)$,

$$\|1_A D_U^{-1} 1_B\| \le (m_0 - e(\alpha))^{-1} e^{-\alpha d}.$$

The estimates are uniform in U. Moreover $\mu(x)/x \to 1$ as $x \downarrow 0$, and consequently $a^{-1}\mu(a\mathsf{m}) \to \mathsf{m}$ uniformly for m in compact subsets of $(0, \infty)$. ■

Proof. Write $R_s f(v) = U(v, v \cdot s) f(v \cdot s)$. This is a self-adjoint unitary involution. Conjugation changes the coefficient of the sth hop to $t_s(\frac{1}{2} - \gamma(u_s(v)))e^{-w_s(v)}$, where $w_s(v) = \varphi(v \cdot s) - \varphi(v)$ and $w_s(v \cdot s) = -w_s(v)$. Its Hermitian part is $B_s R_s$, with

$$B_s(v) = t_s\big(\tfrac{1}{2} \cosh w_s(v) + \gamma(u_s(v)) \sinh w_s(v)\big).$$

The matrix B_s commutes with R_s. For $s = 1, 2, 3$, $w_s = 0$, since the hop stays at the same site. For $s = 0$, it vanishes outside E. Its two eigenvalues are $3(\frac{1}{2}\cosh w \pm \sinh w)$, nonnegative in the stated interval. Hence

$$\operatorname{Re}(e^{\varphi} D_U e^{-\varphi}) = m_0 + L_\rho + (\tfrac{3}{2} - B_0) + B_0(1 - R_0) \ge m_0 + L_\rho - k(\alpha) 1_E.$$

This proves the first estimate.

Choose $\phi(s) = \alpha \min\{\operatorname{dist}(s, B_3), d\}$, where B_3 is the set of third coordinates of B. The weight is zero on B and αd on A. The first estimate makes the conjugated inverse bounded by $(m_0 - e(\alpha))^{-1}$, proving decay.

Let u be the constant unit vector in $\ell^2(W_3)$. There are 16 moving chambers among the 48 chambers, so $\langle u, 1_E u\rangle = 16/48 = 1/3$. Moreover

$$\|(1 - |u\rangle\langle u|) 1_E u\|^2 = \langle u, 1_E u\rangle - |\langle u, 1_E u\rangle|^2 = \tfrac{1}{3} - \tfrac{1}{9} = \tfrac{2}{9}.$$

Writing a vector as $cu + v$, $v \perp u$, and using $L_\rho \ge \lambda_\rho$ on $u^\perp$, gives, for $0 \le k < 3\lambda_\rho/2$,

$$\frac{k}{3} \le h(k) \le \frac{k}{3} + \frac{2k^2}{9\lambda_\rho - 6k}.$$

Indeed the lower bound uses the trial vector u. For the upper bound the quadratic form is bounded below by $-(k/3 + \delta)\|(c, v)\|^2$ when

$$\begin{pmatrix} \delta & -\sqrt{2}k/3 \\ -\sqrt{2}k/3 & \lambda_\rho - 2k/3 + \delta \end{pmatrix} \ge 0; \qquad \delta = \frac{2k^2}{9\lambda_\rho - 6k}$$

satisfies this condition. Since $k(\alpha) = 3\alpha + O(\alpha^2)$, we have $e(\alpha) = \alpha + O(\alpha^2)$. Inversion at zero gives $\mu(x) = x + O(x^2)$ and the claimed uniform limit. □

Lemma 6.2 (Reflection positivity fixes the prefactor)**.** Let T be symmetric on a vector space with a positive semidefinite Hermitian form. If $\langle v, T^{2k} v\rangle \le C\lambda^{2k}$ for all $k \ge 0$, where $0 < \lambda \le 1$, then

$$\langle v, T^{2k} v\rangle \le \lambda^{2k} \langle v, v\rangle, \qquad |\langle v, T^k v\rangle| \le \lambda^k \langle v, v\rangle.$$

■

Proof. Put $a_k = \langle T^k v, T^k v\rangle$. Cauchy–Schwarz gives $a_k^2 \le a_{k-1} a_{k+1}$. If one a_k with $k \ge 1$ vanishes, this inequality forces all such a_k to vanish. Otherwise the ratios a_{k+1}/a_k are nondecreasing. A ratio larger than λ^2 would contradict $a_k \le C\lambda^{2k}$ at large k. Thus $a_k \le \lambda^{2k} a_0$. A final Cauchy–Schwarz inequality gives the second assertion. □

Proposition 6.3 (The lattice charged estimate)**.** Fix $m_0 > 0$ and an infinite-volume torus-limit state ω_ν. Let Θ be reflection in $x_3 = 1/2$ and let τ be translation by e_3. For an even local element F supported in $x_3 \geq 1$, form its reflection class $[F]$. Translation induces a positive self-adjoint contraction T on the resulting Hilbert space.

Let $\mathcal{F} = \{+,-\} \times \{1,\dots,N_f\}$. The weight of a monomial is $n_f = \#\psi_f - \#\bar{\psi}_f$ and $|n|_1 = \sum_f |n_f|$. If F has weight n, then

$$0 \leq \omega_\nu(\tau^k F\,\Theta F) \leq e^{-\mu(m_0)|n|_1 k}\,\omega_\nu(F\,\Theta F), \qquad k \geq 0. \tag{6.ii}$$

If the conditional Gaussian expectation $h_F(U) = \langle F\rangle_U$ vanishes at every U, the same estimate holds with 2 in place of $|n|_1$. ■

Proof. The fixed-spacing states of [QCDM1] are translation invariant and reflection positive at coordinate planes through sites and through links. Thus the form $\langle [G],[F]\rangle = \omega_\nu(F\Theta G)$ is positive, and $\Theta\tau = \tau^{-1}\Theta$ makes translation symmetric. Wick's determinant formula and $\|D_U^{-1}\| \leq m_0^{-1}$ bound $|\omega_\nu(\tau^k F\Theta F)|$ uniformly in k, with a constant depending on F, m_0. Apply Lem. 6.2 with $\lambda = 1$ to obtain a contraction on the quotient. Positivity follows from reflection through $x_3 = 1$, since $\Theta F = \Theta_1(\tau F)$ and $\omega_\nu(\tau F \Theta_1 \tau F) \geq 0$.

For clarity, consider a pair of monomials of total degree $2p$. Wick's formula is a sum of at most $p!$ products of p propagators. A propagator pairs only a quark with an antiquark of the same species. If the monomial on the positive side has weight n, let c_f^+ and c_f^- count its quarks and antiquarks of species f that are paired across the separation. Each pairing confined to this side removes one quark and one antiquark of the same species. Therefore

$$c_f^+ - c_f^- = n_f, \qquad c_f^+ + c_f^- \geq |n_f|.$$

Summing over species shows that at least $|n|_1$ propagators cross. Between the supports of ΘF and $\tau^{2k}F$, their separation in the third coordinate is at least $2k+1$. By Lem. 6.1, for every $\alpha < \mu(m_0)$,

$$\omega_\nu(\tau^{2k} F\Theta F) \leq C(F,\alpha) e^{-\alpha|n|_1(2k+1)}.$$

The finite sum over the monomials only changes $C(F,\alpha)$. Applying Lem. 6.2 removes this constant. Let $\alpha \uparrow \mu(m_0)$ to obtain eq. 6.ii. This removal is performed at each fixed cutoff. The preliminary constant may depend on the cutoff; the final estimate depends on F only through its squared reflection norm, which can subsequently pass to the continuum limit.

For the last assertion subtract the product of the two conditional Gaussian expectations in Wick's formula. The permutations preserving both sets of factors cancel. Every remaining permutation crosses the separation at least twice: if each monomial is neutral, one crossing in each direction is required; otherwise its even nonzero charge already requires at least two crossings. Since $h_F = 0$, the subtracted product vanishes. This gives $C(F,\alpha)e^{-2\alpha(2k+1)}$ and the same argument proves the asserted bound. □

6.2. **Passing the estimate to the enlarged limit.** Let $\mathcal{H}_n^{\mathrm{ext}}$ be the closed span of positive-time tuples of total weight n. A quadratic label has quark number 0 or ± 2, and every label is a sum of weight vectors. For an irreducible representation π of G, write $|\pi|$ for the largest ℓ^1 norm of one of its maximal-torus weights. This is the ℓ^1 norm of its highest weight: the weights lie in the convex hull of its signed-permutation Weyl orbit.

The three estimates below distinguish three notions of charge. The weight $n = (n_f)_{f\in\mathcal{F}}$ records, for each species, the number of ψ_f factors minus the number of $\bar{\psi}_f$ factors. On its weight space, the maximal torus acts by $V(z)v = z^{-n}v$. Taking every phase equal gives the quark number

$$q = \sum_{f\in\mathcal{F}} n_f, \qquad \{Q = q\} = \bigoplus_{\sum_f n_f = q} \mathcal{H}_n^{\mathrm{ext}}.$$

Thus quark number retains only the sum of the species charges. A meson $\bar{\psi}_{+,f}\psi_{-,f}$ has quark number zero but weight of ℓ^1 norm two. The weight estimate below therefore gives a positive lower bound on such weight spaces within the quark-number-zero sector.

The π-isotypic space is the closed span of all copies of the irreducible representation π. It contains the whole multiplet, and may include vectors of weight zero. Since H commutes with G, a spectral subspace of H in this space contains whole copies of π. Applying the weight estimate to a highest-weight vector therefore bounds the energy throughout the multiplet. This is stronger than applying the weight estimate separately to its zero-weight vectors.

Theorem H (Energy and internal charges)**.** Let S^{ext} be an extended volume-first limit and let m be a limit point of the rescaled mass parameters $\mathsf{m}_j = a_j^{-1} m_{0,j} \in [m_-, m_+]$ along its defining sequence. Then

$$H|_{\mathcal{H}_n^{\mathrm{ext}}} \geq |n|_1 \mathsf{m}, \qquad H|_{\mathcal{H}_\pi^{\mathrm{ext}}} \geq |\pi| \mathsf{m}, \qquad H|_{\{Q=q\}} \geq |q| \mathsf{m}.$$

Only even $|n|_1$ and even quark numbers occur. Every nontrivial representation that occurs has $|\pi| \geq 2$. In particular,

$$H \geq 2\mathsf{m} \quad \text{on } ((\mathcal{H}^{\mathrm{ext}})^G)^\perp, \qquad \ker H \subset (\mathcal{H}^{\mathrm{ext}})^G.$$

The spectral projection onto $\ker H$ vanishes on every nonzero-weight space. The charge-two and charge-minus-two spaces are nonzero. ■□

The gap $2\mathsf{m}$ here is on the complement of the G-invariant vectors. In the invariant sector these estimates give only $H \geq 0$. Since the vacuum vector is invariant and has energy zero, the conclusion is

$$\mathbb{C}\Omega \subset \ker H \subset (\mathcal{H}^{\mathrm{ext}})^G.$$

Neither uniqueness of the vacuum nor a positive gap above it within the invariant sector follows from this theorem.

Proof. Pass to a subsequence with $\mathsf{m}_j \to \mathsf{m}$. For a finite combination of positive-time tuples $\mathcal{X} = \sum_i \zeta_i \mathcal{X}_i$, with $\mathcal{X}_i = ((\lambda_{ir}, f_{ir}))_r$, put

$$F_j = \sum_i \zeta_i \prod_r \left(\Phi_j[\lambda_{ir}](f_{ir}) - \omega_{\nu_j}(\Phi_j[\lambda_{ir}](f_{ir})) \right).$$

Here $\Phi_j[0](f)$ means the action-density insertion. The moment–cumulant formula expresses the correlations of these centered polynomials using only blocks of at least two insertions. Thus only the truncated functions already known to converge occur; convergence of the individual one-point functions is unnecessary. If $\mathcal{X}$ has weight n, then F_j has weight n: the mean of a nonzero-weight field vanishes by the torus symmetry, and subtraction does not change a zero weight.

For large j the support lies above the bond plane $x_3 = a_j/2$. Fix t in $\bigcup_k L^{-k}\mathbb{N}$. Then t/a_j is an integer for all large j, and Prop. 6.3 gives

$$\omega_{\nu_j}(\tau^{t/a_j} F_j \Theta_{a_j/2} F_j) \leq e^{-|n|_1 t \mu(a_j \mathsf{m}_j)/a_j} \omega_{\nu_j}(F_j \Theta_{a_j/2} F_j).$$

Reflection in $a_j/2$ differs from reflection in zero by a translation of size a_j. For C_c^1 functions that translation tends to the identity in both $\|\cdot\|_\infty$ and $\|\cdot\|_\sharp$; in the latter case use uniform continuity of the gradient. The supports remain separated. Hence eq. 5.iii, multilinearity and the moment–cumulant formula let both sides pass to the limit. The limiting rate is m by Lem. 6.1, and

$$\langle [\mathcal{X}], e^{-tH}[\mathcal{X}] \rangle \leq e^{-|n|_1 \mathsf{m} t} \|[\mathcal{X}]\|^2.$$

Strong continuity extends this from the dense set of times to all $t \geq 0$.

The maximal-torus projections $\int z^n V(z)\, dz$ have ranges $\mathcal{H}_n^{\mathrm{ext}}$; they are orthogonal, sum to the identity and commute with H. The Laplace transform inequality forces the spectral measure of H on each such space to be supported in $[|n|_1 \mathsf{m}, \infty)$. Indeed positive mass on $[0, c]$ with $c < |n|_1 \mathsf{m}$ would contradict the inequality as $t \to \infty$.

Every spectral projection of H commutes with G. If the isotypic subspace of π had a nonzero spectral part below $|\pi|\mathsf{m}$, that part would contain a copy of π, hence a vector of its highest weight. This contradicts the weight estimate. An even monomial has $|n|_1 \equiv \sum_f n_f \equiv 0 \pmod 2$. The highest weight of any occurring representation is an occurring weight, so its norm is even and is at least two unless the representation is trivial. Compact-group decomposition proves the estimate on the complement of the invariant vectors. The charge estimate follows by summing weight spaces and using $|n|_1 \geq |\sum_f n_f|$. The conclusions about the kernel follow from $\mathsf{m} \geq m_- > 0$, and the last assertion is Prop. 5.19. □

7. Decay of parity-odd meson correlations

The parity argument needs control at coinciding supports only for a conditional expectation over the fermions. The loop-variance estimate of Thm. B supplies that control. The part with a quark propagator joining the two supports is estimated at separated supports.

Write $h_F(U) = \langle F \rangle_U$ and $z(f) = a \sum_x f(x) \operatorname{tr} D^{-1}_{U,+}(x,x)$, where this trace is over spin, colour and the chambers over x, for one flavour. We use the consequence

$$\operatorname{Var}_\nu z(f) \leq \frac{\|f\|_\infty^2}{4\varrho_J^2} \min\left\{ C(R)^{1/N_f}, \frac{C(R)-1}{N_f^2} \right\} \tag{7.i}$$

of Thm. B, for real bounded f supported in a radius-R ball. Its hypotheses and the corresponding restriction on the coupling are assumed throughout this section.

7.1. **Decay of the part joining the supports.** Let $c_h = 3(1+r) = 9/2$ and $G_U^+ = D^{-1}_{U,+}$. For nonnegative functions define

$$\Pi_U(f,h) = a^2 \sum_{x,y} f(x) h(y) \|G_U^+(x,y)\|_{\mathrm{HS}}^2.$$

Lemma 7.1 (A uniform estimate for one quark loop)**.** Fix $R, d > 0$. There is a number $C'(R,d)$, independent of the cutoff and the state, such that, when $a \leq d/4$ and $a\mathsf{m} \leq 2c_h$, the following holds. Suppose $f, h \in C_c^1$ have supports in $B(x_0,R), B(y_0,R)$ and $|x_0 - y_0| \geq 2R + 3d$. For any two meson labels Γ, Γ', the mean of their conditional connected function is bounded in modulus by

$$N_f C'(R,d) \|\Gamma\| \|\Gamma'\| \|f\|_\infty \|h\|_\infty e^{-m_-(|x_0-y_0|-2R-3d)/(2c_h)}. \tag{7.ii}$$

■

Proof. First, for the inverse between arbitrary site sets A, B, conjugating by a bounded weight with nearest-site increments at most α changes the hopping operator by norm at most $c_h(e^\alpha - 1)$. Choosing $\alpha = \log(1 + m_0/(2c_h))$ gives

$$\|1_A G_U^+ 1_B\| \leq \frac{2}{m_0} e^{-\alpha \operatorname{dist}_{\mathrm{graph}}(A,B)}.$$

For $m_0 \leq 2c_h$, $\alpha \geq m_0/(4c_h)$.

Let $d(x) = \operatorname{dist}(x, B(x_0,R))$ and $\eta(x) = \min\{1, \max\{0, 2 - d(x)/d\}\}$. The commutator $[D_{U,+}, \eta]$ is supported in

$$A_d = \{x : d - a \leq d(x) \leq 2d + a\}, \qquad \|[D_{U,+}, \eta]\| \leq c_h a/d.$$

For f supported where $d(x) \leq d$ and a set $B \subset \{d(x) \geq 2d\}$ at Euclidean distance D_0 from A_d, the resolvent commutator identity gives

$$f^{1/2} G_U^+ 1_B = f^{1/2} G_U^+ [D_{U,+}, \eta] G_U^+ 1_B.$$

Take a Hilbert–Schmidt norm in the first factor and an operator norm in the last factor, and square. The preceding decay estimate yields

$$\Pi_U(f, 1_B) \leq \left(\frac{2c_h}{d\mathsf{m}} \right)^2 e^{-\mathsf{m} D_0/(2c_h)} \Pi_U(f, 1_{A_d}). \tag{7.iii}$$

Choose fixed smooth nonnegative functions φ, g such that $\varphi = 1$ on $B(x_0, R)$ and vanishes where $d(x) \geq d/4$, while $g \geq 1$ where $3d/4 \leq d(x) \leq 9d/4$ and vanishes outside $d/2 < d(x) < 5d/2$. Their supports have distance at least $d/4$. For $\Gamma_* = i\tau_1 \otimes N_f^{-1/2}1$, Wick's formula and $G_U^- = (G_U^+)^\dagger$ give

$$S_2^T[\nu]((\Gamma_*, \varphi), (\Gamma_*, g)) = 2 \int \Pi_U(\varphi, g)\, d\nu.$$

The conditional one-point functions vanish for this label. The separated two-point bound of [QCDM1] therefore bounds the nonnegative integral on the right, uniformly in the cutoff and the state.

For two general labels the conditional connected function is

$$-a^2 \sum_{x,y} f(x)h(y)\, \mathrm{tr}[\Gamma G_U(x,y)\Gamma' G_U(y,x)].$$

Decomposing into the two τ_3 blocks and using Cauchy–Schwarz for their Hilbert–Schmidt norms bounds this by $2N_f\|\Gamma\|\|\Gamma'\|\Pi_U(|f|,|h|)$. The transposed charge-conjugation identity gives

$$\|G_U^+(x,y)\|_{\mathrm{HS}} = \|G_U^+(y,x)\|_{\mathrm{HS}}.$$

Apply eq. 7.iii; the separation from $\operatorname{supp} h$ to A_d is at least $|x_0 - y_0| - 2R - 3d$. The displayed separated meson bound then proves eq. 7.ii, with, for example,

$$C'(R,d) = \left(\frac{2c_h}{dm_-}\right)^2 C_2(d/4; R + 3d)\|g\|_\infty,$$

where C_2 is the companion's separated two-point constant. □

7.2. **The parity tail.** Fix a coordinate direction e_μ. Let c and $w > 0$ belong respectively to $\bigcup_k L^{-k}\frac{1}{2}\mathbb{Z}$ and $\bigcup_k L^{-k}\mathbb{Z}$. Let θ_c be reflection in $x_\mu = c$, and θ_0 reflection in the middle plane $x_\mu = c + w/2$. For

$$\mathcal{F} = \sum_{k=1}^{p} c_k(\Gamma_k, f_k), \qquad \|\mathcal{F}\| = \sum_k |c_k|\|\Gamma_k\|\|f_k\|_\infty,$$

assume $\Gamma_k \in \mathfrak{B}$ and that the real functions $f_k \in C_c^1$ are supported in the open slab $c < x_\mu < c + w$ and in a ball $B(x_*, R)$ centered on its middle plane. The parity signs π_Γ are those of [QCDM1]. Put

$$\mathcal{F}^- = \frac{1}{2}\sum_k c_k\big((\Gamma_k, f_k) - \pi_{\Gamma_k}(\Gamma_k, f_k \circ \theta_0)\big), \qquad s_{\mathcal{F}}(t) = \sigma_S(\theta_c(\mathcal{F}^-)^* \cdot \tau_{te_\mu}\mathcal{F}^-).$$

Proposition 7.2 (Decay of the parity-odd part)**.** Let S be a volume-first limit satisfying the hypotheses of Thm. B. Then $s_{\mathcal{F}}$ is continuous, nonnegative and nonincreasing on $(0,\infty)$. For each cutoff of its defining sequence let F_j^- be the polynomial associated to $\mathcal{F}^-$ and set

$$C_{\mathcal{F}} = \limsup_j \int |h_{F_j^-}|^2\, d\nu_j.$$

This number is finite, with

$$C_{\mathcal{F}} \leq \varrho_J^{-2}\|\mathcal{F}\|^2 \min\{N_f^2 C(R)^{1/N_f}, C(R) - 1\}.$$

For every $d, t_0 > 0$ with $t_0 + w \geq 2R + 3d$,

$$\int_{t_0}^{\infty} s_{\mathcal{F}}(t)\, dt \leq (t_0 + w)C_{\mathcal{F}} + N_f C'(R,d)\|\mathcal{F}\|^2 \frac{2c_h}{m_-} e^{-m_-(t_0+w-2R-3d)/(2c_h)}. \tag{7.iv}$$

Consequently $s_{\mathcal{F}}(t) \to 0$ as $t \to \infty$. ■

Proof. At all sufficiently fine cutoffs the two planes are lattice planes and $m = w/a$ is an integer. Let V_{θ_0} be the linear parity transformation on the fermions, using the matrix $\sigma(e_\mu) \otimes \tau_2$. Then $V_{\theta_0}\Phi_\Gamma(f) = \pi_\Gamma \Phi_\Gamma(f \circ \theta_0)$. Thus $V_{\theta_0} F_j^- = -F_j^-$ and its expectation vanishes. Put $A_j = h_{F_j^-}$ and $\Xi_j = [F_j^-]$ in the reflection space at $x_\mu = c$.

For a meson label $\Gamma = 1_{\mathrm{s}} \otimes X \otimes 1_{\mathrm{c}}$, Wick's rule gives

$$h_{\Phi_\Gamma(f)} = -\alpha_\Gamma z(f) - \beta_\Gamma \overline{z(f)}, \quad \alpha_\Gamma = \mathrm{tr}(e_+ X), \quad \beta_\Gamma = \mathrm{tr}(e_- X), \quad |\alpha_\Gamma|, |\beta_\Gamma| \le N_f \|\Gamma\|.$$

Since A_j has mean zero, subtracting the means of these loops does not change their combination. All functions involved are supported in the same ball, which is preserved by θ_0. Minkowski's inequality and eq. 7.i give

$$\|A_j\|_2 \le \varrho_J^{-1} \|\mathcal{F}\| \min\{N_f C(R)^{1/(2N_f)}, (C(R)-1)^{1/2}\}.$$

This proves the stated uniform estimate.

We next retain the sign coming from parity. Define the antilinear conjugation on generators by $\psi^\sharp = (\bar\psi \tau_2)^{\mathsf{T}}$, $\bar\psi^\sharp = (\tau_2 \psi)^{\mathsf{T}}$, and reverse products. The identities on the doubled field give

$$\Theta_0 = \alpha_{(1,-1)} V_{\theta_0} \sharp, \qquad h_{F^\sharp} = \overline{h_F}, \qquad h_{\alpha_S F} = h_F.$$

As $\theta_c = \tau^{-m}\theta_0$ in lattice units, putting $B_j = \tau^{-m}\alpha_{(1,-1)}((F_j^-)^\sharp)$ gives $\Theta_c F_j^- = -B_j$ and $h_{B_j} = \tau^{-m}\overline{A_j}$. Hence

$$s_j(n) := \langle \Xi_j, T^n \Xi_j \rangle = -\langle A_j, \tau^{n+m} A_j \rangle_{L^2(\nu_j)} - R_j(n), \tag{7.v}$$

where $R_j(n)$ is the averaged conditional connected function of $\tau^n F_j^-$ and B_j. The two supports are in radius-R balls whose centers have distance $na + w$. Applying Lem. 7.1 to their finitely many meson terms gives, for $na + w \ge 2R + 3d$,

$$|R_j(n)| \le N_f C'(R,d) \|\mathcal{F}\|^2 e^{-m_-(na+w-2R-3d)/(2c_h)}.$$

Each of the four reflected/unreflected pairings has coefficient $1/4$, so no extra factor is needed.

The sequence $l \mapsto \langle A_j, \tau^l A_j \rangle$ is of positive type, since translation is unitary on $L^2(\nu_j)$. The relevant finite inequality is

$$0 \le \frac{1}{N} \left\| \sum_{l=0}^{N-1} \tau^l A_j \right\|_2^2 = \|A_j\|_2^2 + 2 \sum_{l=1}^{N-1} (1 - l/N) \, \mathrm{Re}\langle A_j, \tau^l A_j \rangle.$$

Let n_0 be the least integer with $n_0 a \ge t_0$ and take $N > m + n_0$. For $l < m + n_0$ bound the real part by $\|A_j\|_2^2$. For $l \ge m + n_0$, the minus sign in eq. 7.v gives

$$\mathrm{Re}\langle A_j, \tau^l A_j \rangle \le -s_j(l-m) + |R_j(l-m)|.$$

Consequently

$$\begin{aligned} 2 \sum_{n=n_0}^{N-m-1} \left(1 - \frac{n+m}{N}\right) s_j(n) &\le \left[1 + 2 \sum_{l=1}^{m+n_0-1} \left(1 - \frac{l}{N}\right)\right] \|A_j\|_2^2 \\ &\quad + 2 \sum_{n=n_0}^{N-m-1} \left(1 - \frac{n+m}{N}\right) |R_j(n)| \\ &\le [2(m+n_0) - 1] \|A_j\|_2^2 + 2 \sum_{n \ge n_0} |R_j(n)|. \end{aligned}$$

Because $0 \le T \le 1$, every $s_j(n)$ is nonnegative. The weights on the left increase to one as $N \to \infty$, whereas the remainder on the right is summable by its exponential bound. Monotone convergence therefore gives

$$\sum_{n \ge n_0} s_j(n) \le (m + n_0 - \tfrac{1}{2}) \|A_j\|_2^2 + \sum_{n \ge n_0} |R_j(n)|.$$

Multiply by a. The first coefficient is at most $w + t_0 + a/2$, and the geometric-series factor in the second term is $a/(1 - e^{-m_- a/(2c_h)}) \to 2c_h/m_-$.

At every time t in $\bigcup_k L^{-k}\mathbb{N}$, $s_j(t/a_j) \to s_{\mathcal{F}}(t)$ by the defining correlation limits. The reflected supports stay separated, so continuity of test-function translation and the companion's two-point bound make $s_{\mathcal{F}}$ continuous. Nonnegativity and monotonicity pass from s_j on the dense set of times to all times. Fix a finite partition $t_0 < t_1 < \cdots < t_p$, with $t_1, \dots, t_p$ in that dense set. For all sufficiently fine cutoffs, monotonicity gives the explicit lower sum

$$a_j \sum_{n \ge n_0} s_j(n) \ge \sum_{i=1}^{p} (t_i - t_{i-1} - a_j) s_j(t_i/a_j).$$

Only finitely many continuum correlation limits are needed on the right. Pass along the defining subsequence, use the upper bound on the left, and then take the supremum over partitions of each fixed interval. Finally let the upper endpoint tend to infinity. Thus the continuum integral is obtained after the finite-cutoff tail bound; no uniform integrable majorant for the correlations has been assumed. The preceding estimates give eq. 7.iv. Finally $(t - t_0)s_{\mathcal{F}}(t) \le \int_{t_0}^{t} s_{\mathcal{F}}(s)\, ds$, proving decay. □

For a real u symmetric about a coordinate plane, the field $\Phi_{\tau_3 \otimes 1}(u)$ is one of these parity-odd combinations. Choose a slightly wider lattice-compatible slab about that plane. Its ordinary two-point correlation with its translate is $s_{\mathcal{F}}$ with a fixed shift in the translation parameter. It therefore tends to zero and has the corresponding integrable tail.

7.3. **Products in a slab.** The same argument applies to products of centered bilinears. The two extra estimates are the moment bound for their conditional expectations and the decay of a permutation cycle joining the two slabs. We give them with constants that do not depend on the cutoff.

Use the slab and reflections above. For a tuple $\mathcal{X} = ((\Gamma_i, f_i))_{i=1}^{n}$ put

$$X_j[\mathcal{X}] = \prod_{i=1}^{n} (\Phi_{\Gamma_i}(f_i) - \omega_{\nu_j}(\Phi_{\Gamma_i}(f_i))), \qquad \pi_{\mathcal{X}} = \prod_i \pi_{\Gamma_i}.$$

Reflection of a tuple means reflection of each test function. Let $\mathcal{X}_1, \dots, \mathcal{X}_p$ have real bounded test functions supported compactly in the slab and in $B(x_*, R)$. Within each tuple their supports have pairwise distance at least $d > 0$. Let $B_{k,j}$ be continuous functions of finitely many links with both endpoints in the slab, with $b_k = \sup_j \|B_{k,j}\|_{L^4(\nu_j)} < \infty$. For coefficients $c_k \in \mathbb{C}$ set

$$F_j = \sum_k c_k B_{k,j} X_j[\mathcal{X}_k], \quad F_j^- = (F_j - V_{\theta_0} F_j)/2, \quad A_j = \mathsf{h}[F_j^-], \quad \Xi_j = [F_j^-].$$

Writing $n_* = \max_k n_k$, define

$$\begin{aligned} \|\mathcal{F}\| &= \sum_k |c_k| b_k \prod_{i=1}^{n_k} \|\Gamma_{k,i}\| \|f_{k,i}\|_\infty, \\ C_{\mathrm{qc}}(d) &= \max\{1, 2c_h/(dm_-)\}, \\ C_1 &= \big(n_*! C_{\mathrm{W}}(4n_*)^{n_*}\big)^2 C_{\mathrm{mb}}(R,d)^{1/2} \|\mathcal{F}\|^2, \\ C_2 &= (2n_*)! \big(C_{\mathrm{qc}}(d) C_{\mathrm{W}}(4n_*)\big)^{2n_*} C_{\mathrm{mb}}(R,d)^{1/2} \|\mathcal{F}\|^2. \end{aligned}$$

These C_1, C_2 are used only in this subsection.

Theorem I (Parity-odd products)**.** Let S be a volume-first limit as in Prop. 7.2, and let the preceding data be given, with $\Gamma_{k,i} \in \mathfrak{B}$. At every sufficiently fine cutoff, $\|A_j\|_2^2 \le C_1$ and $s_j(n) = \langle \Xi_j, T^n \Xi_j \rangle$ is nonnegative and nonincreasing. If $a \le d/4$, $a\mathsf{m}_j \le 2c_h$, and $na + w \ge 2R + 3d$, then

$$s_j(n) = -\langle A_j, \tau^{n + w/a} A_j \rangle - R_j(n), \qquad |R_j(n)| \le C_2 e^{-m_-(na + w - 2R - 3d)/(2c_h)}. \tag{7.vi}$$

For $t_0 > 0$ with $t_0 + w \geq 2R + 3d$, and $n_0 = \lceil t_0/a \rceil$,

$$a \sum_{n \geq n_0} s_j(n) \leq (t_0 + w + a/2)C_1 + (2c_h/m_- + a)C_2 e^{-m_-(t_0+w-2R-3d)/(2c_h)}. \tag{7.vii}$$

For $na > t_0$, $s_j(n)$ is at most the right side divided by $na - t_0$.

If $B_{k,j} = 1$ and all $f_{k,i} \in C_c^1$, define

$$\mathcal{F}^- = \frac{1}{2} \sum_k c_k (\mathcal{X}_k - \pi_{\mathcal{X}_k} \theta_0 \mathcal{X}_k), \qquad s_{\mathcal{F}}(t) = \sigma_S(\theta_c (\mathcal{F}^-)^* \cdot \tau_{te_\mu} \mathcal{F}^-).$$

Here σ_S on a tuple is the sum over partitions into blocks of size at least two of the products of the limiting cumulants, extended sesquilinearly to these combinations. Then $s_{\mathcal{F}}$ is continuous, nonnegative and nonincreasing for $t > 0$, and

$$\int_{t_0}^{\infty} s_{\mathcal{F}}(t)\, dt \leq (t_0 + w)C_1 + \frac{2c_h}{m_-} C_2 e^{-m_-(t_0+w-2R-3d)/(2c_h)}. \tag{7.viii}$$

In particular $s_{\mathcal{F}}(t) \to 0$. ■□

Proof. At fine enough cutoffs both bounding planes are lattice mirrors and $m = w/a$ is an integer. Reflection covariance and centering give

$$V_{\theta_0}(B_{k,j} X_j[\mathcal{X}_k]) = \pi_{\mathcal{X}_k}(\theta_0 B_{k,j}) X_j[\theta_0 \mathcal{X}_k].$$

Conditional expectation commutes with this linear reflection. Thus, with $H_k = B_{k,j} \mathsf{h}[X_j[\mathcal{X}_k]]$, $A_j = \frac{1}{2} \sum_k c_k (H_k - H_k \circ \theta_0)$. Reflection invariance, Hölder, and Prop. 3.8 at $q = 4$ give

$$\|A_j\|_2 \leq \sum_k |c_k| \|B_{k,j}\|_4 \|\mathsf{h}[X_j[\mathcal{X}_k]]\|_4 \leq C_1^{1/2}.$$

Every F_j is neutral and supported in the slab. The algebraic identities with $\sharp$ in the proof of Prop. 7.2 therefore give eq. 7.v for this F_j^- as well. This proves the identity in eq. 7.vi; $0 \leq T \leq 1$ gives nonnegativity and monotonicity. It remains to bound the conditional connected term.

Expand the reflected and translated factors. A pairing of tuples $\mathcal{X}_k, \mathcal{X}_{k'}$ gives two collections of bilinears in balls of radius R with centers $x_* - we_\mu$ and $x_* + nae_\mu$. Their link coefficients retain their L^4 norms. By Lem. 3.7, the conditional connected term is a sum over permutations of $N = n_k + n_{k'}$ indices having at least two edges between the two collections.

For such an edge, with functions $f_l, f_{l'}$, use eq. 7.iii with the ball supporting f_l and the set supporting $f_{l'}$. The annulus lies in $B(x_0, R+3d)$, at distance at least $3d/4$ from $\operatorname{supp} f_l$. Its distance from the other support is at least $na + w - 2R - 3d$. Hence

$$\begin{aligned}(2N_f \Pi_U(|f_l|, |f_{l'}|))^{1/2} \leq C_{\mathrm{qc}}(d) e^{-m_-(na+w-2R-3d)/(4c_h)} \|f_{l'}\|_\infty^{1/2} \\ \times (2N_f \Pi_U(|f_l|, 1_{A_d}))^{1/2}.\end{aligned}$$

Apply this bound to the crossing edges and Lem. 3.7 to the remaining edges. For the two link coefficients use Hölder exponents $4, 4$, and for the N cycle factors use exponent $2N$ each. Their reciprocals sum to one. The fixed points are controlled by Lem. 3.2; Lem. 3.6 controls the internal edges at (R, d) and the crossing edges at $(R + 3d, 3d/4)$. As in the proof of Prop. 3.8, the function norms give their full product. Two crossing edges supply the exponential in eq. 7.vi. There are at most $N!$ permutations, and $N \leq 2n_*$. Summing over k, k' gives C_2: the four reflected pairings have coefficient $1/4$.

The Fejér argument in Prop. 7.2 now gives

$$\sum_{n \geq n_0} s_j(n) \leq (m + n_0 - \tfrac{1}{2}) \|A_j\|_2^2 + \sum_{n \geq n_0} |R_j(n)|.$$

Multiplication by a, together with $a/(1 - e^{-m_- a/(2c_h)}) \leq 2c_h/m_- + a$, proves eq. 7.vii. Monotonicity gives the stated pointwise bound.

For the continuum statement, the reflected and translated tuples have disjoint supports for every $t > 0$. The expectation of their centered product is the sum of products of cumulants over partitions without singleton blocks. Each term therefore converges along the defining sequence. This proves convergence of $s_j(t/a_j)$ on the dense set of lattice-compatible times. Continuity follows by multilinearity, the uniform separated cumulant estimates, and continuity of translations in the supremum norm for C_c^1 test functions. Nonnegativity and monotonicity then hold for every $t > 0$. The finite-partition argument at the end of the proof of Prop. 7.2 passes eq. 7.vii to eq. 7.viii. Finally, $(t - t_0)s_{\mathcal{F}}(t) \le \int_{t_0}^{t} s_{\mathcal{F}}$. □

7.4. **Adding single action densities.** The bound in Prop. 3.11 alone does not give a uniform parity tail. For a single action density we instead keep the near translates together in the Fejér sum. Oddness turns their sum into two test functions separated from the slab.

Lemma 7.3 (Keeping the near translates)**.** Let $A \in L^2(\nu)$ and let τ be the unitary operator induced by an invertible, ν-preserving translation. Let $m \ge 1$ and $n_0 \ge 0$ be integers. Suppose $s(n) = -\langle A, \tau^{n+m} A\rangle - R(n)$, $s(n) \ge 0$, and $\sum_{n \ge n_0} |R(n)| < \infty$. Then

$$\sum_{n \ge n_0} s(n) \le \frac{1}{2} \sum_{|l| < m + n_0} \mathrm{Re}\langle A, \tau^l A\rangle + \sum_{n \ge n_0} |R(n)|.$$

The finite sum on the right is at most $(m + n_0 - \frac{1}{2})\|A\|_2^2$. ■

Proof. Use the positive Fejér sum from the proof of Prop. 7.2, retaining the terms $|l| < m+n_0$ instead of estimating them separately. Substitute the identity for the remaining terms. The coefficients of $s(n)$ increase to one, while the finite sum converges and the remainder is absolutely summable. Monotone convergence proves the first inequality. The second follows from $|\langle A, \tau^l A\rangle| \le \|A\|_2^2$. □

Choose $\chi \in C^1(\mathbb{R}; [0,1])$ with $\chi = 1$ on $(-\infty, 0]$, $\chi = 0$ on $[1, \infty)$, and $|\chi'| \le 2$. For $u \in C_c^1(\mathbb{R}^3)$ and an integer $k > m = w/a$, put

$$\begin{aligned} H(x) &= a \sum_{|l| < k} u(x - lae_\mu), \\ H^+(x) &= -a\chi(x_\mu - c - ka - w) \sum_{l \ge k} u(x - lae_\mu), \\ H^-(x) &= -a\chi(c - ka - x_\mu) \sum_{l \le -k} u(x - lae_\mu). \end{aligned} \tag{7.ix}$$

Lemma 7.4 (The window of an odd function)**.** Suppose $u \circ \theta_0 = -u$ and $\operatorname{supp} u$ is a compact subset of $c < x_\mu < c + w$, with both mirrors lattice-compatible. At every plaquette barycenter, $\sum_{l \in \mathbb{Z}} u(x - lae_\mu) = 0$ and $H = H^+ + H^-$. In particular $O(H) = O(H^+) + O(H^-)$. The two tails are in C_c^1 and have supports in

$$c + ka \le x_\mu \le c + ka + w + 1, \qquad c - ka - 1 \le x_\mu \le c + w - ka,$$

respectively, at distance at least $ka - w$ from the closed slab. If u is supported in a ball of radius R, each tail is supported in a ball of radius $R + w + 1$, and

$$\|H^{\pm}\|_\sharp \le (1 + 80M) w \|u\|_\sharp.$$

■

Proof. At a plaquette barycenter $2x_\mu \in a\mathbb{Z}$. Reflection sends $x - lae_\mu$ to $x - l'ae_\mu$, where $l' = (2x_\mu - 2c - w)/a - l$ is an integer. This bijection of the translates and oddness give the zero sum. Hence H is minus the two omitted tails. The factors χ do not change their values at barycenters: where the positive cutoff differs from one, the sum over $l \ge k$ is already the full, vanishing sum, and likewise for the negative cutoff.

The support claims follow directly from the slab support of u and the cutoffs. Only finitely many translates contribute where the cutoffs are nonzero, giving C^1 regularity and compact support. The transverse support lies in a disc of radius R, and the longitudinal width is at most $w+1$, so the stated ball suffices. At each point at most m translates contribute. Thus $\|H^{\pm}\|_\infty \le w\|u\|_\infty$ and $\operatorname{Lip} H^{\pm} \le w \operatorname{Lip} u + 2w\|u\|_\infty$. The definition $\|u\|_\sharp = \max\{\|u\|_\infty, 40M \operatorname{Lip} u\}$ gives the norm bound. □

Let $F_{0,j}$ be a combination of products as in Thm. I, now with $B_{k,j} = 1$ and C_c^1 test functions. Denote its constants by C_1^0, C_2^0, both zero if the combination is empty. Add a finite combination of single centered action densities with real C_c^1 test functions in the same slab and ball. Its odd part has the form $O(u)$, where $u = \frac{1}{2}\sum_l d_l(h_l - h_l \circ \theta_0)$. For $d' = \min\{d, t_0\}$ and $R' = R + w + 1$ put

$$\overline{C}_p(d'; R') = \max\{1, C_2^\infty(d'; R'), \dots, C_p^\infty(d'; R')\}.$$

These are the separated-cumulant bounds of [QCDM1], in the supremum norm for bilinear tests and the sharp norm for action-density tests. Let B_p be the pth Bell number. If $F_{0,j} = \sum_k c_k X_j[\mathcal{X}_k]$, define

$$D(t_0) = 4(1+80M)w\|u\|_\sharp \Big[\sum_k |c_k| B_{n_k+1} \overline{C}_{n_k+1}(d'; R')^{n_k+1} \prod_i \|f_{k,i}\|_\infty + C_2^\infty(d'; R')\|u\|_\sharp \Big].$$

As in the preceding theorem the bilinear labels belong to the fixed basis $\mathfrak{B}$ used in those bounds.

Theorem J (Parity tail with single action densities)**.** For the enlarged combination, let $s_j(n)$ and $s_{\mathcal{F}}(t)$ be defined as in Thm. I, using parity sign $+1$ for the action density. Under the same cutoff conditions and $t_0 + w \ge 2R + 3d$,

$$a \sum_{n \ge \lceil t_0/a \rceil} s_j(n) \le (t_0 + w + a/2)C_1^0 + D(t_0) + (2c_h/m_- + a)C_2^0 e^{-m_-(t_0+w-2R-3d)/(2c_h)},$$

$$\int_{t_0}^\infty s_{\mathcal{F}}(t)\, dt \le (t_0+w)C_1^0 + D(t_0) + \frac{2c_h}{m_-} C_2^0 e^{-m_-(t_0+w-2R-3d)/(2c_h)}.$$

Both correlations are nonnegative and nonincreasing; $s_{\mathcal{F}}$ is continuous for $t > 0$ and tends to zero. The respective right sides, divided by $na - t_0$ or $t - t_0$, also give pointwise bounds beyond t_0. ■□

Proof. At sufficiently small a, every plaquette meeting an action-density test has all its links inside the slab, because that test has compact support in the open slab. The reflection-space argument therefore applies. The conditional mean of the odd part is $A_j = A_{0,j} + O(u)$, with both summands of mean zero. The action density is a function of the links, so it factors out of conditional expectation. In the conditional connected function all terms containing it cancel. Thus the remainder is precisely that of $F_{0,j}$ and has the bound in Thm. I with C_2^0.

Apply Lem. 7.3 with $k = m + \lceil t_0/a \rceil$. By unitarity and the symmetric range of translates,

$$\frac{a}{2} \sum_{|l|<k} \operatorname{Re}\langle A_j, \tau^l A_j\rangle = \frac{a}{2} \sum_{|l|<k} \operatorname{Re}\langle A_{0,j}, \tau^l A_{0,j}\rangle + \operatorname{Re}\langle A_{0,j}, O(H)\rangle + \tfrac{1}{2}\operatorname{Re}\langle O(u), O(H)\rangle.$$

The first term is at most $(t_0 + w + a/2)C_1^0$. By Lem. 7.4, $O(H)$ is the sum of two tails, each separated by at least t_0 from the slab and with sharp norm at most $(1+80M)w\|u\|_\sharp$.

For either tail $v = H^{\pm}$, expand $\langle A_{0,j}, O(v)\rangle$ using the products in $F_{0,j}^-$ and the conditional conjugation identity. Their means vanish, so this is a centered product with $n_k + 1$ factors. Expand it into cumulants over partitions without singleton blocks. The supports of every block are separated

by at least d', and each lies in a ball of radius R'. There are at most B_{n_k+1} partitions, and at most n_k+1 factors of $\overline{C}_{n_k+1}(d';R')$. Splitting the complex tail into its real and imaginary parts gives

$$|\langle A_{0,j}, O(v)\rangle| \le 2\sum_k |c_k| B_{n_k+1}\overline{C}_{n_k+1}(d';R')^{n_k+1}\prod_i \|f_{k,i}\|_\infty \|v\|_\sharp.$$

Similarly the real and imaginary parts of both tests give

$$|\langle O(u), O(v)\rangle| \le 4C_2^\infty(d';R')\|u\|_\sharp\|v\|_\sharp.$$

After summing the two tails these terms are bounded by $D(t_0)$. Summing the exponential remainder proves the lattice inequality.

The continuum passage is the finite-partition argument of Thm. I. Action-density cumulants are among the defining separated correlations. For their tests, translation is continuous in the sharp norm because the test and its derivative are uniformly continuous. Thus the same argument proves continuity, positivity, monotonicity, the integral bound and decay. □

There is also a version of the Fejér estimate in which one first takes a conditional mean. It separates the part visible to a chosen collection of link observables from its remaining translation correlations.

Lemma 7.5 (Fejér estimate for a conditional mean)**.** In Lem. 7.3, suppose also that $s(n)$ is nonincreasing. Let $q \ge 1$ be an integer and let $\mathcal{G}$ be a sigma-algebra preserved in both directions by τ^q. Put $A^{\mathcal{G}} = \mathsf{E}_\nu[A \mid \mathcal{G}]$ and $c^{\mathcal{G}}(l) = \langle A - A^{\mathcal{G}}, \tau^l(A - A^{\mathcal{G}})\rangle$. For $l \in q\mathbb{Z}$,

$$\langle A, \tau^l A\rangle = \langle A^{\mathcal{G}}, \tau^l A^{\mathcal{G}}\rangle + c^{\mathcal{G}}(l).$$

If i_0 is the least integer with $i_0 q \ge m + n_0$, then

$$\sum_{n \ge n_0+q} s(n) \le (m + n_0 + q/2)\|A^{\mathcal{G}}\|_2^2 + q\sum_{i \ge i_0}\big(|R(iq-m)| + |c^{\mathcal{G}}(iq)|\big),$$

with an infinite right side allowed. ■

Proof. Invariance of ν and $\mathcal{G}$ implies $\mathsf{E}_\nu[\tau^q B \mid \mathcal{G}] = \tau^q \mathsf{E}_\nu[B \mid \mathcal{G}]$: test the identity against a bounded $\mathcal{G}$-measurable function and change variables by τ^q. Orthogonality to $L^2(\mathcal{G})$ then removes the two cross terms, proving the identity. Apply the Fejér sum to $\langle A^{\mathcal{G}}, \tau^{iq} A^{\mathcal{G}}\rangle$. If the remainder sum is finite, the previous argument yields

$$\sum_{i \ge i_0} s(iq-m) \le (i_0 - \tfrac{1}{2})\|A^{\mathcal{G}}\|_2^2 + \sum_{i \ge i_0}\big(|R(iq-m)| + |c^{\mathcal{G}}(iq)|\big).$$

Monotonicity bounds each group of q consecutive terms by q times its first term, and $q(i_0 - \frac{1}{2}) \le m + n_0 + q/2$. This gives the claim. If the remainder sum is infinite the inequality is automatic. □

8. Flavour currents

The conserved lattice current is supported at the two ends of a link. The source estimates in [QCDM1] concern bilinears at one site. We first compare these two insertions at a fixed gauge configuration. The error contains a covariant difference of a fermion propagator away from its source. An energy inequality controls this difference by a positive pseudoscalar two-point function. The source estimates then compare the site insertion with its free value, whose leading term is computed by a Pauli matrix trace.

The construction and all data remain those of the chamber model with $r = 1/2$. We retain r in a few local identities to display the coefficients of the hopping operator. For a neighbouring pair of places $v, v' = v \cdot s$, write

$$h(v,v') = t_s(r - \gamma(u_s(v))), \qquad t^T(v,v') = \bar\psi(v)h(v,v')TU(v,v')\psi(v'),$$

where $T = e_+ \otimes t_+ + e_- \otimes t_-$ acts on species and flavour. Thus T commutes with the gamma matrices and with colour. Its reflection involution is $T^* = \tau_3 T^\dagger \tau_3 = T^\dagger$. The chamber graph, wall directions and transporters are those in section 2. In this section we label sites by $x \in a\mathbb{Z}^3$: the place (x, q) corresponds to $(a^{-1}x, q)$ in eq. 2.i. Set

$$\begin{aligned} \mathfrak{j}^T(v, v') &= t^T(v', v) - t^T(v, v'), \\ j^T_\mu(x) &= \sum_{g:\, g e_1 = e_\mu} \mathfrak{j}^T((x, g), (x, g)\cdot 0), \\ J^T(f) &= ia \sum_{x,\mu} f_\mu(x + \tfrac{1}{2} a e_\mu) j^T_\mu(x). \end{aligned} \tag{8.i}$$

The site sum has eight terms. The factor i is part of the definition of J^T; it determines the sign of its two-point function and its reflection law.

Proposition 8.1 (Lattice conservation)**.** Let δ^T_v be the derivation which fixes the links and satisfies

$$\delta^T_v \psi(u) = \delta_{uv} T \psi(u), \qquad \delta^T_v \bar\psi(u) = -\delta_{uv} \bar\psi(u) T.$$

At every gauge configuration, with $(\operatorname{div} \mathfrak{j}^T)(v) = \sum_{v' \sim v} \mathfrak{j}^T(v, v')$, the conditional Gaussian functional satisfies

$$\langle (\operatorname{div} \mathfrak{j}^T)(v) F \rangle_U = -\langle \delta^T_v F \rangle_U \tag{8.ii}$$

for every polynomial F in finitely many fermion generators. The identity also holds after gauge averaging. At a site,

$$\sum_{v : x_v = x} (\operatorname{div} \mathfrak{j}^T)(v) = \sum_\mu [j^T_\mu(x) - j^T_\mu(x - a e_\mu)].$$

Thus the divergence vanishes in correlations away from the fermion insertions; at an insertion it gives the contact term on the right of eq. 8.ii. ■

Proof. The diagonal mass term is invariant under δ^T_v. Since T commutes with the hopping matrices and the links,

$$\delta^T_v(\bar\psi D_U \psi) = -(\operatorname{div} \mathfrak{j}^T)(v).$$

The changes $\psi(v) \mapsto e^{tT}\psi(v)$ and $\bar\psi(v) \mapsto \bar\psi(v) e^{-tT}$ have cancelling Berezin Jacobians. Differentiating their change-of-variables identity proves eq. 8.ii on a torus. On the infinite lattice the same identity follows from $D_U D_U^{-1} = 1$ and Wick's rule. Summing over the chambers at one site cancels the internal edges and leaves the displayed site divergence. Finally, $\delta^T_v F = 0$ when F has no generator at v. □

Fix $0 < \vartheta < 1$. In this section a test pair at scale d means two nonzero, nonnegative Lipschitz functions f, h such that each support lies in a ball of radius d,

$$\begin{gathered} d \le \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) \le 6d, \\ \int f \ge \vartheta d^3 \|f\|_\infty, \qquad \int h \ge \vartheta d^3 \|h\|_\infty, \qquad \operatorname{Lip} f \le \frac{\|f\|_\infty}{\vartheta d}, \quad \operatorname{Lip} h \le \frac{\|h\|_\infty}{\vartheta d}. \end{gathered} \tag{8.iii}$$

On a torus we also require $20d \le \ell$ and read the supports in one Euclidean ball of radius $5d$. In particular $d \le |x - y| \le 10d$ on the product of the supports. Put

$$k_{\mu\nu}(x) = \frac{\delta_{\mu\nu}|x|^2 - 2x_\mu x_\nu}{|x|^6}, \qquad \mathcal{K}_4(f, h) = \iint f(x) h(y) |x - y|^{-4}\, dx\, dy.$$

The test-pair assumptions give

$$10^{-4} \vartheta^2 d^2 \|f\|_\infty \|h\|_\infty \le \mathcal{K}_4(f, h) \le (4\pi/3)^2 d^2 \|f\|_\infty \|h\|_\infty. \tag{8.iv}$$

Theorem K (Two currents)**.** There are a constant $C'_V(g)$ and a nondecreasing function $\epsilon_V(g,\vartheta;d)$ tending to zero as $d\downarrow 0$ with the following properties. Let $0<d\le d_*:=\min\{1/200,1/(20m_+)\}$ and $16a\le d$, and let (f,h) be a test pair as in eq. 8.iii. The state may be a torus functional or a functional ω_ν with $\nu\in\mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$. For unit vectors e,e', $a_0,b_0\in\{0,3\}$ and Hermitian flavour matrices t,t', put $T=\tau_{a_0}\otimes t$, $T'=\tau_{b_0}\otimes t'$. Then

$$\kappa_2(J^T(fe),J^{T'}(he'))=\frac{\operatorname{tr}(tt')}{2\pi^2}\left[\delta_{a_0b_0}\iint f(x)h(y)\langle e,k(x-y)e'\rangle\,dx\,dy+\varepsilon\mathcal{K}_4(f,h)\right], \tag{8.v}$$

where

$$|\varepsilon|\le\epsilon_V(g,\vartheta;d)+C'_V(g)\left[\left(\frac{a}{\vartheta^4 d}\right)^{1/2}+\frac{a}{\vartheta^3 d}\right]. \tag{8.vi}$$

If $\operatorname{tr}(tt')=0$, the cumulant is exactly zero.

At fixed test pair, these cumulants are bounded along every sequence with $K\to\infty$. Every limit point satisfies eq. 8.v with $|\varepsilon|\le\epsilon_V(g,\vartheta;d)$. This holds on a fixed torus and for infinite-lattice states; it also holds along tori on which the fixed pair is read with $20d\le\ell$. ■□

The assertion about limit points concerns the indicated two-point functions. It does not by itself construct all joint current correlations. For an orthonormal Hermitian flavour basis, $\operatorname{tr}(t_jt_k)=\delta_{jk}$, the leading coefficient is $\delta_{a_0b_0}\delta_{jk}/(2\pi^2)$.

8.1. **Conditional loops and their signs.** We use the resolvent in lattice units in this section:

$$G_U=(D_+[U]+m_0)^{-1}=a^{-1}\mathsf{G}_+.$$

Let $h_+(v,v')=t_s(r-\sigma(u_s(v)))$ be the upper-species hop. Define $Y(v,v')$ by its two nonzero blocks

$$Y(v,v')[v',v]=h_+(v',v)U(v',v),\qquad Y(v,v')[v,v']=-h_+(v,v')U(v,v'),$$

and sum these over the eight chamber edges above a link to obtain $Y_\mu(x)$. For a site vector function F set

$$Y(F)=\sum_{x,\mu}F_\mu(x)Y_\mu(x),\qquad N(F)(x,g)=\sigma(F(x)).$$

For a scalar function f and a unit vector e use $F^{f,e}_\mu(x)=f(x+ae_\mu/2)e_\mu\cdot e$, and abbreviate $Y(fe)=Y(F^{f,e})$, $N(fe)=N(F^{f,e})$. The two loop expectations to be compared are

$$\begin{aligned}E_\nu(f,e;h,e')&=a^2\mathsf{E}_\nu\operatorname{Tr}(G_UY(fe)G_UY(he')),\\ E^{\mathrm{s}}_\nu(f,e;h,e')&=a^2\mathsf{E}_\nu\operatorname{Tr}(G_UN(fe)G_UN(he')).\end{aligned} \tag{8.vii}$$

The trace includes places, spin and colour, and no flavour variable. Subscript 1 denotes the value at $U=1$. We also use the positive form

$$P_\nu(F,H)=2a^2\sum_{u,u'}F(x_u)H(x_{u'})\mathsf{E}_\nu\|G_U(u,u')\|^2_{\mathrm{HS}} \tag{8.viii}$$

for scalar $F,H\ge0$. For a site matrix A acting trivially on colour, write

$$\Phi_A=a\sum_v\bar\psi(v)A(x_v)\psi(v).$$

This is the bilinear in eq. 2.iii, now allowing A to act on spin as well.

Lemma 8.2. The conditional mean of every current is zero. For real f,h,

$$\kappa_2(J^T(fe),J^{T'}(he'))=\operatorname{tr}(t_+t'_+)E_\nu(f,e;h,e')+\operatorname{tr}(t_-t'_-)\overline{E_\nu(f,e;h,e')}. \tag{8.ix}$$

For real site vector functions F,H, choose a Hermitian t with $\operatorname{tr}t^2=1$ and let $N_F=\sigma(F)\otimes e_+\otimes t$, acting as the identity on colour. Then

$$\omega_\nu(\Phi_{N_F})=0,\qquad \kappa_2(\Phi_{N_F},\Phi_{N_H})=-a^2\mathsf{E}_\nu\operatorname{Tr}(G_UN(F)G_UN(H)).$$

For $\Gamma = i\tau_1 \otimes t$ the corresponding scalar bilinears satisfy $\kappa_2(\Phi_\Gamma(F), \Phi_\Gamma(H)) = P_\nu(F, H)$. ■

Proof. Let $C = \sigma_2 \otimes \epsilon$ on spin and colour, where $\epsilon U^{\mathsf{T}} \epsilon^{-1} = U^{-1}$ for $U \in \mathrm{SU}(2)$. Charge conjugation gives

$$G_U(u, u')^{\mathsf{T}} = C^{-1} G_U(u', u) C, \qquad \sigma_2 \sigma_\mu^{\mathsf{T}} \sigma_2 = -\sigma_\mu.$$

Consequently the two orders of $G_U(u, u')$ have equal Hilbert–Schmidt norms and $\mathrm{tr}(\sigma_\mu G_U(v, v)) = 0$. Let $\mathcal{C}$ be C extended by the identity on species and flavour. The change of Grassmann variables $\psi \mapsto (\bar\psi \mathcal{C})^{\mathsf{T}}$, $\bar\psi \mapsto -(\mathcal{C}^{-1}\psi)^{\mathsf{T}}$ preserves the action and its Berezin measure and sends j^T to $-\mathsf{j}^{T^{\mathsf{T}}}$. On the infinite lattice the same invariance follows directly from the covariance identity and Wick's rule.

At fixed U the current mean is a flavour-invariant linear form in T, hence a linear combination of $\mathrm{tr}\, t_+$ and $\mathrm{tr}\, t_-$. It is unchanged by transposition and changed in sign by charge conjugation, so it is zero. Thus the covariance of conditional means makes no contribution. Wick's rule gives $-\mathrm{Tr}(GYGY')$ for two bilinears, while the two factors ia in the currents give $-a^2$. In the lower species $G_- = G_+^*$ and $Y_- = -Y_+^*$; cyclicity of the trace gives the complex conjugate loop. This proves eq. 8.ix.

The vanishing Pauli trace proves the assertion about the site mean; its second cumulant is the stated Wick trace. Finally $i\tau_1$ exchanges the two species. Its two orientations in the loop give twice the squared Hilbert–Schmidt norm, since $G_- = G_+^*$ and the two orders have the same norm. This proves the last identity. □

8.2. **Replacing the two endpoints by a site.** For a chamber function φ, let $\nabla_s \varphi(v) = \varphi(v) - U(v, v \cdot s)\varphi(v \cdot s)$. For a set of sites S write

$$\|\varphi\|_S^2 = \sum_{x_v \in S} |\varphi(v)|^2, \qquad \mathfrak{D}_S(\varphi) = \sum_{x_v \in S} \sum_{s=0}^{3} |\nabla_s \varphi(v)|^2.$$

Lemma 8.3. For a site vector function F supported in S, $Y(F) = N(F) + R(F)$, where

$$|\langle \chi, R(F)\varphi \rangle| \le c_Y \|F\|_\infty (\mathfrak{D}_S(\chi)^{1/2} \|\varphi\|_S + \|\chi\|_S \mathfrak{D}_S(\varphi)^{1/2}), \qquad c_Y = \sqrt{3}(51 + 3r). \tag{8.x}$$

■

Proof. For $v = (x, g)$ with $g e_1 = e_\mu$, the wall-zero hops are $3(r - \sigma_\mu)$ and $3(r + \sigma_\mu)$. Expanding the two endpoint values about v gives

$$\begin{aligned}\langle \chi, Y(v, v \cdot 0)\varphi \rangle = {} & 6\chi(v)^* \sigma_\mu \varphi(v) - 3(\nabla_0 \chi(v))^* (r + \sigma_\mu)\varphi(v) \\ & + 3\chi(v)^* (r - \sigma_\mu) \nabla_0 \varphi(v).\end{aligned}$$

The last two terms sum to at most $3(1 + r)$ times the right-hand expression in eq. 8.x at x. To replace the first term by the sum over all 48 chambers, choose a map from these chambers onto the eight with $g e_1 = e_\mu$, with six preimages each. Put $b(g) = \chi(x, g)^* \sigma_\mu \varphi(x, g)$ and denote the map by π_μ. Then

$$6 \sum_{g:\, g e_1 = e_\mu} b(g) = \sum_g b(g) + \sum_g \big(b(\pi_\mu(g)) - b(g)\big).$$

The first sum is the site insertion. Join each chamber to its image by a shortest gallery using walls $1, 2, 3$. These walls connect the chambers of one site, since their positive quadratic form has precisely the chamber-independent fields as its kernel. Their transporters are the identity. For one step $w \mapsto w \cdot s$ of a gallery,

$$b(w \cdot s) - b(w) = -\chi(w)^* \sigma_\mu \nabla_s \varphi(w) - (\nabla_s \chi(w))^* \sigma_\mu \varphi(w \cdot s).$$

A shortest gallery visits a chamber at most once. There are 48 galleries, so each chamber and each chamber-wall pair occurs at most 48 times. Cauchy–Schwarz therefore gives

$$\left| \sum_g \big(b(\pi_\mu(g)) - b(g)\big) \right| \le 48 (\|\chi\|_x \mathfrak{D}_x(\varphi)^{1/2} + \mathfrak{D}_x(\chi)^{1/2} \|\varphi\|_x).$$

Thus the coefficient is $51 + 3r$ for one direction. Sum over directions using $\sum_\mu |F_\mu(x)| \le \sqrt{3}|F(x)|$ and then apply Cauchy–Schwarz over sites. □

Lemma 8.4 (Energy away from the source)**.** Let $0 \le \eta \le 1$ be a site cutoff with finite support, let δ be the largest difference of η at neighbouring sites, and let A_η contain the places at both ends of every hop where η changes. Suppose $\eta = 1$ on S and its neighbours. If either $(D_+[U] + m_0)\varphi = 0$ or $(D_+[U]^* + m_0)\varphi = 0$ wherever $\eta > 0$, then

$$\mathfrak{D}_S(\varphi) \le C_E \delta \|\varphi\|^2_{A_\eta}, \qquad C_E = 2(1 + r)/r. \tag{8.xi}$$

■

Proof. Write D for either of these two operators without its mass. Its Hermitian part is $\Delta = \sum_s r t_s (1 - R_s)$, where R_s is a self-adjoint unitary and $t_s \ge 3$. For every place v,

$$((D + m_0)(\eta\varphi))(v) = \eta(x_v)((D + m_0)\varphi)(v) + \sum_s D(v, v \cdot s)(\eta(x_{v\cdot s}) - \eta(x_v))\varphi(v \cdot s).$$

The first term is zero where $\eta > 0$ by the equation, and elsewhere by its factor η. Thus $(D+m_0)(\eta\varphi) = [D, \eta]\varphi$. Only wall zero changes the site. Its hop has norm $3(1 + r)$, and $v \mapsto v \cdot 0$ is a bijection, so $\|[D, \eta]\varphi\| \le 3(1 + r)\delta\|\varphi\|_{A_\eta}$. Let T_η be the set of places at which the commutator can be nonzero. It is contained in A_η. Since $m_0 > 0$ and $0 \le \eta \le 1$,

$$\begin{aligned}\langle \eta\varphi, \Delta\eta\varphi\rangle &\le \mathrm{Re}\langle \eta\varphi, (D + m_0)\eta\varphi\rangle \\ &= \mathrm{Re}\langle \eta\varphi, [D, \eta]\varphi\rangle \le \|\varphi\|_{T_\eta}\|[D, \eta]\varphi\| \le 3(1 + r)\delta\|\varphi\|^2_{A_\eta}.\end{aligned}$$

On the other hand this energy is $\frac{1}{2}\sum_{v,s} r t_s |\nabla_s(\eta\varphi)(v)|^2$, which is at least $(3r/2)\mathfrak{D}_S(\varphi)$. □

Lemma 8.5 (Point splitting)**.** Let $16a \le d$ and $\mathrm{dist}(\mathrm{supp}\, f, \mathrm{supp}\, h) \ge d$. Let B_f, B'_f be the sets of sites at distance at most a and $d/4 + 5a$, respectively, from $\mathrm{supp}\, f$, and similarly for h. Then

$$|E_\nu(f, e; h, e') - E^{\mathrm{s}}_\nu(f, e; h, e')| \le C_{\mathrm{split}}(a/d)^{1/2}\|f\|_\infty\|h\|_\infty(P_\nu(1_{B'_f}, 1_{B_h}) + P_\nu(1_{B_f}, 1_{B'_h})), \tag{8.xii}$$

where $C_{\mathrm{split}} = 12(1 + r)c_Y C_E^{1/2}$. The same estimate holds at $U = 1$. ■

Proof. Write $Y(fe) = N(fe) + R_f$ and similarly for h. The difference of traces is

$$\mathrm{Tr}(G_U R_f G_U Y(he')) + \mathrm{Tr}(G_U N(fe) G_U R_h).$$

Let S_f be the site support of $F^{f,e}$; it lies within $a/2$ of $\mathrm{supp}\, f$. Choose a cutoff equal to one within $2a$ of S_f, decreasing linearly to zero over width $d/4$. Its neighbouring-site difference is at most $4a/d$, its transition region lies in B'_f, and it vanishes on B_h.

Expand $Y(he')$ in its nonzero blocks Y'_{pq}. Their norms are at most $3(1 + r)\|h\|_\infty$, and each place is a row and column of at most one such block. The matrix elements of $(G_U R_f G_U)(q, p)$ are $\langle G^*_U(\cdot, q)e_\alpha, R_f G_U(\cdot, p)e_\beta\rangle$. Both columns satisfy the homogeneous equation on the support of the cutoff. For $X_{q,\alpha} = G^*_U(\cdot, q)e_\alpha$ and $\Phi_{p,\beta} = G_U(\cdot, p)e_\beta$, Lems. 8.3 and 8.4 give

$$|\langle X_{q,\alpha}, R_f \Phi_{p,\beta}\rangle| \le 2c_Y\|f\|_\infty(C_E\delta)^{1/2}\|X_{q,\alpha}\|_{B'_f}\|\Phi_{p,\beta}\|_{B'_f}.$$

Set

$$\mathcal{X}_q = \sum_{u \in B'_f} \|G_U(q, u)\|^2_{\mathrm{HS}}, \qquad \mathcal{Y}_p = \sum_{u \in B'_f} \|G_U(u, p)\|^2_{\mathrm{HS}},$$

where a sum over a site set includes all its places. Summing the squares over the four spin-colour basis vectors gives

$$\|(G_U R_f G_U)(q, p)\|_{\mathrm{HS}} \le 2c_Y\|f\|_\infty(C_E\delta)^{1/2}(\mathcal{X}_q\mathcal{Y}_p)^{1/2}.$$

On this four-dimensional fibre the trace norm is at most twice the Hilbert–Schmidt norm. With $c_h = 3(1+r)$,

$$\begin{aligned}|\operatorname{tr}((G_U R_f G_U)(q,p) Y'_{pq})| &\le 4 c_h c_Y \|f\|_\infty \|h\|_\infty (C_E\delta)^{1/2} (\mathcal{X}_q \mathcal{Y}_p)^{1/2} \\ &\le 2 c_h c_Y \|f\|_\infty \|h\|_\infty (C_E\delta)^{1/2} (\mathcal{X}_q + \mathcal{Y}_p).\end{aligned}$$

Each p and q occurs at most once, and $(C_E\delta)^{1/2} \le 2C_E^{1/2}(a/d)^{1/2}$. The first trace is therefore at most

$$C_{\mathrm{split}}(a/d)^{1/2}\|f\|_\infty\|h\|_\infty \sum_{u\in B'_f,\, u'\in B_h} (\|G_U(u,u')\|_{\mathrm{HS}}^2 + \|G_U(u',u)\|_{\mathrm{HS}}^2).$$

For the second trace interchange f, h; the blocks of $N(fe)$ have norm at most $\|f\|_\infty$ and one block per place. Its constant is no larger. Multiply by a^2, integrate, and use the equality of the two Hilbert–Schmidt norms from Lem. 8.2. □

8.3. **Source estimates with a spin matrix.** The site current insertion contains σ_μ. We therefore need the mass source to range in

$$\mathcal{M} = \operatorname{End}(\mathbb{C}^2_{\mathrm{spin}} \otimes \mathbb{C}^2_\tau \otimes \mathbb{C}^{N_f}) \otimes 1_{\mathrm{colour}}.$$

A source $J(x) \in \mathcal{M}$ acts identically at the 48 chambers of a site. Write $\hat J$ for this multiplication operator, $\|J\| = \sup_x \|J(x)\|$, and $\mathsf{n} = -aJ$. The fine operator is $D[U] + m_0 + \hat{\mathsf{n}}$. There are three source domains in this argument:

$$\begin{aligned}\mathscr{O}_k^{\mathcal{M}} &= \{J : \lambda_k \|J\| < r_M/L\}, \\ \mathcal{P}^{\mathcal{M}} &= \{J : \|J\| \le 2\varrho_J\}, \\ \mathcal{O}_j^{\mathrm{lev},\mathcal{M}} &= \{J : \lambda_j \|J\| < R_J\}, \qquad R_J = \min\{r_M, \varrho_S, \varrho'_S/L\}.\end{aligned}$$

The radii are those of [QCDM1, Sections 10 and 11]. Here r_M is chosen from the inverse and walk estimates, and ϱ_S, ϱ'_S are the free-operator radii. The physical radius satisfies $2\varrho_J \le c_J m_-$, where $c_J \le \min\{1/2, \varrho_*, r_M/L\}$; hence $\mathcal{P}^{\mathcal{M}}$ lies strictly inside every $\mathscr{O}_k^{\mathcal{M}}$. The source chain through level k is formed on $\mathscr{O}_k^{\mathcal{M}}$. To estimate the energy made at level j we use the larger level domain $\mathcal{O}_j^{\mathrm{lev},\mathcal{M}} \subset \mathscr{O}_{j-1}^{\mathcal{M}}$. Indeed $\lambda_j = L\lambda_{j-1}$ and $R_J \le r_M$. Only the supremum norm of J enters these domains; no derivative of J is required.

For this proof let $\mathfrak{G}(d)$ consist of bounded site matrix pairs (F, H) whose supports lie in balls of radius $2d$ with centres at distance at most $8d$, and whose supports have separation at least $d/4$. The same notation will be used for scalar or vector pairs after placing them in fixed matrices of norm at most one.

Proposition 8.6. There is a nondecreasing function $A_g(d)$, finite on $(0, 1/200]$ and tending to zero as $d \downarrow 0$, such that for $(F, H) \in \mathfrak{G}(d)$, $16a \le d$, and $m_0 \le 1/2$,

$$\left|\kappa_2(\Phi_F, \Phi_H) + a^2 \operatorname{Tr}(\hat F G_0 \hat H G_0)\right| \le A_g(d) d^2 \|F\|\|H\|. \tag{8.xiii}$$

Here $G_0 = (D[1] + m_0)^{-1}$ includes the full fibre. The estimate holds on admitted tori and in their infinite-lattice limit states. There is also a constant $C_{\mathrm{fr}}(g)$ such that, for nonnegative scalar pairs in $\mathfrak{G}(d)$,

$$P_1(F,H) \le C_{\mathrm{fr}}(g) d^2 \|F\|\|H\|, \qquad P_\nu(F,H) \le (C_{\mathrm{fr}}(g) + A_g(d)) d^2 \|F\|\|H\|. \tag{8.xiv}$$

■

Proof. We first check the operator estimates and local extraction on $\mathcal{M}$. We then carry them through the source chain and the sum over histories, which supplies the cell bound used in the geometric reduction.

The inverse and background maps. On a history, let Ω_1 be the region of the fine fermion variables being integrated. The layered operator on this region has the form

$$\mathcal{K}_i(\mathsf{n}) = (D[U] + m_0)_{\Omega_1\Omega_1} + \mathsf{W}_i + \hat{\mathsf{n}} = \mathcal{K}_i + \hat{\mathsf{n}}.$$

Here W_i is the sum of block-average penalties and s_i is the smoothed local scale of [QCDM1, Section 6]. Let ι embed a site spinor isometrically as a field constant over its 48 chambers; then $\Pi_0 = \iota\iota^*$ is the chamber-constant projection. Multiplication by n commutes with s_i, with scalar site functions, and with the chamber projection Π_0. Moreover, $\iota^*\hat{\mathsf{n}}\iota = \mathsf{n}$ on the chamber-independent subspace. These identities hold because the same matrix acts at each chamber of a site.

The Neumann estimate can be written without a scalar-mass assumption. Put

$$B_i = \mathsf{s}_i^{-1/2}\mathcal{K}_i^{-1}\mathsf{s}_i^{-1/2}, \qquad N_i = \mathsf{s}_i^{1/2}\hat{\mathsf{n}}\mathsf{s}_i^{1/2}.$$

In the weighted row and column kernel norm, $\|B_i\| \le C_{\mathcal{K}}$ and $\|N_i\| \le \sup_x \mathsf{s}_i(x)\|\mathsf{n}(x)\|$. For $i \le j$, this supremum is at most $2r_M$. The choice $r_M \le (4C_{\mathcal{K}})^{-1}$ gives $\|B_iN_i\| \le 2C_{\mathcal{K}}r_M \le 1/2$. These are the source perturbation bounds of [QCDM1, Lem. 6.5 and Section 10]. Thus

$$\mathsf{s}_i^{-1/2}\mathcal{K}_i(\mathsf{n})^{-1}\mathsf{s}_i^{-1/2} = \sum_{q\ge 0}(-B_iN_i)^qB_i, \qquad \|\mathsf{s}_i^{-1/2}\mathcal{K}_i(\mathsf{n})^{-1}\mathsf{s}_i^{-1/2}\| \le 2\|B_i\|. \tag{8.xv}$$

The same calculation after an exponential conjugation gives the weighted inverse bound. The local inverses in the walk construction use the pulled-back site matrix, so their support and decay bounds are unchanged. The heavy block has real part at least $\lambda_\rho + m_0 - \|\mathsf{n}\|$; its bound uses the norm of the perturbation and the commutation with Π_0. Positivity of the complex mass matrix itself is not needed.

For a background column with prescribed coarse data, write $\psi_P(\mathsf{n}) = \mathcal{K}_i(\mathsf{n})^{-1}g_P$. The resolvent identity gives

$$\psi_P(\mathsf{n}) - \psi_P(0) = -\mathcal{K}_i(\mathsf{n})^{-1}\hat{\mathsf{n}}\psi_P(0). \tag{8.xvi}$$

The row twin obeys the corresponding identity, with the order of the factors reversed. For the fluctuation covariance, let A_i be the integrated places at level i and let q_i be the one-step fermion average. Put $T_A = 1_{A_i}q_i\cdots q_1$. The block-kernel parameters are

$$\theta = b_F > 0, \qquad \theta_i = \frac{b_F(1-L^{-1})}{1-L^{-i}} \quad (i \ge 1),$$

as in [QCDM1, Lem. 5.4]. The Schur formula at the start of [QCDM1, Section 6] then reads, for $i \ge 1$,

$$\Gamma_i(\mathsf{n})^{-1} = c_i^{-1} + c_i^{-1}\theta_iT_A\mathcal{K}_{i+1}(\mathsf{n})^{-1}T_A^+\theta_ic_i^{-1}, \qquad c_i = \theta_i + \theta q_{i+1}^+q_{i+1}.$$

At $i = 0$ the identity is $\Gamma_0(\mathsf{n})^{-1} = \mathcal{K}_1(\mathsf{n})^{-1}$. Here $+$ denotes the holomorphic row twin, agreeing with the adjoint at real links. The averages do not depend on n. Substitution of eq. 8.xv therefore gives the same covariance, background, and Gram estimates as [QCDM1, Prop. 6.7], with its fixed enlarged constants. All these objects are holomorphic in n on the stated balls.

The source coordinates and extraction. At zero gauge source the independent coordinates are

$$\mathfrak{c} = (0, \mathsf{n}, (\sigma_i)_{i\le k}), \qquad \mathscr{D}_k^{\mathcal{M}} = \{L^{k+1}\|\mathsf{n}\| < r_M,\ \|\sigma_i\| < B_\sigma\phi_i\ (i \le k)\}.$$

The doubled domain doubles each radius. Since $L^{k+1}\|\mathsf{n}\| < 2r_M$ there, the preceding operator estimates apply through $\mathcal{K}_{k+1}$ and Γ_k. The local matrix coordinates σ_i, the increments, and the extracted zeroth moments already take values in $\mathcal{M}$ in the construction of [QCDM1, Prop. 10.1].

More explicitly, let $k_X(\mathfrak{c})$ be the quadratic kernel at the trivial configuration before extraction, and set

$$\mathrm{ord}_0[k](y) = \tfrac{1}{2}\sum_{y'}[k(y,y') + k(y',y)].$$

On a torus these sums use the lifted paths in the definition of the local moment. The increment is

$$\nu_{k+1}(y;\mathfrak{c}) = \mu_{k+1} - \mathfrak{s}_{k+1}^{-1} \sum_X (\mathrm{ord}_0[k_X(\mathfrak{c})](y) - \mathrm{ord}_0[k_X(0)](y)),$$

with the same restriction to polymers of diameter less than one quarter of the torus side. Let $A_{k+1}(y,z)$ be the moment map of a local mass form, evaluated at the trivial gauge configuration and $\mathsf{n}=0$. It is a linear map on $\mathcal{M}$. On the infinite lattice, constant background fields give

$$\sum_z A_{k+1}(y,z)N = \mathfrak{s}_{k+1}N \qquad (N\in\mathcal{M}). \tag{8.xvii}$$

On a torus, both this row sum and the corresponding column sum differ from $\mathfrak{s}_{k+1}N$ by at most $C_A e^{-4c_t\ell/\lambda_{k+1}}\|N\|$. Indeed, the local pieces which do not wrap around agree with their infinite-lattice counterparts; the remaining pieces are bounded by the exponential walk and polymer tails. For nonzero n, write A_{k+1}^{n} for the moment map. Its change from A_{k+1} has the weighted column-sum bound of [QCDM1, Prop. 10.1, proof], with factor $C_{\mathrm{bg}}^{\mathrm{w}} L^{k+1}\|\mathsf{n}\|/r_M$.

The extraction acts on the entire matrix coefficient. It need not leave a kernel with zero zeroth moment: a varying coefficient ν_{k+1} leaves differences of the form

$$\sum_z A_{k+1}(y,z)[\nu_{k+1}(z) - \nu_{k+1}(y)].$$

The cell estimates retain these differences, the source variation, and the torus error. The extraction forms have coefficients independent of the gauge configuration and commuting with colour, which preserves their joint gauge invariance. The weighted row sums and cell-unit estimates use operator norms of these coefficients, so apply to the larger source space with the same constants.

The source chain is

$$\sigma_0 = 0, \qquad \sigma_{k+1} = L\sigma_k + \nu_{k+1}(\mathfrak{c}_k(J)) - \mu_{k+1}, \qquad \mathfrak{c}_k(J) = (0, -aJ, (\sigma_i(J))_{i\le k}).$$

The coordinate Cauchy estimate in the doubled domain has the same distance as before, with the gauge-source entry omitted:

$$\|\mathfrak{c}-\mathfrak{c}'\|_{k,y} = \sup_x e^{-\kappa_\sigma|x-y|/L^{k+1}} \max\left\{ \frac{L^{k+1}|\mathsf{n}-\mathsf{n}'|(x)}{r_M}, \max_{i\le k}\frac{|\sigma_i-\sigma_i'|(x)}{B_\sigma\phi_i}\right\}.$$

Here x,y are in fine lattice coordinates. In particular

$$|\nu_{k+1}(\mathfrak{c})(y) - \nu_{k+1}(\mathfrak{c}')(y)| \le B_\nu\phi_{k+1}\|\mathfrak{c}-\mathfrak{c}'\|_{k,y}.$$

The induction of [QCDM1, Lem. 10.4] therefore keeps the chain in $\mathscr{D}_k^{\mathcal{M}}$ and gives

$$\|\sigma_k(J)\| \le B_\nu S_\uparrow \phi_k\lambda_k\|J\|/r_M.$$

If J and J' agree outside S, the same induction gives the tail $C_t\phi_k e^{-\kappa_\sigma\,\mathrm{dist}(x,S)/\lambda_k}$, now in physical coordinates. Neither estimate uses a commutation of J with the spin matrices.

Symmetries and the sum over histories. The four source operations are

$$J^\star = \gamma_4 J^\dagger \gamma_4, \quad J^{\mathsf{c}} = C_s J^{\mathsf{T}} C_s^{-1}, \quad J\mapsto VJV^{-1}, \quad J\mapsto S_{\mathsf{t}}(J\circ\mathsf{t}^{-1})S_{\mathsf{t}}^{-1}, \qquad C_s = \sigma_2\otimes 1_\tau\otimes 1_{N_f}.$$

Here $V = e_+\otimes V_1 + e_-\otimes V_2$, with $V_1,V_2\in \mathrm{U}(N_f)$, acts trivially on spin. The map t is a lattice translation followed by a signed coordinate permutation preserving the block partition; S_{t} is its spinor lift from [QCDM1, proof of Prop. 2.5]. The sign ambiguity in the lift disappears in conjugation. They preserve $\mathcal{M}$ and its norm. At real links, for example, write $D_J = D[U] + m_0 - a\hat{J}$. Then

$$D_J^* = \gamma_4 D_{J^\star}\gamma_4, \qquad D_J^{\mathsf{T}} = \mathcal{C}^{-1}D_{J^{\mathsf{c}}}\mathcal{C}.$$

The flavour and isometry identities are obtained by conjugating D_J with their respective lifts. The finite-dimensional integral gives the exact identity $Z[J^\star] = \overline{Z[J]}$. The diameter restriction in local extraction requires the torus qualifications of [QCDM1, Prop. 10.1]; it does not give an exact covariance identity for every local increment.

The exact conjugation identity for the sum of the source energies follows separately, as in [QCDM1, Prop. 10.6, proof]. Sum the new activities and localized normalizers before evaluating their scalar parts. On the background pairs used in the density identity, their sum is the logarithm of the whole step integral. The trivial pair belongs to this domain, and conjugation carries its scalar part to the complex conjugate at the conjugated source. Subtracting the real zero-source value proves the identity for the level sum. The assignment to individual polymers need not be preserved.

The history comparison uses the same fine matrix n in the chain and on the history; only the coordinates σ_i differ. Its distance estimate therefore applies with the same constants. In the history integrals, the fermion covariance and stored background columns are those of eq. 8.xv and eq. 8.xvi. Schur's test, the row bounds, and the Berezin Gram bound depend only on their norms. Products such as $(1-\mathfrak{s}_j^{-1}\sum_y A_j^{\mathsf{n}}(y,z))a_{X,p}(z)$ are estimated in that order, so no commutation of their matrix factors is assumed.

Write $\mathcal{H}$ and $\bar{\mathcal{H}}$ for the composed column and row background maps from the variables before step j to the original fermion variables. They are the maps obtained by successive square completion in [QCDM1, Section 5.3]. In the following formula, the subscript AA restricts both coarse arguments to the variables integrated at that step. The determinant variation used in the local energy is

$$\log\det\Gamma_{j-1}(\mathsf{n})-\log\det\Gamma_{j-1}(0)=\int_0^1 \operatorname{Tr}\left[\Gamma_{j-1}(t\mathsf{n})^{-1}(\bar{\mathcal{H}}\hat{\mathsf{n}}\mathcal{H})_{AA}(t\mathsf{n})\right]dt.$$

Expand the inverse and the two background maps into walks and assign each trace to their joint support. The fibre trace is bounded by $8N_f$ times the product of the operator norms. This proves the same local energy estimate as [QCDM1, Prop. 10.6]. A change of source outside a polymer can be seen only by a walk reaching that change; reserving part of its decay gives the source-distance tail.

At zero gauge source, every variation of the gauge weight and every associated Gaussian or linear term vanishes. The remaining closed fermion loops are gauge invariant. Their first derivative in curvature at the trivial configuration is zero, since a linear form invariant under the adjoint action of SU(2) vanishes. Consequently the history majorants and their sums in [QCDM1, Prop. 10.14] apply with the same constants. The bare density is entire in J, and its coefficients are continuous on the compact link space, as required when that sum is identified with the original partition function. This proves the first-order source and energy bounds needed here. No second-order improvement in ϕ for the enlarged source space is being used.

Reflection and the cell bound. The cell bound requires the correct reflection of a spin matrix. Directly on generators,

$$\Theta_P(\bar\psi(v)J(v)\psi(v))=\bar\psi(\theta v)\gamma(\hat n)J(v)^\dagger\gamma(\hat n)\psi(\theta v).$$

For the reflection lift $S_P=\gamma_4\gamma(\hat n)$ this is the linear reflected source followed by $\star$. The norm is preserved. The operation $\star$ commutes with conjugation by the reflection lift, since $\gamma_4 S_P=-S_P\gamma_4$ and the sign drops out of the conjugation. For a word r in the board reflections, define

$$\mathsf{r}_{\mathrm{lin}}J=S_{\mathsf{r}}(J\circ\mathsf{r}^{-1})S_{\mathsf{r}}^{-1},\qquad \mathsf{r}_\theta J=\begin{cases}\mathsf{r}_{\mathrm{lin}}J, & |\mathsf{r}|\text{ even},\\ (\mathsf{r}_{\mathrm{lin}}J)^\star, & |\mathsf{r}|\text{ odd}.\end{cases}$$

If J is supported strictly inside one cell, the reflected exponentials are even and have disjoint supports. Hence their product is

$$\operatorname{Fold}e^{\Phi_J}=e^{\Phi_{J^{\mathrm{ch}}}},\qquad J^{\mathrm{ch}}=\sum_{\mathsf{r}}\mathsf{r}_\theta J,\qquad \|J^{\mathrm{ch}}\|=\|J\|.$$

The chessboard estimate applies to this product. The partition-function bound just obtained and the separated energy estimate in [QCDM1, Prop. 10.16] then give the cell bound of [QCDM1, Prop. 10.17]. Odd reflection words cause no change in its real energy bound, because $E_K[J^\star]=\overline{E_K[J]}$.

For sources in the central ball of radius $1/4$ this bound takes the form

$$\Psi(J) := e^{-\Lambda(J)}\frac{Z[J]}{Z[0]}, \quad \Psi(0) = 1, \quad |\Psi(J)| \le B'_Z, \quad B'_Z = e^{c_V + \mathscr{C}_*}, \tag{8.xviii}$$

Here $\mathscr{C}_*$ bounds the source count uniformly for supports in the central ball of radius $1/4$, as in [QCDM1, Prop. 10.17], and $\Lambda(J) = \mathsf{N}^{-1}\sum_{\mathsf{r}}(E_K[\mathsf{r}_{\mathrm{lin}}J] - E_K[0])$ and $\mathsf{N} = \ell^3$.

Packing and averaging. A pair in $\mathfrak{G}(d)$ lies in a ball of radius $6d$. Choose a grid of mesh $14d$ inside the ball of radius $1/4 - 7d \ge 0.215$, and move each grid point to the nearest permitted lattice translate of the centre of this pair. The displacement is at most $\sqrt{3}a/2 < d/8$. The translated balls stay inside the cell ball, and their separation is at least $2d - \sqrt{3}a \ge d$. An inscribed cube supplies at least

$$N \ge \left\lfloor \frac{0.43}{14\sqrt{3}d} \right\rfloor^3 \ge (112d)^{-3}$$

points: $0.43/(14\sqrt{3}) > 0.0177$, and $(0.0177 - 1/112)d^{-1} \ge 1$ for $d \le 1/200$.

For separated matrix sources put $\|F\|_\circ = \|F\|/\varrho_J$ and similarly for H. The function $\Psi(z_1F + z_2H)$ is bounded by B'_Z on $|z_1| < 2/\|F\|_\circ$, $|z_2| < 2/\|H\|_\circ$. On the bidisc smaller by $1/(4B'_Z)$, Schwarz's lemma on complex lines gives $|\Psi - 1| \le 1/2$. Its logarithm exists there, and Cauchy's estimate gives

$$|\partial_1\partial_2 \log \Psi(0)| \le 4(B'_Z)^2\|F\|_\circ\|H\|_\circ.$$

Apply this to $\sum_t \varepsilon_t F^t$ and $\sum_t \varepsilon_t H^t$, with independent signs $\varepsilon_t \in \{-1, 1\}$. The supports are disjoint, so these sums have the norms of one pair. Since $\mathsf{E}_\varepsilon \varepsilon_t\varepsilon_{t'} = \delta_{tt'}$, averaging the bilinear derivative retains only matching translates. Translation invariance then yields

$$\kappa_2(\Phi_F, \Phi_H) = \frac{1}{N\mathsf{N}}\sum_{t,\mathsf{r}} \partial_1\partial_2 E_K(F^{t,\mathsf{r}}, H^{t,\mathsf{r}}) + R_{\mathrm{cell}}, \quad |R_{\mathrm{cell}}| \le C_{\mathrm{cell}}(g) d^3 \varrho_J^{-2}\|F\|\|H\|,$$

where $\mathsf{N} = \ell^3$ is the number of unit cells. One may take $C_{\mathrm{cell}}(g) = 4 \cdot 112^3 \exp(2c_V + 2\mathscr{C}_*)$ in the notation of that lemma.

The energy decomposition. At zero gauge source the energy decomposition [QCDM1, Lem. 11.4] has just its fermion and interaction parts. The relative gauge weight is zero, as are the linear gauge source and its Gaussian variations. In particular the term of fourth order in the gauge source is zero. Completing the fermion square at each step is an algebraic identity for every $J \in \mathcal{M}$. The coarse determinant from one step is the input determinant of the next. Write $\mathbb{D}_K(1; J)$ for the final coarse Dirac matrix after these K square completions at the trivial gauge configuration with source J. Their product gives

$$\sum_{j,X} E_j^{\mathrm{F}}(X; J) = \log \frac{\det(D[1] + m_0 - a\hat{J})}{\det(D[1] + m_0)} - \log \frac{\det \mathbb{D}_K(1; J)}{\det \mathbb{D}_K(1; 0)}.$$

This is an identity of logarithmic germs at zero. On the larger level-source domains its continuation is defined by the product of the invertible fluctuation determinants. Separate global logarithms of the two displayed determinants are not required.

For completeness, the small factor in the interaction energy comes from the part of the step potential left after the explicit normalizers and Gaussian terms. On the level-j domain the induction in the proof of that lemma applies to the independent matrix coordinates $(0, -aJ, (\sigma_k)_{k<j})$, with $\|\sigma_k\| < 2B_\sigma\phi(x_k)$, and gives

$$\|\mathcal{V}_i^{\mathrm{rest}}\| \le C_{\mathrm{rest}}\phi(x_{i-1}), \qquad i \le j, \qquad \|\sigma_i\| < 2B_\sigma\phi_i.$$

At zero gauge source its gauge vertices are independent of J, its fermion vertices have the source-step bound proved above, and its closed-loop variations have the weighted trace bound. The localized variation of an older activity has the bound from the preceding induction level. Thus the

scalar output at the trivial configuration, after removing the normalizers, is at most $C_w^{\mathrm{s}}\phi(x_{j-1})$. Subtracting its value at zero source gives

$$|E_j^{\mathrm{I}}(X;J)| \le 2C_w^{\mathrm{s}}\phi(x_{j-1})e^{-\kappa d_j(X)}. \tag{8.xix}$$

This induction also retains holomorphy and locality in the independent coordinates. The matrix-source vertices and covariances are holomorphic and read J on their assigned walks. Scalar localization preserves the parameter dependence of the old arrays; a member assigned to X reads only their restrictions to X. The cluster expansion, its pointed families and the Cauchy integrals defining the remainder therefore have the same properties. These are exactly the coordinate clauses used in the interaction-tail proof of [QCDM1, Lem. 11.4]; the matrix extension above verifies their inputs.

Let $S = \mathrm{supp}(J - J')$ and $D = \mathrm{dist}(X, S) > 0$. Multiply the remainder potential by z. Its scalar output at the trivial pair, before the normalizers are removed, is a holomorphic function $\mathcal{E}(z;\mathfrak{c})$ on the full independent coordinate domain, with

$$|\mathcal{E}(z;\mathfrak{c})| \le K_M(c_{\mathrm{cl}}+1)e^{-\kappa d_j(X)}, \qquad |z| \le R = \frac{c_{\mathrm{cl}}}{4C_{\mathrm{rest}}\phi(x_{j-1})} \ge 2.$$

The two source chains have the same J on X, and their induced coordinates differ there by at most $C_{\mathrm{t}}\phi(x_k)e^{-\kappa_\sigma D/\lambda_j}$. Keep J' fixed and replace all $\sigma_k(J')$ simultaneously by $\sigma_k(J')+\zeta(\sigma_k(J)-\sigma_k(J'))1_X$. Since the chain lies in the inner window $\|\sigma_k(J')\| \le B_\sigma\phi(x_k)$, this curve stays in the coordinate domain for $|\zeta| < 1/\theta$, where $\theta = (C_{\mathrm{t}}/B_\sigma)e^{-\kappa_\sigma D/\lambda_j}$. Locality and Schwarz's lemma bound the difference of the two outputs by

$$C_{\mathrm{tail}}K_M(c_{\mathrm{cl}}+1)e^{-\kappa d_j(X)}e^{-\kappa_\sigma D/\lambda_j}, \qquad C_{\mathrm{tail}} = 2\max\{1, C_{\mathrm{t}}/B_\sigma\}.$$

For $\theta \ge 1$ the two separate bounds suffice; for $\theta = 0$ the outputs agree. There is no sum over k in this interpolation.

The output difference vanishes at $z = 0$, where its scalar potential and quadratic origin read only the zero gauge source. At $z = 1$ it equals the difference of E_j^{I}: the gauge normalizer and Gaussian part agree, and the free fermion normalizer cancels with E_j^{F}. Schwarz's lemma in z now supplies R^{-1}. Thus the constant is at most $4C_{\mathrm{rest}}C_{\mathrm{tail}}K_M(1 + c_{\mathrm{cl}}^{-1})$, already included in the companion's C_I, and

$$|E_j^{\mathrm{I}}(X;J) - E_j^{\mathrm{I}}(X;J')| \le C_I\phi(x_j)e^{-\kappa d_j(X)}e^{-\kappa_\sigma \mathrm{dist}(X,S)/\lambda_j}. \tag{8.xx}$$

If X meets S, use the two bounds eq. 8.xix; the fixed comparison of $\phi(x_{j-1})$ and $\phi(x_j)$ is included in C_I. This explains why the interaction tail, as well as its undifferentiated size, retains a factor ϕ.

Summing the mixed derivatives. For the averaged interaction sum, the level-j source radii are $R_J/(\lambda_j\|F\|)$ and $R_J/(\lambda_j\|H\|)$. At levels with $\lambda_j < d$ the separation $d/4$ gives the tail $e^{-\kappa_\sigma d/(16\lambda_j)}$ and at most $C(d/\lambda_j)^3$ cubes near one support. At levels $\lambda_j \ge d$, use the independent signs for the packed sources before applying Cauchy's estimate. The packed pairs and their board images meet at most $CN\mathsf{N}(d/\lambda_j)^3$ cubes: their union occupies a fixed-volume region in each unit cell and $N \ge cd^{-3}$. The signs have the same source norm as a single member. Dividing by $N\mathsf{N}$ therefore leaves the factor $C(d/\lambda_j)^3$, even at these coarser levels.

After summing the polymers through each cube, Cauchy's estimate and eq. 8.xx bound the mean by $CR_J^{-2}\|F\|\|H\|$ times

$$\sum_j \phi(x_j)\lambda_j^2(d/\lambda_j)^3 \begin{cases} e^{-\kappa_\sigma d/(16\lambda_j)}, & \lambda_j < d, \\ 1, & \lambda_j \ge d. \end{cases} \tag{8.xxi}$$

The two powers of λ_j are the inverse Cauchy radii; the other factor is the number of cubes after averaging. To see its summability, put $t = \mathsf{b}/d$. The scale comparison $x_j \asymp \mathsf{b}/\lambda_j$ and

$\phi(u) = u^{-1/2}(1+\log u)^{p_0}$ give, with fixed comparison constants,

$$\frac{\phi(\mathsf{b}/\lambda)}{\phi(t)} \le \begin{cases} (\lambda/d)^{1/2}(1+\log(d/\lambda))^{p_0}, & \lambda \le d, \\ (\lambda/d)^{1/2}, & \lambda \ge d. \end{cases}$$

All arguments are in the admitted decreasing range of ϕ. Consequently, after division by $d^2\phi(t)$, the terms with $\lambda_j \ge d$ form a geometric sum with ratio $L^{-1/2}$. For $\lambda_j < d$, the sum is bounded by a fixed multiple of

$$\sum_{u\ge 0} L^{u/2}(1+(u+1)\log L)^{p_0} e^{-(\kappa_\sigma/16)L^{u-1}} < \infty.$$

This is the summation in [QCDM1, Lem. 11.6], with the support separation $d/4$. It gives

$$|E^{\mathrm{I}}_{\text{averaged}}| \le C^\natural_M R_J^{-2}\phi(\mathsf{b}/d)d^2\|F\|\|H\|.$$

The last determinant. Put $A(z_1,z_2) = \mathbb{D}_K(1; z_1F + z_2H)$ and $A = A(0,0)$. The real-part bound gives $\|A^{-1}\| \le (c_b\mathsf{m})^{-1}$. Differentiating the inverse formula inserts $-a\hat{F}$ between two background maps. Their squared row sums are $O(a^2)$, so the first derivative satisfies

$$\|A_1\|_1 \le Ca^3\sum_x \|F(x)\| \le Cd^3\|F\|, \qquad \|A_2\|_1 \le Cd^3\|H\|.$$

Here $\|\cdot\|_1$ is the trace norm. The supports lie in balls of radius $2d$, so their site counts are bounded by a constant times $(d/a)^3$. In the mixed derivative the inserted matrices are separated by one free propagator, with bound $Ca^2(16d^{-2} + \mathsf{m}^{-1})$ at separation $d/4$. The same row estimate therefore gives

$$\|A_{12}\|_1 \le C(d^4 + d^6\mathsf{m}^{-1})\|F\|\|H\|.$$

The logarithmic derivative is

$$\partial_1\partial_2 \log\det A(z)\big|_{z=0} = \mathrm{Tr}(A^{-1}A_{12} - A^{-1}A_2A^{-1}A_1).$$

Using the trace norm for the localized derivatives, and the operator norm for A^{-1}, bounds it by

$$C^\natural_{\text{last}}(d^6\mathsf{m}^{-2} + d^4\mathsf{m}^{-1})\|F\|\|H\|,$$

as in the last paragraph of [QCDM1, Lem. 11.5]. No factor proportional to the total volume enters these two traces. The free loop is unchanged by the translates and reflected matrix sources. Thus eq. 8.xiii holds with

$$\alpha_g(d) = C_{\text{cell}}(g)\varrho_J^{-2}d + C^\natural_M R_J^{-2}\sup_{\mathsf{b}\in I_g}\phi(\mathsf{b}/d) + C^\natural_{\text{last}}(d^4{m_-}^{-2} + d^2{m_-}^{-1}), \qquad A_g(d) = \sup_{0<s\le d}\alpha_g(s).$$

This function is finite and tends to zero. The mass labels lie in a compact interval, and the same is true of $\mathsf{b} \in I_g$ at fixed g. On the infinite lattice, pass from the tori by convergence of local Gaussian functionals and by the vanishing of the massive free image terms.

For the positive free form, the propagator estimate in the proof of [QCDM1, Lem. 11.5] gives $\|G_1(u,u')\| \le C'_G(g)|n|^{-2}$, with $n = (x_u - x_{u'})/a$. Its leading term has this bound because $(1+t)e^{-t} \le 1$ and $te^{-t} \le e^{-1}$. The remainder is $O(|n|^{-3})$, and the torus images are bounded by $C'_{\text{im}}(\ell/a)^{-2}$. These constants are included in $C'_G(g)$. The support balls contain at most

$$\frac{4\pi}{3}(2d/a + \sqrt{3}/2)^3 \le 36\pi(d/a)^3$$

sites each, with 48 places per site, and $|n| \ge d/(4a)$. The four-dimensional spin-colour Hilbert–Schmidt norm is at most twice the operator norm. Thus

$$\begin{aligned} P_1(F,H) &\le a^2\, 8C'_G(g)^2\, 48^2(36\pi)^2(d/a)^6(4a/d)^4\|F\|\|H\| \\ &= 2048(36\pi)^2 48^2 C'_G(g)^2 d^2\|F\|\|H\|. \end{aligned}$$

This gives the stated choice of $C_{\mathrm{fr}}(g)$. Apply eq. 8.xiii with $\Gamma = i\tau_1 \otimes t$ and use Lem. 8.2 to obtain the second bound in eq. 8.xiv. □

Theorem L (Comparison with the free current loop)**.** Suppose $16a \le d \le 1/200$ and $m_0 \le 1/2$. Let f, h be real bounded functions, each supported in a ball of radius d, with support separation between d and $6d$. Then

$$|E_\nu(f,e;h,e') - E_1(f,e;h,e')| \le \left(A_g(d) + C_{\mathrm{pt}}(g)(a/d)^{1/2}\right) d^2 \|f\|_\infty \|h\|_\infty, \tag{8.xxii}$$

where $C_{\mathrm{pt}}(g) = 2C_{\mathrm{split}}(2C_{\mathrm{fr}}(g) + A_g(1/200))$. ■□

Proof. The pairs $(1_{B'_f}, 1_{B_h})$ and $(1_{B_f}, 1_{B'_h})$ in Lem. 8.5 belong to $\mathfrak{G}(d)$. Indeed the support enlargement is at most $d/4 + 5a \le 9d/16$, so each enlarged support fits in a ball of radius $2d$, and the remaining separation is at least $d - 9d/16 - a \ge d/4$. The shifted site pair $(F^{f,e}, F^{h,e'})$ also belongs to this class. Apply eq. 8.xiv to the two point-splitting errors, one under ν and one at $U = 1$, and apply eq. 8.xiii to N_F, N_H. Adding the three differences gives eq. 8.xxii. □

8.4. **The free tensor and the current classes.**

Lemma 8.7. For a test pair, $16a \le d$ and $10\mathsf{m}d \le 1$,

$$\left| E_1^{\mathrm{s}}(f,e;h,e') - \frac{1}{4\pi^2} \iint f(x)h(y)\langle e, k(x-y)e'\rangle \, dx\, dy \right| \le C_{\mathrm{cr}} \left(\frac{a}{\vartheta d} + \mathsf{m}d + (d/\ell)^2 \right) d^2 \|f\|_\infty \|h\|_\infty. \tag{8.xxiii}$$

The last term is absent on the infinite lattice. The constant depends only on the fixed model and $m_- \ell_G$, where ℓ_G is the least admitted side. ■

Proof. The free propagator expansion in the proof of [QCDM1, Lem. 11.5] gives, in its light upper block,

$$G_1(n) = v(n)\sigma(n) + s(n) + O(|n|^{-3}), \qquad v(n) = \frac{(1 + m_0|n|)e^{-m_0|n|}}{4\pi|n|^3}, \quad s(n) = \frac{m_0 e^{-m_0|n|}}{4\pi|n|}.$$

The light factor between individual chambers is 48^{-1}; the sum over the two chamber indices cancels its square. The mixed and heavy parts have bounds $C|n|^{-3}$ and $C|n|^{-4}$, hence cost $O(a/d)$ in this loop. Use

$$\mathrm{tr}(\sigma(n)\sigma(F)\sigma(n)\sigma(H)) = 2\big(2(n\cdot F)(n\cdot H) - |n|^2 F\cdot H\big).$$

The sign of the second propagator and the colour factor two give for the v^2 term

$$\frac{(1+t)^2 e^{-2t}}{4\pi^2} \langle F, k(n) H\rangle, \qquad t = m_0|n|.$$

The two vs terms involve three Pauli matrices. Their ratio to the size $4v^2|n|^2|F||H|$ of the leading term is at most $2t$, since

$$\frac{s(n)}{v(n)|n|} = \frac{t}{1+t} \le t.$$

The s^2 term has ratio at most t^2, and $(1+t)^2 e^{-2t} = 1 + O(t^2)$. On the shifted supports,

$$15d/16 \le d - a \le |x-y| \le 10d + a \le 11d, \qquad t \le 11\mathsf{m}d \le 2.$$

The two mixed terms therefore give an $O(\mathsf{m}d)$ error; they cannot be replaced by an $O((\mathsf{m}d)^2)$ error in this estimate.

Homogeneity gives $a^2 k(n) = a^6 k(x-y)$. The discrete error just estimated is bounded by

$$C(\mathsf{m}d + a/d)a^6 \sum_{x,y} |F(x)||H(y)||x-y|^{-4} \le C(\mathsf{m}d + a/d)d^2 \|f\|_\infty \|h\|_\infty.$$

Replacing midpoint values by site values uses

$$|F^{f,e}(x) - f(x)e| \le \tfrac{\sqrt{3}}{2} a \operatorname{Lip} f \le \frac{a}{\vartheta d}\|f\|_\infty,$$

and the same bound for H. For the remaining Riemann sum, $|k(x-y)| \le Cd^{-4}$ and $|\nabla k(x-y)| \le Cd^{-5}$ on the separated supports. The change across a lattice cube is therefore bounded by a constant times

$$a\big(d^{-4}\operatorname{Lip} f\|h\|_\infty + d^{-4}\|f\|_\infty \operatorname{Lip} h + d^{-5}\|f\|_\infty\|h\|_\infty\big).$$

Its support in the product space has volume $O(d^6)$. Summing the cube errors gives $Ca/(\vartheta d)$ times $d^2\|f\|_\infty\|h\|_\infty$. On the torus the antiperiodic image sum is bounded entrywise by $C(m_-\ell_G)(\ell/a)^{-2}$; its contribution to the loop is $C(d/\ell)^2$ at the same normalization. These bounds prove the claim. □

Proof of Thm. K. For $T = \tau_{a_0} \otimes t$ use $t_+ = t$, $t_- = \epsilon_{a_0} t$, with $\epsilon_0 = 1$ and $\epsilon_3 = -1$. Equation eq. 8.ix is therefore $\operatorname{tr}(tt')$ times $E_\nu + \epsilon_{a_0}\epsilon_{b_0}\overline{E_\nu}$. The leading tensor in Lem. 8.7 is real; adding its conjugate multiplies it by $1 + \epsilon_{a_0}\epsilon_{b_0} = 2\delta_{a_0 b_0}$. The remaining error is at most twice the sum of the interacting-to-free difference, the free point-splitting error, and eq. 8.xxiii. Divide by the lower bound in eq. 8.iv. For example, take

$$\epsilon_V(g,\vartheta;d) = \frac{2\cdot 10^4}{c_B \vartheta^2}\left[A_g(d) + C_{\mathrm{cr}}(m_+ d + (d/\ell_G)^2)\right], \qquad c_B = \frac{1}{2\pi^2},$$

and

$$C'_V(g) = \frac{2\cdot 10^4}{c_B}(C_{\mathrm{pt}}(g) + 2C_{\mathrm{split}}C_{\mathrm{fr}}(g) + C_{\mathrm{cr}}).$$

This gives eq. 8.vi; when the flavour trace vanishes the exact loop formula gives zero. At fixed d, f, h, both mesh terms tend to zero with K. The displayed formulas bound the cumulants and every limit point as claimed. If torus sides vary, the assumption $20d \le \ell$ makes the Euclidean pair and its two kernel integrals the same at every cutoff. □

Proposition 8.8 (A nonzero current class)**.** For every sufficiently small fixed $d > 0$ and all sufficiently large K, every nonzero Hermitian flavour matrix t and $T = \tau_{a_0} \otimes t$, $a_0 \in \{0, 3\}$, admit a current supported on the positive side of $P_{1/2} = \{x_3 = a/2\}$ whose class in the lattice reflection-positive Hilbert space is nonzero. More precisely, let $\vartheta_c = 1/500$, $z_d = (5d/8 + a/2)e_3$, and $f(x) = (d/8 - |x - z_d|)_+$. For $\mu = 1, 2, 3$,

$$\omega_\nu(J^T(fe_\mu)\Theta_{P_{1/2}}J^T(fe_\mu)) \ge \tfrac{1}{4}c_B \operatorname{tr}(t^2)\mathcal{K}_4(f, f\circ\theta) > 0. \tag{8.xxiv}$$

The bound is uniform in admitted labels and torus-limit states. It also holds on admitted tori with $20d \le \ell$. ■

Proof. The two tents form a test pair at scale d and ϑ_c. Their centres are $5d/4$ apart and their support radii are $d/8$. Also $\operatorname{Lip} f = 1$, $\|f\|_\infty = d/8$, and $\int f = (\pi/1536)d^3\|f\|_\infty \ge \vartheta_c d^3\|f\|_\infty$. For large K, every endpoint of a bond with midpoint in its support is strictly above $P_{1/2}$.

On generators, reflection reverses the ordered edge and sends $\mathrm{j}^T(v, v')$ to $-\mathrm{j}^{T^*}(\theta v, \theta v')$. Antilinearity also sends i to $-i$. After summing the oriented bonds this gives

$$\Theta_P J^T(f) = J^{T^*}(\varrho \bar{f} \circ \theta), \tag{8.xxv}$$

where ϱ is the linear reflection. The positive form of [QCDM1, Thm. B and Thm. A] therefore evaluates the left side of eq. 8.xxiv by the two-current law at $e = e_\mu$ and $e' = \varrho e_\mu$; the current means are zero.

On these supports $n = x - y$ has $n_3 \ge d$ and $|n_\perp| \le d/4$, so

$$\langle e_\mu, k(n)\varrho e_\mu\rangle \ge \frac{15}{17}|n|^{-4}.$$

Choose d so that $\epsilon_V(g,\vartheta_c;d) < 1/4$, and then K so that the two mesh errors in eq. 8.vi total at most $1/4$. The leading term less the error is at least $(15/17 - 1/2)c_B \operatorname{tr}(t^2)\mathcal{K}_4$, which proves eq. 8.xxiv. The same supports lie in the positive slab of the torus when $20d \le \ell$. □

8.5. **Joint correlations with the flavour currents.** We now include the currents in the family of scalar fields. The estimate needed for this purpose is a bound on the chessboard seminorm of one current. Its conditional mean vanishes, so the fold of a current depends only on propagators between distinct reflected cells. Meson sources give an exponential moment of their squared Hilbert–Schmidt norms.

Write $\mathfrak{g} = \mathfrak{usp}(2N_f) \oplus \mathfrak{usp}(2N_f)$, acting on W as in eq. 5.ii, and put $\|X\| = \max(\|X_+\|, \|X_-\|)$. For a bond $b = \langle x, y\rangle$, let H_b be the sum of the hopping terms of the fermion action across b. Define the derivation δ_X^y by

$$\delta_X^y \widehat{\Psi}(v) = 1_{x_v=y} X \widehat{\Psi}(v), \qquad j_b^X = \delta_X^y H_b, \qquad J^X(F) = a \sum_{x,\mu} F_\mu(x + ae_\mu/2) j^X_{\langle x, x+ae_\mu\rangle}.$$

The derivation fixes the links. With our convention for α_S, it is the derivative of $\alpha_{\exp(-\epsilon X)}$ applied only at y. For Hermitian $T = e_+ \otimes t_+ + e_- \otimes t_-$, put

$$X_T = (\operatorname{diag}(it_+, -i\bar{t}_+), \operatorname{diag}(it_-, -i\bar{t}_-)).$$

Then $J^{X_T} = J^T$ with the normalization in eq. 8.i.

Fix a real basis $\mathfrak{X}$ of $\mathfrak{g}$ with $\|X\| \le 1$ and put $\mathcal{I}^{\mathrm{cur}} = \mathcal{I}^{\mathrm{ext}} \sqcup \mathfrak{X}$. A current entry (X, F) has insertion norm $\|F\|_X = \|X\| \sup_x |F(x)|$, where $|F|$ is the Euclidean norm on $\mathbb{R}^3$. The other insertion norms remain those used in Prop. 5.3. The joint truncated function is the cumulant of these fields. Scalar test functions and vector test functions at current entries are taken with pairwise disjoint supports.

Lemma 8.9 (The currents at a fixed cutoff)**.** Every j_b^X is even, gauge invariant and supported at the endpoints of b. There are $S \in G$ and a Hermitian T with real diagonal flavour blocks such that

$$J^X(F) = \alpha_S J^T(F), \qquad \|T\| = \|X\|.$$

The automorphism α_S preserves each cell algebra and its chessboard seminorm. In particular the conditional mean of $J^X(F)$ is zero.

Let $\|Q\|_1$ be the sum of the absolute values of the coefficients of a Grassmann polynomial in the original generators. Set

$$c_{\mathrm{h}} = 9/2, \qquad c^1_{\mathrm{cur}} = 32 c_{\mathrm{h}} N_f (8N_f)^{3/2}.$$

At every gauge configuration,

$$\|J^X(F)\|_1 \le c^1_{\mathrm{cur}} \|X\| a \sum_b |F(b)|, \qquad \|\Phi[M](f)\|_1 \le 6144 N_f^2 \|M\| a \sum_x |f(x)|.$$

For quadratic polynomials $Q_1, \dots, Q_s$,

$$|\langle Q_1 \cdots Q_s \rangle_U| \le m_0^{-s} \prod_i \|Q_i\|_1.$$

These assertions hold on a torus and on the infinite lattice. ■

Proof. The derivation is linear in X, preserves the degree, and acts only on the generators at y. The hopping action is gauge invariant and the derivation commutes with colour, so j_b^X is gauge invariant and quadratic. For X_T, the derivation is $i\delta_y^T$; differentiating the two oriented hopping terms gives $j_b^{X_T} = i j_\mu^T(x)$.

Every element of the Lie algebra of a compact connected group is conjugate into the Lie algebra of a fixed maximal torus; see [Kna97, Thm. 3.6]. Applied to G, this gives $X = S X_T S^{-1}$ with real diagonal $t_\pm$. Unitarity preserves the norm. The constant substitution α_S preserves H_b. Conjugating

the local derivation therefore gives $\alpha_S j_b^{X_T} = j_b^X$. By Prop. 5.1, α_S commutes with each reflection and preserves the conditional Gaussian functional. Thus

$$\mathrm{Fold}(\alpha_S Q) = \alpha_S \,\mathrm{Fold}(Q), \qquad \|\alpha_S Q\|_{\mathrm{ch}} = \|Q\|_{\mathrm{ch}}.$$

The vanishing mean follows from Lem. 8.2.

Each component of $\widehat{\Psi}$ is a generator times a number of modulus one. The formula for $\phi[M]$ has $16N_f$ components, so $\|\phi[M](v)\|_1 \le \frac{1}{2}(16N_f)^2\|M\|$. There are 48 chambers at a site. For $J^T(F)$ there are eight chamber pairs above each bond, with two hopping blocks of norm at most $a|F(b)|\|T\|c_{\mathrm{h}}$ per pair. An $8N_f$ square matrix has entrywise ℓ^1 norm at most $(8N_f)^{3/2}$ times its operator norm. A unitary symplectic substitution multiplies the coefficient norm of a quadratic polynomial by at most $2N_f$, since each of its two rows has ℓ^1 norm at most $(2N_f)^{1/2}$. These counts give the stated constants. Finally expand $Q_1 \cdots Q_s$ in monomials. Its Gaussian mean vanishes unless there are s generators of each type. Otherwise it is a minor of the covariance; repeated generators give zero, and Hadamard's inequality bounds each remaining minor by $\|D_U^{-1}\|^s \le m_0^{-s}$. □

Lemma 8.10 (Propagators between reflected cells)**.** Let Q be a set of sites in a chessboard cell $\square$, at distance at least δ from its walls. Let P_r project onto the chambers above its reflected copy rQ, and let N be the number of cells. With $\mathsf{G}_+ = a(D_+[U] + m_0)^{-1}$, put

$$N_Q(U) = \sum_{r \ne r'} \|P_{r'}\mathsf{G}_+ P_r\|_2^2, \qquad \mathscr{C}_K^\flat(Q;\delta) = \max_r \mathscr{C}_K(rQ;\delta).$$

Then

$$\mathsf{E}\exp\{N_f \varrho_J^2 N_Q/8\} \le \exp\{\mathsf{N}(c_V|\square| + \mathscr{C}_K^\flat(Q;\delta))\}.$$

■

Proof. Choose a subset A of the cells and let B be its complement. Let χ_A, χ_B be the indicators of the corresponding unions of the reflected sets. Use the meson source

$$J_A = \varrho_J(\chi_A e_{+-} - \chi_B e_{-+}) \otimes 1_{N_f}.$$

Its norm is ϱ_J. Since $\mathsf{G}_- = \mathsf{G}_+^*$, Schur complementation and $\det(1+AB) = \det(1+BA)$ give

$$\det(1 - \mathsf{G}J_A) = \det(1 + \varrho_J^2 T_A T_A^*)^{N_f}, \qquad T_A = \chi_B \mathsf{G}_+ \chi_A.$$

Here $\mathsf{G} = aD_U^{-1}$ on the original, undoubled space. The singular values of $\varrho_J T_A$ are at most $\varrho_J/\mathsf{m} \le 1/4$. Hence

$$\det(1 - \mathsf{G}J_A) \ge \exp\{N_f \varrho_J^2 \|T_A\|_2^2/2\}.$$

The chessboard estimate bounds the partition quotient at J_A by the product of the cell seminorms of the one-sided sources $\varrho_J 1_{rQ} e_{+-}$ or $-\varrho_J 1_{rQ} e_{-+}$. A relative species phase rotates either source by any unit complex number, without changing its seminorm. In eq. 5.iv, average the logarithmic estimate over that phase. The local energy is holomorphic on the disc of radius two and vanishes at zero, so its real part has circle mean zero. Thus each cell seminorm is at most $\exp\{c_V|\square| + \mathscr{C}_K(rQ;\delta)\}$, and

$$\mathsf{E} e^{N_f \varrho_J^2 \|T_A\|_2^2/2} \le \exp\{\mathsf{N}(c_V|\square| + \mathscr{C}_K^\flat(Q;\delta))\}.$$

Now average over all 2^{N} choices of A. Every ordered pair of distinct cells is in $A \times B$ with probability $1/4$, so $\mathrm{avg}_A \|T_A\|_2^2 = N_Q/4$. Jensen's inequality at each configuration, followed by integration in U, proves the assertion. □

Lemma 8.11 (A current in one cell)**.** Suppose that F vanishes at the midpoint of every bond whose endpoints are not both in Q. Then

$$\|J^X(F)\|_{\mathrm{ch}} \le \frac{2\sqrt{2}c_{\mathrm{h}}\|X\|\|F\|_\infty}{\varrho_J} e^{c_V|\square| + \mathscr{C}_K^\flat(Q;\delta)}.$$

■

Proof. By Lem. 8.9, it suffices to treat $J^T(F)$. Write it as $\bar\psi K\psi$. Its two blocks over a chamber edge (v,v') above the bond b are

$$K(v',v)=iaF_\mu(b)h(v',v)TU(v',v),\qquad K(v,v')=-iaF_\mu(b)h(v,v')TU(v,v').$$

The sum of the norms of the hopping blocks in each row and each column is at most $c_{\mathrm h}=3(1{+}r)=9/2$. Schur's test gives $\|K\|\le ak$, $k=c_{\mathrm h}\|T\|\|F\|_\infty$. The reflected copies $J_r=\bar\psi K_r\psi$ have $K_r=P_rK_rP_r$, $\|K_r\|\le ak$, and mean zero at each configuration. Reflection only conjugates these blocks by unitary matrices, possibly taking their adjoints.

Let $\mathbb G=D_U^{-1}$ and $\mathbb G'=\sum_{r\neq r'}P_{r'}\mathbb G P_r$. The conditional mean $\langle\prod_r J_r\rangle_U$ is the coefficient of $\prod_r z_r$ in $\det(1-\mathbb G\sum_r z_rK_r)$. This coefficient is unchanged when $\mathbb G$ is replaced by $\mathbb G'$. Indeed in the logarithmic determinant each square-free word of degree at least two has distinct neighbouring cell indices, including its last and first indices. The degree-one terms vanish by the zero conditional means. The square-free coefficient of the exponential uses only these square-free logarithmic coefficients.

Put $A=\mathbb G'\sum_r z_rK_r$. Its trace is zero. If λ_i are its eigenvalues,

$$|\det(1-A)|=\prod_i|(1-\lambda_i)e^{\lambda_i}|\le\exp\{\tfrac12\sum_i|\lambda_i|^2\}\le e^{\|A\|_2^2/2}.$$

The first inequality follows from $|1-z|^2e^{2\,\mathrm{Re}\,z}\le e^{|z|^2}$; the second follows by triangularizing A unitarily. On $|z_r|=\rho$, the summands have disjoint columns, and hence

$$\|A\|_2^2\le\rho^2a^2k^2\sum_{r\neq r'}\|P_{r'}\mathbb G P_r\|_2^2=2N_f\rho^2k^2N_Q.$$

The last equality uses the two species, their adjoint relation, and the N_f identical flavour copies. Cauchy's coefficient estimate gives

$$|\langle\prod_r J_r\rangle_U|\le\rho^{-\mathsf N}e^{N_f\rho^2k^2N_Q}.$$

Choose $\rho=\varrho_J/(2\sqrt2k)$, apply Lem. 8.10, and take the $\mathsf N$th root of the fold expectation. This proves the cell bound. □

Proposition 8.12 (Joint bounds with currents)**.** Let $n\ge2$, $R,d>0$. Suppose that each test function is real and Lipschitz, its support lies in a ball of radius R, and the supports are pairwise at distance at least d. For labels in $\mathcal I^{\mathrm{ext}}\cup\mathfrak g$,

$$|S_n^T(\mathcal X)|\le[B_{\mathrm{cur}}(R)(1+d^{-1})^{\kappa-1}]^n(n!)^\kappa\prod_i\|f_i\|_{\lambda_i}.$$

The constant is independent of cutoff and volume, and the estimate holds in each fixed-cutoff infinite-volume state. For arbitrary scalar labels $M_i\in\mathscr S$, include the factor $c(N_f)\|M_i\|$ at that entry. One may take

$$\rho_{\mathrm{cur}}=\frac{\varrho_J}{8\sqrt2c_{\mathrm h}},\quad \rho_{\min}^{\mathrm{cur}}=\min(\varrho,\rho_{\mathrm{cur}}),\quad B_{\mathrm{cur}}(R)=C_B^{\mathrm{cur}}\max\{e^{c_V}/\rho_{\min}^{\mathrm{cur}},K_{\mathrm{co}}\}(1+R)^3,$$

where C_B^{cur} depends only on the fixed source and counting constants and $K_{\mathrm{co}}=\max\{162,(165888N_f^2+24c_{\mathrm{cur}}^1)/m_-\}$. ■

Proof. Choose the least $q\ge0$ with $\sqrt3L^{-q}<d$ and put $\eta=L^{-q}/(16n)$. Then $\eta^{-1}\le16\sqrt3Ln(1+d^{-1})$. First suppose $a\le\eta$. For scalar entries use the hat partition of mesh η from Prop. 5.3. For currents use the hat partition of mesh $\eta/2$. Its supports have side η; the endpoints of bonds whose midpoints lie there belong to the concentric cube of side 2η, because $a\le\eta$. Each entry has at most $64(1+R)^3\eta^{-3}$ nonzero pieces.

For each tuple of pieces choose cells of physical side L^{-q} whose walls are at least η from their cubes. In each direction the forbidden wall positions number at most $n(4\eta/a+1)<L^{-q}/a$. The

cell diameter is less than d, so it contains a piece of at most one entry. This remains true for a current piece: a nonzero piece has a bond midpoint in the support of its original entry.

For a scalar piece use its exponential insertion and the one-variable counterterm from Prop. 5.3. For a current piece J_i, use $1 + z_i J_i$ and radius

$$\rho_i = \frac{\varrho_J}{2\sqrt{2}c_{\mathrm{h}}\|f_i\chi_i\|_{\lambda_i}}.$$

Let $F(z)$ be the expectation of this product, multiplied by the exponential of minus the scalar counterterms. Then $F(0) = 1$. Its logarithmic derivative in all variables is the desired cumulant: the square-free coefficient uses only the constant and linear terms of each factor, and the subtracted terms depend on single variables. On a coordinate face indexed by A, the chessboard inequality and Lem. 8.11 give

$$|F(z)| \le \prod_{i\in A} m_i, \qquad m_i = e^{c_V+\mathscr{C}_\eta} \text{ for scalar pieces}, \qquad m_i = 2e^{c_V+\mathscr{C}_\eta} \text{ for currents},$$

where $\mathscr{C}_\eta$ is the supremum of $\mathscr{C}_K(Q;\eta)$ over cubes of side 2η. For currents this uses the seminorm triangle inequality $\|1 + z_i J_i\|_{\mathrm{ch}} \le 1 + |z_i|\|J_i\|_{\mathrm{ch}}$. Cauchy's estimate and the partition formula give

$$|S_n^T(\text{pieces})| \le 2^n n! \prod_i m_i \rho_i^{-1} \le 2^n n! e^{n(c_V+\mathscr{C}_\eta)} \prod_i \frac{(1+80M)\eta^{-1}\|f_i\|_{\lambda_i}}{2\rho_{\min}^{\mathrm{cur}}}.$$

At action-density entries the extra η^{-1} comes from differentiating the hats. At current entries, $2\rho_i^{-1} = \|f_i\chi_i\|_{\lambda_i}/(2\rho_{\mathrm{cur}})$, so the same estimate applies. The source count used in Prop. 5.3 gives $e^{n\mathscr{C}_\eta} \le C_{\mathrm{ct}}^n \eta^{-np/2}$. After summing the pieces we obtain

$$|S_n^T(\mathcal{X})| \le \left[64(1+80M)C_{\mathrm{ct}}\frac{e^{c_V}}{\rho_{\min}^{\mathrm{cur}}}(1+R)^3\right]^n n!\eta^{-n(\kappa-1)}\prod_i \|f_i\|_{\lambda_i}.$$

Use the bound on η^{-1} and $n^n \le e^n n!$.

If $a > \eta$, use the moment estimate in Lem. 8.9, the elementary bound $|O(f)| \le 2a^{-1}\sum_p |f(x(p))|$, and the moment–cumulant formula. Balls of radius R contain at most $81a^{-3}(1+R)^3$ plaquette centres, $27a^{-3}(1+R)^3$ sites and $24a^{-3}(1+R)^3$ bond midpoints. Each insertion is consequently bounded by $K_{\mathrm{co}}a^{-4}(1+R)^3\|f_i\|_{\lambda_i}$ in every product moment. Since $a^{-1} < 16\sqrt{3}Ln(1+d^{-1})$, this gives

$$|S_n^T(\mathcal{X})| \le [2e^4(16\sqrt{3}L)^4 K_{\mathrm{co}}(1+R)^3(1+d^{-1})^4]^n (n!)^5 \prod_i \|f_i\|_{\lambda_i}.$$

Take C_B^{cur} to be the larger of $64(1+80M)C_{\mathrm{ct}}(16\sqrt{3}Le)^{\kappa-1}$ and $2e^4(16\sqrt{3}L)^4$. Since $\kappa \ge 5$, both regimes give the stated estimate. Every current is a local Grassmann polynomial, so its product moments converge along the tori defining a fixed-cutoff state. This passes the estimate to those states. Expansion in Lem. 5.2 gives arbitrary scalar labels. □

Proposition 8.13 (Joint limits with currents)**.** Consider a sequence with cutoff tending to infinity, either on a fixed torus, on growing tori, or in the fixed-cutoff infinite-volume states used to define a volume-first limit. There is a subsequence on which the parameters converge and every joint truncated function of orders $n \ge 2$ for labels in $\mathcal{I}^{\mathrm{cur}}$ converges on pairwise disjoint C_c^1 supports. The resulting family S^{cur} obeys Prop. 8.12; at fixed labels and fixed current components it defines a unique distribution of locally finite order away from the diagonals.

On tuples without currents it is the scalar-bilinear limit of the same subsequence. In particular, if the sequence already defines S^{ext}, that restriction is unchanged. Every volume-first joint limit can also be realized on a sequence of growing tori. These claims include all $2N_f(4N_f-1)$ scalar bilinears, the action density, and all currents J^X, $X \in \mathfrak{g}$. ■

Proof. Fix disjoint closed supports in bounded balls. The bound in Prop. 8.12 gives equicontinuity in the product of their insertion norms. Use a countable dense family of C^1 scalar and vector functions, supports in rational finite unions of closed cubes, and the finite label set $\mathcal{I}^{\mathrm{cur}}$. A diagonal subsequence gives convergence on this family and convergence of the parameters in their compact windows. Equicontinuity gives convergence on all tuples; multilinearity extends the labels. The restrictions are the original truncated functions at every cutoff, proving the compatibility assertion.

For a volume-first sequence enumerate the dense tuples as $\mathcal{X}^{(q)}$. At cutoff K_j choose a side $\ell_j \ge j$ from the torus sequence defining the state so that its truncated functions differ from the state values by at most $1/j$ on the first j tuples. Such a side exists because each truncated function is a polynomial in expectations of local elements; this includes the currents. Uniform equicontinuity extends the resulting convergence from the dense family to every tuple.

For fixed labels and current components, the limiting multilinear form is continuous in C^1 on a product of bounded open sets with disjoint closures. The Schwartz kernel theorem gives a unique distribution of finite order there. Uniqueness makes these distributions agree on overlaps, so they patch off the diagonals. □

All constants introduced here are chosen after the source constants of [QCDM1]. They impose no new small-coupling requirement. The joint compactness statement concerns correlations away from coinciding points; the two-current asymptotic proved above remains a separate, more precise statement about that subfamily.

9. Analytic dependence at fixed volume

A constant source changes the bare coupling or the bare mass. This observation connects the source estimates to analytic dependence on the physical parameters. The partition function itself need not be bounded as the cutoff is removed. The useful quantity is the partition function after the same vacuum energy has been removed from its upper and lower bounds.

Throughout this section the physical torus T_ℓ is fixed. Fix $0 < g \le \gamma$ and an allowed side $\ell \in \mathfrak{L}(g)$, and write

$$I_g = [g^{-2} + \bar{B},\, L(g^{-2} + B_\sharp) - \bar{B}], \qquad W = I_g \times [m_-, m_+].$$

The constants and the cofinal set $\mathfrak{L}(g)$ are those of [QCDM1, Prop. 10.20]. In particular the estimates in this section hold for every $K \ge 1$ and every $(\mathsf{b}, \mathsf{m}) \in W$. We use the chamber operator with $r = 1/2$.

For a source $s = (w, J)$ put

$$Z_K[s] = \int e^{-(\mathsf{b}/a)A(U) + O_K(w)} \det\bigl(D[U] + a\mathsf{m} - a\widehat{J}\bigr)\, dU, \qquad a = L^{-K}.$$

The determinant includes both internal blocks and all flavours. Thus $Z_K[0] > 0$. The source w is a function on plaquette centres and J takes values in the complex span of the physical meson basis $\mathfrak{B}$. The definitions of O_K and the meson insertions are those in section 2. The action-source norm is $\|w\|_\sharp = \max\{\|w\|_\infty, 40M \operatorname{Lip} w\}$. Let

$$\mathcal{P} = \{(w, J) : \|w\|_\sharp \le 2\varrho,\ \|J\|_\infty \le 2\varrho_J\}.$$

The radii are independent of K and ℓ, and $2\varrho_J \le c_J m_-$, $c_J \le 1/2$.

We first recall precisely the source input. It is the combination of the local-energy, history and counting estimates of [QCDM1, Props. 10.6 and 10.14, and the counting estimate preceding Prop. 10.16]. There is a holomorphic energy

$$E_K(s) = E_K(0) + \sum_{j=1}^{K} \sum_{Y \in \mathcal{D}_j} e_j(Y; s), \qquad Z_K[s] = e^{E_K(s)} W_K[s], \tag{9.i}$$

defined on a neighbourhood of $\mathcal{P}$. Here Y is a connected union of blocks of physical side $M\lambda_j$, $\lambda_j = L^{j-K}$, and $d_j(Y)$ is the block-tree length used in the construction. For fixed positive constants,

$$|e_j(Y;s)| \le C_E e^{-\kappa d_j(Y)}, \tag{9.ii}$$

$$|e_j(Y;s) - e_j(Y;s')| \le C_E e^{-\kappa d_j(Y)} e^{-\kappa_\sigma \operatorname{dist}(Y,\operatorname{supp}(s-s'))/\lambda_j}. \tag{9.iii}$$

Distances in this section are physical. The geometry gives $\operatorname{diam} Y \le c_D M\lambda_j(1 + d_j(Y))$. The construction chooses κ so that summing $e^{-\kappa d_j(Y)/2}$ over polymers containing one specified block costs at most C_{cnt}. Finally,

$$|W_K[s]| \le e^{(C+2)\ell^3}, \qquad W_K[0] \ge e^{-C_P(1+\log X_g)\ell^3}, \qquad X_g = L(g^{-2} + B_\sharp). \tag{9.iv}$$

The conjugation of the source makes E_K real for constant real coupling and mass shifts. No positivity at complex sources is used.

9.1. **A zero-free neighbourhood.** Set

$$V_\ell = \exp\{[C + 2 + C_P(1 + \log X_g)]\ell^3\}, \qquad \rho_b = \frac{\varrho}{1+V_\ell}, \qquad \rho_m = \frac{\varrho_J}{1+V_\ell}. \tag{9.v}$$

Let $W_{\mathbb{C}}$ be the union, over $(b,m) \in W$, of the open polydiscs $|b' - b| < \rho_b$, $|m' - m| < \rho_m$.

Proposition 9.1 (Complex parameters)**.** For every K, the partition function $Z_{K,\ell}(b',m') = Z_K[0]$ formed with bare parameters b'/a and am' has no zero on $W_{\mathbb{C}}$. For a real base point $(b,m) \in W$, write $c = (-(b'-b)1, -(m'-m)1)$ and $R_K(b',m') = W_K[c]/W_K[0]$, with the energies and partition functions based at (b,m). Then

$$|R_K - 1| \le (1+V_\ell)\max\left\{\frac{|b'-b|}{2\varrho}, \frac{|m'-m|}{2\varrho_J}\right\} < \frac{1}{2}. \tag{9.vi}$$

On this polydisc the pressure has the decomposition

$$\ell^{-3}\log Z_{K,\ell}(b',m') = \ell^{-3}E_K(c) + \ell^{-3}\log W_K[c].$$

The logarithm is chosen real at the real base point, and

$$|\log R_K| \le \log 2, \qquad \ell^{-3}|\log W_K[c]| \le C + 2 + C_P(1 + \log X_g) + \ell^{-3}\log 2.$$

Consequently every sequence of cutoffs has a subsequence on which $\ell^{-3}\log R_K$ converges locally uniformly on this polydisc, with all parameter derivatives. ■

Proof. The constant source identity $Z_{K,\ell}(b',m') = Z_K[c]$ holds before any integration: the action coefficient becomes b'/a, and the matrix in the determinant becomes $D + am'$. The ratio R_K is holomorphic on the larger polydisc with radii $2\varrho, 2\varrho_J$, $R_K(b,m) = 1$, and $|R_K| \le V_\ell$ by eq. 9.iv. For a nonzero displacement let r be the maximum in eq. 9.vi. Restrict $R_K - 1$ to the complex line through that displacement, rescaled so that its unit disc lies in the larger polydisc. Dividing by $1 + V_\ell$ gives a function of modulus at most one, vanishing at zero. The given displacement corresponds to radius r on this disc. Schwarz's lemma proves eq. 9.vi. Thus R_K has a logarithm with $|\log R_K| \le \sum_{n\ge1} 2^{-n}/n = \log 2$. The remaining bounds follow from the positive real lower bound and the upper bound in eq. 9.iv. Local boundedness gives a normally convergent subsequence; Cauchy's formula gives convergence of every derivative. □

For example, if $p + q \ge 1$, differentiation at a real base point gives

$$\left|S^T_{p+q,K}((0,1)^p,(1,1)^q) - (-\partial_{b'})^p(-\partial_{m'})^q E_K(c(b',m'))\big|_{(b',m')=(b,m)}\right| \le p!q!\log 2\,\rho_b^{-p}\rho_m^{-q}. \tag{9.vii}$$

The RG base point (b,m) is held fixed in this derivative. Here the second label 1 in $(1,1)$ is the constant test function and the first is the identity meson matrix, extended linearly from the chosen basis. The sign comes from the constant-source identity. The vacuum term may diverge with K; this estimate concerns its difference from the cumulant. The radii in eq. 9.v depend on the physical volume. No infinite-volume analyticity follows from this argument.

9.2. **Supports with empty total intersection.** For closed nonempty sets $S_1, \dots, S_n \subset T_\ell$ define

$$d_\cap(S_1, \dots, S_n) = \min_{x \in T_\ell} \max_i \operatorname{dist}(x, S_i).$$

Compactness shows that $d_\cap > 0$ precisely when $\bigcap_i S_i = \emptyset$. Pairwise intersections are allowed. A tuple $X = ((\alpha_i, f_i))_{i=1}^n$ is admissible in this section when $n \geq 2$, $\alpha_i \in \{0\} \cup \mathfrak{B}$, f_i is smooth, and its supports have empty total intersection. We write $d_\cap(X)$ for this distance. As usual $\|f\|_0 = \|f\|_\sharp$ and $\|f\|_\Gamma = \|f\|_\infty$ for $\Gamma \in \mathfrak{B}$; these basis matrices have operator norm at most one.

Lemma 9.2 (Mixed derivatives of the vacuum energy)**.** Let $s_* + \sum_i z_i s_i \in \mathcal{P}$ for $|z_i| \leq r_i$, and suppose the supports of s_i have $d_\cap = d > 0$. There is a constant C_{ov}, independent of K, ℓ and the sources, such that

$$\left| \partial_1 \cdots \partial_n E_K(s_* + \textstyle\sum_i z_i s_i)\big|_{z=0} \right| \leq 2^{n-1} C_{\mathrm{ov}} \ell^3 (1 + d^{-3}) \prod_i r_i^{-1}. \tag{9.viii}$$

■

Proof. Let P_i set $z_i = 0$. A mixed derivative kills every term independent of one variable. It can therefore be estimated by Cauchy's formula applied to $\prod_i (1 - P_i) e_j(Y; s_* + s(z))$. Explicitly, this finite difference is

$$\prod_i (1 - P_i) e_j(Y; s_* + s(z)) = \sum_{A \subset \{1,\dots,n\}} (-1)^{n-|A|} e_j\left(Y; s_* + \sum_{i \in A} z_i s_i\right).$$

The operators $1 - P_i$ commute. We may therefore choose one index i separately for each polymer, apply the source-tail bound to $(1 - P_i)e_j$, and bound the other $n - 1$ differences by their 2^{n-1} terms. This is why empty total intersection suffices: the source that is distant from a small polymer need not be the same for every polymer.

Choose $y \in Y$. Some support S_i has $\operatorname{dist}(y, S_i) \geq d$. If $\operatorname{diam} Y \geq d/2$, half the polymer weight pays for this diameter:

$$e^{-\kappa d_j(Y)/2} \leq e^{\kappa/2} e^{-\kappa d/(4 c_D M \lambda_j)}.$$

Otherwise $\operatorname{dist}(Y, S_i) \geq d/2$, and the source tail supplies $e^{-\kappa_\sigma d/(2\lambda_j)}$. Set

$$\eta = \min\{\kappa/(4 c_D M), \kappa_\sigma/2\}.$$

The derivative of one energy is at most

$$2^{n-1} C_E e^{\kappa/2} \prod_i r_i^{-1} e^{-\eta d/\lambda_j} e^{-\kappa d_j(Y)/2}.$$

Assigning each polymer to one of its blocks and using the polymer count bounds the sum over Y by this expression times $C_{\mathrm{cnt}}(\ell/(M\lambda_j))^3$. Finally

$$\sum_{j=1}^K \lambda_j^{-3} e^{-\eta d/\lambda_j} \leq C_\eta (1 + d^{-3}).$$

Indeed, split at $\lambda_j = d$. The coarse part is a geometric sum bounded by $C d^{-3}$, and the fine part is bounded by $C d^{-3} \sum_{u \geq 0} L^{3u} e^{-cL^u}$; when $d \geq 1$ only the latter estimate is needed. Summing the energy decomposition proves the assertion. □

Let $k_n = \sum_\pi (|\pi| - 1)!$, with the sum over partitions of $\{1, \dots, n\}$, and let $\varrho_{\min} = \min(\varrho, \varrho_J)$.

Proposition 9.3 (Uniform cumulants at complex parameters)**.** For every admissible tuple X the lattice cumulant $S^T_{n,K}(b, m; X)$ is holomorphic on $W_\mathbb{C}$, and

$$|S^T_{n,K}(b, m; X)| \leq \left(\frac{n}{\varrho_{\min}}\right)^n \left[k_n (2V_\ell)^n + 2^{n-1} C_{\mathrm{ov}} \ell^3 (1 + d_\cap(X)^{-3})\right] \prod_i \|f_i\|_{\alpha_i}. \tag{9.ix}$$

For all $p, q \geq 0$,

$$(-\partial_b)^p(-\partial_m)^q S^T_{n,K}(X) = S^T_{n+p+q,K}(X, (0,1)^p, (1,1)^q). \tag{9.x}$$

The tuple on the right has the same $d_\cap$. ■

Proof. Work about a real base point and represent the new parameters by the constant source c of Prop. 9.1. For a zero test function the assertion follows from multilinearity. Otherwise choose $r_i = \varrho_{\alpha_i}/(n\|f_i\|_{\alpha_i})$. The sum of the action-source norms is at most ϱ, and the sum of the meson-source norms is at most ϱ_J. Since $V_\ell \geq 1$, the constant displacement has norms less than $\rho_b \leq \varrho/2$ and $\rho_m \leq \varrho_J/2$. In particular

$$\Big\| c_w + \sum_i z_i w_i \Big\|_\sharp < \tfrac{3}{2}\varrho < 2\varrho, \qquad \Big\| c_J + \sum_i z_i J_i \Big\|_\infty < \tfrac{3}{2}\varrho_J < 2\varrho_J.$$

Thus the whole Cauchy polydisc stays inside $\mathcal{P}$. The triangle inequality gives this source budget even when the supports overlap.

Set $G(z) = W_K[c + s(z)]/W_K[c]$. The zero-free estimate gives $G(0) = 1$ and $|G(z)| \leq 2V_\ell$. As germs at zero,

$$\log Z_K[c + s(z)] - \log Z_K[c] = E_K(c + s(z)) - E_K(c) + \log G(z).$$

The mixed derivative of the last term is

$$\sum_\pi (-1)^{|\pi|-1}(|\pi| - 1)! \prod_{B\in\pi} \partial^B G(0).$$

For each block B of a partition, Cauchy's estimate gives

$$|\partial^B G(0)| \leq 2V_\ell \prod_{i\in B} r_i^{-1}.$$

The product over the blocks contains each inverse radius once and costs $(2V_\ell)^{|\pi|} \leq (2V_\ell)^n$. Summing the partition coefficients therefore bounds the mixed logarithmic derivative by $k_n(2V_\ell)^n \prod_i r_i^{-1}$. This calculation uses the logarithmic germ at zero; the generating function may have zeros elsewhere in its source polydisc. Apply Lem. 9.2 to the energy and use $\prod_i r_i^{-1} \leq (n/\varrho_{\min})^n \prod_i \|f_i\|_{\alpha_i}$. This proves eq. 9.ix.

The finite-dimensional integral is entire in the parameters and the sources. Its partition function is nonzero at zero source throughout $W_\mathbb{C}$; local logarithms therefore give holomorphic cumulants there. Differentiating the constant-source identity proves eq. 9.x. Adding an entry supported on the whole torus adds zero to the maximum defining $d_\cap$, so it leaves that distance unchanged. □

Theorem M (Holomorphic families of continuum correlations)**.** Every sequence $K_j \to \infty$ has a subsequence for which $S^T_{n,K_j}(b, m; X)$ converges for every admissible tuple X, locally uniformly in $(b, m) \in W_\mathbb{C}$ and with all parameter derivatives. Denote the limit by $S_n(b, m; X)$. It obeys eq. 9.ix and eq. 9.x with the cutoff removed. At real parameters its restriction to separated tuples is a continuum limit of the original lattice model on T_ℓ. For each fixed choice of indices it determines a unique distribution on

$$T_\ell^n \setminus \{x_1 = \cdots = x_n\}.$$

This distribution is symmetric and translation invariant, has the flavour and cubic covariance of the lattice correlations, and depends holomorphically on the parameters when tested on admissible tensors. ■□

Proof. Choose a countable collection of products whose factors are finite unions of closed dyadic boxes and have empty total intersection. On each product choose countable dense sets of smooth test functions, in the C^1 norm for an action-density entry and the uniform norm for a meson entry. There are finitely many basis indices at each position. The bound eq. 9.ix and Montel's theorem, followed by a diagonal selection, give locally uniform convergence on every tuple in these sets. Use

one enumeration containing all orders n, all label strings, all these support tuples, and their dense test families. A countable compact exhaustion of $W_{\mathbb{C}}$ makes the local uniform convergence part of this same selection. Thus one subsequence works for all these data.

For an arbitrary admissible tuple the supports have $d = d_{\cap}(X) > 0$. Enlarge them by less than $d/4$ and cover the enlarged sets by sufficiently fine closed dyadic cubes. The resulting sets still have $d_{\cap} \geq d/2$. Approximation within these sets extends the convergence to X uniformly on every compact parameter set. To see the uniformity, write $T_K(f_1, \dots, f_n) = S^T_{n,K}(b, m; X)$ with fixed labels and these enlarged supports. The bound gives a common constant B there. Multilinearity gives

$$|T_K(f_1, \dots, f_n) - T_K(h_1, \dots, h_n)| \leq B \sum_{i=1}^{n} \|f_i - h_i\|_{\alpha_i} \prod_{r<i} \|h_r\|_{\alpha_r} \prod_{r>i} \|f_r\|_{\alpha_r}.$$

The approximating functions may be chosen with bounded norms, so the right-hand side tends to zero uniformly in K and the parameters. Applying the same estimate to two cutoffs proves convergence for the arbitrary tuple.

For a compact parameter set, choose slightly larger compact polydiscs still contained in $W_{\mathbb{C}}$. Cauchy's formula on their boundaries gives convergence of every parameter derivative on the smaller sets. The constant insertions in eq. 9.x also belong to the common extraction: adding a whole-torus support preserves $d_{\cap} > 0$. Hence both sides of the insertion identity converge to the stated identity.

A point outside the total diagonal has two distinct coordinates. It therefore has a product neighbourhood with two disjoint closed factors. On this neighbourhood the estimate defines a continuous multilinear functional on smooth functions, hence a distribution. These local distributions agree on overlaps: products of test functions are dense, and the same limiting functional gives their values. A partition of unity glues them uniquely.

Permutation, flavour and cubic identities hold at each cutoff and pass to the limits. Approximate any torus translation by lattice translations. Translation is continuous in the test-function norms, so the uniform bound gives the full translation identity. At real fixed parameters the same subsequence defines the stated continuum limit. If parameters also converge inside W, local uniform convergence shows that the limit is the value of this same holomorphic family at the limiting parameters. □

Corollary 9.4 (Mass differentiation)**.** For every admissible tuple the function $m \mapsto S_n(b, m; X)$ is real analytic on the allowed real interval and

$$\partial_m S_n(b, m; X) = -S_{n+1}(b, m; X, (1, 1)).$$

The inserted cumulants are bounded uniformly in the cutoff at fixed g, ℓ, n and supports. The same holds for every higher mass derivative and for mixed coupling and mass derivatives. ■□

Proof. Apply the preceding theorem and the insertion identity. The uniform bound for the inserted tuple uses its unchanged $d_{\cap} > 0$. □

Remark 9.5. The theorem fixes correlations on partial diagonals. It does not fix an extension to the total diagonal. If two translation-invariant distributional extensions exist, their difference is supported there. In relative coordinates it is a finite sum of derivatives of the delta distribution with constant coefficients. Thus the remaining ambiguity is local; it cannot be prescribed independently on a proper partial diagonal.

9.3. **Mass and Wilson insertions near the other fields.** Assume $N_f \geq 2$ and choose a Hermitian flavour matrix t with $\operatorname{tr} t = 0$ and $\operatorname{tr} t^2 = 1$. Write $\Gamma^0_t = 1 \otimes t$ and $\Gamma^1_t = i\tau_1 \otimes t$, with the identities on spin and colour understood. In this subsection $a = L^{-K}$ and ν_K is the gauge marginal. All masses are positive and belong to the compact interval fixed in the construction.

The insertion of a constant mass source differentiates a correlation function. An insertion supported near either of the other fields requires a further estimate: separated correlation bounds

do not control it. The fourth Schatten moment in Thm. D supplies this estimate. The Wilson insertion also differentiates the spatial cutoff; this accounts for the factor $1 + a^{1/2}/e$ below.

Put

$$\mathcal{W} = \Delta_+ + a\mathsf{m}, \qquad G_+ = (\mathcal{W} + \mathcal{B}_U)^{-1}, \qquad G_- = G_+^*, \qquad Y = \tfrac{1}{2}(G_+ + G_-),$$

where $\mathcal{B}_U^* = -\mathcal{B}_U$ and $\mathsf{G}_\pm = aG_\pm$. For a bounded function χ define

$$\mathcal{X}_\Gamma(\chi) = \sum_v \chi(x_v)\bar\psi(v)\Gamma(\Delta\psi)(v), \qquad \mathcal{J}_\Gamma(\chi) = \sum_v \chi(x_v)\bar\psi(v)\Gamma(\mathcal{B}_U\psi)(v).$$

Here Δ and $\mathcal{B}_U$ act on the full fermion space; neither expression has a factor a. Set

$$\begin{aligned} T_K(f,\chi,h) &= S^T_{3,K}((\Gamma^1_t, f), (\Gamma^1_1, \chi), (\Gamma^0_t, h)),\\ T^X_K(f,\chi,h) &= \kappa_3(\Phi_{\Gamma^1_t}(f), \mathcal{X}_{\Gamma^1_1}(\chi), \Phi_{\Gamma^0_t}(h)),\\ T^J_K(f,\chi,h) &= \kappa_3(\Phi_{\Gamma^1_t}(f), \mathcal{J}_{\Gamma^1_1}(\chi), \Phi_{\Gamma^0_t}(h)),\\ \Xi_K(f,h) &= S^T_{2,K}((\Gamma^1_t, f), (\Gamma^1_t, h)) - S^T_{2,K}((\Gamma^0_t, f), (\Gamma^0_t, h)). \end{aligned}$$

Lemma 9.6 (The insertion identities)**.** At every configuration,

$$Y = G_-\mathcal{W}G_+ = G_+\mathcal{W}G_-, \qquad Y^{-1} = \mathcal{W} + \mathcal{B}_U^*\mathcal{W}^{-1}\mathcal{B}_U \geq \mathcal{W} > 0.$$

On a torus, for real bounded f, h,

$$\Xi_K(f,h) = 4a^2\mathsf{E}_{\nu_K}\operatorname{Tr}(P_f Y P_h Y) = \mathsf{m}T_K(f,1,h) + T^X_K(f,1,h).$$

If χ has support disjoint from those of f, h, then

$$T^J_K(f,\chi,h) + T^X_K(f,\chi,h) = -\mathsf{m}T_K(f,\chi,h).$$

■

Proof. The sum of $G_+^{-1} = \mathcal{W} + \mathcal{B}_U$ and $G_-^{-1} = \mathcal{W} - \mathcal{B}_U$, multiplied by the two inverses, gives the identities for Y. Multiplication of $(\mathcal{W} + \mathcal{B}_U)\mathcal{W}^{-1}(\mathcal{W} - \mathcal{B}_U)$ proves the inverse formula.

For bilinears at fixed U, the determinant identity gives

$$\kappa^U_n(\bar\psi A_1\psi, \dots, \bar\psi A_n\psi) = -\sum_{\text{cyclic orders}} \operatorname{Tr}(A_{i_1}G\cdots A_{i_n}G).$$

Indeed the coefficient of degree n in $\log\det(1 + GJ)$ has sign $(-1)^{n-1}$, whereas the generating function uses $e^{-\bar\psi J\psi}$. The one-point function of $\Phi_{\Gamma^0_t}$ vanishes because $\operatorname{tr} t = 0$; those of $\Phi_{\Gamma^1_t}$, $\Phi_{\Gamma^1_1}$ and $\mathcal{X}_{\Gamma^1_1}$ vanish because τ_1 is off diagonal in the two propagator blocks. Thus their two- and three-point cumulants are the gauge means of the conditional cumulants. Wick's rule gives

$$\Xi_K = a^2\mathsf{E}_{\nu_K}\operatorname{Tr}(P_f(G_+ + G_-)P_h(G_+ + G_-)),$$

which is the first formula.

For the insertion identity use $\gamma_4 = \tau_1$. Since $\operatorname{tr}\gamma_4 = 0$,

$$\det(e^{\epsilon\gamma_4}(D_{m_0} + J)e^{\epsilon\gamma_4}) = \det(D_{m_0} + J).$$

The derivative at zero replaces the operator by $2\gamma_4\mathcal{W} + \{\gamma_4, J\}$. Differentiation in the two sources uses $\{\gamma_4, \Gamma^1_t\} = 2i\Gamma^0_t$ and $\{\gamma_4, \Gamma^0_t\} = -2i\Gamma^1_t$. It gives the asserted identity because $i\sum_v \bar\psi\gamma_4\mathcal{W}\psi = \mathsf{m}\Phi_{\Gamma^1_1}(1) + \mathcal{X}_{\Gamma^1_1}(1)$. The determinant identity holds before the gauge integral, so it also proves the identity for the full cumulant.

Finally, contraction of $(D_{m_0}\psi)(v)$ with $\bar\psi(w)$ gives δ_{vw}. With all other bilinear insertions away from v, the expectation of $\bar\psi(v)\Gamma(D_{m_0}\psi)(v)$ times their product is $-\operatorname{tr}\Gamma$ times the expectation of that product. This factor is independent of U. Its cumulants of order at least two therefore vanish. Use $D_{m_0} = \mathcal{B}_U + \Delta + a\mathsf{m}$ and sum against χ to obtain the equation of motion. □

Definition 9.7 (Local propagator norms and loops)**.** For a set of sites Q and $\varphi \geq 0$, put

$$n_Q = \|1_Q \mathsf{G}_+ 1_Q\|, \qquad x_\varphi(h) = a^2 \sum_{s=\pm} \operatorname{Tr}(P_\varphi G_s P_{|h|} G_{-s}).$$

For the chamber hops put $Q_s = (r t_s/2)^{1/2}(1 - R_s)$, $0 \leq s \leq 3$, with $r = 1/2$. Define

$$\Delta_\chi = \sum_s Q_s^* P_\chi Q_s, \qquad E_\chi = \sum_s [P_\chi, Q_s^*] Q_s, \qquad x_\varphi^\Delta(h) = a \sum_{s=\pm} \operatorname{Tr}(P_{|h|} G_{-s} \Delta_\varphi G_s).$$

Write $x_Q = x_{1_Q}$ and $x_Q^\Delta = x_{1_Q}^\Delta$. These loops are nonnegative and increase with φ and $|h|$. Moreover $x_\varphi(h) = 2\Pi_U(\varphi, |h|)$.

Choose $e = L^{-n} \leq 1/13$ and let y run over $e\mathbb{Z}^3/\ell\mathbb{Z}^3$. Set

$$c(t) = \begin{cases} \cos^2(\pi t/2), & |t| \leq 1, \\ 0, & |t| \geq 1, \end{cases} \qquad c^\sharp(t) = \begin{cases} 1, & |t| \leq 2, \\ \cos^2(\pi(|t| - 2)/2), & 2 \leq |t| \leq 3, \\ 0, & |t| \geq 3. \end{cases}$$

Define $p_y(x) = \prod_\mu c((x_\mu - y_\mu)/e)$, $\chi_y(x) = \prod_\mu c^\sharp((x_\mu - y_\mu)/e)$, $f_y = f p_y$, $h_y = h p_y$, and $Q_y = \{x : |x - y|_\infty \leq 4e\}$. Differences are periodic. Then $\sum_y p_y = 1$, $\chi_y = 1$ on $\operatorname{supp} p_y$, $\operatorname{Lip} \chi_y \leq 3/e$, and each site belongs to at most 9^3 sets Q_y. If $a \leq e$, all neighbours of $\operatorname{supp} \chi_y$ belong to Q_y. ■

Lemma 9.8 (Bounds for the four rings)**.** The conditional cumulant defining $T_K(f, \chi, h)$ is the sum, over $s = \pm$, of

$$a^3 \operatorname{Tr}(P_f G_{-s} P_\chi G_s P_h G_s), \qquad a^3 \operatorname{Tr}(P_f G_{-s} P_h G_{-s} P_\chi G_s).$$

For T_K^X, replace $a^3 P_\chi$ by $a^2 P_\chi \Delta_+$. The full cumulants are the gauge means of these sums.

Suppose f is supported in Q and write $n = n_Q$, $x = x_Q(h)$, $x^\Delta = x_Q^\Delta(h)$. For $|\xi| \leq 1$ supported in Q, each mass ring is at most $\frac{1}{2}\|f\|_\infty n x$ in modulus. If $0 \leq \chi \leq 1$ is Lipschitz and every neighbour of its support belongs to Q, each Wilson ring is at most

$$\|f\|_\infty \left[n^{1/2} (x^\Delta x)^{1/2} + C_{\mathrm{hop}} a^{1/2} \operatorname{Lip} \chi \{ n (x^\Delta x)^{1/2} + n^{1/2} x \} \right], \qquad C_{\mathrm{hop}} = \sqrt{3}/2.$$

Both assertions hold with f, h exchanged. Finally, if $0 \leq \widetilde{\chi} \leq 1$ vanishes on $\operatorname{supp} h$, and sh consists of the sites at distance at most $3a$ from $\{0 < \widetilde{\chi} < 1\}$, then

$$x_{\widetilde{\chi}}^\Delta(h) \leq C_{\mathrm{en}} \operatorname{Lip} \widetilde{\chi}\, x_{\mathrm{sh}}(h), \qquad C_{\mathrm{en}} = 3.$$

■

Proof. The zero conditional one-point functions noted above remove every proper partition from the third-cumulant formula. In the two cyclic Wick traces, $(i\tau_1)^2 = -1$ cancels its minus sign and τ_1 exchanges the two propagator blocks. The flavour trace is $\operatorname{tr} t^2 = 1$. This gives the four rings, including their signs and powers of a.

For a mass ring insert 1_Q beside P_f, P_ξ and write $P_h = P_{|h|}^{1/2} \operatorname{sgn}(h) P_{|h|}^{1/2}$. Use the operator norm n/a at the propagator between f and ξ, and Hilbert–Schmidt norms at the other two propagators. Twice their product is at most x/a^2. This gives $\|f\|_\infty n x/2$ for either cyclic order.

Since the R_s are self-adjoint involutions, $\Delta_+ = \sum_s Q_s^* Q_s$. Thus $P_\chi \Delta_+ = \Delta_\chi + E_\chi$. Only $s = 0$ moves the physical site, and $t_0 = 3$, so the row $C = ([P_\chi, Q_s^*])_s$ satisfies

$$\|C\| \leq C_{\mathrm{hop}} a \operatorname{Lip} \chi, \qquad \|E_\chi\| \leq \tfrac{3}{2} a \operatorname{Lip} \chi, \qquad \|[P_\chi, \mathcal{B}_U]\| \leq 3a \operatorname{Lip} \chi.$$

Let $q = (P_\chi^{1/2} Q_s)_s$ and $\mathbf{Q} = (Q_s)_s$. The identity $G_{-s} \mathcal{W} G_s = Y$ and $\Delta_\chi \leq \Delta_Q \leq \mathcal{W}$ give

$$\|1_Q G_{-s} q^*\|^2 \leq n/a, \qquad \|(1_Q \otimes 1) \mathbf{Q} G_s 1_Q\|^2 \leq n/a.$$

The squared Hilbert–Schmidt norms with one energy factor are at most x^{Δ}/a; those without it are at most x/a^2. In the part with q^*q, Hölder's inequality gives $a^2(n/a)^{1/2}(x^{\Delta}/a)^{1/2}(x/a^2)^{1/2}$. In the two parts with $C\mathbf{Q}$ it gives, respectively,

$$a^2(n/a)\|C\|(x^{\Delta}/a)^{1/2}(x/a^2)^{1/2}, \qquad a^2(n/a)^{1/2}(x/a^2)\|C\|.$$

These are exactly the three terms in the Wilson bound. Exchanging f, h exchanges the two cyclic orders.

For the energy estimate let $\psi = G_+\phi$, where ϕ is supported on $\operatorname{supp} h$. The real part of $\langle P_{\widetilde{\chi}}\psi, (\mathcal{W} + \mathcal{B}_U)\psi\rangle = 0$ is

$$\langle \psi, (\Delta_{\widetilde{\chi}} + a\mathsf{m}P_{\widetilde{\chi}})\psi\rangle = -\tfrac{1}{2}\langle \psi, [P_{\widetilde{\chi}}, \mathcal{B}_U]\psi\rangle - \operatorname{Re}\langle \psi, E_{\widetilde{\chi}}\psi\rangle.$$

The commutators and the one further hop are supported in sh. Drop the nonnegative mass term and use their norm bounds to obtain $\langle \psi, \Delta_{\widetilde{\chi}}\psi\rangle \le 3a \operatorname{Lip}\widetilde{\chi}\|1_{\mathrm{sh}}\psi\|^2$. The same argument applies to G_-. Sum over $\phi = P_{|h|}^{1/2}e_j$ and multiply by a. □

Lemma 9.9 (A second moment for the joining loop)**.** Suppose $0 \le \varphi_1, \varphi_2 \le 1$ have supports in one ball of radius R', separated by $d_1 > 0$. Then

$$\mathsf{E}_{\nu_K} x_{\varphi_1}(\varphi_2)^2 \le C_{\mathrm{sm}}(g, R', d_1)^2, \qquad C_{\mathrm{sm}} = \frac{4}{\exp(1)\theta_{\mathrm{jl}}} C_{\mathrm{jl}}(R', d_1)^{1/2}.$$

This constant is independent of the side, cutoff and the labels in their fixed intervals. ■

Proof. Use $x_{\varphi_1}(\varphi_2) = 2\Pi_U(\varphi_1, \varphi_2)$ and the $q = 2$ case of Lem. 3.6. □

Proposition 9.10 (The near-support sums)**.** Let f, h be real bounded functions whose supports lie in one ball of radius R and have distance $d > 0$. Set

$$e_0 = \min\{d/64, \sigma_0(g)/8\}, \qquad \epsilon_{\mathrm{nr}}(e) = 8^4(2R+1)^3 C_{\mathrm{S4}} e\ell_g(e)^{p_4},$$

where $\sigma_0, C_{\mathrm{S4}}, p_4, \ell_g$ are the constants and function in Thm. D. There is $C_{\mathrm{nr}} = C_{\mathrm{nr}}(g, R, d)$, independent of the side, cutoff and labels, such that for $a \le e \le e_0$,

$$\sum_y \sup_{\substack{|\xi|\le 1 \\ \operatorname{supp}\xi \subset Q_y}} \big(|T_K(f_y, \xi, h)| + |T_K(f, \xi, h_y)|\big) \le C_{\mathrm{nr}}\epsilon_{\mathrm{nr}}(e)^{1/4}\|f\|_\infty\|h\|_\infty,$$

$$\sum_y \big(|T_K^X(f_y, \chi_y, h)| + |T_K^X(f, \chi_y, h_y)|\big) \le C_{\mathrm{nr}}(1 + a^{1/2}/e)\{\epsilon_{\mathrm{nr}}(e)^{1/8} + \epsilon_{\mathrm{nr}}(e)^{1/4}\}\|f\|_\infty\|h\|_\infty.$$

One may take

$$C_{\mathrm{nr}} = 1458 \max\{4, 12C_{\mathrm{hop}}\} \max\{1, (8C_{\mathrm{en}}/d)^{1/2}\} C_{\mathrm{sm}}(g, 3R, d/4).$$

■

Proof. Consider the pieces of f first and discard zero pieces. Their number is at most $(2R+1)^3 e^{-3}$. Each Q_y has side $8e$, so Thm. D gives

$$\sum_y \mathsf{E}_{\nu_K} n_{Q_y}^4 \le \epsilon_{\mathrm{nr}}(e).$$

For nonnegative random variables Z_y and $0 < p \le 1$, Hölder's inequality on the product of ν_K and counting measure yields

$$\sum_y \mathsf{E}(n_{Q_y}^p Z_y) \le \left(\sum_y \mathsf{E} n_{Q_y}^4\right)^{p/4} \left[\mathsf{E}\left(\sum_y Z_y\right)^2\right]^{1/2}.$$

Indeed the other exponent is $4/(4-p) \le 2$, $\sum_y Z_y^{4/(4-p)} \le (\sum_y Z_y)^{4/(4-p)}$, and monotonicity of probability norms gives the display.

The cubes Q_y lie in the $d/4$-neighbourhood Q' of $\operatorname{supp} f$. Their multiplicity is at most 729, hence

$$\sum_y x_{Q_y}(h) \le 729 x_{Q'}(h), \qquad \sum_y (x^{\Delta}_{Q_y}(h) x_{Q_y}(h))^{1/2} \le 729 (x^{\Delta}_{Q'}(h) x_{Q'}(h))^{1/2}.$$

Take $\widetilde{\chi} = \min\{1, \max\{0, 3 - 8\operatorname{dist}(x, \operatorname{supp} f)/d\}\}$. It equals one on Q', vanishes beyond distance $3d/8$, and has Lipschitz constant $8/d$. Since $a \le d/64$, its shell lies in the $d/2$-neighbourhood Q'' of $\operatorname{supp} f$. The energy inequality gives

$$x^{\Delta}_{Q'}(h) \le (8C_{\mathrm{en}}/d) x_{Q''}(h).$$

Dominate $1_{Q''}$ and $|h|/\|h\|_\infty$ by continuous functions in $[0,1]$, with supports in a ball of radius $3R$ separated by $d/4$. For example use the linear cutoffs $\min\{1, \max\{0, 5 - 8\operatorname{dist}(x, \operatorname{supp} f)/d\}\}$ and $\max\{0, 1 - 8\operatorname{dist}(x, \operatorname{supp} h)/d\}$. Thus Lem. 9.9 bounds $\mathsf{E} x_{Q''}(h)^2$ by $C^2_{\mathrm{sm}} \|h\|^2_\infty$.

Apply the preceding Hölder estimate with $p = 1$ to the mass rings. For the Wilson rings use $p = 1/2$ and $p = 1$, the energy bound and $\operatorname{Lip} \chi_y \le 3/e$. Their four cyclic terms, the multiplicity 729, and the identical estimate with f, h exchanged are bounded by the displayed C_{nr}. This proves both assertions. □

Proposition 9.11 (Recovering the constant insertion)**.** Fix an admitted torus and a subsequence along which all off-diagonal scalar correlations converge to S. For real C^1 functions f, h with disjoint supports, define

$$j_e(S; f, h) = \sum_{y, y'} S_3((\Gamma^1_t, f_y), (\Gamma^1_1, 1 - \chi_y - \chi_{y'}), (\Gamma^0_t, h_{y'})).$$

Then $j_S(f,h) = \lim_{e \downarrow 0} j_e(S; f, h)$ exists, depends only on S, and

$$\begin{aligned} T_K(f, 1, h) &\longrightarrow j_S(f, h), \\ T^X_K(f, 1, h) &\longrightarrow \Xi_S(f, h) - \mathsf{m} j_S(f, h), \\ |j_S - j_e(S)| &\le C_{\mathrm{nr}} \epsilon_{\mathrm{nr}}(e)^{1/4} \|f\|_\infty \|h\|_\infty. \end{aligned}$$

This is also the constant insertion in any analytic family of Thm. M extending S.

If $W_{e,K}$ is the analogous sum with T^J_K in place of S_3, then for $a \le e^2$,

$$|\Xi_K(f,h) + W_{e,K}(f,h)| \le (m_+ + 2) C_{\mathrm{nr}} (\epsilon_{\mathrm{nr}}(e)^{1/8} + \epsilon_{\mathrm{nr}}(e)^{1/4}) \|f\|_\infty \|h\|_\infty.$$

Consequently the iterated limit, first along the cutoff subsequence and then as $e \downarrow 0$, is zero. ■

Proof. For fixed e, the three supports in every nonzero term of j_e are separated by at least e. Indeed $\chi_y = 1$ on the e-neighbourhood of $\operatorname{supp} f_y$; the cutoffs tied to f and h are disjoint because $e \le d/64$. Thus every term has a limit determined by S. Split the constant insertion by $1 = \chi_y + \chi_{y'} + (1 - \chi_y - \chi_{y'})$ and sum over y, y'. The first estimate of Prop. 9.10 bounds the difference between $T_K(f, 1, h)$ and the separated sum uniformly in K. It tends to zero with e. This proves the Cauchy property of j_e, the claimed limit and its error. The insertion identity then gives the Wilson limit. If an analytic family exists, the constant tuple converges by Thm. M; uniqueness of the preceding limit identifies it with j_S.

Apply the equation of motion to the separated middle insertion. The resulting separated mass and Wilson terms cancel the corresponding parts of the exact identity for Ξ_K. Only the two near-support sums remain. If $a \le e^2$, then $1 + a^{1/2}/e \le 2$; Prop. 9.10 gives the last estimate. □

Remark 9.12. The near-support estimates require $a \le e$. Their continuum conclusion takes the cutoff limit before $e \downarrow 0$. They assert neither decay for every $e > 0$ at a fixed lattice spacing nor disappearance of the Wilson insertion: its limit is the displayed $\Xi_S - \mathsf{m} j_S$.

10. Positive masses and continuation

10.1. **Constants independent of a lower mass bound.** The estimates for a step of the flow use $0 < \mathfrak{m}_k \le 1/2$ and $\eta_k \ge c_b = b_F(1-L^{-1})/(b_F(1-L^{-1})+1/2)$. Neither inequality requires a positive lower bound for the physical mass. A lower mass bound enters when the final Berezin integral is estimated relative to its Gaussian part. We record the distinction because the comparison of gauge observables uses the first group of estimates alone.

Definition 10.1 (Constants and weak data)**.** Use the constants of [QCDM1, Prop. 10.20], with the following choices. Increase the flow bound, at its place in the order, so that

$$B_\sharp \ge (1+L^{-1})(c_{2,\mathrm{loop}} + N_f c_2^F) + 2.$$

Use the history mark with dials

$$(\mathsf{w}, c_a, C_{\mathrm{vol}}, c_{\mathrm{Dep}}) = (1/2, 0, 0, c_\star) \quad \text{and share } \eth = 1/2.$$

Here $c_\star$ is the positive rate gained on a created cell at zero loss. The dials lie in the admitted ranges. Choose γ under the original ceilings except

$$\gamma^2 \le \frac{c_b m_-}{8 C_{\mathsf{S}_1} \bar{B}_M}, \qquad \phi(\gamma^{-2}) \le c_*(m_-), \qquad \gamma \le \gamma(2A_\kappa h_*, 5B_s),$$

and except the ceilings formed at the weight $2A_\kappa h_*$ for the last of these. Retain the ceilings at weight h_F and radii $1, 5B_s$. Retain the upper-history and pure-gauge terminal thresholds for the displayed mark and share. Impose in addition

$$(28 C_w \phi(\gamma^{-2}) + 24\bar{B})\gamma^2 \le r_x, \quad C_w \phi(\gamma^{-2}) \le 1,$$
$$4 C_\rho S_3^\sharp \psi_\sharp(\gamma^{-2}) \le 1, \qquad 3\bar{B}_M \gamma \le 1.$$

The function in the third inequality is $\psi_\sharp(x) = x^{-1/2}(1+\log x)^{9\mathfrak{p}_0}$. All constants on the right, and those multiplying these functions, were chosen before the mass interval. Thus these conditions admit a positive γ independently of that interval.

A weak datum is

$$(g, \ell', K', \mathsf{b}', \mathsf{m}'), \qquad 0 < g \le \gamma, \quad \ell' \in \mathfrak{L}(g), \quad \mathsf{b}' \in I_g, \quad 0 < \mathsf{m}' \le 1/2.$$

There is no further lower bound on m'. Its bare mass is $L^{-K'}\mathsf{m}'$, its fine masses are $\mathfrak{m}_k = L^{k-K'}\mathsf{m}'$, and its calibrations are $\mathsf{t}_i = L^{K'-i}\mathsf{b}'$.

For a side $\ell = 2L^m$, $\mathsf{b}_* > 0$ and $\mathsf{m} > 0$, let $\bar{n}$ be the least nonnegative integer such that

$$L^{\bar{n}}\mathsf{b}_* \ge \gamma^{-2} + \bar{B}, \qquad L^{-\bar{n}}\mathsf{m} \le 1/2, \qquad m + \bar{n} \ge m_{\bar{g}}, \quad \bar{g} = (L^{\bar{n}}\mathsf{b}_* - \bar{B})^{-1/2}.$$

For $K > \bar{n}$ the relabelled datum is

$$(\bar{g}, L^{\bar{n}}\ell, \bar{K}, \bar{\mathsf{b}}, \bar{\mathsf{m}}) = (\bar{g}, L^{\bar{n}}\ell, K - \bar{n}, L^{\bar{n}}\mathsf{b}_*, L^{-\bar{n}}\mathsf{m}).$$

■

Lemma 10.2 (The estimates available at a weak datum)**.** At every weak datum the zero-source small-field induction, the step estimates at weight h_F and radii $1, 5B_s$, the upper bounds on the forward mass correction, the second-order expansion with its cubic remainder, and the pointed activity estimates hold with constants independent of a lower mass bound. So do the history estimates, their upper partition bound, the unintegrated lower measure bound, the exact determinant identities, and the step-energy estimate

$$|e_j| \le C_e'(1 + \log x_{j-1}).$$

This assertion does not include a lower partition bound with a constant independent of m'.

At every $1 \le k \le K'$ and real $x \ge \max\{\gamma^{-2}, 3\mathsf{t}_k/4\}$, the small-field weight with coefficient $L^k x$ satisfies

$$\int \mathsf{sf}_k[x]\, dV_k \ge \exp\left[-\left(\tfrac{9}{2}\log x + 3\log 7 + 24(B_3^{\mathrm{cr}})^2\right) n_k\right].$$

The coefficient x_k of the flow is in this range.

The integer $\bar{n}$ of Def. 10.1 exists. Its relabelled datum is weak and has the same bare integral, lattices, calibrations, tests and history terms through level $\bar{K}$ as $(K, \ell, \mathsf{b}_*, \mathsf{m})$. For $\bar{n} \le d < K$, the block field $\mathcal{M}^{K-d}U$ is the level $K-d$ field in both descriptions. ■

Proof. Enlarge the auxiliary mass interval to

$$[m'_-, m'_+] = [\min\{m_-, \mathsf{m}'\}, \max\{m_+, \mathsf{m}'\}].$$

Keep all earlier constants, the radii and γ unchanged, and recompute numbers defined from this interval. The weak datum is a datum for these parameters. We check which conditions can fail and which estimates use them.

The constants before the mass interval are unchanged. The enlarged interval still satisfies $0 < m'_- \le m'_+ \le 1/2$. Of the later conditions only the three omitted ceilings and $2\varrho_J \le c_J m'_-$ can fail. The latter is used to keep the real part of the fine source mass above half the bare mass. It has no role at zero sources. The first omitted ceiling bounds the final mass correction by a fixed fraction of m_-, and bounds $\log(1/\mathfrak{m}_k)$ by $\log x_k$ with a constant fixed from the original interval. All three are used to expand the final Berezin integral at the enlarged weight $2A_\kappa h_*$. That final expansion is the only use of this enlarged weight in the estimates listed here.

We now check the estimates that remain. The background covariance, averaging and reconstruction bounds of [QCDM1, §§5–6] are valid for every $0 < m_0 \le L^{-K'}/2$; their constants depend on the chamber operator, L, b_F and the chosen geometric bounds. The local fermion expansion and the step construction in [QCDM1, §7] are applied at weight h_F. Their Grassmann contractions use the same Gram bounds, and their mass extraction uses $\eta_k \ge c_b$. The retained ceilings $\gamma \le \gamma(h_F, 1)$ and $\gamma \le \gamma(h_F, 5B_s)$ close these steps. No inverse of the physical mass enters these bounds.

The explicit second-order terms, their moment extraction, the identity for their total, and the cubic remainder in [QCDM1, §8] use fine masses in $[0, 1]$ and $\eta_k \ge c_b$. Their constants precede the mass interval. The geometric sum of the cubic remainders and the telescoping one-loop increments give the forward window $|s_k| \le \bar{B}_M/x_k$ under these same ceilings. Only the further comparison of this window with m_- would use the first omitted ceiling.

The pointed estimate in [QCDM1, §9] takes as inputs this step, the window $|s_k| \le 2\bar{B}_M/\mathsf{t}_k$, and the bound $\|E^{(i)}_{\ge 2}\|_i \le C_2\phi(\mathsf{t}_{i-1})^2$. Its pointwise constant is $8M^3C_2$. The leg bounds, cluster counts and mass extraction used in its proof are all at weight h_F; its only mass denominator is η_k, already bounded below by c_b. It therefore applies at the weak datum. This verifies in particular the pointed estimate needed in the history sum; the unpointed estimate alone would not suffice there.

The history construction, its scalar remainders and its upper bound in [QCDM1, §9] use the preceding bounds and the fixed history mark. The constants $C_5^F, C_E^F, C_\flat^F, C_{\mathrm{fs}}, C'_e$ are formed at weight h_F. The upper-bound threshold for the fixed mark is retained. The exact Gaussian determinant identity and the degree-zero part of the history term hold at every positive bare mass. They give the lower bound as measures before it is weakened to a scalar lower partition bound. Positivity of history weights and the displayed step-energy estimate also hold at every positive bare mass. Thus all the assertions made in the statement have the same constants after enlarging the interval.

The estimates deliberately excluded from this argument are the final Gaussian lower estimate, the constants containing $\log(1/m_-)$, the coarse-cutoff moment bounds using $\|(D+m_0)^{-1}\| \le 1/m_0$, and the charged decay estimates whose rates depend on m_-. When any of them is used, its mass-dependent constants must be recomputed. Deleting the matter terms gives the same zero-source

construction without fermions: the relevant conditions on N_f are floors on constants free to increase, so the already chosen constants suffice.

For the small-field integral set $s = x^{-1/2}$. The retained geometric ceilings give

$$s \le \gamma \le \bar{\varepsilon}/4, \qquad s \le 2\mathsf{t}_k^{-1/2} \le \frac{A_0(\log \mathsf{t}_k)^{\mathfrak{p}_0}\mathsf{t}_k^{-1/2}}{4B_3^{\mathrm{cr}}}.$$

The ball of radius s about the trivial block field passes the tests. Its measure is at least $(s^3/7)^{3n_k}$, and the action in the weight is at most $24(B_3^{\mathrm{cr}})^2 s^2 x n_k = 24(B_3^{\mathrm{cr}})^2 n_k$. Multiplying these two bounds gives the stated integral. For $x = x_k$, the flow gives $x_k \ge \mathsf{t}_k - \bar{B} \ge \gamma^{-2}$ and $\bar{B} \le \mathsf{t}_k/4$, so it applies.

Finally the first two conditions defining $\bar{n}$ hold for all sufficiently large integers. In the third condition, $m_{(L^n\mathsf{b}_* - \bar{B})^{-1/2}}$ grows at most like $\log n$: the only growing term is $\log_L R(L^{n+1}\mathsf{b}_*)$, and $R(x) \le L(\log x)^{r_c}$. The left side $m+n$ grows linearly, so $\bar{n}$ exists. Its first condition gives $\bar{g} \le \gamma$ and $\bar{\mathsf{b}} = \bar{g}^{-2} + \bar{B} \in I_{\bar{g}}$; the others give the admitted side and $0 < \bar{\mathsf{m}} \le 1/2$. The identities

$$\ell L^K = (L^{\bar{n}}\ell)L^{\bar{K}}, \qquad L^K \mathsf{b}_* = L^{\bar{K}}\bar{\mathsf{b}},$$
$$L^{-K}\mathsf{m} = L^{-\bar{K}}\bar{\mathsf{m}}, \qquad L^{K-k}\mathsf{b}_* = L^{\bar{K}-k}\bar{\mathsf{b}}$$

identify the bare integral and every intermediate lattice and calibration through level $\bar{K}$. Tests, cubes, history weights and block maps are formed from these data; hence they agree. In particular $n_k = \ell^3 L^{3(K-k)}$, with $n_{\bar{K}} = L^{3\bar{n}}\ell^3$. □

10.2. **Every positive mass label.** We can choose the upper bound for g before choosing a compact positive mass interval. The remaining constants need not stay bounded as the lower endpoint tends to zero. The argument replaces the final Gaussian lower estimate by a determinant inequality valid at every positive bare mass.

Definition 10.3 (Constants for all positive masses)**.** Choose the constants through h_F, the action-source radius ϱ, and

$$0 < c_J \le \min\{1/2, \varrho_*, r_M/L\}$$

in their original order. Choose $\gamma \in (0, 1/2]$ under all the original step, history and source ceilings except the three lower-mass ceilings displayed in Def. 10.1 and their enlarged-weight conditions. Retain the ceilings at weight h_F and radii $1, 5B_s$. Use the original admissible history mark, with its upper history threshold. After these choices take any

$$0 < m_- \le m_+ \le 1/2, \qquad 0 < 2\varrho_J \le c_J m_-.$$

Finally, with $\nu = 8N_f$, define

$$C_P^{(K)} = \max\left\{\tfrac{9}{2}, C_5^F + 3\log 7 + 24(B_3^{\mathrm{cr}})^2 + 1 + \nu \log\frac{2}{c_b m_-}\right\},$$
$$C_P = C_P^{(K)} + C_e', \qquad C_m^w = \log\left(\frac{1}{b_F(1-L^{-1})} + \frac{1+\bar{B}}{4m_-}\right).$$

Form all later insertion and compactness constants from these values. In particular $c_V = C + 2 + C_P(1 + \log X_g)$. The data and admitted sides are otherwise unchanged. ■

Lemma 10.4 (Order of the choices)**.** The choices in Def. 10.3 exist, with a positive upper bound for γ independent of $[m_-, m_+]$. All original hypotheses hold except the three omitted ceilings and the conditions at their enlarged weight. ■

Proof. The dependency check in Lem. 10.2 identifies all uses of the lower mass bound in the construction. The source estimates add $2\varrho_J \le c_J m_-$; we impose it after choosing the interval. Their other radii are ϱ, ϱ_* and r_M, and the domain condition $L^{k+1-K}\|J\|_\infty < r_M$. The source-chain constants and ceilings are formed from these radii and the constants through h_F. The quantities

formed later from ϱ_J, C_P and the mass interval are constants in the resulting estimates; they impose no new ceiling on γ.

The original history mark changes only its fixed upper threshold in the proof of Lem. 10.2. This threshold also precedes the mass interval. Each retained ceiling has a positive right side independent of that interval and either bounds γ directly or bounds a function tending to zero with γ. Their finite conjunction therefore holds on a nonempty interval $(0, \gamma_w)$. The interval and ϱ_J can then be chosen as stated. Finally C_P, C_m^w and every later constant are explicit functions of earlier choices, so introduce no circularity. □

Proposition 10.5 (Positivity of the extracted mass)**.** At every positive-mass datum and every level k of the forward flow, $s_k \geq 0$. ■

Proof. The assertion is immediate for $k = 0$. On the history with no large-field region, put

$$v(U) = \mathfrak{w}_{h_\emptyset}(U) e^{-\beta A(U)} \prod_{j<k} \theta^{-\nu n_{j+1}} \det \Gamma_j[U].$$

The block tower has the same adjoint pair of internal blocks as $D[U] + m_0$. Indeed the real averaging maps commute with the internal projections, and

$$\Gamma_j = \mathbb{D}_j + \theta q^* q, \qquad \mathbb{D}_{j+1} = \theta - \theta^2 q \Gamma_j^{-1} q^*$$

preserve that form. Positive accretivity at $m_0 > 0$ makes these matrices invertible. Each $\det \Gamma_j$ is therefore the squared modulus of a nonzero determinant, so $v \geq 0$.

Completing the Gaussian square at each level gives exactly

$$\check{\tau}^F_{h_\emptyset} = \mathcal{M}_*^k \left(v(U) e^{-\langle \bar{\Psi}, \mathbb{D}_k[U] \Psi \rangle} dU \right).$$

Let $\mathfrak{K}_\ell(V)$ be the degree-two kernel of minus the exponent in its RG representation, and let $\mathfrak{Z} = \mathcal{M}_*^k(v\, dU)$. Comparing degrees zero and two gives the identity of matrix-valued measures

$$\mathcal{M}_*^k(\mathbb{D}_k[U] v(U)\, dU) = \mathfrak{K}_\ell(V) \mathfrak{Z}(dV).$$

Near $V = 1$ the small-field tests pass, the degree-zero density is continuous and nonzero, and $\mathfrak{K}_\ell$ is continuous. Thus $\mathfrak{Z}$ charges every sufficiently small ball B_ϵ about 1, and

$$\mathfrak{K}_\ell(1) = \lim_{\epsilon \downarrow 0} \frac{\int_{\{\mathcal{M}^k U \in B_\epsilon\}} \mathbb{D}_k[U] v(U)\, dU}{\int_{\{\mathcal{M}^k U \in B_\epsilon\}} v(U)\, dU}.$$

The three-square identity of [QCDM1, §8] gives

$$\operatorname{Re} \mathbb{D}_k[U] \geq \frac{\theta_k \mathfrak{m}_k}{\theta_k + \mathfrak{m}_k} 1$$

at every U. A conditional mean with nonnegative weight preserves this inequality, so it holds for $\operatorname{Re} \mathfrak{K}_\ell(1)$.

At the trivial field the free block kernels and background maps are periodizations of exponentially decaying kernels. Activities on polymers that fit inside the torus agree with their infinite-lattice versions. The remaining polymers have diameter at least the side divided by $2M$, minus two, so the polymer decay bounds their contribution to any fixed kernel entry by a quantity tending to zero. It follows that $\mathfrak{K}_\ell(1)$ converges entrywise to an exponentially decaying translation-invariant kernel $\mathfrak{K}_\infty$. The lower quadratic-form bound passes first to finite-support vectors and then to its Fourier symbol at every momentum.

At zero momentum that symbol is the zeroth moment. The mass extraction subtracts the zeroth moment of each activity; their subsequent Gaussian readings preserve its vanishing, by the exact identity used in the forward flow. Consequently the zeroth moment is

$$\frac{\theta_k \mathfrak{m}_k}{\theta_k + \mathfrak{m}_k} + \eta_k^2 s_k.$$

It is real by conjugation. The lower bound therefore gives $\eta_k^2 s_k \geq 0$, and $\eta_k > 0$ proves the claim. □

Lemma 10.6 (The lower partition bound at positive mass)**.** Under Def. 10.3, the forward flow and the upper history estimates retain their original bounds. In particular

$$|s_K| \leq \bar{B}_M / x_K \leq \bar{B}_M g^2, \qquad 0 \leq s_K \leq (2B_F + 1/2)/\mathsf{b}.$$

At every level $1 \leq k \leq K$,

$$Z_K \geq \exp[E_k - C_P(1 + \log x_k)n_k], \qquad \nu \log(1/\theta_k + 1/\mathfrak{m}_k) \leq \nu(\log x_k + C_m^w).$$

■

Proof. The step and mass-flow bounds follow from Lem. 10.2, whose dependency proof applies to the original mark as explained in Lem. 10.4. The induction still gives

$$\left| s_K - \frac{F(K;\mathsf{m})}{\mathsf{b}} \right| \leq C_s \mathsf{b}^{-3/2}(1 + \log \mathsf{b})^{9\mathfrak{p}_0}.$$

The retained cubic-remainder ceiling and the calibration comparison give the stated upper bound $(2B_F + 1/2)/\mathsf{b}$, as in the proof of the forward flow. Positivity is Prop. 10.5.

The determinant tower gives the degree-zero measure of the all-small history as the image of

$$\mathfrak{w}_{h_\emptyset} e^{-\beta A} \frac{\det(D + m_0)}{\det \mathbb{D}_k}\, dU.$$

The exact determinant factorization and accretivity give the lower measure bound with factor $(1/\theta_k + 1/\mathfrak{m}_k)^{-\nu n_k}$. The remaining scalar activity in this term has modulus at most $C_5^F n_k$. At $k = K$,

$$1/\theta_K + 1/\mathsf{m} \leq 1/(c_b \mathsf{m}) \leq 1/(c_b m_-).$$

Integrating the measure bound and using the ball estimate of Lem. 10.2 therefore gives

$$t^F(h_\emptyset^{(K)}) \geq \exp\left[E_K - \left(C_5^F + \nu \log \frac{1}{c_b m_-} + \tfrac{9}{2} \log x_K + 3 \log 7 + 24 (B_3^{\mathrm{cr}})^2 \right) \ell^3 \right].$$

The definition of $C_P^{(K)}$ bounds this below by $e^{E_K - C_P^{(K)}(1+\log x_K)\ell^3}$. All history terms are nonnegative before the comparison, so $Z_K \geq t^F(h_\emptyset^{(k)}) \geq t^F(h_\emptyset^{(K)})$. The energy increments give

$$E_K \geq E_k - C_e'(1 + \log x_k)n_k.$$

Since $n_K \leq n_k$ and $\log x_K \leq \log x_k$, this proves the lower partition bound at every k.

Finally

$$\frac{1}{\theta_k} \leq \frac{x_k}{b_F(1 - L^{-1})}, \qquad \frac{1}{\mathfrak{m}_k} = \frac{\mathsf{t}_k}{\mathsf{bm}} \leq \frac{(1 + \bar{B})x_k}{4m_-}.$$

Here $x_k > 1$, $\mathsf{t}_k \leq x_k + \bar{B}$, and $\mathsf{b} \geq \gamma^{-2} \geq 4$. Taking the logarithm gives the last inequality. □

Proposition 10.7 (Source bounds after the change of constants)**.** The source construction, the intermediate-level generating function bounds, their base-source estimates, and the separated-cumulant estimates hold under Def. 10.3. Use the same source domains, the new C_P in the lower partition bound, and C_m^w in the determinant estimate. ■

Proof. The source-chain steps use $\eta_k \geq c_b$, $L^{k+1-K}\|J\|_\infty < r_M$, and the constants through h_F. On the physical source ball the new choice $2\varrho_J \leq c_J m_-$ still gives a real part at least half the bare mass. At the terminal level their Berezin integral is bounded by the Grassmann norm at h_F. The enlarged-weight Gaussian lower expansion is not used.

The three omitted ceilings entered these source estimates only through the hypotheses of the common partition bound. Its upper bound is retained by Lem. 10.2; its lower bound at every intermediate level is Lem. 10.6. The exact history identities and the lower bound as measures are unchanged. The further estimate of its determinant factor uses the last inequality of Lem. 10.6,

with C_m^w. Thus the source-chain, history and Cauchy arguments in [QCDM1, §10] apply with the stated constants. □

Theorem N (Construction at every positive mass)**.** For the constants of Def. 10.3, the construction of the original scalar family holds on every compact interval $0 < m_- \le \mathsf{m} \le m_+ \le 1/2$ at the same upper bound γ for the coupling. This includes fixed-torus and growing-torus subsequential limits, and the construction which first takes the infinite-volume limit at fixed cutoff. All symmetry, reflection-positivity, reconstruction and leading short-distance conclusions for that family remain valid. The forward mass estimate is

$$0 \le s_K \le (2B_F + 1/2)/\mathsf{b}, \qquad |s_K| \le \bar{B}_M g^2,$$

in addition to the one-loop approximation in Lem. 10.6. The estimates need not be uniform as $m_- \downarrow 0$. ■□

Proof. We check the inputs of the construction rather than requiring the mass-dependent smallness ceilings again. At fixed cutoff the inverse and relative-determinant arguments require only $m_0 > 0$; they therefore construct the same torus-limit gauge states. Their constants may depend on the chosen compact positive interval. The local fermion and flow estimates are supplied by Lem. 10.2; the common upper and lower partition bounds and all source estimates by Lem. 10.6 and Prop. 10.7. The volume-independent cumulant bound has the same form, with

$$B(R) = \max\left\{\frac{C_B e^{c_V}(1+R)^3}{\min\{\varrho, \varrho_J\}}, C_B'(1 + N_f/m_-)(1+R)^3\right\}.$$

The second term is the direct estimate at the finitely many coarse cutoffs; retaining its m_- dependence is essential.

Compactness, centered moments, the reflection inequalities, shrinking-plate bounds and reconstruction then use the same bounds as in [QCDM1, §§12–13]. The leading two-point proof uses the same source energy and free propagator, with constants C_P, ϱ_J, m_- and the corresponding image estimate. Those constants and the resulting short-distance threshold are formed after g; they impose no ceiling on its already chosen upper bound. This verifies nonvanishing and identification of the first label as well. The fixed-cutoff infinite-volume construction and its subsequent continuum extraction use the same local bounds, so the volume-first theorem follows at each of the compact positive intervals. □

Proposition 10.8 (Reaching arbitrary positive labels)**.** Fix the constants through γ in Def. 10.3. Every pair $\mathsf{b} > 0$, $\mathsf{m} > 0$ is reached by scaling from data covered by Thm. N, with these same constants. The chosen mass interval, source radius and the subsequent estimate constants may depend on the pair. ■

Proof. Choose an integer z so large that $\mathsf{b}' = L^z\mathsf{b} > \gamma^{-2} + 4B_\sharp$ and $0 < \mathsf{m}' = L^{-z}\mathsf{m} \le 1/2$. Set $g = (\mathsf{b}' - 3B_\sharp)^{-1/2} < \gamma$. Then $\mathsf{b}' \in I_g$. Now choose $0 < m_- < \mathsf{m}'$, $m_+ = 1/2$, and $0 < 2\varrho_J \le c_J m_-$. These are choices after γ, so Thm. N applies. The identity of bare lattice integrals and the dilation of the test functions are those proved in Prop. 11.28. For fixed tori use the corresponding scaled admitted sides; for infinite-volume limits choose a cofinal sequence of such sides. Scaling preserves bm. □

As $m_- \downarrow 0$ the denominator constants still grow. For instance, with $\varrho_J = c_J m_-/2$,

$$C_P = C_P^0 + \nu \log(1/m_-), \qquad e^{c_V} = e^{C+2+C_P^0(1+\log X_g)} m_-^{-\nu(1+\log X_g)}.$$

Thus the construction for each positive mass does not give uniform estimates at mass zero. The signed-mass continuation uses the different source argument of Thm. P.

10.3. **The mass at scale one.**

Lemma 10.9 (Strict positivity of the zero-mass coefficient)**.** For the chamber operator, $F_\infty > 0$. More precisely, $\sigma_{k-1}(k;0)$ has a positive limit $\bar{s}_1$ and $F_\infty \geq \bar{s}_1$. ■

Proof. In the notation of Prop. B.2, $\sigma_{k-1}(k;0) = F(k;0) - F_{<k-1}(k;0)$ tends to $F^{[0]} - F^{[1]}$. We prove that its lower limit is strictly positive. Work in level-k units and a nonzero Bloch fibre ξ. The curvature covariance of the last slice is the orthogonal projection onto $\mathsf{Fl}_k \ominus \mathsf{Fl}_{k-1}$. The first square in the nonnegative mass form is

$$\theta_k^{-1} \langle |D_k X V_A \chi_c|^2 \rangle_c, \qquad X = RD[1]^{-1}.$$

Gauge covariance gives $XV_{d\lambda}\chi_c = i(Q_k'\lambda)c$. Its component along $\gamma_1\gamma_2$ is therefore gauge invariant. Denote this linear functional of $F = dA$ by $\Lambda(F)$. Since $\|D_k^{-1}\| \leq \theta_k^{-1} + c_1/|\xi|$, the first square bounds a positive constant times

$$\theta_k^{-1}(\theta_k^{-1} + c_1/|\xi|)^{-2} |\Lambda(F)|^2.$$

For a Gaussian with projection covariance P and $F_0 \in \operatorname{ran} P$, $\mathbb{E}|\Lambda(F)|^2 \geq |\Lambda(F_0)|^2/\|F_0\|^2$. We construct such a test curvature.

Let Z be divergence free at level $k-1$ and put

$$B = (dd^* + d^*d)^{-1} (Q_{k-1}^s)^* Z.$$

The commuting identity for Q_{k-1}^s gives $d^*B = 0$ and $d^*dB = (Q_{k-1}^s)^* Z$. Thus dB is orthogonal to Fl_{k-1}, and its energy is $\langle Z, Q_{k-1}^s B \rangle$. It belongs to Fl_k precisely when $Q_k^s B$ has zero curl.

Near $\xi_* = (\pi/2, 0, 0)$ take Z on the two momenta $\xi_a' = \xi$, $\xi_b' = \xi - 2\pi e_1$, with amplitudes $\varpi_a, \varpi_b \in \mathbb{C}^3$ in one colour. There are four linear constraints: the two divergences and two independent components of the coarse curl. On their nullspace the energy is positive definite. The coefficients of the constraints, the energy and Λ are sums of the straight-average, Laplacian, block-average, vertex and propagator symbols. The summable majorants in the proof of Prop. B.2 hold uniformly on a ball about ξ_*. Termwise, the averages tend to their sinc symbols, the light propagator to $(i\gamma \cdot q)^{-1}$, the heavy components to zero and the light vertex to $i\gamma \cdot A$. Hence these finite matrices converge uniformly there. Where the constraint has rank four, the largest quotient $|\Lambda(dB)|^2/\|dB\|^2$ also converges continuously.

At ξ_* choose $\varpi_\alpha = u_\alpha e_2$. The divergences vanish. Only momenta $q = (\xi_\alpha' + 2\pi L l, 0, 0)$ survive the transverse Dirichlet factors. Put $x_\alpha = \xi_\alpha'/(2L)$. In the limit the remaining curl condition and the functional are

$$G_a u_a + G_b u_b = 0, \qquad \Lambda(dB) = H_a u_a + H_b u_b,$$

where

$$G_\alpha = 4 \sin(\xi_\alpha'/2) \sin x_\alpha \sum_{l \in \mathbb{Z}} (\xi_\alpha' + 2\pi L l)^{-4}, \quad H_\alpha = 4 \sin(\xi_\alpha'/2) \sin x_\alpha \sum_{l \in \mathbb{Z}} (\xi_\alpha' + 2\pi L l)^{-5}.$$

Differentiation of $\sum_l (x + \pi l)^{-2} = \csc^2 x$ gives

$$\frac{H_\alpha}{G_\alpha} = \frac{h(x_\alpha)}{2L}, \qquad h(x) = \cot x \frac{2 + \cos^2 x}{1 + 2\cos^2 x}.$$

Take $u_a = G_b$, $u_b = -G_a$. Then

$$\Lambda(dB) = \frac{G_a G_b}{2L} (h(\pi/(4L)) - h(-3\pi/(4L))) \neq 0,$$

since $G_a G_b \neq 0$ and h is odd and positive on $(0, \pi/2)$. The four constraints have rank four: the divergences constrain the two first components, and the other curl equation acts separately on the third components. Thus the limiting quotient is positive on a ball about ξ_*. Uniform convergence makes the finite-k quotient bounded below on a smaller ball for all sufficiently large k. Integration

over this ball gives a positive lower limit for the first square and hence for the slice. All slices are nonnegative, so $F_\infty \geq \bar{s}_1 > 0$. □

Proposition 10.10 (Prescribing the mass at scale one)**.** Use the constants of Def. 10.3 and write $A = 2B_F + 1/2$. At every admitted datum,

$$m \leq m + s_K \leq m + A/b, \qquad |bs_K - F(K;m)| \leq C_s b^{-1/2}(1+\log b)^{9p_0}.$$

Moreover,

$$b(m+s_K) \geq \max\{bm, bm + F_\infty - B_5(m + L^{-K/4}) - C_s b^{-1/2}(1+\log b)^{9p_0}\}.$$

At fixed K, b, every $m' \in [m_- + A/b, m_+]$ is $m + s_K$ for at least one $m \in [m_-, m_+]$, with $0 \leq m' - m \leq A/b$. For any such choice,

$$\left| m - \left(m' - \frac{F(K;m')}{b} \right) \right| \leq B_5 A b^{-2} + C_s b^{-3/2}(1+\log b)^{9p_0}.$$

The continuum compactness and correlation conclusions of Thm. N apply to these chosen labels. ■

Proof. The first two bounds are the forward mass estimates of Lem. 10.6 and Thm. N. By Prop. B.2, $|F(K;m) - F(K;0)| \leq B_5 m$ and $|F(K;0) - F_\infty| \leq B_5 L^{-K/4}$. Combine them with the first display and Lem. 10.9 for the lower bound.

The mass flow is real analytic in m on the fixed interval. Its value $m + s_K$ at the left endpoint is at most $m_- + A/b$ and at the right endpoint is at least m_+. The intermediate value theorem therefore supplies the chosen label. Finally $|F(K;m) - F(K;m')| \leq B_5|m - m'|$ by Prop. B.2. Use $|m - m'| \leq A/b$ and the one-loop remainder to obtain the last inequality. The labels remain in a compact positive interval; their convergent subsequences and the simultaneous correlation subsequences are those of Thm. N. □

Remark 10.11. With γ fixed, the estimates are not uniform as $m_- \downarrow 0$. Choosing $2\varrho_J = c_J m_-$ gives $C_P = C_P^0 + \nu \log(1/m_-)$ and

$$e^{c_V(g)} = e^{C+2+C_P^0(1+\log X_g)} m_-^{-\nu(1+\log X_g)}.$$

The source radius also tends to zero. These are consequences of the positive-mass partition bound; they do not require γ to decrease with m_-.

At fixed real coupling and fixed volume, one subsequence gives meromorphic continuations of all separated correlations in the mass, with a common discrete exceptional set. We prove that this set misses a neighbourhood of a longer positive real mass interval. Its width is at least a constant times $[\ell^3(1+\log \ell)]^{-1}$; see Thm. O and Prop. 10.16. The proof first makes the vacuum normalization independent of the real mass used to define it. Reflection into small cells then gives a differential inequality on the real axis, and a harmonic majorant extends its control to complex masses. The final two propositions retain the short-distance kernels and the volume-independent real-mass cumulant bounds. The continuum tests in this section are C^1 at the action index and continuous at the scalar-bilinear indices; the finite-cutoff estimates also apply to Lipschitz tests.

10.4. **A common normalization as the mass varies.** We now fix the real coupling $b \in I_g$ and continue in the mass alone. Write $I = [m_-, m_+]$ and

$$U_r = \{z \in \mathbb{C} : \operatorname{dist}(z, I) < r\}, \qquad r_J = r_M/L.$$

The source radii satisfy

$$2\varrho_J \leq \tfrac{1}{2} r_J, \qquad 2\varrho_J \leq \tfrac{1}{2} m_-.$$

The torus, g, and b remain fixed in every complex continuation below. All cutoff limits are subsequential.

For a real centre $x \in I$, denote its source energy by $E_K^{(x)}(s)$. Put

$$q_z^{(x)} = (0, -(z-x)1), \qquad Z_K(z;t) = Z_K^{(x)}[q_z^{(x)} + \textstyle\sum_i t_i s_i].$$

Here s_i is the source of the entry (α_i, f_i). The right side is independent of the centre: in its determinant the mass matrix is

$$ax - a(J - (z-x)1) = az - aJ.$$

Thus $Z_K(z;t)$ is the partition function at bare mass az, with the indicated insertions. It is entire in z, t. Write

$$W_K^{(x)}[s] = e^{-E_K^{(x)}(s)} Z_K^{(x)}[s].$$

This is the normalized partition function denoted by W_K above. When a centre is fixed, we suppress its superscript. The energies at two centres need a separate comparison.

Choose $n_\pi \geq 1$ with $(1 + 1/n_\pi)^{p_0} \leq 4/3$, and put

$$\pi_j = (1 + (K-j)/n_\pi)^{p_0}, \qquad \|w\|_{\sharp,j} = \max\{\|w\|_\infty, 40M\lambda_j\pi_j \operatorname{Lip} w\}.$$

Here p_0 is the exponent already fixed in section 4; no new RG weight is chosen. We use the source construction of [QCDM1] on

$$\mathscr{O}_j = \{(w,J) : \|w\|_{\sharp,j} < 4\varrho,\ \lambda_j\|J\|_\infty < r_J\}, \qquad \lambda_j = L^{j-K}.$$

In particular its local energies are holomorphic on $\mathscr{O}_{j-1}$, with the size and tail bounds recalled above. The history bound applies on

$$\mathcal{P}_j^{\#} = \{(w,J) : \|w\|_\sharp < 2\varrho,\ \lambda_j\|J\|_\infty \leq r_J/2\}.$$

We will also use the almost-additivity estimate at a base source and its bilinear improvement from [QCDM1, Props. 10.15 and 10.18]. For bilinear sources $s_0 = (0, J_0)$, $s_i = (0, J_i)$ with

$$\|J_0\|_\infty \leq r_J/2, \qquad \|J_i\|_\infty \leq \epsilon \leq r_J/8,$$

whose supports have mutual distance at least δ and lie in balls of radii R_i, that improvement is

$$\left| E_K\left(s_0 + \sum_i s_i\right) - \sum_i E_K(s_0 + s_i) + (N-1)E_K(s_0) \right| \leq C_\flat \sum_i \left(1 + (R_i/\delta)^3\right). \tag{10.i}$$

The constant $C_\flat$ is independent of the cutoff and side. At the fine scales this bound uses the separation exponential. At the coarse scales Schwarz's lemma in the bilinear source adds a factor $\lambda_{j-1}/4$. This makes the sum independent of the number of coarse scales. That improvement is needed when the cells below become small.

Proposition 10.12 (Compatibility of the vacuum energies)**.** Let $x, x' \in I$, $\zeta = x' - x$, and $s_\zeta = (w, J - \zeta 1)$ for $s = (w, J)$. For $1 \leq k \leq K$, if $s, s_\zeta \in \mathscr{O}_k$, then

$$E_k^{(x)}(s_\zeta) + \mathfrak{f}_k^{(x)} = E_k^{(x')}(s) + \mathfrak{f}_k^{(x')}.$$

Here $\mathfrak{f}_k^{(x)}$ is the real finite-size constant in the empty-history scalar normalization of [QCDM1, Lem. 9.15]; $|\mathfrak{f}_k^{(x)}| \leq C_{\mathrm{fs}}$. ∎

Proof. The two bare densities agree by the mass-source identity. Their block integration also agrees before it is expressed in terms of extracted coefficients. Indeed the fermion block kernel reads only b_F and the block averages, while the gauge chart integral and the density test read the calibrations $\mathfrak{t}_j = L^{K-j}b$. None of these depends on the mass centre or on a coefficient extracted by the flow.

More explicitly, if d_j is the density of the empty-history term at level j, the next density is

$$d_{j+1}(W) = \chi_{j+1}(W)\, \mathsf{I}_j[\mathrm{Ber}_j(\mathrm{Ker}_j\, d_j)]\,(W).$$

Here χ_{j+1} is the density-test indicator, I_j integrates the gauge chart, Ber_j integrates the old Grassmann variables, and Ker_j is the fermion block kernel. The density is zero when W fails the

test. This is the exact tested chart and block-integration identity of [QCDM1, Lem. 7.1 and Prop. 10.1]. The two inputs have $d_0 = d_0'$; linearity and the identical kernel, test and chart give $d_j = d_j'$ at every subsequent level.

We compare the scalar coefficients at the trivial block field. In the extracted form of the empty-history density all fermion quadratic terms and activities with legs have positive Grassmann degree. Thus its degree-zero coefficient, at a tested block field V, is

$$g_0(V) = \exp\left[E_k^{(x)}(s_\zeta) - A_k(U; s_\zeta) + \sum_{j\le k}\sum_X E_0^{(j)}(X; \mathbf{p}_X^{(j)}(U); s_\zeta)\right], \tag{10.ii}$$

where U is any background in the minimal orbit over V. Gauge invariance makes this expression independent of that choice. The analogous formula defines g_0'.

The tested block fields form an open set containing 1. On a neighbourhood of 1, the canonical critical-background theorem of [YM3, Thm. B] provides a continuous representative $V \mapsto U^\circ(V)$ of the minimal orbit. To apply it, write $V = e^{iB_V}$ bond by bond using the principal logarithm. Write r_{B} for the positive logarithmic row radius in that theorem. For $\|V-1\| < \min\{1, 2r_{\mathrm{B}}/\pi\}$, $\|B_V\| \le (\pi/2)\|V-1\| < r_{\mathrm{B}}$. The logarithm is continuous and the representative is holomorphic in B_V on this ball. Their composition has the required continuity. The geometric hypotheses are those already verified in the tested-block-field lemma of [QCDM1]. Its restricted pairs depend continuously on V; the scalar activities are holomorphic on an open pair domain containing them. The action and the currents entering those pairs are continuous functions of the link matrices. There are only finitely many polymers at fixed cutoff and volume. Therefore eq. 10.ii is continuous near 1, and so is g_0'.

The exact density identity gives equal integrals of g_0 and g_0' over every Borel subset of the block-field space. The reference measure there is Haar measure and charges every nonempty open set. Two continuous densities agreeing almost everywhere must consequently agree throughout this neighbourhood. At $V = 1$ we may take $U = 1$. The action vanishes, and the scalar normalization in the source step gives

$$g_0(1) = \exp(E_k^{(x)}(s_\zeta) + \mathfrak{f}_k^{(x)}), \qquad g_0'(1) = \exp(E_k^{(x')}(s) + \mathfrak{f}_k^{(x')}).$$

Their equality determines the difference of the exponents modulo $2\pi i\mathbb{Z}$.

The common domain $\{s : s, s_\zeta \in \mathscr{O}_k\}$ is convex. The difference of the two exponents is continuous there and takes values in $2\pi i\mathbb{Z}$, hence is constant. If the domain is nonempty, its mass inequalities imply $|\zeta| < 2r_J/\lambda_k$; it therefore contains $s_* = (0, \zeta 1/2)$. The two sources at this point are $(0, \zeta 1/2)$ and $(0, -\zeta 1/2)$. They are fixed by source conjugation, so both energies and both finite-size constants are real. Their difference is therefore zero. This fixes the branch and proves the identity. □

Proposition 10.13 (A common normalizer)**.** For every $K \ge 1$ there is a holomorphic function

$$\mathcal{E}_K(z) = E_K^{(x)}(q_z^{(x)}) + \mathfrak{f}_K^{(x)}, \qquad |z - x| < r_J, \quad x \in I,$$

on U_{r_J}. It satisfies $\mathcal{E}_K(\bar z) = \overline{\mathcal{E}_K(z)}$. Set

$$M_0 = \exp\Big(\max\{C+2, C_P(1+\log X_g)\}\ell^3\Big), \qquad h_K(z) = Z_K(z)e^{-\mathcal{E}_K(z)}.$$

Then

$$|h_K(z)| \le M_0 e^{C_{\mathrm{fs}}} \quad (\mathrm{dist}(z, I) \le 2\varrho_J), \qquad h_K(x) \ge (M_0 e^{C_{\mathrm{fs}}})^{-1} \quad (x \in I). \tag{10.iii}$$

For $0 < r < 2\varrho_J$, put

$$\rho_{n,r} = \min\{\varrho/n, 2\varrho_J - r\}, \qquad M_{n,d} = M_0 \exp\Big[nC_{\mathrm{count}}(1 + d^{-3})\ell^3\Big].$$

Here C_{count} is the torus source-count constant. If $n \ge 2$, the supports of $X = ((\alpha_i, f_i))_{i=1}^n$ are separated by at least d, and $\|f_i\|_{\alpha_i} \le \rho_{n,r}$, then there are holomorphic functions $\mathcal{E}_{K,i}(z, t_i)$ on

$U_r \times \{|t_i| < 1\}$, zero at $t_i = 0$, such that

$$\left| Z_K(z;t) e^{-\mathcal{E}_K(z) - \sum_i \mathcal{E}_{K,i}(z,t_i)} \right| \le M_{n,d} e^{C_{\mathrm{fs}}}. \tag{10.iv}$$

■

Proof. If $|z-x|, |z-x'| < r_J$, apply Prop. 10.12 to $s = q_z^{(x')}$. Its shifted source is $q_z^{(x)}$. Thus the displayed definitions of $\mathcal{E}_K$ agree. Each is holomorphic on its disc, and the discs cover U_{r_J}. Source conjugation gives the stated reality identity. The upper bound follows from eq. 9.iv at $q_z^{(x)}$, and the lower bound at $x \in I$ follows from the zero-source lower bound at that centre. The factor $e^{C_{\mathrm{fs}}}$ accounts for $\mathfrak{f}_K^{(x)}$.

Define locally

$$\mathcal{E}_{K,i}(z,t_i) = E_K^{(x)}(q_z^{(x)} + t_i s_i) - E_K^{(x)}(q_z^{(x)}).$$

For $|z-x| < r$, $|t_i| < 1$, every partial sum of these sources lies in the interior of $\mathcal{P}$: the action norm is less than $n\rho_{n,r} \le \varrho$, and disjointness of the supports gives the matrix-source norm

$$|z-x| + \max_i |t_i| \, \|f_i\|_\infty < 2\varrho_J.$$

Apply Prop. 10.12 with and without $t_i s_i$, and subtract. The finite-size constants cancel, proving that $\mathcal{E}_{K,i}$ is independent of the chosen centre.

Almost additivity at the base $q_z^{(x)}$ gives

$$\left| E_K^{(x)}\left(q_z^{(x)} + \sum_i t_i s_i\right) - E_K^{(x)}(q_z^{(x)}) - \sum_i \mathcal{E}_{K,i}(z,t_i) \right| \le \sum_i \mathscr{C}_K(\operatorname{supp} s_i; d) \le n C_{\mathrm{count}} (1 + d^{-3}) \ell^3.$$

Combining this with the partition upper bound proves eq. 10.iv. □

10.5. **A lower bound on a longer real mass interval.** The common normalizer controls possible zeros uniformly on compact subsets of $U_{2\varrho_J}$. To obtain a useful zero-free neighbourhood we first bound the normalized partition function on a longer real interval.

Fix $0 < \epsilon \le \varrho_J/2$, and set

$$R_\epsilon = 2\varrho_J - 2\epsilon, \qquad c_g = C_P(1 + \log X_g),$$
$$c_\epsilon = 512 \epsilon^{-1} L^3 e^{1+19C_\flat}, \qquad A_\epsilon = (C + 3 + c_g) e^{c_\epsilon R_\epsilon},$$
$$K_\epsilon = 1 + \left\lceil \tfrac{1}{3} \log_L A_\epsilon \right\rceil.$$

These constants are independent of the side and cutoff.

Proposition 10.14 (The normalized partition function at real masses)**.** Let $x_0 \in I$, $K \ge K_\epsilon$, and

$$h_{K,x_0}(x) = Z_K(x) e^{-E_K^{(x_0)}(q_x^{(x_0)})}.$$

For real $|x - x_0| \le R_\epsilon$,

$$h_{K,x_0}(x) \ge e^{-A_\epsilon \ell^3}, \qquad |(\log h_{K,x_0})'(x)| \le c_\epsilon A_\epsilon \ell^3. \tag{10.v}$$

■

Proof. Write $E = E_K^{(x_0)}$, $q_x = q_x^{(x_0)}$, $h = h_{K,x_0}$, and $u(x) = -\ell^{-3} \log h(x)$. All these real masses exceed $m_-/2$. The positive real-mass density gives $Z_K(x) > 0$, and $E(q_x)$ is real, so $h > 0$ and u is real analytic.

For $1 \le i \le K-1$, divide the torus into cells of physical side $\lambda = L^{-i}$. Their lattice side $\lambda_{\mathrm{l}} = L^{K-i}$ is odd and at least L; the number of cells in each direction is even. Let $N_{\mathrm{c}} = (\ell/\lambda)^3$. In a cell centred at y^c, in lattice coordinates, put

$$\eta_c(y) = \prod_{\nu=1}^{3} \tau(y_\nu - y_\nu^c), \qquad \tau(v) = \min\left\{1, \max\left\{0, \frac{3\lambda_{\mathrm{l}}/8 - |v|}{\lambda_{\mathrm{l}}/4}\right\}\right\}.$$

These functions are carried to one another by wall reflections. Their supports have mutual distance at least $\lambda/4$, and each lies in a ball of radius $3\sqrt{3}\lambda/8$. Consequently $1 + (R/\delta)^3 \le 19$.

For $|t| \le \epsilon$, apply the chessboard inequality to $e^{\Phi(q_x + t\eta_c 1)}$ in one cell and to $e^{\Phi(q_x)}$ in the others, with each constant source restricted to its own cell. Since q_x is real and scalar, its fold is the same constant source. Thus

$$|Z[q_x + t\eta_c 1]| \le Z[q_x + (t\eta)^{\mathrm{ch}}]^{1/N_{\mathrm{c}}} Z[q_x]^{1-1/N_{\mathrm{c}}},$$

where the coefficient in a reflected cell is t or $\bar{t}$. The folded source has norm at most $R_\epsilon + \epsilon < 2\varrho_J$. Use eq. 10.i and the partition upper bound. With

$$\Lambda(t) = N_{\mathrm{c}}^{-1} \sum_{c'} [E(q_x + t\eta_{c'} 1) - E(q_x)],$$

source conjugation identifies the real parts of the odd and even reflected terms. We obtain

$$\psi_c(t) = e^{-\Lambda(t)} \frac{Z[q_x + t\eta_c 1]}{Z[q_x]}, \qquad \psi_c(0) = 1,$$

$$|\psi_c(t)| \le \exp\left[19C_\flat + (C + 2 + u(x))\lambda^3\right] =: B_1(x, \lambda).$$

These functions are holomorphic past $|t| = \epsilon$, so $|\psi_c'(0)| \le B_1/\epsilon$.

For a site function v, define the linear functional

$$\mathcal{V}(v) = \frac{d}{dt} \log W_K[q_x + tv1]\Big|_{t=0}.$$

For one offset o of the board, $\sum_c \psi_c'(0) = \mathcal{V}(\eta^{(o)})$, where $\eta^{(o)} = \sum_c \eta_c$. Hence $|\mathcal{V}(\eta^{(o)})| \le N_{\mathrm{c}} B_1/\epsilon$. Summing over the λ_{l}^3 offsets gives

$$\sum_o \eta^{(o)} = \sigma^3, \qquad \sigma = \sum_{v \in \mathbb{Z}} \tau(v) \ge \lambda_{\mathrm{l}}/8.$$

Since $q_x + (0, t1) = q_{x-t}$, this yields

$$|u'(x)| \le 512\epsilon^{-1}\lambda^{-3} \exp\left[19C_\flat + (C + 2 + u(x))\lambda^3\right].$$

Set $Y = C + 2 + \max\{u, 1\}$. At a point where $Y \le A_\epsilon$, choose $i = \lceil \frac{1}{3} \log_L Y \rceil$. Then

$$1 \le i \le K - 1, \qquad \lambda^3 Y \le 1 < L^3 \lambda^3 Y.$$

The preceding bound gives $|u'| \le c_\epsilon Y$ and $|Y'| \le c_\epsilon Y$ almost everywhere. At x_0, the zero-source floor gives $u(x_0) \le c_g$, so $Y(x_0) \le C + 3 + c_g$. Integrate the differential inequality on the maximal interval where $Y \le A_\epsilon$. If its right endpoint were less than $x_0 + R_\epsilon$, then

$$Y(x) \le (C + 3 + c_g) e^{c_\epsilon (x - x_0)} < A_\epsilon$$

at that endpoint, contradicting maximality and continuity. The same argument to the left gives $Y \le A_\epsilon$ throughout $|x - x_0| \le R_\epsilon$. The two assertions now follow from $u \le A_\epsilon$ and $|(\log h)'| = \ell^3 |u'|$. □

Lemma 10.15 (A complex neighbourhood from a real derivative bound)**.** Let h be holomorphic near $\{|z - x| \le r\}$, where x is real, and suppose

$$h(\bar{z}) = \overline{h(z)}, \qquad h > 0 \text{ on } [x - r, x + r],$$

$$\log|h(z)/h(x)| \le W, \qquad \log(h(x')/h(x)) \le \Lambda |x' - x| \quad (x' \in [x - r, x + r]).$$

For $z = x + \xi + i\eta$, $|z - x| \le r/2$, $\eta \ne 0$,

$$\log|h(z)/h(x)| \le \frac{16W}{3\pi} \frac{|\eta|}{r} + \Lambda|\xi| + \frac{2}{\pi} \Lambda |\eta| \log \frac{r}{|\eta|}. \tag{10.vi}$$

Consequently, if $0 < \rho \le r/e$ and

$$\rho\left[\frac{16W}{3\pi r} + \Lambda + \frac{2}{\pi}\Lambda\log(r/\rho)\right] \le 1,$$

then $|h(z)| \ge h(x)/2$ for $|z-x| \le \rho/(2(e+1))$. ■

Proof. By conjugation it suffices to consider $\eta > 0$. On the upper unit half-disc put

$$\zeta = (z-x)/r, \qquad u(\zeta) = \log|h(x+r\zeta)/h(x)|.$$

This function is subharmonic and bounded above, including when h has zeros. The harmonic measure of the semicircle is

$$\omega(\zeta) = \frac{2}{\pi}\arctan\frac{2\operatorname{Im}\zeta}{1-|\zeta|^2}.$$

The fractional linear map $(1+\zeta)/(1-\zeta)$ sends the half-disc to the first quadrant and its semicircle to the positive imaginary axis, giving this formula. For $|\zeta| \le 1/2$, $\omega(\zeta) \le 16\operatorname{Im}\zeta/(3\pi)$. The second harmonic function

$$q(\zeta) = \operatorname{Re}\left(\zeta + \frac{2i}{\pi}\zeta\log\zeta\right) = \operatorname{Re}\zeta\left(1 - \frac{2}{\pi}\arg\zeta\right) + \frac{2}{\pi}\operatorname{Im}\zeta\log\frac{1}{|\zeta|}$$

uses the principal logarithm. It extends continuously to the diameter with value $|s|$, and on the semicircle its value is $\cos t(1-2t/\pi) \ge 0$. The boundary assumptions therefore give $u \le W\omega + \Lambda r q$ on the half-disc by the maximum principle. The two endpoints cause no difficulty because u is bounded above. Bound $|1 - 2\arg\zeta/\pi| \le 1$ and $\log(1/|\zeta|) \le \log(1/\operatorname{Im}\zeta)$. This proves eq. 10.vi.

Since $y\log(r/y)$ increases for $0 < y \le r/e$, the hypothesis on ρ gives $|h| \le e\,h(x)$ on the disc of radius ρ. Schwarz's lemma applied to $h - h(x)$, bounded there by $(e+1)h(x)$, gives

$$|h(z) - h(x)| \le (e+1)h(x)|z-x|/\rho.$$

The asserted lower bound follows. □

Set

$$I_\epsilon = [m_- - 2\varrho_J + 3\epsilon,\ m_+ + 2\varrho_J - 3\epsilon], \quad W_\ell = (C + 2 + A_\epsilon)\ell^3, \quad \Lambda_\ell = c_\epsilon A_\epsilon \ell^3.$$

Let ρ_ℓ be the largest number in $(0, \epsilon/e]$ such that

$$\rho_\ell\left[\frac{16W_\ell}{3\pi\epsilon} + \Lambda_\ell + \frac{2}{\pi}\Lambda_\ell\log(\epsilon/\rho_\ell)\right] \le 1, \qquad w_\ell = \frac{\rho_\ell}{2(e+1)}. \tag{10.vii}$$

The left side is increasing on this interval and tends to zero at zero. Thus ρ_ℓ exists. Define

$$\Omega_\epsilon = \{z : \operatorname{dist}(z, I_\epsilon) < w_\ell\}, \qquad \rho_n^\epsilon = \min\{\varrho/n, 2\epsilon\}.$$

All real points of Ω_ϵ exceed $m_-/2$.

Proposition 10.16 (A zero-free neighbourhood of the longer interval)**.** For $K \ge K_\epsilon$ and $z \in \Omega_\epsilon$,

$$|Z_K(z)| \ge \tfrac{1}{2}e^{-A_\epsilon\ell^3 - C_{\mathrm{fs}}}|e^{\mathcal{E}_K(z)}| > 0. \tag{10.viii}$$

If X has $n \ge 2$ entries with pairwise separation at least $d > 0$, then

$$|S^T_{n,K}(z;X)| \le \left[\frac{2(2e^{A_\epsilon\ell^3}M_{n,d}+1)}{\rho_n^\epsilon}\right]^n \prod_i \|f_i\|_{\alpha_i}. \tag{10.ix}$$

There is $c_w > 0$, depending on the fixed constants, g and ϵ, but not on the side, such that

$$w_\ell \ge \frac{c_w}{\ell^3(1+\log\ell)}.$$

■

Proof. For $z \in \Omega_\epsilon$, choose $x \in I_\epsilon$ with $|z-x| < w_\ell$, and let $x_0 \in I$ be nearest to x. Then

$$|x - x_0| \le R_\epsilon - \epsilon, \qquad \{|\zeta - x| \le \epsilon\} \subset \{|\zeta - x_0| \le R_\epsilon\}.$$

On this disc h_{K,x_0} is holomorphic, $|h_{K,x_0}| \le e^{(C+2)\ell^3}$, and $h_{K,x_0}(x) \ge e^{-A_\epsilon \ell^3}$. On its real diameter Prop. 10.14 gives the logarithmic derivative bound Λ_ℓ. Apply Lem. 10.15 with $r = \epsilon$, $W = W_\ell$, and $\Lambda = \Lambda_\ell$. It gives

$$|h_{K,x_0}(z)| \ge \tfrac{1}{2} e^{-A_\epsilon \ell^3}.$$

Since $\mathcal{E}_K(z) = E_K^{(x_0)}(q_z^{(x_0)}) + \mathfrak{f}_K^{(x_0)}$, this is eq. 10.viii.

Also $|z - x_0| < R_\epsilon$. For $\|f_i\|_{\alpha_i} \le \rho_n^\epsilon$, the local insertion bound used in Prop. 10.13 applies at this centre with radius $R_\epsilon = 2\varrho_J - 2\epsilon$. The holomorphic function

$$\Psi(t) = \frac{Z_K(z;t) e^{-E_K^{(x_0)}(q_z^{(x_0)}) - \sum_i \mathcal{E}_{K,i}(z,t_i)}}{h_{K,x_0}(z)}$$

has $\Psi(0) = 1$ and $|\Psi| \le 2e^{A_\epsilon \ell^3} M_{n,d}$ on $\max_i |t_i| < 1$. On each line through zero, Schwarz's lemma bounds $|\Psi(t) - 1|$ by $(2e^{A_\epsilon \ell^3} M_{n,d} + 1) \max_i |t_i|$. On the polydisc of radius $[2(2e^{A_\epsilon \ell^3} M_{n,d} + 1)]^{-1}$, its logarithm has modulus at most $\log 2$. For $n \ge 2$, its full mixed derivative equals that of $\log Z_K(z;t)$: each subtracted insertion energy depends on only one t_i. Cauchy's inequality and multilinearity give eq. 10.ix.

For the width, put

$$\alpha = \frac{16(C + 2 + A_\epsilon)}{3\pi\epsilon} + c_\epsilon A_\epsilon, \quad \beta = \frac{2c_\epsilon A_\epsilon}{\pi}, \quad L_\ell = \max\{1, \log(2\epsilon\ell^3(\alpha + \beta))\}.$$

The defining expression in eq. 10.vii is $\rho\ell^3[\alpha + \beta \log(\epsilon/\rho)]$. For $\rho_* = [2\ell^3(\alpha + 2\beta L_\ell)]^{-1}$,

$$\log(\epsilon/\rho_*) \le L_\ell + \log(2L_\ell) \le 2L_\ell.$$

Thus its value at ρ_* is at most one when $\rho_* \le \epsilon/e$; if $\rho_* > \epsilon/e$, its value at ϵ/e is less than $1/2$. It follows that

$$\rho_\ell \ge \min\{\epsilon/e, \rho_*\}.$$

For $\ell \ge 1$,

$$L_\ell \le a_w + 3\log \ell \le 3a_w(1 + \log \ell), \qquad a_w = 1 + |\log(2\epsilon(\alpha + \beta))|.$$

One may therefore take

$$c_w = \frac{1}{2(e+1)} \min\{\epsilon/e, [2(\alpha + 6\beta a_w)]^{-1}\}.$$

□

10.6. **A common meromorphic family.**

Theorem O (Continuation in the mass)**.** Fix $b \in I_g$ and $0 < \epsilon \le \varrho_J/2$. Every sequence $K_j \to \infty$ has a subsequence and a holomorphic function h on $U_{2\varrho_J}$, not identically zero, with the following properties.

For every separated tuple X of order $n \ge 2$, $S^T_{n,K_j}(z;X)$ converges locally uniformly off the discrete set $\mathcal{Z} = \{h = 0\}$ to a meromorphic function $\mathcal{S}_n(z;X)$ on $U_{2\varrho_J}$, and $h^n \mathcal{S}_n(\cdot;X)$ is holomorphic there. The same h, subsequence and exceptional set work for all orders, labels and tuples. Moreover

$$h_{K_j} \longrightarrow h \quad \text{locally uniformly on } U_{2\varrho_J}.$$

The set $\mathcal{Z}$ is symmetric under conjugation and is the set of accumulation points, in this open domain, of zeros of Z_{K_j}.

The set $\mathcal{Z}$ is disjoint from Ω_ϵ. On Ω_ϵ the limits are holomorphic, obey eq. 10.ix, and converge with all mass derivatives. If $z_j \to z \in \Omega_\epsilon$, then

$$S^T_{n,K_j}(z_j;X) \longrightarrow \mathcal{S}_n(z;X).$$

At $m \in I$ these values are continuum limits with labels (b, m). Every continuum limit at fixed b, obtained with masses $m_j \to m \in I$, occurs this way after a subsequence. On all real masses in Ω_ϵ the centered limiting functional is reflection positive on the separated polynomial words used in its construction. It has the permutation, translation, cubic, flavour and charge-conjugation symmetries of the finite lattice theory.

Two such meromorphic families which agree on a subset of I having an accumulation point agree throughout $U_{2\varrho_J}$. This is a uniqueness assertion for analytic continuation from those values; it does not assert that distinct cutoff subsequences have the same limits. ■□

Proof. The functions h_K are uniformly bounded on $U_{2\varrho_J}$ by eq. 10.iii. We construct bounded holomorphic numerators for the cumulants on the same domain. Fix $0 < r < 2\varrho_J$, a separated tuple X, and first rescale its entries so that $\|f_i\|_{\alpha_i} \le \rho_{n,r}$. Put

$$\Psi_K(z,t) = Z_K(z;t)e^{-\mathcal{E}_K(z)-\sum_i \mathcal{E}_{K,i}(z,t_i)}.$$

It is bounded by $M_{n,d}e^{C_{\mathrm{fs}}}$ on $U_r \times \{\max_i |t_i| < 1\}$, and $\Psi_K(z,0) = h_K(z)$. Cauchy's formula on smaller circles, followed by their radius limit, gives the same bound for $\partial_B \Psi_K(z,0)$, $B \subset \{1,\dots,n\}$. The logarithmic derivative formula used in Prop. 9.3 gives, where $h_K \neq 0$,

$$S^T_{n,K}(z;X) = \frac{N_K(z;X)}{h_K(z)^n},$$

$$N_K(z;X) = \sum_\pi (-1)^{|\pi|-1}(|\pi|-1)!\, h_K(z)^{n-|\pi|} \prod_{B\in\pi} \partial_B \Psi_K(z,0).$$

The one-variable insertion energies have zero full mixed derivative because $n \ge 2$. Each numerator is holomorphic. The formulas obtained on different U_r's agree away from the discrete zeros of Z_K, hence everywhere by continuity. They therefore define one holomorphic $N_K(\cdot;X)$ on $U_{2\varrho_J}$. For the original entries its bound on U_r is

$$|N_K(z;X)| \le k_n \left(\frac{M_{n,d}e^{C_{\mathrm{fs}}}}{\rho_{n,r}}\right)^n \prod_i \|f_i\|_{\alpha_i}.$$

Apply Montel's theorem to h_K and these numerators on a countable compact exhaustion of $U_{2\varrho_J}$. Use one enumeration of all orders, basis labels and dense smooth test families with supports in fixed separated closed sets. The same diagonal selection gives convergence for every member of this enumeration. The displayed multilinear bound extends convergence to arbitrary separated tuples: enlarge their supports slightly while keeping positive separation, approximate in the action C^1 norm and meson uniform norm, and telescope the difference of the multilinear forms one entry at a time. The resulting error tends to zero uniformly on each compact mass set and uniformly in K.

Denote the limits by h, N. On I, eq. 10.iii gives $h \ge (M_0 e^{C_{\mathrm{fs}}})^{-1}$, so $h \not\equiv 0$. Its zeros are discrete. On a compact set avoiding them, h_{K_j} is eventually bounded away from zero, and hence

$$S^T_{n,K_j} \longrightarrow \mathcal{S}_n = N/h^n$$

locally uniformly there. Source conjugation makes h_K real under conjugation, and the same is true of h. If $h(z) \neq 0$, uniform convergence excludes nearby zeros of h_{K_j} for all sufficiently large j. Conversely, on a small circle around a zero of h and containing no other zero, Rouché's theorem gives zeros of h_{K_j} inside for every sufficiently large j. Since h_K and Z_K have the same zeros, this proves the assertion about $\mathcal{Z}$.

By eq. 10.viii,

$$|h_K(z)| \ge \tfrac{1}{2} e^{-A_\epsilon \ell^3 - C_{\mathrm{fs}}} \quad (z \in \Omega_\epsilon,\ K \ge K_\epsilon).$$

Thus h has no zero there. The cumulant bound passes to the limit; Cauchy's formula on slightly larger compact sets gives convergence of all mass derivatives. Local uniform convergence and

continuity give convergence at moving masses. In particular a continuum limit already obtained at $m_j \to m \in I$ is preserved when this further extraction is made.

For reflection positivity no convergence of one-point functions is required. At a real positive mass, center each lattice insertion by its own mean. Every centered moment is the sum, over partitions into blocks of size at least two, of products of cumulants. It therefore converges by the preceding result. On a positive-time polynomial the finite lattice reflection form is nonnegative, where each monomial has internally separated supports. Approximate a coordinate reflection plane by a bond plane at distance at most a; the reflected tests differ by a translation of at most $2a$. For smooth tests supported away from the plane, their norms converge and all separations in the reflection form stay bounded below. The uniform cumulant bound and multilinearity pass the inequality to the limit. This proves reflection positivity of the centered functional at every real mass in Ω_ϵ.

The lattice permutation and internal identities hold for the same finite integrals at these masses and pass to the limits. Lattice translations approximate all torus translations in the test norms, as in Thm. M; cubic covariance passes directly. Equivalently, the finite-cutoff identities are identities of rational functions of the mass and hold throughout their common domain of definition. Finally the identity theorem, applied after clearing denominators, proves the last assertion. □

10.7. **Correlation estimates at the continued masses.** The continuation preserves the two leading short-distance kernels. We first estimate the change of the source energy. The estimate uses Schwarz's lemma in the constant mass source; its scale factor is needed before the sum over levels.

Use the test pairs of eq. 8.iii, with $0 < d \le 1/200$, and put

$$\mathcal{K}_q(f,h) = \iint f(x)h(y)|x-y|^{-q}\,dx\,dy, \qquad H_d = 1 + \log_L(1/d).$$

Thus

$$\mathcal{K}_q(f,h) \ge 10^{-q}\vartheta^2 d^{6-q}\|f\|_\infty\|h\|_\infty. \tag{10.x}$$

Only action-density and scalar-bilinear indices are used below.

Lemma 10.17 (Changing the mass in the mixed energy)**.** Fix a centre $x_0 \in I$ and $|\zeta| < 2\varrho_J$, and set $q[\zeta] = (0, -\zeta 1)$. Let $(F_\nu, H_\nu)_{\nu=1}^N$ be translates and coordinate mirror images of one test pair, including the induced action on its labels. Suppose the pairs have mutual distance at least d and $N \ge (\ell/(96d))^3$. Write

$$Q_K(\zeta) = \frac{1}{N}\sum_{\nu=1}^{N} \partial_{t_1}\partial_{t_2} E_K^{(x_0)}(q[\zeta] + t_1F_\nu + t_2H_\nu)\Big|_{t=0}.$$

There are constants A_B, A_0, independent of K, ℓ, b, x_0, such that

$$|Q_K(\zeta) - Q_K(0)| \le A_B|\zeta|d^3H_d\|f\|_\infty\|h\|_\infty$$

at two bilinear indices, and

$$|Q_K(\zeta) - Q_K(0)| \le A_0\left(\frac{40M}{\vartheta\varrho}\right)^2 |\zeta|dH_d^{1+2p_0}\|f\|_\infty\|h\|_\infty$$

at two action-density indices. Here p_0 is the exponent in the RG weight $\phi(t) = t^{-1/2}(1+\log t)^{p_0}$. ■

Proof. Write $E = E_K^{(x_0)}$ and

$$e_j^\zeta(X;s) = e_j(X; q[\zeta]+s) - e_j(X;s).$$

The desired difference is the sum over j, X of the mean mixed derivatives of e_j^ζ. For a fixed level let $f_1 = f, f_2 = h$, and take source radii

$$r_i = \frac{r_J}{2\lambda_{j-1}\|f_i\|_\infty} \quad \text{at a bilinear index,} \qquad r_i = \frac{2\varrho}{\|f_i\|_{\sharp,j-1}} \quad \text{at the action index.}$$

For sources F, H with the disjoint supports and norms of the given pair, $s(t) = t_1 F + t_2 H$ lies in the level source domain when $|t_i| < r_i$. Indeed disjoint supports make the norm of the sum the maximum of the two norms. Adding $q[\zeta']$, with $|\zeta'| < r_J/(2\lambda_{j-1})$, still gives $\lambda_{j-1}\|J\|_\infty < r_J$ and $\|w\|_{\sharp,j-1} < 2\varrho$.

If s, s' are two such sources, the function

$$\zeta' \longmapsto e_j^{\zeta'}(X; s) - e_j^{\zeta'}(X; s')$$

vanishes at zero. The local source-tail estimate bounds it on this mass disc by

$$2C_E e^{-\kappa d_j(X)} \exp\left[-\kappa_\sigma \operatorname{dist}(X, \operatorname{supp}(s - s'))/L^j\right].$$

Schwarz's lemma therefore multiplies this bound by $2\lambda_{j-1}|\zeta|/r_J$. Apply the estimate with s' obtained by setting either t_1 or t_2 equal to zero. The finite-difference formula and Cauchy's estimate on the two source circles give

$$\begin{aligned} |\partial_{12} e_j^\zeta(X; F, H)| \le{}& \frac{8C_E\lambda_{j-1}|\zeta|}{r_J r_1 r_2} e^{-\kappa d_j(X)} \\ &\times \exp\left[-\frac{\kappa_\sigma}{L^j}\max\{\operatorname{dist}(X, \operatorname{supp} F), \operatorname{dist}(X, \operatorname{supp} H)\}\right]. \end{aligned} \tag{10.xi}$$

The distances in this local display are in the original lattice units. Their physical scale is λ_j.

The energy-count estimate of [QCDM1, Prop. 10.16 and its proof] sums the last two factors over polymers. At $\lambda_j \le d$, each support lies in a radius-d ball and $\#_j^+(\operatorname{supp} F_\nu) \le 1728(d/\lambda_j)^3$. Separation retains $e^{-\kappa_E d/\lambda_j}$, where $\kappa_E > 0$ is the separation rate in the cited counting estimate; it is chosen at most $\min\{\kappa_\sigma/8, \kappa/(4c_{\mathcal{D}}M)\}$. Consequently the level contribution to the mean is at most

$$\frac{8 \cdot 1728 C_{\mathscr{C}}\lambda_{j-1}|\zeta|}{r_J r_1 r_2}(d/\lambda_j)^3 e^{-\kappa_E d/\lambda_j}.$$

At $\lambda_j > d$, give the pairs independent signs ϵ_ν. The two sums $F_\epsilon = \sum_\nu \epsilon_\nu F_\nu$ and $H_\epsilon = \sum_\nu \epsilon_\nu H_\nu$ retain the norms and disjointness required in eq. 10.xi. The sign average of any bilinear form at these two sums is its sum over the diagonal pairs. The total number of level cubes is bounded by

$$(\ell/\lambda_j)^3 \le 96^3 N (d/\lambda_j)^3.$$

The mean contribution of this level is therefore at most

$$\frac{8 \cdot 96^3 C_{\mathscr{C}}\lambda_{j-1}|\zeta|}{r_J r_1 r_2}(d/\lambda_j)^3.$$

This use of the total cube count avoids multiplying the coarse-scale bound by N.

For bilinears,

$$(r_1 r_2)^{-1} = 4\lambda_{j-1}^2 r_J^{-2}\|f\|_\infty\|h\|_\infty, \qquad \lambda_{j-1}^3(d/\lambda_j)^3 = d^3/L^3.$$

There are at most H_d coarse levels. The fine levels have distinct integers $p \ge 0$ with $d/\lambda_j \in [L^p, L^{p+1})$; their exponentials have a bounded sum. For example one may take

$$C'_\kappa = \sum_{p \ge 0} L^{3p} e^{-\kappa_E L^p}, \qquad A_B = \frac{2^{25} C_{\mathscr{C}}(1 + C'_\kappa)}{L^3 r_J^3}.$$

For action sources let $\pi_{j-1} = (1 + (K - j + 1)/n_\pi)^{p_0}$, the weight in the level source norm. The test-pair Lipschitz bound gives

$$\|f\|_{\sharp,j-1} \le \frac{40M\pi_{j-1}}{\vartheta}\max\{1, \lambda_{j-1}/d\}\|f\|_\infty.$$

At the coarse levels,

$$\pi_{j-1} \le (3 + \log_L(1/d))^{p_0}, \qquad \lambda_{j-1}(d/\lambda_j)^3 \max\{1, \lambda_{j-1}/d\}^2 \le d/L.$$

At the fine level indexed by p, the corresponding bounds are $((3+\log_L(1/d))(1+p))^{p_0}$ and dL^{2p+1}. Thus the fine-scale sum is paid by

$$C''_\kappa = \sum_{p\ge 0} L^{2p}(1+p)^{2p_0} e^{-\kappa_E L^p}.$$

One may take

$$A_0 = \frac{2^{21} 9^{p_0} C_{\mathscr{C}}(1+LC''_\kappa)}{r_J}.$$

Substitution proves the two estimates. □

The next estimate keeps the partition function normalized at the shifted real mass. Its loss is proportional to the volume of an occupied cell, rather than to the volume of the torus.

Lemma 10.18 (A cell at a shifted real mass)**.** Let $x_0 \in I$, $|x-x_0| < 2\varrho_J$, and x be real. Put $q = q_x^{(x_0)}$, $E = E_K^{(x_0)}$, and

$$u(x) = -\ell^{-3}\log(Z_K(x)e^{-E(q)}).$$

Consider a reflection board with N cells of physical volume $|\square|$. Suppose $s=(w,J)$ is supported in one cell, at distance at least $\delta>0$ from its walls, and

$$\|w\|_\sharp < 2\varrho, \qquad \|J\|_\infty \le 2\varrho_J - |x-x_0|.$$

If r_{lin} denotes the linear part of each board reflection on sources, set

$$\lambda_x(s) = \frac{1}{\mathsf{N}}\sum_r \{E(q+r_{\mathrm{lin}}s) - E(q)\}.$$

Then

$$\left|\frac{Z[q+s]}{Z[q]}\right| \le \exp\left[\operatorname{Re}\lambda_x(s) + (C+2+u(x))|\square| + \mathscr{C}_K(\operatorname{supp} s;\delta)\right].$$

For a sum $s=\sum_\square s_\square$ satisfying these conditions in its respective cells, the right side is the product of the displayed factors for the occupied cells. ■

Proof. The constant scalar source q is real and is fixed by reflection. Its exponential is the product of the even factors $e^{\Phi(q1_\square)}$ in the cell algebras. Apply the chessboard inequality at the original positive mass x_0, using $e^{\Phi(q1_\square)}e^{O(w)+\Phi(J)}$ in the occupied cell and $e^{\Phi(q1_\square)}$ elsewhere. Folding an unoccupied cell gives $e^{\Phi(q)}$. Folding the occupied cell gives the source $q+s^{\mathrm{ch}}$, where the images of s have conjugated coefficients after an odd number of reflections. Hence

$$\left|\frac{Z[q+s]}{Z[q]}\right| \le \left(\frac{Z[q+s^{\mathrm{ch}}]}{Z[q]}\right)^{1/\mathsf{N}}.$$

The folded numerator is nonnegative. Its source lies in $\mathcal{P}$: folding preserves the norm of w, and the matrix norm is at most $|x-x_0|+\|J\|_\infty$. The history bound and base-source almost additivity give

$$Z[q+s^{\mathrm{ch}}] \le \exp\left[\operatorname{Re}E(q+s^{\mathrm{ch}}) + (C+2)\ell^3\right],$$

$$\left|E(q+s^{\mathrm{ch}}) - E(q) - \sum_r\{E(q+r_\theta s)-E(q)\}\right| \le \mathsf{N}\mathscr{C}_K(\operatorname{supp} s;\delta).$$

The reflected supports have mutual distance at least 2δ. The reflected cube count costs at most 27; this factor is included in $\mathscr{C}_K$, as in the one-cell estimate of [QCDM1, Prop. 10.17]. Since $q^\star = q$ and $E(v^\star) = \overline{E(v)}$, the real part of the sum is $\mathsf{N}\operatorname{Re}\lambda_x(s)$. Use $Z[q] = e^{E(q)-u(x)\ell^3}$ and $\ell^3/\mathsf{N} = |\square|$, then take the N-th root.

For several occupied cells use their respective factors in the same chessboard inequality. Each cell contributes $(Z[q]/Z[0])^{1/\mathsf{N}}$; their product is $Z[q]/Z[0]$. For an occupied cell its remaining factor is bounded by the single-cell computation. Dividing out the common product proves the multiple-cell bound. □

Proposition 10.19 (The two kernels at the continued masses)**.** Let $c_2 = 9/(16\pi^2)$, $c_B = 1/(2\pi^2)$ and fix ϵ, ϑ as above. Require the test pair (f, h) to be of class C^1. There are functions $\epsilon_0(d), \epsilon_B(d)$, tending to zero as $d \downarrow 0$, and constants D_0, D_B such that every family in Thm. O satisfies, on Ω_ϵ,

$$\mathcal{S}_2(z;(0,f),(0,h)) = c_2 b^{-2} \mathcal{K}_6(f,h)(1+\eta_0),$$
$$\mathcal{S}_2(z;(\Gamma,f),(\Gamma',h)) = c_B \mathcal{K}_4(f,h)(\delta_{\Gamma\Gamma'} + \eta_B),$$

where

$$|\eta_0| \le \epsilon_0(d) + D_0\{d + \operatorname{dist}(z,I) d H_d^{1+2p_0}\},$$
$$|\eta_B| \le \epsilon_B(d) + D_B\{d + \operatorname{dist}(z,I) d H_d\}.$$

The functions are uniform in the admitted side and mass centre; the displayed constants may depend on the side. At real $z \in \Omega_\epsilon$, the same conclusion holds with constants independent of the side. In particular the short-distance coefficient of the action two-point function determines b^{-2} throughout Ω_ϵ. ∎

Proof. We prove uniform finite-cutoff bounds and then pass to the subsequence. Choose the centre $x_0 \in I$ nearest to z and put $\zeta = z - x_0$. On Ω_ϵ, $|\zeta| < R_\epsilon$. For a source associated to one entry define

$$\|F\|_\circ^\epsilon = \begin{cases} \|f\|_\sharp/\varrho, & \alpha = 0, \\ \|f\|_\infty/\epsilon, & \alpha \in \mathfrak{B}. \end{cases}$$

For the test pair, $\|f\|_\sharp \le 40M\|f\|_\infty/(\vartheta d)$. On $|t_i| < 2/\|F_i\|_\circ^\epsilon$, the source $q + t_1F + t_2H$ lies in $\mathcal{P}$: the extra bilinear norm is less than 2ϵ, while $|\zeta| + 2\epsilon < 2\varrho_J$.

Write $E = E_K^{(x_0)}$. At a nonzero partition function,

$$S_{2,K}^T(z;F,H) = \partial_{12} E(q;F,H) + \mathcal{B}_\zeta(F,H), \qquad \mathcal{B}_\zeta(F,H) = \partial_{12} \log W_K[q + t_1F + t_2H]\big|_0.$$

Both terms are bilinear in F, H. The zero-free estimate, the finite-size normalization and the history upper bound imply

$$\left| \frac{W_K[q + t_1F + t_2H]}{W_K[q]} \right| \le B_c, \qquad B_c = 2e^{(C+2+A_\epsilon)\ell^3 + 2C_{\mathrm{fs}}}.$$

For a holomorphic function ψ on the indicated bidisc with $\psi(0) = 1$ and $|\psi| \le B$, Schwarz's lemma on complex lines gives $|\psi - 1| \le 1/2$ on the bidisc smaller by $1/(4B)$. Its logarithm and Cauchy's formula therefore give

$$|\partial_{12} \log \psi(0)| \le 4B^2 \|F\|_\circ^\epsilon \|H\|_\circ^\epsilon.$$

This applies to $\mathcal{B}_\zeta$ with $B = B_c$.

Take a torus grid with mesh at least $12d$, rounded to the lattice. For $8a \le d$, the radius-$5d$ balls containing its translated pairs are at distance at least d. There are $N \ge (\ell/(24d))^3$ such pairs. Translation invariance holds at complex mass: the finite integrals and their insertion derivatives are polynomials in z, so the real-mass identity extends as an identity of rational functions. The sign average used in the proof of Lem. 10.17 now yields

$$S_{2,K}^T(z;F,H) = Q_K(\zeta) + \mathcal{R}_c, \qquad |\mathcal{R}_c| \le 4B_c^2 (24d/\ell)^3 \|F\|_\circ^\epsilon \|H\|_\circ^\epsilon. \tag{10.xii}$$

For real x with $\operatorname{dist}(x, I) \le R_\epsilon$ one can replace the volume loss by a cell loss. The lower bound of Prop. 10.14 gives $u(x) \le A_\epsilon$. Translate the pair into the radius-$1/4$ ball of a unit cell. The cell packing of [QCDM1, Lem. 11.2 and its proof] has at least $(96d)^{-3}$ translates inside this ball. For their two sums with signs, Lem. 10.18 applies with margin $1/4$ and gives

$$\left| e^{-\lambda_x(t_1F + t_2H)} \frac{Z[q + t_1F + t_2H]}{Z[q]} \right| \le B_r, \qquad B_r = e^{C+2+A_\epsilon + \mathscr{C}_*},$$

where $\mathscr{C}_*$ is the fixed upper bound for $\mathscr{C}_K(S;1/4)$ when S lies in that ball. The logarithmic derivative subtracts the board average of $\partial_{12}E$. Averaging the signs retains its diagonal pairs, now including their images in all ℓ^3 cells. Thus

$$S^T_{2,K}(x;F,H) = Q_K(x-x_0) + \mathcal{R}_r, \qquad |\mathcal{R}_r| \le 4B_r^2(96d)^3\|F\|_\circ^\epsilon\|H\|_\circ^\epsilon. \tag{10.xiii}$$

Here $N \ge (\ell/(96d))^3$, so Lem. 10.17 applies to both packings.

For clarity, the estimate of $Q_K(0)$ uses the free-template and interaction-error estimates in [QCDM1, proofs of Lems. 11.5 and 11.6], before adding the partition-function remainder in its two-point theorem. Their proofs apply to either packing just used: at fine levels they use the separated pair alone, and at coarse levels they use precisely $(\ell/\lambda_j)^3 \le 96^3 N(d/\lambda_j)^3$. The fourth-order source term has zero mixed derivative at these indices. The free terms give $c_2 b^{-2}\mathcal{K}_6$ and $c_B\delta_{\Gamma\Gamma'}\mathcal{K}_4$. After division by eq. 10.x, the interaction errors are bounded by constants times $\phi(b/d)(1+\log(b/d))^{2p_0}$ and $\phi(b/d)$, respectively. The free remainders contain powers of d, d/ℓ, and the positive mass bounds, and hence tend to zero uniformly for admitted ℓ. Their mesh terms tend to zero for each fixed test pair as $K\to\infty$. Denote the resulting cutoff-independent remainders by $\epsilon_0(d), \epsilon_B(d)$.

Add the change $Q_K(\zeta) - Q_K(0)$ from Lem. 10.17 and the appropriate packing remainder. For the action term the denominator is at least

$$c_2\, 10^{-6}\vartheta^2 X_g^{-2}\|f\|_\infty\|h\|_\infty;$$

for the bilinear term it is at least $c_B 10^{-4}\vartheta^2 d^2\|f\|_\infty\|h\|_\infty$. The mass-shift terms therefore have the displayed orders $|\zeta|dH_d^{1+2p_0}$ and $|\zeta|dH_d$. Each packing remainder has relative order d. Its constant depends on ℓ in eq. 10.xii, and is independent of ℓ in eq. 10.xiii. Absorb these explicit factors into D_0, D_B.

Finally take the cutoff subsequence defining the holomorphic family. The tests are separated and of class C^1, so the cumulants converge; all mesh remainders vanish. The bounds survive and their errors tend to zero as $d\downarrow 0$. The coefficient conclusion follows after dividing the first formula by $c_2\mathcal{K}_6$. □

Proposition 10.20 (Uniform real-mass cumulant bounds)**.** Let $n\ge 2$, and let the real Lipschitz tests f_i have pairwise distance at least $d>0$, each support lying in a ball of radius R. For

$$K \ge K_\epsilon, \qquad L^K \ge 16n, \qquad x\in\mathbb{R}, \quad \operatorname{dist}(x,I)\le R_\epsilon,$$

one has

$$|S^T_{n,K}(x;(\alpha_i,f_i)_i)| \le (1+d^{-1})^3 B_\epsilon(R)^n (n!)^{5+p/2}\prod_i \|f_i\|_{\alpha_i},$$

where

$$B_\epsilon(R) = \frac{C_B e^{C+2+A_\epsilon}(1+R)^3}{\min\{\varrho,\epsilon\}}, \qquad p = \frac{432 C_{\mathscr{C}}}{\log L}.$$

The constants $C_B, C_{\mathscr{C}}$ are those of the separated-cumulant bound in [QCDM1, Thm. F]. This bound is independent of the side, cutoff, coupling within I_g, and mass in the displayed interval. For tests in the continuum class specified at the start of this section, it passes to every continued family at its real masses. ■

Proof. Choose $x_0\in I$ nearest to x, and use $q = q_x^{(x_0)}$. Then $Z[q]>0$ and $u(x)\le A_\epsilon$. Set $\eta = (16n)^{-1}$. The product hat partition of unity at mesh η splits each f_i into at most $(2R/\eta+3)^3$ pieces $f_i\chi_a$. Each piece is supported in a cube of side 2η, with insertion norm at most $(1+80M/\eta)\|f_i\|_{\alpha_i}$.

Fix one piece from every insertion. The board selection in the proof of [QCDM1, Thm. F] leaves distance at least η between these supports and the walls of unit cells. To recall why the choice is available, in each coordinate a support forbids at most $4\eta L^K+1$ half-integral wall positions. The n supports therefore forbid fewer than L^K positions because $L^K\ge 16n$. The resulting board has an even number of slabs on every admitted torus.

Write s_i for the source of the chosen piece and choose radii

$$\rho_i = \begin{cases} 2\varrho/\|f_i\chi_{a_i}\|_\sharp, & \alpha_i = 0, \\ 2\epsilon/\|f_i\chi_{a_i}\|_\infty, & \alpha_i \in \mathfrak{B}. \end{cases}$$

Group $s(z) = \sum_i z_i s_i$ by its cells, and put

$$\Lambda_x(z) = \sum_{\square} \lambda_x(s_\square(z)), \qquad G(z) = e^{-\Lambda_x(z)} Z[q + s(z)]/Z[q].$$

Disjoint supports ensure that every group has action norm less than 2ϱ and matrix norm less than $2\epsilon \le 2\varrho_J - |x - x_0|$. For a coordinate face indexed by B, at most $|B|$ cells are occupied. Lem. 10.18 and subadditivity of the support count give

$$|G(z_B)| \le \exp\left[|B|(C + 2 + A_\epsilon) + \sum_{i\in B} \mathscr{C}_K(\operatorname{supp} s_i; \eta)\right].$$

The refined cube count from [QCDM1, proof of Thm. F] gives

$$\mathscr{C}_K(\operatorname{supp} s_i; \eta) \le 216 C_{\mathscr{C}}(1 + \log_L(1/\eta)) + C.$$

Cauchy's formula on each face consequently bounds $\partial_B G(0)$ by this exponential times $\prod_{i\in B} \rho_i^{-1}$.

Differentiate $\log Z = \Lambda_x + \log G + \log Z[q]$:

$$S^T_{n,K} = \partial_{[n]}\Lambda_x + \sum_{\pi} (-1)^{|\pi|-1}(|\pi| - 1)! \prod_{B\in\pi} \partial_B G(0).$$

The sum of the absolute partition coefficients is at most $2^n n!$. Thus its contribution is bounded by

$$C^n e^{n(C+2+A_\epsilon)} (16n)^{np/2} 2^n n! \prod_i \rho_i^{-1}.$$

The full mixed derivative of Λ_x is zero unless all the pieces occupy one cell. In that case its reflected energy derivatives obey [QCDM1, Prop. 10.18(i)], since every partial source remains in $\mathcal{P}$. It bounds this term by

$$2^{n-1} \prod_i \rho_i^{-1} \sup_Q \mathscr{C}_K(Q; d) \le C^n (1 + d^{-1})^3 \prod_i \rho_i^{-1},$$

where Q ranges over cubes of side 2η. The reflected count, including its factor 27, is the one already used in Lem. 10.18.

Sum over the choices of pieces. Their number and norm enlargement cost at most

$$(32nR + 3)^{3n}(1 + 1280Mn)^n.$$

The inequality $n^n \le e^n n!$ turns these four powers of n into $(n!)^4$. The partition sum contributes one further factorial and the logarithmic count contributes $(n!)^{p/2}$. Absorbing fixed constants gives the stated $B_\epsilon(R)$. No estimate used the torus side. For the stated continuum tests, passing along the cutoff subsequence proves the last assertion; no one-point limit is involved. □

10.8. **Zero and negative mass labels.** The larger matrix-source domain permits a continuation across zero. The measure used in the chessboard estimate still has a positive reference mass. The change of mass is included in the cell insertions. A differential inequality then proves that the partition function at the changed mass stays positive.

Here the constants have their original order. Choose

$$I_0 = [r_J/8, 1/2], \qquad r_J = r_M/L,$$

as the positive mass interval before choosing γ. The radius $r_M \le 1$ was fixed earlier, so this choice is admissible. For full cutoff convergence impose also the conditions of Def. 11.1. They are choices at their respective places in the same order. Put

$$\epsilon_o = r_J/16, \qquad R_o = 7r_J/16, \qquad I^\sharp = [-r_J/4, 1/2]. \tag{10.xiv}$$

For real x, write $x_0(x)$ for the nearest point of I_0. Then $|x - x_0(x)| \le 3r_J/8 = R_o - \epsilon_o$ on $I^\sharp$. Use the common energy $\mathcal{E}_K$ of Prop. 10.13, and write $q_z^{(x_0)} = (0, -(z - x_0)1)$ as before. Thus

$$E_K^{(x_0)}(q_z^{(x_0)}) = \mathcal{E}_K(z) - \mathfrak{f}_K^{(x_0)}, \qquad |\mathfrak{f}_K^{(x_0)}| \le C_{\mathrm{fs}}.$$

All assertions in this subsection concern the original indices $\mathcal{I} = \{0\} \cup \mathfrak{B}$.

Set

$$c_g = C_P(1 + \log X_g), \qquad c_o = 512\epsilon_o^{-1}L^3 e^{1+19C_\flat},$$
$$A^\sharp = (C + 3 + c_g)e^{c_o R_o}, \qquad K_\sharp = 1 + \left\lceil \tfrac{1}{3}\log_L A^\sharp \right\rceil.$$

These numbers are independent of the side, cutoff, and labels. Here $C_\flat$ is the constant in eq. 10.i. Their dependence on g enters through $\log X_g$.

Lemma 10.21 (The larger source domain)**.** Fix a positive reference mass $x_0 \in I_0$. If

$$\|w\|_\sharp < 2\varrho, \qquad |z - x_0| + \|J\|_\infty \le r_J/2,$$

then $q_z^{(x_0)} + (w, J) \in \mathcal{P}_K^\#$ and

$$Z_K(z; w, J) = e^{E_K^{(x_0)}(q_z^{(x_0)} + (w,J))} Z_K^{\mathrm{sub}}[q_z^{(x_0)} + (w, J)], \qquad |Z_K^{\mathrm{sub}}| \le e^{(C+2)\ell^3}.$$

The factor Z_K^{sub} is holomorphic in the interior of that source domain. In particular

$$|Z_K(z)| \le e^{(C+2)\ell^3 + C_{\mathrm{fs}}} |e^{\mathcal{E}_K(z)}| \quad \text{if } \operatorname{dist}(z, I_0) \le r_J/2.$$

■

Proof. The matrix source is $J-(z-x_0)1$, of norm at most $r_J/2$, and $\lambda_K = 1$. The estimate is the final-level case of [QCDM1, Prop. 10.15]. The identity follows directly from $ax_0 - a(J-(z-x_0)1) = az - aJ$. The energy identity from Prop. 10.13 proves the last bound. Source conjugation also shows that $E_K^{(x_0)}(q_x^{(x_0)})$ is real for real x with $|x - x_0| < r_J$. □

Theorem P (A positive partition function across zero)**.** Let $K \ge K_\sharp$, $\ell \in \mathfrak{L}(g)$ and $\mathsf{b} \in I_g$. For $x_0 \in I_0$ define

$$h_{x_0}(z) = Z_K(z) e^{-E_K^{(x_0)}(q_z^{(x_0)})}.$$

For real $|x - x_0| \le R_o$,

$$e^{-A^\sharp \ell^3} \le h_{x_0}(x) \le e^{(C+2)\ell^3}, \qquad |(\log h_{x_0})'(x)| \le c_o A^\sharp \ell^3.$$

Consequently $Z_K(x) > 0$ whenever $\operatorname{dist}(x, I_0) \le R_o$, and

$$e^{\mathcal{E}_K(x) - A^\sharp\ell^3 - C_{\mathrm{fs}}} \le Z_K(x) \le e^{\mathcal{E}_K(x) + (C+2)\ell^3 + C_{\mathrm{fs}}}, \qquad \left|\frac{d}{dx}\log(Z_K(x)e^{-\mathcal{E}_K(x)})\right| \le c_o A^\sharp \ell^3.$$

■□

Proof. Work at the positive reference mass ax_0. Write $E = E_K^{(x_0)}$, $q_x = q_x^{(x_0)}$ and $h = h_{x_0}$. At every real mass the gauge weight is

$$e^{-\beta A(U)} |\det(D_+[U] + ax)|^{2N_f} \ge 0.$$

Thus h is nonnegative and real analytic on $[x_0 - R_o, x_0 + R_o]$. On an interval where $h > 0$, put $u = -\ell^{-3}\log h$. At the reference mass the lower partition bound gives $u(x_0) \le c_g$; the upper bound of Lem. 10.21 gives $u \ge -(C + 2)$.

Fix such an x, an integer $1 \le i \le K - 1$, and $\lambda = L^{-i}$. Use the board and bumps from the proof of Prop. 10.14, with cell side L^{K-i} in lattice units. There are $N_c = (\ell/\lambda)^3$ cells. Distinct bump supports have distance at least $\lambda/4$ and lie in balls of radius $3\sqrt{3}\lambda/8$. For $|t| \le \epsilon_o$, put in one cell the even element

$$\exp \Phi\big(-(x - x_0)1_c + t\eta_c 1\big),$$

and in every other cell c' put $\exp\Phi(-(x-x_0)1_{c'})$. Their product is the mass shift q_x with the extra insertion $t\eta_c 1$. The chessboard estimate at the positive reference mass gives

$$|Z[q_x+t\eta_c 1]| \le Z[q_x+(t\eta)^{\mathrm{ch}}]^{1/N_c} Z[q_x]^{1-1/N_c}.$$

The reflected coefficients in $(t\eta)^{\mathrm{ch}}$ are t or $\bar t$. The real scalar mass shift is unchanged by reflection. Since $|x-x_0|+\epsilon_o \le r_J/2$, Lem. 10.21 applies to the full reflected source. Almost additivity at the base q_x gives

$$\operatorname{Re} E(q_x+(t\eta)^{\mathrm{ch}}) \le E(q_x) + N_c \operatorname{Re}\Lambda(t) + 19 C_\flat N_c, \quad \Lambda(t) = N_c^{-1}\sum_{c'}[E(q_x+t\eta_{c'}1)-E(q_x)].$$

The factor 19 bounds $1+(3\sqrt{3}/2)^3$.

The holomorphic function

$$\psi_c(t) = e^{-\Lambda(t)} Z[q_x+t\eta_c 1]/Z[q_x]$$

satisfies $\psi_c(0)=1$ and

$$|\psi_c(t)| \le \exp[19C_\flat + (C+2+u(x))\lambda^3] \quad (|t|\le\epsilon_o).$$

Cauchy's inequality bounds $|\psi_c'(0)|$ by this number divided by ϵ_o. Let

$$V(v) = \frac{d}{dt}\log Z_K^{\mathrm{sub}}[q_x+tv1]\Big|_{t=0}.$$

This is linear in v, and $(\log h)'(x) = -V(1)$. Summing over cells cancels the derivatives of Λ. Summing in turn over all board offsets uses the bump identity $\sum_o \eta^{(o)} = \sigma^3$, with $\sigma \ge L^{K-i}/8$, proved in Prop. 10.14. It follows that

$$|u'(x)| \le 512\epsilon_o^{-1}\lambda^{-3} e^{19C_\flat + (C+2+u(x))\lambda^3}.$$

Put $Y = C+2+\max\{u,1\}$. As long as $h>0$ and $Y \le A^\sharp$, choose $i = \lceil \frac{1}{3}\log_L Y\rceil$. Then $1\le i\le K-1$, $\lambda^3 Y \le 1 < L^3\lambda^3 Y$, and hence

$$|u'| \le c_o Y.$$

On this interval Y is Lipschitz and $|Y'| \le c_o Y$ almost everywhere. Starting at x_0 therefore gives

$$Y(x) \le (C+3+c_g)e^{c_o|x-x_0|}.$$

For completeness, let T be the first possible right endpoint of the interval on which $h>0$ and $Y\le A^\sharp$. The last estimate gives $h(T) \ge \exp[-(C+3+c_g)e^{c_o(T-x_0)}\ell^3] > 0$. If $T < x_0+R_o$, it also gives $Y(T) < A^\sharp$, so continuity extends the interval past T. This is impossible. The same argument to the left reaches x_0-R_o. Thus $u\le A^\sharp$ and $|u'|\le c_o A^\sharp$ throughout. The energy identity and $|\mathfrak{f}_K^{(x_0)}| \le C_{\mathrm{fs}}$ give the remaining conclusions. □

Corollary 10.22 (A complex neighbourhood of the window)**.** Set

$$W_\ell = (C+2+A^\sharp)\ell^3, \qquad \Lambda_\ell = c_o A^\sharp \ell^3.$$

Let ρ_ℓ be the largest $\rho \in (0,\epsilon_o/e]$ with

$$\rho\left[\frac{16 W_\ell}{3\pi\epsilon_o} + \Lambda_\ell + \frac{2}{\pi}\Lambda_\ell \log(\epsilon_o/\rho)\right] \le 1,$$

and put $w_\ell = \rho_\ell/(2(e+1))$ and $\mathcal{O}^\sharp = \{z : \operatorname{dist}(z, I^\sharp) < w_\ell\}$. There is $c_w>0$, independent of $K,\ell,\mathfrak{b}$, such that

$$w_\ell \ge \frac{c_w}{\ell^3(1+\log\ell)}, \qquad |Z_K(z)| \ge \tfrac{1}{2}e^{-A^\sharp\ell^3 - C_{\mathrm{fs}}}|e^{\mathcal{E}_K(z)}| > 0 \quad (z\in\mathcal{O}^\sharp, K\ge K_\sharp).$$

More precisely, if $x\in I^\sharp$, $|z-x|<w_\ell$ and $x_0 = x_0(x)$, then $|h_{x_0}(z)| \ge \frac{1}{2}e^{-A^\sharp\ell^3}$. ■□

Proof. Apply Lem. 10.15 to h_{x_0} on the disc of radius ϵ_o about x. This disc lies in $|z-x_0|\le R_o$, because $|x-x_0|\le R_o-\epsilon_o$. Its upper bound is $e^{(C+2)\ell^3}$, its value at the centre is at least $e^{-A^\sharp\ell^3}$, and its logarithmic derivative on the real diameter is bounded by Λ_ℓ. These are exactly the hypotheses of that lemma.

Here is an explicit lower bound for the radius. Put

$$\alpha=\frac{16(C+2+A^\sharp)}{3\pi\epsilon_o}+c_oA^\sharp,\qquad \beta=2c_oA^\sharp/\pi,\qquad A_w=1+|\log(2\epsilon_o(\alpha+\beta))|.$$

The function $\rho\ell^3[\alpha+\beta\log(\epsilon_o/\rho)]$ increases on $(0,\epsilon_o/e]$ and tends to zero at zero. The computation in Prop. 10.16 gives

$$\rho_\ell\ge\min\left\{\frac{\epsilon_o}{e},\frac{1}{2\ell^3(\alpha+2\beta L_\ell)}\right\},\qquad L_\ell=\max\{1,\log(2\epsilon_o\ell^3(\alpha+\beta))\}.$$

Since $L_\ell\le 3A_w(1+\log\ell)$, one may take

$$c_w=\frac{1}{2(e+1)}\min\{\epsilon_o/e,[2(\alpha+6\beta A_w)]^{-1}\}.$$

Finally $w_\ell<\epsilon_o$; thus $\mathcal{O}^\sharp$ is contained in the open R_o-neighbourhood of I_0. The common energy identity proves the bound for Z_K. □

Proposition 10.23 (Cumulants across zero)**.** Let $n\ge 2$. Suppose the test functions are real and Lipschitz, their supports are pairwise disjoint, and two supports have distance at least $d>0$. For $K\ge K_\sharp$ and $z\in\mathcal{O}^\sharp$,

$$|S^T_{n,K}(z;X)|\le B^w_n(d)e^{n(C+2+A^\sharp)\ell^3}\prod_i\|f_i\|_{\alpha_i},\quad B^w_n(d)=\frac{2^{n-1}C'_{\mathscr{C}}(1+d^{-3})+6^n}{\min\{\varrho/n,\epsilon_o\}^n}.$$

Here $C'_{\mathscr{C}}$ is the whole-torus support-count constant in the cube-counting estimate of [QCDM1, §10]. ■

Proof. If some f_i is zero, its source is zero and the generating function is independent of the corresponding source variable. The cumulant and the right-hand side of the asserted bound then both vanish. We may therefore assume that every $\|f_i\|_{\alpha_i}$ is positive. Choose $x\in I^\sharp$ and $x_0=x_0(x)$ as in Cor. 10.22. Let $q=q_z^{(x_0)}$ and let s_i be the source of entry i. Use radii

$$r_i=\min\{\varrho/n,\epsilon_o\}/\|f_i\|_{\alpha_i}.$$

On their closed polydisc the action norm of $s(y)=\sum_i y_is_i$ is at most ϱ. Its matrix norm is at most ϵ_o, because the supports are disjoint. Thus $q+s(y)$ lies in the interior of $\mathcal{P}^{\#}_K$. Write

$$g(y)=Z^{\mathrm{sub}}_K[q+s(y)]/Z^{\mathrm{sub}}_K[q].$$

Then $g(0)=1$ and $|g|\le B=2e^{(C+2+A^\sharp)\ell^3}$. Schwarz's lemma on lines through the origin gives $|g-1|\le(B+1)\max_i|y_i|/r_i$. On the smaller polydisc of radii $r_i/(2(B+1))$, $|g-1|\le 1/2$ and $|\log g|\le\log 2$. Cauchy's formula therefore gives

$$|\partial_{[n]}\log g(0)|\le 6^ne^{n(C+2+A^\sharp)\ell^3}\prod_i r_i^{-1}.$$

The derivative of $E(q+s(y))$ is bounded, by the base-source estimate of [QCDM1, Prop. 10.18(i)], by

$$2^{n-1}\prod_i r_i^{-1}\,C'_{\mathscr{C}}(1+d^{-3})\ell^3.$$

This uses the two separated supports; the others need only be disjoint for the matrix-norm bound. Add the two estimates and use $\ell^3\le e^{n(C+2+A^\sharp)\ell^3}$. □

Lemma 10.24 (A cell at a changed real mass)**.** The cell bounds of Lem. 10.18 hold for real $|x - x_0| \le R_o$, with $u(x) \le A^\sharp$, whenever

$$\|w\|_\sharp < 2\varrho, \qquad \|J\|_\infty \le r_J/2 - |x - x_0|.$$

■

Proof. Use the positive reference mass ax_0 in the chessboard inequality and put the mass shift into every cell element, as in Thm. P. The reflected source now lies in $\mathcal{P}_K^\#$ by Lem. 10.21. Its upper bound is $e^{\operatorname{Re} E + (C+2)\ell^3}$, and its denominator is positive by Thm. P. The base-source almost-additivity estimate is [QCDM1, Prop. 10.18(i)]; the reflected support count still costs at most 27. These are the three inputs in the proof of Lem. 10.18, and give its two inequalities with the present radius. □

Proposition 10.25 (Bounds independent of volume)**.** Suppose the supports have pairwise distance at least $d > 0$, each lies in a ball of radius R, $K \ge K_\sharp$, and $L^K \ge 16n$. For every $x \in I^\sharp$,

$$|S_{n,K}^T(x; X)| \le (1 + d^{-1})^3 B^w(R)^n (n!)^{5+p/2} \prod_i \|f_i\|_{\alpha_i}, \qquad B^w(R) = \frac{C_B e^{C+2+A^\sharp}(1+R)^3}{\min\{\varrho, \epsilon_o\}}.$$

The constants are independent of K, ℓ, b, x in these ranges. ■

Proof. Repeat the partition into hats and the board choice in Prop. 10.20, with $\eta = (16n)^{-1}$ and with matrix source radius $2\epsilon_o$ in each occupied cell. The available radius is sufficient, since $r_J/2 - |x - x_0(x)| \ge 2\epsilon_o$. Lem. 10.24 bounds each coordinate face of the normalized generating function, with $C + 2 + A^\sharp$ in place of $C + 2 + A_\epsilon$. The full mixed derivative of its logarithm is the partition sum displayed in the proof of Prop. 10.20. Its absolute coefficients sum to at most $2^n n!$. The reflected energy derivative has the same $C'_{\mathscr{C}}(1 + d^{-3})$ bound. The count of pieces costs $(32nR + 3)^{3n}$, the action norm costs $1 + 1280Mn$ per entry, and the logarithmic support count costs $(16n)^{np/2}$. Using $n^n \le e^n n!$ gives the displayed exponent and constant. Every estimate is independent of the side. □

Proposition 10.26 (Limits at real masses in the window)**.** From every sequence of data with $K_j \to \infty$, $\mathsf{b}_j \in I_g$ and $x_j \in I^\sharp$, one may extract a subsequence on which the labels and all separated cumulants converge. This holds on a fixed admitted torus and on growing admitted tori $\ell_j \to \infty$. The resulting families have permutation symmetry, the reality and internal symmetry identities, and the spatial symmetries inherited from the lattice. Their centered moments are reflection positive. On $\mathbb{R}^3$ they are translation invariant and O(3) covariant, define distributions off the diagonals, and have the reconstruction properties of the original scalar family. The bounds are those of Props. 10.23 and 10.25. ■

Proof. For each fixed order, the cutoff eventually satisfies $L^K \ge 16n$. The multilinear bounds give compactness on a countable family of smooth tests dense in the indicated norms. Take one diagonal subsequence for every order, index and tuple, and for the compact label rectangle. Slightly enlarge each separated support before approximation. The multilinear telescoping estimate extends convergence to all the tests. This works on the torus by Prop. 10.23 and on growing tori by Prop. 10.25.

At finite cutoff each correlation is a rational function of the mass, whose denominator is a power of Z_K. Numerators are obtained by differentiating the finite determinant integral. The symmetry identities hold at all positive masses, so are identities of these rational functions. They hold throughout $I^\sharp$, where Thm. P excludes poles. For a conjugation identity, conjugate the coefficients of the rational function and use that the mass is real.

Finite-torus reflection positivity holds at every real bare mass: the mass term belongs to the two reflected halves of the action and does not change the positive crossing term in the reflection decomposition. The normalization is positive by Thm. P. Center the lattice insertions by their

own means. A centered moment is a finite sum of products of cumulants of order at least two; hence it converges, and the reflection inequality passes to the limit. Moving a bond plane to a fixed plane changes smooth tests by translations tending to zero, controlled by the same bounds. The spatial and internal identities pass in the same way. The rotation and reconstruction arguments for the original scalar family in [QCDM1, §§12–13] use these bounds and reflection inequalities, and therefore apply to the limits just obtained. The nonzero two-point functions follow from Prop. 10.28 below. □

Proposition 10.27 (Holomorphic families about the window)**.** On a fixed torus and at fixed b, every cutoff sequence has a subsequence on which all separated cumulants converge locally uniformly on $\mathcal{O}^\sharp$, with all mass derivatives. The same subsequence works on the union with the positive-mass neighbourhood of Thm. O. Its values at $x \in I^\sharp$ are the limits of Prop. 10.26. Every limit obtained at moving real masses $x_j \to x$ is the value of such a holomorphic family after further extraction. ■

Proof. The partition functions have no zeros on $\mathcal{O}^\sharp$, and Prop. 10.23 bounds all cumulants there uniformly in K. Apply Montel's theorem and the dense-test diagonal argument of Thm. O. On the union use the larger of the two available bounds; the domains overlap along I_0, so their limits agree. Cauchy's formula gives convergence of derivatives. Local uniform convergence at moving masses gives the last assertion. Positivity and the symmetry identities at real masses follow as in Prop. 10.26. □

Proposition 10.28 (The short-distance laws across zero)**.** Let $0 < \vartheta < 1$, $0 < d \le 1/200$, $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ and $\Gamma, \Gamma' \in \mathfrak{B}$. At $K \ge \max\{K_\sharp(g), s(d)+1\}$, the action and scalar-bilinear two-point laws of Prop. 10.19 hold on $\mathcal{O}^\sharp$. Their errors tend to zero with the distance, uniformly over real masses in $I^\sharp$ and admitted volumes. In particular the action law still identifies b, and both leading two-point functions are nonzero. ■

Proof. Use $\epsilon_1 = \epsilon_o/2$ as the bilinear radius in the proof of Prop. 10.19. At a nearest reference mass x_0, write $\zeta = z - x_0$. On $\mathcal{O}^\sharp$ one has $|\zeta| < R_o$. The two test sources have total matrix norm less than $2\epsilon_1 = \epsilon_o$, since their supports are disjoint. Thus the shifted source lies in $\mathcal{P}_K^{\#}$. Lem. 10.21 and Cor. 10.22 give the ratio bound

$$B_c = 2e^{(C+2+A^\sharp)\ell^3 + 2C_{\mathrm{fs}}}.$$

At real $x \in I^\sharp$, the cell estimate gives instead

$$B_r = e^{C+2+A^\sharp+\mathscr{C}_*}.$$

The energy comparison of Lem. 10.17 remains valid for $|\zeta| < r_J/2$. Its Schwarz estimate uses the disc $|\zeta'| < r_J/(2\lambda_{j-1})$, so the proof and its constants do not require the smaller radius $2\varrho_J$.

For explicit bounds, replace the four packing constants in Prop. 10.19 by

$$P_c^B = \frac{4\cdot 10^4 \cdot 24^3 B_c^2}{c_B \vartheta^2 \epsilon_1^2 \ell^3}, \quad P_c^0 = \frac{4\cdot 10^6 \cdot 24^3 (40M)^2 X_g^2 B_c^2}{c_2 \vartheta^4 \varrho^2 \ell^3},$$
$$P_r^B = \frac{4\cdot 10^4 \cdot 96^3 B_r^2}{c_B \vartheta^2 \epsilon_1^2}, \quad P_r^0 = \frac{4\cdot 10^6 \cdot 96^3 (40M)^2 X_g^2 B_r^2}{c_2 \vartheta^4 \varrho^2}.$$

The action error is bounded by

$$\epsilon_0(d) + D_0 \operatorname{dist}(z, I_0) d (1 + \log_L(1/d))^{1+2\mathfrak{p}_0} + P_c^0 d;$$

the bilinear error is bounded by

$$\epsilon_B(d) + \frac{C_L a}{\vartheta^3 d} + D_B \operatorname{dist}(z, I_0) d (1 + \log_L(1/d)) + P_c^B d.$$

At real masses replace P_c^0, P_c^B by P_r^0, P_r^B. Here the base errors and D_0, D_B are those defined in Prop. 10.19. The derivation uses exactly its packing and source comparison, with the ratio bounds

just proved. Lattice translation invariance needed in real packing extends from positive masses as an identity of rational functions, as in Prop. 10.26.

For C^1 test pairs the two identities pass to a continuum limit. The mesh error disappears, and every remaining error tends to zero as $d \downarrow 0$. This also holds for convergent labels (b_j, x_j), since the main term depends continuously on b_j^{-2} and the error on $\operatorname{dist}(x_j, I_0)$. The same lower bounds for the two kernels used in Prop. 10.19 prove nonvanishing and recover b^{-2} from the normalized action two-point function. □

11. Convergence at fixed volume

We compare cutoffs K and $K+1$ at the same physical coupling and mass. After one additional averaging on the finer lattice, their block fields occupy the same lattice. The proof restarts both flows at an intermediate scale, shows that they forget their earlier activities, and compares the two restarted densities on this common space. Local terms in the source energy disappear from mixed derivatives at disjoint supports. We first prove the comparison at positive mass labels. We then apply it to derivatives of the pressure and use the analytic continuation of section 10 to obtain convergence at the other real mass labels.

This section concerns the original index set $\mathcal{I}$ of the action density and scalar bilinears. The cutoff comparison uses positive mass labels; its continuation gives full convergence at the real mass labels specified below. The enlarged current family has only the subsequential construction already proved above.

11.1. **The comparison.**

Definition 11.1. Fix $N_f \geq 1$, the chamber operator with $r = 1/2$, and an admissible choice of the constants in [QCDM1], including its source estimates and the five requirements below. Let $0 < g \leq \gamma$, $\ell \in \mathfrak{L}(g)$, and set

$$\mathsf{W}(g) = \bar{I}(g) \times [\mathsf{m}_-, \mathsf{m}_+], \qquad 0 < \mathsf{m}_- \leq \mathsf{m}_+.$$

The two data, denoted by A, B, have cutoffs $K, K+1$ and a common label $(\mathsf{b}, \mathsf{m}) \in \mathsf{W}(g)$. Thus

$$\beta^{\mathrm{A}} = L^K \mathsf{b}, \quad m_0^{\mathrm{A}} = L^{-K} \mathsf{m}, \qquad \beta^{\mathrm{B}} = L^{K+1} \mathsf{b}, \quad m_0^{\mathrm{B}} = L^{-K-1} \mathsf{m}.$$

We write $k^{\mathrm{A}} = k$, $k^{\mathrm{B}} = k+1$. These levels have the same spacing and calibration, $\mathsf{t}_k^{\mathrm{A}} = \mathsf{t}_{k+1}^{\mathrm{B}} = L^{K-k} \mathsf{b}$. Their block-field spaces, Haar measures, gauge groups, averaging and gauge-fixing factors agree. Their reconstructed backgrounds, field cuts and density tests can differ, since they use different bare lattices. Sources on T_ℓ are sampled on the sites or plaquette barycentres of each bare lattice.

For an n-tuple define

$$\begin{aligned} \phi_2(n) &= \frac{1}{2(6n+11)}, \\ \phi_1(n) &= \frac{c_\#}{10} \phi_2(n), \\ s &= \lceil \phi_2(n) K \rceil, \\ u &= \lceil \phi_1(n) K \rceil, \\ t_0 &= K - s, \\ t_1 &= K - u. \end{aligned} \tag{11.i}$$

The number $c_\#$ is fixed in Def. 11.5. For a depth d, put

$$x_{[d]} = L^d \mathsf{b}, \quad N_{[d]} = \ell^3 L^{3d}, \quad \mathsf{y}_{[d]} = (\log x_{[d]})^{2\mathfrak{p}_0}, \quad \bar{\mathsf{y}}_{[d]} = (\log(L^d X_g))^{2\mathfrak{p}_0}.$$

The common spaces at depths s, u are $\mathcal{Y} = \mathcal{V}(\Lambda_{s,\ell})$ and $\mathcal{X} = \mathcal{V}(\Lambda_{u,\ell})$, with Haar probability measures. Write

$$\mathsf{In}_{[u]} = \{V \in \mathcal{X} : \varpi(V) < 2\varepsilon(x_{[u]})\}, \qquad W_k = \mathcal{M}^{k-t_0} W \quad (W \in \mathcal{Y}).$$

An index (k) on a gauge object denotes level $k^\bullet$ of $\bullet \in \{\mathrm{A},\mathrm{B}\}$. In particular, $\mathsf{Cut}^\bullet_{(k)}$, $\chi^\bullet_{(k)}$, $\mathfrak{G}_{(k)}$ and $\mathsf{wab}^\bullet(\mathsf{c},W)$ are its field cut, fluctuation test, gauge-fixing factor and weighted background action. The last is linear in c. The weights $\mathsf{k}^\bullet$ on $\mathcal{Y}$, including their top factor $\mathbf{1}_{\mathsf{In}_{[u]}}(W_{t_1})$, are specified in the comparison of tests below.

A tuple is $X=((\alpha_i,f_i))_{i\le n}$, where $\alpha_i\in\mathcal{I}$ and f_i is nonzero, real and Lipschitz on T_ℓ. Set

$$\|f\|_\sharp=\max\{\|f\|_\infty,40M\,\mathrm{Lip}\,f\},\qquad s(z)=\sum_i z_i\mathbf{s}_i=(w_z,J_z),$$

where $\mathbf{s}_i$ is its physical source. With the source radius ϱ_{α_i} and the constant $0<c_\lambda\le 1/8$ of Prop. 11.13, let

$$R_i=\frac{(1-c_\lambda)\varrho_{\alpha_i}}{2n\|f_i\|_\sharp},\qquad \varrho_i=\frac{\varrho_{\alpha_i}}{n\|f_i\|_\sharp},\qquad \mathbb{Q}=\{z:|z_i|<2R_i\}. \tag{11.ii}$$

Then $\|w_z\|_\sharp<(1-c_\lambda)\varrho$ and $\|J_z\|_\infty<(1-c_\lambda)\varrho_J$ on $\mathbb{Q}$. Here s denotes a depth in numerical bounds, and a source only when it is an argument, as in Φ_s.

The additional requirements on the construction constants are the following. The constants on the right are defined in the proofs indicated, before the constant chosen on the left.

(i) The bounds for fermion terms satisfy

$$C^{\square,\mathrm{s}}_{\mathrm{leg}}\ge c^{\mathrm{leg}}_{\mathrm{o}}+N_{\mathrm{u}}c^{\mathrm{leg}}_{\mathrm{u}},\qquad C^{\mathrm{s}}_{\mathrm{leg}}\ge\phi(\tfrac12 e^{18\mathfrak{p}_0})C^{\square,\mathrm{s}}_{\mathrm{leg}}.$$

These are chosen after C^{s}_w and before $C^{\mathrm{s}}_{\mathrm{lay}}$, as in the proof of Prop. 11.13.

(ii) The history-summation threshold uses $c_{\mathrm{tl}}=(c_\star+3c_\Theta)/4$ and $\mathsf{w}_{\mathrm{tl}}=c_\Theta/c_{\mathrm{tl}}$: $\log\gamma^{-2}\ge\mathsf{Y}_{\mathrm{fl}}$ when $c_a=0$, and $\log\gamma^{-2}\ge\mathsf{Y}_{\mathrm{f2}}$ when $c_a>0$. The thresholds are those of the history bound with $(c_{\mathrm{tl}},C,C^F_\flat+C'_{\mathfrak{E}},\eth)$, and also (C_{vol},c_a) in the second case. The respective initial conditions on $\mathfrak{p}_0$ are retained.

(iii) For free earlier activities require

$$K^F_0\delta(\gamma^{-2})+b^{\mathrm{blk}}(\gamma)\le\tfrac14 c_{\mathrm{cl}},\qquad b^{\mathrm{ms}}(\gamma)\le\tfrac{3}{16}\varepsilon_\flat,\qquad b^{\mathrm{lay}}(\gamma)+b^{\mathrm{pt}}(\gamma)\le\tfrac14\varepsilon_\flat,$$

where

$$b^{\mathrm{pt}}(\gamma)=2\Theta^\sharp(\gamma)C_{\mathrm{tab}}S_\omega C^{\mathrm{pt}}_2\phi(\gamma^{-2})^2,\qquad C^{\mathrm{pt}}_2=8M^3C_2.$$

These functions tend to zero with γ; the estimates are proved in Prop. 11.14.

(iv) At the original choice of A_0, after A_1, require

$$(1-L^{-3/2})A_0>2B_3C_HA_1,\qquad C_H=4C_{\mathrm{db}}\max\{2,K^{\mathrm{s}}_B(\rho_s)\}.$$

This is used in the comparison of tests.

(v) At the choice of $C_{\mathscr{C}}=27C^0_{\mathscr{C}}$, after $C_{\mathfrak{E}}$, require

$$C_{\mathscr{C}}\ge C_{\mathfrak{E}}C^\natural_{\mathscr{C}},\quad C^\natural_{\mathscr{C}}=e^{\kappa/2}C_{\mathrm{cnt}}S_\sigma,\quad S_\sigma=\sum_{t\ge0}(2t+3)^3e^{-\kappa_\sigma Mt/2}.$$

This supplies the contact-term estimate of Prop. 11.12.

After each choice, subsequent constants are recomputed in their original order. Conditions (i), (iv), (v) increase freely chosen constants; (ii), (iii) decrease the final choice of γ. They are requirements on the construction, distinct from the large-cutoff requirements in Def. 11.5. ∎

The gauge comparison uses Prop. A.5 and Prop. A.9. Their two-lattice setting applies: the admissible constants of [QCDM1] restrict to the gauge constants, and $\gamma^{-2}\le\mathsf{b}\le X_g\le L(X_g+\bar B)$ places the label in the interval allowed on the side ℓ. For $K\ge2$ the two partner lattices are therefore defined; the top-level hypotheses hold once $L^u\mathsf{b}\ge\Theta_\star$. These gauge objects use neither the source radius nor the scalar source-activity bound chosen in that companion. Here those quantities always have the values for the theory with fermions.

11.2. **Restarting at a common depth.**

Definition 11.2. Assume $1 \le u < s < K$. At level $t_1^\bullet$, let $\Phi_s^\bullet$ be the exponent of the original history with no large-field regions, and $c^\bullet(s) = E^\bullet_{t_1^\bullet}(s)$ its extracted scalar energy. Its total after Berezin integration is

$$T^\bullet(z) = \int \mathfrak{w}_{h_\emptyset^{(t_1^\bullet)}}(U) e^{-A(\beta^\bullet - a_{K^\bullet}^{-1} w_z, U)} \det(D[U] + m_0^\bullet - a_{K^\bullet}\hat{J}_z)\, dU. \tag{11.iii}$$

It is entire in z.

At level $t_0^\bullet$, restart from the density

$$\begin{aligned}\hat{d}^\bullet_{t_0}(s)(dW) = \chi^\bullet_{t_0}(T_\ell)(W) \exp\big[-\mathsf{wab}^\bullet(\mathsf{c}_w, W) - \langle \bar\Psi, \mathbb{D}^\bullet_{t_0^\bullet}(\mathsf{U}^\bullet_{t_0^\bullet}(W); m_0^\bullet - a_{K^\bullet}\hat{J})\Psi\rangle \\ - L^{-t_0^\bullet}\mathsf{S}_1(\hat\Sigma_w \psi^{\mathrm{bg}})\big]\, dW,\end{aligned} \tag{11.iv}$$

where

$$\mathsf{c}_w = L^s(\mathsf{b} - w), \qquad \hat\Sigma_w(x) = \frac{F^{\mathrm{UV}}}{L^s(\mathsf{b} - w(x))}, \qquad F^{\mathrm{UV}} = 3\Sigma(r) \tag{11.v}$$

and F^{UV} is given by Prop. B.2. The density test is that of lattice $\bullet$ at its own bare plaquettes. There are no activities or extracted scalar terms at the start. The function $\hat\Sigma_w$ is the same on both lattices and is independent of J, K, t_0.

The density test implies $\varpi(W) < 2\varepsilon(L^s\mathsf{b}) \le \varepsilon_{\mathcal{W}}\alpha_0 \le \bar\varepsilon$. The background in eq. 11.iv is therefore defined. Moreover $L^{t_0^\bullet}\mathsf{c}_w = \beta^\bullet - a_{K^\bullet}^{-1} w$, so its action has the required physical source normalization.

The zero-source restart is also made on $\mathbb{Z}^3$, using these local formulas without the torus density test. Its activities determine its own extracted numbers $(\hat x_i, \hat s_i, \hat e_i, \hat b_i, \hat\mu_i)$. In particular $\hat x_{t_0^\bullet} = L^s\mathsf{b}$ and $\hat s_{t_0^\bullet} = F^{\mathrm{UV}}/(L^s\mathsf{b})$. At sources, the chain starts at $\hat\sigma_{t_0^\bullet}(s) = \hat\Sigma_w - F^{\mathrm{UV}}/(L^s\mathsf{b})$ and satisfies

$$\hat\sigma_{k+1}(s)(x) = L\hat\sigma_k(s)(x) + \hat\nu_{k+1}(w, -a_{K^\bullet}J, (\hat\sigma_i(s))_{t_0^\bullet \le i \le k})(y_{k+1}(x)) - \hat\mu_{k+1}.$$

The same construction is made on every allowed torus, with the same infinite-lattice extraction. Thus its extracted numbers are independent of the side.

At level $t_1^\bullet$, denote its exponent by $\hat\Phi_s^\bullet$, its total by $\hat T^\bullet(z)$, and its extracted energy by

$$\hat c^\bullet(s) = \sum_{t_0^\bullet < j \le t_1^\bullet} \left(\hat e_j n_j + \sum_{X \in \mathcal{D}_j} \hat e_j(X; s) \right).$$

The contact term is $\mathfrak{c}^\bullet(z) = c^\bullet(s(z)) - \hat c^\bullet(s(z))$.

Let $\hat{\mathsf{k}}^\bullet$ be the product, read on $\mathcal{Y}$, of all tests and gauge-fixing factors along the empty history from $t_0^\bullet$ to $t_1^\bullet$. Let $\mathbb{M}^\bullet(W; \hat\Sigma_w)$ be the matrix of the quadratic form $L^{-t_0^\bullet}\mathsf{S}_1(\hat\Sigma_w\psi^{\mathrm{bg}})$, and put

$$\hat{\mathbb{D}}^\bullet(W; s) = \mathbb{D}^\bullet_{t_0^\bullet}(\mathsf{U}^\bullet_{t_0^\bullet}(W); m_0^\bullet - a_{K^\bullet}\hat J) + \mathbb{M}^\bullet(W; \hat\Sigma_w),$$

$$\hat F_z^\bullet(W) = \mathbf{1}_{\mathsf{Cut}^\bullet_{(t_0)}}(W) e^{-\mathsf{wab}^\bullet(\mathsf{c}_{w_z}, W)} \det \hat{\mathbb{D}}^\bullet(W; s(z)), \qquad T^{\mathrm{sf},\bullet}(z) = \int \mathsf{k}^\bullet \hat F_z^\bullet\, dW. \tag{11.vi}$$

An undefined tower in this formula is assigned value zero. For $s \ge s_F$, Prop. 11.7 proves that it is defined throughout the field cut and that $\hat F_z^\bullet$ is bounded and Borel. The density-test set lies in this field cut, so the Berezin integral of the starting density is $\chi^\bullet_{t_0}(T_\ell)\hat F_z^\bullet\, dW$. The restarted flow requires $u \ge u_*$; the field-cut comparison also requires $L^u\gamma^{-2} \ge \Theta_\star$. ■

Definition 11.3. Fix one lattice and suppress $\bullet$. Let $t_0 < t_b \le t_1$. For $j \le t_b$, set $\hat E^{(j)} = 0$ when $j \le t_0$, and let $\Delta^{(j)}$ be $E^{(j)} - \hat E^{(j)}$ after subtracting the value of its degree-zero part at the trivial pair. Use the same subtraction pointer by pointer. Let $d_b(s)$ be the sum of those subtracted values with the opposite sign, and put

$$\hat w_b = L^{-(K-t_b)}(x_{t_b} - \hat x_{t_b}), \qquad \delta_{t_b}(s) = s_{t_b} + \sigma_{t_b}(s) - \hat s_{t_b} - \hat\sigma_{t_b}(s).$$

Then $\hat{w}_b$ is real, and the two exponents obey

$$\hat{\Phi}_s^{(t_b)} - \Phi_s^{(t_b)} = d_b(s) + a_K^{-1} A(\hat{w}_b, \mathsf{U}) + L^{-t_b} \mathsf{S}_1(\delta_{t_b}(s)\psi^{\mathrm{bg}}) - \sum_{j \le t_b} \sum_X \Delta^{(j)}(X; s). \tag{11.vii}$$

For $\lambda \in \mathbb{C}$, define the Grassmann-valued measure

$$\rho_{s,\lambda}^{(t_b)}(dV) = e^{E_{t_b}(s)} \mathsf{Stay}(V) e^{\Phi_s^{(t_b)} + \lambda(\hat{\Phi}_s^{(t_b)} - \Phi_s^{(t_b)})} \mathscr{M}_{h_\emptyset}^{\mathrm{fr}}(dV). \tag{11.viii}$$

At $\lambda = 0$ this is the original term. At $\lambda = 1$ it is the restarted term multiplied by $e^{E_{t_b}(s) - \hat{c}_b(s)}$. Its earlier activities are $E^{(j)} - \lambda\Delta^{(j)}$, its source is $(w + \lambda\hat{w}_b, J)$, and its mass term uses $s_{t_b} + \sigma_{t_b}(s) - \lambda\delta_{t_b}(s)$.

For $k \ge t_b$, retain the histories

$$\mathcal{H}_k^{(t_b)} = \{h \in \mathcal{H}_k : h|_{t_b} = h_\emptyset^{(t_b)}\}.$$

Their terms are the images of eq. 11.viii under the subsequent weighted kernels and partial Berezin integrals, with the original extracted numbers. Thus the terms depend only on this starting measure. A density $g_h[s, \lambda]$ with respect to $\mathscr{M}_h^{\mathrm{fr}}$, when it exists, is understood coefficient by coefficient. Any two such densities agree almost everywhere; their integrals, integrated fibre norms and total Berezin integrals agree.

The difference families are required to have the following form.

(F1) Each pointed term is even, local, jointly gauge invariant, Borel, holomorphic on the fixed complex domains of regular pairs and holomorphic in the source. It is invariant under $\mathrm{U}(N_f) \times \mathrm{U}(N_f)$ acting jointly on the fermions and J, by conjugation on the latter.

(F2) Its scalar part vanishes at the trivial pair and satisfies the pointed sum rule.

(F3) On an open source set containing $\mathcal{P}$, it is a free family in the sense of Def. C.2. Its fermion terms are obtained from $F_j(s) - \hat{F}_j(s), G_j - \hat{G}_j, \mu_j - \hat{\mu}_j$, with the corresponding infinite-lattice families at zero source. Here F_j, G_j are the terms before extraction, at the source coordinates and at zero respectively. The hatted data are zero for $j \le t_0$. All these families are invariant under the common scalar phase on fermions. Let their sizes be $r_j^\psi(s)$ and $r_j^0(s)$.

For a pointed family define

$$\|G\|_j^{\mathrm{pt}} = \sup_\square \sum_{X \supset \square} e^{\kappa d_j(X)} \sum_{z \in X} \sup_{\mathsf{U}} \|G(X; \mathsf{U}, \cdot; z)\|_{h^{(j)}}, \qquad |G_0|_j = \sup_{X,z} e^{(\kappa+1)d_j(X)} \sup |G_0(X; \cdot; z)|.$$

The suprema are on the fixed domains of regular complex pairs; in particular $|\Delta_0^{(j)}|_j = r_j^0$. The lag bounds with $\Lambda > 0$ at level t_b are

$$\sum_{j \le t_b} \omega_{k+1-j} r_j^\psi(s) \le \Lambda^{-1} C_w^{\mathrm{s}} \phi_k, \qquad t_b \le k \le K, \tag{11.ix}$$

$$\sum_{j \le t_b} L^{-(k-j)} |\Delta_0^{(j)}(\cdot; s)|_j \le \Lambda^{-1} E_0^{\mathrm{s}}, \qquad t_b \le k \le K, \tag{11.x}$$

$$\|\delta_{t_b}(s)\| \le \Lambda^{-1} B_\sigma \phi_{t_b}, \qquad |\hat{w}_b| \le \Lambda^{-1} \varrho. \tag{11.xi}$$

Here $\omega_j = C_\omega L^{j/4}$ and $\phi_k = \phi(x_k)$. ■

Definition 11.4. At zero source insert, in each restarted step $t_0 \le k < t_1$, the factor

$$\mathsf{m}_{k,\lambda}^\bullet(V) = \prod_{b \in \mathfrak{X}_{(k)}^+} (1 + \lambda \bar{\chi}_{(k),b}^\bullet(V)),$$

where $\bar{\chi}_{(k),b}^\bullet$ indicates that the fluctuation is defined and has modulus at least $\delta(x_{(k)})/2$. No mark is inserted after level $t_1^\bullet$. Its total is

$$Z_\lambda^\bullet = \int_{\mathcal{Y}} \hat{\mathsf{k}}^\bullet(W) \prod_{t_0 \le k < t_1} \mathsf{m}_{k,\lambda}^\bullet(W_k) \hat{F}_0^\bullet(W)\, dW. \tag{11.xii}$$

This is a polynomial in λ, $Z_0^\bullet = \hat{T}^\bullet(0)$, and

$$\partial_\lambda Z_\lambda^\bullet|_0 = \sum_{t_0 \le k < t_1} \sum_{b \in \mathfrak{X}^+_{(k)}} \int \hat{\mathsf{k}}^\bullet(W) \bar{\chi}^\bullet_{(k),b}(W_k) \hat{F}_0^\bullet(W)\, dW.$$

The integral identities, including the equality at zero, are proved in Lem. 11.16. ∎

11.3. **Constants and the convergence theorem.**

Definition 11.5. Let $c_\#$ be the largest $c \in (0, 1/8]$ for which

$$L^c \le 11/10, \qquad \sum_{j \ge 1} L^{-j(1/4-c)}(1+j)^{\mathfrak{p}_0} \le \tfrac{3}{2} S_{1/4}, \qquad S_{1/4} = \sum_{j \ge 0} L^{-j/4}(1+j)^{\mathfrak{p}_0}.$$

It is positive: at $c = 0$ the sum is $S_{1/4} - 1$, and it is continuous and increasing for $c < 1/4$. Put

$$c_n = \tfrac{1}{2} c_\# \phi_2(n) = \frac{c_\#}{4(6n+11)}, \qquad P_n = \frac{2}{\phi_1(n)} = \frac{40(6n+11)}{c_\#}.$$

These depend only on $L, \mathfrak{p}_0, n$.

The preceding construction constants determine the restart threshold u_*, the constants C_Λ, p_Λ in the lag estimate, and C_D, K_F, s_F in the comparison of the starting operators. They also determine

$$C'_E = \frac{C_{\mathfrak{E}} C_{\mathrm{cnt}}}{M^3(L^3-1)}, \quad 0 < c_\lambda \le 1/8, \quad \theta_{\mathrm{dis}} = \tfrac{1}{2}(c_\star - c_\Theta) > 0, \quad \theta_{\mathrm{tl}} = \tfrac{1}{2}\theta_{\mathrm{dis}}, \quad \kappa_{\mathrm{t}} = \frac{c_{\mathrm{cl}}}{128 M^3}.$$

For the chamber operator $\nu = 8N_f$ and $\nu_\natural = 48\nu$. The number of generators per bare site uses $\nu_\natural$; the block Dirac matrices use ν. Set

$$\mathsf{M}_{g,\ell} = \exp\{[C + 2 + C_P(1 + \log X_g)]\ell^3\}.$$

The constants C, C_P are the upper and lower partition bounds, with the source constants chosen as in Def. 11.1.

At a cutoff K, define the following errors:

$$R^F = \mathsf{M}_{g,\ell} \exp[-\theta_{\mathrm{dis}} A_0^2 (\log(\tfrac{1}{2} L^u \gamma^{-2}))^{2\mathfrak{p}_0}], \qquad \hat{R}^F = 2eR^F, \tag{11.xiii}$$

$$\Lambda = C_\Lambda^{-1}(K+1)^{-p_\Lambda} L^{c_\#(s-u)}, \qquad r_\lambda = \tfrac{1}{2}(3L^{1/2})^{-u} c_\lambda \Lambda, \tag{11.xiv}$$

$$\epsilon_{\mathrm{ro}} = \frac{2(4\mathsf{M}_{g,\ell})^{n+1} + 2C'_E N_{[u]}}{r_\lambda}, \tag{11.xv}$$

$$\epsilon_{\mathrm{cl}} = 4\mathsf{K}_{\mathrm{w}} c_\rho^2 A_0^2 (1 + 2\varrho\gamma^2) \bar{\mathsf{y}}_{[K]} N_{[s]} K^{1/2} L^{-(K-s)/2}. \tag{11.xvi}$$

The last bound uses the gauge comparison constants $\mathsf{K}_{\mathrm{w}}, c_\rho$. For the fermion determinant put

$$\bar{\varepsilon}_{\mathrm{r}} = 2c_\rho \varepsilon(L^s \gamma^{-2}), \qquad \varepsilon(x)^2 = A_0^2 (\log x)^{2\mathfrak{p}_0}/x, \tag{11.xvii}$$

$$\delta_F = K_F \left[L^{-(K-s)} + \bar{\varepsilon}_{\mathrm{r}} (N_{[s]} L^{-(K-s)})^{1/3} \right], \qquad \epsilon_F = \nu N_{[s]} \frac{8L^s}{\mathsf{m}_-} \delta_F. \tag{11.xviii}$$

Finally define

$$\Lambda_{\mathrm{mk}} = \exp[\kappa_{\mathrm{t}} A_1^2 (\log(L^u \gamma^{-2}))^{2\mathfrak{p}_0}], \qquad \epsilon_{\mathrm{mk}} = 2e\mathsf{M}_{g,\ell} \Lambda_{\mathrm{mk}}^{-1/2}, \tag{11.xix}$$

$$\epsilon_{\mathrm{sh}} = 2(\epsilon_{\mathrm{mk}} + 3sN_{[s]} \hat{R}^F), \qquad C_\psi = \tfrac{3}{2} c_\rho^2 A_0^2 \gamma^2 \bar{\mathsf{y}}_{[s]} + \frac{8\nu C_D}{\mathsf{m}_-}. \tag{11.xx}$$

The last C_ψ is a bound at this cutoff, distinct from the fixed activity constant with that name.

For a tuple, write $\varrho_{\min} = \min\{\varrho, \varrho_J\}$ and

$$\hat{r}_i = \min\{R_i, (4nN_{[s]} C_\psi \|f_i\|_\sharp)^{-1}\}, \qquad \hat{\mathbb{D}} = \{z : |z_i| \le \hat{r}_i\}.$$

For $\bullet = \mathrm{A}, \mathrm{B}$, set

$$s_i^{\bullet} = \begin{cases} 6\ell^3 L^{4K^{\bullet}} \|f_i\|_\infty, & \alpha_i = 0, \\ 2\nu_\natural \ell^3 L^{3K^{\bullet}} \|f_i\|_\infty / \mathsf{m}_-, & \alpha_i \in \mathfrak{B}. \end{cases}$$

When $n \geq 2$ and $\mathsf{d} > 0$, let

$$\epsilon_{\mathrm{mx}} = 2^n C_{\mathscr{C}} \ell^3 L^{3u} e^{-\kappa_{\mathrm{E}} \mathsf{d} L^u} \prod_i \varrho_i^{-1}.$$

The proof uses the following conditions on K:

(k0) $1 \leq u < s < K$.
(k1) $L^u \gamma^{-2} \geq \Theta_\star$, $u \geq u_*$, $s \geq s_F$, and $e^{-c_{\mathrm{t}} L^u} \leq L^{-c_\#(s-u)}$.
(k2) $r_\lambda \geq 16 \mathsf{M}_{g,\ell}$.
(k3) $\hat{R}^F \leq 1/40$, $\epsilon_{\mathrm{cl}} + \epsilon_F \leq 1/40$, and $\epsilon_{\mathrm{sh}} \leq 1/40$.
(k4) $256 \mathsf{K}_u c_\rho^2 A_0^2 N_{[s]} L^{-(K-s)} \leq \gamma_{\mathrm{r}} A_1^2$.
(k5) $\kappa_{\mathrm{E}} \mathsf{d} L^u (L-1) \geq \log(2L^3)$.
(k6) $K \geq \max\{K_{\mathrm{tt}}(n), K_{\mathrm{mk}}(g,\ell,n)\}$.

Here K_{tt} is the threshold for the top test, and $K_{\mathrm{mk}} = \max\{K_{\mathrm{rs}}, K_\flat\}$ is the threshold for the marked restart. They are obtained in Props. 11.10 and 11.11; their conditions involve only the already fixed constants and the displayed quantities at a cutoff. Thus none of these definitions changes an earlier construction constant. Only (k5) depends on d.

Let $K_n(g,\ell)$ be the least integer at least two such that (k0)–(k4), (k6), and $\kappa_{\mathrm{E}} L^{\phi_1(n)K/2}(L-1) \geq \log(2L^3)$ hold for every $K \geq K_n(g,\ell)$. Set

$$\sigma_K = \max\{6\ell^3 L^{4(K+1)}, 2\nu_\natural \ell^3 L^{3(K+1)}/\mathsf{m}_-\}, \quad \hat{\varrho}_K = \max\left\{\frac{2n}{(1-c_\lambda)\varrho_{\min}}, 4nN_{[s]}C_\psi\right\},$$

and

$$\begin{aligned} \mathsf{B}_n(K) &= 20R^F(4n\sigma_K)^n + 2\epsilon_{\mathrm{ro}}\left(\frac{2n}{(1-c_\lambda)\varrho_{\min}}\right)^n \\ &\quad + [8(\epsilon_{\mathrm{cl}}+\epsilon_F) + 12\epsilon_{\mathrm{sh}} + 16\hat{R}^F]\hat{\varrho}_K^n \\ &\quad + 2^{n+2} C_{\mathscr{C}} \ell^3 L^{3u} e^{-\kappa_{\mathrm{E}} L^{\phi_1(n)K/2}} (n/\varrho_{\min})^n, \\ C_n(g,\ell) &= \sup_{K \geq K_n(g,\ell)} L^{c_n K} \mathsf{B}_n(K). \end{aligned} \tag{11.xxi}$$

The threshold exists and this supremum is finite by Lem. 11.20. Both are uniform in the labels in $\mathsf{W}(g)$. ∎

Theorem Q. Use the chamber model and constants of Def. 11.1, and write $\partial_z^n = \partial_{z_1} \cdots \partial_{z_n}|_{z=0}$.

(i) For any n-tuple and $K \geq 2$ satisfying (k0)–(k4), (k6),

$$\begin{aligned} &\left| S^T_{n;K+1}(X) - S^T_{n;K}(X) - \partial_z^n(\mathfrak{c}^{\mathrm{B}} - \mathfrak{c}^{\mathrm{A}})(0) \right| \\ &\quad \leq 2\epsilon_{\mathrm{ro}} \prod_i R_i^{-1} + [8(\epsilon_{\mathrm{cl}}+\epsilon_F) + 12\epsilon_{\mathrm{sh}} + 16\hat{R}^F] \prod_i \hat{r}_i^{-1} \\ &\qquad + 10R^F \left[\prod_i (4n s_i^{\mathrm{A}}) + \prod_i (4n s_i^{\mathrm{B}}) \right]. \end{aligned} \tag{11.xxii}$$

The suppressed labels are $\ell, L^K\mathsf{b}, L^{-K}\mathsf{m}$ at cutoff K, and the corresponding labels at cutoff $K+1$.

(ii) If $n \geq 2$, the supports are pairwise disjoint, two are at distance at least $\mathsf{d} > 0$, and (k5) holds, then $|\partial_z^n \mathfrak{c}^{\bullet}(0)| \leq 2\epsilon_{\mathrm{mx}}$.

(iii) Under the support assumptions in (ii), for

$$K \geq P_n \max\{0, \log_L(1/\mathsf{d})\} + K_n(g,\ell),$$

one has

$$|S^T_{n;K+1}(X) - S^T_{n;K}(X)| \leq C_n(g,\ell) L^{-c_n K} \prod_i \|f_i\|_\sharp. \tag{11.xxiii}$$

The constants c_n, P_n depend only on $L, \mathfrak{p}_0, n$; K_n, C_n also depend on the construction constants, g, ℓ. None depends on the label in $\mathsf{W}(g)$, the tuple, or its separation. The theorem holds for every $N_f \geq 1$ and every index in $\mathcal{I}$. ■□

The proof follows the gauge comparison of Prop. A.20. First discard histories with a large-field region before depth u. On each lattice replace the remaining total $T^\bullet$ by its restarted total $\hat{T}^\bullet$, keeping the extracted contact term. The loss of memory between depths s and u controls this replacement. Finally compare the two starts on $\mathcal{Y}$: the gauge backgrounds and block Dirac operators are close, and the failures of the two sets of tests have small total mass. All logarithmic derivatives are taken after the fermions have been integrated. The proof and the summation of its errors appear after these estimates have been established.

Corollary 11.6. Under the hypotheses of Thm. Q, let $n \geq 2$ and let the tuple have pairwise disjoint supports. Choose $\mathsf{d} > 0$ smaller than the distance between two of them. For integers $K' \geq K \geq P_n \max\{0, \log_L(1/\mathsf{d})\} + K_n(g,\ell)$,

$$|S^T_{n;K'}(X) - S^T_{n;K}(X)| \leq \frac{C_n(g,\ell)}{1 - L^{-c_n}} L^{-c_n K} \prod_i \|f_i\|_\sharp. \tag{11.xxiv}$$

Consequently the full sequence has a limit $S_n^{(\ell,\mathsf{b},\mathsf{m})}(X)$, uniformly for $(\mathsf{b},\mathsf{m}) \in \mathsf{W}(g)$, and the limit is continuous in these labels. If the labels converge while $K \to \infty$, the same sequence converges to the limit at the limiting label. On the smooth disjoint-support test space, this is the unique fixed-torus continuum family at that label from [QCDM1]; it inherits all the properties proved there. ■□

Proof. Sum eq. 11.xxiii over the intervening cutoffs. The geometric series gives eq. 11.xxiv, including its uniformity in the labels. At a finite cutoff the normalized integrals are continuous in the positive labels, so their uniform limit is continuous. Uniform convergence followed by continuity proves the assertion for varying labels. The subsequential construction of [QCDM1] must agree with this limit on every test tuple, which proves the last assertion. □

11.4. **Estimates used in the cutoff comparison.** The following estimates separate the comparison into its three tasks: compare the two starting quadratic forms, control the histories removed at either end, and propagate the difference between the two flows. Throughout, the datum, depths, errors and conditions (k0)–(k2) are those of Defs. 11.1 and 11.5. The constants are uniform in the label $(\mathsf{b},\mathsf{m}) \in \mathsf{W}(g)$.

Proposition 11.7 (The two starting operators)**.** There are $C_D, K_F \geq 1$ and an integer $s_F \geq 1$, determined after γ by the construction constants, with $L^{s_F}\gamma^{-2} \geq \Theta_\star$, for which the following holds. Assume (k0) and $s \geq s_F$. Let w and J be Lipschitz on T_ℓ, with J taking values in the complex span of $\mathfrak{B}$, and

$$\|w\|_\sharp < \varrho, \qquad \|J\|_\sharp := \max\{\|J\|_\infty, 40M \operatorname{Lip} J\} < \varrho_J.$$

The sources are sampled as in Def. 11.1. The operators $\hat{\mathbb{D}}^\bullet(W; w, J)$ of Def. 11.2 act on $\ell^2(\Lambda_{s,\ell}) \otimes \mathsf{F}$, with the inner product in depth-s units.

(a) On $\mathsf{Cut}^\bullet_{(t_0)}$, the operator is independent of the representative of the minimal orbit, bounded and Borel in W, and holomorphic in the sources. Moreover,

$$\operatorname{Re} \hat{\mathbb{D}}^\bullet(W; w, J) \geq \tfrac{1}{8} L^{-s} \mathsf{m}.$$

(b) For two sources in the stated domain,

$$\|\hat{\mathbb{D}}^{\bullet}(W;w,J)-\hat{\mathbb{D}}^{\bullet}(W;w',J')\|\le C_D L^{-s}(\|J-J'\|_\infty+\|w-w'\|_\infty).$$

(c) On the intersection of the two field cuts,

$$\|\hat{\mathbb{D}}^{\mathrm{B}}(W;w,J)-\hat{\mathbb{D}}^{\mathrm{A}}(W;w,J)\|\le K_F\left[L^{-(K-s)}+\bar{\varepsilon}_{\mathrm{r}}\min\{(N_{[s]}L^{-(K-s)})^{1/2},1\}\right]\le\delta_F.$$

(d) If w is real and $\gamma_4 J(x)^{\dagger}\gamma_4=J(x)$ for every x, then $\det\hat{\mathbb{D}}^{\bullet}(W;w,J)>0$.

■

Proof. Cor. A.44 gives the constants and all four assertions. Its lower bound is $2L^{-s}\mathsf{m}/9$, hence implies the one stated here. The last bound in (c) uses $\min\{x^{1/2},1\}\le x^{1/3}$ for $x\ge 0$. □

Proposition 11.8 (Discarding large-field histories)**.** Assume (k0). For either lattice, its original partition function $Z^{\bullet}$ and the total $T^{\bullet}$ of Def. 11.2 satisfy

$$0\le Z^{\bullet}-T^{\bullet}(0)\le R^F Z^{\bullet}.$$

■

Proof. Apply Prop. B.43 at $t=t_1^{\bullet}$. The calibration is $L^u\mathsf{b}\ge L^u\gamma^{-2}$; $\mathsf{Q}_\ell\le\mathsf{M}_{g,\ell}$ gives precisely the error R^F. This application is recorded in appendix B.18. □

Proposition 11.9 (Comparing the totals at depth u)**.** Assume (k0)–(k2), and let X be any n-tuple. The functions $T^{\bullet},\hat{T}^{\bullet},\mathfrak{c}^{\bullet}$ are holomorphic on $\mathbb{Q}$, and $T^{\bullet}(0),\hat{T}^{\bullet}(0)>0$. Their logarithms near zero satisfy

$$\left|\partial_{z_1}\cdots\partial_{z_n}[\log T^{\bullet}-\log\hat{T}^{\bullet}-\mathfrak{c}^{\bullet}](0)\right|\le\epsilon_{\mathrm{ro}}\prod_{i=1}^{n}R_i^{-1}.$$

Here C_E' in ϵ_{ro} is determined after γ by the construction constants. ■

Proof. The holomorphic interpolation and Cauchy estimate are Prop. B.42 and appendix B.18; the positive lower bound for the restarted total is Prop. B.31. □

Proposition 11.10 (Removing the top test)**.** There is $K_{\mathrm{tt}}(n)$, determined after γ by the construction constants and n, such that (k0)–(k2) and $K\ge K_{\mathrm{tt}}(n)$ imply, for either lattice,

$$\int_{\mathcal{Y}}|\mathsf{k}^{\bullet}-\hat{\mathsf{k}}^{\bullet}|\hat{F}_0^{\bullet}\,dW\le R^F\hat{T}^{\bullet}(0)\le\hat{R}^F\hat{T}^{\bullet}(0).$$

The weights and starting density are those of Def. 11.2 and the comparison of tests. ■

Proof. This is Prop. B.72. Its depth condition follows from the definition of $K_{\mathrm{tt}}(n)$; (k1) supplies $s\ge s_F$, $u\ge u_*$, and $L^u\gamma^{-2}\ge\Theta_\star$. □

Proposition 11.11 (The marked restart)**.** There is $K_{\mathrm{mk}}(g,\ell,n)$, determined after γ by the construction constants, g,ℓ,n, with the following property. Assume (k0)–(k2) and $K\ge K_{\mathrm{mk}}(g,\ell,n)$. The marked restart of Def. 11.4 is defined for $|\lambda|<2\Lambda_{\mathrm{mk}}^{1/2}$ through $t_1^{\bullet}$. At $t_0^{\bullet}<k\le t_1^{\bullet}$, its empty-history term has the exact measure form of Prop. B.56, with no activity at a level at most $t_0^{\bullet}$, and with

$$|\hat{x}_i(\lambda)-\mathsf{t}_i|\le\bar{B},\qquad|\hat{s}_i(\lambda)|\le\bar{B}_M/\mathsf{t}_i,$$
$$\|\hat{F}_0^{(j)}(\lambda)\|_j\le C_w\phi(\mathsf{t}_{j-1}),\qquad\|\hat{E}_{\ge 2}^{(j)}(\lambda)\|_j\le C_2\phi(\mathsf{t}_{j-1})^2.$$

These bounds hold for the activities and their pointed terms. The locality, symmetry, holomorphy, scalar-array and later-level reading properties are those specified in Prop. B.56; the normalizers use the actual pair $(\mathsf{t}_{j-1},\hat{x}_{j-1}(\lambda))$. The numbers and families are holomorphic in this disc, real at real λ with the corresponding conjugation identity for the families, and equal to the unmarked restart at zero. Moreover,

$$Z^{\bullet}_{\Lambda_{\mathrm{mk}}^{1/2}}\le 2e\,\mathsf{M}_{g,\ell}\hat{T}^{\bullet}(0).$$

■

Proof. The construction and its domain are Cor. B.64. The total is bounded in Prop. B.63 by the slightly stronger quantity $e\,\mathsf{M}_{g,\ell}\hat{T}^{\bullet}(0)$. □

Proposition 11.12 (Source energies and contact terms)**.** Assume (k0) and $u \geq u_*$. For $t_0^\bullet < j \leq t_1^\bullet$ and $X \in \mathcal{D}_j$, the restarted source energy $\hat{e}_j(X;\cdot)$ is holomorphic on $\mathscr{O}_{j-1}$, vanishes at zero, and obeys

$$|\hat{e}_j(X;s)| \leq C_{\mathfrak{E}} e^{-\kappa d_j(X)},$$

$$|\hat{e}_j(X;s) - \hat{e}_j(X;s')| \leq C_{\mathfrak{E}} e^{-\kappa d_j(X)} \exp\left[-\frac{\kappa_\sigma \operatorname{dist}(X, \operatorname{supp}(s-s'))}{L^j}\right] \quad (s, s' \in \mathscr{O}_{j-1}).$$

These are the constants of the original source induction. Let $n \geq 2$, let $\mathbf{s}_1, \dots, \mathbf{s}_n$ have pairwise disjoint supports, and suppose the first two supports have physical distance at least $\mathsf{d} > 0$. If $s(z) = \sum_i z_i \mathbf{s}_i \in \mathcal{P}$ for $|z_i| \leq \varrho_i$, then

$$\left|\partial_{z_1} \cdots \partial_{z_n} \hat{c}^{\bullet}(s(z))\big|_{z=0}\right| \leq 2^{n-1} \prod_i \varrho_i^{-1} (\mathscr{C}_{t_1^\bullet} - \mathscr{C}_{t_0^\bullet})(\operatorname{supp} \mathbf{s}_1; \mathsf{d}).$$

The right side is at most the same expression with $\mathscr{C}_{t_1^\bullet}$ in place of the difference, where

$$\mathscr{C}_k(S;\mathsf{d}) = C_{\mathscr{C}} \sum_{j \leq k} \#_j^+(S) e^{-\kappa_{\mathrm{E}} \mathsf{d}/\lambda_j}.$$

■

Proof. Use Prop. B.34(c) for the energies and Prop. B.36 for their mixed derivatives. The required floor on $C_{\mathscr{C}}$ is part of Def. 11.1. □

Proposition 11.13 (Source histories above a base)**.** Choose $0 < c_\lambda \leq 1/8$ as in appendix C.2, after γ and from the construction constants, so that

$$c_\lambda E_0^{\mathrm{s}} \leq c_0^\Delta, \qquad 2\sqrt{L} c_\lambda C_w^{\mathrm{s}} \leq c_\psi^\Delta, \qquad c_0^\Delta \leq M^{-3}\mathsf{B}.$$

Assume (k0), $u \geq u_*$, and the setting of Def. 11.3 on either lattice. Suppress $\bullet$ in the level indices. Let $t_0 < t_b \leq t_1$, $\Lambda_b \geq 1$, and suppose the differences have the format (F1)–(F3) of that definition, $d_b(s) = d_b$ is constant on $\mathcal{P}$, and eq. 11.ix–eq. 11.xi hold there with $\Lambda = \Lambda_b$. Set

$$\rho_b = \tfrac{1}{2}(3\sqrt{L})^{-(K^\bullet - t_b)} c_\lambda \Lambda_b, \qquad \mathcal{D}_b = \{(s,\lambda) : s = (w,J) \in \mathcal{P},\ \|w\|_\sharp \leq (2 - c_\lambda)\varrho,\ |\lambda| \leq \rho_b\}.$$

Then $(w + \lambda \hat{w}_b, J) \in \mathcal{P}$ throughout $\mathcal{D}_b$. For the histories and measures of Def. 11.3, the following assertions hold.

(B1) The total Berezin mass is preserved:

$$\operatorname{Ber} \rho_{s,\lambda}^{(t_b)}(\mathcal{V}_{t_b}) = \sum_{h \in \mathcal{H}_{K^\bullet}^{(t_b)}} \operatorname{Ber} \int g_h[s,\lambda]\, d\mathscr{M}_h^{\mathrm{fr}}.$$

The densities may be chosen as the representatives in Prop. C.21; coefficient by coefficient they are pointwise holomorphic on the interior of $\mathcal{D}_b$. They represent the given measures, so their values on the exceptional null set do not change this identity.

(B2) At each $t_b \leq k \leq K^\bullet$, every history has the source history identity of [QCDM1, Def. 10.11], with the following changes. There are no pieces, generated boundary terms or edge terms at levels at most t_b, and no stored layers below t_b. The action weight is $\beta_k - a_{K^\bullet}^{-1}(w + \lambda\hat{w}_b)$. The mass chain begins at $\sigma_{t_b}(s) - \lambda\delta_{t_b}(s)$ and obeys $\|\sigma_n^h(s,\lambda)\| \leq 2B_\sigma \phi_n$. The early activities are $E^{(j)}(s) - \lambda\Delta^{(j)}(s)$; the later ones are the actual step outputs. Their frame sizes and insertion sizes obey

$$\mathsf{sz}_j^\natural \leq \tfrac{64}{51} b^{\mathrm{tot,s}}(\gamma), \qquad \bar{v}_j^{\mathrm{s}} \leq b^{\mathrm{blk,s}}(\gamma).$$

The scalar and fermion layer estimates retain their source constants. The vacuum term has the form

$$-\mathcal{E}^F_{k,\mathcal{S}} = E_k(s,\lambda) - \sum_{t_b<n\le k} f_n(\mathcal{S}) + \sum_{t_b<n\le k} g_n(\mathcal{S};s,\lambda),$$

where the g_n are holomorphic in the interior and $|g_n| \le C'_{\mathfrak{E}} \#_n(\tilde{Z}_n)$. The regular-sum estimate retains the source constant C^F_E.

(B3) The sum of the integrated fibre norms satisfies

$$\sum_{h\in\mathcal{H}^{(t_b)}_{K^\bullet}} \int \|g_h[s,\lambda]\|_{h_F}\, d\mathscr{M}^{\mathrm{fr}}_h \le e^{\operatorname{Re} E_{K^\bullet}(s,\lambda)+(C+2)\ell^3}.$$

(B4) For $t_b \le k \le K^\bullet$,

$$E_k(s,\lambda) = E_{t_b}(s) + \lambda d_b + \sum_{t_b<j\le k} \left(e_j n_j + \sum_{X\in\mathcal{D}_j} e_j(X;s,\lambda)\right).$$

The energies $e_j(X;\cdot,\cdot)$ are holomorphic in the interior of $\mathcal{D}_b$, equal to $e_j(X;s)$ at $\lambda=0$, and bounded by $C_{\mathfrak{E}} e^{-\kappa d_j(X)}$.

(B5) For $t_b < t \le K^\bullet$,

$$\sum_{\substack{h\in\mathcal{H}^{(t_b)}_{K^\bullet}\\ Z_t(h)\neq\varnothing}} \int \|g_h[s,\lambda]\|_{h_F}\, d\mathscr{M}^{\mathrm{fr}}_h \le e^{-\theta_{\mathrm{tl}} p_0(g_t)^2} e^{\operatorname{Re} E_{K^\bullet}(s,\lambda)+(C+2)\ell^3}, \qquad \theta_{\mathrm{tl}} = \tfrac14(c_\star - c_\Theta).$$

■

Proof. Since $\hat{w}_b$ is constant, the last lag bound gives

$$|\lambda|\,|\hat{w}_b| \le \tfrac12 c_\lambda \varrho, \qquad \|w+\lambda\hat{w}_b\|_\sharp < 2\varrho.$$

The statements are Cor. C.23, applying Prop. C.21 to the free differences $(-\Delta^{(j)}, -\delta_{t_b}, \hat{w}_b, d_b)$. In particular Lem. C.18 proves that the early scalar and fermion excesses fit within the original source bounds. The measure identity uses the Borel representatives constructed in Lems. C.5, C.8 and C.9; it requires weighted almost-everywhere invariance, as proved there. □

Proposition 11.14 (Histories with a prescribed past)**.** Assume the constants and geometric hypotheses of appendix C.11, including (Γ^{F}) in Rem. C.30. Let $1 \le t_b < K$, and let a flow satisfying all the hypotheses of Def. C.31 be given, with its base t_1 there replaced by t_b. Thus the pointed families on the torus and on $\mathbb{Z}^3$ have the invariance, holomorphy, zero cell moment and scalar normalization of (G1); above t_b they are the actual step outputs with the coefficient, mass, scalar and pointed bounds of (G2). Before t_b they split as $R+D$ as in (G3), with

$$\Lambda^\psi_k(D) = \sum_{j\le t_b} L^{(k+1-j)/4}\epsilon^\psi_j(D) \le \lambda_0 C_2 \phi(\mathfrak{t}_k)^2, \qquad \Lambda^0_k(D) = \sum_{j\le t_b} L^{-(k-j)}\mathfrak{g}_j(D) \le \lambda'_0(1+E_0)$$

for every $t_b \le k \le K$. The constants $\lambda_0, \lambda'_0 \in (0,1]$ are chosen before h_F, and C_Σ, C'_Σ after γ, as in Rem. C.30.

(a) With all later families, coefficients and boundary terms fixed, each frame member is affine in D. Its linear part and the differences of the step outputs from those with D removed are bounded in the respective step norms by

$$\mathfrak{d}_k(D) = C_\Sigma \Lambda^\psi_k(D) + C'_\Sigma \Lambda^0_k(D).$$

The outputs, pointed activities and extracted numbers are holomorphic in the input members.

(b) Every $h \in \mathcal{H}_k^{[t_b]}$, $t_b \le k \le K$, has the history identity and bounds of Prop. C.38, with

$$\beta_h = L^{t_b} x_{t_b} - \sum_{t_b<j\le k} L^j b_j \varphi_j, \qquad E_k = E_{t_b} + \sum_{t_b<j\le k} e_j n_j, \qquad f_n = 0 \ (n \le t_b).$$

In particular there is no generated boundary or edge term at a level at most t_b; $\mathsf{sz}_k^{\natural} \le \varepsilon_b$; and the boundary terms agree for equal excerpts under the torus condition of Lem. C.35. The scalar, fermion and integrated bounds of Lem. C.37 hold with

$$E_0 \longmapsto E_0 + \mathfrak{d}^0, \qquad C_{\mathrm{leg}} \longmapsto C_{\mathrm{leg}} + C_{\mathrm{leg}}^{\square}\mathfrak{d}^{\psi}, \qquad \mathfrak{d}^a = \max_{t_b\le k\le K} \Lambda_k^a(D).$$

Here $\mathfrak{d}^0 \le \lambda_0'(1+E_0)$ and $\mathfrak{d}^{\psi} \le \lambda_0 C_2$. The edge mass bound uses $4\bar{B}_M/\mathsf{t}_j$. The integrated estimate uses $\operatorname{Re} E_k$; its action profile can be β_h, or the constant $L^{t_b}x_{t_b}$, with the action correction of Lem. C.37. No positivity is asserted for this complex flow.

(c) If the flow depends holomorphically on a parameter on a domain where (G1)–(G3) hold, the frame members, step outputs, extracted numbers and bounded Borel coefficient functions are holomorphic there. The real x_i, hence also $b_i = L^{-1}x_{i-1} - x_i$, are constant on each connected component.

■

Proof. The hypotheses have been stated in the precise form used by the induction. The sizes in Def. C.24 take the maximum over the two lattices and place the supremum over regular pairs inside the pointer sum. Thus the two displayed sums are exactly the scalar and fermion lag sums for D. Joint gauge invariance and translation covariance imply that the quadratic moment commutes with colour and is periodic; its zero cell sum is the explicit cancellation hypothesis in (G1). No additional cancellation is inferred from a norm bound.

The three conclusions are Lem. C.36, Lem. C.37 and Prop. C.38, and the parameter assertion of Prop. C.38, respectively. □

Proposition 11.15 (The restarted total at zero source)**.** Assume (k0), $u \ge u_*$, $s \ge s_F$, and $L^u\gamma^{-2} \ge \Theta_\star$. For either lattice, suppress $\bullet$ in the level indices.

(a) The zero-source restart of Def. 11.2, with starting mass $F^{\mathrm{UV}}/(L^s\mathsf{b})$, continues through $K^\bullet$ on the torus and on $\mathbb{Z}^3$. It extracts its own real numbers from the infinite-lattice activities, with

$$\hat{x}_j = L^{-1}\hat{x}_{j-1} - \hat{b}_j, \qquad \hat{s}_j = L\hat{s}_{j-1} + \hat{\mu}_j, \qquad \hat{E}_k = \sum_{t_0<j\le k} \hat{e}_j n_j.$$

The exact measure form, coefficient and mass windows, locality, symmetries and scalar bounds are those of Prop. B.26. There are no activities at levels at most t_0, and

$$\|\hat{F}_0^{(j)}\|_j \le C_w\phi(\mathsf{t}_{j-1}), \qquad \|\hat{E}_{\ge 2}^{(j)}\|_j \le C_2\phi(\mathsf{t}_{j-1})^2.$$

These bounds also hold pointer by pointer. At the trivial configuration on $\mathbb{Z}^3$, the explicit part is $T^{(j)}(\hat{y})$, with $\hat{y}_i = 0$ for $i < t_0$ and $\hat{y}_i = 1/\hat{x}_i$ for $i \ge t_0$; its remainder is bounded by $C_3\phi(\mathsf{t}_{j-1})^3$. Through t_1 this continuation is the restart already defined.

(b) Start each history $h \in \mathcal{H}_k^{(t_0)}$ from $\hat{d}_{t_0}^{\bullet}(0)$, multiplied by the density test $\chi_{t_0}^{\bullet}(T_\ell)$, and apply its later weights and partial fermion kernels. If $\hat{g}_h$ is its density, then for $t_0 < t \le K^\bullet$,

$$\sum_{\substack{h\in\mathcal{H}_{K^\bullet}^{(t_0)}\\ Z_t(h)\neq\varnothing}} \int \|\hat{g}_h\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}} \le e^{-\theta_{\mathrm{dis}}p_0(g_t)^2} e^{\hat{E}_{K^\bullet}+(C+2)\ell^3}.$$

(c) The starting density $\hat{F}_0^{\bullet}$ is nonnegative, and

$$\hat{T}^{\bullet}(0) \ge e^{\hat{E}_{K^\bullet} - C_P(1+\log X_g)\ell^3}.$$

For $t_1 \le k \le K^\bullet$, each history above t_1 has total Berezin mass equal to the integral of $\hat F_0^\bullet$ against a nonnegative Borel weight on $\mathcal{Y}$ containing $\hat{\mathsf{k}}^\bullet$. Here

$$\hat{\mathsf{k}}^\bullet \chi^\bullet_{t_0}(T_\ell) = \hat{\mathsf{k}}^\bullet = \hat{\mathsf{k}}^\bullet \mathbf{1}_{\mathsf{Cut}^\bullet_{(t_0)}}.$$

These masses sum to $\hat T^\bullet(0)$ at every such level. In particular this total is at least the mass of the final empty history.

Consequently, for $t_0 < t \le t_1$, the histories above the start in which a large-field region has occurred by level t have combined mass at most

$$e^{-\theta_{\mathrm{dis}} p_0(g_t)^2} \mathsf{M}_{g,\ell} \hat T^\bullet(0).$$

■

Proof. The continuation is Prop. B.26; its history estimate is Prop. B.28. Positivity and consistency of the Berezin masses are proved in Lems. B.29 and B.30, and their lower bound in Prop. B.31. The support identities follow from Lem. B.25. To obtain the last estimate, extend the histories to $K^\bullet$. Consistency preserves their masses, and nesting of the large-field regions identifies their extensions with the histories $Z_t(h) \neq \varnothing$. Use (b), followed by (c), and $\log \mathsf{M}_{g,\ell} = (C+2+C_P(1+\log X_g))\ell^3$. □

11.5. **Comparing the totals.**

Lemma 11.16. Under (k0), put

$$\rho^\bullet_z(U) = e^{-A(\beta^\bullet - a_{K^\bullet}^{-1} w_z, U)} \det(D[U] + m_0^\bullet - a_{K^\bullet} \hat J_z).$$

Then

$$T^\bullet(z) = \int \mathfrak{w}_{h_\emptyset} \rho^\bullet_z \, dU, \qquad Z^\bullet[s(z)] - T^\bullet(z) = \int \Big(\sum_{h \neq h_\emptyset} \mathfrak{w}_h \Big) \rho^\bullet_z \, dU.$$

The weights are nonnegative bounded Borel functions, independent of the source. If $u \ge u_*$ and $L^u \gamma^{-2} \ge \Theta_\star$, then

$$\hat{\mathsf{k}}^\bullet \chi^\bullet_{t_0}(T_\ell) = \hat{\mathsf{k}}^\bullet = \hat{\mathsf{k}}^\bullet \mathbf{1}_{\mathsf{Cut}^\bullet_{(t_0)}}, \qquad \hat T^\bullet(z) = \int \hat{\mathsf{k}}^\bullet \hat F^\bullet_z \, dW$$

for $s(z) \in \mathscr{O}_{t_1^\bullet}$. The integrand is bounded and Borel. If also $s \ge s_F$, then $\hat F_0^\bullet \ge 0$, and the marked total in eq. 11.xii is the total of the marked flow wherever that flow is defined. Its polynomial coefficients are nonnegative, its value at zero is $\hat T^\bullet(0)$, and its linear coefficient is the sum in Def. 11.4. ■

Proof. The exact history identity of [QCDM1] gives the first two formulas. Sources change the density, leaving the tests and gauge-fixing factors unchanged.

In one step along the empty history, the measure is multiplied by the tests, the gauge-fixing factor and the fermion kernel, partially Berezin-integrated, then pushed forward by $\mathcal{M}$. The fermion kernel has total Berezin integral one, and pushforward preserves total mass. Repeating these identities leaves precisely the product $\hat{\mathsf{k}}^\bullet$ against the Berezin integral of the starting density. The first fluctuation test and the density test at the next level imply the density test at the starting level, by the test-descent estimate used in the original step. Thus $\hat{\mathsf{k}}^\bullet$ vanishes unless that starting test passes. Its support lies in the field cut, as observed after eq. 11.vi. This proves the second pair of formulas.

Each test is Borel, each gauge-fixing factor is bounded and Borel, and the iterated averaging maps are Borel. The finite product $\hat{\mathsf{k}}^\bullet$ has these properties. The starting Grassmann density is bounded and Borel by Lem. B.12; its Berezin integral has the same properties. This proves the assertion about the integrand without requiring continuity of the averaging map.

The argument also applies with the mark factors inserted. At zero source the starting Dirac determinant is positive by Prop. 11.7; the action is real and nonnegative on the field cut. Thus

$\hat F_0^\bullet \ge 0$. Each coefficient of $\prod_{k,b}(1+\lambda\bar\chi^\bullet_{(k),b})$ is an elementary symmetric function of finitely many values in $\{0,1\}$. Integrating it against $\hat{\mathsf k}^\bullet \hat F_0^\bullet\, dW$ gives a finite nonnegative coefficient. The constant and linear coefficients are the stated ones. □

Lemma 11.17. Assume (k0) and $R^F \le 1/10$. On $\mathbb D^\bullet = \{z : |z_i| \le (4ns_i^\bullet)^{-1}\}$,

$$|\rho_z^\bullet(U) - \rho_0^\bullet(U)| \le (e^{1/2}-1)\rho_0^\bullet(U).$$

Both $Z^\bullet[s(z)]$ and $T^\bullet(z)$ are nonzero there, and

$$|S^T_{n;K^\bullet}(X) - \partial_z^n \log T^\bullet(0)| \le 10R^F \prod_i (4ns_i^\bullet).$$

■

Proof. Suppress $\bullet$. Since $0 \le a_p \le 2$ and there are $3\ell^3 L^{3K}$ plaquettes,

$$|a_K^{-1}A(w_z,U)| \le 6\ell^3 L^{4K}\|w_z\|_\infty \le \sum_{\alpha_i=0} |z_i| s_i.$$

Accretivity gives $\|(D[U]+m_0)^{-1}\| \le L^K/\mathsf m_-$. For $Y = a_K(D[U]+m_0)^{-1}\hat J_z$, the chosen polydisc gives $\|Y\| \le 1/2$. The logarithmic series therefore gives

$$|\operatorname{tr}\log(1-Y)| \le 2\nu_\natural \ell^3 L^{3K}\|Y\| \le \sum_{\alpha_i\in\mathfrak B} |z_i| s_i.$$

Each sum is at most $1/4$, so $\rho_z = \rho_0 e^a$ with $|a| \le 1/2$.

The nonnegative weights of Lem. 11.16 and the discarded history bound Prop. 11.8 imply

$$|Z[s(z)] - Z[0]| \le (e^{1/2}-1)Z[0], \qquad |Z[s(z)] - T(z)| \le e^{1/2}R^F Z[0].$$

Hence $|Z[s(z)]| \ge (2-e^{1/2})Z[0]$, and $q = (Z-T)/Z$ satisfies $|q| \le 4.7R^F \le 1/2$. Thus $T = Z(1-q) \ne 0$. The principal logarithm $-\log(1-q)$ is bounded by $10R^F$. Near zero it differs from $\log Z - \log T$ by a constant, and Cauchy's inequality proves the claim. □

Lemma 11.18. Assume (k0)–(k4), (k6). The two restarted totals have no zero on $\hat{\mathbb D}$, and their ratio has a holomorphic logarithm $\mathsf L$, on a neighbourhood of that polydisc, with

$$\sup_{\hat{\mathbb D}} |\mathsf L| \le 8(\epsilon_{\mathrm{cl}} + \epsilon_F) + 12\epsilon_{\mathrm{sh}} + 16\hat R^F.$$

In particular,

$$|\partial_z^n(\log \hat T^{\mathrm B} - \log \hat T^{\mathrm A})(0)| \le [8(\epsilon_{\mathrm{cl}} + \epsilon_F) + 12\epsilon_{\mathrm{sh}} + 16\hat R^F] \prod_i \hat r_i^{-1}.$$

■

Proof. On $\hat{\mathbb D}$, both source norms are at most one half of their admitted radii. Thus Prop. 11.7 applies along every segment $s(tz)$, $0 \le t \le 1$.

Variation in the source. On $\mathsf{Cut}^\bullet_{(t_0)}$, write $\hat F_z^\bullet = \hat F_0^\bullet e^{\psi_z^\bullet}$, where

$$\psi_z^\bullet(W) = \mathsf{wab}^\bullet(L^s w_z, W) + \int_0^1 \operatorname{tr}[\hat{\mathbb D}^\bullet(W; s(tz))^{-1}\partial_t \hat{\mathbb D}^\bullet(W; s(tz))]\,dt.$$

The background-action bound is

$$|\mathsf{wab}^\bullet(L^s w_z, W)| \le \tfrac{3}{2} c_\rho^2 A_0^2 \gamma^2 \bar{\mathsf y}_{[s]} N_{[s]} \|w_z\|_\infty.$$

The operator comparison gives

$$\|\hat{\mathbb D}^{-1}\| \le 8L^s/\mathsf m_-, \qquad \|\partial_t \hat{\mathbb D}\| \le C_D L^{-s}(\|J_z\|_\infty + \|w_z\|_\infty).$$

The matrix dimension is $\nu N_{[s]}$. Therefore $|\psi_z^\bullet| \le N_{[s]} C_\psi \sum_i |z_i| \|f_i\|_\infty \le 1/4$. Both $\hat F_z^\bullet$ and $\hat F_0^\bullet$ vanish off the field cut.

Comparison on the common support. Where both k^{A} and k^{B} are positive, they equal the common product of gauge-fixing factors. Also $\varpi(W) < \varepsilon_{\mathrm{r}} = 2c_\rho \varepsilon(L^s \mathsf{b})$. The weighted gauge-action comparison Prop. A.9 applies to $\mathsf{c}_{w_z} = L^s(\mathsf{b} - w_z)$. Its Lipschitz norm is at most $L^s(\mathsf{b}+2\varrho)$. It gives

$$|\mathsf{wab}^{\mathrm{A}}(\mathsf{c}_{w_z}, W) - \mathsf{wab}^{\mathrm{B}}(\mathsf{c}_{w_z}, W)| \le \mathsf{K}_{\mathrm{w}} L^s(\mathsf{b}+2\varrho)\varepsilon_{\mathrm{r}}^2 N_{[s]} K^{1/2} L^{-(K-s)/2} \le \epsilon_{\mathrm{cl}}.$$

For the determinant use $\mathbb{D}_t = (1-t)\hat{\mathbb{D}}^{\mathrm{A}} + t\hat{\mathbb{D}}^{\mathrm{B}}$. By Prop. 11.7, its real part is at least $L^{-s}\mathsf{m}/8$, so its inverse has norm at most $8L^s/\mathsf{m}_-$. Hence

$$\det \hat{\mathbb{D}}^{\mathrm{B}} = \det \hat{\mathbb{D}}^{\mathrm{A}} e^{\eta_z}, \qquad \eta_z = \int_0^1 \mathrm{tr}[\mathbb{D}_t^{-1}(\mathbb{D}_1 - \mathbb{D}_0)]dt, \qquad |\eta_z| \le \epsilon_F.$$

Thus $\hat{F}_z^{\mathrm{B}} = \hat{F}_z^{\mathrm{A}} e^{\xi_z}$ on the common support, with $|\xi_z| \le \epsilon_{\mathrm{cl}} + \epsilon_F$.

The mass outside the common support. The test comparison Lem. B.50 gives, in either direction,

$$\mathbf{1}_{\{\mathsf{k}^\circ=0\}}\mathsf{k}^\bullet \le \sum_{k,b} \hat{\mathsf{k}}^\bullet \bar{\chi}^\bullet_{(k),b}(W_k) + \sum_{k,b} |\mathsf{k}^\bullet - \hat{\mathsf{k}}^\bullet|.$$

There are at most $3sN_{[s]}$ terms. Integrate against $\hat{F}_0^\bullet \ge 0$. The first sum is $\partial_\lambda Z_\lambda^\bullet|_0$. Nonnegativity of the polynomial coefficients implies

$$\partial_\lambda Z_\lambda^\bullet|_0 \le \frac{Z_r^\bullet}{r} \le \epsilon_{\mathrm{mk}} \hat{T}^\bullet(0), \qquad r = \Lambda_{\mathrm{mk}}^{1/2},$$

by Prop. 11.11. The top-test estimate Prop. 11.10 bounds the second sum by $3sN_{[s]}\hat{R}^F\hat{T}^\bullet(0)$, and also gives $T^{\mathrm{sf},\bullet}(0) \ge (1-\hat{R}^F)\hat{T}^\bullet(0)$. Using $|\psi_z^\bullet| \le 1/4$, we conclude that

$$\int \mathbf{1}_{\{\mathsf{k}^\circ=0\}}\mathsf{k}^\bullet |\hat{F}_z^\bullet|\, dW \le \epsilon_{\mathrm{sh}} \int \mathsf{k}^\bullet |\hat{F}_z^\bullet|\, dW.$$

Here $e^{1/2}/(1-\hat{R}^F) < 2$, by (k3).

Lemma B.37 now applies, with $\epsilon = \epsilon_{\mathrm{cl}} + \epsilon_F$ and $\delta = \epsilon_{\mathrm{sh}}$: the weights and integrands are bounded and Borel, the zero-source integral is positive by Prop. 11.9, and both errors are at most $1/40$. It gives

$$\left|\frac{T^{\mathrm{sf,B}}(z)}{T^{\mathrm{sf,A}}(z)} - 1\right| \le \epsilon_3 := 4(\epsilon_{\mathrm{cl}} + \epsilon_F) + 6\epsilon_{\mathrm{sh}} \le 1/4,$$

$$|T^{\mathrm{sf,A}}(z)| \ge e^{-1/4}\cos(1/4) T^{\mathrm{sf,A}}(0) \ge \tfrac{3}{4} T^{\mathrm{sf,A}}(0).$$

Restoring the original restart tests. The top-test bound also gives

$$|\hat{T}^\bullet(z) - T^{\mathrm{sf},\bullet}(z)| \le e^{1/4}\hat{R}^F\hat{T}^\bullet(0).$$

Writing $a = T^{\mathrm{sf,A}}(0)$, the preceding bounds give $T^{\mathrm{sf,B}}(0) \le (1+\epsilon_3)a$ and $|T^{\mathrm{sf,B}}(z)| \ge 9a/16$. Consequently

$$\left|\frac{\hat{T}^{\mathrm{A}}}{T^{\mathrm{sf,A}}} - 1\right| \le 2\hat{R}^F, \qquad \left|\frac{\hat{T}^{\mathrm{B}}}{T^{\mathrm{sf,B}}} - 1\right| \le 3\hat{R}^F.$$

All three ratios lie within distance $1/4$ of one. Their principal logarithms therefore define

$$\mathsf{L} = \mathrm{Log}\,\frac{\hat{T}^{\mathrm{B}}}{T^{\mathrm{sf,B}}} + \mathrm{Log}\,\frac{T^{\mathrm{sf,B}}}{T^{\mathrm{sf,A}}} - \mathrm{Log}\,\frac{\hat{T}^{\mathrm{A}}}{T^{\mathrm{sf,A}}}.$$

The inequality $|\mathrm{Log}(1+x)| \le 4|x|/3$ gives the stated bound.

For completeness these integrals are holomorphic on $\mathbb{Q}$. At each field the integrand is holomorphic, and the source-variation bound above gives the common integrable majorant

$$\mathsf{k}^\bullet \hat{F}_0^\bullet \exp\{N_{[s]} C_\psi \sum_i 2R_i \|f_i\|_\infty\}.$$

Integration preserves holomorphy. Nonvanishing and the ratio bounds on the compact polydisc extend to a neighbourhood of it. Cauchy's inequality now proves the derivative estimate. □

Lemma 11.19. Assume (k0), (k1), (k5), $n \geq 2$, and pairwise disjoint supports, two separated by at least $\mathsf{d} > 0$. Then $|\partial_z^n \mathfrak{c}^\bullet(0)| \leq 2\epsilon_{\mathrm{mx}}$. ■

Proof. Relabel the separated supports as the first two. The original and restarted source-energy bounds each give

$$2^{n-1} \prod_i \varrho_i^{-1} C_{\mathscr{C}} \sum_{m \geq u} \ell^3 L^{3m} e^{-\kappa_{\mathrm{E}} \mathsf{d} L^m}$$

for the mixed derivative; the restarted bound is Prop. 11.12. By (k5), each summand is at most half the preceding one. Each derivative is therefore bounded by ϵ_{mx}. Their difference is the derivative of $\mathfrak{c}^\bullet$. □

11.6. **Summing the errors.**

Lemma 11.20. The threshold $K_n(g,\ell)$ and constant $C_n(g,\ell)$ in Def. 11.5 are finite. If $K \geq P_n \max\{0, \log_L(1/\mathsf{d})\} + K_n(g,\ell)$, then all (k0)–(k6) hold and $\mathsf{d} L^u \geq L^{\phi_1(n)K/2}$. ■

Proof. Abbreviate $\phi_i(n)$ by ϕ_i. We have

$$\phi_1 K \leq u \leq \phi_1 K + 1, \quad \phi_2 K \leq s \leq \phi_2 K + 1, \quad \phi_2 \leq 1/34, \quad \phi_1 = c_\# \phi_2/10, \quad c_n = c_\# \phi_2/2.$$

Also $3 \leq L^{1/2}$, $\gamma^{-2} \geq 2$, and $\mathfrak{p}_0 \geq 1$. The factors $\bar{\mathsf{y}}_{[s]}$, $\bar{\mathsf{y}}_{[K]}$ grow at most as powers of K. From the definition of ε,

$$\bar{\varepsilon}_{\mathrm{r}} \leq 2 c_\rho A_0 \gamma L^{-s/2} \bar{\mathsf{y}}_{[s]}^{1/2}.$$

Both discarded-history and marked-history errors contain $\exp[-c(\phi_1 K \log L)^{2\mathfrak{p}_0}]$, which decays faster than every L^{-aK}.

The depths imply (k0) for large K, and (k1) follows from $u, s \to \infty$ and the faster decay of $e^{-c_{\mathrm{t}} L^u}$. For (k2),

$$r_\lambda \geq \tfrac{1}{2} c_\lambda C_\Lambda^{-1} (K+1)^{-p_\Lambda} L^{c_\# s - (1+c_\#)u} \longrightarrow \infty.$$

Indeed its exponential rate is $[1 - (1+c_\#)/10] c_\# \phi_2 > 0$. The remaining errors satisfy, up to fixed constants,

$$\epsilon_{\mathrm{cl}} \leq K^{1/2} \bar{\mathsf{y}}_{[K]} L^{(7s-K)/2}, \qquad \epsilon_F \leq L^{5s-K} + \bar{\mathsf{y}}_{[s]}^{1/2} L^{(29s-2K)/6}.$$

They tend to zero since $7\phi_2 < 1$, $5\phi_2 < 1$, and $29\phi_2 < 2$. This proves (k3). Condition (k4) uses $N_{[s]} L^{-(K-s)} = \ell^3 L^{4s-K} \to 0$. The test and mark thresholds give (k6). They were defined using (k0), (k1) and the displayed marked radius, before (k6); thus there is no circular dependence on K_n. Finally $L^{\phi_1 K/2} \to \infty$, proving the existence of K_n.

If $\mathsf{d} \geq 1$, the asserted separation bound is immediate. If $\mathsf{d} < 1$, then $\phi_1 P_n = 2$ gives $\phi_1 K \geq 2 \log_L(1/\mathsf{d})$, and hence $\mathsf{d} L^u \geq L^{\phi_1 K/2}$. The extra condition in the definition of K_n now implies (k5).

It remains to bound $L^{c_n K} \mathsf{B}_n(K)$. The first term is bounded because its discarded-history factor decays faster than the exponential Cauchy cost L^{4nK}. For the readout term,

$$\epsilon_{\mathrm{ro}} \leq C (K+1)^{p_\Lambda} L^{(4+c_\#)u - c_\# s},$$

and

$$(4+c_\#)u - c_\# s \leq -\tfrac{1}{10}(6 - c_\#) c_\# \phi_2 K + 4 + c_\# \leq -c_n K - \tfrac{7}{80} c_\# \phi_2 K + 5.$$

This has a strict exponential margin beyond c_n.

Next $\hat{\varrho}_K \leq C \ell^3 L^{3s} (1 + \bar{\mathsf{y}}_{[s]})$. The three possible exponential costs from $(\epsilon_{\mathrm{cl}} + \epsilon_F) \hat{\varrho}_K^n$ are therefore

$$L^{((6n+7)s-K)/2}, \qquad L^{(3n+5)s-K}, \qquad L^{((18n+29)s-2K)/6},$$

times powers of K. Their rates of decay in K are

$$\frac{6n+15}{2(12n+22)}, \qquad \frac{9n+17}{12n+22}, \qquad \frac{2n+5}{4(6n+11)},$$

each strictly larger than c_n. The other terms in the third line of eq. 11.xxi decay faster than every exponential, even after multiplication by $\hat{\varrho}_K^n$. The contact term is bounded by a constant times $L^{3\phi_1 K+3+c_n K} e^{-\kappa_{\mathrm{E}} L^{\phi_1 K/2}}$, which is bounded as well. Finitely many smaller cutoffs cause no difficulty. Every bound uses only the stated constants, n, g, ℓ, proving the claimed uniformity. □

Proof of Thm. Q. For each lattice, Lem. 11.17 replaces the cumulant by $\partial_z^n \log T^\bullet(0)$ with its discarded-history error. The readout estimate Prop. 11.9 gives

$$\left|\partial_z^n(\log T^\bullet - \log \hat{T}^\bullet - \mathfrak{c}^\bullet)(0)\right| \le \epsilon_{\mathrm{ro}} \prod_i R_i^{-1}.$$

Subtract the two identities and apply Lem. 11.18 to the remaining difference of restarted logarithms. This is eq. 11.xxii. The second assertion is Lem. 11.19.

For the stated threshold, Lem. 11.20 applies. The bounds

$$R_i^{-1} \le \frac{2n\|f_i\|_\sharp}{(1-c_\lambda)\varrho_{\min}}, \quad \hat{r}_i^{-1} \le \hat{\varrho}_K \|f_i\|_\sharp, \quad s_i^\bullet \le \sigma_K \|f_i\|_\sharp, \quad \varrho_i^{-1} \le \frac{n}{\varrho_{\min}} \|f_i\|_\sharp$$

put the right side, including both contact terms, below $\mathsf{B}_n(K) \prod_i \|f_i\|_\sharp$. The definition of $C_n(g,\ell)$ proves eq. 11.xxiii. □

11.7. **Three mass derivatives of the pressure.** This subsection concerns a fixed admitted torus T_ℓ and $N_f \ge 1$. Use the constants and adjacent-cutoff comparison of Def. 11.1, Thm. Q, and Cor. 11.6. For $W = I_g \times [m_-, m_+]$ put

$$p_K = \ell^{-3} \log Z_K, \qquad y_K = -\partial_m p_K, \qquad y_K^{(2)} = -\partial_m^3 p_K = \ell^{-3} S_{3,K}^T((1,1),(1,1),(1,1)). \quad (11.\mathrm{xxv})$$

The last identity follows by inserting a constant matrix source. We shall prove convergence of $y_K^{(2)}$. The contact energy in the adjacent-cutoff comparison does not disappear from this derivative. Its third derivative is the difference of two free determinants, up to a summable remainder. This is the cancellation needed below.

Lemma 11.21 (The third derivative of the free determinant)**.** Let $\nu = 8N_f$, $\nu_{\mathrm{c}} = 48\nu$, and let $\widehat{D}(p)$ be the chamber symbol, including flavour and colour. Define

$$\Phi_K^{\mathrm{fr}}(m) = \ell^{-3} \partial_m^3 \log\det(D[1] + am), \qquad a = L^{-K}.$$

There are constants $C_{\mathrm{fr}}, C_{\mathrm{pr}}$ independent of K, m, ℓ such that, for every fixed admitted side,

$$|\Phi_K^{\mathrm{fr}}(m) - \Phi_\infty^{\mathrm{fr}}(m)| \le C_{\mathrm{fr}} a^{1/2}, \qquad 0 < m \le 1/2,$$

once K exceeds a fixed symbol threshold. Here

$$\Phi_\infty^{\mathrm{fr}}(m) = 2\int_{[-\pi,\pi]^3} \operatorname{tr} \widehat{D}(p)^{-3} \frac{dp}{(2\pi)^3} + 2\nu\ell^{-3} \sum_{k \in \Lambda_\ell^*} \frac{m^3 - 3m|k|^2}{(m^2+|k|^2)^3},$$

$$\Lambda_\ell^* = (2\pi/\ell)(\mathbb{Z} + 1/2)^3.$$

The integral and series converge absolutely. On every disc $|m' - m| \le m/(2C_{\mathrm{pr}})$, with $m \in [m_-, m_+]$, Φ_K^{fr} is holomorphic and bounded by a constant $C_{\mathrm{fr},1}(\ell, m_+)$ independent of K and m. ■

Proof. We give the symbol and summation estimates, since their cancellation is essential. The free-symbol calculation in [QCDM1, §6] decomposes the chamber space into the constant chamber vector and its orthogonal complement. Let ι be the isometric inclusion of the first space. The heavy block has real part at least λ_ρ, the two couplings are $O(|p|)$, and the light Schur complement is $m + i\gamma \cdot p + O(|p|^2)$. Away from $p = 0$ the real part has a positive gap. The Schur formula and the resolvent identity therefore give

$$\|(\widehat{D}(p) + \mu)^{-1}\| \le \frac{C_{\mathrm{pr}}}{|p| + \mu}, \quad 0 \le \mu \le 1,$$

and, for $|ak|$ sufficiently small,

$$\|(a^{-1}\widehat{D}(ak)+m)^{-1}-\iota(m+i\gamma\cdot k)^{-1}\iota^*\|\le C_{\mathrm{lt}}a.$$

All constants depend only on the chamber operator. For completeness, the error in the light block is $O(|p|^2)$ and multiplication by its two inverses costs $O((|p|+\mu)^{-2})$. Each off-diagonal inverse block has one additional factor $|p|$ from its coupling; the heavy-heavy block has a bounded regular part. These bounds prove the displayed estimates uniformly in μ. Differentiating the same block formula proves the corresponding symbol estimates.

In particular, at mass zero,

$$\widehat{D}(p)^{-1}=\iota(i\gamma\cdot p)^{-1}\iota^*+R(p),\qquad |\partial^\alpha R(p)|\le C_\alpha|p|^{-|\alpha|}.$$

The leading trace of the cube vanishes: $\operatorname{tr}(i\gamma\cdot p)^{-3}=0$. Thus $F(p)=\operatorname{tr}\widehat{D}(p)^{-3}$ satisfies

$$|\partial^\alpha F(p)|\le C_F|p|^{-2-|\alpha|},\qquad |\alpha|\le 2.$$

This proves integrability at the origin.

Antiperiodic Fourier diagonalization gives, with $\Lambda^*_{\ell,a}=\Lambda^*_\ell\cap[-\pi/a,\pi/a)^3$,

$$\Phi^{\mathrm{fr}}_K(m)=2\ell^{-3}\sum_{k\in\Lambda^*_{\ell,a}}\operatorname{tr}(a^{-1}\widehat{D}(ak)+m)^{-3}.$$

The midpoint sum for F differs from its integral by $O(a)$. To check the singular point, integrate the bound $C_F|p|^{-2}$ on the eight cubes next to zero. Their total is $O(a)$. On the other cubes Taylor's theorem and the cancellation of the linear midpoint term give their volume times $Ca^2\sup|p|^{-4}$; summing radial shells again gives $O(a)$.

Subtract the mass-zero trace and split the momentum sum at $R=a^{-1/2}$. Below R, the preceding inverse comparison and the identity for the difference of cubes bound the difference from the continuum expression by $Ca|k|^{-2}$. Its sum is $O(aR)$, because $\ell^{-3}\sum_{0<|k|\le R}|k|^{-2}\le CR$. Above R, differentiate the trace cube in the mass:

$$\partial_m\operatorname{tr}(a^{-1}\widehat{D}(ak)+m)^{-3}=-3\operatorname{tr}(a^{-1}\widehat{D}(ak)+m)^{-4}.$$

The resolvent bound gives $Cm|k|^{-4}$ for the mass difference. The continuum trace has the same bound, and its omitted tail is $O(R^{-1})$. Finally

$$\operatorname{tr}(m+i\gamma\cdot k)^{-3}=\nu\frac{m^3-3m|k|^2}{(m^2+|k|^2)^3}.$$

Combining $O(a)$, $O(aR)$ and $O(R^{-1})$ proves the claimed $a^{1/2}$ rate. The lattice momentum estimates are uniform in the admitted sides; the antiperiodic shift removes the zero mode.

For complex m', expand the resolvent about real m. Put $S_\ell=\ell^{-3}\sum_{k\in\Lambda^*_\ell}|k|^{-4}<\infty$. For $n\ge 1$, the resolvent bound gives

$$|\partial_m^n\Phi^{\mathrm{fr}}_K(m)|\le(n+2)!\nu_{\mathrm{c}}C_{\mathrm{pr}}^{n+3}m^{1-n}S_\ell.$$

The Taylor series converges for $|m'-m|<m/C_{\mathrm{pr}}$. On the half-radius disc its tail is bounded using $\sum_{n\ge1}(n+1)(n+2)2^{-n}=14$. The zeroth term is bounded by the mass-zero midpoint sum and $6\nu_{\mathrm{c}}C_{\mathrm{pr}}^4m_+S_\ell$. These estimates give the stated $C_{\mathrm{fr},1}$. □

Lemma 11.22 (Constant sources and fermion normalizers)**.** At level j let $L_j^F(m')$ be the holomorphic logarithm of the free fermion fluctuation determinant. It exists on

$$|m'-m|<\max\{r_J/\lambda_{j-1},m/2\}.$$

At the constant source $v_\zeta=(0,\zeta 1)$ one has

$$\sum_X e_j(X;v_\zeta)=L_j^F(m-\zeta)-L_j^F(m)+e_j^{\mathrm{fl}}(\zeta),$$

where

$$\ell^{-3}|\partial_\zeta^3 e_j^{\mathrm{fl}}(0)| \le C_{\mathrm{tf}}\phi_{j-1}, \qquad C_{\mathrm{tf}} = 12C_w^s(Mr_J)^{-3}.$$

The restarted source energies have the same free determinant, with a remainder bounded by $C_{\mathrm{tf}}\widehat{\phi}_{j-1}$.

For $|m'-m| < m/2$, the determinant identity for the tower on the empty history gives

$$\sum_{j\le k}[L_j^F]_m^{m'} = [\log\det(D[1]+am')]_m^{m'} - [\log\det \mathbb{D}_k(1;am')]_m^{m'}.$$

At its terminal level,

$$\ell^{-3}|\partial_{m'}^3 \log\det \mathbb{D}_K(1;am')| \le 128\nu c_1 m^{-4}, \qquad |m'-m| \le m/4,$$

where $c_1 = \max\{c_{\mathrm{disc}}+1, b_F\}$. ■

Proof. There are two domains for the fluctuation operator. If $\lambda_{j-1}|m'-m| < r_J$, the source fluctuation estimate applies, since $L\lambda_{j-1}|m'-m| < r_M$. If $|m'-m| < m/2$, accretivity forms the entire tower. The two discs have the same centre; their union is the displayed disc. The determinant is nonzero and holomorphic there, and hence has a holomorphic logarithm.

At the trivial pair write $D^{(j)}(X;c)$ for the degree zero part of the new activity after removing its Gaussian contribution and the two normalizer logarithms. The source step in [QCDM1, §10] bounds

$$\sum_{X\supset\square} e^{\kappa d_j(X)}|D^{(j)}(X;c)| \le C_w^s\phi_{j-1}$$

when the gauge source is constant: its Lipschitz contribution vanishes. Assign each X to one of its cubes. Summing over the $(\ell/(M\lambda_j))^3$ cubes bounds the sum by this number times $C_w^s\phi_{j-1}$.

The gauge normalizer is unchanged by the source. The Gaussian contribution depends only on the gauge source, so is unchanged at v_ζ. The localized fermion normalizer logarithms sum to

$$\int_0^1 \mathrm{Tr}[\Gamma_{j-1}(t)^{-1}\partial_t\Gamma_{j-1}(t)]\,dt = L_j^F(m-\zeta) - L_j^F(m).$$

The equality is exact: the segment is inside the zero-free disc just established, so its logarithm is continued there. The remaining difference of the two $D^{(j)}$ sums is holomorphic for $|\zeta| < r_J/\lambda_j$ and bounded by $2C_w^s\phi_{j-1}(\ell/(M\lambda_j))^3$. Cauchy's third-derivative bound cancels the factor λ_j^{-3} and gives $C_{\mathrm{tf}}\ell^3\phi_{j-1}$. The restarted chain uses the same tower and its normalizers; its step estimates have $\widehat{\phi}$ in place of ϕ. This proves the restarted assertion. For a constant gauge source the same decomposition holds with an additional function of that source alone. This observation will also be used below.

Iterate the Schur determinant identity. Its factors $\theta^{-\dim H_j}$ do not depend on m', so their logarithmic increments cancel and give the tower formula. On the empty history at level K no field is stored. Writing $T_K = q_K\cdots q_1$, one has

$$\mathbb{D}_K^{-1} = T_K(D[1]+am')^{-1}T_K^* + \theta_K^{-1}, \qquad \|T_K\|^2 = a.$$

Thus

$$\partial_{m'}\log\det\mathbb{D}_K = a\,\mathrm{Tr}[\mathbb{D}_K T_K(D[1]+am')^{-2}T_K^*].$$

Since $\|\mathbb{D}_K\| \le c_1$, its dimension is $\nu\ell^3$, and $\|(D[1]+am')^{-1}\| \le (a\,\mathrm{Re}\,m')^{-1}$, the normalized derivative is at most $4\nu c_1 m^{-2}$ on $|m'-m| < m/2$. Two further Cauchy derivatives on circles of radius $m/4$ give $128\nu c_1 m^{-4}$. □

Lemma 11.23 (The two free operators at the comparison scale)**.** Use the adjacent data A, B and depths $s = \lceil K/58\rceil$, $u = \lceil c_\# K/580\rceil$. At the trivial configuration let D_ζ^A, D_ζ^B be their common-space operators at depth s, with source $L^{-s}\zeta 1$. For $|\zeta| < m/2$,

$$\mathrm{Re}\,D_\zeta^\bullet \ge \tfrac{1}{2}c_b L^{-s}m, \quad \|D_\zeta^B - D_\zeta^A\| \le K_c L^{s-K}, \quad \det D_\zeta^\bullet = \det\mathbb{D}_{t_0^\bullet}(1;a_{K^\bullet}(m-\zeta)).$$

Consequently

$$\varphi^{\mathrm{bl}}(\zeta) = \int_0^1 \operatorname{Tr}\left[((1-\tau)D_\zeta^A + \tau D_\zeta^B)^{-1}(D_\zeta^B - D_\zeta^A)\right] d\tau$$

is a holomorphic logarithm of the determinant ratio, and

$$\ell^{-3}|\partial_\zeta^3 \varphi^{\mathrm{bl}}(0)| \le C_{\mathrm{dp}} L^{5s-K}, \qquad C_{\mathrm{dp}} = 96\nu K_c c_b^{-1} m_-^{-4}.$$

■

Proof. The identification with the tower is the exact composition of its averages in Cor. A.44. At the trivial field this tower exists by accretivity throughout $|\zeta| < m/2$. In the Schur form set $\varphi = a_0 \mathcal{K}^{-1} R^* \chi$. Then

$$\langle \chi, D_\zeta^\bullet \chi\rangle = a_0 \|\chi - R\varphi\|^2 + \langle \varphi, (D^\xi[1] + L^{-s}(m-\zeta))\varphi\rangle.$$

Here $a_0 \ge a_-$ and $\|R\| \le 1$. With $\mu = L^{-s}m/2$, minimizing $a_0\|\chi - v\|^2 + \mu\|v\|^2$ gives $a_0\mu/(a_0+\mu) \ge c_b\mu$. The comparison Prop. A.41 applies with zero curvature and constant source; its gradient contribution vanishes. It gives the norm difference above.

Every convex interpolation is accretive by the same floor. Its inverse has norm at most $2L^s/(c_b m)$, and its dimension is $\nu\ell^3 L^{3s}$. Hence $|\varphi^{\mathrm{bl}}| \le 2\nu\ell^3 K_c c_b^{-1} m^{-1} L^{5s-K}$. Differentiating the determinant along τ identifies its exponential with the ratio. Cauchy's estimate with radius $m/2$ gives the stated third derivative. □

Proposition 11.24 (Cancellation of the contact derivative)**.** Let $c^\bullet$ be the contact energy of Def. 11.2. For all sufficiently large K, uniformly on W,

$$\ell^{-3}\partial_\zeta^3 c^\bullet[v_\zeta]\big|_0 = -\Phi^{\mathrm{fr}}_{K\bullet}(m) - \ell^{-3}\partial_\zeta^3 \log\det D_\zeta^\bullet\big|_0 + r^\bullet,$$
$$|r^\bullet| \le 2C_{\mathrm{tf}} S_\phi \phi(\tfrac{1}{2}L^{u+1}\gamma^{-2}), \qquad S_\phi = (1 - 2L^{-1/2})^{-1}.$$

In particular

$$|(y^{(2)}_{K+1} + \Phi^{\mathrm{fr}}_{K+1}) - (y^{(2)}_K + \Phi^{\mathrm{fr}}_K)| \le \ell^{-3} N_f^{9/2} C_3 L^{-c_3 K} + C_{\mathrm{dp}} L^{5s-K} + 4C_{\mathrm{tf}} S_\phi \phi(\tfrac{1}{2}L^{u+1}\gamma^{-2}), \qquad c_3 = c_\#/116.$$

■

Proof. The original and restarted coefficients satisfy $x_i, \widehat{x}_i \ge \frac{1}{2}L^{K^\bullet - i}\gamma^{-2}$. The constant ceiling on γ implies $\phi(Lx) \le 2L^{-1/2}\phi(x)$ for $x \ge \gamma^{-2}/2$. Indeed the logarithmic factor in this ratio is at most $e^{1/3} < 2$. It follows that each sum of ϕ_i ending at depth $u+1$ is at most $S_\phi \phi(\frac{1}{2}L^{u+1}\gamma^{-2})$.

The contact energy is the original energy through $t_1^\bullet$ less the restarted energy from $t_0^\bullet$ to $t_1^\bullet$. Apply Lem. 11.22. The terms $L_j^F(m-\zeta) - L_j^F(m)$ cancel for $t_0^\bullet < j \le t_1^\bullet$. The remaining sum through $t_0^\bullet$ telescopes by Lem. 11.22 to the fine free determinant less the determinant of $D_\zeta^\bullet$. Its third ζ derivative is minus the third m derivative of the fine determinant. The two sums of fluctuation remainders obey the first displayed bound by the geometric estimate above. Subtract the identities for A, B and use Lem. 11.23 to control the block determinant ratio.

To apply Thm. Q, expand the flavour identity in the fixed orthonormal Hermitian basis: $1 = \sum_j \operatorname{tr}(t_j) t_j$. There are N_f^2 terms and $\sum_j |\operatorname{tr} t_j| \le N_f^{3/2}$. The three constant entries are an admissible tuple for the comparison with contact terms retained. Its error is at most $C_3 L^{-c_3 K}$ per basis triple. Multilinearity therefore costs at most $N_f^{9/2}$. Divide by ℓ^3 and combine the three bounds. □

Lemma 11.25 (A complex neighbourhood independent of the cutoff)**.** Let ρ_b, ρ_m be the radii in Prop. 9.3 and put

$$\rho_b^\flat = \rho_b/2, \qquad \rho_m^\flat = \min\{\rho_m/2, m_-/(4C_{\mathrm{pr}})\}.$$

Let $W^\flat$ be the union of the open polydiscs of these radii about W. Every $y_K^{(2)}$ is holomorphic there and bounded by the cutoff-independent number

$$C_{\mathrm{pd}} = C_{\mathrm{fr},1} + 128\nu c_1 m_-^{-4} + 48 C_w^s (LMr_J)^{-3} S_\phi \phi(\tfrac{1}{2}L\gamma^{-2}) + \frac{48\log 2}{\ell^3 \rho_m^3}.$$

■

Proof. About a real centre (b_0, m_0) write $b' = b_0 + t$, $m' = m_0 + \zeta$. The zero-free estimate of Prop. 9.3 gives

$$\log Z_K(b', m') = E_K(-t1, -\zeta 1) + \log Z_K^{\mathrm{sub}}[0] + \log \psi(t, \zeta), \qquad |\psi - 1| \le 1/2,$$

on the original polydisc. Use the principal logarithm of ψ, whose modulus is at most $\log 2$. Its third mass derivative on the smaller polydisc is at most $48\log 2/\rho_m^3$.

For the energy use Lem. 11.22 with constant gauge source $-t$. The additional Gaussian term depends only on t, so its mass derivatives vanish. The fermion normalizer sum telescopes, giving a free determinant term bounded by $C_{\mathrm{fr},1}$ and a terminal term bounded by $128\nu c_1 m_-^{-4}$. The remaining degree-zero sum at level j is bounded by $C_w^s \phi_{j-1}(\ell/(M\lambda_j))^3$. The circle of radius $r_J/(2\lambda_{j-1})$ about ζ remains in its domain, since $|\zeta| \le \rho_m/2 \le r_J/16$. Cauchy's estimate gives $48 C_w^s (LMr_J)^{-3}\phi_{j-1}$ after division by ℓ^3. Sum by the same geometric estimate. These four contributions are C_{pd}. □

Lemma 11.26 (From a real segment to a half-disc)**.** Suppose f is holomorphic on $|z - x_0| < R$, bounded there by M, and bounded by $\varepsilon \le M$ on a real interval of length 2δ, whose centre lies within δ of x_0, where $0 < \delta \le R/8$. Then, for $|z - x_0| \le R/2$,

$$|f(z)| \le M^{1-\omega}\varepsilon^\omega, \qquad \omega = \frac{\log(13/11)}{\log(2R/\delta)}.$$

■

Proof. If x_1 is the centre of the segment, use $J(w) = x_1 + \frac{\delta}{2}(w + w^{-1})$. It maps $|w| > 1$ onto the complement of the segment. For $1 \le |w| \le \rho_1 = (2R - 3\delta)/\delta$ its image is inside $|z - x_1| < R - \delta$, hence inside the original disc. On the annulus the maximum principle applied to $\log|f \circ J|$ bounds it by the harmonic interpolation of $\log\varepsilon$ on $|w| = 1$ and $\log M$ on $|w| = \rho_1$. At a point with $|z - x_0| \le R/2$, $|J^{-1}(z)| < (R + 3\delta)/\delta$. Its harmonic weight on the inner circle is therefore at least

$$\frac{\log((2R - 3\delta)/(R + 3\delta))}{\log((2R - 3\delta)/\delta)} \ge \frac{\log(13/11)}{\log(2R/\delta)}.$$

The same bound on the segment is immediate. □

Theorem R (Convergence of the differentiated condensate)**.** On every fixed admitted torus, $y_K^{(2)}$ converges along the full sequence of cutoffs, uniformly on W. More precisely, for $K \ge K_*(g, \ell)$,

$$\sup_W |y_{K+1}^{(2)} - y_K^{(2)}| \le C(g, \ell)(1 + K)^{p_0} L^{-c_y K}, \qquad c_y = c_\#/1160.$$

It also converges on the union of the closed discs of radius $\rho_b^\flat/2$ about I_g, uniformly for real $m \in [m_-, m_+]$, with all b derivatives on W. If $m_- < m_+$, it converges uniformly on the union $W^\natural$ of the closed polydiscs of radii $\rho_b^\flat/2, \rho_m^\flat/2$ about W. Its limit is holomorphic on the interior, and all parameter derivatives converge on W. In particular

$$\ell^{-3} S_{p+q,K}^T((1,1)^q, (0,1)^p), \qquad q \ge 3,$$

converges for every $p \ge 0$. All these assertions concern a fixed side; their constants may depend on it. ■□

Proof. By Lem. 11.21, the increment of the free term is at most $2C_{\mathrm{fr}}L^{-K/2}$. Add this to Prop. 11.24. Since

$$5s - K \le 5 - 53K/58, \qquad u \ge c_{\#}K/580 = 2c_yK,$$

each of its three errors is at most a constant times $(1+K)^{p_0}L^{-c_yK}$. Their series converges, proving the real assertion with a tail of the same order.

The increments are bounded by $2C_{\mathrm{pd}}$ on $W^\flat$. Apply Lem. 11.26 in b with

$$\delta_b = \min\{\rho_b^\flat/8, |I_g|/2\}, \qquad h_b = \frac{\log(13/11)}{\log\max\{16, 4\rho_b^\flat/|I_g|\}}.$$

Choose the real segment inside I_g with centre nearest the base point. The complex increment is bounded by $(2C_{\mathrm{pd}})^{1-h_b}\varepsilon_K^{h_b}$, where ε_K is the lesser of its real bound and $2C_{\mathrm{pd}}$. This series is summable.

If $m_- < m_+$, repeat in m with

$$h_m = \frac{\log(13/11)}{\log\max\{16, 4\rho_m^\flat/(m_+ - m_-)\}}.$$

The resulting exponent is $h_bh_m > 0$. Thus the series converges uniformly on $W^\natural$, at rate at most $C(1+K)^{p_0}L^{-h_bh_mc_yK}$. Uniform limits are holomorphic in the interior. Cauchy's formula on the half-radius polydiscs gives convergence of each derivative; its cost is $p!(q-3)!(2/\rho_b^\flat)^p(2/\rho_m^\flat)^{q-3}$. The constant-source identity identifies these derivatives with the stated cumulants. □

The radii and constants in this subsection are formed after the construction constants, g and ℓ; they impose no additional mass-dependent ceiling on γ. The theorem proves convergence of the second mass derivative of y_K. It makes no assertion that y_K itself converges without a subtraction.

11.8. **The subtracted scalar one-point function.** The preceding theorem controls the second mass derivative of y_K, leaving an affine function of the mass undetermined. The next result identifies its divergent part. There are only three cutoff scales: a_K^{-2}, a_K^{-1} and K. Their coefficients are independent of the physical torus and of the auxiliary choices in the construction.

Retain the chamber operator with $r = 1/2$ and $N_f \ge 1$. Write $I_g = \bar I(g) = [b_-(g), b_+(g)]$, where

$$b_-(g) = g^{-2} + \bar B, \qquad b_+(g) = L(g^{-2} + B_\sharp) - \bar B, \qquad W(g) = I_g \times [m_-, m_+].$$

The notation b, m is the notation of eq. 11.xxv; it denotes the labels b, m used elsewhere. In particular $y_{K,\ell} = -\ell^{-3}\partial_m \log Z_{K,\ell}$.

Theorem S (Convergence of the subtracted condensate)**.** Choose the construction and source constants as in [QCDM1, Prop. 10.20], including its gauge-admission conditions, and impose all five additional requirements in Def. 11.1. Impose also the fermion-weight floor eq. 14.viii and the field-scale choices of Rem. 14.9, including its ceilings defining γ_{e}. Choose the mass window with

$$0 < Lm_- < m_+ \le \tfrac{1}{2}.$$

Fix $0 < g \le \gamma_{\mathrm{e}}$ and one physical torus T_ℓ, $\ell \in \mathfrak{L}(g)$.

There are real constants $\mathsf{A}_0, \mathsf{A}_1, \mathsf{A}_2, w_1, w_2$ such that, with $a_K = L^{-K}$,

$$\mathsf{br}_K(b, m) = \mathsf{A}_0 a_K^{-2} + (\mathsf{A}_1 m + \mathsf{A}_2/b)a_K^{-1} + (w_1 m/b + w_2/b^2)K, \tag{11.xxvi}$$

the following assertions hold.

(a) The free coefficients depend only on the chamber operator and N_f:

$$\mathsf{A}_0 = -\int_{[-\pi,\pi]^3} \operatorname{tr}\widehat D(p)^{-1}\,\frac{\mathrm{d}^3p}{(2\pi)^3}, \qquad \mathsf{A}_1 = \int_{[-\pi,\pi]^3} \operatorname{tr}\widehat D(p)^{-2}\,\frac{\mathrm{d}^3p}{(2\pi)^3}.$$

Here $\widehat D$ is the mass-zero chamber symbol of Lem. 11.21, including colour and flavour. Both integrals are absolutely convergent. The remaining three coefficients depend only on this

operator, N_f and L. All five are independent of g, ℓ, K, b, m, and are the same for any admissible auxiliary choices with the same L.

The interaction coefficients are determined by the field-scale coefficients q_n of Def. 15.1:

$$\mathsf{A}_2 = -\lim_{K\to\infty} \frac{L^{-K}\mathsf{q}_1(K;b;m_1,m_2)}{(L-1)(m_2-m_1)},$$
$$w_2 = -\lim_{K\to\infty} \frac{\mathsf{q}_2(K;b;m_1,m_2)}{m_2-m_1},$$
$$w_1 = -\lim_{K\to\infty} \frac{2}{m_3-m_1}\left(\frac{\mathsf{q}_1(K;b;m_2,m_3)}{m_3-m_2} - \frac{\mathsf{q}_1(K;b;m_1,m_2)}{m_2-m_1}\right).$$

These limits have the same values for every label in any admitted window and every $m_- \le m_1 < m_2 < m_3 \le m_+$.

(b) Put $c_y = c_\#/1160$, $q = 8\mathfrak{p}_0 + 1$ and $S_q = \sum_{j\ge 0}(1+j)^q L^{-c_y j}$. There are $K_\Xi = K_\Xi(g,\ell)$ and $C_\Xi = C_\Xi(g,\ell) < \infty$ such that

$$\sup_{W(g)} |(y_{K+1} - \mathsf{br}_{K+1}) - (y_K - \mathsf{br}_K)| \le C_\Xi (1+K)^q L^{-c_y K} \qquad (K \ge K_\Xi).$$

Consequently $y_K - \mathsf{br}_K$ converges along the full cutoff sequence, uniformly on $W(g)$, to a continuous real function $y^\flat_\infty$, and

$$\sup_{W(g)} |y_K - \mathsf{br}_K - y^\flat_\infty| \le C_\Xi S_q (1+K)^q L^{-c_y K}.$$

(c) The convergence extends to a complex neighbourhood independent of the cutoff. More precisely, use the radii $\rho^\flat_b, \rho^\flat_m$ of Lem. 11.25 and put

$$\vartheta = 1 - \left(\frac{4+c_y}{4+2c_y}\right)^{1/2}, \qquad \omega = (1-\vartheta)^2, \qquad \delta_b = \tfrac{1}{2}(b_+ - b_-), \quad \delta_m = \tfrac{1}{2}(m_+ - m_-).$$

For $z = b, m$, let $R'_z > 1$ solve $\frac{\delta_z}{2}(R'_z - R'^{-1}_z) = \rho^\flat_z/2$, and set

$$r_z = \frac{\delta_z}{2}\frac{(R'^{\vartheta}_z - 1)^2}{R'^{\vartheta}_z}, \qquad W^\sharp = \{(b', m') : \operatorname{dist}(b', I_g) < r_b, \operatorname{dist}(m', [m_-, m_+]) < r_m\}.$$

Every y_K and br_K is holomorphic on $W^\sharp$. For $K \ge K_\Xi$,

$$\sup_{W^\sharp} |(y_{K+1} - \mathsf{br}_{K+1}) - (y_K - \mathsf{br}_K)| \le C'_\Xi(g,\ell)(1+K)^q L^{-c_y K/2}.$$

The limit is holomorphic there and real on the real window. Every derivative with respect to b and m converges uniformly on $W(g)$. In particular,

$$\partial_m y_K - \mathsf{A}_1 L^K - w_1 K/b$$

converges uniformly on that window.

The thresholds, rates, radii and limit may depend on g and ℓ. No passage to infinite volume is asserted here. ■□

The proof is in §15. The contact terms in the adjacent-cutoff comparison are retained. After integration in the mass, they are differences of vacuum energies. A common field scale expands those energies through degree six with summable error. Removing the Gaussian tests makes the coefficients homogeneous in $1/b$. The mass-derivative estimate and exact lattice rescaling then determine the three cutoff scales in eq. 11.xxvi.

11.9. **Continuation of cutoff convergence.**

Lemma 11.27 (Propagation along a thin complex neighbourhood)**.** Let $a < p < b$, $0 < w \le b - p$, and $G = \{z : \operatorname{dist}(z, [a, b]) < w\}$. The harmonic measure ω of $[p, b]$ in $G \setminus [p, b]$ satisfies

$$\omega(x) \ge \tfrac{1}{5}\, 3^{-4(p-x)/w} \qquad (a \le x < p).$$

If F is holomorphic on G, $|F| \le M$ there and $|F| \le m$ on $[p, b]$, then

$$|F(x)| \le M^{1-\omega_*} m^{\omega_*} \quad (a \le x \le b), \qquad \omega_* = \tfrac{1}{5}\, 3^{-4(p-a)/w}.$$

■

Proof. In the disc $|z - p| < w$, slit along $[p, p + w)$, compare ω with the harmonic measure of the slit. After setting $\zeta = (z - p)/w$ and $s = \sqrt{\zeta}$, with $0 < \arg s < \pi$, this is the harmonic measure of the diameter of the upper half-disc. At $\zeta = -\tau^2$, $0 < \tau \le 1/\sqrt{2}$, its value is

$$1 - \frac{2}{\pi} \arctan \frac{2\tau}{1 - \tau^2} \ge 1 - \frac{2}{\pi} \arccos(1/3) > 1/5.$$

The half-disc formula was computed in Lem. 10.15. The maximum principle gives the same lower bound for ω on $[p - w/2, p)$. For a centre $y \le p - w/2$, the disc $|z - y| < w/2$ lies in $G \setminus [p, b]$. Harnack's inequality gives a factor at worst $1/3$ for a step of length at most $w/4$. A chain from $p - w/2$ to x has fewer than $4(p - x)/w$ such steps, proving the displayed lower bound.

Apply the subharmonic maximum principle to

$$\log |F| - (1 - \omega) \log M - \omega \log m.$$

Its boundary upper limits are nonpositive, so $|F| \le M^{1-\omega} m^{\omega}$ inside. If $m \le M$, use $\omega \ge \omega_*$ on the real interval to the left of p; if $m > M$, the bound $|F| \le M$ is stronger. On $[p, b]$ use $|F| \le \min\{m, M\}$. □

Theorem T (Full cutoff convergence at real masses)**.** Impose the cutoff-comparison hypotheses of Def. 11.1. Keep g, ℓ, b fixed. Let X have pairwise disjoint real Lipschitz test supports, with two supports at distance at least $d > 0$. Put

$$\omega_\ell = \tfrac{1}{5}\, 3^{-3r_J/(2w_\ell)}, \qquad C_n^w(d) = \frac{[2B_n^w(d) e^{n(C+2+A^\sharp)\ell^3}]^{1-\omega_\ell} C_n(g, \ell)^{\omega_\ell}}{1 - L^{-c_n \omega_\ell}}.$$

Here $c_n, C_n(g, \ell)$ and $K_n(g, \ell; d)$ are the cutoff-comparison constants in Thm. Q and Cor. 11.6. For $x \in I^\sharp$ and $K' \ge K \ge \max\{K_\sharp, K_n(g, \ell; d)\}$,

$$|S^T_{n,K'}(x; X) - S^T_{n,K}(x; X)| \le C_n^w(d) L^{-c_n \omega_\ell K} \prod_i \|f_i\|_\sharp.$$

The full sequence converges locally uniformly on $\mathcal{O}^\sharp$ to a holomorphic function, with all mass derivatives. It also converges at moving masses $x_j \to x \in I^\sharp$. The rate is independent of the mass and coupling within the stated ranges, but depends on the fixed torus:

$$\omega_\ell \ge \tfrac{1}{5} \exp[-\tfrac{3}{2}(\log 3)(r_J/c_w)\ell^3(1 + \log \ell)].$$

■□

Proof. By Cor. 10.22 and Prop. 10.23, the increment $F_K = S^T_{n,K+1}(\cdot; X) - S^T_{n,K}(\cdot; X)$ is holomorphic on $\mathcal{O}^\sharp$ and bounded there by

$$2B_n^w(d) e^{n(C+2+A^\sharp)\ell^3} \prod_i \|f_i\|_\sharp.$$

On the positive interval I_0, Thm. Q bounds the same increment by $C_n(g, \ell) L^{-c_n K} \prod_i \|f_i\|_\sharp$. Apply Lem. 11.27 with $a = -r_J/4$, $p = r_J/8$, $b = 1/2$ and $w = w_\ell$. Since $p - a = 3r_J/8$, its exponent is ω_ℓ. Summing the resulting geometric estimate proves the displayed cutoff bound. Zero test functions cause no difficulty by multilinearity.

The sequence is locally bounded on the connected domain $\mathcal{O}^\sharp$ and converges on I_0, which has accumulation points there. Vitali's theorem gives local uniform convergence throughout the domain. Cauchy's formula gives convergence of all derivatives, and local uniform convergence gives the assertion at moving masses. The final inequality follows from the lower bound for w_ℓ in Cor. 10.22. □

Proposition 11.28 (Scaling to any real mass label)**.** Let $\mathsf{b} > 0$ and $\mathsf{m} \in \mathbb{R}$. Choose an integer z with

$$L^z \geq \max\left\{\frac{\gamma^{-2} + 4B_\sharp + 1}{\mathsf{b}}, 2|\mathsf{m}|, \frac{4|\mathsf{m}|}{r_J}\right\}.$$

Then

$$\mathsf{b}' = L^z\mathsf{b}, \qquad \mathsf{m}' = L^{-z}\mathsf{m}, \qquad g_z = (\mathsf{b}' - 3B_\sharp)^{-1/2}$$

satisfy $g_z < \gamma$, $\mathsf{b}' \in I_{g_z}$ and $\mathsf{m}' \in I^\sharp$. The product bm is unchanged. For every $\ell' \in \mathfrak{L}(g_z)$ such that $\ell = L^{-z}\ell'$ is an admitted torus side, the model at $(\ell, K, \mathsf{b}, \mathsf{m})$ is the same finite lattice integral as the model at $(\ell', K - z, \mathsf{b}', \mathsf{m}')$. Thus the fixed-torus conclusions above transfer to these sides. The corresponding growing-torus subsequential conclusions transfer as well. ■

Proof. The inequalities give $\mathsf{m}' \leq 1/2$ if the mass is positive and $\mathsf{m}' \geq -r_J/4$ if it is negative. The zero mass stays zero. Also $\mathsf{b}' > \gamma^{-2} + 4B_\sharp$, whence $g_z < \gamma$ and

$$\mathsf{b}' = g_z^{-2} + 3B_\sharp \in [g_z^{-2} + 3B_\sharp, Lg_z^{-2}] \subset I_{g_z}.$$

The two descriptions have the same number of sites and the same bare parameters:

$$L^{K-z}\mathsf{b}' = L^K\mathsf{b}, \qquad L^{-(K-z)}\mathsf{m}' = L^{-K}\mathsf{m}.$$

The insertion formulas therefore give the scaling identity directly. For $f^{[z]}(x) = f(L^{-z}x)$,

$$O_{K-z}(f^{[z]}) = L^{-z}O_K(f), \qquad \Phi_{K-z}(f^{[z]}) = L^z\Phi_K(f).$$

Apply Props. 10.26 and 10.28 and Thm. T at the primed parameters and carry each multilinear form back by these identities. In the two-point laws $\mathcal{K}_6$ is unchanged by this dilation and $\mathcal{K}_4$ is multiplied by L^{2z}, which gives the original leading coefficients after scaling back. Growing tori use the uniform bound of Prop. 10.25 and the same diagonal extraction. □

The continuation and scaling prove these conclusions for the mass label m. They supply no assertion that the physical spectral mass vanishes when $\mathsf{m} = 0$. The full cutoff theorem here keeps the torus fixed; its rate does not give convergence uniformly in growing volume.

12. Laws of the block fields

Fix $0 < g \leq \gamma$, $\ell \in \mathfrak{L}(g)$, the physical torus T_ℓ and the compact positive label rectangle $\mathsf{W}(g) = \bar{I}(g) \times [m_-, m_+]$. Throughout this section we impose the constant choices of Def. 11.1, including its five additional floors and ceilings. The block map $\mathcal{M}$ is the same Borel gauge-covariant map at every cutoff. At depth d its configuration space $\mathcal{V}_{[d]}$ is a finite product of copies of SU(2) on the lattice of physical spacing L^{-d} in T_ℓ. For $K \geq d$ let

$$\mu_d^{F,K} = (\mathcal{M}^{K-d})_*\nu_K. \tag{12.i}$$

The superscript F distinguishes the matter law from the pure-gauge law used below.

Theorem U (Convergence of the block laws)**.** At the preceding data there are numbers τ_K, independent of the labels, such that $\sum_K \tau_K < \infty$ and, for all sufficiently large K,

$$\mathrm{TV}(\mu_d^{F,K}, \mu_d^{F,K+1}) \leq \tau_K \quad \left(0 \leq d \leq d_K := \left\lfloor \tfrac{1}{4}\log_L K \right\rfloor\right), \qquad \tau_K \leq (8\mathsf{M}_{g,\ell} + 26)K^{-2}.$$

Here $\mathrm{TV}(P,Q)=\sup_E|P(E)-Q(E)|$. For each fixed d, the full sequence converges in total variation to a gauge-invariant probability measure μ_d^F, uniformly on $\mathsf{W}(g)$, with

$$\mathrm{TV}(\mu_d^{F,K},\mu_d^F)\le\sum_{j\ge K}\tau_j$$

once K is sufficiently large and $K\ge L^{4d}$. The limit depends continuously on the labels in total variation and satisfies $\mathcal{M}_*\mu_{d+1}^F=\mu_d^F$.

On the compact product $\mathcal{V}_{\mathrm{thr}}=\prod_{d\ge0}\mathcal{V}_{[d]}$ there is exactly one probability measure μ_{thr}^F whose coordinates through depth D have law

$$(V\mapsto(\mathcal{M}^{D-d}V)_{d\le D})_*\mu_D^F.$$

It is concentrated on the compatible sequences $V_d=\mathcal{M}V_{d+1}$. The laws obtained from cutoff K by using $V_d=\mathcal{M}^{K-d}U$ for $d\le K$ and any fixed configurations for $d>K$ converge weakly to μ_{thr}^F along the full sequence. ■□

The proof compares the measures before taking derivatives in sources. Its three operations on each lattice are to discard histories with an early large-field region, change to the restarted history, and sum the histories above its base. The two restarted measures can then be compared on one configuration space. We carry a bounded Borel test function through each operation; this gives total variation.

12.1. **Normalization and positive measures.**

Lemma 12.1. The laws in eq. 12.i are gauge invariant and satisfy $\mathcal{M}_*\mu_{d+1}^{F,K}=\mu_d^{F,K}$ for $K\ge d+1$. For gauge-invariant probability measures, total variation can be tested on gauge-invariant Borel functions with values in $[0,1]$. Borel image maps contract total variation.

The following normalization bounds will be used. Write $x(f)=\int f\,dx$.

(1) If x,y are positive measures, $x(1)>0$, and $0\le(x-y)(f)\le(x-y)(1)\le Rx(1)$ for $0\le f\le1$, with $R<1$, then $y(1)>0$ and the normalized integrals differ by at most $R/(1-R)$.
(2) If a,p are positive measures, $a(1)>0$, $c\ne0$, and $|cp(f)-a(f)|\le\varepsilon a(1)$ for all Borel $|f|\le1$, where $\varepsilon\le1/2$, then $p(1)>0$ and the normalized integrals differ by at most 4ε.
(3) If x,y are positive measures, $x(1)>0$, and $|x(f)-y(f)|\le\eta x(1)$ for $|f|\le1$, where $\eta<1$, then $y(1)>0$ and the normalized integrals differ by at most 2η.

The first bound also holds on a class of tests containing 1 when its displayed inequalities hold on that class. ■

Proof. Lift a gauge transformation at depth d to one constant on each block of the original lattice. Gauge covariance of $\mathcal{M}^{K-d}$ and invariance of the bare density and Haar measure give gauge invariance of its image. Composition of the block maps gives consistency. For a Borel set E, average $1_E(V^v)$ over the compact group of gauge transformations. Fubini gives a Borel invariant function $f_E\in[0,1]$ with $P(f_E)=P(E)$ and $Q(f_E)=Q(E)$. Conversely, integration of the level sets of any $[0,1]$-valued test bounds its integral difference by total variation. The contraction assertion follows by taking inverse images of Borel sets.

For (i) put $\delta=x-y$. The numerator of the difference is $x(f)\delta(1)-\delta(f)x(1)$, of modulus at most $\delta(1)x(1)$, and $y(1)\ge(1-R)x(1)$. For (ii), $|cp(1)|\ge(1-\varepsilon)a(1)>0$; subtract the normalized fractions to get $2\varepsilon/(1-\varepsilon)\le4\varepsilon$. For (iii), use $|y(f)|\le y(1)$ when subtracting the fractions with denominator $x(1)y(1)$. Each of the two terms is at most η. □

12.2. **The comparison at a growing depth.** Use the two cutoffs $K^{\mathrm{A}}=K$, $K^{\mathrm{B}}=K+1$. The depths of the cutoff comparison at $n=1$ are $s=\lceil K/34\rceil$ and $u=\lceil c_{\#}K/340\rceil$; the corresponding

levels are $t_0^\bullet = K^\bullet - s$ and $t_1^\bullet = K^\bullet - u$. The common spaces at these depths are $\mathcal{Y}$ and $\mathcal{X}$. Retain the interpolation radius r_λ and the errors $R^F, \varepsilon_{\mathrm{sh}}, \varepsilon_{\mathrm{cl}}, \varepsilon_F$ from Def. 11.5 at $n = 1$. For $d' \geq 0$ put

$$\begin{aligned} N_{[d']} &= \ell^3 L^{3d'}, \\ R_{d'} &= \mathsf{M}_{g,\ell} \exp[-\theta_{\mathrm{tl}} A_0^2 (\log(L^{d'}\gamma^{-2}))^{2\mathfrak{p}_0}], \\ e_{d'} &= \frac{\exp[(C + 2 + C_P(1 + \log(L^{d'} X_g + \bar{B})))N_{[d']}]}{r_\lambda - 1}, \\ \eta_K &= 3R^F + 2\varepsilon_{\mathrm{sh}} + 2(\varepsilon_{\mathrm{cl}} + \varepsilon_F), \\ \tau_K &= 8R_{d_K} + 8e_{d_K} + 2\eta_K. \end{aligned}$$

Take K large enough for the cutoff-comparison estimates, $r_\lambda \geq 16\mathsf{M}_{g,\ell} \geq 16$, and

$$d_K \leq u - 1, \qquad 8R_{d_K} \leq 1, \qquad 2e_{d_K} \leq 1. \tag{12.ii}$$

The proof below verifies that these conditions hold eventually, uniformly in the labels.

Lemma 12.2 (Histories above the restart)**.** For $t_1^\bullet \leq k \leq K^\bullet$, the interpolation at zero source satisfies

$$\sum_{h \in \mathcal{H}_k^{(t_1^\bullet)}} \int \|g_h[0,\lambda]\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}} \leq \exp[\operatorname{Re} E_k(0,\lambda) + (C+2)n_k], \qquad |\lambda| \leq r_\lambda.$$

The energy is holomorphic for $|\lambda| < r_\lambda$, and $E_k(0,0) = E_k$ of the original flow. ■

Proof. Apply the free-past source induction Prop. C.21 at base $t_1^\bullet$, using Cor. C.23. The difference perturbation has the four lag bounds with Λ, by Prop. B.23. At zero source those bounds hold at radius $c_\lambda \Lambda$, and $r_\lambda \leq R_K/2 \leq R_k/2$. Thus every point of the stated closed disc lies in the open induction domain. Its history estimate, after dropping the nonnegative mark, gives the displayed bound. The densities are those of Lem. C.63, which agree almost everywhere with the induction densities. The energy conclusion is the holomorphy and zero-parameter clause of that induction. □

Lemma 12.3 (The restarted term is positive)**.** Let $\hat{\theta}^\bullet$ be the restarted term at level $t_1^\bullet$, and let $\nu_\lambda^\bullet$ be the scalar Berezin integral of the interpolated measure at that level and zero source. Then

$$\operatorname{Ber} \hat{\theta}^\bullet = (\mathcal{M}^{s-u})_*(\hat{\mathsf{k}}^\bullet \hat{F}_0^\bullet\, dW), \qquad \nu_1^\bullet = e^{E_{t_1^\bullet} - \hat{c}^\bullet(0)} \operatorname{Ber} \hat{\theta}^\bullet.$$

These are positive measures and

$$\nu_1^\bullet(\mathcal{X}) \geq \tfrac{5}{8} \exp\left[\operatorname{Re} E_{K^\bullet}(0,1) - C_P(1 + \log x_{K^\bullet})\ell^3\right].$$

■

Proof. At a step along the history without large fields the fermion kernel integrates to one in the new generators. Consequently its scalar Berezin integral pushes forward the old scalar measure multiplied by the passing-test indicator and the gauge-fixing factor. Iterating from $t_0^\bullet$ to $t_1^\bullet$ and using the common block map gives the first identity. The initial cutoff indicator is absorbed in $\hat{\mathsf{k}}^\bullet$, as in Lem. 11.16. Both factors in this density are nonnegative, and its mass is $\hat{T}^\bullet(0) > 0$. The interpolation at $\lambda = 1$ is exactly the restarted term times the displayed energy factor. That factor is positive: both zero-source energies are real. Finally the lower bound in Prop. B.41(d), with the readout of Prop. B.42, gives the claimed mass after this energy factor is included. □

Lemma 12.4 (Discarding large-field histories)**.** Let $\varsigma_0 = e^{-\beta A} \det(D + m_0)$, $Z = \int \varsigma_0\, dU$, and $m_t(F) = \int \mathfrak{w}_{h_\emptyset^{(t)}} F \varsigma_0\, dU$. For gauge-invariant Borel $0 \leq F \leq 1$,

$$0 \leq \int F\varsigma_0\, dU - m_t(F) \leq Z - m_t(1) \leq \mathsf{M}_{g,\ell} e^{-\theta_{\mathrm{dis}} A_0^2 (\log \mathfrak{t}_t)^{2\mathfrak{p}_0}} Z.$$

■

Proof. The exact history decomposition holds with the density $F\varsigma_0$. Its terms are nonnegative and bounded by the corresponding terms with $F = 1$. The histories whose restriction at level t is not $h_\emptyset^{(t)}$ are precisely those discarded in Prop. B.43(a). Summing their masses gives the bound. □

Lemma 12.5 (Changing to the restart)**.** Let $0 \le d' \le u-1$, $2e_{d'} \le 1$, and $k = K^\bullet - d'$. The interpolated scalar measure on $\mathcal{V}_{[d']}$ along $h' = h_\emptyset^{(k)}$ is

$$J_\lambda^\bullet = (\mathcal{M}^k)_*(\mathfrak{w}_{h'}\varsigma^{0,\lambda}\,dU),$$

where $\varsigma^{0,\lambda}$ is the free-past density of Lem. C.7. At $\lambda = 0$ this is a positive measure with

$$J_0^\bullet(1) \ge e^{E_k - C_P(1+\log x_k)N_{[d']}}.$$

At $\lambda = 1$ it is the image under $\mathcal{M}^{u-d'}$ of $\nu_1^\bullet$ times the nonnegative weight of the steps from $t_1^\bullet$ to k. It is positive, and

$$\left|\frac{J_1^\bullet(f)}{J_1^\bullet(1)} - \frac{J_0^\bullet(f)}{J_0^\bullet(1)}\right| \le 4e_{d'} \qquad (|f| \le 1).$$

■

Proof. The free-past construction Lem. C.63 at base $t_1^\bullet$ gives the same interpolated history terms as Lem. 12.2. Along h' there are no proper or external bonds below level k, no new fluctuation variable, and all free coordinates are the block field $\mathcal{M}^k U$. Their reference measure is Haar measure on $\mathcal{V}_k$. This remains true at $k = K^\bullet$. Consequently the scalar Berezin integral of this history is the displayed image measure.

At $\lambda = 0$, $\varsigma^{0,0} = \varsigma_0$, giving positivity and the lower mass bound from the partition estimate of [QCDM1]. Factor the history weight as

$$\mathfrak{w}_{h'}(U) = \mathfrak{w}_{h_\emptyset^{(t_1)}}(U)\mathfrak{w}_{h'}^\circ(\mathcal{M}^{t_1}U).$$

By Lem. C.12, $\mathfrak{w}_{h'}^\circ$ is a product of passing-test indicators and gauge-fixing factors. It is nonnegative. At $\lambda = 1$ the image at t_1 is $\nu_1^\bullet$, positive by Lem. 12.3; hence $J_1^\bullet$ is positive as claimed.

For bounded Borel f, $J_\lambda^\bullet(f)$ is entire in λ. Indeed the initial density is entire at each configuration and uniformly bounded on each compact parameter set. Dominated convergence and Morera's theorem justify integration. By Lem. 12.2,

$$H(\lambda) = e^{-E_k(0,\lambda)}J_\lambda^\bullet(f) \quad \text{satisfies} \quad |H(\lambda)| \le e^{(C+2)N_{[d']}}, \quad |\lambda| < r_\lambda.$$

Here the Berezin integral is bounded by the fibre norm because $h_F \ge 1$. Cauchy's estimate on discs centered on $[0,1]$, with radii increasing to $r_\lambda - 1$, gives

$$|H(1) - H(0)| \le \frac{e^{(C+2)N_{[d']}}}{r_\lambda - 1}.$$

The lower mass bound and $x_k \le L^{d'}\mathsf{b} + \bar{B} \le L^{d'}X_g + \bar{B}$ make this at most $e_{d'}e^{-E_k}J_0^\bullet(1)$. Apply Lem. 12.1(ii) to $a = e^{-E_k}J_0^\bullet$ and $cp = e^{-E_k(0,1)}J_1^\bullet$. □

Lemma 12.6 (Summing above the restart)**.** Under the preceding conditions and $8R_{d'} \le 1$, put

$$Q^\bullet = (\hat{T}^\bullet(0))^{-1}(\mathcal{M}^{s-d'})_*(\hat{\mathsf{k}}^\bullet\hat{F}_0^\bullet\,dW).$$

Then for Borel $0 \le f \le 1$,

$$\left|\frac{J_1^\bullet(f)}{J_1^\bullet(1)} - Q^\bullet(f)\right| \le 2R_{d'}.$$

■

Proof. The exact image identity of Prop. C.10, applied at $\lambda = 1$, sums the histories above $t_1^\bullet$ to $(\mathcal{M}^{u-d'})_* \nu_1^\bullet$. Each summand is positive: its weight factors over $\nu_1^\bullet$ as in the preceding proof. Its mass is preserved by summing its descendants. Thus removing h' leaves a positive measure, whose mass is the sum over final histories having a large-field region at level k. The marked estimate Prop. 11.13(B5) bounds this by

$$e^{-\theta_{\mathrm{t1}} A_0^2 (\log(L^{d'}\mathsf{b}))^{2\mathfrak{p}_0}} e^{\operatorname{Re} E_{K^\bullet}(0,1) + (C+2)\ell^3}.$$

Divide by the lower bound in Lem. 12.3. The discarded fraction is at most $8R_{d'}/5 \le 1/5$. By Lem. 12.1(i), normalization changes the integrals by at most $2R_{d'}$. The normalized full image is $Q^\bullet$, by the two identities in Lem. 12.3. □

Lemma 12.7 (The two restarted densities)**.** At the cutoff-comparison thresholds,

$$\int_{\mathcal{Y}} |\hat{\mathsf{k}}^{\mathrm{A}} \hat{F}_0^{\mathrm{A}} - \hat{\mathsf{k}}^{\mathrm{B}} \hat{F}_0^{\mathrm{B}}|\, dW \le \eta_K \hat{T}^{\mathrm{A}}(0).$$

Consequently $|Q^{\mathrm{A}}(f) - Q^{\mathrm{B}}(f)| \le 2\eta_K$ for $|f| \le 1$, at every depth $d' \le s$. ■

Proof. Insert the passing-test weights $\mathsf{k}^\bullet$ and split the integrand into the two differences $|\hat{\mathsf{k}}^\bullet - \mathsf{k}^\bullet| \hat{F}_0^\bullet$ and the middle difference $|\mathsf{k}^{\mathrm{A}} \hat{F}_0^{\mathrm{A}} - \mathsf{k}^{\mathrm{B}} \hat{F}_0^{\mathrm{B}}|$. The first and last integrals are bounded by $R^F \hat{T}^\bullet(0)$, by Prop. 11.10. Where both weights are positive, they agree, and the two densities differ by a factor e^{ξ_0} with $|\xi_0| \le \varepsilon := \varepsilon_{\mathrm{cl}} + \varepsilon_F \le 1/40$. The corresponding integral is at most $e^{1/40}(1 + R^F)\varepsilon \hat{T}^{\mathrm{A}}(0)$. Where only one weight is positive, the marked-tail comparison in the proof of Lem. 11.18 bounds its mass by $\varepsilon_{\mathrm{sh}} \hat{T}^\bullet(0)/2$. This estimate uses the nonnegative coefficients of the mark polynomial and includes the $3sN_{[s]}$ test terms.

Let D denote the integral to be bounded and $b = R^F + \varepsilon_{\mathrm{sh}}/2 < 1/50$. Since $\hat{T}^{\mathrm{B}}(0) \le \hat{T}^{\mathrm{A}}(0) + D$,

$$D \le (2b + e^{1/40}(1 + R^F)\varepsilon)\hat{T}^{\mathrm{A}}(0) + bD.$$

Thus $D \le \frac{50}{49}(2R^F + \varepsilon_{\mathrm{sh}} + 1.04\varepsilon)\hat{T}^{\mathrm{A}}(0) \le \eta_K \hat{T}^{\mathrm{A}}(0)$. Both densities live on the same Haar space $\mathcal{Y}$ and use the same image map to depth d'. Apply Lem. 12.1(iii); the cutoff thresholds give $\eta_K < 1$. □

12.3. **Summation and the compatible law.**

Proof of Thm. U. Put $d' = d_K$ and first test at this depth. For a gauge-invariant Borel $f \in [0,1]$, Lem. 12.4 and Lem. 12.1(i) compare the original law to the normalized $J_0^\bullet$ with error at most $8R_{d'}/7$. The next two lemmas give errors $4e_{d'}$ and $2R_{d'}$ in passing to $Q^\bullet$. The two restarted laws differ by at most $2\eta_K$. Adding the three errors on each lattice gives

$$2(8/7 + 2)R_{d'} + 8e_{d'} + 2\eta_K \le \tau_K.$$

Gauge averaging proves the total-variation bound at depth d'. Composing the test with $\mathcal{M}^{d'-d}$ proves it at every $d \le d'$.

We verify summability and the existence of the thresholds. Since $d_K \ge \frac{1}{4}\log_L K - 1$ and $2\mathfrak{p}_0 > 1$,

$$R_{d_K} \le \mathsf{M}_{g,\ell} K^{-2}$$

for all sufficiently large K. Moreover $L^{d_K} \le K^{1/4}$ and $N_{[d_K]} \le \ell^3 K^{3/4}$. The interpolation-radius bound of Lem. 11.20, at $n = 1$, has the form

$$r_\lambda \ge c(K+1)^{-p_\Lambda} L^{a_1 K}, \qquad a_1 = \left(1 - \frac{1 + c_\#}{10}\right)\frac{c_\#}{34} > 0,$$

where $c > 0$ depends only on the fixed data. Hence

$$e_{d_K} \le C(K+1)^{p_\Lambda} \exp\left[(C + 2 + C_P(1 + \log(K^{1/4} X_g + \bar{B})))\ell^3 K^{3/4} - a_1 K \log L\right].$$

Its positive exponent is $o(K)$, so this is at most K^{-2} eventually. The same comparison estimates give

$$\begin{aligned} R^F &\le Ce^{-cK^{2\mathfrak{p}_0}}, \\ \varepsilon_{\mathrm{sh}} &\le C(1+KL^{3K/34})e^{-cK^{2\mathfrak{p}_0}}, \\ \varepsilon_{\mathrm{cl}} &\le CK^{1/2}(K\log L+\log X_g)^{2\mathfrak{p}_0}L^{(7s-K)/2}, \\ \varepsilon_F &\le C\left[L^{5s-K}+(s\log L+\log X_g)^{\mathfrak{p}_0}L^{(29s-2K)/6}\right]. \end{aligned}$$

Here $s \le K/34+1$, so $7s-K \le 7-27K/34$, $5s-K \le 5-29K/34$, and $29s-2K \le 29-39K/34$. Each of these four errors is eventually at most K^{-2}. Also $d_K \le u-1$ eventually. This proves eq. 12.ii, all required cutoff thresholds, and $\tau_K \le (8\mathsf{M}_{g,\ell}+26)K^{-2}$. Every bound is uniform in the label rectangle.

For fixed d, telescope the adjacent-cutoff inequality from K to $K' > K$, with $K \ge L^{4d}$. Completeness of the space of finite signed measures in variation norm, and closedness of its probability measures, give a limit μ_d^F and the asserted tail estimate. Gauge invariance passes to this limit. Borel pushforward contracts total variation, so consistency passes as well. At fixed K the normalized bare density is continuous on the compact product of the label rectangle and the link space, with positive normalization. It therefore depends continuously on the labels in L^1. Its block images are continuous in total variation, and their uniform limit has the same property.

For each D, push μ_D^F to the coordinates through D by $V \mapsto (\mathcal{M}^{D-d}V)_{d\le D}$. Consistency gives a projective family of probability measures on compact metric spaces. The countable-product extension theorem supplies its unique Borel probability measure. Each relation $V_d = \mathcal{M}V_{d+1}$ is Borel and has probability one; their countable intersection does too. For a continuous cylinder test on coordinates through D, its expectation at cutoff $K \ge D$ is an integral against $\mu_D^{F,K}$ of a bounded Borel function. Total variation gives convergence even though the block map need not be continuous. Continuous cylinder functions are uniformly dense in the continuous functions on the compact product, by Stone–Weierstrass. This proves weak convergence of the cutoff laws and finishes the theorem. □

13. Recovering parameters from block fields

The mean and variance of a background action provide observables that recover the coupling and flavour number. We first establish the pure-gauge comparison, then compute the change caused by the fermion determinant.

13.1. **A background statistic under Wilson's measure.** We first establish the pure-gauge estimates used to distinguish the coupling and flavour number. The statistic is the action of the background determined by a block field. Its definition is independent of the label at which the law is evaluated. This independence will let us differentiate its mean.

Use the pure-gauge constants admitted in [QCDM1, Def. 10.19 and Prop. 10.20], including the source and tail conditions for the small-field chain in [YM3, §6.12]. Fix $\ell = 2L^m$ and $J = [b_-, b_+]$ with

$$b_- \ge \gamma^{-2}, \qquad MR(b_+) \le L^m.$$

All constants in this subsection may depend on these data, but not on $s \in J$, K or d. The spaces of block fields and their averaging map are those of eq. 12.i. Write $N_d = \ell^3 L^{3d}$ and

$$O(U) = L^K \sum_p a_p(U), \qquad \nu_{K,s}^{\mathrm{W}}(dU) = Z_{K,s}^{-1} e^{-sO(U)}dU.$$

The law of $V_d = \mathcal{M}^{K-d}U$ is denoted by $\mu_d^{\mathrm{W,K}}[s]$.

Definition 13.1 (The background statistic)**.** Let $d_\natural \geq 1$ be the least integer with $L^{d_\natural} b_- \geq \mathfrak{t}_\natural$, the lower threshold for the field cuts. For $K > d \geq d_\natural$ put $t = K - d$. On $\{\varpi(V) < \bar{\varepsilon}\}$ let $U_t(V)$ be a background over V: a regular critical point of the Wilson action on $\{U : \mathcal{M}^t U = V\}$. Backgrounds form one orbit of the gauge transformations preserving this fibre. Thus their Wilson actions agree. Define

$$\begin{aligned}
\mathrm{In}_t[s] &= \{V : \varpi(V) < 2\varepsilon(L^d s)\},\\
\mathrm{Cut}_t[s] &= \{V : \varpi(V) < \bar{\varepsilon}, \quad L^{2t}\varpi(U_t(V)) < c_\rho \varepsilon(L^d s)\},\\
T_d^K(V) &= O(U_t(V)) 1_{\mathrm{Cut}_t[b_-]}(V),\\
X_{K,d,s} &= \mathsf{E}_{K,s}(O - \mathsf{E}_{K,s} O \mid V_d).
\end{aligned}$$

Here $\varepsilon(x) = A_0 x^{-1/2}(\log x)^{\mathfrak{p}_0}$, and ϖ is the largest plaquette deviation from the identity. The product in the definition of T_d^K means that the function is zero off the cut; no background is chosen there. ■

The background and cut properties used here are those of [YM3, Prop. 4.58] and [YM3, Lem. 6.33]. In particular the cuts are open and gauge invariant, the background action is continuous on its domain, and

$$\mathrm{In}_t[s] \subset \mathrm{Cut}_t[s] \subset \mathrm{Cut}_t[b_-].$$

The last inclusion follows because ε decreases on $[\gamma^{-2}, \infty)$. Consequently T_d^K is Borel, gauge invariant and independent of s.

Proposition 13.2 (Variance of the conditional action)**.** There are $d_{\mathrm{G}} \geq 2$, K_{G}, K_T and $Q' > 0$ such that, for $s \in J$ and $K > d \geq d_{\mathrm{G}}$,

$$0 \leq T_d^K \leq Q' L^{4d}, \tag{13.i}$$

$$\left|\mathsf{E}_{K,s} X_{K,d,s}^2 - 3 s^{-2} N_d\right| \leq K_{\mathrm{G}} L^{-d/3} N_d, \tag{13.ii}$$

$$\mathsf{E}_{K,s} |X_{K,d,s} - (T_d^K - c_s)|^2 \leq K_T L^{-d/3} N_d \tag{13.iii}$$

for a real number $c_s = c_s(K, d)$. The bound eq. 13.i holds for every $K > d \geq d_\natural$. ■

The coefficient 3 counts the Gaussian directions integrated at each step. To recover it for the conditional action, we compare the flow between depths d and d' with flows started with zero earlier terms. A second comparison, using three choices of the source in the earlier terms, controls their contribution in L^2.

13.1.1. *Earlier terms on a small background.* Fix $s \in J$ for the moment. At level t write $u = K - t$, $x_t = L^u s$ and

$$\epsilon_u^\sharp = \frac{2 c_\rho \varepsilon(L^u s)}{\varepsilon_{\mathcal{W}}}, \quad \Xi_u = \frac{c_A \alpha_0}{M \epsilon_u^\sharp}, \quad \Xi_u' = \frac{L^{m+u}}{2M}.$$

The constants c_A, C_A are those of the quadratic bound for an earlier local term, and $\varepsilon_{\mathcal{W}}$ is the radius in the background theorem [YM3, Prop. 4.58]. The entropy constants are $C_{E,p} = 2048 \sum_{n \geq 0} (n+2)^p e^{-n}$. Set

$$\begin{aligned}
k_u^\sharp &= C_A M^2 C_{E,2} (\epsilon_u^\sharp / \alpha_0)^2 + 2 C_{E,0} (e^{1-\Xi_u} + e^{3-\Xi_u'}),\\
m_u^\sharp &= k_u^\sharp \frac{L}{L-1} E_0 N_u.
\end{aligned}$$

These are independent of the source. Uniformly for $s \in J$, $m_u^\sharp = O((1+u)^{2\mathfrak{p}_0} L^{-u}) N_u$.

Lemma 13.3 (Earlier actions at a background)**.** For $V \in \mathrm{Cut}_t[s]$ and $\Xi_u \ge 2$, an array of earlier families $D^{(i)}$, $1 \le i \le t$, with zero counterterms and empty slot satisfies

$$|S[D](U_t(V))| \le k_u^\sharp N_u \sum_{i=1}^{t} L^{-(t-i)} \|D^{(i)}\|_{i,\kappa+1}.$$

In particular the earlier action of the chain is bounded by $m_u^\sharp$ on the full source disc. ■

Proof. The background theorem at radius $\epsilon_u^\sharp$ puts $U_t(V)$ in its regularity class. At level $i = t - l$ its radius is $\epsilon_u^\sharp L^{-2l}$. In the sum

$$S[D](U) = \sum_{i,z} \sum_{X \ni z} (D^{(i)}(X, U, z) - D^{(i)}(X, \mathbf{1}, z)),$$

call X short if $1 + d_i(X) \le \Xi_u L^{2l}$ and $1 + d_i(X) < \Xi_u' L^l - 2/M$. Such a domain does not wrap around the torus. The quadratic bound [YM3, eq. 7.xxv], and its proof gives

$$|D^{(i)}(X, U, z) - D^{(i)}(X, \mathbf{1}, z)| \le C_A \|D^{(i)}\|_{i,\kappa+1} e^{-(\kappa+1)d_i(X)} (\epsilon_u^\sharp/\alpha_0)^2 L^{-4l} M^2 (1 + d_i(X))^2.$$

Entropy and the $L^{3l} N_u$ sites give the first term of $k_u^\sharp L^{-l} N_u \|D^{(i)}\|$. For the other domains use twice the norm and the entropy tail at $D_l = \min\{\Xi_u L^{2l}, \Xi_u' L^l - 2/M\} - 1$. Since $\Xi_u \ge 2$ and $\Xi_u' \ge 4$,

$$L^{4l} e^{-D_l} \le e^{1-\Xi_u} + e^{3-\Xi_u'}.$$

This gives the second term. Sum over i; the chain has $\|E^{(i)}\|_{i,\kappa+1} \le E_0$. Finally $\varepsilon(L^u s)^2 = O((1+u)^{2\mathfrak{p}_0} L^{-u})$ and $e^{-x} \le 2x^{-2}$ give the asserted decay. □

Lemma 13.4 (The trivial background)**.** At the trivial block field the background can be chosen to be $\mathbf{1}$. Every weighted Wilson action and every earlier action $S[D]$ vanish at all backgrounds over it. ■

Proof. The background can be chosen to be $\mathbf{1}$: averaging preserves $\mathbf{1}$, which minimizes the nonnegative Wilson action on its fibre. Every other background over it is gauge equivalent to it. Thus every weighted Wilson action and every earlier action $S[D]$ vanish there, including arrays with counterterms and a slot. In the local-term formula each bracket is a value minus itself. □

We will use this fact when evaluating the scalar normalization of a step.

13.1.2. *Conditional means and the discarded fields.* Let $\varrho > 0$ denote the pure-gauge source radius. In this subsection the source is the constant z, $|z| < 2\varrho$. Let $E^{(i)}[z]$ be the chain families and set

$$c_t(z) = \sum_{i=1}^{t} c_i^{\mathrm{step}}(z),$$
$$\psi_t(z, V) = S[(E^{(i)}[z], 0)_{i \le t}](U_t(V)),$$
$$\varphi_t(z, V) = zO(U_t(V)) + \psi_t(z, V) - \psi_t(0, V).$$

The ordinary bound on earlier actions gives $|\psi_t| \le C_{\mathrm{old}} L E_0 N_u/(L-1)$; the preceding lemma improves this to $m_u^\sharp$ when $\Xi_u \ge 2$. We choose Q' large enough that

$$Q' \ge \max\left\{ \varrho^{-1},\ \tfrac{3}{2} c_\rho^2 \ell^3 \sup_{u \ge 1} \varepsilon(L^u s)^2,\ \varrho^{-1} C_{\mathrm{old}} \tfrac{L}{L-1} E_0 \ell^3 \right\}, \qquad Q'' = \max\{2Q', 6\ell^3\}.$$

The field-cut action bound proves eq. 13.i. All these choices can be made uniformly on J.

The discarding estimate [YM3, Prop. 6.37] gives

$$\rho(u) = Q_\ell L^{3u} e^{-\theta_0 u^2}, \qquad \theta_0 = c' A_0^2 (\log L)^{2\mathfrak{p}_0},$$

where one may take

$$Q_\ell = 2M^{-3} \ell^3 \exp[(C_{\mathrm{maj}}^\Sigma + 2 + c_{\mathrm{low}} + \tfrac{9}{2} \log s)\ell^3].$$

Increase a depth $u_\flat \geq d_\natural$ so that ρ decreases and is at most $1/4$ above it. Put

$$\begin{aligned}\omega_q(d) &= \sum_{u\geq d}(u-d+1)L^{4q(u+1)}\rho(u),\\ \eta(d,d') &= 1152(\omega_3(d)+L^{8d}\rho(d')),\\ C_1 &= \varrho^{-2}E_0\left(C_{E,0}L^{-3}+\tfrac{1}{2}C_{\mathrm{old}}\tfrac{L}{L-1}\right),\\ \eta_1(d,d') &= 2\rho(d')(C_1N_d+43Q'^2L^{8d'})+Q''^2\eta(d,d').\end{aligned}$$

Both $L^{cd}\omega_q(d)$ and $L^{cd}\rho(\lceil d/3\rceil)$ tend to zero for every fixed c.

Lemma 13.5 (Two depths of the small-field chain)**.** For $u_\flat \leq d' < d < K$, set $t = K-d$ and $t' = K-d'$. There are positive probability measures $\tilde\nu$ on $\mathrm{Cut}_t[s]$ and $\hat\nu$ on $\mathrm{In}_{t'}[s]$ such that, with

$$Y = \partial_z\varphi_t(0,\cdot),\quad \tilde Y = Y-\tilde\nu(Y),\quad Y_2 = \partial_z^2\psi_t(0,\cdot),$$

extended by zero before centering off their cut, $\tilde Y$ is real almost everywhere for $\tilde\nu$ and

$$\tilde\nu(\tilde Y^2) = c''_{t,t'}(0)+(\mathrm{Log}\,\hat J)''(0)-\tilde\nu(Y_2), \tag{13.iv}$$

$$|(\mathrm{Log}\,\hat J)''(0)| \leq 43Q'^2L^{8d'},\qquad \tilde\nu(\tilde Y^2)\leq C_1N_d+43Q'^2L^{8d'}, \tag{13.v}$$

$$|\mathsf{E}_{K,s}X^2_{K,d,s}-\tilde\nu(\tilde Y^2)| \leq \eta_1(d,d'), \tag{13.vi}$$

$$\int |X_{K,d,s}-\mathrm{Re}\,\tilde Y|^2 d\mu_d^{\mathrm{W,K}}[s] \leq \eta_1(d,d')+Q''^2\eta(d,d'). \tag{13.vii}$$

Here $\hat J(z) = \hat\nu(e^{\varphi_{t'}(z)})$ and

$$c_{t,t'}(z) = \sum_{t<j\leq t'}\sum_{X,z'\in X}E^{(j)}[z](X,\mathbf{1},z').$$

The same estimate for the conditional variance at $d=K$ holds with $c''_{0,t'}(0)$ and error $43Q'^2L^{8d'}+12Q''^2L^{8d}\rho(d')$. ∎

Proof. The small-field weight $w_j \geq 0$ is the product of the field-cut factors through level $j-1$ and $1_{\mathrm{In}_j[s]}(\mathcal{M}^jU)$. For nonnegative gauge-invariant F, the discarding theorem gives

$$\int Fw_je^{-sO}\,dU \leq \int Fe^{-sO}\,dU,\qquad 0\leq Z-\int w_je^{-sO}\,dU\leq\rho(K-j)Z.$$

The weighted mass is positive. Indeed its image at level j has an exponential density on the nonempty open set $\mathrm{In}_j[s]$. Positivity of the image makes that density positive almost everywhere on this set. The normalized unweighted functional is a convex combination of the normalized weighted functional and a positive functional of mass one. Thus inserting w_j changes a bounded mean by at most $2\rho(K-j)\|F\|_\infty$ and a second cumulant by at most $6\rho(K-j)\|F_1\|_\infty\|F_2\|_\infty$.

Factor $w_{t'}$ into the factors below t and a Borel function $H_{t,t'}(\mathcal{M}^tU)$ of the later block fields. The latter has no source dependence. The recursion of the chain Prop. A.12 gives the exact identity

$$\int G(\mathcal{M}^tU)w_{t'}e^{-sO+zO}\,dU = e^{c_t(z)-c_t(0)}\int G(V)e^{\varphi_t(z,V)}\,d\sigma_{t,t'}(V), \tag{13.viii}$$

where

$$\sigma_{t,t'} = (\mathcal{M}^t)_*(w_{t'}e^{-sO}dU) = e^{c_t(0)}1_{\mathrm{Cut}_t[s]}H_{t,t'}e^{\Phi^0_t}\,dV.$$

Normalize this measure to obtain $\tilde\nu$; the analogous image at t' gives $\hat\nu$. The masses coincide and are positive. The source results Lem. A.15 and Lem. A.16 give holomorphy on $|z|<2\varrho$ and

$$|c_{t,t'}(z)|\leq 2E_0C_{E,0}L^{-3}N_d.$$

The difference $c_{t'} - c_t - c_{t,t'}$ has no source dependence. At $G = 1$, the two readings of the left side of eq. 13.viii therefore yield

$$\tilde{\nu}(e^{\varphi_t(z)}) = e^{c_{t,t'}(z) - c_{t,t'}(0)} \hat{J}(z).$$

Differentiating twice proves eq. 13.iv. The first derivative of eq. 13.viii, tested against the imaginary part of $c_t'(0) + Y$, shows that this function is real almost everywhere for $\sigma_{t,t'}$. Hence $\tilde{Y}$ is real.

The source bound at t' gives $|\varphi_{t'}| \le 1/4$ on $|z| < (8Q'L^{4d'})^{-1}$. Consequently $|\hat{J} - 1| \le e^{1/4} - 1$ and $|\operatorname{Log} \hat{J}| < 0.3343$ there. Cauchy's inequality gives $2(0.3343)(8Q'L^{4d'})^2 < 43Q'^2L^{8d'}$. Apply Cauchy's inequality also to $c_{t,t'}$ and ψ_t on the disc of radius 2ϱ to obtain the second bound in eq. 13.v.

We give the descent argument because the bound $|O| \le 6\ell^3 L^{4K}$ alone would not be uniform in K. Write $a = X_{K,d,s}$ and $Y^{(j)} = \partial_z \varphi_j(0)$, extended by zero off $\operatorname{Cut}_j[s]$; put $Y^{(0)} = O$ and $w_0 = 1$. Cauchy's inequality gives $|Y^{(j)}| \le Q'' L^{4(K-j)}$. Differentiate eq. 13.viii at adjacent levels. Under w_{i+1} the cumulants of $F(V_d)$ with $Y^{(i)}$ and with $Y^{(i+1)}$ agree. Comparing consecutive weights and summing gives

$$\left| \int F a \, d\mu_d^{\mathrm{W,K}}[s] - \operatorname{cum}_{w_j}(F(V_d), Y^{(j)}) \right| \le \delta_j \|F\|_\infty, \quad \delta_j = 24Q'' \sum_{u=K-j}^{K-1} \rho(u) L^{4(u+1)}.$$

This holds for $j \le t$. At $j = t$, change the weight once more to $w_{t'}$. The right side is then $\delta_* \|F\|_\infty$, where

$$\delta_* \le 24Q''(\omega_1(d) + \rho(d')L^{4d}),$$

and the weighted cumulant is $\tilde{\nu}(F\tilde{Y})$.

For $\lambda \ge 8Q'' L^{4(K-j)}$, use $F = 1_{\{\pm a > \lambda\}}$. Weighted probabilities are at most $(1-\rho)^{-1} \le 2$ times unweighted probabilities. Absorbing the cumulant term gives $\mu_d^{\mathrm{W,K}}[s](|a| > \lambda) \le 4\delta_j/\lambda$. The shells

$$8Q'' L^{4u} < |a| \le 8Q'' L^{4(u+1)}, \qquad d \le u < K,$$

therefore satisfy

$$\int a^2 1_{\{|a| > 8Q'' L^{4d}\}} \, d\mu_d^{\mathrm{W,K}}[s] \le 768 Q''^2 \omega_3(d).$$

On the complementary set put $F = a 1_{\{|a| \le \Lambda\}}$, $\Lambda = 8Q'' L^{4d}$, and $x^2 = \int F^2 d\mu_d^{\mathrm{W,K}}[s]$. Cauchy–Schwarz and the descent give $x^2 \le Ax + B_2$, with

$$A^2 = (1 - \rho(d'))^{-1} \tilde{\nu}(\tilde{Y}^2), \qquad B_2 = 192 Q''^2 (\omega_2(d) + L^{8d} \rho(d')).$$

Hence $x^2 \le A^2 + 2B_2$. Since $2 \cdot 192 + 768 = 1152$ and $(1-\rho)^{-1} \le 1 + 2\rho$,

$$\int a^2 d\mu_d^{\mathrm{W,K}}[s] \le (1 + 2\rho(d'))\tilde{\nu}(\tilde{Y}^2) + Q''^2 \eta(d, d').$$

Finally apply the descent to $F = \operatorname{Re} \tilde{Y}$ and the weight comparison to F^2. Expanding $\int (a - F)^2$ gives

$$\int (a - F)^2 d\mu_d^{\mathrm{W,K}}[s] \le 2\rho(d')\tilde{\nu}(\tilde{Y}^2) + 2Q''^2 \eta(d, d').$$

The same expansion bounds the variance from below. Together with eq. 13.v these prove eq. 13.vi–eq. 13.vii. For $d = K$, compare the variance of O directly under the original and weighted laws, then use the identity at t'. □

13.1.3. *The Gaussian count and cancellation of earlier sources.*

Lemma 13.6 (The constant source at one step)**.** At level k, put $u = K - k$. Let $T_k(z)$ be the scalar constant of the step started with zero earlier terms, with its source-independent part removed. It is holomorphic on $|z| < 2\varrho$. The following identity holds on $|z| < \min\{2\varrho, \sigma_\star s\}$, and its remainder r_k is holomorphic on the larger disc $|z| < \sigma_\star s$:

$$T_k(z) - T_k(0) = -3(N_u - N_{u-1}) \operatorname{Log}(1 - z/s) + r_k(z) - r_k(0), \tag{13.ix}$$

$$|r_k(z)| \le C_{E,0} \epsilon_u^F N_{u-1}, \qquad \epsilon_u^F = K_M^f \left(\delta(L^u s) + e^{-\gamma_2 \mathfrak{b}(L^u s)^2/2} \right), \quad |z| < \sigma_\star s. \tag{13.x}$$

For the actual chain,

$$|(c_{k+1}^{\mathrm{step}})''(0) - 3s^{-2}(N_u - N_{u-1})| \le C(1+u)^{\mathfrak{p}_0} L^{-u/2} N_u.$$

The latter bound is uniform over systems of earlier terms of size at most E_0, holomorphic and local in the source. ∎

Proof. The constant source changes the action coefficient from $x_k = L^u s$ to $x = x_k(1 - z/s)$; the cut and gauge fixing remain at x_k. The complex-coefficient step is [YM3, Prop. 4.54], [YM3, Prop. 4.56], [YM3, eq. 4.cclxv], and [YM3, eq. 4.ccxci]. Evaluate it at the trivial block field, where every earlier action vanishes. The Gaussian normalization has exponent

$$\tfrac{1}{2} n_{\mathcal{O}} - \tfrac{3}{2} \#\mathfrak{X}^+, \qquad n_{\mathcal{O}} = \#\mathfrak{X}^+ = 3(N_u - N_{u-1}).$$

It is therefore $-3(N_u - N_{u-1}) \operatorname{Log}(x/x_k)$. Equivalently there are $6(N_u - N_{u-1})$ integrated directions after gauge fixing; this is the dimension count and logarithm in [YM3, §4.9.2] and [YM3, eq. 4.cclxxvii]. The one-step estimate on $|x - x_k| < \sigma_\star x_k$ bounds each remaining local term by $\epsilon_u^F e^{-(1+4\beta_s)\kappa d(X)}$. Entropy and the N_{u-1} sites give eq. 13.x. The logarithms agree up to a constant multiple of $2\pi i$, which disappears on subtracting their values at zero.

For completeness, the effect of earlier terms is quantitative. The one-step Lipschitz estimate Prop. A.13 bounds the difference of the output families by $K^d K_\Lambda \mathsf{s}_k \sum_{i \le k} L^{-(k-i)} \Delta_i$. Against the zero start, $\Delta_i \le E_0$ and

$$K^d K_\Lambda \mathsf{s}_k \frac{L}{L-1} \le \frac{13}{72} L^{-u/2} (1 + u c_{\log})^{\mathfrak{p}_0}.$$

The scalar difference is the sum of this family at the trivial field; the normalizations agree. Entropy followed by Cauchy's inequality on $|z| < 2\varrho$ bounds its second derivative by

$$\frac{13 E_0 C_{E,0}}{144 \varrho^2 L^3} N_u L^{-u/2} (1 + u c_{\log})^{\mathfrak{p}_0}.$$

Cauchy's inequality on $|z| < \sigma_\star s$ bounds the second derivative of r_k by $2 C_{E,0} \epsilon_u^F (\sigma_\star s)^{-2} N_{u-1}$. These two bounds prove the assertion, since $\epsilon_u^F = O((1+u)^{\mathfrak{p}_0} L^{-u/2})$. □

Lemma 13.7 (Three choices of the earlier source)**.** In Lem. 13.5, suppose also $\Xi_d \ge 2$ and write

$$\tilde{Y} = \tilde{A} + P, \qquad \tilde{A} = O(U_t) - \tilde{\nu}(O(U_t)), \quad P = \partial_z \psi_t(0) - \tilde{\nu}(\partial_z \psi_t(0)).$$

Put $v_{\mathrm{cl}} = \sum_{k=t}^{t'-1} T_k''(0)$ and

$$e_I = \frac{13 E_0 C_{E,0}}{144 \varrho^2 L^3} \sum_{u=d'+1}^{d} N_u L^{-u/2} (1 + u c_{\log})^{\mathfrak{p}_0} + 43 Q'^2 L^{8d'} + \frac{m_d^\sharp}{2\varrho^2}.$$

For $\theta = 1, 0, -1$,

$$|\tilde{\nu}((\tilde{A} + \theta P)^2) - v_{\mathrm{cl}}| \le e_I.$$

Consequently $\tilde{\nu}(P^2) \le 2e_I$, $|\tilde{\nu}(\tilde{A}P)| \le e_I/2$, and

$$\int |X_{K,d,s} - \tilde{A}|^2 d\mu_d^{\mathrm{W,K}}[s] \le 4e_I + 8\rho(d')(m_d^\sharp/\varrho)^2 + 2\eta_1 + 2Q''^2 \eta.$$

∎

Proof. Run the flow above t with earlier families $E^{(i)}[\theta z]$, $i \le t$. These are admissible earlier families: the source disc is invariant under multiplication by θ, and their norm, locality and holomorphy are unchanged. All three flows agree at $z = 0$. The factor $H_{t,t'}$ is independent of the source and factors over the intermediate levels. Applying the flow recursion and dividing by its positive value at zero therefore gives

$$\tilde{\nu}\left(e^{zO(U_t)+\psi_t(\theta z)-\psi_t(0)}\right) = e^{c^{\theta}_{t,t'}(z)-c^{\theta}_{t,t'}(0)}\hat{J}_\theta(z).$$

The source estimate at t' gives $|(\operatorname{Log}\hat{J}_\theta)''(0)| \le 43Q'^2L^{8d'}$, for every θ. Differentiating twice yields

$$\tilde{\nu}((\tilde{A}+\theta P)^2) = (c^{\theta}_{t,t'})''(0) + (\operatorname{Log}\hat{J}_\theta)''(0) - \theta^2\tilde{\nu}(Y_2).$$

The preceding one-step comparison bounds the first term's difference from v_{cl} by the sum in e_I. The earlier-action bound and Cauchy's inequality give $|\tilde{\nu}(Y_2)| \le m_d^\sharp/(2\varrho^2)$. This proves the first assertion. The identities

$$P^2 = \tfrac{1}{2}((\tilde{A}+P)^2 + (\tilde{A}-P)^2) - \tilde{A}^2, \qquad 4\tilde{A}P = (\tilde{A}+P)^2 - (\tilde{A}-P)^2$$

cancel v_{cl}. The centered function $\tilde{Y}$ is real for the common positive law by Lem. 13.5. Since $\tilde{A}$ is real, so are $P = \tilde{Y} - \tilde{A}$ and $\tilde{A} + \theta P$. Finally $|P| \le m_d^\sharp/\varrho$. Changing from the weighted law costs at most $4\rho(d')\sup|P|^2$; combine this with eq. 13.vii and $|a+b|^2 \le 2|a|^2 + 2|b|^2$. □

Proof of Prop. 13.2. Choose all bounds uniformly over $s \in J$. In the discarding bound use b_+; in the field radii and inverse powers of s use b_-. These choices cover the same cuts and source discs throughout the interval.

Sum Lem. 13.6 from $t = K - d$ to $t' = K - d'$. The main terms telescope to $3s^{-2}(N_d - N_{d'})$. By Lems. 13.3 and 13.5, the difference from $\mathsf{E}_{K,s}X^2_{K,d,s}$ is at most

$$\frac{m_d^\sharp}{2\varrho^2} + 43Q'^2L^{8d'} + \eta_1(d,d') + C\sum_{u=d'+1}^{d}(1+u)^{\mathfrak{p}_0}L^{-u/2}N_u.$$

Add $3b_-^{-2}N_{d'}$ and take $d' = \lceil d/3\rceil$. After division by N_d, the endpoint term is $O(L^{-d/3})$, the earlier-action term is $O((1+d)^{2\mathfrak{p}_0}L^{-d})$, the sum is $O((1+d)^{\mathfrak{p}_0}L^{-d/2})$, and the discarded terms decay faster than any fixed power of L^{-d}. This proves eq. 13.ii.

By Lem. 13.7, the conditional action differs in L^2 from the centered background action on $\mathrm{Cut}_t[s]$ by $O(L^{-d/3})N_d$. The weighted law at t is carried by $\mathrm{In}_t[s]$, so discarding gives

$$\mu_d^{\mathrm{W,K}}[s](\mathrm{In}_t[s]^c) \le \rho(d).$$

The statistic T_d^K differs from that background action only in $\mathrm{Cut}_t[b_-] \setminus \mathrm{Cut}_t[s]$, outside this inner set. Its additional squared error is at most $2Q'^2L^{8d}\rho(d)$. This proves eq. 13.iii, with $c_s = \tilde{\nu}(O(U_t))$. Increasing d_{G} ensures $d' \ge u_\flat$ and $\Xi_d \ge 2$ throughout. □

13.2. **The pure-gauge block laws.** We use the same configuration spaces and Borel averaging map for the pure-gauge comparison. The estimate below follows from the comparison of weighted densities in Prop. A.17, and its proof. We give the passage to arbitrary Borel tests, which is needed for the statistic T_d^K.

Proposition 13.8 (Pure-gauge block laws)**.** In the setting of Def. 13.1, there are C_{bf} and K_{bf}, uniform over $s \in J$, such that, for $K \ge K_{\mathrm{bf}}$ and $0 \le d \le \lceil K/1260\rceil$,

$$\mathrm{TV}(\mu_d^{\mathrm{W,K+1}}[s], \mu_d^{\mathrm{W,K}}[s]) \le C_{\mathrm{bf}}\ell^3K^{2\mathfrak{p}_0+1/2}L^{-K/10}.$$

Here and below $\mathrm{TV}(\mu,\nu) = \sup_E|\mu(E) - \nu(E)|$. For each d, these laws converge in total variation to a Borel probability measure $\mu_d^{\mathrm{W}}[s]$, uniformly over $s \in J$, with error at most

$$\tau_J(K) = C_{\mathrm{bf}}\ell^3\sum_{j\ge K} j^{2\mathfrak{p}_0+1/2}L^{-j/10}$$

under the same depth condition. The limits satisfy $\mathcal{M}_*\mu^{\mathrm{W}}_{d+1}[s] = \mu^{\mathrm{W}}_d[s]$. ■

Proof. Use the two lattices (ℓ, K) and $(\ell, K+1)$, put $u = \lceil K/1260 \rceil$ and $v = \lceil K/9 \rceil$, and take the constant auxiliary test 1 at zero source in the comparison of Prop. A.17. For a bounded gauge-invariant Borel function F at depth $d \le u$, replace it by $F \circ \mathcal{M}^{u-d}$ at depth u. This preserves its supremum bound and gauge invariance. It therefore suffices to work at $d = u$.

For either lattice let Λ_0 be the unnormalized Wilson expectation of $F(V_u)$, Λ_1 its expectation with the small-field weight through depth u, and Λ_2 the expectation obtained by starting at depth v with zero earlier terms and running to depth u. All three are positive functionals of positive mass. For Λ_1 this follows from the history image identity. For Λ_2 it follows from the positive start and positive chart kernels; its final exponential density is nonzero on the common inner set, an open set containing the identity.

There are three comparisons. First, discarding gives $0 \le \Lambda_0 - \Lambda_1$ and $(\Lambda_0 - \Lambda_1)(1) \le R\Lambda_0(1)$. For $R \le 1/2$, their normalized means differ by at most $4R\|F\|_\infty$. Second, the two densities of the chain and the zero start on the final inner set are $p_1 = e^c e^{\Phi}$ and $p_2 = e^{\hat{c}} e^{\hat{\Phi}}$. The forgetting bound is $|\Phi - \hat{\Phi}| \le \epsilon_{\mathrm{fg}} \le 1/8$. Since both densities are positive almost everywhere,

$$|p_1 - \varkappa p_2| \le (e^{\epsilon_{\mathrm{fg}}} - 1)\varkappa p_2, \qquad \varkappa = |e^{c-\hat{c}}| > 0.$$

Normalization cancels $\varkappa$ and costs at most $5\epsilon_{\mathrm{fg}}\|F\|_\infty$.

Third, write the two zero-start functionals on the common configuration space at depth v. With W_u the block field obtained by averaging W down to depth u, they are integrals of $F(W_u)k^\bullet(W)\hat{F}^\bullet_0(W)$. The proof of Prop. A.17 gives

$$\int |k^A \hat{F}^A_0 - k^B \hat{F}^B_0| \, dW \le (2\epsilon_{\mathrm{cl}} + 3\epsilon_{\mathrm{sh}})\Lambda^A_2(1).$$

This is an estimate of the densities, so it applies to bounded Borel F. When $2\epsilon_{\mathrm{cl}} + 3\epsilon_{\mathrm{sh}} \le 1/4$, the normalized means differ by at most $(8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}})\|F\|_\infty$. Combining the five comparisons from the original law on one lattice through both zero starts to the original law on the other gives

$$|\mu^{\mathrm{W,K+1}}_u[s](F) - \mu^{\mathrm{W,K}}_u[s](F)| \le (8R + 12\epsilon_{\mathrm{fg}} + 8\epsilon_{\mathrm{cl}} + 12\epsilon_{\mathrm{sh}})\|F\|_\infty. \tag{13.xi}$$

The normalization estimates just used follow directly by writing the difference of two ratios: if $|\Lambda'(F) - \Lambda(F)| \le \epsilon\Lambda(1)\|F\|_\infty$ and $\epsilon \le 1/2$, their normalized difference is at most $4\epsilon\|F\|_\infty$.

The bounds of Prop. A.17 and the proof of Prop. A.20 are uniform on J. They give

$$12\epsilon_{\mathrm{fg}} \le 12 C_{\mathrm{old}} C_{\mathrm{fg}} \tfrac{L}{L-1} E_0 L^4 \ell^3 L^{-K/10},$$
$$8\epsilon_{\mathrm{cl}} \le 32 K_w c_\rho^2 A_0^2 (1 + 2\varrho/b_-)(\log(Lb_+))^{2\mathfrak{p}_0} L^{7/2} \ell^3 K^{2\mathfrak{p}_0 + 1/2} L^{-K/9}.$$

The other two errors decay faster than any exponential in K. Increase a common K_{bf} to ensure all comparison smallness conditions and to absorb these errors into the same majorant. This proves the claimed increment for gauge-invariant tests.

Each block law is gauge invariant. For any Borel set E on its finite product of compact groups, the gauge average

$$\bar{1}_E(V) = \int 1_E(V^h)\, dh$$

is Borel, gauge invariant, between zero and one, and has the same integral as 1_E. Applying eq. 13.xi to $\bar{1}_E$ gives the estimate for all Borel sets. The increments are summable uniformly on J. Their sum is a finite signed measure in total variation, so the Cauchy sequence has a countably additive limit; positivity and mass one pass to the limit. Summing the increments gives $\tau_J(K)$. Finally pushforward by a Borel map contracts total variation. Apply this to the finite-cutoff identity $\mathcal{M}_*\mu^{\mathrm{W,K}}_{d+1} = \mu^{\mathrm{W,K}}_d$. □

Corollary 13.9 (The pure-gauge thread law)**.** There is a unique Borel probability measure $\mu^{\mathrm{W}}_{\mathrm{thr}}[s]$ on $\prod_{d\ge 0}\mathcal{V}_{[d]}$ whose first $d+1$ coordinates have the law

$$(V\mapsto(\mathcal{M}^dV,\dots,\mathcal{M}V,V))_*\mu^{\mathrm{W}}_d[s].$$

It is carried by $V_d=\mathcal{M}V_{d+1}$ for every d. ■□

Proof. The finite-dimensional laws are consistent by Prop. 13.8. Kolmogorov's extension theorem on these compact metric spaces gives existence and uniqueness. The graph of $\mathcal{M}$ is Borel, and each compatibility identity has probability one. Their countable intersection therefore has probability one. □

13.3. **Variance of the background statistic with fermions.** We now estimate the same statistic under the gauge law with fermions. The proof groups histories by the first level at which a large-field region appears. On each remaining small-field portion, an exponential of T_d^K changes one running coefficient. Holomorphy controls that change; the large-field bound makes the sum over portions convergent.

For this subsection take the constants of section 2, with the additional history mark

$$(\mathsf{w},c_a,C_{\mathrm{vol}},c_{\mathrm{Dep}})=(1/2,0,0,c_\star),\qquad \eth=1/2,$$

chosen before γ, and impose the associated ceiling on γ in the partition lower bound of [QCDM1, Thm. E]. Fix $0<g\le\gamma$, $\ell=2L^m\in\mathfrak{L}(g)$, $b\in\bar{I}(g)$ and $\mathsf{m}\in[m_-,m_+]$. In Def. 13.1, one may replace b_- by any $b_\flat\in[\gamma^{-2},b]$ with $MR(b_\flat)\le L^m$, and $d_\natural$ by any $n\ge 1$ satisfying $L^nb_\flat\ge\mathsf{t}_\natural$. This defines the same T_d^K at that lower label.

Lemma 13.10 (Two exponential bounds)**.** Let ν be a positive measure of mass at most one and X a real measurable function. Suppose $s,u_0>0$, $B\ge 1$, $s^2u_0^2\ge 2B$, and

$$\int e^{\pm uX}d\nu\le e^{u^2s^2/2+B},\qquad 0\le u\le u_0.$$

Then $\int X^2d\nu\le 2s^2(B+10)$ and $(\int X^4d\nu)^{1/2}\le 2s^2(B+12)$. ■

Proof. Markov's inequality, with $u=x/s^2$ when $x\le s^2u_0$ and $u=u_0$ otherwise, gives

$$\nu(|X|>x)\le\begin{cases}2e^{B-x^2/(2s^2)}, & x\le s^2u_0,\\ 2e^{B-u_0x/2}, & x>s^2u_0.\end{cases}$$

Also use $\nu(|X|>x)\le 1$ below $s\sqrt{2B}$. Put $w=s^2u_0^2/2\ge B$. Integration of these three bounds against $2x\,dx$ gives at most $2Bs^2+4s^2+16s^2$. Against $4x^3\,dx$ it gives at most

$$s^4\big(4B^2+16(B+1)+8(4B+60)\big)\le 4s^4(B+12)^2.$$

Here $e^{B-w}\le 1$, $we^{B-w}\le B$ and $u_0^{-2}\le s^2/(2B)$ bound the last integrals. □

We need the density identity at real points of the coefficient polydiscs, as well as the norm estimates there. We record why the same step supplies both.

Lemma 13.11 (A real change of the running coefficients)**.** Let (x_i,s_i) be the coefficients and shifts of the canonical flow at a fixed cutoff. At a level $1\le k<K$ suppose real numbers $(x_i',s_i')_{i\le k}$ lie in its windows and satisfy $|x_i'-x_i|\le(2/3)r_xx_i$. Use the activities obtained by the steps at these numbers, with the original fine masses. Let d_k be their standard level-k density on the density-test set, extended by zero outside it, with any real scalar normalization E'. Then the next density is

$$d_{k+1}=\chi_{k+1}\mathsf{I}[\mathrm{Ber}_k(\mathrm{Ker}_k\,d_k)].$$

It is bounded and Borel and has the standard form with

$$x'_{k+1}=L^{-1}x'_k-b'_{k+1},\qquad s'_{k+1}=Ls'_k+\mu'_{k+1},\qquad E'\longmapsto E'+n_{k+1}e'_{k+1}.$$

The output activities retain the symmetries, locality, holomorphy, zero moments and closed norm bounds of the one-step theorem. The density identity also carries bounded Borel scalar factors in the bare density. If d_k is the actual history density obtained from a bare density $\varsigma e^{-\beta A - S_F}$ with $\varsigma \geq 0$, integrating all remaining Grassmann variables gives its positive history measure. This positivity assertion uses the history identity with that bare density. ■

Proof. The history weight at level $k+1$ is the previous weight times the next density test, the two intermediate tests, the gauge-fixing factor and the fluctuation test. The intermediate tests equal one on the support of the next density and fluctuation tests. The full fermion kernel acts by a linear Berezin integral; thus any scalar Borel factor is carried along. The chart image formula gives the displayed density coordinate by coordinate. Its coefficients are bounded and Borel on the image domain and vanish outside the next density-test set. Each full kernel has Berezin integral one in the variables it introduces. Integrating the final variables therefore recovers the history weight times the original scalar density and fermion determinant.

The given point lies in the coefficient polydiscs of the holomorphic one-step construction. Indeed $x_i \leq (5/4)\mathsf{t}_i$, while the shift window gives $|s_i'| \leq 2\bar{B}_M/\mathsf{t}_i \leq (5/2)\bar{B}_M/x_i \leq A\phi(x_i)$, since $A \geq 5\bar{B}_M$. The windows and the ceiling $4\bar{B}\gamma^2 \leq \sigma_\star$ also give $|x_i'/\mathsf{t}_i - 1| \leq \sigma_\star/4$. At each chart point the scalar part of d_k is the exponential of the Wilson action at x_k' and the earlier degree-zero action, times a constant. It is positive and bounded away from zero on the chart. The remaining Grassmann-valued factor is bounded and Borel. Hence the chart formula for that coefficient applies to it.

Rotate the generators by the site factor and shift by the background. Both substitutions have Berezin Jacobian one. The quadratic form and the full kernel give the new quadratic form, the fermion fluctuation Gaussian and its normalization. The other terms are their values at the new background plus their variations: the shift gives the mass vertex; the old activities give the scalar and positive-degree old vertices; the gauge terms use the pair (t_k, x_k'). The potential remains below the same cluster threshold on these polydiscs. Extracting its vacuum, action and mass coefficients therefore gives exactly $e_{k+1}', b_{k+1}', \mu_{k+1}'$ and the stated new density.

For the applicability of the history norm, the torus symmetries must hold also at this point. Translation by a vector of $LM\mathbb{Z}^3$ carries the charts, pivot bonds, site factors, walk expansions, local terms and filing cells to their translates; the seam map multiplies each local mass bilinear by $u(x)^2 = 1$. The coefficients and masses are scalars independent of the site. Thus induction from the empty family at level zero gives the pointed translation covariance on every domain, including those that wrap around the torus. On nonwrapping domains the same formulas are the restrictions of the infinite-lattice activities. The field-independent shares assigned to wrapping rows are translated with their rows.

Charge conjugation is preserved by the same algebraic operations. For an isometry, apply the chart image identity to the two densities at a pair of covariance. After extracting the local coefficients their quotient is an exponential of a real scalar, a multiple of the background action and a multiple of the mass bilinear; each is carried by the isometry. Starting with the covariant bare density proves covariance at every later level. The gauge Gaussian part uses the pointed sum rule at the actual coefficient x_i', which lies in its disc, so it has the same property. The extraction makes the quadratic zeroth moment vanish. The reblocking estimate then applies with this zero moment, holomorphy and joint gauge invariance. The closed norm bounds and the degree-zero array bound consequently hold with the constants of the original step. These are exactly the inputs used by the history norm bound. □

Proposition 13.12 (Variance with fermions)**.** There are $d_2 \geq n$ and $C_v < \infty$, depending on the fixed constants, ℓ, b and n, such that

$$\mathrm{Var}_{\mu_d^{F,K}[b,\mathsf{m}]} T_d^K \leq C_v N_d, \qquad K > d \geq d_2.$$

They are independent of K, d and $\mathsf{m} \in [m_-, m_+]$. ■

Proof. Put $k_0 = K - d$ and $\mathcal{T} = T_d^K(\mathcal{M}^{k_0}U)$. Let $m_j(F)$ be the unnormalized integral of F on the history having no large-field region through level j. This is a positive functional on nonnegative gauge-invariant Borel functions, and $m_j(1) \le Z$. By Lem. 13.11, it is also the total Berezin integral of the corresponding term with scalar factor F.

The density-test set at level k_0 lies in $\mathrm{In}_{k_0}[b_\flat]$. On it $T_d^K = L^K A(U_{k_0})$. Thus multiplying the bare density by $e^{-u\mathcal{T}}$ changes the level-k_0 coefficient from x_{k_0} to $x_{k_0} + v$, $v = uL^d$, and leaves all earlier activities unchanged.

Continue the flow with this new coefficient. It is holomorphic for $|v| < \rho_v = r_x x_{k_0}/4$, and induction gives

$$|x_j(v) - x_j| \le \tfrac{7}{12} r_x x_j, \qquad |s_j(v)| \le \bar{B}_M/x_j, \qquad |e_j(v) - e_j| \le C_e^o := L^3. \tag{13.xii}$$

Here are the bounds closing the induction. The gauge and fermion one-loop parts of b_j depend only on the calibration and fine mass. The remaining part has size at most $C_w\phi(x_{j-1})$, so

$$|x_j(v) - x_j| \le L^{-(j-k_0)}|v| + \frac{2L}{L-1} C_w\phi(\gamma^{-2}) < \left(\tfrac{1}{2} + \tfrac{1}{12}\right) r_x x_j.$$

We used $x_{k_0} \le 2L^{j-k_0}x_j$ and $26C_w\phi(\gamma^{-2}) \le r_x\gamma^{-2}$. The one-loop mass identity, valid for every nonzero coefficient sequence, gives $s_j(v) = G_j(x(v)) + \sum_{i\le j} L^{j-i}\rho_i(v)$. Its slices are nonnegative and $|x_i(v)| \ge (11/12)x_i$, so

$$|G_j(x(v))| \le \frac{24B_F}{11x_j}, \qquad \sum_{i\le j} L^{j-i}|\rho_i(v)| \le \frac{1}{4x_j}.$$

This proves the shift bound. Finally the vacuum coefficient has the form

$$e_j = e_j^{\mathrm{YM},0} + e_j^{F,0} - 3(L^3-1)\,\mathrm{Log}(x_{j-1}/\mathtt{t}_{j-1}) + \tilde{e}_j, \qquad |\tilde{e}_j| \le C_w\phi(\mathtt{t}_{j-1}).$$

The first two terms are unchanged, and the same remainder bound holds at v. Since $|x_{j-1}(v)/x_{j-1} - 1| \le 7/96$,

$$|e_j(v) - e_j| \le 3(L^3-1)\frac{7}{89} + 2C_w\phi(\gamma^{-2}) \le L^3.$$

At each stage eq. 13.xii lies strictly within the polydiscs required for the next step.

For real $|v| \le 1/2$ the moved coefficients also remain in the original windows. Indeed $|b_i(v)| \le B_\sharp - 1/2$, and the added jump v at k_0 gives

$$|x_j(v) - \mathtt{t}_j| \le \tfrac{1}{2} + (B_\sharp - \tfrac{1}{2})\frac{L}{L-1} = \bar{B} - \frac{1}{2(L-1)} < \bar{B}.$$

Thus Lem. 13.11 identifies the iterated density with the actual weighted history term, rather than only a formal analytic continuation. All inputs of the history norm bound have been retained: the closed activity norm, degree-zero array norm, quadratic zero moment, gauge and flavour invariance, translation covariance and reality. Consequently, for $k_0 \le t' \le K$,

$$m_{t'}(e^{-u\mathcal{T}}) \le \exp(E_{t'}(v) + C_E^F n_{t'}), \qquad |u|L^d \le 1/2,$$

where $E_{t'}(v) = E_{k_0} + \sum_{k_0<j\le t'} n_j e_j(v)$. The positive Wilson action can be discarded in this upper bound, and the remaining Haar measure has mass one.

Put $d' = K - t'$ and

$$\begin{aligned} B_{d'} &= \left[1 + C_E^F + C_P(1 + \log(L^{d'}b + \bar{B}))\right] N_{d'}, \\ f_{t'}(v) &= E_{t'}(v) - E_{t'}, \qquad c_{t'} = -L^d f'_{t'}(0), \\ C_{\mathrm{cen}} &= \frac{8C_e^o}{(L^3-1)r_x}. \end{aligned}$$

The partition lower bound gives $Z \geq \exp(E_{t'} - C_P(1+\log x_{t'})n_{t'})$. On $|v| < \rho_v$, $|f_{t'}(v)| \leq C_e^o N_d/(L^3-1)$ and $f_{t'}(0)=0$. Moreover $\rho_v \geq r_x L^d b/8 \geq 3/2$. Cauchy's estimates therefore imply, for $|u| \leq u_0 = L^{-d}/2$,

$$Z^{-1} m_{t'}(e^{-u(\mathcal{T}-c_{t'})}) \leq \exp(u^2\sigma_d^2/2 + B_{d'}), \tag{13.xiii}$$

$$\sigma_d^2 = \frac{256 C_e^o N_d}{(L^3-1) r_x^2 b^2},$$

$$|c_{t'}| \leq C_{\mathrm{cen}} N_d/b, \qquad |c_{t'} - c_K| \leq C_{\mathrm{cen}} N_{d'}/b. \tag{13.xiv}$$

For the last estimate apply Cauchy's inequality to $f_K - f_{t'}$, whose supremum is at most $C_e^o N_{d'}/(L^3-1)$. The augmented variance parameter

$$\sigma_{d,d'}^2 = \sigma_d^2 + 8L^{2d} B_{d'}$$

satisfies $\sigma_{d,d'}^2 u_0^2 \geq 2B_{d'}$, as required in Lem. 13.10.

Group all histories into the history with no large fields, the histories whose first nonempty outcome occurs at $t'+1$, $k_0 \leq t' < K$, and those with such an outcome at or before k_0. Write their positive functionals as m_K, $\lambda_{t'}$ and λ_-, respectively. The history identity gives

$$Z\mathsf{E}F = m_K(F) + \sum_{t'=k_0}^{K-1} \lambda_{t'}(F) + \lambda_-(F), \qquad 0 \leq \lambda_{t'}(F) \leq m_{t'}(F) \quad (F \geq 0).$$

A first nonempty outcome creates at least one cell. The chosen mark is therefore at least $\frac{1}{2}c_\star A_0^2 (\log(L^{d'-1}b))^{2\mathfrak{p}_0}$. The marked history sum and partition lower bound give

$$\lambda_{t'}(1)/Z \leq \min\{1, R_{d'}\}, \qquad \lambda_-(1)/Z \leq R_{d+1},$$

$$R_j = \exp\left[C_{\mathrm{lf}} - \tfrac{1}{2}c_\star A_0^2 (\log(L^{j-1}b))^{2\mathfrak{p}_0}\right],$$

$$C_{\mathrm{lf}} = \left[C_{\mathrm{maj}}^\dagger + 2 + C_P(1+\log(b+\bar{B}))\right]\ell^3.$$

This estimate bounds the final history masses; it does not require each intermediate gauge-fixing factor to be at most one.

Apply Lem. 13.10 to $Z^{-1}m_{t'}$ and $\mathcal{T} - c_{t'}$, using eq. 13.xiii. At $t' = K$ this gives

$$Z^{-1} m_K((\mathcal{T}-c_K)^2) \leq 2\sigma_{d,0}^2(B_0+10).$$

For the other layers, Cauchy–Schwarz, the fourth-moment bound and eq. 13.xiv give

$$Z^{-1}\lambda_{t'}((\mathcal{T}-c_K)^2) \leq \sqrt{8}\min\{1,R_{d'}\}^{1/2}\left[2\sigma_{d,d'}^2(B_{d'}+12) + C_{\mathrm{cen}}^2 N_{d'}^2/b^2\right].$$

Finally $0 \leq \mathcal{T} \leq (3/8)\ell^3 L^{4d}$, so the last portion is bounded by

$$Z^{-1}\lambda_-((\mathcal{T}-c_K)^2) \leq \left((3/8)\ell^3 L^{4d} + C_{\mathrm{cen}} N_d/b\right)^2 R_{d+1}.$$

Choose d_2 so this is at most one for $d \geq d_2$. All three series

$$\sum_{j\geq 1}(B_j+12)\sqrt{\min\{1,R_j\}}, \quad \sum_{j\geq 1} N_j^2\sqrt{\min\{1,R_j\}}, \quad \sum_{j\geq 1} B_j(B_j+12)\sqrt{\min\{1,R_j\}}$$

converge, since $2\mathfrak{p}_0 > 1$. Summing the layer bounds and using $L^{2d} \leq N_d$ gives $\mathsf{E}(\mathcal{T}-c_K)^2 \leq C_v N_d$, and hence the variance bound. Every constant used above is uniform over the stated mass interval. □

13.4. **The coefficient determined by the block average.**

Definition 13.13 (The coefficient κ)**.** At bare mass $m_0 \in [0,1]$, let $\widehat{W}_{m_0}$ be the translation-invariant operator on exact curvatures for which the quadratic term of the negative logarithmic fermion determinant is

$$\tfrac{1}{2}\sum_{c=1}^{3}\langle dA^c, \widehat{W}_{m_0} dA^c\rangle.$$

Thus $\widehat{W}_{m_0} = N_f \widehat{W}^{\mathrm{f}}_{m_0}$, where the one-flavour polarization kernel satisfies $\Pi = d^* \widehat{W}^{\mathrm{f}}_{m_0} d$. Let Π^{tp}_l be the orthogonal projection onto the background curvatures of the l-fold straight average, and let τ_l denote trace per site of $L^l\mathbb{Z}^3$. Put

$$\kappa(l,\mu) = \frac{1}{2L^l}\tau_l(\Pi^{\mathrm{tp}}_l \widehat{W}_{L^{-l}\mu}), \qquad l \geq 1, \quad 0 \leq \mu \leq L^l.$$

The curvature slice is $\mathcal{C}_{(j+1)} = \Pi^{\mathrm{tp}}_j - \Pi^{\mathrm{tp}}_{j+1}$.

Here is the limiting coefficient of the block average. For $s(y) = (e^{iy}-1)/(iy)$, with $s(0)=1$, let $\hat{B}(k)$ be the diagonal matrix with entries

$$\hat{B}_\mu(k) = s(k_\mu)\prod_\nu s(k_\nu).$$

Define also

$$P(k) = 1 - kk^{\mathsf{T}}/|k|^2, \qquad G(p) = (e^{ip_\mu}-1)_\mu.$$

For $p \in [-\pi,\pi]^3 \setminus \{0\}$ and $a = 1,2$ set

$$M_a(p) = \sum_{n\in\mathbb{Z}^3} \hat{B}(k_n)P(k_n)\hat{B}(k_n)^* |k_n|^{-1-a}, \qquad k_n = p + 2\pi n.$$

Let $J_p : G(p)^\perp \hookrightarrow \mathbb{C}^3$ be the inclusion and put

$$\mathrm{Del}(p) = J_p(J_p^* M_1(p) J_p)^{-1} J_p^*, \qquad I_B = \frac{1}{2}\int_{[-\pi,\pi]^3} \mathrm{tr}(\mathrm{Del}(p)M_2(p))\frac{d^3p}{(2\pi)^3}.$$

The following lemma justifies the inverse and integral. ■

Theorem V (The fermion contribution to the background action)**.** Use the constants of Def. 10.1. Fix an admissible side $\ell = 2L^m$, an interval $J = [b_-, b_+]$ as in Def. 13.1, $\mathsf{b}_* \in J$ and $\mathsf{m} > 0$. Let $\mathbb{E}^{\mathrm{F}}_K$ and $\mathbb{E}^{\mathrm{W}}_K$ denote the gauge expectations with and without fermions at $\beta = L^K \mathsf{b}_*$, with bare fermion mass $m_0 = L^{-K}\mathsf{m}$. There are $d_2 \geq d_\natural$ and $\mathsf{K}_{\mathrm{S}} < \infty$, depending on these fixed data but independent of K, d, such that

$$\left|\mathbb{E}^{\mathrm{F}}_K T^K_d - \mathbb{E}^{\mathrm{W}}_K T^K_d + 3\mathsf{b}_*^{-2}\ell^3 L^{2d}\kappa(K-d, L^{-d}\mathsf{m})\right| \leq \mathsf{K}_{\mathrm{S}} L^{7d/4}, \qquad K > d \geq d_2.$$

Here κ is defined in Def. 13.13, satisfies $|\kappa(l,\mu)| \leq C_{\mathrm{kp}} N_f$, and tends jointly to $N_f I_B/4$ as $l \to \infty, \mu \to 0$, with $0 < I_B < \infty$. Consequently, for any $K_0(d) > d$ with $K_0(d) - d \to \infty$,

$$\lim_{d\to\infty} \sup_{K \geq K_0(d)} \left|L^{-2d}(\mathbb{E}^{\mathrm{F}}_K T^K_d - \mathbb{E}^{\mathrm{W}}_K T^K_d) + \tfrac{3}{4}\mathsf{b}_*^{-2}\ell^3 N_f I_B\right| = 0.$$

■□

The proof is given after Lem. 13.26. The scale L^{2d} is smaller than the order L^{3d} of the mean itself, but larger than its standard deviation, which is bounded by a constant times $L^{3d/2}$ by Prop. 13.12. This separation of powers will distinguish the flavour number from the background gauge field.

Corollary 13.14 (Upper and lower bounds for the mean shift)**.** In the theorem's setting put $\kappa_0 = \frac{3}{4}\mathsf{b}_*^{-2}\ell^3 N_f I_B$. There is $\kappa_+ < \infty$ such that

$$|\mathbb{E}_K^{\mathrm{F}} T_d^K - \mathbb{E}_K^{\mathrm{W}} T_d^K| \le \kappa_+ L^{2d} \qquad (K > d \ge d_\natural).$$

There are d_3, l_0 such that

$$\mathbb{E}_K^{\mathrm{W}} T_d^K - \mathbb{E}_K^{\mathrm{F}} T_d^K \ge \tfrac{1}{4}\kappa_0 L^{2d} \qquad (d \ge d_3,\ K \ge d + l_0).$$

■□

Proof. For $d \ge d_2$ use the theorem and the bound on κ. For the finitely many smaller depths use $0 \le T_d^K \le \mathsf{Q}' L^{4d}$ and increase κ_+ by at most $\mathsf{Q}' L^{2d_2}$. Choose l_0, μ_0 so that $\kappa(l,\mu) \ge N_f I_B/8$ for $l \ge l_0$, $0 \le \mu \le \mu_0$. Then choose $d_3 \ge d_2$ with $L^{-d_3}\mathsf{m} \le \mu_0$ and $\mathsf{K}_{\mathrm{S}} L^{-d_3/4} \le \kappa_0/4$. The leading term in the theorem is at least $\kappa_0 L^{2d}/2$, leaving the stated lower bound. □

13.5. **Comparing the two flows.** We compare the gauge marginal with fermions to Wilson's measure at the same bare coupling. The histories, tests and calibrations are the same. The fermions change the running coefficients and the activities. To find the resulting change in the mean of T_d^K, we insert $e^{\zeta T_d^K}$ and differentiate the vacuum energies produced after the insertion. The final integral will be estimated by positivity and convexity.

Use the constants of Def. 10.1 and the interval $J = [b_-, b_+]$ of Def. 13.1. Fix $\mathsf{b}_* \in J$ and $\mathsf{m} > 0$, and relabel as in Lem. 10.2. Throughout this subsection put

$$t = K - d, \qquad d' = \lceil d/8 \rceil, \qquad t' = K - d', \qquad N_u = \ell^3 L^{3u}, \qquad n_k = N_{K-k},$$

and assume $K > d$ and $d' \ge \max\{\bar{n}, d_\natural\}$. The two source radii used below are

$$\rho' = \tfrac{1}{4} r_x \mathsf{b}_*, \qquad \zeta_0 = \tfrac{1}{2} L^{-d}.$$

The bare densities and expectations are

$$\varsigma^{\mathrm{F}} = e^{-\beta A}\det(D + m_0), \qquad \varsigma^{\mathrm{W}} = e^{-\beta A}, \qquad \mathbb{E}_K^\sharp F = (Z^\sharp)^{-1}\int F \varsigma^\sharp \, dU, \quad \sharp \in \{\mathrm{F}, \mathrm{W}\},$$

where $\beta = L^K \mathsf{b}_*$ and $m_0 = L^{-K}\mathsf{m}$. All constants in this subsection are independent of K, d. Dependence on m will be displayed where it enters.

Lemma 13.15 (The flow without fermions)**.** The small-field step of [QCDM1, §7], with all fermion terms omitted, defines coefficients and activities

$$x_0^{\mathrm{W}} = \beta, \qquad x_{k+1}^{\mathrm{W}} = L^{-1} x_k^{\mathrm{W}} - b_{k+1}^{\mathrm{W}}, \qquad E_k^{\mathrm{W}} = \sum_{j \le k} e_j^{\mathrm{W}} n_j.$$

Write $b_j^{\mathrm{W}} = e_{\mathsf{N},j} + b_j^{r,\mathrm{W}}$. At every level of a weak datum,

$$|x_k^{\mathrm{W}} - \mathsf{t}_k| \le \bar{B}, \qquad |b_{k+1}^{r,\mathrm{W}}| \le C_w \phi(x_k^{\mathrm{W}}), \qquad \|F_0^{\mathrm{W},(j)}\|_j \le C_w \phi(\mathsf{t}_{j-1}).$$

The degree-zero activities have the bound E_0 and the background variation bound of the step. The term of the history without large fields is

$$e^{E_k^{\mathrm{W}}}\mathsf{Stay}(V)\exp\left[-L^k x_k^{\mathrm{W}} A(\mathsf{U}_k(V)) + \sum_{j \le k}\sum_X E_0^{\mathrm{W},(j)}(X; \mathsf{U}_k(V))\right] \mathscr{M}_{h_\emptyset}^{\mathrm{fr}}(dV).$$

As measures it lies between $e^{E_k^{\mathrm{W}} \pm C_5^{\mathrm{W}} n_k}\mathsf{sf}_k[x_k^{\mathrm{W}}]\, dV_k$, and $|e_j^{\mathrm{W}}| \le C'_{e,\mathrm{W}}(1 + \log x_{j-1}^{\mathrm{W}})$.

These activities and coefficients are holomorphic in the earlier coefficients on their step polydiscs. The bounds for the activities and the extracted step numbers hold at $|x_i' - x_i^{\mathrm{W}}| \le (2/3) r_x x_i^{\mathrm{W}}$. There is also a density identity for a restart. Suppose $G \ge 0$ is bounded and Borel on the level-n_0 fields, and its history term has the displayed form at level n_0 with coefficient x_{n_0}'' and vacuum energy E_{n_0}'', and has a bounded Borel density against dV_{n_0}. Keep the original coefficients below n_0 and apply

the same steps thereafter. If $|x''_k - \mathfrak{t}_k| \le \bar{B}$ for $n_0 \le k \le n_1$, the history terms of $G(\mathcal{M}^{n_0}U)e^{-\beta A}$ have this form at every such level, with

$$x''_{k+1} = L^{-1}x''_k - b''_{k+1}, \qquad E''_k = E''_{n_0} + \sum_{n_0 \le j < k} e''_{j+1} n_{j+1},$$

and satisfy the same two measure bounds. ■

Proof. In the combined step the fermion normalization becomes one, the mass and positive-degree vertices disappear, and the remaining integral is the gauge Gaussian integral with the gauge vertices and the variations of the old scalar terms. The Gaussian uses x_k^{W}; the tests, charts and gauge fixing use $\mathfrak{t}_k$. This is the gauge step in the chosen calibration frame. Every estimate for the omitted vertices is replaced by zero. The proof of the small-field induction in [QCDM1, §7] therefore applies, including its holomorphic dependence and the scalar array bound. More explicitly, the new fluctuation family has norm at most $\frac{1}{2}C_w\phi(\mathfrak{t}_k)$, and its extracted coupling has modulus at most $C_b\phi(\mathfrak{t}_k)$. The inequalities $C_w \ge 2\sqrt{2}C_b$ and $x_k^{\mathrm{W}} \le (1+\bar{B}\gamma^2)\mathfrak{t}_k$ give the claimed remainder bound. Moreover

$$|e_{\mathsf{N},j}| \le (1+L^{-1})c_{2,\mathrm{loop}}, \qquad |b_j^{\mathrm{W}}| \le B_\sharp - 1,$$

by Def. 10.1. Summing the coefficient recursion gives $|x_k^{\mathrm{W}} - \mathfrak{t}_k| \le \bar{B}$.

The scalar array consists of the gauge normalization, the extracted action term, the Gaussian correction and the fluctuation family. On the coefficient polydiscs their polymer bounds are the bounds used for the degree-zero array in [QCDM1, §7]; the closed fermion loops and the fermion coupling coefficient are absent. These bounds give E_0 at rate $\kappa+1$ and the corresponding background variation. The lower and upper measure bounds are then the degree-zero history estimate of [QCDM1, Thm. E] with the determinants omitted. The vacuum-energy estimate is obtained from the same step.

For the restart, the two window bounds imply

$$|x''_i - x_i^{\mathrm{W}}| \le 2\bar{B} \le \tfrac{1}{2}r_x\gamma^{-2} \le \tfrac{2}{3}r_x x_i^{\mathrm{W}}.$$

Thus all the required activities are in the preceding polydiscs and retain their bounds. The history weights multiply at each step. For every bounded Borel bare density, the next history term is consequently the image of the current term times the next tests and gauge-fixing factor. The chart formula uses its density and its standard form, so it also applies to the restarted term. Induction gives the displayed recursion and a bounded Borel density at each level. Applying the scalar array bound to this density gives the two measure inequalities. □

Lemma 13.16 (Discarding histories)**.** Let $\mathfrak{w}$ be the weight of the history with no large-field region through level t'. Let $\widetilde{\nu}^\sharp$ be the normalized image of $\mathfrak{w}\varsigma^\sharp\, dU$ under $\mathcal{M}^t$. There are constants $C^\sharp_{\mathrm{dis}}$, independent of K, d, such that, with

$$\rho^\sharp(d') = \exp\left[C^\sharp_{\mathrm{dis}} - \tfrac{1}{2}c_\star A_0^2 (d'\log L)^{2\mathfrak{p}_0}\right],$$

$$0 \le Z^\sharp - \int \mathfrak{w}\varsigma^\sharp\, dU \le \rho^\sharp(d')Z^\sharp, \qquad |\mathbb{E}^\sharp_K F(V_d) - \widetilde{\nu}^\sharp(F)| \le 2\rho^\sharp(d')\|F\|_\infty$$

for every bounded gauge-invariant Borel F. Both the history at level t and $\widetilde{\nu}^\sharp$ are carried by

$$\mathrm{In}_t[\mathsf{b}_*] \subset \mathrm{Cut}_t[\mathsf{b}_*] \subset \mathrm{Cut}_t[b_-].$$

On this set $T_d^K(V) = L^K A(\mathsf{U}_t(V))$. ■

Proof. The weights are nonnegative. A history with a nonempty large-field region by level t' has a created cell at some level $j \le t'$. The mark counts this cell, and $\mathfrak{t}_j \ge L^{d'}$, so its value is at least $\frac{1}{2}c_\star A_0^2(d'\log L)^{2\mathfrak{p}_0}$. Apply the marked history sum at the last level $\bar{K} = K - \bar{n}$. In the fermion

model its upper bound is $\exp[E_{\bar K}+(C+2)n_{\bar K}]$. The unintegrated lower measure bound and the ball integral of Lem. 10.2 give a lower partition bound with

$$C_{\mathrm{dis}}^{\mathrm{F}}=\Big[C+2+C_5^F+\nu\log\left(\frac{1}{b_F(1-L^{-1})}+\frac{L^{\bar n}}{\mathsf m}\right)+\tfrac{9}{2}\log(L(\bar g^{-2}+B_\sharp))\\ +3\log 7+24(B_3^{\mathrm{cr}})^2\Big]n_{\bar K}.$$

Here $n_{\bar K}=L^{3\bar n}\ell^3$ is independent of K. In the gauge model the marked sum and the partition lower bound of [YM3, Thm. C], at the same mark and share, give

$$C_{\mathrm{dis}}^{\mathrm{W}}=(\mathsf C_{\mathrm{maj}}+2+C_{\mathrm{st}}+\tfrac{9}{2}\log\bar{\mathsf b})n_{\bar K}.$$

Taking the quotient proves the lost-mass estimate.

The history identity with the insertion $F(\mathcal M^tU)$ bounds the lost numerator by $\rho^\sharp Z^\sharp\|F\|_\infty$. Normalizing the retained numerator costs the same amount. This gives the factor two. The argument uses positivity of the complete history measures, without bounding individual gauge-fixing factors by one.

The indicator of the history through t' vanishes unless the history through t passes its tests. Its block field then has plaquette deviation less than $2\varepsilon(\mathsf t_t)$, where $\mathsf t_t=L^d\mathsf b_*$. The cut inclusions in Def. 13.1 prove the carrier assertion and the identity for T_d^K. □

Lemma 13.17 (Inserting the background action)**.** For $|\zeta|\le\rho'$ keep the coefficients below t, replace x_t by $x_t-\zeta L^d$, keep s_t, and continue the step. Denote the resulting coefficients by x'_k,s'_k and the extracted numbers by $b_{k+1}(\zeta),\mu_{k+1}(\zeta),e_{k+1}(\zeta)$. Restart the gauge flow in the same way at $x_t^{\mathrm W}-\zeta L^d$. For $t\le k\le t'$,

$$|x'_k-x_k|\le\tfrac23 r_xx_k,\qquad |s'_k|\le 2\bar B_M/\mathsf t_k,\qquad |x_k^{\mathrm W}(\zeta)-x_k^{\mathrm W}|\le\tfrac23 r_xx_k^{\mathrm W}.$$

The coefficients and step energies are holomorphic for $|\zeta|<\rho'$, continuous on the closed disc, and real on its real interval. The gauge normalization coefficient $e_{\mathsf N,j}$ and the fermion one-loop coefficient b_j^F are constant in ζ. In particular

$$x'_k-x_k=-\zeta L^{K-k}-\sum_{t<j\le k}L^{-(k-j)}(b_j^r(\zeta)-b_j^r(0)),\qquad |b_j^r(\zeta)|\le C_w\phi(x_{j-1}),$$

and the analogous formula holds without fermions.

For real $|\zeta|\le\zeta_0$, both restarted coupling windows satisfy $|x_k(\zeta)-\mathsf t_k|\le\bar B-1/2$. The history terms of $e^{\zeta T_d^K\circ\mathcal M^t}\varsigma^\sharp$ are precisely these restarted densities. Their vacuum energies are $E_t^\sharp+\sum_{t\le j<k}e_{j+1}^\sharp(\zeta)n_{j+1}$. ■

Proof. Assume the bounds through level k. The data lie in the step polydiscs of radius one: indeed

$$2\bar B_M/\mathsf t_i\le 2\bar B_M\gamma\mathsf t_i^{-1/2}\le\tfrac23\mathsf t_i^{-1/2}\le x_i^{-1/2}.$$

The step therefore supplies its original constants. Its normalization coefficient depends only on the level and L, and b_j^F depends only on the fine mass. Subtracting the recursions gives the displayed formula. The jump has modulus at most $0.26r_xx_k$, since $\bar B\le x_k/192$. The remaining sum is at most

$$2C_w\phi(x_k)\sum_{j\ge0}L^{-j}\le\tfrac{13}{6}C_w\phi(x_k)\le\tfrac{13}{168}r_xx_k.$$

Here $x_{j-1}\ge x_k$ for $j\le k$ and ϕ decreases on the admitted range. Their sum is below $(2/3)r_xx_k$.

The one-loop mass increments telescope at these complex coefficients, as in [QCDM1, §8]. Writing their cumulative value as $G_k(x')$ gives

$$s'_k=G_k(x')+\sum_{j\le k}L^{k-j}\rho_j,\qquad |G_k(x')|\le 4B_F/\mathsf t_k,\quad |\rho_j|\le C_\rho\phi_\sharp(x_{j-1})^3.$$

The geometric estimate for the last sum, together with $x_{j-1} \geq 3\mathsf{t}_{j-1}/4$, bounds it by $0.4/\mathsf{t}_k$ under Def. 10.1. Since $\bar{B}_M = 3B_F + 1$, this closes the shift window. The gauge induction uses the same proof with $s = b^F = 0$. Composition gives holomorphy and reality.

For $|\zeta| \leq \zeta_0$ the jump is at most $1/2$ in each coupling window, and every extracted coupling has modulus at most $B_\sharp - 1$. Thus

$$|x'_k - \mathsf{t}_k| \leq \tfrac{1}{2} + (B_\sharp - 1)\frac{L}{L-1} \leq \bar{B} - \tfrac{1}{2},$$

and likewise in the gauge flow. At level t the carrier identity of Lem. 13.16 gives

$$e^{\zeta T_d^K(V)} e^{-L^t x_t A(\mathsf{U}_t(V))} = e^{-L^t(x_t - \zeta L^d)A(\mathsf{U}_t(V))}.$$

The activities at this level read the earlier coefficients, so they are unchanged. The source factor is bounded and Borel. The step's image-density identity then applies at every later level by Lem. 13.11: it reads the standard density, its coefficients in the polydiscs, and the activities with their closed bounds. It does not require the earlier coefficients to obey the unperturbed recursion. These bounds hold also on all comparison tori, as the holomorphic construction is made on the infinite lattice and its restrictions. Induction proves the density identity with fermions. For the gauge density apply the restart clause of Lem. 13.15, with $G = e^{\zeta T_d^K}$. □

Lemma 13.18 (The mean and the vacuum energies)**.** For $\sharp \in \{\mathrm{F}, \mathrm{W}\}$ put

$$c_\sharp(\zeta) = \sum_{t \leq k < t'} e_{k+1}^\sharp(\zeta) n_{k+1}, \qquad f_\sharp(\zeta) = \log \widetilde{\nu}^\sharp(e^{\zeta T_d^K}) \quad (\zeta \in \mathbb{R}).$$

There is C_{fc} depending only on the fixed construction constants such that, with

$$R_{\mathrm{fc}} = C_{\mathrm{fc}}(1 + d' + |\log \mathsf{b}_*| + \log_+(1/\mathsf{m}))N_{d'},$$
$$C_{\mathrm{cx}} = L^3 + 6K_M c_{\mathrm{cl}}, \qquad \bar{S} = 128 C_{\mathrm{cx}} N_d/(r_x^2 \mathsf{b}_*^2),$$

we have $f_\sharp(0) = 0$ and

$$f_\sharp(\zeta) \leq c_\sharp(\zeta) - c_\sharp(0) + R_{\mathrm{fc}} \quad (|\zeta| \leq \zeta_0).$$

The function $c_\sharp$ is holomorphic on $|\zeta| < \rho'$, with

$$|c_\sharp(\zeta) - c_\sharp(0)| \leq C_{\mathrm{cx}} N_d, \qquad |c''_\sharp(\zeta)| \leq \bar{S} \quad (|\zeta| \leq \rho'/2).$$

If $(2R_{\mathrm{fc}}/\bar{S})^{1/2} \leq \zeta_0 \leq \rho'/2$, then

$$|\widetilde{\nu}^\sharp(T_d^K) - c'_\sharp(0)| \leq (2R_{\mathrm{fc}}\bar{S})^{1/2}.$$

■

Proof. By Lem. 13.17, the mass of the retained tilted history is its vacuum exponential times a final integral $I^\sharp(\zeta)$. Thus

$$f_\sharp(\zeta) = c_\sharp(\zeta) - c_\sharp(0) + \log \frac{I^\sharp(\zeta)}{I^\sharp(0)}.$$

In the fermion model the degree-zero history bound gives $e^{\mathsf{rg}} \mathsf{sf}'_{t'} dV_{t'}$ with $|\mathsf{rg}| \leq C_5^F n_{t'}$ and $0 \leq \mathsf{sf}'_{t'} \leq 1$, before the remaining fermion determinant is restored. The exact determinant identity of [QCDM1, §9] gives $0 < \det \mathbb{D}_{t'} \leq \theta_{t'}^{\nu n_{t'}}$, where $b_F(1 - L^{-1}) \leq \theta_{t'} \leq b_F$. Consequently

$$I^{\mathrm{F}}(\zeta) \leq e^{(\nu \log(1+b_F) + C_5^F) n_{t'}}.$$

At zero source the lower measure bound and the ball integral in Lem. 10.2 give

$$I^{\mathrm{F}}(0) \geq \exp\left[-\left(C_5^F + \nu \log(1/\theta_{t'} + 1/\mathfrak{m}_{t'}) + \tfrac{9}{2}\log x_{t'} + 3\log 7 + 24(B_3^{\mathrm{cr}})^2\right) n_{t'}\right].$$

Here $\mathfrak{m}_{t'} = L^{-d'}\mathsf{m}$, $x_{t'} \leq \frac{3}{2} L^{d'} \mathsf{b}_*$ and $n_{t'} = N_{d'}$. Their logarithms are bounded by a constant times $1 + d' + |\log \mathsf{b}_*| + \log_+(1/\mathsf{m})$. Without fermions, Lem. 13.15 gives the same upper and lower estimates

with the determinant terms omitted. For the lower estimate $x_{t'}^{\mathrm{W}} \ge \gamma^{-2}$ and $x_{t'}^{\mathrm{W}} \ge 3\mathsf{t}_{t'}/4$, so the ball integral applies. This proves the first inequality.

The parts of a vacuum energy independent of ζ cancel in its difference. The Gaussian part is

$$g_k(x) = -3(L^3-1)\,\mathrm{Log}(x/\mathsf{t}_k).$$

Its coefficient is half the number of integrated nongauge directions per level-$(k+1)$ site; see Lem. 13.6. The test contribution has modulus at most $K_M c_{\mathrm{cl}}$, and the cluster contribution at most $2K_M c_{\mathrm{cl}}$. Since $|x'_k/x_k - 1| \le 1/12$, the difference of the Gaussian terms is at most L^3. Hence $|e^\sharp_{k+1}(\zeta) - e^\sharp_{k+1}(0)| \le C_{\mathrm{cx}}$. Now $\sum_{t\le k<t'} n_{k+1} \le N_d$, which proves the holomorphic bound. Cauchy's estimate on discs of radius $\rho'/2$ gives the stated second-derivative bound.

Finally $f_\sharp$ is smooth and convex because T_d^K is bounded. For $0 < h \le \zeta_0$, $\pm h f'_\sharp(0) \le f_\sharp(\pm h)$. Taylor's formula for $c_\sharp$ therefore gives

$$|f'_\sharp(0) - c'_\sharp(0)| \le \tfrac12 h\bar S + R_{\mathrm{fc}}/h.$$

Use $h = (2R_{\mathrm{fc}}/\bar S)^{1/2}$ and $f'_\sharp(0) = \widetilde\nu^\sharp(T_d^K)$. □

13.6. **The scalar part of the step.** We isolate the terms of second order which remain after subtracting the two vacuum energies. The expansion uses five parameters: the gauge fluctuation, the old fermion terms, the gauge potential, the mass term, and the difference of the old scalar terms. Their respective weights are $1, 2, 1, 2, 1$.

Lemma 13.19 (The deformed potential)**.** Use the restarted flows of Lem. 13.17 at a level $t \le k < t'$ and $|\zeta| \le \rho'$. All expressions in this lemma are evaluated at the trivial background on the infinite lattice. Write

$$\mathbf{N}_\natural^{\mathrm{F},(j)} = \mathbf{N}^{\mathrm{F},(j)} - \mathbf{c}_j^{\mathrm{F}}, \qquad \Delta^{(j)} = (\mathbf{G}^{(j)} - \mathbf{G}^{\mathrm{W},(j)}) + (F_0^{(j)} - F_0^{\mathrm{W},(j)}),$$

where $\mathbf{c}_j^{\mathrm{F}}$ is b_j^F times the level-j action. Then $E_0^{(j)} = E_0^{\mathrm{W},(j)} + \mathbf{N}_\natural^{\mathrm{F},(j)} + \Delta^{(j)}$. Let $\mathcal{V}^{\mathrm{N}}$ and $\mathcal{V}^\delta$ be the variations of the last two families from the background U to the field $\mathsf{U}'(B)$ of the step. Let $\mathcal{V}_x^{0,\mathrm{W}}$ be the gauge potential at (t_k, x) plus the variations of the old gauge activities. Define

$$\begin{aligned}\mathcal{V}(z) = {} & z_3 \mathcal{V}^{0,\mathrm{W}}_{x'_k}(B) + \mathcal{V}^{\mathrm{N}}(z_1 B) + z_5 \mathcal{V}^\delta(z_1 B) \\ & + \mathcal{V}_k^{\mathrm{F}}(z_1 B) + z_2 \mathcal{V}^{\mathrm{old}}_{\ge 2}(z_1 B) + z_4 \mathcal{V}^s(z_1 B), \qquad z \in \mathbb{C}^5.\end{aligned}$$

The remaining arguments are the fermion variables of the step. At $z = (1,1,1,1,1)$ this is its full potential.

Put $c'_\diamond = c_{\mathrm{cl}}/6$ and $\delta_k = \delta(\mathsf{t}_k)$. Choose

$$r'_1 = \min\left\{ \frac{c'_\diamond}{C_V h_F^2 \mathsf{C}_{\mathcal{A}} \delta_k}, \frac{\beta_0}{2\mathsf{C}_{\mathcal{A}}\delta_k}, \frac{c'_\diamond a_{\mathrm{an}}}{c_h K_{\mathrm{old}} \mathsf{C}_{\mathcal{A}} \delta_k}, \frac{\mathsf{s}^\star \alpha_0}{(1+\beta_s)\delta_k}, \frac{\varepsilon_\natural}{\mathsf{C}_{\mathcal{A}}\delta_k}, \frac{\min\{1, \delta(\gamma^{-2})\}}{\delta_k} \right\}.$$

The radii r'_2, r'_3, r'_4 are the old-fermion, gauge and mass radii of the second-order step expansion in [QCDM1, §8], with $c'_\diamond$ in place of its cluster allowance. Set

$$C_\Delta = 3K_\Lambda \mathsf{C}_{\mathcal{A}} \mathsf{s}^\star C_w + 4(2\bar B + 9)/\sigma_\star, \qquad r'_5 = \frac{c'_\diamond}{C_\Delta \phi(\mathsf{t}_k)}.$$

There is a constant C_{pd}, formed from the displayed constants, such that

$$(r'_i)^{-1} \le C_{\mathrm{pd}} \phi(\mathsf{t}_k)^{w_i}, \qquad w = (1,2,1,2,1).$$

Assume $C_{\mathrm{pd}}\phi(\mathsf{t}_k) \le 1/2$. Then $r'_i \ge 2$. On $|z_i| \le r'_i$, the scalar and fermion parts of the potential each have norm at most $3c'_\diamond$. The vacuum density $\hat e(z)$ of its cluster expansion is holomorphic and its Taylor coefficients satisfy

$$|\hat e_n| \le 2K_M c_{\mathrm{cl}} \prod_i (r'_i)^{-n_i}.$$

Write $\hat{e}_k^{\mathrm{W}}(x)$ for the gauge cluster density and $\mathsf{e}_k^{\mathrm{t}}(x)$ for the test contribution. With g_k as in Lem. 13.18,

$$e_{k+1}(\zeta) - e_{k+1}^{\mathrm{W}}(\zeta) = e_{k+1}^{\mathrm{F},0} + g_k(x_k') - g_k(x_k^{\mathrm{W}}) + \mathsf{e}_k^{\mathrm{t}}(x_k') - \mathsf{e}_k^{\mathrm{t}}(x_k^{\mathrm{W}}) \\ + \hat{e}(1,1,1,1,1) - \hat{e}_k^{\mathrm{W}}(x_k^{\mathrm{W}}), \qquad \hat{e}(0,0,1,0,0) = \hat{e}_k^{\mathrm{W}}(x_k').$$

Here and below $x_k^{\mathrm{W}} = x_k^{\mathrm{W}}(\zeta)$ when the argument is omitted. The coefficients $\hat{e}_n$ vanish if $n_1 = 0, n_5 \geq 1$, and also for $n = e_1, e_1 + e_5, e_1 + e_3$. ■

Proof. The identities for the scalar arrays follow by separating the fermion normalization and its extracted action term. The remaining gauge normalization and extraction terms are formed at the calibrations and agree in the two flows. The potential identity follows from the exact step.

The radius bounds use $\delta_k \leq A_1 \phi(\mathsf{t}_k)$,

$$\sum_{j \leq k} C_7 L^{(k+1-j)/4} \epsilon_j^{\psi} \leq C_7 C_2 S_\omega \phi(\mathsf{t}_k)^2, \qquad |s_k'| \leq 2\bar{B}_M \phi(\mathsf{t}_k)^2.$$

Thus C_{pd} can be taken as the maximum of one and the coefficients obtained by substituting these three bounds into the inverse radii. It depends only on the constants up to γ. The first, second, third and fourth usual vertices are bounded by $c_\diamond'$ on these discs, just as in the second-order expansion of [QCDM1, §8].

For the two additional scalar variations, put $\delta_\flat = \max\{1, |z_1|\}\delta_k$. The last radius condition gives $\delta_\flat \leq \min\{1, \delta(\gamma^{-2})\}$. Since δ decreases continuously to zero, choose $\mathsf{t}_\flat \in [\gamma^{-2}, \mathsf{t}_k]$ with $\delta(\mathsf{t}_\flat) = \delta_\flat$. The background-variation estimate of the step applies there: the fourth and fifth radius conditions give

$$(1 + \beta_s)\delta(\mathsf{t}_\flat)/\alpha_0 \leq \mathsf{s}^\star, \qquad \mathsf{C}_{\mathcal{A}} \delta(\mathsf{t}_\flat) \leq \varepsilon_{\mathfrak{h}}.$$

The other geometric conditions hold at every $\mathsf{t}_\flat \geq \gamma^{-2}$. For the explicit normalization families this estimate is $\mathsf{s}_a(\mathsf{t}_\flat) K_{\mathrm{old}}$, where $\mathsf{s}_a(x) = c_h \delta(x)/a_{\mathrm{an}}$. The third radius condition bounds it by $c_\diamond'$.

The difference of the fluctuation families contributes at most

$$3K_\Lambda \mathsf{C}_{\mathcal{A}} \mathsf{s}^\star C_w \phi(\mathsf{t}_k),$$

using $\sum_{j \leq k} L^{-(k-j)} \leq 3/2$. For the Gaussian families, write $\tau_i = x_i/\mathsf{t}_i$. Their total variation is holomorphic on $|\tau_i - 1| < \sigma_\star$ and bounded there by $\mathsf{s}_a(\mathsf{t}_\flat) K_{\mathrm{old}} \leq 1$. Both flows are in the half-radius polydisc. Their common jump cancels when they are subtracted; hence $|x_i' - x_i^{\mathrm{W}}| \leq 2\bar{B} + 9$. Cauchy's estimate and $\sum_{i<k} \mathsf{t}_i^{-1} \leq 2\phi(\mathsf{t}_k)^2$ give the remaining bound $4(2\bar{B} + 9)\phi(\mathsf{t}_k)^2/\sigma_\star$. These two contributions prove $\|\mathcal{V}^\delta(z_1 B)\| \leq C_\Delta \phi(\mathsf{t}_k)$. The r_5' condition completes the potential bound. The cluster theorem and Cauchy's estimate now give the stated bound on $\hat{e}_n$.

The vacuum energy is the sum of its gauge normalization, fermion normalization, Gaussian, test and cluster parts. The first is common to the two flows and cancels. This proves the vacuum-energy identity. At $z = (0,0,1,0,0)$ the Grassmann integral factors and equals one, giving the gauge identity.

Finally $\mathcal{V}^{\mathrm{N}}, \mathcal{V}^\delta$ and $\mathcal{V}_k^{\mathrm{F}}$ vanish at $B = 0$. Thus no term with $n_1 = 0, n_5 \geq 1$ occurs. With $z_2 = z_3 = z_4 = 0$, the substitution $B \mapsto -B$ changes z_1 to $-z_1$ and preserves the Gaussian, tests and integration domain. This removes e_1 and $e_1 + e_5$. The coefficient $e_1 + e_3$ is linear in the first derivative in B of $\mathcal{V}^{\mathrm{N}} + \mathcal{V}_k^{\mathrm{F}}$. The remaining integrand is invariant under constant colour rotations. Averaging that derivative over these rotations gives zero, since the adjoint representation has no invariant vector. This removes $e_1 + e_3$. □

Lemma 13.20 (The scalar contraction of a total)**.** At the trivial pair let μ be the normalized Berezin Gaussian with covariance $\Gamma_k(\mathbf{1})^{-1}$. Let $\mathcal{G} = LM\mathbb{Z}^3$ act in level-k units. Choose a cube Λ of L^3 level-k cells and let B_R be its R^3 translates with indices in $\{0, \dots, R-1\}^3$. For a formal even

total P with no scalar part, put

$$(\mathcal{J}_k[P])_0 = \lim_{R\to\infty} (R^3M^3)^{-1}\mu(P_{B_R}),$$

when the limit exists. The subscript restricts monomials to places in B_R, and P is read at $\Psi_k = W$.

Suppose v_Y are even local vertices, independent of the gauge fluctuation, which vanish at $W = 0$, commute with $\mathcal{G}$ and satisfy

$$S_v = \sup_{\square} \sum_{Y\supset\square} e^{(\kappa-1)d_k(Y)}\|v_Y\| < \infty.$$

Let $\bar v$ be the total of their parts in W alone, and $a^v(X)$ the coefficient of z in the step expansion at potential zv. Then $\|a^v\| \le K_M S_v$, and

$$M^{-3} \sum_{X \text{ filed at } c} a^v(X)_0 = M^{-3} \sum_{Y \text{ filed in } \Lambda} \mu(v_Y)_0 = (\mathcal{J}_k[\bar v])_0.$$

In particular the limit exists and is bounded by $K_M S_v$. For the old totals P_j of the second-order step,

$$|(\mathcal{J}_k[\sum_{j\le k} P_j])_0| \le K_M c_\psi \sum_{j\le k} C_7 L^{(k+1-j)/4}\|P_j\|_j.$$

The same estimate holds with the norms of any representing families in place of $\|P_j\|_j$.

The scalar contractions of the mass total $\mathsf{S}_1(\mathsf{H}_{0,k}\Psi_k)$ and of the second-order total $\mathscr{T}_k$ also exist. For the latter, the contribution of a slice A_i, $i < k$, is the slice mean of the density per level-$(k+1)$ site of

$$\operatorname{Tr}\big((\delta^2_{A_i}\mathbb{D}_k)_{AA}\Gamma_k^{-1}\big) - \tfrac12 \operatorname{Tr}\big[((\delta_{A_i}\mathbb{D}_k)_{AA}\Gamma_k^{-1})^2\big].$$

■

Proof. The linear coefficient of the cluster expansion has the maps used in the identity of totals in [QCDM1, §8]. If $Y^\sharp$ is the enlargement of Y to the next level, these maps satisfy

$$a^v(X) = \sum_{Y:Y^\sharp\subset X} \mathcal{L}_{X,Y}(v_Y), \qquad \sum_{X\supset Y^\sharp} \mathcal{L}_{X,Y}(v_Y) = \mu(v_Y),$$

and $\sum_{X\supset Y^\sharp} \|\mathcal{L}_{X,Y}(v_Y)\| \le K_E C_E w_Y(v_Y)$. The weights w_Y are summable over polymers filed at a fixed cell. Indeed their input rate is at most $(1-2\delta_{\mathrm{w}})\kappa \le \kappa - 1$; the remaining exponential factors are those of the cluster count. The count and Cauchy's estimate give $\|a^v\| \le K_M S_v$.

The group $\mathcal{G}$ acts freely on pairs (X, Y) with $Y^\sharp \subset X$. Each orbit has exactly one representative with X filed at c, and exactly one with Y filed in Λ. Translation covariance makes $\mathcal{L}_{X,Y}(v_Y)_0$ constant on an orbit. The preceding absolute bound allows these two sums over orbit representatives to be equated. This proves the first equality.

It remains to identify the limit without dropping boundary terms. Putting generators outside B_R equal to zero does not increase their norm. Gram's inequality then gives

$$\left|\mu(\bar v_{B_R}) - \sum_{Y \text{ filed in } B_R} \mu(v_Y)_0\right| \le 2 \sum_{Y\in\mathcal{Y}_R} \|v_Y\|,$$

where $\mathcal{Y}_R$ consists of polymers meeting both B_R and its complement. A polymer meeting the nth cell layer from the boundary has $d_k(Y) > n - 2$. There are at most $6(RL)^2$ cells in each layer, so

$$\sum_{Y\in\mathcal{Y}_R} \|v_Y\| \le 6(RL)^2 e^{2(\kappa-1)} S_v \sum_{n\ge0} e^{-(\kappa-1)n} = O(R^2).$$

The interior sum is R^3 times the sum filed in Λ. Division by R^3M^3 proves the limit.

Apply this argument to the old-activity variation at $B = 0$. Every term contains a W leg, since it is the old total at $\mathsf{H}_k\Psi_{k+1} + W$ minus that at $\mathsf{H}_k\Psi_{k+1}$. Its vertex norm is at most $c_\psi \sum_j C_7 L^{(k+1-j)/4}\|F^{(j)}\|_j$, and its part in W alone is $\sum_j P_j$. The resulting scalar contraction is

linear and reads only these totals. Taking representatives whose norms decrease to $\|P_j\|_j$ gives the quotient-norm estimate. The mass vertex is treated in the same way, using $\mathsf{H}_{0,k}\mathsf{H}_k = \mathsf{H}_{0,k+1}$. The identity $\mathscr{T}_k = \sum_{j\le k} T^{(j)} - L^{-k} G_k \mathsf{S}_1(\mathsf{H}_{0,k}\Psi_k)$ therefore gives its scalar contraction.

For its value, apply Wick's rule in B_R to the quadratic and quartic fermion terms of each slice. The square of the single trace vanishes: the free covariance is scalar in colour and the first vertex has colour trace zero. The other contractions are the two traces displayed above. Their kernels decay exponentially in the distances among their places. After division by R^3M^3, replacing the finite-box sums by a sum with one place in Λ costs $O(R^{-1})$. Their limits are the stated trace densities. □

Lemma 13.21 (A row bound for the quadratic term)**.** At the trivial configuration of a weak datum let $u = \mathsf{t}_k^{-1/2} B$ be the unscaled fluctuation on the nonpivot bonds. Let Q be the symmetric kernel of the quadratic term in

$$\log\det \Gamma_k(\mathsf{U}'(u)) + \sum_{j\le k}\sum_{X} \mathbf{N}_{\natural}^{\mathrm{F},(j)}(X;\mathsf{U}'(u)),$$

so the term is $\sum_{i,j} Q(i,j) u_i u_j$, where $i = (b,a)$ includes the colour coordinate. Then

$$\sup_i \sum_j |Q(i,j)| \le C_{\mathrm{ker}},$$

uniformly in the torus, level and fine mass. One may take

$$\delta_\star = \min\left\{ \frac{\mathsf{s}^\star \alpha_0}{1+\beta_s}, \frac{\varepsilon_{\mathfrak{h}}}{\mathsf{C}_{\mathcal{A}}}, 1, \delta(\gamma^{-2}) \right\}, \qquad S_D = \sum_{n\in\mathbb{Z}^3} e^{-2\alpha_D M(|n|_\infty - 1)_+},$$

$$C_{\mathrm{ker}} = \frac{36 C_V h_F^2}{\beta_0} + \frac{144 S_D C_D^2 C_V^2 h_F^4}{\Gamma_F^4} + \frac{36 C_E c_h K_{\mathrm{old}}}{a_{\mathrm{an}} \mathsf{C}_{\mathcal{A}}^2 \delta_\star}.$$

Here C_D, α_D are the free covariance bounds, Γ_F is its Grassmann weight, and $C_E = 4096$ is the polymer-count constant. Thus C_{ker} is formed from constants up to γ. ■

Proof. We first record a Cauchy estimate which controls a whole row. If f is holomorphic and bounded by M on $\{|u_i| < r\}$, then

$$\sum_j |\partial_i \partial_j f(0)| \le 4M/r^2.$$

Indeed apply the two-variable estimate to $f(se_i + tv)$ on $|s|, |t| < r/2$, choosing $|v_j| = 1$ so that the terms $v_j \partial_i \partial_j f(0)$ have the same phase. A polymer reading the bond b contains the cell of one of its endpoints. Its sum is therefore bounded by two sums over polymers containing a fixed cell.

Put the external fermion variables equal to zero in $\mathcal{V}_k^{\mathrm{F}}(Y;u)$ and call the result $\mathfrak{v}_Y(u)$. With $V = \sum_Y \mathfrak{v}_Y$ and μ the free Berezin Gaussian, the exact step gives

$$\det \Gamma_k(\mathsf{U}'(u)) / \det \Gamma_k(\mathbf{1}) = \mu(e^{V(u)}).$$

The localized variations $\mathcal{V}^{\mathrm{N}}(Y;u)$ sum to the change in the old normalization array for small real u. Gauge invariance permits the chart background to be replaced by $\mathsf{U}'(u)$. Holomorphy extends this identity near zero. Thus $Q = Q^{\mathrm{F}} + Q^{\mathrm{N}}$, from $\log \mu(e^V)$ and $\sum_Y \mathcal{V}^{\mathrm{N}}(Y;u)$. As $V(0) = 0$, their fermion part through order two is

$$\mu(V_1) + \mu(V_2) + \tfrac{1}{2}\{\mu(V_1^2) - \mu(V_1)^2\}.$$

Let v_Y be the supremum of the vertex norm for $\sup_b |u(b)| \le \beta_0$. Then

$$\sum_{Y \supset \square} e^{\kappa d_k(Y)} v_Y \le C_V h_F^2 \beta_0.$$

The substitution removing external variables contracts the norm, and Gram's inequality bounds $|\mu(F)|$ by $\|F\|$. The coordinate polydisc of radius $r_F = \beta_0/3$ lies in this domain, since the three Pauli

matrices have norm one. The row estimate applied to $\mu(\mathfrak{v}_Y(u))$, with the factor $1/2$ from its quadratic kernel and the two endpoint cells, bounds the rows of $\mu(V_2)$ by $4C_V h_F^2\beta_0/r_F^2 = 36C_V h_F^2/\beta_0$.

For quadratic elements F, F' supported in Y, Y', Wick's rule and covariance decay give

$$|\mu(F;F')| \le 2C_D^2\Gamma_F^{-4} e^{-2\alpha_D \operatorname{dist}_k(Y,Y')}\|F\|\|F'\|.$$

Write $x_{Y,i} = \partial_i \mathfrak{v}_Y(0)$ and choose phases v_j in the row as before. Cauchy's estimate bounds both $\|x_{Y,i}\|$ and $\|\sum_j v_j x_{Y,j}\|$ by v_Y/r_F. File Y' at a cell nearest to Y. The covariance sum over such cells is at most $\#Y S_D$, and $\#Y \le 8(d_k(Y)+1) \le 8e^{\kappa d_k(Y)}$. Summing first over Y' and then over the two endpoint cells of Y gives

$$\sum_j |\mu(\partial_i V_1; \partial_j V_1)| \le \frac{32 S_D C_D^2 (C_V h_F^2\beta_0)^2}{\Gamma_F^4 r_F^2}.$$

Half this quantity is the quadratic-kernel bound, giving the second term of C_{ker}.

For Q^{N} choose $\mathfrak{t}_\star \ge \gamma^{-2}$ with $\delta(\mathfrak{t}_\star) = \delta_\star$. The radius and background conditions checked in Lem. 13.19 apply at this value. Each normalization variation is holomorphic for $|u(b)| < \mathsf{C}_{\mathcal{A}}\delta_\star$ and bounded by

$$\frac{c_h \delta_\star K_{\mathrm{old}}}{a_{\mathrm{an}}} e^{-(1-2\delta_{\mathrm{w}})\kappa d_k(Y)}.$$

Use the coordinate radius $r_N = \mathsf{C}_{\mathcal{A}}\delta_\star/3$. The row Cauchy estimate, the two endpoint cells and the quadratic factor $1/2$ give the last term of C_{ker}, since the polymer sum at this rate is at most C_E. Adding the three estimates proves the assertion. □

13.7. **The terms of second order.** For an operator $X(U)$ of the fermion tower, or for $X(U) = D[U] + m_0$, write $\mathbb{E}_i\langle \delta^2 \log\det X\rangle$ for the density per level-$(k+1)$ site of the quadratic term of its logarithmic determinant, averaged over the slice A_i at coefficient x_i'. This notation also applies to $X = D + m_0$. The densities are justified below by their absolutely convergent kernels.

Lemma 13.22 (The scalar coefficients)**.** In the setting of Lem. 13.19, put $c_k^F = \sum_{j\le k} L^{-(k-j)} b_j^F$. Then

$$\begin{aligned}
\hat e_{2e_1} &= \mathbb{E}_k\langle \delta^2 \log\det \mathcal{K}_{k+1}\rangle - 3(L^3-1)c_k^F/x_k' + \mathfrak{t}_k, && |\mathfrak{t}_k| \le C_{\chi 0}\phi(\mathfrak{t}_k)^4,\\
\hat e_{e_2} + \hat e_{e_4} &= (\mathcal{J}_k[\mathscr{T}_k])_0 + \mathfrak{r}_k, && |\mathfrak{r}_k| \le C_{\mathrm{rem}}\phi_\sharp(\mathfrak{t}_k)^3,\\
\sum_{|n|_w\ge 3} |\hat e_n| &\le C_{\mathrm{w}3}\phi(\mathfrak{t}_k)^3.
\end{aligned}$$

The constants may be chosen as

$$\begin{aligned}
C_{\mathrm{w}3} &= 2K_M c_{\mathrm{cl}} C_{\mathrm{pd}}^3 \sum_{w\ge 3}(w+1)^5 2^{3-w},\\
C_{\chi 0} &= 6K_M c_{\mathrm{cl}} C_{\mathrm{pd}}^2 + 24 L^3 C_{\mathrm{ker}} C_E K_{\mathrm{cp}},\\
C_{\mathrm{rem}} &= 2K_M c_\psi C_7 C_3 S_\omega + 12(4/3)^{3/2} K_M c_\psi' c_{\mathsf{S}} h_F^2 C_\rho S_3^\sharp.
\end{aligned}$$

■

Proof. For $2e_1$ only $\mathcal{V}^{\mathrm{N}}(z_1 B)$ and $\mathcal{V}_k^{\mathrm{F}}(z_1 B)$ remain. These vertices depend on the fine masses and the calibration, but not on the running coefficients. Let $\hat e_{2e_1}(x)$ denote this coefficient with Gaussian coefficient x. Set

$$\vartheta(x) = \mathbb{E}_k^x\langle \delta^2 \log\det \mathcal{K}_{k+1}\rangle - 3(L^3-1)c_k^F/x = \vartheta_1/x.$$

We first compare these quantities for real $|x - x_k| \le r_x x_k$, where $x \ge 3\mathfrak{t}_k/4$.

Take tori whose sides N, in level-$(k+1)$ units, tend to infinity. Terms on polymers which do not wrap around the torus agree with the infinite-lattice terms. The exponential polymer norm bounds the contribution of all other polymers, per site, by

$$4M^{-3}K_M c_{\mathrm{cl}}(r_1')^{-2}e^{-\kappa(N/(2M)-2)}.$$

To see the count, file each infinite-lattice polymer at one of its cells. Each nonwrapping torus polymer has exactly one copy filed in a fixed fundamental domain. A wrapping polymer, or one too large to fit, has size at least $N/(2M)-2$. Summing the norm at the N^3/M^3 cells gives the displayed bound after division by N^3.

On each torus the exact fermion integral and the determinant identities of the tower telescope the normalization terms. The scalar quotient whose logarithm is being expanded is

$$\nu_{\mathbb{T}}(e^{f_{\mathbb{T}}(z_1 B)}), \qquad f_{\mathbb{T}}(B)=\log\frac{\det \mathcal{K}_{k+1}[\mathsf{U}'(B)]}{\det \mathcal{K}_{k+1}[\mathbf{1}]}-\sum_{j\le k} b_j^F L^j A(\mathsf{U}'(B)),$$

where $\nu_{\mathbb{T}}$ is the normalized gauge Gaussian with the tests. The first variation vanishes by constant colour invariance. Thus its quadratic coefficient is the mean of the quadratic form $Q_{\mathbb{T}}(A_k)$ obtained from $f_{\mathbb{T}}$. The second derivative of the chart contributes through the vanished first variation and hence gives zero.

Without the tests the curvature covariance is $L^{-k}x^{-1}\mathcal{C}_{(k+1)}\otimes 1$. The kernels of the logarithmic determinant decay exponentially; those of the action have finite range. Periodization and dominated convergence therefore give the infinite-lattice mean $\vartheta(x)$, as detailed in Lem. 13.23. Here $\tau_{k+1}(\mathcal{C}_{(k+1)})=2(L^3-1)$, which accounts for the action coefficient. The comparison of tested and untested second moments costs at most $C_E K_{\mathrm{cp}} x^{-1}e^{-\gamma_{\mathrm{tail}}\mathfrak{b}(\mathsf{t}_k)^2/2}$ per matrix entry. The row bound of Lem. 13.21 and the at most $9L^3$ bond-colour coordinates per next-level site give

$$|\hat{e}_{2e_1}(x)-\vartheta(x)|\le 12L^3 C_{\mathrm{ker}}C_E K_{\mathrm{cp}}\phi(\mathsf{t}_k)^2 e^{-\gamma_{\mathrm{tail}}\mathfrak{b}(\mathsf{t}_k)^2/2}.$$

The difference is holomorphic on $|x-x_k|<r_x x_k$. The coefficient bound and the real estimate at x_k bound it on this disc by $C_{\chi 0}\phi(\mathsf{t}_k)^2$. On the real diameter it has the preceding exponential gain. The two-constants estimate used in the second-order expansion of [QCDM1, §8], with harmonic-measure lower bound c_{hm} on the two-thirds disc, gives

$$|\hat{e}_{2e_1}(x)-\vartheta(x)|\le C_{\chi 0}\phi(\mathsf{t}_k)^2 e^{-c_{\mathrm{hm}}\gamma_{\mathrm{tail}}\mathfrak{b}(\mathsf{t}_k)^2/2}\le C_{\chi 0}\phi(\mathsf{t}_k)^4.$$

The last inequality is the imposed Gaussian-tail ceiling. It applies at $x=x_k'$.

For e_2, the gauge fluctuation is absent. The identity of scalar totals in Lem. 13.20 gives $\hat{e}_{e_2}=(\mathcal{J}_k[\sum_{j\le k}E_{\ge 2}^{(j)}])_0$. For e_4 it gives $\hat{e}_{e_4}=-L^{-k}s_k'\sigma_k$, where $\sigma_k=(\mathcal{J}_k[\mathsf{S}_1(\mathsf{H}_{0,k}\Psi_k)])_0$ and $|L^{-k}\sigma_k|\le 12K_M c_\psi' c_{\mathsf{S}} h_F^2$. Use the identities of totals

$$E_{\ge 2}^{(j)}=T^{(j)}+R^{(j)}, \qquad \sum_{j\le k}T^{(j)}=\mathscr{T}_k+L^{-k}G_k(x')\mathsf{S}_1(\mathsf{H}_{0,k}\Psi_k).$$

The remainder totals have vanishing quadratic cell sums and representatives of norm at most $2C_3\phi(\mathsf{t}_{j-1})^3$. Their scalar contraction is bounded by the first term of $C_{\mathrm{rem}}\phi_\sharp(\mathsf{t}_k)^3$. The mass remainder is bounded using

$$|s_k'-G_k(x')|\le (4/3)^{3/2}C_\rho S_3^\sharp\phi_\sharp(\mathsf{t}_k)^3.$$

This proves the second estimate.

Finally there are at most $(w+1)^5$ multi-indices of weight w. The coefficient bound of Lem. 13.19 and $C_{\mathrm{pd}}\phi(\mathsf{t}_k)\le 1/2$ sum to the stated C_{w3} bound. □

Lemma 13.23 (Transport through the fermion tower)**.** For an operator X commuting at the trivial background with translations of $L^m\mathbb{Z}^3$, $m\ge i+1$, let $\mathfrak{M}_i[X]$ be n_m times the density per level-m

site of the slice-i mean of its quadratic logarithmic determinant. This quantity does not change on passing to a coarser common period. At the relabelled datum,

$$\begin{aligned} n_{k+1}(\mathcal{J}_k[\mathscr{T}_k])_0 &= \sum_{i<k} \mathfrak{M}_i[\Gamma_k], \\ \mathfrak{M}_i[\mathcal{K}_{n+1}] &= \mathfrak{M}_i[\mathcal{K}_n] + \mathfrak{M}_i[\Gamma_n], \qquad & \mathfrak{M}_i[\mathcal{K}_1] = \mathfrak{M}_i[\Gamma_0], \\ \mathfrak{M}_i[D+m_0] &= \mathfrak{M}_i[\mathcal{K}_n] + \mathfrak{M}_i[\mathbb{D}_n], \\ \mathfrak{M}_i[D+m_0] &= -\tfrac{3}{2}(L^i x_i)^{-1} n_{i+1}\tau_{i+1}(\mathcal{C}_{(i+1)}\widehat{W}_{m_0}). \end{aligned}$$

The identities hold whenever their displayed levels exist in the tower. Moreover

$$\left|\sum_{i<n} \mathfrak{M}_i[\mathbb{D}_n]\right| \le n_n C'_M C_{\mathscr{T}}/\mathtt{t}_n,$$

where

$$\begin{aligned} C_M &= \nu_0^2 C_{\mathrm{el}} c_z + (b_F(1-L^{-1}))^{-1}, \qquad & C'_M &= 2(C_M + 2C_M^2), \\ C_{\mathscr{T}} &= C_7 C_T h_F^4 + 4c_{\mathsf{S}} h_F^2 B_F, & c_z &= \sup_{s\ge 1} s^{-1} \sum_{|z|_\infty \le s} (1+|z|)^{-2}. \end{aligned}$$

Here $\nu_0 = 48$ is the number of chamber places per site. The last sum is the full contraction of $\mathscr{T}_n$ with $\mathbb{D}_n(\mathbf{1})^{-1}$, multiplied by n_n. ∎

Proof. For a slice $i < k$, the linear block average of level k is zero. Its second-order displacement is a colour vector quadratic in the slice and has zero mean. Consequently the slice derivatives of $\Gamma_k = (\mathbb{D}_k)_{AA} + \theta q_{k+1}^+ q_{k+1}$ in the Wick contraction agree with those of $(\mathbb{D}_k)_{AA}$. The first trace vanishes by its colour trace. The two surviving contractions in Lem. 13.20 are therefore the quadratic logarithmic determinant of Γ_k. This proves the first identity.

On every compatible torus the Schur-complement identities give, up to constants independent of the field,

$$\log\det \mathcal{K}_{n+1} = \log\det \mathcal{K}_n + \log\det \Gamma_n, \qquad \log\det(D+m_0) = \log\det \mathcal{K}_n + \log\det \mathbb{D}_n,$$

and $\Gamma_0 = \mathcal{K}_1$. We justify the passage of their quadratic terms to infinite-lattice densities. At the fixed positive bare mass, the free inverse G_{m_0}, the tower inverses and their first two field derivatives have exponentially decaying kernels. For the remaining coarse covariance, the exact identity is

$$\mathbb{D}_n(\mathbf{1})^{-1} = T_n G_{m_0} T_n^* + \theta_n^{-1}.$$

It holds as a bounded-operator identity on ℓ^2: multiply it on either side by $\mathbb{D}_n = \theta_n - \theta_n^2 T_n \mathcal{K}_n^{-1} T_n^*$, using $\mathcal{K}_n = D + m_0 + \theta_n T_n^* T_n$. The products reduce to the identity. Since T_n is block-local, this also proves exponential decay of this inverse at the fixed mass.

For any of these operators, the coefficient kernel of

$$Q_X(A) = \mathrm{Tr}(X^{-1}\delta_A^2 X) - \tfrac{1}{2}\,\mathrm{Tr}((X^{-1}\delta_A X)^2)$$

has the form $q_X(b,a;b',a') = \sum_v y(v;b,a;b',a')$, with $|y| \le Ce^{-c(|x_v - x_b| + |x_v - x_{b'}|)}$. It is covariant under the common period. If Λ_m contains the bonds starting in one level-m block, the slice mean per block is the absolutely convergent sum

$$d_\infty[X] = L^{-i} x_i^{-1} \sum_{b\in\Lambda_m} \sum_{b',a} q_X(b,a;b',a) C_\infty(b,b').$$

The torus kernels are signed periodizations of the same kernels. Terms winding around a torus of side H have total coefficient sum at most $CH^3 e^{-cH/2}$. The nonwinding terms give the displayed sum with the periodic covariance $C_{\mathbb{T}}$. These covariances are uniformly bounded and converge pointwise to C_∞. Dominated convergence proves convergence of their densities. The determinant identities therefore pass to $\mathfrak{M}_i = n_m d_\infty$.

The free determinant has quadratic term $-\frac{1}{2}\sum_c \langle dA^c, \widehat{W}_{m_0} dA^c\rangle$. The slice curvature covariance is $L^{-i}x_i^{-1}\mathcal{C}_{(i+1)}$ in each of the three colours. This gives the fourth identity and its sign.

For the final bound, the entries of T_n have norm at most L^{-2n} and there are $\nu_0 L^{3n}$ fine places in a block. The bound $|G_{m_0}(v,v')| \le C_{\mathrm{el}}(1+|x_v - x_{v'}|)^{-2}$ gives, uniformly in $m_0 > 0$,

$$|(T_n G_{m_0} T_n^*)(v,v')| \le L^{-4n}(\nu_0 L^{3n})(\nu_0 C_{\mathrm{el}} c_z L^n) = \nu_0^2 C_{\mathrm{el}} c_z.$$

Thus the coarse covariance entries are bounded by C_M. A full contraction of a product of unit fibre vectors is bounded by C_M in degree two and by $2C_M^2$ in degree four. It vanishes unless the barred and unbarred degrees agree.

File representatives of $T^{(j)}$ in the level-n cells and read them at level n through the background maps. The contribution to one cell has norm at most $C_7 L^{(n-j)/4} C_T h_F^4/\mathsf{t}_{j-1}$. The mass term in the identity of $\mathscr{T}_n$ contributes at most $4c_{\mathsf{S}} h_F^2 B_F/\mathsf{t}_n$. Since $\sum_{j\le n} L^{(n-j)/4}/\mathsf{t}_{j-1} \le \mathsf{t}_n^{-1}$, their sum in a cell has norm at most $C_{\mathscr{T}}/\mathsf{t}_n$. Its degree-two and degree-four parts sum to $\mathscr{T}_n$. The contraction commutes with cell translations, and the coefficient sums are absolutely convergent. Hence the density is M^{-3} times the contraction in one cell. The preceding covariance bound proves the last inequality. The same Wick calculation as for Γ_k identifies this contraction with $\sum_{i<n} \mathfrak{M}_i[\mathbb{D}_n]$. □

Lemma 13.24 (The free coefficient)**.** The number I_B is finite and positive. There is C_{kp} depending only on the fixed chamber operator such that

$$|\kappa(l,\mu)| \le C_{\mathrm{kp}} N_f, \qquad \kappa(l,\mu) \longrightarrow \tfrac{1}{4} N_f I_B \quad (l\to\infty,\ \mu\to 0).$$

The convergence is joint in these two limits. ■

Proof. We first control the matrices in the definition. The bound $|s(y)| \le \min\{1, 2/|y|\}$ makes the sum over $n \neq 0$ in M_a bounded by $\pi^{-1-a}((7/3)^3 - 1)$. Indeed a nonzero coordinate of n gives a factor $4/(\pi^2 n_\nu^2)$ and $|k_n| \ge \pi$. On the central cube $|s(p_\nu)| \ge 2/\pi$. If $w \perp G(p)$, then

$$p^{\mathsf{T}} \hat{B}(p)^* w = i \prod_\nu \overline{s(p_\nu)} G(p)^* w = 0.$$

Thus the $n = 0$ term alone gives

$$w^* M_1(p) w \ge (2/\pi)^8 |w|^2 |p|^{-2}, \qquad w^* M_2(p) w \ge (2/\pi)^8 |w|^2 |p|^{-3}.$$

It follows that $\|\operatorname{Del}(p)\| \le (\pi/2)^8 |p|^2$ and $\operatorname{tr}(\operatorname{Del} M_1) = 2$. The corresponding trace with M_2 is positive and bounded by

$$(\pi/2)^8 |p|^2 \left(3|p|^{-3} + 3\pi^{-3}(7/3)^3\right).$$

This is integrable in three dimensions and proves $0 < I_B < \infty$.

For completeness, M_1 is invertible off the coordinate axes. Its kernel consists of w for which $\hat{B}(k_n)^* w$ is parallel to k_n for every n. If two coordinates of p are nonzero, varying one alias index changes the squared ratio of these coordinates. The relations $i\overline{G_\mu(p)} w_\mu = \lambda_n k_{n,\mu}^2$ then force all components of w to vanish. If $p = p_1 e_1 \neq 0$, only $n_2 = n_3 = 0$ contribute, and each summand is proportional to $\operatorname{diag}(0,1,1)$. The kernel is then the span of $G(p)$. In particular, off the axes the formula for Del is

$$M_1^{-1} - M_1^{-1} G (G^* M_1^{-1} G)^{-1} G^* M_1^{-1}.$$

We next compute the finite-scale projection. Write $q = L^l$, $k_n = q^{-1}(p + 2\pi n)$ and sum over $n \in \{-(q-1)/2, \dots, (q-1)/2\}^3$. Let $\hat{s}(k) = (e^{ik_\mu} - 1)_\mu$ and $P^{\mathrm{lat}}(k) = 1 - \hat{s}(k)\hat{s}(k)^*/|\hat{s}(k)|^2$. The straight-average symbol is diagonal, with

$$\hat{q}_{l,\mu}(k) = q^{-3} \sum_{x \in B^l(0)} e^{ik\cdot x} \sum_{j=0}^{q-1} e^{ik_\mu j}.$$

The background curvatures are the image of $G(p)^{\perp}$ under

$$J_p^{(l)}\lambda = (|\hat{s}(k_n)|^{-2}\hat{d}(k_n)\hat{q}_l(k_n)^*\lambda)_n.$$

To verify this, their orthogonal complement is $d(\ker \mathbf{Q}_l^s)$. Orthogonality means $d^*da = (\mathbf{Q}_l^s)^*\lambda$. The telescoping identity for a straight average of a gradient gives $\hat{q}_{l,\mu}(k)\hat{s}_\mu(k) = G_\mu(p)\hat{A}_l(k)$. Since $\hat{A}_l(k_0) \neq 0$, the right side is transverse in every alias exactly when $\lambda \perp G(p)$. Solving for the curvature gives $J_p^{(l)}$.

Put

$$M_1^{(l)}(p) = \sum_n \hat{q}_l(k_n)P^{\mathrm{lat}}(k_n)\hat{q}_l(k_n)^*|\hat{s}(k_n)|^{-2}.$$

Then $J_p^{(l)*}J_p^{(l)} = J_p^* M_1^{(l)} J_p$. The $n = 0$ term is positive on $G(p)^{\perp}$, so the background projection is $J_p^{(l)}(J_p^* M_1^{(l)} J_p)^{-1} J_p^{(l)*}$ and has rank two. Denote its diagonal alias blocks by $\Pi_{nn}(p)$. They are positive and their traces sum to two. In particular

$$\tau_l(\Pi_l^{\mathrm{tp}}) = 2, \qquad \tau_{j+1}(\mathcal{C}_{(j+1)}) = 2(L^3 - 1),$$

and, for any translation-invariant curvature operator Y,

$$\tau_l(\Pi_l^{\mathrm{tp}}Y) = \int \sum_n \mathrm{tr}(\Pi_{nn}(p)Y(k_n))\frac{d^3p}{(2\pi)^3}.$$

The geometric sums in $\hat{q}_l$ give, for each fixed alias n as $l \to \infty$,

$$L^{-l}\hat{q}_l(k_n) \longrightarrow \omega_n \hat{B}(p + 2\pi n), \qquad L^{2l}|\hat{s}(k_n)|^2 \longrightarrow |p + 2\pi n|^2,$$

where ω_n is a common phase. They also give

$$|L^{-l}\hat{q}_{l,\mu}(k_n)| \le \min\{1, \pi/|p_\mu + 2\pi n_\mu|\}\prod_\nu \min\{1, \pi/|p_\nu + 2\pi n_\nu|\}, \quad L^{2l}|\hat{s}(k_n)|^2 \ge (4/\pi^2)|p + 2\pi n|^2.$$

These bounds follow from $|\sin(\theta/2)| \ge |\theta|/\pi$ on $[-\pi, \pi]$.

The polarization calculation in Lem. 14.29 gives on exact curvatures

$$\|\widehat{W}_{m_0}(k)\| \le C_{\widehat{W}} N_f |k|^{-1}, \qquad \|\widehat{W}_{m_0}(k) - \Upsilon(k; m_0)\| \le C_{\widehat{W}} N_f (1 + |\log|k||),$$

uniformly for $0 \le m_0 \le 1$. Here

$$\Upsilon(k; m) = \frac{N_f}{2\pi}\int_{2m}^{\infty} \frac{1 + 4m^2/u^2}{u^2 + |k|^2}\, du.$$

Indeed $\hat{\Pi} = \hat{d}^*\widehat{W}^{\mathrm{f}}\hat{d}$, and $\hat{d}$ has lower singular value $(2/\pi)|k|$ on the transverse space. The remainder bound for $\hat{\Pi}$ in Lem. 14.29, divided by this square, gives the second estimate near zero. The difference between $\hat{d}^*\hat{d}$ and $|k|^2 - kk^{\mathsf{T}}$ is $O(|k|^3)$ and gives the same bound. Away from zero compactness gives a uniform bound. Changing bond or plaquette placements conjugates by diagonal phases and does not change the norm of the difference from a scalar. The integral formula shows homogeneity $\Upsilon(\epsilon k; \epsilon m) = \epsilon^{-1}\Upsilon(k; m)$. Differentiation in m gives $\partial_m \Upsilon \le 0$; the boundary term is $-4/(4m^2 + |k|^2)$ and the integral term is at most $8m(4m^2 + |k|^2)^{-1}\int_{2m}^{\infty} u^{-2}\, du$. Thus $0 \le \Upsilon(k; m) \le N_f/(4|k|)$. The first operator bound follows from the second.

Since $|k_n| \ge L^{-l}\min\{|p|, \pi\}$ and $\sum_n \mathrm{tr}\,\Pi_{nn} = 2$, the trace formula gives $|\kappa(l, \mu)| \le C_{\widehat{W}} c_{\mathrm{mi}} N_f$, where $c_{\mathrm{mi}} = \int(|p|^{-1} + \pi^{-1})\, d^3p/(2\pi)^3$. For the limit, replace $\widehat{W}_{L^{-l}\mu}(k_n)$ by $L^l\Upsilon(p + 2\pi n; \mu)$. The resulting error in κ is bounded by

$$C_{\widehat{W}} N_f L^{-l}\int(3 + l\log L + |\log|p||)\frac{d^3p}{(2\pi)^3},$$

which tends to zero uniformly in μ. The remaining integrand is

$$\tfrac{1}{2}\,\mathrm{tr}[(J_p^* M_1^{(l)} J_p)^{-1} J_p^* M_\Upsilon^{(l)} J_p],$$

where $M_{\Upsilon}^{(l)}$ is $M_1^{(l)}$ with the factor $\Upsilon(p+2\pi n;\mu)$ in each summand. The alias bounds give, jointly as $l\to\infty$ and $\mu\to 0$,

$$L^{-4l}M_1^{(l)} \longrightarrow M_1, \qquad L^{-4l}M_{\Upsilon}^{(l)} \longrightarrow (N_f/4)M_2.$$

The phases cancel. The compressed inverse converges by its proved positivity. The trace is bounded by $(N_f/2)(|p|^{-1}+\pi^{-1})$. Dominated convergence therefore gives $\kappa(l,\mu)\to(N_f/8)\int \operatorname{tr}(\operatorname{Del} M_2) = N_f I_B/4$. □

13.8. **Summing the change in the vacuum energy.**

Lemma 13.25 (The difference of one step)**.** In the setting of Lem. 13.19, put

$$v_{k+1}(\zeta) = \mathbb{E}_k\langle \delta^2 \log\det \mathcal{K}_{k+1}\rangle + \sum_{i<k}\mathbb{E}_i\langle \delta^2 \log\det \Gamma_k\rangle.$$

Then

$$e_{k+1}(\zeta) - e_{k+1}^{\mathrm{W}}(\zeta) = e_{k+1}^{\mathrm{F},0} + v_{k+1}(\zeta) + \mathrm{Rem}_{k+1}(\zeta), \qquad |\mathrm{Rem}_{k+1}(\zeta)| \le C_{\mathrm{vac}}\phi_\sharp(\mathfrak{t}_k)^3.$$

The remainder is holomorphic on $|\zeta|<\rho'$. One may take

$$\begin{aligned} C_{\mathrm{vac}} &= C_{\mathfrak{g}} + C_{\chi 0} + C_{\mathrm{rem}} + C_{\mathrm{w3}} + C_{\mathrm{eq}},\\ C_{\mathfrak{g}} &= 3(L^3-1)(16(2B_\sharp+6)^2 + 24C_w),\\ C_{\mathrm{eq}} &= 8(2\bar B+9)K_M(K_0^\sharp A_1+4)/r_x. \end{aligned}$$

■

Proof. Subtract the gauge-only value $\hat e(0,0,1,0,0)$ from $\hat e(1,1,1,1,1)$. Among the remaining indices of weight at most two, the cancellations of Lem. 13.19 leave precisely $2e_1, e_2, e_4$. Apply Lems. 13.22 and 13.23 to these coefficients. In addition to their remainders, there remain

$$\mathfrak{g}_k = g_k(x_k') - g_k(x_k^{\mathrm{W}}) - 3(L^3-1)c_k^F/x_k'$$

and the differences of the test and gauge cluster densities at x_k', x_k^{W}.

The two coefficient recursions give

$$x_k^{\mathrm{W}} - x_k' = c_k^F + \Delta r_k, \qquad \Delta r_k = \sum_{j\le k} L^{-(k-j)}(b_j^r - b_j^{r,\mathrm{W}}).$$

The bounds are $|c_k^F|\le 2B_\sharp$ and $|\Delta r_k|\le 6C_w\phi(\mathfrak{t}_k)$. For $y=(c_k^F+\Delta r_k)/x_k'$ we have $|y|\le 1/2$ under the coupling ceilings. Thus

$$|\mathfrak{g}_k| \le 3(L^3-1)(|y|^2+4|\Delta r_k|/\mathfrak{t}_k) \le C_{\mathfrak{g}}\phi_\sharp(\mathfrak{t}_k)^3.$$

Here we used $\mathfrak{t}^{-2}, \phi(\mathfrak{t})/\mathfrak{t} \le \phi_\sharp(\mathfrak{t})^3$. This is the cancellation between the extracted fermion coupling and the change in the Gaussian normalization.

The test and gauge cluster densities are holomorphic in $|x - x_k^{\mathrm{W}}(0)| < r_x x_k^{\mathrm{W}}(0)$ and bounded there by $2K_M\phi(\mathfrak{t}_k)^2$ and $K_M(K_0^\sharp A_1+2)\phi(\mathfrak{t}_k)$, respectively. The segment between x_k' and x_k^{W} has length at most $2\bar B+9$. It stays at distance at least $r_x\mathfrak{t}_k/8$ from the boundary: $k<t'$ gives $\mathfrak{t}_k \ge L\gamma^{-2}$, and the ceiling $24\bar B\gamma^2 \le r_x$, with $L\ge 13$ and $\bar B\ge 2$, gives $\mathfrak{t}_k \ge 8(2\bar B+9)/r_x$. Cauchy's estimate therefore bounds their two differences by $C_{\mathrm{eq}}\phi(\mathfrak{t}_k)/\mathfrak{t}_k$. The remaining three bounds are those of Lem. 13.22. Their sum proves the assertion. □

Lemma 13.26 (The derivative of the vacuum-energy difference)**.** Suppose $d'\ge \max\{\bar n, d_\natural\}$ and $C_{\mathrm{pd}}\phi(L^{d'+1}\mathsf{b}_*)\le 1/2$. Then the sums in Lem. 13.18 satisfy

$$c_{\mathrm{F}}'(0) - c_{\mathrm{W}}'(0) = -3\mathsf{b}_*^{-2}\ell^3 L^{2d}\kappa(K-d, L^{-d}\mathsf{m}) + \mathcal{E}_d,$$

where

$$|\mathcal{E}_d| \le C_{\mathcal{E}}\ell^3\big[(N_f+1)L^{2d'}\mathsf{b}_*^{-2} + N_f L^d \mathsf{b}_*^{-3} + \mathsf{b}_*^{-5/2}L^{3d/2}(1+d\log L+\log \mathsf{b}_*)^{9\mathfrak{p}_0}\big].$$

One may take

$$C_{\mathcal{E}} = \max\{3C_{\mathrm{kp}} + 4C'_M C_{\mathscr{T}}/r_x, 9C_{\mathrm{kp}}C_{\mathrm{dv}}, 8C_{\mathrm{vac}}/r_x\}, \qquad C_{\mathrm{dv}} = 24C_w/r_x + 6\bar{B}.$$

■

Proof. Sum Lem. 13.25 with weights n_{k+1} over $t \le k < t'$. The free fermion normalizations are independent of ζ. Cauchy's estimate gives

$$\sum_{t\le k<t'} n_{k+1}|\operatorname{Rem}'_{k+1}(0)| \le \frac{8C_{\mathrm{vac}}}{r_x}\ell^3\mathsf{b}_*^{-5/2}L^{3d/2}(1+d\log L+\log\mathsf{b}_*)^{9\mathfrak{p}_0}.$$

This follows by setting $u = K-k$, using $n_{k+1} = \ell^3 L^{3u-3}$ and $1/\rho' = 4/(r_x\mathsf{b}_*)$, and summing the resulting geometric progression.

By Lem. 13.23, the quadratic terms satisfy

$$n_{k+1}v_{k+1} = \mathfrak{M}_k[\mathcal{K}_{k+1}] + \sum_{i<k}\mathfrak{M}_i[\Gamma_k].$$

Terms with $i < t$ read only unchanged slice coefficients and are independent of ζ. For the other terms, the determinant identities telescope:

$$\sum_{t\le i<t'}\left(\mathfrak{M}_i[\mathcal{K}_{i+1}] + \sum_{i<k<t'}\mathfrak{M}_i[\Gamma_k]\right) = \sum_{t\le i<t'}(\mathfrak{M}_i[D+m_0] - \mathfrak{M}_i[\mathbb{D}_{t'}]).$$

The coarse-fermion sum is holomorphic and, including the constant terms $i < t$, is bounded by $n_{t'}C'_M C_{\mathscr{T}}/\mathsf{t}_{t'}$. Its derivative consequently has modulus at most $4C'_M C_{\mathscr{T}}\ell^3 L^{2d'}/(r_x\mathsf{b}_*^2)$.

It remains to differentiate the free polarization term. Cauchy's estimate in the coefficient recursion gives

$$\partial_\zeta x'_i(0) = -L^{K-i}(1+\epsilon_i), \qquad |\epsilon_i| \le 12C_w/(r_x\mathsf{t}_i).$$

Also $x_i^{-2} = \mathsf{t}_i^{-2}(1+\eta_i)$ with $|\eta_i| \le 3\bar{B}/\mathsf{t}_i$. Hence

$$\partial_\zeta(1/x'_i)(0) = L^{K-i}\mathsf{t}_i^{-2}(1+\epsilon'_i), \qquad |\epsilon'_i| \le C_{\mathrm{dv}}/\mathsf{t}_i.$$

Since $L^{-i}L^{K-i}\mathsf{t}_i^{-2} = L^{-K}\mathsf{b}_*^{-2}$, the leading part has a coefficient independent of i. Using the projection difference for the curvature slice,

$$\sum_{t\le i<t'} n_{i+1}\tau_{i+1}(\mathcal{C}_{(i+1)}\widehat{W}_{m_0}) = 2n_tL^t\kappa(t,L^tm_0) - 2n_{t'}L^{t'}\kappa(t',L^{t'}m_0).$$

The first term gives the stated leading term. The second is bounded by $3C_{\mathrm{kp}}N_f\ell^3L^{2d'}\mathsf{b}_*^{-2}$. For the errors ϵ'_i, use

$$|n_{i+1}\tau_{i+1}(\mathcal{C}_{(i+1)}\widehat{W}_{m_0})| \le 4C_{\mathrm{kp}}N_f\ell^3L^{3(K-i)}L^i.$$

Their sum is at most $9C_{\mathrm{kp}}C_{\mathrm{dv}}N_f\ell^3L^d\mathsf{b}_*^{-3}$. These estimates give $C_{\mathcal{E}}$. □

Proof of Thm. V. Choose $d_2 \ge \max\{2, d_\natural\}$ so that, for every $d \ge d_2$, the following hold:

$$d' \ge \max\{\bar{n}, d_\natural\}, \qquad C_{\mathrm{pd}}\phi(L^{d'+1}\mathsf{b}_*) \le 1/2,$$
$$(2R_{\mathrm{fc}}/\bar{S})^{1/2} \le \zeta_0 \le \rho'/2, \qquad 2(\rho^{\mathrm{F}}(d')+\rho^{\mathrm{W}}(d'))\mathsf{Q}'L^{4d} \le L^{7d/4}.$$

These are conditions on d, not further restrictions on γ. They hold eventually: the ratio of the left source radius to ζ_0 is bounded by a constant times $(1+d)^{1/2}L^{-5d/16+3/2}$, while the discarded-history exponent decreases quadratically in d or faster. Here $\mathfrak{p}_0 \ge 1$.

The statistic bound $0 \le T_d^K \le \mathsf{Q}' L^{4d}$ and Lem. 13.16 control the discarded histories. For the retained histories, Lem. 13.18 bounds each error in replacing the mean by $c'_\sharp(0)$ by

$$(2R_{\mathrm{fc}}\bar{S})^{1/2} \le C_{\mathrm{conv}} \ell^3 \mathsf{b}_*^{-1} (1+d)^{1/2} L^{27d/16},$$

where C_{conv} may depend on $1 + |\log \mathsf{b}_*| + \log_+(1/\mathsf{m})$. Apply Lem. 13.26. Since $27/16 < 7/4$, $3/2 < 7/4$, and $2d' \le d/4 + 2$, all the errors, including their polynomial factors in d, are at most $\mathsf{K}_{\mathrm{S}} L^{7d/4}$ for a fixed constant. This proves the estimate uniformly for $K > d \ge d_2$.

The bounds and joint limit for κ were proved in Lem. 13.24. After division by L^{2d}, the error just proved tends to zero. If $K \ge K_0(d)$ and $K_0(d) - d \to \infty$, then $K - d \to \infty$ uniformly over this range and $L^{-d}\mathsf{m} \to 0$. The joint limit of κ therefore gives the asserted uniform limit of the means. □

Remark 13.27 (Dependence of the constants). The free-kernel constants are fixed with the chamber operator, L and b_F. The bounds of the step, the two scalar histories and their variations are formed after h_F and before the mass interval. The history mark and its thresholds precede γ, whose additional ceilings are listed in Def. 10.1. The constants C_{pd} and C_{ker} are formed after γ; they impose only an eventual depth requirement through d_2. The relabelling, discarding constants, d_2 and K_{S} may depend on the given positive physical mass. Thus the proof does not assert uniformity of these latter constants as that mass tends to zero.

Remark 13.28 (A finite set of flavour numbers). For a fixed finite set $F \subset \{1, 2, \dots\}$, choose each constant free upward to meet its floors for every $N_f \in F$, and each constant free downward to meet its ceilings for every $N_f \in F$, in the original order. All conditions read earlier constants alone, so this gives a common admissible assignment. Its gauge constants, label interval, cuts and statistic T_d^K are the same for every $N_f \in F$. The mean comparison holds for each of them, with limit proportional to its own N_f and with its own eventual depth and error constant.

13.9. **Recovering the coupling and the flavour number.** The same sequence of gauge-field observables distinguishes the coupling and the number of fermion flavours. First invert the pure-gauge mean of T_d^K. Its slope has order L^{3d}, so the fermion correction of order L^{2d} still permits recovery of the coupling. Once this label is known, the normalized correction recovers N_f.

Definition 13.29 (A common parameter range)**.** Fix a nonempty finite set $\mathcal{N}_f \subset \{1, 2, \dots\}$. Choose the constants to satisfy both Defs. 10.1 and 11.1 for each $N_f \in \mathcal{N}_f$, retaining all the positive-mass conditions of the original construction. At each stage take a constant free upward above all its finitely many floors, or a constant free downward below all its finitely many ceilings. Determined constants are evaluated at each N_f. Use common gauge constants and mass interval $[m_-, m_+]$.

Fix $0 < g \le \gamma$, $\ell = 2L^m \in \mathfrak{L}(g)$ and

$$J_g = [b_-, b_+] = [g^{-2} + \bar{B}, X_g - \bar{B}], \qquad X_g = L(g^{-2} + B_\sharp), \qquad \mathsf{W}(g) = J_g \times [m_-, m_+].$$

Form T_d^K and $d_\natural$ from Def. 13.1 at this interval. We write $\mu_d^{F,K}[N_f; b, \mathsf{m}]$, $\mu_d^F[N_f; b, \mathsf{m}]$ and $\mu_{\mathrm{thr}}^F[N_f; b, \mathsf{m}]$ for the block and thread laws, making the flavour number explicit. For $N_f = 0$ these denote the corresponding Wilson laws, which are independent of m. ■

Lemma 13.30 (Compatibility of the estimates)**.** The choices in Def. 13.29 exist. They give one statistic T_d^K for all the indicated laws, to which the pure-gauge estimate Prop. 13.2, the fermion variance bound Prop. 13.12, the mean comparison Thm. V, and the total-variation convergence Thm. U and Prop. 13.8 all apply. ■

Proof. Every additional floor or ceiling in Def. 11.1 reads constants already chosen at its position in the order. The same holds for the four extra coupling ceilings and the larger $B_\sharp$ in Def. 10.1. The second history mark, with dials $(1/2, 0, 0, c_\star)$ and share $1/2$, is fixed before γ. It does not replace the mark used in the cutoff comparison. In particular that mark's rate θ_{tl} and its associated ceiling

remain unchanged. Both sets of thresholds can be imposed on γ. Finitely many flavour values therefore give a consistent choice in the original order, as in Rem. 13.28.

The gauge constants, $B_\sharp$, $\bar B$, windows, admissible sides, mass interval and cuts are common. Moreover $b_- \ge \gamma^{-2}$ and

$$b_+ - b_- = (L-1)g^{-2} + (L-3)\bar B > 0.$$

The side condition defining $\mathfrak{L}(g)$ gives $MR(b) \le L^m$ for every $b \le L(X_g + \bar B)$; see Cor. A.25. Hence J_g is an admissible interval for Def. 13.1. For any $b \in J_g$, the variance proof may use $b_\flat = b_-$ and $n = d_\natural$: all its conditions, $b_\flat \le b$, $MR(b_\flat) \le L^m$ and $L^n b_\flat \ge \mathfrak{t}_\natural$, hold. Its cut is precisely

$$\{V : \varpi(V) < \bar\varepsilon, L^{2(K-d)}\varpi(U_{K-d}(V)) < c_\rho \varepsilon(L^d b_-)\}.$$

On this cut both definitions give $L^K A(U_{K-d}(V))$, and off it both give zero. The background action is constant on its gauge orbit. Thus the statistics in all four estimates agree as Borel functions, not only almost everywhere for a particular law. The remaining hypotheses are the constant choices just checked. □

Lemma 13.31 (The slope of the pure-gauge mean)**.** There is $d_{\mathrm W} \ge d_\natural$, depending only on the common constants, g and ℓ, such that for $K > d \ge d_{\mathrm W}$ the function

$$m_d^K(s) = \mathbb{E}_{K,s} T_d^K(V_d), \qquad s > 0,$$

is continuously differentiable and satisfies on J_g

$$2.65 s^{-2} N_d \le -(m_d^K)'(s) \le 3.35 s^{-2} N_d, \qquad \mathrm{Var}_{\mu_d^{\mathrm W,K}[s]} T_d^K \le 3.62 s^{-2} N_d.$$

Its inverse on $[m_d^K(b_+), m_d^K(b_-)]$ has Lipschitz constant $b_+^2/(2.65 N_d)$. ■

Proof. The bare action and T_d^K are bounded at fixed cutoff, and T_d^K is independent of s. Differentiating $e^{-sO}/\int e^{-sO} dU$ gives $-(m_d^K)'(s) = \mathrm{Cov}_{K,s}(T_d^K, O)$. Put $X = \mathbb{E}_{K,s}(O - \mathbb{E}_{K,s} O \mid V_d)$. The tower property gives

$$-(m_d^K)'(s) = \mathbb{E}_{K,s}[(T_d^K - c_s)X]$$

for the constant c_s in Prop. 13.2. Choose $d_{\mathrm W}$ large enough that its two errors, after multiplication by b_+^2, are at most $1/10$ and $1/50$. Then

$$|\mathbb{E}X^2 - 3 s^{-2} N_d| \le 0.1 s^{-2} N_d, \qquad \mathbb{E}|X - (T_d^K - c_s)|^2 \le 0.02 s^{-2} N_d.$$

Cauchy–Schwarz bounds the covariance error by $\sqrt{3.1/50}\, s^{-2} N_d < 0.25 s^{-2} N_d$. This gives the two slope bounds. Similarly

$$\mathrm{Var}\, T_d^K \le \mathbb{E}(T_d^K - c_s)^2 \le (\sqrt{3.1} + \sqrt{1/50})^2 s^{-2} N_d < 3.62 s^{-2} N_d.$$

The slope is bounded away from zero by $2.65 b_+^{-2} N_d$, proving the inverse bound. □

Definition 13.32 (The estimators)**.** For each $d \ge 0$, choose the least $K_d^\sharp > d$ such that the total-variation tail bounds of Thm. U for every $N_f \in \mathcal{N}_f$ and of Prop. 13.8 are at most L^{-d} at depth d. Require also their cutoff thresholds, $K_d^\sharp \ge L^{4d}$ and $\lceil K_d^\sharp/1260 \rceil \ge d$. All these requirements persist at larger cutoffs, so $K_d^\sharp$ exists and $K_d^\sharp \ge 2d$.

For $d \ge d_{\mathrm W}$ put $T_d^\sharp = T_d^{K_d^\sharp}$ and $m_d^\sharp = m_d^{K_d^\sharp}$. Let π_d be the nearest-point projection onto the real interval $[m_d^\sharp(b_+), m_d^\sharp(b_-)]$ and define

$$\hat b_d = (m_d^\sharp)^{-1}(\pi_d(T_d^\sharp)).$$

For the finitely many smaller depths put $\hat b_d = b_-$. On $\mathcal{V}_{\mathrm{thr}}$ define

$$\hat b(V) = \lim_{d\to\infty} \hat b_d(V_d), \qquad \Delta_d(V) = L^{-2d}(T_d^\sharp(V_d) - m_d^\sharp(\hat b(V))), \qquad \hat N(V) = -\frac{4\hat b(V)^2}{3\ell^3 I_B} \lim_{d\to\infty} \Delta_d(V).$$

Where the first limit fails, put $\hat{b} = b_-$; where the second fails to exist as a finite real number, use zero for that limit. Put $\Delta_d = 0$ for $d < d_{\mathrm{W}}$.

These are Borel functions: $T_d^\sharp$ is Borel, $m_d^\sharp$ has a continuous inverse, and the convergence sets of sequences of real Borel functions are Borel. They depend on the common constants, g, ℓ and $\mathcal{N}_f$, but not on the unknown labels b, m or the particular N_f. ■

The definition of Δ_d uses the limiting estimate $\hat{b}$, rather than the depth-d estimate. Substituting $\hat{b}_d$ would remove the same mean shift which is used to recover the flavour number.

Lemma 13.33 (The pure-gauge thread limit)**.** The maps thr_K which retain all block fields of depths $d \le K$, and use fixed configurations after depth K, push Wilson's measures to laws converging weakly to $\mu_{\mathrm{thr}}^{\mathrm{W}}[b]$ along the full sequence $K \to \infty$. ■

Proof. The coordinates through depth D are the image of $\mu_D^{\mathrm{W},K}[b]$ under $\iota_D(V) = (\mathcal{M}^{D-d}V)_{d\le D}$. For a continuous function Ψ_D of these finitely many coordinates, the difference from its limiting integral is at most $2\|\Psi_D\|_\infty \tau_{J_g}(K)$ by Prop. 13.8, since $\Psi_D \circ \iota_D$ is bounded and Borel. The limit is the corresponding integral for Cor. 13.9. Continuous cylinder functions form an algebra containing constants and separating points of the compact product $\mathcal{V}_{\mathrm{thr}}$. Stone–Weierstrass and mass one extend convergence to every continuous function. □

Theorem W (Parameters as functions of the block fields)**.** In the common setting of Def. 13.29, the Borel functions of Def. 13.32 recover

$$(\hat{b}, \hat{N}) = (b, N_f) \quad \mu_{\mathrm{thr}}^F[N_f; b, \mathsf{m}]\text{-almost surely}$$

for every $N_f \in \mathcal{N}_f \cup \{0\}$ and $(b, \mathsf{m}) \in \mathsf{W}(g)$.

More precisely, for each positive N_f there are $d_{\mathrm{F}}, c_{\mathrm{F}}$ such that almost surely $|\hat{b}_d(V_d) - b| \le c_{\mathrm{F}} L^{-d}$ for all $d \ge d_0(V)$. Under the pure-gauge law, for every $\epsilon > 0$ this bound can be replaced by $L^{-(3/2-\epsilon)d}$. In both cases

$$\Delta_d \longrightarrow -\tfrac{3}{4} b^{-2} \ell^3 N_f I_B \quad \text{almost surely.}$$

Consequently the laws at distinct pairs (b, N_f) are mutually singular.

At a fixed b and positive N_f, the laws at two masses $m_- \le \mathsf{m} < \mathsf{m}' \le m_+$ are mutually absolutely continuous. Writing these laws as P, Q, for every Borel set E,

$$Q(E) \le (1 + M_g)^3 P(E)^{2^{-n_{\mathrm{ch}}}}, \qquad P(E) \le (1 + M_g)^3 Q(E)^{2^{-n_{\mathrm{ch}}}},$$

with the constants of Thm. E. ■□

Proof. At $K = K_d^\sharp$ each block law is within L^{-d} in total variation of its limit. This transfers probabilities of events; it avoids multiplying a total-variation error by the growing supremum of $T_d^\sharp$.

The inverse bound and the contraction property of π_d give, for every block field,

$$|\hat{b}_d - b| \le \frac{b_+^2}{2.65 N_d} |T_d^\sharp - m_d^\sharp(b)|.$$

For positive N_f, take d large enough for Prop. 13.12 and Cor. 13.14. The variance is at most $C_v N_d$ and the difference between the fermion mean and $m_d^\sharp(b)$ is at most $\kappa_+ L^{2d}$. Chebyshev's inequality therefore bounds the event where the difference from the fermion mean exceeds $\kappa_+ L^{2d}$ by $C_v \ell^3 \kappa_+^{-2} L^{-d}$. Off this event the inverse inequality gives $|\hat{b}_d - b| < c_{\mathrm{F}} L^{-d}$ with $c_{\mathrm{F}} = b_+^2 \kappa_+ \ell^{-3}$. Hence the limiting block law satisfies

$$\mu_d^F(|\hat{b}_d - b| > c_{\mathrm{F}} L^{-d}) \le (1 + C_v \ell^3 \kappa_+^{-2}) L^{-d}.$$

These probabilities are summable. Since the thread law has these block marginals, Borel–Cantelli gives the asserted rate and $\hat{b} = b$ almost surely.

Without fermions the mean is exactly $m_d^\sharp(b)$ and the variance is at most $3.62b^{-2}N_d$. The same argument gives

$$\mu_d^{\mathrm{W}}(|\hat{b}_d - b| > L^{-(3/2-\epsilon)d}) \le 0.52 b_+^4 b^{-2}\ell^{-3}L^{-2\epsilon d} + L^{-d},$$

because $3.62/2.65^2 < 0.52$. Borel–Cantelli for $\epsilon = 1/j$, $j \ge 1$, gives the statement for every positive ϵ on one event of probability one.

On the event $\hat{b} = b$, in the fermion model write

$$\Delta_d = L^{-2d}(T_d^\sharp - \mathbb{E}_K^{\mathrm{F}} T_d^K) + L^{-2d}(\mathbb{E}_K^{\mathrm{F}} T_d^K - \mathbb{E}_K^{\mathrm{W}} T_d^K).$$

The second term tends to $-3b^{-2}\ell^3 N_f I_B/4$ by Thm. V, since $K_d^\sharp \ge 2d$. For each $j \ge 1$, Chebyshev and total variation bound the probability that the first term exceeds $1/j$ in modulus by $(j^2 C_v \ell^3 + 1)L^{-d}$. Borel–Cantelli, followed by a countable intersection over j, makes this term tend to zero almost surely. For Wilson's law the same proof uses the variance constant $3.62b^{-2}$ and has no mean shift. This proves the assertions about Δ_d and $\hat{N}$.

Each law therefore assigns mass one to the Borel set $\{\hat{b} = b, \hat{N} = N_f\}$. These sets are disjoint for different pairs, proving mutual singularity. For the two-mass assertion, the finite-cutoff inequalities and their passage to weak limits in Prop. 3.13 apply to the Borel maps thr_K into the compact metric product. The full weak convergence of these laws is Thm. U. It identifies the limiting pair with P, Q and gives both displayed inequalities. □

Corollary 13.34 (Other realizations of the block fields)**.** Let Y be Polish, let f_K be Borel maps from the cutoff configurations to Y, and suppose a continuous map $\varphi : Y \to \mathcal{V}_{\mathrm{thr}}$ satisfies $\varphi \circ f_K = \mathrm{thr}_K$. Every weak limit P of the laws pushed forward by f_K at (b, N_f, m) satisfies

$$\varphi_* P = \mu_{\mathrm{thr}}^F[N_f; b, \mathsf{m}].$$

Limits corresponding to distinct (b, N_f) are mutually singular on Y. ■□

Proof. For each bounded continuous h on the thread space, $h \circ \varphi$ is bounded and continuous on Y. Weak convergence and the factorization give convergence to its integral against $\varphi_* P$. The full thread limit is already fixed by Thm. U and Lem. 13.33, so uniqueness of weak limits proves the identity. The inverse images of the disjoint full-measure sets in Thm. W then prove singularity. □

14. The next short-distance coefficient

The leading action-density correlation is Gaussian. Its first correction contains both a pure-gauge term and the fermion polarization. We prove that their sum has coefficient $(8 - 2N_f)/(3\pi)$ and control the remainder uniformly in the volume. The cutoff comparison is needed to turn the estimate in a finite cutoff window into a continuum law.

Definition 14.1 (Test pairs)**.** For $d > 0$ and $0 < \vartheta < 1$, let $\mathfrak{T}_{\mathbb{R}^3}(d; \vartheta)$ consist of the pairs of nonzero nonnegative Lipschitz functions f, h for which there are $x_f, x_h \in \mathbb{R}^3$ such that

$$\begin{gathered}
\operatorname{supp} f \subset \overline{B}(x_f, d), \qquad \operatorname{supp} h \subset \overline{B}(x_h, d), \\
d \le \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) \le 6d, \\
\int f \ge \vartheta d^3 \|f\|_\infty, \qquad \int h \ge \vartheta d^3 \|h\|_\infty, \\
\operatorname{Lip} f \le (\vartheta d)^{-1}\|f\|_\infty, \qquad \operatorname{Lip} h \le (\vartheta d)^{-1}\|h\|_\infty.
\end{gathered}$$

■

14.1. **Separated pairs and the short-distance scale.** The next coefficient is measured by two nonnegative functions whose supports have comparable diameters and separation. Use $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ from Def. 14.1.

Definition 14.2 (The torus class and the witness depth)**.** For $20d \le \ell$, define $\mathfrak{T}_{T_\ell}(d;\vartheta)$ by the same support, separation, integral and Lipschitz conditions as $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, with the distance and integration on T_ℓ. Write $\mathfrak{T}(d;\vartheta)$ when the underlying space is specified. For either space put

$$\mathfrak{f} = \|f\|_\infty \|h\|_\infty, \qquad \mathcal{K}_q(f,h) = \iint f(x)h(y)|x-y|^{-q}\,dx\,dy, \qquad \mathcal{K}_{\text{free}} = \mathcal{K}_6.$$

Here the torus distance is used in the torus integral. The constants in the action-density expansion are

$$c_2 = \frac{9}{16\pi^2}, \qquad \mathfrak{c}_1(N_f) = \frac{8-2N_f}{3\pi}.$$

For the admitted reference coupling g, put

$$X_g = L(g^{-2} + B_\sharp), \qquad \mathsf{b}_- = g^{-2} + \bar{B}, \qquad \mathsf{b}_+ = X_g - \bar{B}, \qquad \bar{x}_s = L^s(X_g + \bar{B}).$$

Retain the witness constant $\mathsf{c}_W \ge 1$ of the lattice construction, chosen so that $C_W(s) := \mathsf{c}_W(\log \bar{x}_s)^{2\mathfrak{p}_0} > 4L^2M$. The witness depth is

$$s(d) = \min\{s \in \mathbb{Z}_{\ge 0} : L^s d \ge C_W(s)\}.$$

The minimum exists since an exponential eventually exceeds the polynomial in s on the right. In particular,

$$K > s(d) \quad \Longrightarrow \quad a_K < \frac{d}{4L^3M} \le \frac{d}{8}.$$

■

For either space, the kernel integral satisfies

$$10^{-q}\vartheta^2 d^{6-q}\mathfrak{f} \le \mathcal{K}_q(f,h) \le (4\pi/3)^2 d^{6-q}\mathfrak{f}. \tag{14.i}$$

Indeed the distance between the supports is at least d, and their diameters are at most $2d$, so every distance in the integral is at most $10d$. The lower bound uses the two integral lower bounds. The upper bound uses the support volumes and their separation. On the torus the balls of radius d have Euclidean volume because $20d \le \ell$. In particular, $\mathcal{K}_5(f,h) > 0$.

These classes contain smooth pairs for every $0 < \vartheta < 1$ and every allowed d. Indeed, [QCDM1, Lem. A.3] constructs a smooth function supported in $B(2de_3, d)$ whose pair with its reflection across $x_3 = 0$ has the required bounds. The pair also defines a torus pair when $20d \le \ell$, since its supports and their distances fit in one coordinate ball.

Throughout this section the model is the $SU(2)$ chamber model with $r = 1/2$ and $N_f \ge 1$. The positive mass window and the two conditions on the source construction are those of section 2. The additional floor on h_F and the ceiling γ_{e} are fixed in Rem. 14.9. All constants below are chosen in that order, before a datum.

14.2. **The next coefficient.** The pure gauge contribution to the next coefficient is $8/(3\pi)$. The fermion polarization subtracts $2N_f/(3\pi)$. We first keep the cutoff in a polynomial window, then use cutoff convergence and a comparison of different tori to remove that restriction.

Proposition 14.3 (The expansion at finite cutoff)**.** Assume the constants of Rem. 14.9, including eq. 14.viii. Let $0 < g \le \gamma_{\mathrm{e}}$, $0 < \vartheta < 1$ and $P > 1$. There are $C_{\mathrm{lat}} \ge 1$,

$$0 < d_{\mathrm{e}} \le \min\{d_1, d_{\mathrm{T}}\}, \qquad d_1 = \min\{(L\mathsf{m}_+)^{-2}, 1/64\},$$

and a nondecreasing function ϵ_1 on $(0, d_{\mathrm{e}}]$ tending to zero at zero, such that

$$\begin{aligned}\big|S_2^T((0,f),(0,h)) - c_2\mathsf{b}^{-2}[\mathcal{K}_{\mathrm{free}} + \mathfrak{c}_1(N_f)\mathsf{b}^{-1}\mathcal{K}_5](f,h)\big| \\ \le \left[\epsilon_1(d)\mathsf{b}^{-3}d + C_{\mathrm{lat}}\mathsf{b}^{-2}\frac{a_K}{d}\right]\mathfrak{f}.\end{aligned} \tag{14.ii}$$

This holds for every datum $(g, \ell, K, \mathsf{b}, \mathsf{m})$, every $0 < d \le d_{\mathrm{e}}$ with

$$K > s(d), \qquad L^K \le d^{-P},$$

and every pair in $\mathfrak{T}_{T_\ell}(d;\vartheta)$. It also holds for $S_2^T[\nu]$, with $\nu \in \mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$ and $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$. Here d_{T} is the threshold of Prop. 14.43. The constants and ϵ_1 depend only on g, ϑ, P and the chosen construction constants. ■

Proposition 14.4 (The expansion on a fixed torus)**.** Assume also the floors and ceilings of Def. 11.1. Fix $0 < g \le \gamma_{\mathrm{e}}$, $\ell \in \mathfrak{L}(g)$ and $0 < \vartheta < 1$. There are $0 < d_2 \le \min\{d_1, \ell/20\}$ and a nondecreasing $\epsilon_2(d) \to 0$ as $d \to 0$, independent of the label in $\mathsf{W}(g)$, such that the following hold for the continuum limit $S^{(\ell,\mathsf{b},\mathsf{m})}$ of Cor. 11.6.

(a) For $0 < d \le d_2$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$,

$$\begin{aligned}\big|S_2((0,f),(0,h)) - c_2\mathsf{b}^{-2}[\mathcal{K}_{\mathrm{free}} + \mathfrak{c}_1(N_f)\mathsf{b}^{-1}\mathcal{K}_5](f,h)\big| \\ \le \epsilon_2(d)\mathsf{b}^{-3}d\,\mathfrak{f}.\end{aligned} \tag{14.iii}$$

(b) Along such pairs with $d \to 0$,

$$\begin{aligned}\mathsf{b}^{-2} &= \lim \frac{S_2((0,f),(0,h))}{c_2\mathcal{K}_{\mathrm{free}}(f,h)}, \\ \mathfrak{c}_1(N_f) &= \lim \frac{\mathsf{b}^3 S_2((0,f),(0,h)) - c_2\mathsf{b}\mathcal{K}_{\mathrm{free}}(f,h)}{c_2\mathcal{K}_5(f,h)}.\end{aligned} \tag{14.iv}$$

Consequently $N_f = 4 - \frac{3\pi}{2}\mathfrak{c}_1(N_f)$ is determined by these two-point functions.

(c) Suppose S' is another such limit on the same torus with the same first label b and with $N_f' \ne N_f$. Its other admitted constants, reference coupling and mass may differ. Let d_2', ϵ_2' be its threshold and error. Then the two-point functions differ at every pair of the class whenever

$$d \le \min\{d_2, d_2'\}, \qquad \epsilon_2(d) + \epsilon_2'(d) < \frac{3\vartheta^2}{8\pi^3}10^{-5}.$$

In particular this holds at all sufficiently small d.

The threshold and error here may depend on ℓ. ■

Theorem X (The continuum coefficient, uniformly in the volume)**.** Assume the hypotheses of Prop. 14.4. Fix $0 < g \le \gamma_{\mathrm{e}}$ and $0 < \vartheta < 1$. Let d_2^G, ϵ_2^G be the threshold and error there on the least torus T_{ℓ_G}, and let $d_{\mathrm{T}}, \epsilon_{\mathrm{T}}$ be those of Prop. 14.43. Set

$$d_4 = \min\{d_2^G, d_{\mathrm{T}}\}, \qquad \epsilon_4(d) = \epsilon_2^G(d) + \epsilon_{\mathrm{T}}(d).$$

These depend only on g, ϑ and the construction constants; ϵ_4 is nondecreasing and tends to zero.

(a) For every $\ell \in \mathfrak{L}(g)$, every label $(\mathsf{b},\mathsf{m}) \in \mathsf{W}(g)$, $0 < d \le d_4$, and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$, eq. 14.iii holds with ϵ_4 in place of ϵ_2. For C^1 pairs it therefore holds for every continuum limit with this label. For all pairs in the class, the lattice two-point functions converge to this value whenever $K_j \to \infty$ and $(\mathsf{b}_j, \mathsf{m}_j) \to (\mathsf{b},\mathsf{m})$ on the fixed torus.

(b) Let S be a continuum limit on $\mathbb{R}^3$ along growing tori, or a volume-first continuum limit, at g, with label $(\mathsf{b},\mathsf{m}) \in \mathsf{W}(g)$. For $0 < d \le d_4$ and $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ of class C^1,

$$\begin{aligned}\big|S_2((0,f),(0,h)) - c_2\mathsf{b}^{-2}[\mathcal{K}_{\mathrm{free}} + \mathfrak{c}_1(N_f)\mathsf{b}^{-1}\mathcal{K}_5](f,h)\big| \\ \le \epsilon_4(d)\mathsf{b}^{-3}d\,\mathfrak{f}.\end{aligned} \tag{14.v}$$

Moreover its difference from

$$S_2^{(\ell_G,\mathsf{b},\mathsf{m})}((0,f^{\langle\ell_G\rangle}),(0,h^{\langle\ell_G\rangle}))$$

has modulus at most $\epsilon_{\mathrm{T}}(d)\mathsf{b}^{-3}d\mathfrak{f}$. The superscript denotes periodization after placing the two supports in one fundamental domain.

(c) The identities eq. 14.iv hold for S in (b) along C^1 test pairs. Thus these two-point functions determine b and N_f. If S' has the same first label and a different number of flavours, the two-point functions differ on every C^1 test pair at every $d \le \min\{d_4, d_4'\}$ satisfying

$$\epsilon_4(d) + \epsilon_4'(d) < \frac{3\vartheta^2}{8\pi^3}10^{-5}.$$

(d) For the limits in (b) the following two comparisons hold.

(i) Choose an admitted pure-theory construction as in Prop. D.70(c), with blocking factor $\bar{L}$, constants $B_\sharp^0, \bar{B}^0$, and ceilings $\gamma_{\mathrm{e}}^0, \gamma_W^{\mathrm{sd}}$. Use one constant assignment satisfying the numerical hypotheses of [YM3, Prop. 4.163], and require the full hypotheses of its history induction [YM3, Prop. 4.165] and [YM3, Prop. 4.166] at every running sequence, radius profile, lattice and step datum used in the pure comparison. In particular, the running sequences are admissible and a torus of side $2\bar{L}^m$ satisfies $M^0R^0(x_K) \le \bar{L}^m$, with the pure construction constants; thus [YM3, Cor. 4.167] applies. Let

$$0 < g_0 \le \bar{L}^{-1/4}\min\{\gamma_{\mathrm{e}}^0, \gamma_W^{\mathrm{sd}}\}.$$

Suppose S^0 is a pure-theory limit along growing tori at g_0, with first-passage labels $\bar{L}^{-K_j}\beta_j \to \mathsf{b}$. There are $d^0 > 0$ and $\epsilon^0(d) \to 0$, depending only on g_0, ϑ and the admitted extended list of pure-theory constants, such that

$$\begin{aligned}|S_2^0(f,h) - c_2\mathsf{b}^{-2}[\mathcal{K}_{\mathrm{free}} + \tfrac{8}{3\pi}\mathsf{b}^{-1}\mathcal{K}_5](f,h)| \\ \le \epsilon^0(d)\mathsf{b}^{-3}d\mathfrak{f}\end{aligned}$$

for C^1 pairs in $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, $d \le d^0$. Hence $S_2((0,f),(0,h)) \ne S_2^0(f,h)$ whenever

$$d \le \min\{d_4, d^0\}, \qquad \epsilon_4(d) + \epsilon^0(d) < \frac{3\vartheta^2}{8\pi^3}10^{-5}.$$

Such pure limits exist if

$$\mathsf{b} \in I(g_0) = (g_0^{-2} + \bar{B}^0,\ \bar{L}(g_0^{-2} + B_\sharp^0) - \bar{B}^0].$$

In particular, if

$$\mathsf{b} \ge 2\bar{B}^0 + \bar{L}^{1/2}\max\{(\gamma_{\mathrm{e}}^0)^{-2}, (\gamma_W^{\mathrm{sd}})^{-2}\},$$

one can take $g_0 = (\mathsf{b} - 2\bar{B}^0)^{-1/2}$.

(ii) For a C^1 pair in $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$, $d \le d_4$, put $f_\lambda = f(\cdot/\lambda)$, $h_\lambda = h(\cdot/\lambda)$, $0 < \lambda \le 1$. Then

$$\begin{aligned}|S_2((0,f_\lambda),(0,h_\lambda)) - c_2\mathsf{b}^{-2}\mathcal{K}_{\mathrm{free}}(f,h) - \lambda c_2\mathfrak{c}_1(N_f)\mathsf{b}^{-3}\mathcal{K}_5(f,h)| \\ \le \epsilon_4(\lambda d)\mathsf{b}^{-3}\lambda d\mathfrak{f}.\end{aligned} \tag{14.vi}$$

Thus the quotient by λ after subtraction of the leading term tends to

$$c_2\mathfrak{c}_1(N_f)\mathsf{b}^{-3}\mathcal{K}_5(f,h).$$

If $N_f \ne 4$, then

$$|S_2((0,f_\lambda),(0,h_\lambda)) - S_2((0,f_{\lambda/2}),(0,h_{\lambda/2}))| \ge \left[\frac{3\vartheta^2}{16\pi^3}10^{-5} - \frac{3}{2}\epsilon_4(\lambda d)\right]\mathsf{b}^{-3}\lambda d\mathfrak{f} > 0$$

whenever $\epsilon_4(\lambda d) < \vartheta^2 10^{-5}/(8\pi^3)$. If $N_f = 4$, eq. 14.vi instead gives an error $o(\lambda)$ after subtraction of the leading term.

■□

The proofs are in sections 14.28 to 14.30, after the source-energy estimates and coefficient calculations.

14.3. **The proof and its estimates.** The proof first replaces the correlation by mixed derivatives of the source energies, with an error smaller than the desired correction (Lem. 14.12). One complex field scale then separates their terms of orders four and six from the higher terms (Prop. 14.26). The order-four term is the free template. The order-six term is a constrained Gaussian coefficient; comparison of the constrained and unconstrained kernels identifies its pure-gauge and fermion contributions (Prop. 14.42). Finally, the comparison of two volumes in Prop. 14.43 and the cutoff convergence in Cor. 11.6 give the continuum result, as proved in section 14.30.

The source-derivative bounds are proved below. The pure-gauge Gaussian calculations used in the coefficient identification are in appendix D.

14.4. **Histories with no large-field region.** For an action source $(w, 0)$, the source enters the bare density. The calibrations, tests, history weights and coordinate maps are those at zero source. This permits an exact summation of all extensions of a history before estimating its effective density.

Lemma 14.5. At a datum with the prescribed calibrations, let $1 \le k \le K$ and let $h_{\emptyset}^{(k)}$ be the sequence of empty outcomes through k. Its regions satisfy

$$\Omega_j = \Lambda_j = T_\ell, \qquad Z_j = S_j^{\mathrm{pr}} = \varnothing, \qquad 1 \le j \le k.$$

Its layers below k are empty and its last layer is T_ℓ. Its generators are exactly those of Gr_k, and both its coordinates and its free coordinates are level-k block fields. On the domain of its coordinate map, $\xi_h^{\mathrm{co}\prime}(U) = \mathcal{M}^k U$; the domain contains $\{\mathfrak{w}_h \neq 0\}$. It has no outer bonds. Its evaluation configurations over a block field V are the adapted elements of the collared background over V.

For $h \in \mathcal{H}_{k'}$, $k' \ge k$, the following are equivalent:

$$Z_k(h) = \varnothing, \qquad Z_j(h) = \varnothing \ (j \le k), \qquad h|_k = h_{\emptyset}^{(k)}.$$

For every complex Lipschitz function w,

$$\sum_{\substack{h \in \mathcal{H}_K \\ Z_k(h) = \varnothing}} \mathrm{Ber}\, \check{\tau}_h^F[w, 0](\mathsf{X}_h^{\mathrm{fr}}) = \mathrm{Ber}\, \check{\tau}_{h_{\emptyset}^{(k)}}^F[w, 0](\mathcal{V}_k).$$

Every term here is entire in the finitely many values of w at bare plaquette barycentres. ■

Proof. Empty sets are admitted outcomes. Their hulls and enlargements are empty, so starting from the empty input region gives $\Omega_j^c = Z_j = \varnothing$ and hence $\Lambda_j = T_\ell$ at every successive step. The prescribed regions lie in $\Omega_j \setminus \Lambda_j$, so are empty. It follows that every earlier layer is empty and all remaining fermion places are at level k.

The determining bonds below k have no endpoint in the next full region and hence are absent. The external fluctuation bonds are absent for the same reason. At level k the determining bonds are all the bonds of that lattice. The coordinate product therefore reduces to $\mathcal{V}_k$, and the coordinate map to $\mathcal{M}^k$. The support-domain property of the canonical coordinates gives the stated inclusion. Every bare bond is an inner bond, leaving no prescribed outer value in the definition of the evaluation configurations.

The large-field regions increase with the level. Thus $Z_k = \varnothing$ implies $Z_j = \varnothing$ for $j \le k$. Conversely the defining formula for each Z_j contains the enlarged plaquette sets, fluctuation sets and the three pieces of its recursive outcome. If Z_j is empty, each such set is empty. The recursion then starts and ends with empty collar and rim, so its bond set is empty as well. The outcome is the empty one. This proves the equivalences.

Set

$$\varsigma_w(U) = e^{-A(\beta - a_K^{-1} w, U)} \det(D[U] + m_0).$$

It is continuous, gauge invariant and bounded on the compact bare configuration space. It is entire in the source values, uniformly bounded in U on their bounded sets. The exact Berezin and coordinate identities for a history give

$$\operatorname{Ber} \check{\tau}_h^F[w, 0](\mathsf{X}_h^{\mathrm{fr}}) = \int \mathfrak{w}_h(U) \varsigma_w(U)\, dU.$$

The weight is bounded and source independent, so each term is entire.

The gauge history consistency theorem applies at the calibration thresholds: $\mathtt{t}_j \geq \gamma^{-2} \geq e^{18\mathfrak{p}_0}$, $t \mapsto A_0 t^{-1/2} (\log t)^{\mathfrak{p}_0}$ decreases there, and its value at γ^{-2} satisfies

$$\varepsilon(\gamma^{-2}) = A_0 \gamma (\log \gamma^{-2})^{\mathfrak{p}_0} \leq \min\{\bar{\varepsilon}/2, \varepsilon_\flat\}.$$

These are among the history ceilings checked in the proof of [YM3, Lem. 5.58]. Summing the coordinate measures over all extensions of $h_\emptyset^{(k)}$ therefore gives its block pushforward. This pushforward preserves total mass. The equivalences just proved identify the extensions with $Z_k(h) = \varnothing$, giving the asserted equality. □

14.5. **Source derivatives and their spatial weights.** The estimates below distinguish the scale of a source from the scale at which an activity is read. Write

$$a_l = L^l a_K, \qquad D_i = \{|\lambda| < \bar{\lambda}_i\}, \qquad q_n = \min\{2L^{-n}(1 + n \log L)^{2\mathfrak{p}_0}, 2^{10/9} L^{-8n/9}\}.$$

The radii in Lemma 14.26 satisfy $(\bar{\lambda}_i / \bar{\lambda}_{l-1})^2 \leq q_{i+1-l} < 1/4$. All source derivatives are evaluated at zero source. A direction $\mathtt{g}$ is specified on bare plaquettes; it acts by $w = z\mathtt{g}$. Equivalently, extend these finitely many values to a Lipschitz function on the torus as in Lemma 14.15. For a plaquette p, write δ_p for its coordinate direction. Linearity gives $\partial_{\mathtt{g}} = \sum_p \mathtt{g}(p) \partial_{\delta_p}$.

Definition 14.6 (Weights and source responses)**.** (1) Choose $c' \geq 18$ so that every polymer X has diameter at most $c' M (1 + d_l(X))/2$ in the units of its level, and put

$$\kappa_{\mathrm{s}} = \min\{\kappa/(8c'),\, 1/c',\, 1/4\}, \qquad \delta_l(Y; S) = \frac{\operatorname{dist}(Y, S)}{M a_l}.$$

Here polymers and cells denote their closed sets of points; distances on a torus are torus distances. The diameter assertion follows by joining the centres of its at most $8(1 + d_l(X))$ cubes along a wall-connected chain: their union has diameter at most $(8 d_l(X) + 7 + \sqrt{3}) M \leq 9M(1 + d_l(X))$.

A weight system is a finite family $\mathscr{S} = ((S_s, \kappa_{(s)}))_s$ of nonempty sets and positive rates with $\sum_s \kappa_{(s)} \leq 2\kappa_{\mathrm{s}}$. Define

$$W_l(Y) = \exp\left(\sum_s \kappa_{(s)} \delta_l(Y; S_s) \right), \qquad \mathcal{C}_l(\mathscr{S}) = \sum_c W_l(c)^{-1/2},$$

where c runs over the cells of level l. A single set S means rate κ_{s}; two sets mean that rate on each. The union of two systems multiplies their weights.

For a family F at level l and $D \subset \mathbb{C}$, set

$$\|F\|_{l;D;\mathscr{S}} = \sup_{\square} \sum_{X \supset \square} e^{\kappa d_l(X)} W_l(X) \sup_{\lambda \in D} \sup_P \|F(X; P, \cdot; \lambda)\|_{h^{(l)}}. \tag{14.vii}$$

The last supremum is over the regular complex pairs on X at (α_0, α_1). Thus both suprema inside the sum may use a different pair and a different λ for each polymer. For the scalar part, local on X, also use

$$\|F_0\|^{\mathrm{arr}}_{l;D;\mathscr{S}} = \sup_X e^{(\kappa+1) d_l(X)} W_l(X) \sup_{\lambda \in D} \|F_0(X; \lambda)\|_X.$$

Point $F_0(X)$ equally at the sites of X. This array norm bounds the pointed norm at rate $\kappa+1$, polymer by polymer.

For a step direction $\delta=(\delta^{\mathrm{p}},\delta^{\mathrm{o}})$, put the weight $W_{i+1}(Y)$ on each term at Y in its scalar, fermionic and quadratic-origin norms. Denote the resulting sizes by $\epsilon_{V;\mathscr{S}}$, $\epsilon_{F;\mathscr{S}}$ and $\|\cdot\|_{\star;\mathscr{S}}$. The distances in these weights are in the units of the *output* level $i+1$. For a scalar- or matrix-valued function c on bare sites, set

$$\langle c\rangle_{l;D;S}=\sup_{\square} e^{\kappa_{\mathrm{s}}\delta_l(\square;S)}(M^3L^{3l})^{-1}\sum_{x\in\square}\sup_{\lambda\in D}|c(x;\lambda)|.$$

The matrix norm is the operator norm.

If g is supported in a closed ball of radius $r\ge a_K$, define

$$e_l[\mathsf{g}]=\|\mathsf{g}\|_\infty a_l^{-1}\min\{1,(r/a_l)^3\}.$$

For the separated pair (f,h) at distance scale $\mathsf{d}\ge a_K$, write

$$\mathfrak{f}=\|f\|_\infty\|h\|_\infty,\qquad e_l[f,h]=\mathfrak{f}\,a_l^{-2}\min\{1,(\mathsf{d}/a_l)^{5/2}\},\qquad \mathfrak{l}_l=1+\log^+(a_l/\mathsf{d}).$$

Use $r=\mathsf{d}$ for f,h, and $r=a_K$ for δ_p. The choice $r=\infty$, with S equal to the whole space, imposes no support restriction and gives weight one and $e_l[\mathsf{g}]=\|\mathsf{g}\|_\infty a_l^{-1}$.

(2) Let $\widetilde{E}_i[\mathcal{V};\mathcal{V}^{\mathrm{o}}]$ denote the activity obtained by the step from level i, before removing its vacuum and mass numbers. Its three origins are the restriction of $\mathcal{V}^{\mathrm{o}}$ to the tested real fluctuations, the scalar part $\mathcal{V}_0$, and $\mathcal{V}_{\ge2}$. Include the Gaussian cluster sum of $\mathcal{V}^{\mathrm{o}}$ at threshold ∞; do not include the localised logarithms of the normalisers, which do not depend on the source. Assume $\|\mathcal{V}^{\mathrm{o}}\|_{\star,\lambda_V}\le w_*/2$. The tests and Gaussian covariance are those at $(\mathsf{t}_i,\xi_i(\lambda))$ in Lemma 14.26.

At zero source the base point is $(\mathfrak{V}_i(\lambda),\mathcal{V}^{\mathrm{q}})$. Write $D\widetilde{E}_i$ and $D^2\widetilde{E}_i$ for its first and second derivatives there. A direction has a potential component and a quadratic-origin component, as in Lemma 14.16.

(3) The direct source direction $\mathcal{V}_i^w[\mathsf{g}]$ is the variation of

$$a_K^{-1}\sum_p \mathsf{g}(p)\big[a_p(\mathsf{U}'(\lambda B))-a_p(\mathsf{U})\big].$$

Its terms of fluctuation orders one and at least three belong to the potential; its quadratic term belongs to the quadratic origin and is defined at every real fluctuation. For $i\ge1$, localise the term at p by the shift parameters on the polymers containing $X(p)$, the union of cells required to read that plaquette. At $i=0$, take $X(p)$ to be the union of cubes containing a corner of p; there is no shift system.

The mass direction is

$$\mathcal{V}_i^{\mathrm{s}}[c]=-L^{-i}\left[\mathsf{S}_1(c\,\mathsf{H}_{0,i}(\mathsf{U}')\widetilde{\Psi})-\mathsf{S}_1(c\,\mathsf{H}_{0,i+1}(\mathsf{U})\Psi_{i+1})\right].$$

Write and localise its legs by the mass-variation rule of the step. For an old family F at level $l\le i$, write $\mathcal{V}_i^{\mathrm{old}}[F]$ for its variation at λB. For F_0, apply the scalar bracket localisation to the bare array consisting of F_0, pointed equally at its sites, with all other entries zero.

For $F_{\ge2}$ and $i-l<n_{\mathrm{lag}}$, use the usual fermionic variation. At $i-l\ge n_{\mathrm{lag}}$, first use the long-lag decomposition of [QCDM1, Lem. 7.9], with the cell moment retained as in Lem. 14.13. If $t(\mu)$ is one of its terms when the shift is multiplied by μ, take

$$[t(1)-t(0)](\widetilde{\Psi})+t(0)(\widetilde{\Psi})-t(0)(\mathsf{H}_i(\mathsf{U})\Psi_{i+1}),\qquad \widetilde{\Psi}=\mathsf{H}_i(\mathsf{U})\Psi_{i+1}+W.$$

Then localise each term by the shift parameters. Keep a cell's local mass term $\bar\Phi(z_c)m_c\Phi(z_c)$ whole, including on a torus. File it at the cube containing c, enlarged successively by the walks of both legs and by the cubes of the finite differences. This is the filing estimated in Lemma 14.18.

(4) The first response is defined recursively. Put $\mathfrak{N}_0[\mathsf{g}] = 0$ and

$$\delta_i[\mathsf{g}] = \mathcal{V}_i^{w}[\mathsf{g}] + \mathcal{V}_i^{\mathrm{s}}[\lambda^4 \mathfrak{N}_i[\mathsf{g}]] + \sum_{l=1}^{i} \mathcal{V}_i^{\mathrm{old}}[\mathbf{P}^{(l)}[\mathsf{g}]], \qquad \widetilde{\mathbf{P}}^{(i+1)}[\mathsf{g}] = D\widetilde{E}_i[\delta_i[\mathsf{g}]].$$

Let $\mathfrak{e}_{i+1}[\mathsf{g}](X)$ be its scalar value at the trivial configuration. Read the quadratic kernels at level $i+1$, and split them by the mass-extraction rule. If $\mathsf{k}_{X,p}$ is a piece, define

$$\mathfrak{n}_{i+1}[\mathsf{g}](y) = [\lambda^4] \left(-\mathfrak{s}_{i+1}^{-1} {\sum_{X,p}}' \operatorname{ord}_0[\mathsf{k}_{X,p}](y) \right).$$

The prime keeps the pieces of diameter less than one quarter of the torus side; on $\mathbb{Z}^3$ it keeps all pieces. Set

$$\begin{aligned} \mathbf{P}^{(i+1)}[\mathsf{g}] &= \widetilde{\mathbf{P}}^{(i+1)}[\mathsf{g}] - \mathfrak{e}_{i+1}[\mathsf{g}] + L^{-(i+1)} \mathsf{S}_1(\lambda^4 \mathfrak{n}_{i+1}[\mathsf{g}]\,\cdot), \\ \mathfrak{N}_{i+1}[\mathsf{g}](x) &= L\mathfrak{N}_i[\mathsf{g}](x) + \mathfrak{n}_{i+1}[\mathsf{g}](y_{i+1}(x)). \end{aligned}$$

Here $y_{i+1}(x)$ is the site whose block contains x. The added mass term is filed as the forms of the mass-extraction rule at the coefficients of λ^4.

We specify its units because the long-lag estimates use their cell sums. Let k_X be the quadratic kernel of $\widetilde{\mathbf{P}}^{(l)}$, let $a_{X,p} = \operatorname{ord}_0 \mathsf{k}_{X,p}$ and $a_X = \sum_p a_{X,p}$. The units are

$$\mathsf{k}_X - \operatorname{diag} a_X \quad \text{at the cell of } X, \qquad \bar{\Phi}(y) a_{X,p}(y) \Phi(y) \quad \text{at the cell of } y,$$

and, for every retained piece,

$$P_{z;X,p} = -\mathfrak{s}_l^{-1} L^{-l} \mathsf{S}_1(\lambda^4 a_{X,p}^{[4]}(z) 1_z \psi^{\mathrm{bg}}) \quad \text{at the cell of } z.$$

The symmetrised zeroth moment is $a_X(y) = \frac{1}{2} \sum_{y'} (\mathsf{k}_X(y,y') + \mathsf{k}_X(y',y))$. Consequently the absolute coefficients used in moving one diagonal term sum to one. The units of each polymer have zero cell sum before the moved terms and forms are included; their complete cell sum is

$$m_c = \sum_{z \in c} \left[\sum_{X,p} a_{X,p}(z;\lambda) - \mathfrak{s}_l^{-1} \sum_{y} A_l(y,z) {\sum_{X,p}}' \lambda^4 a_{X,p}^{[4]}(z) \right].$$

Here A_l is the moment map in the mass-extraction rule.

(5) Omitting $\mathcal{V}_i^{\mathrm{s}}[\lambda^4 \mathfrak{N}_i[\mathsf{g}]]$ from this recursion defines the reduced system $\mathbf{P}^{\mathrm{red}}$, $\mathfrak{e}^{\mathrm{red}}$ and $\mathfrak{n}^{\mathrm{red}}$. Put $\mathbf{P}^{\mathrm{rsp}} = \mathbf{P} - \mathbf{P}^{\mathrm{red}}$. All three depend on the same source direction.

(6) The mixed response is

$$\begin{aligned} \widetilde{\mathbf{Q}}^{(i+1)}[f,h] &= D\widetilde{E}_i \left[\sum_{l=1}^{i} \mathcal{V}_i^{\mathrm{old}}[\mathbf{Q}^{(l)}[f,h]] \right] + D^2 \widetilde{E}_i[\delta_i[f], \delta_i[h]], \\ \mathbf{Q}^{(i+1)}[f,h] &= \widetilde{\mathbf{Q}}^{(i+1)}[f,h] - \mathfrak{e}_{i+1}[f,h], \end{aligned}$$

where $\mathfrak{e}_{i+1}[f,h](X)$ is the scalar value at the trivial configuration. Its quadratic units are its polymers, each at its cell: no diagonal terms are moved.

For $\mathbf{P}^{\mathrm{rsp}}$, subtract the units of the full and reduced systems using the same linear kernel splitting. Thus its units are $\mathsf{k}_X^{\mathrm{rsp}} - \operatorname{diag} a_X^{\mathrm{rsp}}$ and the moved terms $\bar{\Phi}(y) a_{X,p}^{\mathrm{rsp}}(y) \Phi(y)$. There are no forms, since $\mathfrak{n}_l = \mathfrak{n}_l^{\mathrm{red}}$ by Lemma 14.20. Its cell sum is therefore

$$m_c^{\mathrm{rsp}} = \sum_{z \in c} \sum_{X,p} a_{X,p}^{\mathrm{rsp}}(z;\lambda).$$

Linearity of this filing gives $\mathcal{V}_i^{\mathrm{old}}[\mathbf{P}^{\mathrm{rsp}}] = \mathcal{V}_i^{\mathrm{old}}[\mathbf{P}] - \mathcal{V}_i^{\mathrm{old}}[\mathbf{P}^{\mathrm{red}}]$ term by term.

The same definitions apply to the infinite-lattice families whose fitting parts give the torus families. A subscript ∞ will distinguish these families when necessary. ■

14.6. **Derivatives joining different cells.** The transfer between torus sizes also uses source derivatives of order at most sixteen. Only derivatives whose sources occupy different cells require an additional estimate.

Definition 14.7 (Sources in the comparison cells)**.** On the least torus T_{ℓ_G} use the board with cell side $\ell_G/2 = L^{m_{\mathfrak{L}}(g)}$ and centre $o \in a_K\mathbb{Z}^3$ for cell 0. Its side in lattice units, $L^{m_{\mathfrak{L}}(g)+K}$, is odd and at least three, so its walls are coordinate lattice mirrors through links. There are eight cells on T_{ℓ_G} and $N_\ell = 8(\ell/\ell_G)^3$ on T_ℓ. Let θ_c reflect cell 0 to c, using one wall reflection in each differing coordinate. It is an involution. The balls

$$\mathbb{B}_* = \overline{B}(o, 1/4), \qquad \mathbb{B}_*^{(c)} = \theta_c \mathbb{B}_*$$

in different cells have distance at least $\ell_G/2 - 1/2 \geq \ell_G/4$.

Let $\mathfrak{J}$ consist of at most sixteen real source directions, at most two in a cell. Those in cell c are members of a pair $(f_c, h_c) \in \mathfrak{T}(\mathsf{d}; \vartheta)$, with $\mathsf{d} \geq a_K$. Each support S_s lies in a closed ball of radius d; supports in different cells have distance at least $\ell_G/4$. Write D_{x} for the least of these distances, so $\mathsf{d} \leq \ell_G/32 < D_{\mathrm{x}}$.

For $\mathsf{J} \subset \mathfrak{J}$, let J^c be its part in cell c. Say that J joins cells if at least two sets J^c are nonempty. Define

$$\partial_{\mathsf{J}} = \prod_{s \in \mathsf{J}} \partial_{z_s} \Big|_{z=0}, \qquad w = \sum_s z_s \mathsf{g}_s,$$

and let $\mathrm{Part}(\mathsf{J})$ be its partitions into nonempty sets. The sizes factor over cells:

$$\widehat{e}_l[\{\mathsf{g}\}] = e_l[\mathsf{g}]\mathfrak{l}_l^{4\mathfrak{p}_0+1}, \quad \widehat{e}_l[\{f_c, h_c\}] = e_l[f_c, h_c], \quad \widehat{e}_l[\emptyset] = 1, \qquad \widehat{e}_l[\mathsf{J}] = \prod_c \widehat{e}_l[\mathsf{J}^c].$$

Put $\kappa_{\mathrm{B}} = \kappa_{\mathrm{s}}/8$ and

$$w_l^{\mathsf{J}}(Y) = \exp\left(\kappa_{\mathrm{B}} \sum_{s \in \mathsf{J}} \delta_l(Y; S_s)\right).$$

This is a weight system since $\#\mathsf{J}\,\kappa_{\mathrm{B}} \leq 2\kappa_{\mathrm{s}}$. Use the norm

$$\|F\|_{l;\mathsf{J}} = \max\{\|F\|_{l;\{1\};\mathscr{S}_{\mathsf{J}}}, \|F_0\|^{\mathrm{arr}}_{l;\{1\};\mathscr{S}_{\mathsf{J}}}\},$$

and also the weaker norm with one pair for the whole sum:

$$\|F\|^{\mathrm{pair}}_{l;\mathsf{J}} = \sup_{Z,P} \sup_{\square \subset Z} \sum_{\square \subset Y \subset Z} e^{\kappa d_l(Y)} w_l^{\mathsf{J}}(Y) \|F(Y; P|_Y, \cdot; 1)\|_{h^{(l)}}.$$

Here Z may be the whole torus, and P is a regular complex pair on Z. Restrictions remain regular pairs, so $\|F\|^{\mathrm{pair}}_{l;\mathsf{J}} \leq \|F\|_{l;\mathsf{J}}$. The corresponding step size is $\epsilon_{\mathsf{J}}(\delta) = \epsilon_{\mathscr{S}_{\mathsf{J}}}(\delta)$ of Lemma 14.16, with the weight at level $i+1$. For one source, or a pair in one cell, this size is bounded by the size with rate κ_{s} on its supports.

At $\lambda = 1$ let $D^r \widetilde{E}_i[\delta_1, \dots, \delta_r]$, $r \leq 16$, be the mixed derivative at zero of

$$\widetilde{E}_i\left[\mathfrak{V}_i(1) + \sum_a t_a \delta_a^{\mathrm{p}}; \mathcal{V}^{\mathrm{q}} + \sum_a t_a \delta_a^{\mathrm{o}}\right].$$

Set $\mathbf{R}^{(l)}[\{\mathsf{g}\}] = \mathbf{P}^{(l)}[\mathsf{g}]$ and $\mathbf{R}^{(l)}[\{f_c, h_c\}] = \mathbf{Q}^{(l)}[f_c, h_c]$. Use $\delta_i[\{\mathsf{g}\}] = \delta_i[\mathsf{g}]$; for $\#\mathsf{J} \geq 2$, put

$$\delta_i[\mathsf{J}] = \sum_{l=1}^{i} \mathcal{V}_i^{\mathrm{old}}[\mathbf{R}^{(l)}[\mathsf{J}]].$$

Thus $\delta_0[\mathsf{J}] = 0$ for these sets. For J joining cells, define recursively

$$\widetilde{\mathbf{R}}^{(i+1)}[\mathsf{J}] = \sum_{\pi \in \mathrm{Part}(\mathsf{J})} D^{\#\pi} \widetilde{E}_i[\delta_i[b] : b \in \pi],$$
$$\mathbf{R}^{(i+1)}[\mathsf{J}] = \widetilde{\mathbf{R}}^{(i+1)}[\mathsf{J}] - \mathfrak{e}_{i+1}[\mathsf{J}].$$

Only the scalar value at the trivial configuration is removed. The quadratic units of $\mathbf{R}$ are its polymers. Finally set

$$\eta_{\mathrm{x}} = \min\{7\kappa_{\mathrm{s}}/8,\, 2c_E/c'\}, \quad c_E = \log 18432, \quad c_1 = \eta_{\mathrm{x}}/M, \quad c_{\mathrm{x}} = c_1/2, \quad \varsigma_l^{\mathrm{x}} = e^{-c_1 D_{\mathrm{x}}/a_l}.$$

These constants depend only on κ, M, c'. Choose them, and the constants in Lemma 14.25, after c_λ, ρ and before γ_{e}. ∎

Lemma 14.8 (Reading activities at a background)**.** Use the calibration frame and both admissibility conditions of the construction. Let $1 \le k \le K$, $a_k = L^k a_K$, and

$$\mathcal{S}_k = \mathcal{M}^k\{\mathfrak{w}_{h_\emptyset^{(k)}} \neq 0\}.$$

For $V \in \mathcal{S}_k$ let $\mathsf{Ev}_k(V)$ be its evaluation configurations of Lem. 14.5. Give the formal background legs of level $n \le k$ the weights $h^{(k)}(n) = C_H L^{-(k-n)} h_F$. Here C_H is the fermion background row constant. Write $F \circ \mathsf{H}(\mathsf{U})$ for substitution of

$$\Psi_n^{\mathrm{bg}} = \mathsf{H}_{n,k}(\mathsf{U})\Psi_k, \qquad \bar{\Psi}_n^{\mathrm{bg}} = \bar{\Psi}_k \bar{\mathsf{H}}_{k,n}(\mathsf{U}).$$

For $V \in \mathcal{S}_k$ and $\mathsf{U} \in \mathsf{Ev}_k(V)$ the following hold.

(1) For every level $1 \le l \le k$ and polymer $X \in \mathcal{D}_l$, the pair of U on X lies in the regular core at radii $\alpha_0/2, \alpha_1/2$. These pairs agree under restriction between levels and polymers. At every bare plaquette,

$$|\partial\mathsf{U}(p) - 1| < \tfrac{1}{2}\alpha_0 L^{-2k}, \qquad |a_p(\mathsf{U})| < \tfrac{1}{8}\alpha_0^2 L^{-4k}.$$

(2) For $n < k$, every row satisfies

$$\sum_y \|\mathsf{H}_{n,k}(\mathsf{U})(x,y)\| \le \tfrac{1}{2} C_H L^{-(k-n)}.$$

The same bound holds for the barred row maps, and $\mathsf{H}_{k,k} = 1$. Thus the weighted row inequality $h_F \sum_y \|\mathsf{H}_{n,k}(x,y)\| \le h^{(k)}(n)/2$ holds also for $n = k$.

(3) If F_p is the degree-p part of $F \in \mathrm{Gr}_k^{\mathrm{bg}}(T_\ell)$, then

$$\|F \circ \mathsf{H}(\mathsf{U})\|_{h_F} \le \sum_p 2^{-p} \|F_p\|_{h^{(k)}}.$$

(4) For a family of terms $G(X; q, \cdot; \lambda)$ over polymers at levels $l(X) \le k$,

$$\sup_{\lambda \in D} \Big\| \sum_X G(X; \mathbf{p}_X^{(l(X))}(\mathsf{U}), \cdot; \lambda) \circ \mathsf{H}(\mathsf{U}) \Big\|_{h_F}$$
$$\le \sup_{\lambda \in D} \sum_X \|G(X; \mathbf{p}_X^{(l(X))}(\mathsf{U}), \cdot; \lambda)\|_{h^{(k)}}$$
$$\le \sum_X \sup_{\lambda \in D} \sup_q \|G(X; q, \cdot; \lambda)\|_{h^{(k)}}.$$

For a level-k family the middle expression evaluates all terms at restrictions of one regular pair P on the torus. With the source norms of Def. 14.6, if S lies in a closed ball of radius $r \ge a_K$, it is bounded by

$$e^{\kappa/16} C_{\mathcal{C}} (1 + r/a_k)^3 \|G\|^{\mathrm{o}}_{k;D;S},$$

and the last expression by the same factor times $\|G\|_{k;D;S}$. For two supports S_f, S_h the corresponding bounds are $e^{\kappa/8}\mathcal{C}'_k\|G\|^{\circ}_{k;D;S_f,S_h}$ and $e^{\kappa/8}\mathcal{C}'_k\|G\|_{k;D;S_f,S_h}$.

(5) For a scalar- or matrix-valued function $c(x;\lambda)$ on bare sites,

$$\begin{aligned}&\sup_{\lambda\in D}\|L^{-k}\mathsf{S}_1(c(\cdot;\lambda)\mathsf{H}_{0,k}(\mathsf{U})\Psi_k)\|_{h_F}\\&\quad\le \tfrac{1}{4}c_{\mathsf{S}}h_F^2\sum_{\square}\frac{1}{M^3L^{3k}}\sum_{x\in\square}\sup_{\lambda\in D}|c(x;\lambda)|\\&\quad\le \tfrac{1}{4}C_{\mathcal{C}}(1+r/a_k)^3c_{\mathsf{S}}h_F^2\langle c\rangle_{k;D;S}.\end{aligned}$$

Here $|\cdot|$ is the operator norm for matrix values, $c_{\mathsf{S}}=M^3\nu_P\nu C_H^2$, and ν_P is the number of places at a bare site.

The same assertions hold almost everywhere on the support of the level-k density test against the coordinate reference measure. There the regular radii are α_0,α_1 and the plaquette bounds are α_0L^{-2k} and $\alpha_0^2L^{-4k}/2$, respectively. ■

Proof. The calibration frame is admitted at every level. The background part of its induction theorem gives nonempty evaluation sets on $\mathcal{S}_k$, with pairs in the stated half-radius cores on every polymer in $\Lambda_l=T_\ell$. For a bare plaquette p, the union $X(p)$ of the cubes containing its corners is one, two or four cubes and is an admitted polymer. All four bonds lie in its interior bond set. The core's plaquette condition there gives $|\partial\mathsf{U}(p)-1|<\alpha_0L^{-2k}/2$. For SU(2), $|a_p(\mathsf{U})|=|\det(\partial\mathsf{U}(p)-1)|/2$, giving the action bound. Restrictions are the restrictions of one bare field and its level currents, so have the required consistency. The almost-everywhere clause of the same background theorem gives the full-radius cores outside a Borel null set on the density-test support. The argument at $X(p)$ then gives the stated full-radius plaquette bounds.

Each evaluation configuration is real. The ceiling on α_0 makes the plaquette bound at most ϵ_*L^{-2k}. There are no lower layers and the mass satisfies $m_0=L^{-K}\mathsf{m}\le L^{-K}/2$. These are the hypotheses of the fermion background row estimate, which gives

$$\sum_y\|\mathsf{H}_{n,k}(x,y)\|\le(1+C_\Sigma L^{3/2})C_H^{\circ}L^{-(k-n)}\le\tfrac{1}{2}C_HL^{-(k-n)}.$$

The barred map has the same estimate. At $n=k$ use $C_H\ge 2$.

Give each formal leg weight $h^{(k)}(n)/2$. The row estimates show that substitution contracts the exterior-algebra norm into the norm of weight h_F. That norm is the sum of the norms of the homogeneous parts, and halving the weights multiplies the degree-p norm by 2^{-p}. This proves (iii). Apply it term by term and use the triangle inequality to obtain (iv). At level k, the torus itself is an admitted polymer, so its regular pair restricts to all the pairs in this sum. The source-weight summation estimate of Lem. 14.24 therefore applies both with the common-pair supremum outside the sum and with the separate suprema inside. Its cell count is at most $C_{\mathcal{C}}(1+r/a_k)^3$, since distance to S is at least distance to any ball containing S.

Finally a bare mass monomial at one place has norm at most $\nu|c(x)|C_H^2L^{-2k}h_F^2$. Include L^{-k} and the ν_P places at each of the M^3L^{3k} bare sites in a cell. This gives $c_{\mathsf{S}}h_F^2$ times its cell mean. Its degree is two, so substitution gains the factor $1/4$. The weighted mean in Def. 14.6 bounds this cell mean by $e^{-\kappa_{\mathrm{s}}\delta_k(\square;S)}\langle c\rangle_{k;D;S}$. Summing cells and using the same cell count proves (v). □

14.7. **Constants for the complex field scale.** The field-scale argument uses a larger analytic disc than the local parameter disc of a single step. We choose its constants in an order that leaves the feedback contractions available.

Remark 14.9 (Order of the constants). Retain the order of the constants in the lattice construction. The following additional choices are made before selecting the datum.

(1) With the gauge constants fixed, choose $\mathsf{t}_\flat$ so that for every $\mathsf{t} \ge \mathsf{t}_\flat$,

$$\mathsf{t} \ge e^{2\mathfrak{p}_0}, \qquad \mathsf{C}_{\mathcal{A}}\delta(\mathsf{t}) \le \varepsilon_P, \qquad \mathfrak{b}(\mathsf{t})^2 \ge M^3, \qquad (1+\beta_s)\delta(\mathsf{t}) \le \mathsf{s}^\star\alpha_0.$$

Here $\delta(\mathsf{t}) = A_1\mathsf{t}^{-1/2}(\log \mathsf{t})^{\mathfrak{p}_0}$, $\mathfrak{b}(\mathsf{t}) = A_1(\log \mathsf{t})^{\mathfrak{p}_0}$, ε_P is the vertex radius, and $\mathsf{s}^\star$ is the admissible scalar-localisation ceiling. These conditions hold eventually and depend only on constants chosen before the coupling ceiling.

(2) Choose the source-weight constants before h_F: κ_{s}, $C_{\mathcal{C}} \ge 8M^3$, $C_{\mathrm{rd}}, C_{\mathrm{mo}} \ge 1$, $K_{\mathfrak{n}} = 3c_{\mathsf{S}}c_b^{-2}C_{\mathrm{mo}}$, the torus-defect constants, and the sums $S_{\mathrm{tor}}, S_\diamond$ in Lemmas 14.18–14.21. The direct source constants are

$$A_{\mathrm{src}} = 3\max\{\alpha_0^2,\, 2^{13}\mathsf{c}_{\mathrm{os}}\alpha_0 C_{\mathrm{cf}}/c_h\}(1+2M^3c_{\mathrm{cl}}/w_*), \qquad C_{\mathrm{src}} = 3^5 C_\Sigma M^3 A_{\mathrm{src}}.$$

The mass-mean constant is $C_{\mathrm{mn}} = 37$. Also choose

$$C_{\mathrm{Lip}} = 78\pi e\,\kappa_{\mathrm{s}}^{-4}, \qquad \Sigma_{\mathrm{w}} = \sum_{u\ge 0} L^{2u}q_{u+1}^2 e^{-\kappa_{\mathrm{s}}(L^u-1)}, \qquad C_{\mathfrak{l}} = \sup_{t\ge 1} t^{-7/2}(1+\log t)^{8\mathfrak{p}_0+2}.$$

These constants do not depend on h_F.

Increase h_F to satisfy, in addition to its previous floors,

$$3K_\mu^{\mathrm{s}}K_M C_{\mathrm{rd}}C_{\mathcal{C}}(1+2C_{\mathrm{mo}})S_\diamond\, c'_\psi(h_F) \le \frac{1}{4}, \qquad c'_\psi(h) = \frac{2A_\kappa\Gamma_F}{h} + \frac{(c_h/a_{\mathrm{an}} + C_G)c_{\mathrm{cl}}}{4C_V h^2}. \tag{14.viii}$$

Every factor other than $c'_\psi(h_F)$ is already fixed, and $c'_\psi(h) \to 0$. Choose C^+ from the reduction before the lower mass endpoint.

(3) After the original coupling ceiling γ, put

$$\begin{aligned} C_V^\lambda &= (4K_P + c_h\mathsf{C}_{\mathcal{A}}K_{\mathrm{old}}/a_{\mathrm{an}})A_1 2^{\mathfrak{p}_0+1},\\ C_F^\lambda &= C_V h_F^2 \mathsf{C}_{\mathcal{A}}A_1 2^{\mathfrak{p}_0+1} + 2c'_\psi c_{\mathsf{S}} h_F^2 \bar{B}_M,\\ C_\eta &= 8\big[K_M(C_V^\lambda + C_F^\lambda + 2) + 6Lc_{\mathsf{S}}h_F^2\bar{B}_M\big]. \end{aligned}$$

Choose $c_\lambda \ge 1$ so large that

$$\begin{gathered} c_\lambda \ge 4\bar{B}/r_x, \qquad A_1 2^{\mathfrak{p}_0+1}c_\lambda^{-1/2} \le \delta(\mathsf{t}_\flat),\\ \beta_\lambda := \mathsf{C}_{\mathcal{A}}A_1 2^{\mathfrak{p}_0+1}c_\lambda^{-1/2} \le \min\{\beta_0/2,\, c_{\mathrm{cl}}/(4C_V h_F^2),\, a_{\mathrm{an}}/(3c_h)\},\\ (C_V^\lambda + C_F^\lambda + C_\eta)c_\lambda^{-1/2} \le c_{\mathrm{cl}}/4,\\ 2K_M^\bullet(C_V^\lambda + C_F^\lambda + C_\eta + 2)c_\lambda^{-1/2} \le 1/2, \end{gathered}$$

and

$$6K_\mu^{\mathrm{s}}K_M C_{\mathrm{rd}}C_{\mathcal{C}}(1+2C_{\mathrm{mo}})S_\diamond c_h\beta_\lambda/a_{\mathrm{an}} \le 1/4. \tag{14.ix}$$

Put $\eta = C_\eta c_\lambda^{-1/2}$ and $s_\lambda = c_h\beta_\lambda/a_{\mathrm{an}} \le 1/3$. The two inequalities eq. 14.viii and eq. 14.ix give the contraction factor $\theta \le 1/2$ in Lemma 14.21.

Next set

$$C_{\mathrm{fb}} = 43LC_{\mathrm{mn}}K_{\mathfrak{n}}K_\mu^{\mathrm{s}}K_M c'_\psi S_\diamond, \qquad \rho = \min\{1/2, C_{\mathrm{fb}}^{-1/2}\}. \tag{14.x}$$

Choose the constants for sources joining cells after these. Finally choose $0 < \gamma_{\mathrm{e}} \le \gamma$ below the pure-theory comparison ceiling and so that

$$\gamma_{\mathrm{e}}^{-2} \ge 2\mathsf{t}_\flat, \qquad \log\gamma_{\mathrm{e}}^{-2} \ge \mathsf{Y}_{\mathrm{e}},$$

and, for every $x \ge \gamma_{\mathrm{e}}^{-2}$,

$$x(\log x)^{-2\mathfrak{p}_0} \ge c_\lambda/\rho^2, \qquad C_w\phi(x) \le 1/2, \qquad e^{-\gamma_{\mathrm{tail}}\mathfrak{b}(x/2)^2/2} \le c_\lambda^{-1/2}.$$

All three conditions hold eventually. For the radii used below this gives

$$(\rho\bar{\lambda}_j)^2 \ge 2, \qquad \bar{\lambda}_j^2 \ge 8, \qquad C_{\mathrm{fb}}\bar{\lambda}_j^{-2} \le 1/2.$$

(4) After g choose $\ell_G, X_g, \mathsf{b}_\pm$ and the constants of the empty-history reduction. After (g,ϑ) choose the response bounds $A^{\mathrm{red}}, A_c, C^{\mathfrak{N}}_{\mathrm{wr}}, B_f, A_{\mathfrak{N}}, A^{\mathrm{rsp}}, A_{\mathbf{P}}, A'_{\mathbf{P}}, A_{\mathbf{Q}}$, then $C_{\mathfrak{e},1}, C_{\mathfrak{e},2}, C_{\mathrm{iii}}, C^{\mathrm{ii}}_m$, the bounds for sources joining cells, and finally $C_{\mathrm{T}}, \mathsf{d}_{\mathrm{R}}, C_{\mathrm{R}}, \epsilon_{\mathrm{R}}$. The transfer between sides is chosen at this stage. The comparison with lattice perturbation theory additionally depends on the cutoff-window exponent $P > 1$ and is chosen after (g,ϑ,P). No constant chosen at an earlier stage depends on a later choice.

14.8. **Slab operators for chamber fermions.** The comparison between two torus sizes requires an estimate for unnormalized integrals. We prove it by writing a folded insertion as the trace of a power of one positive operator. The operator depends on the cross-section and the cell, but not on the number of cells in the direction being shortened.

Work in lattice units with the chamber operator at $r = 1/2$, $\beta \geq 0$, and $m_0 > 0$. Gauge fields are periodic and fermions antiperiodic on a rectangular torus with even side lengths. Write

$$\Omega(F) = \int dU \int d\bar\psi\, d\psi\, e^{-\beta A - \bar\psi (D[U]+m_0)\psi} F.$$

The determinant is positive and $Z = \Omega(1) > 0$, by [QCDM1, Prop. 2.1]. The proof of that proposition uses unitarity of the nearest-neighbour shifts and the two adjoint chiral blocks; it applies without change to these rectangular antiperiodic tori.

Fix a coordinate direction and an odd integer $\lambda \geq 3$. Cut the torus at parallel planes, separated by λ, that bisect gauge links. A slab contains λ layers of sites, all chambers at those sites, and the gauge links between them. Every link, plaquette and chamber hop is either internal to one slab or crosses exactly one wall. For chamber hops this uses the fact that only $s = 0$ changes the site.

On each bisected link substitute $U_b = V_b'^{-1} V_b$, with V_b, V_b' independent Haar variables. This substitution preserves the integral and assigns one half-link variable to each neighbouring slab. Let J_{sl} denote integration over the gauge and Grassmann variables of one model slab, including its half links, with the exponential of the internal action. A slab element is an even Grassmann polynomial in these generators with bounded Borel coefficients in its gauge variables.

The wall contribution to the exponent has the form

$$-\beta n_\times + Y + X_F, \qquad Y = \frac{\beta}{2} \sum_{\substack{p \text{ cut} \\ a,b}} A^p_{ab} \Theta(A^p_{ab}),$$

where $n_\times$ counts the cut plaquettes and A^p is transport along the half boundary of p, including the half links. For a crossing chamber v, with the normal n pointing toward v, put $\chi = V_b \psi(v)$, $\bar\chi = \bar\psi(v) V_b^{-1}$. Its pair of fermion hops contributes

$$\sum_{i,j} \bar\chi_i \big(t_s + \tfrac{1}{2} t_s \gamma(n)\big)_{ij} \Theta \bar\chi_j + \sum_{i,j} \chi_j \big(t_s - \tfrac{1}{2} t_s \gamma(n)\big)_{ij} \Theta \chi_i.$$

Both matrices have eigenvalues $t_s/2, 3t_s/2$. These are the half-link and crossing identities in the proof of [QCDM1, Thm. B]; they are identities at a wall on the covering lattice, so they apply at every wall of this partition.

Diagonalize the positive matrices and expand the gauge exponential. Products of terms $G\Theta G$, with G homogeneous, have the same form. The fermion exponential is a finite sum and the gauge exponential converges uniformly. We obtain

$$e^{Y+X_F} = \sum_{\alpha \in I} c_\alpha G_\alpha \Theta G_\alpha, \qquad c_\alpha \geq 0, \qquad \sum_{\alpha \in I} c_\alpha \sup \|G_\alpha\|^2 < \infty. \tag{14.xi}$$

Here I is countable, each G_α is homogeneous, and the norm is any fixed coefficient norm on the finite-dimensional Grassmann algebra at the wall. The same family is used at all walls by translation. It depends on the transverse torus and the model parameters, not on the number of slabs.

For a slab element X, define a matrix on $\ell^2(I)$ by

$$\mathfrak{Y}[X]_{\alpha\alpha'} = (c_\alpha c_{\alpha'})^{1/2} J_{\mathrm{sl}}((\Theta G_\alpha)^{\mathrm{left}} X G_{\alpha'}^{\mathrm{right}}). \tag{14.xii}$$

The superscripts specify the two faces of the slab. Let X^Θ be reflection of X in its middle plane, with the antilinear reversal of Grassmann products. This middle plane passes through sites; only reflection invariance, not a positivity assertion at that plane, is used below.

Lemma 14.10 (The slab trace)**.** The operator $\mathfrak{Y}[X]$ is Hilbert–Schmidt and $\mathfrak{Y}[X^\Theta] = \mathfrak{Y}[X]^*$. For an even number $q \geq 2$ of slabs and even slab elements $X_1, \dots, X_q$, the product below is trace class and

$$\Omega\left(\prod_{t=1}^{q} X_t\right) = e^{-\beta n_\times q} \operatorname{Tr}(\mathfrak{Y}[X_1] \cdots \mathfrak{Y}[X_q]). \tag{14.xiii}$$

The operators in this identity do not depend on q. ■

Proof. Expand eq. 14.xi at each wall, with G_{α_t} in slab t and ΘG_{α_t} in slab $t+1$. The product of crossing factors can be ordered as

$$G_{\alpha_1}[\Theta G_{\alpha_1} G_{\alpha_2}] \cdots [\Theta G_{\alpha_{q-1}} G_{\alpha_q}] \Theta G_{\alpha_q}.$$

The bracket at position t belongs to slab t. Moving the last factor to the front costs $(-1)^{p(\alpha_q)}$, since the preceding product has the same parity. That last factor is initially in slab $q+1$. When expressed in the generators of slab 1, the antiperiodic boundary condition contributes the same sign. The two signs cancel.

A bracket with unequal boundary parities is odd and integrates to zero. All other brackets are even, commute, and integrate separately over the slabs. Allocating half of each c_{α_t} to each adjacent slab gives exactly the matrix product in eq. 14.xiii. The scalar factors from the walls give $e^{-\beta n_\times q}$.

Reflection in the slab's middle plane exchanges its faces and preserves its internal action and Haar measure. It sends the Berezin top monomial to its conjugate with sign $+1$: the determinants of the two spin matrices cancel. Consequently $J_{\mathrm{sl}}(F^\Theta) = \overline{J_{\mathrm{sl}}(F)}$. On entries that can be nonzero the two boundary parities agree, so the face exchange gives $\mathfrak{Y}[X^\Theta]_{\alpha\alpha'} = \overline{\mathfrak{Y}[X]_{\alpha'\alpha}}$.

The summability in eq. 14.xi also justifies these rearrangements. On the finite-dimensional slab algebra, integration against the internal weight is a bounded linear functional. Thus, for a constant depending on the fixed slab and X,

$$|\mathfrak{Y}[X]_{\alpha\alpha'}| \leq C_X (c_\alpha c_{\alpha'})^{1/2} \sup \|G_\alpha\| \sup \|G_{\alpha'}\|.$$

Its squared entries are summable by eq. 14.xi. This proves the Hilbert–Schmidt assertion. Products of at least two such operators are trace class. Finite truncations of I converge in Hilbert–Schmidt norm; their products converge in trace norm. The absolutely convergent expansion of the original integral has the same limit, proving the trace identity. □

14.9. **The unnormalized two-volume inequality.** Let $\lambda \geq 3$ be odd, let $\mathsf{n}_\nu \geq 2$ be even integers, and put

$$\mathfrak{G} = \lambda(\mathsf{n}_1\mathbb{Z} \times \mathsf{n}_2\mathbb{Z} \times \mathsf{n}_3\mathbb{Z}), \qquad \mathsf{N} = \mathsf{n}_1\mathsf{n}_2\mathsf{n}_3.$$

For $o \in \mathbb{Z}^3$, the board cells are

$$\square_{\mathbf{k}} = \{x : |x_\nu - o_\nu - \lambda \mathbf{k}_\nu| \leq (\lambda-1)/2,\ \nu = 1,2,3\}, \qquad \mathbf{k} \in \prod_\nu \mathbb{Z}/\mathsf{n}_\nu\mathbb{Z}.$$

A cell element has generators only at chambers whose sites belong to that cell, is even, and has bounded Borel coefficients depending only on links with both endpoints in the cell. Its mirror copies $G^{[\mathbf{k}']}$ are obtained by reflection at the intervening cell walls and even translations. Write

$$\operatorname{Fold} G = \prod_{\mathbf{k}'} G^{[\mathbf{k}']}, \qquad \|G\|_{\mathrm{ch},\mathsf{n}} = \left(\frac{\Omega_{\mathsf{n}}(\operatorname{Fold} G)}{Z_{\mathsf{n}}}\right)^{1/\mathsf{N}}.$$

Proposition 14.11 (Two-volume chessboard estimate)**.** For the chamber model with $\beta \geq 0$, $m_0 > 0$, and antiperiodic fermions, the following hold.

(C0) Mirror copies are well defined on the even cell algebra, preserve products, and compose according to their destination cells.

(C1) Folded integrals are nonnegative, and

$$\left|\Omega_{\mathsf{n}}\left(\prod_{\mathbf{k}} G_{\mathbf{k}}\right)\right| \leq \prod_{\mathbf{k}} \Omega_{\mathsf{n}}(\operatorname{Fold} G_{\mathbf{k}})^{1/\mathsf{N}}.$$

In particular $|\langle G\rangle_{\mathsf{n}}| \leq \|G\|_{\mathrm{ch},\mathsf{n}}$ for one cell element.

(C3) Let another board have the same λ and cell zero, with even $2 \leq \mathsf{n}'_\nu \leq \mathsf{n}_\nu$. For one fixed cell element G_0,

$$0 \leq \Omega_{\mathsf{n}}(\operatorname{Fold} G_0) \leq \Omega_{\mathsf{n}'}(\operatorname{Fold} G_0)^{\prod_\nu \mathsf{n}_\nu/\mathsf{n}'_\nu}.$$

The physical spacing, β, m_0, and the cell element are the same in the two integrals.

■

Proof. Two parallel reflections compose to a translation up to a parity sign. Two orthogonal reflections anticommute on a generator, and translation by a period changes its sign. All these signs disappear on even elements. This proves (C0), including independence of the route through the cells.

For (C1), group the cell elements into slabs in direction 1, and denote their products by X_t. Apply Lem. 14.10 and Schatten Hölder:

$$\left|\operatorname{Tr}\prod_{t=1}^{\mathsf{n}_1} \mathfrak{Y}[X_t]\right| \leq \prod_t \|\mathfrak{Y}[X_t]\|_{\mathsf{n}_1}.$$

All operators are Hilbert–Schmidt, hence belong to the Schatten class of even order $\mathsf{n}_1 \geq 2$. Writing $Y = \mathfrak{Y}[X_t]$, its factor obeys

$$\|Y\|_{\mathsf{n}_1}^{\mathsf{n}_1} = \operatorname{Tr}(Y^*Y)^{\mathsf{n}_1/2}.$$

The trace formula identifies this with $e^{\beta n_\times \mathsf{n}_1}$ times the integral obtained by filling direction 1 with X_t and X_t^Θ alternately. It is nonnegative. The scalar exponential cancels the one in eq. 14.xiii. Repeat in directions 2 and 3. This gives the displayed inequality and nonnegativity of each fully folded integral. Taking all but one cell element equal to 1 and dividing by $Z_{\mathsf{n}} > 0$ gives the normalized assertion.

For (C3), first keep $\mathsf{n}_2, \mathsf{n}_3$ fixed. Fold G_0 in those two directions within one slab, and call this slab element X. The fold in direction 1 has alternating slabs X, X^Θ. Therefore

$$\Omega_{\mathsf{n}}(\operatorname{Fold} G_0) = e^{-\beta n_\times \mathsf{n}_1} \operatorname{Tr} B^{\mathsf{n}_1/2}, \qquad B = \mathfrak{Y}[X]^*\mathfrak{Y}[X] \geq 0.$$

The operator B is trace class and independent of n_1. If $k \geq k' \geq 1$ and b_i are its eigenvalues, then

$$\operatorname{Tr} B^k = \sum_i (b_i^{k'})^{k/k'} \leq \left(\sum_i b_i^{k'}\right)^{k/k'}.$$

For completeness, when the sum on the right is positive, divide by it: the resulting nonnegative numbers sum to one, and each has its k/k'-th power at most itself. When the sum is zero the claim is immediate. Apply this with $k = \mathsf{n}_1/2$, $k' = \mathsf{n}'_1/2$. The scalar exponential has exactly the same power, so it gives the desired inequality when only the first side changes. Next shorten the second side and then the third, using the already shortened transverse tori. The powers multiply to $\prod_\nu \mathsf{n}_\nu/\mathsf{n}'_\nu$. □

In physical coordinates the board may have any offset in $a_K\mathbb{Z}^3$. In the side comparison of Prop. 14.43, its cell side is $\ell_G/2$, so $\lambda = \ell_G/(2a_K)$ is an odd power of L and at least 3. The least torus has two cells in each direction; on $\ell = 2L^m$ the number is $2L^{m-m_{\mathfrak{L}}(g)}$. Both boards therefore satisfy (C3), with the same cell zero and exponent $(\ell/\ell_G)^3$. For a bounded real cell function Y,

$$\|Y\|_{\mathrm{ch},\ell} \le \|Y\|_{\mathrm{ch},\ell_G} \left(\frac{Z_{\ell_G}^{(\ell/\ell_G)^3}}{Z_\ell} \right)^{1/N_\ell}, \qquad N_\ell = 8(\ell/\ell_G)^3.$$

The partition-function ratio is retained here: (C3) compares unnormalized integrals and does not assert monotonicity of their normalized ratios.

14.10. **Reduction to the mixed source energies.** At zero source, positivity gives a lower bound for the contribution with no large-field region. At a complex source, the history estimate gives an upper bound for its complement. We use these bounds at a level whose remaining volume grows only logarithmically as the separation decreases. Averaging over many translates then removes the volume cost from differentiating the complementary contribution.

Here are the constants used in this reduction. Retain the source-history rate $c = (c_\star + c_\Theta)/2$, the share $\eth \in (0,1)$ and $c_\eth = (1-\eth)c$. Set

$$C^+ = \max\{C, \mathsf{C}_{\mathrm{maj}} + \lambda_L\}, \quad \mathsf{C}_{\mathrm{maj}} = 48L^3 C_E^F + C'_\star, \quad \lambda_L = (6L^3+1)\log 2.$$

The constant C is the source-history norm constant, enlarged if necessary before the mass interval is chosen. Impose the history tail threshold

$$\log \gamma_{\mathrm{e}}^{-2} \ge \mathsf{Y}_{\mathrm{e}} = \max\{\mathsf{Y}_{\mathrm{b}}, 2\log(1+\bar{B})\},$$

formed at $(c, \mathsf{C}_{\mathrm{maj}}, C_\flat^F + C'_{\mathfrak{E}}, \eth)$. All its parameters precede γ_{e} in the choice of constants.

For fixed $g \le \gamma_{\mathrm{e}}$ write

$$\ell_G = \ell_G(g), \qquad X_g = L(g^{-2} + B_\sharp), \qquad \mathsf{b}_- = g^{-2} + \bar{B}, \qquad \mathsf{b}_+ = X_g - \bar{B}.$$

Thus $\mathsf{b}_\pm$ are the ends of the label interval. With the empty-history lower-bound constant C_P, put

$$\begin{aligned} c_V^+(m) &= C^+ + 2 + C_P(1 + \log(X_g L^m)), \\ \mathsf{Q}_{K-m} &= \exp\{c_V^+(m)\ell_G^3 L^{3m}\}, \\ \mathsf{Q} = \mathsf{Q}_K, &\qquad M_0 = \exp\{(C^+ + 2)\ell_G^3\}. \end{aligned}$$

For $0 < \mathsf{d} \le \mathsf{Q}^{-8}$ define

$$m_*(\mathsf{d}) = \max\{m \ge 0 : 8c_V^+(m)\ell_G^3 L^{3m} \le \log(1/\mathsf{d})\}, \tag{14.xiv}$$

$$\omega_{\mathrm{r}}(\mathsf{d}) = \min\left\{1,\ 2M^{-3}\ell_G^3 L^{3m_*} e^{-c_\eth A_0^2 (\log(\mathsf{b}_- L^{m_*}))^{2\mathfrak{p}_0}}\right\}. \tag{14.xv}$$

The constants from Lemmas 14.18 and 14.21–14.24 determine

$$\begin{aligned} C_{\mathrm{T}} &= \max\{1, (8e^{\kappa/16}A_{\mathbf{P}} + e^{\kappa/8}A_{\mathbf{Q}})C_{\mathcal{C}}[1 + C_{\mathrm{rd}}C_{\mathcal{C}}(1+2C_{\mathrm{mo}})S_\diamond] \\ &\qquad + 2C_{\mathcal{C}}A_{\mathfrak{N}} + 81\alpha_0^2/8\}, \\ \mathsf{d}_{\mathrm{R}} &= \min\{\mathsf{d}_0(g,\vartheta),\ 1/400,\ e^{-8c_V^+(1)\ell_G^3 L^3}\}, \\ C_{\mathrm{R}} &= \max\{C_{\epsilon,2} + 2C_{\mathrm{T}}^2(1+C_{\mathrm{I}}),\ 7\cdot 24^3\cdot 1600 M^2 \mathsf{Q}^3 \varrho^{-2}\vartheta^{-2}\ell_G^{-3}\}, \\ \epsilon_{\mathrm{R}}(\mathsf{d}) &= \mathsf{b}_+^3 C_{\mathrm{R}} \sup_{0<\mathsf{d}'\le\mathsf{d}} [\omega_{\mathrm{r}}(\mathsf{d}') + \mathsf{d}']. \end{aligned}$$

Choose these constants after the source-derivative constants and before the threshold in the final short-distance theorem.

Lemma 14.12 (Reduction to source energies)**.** Assume the two admissibility conditions for the source flow and the floor on h_F in 14.9. Let $0 < \mathsf{d} \le \mathsf{d}_{\mathrm{R}}$ and consider a datum on T_{ℓ_G} with $K > s(\mathsf{d})$ and a pair $(f,h) \in \mathfrak{T}(\mathsf{d};\vartheta)$. Put $k = K - m_*(\mathsf{d})$.

The function ϵ_{R} is nondecreasing and tends to zero at zero. Moreover,

$$1 \le k \le K-1, \qquad a_k \ge \mathsf{d}, \qquad a_k^{-3} \le \frac{\log(1/\mathsf{d})}{128}, \qquad \mathsf{Q}_k \le \mathsf{d}^{-1/8}, \qquad \mathsf{d}^{1/4}\log(1/\mathsf{d})^{3/2} \le 1.$$

The following statements hold.

(1) Split the full history sum as $Z[w,0] = \mathsf{A}_k[w] + \mathsf{B}_k[w]$, according as $Z_k(h) = \emptyset$ or $Z_k(h) \neq \emptyset$. Both terms are entire in the finitely many plaquette values of w. On the source class $\mathcal{P}$,

$$\mathsf{A}_k[w] = e^{E_k(w)}\mathsf{I}_k[w], \qquad \mathsf{I}_k[w] = \operatorname{Ber}\int \mathsf{g}_k e^{\mathsf{T}_k(w)}\, d\mathscr{M}_k.$$

Here $\mathscr{M}_k$ is the measure for the empty history at level k, and g_k is its zero-source density times e^{-E_k}. At the background U of its block field,

$$\begin{aligned}\mathsf{T}_k(w) = {} & a_K^{-1}\sum_p w(x(p))a_p(\mathsf{U}) - L^{-k}\mathsf{S}_1(\sigma_k(w,0)\,\cdot) \\ & + \sum_{j\le k}\sum_X [E^{(j)}(X;\mathsf{U},\bar\Psi^{\mathrm{bg}},\Psi^{\mathrm{bg}};(w,0)) - E^{(j)}(X;\mathsf{U},\bar\Psi^{\mathrm{bg}},\Psi^{\mathrm{bg}};0)].\end{aligned}$$

This element of Gr_k is independent of the chosen member of the minimal orbit. It is even, vanishes at zero source, and is holomorphic for $\|w\|_\sharp < 4\varrho$.

(2) On $\mathcal{P}$,

$$|e^{-E_K(w)}\mathsf{A}_k[w]| \le M_0, \qquad |e^{-E_K(w)}\mathsf{B}_k[w]| \le \omega_{\mathrm{r}}(\mathsf{d})M_0.$$

At zero source, $e^{-E_K}\mathsf{A}_k[0] \ge M_0/\mathsf{Q}$ and $\mathsf{B}_k[0] \ge 0$. For disjointly supported complex Lipschitz directions F, H, define

$$\mathsf{r}_1(F) = \partial_F \log(1+\mathsf{B}_k/\mathsf{A}_k)\big|_0, \qquad \mathsf{r}_2(F,H) = \partial_F\partial_H \log(1+\mathsf{B}_k/\mathsf{A}_k)\big|_0.$$

With $\|F\|_\circ = \|F\|_\sharp/\varrho$,

$$|\mathsf{r}_1(F)| \le 2\omega_{\mathrm{r}}\mathsf{Q}^2\|F\|_\circ, \qquad |\mathsf{r}_2(F,H)| \le 7\omega_{\mathrm{r}}\mathsf{Q}^3\|F\|_\circ\|H\|_\circ.$$

(3) Let $\mathcal{S}_k$ be the actual image set in Lemma 14.8. For an Gr_k-valued function F on $\mathcal{S}_k$ for which $\mathsf{g}_k F$, extended by zero, is measurable for the completion of $\mathscr{M}_k$ and has integrable fibre norm, define

$$\langle F\rangle_k = \frac{\operatorname{Ber}\int \mathsf{g}_k F\, d\mathscr{M}_k}{\operatorname{Ber}\int \mathsf{g}_k\, d\mathscr{M}_k}.$$

If F is bounded on $\mathcal{S}_k$, then

$$|\langle F\rangle_k| \le \mathsf{Q}_k \sup_{\mathcal{S}_k}\|F\|_{h_F}.$$

For any finite set of Lipschitz directions F_i, the functions $e^{\mathsf{T}_k(\sum_i z_i F_i)}$ and all their derivatives at zero belong to this class locally in z. Their integrals are holomorphic and may be differentiated under the integral.

(4) Writing $\|\cdot\| = \sup_{\mathcal{S}_k}\|\cdot\|_{h_F}$,

$$\|\mathsf{T}_k[f]\| \le C_{\mathrm{T}} e_k[f]\mathfrak{l}_k^{4\mathfrak{p}_0+1}, \qquad \|\mathsf{T}_k[f,h]\| \le C_{\mathrm{T}} e_k[f,h].$$

The first bound also holds for h. All families are evaluated at one regular pair on the torus before summing them; in particular, the bounds with one pair outside the polymer sum are sufficient.

(5) There is a set $\mathcal{T}$ of $N \geq (\ell_G/(24\mathsf{d}))^3$ lattice translations such that balls of radius $8\mathsf{d}$ containing the translated pairs are mutually at distance at least d. For every such set,

$$S_2^T((0,f),(0,h)) = \frac{1}{N}\sum_{t\in\mathcal{T}} \partial_{z_1}\partial_{z_2} E_K(z_1 f^t + z_2 h^t)\Big|_{z=0} + \mathsf{R},$$

where the arguments of E_K denote the action-source slot and

$$|\mathsf{R}| \leq C_{\mathrm{R}}[\omega_{\mathrm{r}}(\mathsf{d}) + \mathsf{d}]\mathsf{d}\,\mathfrak{f} \leq \epsilon_{\mathrm{R}}(\mathsf{d})\mathsf{b}^{-3}\mathsf{d}\,\mathfrak{f}.$$

■

Proof. The witness-depth estimate gives $L^K\mathsf{d} > 4L^3M$, hence $a_K < \mathsf{d}/(4L^3M)$. The defining set in eq. 14.xiv contains $0, 1$, is finite, and contains every fixed integer when d is sufficiently small. Thus $m_* \geq 1$ and $m_* \to \infty$. Since $8c_V^+(m)\ell_G^3 \geq 128$,

$$L^{3m_*} \leq \frac{\log(1/\mathsf{d})}{128} \leq \mathsf{d}^{-3}, \qquad \mathsf{Q}_k \leq \mathsf{d}^{-1/8}.$$

It follows that $a_k \geq \mathsf{d}$ and $L^{m_*} \leq \mathsf{d}^{-1} < L^K$, so $1 \leq k \leq K-1$. Also $\mathsf{d} \leq \mathsf{Q}^{-8} \leq e^{-128}$. For $v \geq 128$, $v^{3/2}e^{-v/4} \leq 1$, which proves the remaining numerical bound. If $u = m_* \log L$, the logarithm of the second entry in eq. 14.xv is at most

$$\log(2M^{-3}\ell_G^3) + 3u - c_{\mathfrak{d}}A_0^2 u^{2\mathfrak{p}_0}.$$

It tends to $-\infty$ since $2\mathfrak{p}_0 \geq 2$. This proves the asserted properties of ϵ_{R}.

The history sum is finite and each term is entire. Lemma 14.5 identifies $Z_k(h) = \emptyset$ with restriction to the empty history, and sums all its extensions back to that history at level k. Its source-dependent density is therefore the density in (1). Indeed, the action source enters linearly, the remaining exponent is even, and all its terms commute. Subtracting the zero-source exponent gives exactly $\mathsf{T}_k(w)$.

We check that this identity can be evaluated on $\mathcal{S}_k$. Two backgrounds in the minimal orbit differ by a residual gauge transformation equal to one at the nodes, including every site of level k. Each plaquette action is gauge invariant; the tower and its legs transform covariantly; the activities are jointly gauge invariant; and the mass coefficient acts trivially on colour. Thus their values after reading the legs at level k are the same element of Gr_k at both backgrounds. Every $V \in \mathcal{S}_k$ has an adapted background by Lemma 14.8. Moreover it passes the level-k density test: the nonzero empty-history weight contains the factor $\chi_k(T_{\ell_G})(V) \in \{0,1\}$. The source flow is holomorphic on $\|w\|_\sharp < 4\varrho$ at the fixed regular-pair radii. Lemma 14.8 places these backgrounds in that domain. This proves (1), including the claimed holomorphy.

For each top history let $\mathsf{g}_h[w]$ denote its unnormalised density. Berezin integration and $h_F \geq 1$ give

$$|\mathsf{A}_k[w]| + |\mathsf{B}_k[w]| \leq \sum_h \int \|\mathsf{g}_h[w]\|_{h_F} \leq e^{\operatorname{Re} E_K(w) + (C+2)\ell_G^3} \leq |e^{E_K(w)}|M_0.$$

For the complementary sum, the history tail estimate at levels (K,k) gives the more precise bound

$$e^{-\operatorname{Re} E_K(w)} \sum_{Z_k(h)\neq\emptyset} \int \|\mathsf{g}_h[w]\|_{h_F} \leq 2M^{-3}\#\mathbb{T}^{(k)} e^{-c_{\mathfrak{d}}p_0(g_k)^2}\Pi_K(\mathsf{C}_{\mathrm{maj}} + \lambda_L, c_{\mathfrak{d}}). \tag{14.xvi}$$

Its hypotheses hold for the calibration frame: the increments are zero, the flow bound is $B_\sharp$, and $\log \mathsf{t}_K = \log \mathsf{b} \geq \log \gamma_{\mathrm{e}}^{-2} \geq \mathsf{Y}_{\mathrm{e}}$. The required age floor is the chosen floor when $c_a = 0$; when $c_a > 0$, it follows from the stronger age floor used for the source-history sum. The source pair has parameters $(c, \mathsf{C}_{\mathrm{maj}}, C_\flat^F + C'_{\mathfrak{E}})$. Finally,

$$\#\mathbb{T}^{(k)} = \ell_G^3 L^{3m_*}, \qquad \mathsf{t}_k \geq \mathsf{b}_- L^{m_*}, \qquad \Pi_K(\mathsf{C}_{\mathrm{maj}} + \lambda_L, c_{\mathfrak{d}}) \leq M_0.$$

The last inequality uses the product threshold included in Y_{e}. Combining eq. 14.xvi with the preceding ordinary norm bound proves the estimate with ω_{r}. At zero source every history contribution is

nonnegative. The fully empty top history is one of the extensions being summed, so its lower bound gives

$$e^{-E_K}\mathsf{A}_k[0] \geq e^{-C_P(1+\log x_K)\ell_G^3} \geq e^{-C_P(1+\log X_g)\ell_G^3} = M_0/\mathsf{Q}.$$

For later use we give the derivative estimate explicitly. The sharp norm of a sum of continuous functions with disjoint supports is at most the largest sharp norm of its summands when their coefficients have modulus at most one. For points in different supports, follow a shortest segment between them. The functions vanish at its first exit from the first support and its last entry into the second. Their difference is consequently bounded by the larger Lipschitz constant times the segment length. Points in the same support, or outside all supports, give the same conclusion. For coefficients of modulus one equality holds.

Set $R_1 = \|F\|_\circ^{-1}$ and $R_2 = \|H\|_\circ^{-1}$, omitting a zero direction. The bidisc $|z_i| \leq R_i$ at $w = z_1F + z_2H$ lies in the interior of $\mathcal{P}$. On it put $a = e^{-E_K}\mathsf{A}_k$ and $b = e^{-E_K}\mathsf{B}_k$. Cauchy's inequalities give

$$|a_i| \leq M_0/R_i, \quad |b_i| \leq \omega_{\mathrm{r}} M_0/R_i, \quad |a_{12}| \leq M_0/(R_1R_2), \quad |b_{12}| \leq \omega_{\mathrm{r}} M_0/(R_1R_2).$$

At zero, $a_0 \geq M_0/\mathsf{Q}$ and $0 \leq b_0 \leq \omega_{\mathrm{r}} M_0$. Differentiate $\log(a+b) - \log a$:

$$\mathsf{r}_1 = \frac{a_0b_1 - a_1b_0}{a_0(a_0+b_0)}$$

and

$$\mathsf{r}_2 = \frac{b_{12}}{a_0+b_0} - \frac{a_{12}b_0}{a_0(a_0+b_0)} + a_1a_2\left(\frac{1}{a_0^2} - \frac{1}{(a_0+b_0)^2}\right) - \frac{a_1b_2 + b_1a_2 + b_1b_2}{(a_0+b_0)^2}.$$

The displayed bounds, $\omega_{\mathrm{r}} \leq 1$ and $\mathsf{Q} \geq 1$ give respectively the constants 2 and 7 in (2).

We next justify the quotient and differentiation in (3). The set $\mathcal{S}_k$ is the image, under the Borel block map, of the Borel set where the history weight is nonzero. It is analytic, hence universally measurable. The density measure is supported there: every Borel set disjoint from $\mathcal{S}_k$ has zero pushforward measure. Inner regularity then gives zero mass to its complement in the completed measure. This argument applies to the norm measure as well, so extending a function by zero causes no ambiguity. At level k the norm and empty-history lower bounds give

$$\int \|\mathsf{g}_k\|_{h_F}\, d\mathscr{M}_k \leq e^{(C+2)\#\mathbb{T}^{(k)}}, \qquad \operatorname{Ber}\int \mathsf{g}_k\, d\mathscr{M}_k \geq e^{-C_P(1+\log x_k)\#\mathbb{T}^{(k)}}.$$

Since $x_k \leq X_g L^{m_*}$, their ratio is at most Q_k. Submultiplicativity of the fibre norm proves the quotient bound.

For differentiation let $G(z,V) = \mathsf{g}_k(V)e^{\mathsf{T}_k(\sum z_iF_i)(V)}$ on $\mathcal{S}_k$ and zero elsewhere. On a sufficiently small closed polydisc it is pointwise holomorphic and its norm integral is uniformly bounded by the source-density estimate. Apply Cauchy's formula on the boundary of a slightly larger polydisc. Tonelli's theorem makes the boundary norm integral finite, and Fubini's theorem permits its integration in V first. The Cauchy integrals therefore give both a holomorphic integral and its differentiated coefficients. The same boundary estimate proves integrability of the derivatives. No pointwise dominating function in a neighbourhood of zero is required.

We prove (4) using the exact derivative identities of Lemma 14.15 at $\lambda = 1$:

$$\mathsf{T}_k[f] = \mathsf{T}_k^w[f] + \sum_{l\leq k}\sum_X \mathbf{P}^{(l)}[f](X) - L^{-k}\mathsf{S}_1(\mathfrak{N}_k[f]\,\cdot),$$

$$\mathsf{T}_k[f,h] = \sum_{l\leq k}\sum_X \mathbf{Q}^{(l)}[f,h](X).$$

These hold on $\mathcal{S}_k$ by the density-test check above. The background row substitution in Lemma 14.8 has norm at most one. For any family G read at level k and the pair P of this background on the torus, it gives

$$\left\| \sum_{X} G(X; P|_X, \cdot) \circ \mathsf{H}(\mathsf{U}) \right\|_{h_F} \le \sum_{X} \|G(X; P|_X, \cdot)\|_{h^{(k)}}.$$

Thus the long-lag estimate, with the pair outside the sum, applies directly; it need not be replaced by a stronger termwise supremum.

The support ball contains at most $81(\mathsf{d}/a_K)^3$ bare plaquette barycentres, and background regularity gives $|a_p(\mathsf{U})| \le \alpha_0^2 L^{-4k}/8$. Hence

$$\|\mathsf{T}_k^{w}[f]\| \le (81\alpha_0^2/8)\|f\|_\infty \mathsf{d}^3 a_k^{-4} = (81\alpha_0^2/8) e_k[f].$$

Lemma 14.22, followed by the weighted total estimate in Lemma 14.8, bounds the old first responses by

$$8e^{\kappa/16} C_{\mathcal{C}} [1 + C_{\mathrm{rd}} C_{\mathcal{C}} (1 + 2C_{\mathrm{mo}}) S_\diamond] A_{\mathbf{P}} e_k[f] \mathfrak{l}_k^{4\mathfrak{p}_0+1}.$$

Lemma 14.21 bounds the mass term by $2C_{\mathcal{C}} A_{\mathfrak{N}} e_k[f] \mathfrak{l}_k^{4\mathfrak{p}_0+1}$. For the mixed response use Lemma 14.23 and the same total estimate, with factor $e^{\kappa/8} C_{\mathcal{C}}$; the count factor is bounded by $C_{\mathcal{C}}$ since $a_k \ge \mathsf{d}$. The definition of C_{T} now proves (4).

It remains to compare the second derivative of $\log Z$ with that of E_K. For a translated pair, write

$$\partial_f \partial_h (\log Z - E_K) = -\Delta^{(1)} + \Delta^{(2)} + \mathsf{r}_2(f,h), \qquad \Delta^{(1)} = \partial_f \partial_h (E_K - E_k), \quad \Delta^{(2)} = \partial_f \partial_h \log \mathsf{I}_k.$$

Lemma 14.24 and its geometric sum over $j > k$ yield

$$|\Delta^{(1)}| \le C_{\mathfrak{e},2} \mathfrak{f} \, \mathsf{d}^{5/2} a_k^{-9/2}.$$

By (3), evenness and $\mathsf{T}_k(0) = 0$,

$$\Delta^{(2)} = \langle \mathsf{T}_k[f,h] + \mathsf{T}_k[f]\mathsf{T}_k[h] \rangle_k - \langle \mathsf{T}_k[f] \rangle_k \langle \mathsf{T}_k[h] \rangle_k.$$

For $a_k \ge \mathsf{d}$,

$$e_k[f] e_k[h] \mathfrak{l}_k^{8\mathfrak{p}_0+2} = (\mathsf{d}/a_k)^{7/2} (1 + \log(a_k/\mathsf{d}))^{8\mathfrak{p}_0+2} e_k[f,h] \le C_{\mathfrak{l}} e_k[f,h].$$

It follows that

$$|\Delta^{(2)}| \le 2\mathsf{Q}_k^2 C_{\mathrm{T}}^2 (1 + C_{\mathfrak{l}}) e_k[f,h].$$

To bound the averaged r_2, put $F_\varepsilon = \sum_t \varepsilon_t f^t$ and $H_\varepsilon = \sum_t \varepsilon_t h^t$ for independent signs ε_t. The supports of all the summands, including the two within each pair, are disjoint. The sharp norm observation gives $\|F_\varepsilon\|_\sharp = \|f\|_\sharp$ and $\|H_\varepsilon\|_\sharp = \|h\|_\sharp$. Bilinearity and averaging the signs remove the cross terms:

$$\frac{1}{2^N} \sum_{\varepsilon} \mathsf{r}_2(F_\varepsilon, H_\varepsilon) = \sum_t \mathsf{r}_2(f^t, h^t).$$

Consequently, using $\|f\|_\sharp \le 40M\|f\|_\infty/(\vartheta \mathsf{d})$,

$$\left| \frac{1}{N} \sum_t \mathsf{r}_2(f^t, h^t) \right| \le 7 \cdot 24^3 \cdot 1600 M^2 \mathsf{Q}^3 \varrho^{-2} \vartheta^{-2} \ell_G^{-3} \omega_{\mathrm{r}} \mathsf{d} \, \mathfrak{f}.$$

Such translations exist. Set $\delta = a_K \lceil 17\mathsf{d}/a_K \rceil$ and $n_\delta = \lfloor \ell_G/\delta \rfloor$. Then $17\mathsf{d} \le \delta < 18\mathsf{d}$. Different points of $\delta\{0, \dots, n_\delta - 1\}^3$ have torus distance at least δ: in a differing coordinate the distance is $\min\{j\delta, \ell_G - j\delta\} \ge \delta$. The pair lies in a ball of radius $5\mathsf{d}$ about the midpoint of its two supporting ball centres, since those centres have distance at most $8\mathsf{d}$. The balls of radius $8\mathsf{d}$ about their translates therefore have distance at least d. Since $\ell_G \ge 2$ and $\mathsf{d} \le 1/400$,

$$n_\delta \ge \ell_G/(18\mathsf{d}) - 1 \ge \ell_G/(24\mathsf{d}).$$

Finally,

$$\mathsf{Q}_k^2 e_k[f,h] \le \mathfrak{f} \, \mathsf{d}^{9/4} \log(1/\mathsf{d})^{3/2} \le \mathfrak{f} \, \mathsf{d}^2,$$

and the same last bound holds for $\mathfrak{f}\,\mathsf{d}^{5/2}a_k^{-9/2}$. Average the identity for $\log Z - E_K$ over these translations; translation invariance leaves S_2^T unchanged. The three preceding estimates give the first remainder bound in (5). The second follows from $\mathsf{b} \le \mathsf{b}_+$. All constants are independent of $K, \mathsf{b}, \mathsf{m}$, the position of the pair and the chosen packing. □

14.11. **Reading a nonzero cell moment.** The quadratic cell moment need not vanish during source differentiation. We record the corresponding form of the old-activity estimate.

Lemma 14.13. Let $1 \le j \le k$ and $n = k - j$. Let F be an even family with legs of levels at most j, holomorphic and jointly gauge invariant on the whole regular pair core, with norm at most ϵ. If F is a sum of summands, use the sum of their norms and file each summand at a cell of its polymer. At the trivial pair, read its quadratic part in the level-j field Φ and write it as units β_u filed at cells $c(u)$, with

$$\sum_{u:c(u)=c} \sum_{y,y'} \|\beta_u(y,y')\|(1 + |y - z_c|_j + |y' - z_c|_j) \le C h_F^{-2} \epsilon.$$

Put $m_c = \sum_{u:c(u)=c} \sum_{y,y'} \beta_u(y,y')$. The units may move diagonal terms from the cell of a polymer X to the cell of y. Require each such term to be a linear combination of row and column sums at y of the quadratic kernels of $F_1(X)$ and $G(X)$, with sum of coefficient moduli at most two. Here $F = F_1 + F_2$, the families F_1, G have the same structural properties and norm bounds, and their background maps may use different fine mass matrices in the common admitted mass ball.

There is a constant C_ω, independent of the coupling and h_F, such that the local reading at level k satisfies

$$\|F_{\ge 2}\|_{\mathrm{read},k} \le C_\omega L^{n/4} \epsilon + C_\omega L^n h_F^2 \sup_c \|m_c\|. \tag{14.xvii}$$

Here the norm is the sum of localized terms meeting a level-k cube, with weight $e^{\kappa d_k(Y)}$, followed by the supremum over cubes. Degree at least four and the quadratic difference from its transported trivial-pair value have bound $C_\omega L^{-n}\epsilon$. For

$$n \ge n_{\mathrm{lag}} := \min\{n : 2(2A_\kappa C_H)^2 L^{-n} \le 1\},$$

the same estimates hold at the final-generator weight $C_H h_F$. At smaller lags the background legs remain formal at their own levels; only the fluctuation images are expanded in the step. The quadratic part is bounded whole, without this subtraction. The assertions are uniform on the constant smaller-radius systems and on the shifted pairs whose amplitude satisfies $|\mu| c_h \sup|B| \le c_{\mathrm{os}} a$, strictly at lag zero, at pair radius a.

On a torus the cell moments are formed piece by piece with the antiperiodic seam in the transport. If the family, its moved terms and its paths are the periodic images of an infinite-lattice family F^∞ wherever they fit, assume also that $F^\infty, F_1^\infty, G^\infty$ obey the same norm and unit bounds. Read them at the periodic lifts of the fine masses and require covariance under period translations, with the antiperiodic seam signs. The same bound then holds with the moments m_c^∞, which are independent of the chosen lift. The extra terms are filed on polymers that wrap around the torus. ■

Proof. The proof of [QCDM1, Lem. 7.9] separates the higher-degree part, the quadratic variation with the pair, and the quadratic value at the trivial pair. Its first two estimates use holomorphy, gauge invariance and the norm bound; they do not use the vanishing moment. Thus only the last step changes.

First note the weights when both stages of a leg are expanded into their local and walk terms. At the final weight $C_H h_F$, a leg costs at most $(2A_\kappa C_H)^2 L^{-n}$ times its old weight. At $n \ge n_{\mathrm{lag}}$, every leg beyond the fourth has a spare factor $1/2$. This pays the polynomial counts in the number of legs. Four legs and the L^{3n} old cells give $CL^{-n}\epsilon$. The same two-stage expansion in the quadratic gauge comparison gives its L^{-n} bound. Finitely many smaller lags are bounded by the plain cell count and absorbed into C_ω.

Split each moved diagonal term into the same two-walk pieces as the kernel from which it is formed. Such a piece is supported in the polymer enlarged by the two walks, and its site y belongs to that enlarged polymer. Place it with opposite signs in the original unit and the unit at the cell of y. The row and column sums obey the same positive walk bounds as the kernel itself, at either of the admitted fine masses. In particular, if Z_p is its enlarged polymer and m_p its constant row-sum contribution,

$$\sum_p \|m_p\|_{\mathrm{tr}} e^{\kappa(d_j(Z_p)-d_j(X))} \le C A_\kappa^2 h_F^{-2}(\|F(X)\| + \|F_1(X)\| + \|G(X)\|).$$

The same estimate holds for the absolute double sum of its kernel. The walk weights also pay the number of cells from which a moved piece can reach a fixed cell.

Transport both legs to z_c and write $\Phi(y) = \Phi(z_c) + [\Phi(y) - \Phi(z_c)]$. The terms with a difference are exactly those estimated in [QCDM1, Lem. 7.9]: the 3/4 Hölder gain and the first absolute moment give $C\epsilon L^{-11n/4}$ per old cell, hence $C\epsilon L^{n/4}$ per new cube. The terms without a difference now sum to $\bar\Phi(z_c) m_c \Phi(z_c)$ instead of zero. Their norm at a cell is at most $\nu\|m_c\|(C_H L^{-n} h_F)^2$. There are L^{3n} cells, so their sum has the second bound in eq. 14.xvii. To localize this member, expand its two legs at level k and file it at the cube of c enlarged successively by their walk polymers. The weighted walk sums and the counts of reached cubes cost only fixed factors.

The sum over polymers filed in a cube and the sum over units filed there differ only by moved pieces crossing its boundary. Exponential localization leaves $O(L^{2n})$ such pieces, each with the two-leg factor L^{-2n}. Their bound $C\epsilon$ is absorbed in the first term. The hull estimate $d_k(X^{\mathrm{up}}) \le L^{-n} d_j(X)$ and the spare old rate pay the path and diameter factors, as in the cited proof. This also proves the assertions for summands and for the larger final-generator weight.

On the torus, nonwrapping pieces and their lifts have the same constant row sums: the two seam signs cancel at z_c. Hence $m_c^{\mathrm{tor}} - m_c^\infty$ is the sum of wrapping pieces minus the sum of nonfitting lifted pieces. Each has $d_j(Z_p) \ge \ell_j/(2M) - 2$. The preceding walk estimate bounds this difference by $2C_{\mathrm{wr}} h_F^{-2} \epsilon e^{-\kappa(\ell_j/(2M)-2)}$. File its pieces at their coarse hulls, or the torus images of those hulls; these polymers wrap. Their localized sum is at most

$$C\epsilon L^n \sum_{d \ge \ell_j/(2M)-2} (1+d)^3 e^{-\kappa(1-L^{-n})d} \le C\epsilon,$$

since $\ell_j/(2M) \ge L^n$. The small lags are again covered by the plain count. This proves the torus assertion. The pair and shift domains are precisely those in the gauge comparison; reducing the radii rescales its numerator and its holomorphic radius by the same factor. □

14.12. **One complex scale for all gauge fluctuations.** We now multiply every gauge fluctuation by the same complex number λ. Fermion fluctuations are left unchanged. The coupling and mass numbers removed at each step are the numbers of the original flow. Their dependence on λ is placed in the action and mass term explicitly. This choice makes the composition recover the original bare action with inverse coupling β/λ^2.

Write $\mathsf{t}_i = L^{-i}\beta = \mathsf{b}/a_i$ and define

$$\xi_i(\lambda) = \mathsf{t}_i + \lambda^2(x_i - \mathsf{t}_i), \qquad \tau_i(\lambda) = \xi_i(\lambda)/\mathsf{t}_i, \qquad \bar\lambda_i^2 = \frac{2x_i}{c_\lambda (\log x_i)^{2\mathfrak{p}_0}}. \tag{14.xviii}$$

Let $D_i = \{|\lambda| < \bar\lambda_i\}$. The action at level i is $\lambda^{-2} A(L^i \xi_i(\lambda) - \lambda^2 a_K^{-1} w, \cdot)$; the mass term is $-L^{-i}\mathsf{S}_1(\lambda^2(s_i + \sigma_i(w,0))\,\cdot)$. The expression for the action is used through its fluctuation expansion at $\lambda = 0$.

The step keeps the tests and thresholds at t_i. Its Gaussian covariance is $\xi_i(\lambda)^{-1} C_i$, and its quadratic origin is

$$\mathcal{V}^{\mathrm{q}} = -\frac{1-\tau_i(\lambda)}{2} (\mathsf{t}_i^{1/2} B_i, \widehat{\mathsf{G}}\mathsf{t}_i^{1/2} B_i)_1.$$

Thus the action Hessian has coefficient $\xi_i(\lambda)$, while the gauge fixing has its calibration coefficient. The Gaussian normalisation introduces the scalar factor $\tau_i(\lambda)^{-\dim B_i/2}$.

Use the unscaled vertex functions $\mathsf{v}^{\mathrm{J}}(Y;P,\mathcal{A})$ and $\mathsf{v}^{\natural}(Y;P,\mathcal{A})$, $\natural = \mathrm{G},\mathrm{A}$, which do not depend on the calibration. For $B = \mathsf{t}_i^{1/2} B_i$ set

$$\begin{aligned} \mathbb{V}^{\mathrm{J},\lambda}(Y;P,B) &= \mathsf{v}^{\mathrm{J}}(Y;P,\lambda\mathsf{t}_i^{-1/2}B), \\ \mathbb{V}^{\natural,\lambda}(Y;P,B) &= -\mathsf{t}_i\lambda^{-2}\mathsf{v}^{\natural}(Y;P,\lambda\mathsf{t}_i^{-1/2}B), \qquad \natural = \mathrm{G},\mathrm{A}, \\ \mathcal{V}^{\mathrm{g},\lambda} &= \mathbb{V}^{\mathrm{J},\lambda} + \mathbb{V}^{\mathrm{G},\lambda} + \tau_i(\lambda)\mathbb{V}^{\mathrm{A},\lambda}. \end{aligned} \tag{14.xix}$$

The quotients extend holomorphically to zero, with value zero, since the two unscaled functions vanish to order three. The other vertices are the fermion vertex, the mass variation and all old-activity variations at

$$\mathsf{U}'(\lambda B) = e^{i\mathcal{A}(\lambda\mathsf{t}_i^{-1/2}B;\mathsf{U})}\mathsf{U}.$$

At nonzero source also include the linear, quadratic and higher-order terms of the source weight. Their sum is the potential $\mathfrak{V}_i(\lambda)$, with its quadratic source term assigned to the quadratic origin.

From the new activity remove the following quantities. Remove its vacuum energy and source energies at λ, including the scalar Gaussian factor $-\frac{1}{2}(\dim B_i/n_{i+1})\operatorname{Log}\tau_i(\lambda)$. Remove the original coupling shift $b_{i+1} = e_{\mathsf{N},i+1} + b^F_{i+1} + b^r_{i+1}$, and the original source mass shift multiplied by λ^2. Everything else remains in the activity. In particular, the scalar term $(b^r_{i+1}(\lambda) - b^r_{i+1})\mathsf{W}_1$ remains. If $\widehat{Z}_{i+1}(\lambda)$ is the cell-mean zeroth moment before removing the local mass term, the moment left afterwards is

$$Z_{i+1}(\lambda) = \widehat{Z}_{i+1}(\lambda) - \lambda^2\widehat{Z}_{i+1}(1). \tag{14.xx}$$

Indeed $\widehat{Z}_{i+1}(1) = -\mathfrak{s}_{i+1}\mu_{i+1}1$. These rules imply

$$\xi_{i+1}(\lambda) = L^{-1}\xi_i(\lambda) - \lambda^2 b_{i+1}.$$

Proposition 14.14 (The flow at a common field scale)**.** Assume $g \le \gamma_{\mathrm{e}}$ and the constant choices in 14.9. For a datum on the least torus, the preceding construction has the following properties.

(1) At zero source the step from level i is defined on D_i. Its covariance, potentials and activities are holomorphic there. The potentials satisfy $\mathfrak{V}_i(-\lambda,-B) = \mathfrak{V}_i(\lambda,B)$; the integrated activities, their pointed terms, vacuum energies and field-independent parts are even in λ. For every $\lambda_\circ \in [1,\bar{\lambda}_j)$ there are

$$4\varrho = \varepsilon_0(\lambda_\circ) \ge \cdots \ge \varepsilon_j(\lambda_\circ) > 0$$

such that steps $1,\dots,j$ are holomorphic in (λ,w) for $|\lambda| \le \lambda_\circ$ and $\|w\|_\sharp < \varepsilon_j(\lambda_\circ)$. The radii may depend on the datum. At $\lambda = 1$ the construction equals the original source flow, polymer by polymer and pointer by pointer. At that value it is also defined for every source in $\mathcal{P}$ by the original flow. At $\lambda = 0$ the potential vanishes; the gauge integral has only its calibration tests and the fermion integral is Gaussian at the background.

(2) Put $\eta = C_\eta c_\lambda^{-1/2}$ and use the norm with the supremum over λ inside the polymer sum. For each $1 \le i \le j$,

$$\|E^{(i)}_{\ge 2}\|_{i;D_{i-1}} \le \eta/2, \qquad \|F^{(i)}_0\|_{i;D_{i-1}} \le 1, \qquad \|\widehat{Z}_i(\lambda)\|_{\mathrm{tr}} \le C^{\psi}_{am}h_F^{-2}\eta. \tag{14.xxi}$$

The scalar bound also holds in the array norm at rate $\kappa+1$ and at every pointer. These bounds hold for both the torus and infinite-lattice families. $E^{(i)}_{\ge 2}$ and $\widehat{Z}_i$ vanish at $\lambda = 0$. At every zero-source step,

$$\epsilon_V + \epsilon_F \le c_{\mathrm{cl}}/4, \qquad \|\mathcal{V}^{\mathrm{q}}\|_{\star,\lambda_V} \le 5w_*/64.$$

The joined integral and its Gaussian normalising factor have logarithms given by their convergent cluster sums. Their scalar parts are nonzero, and the normalised step integral is therefore defined throughout this domain.

(3) For real $0 < \lambda \le 1$ and real sources in the indicated range, the steps compose exactly at every block field passing its density test. Up to a nonzero scalar independent of field and source, the level-j form is the composite chart integral over the calibration boxes for B_i, $i < j$, with each fluctuation entering the chart as λB_i. Its integrand contains the chart Jacobians and elimination factors, the gauge fixings with coefficients t_i/λ^2, and

$$e^{-A(\beta/\lambda^2 - a_K^{-1} w, U)} T^F_{h\emptyset}[U](e^{-S_F}).$$

There is no residual counterterm:

$$\lambda^{-2} L^i \xi_i(\lambda) + \sum_{l \le i} L^l b_l = \lambda^{-2}\beta. \tag{14.xxii}$$

At the same bare configuration U, with $\theta = b_F$,

$$T^F_{h\emptyset}[U](e^{-S_F}) = \prod_{i<j} \theta^{-\dim H_{i+1}} \det \Gamma_i[U]\, e^{-\langle \bar\Psi_j, \mathbb{D}_j[U]\Psi_j\rangle}.$$

Its value at $\Psi_j = 0$ is $\theta_j^{-\dim H_j} \det \mathcal{K}_j[U; m_0]$; its Berezin integral over Ψ_j is $\det(D[U] + m_0)$.

■

Proof. We first record the scale comparisons. The windows $|x_i - \mathsf{t}_i| \le \bar B \le \mathsf{t}_i/4$ give $x_i \ge x_j \ge x_K \ge g^{-2}$ for $i \le j$. Since $x(\log x)^{-2\mathfrak{p}_0}$ increases for $x \ge e^{2\mathfrak{p}_0}$, the discs D_i decrease with i. For $1 \le l \le i$ and $n = i + 1 - l$,

$$\frac{3}{5} L^n x_i \le x_{l-1} \le \frac{5}{3} L^n x_i.$$

Using $\log x_i \ge 18\mathfrak{p}_0$, we obtain

$$\left(\frac{\bar\lambda_i}{\bar\lambda_{l-1}}\right)^2 \le 2L^{-n}(1 + n\log L)^{2\mathfrak{p}_0}, \qquad \left(\frac{\bar\lambda_i}{\bar\lambda_{l-1}}\right)^2 \le 2^{10/9} L^{-8n/9} < 1/4. \tag{14.xxiii}$$

For the second inequality use $(1 + (\log 2 + n\log L)/(18\mathfrak{p}_0))^{2\mathfrak{p}_0} \le 2^{1/9} L^{n/9}$ and $L \ge 13$. The same windows give

$$\bar\lambda_i \delta(\mathsf{t}_i) \le A_1 2^{\mathfrak{p}_0+1} c_\lambda^{-1/2}, \qquad \bar\lambda_i^{-2} \le c_\lambda (\log x_i)^{2\mathfrak{p}_0} a_i/\mathsf{b}.$$

Thus the fluctuation on the field polydisc satisfies $\sup |\lambda \mathsf{t}_i^{-1/2} B| < \beta_\lambda$.

The scaled coefficient stays in the coefficient disc of the one-step expansion:

$$|\xi_i(\lambda) - x_i| \le (1 + \bar\lambda_i^2)\bar B \le 2\bar\lambda_i^2 \bar B \le r_x x_i.$$

Moreover,

$$|\tau_i(\lambda) - 1| \le \frac{5}{2}\bar B/c_\lambda \le \frac{5}{8} r_x \le \min\{5\sigma_\star/32,\, 5/64\}.$$

Hence $|\tau_i| \le 3/2$, $|\tau_i^{-1}| < 2$ and $|\xi_i| > \mathsf{t}_i/2$. The complete and decoupled covariances are the corresponding calibration covariances multiplied by τ_i^{-1}. They retain their decay bounds, and

$$\|\mathcal{V}^{\mathrm{q}}\|_{\star,\lambda_V} \le |1 - \tau_i| K_{\mathsf{G}} \le 5\sigma_\star K_{\mathsf{G}}/32 \le 5w_*/64.$$

The covariance, the quadratic origin and its Gaussian cluster sum are holomorphic in the coefficient on this disc, and therefore even and holomorphic in λ. Also $\operatorname{Re}\tau_i > 0$, so its scalar logarithm is defined.

We next establish the vertex bounds on the whole field polydisc. This requires the fixed calibration $\mathsf{t}_\flat$, rather than extrapolation from the real-coupling estimate. Let

$$r_\flat = \mathsf{C}_{\mathcal{A}} \mathfrak{b}(\mathsf{t}_\flat), \quad r = \mathsf{C}_{\mathcal{A}} \mathfrak{b}(\mathsf{t}_i), \quad \mu = \lambda(\mathsf{t}_\flat/\mathsf{t}_i)^{1/2}, \quad R_i = \delta(\mathsf{t}_\flat)/\delta(\mathsf{t}_i).$$

Then $r_\flat \le r$, $R_i \ge \bar\lambda_i$, and $|\mu| r/r_\flat = |\lambda|/R_i < 1$. At $\mathfrak{t}_\flat$ the gauge vertices are holomorphic on the polydisc of radius $r_\flat$ and satisfy

$$|\mathbb{V}^\natural[\mathfrak{t}_\flat](Y;P,B')| \le K_P \delta(\mathfrak{t}_\flat) e^{-(\bar\kappa-2)d_i(Y)} Q_Y(B'), \qquad Q_Y(B') = 1 + M^{-3} \sum_{b\in \mathfrak{X}^{\circ+}\langle Y\rangle} |B'(b)|^2.$$

All vanish at zero; the G, A vertices vanish to order three.

For clarity, the Schwarz estimates used to rescale this bound are

$$\begin{aligned} |f(\mu B)| &\le \frac{|\mu| r}{r_\flat} K Q_Y(B) && \text{if } f(0)=0,\ |f| \le K Q_Y, \\ |f(\mu B)| &\le \frac{|\mu| r}{r_\flat} K && \text{if } f(0)=0,\ |f| \le K, \\ |\mu^{-2} f(\mu B)| &\le \frac{|\mu| r}{r_\flat} K(1 + M^3 r_\flat^{-2})(Q_Y(B)-1) && \text{if } f = O(B^3),\ |f| \le K Q_Y. \end{aligned} \tag{14.xxiv}$$

For the first two, apply Schwarz's lemma to $f(\zeta r_\flat B/r)$ on the unit disc. For the third, let $\rho_B = \max_b |B(b)|$ and $R = r_\flat/\rho_B$. The maximum principle for $\zeta^{-3} f(\zeta B)$ gives

$$|\mu^{-2} f(\mu B)| \le |\mu| K[R^{-3} + R^{-1}(Q_Y(B)-1)].$$

Now use $R^{-1} \le r/r_\flat$ and $R^{-2} \le M^3 r_\flat^{-2}(Q_Y(B)-1)$. The case $B=0$ follows by continuity.

The unscaled vertex definitions imply

$$\mathbb{V}^{\mathrm{J},\lambda} = \mathbb{V}^{\mathrm{J}}[\mathfrak{t}_\flat](\mu B), \qquad \mathbb{V}^{\natural,\lambda} = \mu^{-2}\mathbb{V}^{\natural}[\mathfrak{t}_\flat](\mu B).$$

Since $M^3 r_\flat^{-2} \le 1/9$, (because $\mathfrak{b}(\mathfrak{t}_\flat)^2 \ge M^3$ and $\mathsf{C}_{\mathcal{A}} \ge 3$), eq. 14.xxiv gives

$$|\mathcal{V}^{\mathrm{g},\lambda}(Y)| \le 4 K_P A_1 2^{\mathfrak{p}_0+1} c_\lambda^{-1/2} Q_Y(B) e^{-(\kappa+1)d_i(Y)}.$$

The quotients at $\mu = 0$ are jointly holomorphic: Cauchy's formula for $\zeta^{-2} f(\zeta B)$ proves this also in the pair variables. The vertices retain joint gauge invariance, translation covariance and reality.

The bound $\beta_\lambda \le \beta_0/2$ places the completed field in the regular complex domain for the fermion tower. At a step after level i, it is of the form $e^{iX}U_1$ with $|X| \le \alpha_1 L^{-i} \le \alpha_1^* L^{-(i+1)}$. For the first step this uses $2C_{\mathrm{cf}}\beta_0 \le \min\{r_P^\circ, \alpha_1/2\}$; for later steps use $c_h\beta_0 \le \alpha_1/2$. Thus the fluctuation operators, covariances, background maps and normalisers are holomorphic with their regular complex-domain bounds. The real part of each fluctuation operator keeps a positive floor.

The variation estimates consequently give

$$\begin{aligned} \|\mathcal{V}^{F,\lambda}\| &\le C_V h_F^2 \beta_\lambda, \\ \|\mathcal{V}^{\mathrm{s},\lambda}\| &\le (c_\psi + C_G\beta_\lambda) c_{\mathsf{S}} h_F^2 |\lambda|^2 \|s_i + \sigma_i\|_\infty, \\ \|\mathcal{V}^{\mathrm{old},\lambda}_{\ge 2}\| &\le [c_\psi + (c_h/a_{\mathrm{an}} + C_G)\beta_\lambda] \varsigma_i(\lambda), \\ |\mathcal{V}^{\mathrm{old},\lambda}_0| &\le c_h \beta_\lambda K_{\mathrm{old}}/a_{\mathrm{an}}. \end{aligned}$$

Here ς_i is the sum of the old fermionic families read at the step. We justify the scalar estimate separately. Its bracket localisation is evaluated at $(P, \lambda \mathfrak{t}_i^{-1/2} B)$. The same point lies in the domain at $\mathfrak{t}_\flat$. The scalar localisation has the same value there, since the calibration enters its definition only through the domain. The absolute system remains admissible: only its radius condition reads the calibration, and that condition is precisely $(1+\beta_s)\delta(\mathfrak{t}_\flat) \le \mathsf{s}^\star \alpha_0$. Its other properties are those of the original step. At $\mathfrak{t}_\flat$ its bound is

$$K_\Lambda \frac{(1+\beta_s)\delta(\mathfrak{t}_\flat)}{\alpha_0} N_{\mathfrak{a},\omega}(\mathfrak{f}_{\circ,\lambda}) e^{-(1-2\delta_{\mathrm{w}})\kappa d_i(Y)}, \qquad \omega_n = L^{-(n-1)}.$$

It vanishes at zero fluctuation. Apply the second estimate in eq. 14.xxiv to obtain

$$|\mathcal{V}_0^{\mathrm{old},\lambda}(Y;P,B)| \le |\lambda|\frac{c_h\delta(\mathtt{t}_i)}{a_{\mathrm{an}}}K_{\mathrm{old}}e^{-(1-2\delta_{\mathrm{w}})\kappa d_i(Y)} \le \frac{c_h\beta_\lambda}{a_{\mathrm{an}}}K_{\mathrm{old}}e^{-(1-2\delta_{\mathrm{w}})\kappa d_i(Y)}. \tag{14.xxv}$$

The array bound needed here follows from eq. 14.xxi: the fixed normaliser terms and counterterms do not read the coefficient, the Gaussian term stays in its coefficient disc, and the fluctuation family has norm at most one. Joint holomorphy follows from the holomorphy of the localisation in the field and in the array parameters, the uniform bound, and composition with $(\lambda, B) \mapsto (\lambda, \mu B)$.

At zero source the mass window gives $|\lambda^2 s_i| \le 2\bar{B}_M/c_\lambda$. Therefore

$$\epsilon_V \le C_V^\lambda c_\lambda^{-1/2}, \qquad \epsilon_F \le C_F^\lambda c_\lambda^{-1/2} + c'_\psi \varsigma_i(\lambda). \tag{14.xxvi}$$

We now close the induction in (2). For an old family at level $l \le i$, put $n = i+1-l$. Its long-lag estimate, with its polymers as units, is

$$\varsigma_i(\lambda) \le \sum_{l\le i} C_\omega[L^{n/4}\|E_{\ge 2}^{(l)}(\lambda)\|_l + L^n h_F^2 M^3 \|Z_l(\lambda)\|_{\mathrm{tr}}].$$

Apply Lemma 14.13 with $j = l$, $k = i+1$, and the original polymer units. These families are even, holomorphic and gauge invariant on the whole regular core; the background walk estimates give the required first absolute moment. At long lags use its transported-leg decomposition, including its final-generator weight $C_H h_F$; at shorter lags use its direct reading. No diagonal term is moved in these units, so the additional row and column conditions are vacuous. Translation covariance gives the cell sum $M^3 Z_l(\lambda)$. On the torus the infinite-lattice cell sum is used and the remainder is included in C_ω.

The induction hypothesis, evenness, and Schwarz's lemma in λ^2 give

$$\|E_{\ge 2}^{(l)}(\lambda)\|_l \le \eta L^{-n}(1 + n\log L)^{2\mathfrak{p}_0}.$$

Expand $\widehat{Z}_l = \sum_{m\ge 1} z_m \lambda^{2m}$. Its coefficient bound is $\|z_m\|_{\mathrm{tr}} \le C_{am}^\psi h_F^{-2}\eta\bar{\lambda}_{l-1}^{-2m}$. The order-two term cancels in eq. 14.xx. With $q = (\max\{1, |\lambda|\}/\bar{\lambda}_{l-1})^2 < 1/4$,

$$\|Z_l(\lambda)\|_{\mathrm{tr}} \le 2C_{am}^\psi h_F^{-2}\eta \sum_{m\ge 2} q^m \le 12 C_{am}^\psi h_F^{-2}\eta L^{-2n}(1+n\log L)^{4\mathfrak{p}_0}.$$

Consequently,

$$\varsigma_i(\lambda) \le (1 + 12M^3 C_{am}^\psi) C_\omega S_\omega \eta \le 3K_\mu C_\omega S_\omega \eta.$$

Here $K_\mu \ge 1 + 4M^3 C_{am}^\psi$. The original contraction $2\sqrt{2}\theta_E C_\omega S_\omega \le 1/2$, with $\theta_E = \sqrt{2}K_\mu K_M c'_\psi$, yields $K_M c'_\psi \varsigma_i \le 3\eta/8$. Together with eq. 14.xxvi this proves $\epsilon_V + \epsilon_F \le c_{\mathrm{cl}}/4$.

Apply the holomorphic expansion with two independent parameters (λ, x'), where $|x' - x_i| < r_x x_i$. Replace $\tau_i(\lambda)$ in the action vertex by $x'/\mathtt{t}_i$. All the preceding estimates are uniform there, since $|x'/\mathtt{t}_i| < 3/2$. After forming the expansion, evaluate at $x' = \xi_i(\lambda)$, an interior point of the disc. The new activity is the logarithm of the joined integral: the logarithm of the normalised interacting quotient plus the logarithm of the tests alone. The latter is scalar. The Gaussian quadratic-origin cluster sum is kept separately and restored in the scalar normalisation. Thus neither the tests nor their Gaussian factor have been dropped. The cluster representations give their nonvanishing scalar parts.

The expansion and the tail ceiling imply

$$\|\widetilde{E}_{\ge 2}^{(i+1)}\|_{i+1;D_i} \le K_M(C_V^\lambda + C_F^\lambda + 1)c_\lambda^{-1/2} + 3\eta/8.$$

The removed local mass has norm at most $6Lc_{\mathsf{S}}h_F^2\bar{B}_M/c_\lambda$: indeed $|\mu_{i+1}| \le 3L\bar{B}_M/x_i$ and $|\lambda|^2 \le 2x_i/c_\lambda$. The definition of C_η therefore gives $\|E_{\ge 2}^{(i+1)}\| \le \eta/8 + 3\eta/8 = \eta/2$. The moment estimate gives $\|\widehat{Z}_{i+1}\|_{\mathrm{tr}} \le C_{am}^\psi h_F^{-2}\eta$. At $\lambda = 0$ the potential is zero. Only the scalar test activity remains, so both fermionic quantities vanish.

The scalar and pointed bounds require their own admission check. Put $\bar\epsilon_V = C_V^\lambda c_\lambda^{-1/2}$, $\bar\epsilon_F = (C_F^\lambda + C_\eta)c_\lambda^{-1/2}$ and $p_i = e^{-\gamma_{\mathrm{tail}}\mathfrak{b}(\mathsf{t}_i)^2/2}$. The four scalar vertex families have norms bounded by

$$K_P|\lambda|\delta(\mathsf{t}_i), \quad \frac{10}{9}K_P|\lambda|\delta(\mathsf{t}_i), \quad \frac{10}{9}K_P|\lambda|\delta(\mathsf{t}_i), \quad c_h\beta_\lambda K_{\mathrm{old}}/a_{\mathrm{an}}.$$

Their admission sizes satisfy

$$\mathsf{n}^{\mathrm{int}} \le \frac{34}{9}K_P|\lambda|\delta(\mathsf{t}_i), \qquad \mathsf{n}' \le c_h\beta_\lambda K_{\mathrm{old}}/a_{\mathrm{an}}, \qquad \mathsf{n}^{\mathrm{int}} + \mathsf{n}' \le \bar\epsilon_V \le c_{\mathrm{cl}}/4 = w_*/4.$$

They are holomorphic on the field polydisc, the first and fourth vanish at zero, and the middle two vanish to order three. The pointed scalar expansion therefore gives

$$|\mathbf{F}^{\mathrm{v}}(X,P,z;\lambda)| \le K_{\mathbf{M}}^{\mathrm{v}}\left(\frac{9}{32}K_P^{-1}\mathsf{n}^{\mathrm{int}} + \frac{1}{9}K_\Lambda^{-1}\mathsf{n}' + p_i\right)e^{-\kappa_+ d_{i+1}(X)} \le K_{\mathbf{M}}^{\mathrm{v}}(\bar\epsilon_V + p_i)e^{-\kappa_+ d_{i+1}(X)}. \tag{14.xxvii}$$

The fermion insertion at a fixed pointer is bounded by the fermion potential termwise, so its pointed activity has norm at most $K'_M\bar\epsilon_F e^{-\kappa_+ d_{i+1}(X)}$. Summing with $\sup_\square \sum_{X\supset\square} e^{-(\kappa_+-\kappa)d_{i+1}(X)} \le C_E$ gives $K_M^\bullet(\bar\epsilon_V + \bar\epsilon_F + p_i)$, where $K_M^\bullet = K_M + C_E K_{\mathbf{M}}^{\mathrm{v}}$. The same bound holds in the array norm because $\kappa_+ \ge \kappa + 1$, and on the infinite lattice. All suprema over λ remain inside these sums.

Removing the scalar value at the trivial configuration costs at most a factor two, for the family and at every pointer. Translation covariance under $M\mathbb{Z}^3$ identifies the value removed on a cube with the sum of its site values. Removing the fixed coupling term adds at most $C_w\phi(x_i)$: the normalised action family has norm at most one in both norms. Hence

$$\|F_0^{(i+1)}\|_{i+1;D_i} \le 2K_M^\bullet(C_V^\lambda + C_F^\lambda + C_\eta + 2)c_\lambda^{-1/2} + C_w\phi(x_i) \le 1.$$

This completes the induction.

All expansions converge uniformly with these bounds. Their summands are jointly holomorphic in (λ, x', P), as are the scalar localisations and the pointed terms. The latter can also be formed by inserting a pointer parameter into the same expansion. For the scalar pointed terms, the four admitted families are jointly holomorphic and obey the admission bounds uniformly, so the scalar pointing theorem applies with the same extra parameters. The substitution $(\lambda, B) \mapsto (-\lambda, -B)$ fixes the covariance, tests and quadratic origin and interchanges the potentials at λ and $-\lambda$. The expansion and its pointing commute with this orthogonal change of variables. Thus their outputs are even in λ. At $\lambda = 1$ all the covariance, vertex, extraction and pointing rules coincide with those of the original step. Induction proves equality there.

For completeness, we construct the source neighbourhood. Its relative weight at step i is

$$v_i^\lambda = -\lambda^2 L^{K-i}w/\xi_i(\lambda), \qquad |v_i^\lambda| \le 2\lambda_\circ^2\|w\|_\infty/\mathsf{b}.$$

The quadratic source origin has norm $\le C_v^{\mathrm{q}} \sup|v_i^\lambda|$. The linear term has size at most

$$2\alpha_0 C_{\mathrm{lin}} c_\lambda^{-1/2}\|w\|_\sharp.$$

The higher source vertices and $\lambda^2\sigma_i(w,0)$ are holomorphic and vanish at $w = 0$. On a fixed finite datum they are bounded on $|\lambda| \le \lambda_\circ$, $\|w\|_\sharp < 4\varrho$.

Suppose the preceding steps are formed on a source ball of radius ε_i. Their activities are uniformly bounded there by the expansion estimate. The differences between the next potential and its zero-source value, and between its quadratic origin and $\mathcal{V}^{\mathrm{q}}$, therefore have a finite bound $S_i(\lambda_\circ)$. This includes the old-family variations, read by their finite-lag estimates. Schwarz's lemma permits the choice

$$\varepsilon_{i+1} = \min\left\{\frac{\varepsilon_i}{2}, \frac{\varepsilon_i}{S_i}\min\{c_{\mathrm{cl}}/4, w_*/8\}\right\},$$

with the second entry omitted if $S_i = 0$. On this smaller ball the potential size is at most $c_{\mathrm{cl}}/2$ and the quadratic-origin norm at most $5w_*/64 + w_*/8 < w_*/4$. The expansion is thus jointly

holomorphic. Its parity follows as before, including the linear source term, which is odd in both λ and B. Starting with $\varepsilon_0 = 4\varrho$ proves (1).

It remains to verify the exact identity in (3). Let $0 < \lambda \le 1$ be real and let W pass the output density test. For B in the calibration box, λB remains in that box. The chart at λB is defined, its fluctuation test equals one, and its input block field $V_W(\lambda B)$ passes the input density test. The vertex sum identities hold at this point. In the scaled fluctuation coordinates they read

$$\sum_Y \mathcal{V}^{\mathrm{g},\lambda}(Y; P_W, B) = \mathsf{J}_W(\lambda B) - \frac{\mathsf{t}_i}{\lambda^2} G_W(\lambda B) - \frac{\xi_i(\lambda)}{\lambda^2}[\mathbb{F}_W(\lambda B) - \mathbb{F}_W(0)] + \frac{1}{2}(B, \widehat{\mathsf{G}}_W B)_1 + \frac{\tau_i(\lambda)}{2}(B, \mathsf{B}^\circ B)_1.$$

With $\mathsf{A}^\circ = \mathsf{B}^\circ + \widehat{\mathsf{G}}_W$ and $C^\circ = (\mathsf{A}^\circ)^{-1}$,

$$e^{-\frac{1}{2}(B,\widehat{\mathsf{G}}_W B)_1 - \frac{\tau}{2}(B,\mathsf{B}^\circ B)_1}\, dB = \tau^{-\dim B_i/2} \left(\int e^{-\frac{1}{2}(B_1, \mathsf{A}^\circ B_1)_1}\, dB_1 \right) \cdot d\mu_{\tau^{-1}C^\circ}(B) e^{\mathcal{V}^{\mathrm{q}}(B)}.$$

Thus the gauge part has precisely the announced Jacobian, gauge fixing and action weight, including its scalar Gaussian factor.

The fermion substitution and shift give the fermion vertex and its normaliser at the same field $\mathsf{U}'(\lambda B)$. The old-family and mass variations add to the differences between their values at that field and the background. The bracket sum identity ensures this also for the scalar localisation. At nonzero source the terms of orders one, two and at least three add to the full source weight on the fibre. The Gaussian quadratic-origin sum is restored to the level-$i+1$ scalar activity and energy, including at nonzero source. The remaining joined integral is the exponential of the new activity sum. Removing the prescribed numbers changes only their placement in the level-$i+1$ form, and

$$\lambda^{-2} L^i \xi_i(\lambda) = \lambda^{-2} L^{i+1} \xi_{i+1}(\lambda) + L^{i+1} b_{i+1}.$$

This proves the one-step identity up to a nonzero constant independent of field and source.

At level zero the form is $e^{-A(\beta/\lambda^2 - a_K^{-1} w, U) - S_F(U)}$. Fubini's theorem composes the finite chart integrals. Each chart lies in the fibre over its output block field, so all kernels are evaluated at one bare configuration U. Their Berezin integrals compose to $T^F_{h\emptyset}[U]$. Completing the square at each step has Berezin Jacobian one and produces $\theta^{-\dim H_{i+1}} \det \Gamma_i[U]$. The determinant identity in Proposition 14.37 gives its value at $\Psi_j = 0$; its hypothesis $m_0 \le L^{-j}$ follows from $m_0 = L^{-K}\mathsf{m} \le L^{-K}/2$. Integrating the remaining generators gives the original bare determinant. Finally the coupling identities telescope to eq. 14.xxii, while the mass members start at $s_0 = \sigma_0 = 0$. This proves (3). □

14.13. **The response recursion gives the source derivatives.** The response families need not coincide polymer by polymer with derivatives of the original flow. Their sums do coincide after reading all legs at the current level. This is the identity needed for source energies and observables.

Lemma 14.15 (Derivatives of the flow)**.** In the setting of Proposition 14.14, the objects of Definition 14.6 at level l are holomorphic and even on D_{l-1}. The input directions satisfy $\delta_i[\mathsf{g}](-\lambda, -B) = \delta_i[\mathsf{g}](\lambda, B)$. On $|\lambda| < \bar{\lambda}_j$, for $l \le j$, the following identities hold.

(1) For every direction g,

$$\sum_X \partial_z \mathfrak{E}_l(X; z\mathsf{g}, 0)|_{z=0} = \sum_X \mathfrak{e}_l[\mathsf{g}](X).$$

The derivative of the field-dependent activities through level l, together with the mass term, is

$$\sum_{l'\le l}\sum_{X}\mathbf{P}^{(l')}[\mathsf{g}](X)-L^{-l}\mathsf{S}_1(\lambda^4\mathfrak{N}_l[\mathsf{g}]\,\cdot). \tag{14.xxviii}$$

This is an identity of functions of the fields at level l, at backgrounds of real block fields passing the density test, including the trivial block field. All formal legs of lower levels are read at level l. The mass coefficients satisfy

$$\lambda^2\partial_{\mathsf{g}}\sigma_l(x)=\lambda^4\mathfrak{N}_l[\mathsf{g}](x)-\sum_{l'\le l}L^{l-l'}(\lambda^4\mathfrak{n}_{l'}[\mathsf{g}]-\lambda^2\partial_{\mathsf{g}}\nu_{l'})(y_{l'}(x)).$$

The direct action-source term stands beside eq. 14.xxviii; its derivative is $a_K^{-1}\sum_p \mathsf{g}(p)a_p(\mathsf{U})$.

(2) For the pair (f,h),

$$\mathfrak{n}_l^{\lambda}=\sum_X\partial_{z_1}\partial_{z_2}\mathfrak{E}_l(X;z_1f+z_2h,0)|_{z=0}=\sum_X\mathfrak{e}_l[f,h](X).$$

The mixed derivative of the field-dependent terms is $\sum_{l'\le l}\sum_X\mathbf{Q}^{(l')}[f,h](X)$, in the same sense as (1). The mixed derivative of the direct action-source term is zero.

(3) On $\mathbb{Z}^3$, in the constant direction 1, $\mathfrak{n}_l^{\infty}[1]$ and $\mathfrak{N}_l^{\infty}[1]$ are invariant under translations by $M\mathbb{Z}^3$ in level-l units. For the constant-source chain,

$$\partial_c\sigma_k^{\mathrm{c}}(0)=\sum_{l\le k}L^{k-l}\partial_c\nu_l^{\infty},\qquad \partial_c\nu_l^{\infty}=-\mathfrak{s}_l^{-1}\sum_{X,p}\mathrm{ord}_0[\mathsf{k}_{X,p}](y).$$

In the last expression the kernels are those of $\widetilde{\mathbf{P}}^{\infty(l)}[1]$ at $\lambda=1$. The expression has the same value at every site y.

■

Proof. The recursion is first justified level by level. The derivatives of the expansion in Lemma 14.16 are defined and holomorphic for directions of finite size. The direct source direction has finite size by Lemma 14.19; the old scalar and fermionic variations have finite size by Lemma 14.18. The moments of a quadratic family at a site are bounded by $4A_\kappa^2h_F^{-2}$ times its fibre norm at the trivial pair. Cauchy's estimate on D_{l-1} bounds their λ^4 coefficient. Thus $\mathfrak{n}_l,\mathfrak{N}_i$ are finite, as are their mass-variation directions. Removing scalar values costs at most another copy of the norm, and the extracted forms cost at most $K_\mu^{\mathrm{s}}-K_\mu$ copies, by the piecewise mass-extraction estimate. The coefficient bound $|\lambda^4a_{X,p}^{[4]}|\le\sup_{D_{l-1}}|a_{X,p}|$ is sufficient here. These estimates involve only preceding levels, so they close a finite induction without using the quantitative response bounds to be proved later.

All localisations and background walk terms are holomorphic in λ; finite differences and moment extraction are linear operations. The expansion commutes with $B\mapsto -B$. Its input directions at $(-\lambda,-B)$ equal those at (λ,B), because the direct vertex reads λB, the mass direction has the factor λ^4, and the preceding families are even. Induction proves the asserted holomorphy and parity.

An arbitrary plaquette direction can be inserted into the original Lipschitz source flow. Define

$$\varphi_p(x)=\max\{0,1-2\sqrt{2}\,|x-x(p)|/a_K\},\qquad \widehat{\mathsf{g}}=\sum_p\mathsf{g}(p)\varphi_p.$$

Distinct barycentres have distance at least $a_K/\sqrt{2}$, so the open supports of these tents are disjoint. Thus

$$\widehat{\mathsf{g}}(x(p))=\mathsf{g}(p),\quad \|\widehat{\mathsf{g}}\|_\infty=\|\mathsf{g}\|_\infty,\quad \mathrm{Lip}\,\widehat{\mathsf{g}}\le 2\sqrt{2}\,a_K^{-1}\|\mathsf{g}\|_\infty,\quad \|z\widehat{\mathsf{g}}\|_\sharp\le 80\sqrt{2}\,Ma_K^{-1}|z|\|\mathsf{g}\|_\infty.$$

The source neighbourhood in Proposition 14.14 therefore admits this direction. Differentiating there shows that the actual source-flow derivatives are holomorphic and even in λ. Since $\lambda_\circ<\bar\lambda_j$ was arbitrary, this holds on the claimed disc.

We prove (1) by induction on l. The assertion is empty at level zero. First, the usual chain rule applies to the cluster expansion with source-dependent origins. Indeed, write their joint input as $\mathcal{W}(z) = \mathcal{W}(0) + z\dot{\mathcal{W}} + z^2 R(z)$. The expansion at $\mathcal{W}(0) + t\dot{\mathcal{W}} + sR(z)$ is holomorphic in (t, s, z). On $(t, s) = (z, z^2)$, differentiation at zero gives $D\widetilde{E}_i[\dot{\mathcal{W}}]$.

Second, totals depend only on totals of the directions. The sum of $\widetilde{E}_i[\mathcal{V}; \mathcal{V}^{\mathrm{o}}]$ is the logarithm of the joined integral with weight $e^{\mathcal{V}^{\mathrm{o}}}$, tests, and $e^{\mathcal{V}}$. The Gaussian cluster sum restores precisely the normalising denominator of the joined integral. This sum therefore reads $\sum_Y (\mathcal{V}(Y) + \mathcal{V}^{\mathrm{o}}(Y))$ alone. Two inputs with the same total have equal exponentials of their activity sums. Their continuous logarithms agree at the base point, so their difference, a constant in $2\pi i\mathbb{Z}$, is zero on the common parameter disc. Their first and mixed derivatives agree. Reading the formal legs at level $i+1$ is a unital Grassmann-algebra homomorphism which fixes the fluctuation generators. It commutes with their Berezin integral, the exponential and the logarithm. Thus the same assertion holds after reading the legs.

The localisations in Definition 14.6 have the required totals. In degree zero the bracket localisation sums to the variation of the scalar array; its equal shares sum to the original family. For fermionic terms, the finite differences sum to the term at full shift, and the terms in the long-lag decomposition sum to the family at every shift amplitude. Consequently the three directions sum respectively to the direct source variation, the mass variation, and the variation of the old family total. This is first an identity at real $0 < \lambda \leq 1$, where λB passes the tests, and then on D_i by holomorphy.

At the background of a block field W passing the output test, the field $\mathsf{U}'(\lambda B)$ for real tested fluctuation is, up to the site gauge factor, the background of its input block field. That input block field passes its density test. The same is true at $B = 0$. Joint gauge invariance and the induction hypothesis therefore identify the derivatives of the actual field-dependent exponent with the response totals at both ends of the variation. It follows that the derivative of the actual step input has the total of $\delta_i[\mathrm{g}]$. The preceding logarithm argument identifies the total of its new activity derivative with $\widetilde{\mathbf{P}}^{(i+1)}[\mathrm{g}]$. At the trivial block field this proves the identity for the scalar source energy.

The mass bookkeeping is exact even though its extracted coefficients differ from the response coefficients. The actual derivative of the new activity has total

$$\mathbf{P}^{(i+1)}[\mathrm{g}] + L^{-(i+1)}\mathsf{S}_1((\lambda^2 \partial_{\mathrm{g}} \nu_{i+1} - \lambda^4 \mathfrak{n}_{i+1}[\mathrm{g}])\,\cdot).$$

The mass term has derivative $-L^{-(i+1)}\mathsf{S}_1(\lambda^2 \partial_{\mathrm{g}} \sigma_{i+1}\,\cdot)$, where $\partial_{\mathrm{g}} \sigma_{i+1} = L\partial_{\mathrm{g}} \sigma_i + \partial_{\mathrm{g}} \nu_{i+1} \circ y_{i+1}$. The terms with $\partial_{\mathrm{g}} \nu_{i+1}$ cancel. The term added with $\mathfrak{n}_{i+1}$ is cancelled by the recurrence $\mathfrak{N}_{i+1} = L\mathfrak{N}_i + \mathfrak{n}_{i+1} \circ y_{i+1}$. Mass terms carry exactly between levels: $L^{-l}\mathsf{S}_1(c\,\cdot)$ becomes $L^{-i}\mathsf{S}_1(L^{i-l}c\,\cdot)$ at level i. This proves eq. 14.xxviii and the displayed coefficient identity. The argument also shows that the total derivative depends only on the plaquette values of the source, so any Lipschitz extension gives the same result.

For (2), the direct weight is linear in w. The two-variable chain rule gives the mixed derivative of the new activity as $D\widetilde{E}_i$ of the mixed input derivative, plus $D^2\widetilde{E}_i$ of its two first derivatives. The latter have the totals of $\delta_i[f], \delta_i[h]$. By induction, the mixed input derivative has the total of $\sum_{l \leq i} \mathcal{V}_i^{\mathrm{old}}[\mathbf{Q}^{(l)}[f, h]]$. The scalar part at the trivial configuration is $\mathfrak{e}_{i+1}[f, h]$. All mixed derivatives of the mass shifts cancel between the activity and the carried mass term, exactly as above. No mass coefficient is moved in the mixed response recursion. This proves (2).

We give the infinite-lattice assertion separately; translation covariance alone would only give a cell-periodic moment. The translations by $ML^{i+1}\mathbb{Z}^3$ preserve the cubes at all preceding levels, the shift systems, the chosen paths, and every localisation rule. At the trivial configuration they preserve the structure and tests as well. For the constant direction, induction therefore gives covariance of $\delta_i^{\infty}[1]$, $\widetilde{\mathbf{P}}^{\infty(i+1)}[1]$ and $\mathbf{P}^{\infty(i+1)}[1]$, including pieces and units. The moments and $\mathfrak{N}^{\infty}$ have the

asserted cell periodicity. The constant chain itself satisfies

$$\sigma_k^{\mathrm{c}}(c) = \sum_{l \le k} L^{k-l}(\nu_l^{\infty}(c) - \mu_l)$$

on its source disc. Differentiation gives the first formula of (3).

To identify the moment at every site, fix l and compare increasing tori T_ℓ at the same cutoff and labels. At $\lambda = 1$ form the response recursion on T_ℓ using the infinite-lattice numbers $\mathfrak{n}_{l'}^{\infty}[1]$ in its mass forms. They define functions on the torus by the cell periodicity just proved. Put the forms at the images of their polymers; cells may wrap around. In old variations keep the infinite-lattice cell sum at the cell, and file the remainder at its possibly wrapping polymers. The proof of (1) applies to this recursion because the cancellation is valid for any numbers moved. The derivative expansion at $\lambda = 1$ is admissible on every such torus: its zero-source potential has size below $c_{\mathrm{cl}}/2$; using half the parameter radius leaves size below c_{cl} and quadratic-origin norm below $(5/64 + 1/4)w_* < w_*/2$. Its direction norms are finite uniformly in ℓ at the finitely many levels considered. The image of a polymer has no larger size, and the cell decomposition has the same long-lag bound.

Let $K_l^{\mathrm{fl},\ell}$ and $K_l^{\mathrm{rsp},\ell}$ be the quadratic kernels, read at level l and at the trivial configuration, of the actual step derivative and this response total. The preceding identity gives $K_l^{\mathrm{fl},\ell} = K_l^{\mathrm{rsp},\ell}$. The finite-torus mass rule omits pieces of diameter at least $\ell/4$ and treats wrapping pieces separately. Its value differs from $-\mathfrak{s}_l^{-1} \operatorname{ord}_0 K_l^{\mathrm{fl},\ell}(y)$ by a quantity tending to zero. Indeed, the first omitted polymers have $d_l \ge \ell_l/(2c'M) - 1$, where $\ell_l = \ell/a_l$; wrapping pieces have $d_l \ge \ell_l/(2M) - 2$. Cauchy's estimate on the constant-source disc $|c| < 2\varrho$ bounds their derivatives. Half the kernel-piece decay pays for the moment sum; the remaining half gives a bound by a datum-dependent constant times $e^{-\kappa\ell_l/(4c'M)}$. The torus comparison for the constant-source chain also gives

$$|\partial_c \nu_l^{(\ell)}(y;0) - \partial_c \nu_l^{\infty}| \le \frac{C'_{\mathrm{tor}}}{\varrho}[\phi(x_l) + L\phi(x_{l-1})]e^{-c_{\mathfrak{t}}\ell/(2a_l)}.$$

This follows by applying Cauchy's estimate to $\sigma_{l'}^{(\ell)} - \sigma_{l'}^{\mathrm{c}}$ and using $\nu_{l'} - \mu_{l'} = \sigma_{l'} - L\sigma_{l'-1}$.

For a polymer that fits in T_ℓ, the response objects, pieces and units are the images of the infinite-lattice ones. This follows inductively from locality of the expansion derivative and of each shift and walk term; the forms were chosen to be those images. At fixed sites their kernels consequently converge to the infinite-lattice kernel. The background maps at the trivial configuration are antiperiodic sums of their infinite-lattice maps and converge locally, with uniform decay. The sums over wrapping polymers tend to zero, bounded by $e^{-\kappa(\ell_l/(2M)-2)}$ times their uniform norm. The off-diagonal kernel bounds dominate the moment sums. Therefore

$$\operatorname{ord}_0 K_l^{\mathrm{rsp},\ell}(y) \longrightarrow \operatorname{ord}_0 K_l^{\mathrm{rsp},\infty}(y).$$

Combining the limits gives

$$\partial_c \nu_l^{\infty} = -\mathfrak{s}_l^{-1} \operatorname{ord}_0 K_l^{\mathrm{rsp},\infty}(y) = -\mathfrak{s}_l^{-1} \sum_{X,p} \operatorname{ord}_0[\mathsf{k}_{X,p}](y).$$

The last sum is absolutely convergent by the piecewise moment bound. The left side is independent of y, which proves the final assertion. □

14.14. **Differentiating one expansion.**

Lemma 14.16 (First and mixed step derivatives)**.** Let $0 \le i < j$, $\lambda \in D_i$ and $D \subset D_i$. A direction $\delta = (\delta^{\mathrm{p}}, \delta^{\mathrm{o}})$ consists of a potential on the field polydisc of the step, with scalar and fermionic local terms, and a scalar quadratic-origin family defined at every real fluctuation. Set

$$\epsilon(\delta) = \epsilon_V(\delta^{\mathrm{p}}) + \epsilon_F(\delta^{\mathrm{p}}) + 2(c_{\mathrm{cl}}/w_*)\|\delta^{\mathrm{o}}\|_{\star,\lambda_V}.$$

The expansion at $(\mathfrak{V}_i(\lambda)+t\delta^{\mathrm{p}},\mathcal{V}^{\mathrm{q}}+t\delta^{\mathrm{o}})$ is holomorphic for $|t|<3c_{\mathrm{cl}}/(4\epsilon(\delta))$. Its first derivative is linear in δ, and its mixed derivative is bilinear in the two directions. For each output polymer X,

$$\|D\widetilde{E}_i[\delta](X)\|\le\frac{3}{2}K'_M\epsilon(\delta_X)e^{-\kappa_+ d_{i+1}(X)},$$
$$\|D^2\widetilde{E}_i[\delta_1,\delta_2](X)\|\le 8c_{\mathrm{cl}}^{-1}K'_M\epsilon(\delta_{1,X})\epsilon(\delta_{2,X})e^{-\kappa_+ d_{i+1}(X)}.$$

Here δ_X restricts both components to input polymers Y with $Y^\sharp\subset X$.

For weight systems as in Definition 14.6, put

$$\epsilon_{\mathscr{S}}(\delta)=\epsilon_{V;\mathscr{S}}(\delta^{\mathrm{p}})+\epsilon_{F;\mathscr{S}}(\delta^{\mathrm{p}})+2(c_{\mathrm{cl}}/w_*)\|\delta^{\mathrm{o}}\|_{\star;\mathscr{S}},$$

with the weights measured at output level $i+1$. Then

$$\|D\widetilde{E}_i[\delta]\|_{i+1;D;\mathscr{S}}\le\frac{3}{2}K_M\sup_{\lambda\in D}\epsilon_{\mathscr{S}}(\delta),$$
$$\|D^2\widetilde{E}_i[\delta_1,\delta_2]\|_{i+1;D;\mathscr{S}_1\cup\mathscr{S}_2}\le 8c_{\mathrm{cl}}^{-1}K_M\sup_{\lambda\in D}\epsilon_{\mathscr{S}_1}(\delta_1)\sup_{\lambda\in D}\epsilon_{\mathscr{S}_2}(\delta_2).$$

The scalar parts obey the same bounds in the array norm, with K'_M instead of K_M. The derivatives retain holomorphy in all input parameters, locality on X, and joint gauge invariance when the directions have these properties. Their scalar parts are then local terms at the fixed radii (α_0,α_1). All assertions hold on $\mathbb{Z}^3$ with the same constants. ■

Proof. The base potential has size at most $c_{\mathrm{cl}}/4$ and the quadratic origin has norm at most $5w_*/64$. Since $c_{\mathrm{cl}}=w_*$, on the indicated t-disc the perturbed potential has size below c_{cl} and the quadratic origin has norm below $(5/64+3/8)w_*=29w_*/64<w_*/2$. Thus both the joined expansion and the quadratic-origin Gaussian cluster sum exist and are holomorphic. The same statement holds in several parameters entering the origins linearly.

Write $p=e^{-\gamma_{\mathrm{tail}}\mathfrak{b}(\mathfrak{t}_i)^2/2}$. The tail ceiling and $C_\eta c_\lambda^{-1/2}\le c_{\mathrm{cl}}/4$, $C_\eta\ge 16K_M\ge 16$, imply $p\le c_{\mathrm{cl}}/64$. The joined activity minus its Gaussian part is bounded at X by $K'_M(\epsilon_V+\epsilon_F+p)e^{-\kappa_+ d_{i+1}(X)}$. The Gaussian part is bounded by $K_\Pi\|\mathcal{V}^{\mathrm{q}}+t\delta^{\mathrm{o}}\|_{\star,\lambda_V}e^{-\kappa_+ d_{i+1}(X)}$. Combining the base sizes and the perturbation size gives the common bound

$$\frac{71}{64}K'_M c_{\mathrm{cl}}e^{-\kappa_+ d_{i+1}(X)}.$$

Locality says that the activity at X sees only δ_X. Apply Cauchy's estimate on its disc of radius $3c_{\mathrm{cl}}/(4\epsilon(\delta_X))$. The factor is $71/48\le 3/2$. For two directions use the bidisc with radii $3c_{\mathrm{cl}}/(8\epsilon(\delta_{a,X}))$. The sum of the two perturbation sizes remains below $3c_{\mathrm{cl}}/4$, and the resulting factor is $71/9\le 8$. Zero-size directions follow by continuity. Linearity and bilinearity follow by differentiating the holomorphic expansion in its linear parameters.

For $Y^\sharp\subset X$, one has $\mathrm{dist}(Y,S)\ge\mathrm{dist}(X,S)$. Consequently

$$\epsilon(\delta_X)\le W_{i+1}(X)^{-1}\epsilon_{\mathscr{S}}(\delta).$$

For two directions the weights multiply. The polymer estimates are uniform in λ; take the supremum at each polymer before summing. The geometric sum $\sup_{\square}\sum_{X\supset\square}e^{-(\kappa_+-\kappa)d_{i+1}(X)}\le C_E$ gives $K_M=C_EK'_M$ and the asserted family bounds. Since $\kappa_+\ge\kappa+1$ and a scalar part has at most its fibre norm, the array bounds follow without the geometric sum.

The structure, covariances, quadratic origin and base potential are holomorphic in the pair. Adjoin the pair variables to the perturbation parameters in the uniformly convergent expansion. Its locality shows that it reads only the restriction of that pair to X. Cauchy integration in the perturbation parameters preserves both properties. Gauge covariance of the expansion similarly passes to its derivatives; in degree zero it is strict gauge invariance. These properties and the displayed bounds are the defining conditions for scalar local terms at the fixed radii.

On $\mathbb{Z}^3$, each polymer calculation is the calculation in a torus into which that polymer fits. The scaled-flow induction gives the same input sizes there, and the proof uses only local data and uniform polymer sums. Thus the constants are unchanged. □

14.15. **Taking finite differences inside a term sum.** The long-lag estimates control a sum at one pair. Moving a supremum inside that sum would lose this estimate. The following form of Cauchy's formula keeps the sum inside the supremum when the shift is varied and localised.

Lemma 14.17 (Amplitude differences and localisation)**.** Let $t(\mu;s)$ be terms holomorphic for $|\mu|<R$ and $|s|<e^{\bar\kappa}$, where s are the shift parameters of the step from level i. The term $t(\mu;s)$ is read at the pair obtained by substituting the shift $\mu\mathfrak{h}(s)$; this substitution is its only dependence on μ,s. Each term is filed at $Y_t\in\mathcal{D}_i$, and depends on the shift only at bonds of Y_t. All terms use the restrictions of one shift system on a polymer containing them. Let $R_t\subset Y_t$ be the cubes containing endpoints of the bonds used by t. For a set of cubes Γ, let Δ_Γ be the finite difference with the other parameters zero; file it at the enlargement $Y_{t,\Gamma}$ prescribed by the step. For nonnegative numbers w_t, $1<\rho<R$ and $r=e^{\bar\kappa}/2$,

$$\begin{aligned}\sum_{t,\Gamma:\,Y_{t,\Gamma}\supset\square} & e^{(\kappa-1)d_i(Y_{t,\Gamma})}w_t\|\Delta_\Gamma[t(1;\cdot)-t(0;\cdot)]\| \\ &\le \frac{1.19}{\rho-1}\sup_{\square'}\sup_{|\mu|=\rho,\ |s|\le r}\sum_{t:\,Y_t\supset\square'} e^{(\kappa-1/2)d_i(Y_t)}w_t\|t(\mu;s)\|.\end{aligned} \tag{14.xxix}$$

The same estimate holds for the sums over all terms, without the conditions involving $\square,\square'$. ■

Proof. The term with $\Gamma=\emptyset$ vanishes, since the shift is zero at $s=0$. Cauchy's formula on $|\mu|=\rho$ and $|s_b|=r$ for $b\in\Gamma$ writes the difference as the angular average of $t(\rho e^{i\varphi};re^{i\theta_\Gamma},0)$ multiplied by

$$(\rho e^{i\varphi}-1)^{-1}\prod_{b\in\Gamma}(re^{i\theta_b}-1)^{-1}.$$

Its norm is at most $(\rho-1)^{-1}(r-1)^{-\#\Gamma}$ times the angular average of $\|t\|$. The parameter floor $\bar\kappa\ge(\kappa+2)c_0+\log C_{\mathrm{g}}+9\ge\kappa+3$ gives $r-1\ge e^{\bar\kappa-1}$. The finite difference vanishes unless every wall-connected component of Γ meets R_t.

Use one auxiliary probability law for all the terms. Let $\mu=\rho e^{i\varphi}$ have uniform angle, and independently let each s_b be zero with probability $1-p$ and have uniform angle on $|s_b|=r$ with probability $p=e^{-4}$. For an admissible Γ, let E^t_Γ be the event that the support of s meets the union of Γ, its wall neighbours and R_t exactly in Γ. On this event the support components meeting R_t are precisely those of Γ. The component property of the shift therefore gives $t(\mu;s)=t(\mu;s1_\Gamma)$. The events E^t_Γ are disjoint, and

$$\mathbb{P}(E^t_\Gamma)\ge p^{\#\Gamma}(1-p)^{6\#\Gamma+\#R_t}.$$

Conditioned on this event, the active angles remain uniform. Thus the angular average above equals the integral over E^t_Γ divided by its probability.

The enlargement estimate gives

$$e^{(\kappa-1)d_i(Y_{t,\Gamma})}\le e^{(\kappa-1)d_i(Y_t)}e^{(\kappa+2)c_0\#\Gamma}.$$

Put

$$y=e^{(\kappa+2)c_0}(r-1)^{-1},\qquad y_1=\frac{y}{p(1-p)^6}.$$

Since $C_{\mathrm{g}}y\le e^{-8}$ and $p(1-p)^6\ge0.0163$, $C_{\mathrm{g}}y_1\le0.0206$. For one term, summing over its disjoint events therefore bounds its total by

$$(\rho-1)^{-1}(1-p)^{-\#R_t}e^{(\kappa-1)d_i(Y_t)}w_t\int\|t(\mu;s)\|\,d\mathbb{P}.$$

Here $\#R_t \le 8(d_i(Y_t)+1)$ and $(1-p)^{-8} \le e^{0.148}$, so

$$(1-p)^{-\#R_t} e^{(\kappa-1)d_i(Y_t)} \le e^{0.148} e^{(\kappa-1/2)d_i(Y_t)}.$$

For terms with $Y_t \supset \square$, sum under this integral and use the displayed supremum in eq. 14.xxix.

If $\square \notin Y_t$ but $Y_{t,\Gamma} \supset \square$, the component of Γ reaching $\square$ has some size q. There are at most C_{g}^q possible connected sets of that size containing $\square$, and at most q choices for its cube in $R_t \subset Y_t$. Retain the factor y_1^q in the preceding event estimate. The extra count is

$$\sum_{q\ge 1} q(C_{\mathrm{g}} y_1)^q \le 0.0215.$$

Consequently the total factor is at most $e^{0.148}(1+0.0215) \le 1.19$. For the unrestricted sum there is no extra count, and $e^{0.148} \le 1.19$ suffices. The proof sums the terms before taking suprema over the common parameters throughout. □

14.16. **Reading and varying an old response.** The estimate for an old response has three terms. Terms of degree at least four, and the field-dependent part of the quadratic term, gain four powers of the scale ratio. A difference of quadratic legs gains $2+\alpha$ powers, where $\alpha = 3/4$. The remaining local quadratic term is measured by its cell sum. We keep this last term explicitly until the moment equation controls it.

Let $1 \le l \le i < K$ and $n = i+1-l$. Use a weight system $\mathscr{S}$ and put

$$\mathcal{N} = \min\{M^3 L^{3n}, \mathcal{C}_l(\mathscr{S})\}.$$

Assume that a family F at level l has the following properties. Its scalar terms are local on the whole regular-pair core, at (α_0, α_1), and are pointed equally at their sites. Its fermionic part is an absolutely convergent sum $F_{\ge 2}(X) = \sum_\sigma F_\sigma(X)$, each summand filed at a cell of X, holomorphic and jointly gauge invariant on that core.

At the trivial pair, read the quadratic kernels in the level-l field Φ. Their units are obtained by moving diagonal terms from the cell of a summand to the cell of their site y. Each moved term is a fixed linear combination of row and column sums at y of the quadratic kernels of two families F_1, G with the same holomorphy and invariance, with sum of absolute coefficients at most two. The decomposition is $F = F_1 + F_2$, with $F_2 = 0$ allowed. Kernels of G may be read at a second admitted fine mass, provided their background maps have the same bounds. Attach each moved term to the summand from which it is moved.

In the norm of $F_{\ge 2}$, sum the norms of its summands before the polymer sum. For a summand to which moved terms are attached, let $\|F^{\mathrm{tr}}_{\sigma,2}(X)\|$ include the trivial-pair quadratic norms of $F_\sigma(X)$, $F_1(X)$ and $G(X)$. The norm $\|F_2^{\mathrm{tr}}\|_{l;D;\mathscr{S}}$ uses these quantities. Write m_c for the sum of the quadratic units filed at the cell c; on a torus include the seam transport in every piece and write m_c^{tor}.

For the output we also use the norm

$$\|H\|^{\mathrm{out}}_{i+1;D;\mathscr{S}} = \sup_{\lambda\in D} \sup_{Z,P} \sup_{\square\subset Z} \sum_{\square\subset Y\subset Z} e^{\kappa d_{i+1}(Y)} W_{i+1}(Y) \|H(Y;P|_Y;\lambda)\|_{h^{(i+1)}}.$$

Here P is one triple in the regular-pair core on Z, whose restrictions are used in the sum. This norm is bounded by the strong family norm of Definition 14.6.

There are two readings. In reading (R), read at level $i+1$ with no shift, so the filing level is $i_{\mathrm{f}} = i+1$ and its lag is $n_{\mathrm{f}} = n$. In variation (V), read at level i in a step to level $i+1$, so $i_{\mathrm{f}} = i$ and $n_{\mathrm{f}} = n-1$. Substitute the shift $\mu\mathfrak{h}_Y(s;P,B')$, where $B' = \lambda \mathsf{t}_i^{-1/2} B$, at every parameter of the shift polydisc and every amplitude satisfying

$$|\mu| c_h \sup|B'| \le \mathsf{c}_{\mathrm{os}}\alpha_0,$$

strictly when $n_{\mathrm{f}} = 0$. In both cases use the weight W_{i+1}, and put $\pi_{\mathrm{f}}(Y) = (1+\operatorname{diam}_{i_{\mathrm{f}}} Y/M)^4$.

At $n_{\mathrm{f}} \ge n_{\mathrm{lag}}$, where n_{lag} is the least integer with $2(2A_\kappa C_H)^2 L^{-n_{\mathrm{lag}}} \le 1$, write the legs first at level l, then at the filing level, using their local and walk terms. Transport the legs of each quadratic

term to the centre of its cell. For a summand σ of X, let $T_{\mathrm{b}}(\sigma)$ contain its terms of degree at least four and the difference between its quadratic term and its value at the trivial pair. Let $T_{\mathrm{c}}(\sigma)$ contain its trivial-pair quadratic terms with at least one difference of legs, including both copies of each moved piece attached to the summand. Let $T_{\mathrm{d}}(c)$ contain the terms of $\bar\Phi(z_c)m_c\Phi(z_c)$. File each term at the hull of its original polymer, walk sets and transport paths, enlarged by the at most eight cubes used for its regular gauge. A cell term is filed at the cube of c, enlarged by the walks of its two legs.

Lemma 14.18 (Lag estimates)**.** There are constants $C_{\mathrm{rd}}, C_{\mathrm{mo}} \ge 1$, independent of h_F, coupling, field scale, level, side and source, with the following properties.

(R0) Subtract the trivial-pair value of each scalar term and read it at any level $i+1 \le K$, including K. File it at the hull $Y(X)$, enlarged by the regular-gauge cube when that cube is used. On the whole output core,

$$|F_0(X;\rho_X P) - F_0(X;\mathbf{1},0)| \le C_{\mathrm{rd}}(1+d_l(X))^4 L^{-4n}\|F_0(X)\|_X.$$

The resulting family has strong weighted norm at most $C_{\mathrm{rd}}\mathcal{N}L^{-4n}\|F_0\|^{\mathrm{arr}}_{l;D;\mathscr{S}}$. For $D \subset D_i$, its scalar variation has size

$$\epsilon_{V;\mathscr{S}}(\mathcal{V}_i^{\mathrm{old}}[F_0]) \le 2s_\lambda C_{\mathrm{rd}}\mathcal{N}L^{-4n}\|F_0\|^{\mathrm{arr}}_{l;D;\mathscr{S}}, \qquad s_\lambda = c_h\beta_\lambda/a_{\mathrm{an}},$$

at rate $(1-2\delta_{\mathrm{w}})\kappa \ge \lambda_V$.

(T) At $n_{\mathrm{f}} \ge n_{\mathrm{lag}}$, at one pair and at either weight h_F or $C_H h_F$ on the filing-level generators,

$$\sum_{t\in T_{\mathrm{b}}(\sigma)} e^{\kappa d_{i_{\mathrm{f}}}(Y_t)}\pi_{\mathrm{f}}(Y_t)\|t\| \le C_{\mathrm{rd}}L^{-4n}e^{\kappa d_l(X)/2}\|F_\sigma(X)\|,$$

$$\sum_{t\in T_{\mathrm{c}}(\sigma)} e^{\kappa d_{i_{\mathrm{f}}}(Y_t)}\pi_{\mathrm{f}}(Y_t)\|t\| \le C_{\mathrm{rd}}L^{-(2+\alpha)n}e^{\kappa d_l(X)/2}\|F^{\mathrm{tr}}_{\sigma,2}(X)\|,$$

$$\sum_{t\in T_{\mathrm{d}}(c)} e^{\kappa d_{i_{\mathrm{f}}}(Y_t)}\pi_{\mathrm{f}}(Y_t)\|t\| \le C_{\mathrm{rd}}L^{-2n}h_F^2\|m_c\|.$$

Here $\alpha = 3/4$.

(M) For every $\lambda \in D$, site y and cell c,

$$W_l(y)\sum_{X,p}\|\operatorname{ord}_0 \mathsf{k}_{X,p}(y)\|_{\mathrm{tr}} \le C_{\mathrm{mo}}h_F^{-2}\|F_2^{\mathrm{tr}}\|_{l;D;\mathscr{S}}, \qquad W_l(c)\|m_c\|_{\mathrm{tr}} \le C_{\mathrm{mo}}h_F^{-2}\|F_2^{\mathrm{tr}}\|_{l;D;\mathscr{S}}.$$

These estimates apply separately to F, F_1, G. The units also have the first absolute moment bound

$$\sum_{\mathfrak{u}\text{ at }c}\sum_{y,y'}\|\beta_{\mathfrak{u}}(y,y')\|(1+|y-z_c|_l+|y'-z_c|_l) \le C_{\mathrm{mo}}h_F^{-2}\varepsilon_{\mathrm{f}},$$

where ε_{f} bounds their unweighted family norms. Restricting the displayed ord_0 moment sum over X,p to pieces whose polymer Z_p has $d_l(Z_p) \ge d_*$ gives the additional factor $e^{-\kappa d_*/2}$. If the kernels are holomorphic on $|\lambda| < \bar\lambda$, the same estimates hold for their λ^4 moments with $\bar\lambda^4\sum\|a^{[4]}_{X,p}(y)\|_{\mathrm{tr}}$ in place of the moment sum.

(R2) Put

$$\mathsf{B}_{\mathrm{R}} = L^{-4n}\|F_{\ge 2}\|_{l;D;\mathscr{S}} + L^{-(2+\alpha)n}\|F_2^{\mathrm{tr}}\|_{l;D;\mathscr{S}} + L^{-2n}h_F^2\sup_c W_l(c)\sup_{\lambda\in D}\|m_c\|.$$

The fermionic family read at level $i+1$ has norm at most $C_{\mathrm{rd}}\mathcal{N}\mathsf{B}_{\mathrm{R}}$. For $n \ge n_{\mathrm{lag}}$ this is the $\|\cdot\|^{\mathrm{out}}$ norm. For $n < n_{\mathrm{lag}}$ it is the strong family norm.

(V2) For $D \subset D_i$ and $n_{\mathrm{f}} \ge n_{\mathrm{lag}}$, the variation is the sum of $t(1)-t(0)$ at $\widetilde\Psi = \Psi_i^{\mathrm{bg}} + W$ and $t(0)(\widetilde\Psi) - t(0)(\Psi_i^{\mathrm{bg}})$, followed by the finite differences in the shift parameters. At every lag the variation has size

$$\epsilon_{F;\mathscr{S}}(\mathcal{V}_i^{\mathrm{old}}[F_{\ge 2}]) \le c'_\psi C_{\mathrm{rd}}\mathcal{N}\mathsf{B}_{\mathrm{R}}$$

in the summed norm at rate $\kappa - 1$. When $n_{\mathrm{f}} \geq n_{\mathrm{lag}}$ one may replace c'_{ψ} by $s_{\lambda} + c_{\psi}$.

(H) The full, reduced and feedback first-response families and the mixed-response families have the hypotheses above, with the units specified in Definition 14.6. The families joining cells have them at $D = \{1\}$, with their polymers as units.

For lags below n_{lag}, no leg expansion is needed: reading keeps each summand at the hull of its polymer, and variation uses the direct fermionic variation. The constants in the proof cover these lags as well. On a torus all cell sums and distances are those of the torus, with the seam in the transports. ∎

Proof. We give constants, then prove the scalar and fermionic estimates separately. All estimates are at one $\lambda \in D$ until the last supremum is taken.

Constants and weights. Let

$$C_4 = \sum_{v \in \mathbb{Z}^3} (1 + |v|_{\infty})^{-4}, \qquad c_{\mathrm{ft}} = \min\{\kappa/(16C_t), 1/(48C_t)\},$$
$$P_{\mathrm{ft}} = e(6/(ec_{\mathrm{ft}}))^6, \qquad C' = \sup_x \sum_y e^{-\alpha_D d(x,y)/2}.$$

Thus $u^6 \leq P_{\mathrm{ft}} e^{c_{\mathrm{ft}} u - 1}$ for $u \geq 0$. Use the constants of the localized leg bounds, the regular gauge construction and scalar localization, and set

$$\begin{aligned}
C_{\mathrm{ft}} &= 2^{10}(1 + 8C_{\mathrm{nb}} M^3 A_{\kappa})^2 (A_{\kappa} C_H)^8 \\
&\quad \times \left[1 + C_{\mathrm{gf}} + \nu + C_t\{C_H(\alpha) + 4C_H(c_T \epsilon_* + C_Q L \alpha_1)\}\right]^2, \\
C_{\mathrm{pt}} &= 4e^{\kappa/8} e^{\kappa c_0} M^2 P_{\mathrm{ft}} C_{\mathrm{ft}} L^4, \\
C^{(2)}_{\mathrm{gf}} &= (6C_{\mathrm{lg}} C_t M/\rho_{\mathrm{A}})^2 + (8C_t M/s_Q)^4 + L^{4n_{\mathrm{lg}}}, \\
C^{0}_{\mathrm{rd}} &= 2C_{\Sigma} M^3 (8e\, \mathsf{C}_{\mathrm{oc}} C_{\mathrm{E},2} L^8 + 8L^{11}), \\
C^{\mathrm{R0}}_{\mathrm{rd}} &= 2232 e^{1/2} e^{\kappa c_0} C_{\mathrm{E},0} C^{(2)}_{\mathrm{gf}}, \\
C^{\psi}_{\mathrm{rd}} &= 2^7 C_4 e^{\kappa/16} L^{4n_{\mathrm{lag}}} C_{\mathrm{pt}}, \\
C_{\mathrm{rd}} &= \max\{C^0_{\mathrm{rd}}, C^{\mathrm{R0}}_{\mathrm{rd}}, C^{\psi}_{\mathrm{rd}}\}, \\
C_{\mathrm{mo}} &= 2^{14} e^{\kappa/8} A_{\kappa}^2 C'^2 M^3 (1 + 3M)(1 + c'M).
\end{aligned}$$

Here $\nu = 8N_f$, $c_T = 64L^2$, $\alpha = 3/4$ and $C_{\mathrm{E},0} \leq C_{\mathrm{E},2}$ are the polymer-counting constants with counting rate $\kappa_{\mathcal{D}} = \log 9216$. These constants precede the choice of h_F.

Write $d = d_l(X)$. If $Z \subset X$, the diameter estimate $\operatorname{diam}_l X \leq c' M(1 + d)/2$ and the restrictions $\kappa_{\mathrm{s}} c' \leq \min\{\kappa/8, 1\}$ imply

$$\begin{aligned}
W_l(Z) &\leq e^{\kappa_{\mathrm{s}} c'(1+d)} W_l(X), \\
W_l(X)^{-1/2} &\leq \min\{e^{\kappa(1+d)/16}, e^{(1+d)/2}\} W_l(Z)^{-1/2}.
\end{aligned} \tag{14.xxx}$$

If $Y \supset Z$, distances at the output level decrease by L^{-n}. Consequently,

$$W_{i+1}(Y) \leq W_l(Z)^{L^{-n}} \leq W_l(Z) W_l(Z)^{-1/2}.$$

The last inequality uses $L^{-n} \leq 1/2$.

The scalar variation. Use the scalar localization of the bare array whose only nonzero member is $G^{(l)} = F_0$, pointed equally at its sites, with $e_l = 0$. Proposition 14.14 places $(P, \lambda \mathsf{t}_i^{-1/2} B)$ in the admitted domain at $\mathsf{t}_\flat$. A localized potential with index $\mathfrak{i}$ therefore has bound

$$2\mathsf{C}_{\Phi} \mathsf{s}_{\mathfrak{a}}(\mathsf{t}_\flat) \|f_{\mathfrak{i}}\| e^{-(\bar{\kappa}-1)\mathfrak{m}_i(X_{\mathfrak{i}}, Y)}, \qquad \mathsf{s}_{\mathfrak{a}}(\mathsf{t}_\flat) = (1 + \beta_s)\delta(\mathsf{t}_\flat)/\alpha_0.$$

We check the sum of the weighted data in this estimate.

For a narrow index (l,z), which occurs only if $l<i$, the domain $X'(z)$ is the union of at most eight cubes about one corner at level i. It has $d_i(X'(z))=0$ and contains at most $8M^3L^{3(n-1)}$ level-l sites. The scalar localization estimate gives

$$\|f_{(l,z)}\| \le \mathsf{C}_{\mathrm{oc}} L^{8-4n} \sum_{X_1 \ni z} (d_l(X_1)+2)^2 \|F_0(X_1,\cdot,z)\|_{X_1}.$$

In this sum $W_l(z) \le e^{1+d_l(X_1)} W_l(X_1)$. The unit of decay between $\kappa+1$ and κ and the polymer count give

$$W_{i+1}(Y)\|f_{(l,z)}\| \le e\,\mathsf{C}_{\mathrm{oc}} C_{\mathrm{E},2} L^{8-4n} W_l(c(z))^{-1/2} \|F_0\|^{\mathrm{arr}}_{l;D;\mathscr{S}}.$$

Either count the sites in $X'(z)$ or use $\sum_z W_l(c(z))^{-1/2} \le M^3 \mathcal{C}_l(\mathscr{S})$. The total for one narrow domain is bounded by

$$\mathsf{B}_{\mathrm{nr}} = 8e\,\mathsf{C}_{\mathrm{oc}} C_{\mathrm{E},2} M^3 L^8 \mathcal{N} L^{-4n} \|F_0\|^{\mathrm{arr}}_{l;D;\mathscr{S}}.$$

For a wide index (l,z,X), use

$$W_{i+1}(Y) \le W_l(X) e^{(1+d)/2} W_l(c)^{-1/2}, \qquad c \subset X.$$

The shares of X add to $F_0(X)$. The remaining decay pays $e^{d/2}$, while $W_l(c)^{-1/2} \le \min\{1, \mathcal{C}_l(\mathscr{S})\} \le \mathcal{N}$. Thus one wide domain contributes at most $\mathsf{B}_{\mathrm{wd}} e^{-\kappa d}$, where

$$\mathsf{B}_{\mathrm{wd}} = e^{1/2} \mathcal{N} \|F_0\|^{\mathrm{arr}}_{l;D;\mathscr{S}}.$$

There are no such domains with $d < \max\{2L^{n-2}-2, 0\}$. The localization sum uses $\varpi_m = L^{3m} e^{-(2L^{m-1}-2)} \le L^{7-4m}$ for $m \ge 1$. Indeed, $7(m-1)\log L \le 2L^{m-1}-2$ for $L \ge 13$. It follows that

$$\begin{aligned} W_{i+1}(Y) &\sum_{\mathfrak{i}} \|f_{\mathfrak{i}}\| e^{-(\bar\kappa-1)\mathfrak{m}_i(X_{\mathfrak{i}},Y)} \\ &\le C_\Sigma (\mathsf{B}_{\mathrm{nr}} + L^{11-4n} \mathsf{B}_{\mathrm{wd}}) e^{-(1-2\delta_{\mathrm{w}})\kappa d_i(Y)} \\ &\le \tfrac{1}{2} C^0_{\mathrm{rd}} \mathcal{N} L^{-4n} \|F_0\|^{\mathrm{arr}}_{l;D;\mathscr{S}} e^{-(1-2\delta_{\mathrm{w}})\kappa d_i(Y)}. \end{aligned}$$

When $l=i$, there are no narrow indices and the same bound uses only B_{wd}.

The localized variation is holomorphic in the amplitude ζ of B', vanishes at $\zeta=0$, and is admitted for $|\zeta| < \mathsf{R}_i/|\lambda|$, where $\mathsf{R}_i = \delta(\mathsf{t}_\flat)/\delta(\mathsf{t}_i) > |\lambda|$. Schwarz's lemma supplies $|\lambda|/\mathsf{R}_i$. The remaining scalar prefactor is

$$\begin{aligned} 2\mathsf{C}_\Phi \mathsf{s}_{\mathfrak{a}}(\mathsf{t}_\flat) \frac{|\lambda|}{\mathsf{R}_i} &= \frac{3(1+\beta_s) c_h \mathsf{C}_{\mathcal{A}} |\lambda| \delta(\mathsf{t}_i)}{a_{\mathrm{an}}} \\ &\le 3(1+\beta_s) s_\lambda \le 3.6 s_\lambda. \end{aligned}$$

We used $2a_{\mathrm{an}} = \mathsf{c}_{\mathrm{os}}\alpha_0$ and the field-scale bound $\mathsf{C}_{\mathcal{A}}|\lambda|\delta(\mathsf{t}_i) \le \beta_\lambda$. The result is $1.8 s_\lambda C^0_{\mathrm{rd}}$ times the displayed array bound. This proves the scalar variation with the asserted constant $2C_{\mathrm{rd}}$ and rate.

The scalar reading. Fix X, its filing cell c, and put $D' = M(1+d)$. Let $E(\mathbf{q}) = F_0(X;\mathbf{q})$ and let $\mathbf{p}_*$ be the pair obtained by reading an output triple P to X. Reading maps the full output core to the full core on X. If $2(C_t D' + M + 2) > s_Q L^n$, the elementary estimate $|E(\mathbf{p}_*) - E(\mathbf{1},0)| \le 2\|F_0(X)\|_X$ gives

$$|E(\mathbf{p}_*) - E(\mathbf{1},0)| \le 2(8C_t M/s_Q)^4 (1+d)^4 L^{-4n} \|F_0(X)\|_X.$$

If this inequality of lengths fails but $n < n_{\mathrm{lg}}$, the same elementary estimate gives

$$2L^{4n_{\mathrm{lg}}} L^{-4n} \|F_0(X)\|_X.$$

In the remaining case the regular-gauge cube Q contains the bonds of X and their neighbours. The at most eight output cubes meeting Q are included in $Y(X)$. The regular gauge and its complex continuation carry $\mathbf{p}_*$ to a pair $\mathbf{q}_1$ without changing E. The radial disc through that pair is

$$\zeta \longmapsto (e^{i\zeta L^{-l} A_{Q,1}}, \zeta J_{Q,1}), \qquad |\zeta| \le \rho_{\mathrm{A}}/\beta_2, \qquad \beta_2 = 6 C_{\mathrm{lg}} C_t L^{-2n} D'.$$

It lies in the admitted pair domain. Subtract $E(\mathbf{1},0)$ along this disc to obtain $g(\zeta)$. Then $|g| \le 2\|F_0(X)\|_X$ and $g(0)=0$. Also $g'(0)=0$: the differential at the trivial pair is an invariant complex linear form on the Lie-algebra valued pair, and averaging its argument under constant SU(2) conjugations gives zero. Schwarz's lemma to second order gives

$$|g(1)| \le 2(6C_{\mathrm{lg}}C_t M/\rho_{\mathrm{A}})^2(1+d)^2 L^{-4n}\|F_0(X)\|_X.$$

If $\beta_2 > \rho_{\mathrm{A}}$, the same bound follows from the elementary estimate instead. All three cases are bounded by $2C_{\mathrm{gf}}^{(2)}(1+d)^4 L^{-4n}\|F_0(X)\|_X$. The composition with pair restriction is defined on the whole output core, also when $i+1=K$.

For the family norm, $d_{i+1}(Y(X)) \le c_0 + L^{-n}d$. If $Y(X)$ contains a fixed output cube, X meets one of at most $9L^{3n}$ old cells, including the possible regular-gauge enlargement. Alternatively sum their weights, giving at most $9\mathcal{C}_l(\mathscr{S})$. For such a cell c'', eq. 14.xxx gives

$$W_{i+1}(Y(X)) \le W_l(X)\min\{1, e^{(1+d)/2}W_l(c'')^{-1/2}\}.$$

Use $(1+d)^4 e^{-d/2} \le 124$. The polymer summand is at most

$$248 e^{1/2} e^{\kappa c_0} C_{\mathrm{gf}}^{(2)} L^{-4n}\min\{1, W_l(c'')^{-1/2}\} e^{-\kappa(1-L^{-n})d}\|F_0\|^{\mathrm{arr}}_{l;D;\mathscr{S}}.$$

Since $\kappa(1-L^{-n}) \ge 12\kappa/13 > \kappa_{\mathcal{D}}+2$, its polymer sum is bounded by $C_{\mathrm{E},0}$. The cell count proves (R0), with $9\cdot 248 = 2232$ as in $C_{\mathrm{rd}}^{\mathrm{R0}}$.

Terms from the fermionic reading. Suppose first that $n_{\mathrm{f}} \ge n_{\mathrm{lag}}$. Expand each leg at level l and then at the filing level, retaining the local term and every walk term. The localized leg estimate, including the factors $e^{|\omega'|}$ and $e^{|\omega|}$ for the two stages and the positive growth in polymer length, is

$$(2A_\kappa C_H)^2 L^{-n_{\mathrm{f}}} h^{(l)}(n').$$

For a term of degree $2p \ge 4$, the definition of n_{lag} bounds the product of these estimates by

$$(2A_\kappa C_H)^8 L^{-4n_{\mathrm{f}}} 2^{-(2p-4)}$$

times its original coefficient norm. This proves the four powers of scale for the first part of T_{b}.

For the quadratic term minus its trivial-pair value, transport its coefficient and both legs to the centre of the filing cell. The regular-gauge estimate for the coefficient is $2\mathsf{B}_E C_{\mathrm{gf}} L^{-2n_{\mathrm{f}}} D'^2$, where now

$$D' = M(1+d) + 9M_w \sum_{\omega'}(|\omega'|+1).$$

Together with two legs this again gives $L^{-4n_{\mathrm{f}}}$. The coefficient is estimated on all pairs, but the second-stage legs must remain at one pair. To respect this order, sum the first-stage rows with their individual suprema before estimating the coefficient. There are at most $8C_{\mathrm{nb}}M^3(1+d)$ first cubes of a row. Each leg therefore costs at most the additional factor $(1+8C_{\mathrm{nb}}M^3A_\kappa)(1+d)$. Its square is included in C_{ft}, and $(1+d)^2 \le e^{\kappa d/11}$ because $\kappa > 22$. The remaining estimates keep one common pair.

For T_{c}, a transported difference of legs has the bound

$$[2C_H(\alpha) + 4C_H(c_T\epsilon_* + C_Q L\alpha_1)]L^{-(1+\alpha)n_{\mathrm{f}}}.$$

In the local and walk decomposition one may use

$$5L^{-2n_{\mathrm{f}}} + 2A_\kappa[C_H(\alpha) + 4C_H(c_T\epsilon_* + C_Q L\alpha_1)]L^{-(1+\alpha)n_{\mathrm{f}}}.$$

The other leg contributes $L^{-n_{\mathrm{f}}}$. Transport paths have length at most $C_t D'$; a moved site term needs at most $3M$ within its cell. The row and column sums defining moved pieces obey the double-kernel

bound $4A_\kappa^2 h_F^{-2}$. Their coefficients have absolute sum at most two. Counting the removed and inserted copies therefore gives

$$\sum_p e^{\kappa \sum \Delta^+} \sum_y \|a_{X,p}(y)\|_{\mathrm{tr}} \le 16 A_\kappa^2 h_F^{-2} \|F^{\mathrm{tr}}_{\sigma,2}(X)\|.$$

Thus these pieces have the same $2+\alpha$ gain. For T_{d} the two legs give directly

$$\nu \|m_c\| (2A_\kappa C_H^2 L^{-n_{\mathrm{f}}} h_F)^2.$$

We include the filing and diameter costs explicitly. The hull of a term has length at most $L^{-n_{\mathrm{f}}} d + \sum \Delta^+$, before its regular-gauge enlargement. The latter costs $e^{\kappa c_0}$. Transports cost at most

$$e^{\kappa(1+d)/8} e^{\sum_{\omega'} (|\omega'|+1)/8}.$$

For all three types of terms, including the cell terms,

$$\pi_{\mathrm{f}}(Y_t) D'^2 \le M^2 u^6, \qquad u = 5 + 2C_t(1+d) + 6C_t \sum_{\omega'} (|\omega'|+1) + 2 \sum_{\omega} (|\omega|+1).$$

The rounding error of two per walk is paid by $6C_t(1-L^{-n_{\mathrm{f}}})(|\omega'|+1) \ge 72/13$. For a cell term the formula at $d=0$ also covers its diameter $6 + 2\sum_\omega (|\omega|+1)$. The definition of P_{ft} now gives

$$M^2 u^6 \le M^2 P_{\mathrm{ft}} e^{\kappa d/8} \prod_{\omega'} e^{(|\omega'|+1)/8} \prod_{\omega} e^{(|\omega|+1)/24}.$$

The shares of the original polymer decay used here are $1/13 + 1/8 + 1/8$, and, for the coefficient supremum, another $1/11$. Their sum is less than $1/2$. The walk factors use only $1/4$ of a first-stage walk rate and $1/24$ of a second-stage rate. Their remaining factor $e^{7/24}$ per leg is absorbed by the degree sum, since $e^{7/12} < 2$. Consequently the three estimates in (T) hold first with C_{pt} on the right. This constant includes the factor L^4 needed to replace $n_{\mathrm{f}} \ge n-1$ by n.

Summing the reading and taking variations. Lemma 14.17 applies under the sum over the terms just constructed. In particular, its factor 1.19 does not interchange the common-pair supremum with that sum.

For a long-lag reading use $W_{i+1}(Y_t) \le W_l(X) W_l(X)^{-1/2}$ or the corresponding cell inequality. To replace $W_l(X)^{-1/2}$ by the weight of its filing cell costs at most $e^{\kappa(1+d)/16}$. This is paid by the decay still left above. There are two ways to sum over the cells. Dropping the requirement that the output contains a specified cube gives $\mathcal{C}_l(\mathscr{S})$. Dropping the weights instead, and using the factor $\pi_{\mathrm{f}}(Y_t)$, gives at most $16 C_4 L^{3n_{\mathrm{f}}}$. The resulting constant is $16 C_4 e^{\kappa/16} C_{\mathrm{pt}} \mathcal{N}$, bounded by $C_{\mathrm{rd}} \mathcal{N}$. This proves (R2). The supremum over λ and the common pair is taken only after the sum over the output polymers contained in that pair's domain.

For the variation, the shift-amplitude disc has radius

$$R = \frac{\mathsf{c}_{\mathrm{os}} \alpha_0}{c_h \sup |B'|} > \frac{2}{s_\lambda} \ge 6.$$

Holomorphy follows term by term from the localized legs. The weight $C_H h_F$ on the filing-level field covers both its background contribution of weight $C_H L^{-1} h_F$ and its fluctuation contribution of weight Γ_F. The sum of the decomposed differences is the original variation, first at real admitted fields and then throughout the disc by holomorphy. For $t(1) - t(0)$, use the amplitude-difference lemma with weights $W_{i+1}(Y_t)$ and $\rho = 1 + s_\lambda^{-1}$. Its cost is $1.19 s_\lambda$. For $t(0)(\Psi_i^{\mathrm{bg}} + W) - t(0)(\Psi_i^{\mathrm{bg}})$, there is no shift-parameter dependence. The Grassmann substitution estimate uses

$$h = C_H h_F, \quad h_a = C_H L^{-1} h_F, \quad h_b = \Gamma_F, \quad h_b' = C_H h_F (1 - L^{-1}).$$

It costs $h_b / h_b' \le c_\psi$. The total cost is at most $1.19(s_\lambda + c_\psi)$. Use the preceding cell count at $n_{\mathrm{f}} = n-1$. The inequalities $1.19 \le 2$ and $2 \cdot 16 C_4 e^{\kappa/16} C_{\mathrm{pt}} \le C_{\mathrm{rd}}$ prove the stronger assertion in (V2).

For a lag below n_{lag} retain the unexpanded reading and file it at the hull of its original polymer. Its least degree is two. The two legs give $L^{-2n} \le L^{2n_{\mathrm{lag}}} L^{-4n}$. The hull uses $e^{\kappa d/13}$; its diameter factor is bounded by $P_{\mathrm{ft}} e^{\kappa d/8}$. For the direct variation split the legs into their background and fluctuation parts and use the same amplitude-difference lemma. The fluctuation, amplitude and chain costs are at most

$$1.19(c_\psi + s_\lambda + C_G \beta_\lambda) \le 1.19 c'_\psi .$$

The amplitude radius here is $2a_{\mathrm{an}} L^{n_{\mathrm{f}}} / (c_h \sup |B'|) > 2/s_\lambda$. The remaining chain dependence is paid by the decay at rate κ. The unweighted cell recount is 319, and hence the total is at most $380 L^{4n_{\mathrm{lag}}} M^3 L^{3n} L^{-4n}$ times c'_ψ and the old norm. For the weighted count it is at most

$$2 e^{\kappa/16} e^{1/2} c'_\psi \mathcal{C}_l(\mathscr{S}) L^{4n_{\mathrm{lag}}} L^{-4n}$$

times that norm. If $n_{\mathrm{f}} \ge 1$, the hull pays the additional weight with half the source distance decay; if $n_{\mathrm{f}} = 0$, use $e^{-d/2}$. Both bounds are already included in C^{ψ}_{rd}. Thus the same constants cover the short lags.

Moments and moved terms. A kernel piece has support $Z_p \times Z_p$, with $Z_p \supset X$ and $d_l(Z_p) \le d + \sum \Delta^+$. For $y \in Z_p$, $W_l(y) \le W_l(X) e^{\kappa(1+d_l(Z_p))/8}$. The two localized legs, summed at rate $\kappa/8$ with their walk factors, give the double-kernel bound $4 A_\kappa^2 h_F^{-2}$ times the original norm. For the cell containing y, the walk recount is

$$(1 - e^{-1}) \sum_{t \ge 0} (2t+5)^3 e^{-t} < 319.$$

Including both row and column sums gives $2552 e^{\kappa/8} A_\kappa^2 h_F^{-2} \|F_2^{\mathrm{tr}}\|_{l;D;\mathscr{S}}$. The cell sum consists of the original summands, their removed diagonal pieces and the pieces inserted at the cells of their sites. It is bounded by

$$(4 + 5104 M^3 A_\kappa^2) e^{\kappa/8} h_F^{-2} \|F_2^{\mathrm{tr}}\|_{l;D;\mathscr{S}}.$$

For the first absolute moment, the distance within a summand's support is at most $c' M (1 + d_l(Z_p))/2$. Use $1 + d_l(Z_p) \le e^d e^{\sum \Delta^+}$. An inserted site piece is at distance at most $3M/2$ from its cell centre. The two contributions are bounded by

$$[36(1 + c' M) A_\kappa^2 + 10208 A_\kappa^2 M^3 (1 + 3M)] h_F^{-2} \varepsilon_{\mathrm{f}}.$$

The moved terms of a fixed X also obey

$$4 C'^2 h_F^{-2} (\|F_1(X)\| + \|G(X)\|) e^{-\alpha_D d(X,y)/2}.$$

These estimates are all bounded by the stated C_{mo}, since $36 + 10208 < 2^{14}$.

For the tail $d_l(Z_p) \ge d_*$ use

$$1_{\{d_l(Z_p) \ge d_*\}} \le e^{-\kappa d_*/2} e^{\kappa d/2} e^{\kappa \sum \Delta^+/2}.$$

Together with the weight already used, the total rate is at most $5\kappa/8 \le \kappa + 2$. The same localized-leg constant and the same cell recount therefore prove the tail assertion. Apply Cauchy's formula for the fourth coefficient on $|\lambda| = r < \bar{\lambda}$ under this absolutely convergent sum, and then let r increase to $\bar{\lambda}$. This proves the coefficient version.

Applicability and the torus. The differentiated step functions in Lemma 14.16 are holomorphic and invariant on the full core. Each direct, mass and old variation has these properties term by term. The transports are gauge covariant, and the trivial-pair value is invariant under constant gauge transformations. The scalar arrays are pointed by equal shares. For P and P_{red}, the moved pieces use $F_1 = \widetilde{P}$, $G = 0$, and the row and column coefficients $1/2, 1/2$. The fourth coefficients are bounded by the Cauchy estimate just proved. For P_{rsp}, use $F_1 = \widetilde{P}_{\mathrm{rsp}}$, $G = 0$; there is no order-four subtraction, by Lemma 14.20. The families Q and R move no diagonal terms. The degree assertions in that lemma use only the first two parts of its proof, and do not depend on the applicability clause

proved here. The families joining cells have the same termwise invariance and locality. This proves (H).

On a torus, retain the seam in every transport and use the torus distance in every count. The terms with no leg difference then give m_c^{tor}, as required. For an infinite-lattice scalar variation each localization index fits in a sufficiently large torus; its definition and its bounds agree there. Passing back index by index, and using the absolute bounds proved above, gives the same variation on the infinite lattice. □

14.17. **The direct source and mass directions.** The direct source estimate counts the plaquettes in its support. The mass estimate uses the mean of its coefficient over a cell. Both estimates retain this spatial information through localization.

Lemma 14.19. Let $0 \le i < j$, $\lambda \in D_i$ and $s_\lambda = c_h\beta_\lambda/a_{\mathrm{an}} \le 1/3$. Let C_{cf} be the completed-field constant of the first step, and put

$$a_{\mathrm{src}} = \max\{\alpha_0^2, 2^{13}\mathsf{c}_{\mathrm{os}}\alpha_0 C_{\mathrm{cf}}/c_h\},$$
$$A_{\mathrm{src}} = 3a_{\mathrm{src}}(1 + 2M^3 c_{\mathrm{cl}}/w_*), \qquad C_{\mathrm{src}} = 3^5 C_\Sigma M^3 A_{\mathrm{src}}.$$

Then:

(1) For a bare plaquette p,
$$\epsilon(\mathcal{V}_i^w[\delta_p]) \le A_{\mathrm{src}} s_\lambda a_K^3 a_i^{-4}.$$
If $i \ge 1$, its terms are the finite differences in the shift parameters, filed at polymers Y containing the cube set $X(p)$, with $d_i(X(p)) = 0$. If $i = 0$, it is one term at the cubes containing a corner of p, again with $d_0(X(p)) = 0$.

(2) Let g be supported on plaquettes whose barycentres lie in a closed ball of radius $r \ge a_K$, and let the nonempty closed set S contain these barycentres and lie in that ball. Then
$$\epsilon_S(\mathcal{V}_i^w[\mathsf{g}]) \le C_{\mathrm{src}} s_\lambda e_i[\mathsf{g}].$$
This also holds at $r = \infty$, with S the entire torus or $\mathbb{R}^3$.

(3) Let $\varkappa$ be a function on the bare sites with scalar values or values in the matrices on $\mathbb{C}_{\mathrm{s}}^2 \otimes \mathbb{C}_\tau^2 \otimes \mathbb{C}^{N_f}$, acting trivially on colour. For a matrix coefficient use its operator norm in the weighted cell mean $\langle\varkappa\rangle_{i;\{\lambda\};S}$. For every nonempty S,
$$\epsilon_S(\mathcal{V}_i^{\mathrm{s}}[\varkappa]) \le C_{\mathrm{mn}} c'_\psi c_{\mathsf{S}} h_F^2 \langle\varkappa\rangle_{i;\{\lambda\};S}, \qquad C_{\mathrm{mn}} = 37.$$

■

Proof. The weight in the level-i action is $\beta_i - a_K^{-1}w$. Its direction at δ_p therefore has sign + and is obtained from
$$a_K^{-1}[a_p(\mathsf{U}'(\mu\lambda B)) - a_p(\mathsf{U})]$$
by taking its terms of orders 1, 2 and at least 3 in μ, at $\mu = 1$. The first and third terms belong to the scalar potential; the quadratic term belongs to the quadratic origin. Write $\mathcal{A} = \lambda\mathsf{t}_i^{-1/2}B$. On the step polydisc, $\sup|\mathcal{A}| < \beta_\lambda \le \beta_0/2$.

Suppose first that $i \ge 1$. On $X(p)$, the function $f_p = a_K^{-1}a_p$ is local on the regular-pair core and satisfies
$$\|f_p\|_{X(p)} \le \tfrac{1}{2}\alpha_0^2 a_K^{-1} L^{-4i} = \tfrac{1}{2}\alpha_0^2 a_K^3 a_i^{-4}.$$
Its localization is the finite difference $\psi_Y(\mu)$ at the polymers $Y \supset X(p)$. The sum of these differences is f_p at the shifted background minus f_p at the background. For $|\mu| \le 2$ the shift remains admitted, and its term radius is
$$\varrho_{\mathrm{t}} = \frac{\mathsf{c}_{\mathrm{os}}\alpha_0}{2c_h\beta_\lambda} = s_\lambda^{-1} \ge 3.$$
The localized function is holomorphic near this disc, vanishes at $\mu = 0$, and obeys
$$|\psi_Y(\mu)| \le \alpha_0^2 s_\lambda a_K^3 a_i^{-4} e^{-(\bar\kappa - 1)\mathfrak{m}_i(X(p),Y)}.$$

Cauchy's estimate on $|\mu| = 2$ bounds the linear, quadratic and remaining terms at $\mu = 1$ by, respectively, $1/2$, $1/4$ and $1/4$ of this expression. The quadratic coefficient is a quadratic form in $\mathcal{A}$, defined at all fluctuations.

The cubes of $X(p)$ have a common corner. Starting there, add the other cubes of Y in wall-connected order; one edge for each added cube extends an edge tree meeting all cubes. Thus $d_i(Y) \le \mathfrak{m}_i(X(p), Y)$. Since $\bar{\kappa} \ge \kappa + 3$ and $\lambda_V \le (1 - 2\delta_{\mathrm{w}})\kappa$, the localization decay pays the factor $e^{\lambda_V d_i(Y)}$.

At $i = 0$ there is no shift system. The field is $e^{iB'}\mathsf{U}$, where B' is the completed fluctuation. Its pivot values depend only on the pair and fluctuation in $X(p)$ when their pivot bond belongs to $X(p)$. The entire expression therefore belongs to this one set. For $|\mu| \le 2$, the completed-field estimate gives

$$\sup |B'(\mu\mathcal{A})| \le \varepsilon_{B'} := 4C_{\mathrm{cf}}\beta_\lambda \le 1.$$

The four factors Z_s of the original plaquette have norm at most 2; their perturbed versions have norm at most $2e^{\varepsilon_{B'}}$ and $|Z'_s - Z_s| \le 2(e^{\varepsilon_{B'}} - 1)$. For the inverse factors use $U^{-1} = JU^TJ^{-1}$, with J unitary, since $\det U = 1$. Telescoping the product of four factors gives

$$|\partial(e^{iB'}\mathsf{U})(p) - \partial\mathsf{U}(p)| \le 64e^{3\varepsilon_{B'}}(e^{\varepsilon_{B'}} - 1) \le 2^{12}\varepsilon_{B'}.$$

As $a_p = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$, the function $\psi(\mu)$ above obeys

$$\begin{aligned} |\psi(\mu)| &\le 2^{14}C_{\mathrm{cf}}\beta_\lambda a_K^{-1} \\ &= 2^{13}(\mathsf{c}_{\mathrm{os}}\alpha_0 C_{\mathrm{cf}}/c_h)s_\lambda a_K^{-1} \\ &\le a_{\mathrm{src}}s_\lambda a_K^3 a_0^{-4}. \end{aligned}$$

It vanishes at zero. The same Cauchy estimate gives the three fractions $1/2, 1/4, 1/4$.

At every level the two scalar parts, after the decay weight is inserted, cost at most $3a_{\mathrm{src}}s_\lambda a_K^3 a_i^{-4}/4$. If $\mathcal{Q}_Y$ is the quadratic part, its bound on the field polydisc of radius $r_{\mathrm{f}} \ge 1$ gives at real fluctuations

$$|\mathcal{Q}_Y(B)| \le M^3 \sup|\mathcal{Q}_Y| \left(1 + M^{-3}\sum_b |B(b)|^2\right).$$

Its origin norm therefore costs at most M^3 times the supremum. Combining the potential and origin norms gives

$$\epsilon(\mathcal{V}_i^w[\delta_p]) \le a_{\mathrm{src}}s_\lambda a_K^3 a_i^{-4}\left(\frac{3}{4} + \frac{M^3 c_{\mathrm{cl}}}{2w_*}\right),$$

which proves the first assertion.

For the second, the set $X(p)$ contains the square of p, hence its barycentre. Every polymer carrying this plaquette contribution meets S, so its source weight is 1. There are at most $3(2r/a_K + 1)^3 \le 81(r/a_K)^3$ bare plaquettes with barycentre in the ball, and at most $12M^3L^{3i}$ plaquettes have a prescribed set $X(p)$. At $i \ge 1$, sum the localized terms using the scalar localization estimate, with data

$$b_{i,X} = a_{\mathrm{src}}s_\lambda a_K^3 a_i^{-4} \sum_{p:X(p)=X} |\mathsf{g}(p)|.$$

These data vanish unless $d_i(X) = 0$. The sum is bounded by $C_\Sigma \sup_X b_{i,X} e^{-(1-2\delta_{\mathrm{w}})\kappa d_i(Y)}$. At $i = 0$ the same estimate holds directly, since the terms have size zero and $C_\Sigma \ge 1$. The two plaquette counts and

$$a_K^3 a_i^{-4} \min\{M^3L^{3i}, (r/a_K)^3\} \le M^3 a_i^{-1} \min\{1, (r/a_i)^3\}$$

give $\sup_X b_{i,X} \le 81M^3 a_{\mathrm{src}}s_\lambda e_i[\mathsf{g}]$. Use the preceding potential and origin factors to obtain the bound with C_{src}. For $r = \infty$ only the count per set is needed.

Finally consider the mass direction. The usual mass-variation proof is linear in its coefficient at each place v. Before counting which cubes its walks reach, a place contributes its operator

norm times $L^{-i}\nu(C_H L^{-i} h_F)^2$. There are $\nu_P M^3 L^{3i}$ places per cube and $c_{\mathsf{S}} = M^3 \nu_P \nu C_H^2$. Thus the constant coefficient in that estimate can be replaced by the mean of $|\varkappa|$ over the bare sites of the cube. The part with a fluctuation leg costs at most $c_\psi c_{\mathsf{S}} h_F^2/2$ before its walks are counted. The change from U to U' costs at most $C_G \beta_\lambda c_{\mathsf{S}} h_F^2$. A matrix coefficient has trace norm at most ν times its operator norm, so the same constants apply.

At $i = 0$ the variation is explicitly

$$-\sum_v \big[\bar\psi^{\mathrm{bg}}(v)\varkappa(x_v)W(v) + \bar W(v)\varkappa(x_v)\psi^{\mathrm{bg}}(v) + \bar W(v)\varkappa(x_v)W(v)\big], \qquad \psi^{\mathrm{bg}} = H_0(\mathsf{U})\Psi_1.$$

There is no change-of-background part and no walk for the identity image of W; the same estimate applies place by place.

The terms originating in a cube c are summable at rate $\kappa + 2$. If their polymer also contains a cube $\square$, lowering the rate to κ gives $e^{-2\,\mathrm{dist}(c,\square)/M}$. The sum of these factors over c is at most 37. For the source weight,

$$e^{\kappa_{\mathrm{s}}\delta_{i+1}(Y;S)} \le e^{\kappa_{\mathrm{s}}\sqrt{3}/L} e^{\kappa_{\mathrm{s}}\delta_i(c;S)} \le 2e^{\kappa_{\mathrm{s}}\delta_i(c;S)}.$$

The weighted mean over c is bounded by $\langle\varkappa\rangle_{i;\{\lambda\};S}$. Consequently

$$\begin{aligned}\epsilon_S(\mathcal{V}_i^{\mathrm{s}}[\varkappa]) &\le 2\left(\frac{37}{2}c_\psi + C_G\beta_\lambda\right) c_{\mathsf{S}} h_F^2 \langle\varkappa\rangle_{i;\{\lambda\};S} \\ &\le 37 c'_\psi c_{\mathsf{S}} h_F^2 \langle\varkappa\rangle_{i;\{\lambda\};S}.\end{aligned}$$

□

14.18. **Orders at zero field scale.** Here the order of a function means its joint Taylor order at $\lambda = 0$ and at the trivial pair, in λ and the coordinates of that pair, at fixed fluctuation B. In particular, the fluctuation enters the members as λB.

Lemma 14.20. The response families have the following orders.

(1) The scalar parts of $\delta_i[\mathsf{g}]$, $\widetilde{\mathbf{P}}^{(i+1)}[\mathsf{g}]$, $\widetilde{\mathbf{P}}^{\mathrm{red},(i+1)}[\mathsf{g}]$, $\mathbf{P}^{(i+1)}[\mathsf{g}]$ and $\mathbf{P}^{\mathrm{red},(i+1)}[\mathsf{g}]$ have order at least 2; their parts with fermion legs have order at least 3. The member $\mathcal{V}_i^{\mathrm{s}}[\lambda^4\mathfrak{N}_i[\mathsf{g}]]$ has order at least 4, and the parts with legs of its image under $D\widetilde{E}_i$ have order at least 5. The scalar parts of $\mathbf{P}^{\mathrm{rsp},(i+1)}[\mathsf{g}]$, $\widetilde{\mathbf{Q}}^{(i+1)}[f,h]$ and $\mathbf{Q}^{(i+1)}[f,h]$ have order at least 4, and their parts with legs have order at least 5.

(2) The integrated response families, including the numbers $\mathfrak{e}_{i+1}$, are even in λ. The members $\delta_i[\mathsf{g}]$ are invariant under $(\lambda, B) \mapsto (-\lambda, -B)$. All these families and members vanish at $\lambda = 0$ at every pair. The numbers $\mathfrak{n}_i[\mathsf{g}]$ and $\mathfrak{N}_i[\mathsf{g}]$ are independent of λ. At the trivial pair, the parts with legs of $\widetilde{\mathbf{P}}$ and $\widetilde{\mathbf{P}}^{\mathrm{red}}$ start at order λ^4; those of $\mathbf{P}^{\mathrm{rsp}}$ and $\widetilde{\mathbf{Q}}$ start at order λ^6. The values $\mathfrak{e}_{i+1}[f,h](X)$ start at order λ^4. Hence $\mathfrak{n}_j^\lambda$ is holomorphic, even and of order at least λ^4.

(3) One has $\mathfrak{n}_l[\mathsf{g}] = \mathfrak{n}_l^{\mathrm{red}}[\mathsf{g}]$. Let F be either $\mathbf{P}^{(l)}[\mathsf{g}]$ or $\mathbf{P}^{\mathrm{red},(l)}[\mathsf{g}]$, $1 \le l \le j$, with its prescribed units. Write $a_{X,p}(z;\lambda)$ for the moments of its pre-extraction kernels and $a^{[4]}_{X,p}(z)$ for their fourth coefficients. The latter coefficients agree in the full and reduced systems. The cell sum is

$$m_c = \sum_{z\in c}\left[\sum_{X,p} a_{X,p}(z;\lambda) - \mathfrak{s}_l^{-1}\sum_y A_l(y,z)\lambda^4 {\sum_{X,p}}' a^{[4]}_{X,p}(z)\right]. \tag{14.xxxi}$$

The prime denotes the pieces read by the extraction rule. On $\mathbb{Z}^3$, m_c has order at least λ^6. On a torus write $m_c = m_c^{(6)} + \lambda^4 m_c^{[4]}$, where $m_c^{(6)}$ is even and has order at least 6. Put

$$c_{\mathrm{def}} = \min\{4c_{\mathrm{t}}, \kappa/(8c'M)\}, \qquad C_{\mathrm{def}} = M^3 C_{\mathrm{mo}}(c_b^{-2}C_A + e^{\kappa/2}).$$

For weights of one or two sets and $\lambda \in D_{l-1}$,

$$h_F^2 W_l(c)|\lambda|^4 \|m_c^{[4]}\| \le C_{\mathrm{def}} e^{-c_{\mathrm{def}}\ell_G/a_l} \left(\frac{|\lambda|}{\bar{\lambda}_{l-1}}\right)^4 \|F_2^{\mathrm{tr}}\|_{l;D_{l-1};\mathscr{S}},$$

$$h_F^2 W_l(c) \sup_{D_{l-1}} \|m_c^{(6)}\| \le 2C_{\mathrm{mo}} \|F_2^{\mathrm{tr}}\|_{l;D_{l-1};\mathscr{S}}.$$

The cell sums of $\mathbf{P}^{\mathrm{rsp},(l)}[\mathsf{g}]$ and $\mathbf{Q}^{(l)}[f,h]$ have no extraction forms and have order at least λ^6 on both lattices.

■

Proof. We prove the orders, parity and equality of fourth moments together by induction. At level zero there are no response families and $\mathfrak{N}_0 = 0$. Every potential member vanishes at $\lambda = 0$ and depends on λ and B through λB and λ^2; it therefore has joint order at least one. The direct weight direction has order at least two, since $a_p(\mathbf{1}) = 0$ and $a_p = \frac{1}{2}\det(\partial U(p) - \mathbf{1})$ is quadratic in $\partial U(p) - \mathbf{1}$. Each of its three amplitude parts retains this joint order. So do the finite differences used to localize it: each partial shift is the original background when $\lambda B = 0$. At level zero the completed-field formula gives the same conclusion.

The mass direction has order at least four because of its factor λ^4. Some of its terms linear in W have exactly that order, so one cannot claim order five before integration. Its part of degree two in the new generators vanishes when $\mathsf{U}' = \mathsf{U}$, by $H_{0,i}(\mathsf{U})H_i(\mathsf{U}) = H_{0,i+1}(\mathsf{U})$, and hence has order at least five. The parts linear in W have Gaussian mean zero; to contribute after expansion they require another member with a W leg, of order at least one. The $\bar{W}W$ part has degree zero in the new generators, and requires another member with legs to contribute to their positive-degree part. Thus the parts with legs of the image under $D\widetilde{E}_i$ have order at least five.

To justify this order count through the expansion, use its derivative formulas from Lemma 14.16. They are sums of Gaussian means of the direction, or the two directions, multiplied by potential members. The Gaussian covariances and quadratic origin are holomorphic near $\lambda = 0$ and the trivial pair; the tests depend on the fluctuation. The derivative bounds give uniform convergence on a smaller polydisc in λ and the pair. Taylor coefficients therefore pass through the integrals and sums. A product whose factors have orders $k_1, k_2, \dots$ has no coefficient below their sum. A scalar direction can produce legs only in the presence of another member with legs, of order at least one.

Old-family variation also preserves joint order. Its maps of the pair and λB send zero and the trivial pair to the trivial pair, and its finite differences are linear combinations of these compositions. Removing a scalar value at the trivial pair preserves the orders. The induction hypothesis now proves the first response orders in the first assertion.

The full and reduced systems differ, before extraction, by

$$\widetilde{\mathbf{P}}^{\mathrm{rsp},(i+1)}[\mathsf{g}] = D\widetilde{E}_i\left[\mathcal{V}_i^{\mathrm{s}}[\lambda^4\mathfrak{N}_i[\mathsf{g}]] + \sum_{l\le i}\mathcal{V}_i^{\mathrm{old}}[\mathbf{P}^{\mathrm{rsp},(l)}[\mathsf{g}]]\right].$$

This equality holds polymer by polymer with the specified units. At preceding levels the full and reduced fourth moments agree, so their forms agree. Their remaining units differ by the kernel difference minus its diagonal moment at the original cell, and the differences of the moved pieces at their destination cells. The piece rule and variation are linear in these units. The induction hypothesis and the mass-direction count above therefore give orders at least four and five for the scalar and fermionic parts of this difference.

The expansion commutes with $B \mapsto -B$. Direct directions depend on λB; the mass coefficient is $\lambda^4\mathfrak{N}_i$; old variations of even families have the same simultaneous symmetry. Thus their integrated families are even in λ. At the trivial pair, joint order is order in λ. The fermionic feedback part, of order at least five and even, starts at order six. Its fourth moments vanish. Hence the full and reduced fourth moments agree at level $i+1$, as do their extraction forms. Removing the scalar

value then gives the claimed orders of $\mathbf{P}^{\mathrm{rsp},(i+1)}$. The form added to $\mathbf{P}$ itself has order four and does not change its asserted lower orders.

The new contribution to $\widetilde{\mathbf{Q}}$ is $D^2\widetilde{E}_i[\delta_i[f],\delta_i[h]]$. Each direction has order at least two. A term with legs uses a direction's part with legs or a further member with legs, and so has order at least five. The old $\mathbf{Q}$ variations preserve the inductive orders. This proves the first assertion and the equality of fourth moments.

At $\lambda = 0$ the fine field equals its background, so the direct weight direction vanishes. The mass direction vanishes by its factor, and old variations vanish by the induction hypothesis at every pair, including their zero-shift differences involving W. Linearity and bilinearity of the expansion derivatives then show that all new response families vanish there. The extraction numbers are fourth Taylor coefficients and do not depend on λ. Parity upgrades the joint orders at the trivial pair from three to four and from five to six. The scalar mixed values have order at least four, and their absolutely convergent sum is n_j^{λ}. This proves the second assertion.

It remains to establish the cell formula and the torus bounds. The units at the original cell consist of a kernel minus its diagonal moment and have total moment zero. The moved piece at z contributes $a_{X,p}(z;\lambda)$ at the cell of z. The extraction form contributes $-\mathfrak{s}_l^{-1}A_l(y,z)\lambda^4 a^{[4]}_{X,p}(z)$ at y. Summing gives eq. 14.xxxi. On $\mathbb{Z}^3$ every piece is read and $\sum_y A_l(y,z) = \mathfrak{s}_l$. The cell sum is consequently the sum of $a_{X,p}(z;\lambda) - \lambda^4 a^{[4]}_{X,p}(z)$, which starts at order six.

On the torus its fourth coefficient is

$$m_c^{[4]} = \sum_{z\in c}\Big[\sum_{X,p}{}'\left(1 - \mathfrak{s}_l^{-1}\sum_y A_l(y,z)\right)a^{[4]}_{X,p}(z) + \sum_{X,p}{}'' a^{[4]}_{X,p}(z)\Big].$$

The double prime denotes the omitted pieces whose original polymer has diameter at least $\ell_G/4$. Pieces whose expanded polymer wraps around the torus remain in the first sum. The matrix map $\Lambda_z N = N - \mathfrak{s}_l^{-1}\sum_y A_l(y,z)N$ satisfies

$$\|\Lambda_z N\| \le c_b^{-2} C_A e^{-4c_t\ell_G/a_l}\|N\|.$$

Cauchy's formula for the fourth coefficient and the moment bound of Lemma 14.18 give, at each site,

$$h_F^2 W_l(z)|\lambda|^4\left\|\sum_{X,p}{}' a^{[4]}_{X,p}(z)\right\| \le \left(\frac{|\lambda|}{\bar{\lambda}_{l-1}}\right)^4 C_{\mathrm{mo}}\|F_2^{\mathrm{tr}}\|_{l;D_{l-1};\mathscr{S}}.$$

For an omitted piece its expanded polymer has

$$d_l(Z_p) \ge t_G := \frac{\ell_G}{2c'Ma_l} - 1.$$

Apply the tail version of the same moment bound before Cauchy's formula. It gives the extra factor

$$e^{-\kappa t_G/2} = e^{\kappa/2}e^{-\kappa\ell_G/(4c'Ma_l)} \le e^{\kappa/2}e^{-2c_{\mathrm{def}}\ell_G/a_l}.$$

If $t_G < 0$, this factor exceeds one and the ordinary moment bound gives the same inequality. There are M^3 sites in c, and $W_l(z) \ge W_l(c)$. Summing proves the first torus estimate. The moment estimate for m_c and Cauchy's bound $|\lambda|^4\|m_c^{[4]}\| \le \sup_{D_{l-1}}\|m_c\|$ prove the second.

The feedback families have no forms, since the full and reduced forms agree. Their moments are the differences of the two sets of moments, whose fourth coefficients agree. They therefore start at order six. The $\mathbf{Q}$ families have only their scalar values removed; their fermionic parts at the trivial pair already start at order six. This proves the final assertion. □

14.19. **The fourth moment and its accumulated mass.** The reduced response controls the extracted fourth coefficient. The full response is needed to compare its accumulation with the constant-source mass chain. That comparison bounds the constant part before the Lipschitz variation of a test function is estimated.

Let $c_{\mathrm{def}}, C_{\mathrm{def}}$ be as in Lemma 14.20, and set

$$S_{\mathrm{tor}} = 3C_{\mathrm{def}} \sum_{n\ge 1} L^{2n} q_n^2 e^{-2Mc_{\mathrm{def}}L^n},$$
$$S_\diamond = \sum_{n\ge 1} (L^{n/2} q_n + 3L^{(2-\alpha)n} q_n^2 + 3L^{5n/2} q_n^3) + S_{\mathrm{tor}},$$
$$K_{\mathfrak{n}} = 3c_{\mathsf{S}} c_b^{-2} C_{\mathrm{mo}}.$$

These sums converge. Indeed, $q_n \le 2^{10/9} L^{-8n/9}$ gives exponents $-7n/18$, $-19n/36$ and $-n/6$ in the three ordinary sums; in the torus sum $L^{2n} q_n^2 \le 4(1+n\log L)^{4\mathfrak{p}_0}$.

Choose $C_{\mathcal{C}} \ge 8M^3$ to satisfy, for $n = i+1-l \ge 0$ and the directions and sets of Lemma 14.19,

$$e_l[\mathsf{g}] \min\{M^3 L^{3n}, \mathcal{C}_l(S)\} \le C_{\mathcal{C}} L^{4n} e_{i+1}[\mathsf{g}], \tag{14.xxxii}$$
$$e_l[f,h] \min\{M^3 L^{3n}, \mathcal{C}_l(S_f, S_h)\} \le C_{\mathcal{C}} L^{9n/2} e_{i+1}[f,h]. \tag{14.xxxiii}$$

Require also $\mathcal{C}_l(S) \le C_{\mathcal{C}}(1 + r/a_l)^3$ for S in a radius-r ball, and

$$\mathcal{C}'_l := \sum_{\square} e^{-\kappa_{\mathrm{s}}(\delta_l(\square;S_f)+\delta_l(\square;S_h))} \le \begin{cases} C_{\mathcal{C}}, & a_l \ge \mathsf{d}, \\ C_{\mathcal{C}}(\mathsf{d}/a_l)^3 e^{-\kappa_{\mathrm{s}}\mathsf{d}/(2Ma_l)}, & a_l < \mathsf{d}. \end{cases}$$

These conditions are simultaneous on every torus and on $\mathbb{Z}^3$; their verification is included below. They require only a constant depending on M, κ_{s}. Finally put

$$\theta = 3K_\mu^{\mathrm{s}} K_M C_{\mathrm{rd}} C_{\mathcal{C}} (1 + 2C_{\mathrm{mo}}) S_\diamond (2s_\lambda + c'_\psi) \le \frac{1}{2}.$$

The last inequality is one of the parameter choices in the field-scale construction: the c'_ψ term is at most $1/4$ after choosing h_F, and the $2s_\lambda$ term is at most $1/4$ after choosing c_λ.

Lemma 14.21. For $1 \le i \le j$ the following estimates hold.

(1) The reduced response and its pre-extraction version satisfy

$$\|\mathbf{P}^{\mathrm{red},(i)}[\mathsf{g}]\|_{i;D_{i-1};S} \le A^{\mathrm{red}} e_i[\mathsf{g}], \qquad A^{\mathrm{red}} = 6K_\mu^{\mathrm{s}} K_M C_{\mathrm{src}} L^4.$$

This holds in the scalar array norm and in the fermionic norm that sums the individual summands at each polymer, including each extraction form at its own polymer.

(2) At each level-i site y,

$$|\mathfrak{n}_i[\mathsf{g}](y)| \le K_{\mathfrak{n}} (c_{\mathsf{S}} h_F^2)^{-1} A^{\mathrm{red}} \bar{\lambda}_{i-1}^{-4} e_i[\mathsf{g}] e^{-\kappa_{\mathrm{s}}\delta_i(y;S)}.$$

Here and below matrix absolute values are operator norms.

(3) Define

$$C_{\mathrm{fb}} = 43 L C_{\mathrm{mn}} K_{\mathfrak{n}} K_\mu^{\mathrm{s}} K_M c'_\psi S_\diamond.$$

Use ρ from eq. 14.x; the parameter choice $(\rho\bar{\lambda}_K)^2 \ge 2$ implies $C_{\mathrm{fb}} \bar{\lambda}_i^{-2} \le 1/2$. In the constant direction on $\mathbb{Z}^3$,

$$|\mathfrak{N}_i^\infty[1](x)| \le \frac{A_c}{\mathsf{bt}_i}, \qquad A_c = 2\left(\frac{\bar{B}_M}{4\varrho} \sup_{\bar{I}(g)} \mathsf{b} + C_r\right).$$

The finite number C_r is specified in the proof. The function is invariant under the level-i translations by $M\mathbb{Z}^3$.

(4) On the least torus, for any representative $\hat{x}$ of a bare site x,

$$|\mathfrak{N}_i[1](x) - \mathfrak{N}_i^{\infty}[1](\hat{x})| \le \frac{C_{\mathrm{wr}}^{\mathfrak{N}}}{\mathsf{bt}_i}.$$

The constant is given below and depends on g and the fixed constants.

(5) For a direction of the test pair, on D_i,

$$c_{\mathsf{S}} h_F^2 |\lambda|^4 \langle \mathfrak{N}_i[f] \rangle_{i;\{\lambda\};S_f} \le A_{\mathfrak{N}} e_i[f] \mathfrak{l}_i^{4\mathfrak{p}_0+1}.$$

The same holds for h; $A_{\mathfrak{N}}$ depends on g, ϑ and the fixed constants.

■

Proof. The spatial counts. Cells have spacing Ma_l. Summing their exponential distance weights from a radius-r ball gives $\mathcal{C}_l(S) \le C(1 + r/a_l)^3$. On a torus, the weight of a cell is at most the sum of weights of its translates in $\mathbb{R}^3$ relative to one representative of the ball. Thus the same constant works on every torus. For eq. 14.xxxii, if $a_l \le a_{i+1} \le r$, the size ratio is L^n and the geometric count is at most $M^3 L^{3n}$. If $r \le a_l$, the ratio is L^{4n} and the weighted count is at most $8C$. If $a_l < r < a_{i+1}$, the ratio is $L^n (a_{i+1}/r)^3$ and the count is at most $8C(r/a_l)^3$. At $r = \infty$ the first case applies.

For the pair, if $a_l \ge \mathsf{d}$, the size ratio is $L^{9n/2}$ and the count is at most $8C$. If $a_l < \mathsf{d}$, the ratio is at most $L^{2n} + L^{9n/2}(a_l/\mathsf{d})^{5/2}$. The distance between the supports is at least d. Split the two-set weight in half: one half gives $e^{\sqrt{3}\kappa_{\mathrm{s}}/4} e^{-\kappa_{\mathrm{s}} \mathsf{d}/(4Ma_l)}$; the other sums about S_f to at most a constant times $(\mathsf{d}/a_l)^3$. The boundedness of $t^5 e^{-\kappa_{\mathrm{s}} t/(4M)}$ proves eq. 14.xxxiii; when $a_{i+1} \le \mathsf{d}$, use also $L^{2n} \le (\mathsf{d}/a_l)^2$. For $\mathcal{C}_l'$, the corresponding split gives $e^{-\kappa_{\mathrm{s}} \mathsf{d}/(2Ma_l)}$, and a remaining cell count at most $16C(\mathsf{d}/a_l)^3$. For $a_l \ge \mathsf{d}$ its count is at most $8C$. Taking the largest of these constants proves all requirements on $C_{\mathcal{C}}$.

The reduced response. Proceed by induction on i on either lattice. There is no family at level zero. The direction at the step from i is

$$\mathcal{V}_i^{w}[\mathsf{g}] + \sum_{1 \le l \le i} \mathcal{V}_i^{\mathrm{old}}[\mathbf{P}^{\mathrm{red},(l)}[\mathsf{g}]].$$

The direct term gives at most $\frac{3}{2} K_M C_{\mathrm{src}} L^4 e_{i+1}[\mathsf{g}]$ after applying $D\widetilde{E}_i$, by Lemmas 14.16 and 14.19.

For an old family $F = \mathbf{P}^{\mathrm{red},(l)}[\mathsf{g}]$, put $n = i + 1 - l$. It is holomorphic on D_{l-1}. By Lemma 14.20, it is even and vanishes at zero; at the trivial pair its fermionic summands, including its forms, start at order four. Schwarz's lemma, applied separately to each polymer and summand and using $(\bar{\lambda}_i/\bar{\lambda}_{l-1})^2 \le q_n$, gives

$$\|F_0\|_{l;D_i;S}^{\mathrm{arr}}, \ \|F_{\ge 2}\|_{l;D_i;S} \le q_n A^{\mathrm{red}} e_l[\mathsf{g}],$$
$$\|F_2^{\mathrm{tr}}\|_{l;D_i;S} \le 3 q_n^2 A^{\mathrm{red}} e_l[\mathsf{g}].$$

For the second line, the pre-extraction summand is counted again as F_1 when its moved pieces are attached; the forms are counted once. The two inductive norms bound their sum by $2A^{\mathrm{red}} e_l$, hence by the displayed factor three.

The order-six cell remainder gains q_n^3. The torus fourth-coefficient defect gains $q_n^2 e^{-c_{\mathrm{def}} \ell_G / a_l} \le q_n^2 e^{-2Mc_{\mathrm{def}} L^n}$. Thus

$$h_F^2 \sup_c W_l(c) \sup_{D_i} \|m_c\| \le (6C_{\mathrm{mo}} q_n^3 + 3C_{\mathrm{def}} q_n^2 e^{-2Mc_{\mathrm{def}} L^n}) A^{\mathrm{red}} e_l[\mathsf{g}].$$

On $\mathbb{Z}^3$ omit the second term. The lag estimates and eq. 14.xxxii therefore give

$$\epsilon_S(\mathcal{V}_i^{\mathrm{old}}[F]) \le C_{\mathrm{rd}} C_{\mathcal{C}} L^{4n} [(2s_\lambda + c_\psi') L^{-4n} q_n + 3c_\psi' L^{-(2+\alpha)n} q_n^2$$
$$+ c_\psi' L^{-2n} (6C_{\mathrm{mo}} q_n^3 + 3C_{\mathrm{def}} q_n^2 e^{-2Mc_{\mathrm{def}} L^n})] A^{\mathrm{red}} e_{i+1}[\mathsf{g}].$$

Summing over l is bounded by $\theta A^{\mathrm{red}} e_{i+1}/(3K_\mu^{\mathrm{s}} K_M)$. Here $q_n \le L^{n/2} q_n$ and $L^{2n} q_n^3 \le L^{5n/2} q_n^3$, so all terms are included in $S_\diamond$. The derivative estimate now gives

$$\|\widetilde{\mathbf{P}}^{\mathrm{red},(i+1)}[\mathsf{g}]\| \le \left(\tfrac{3}{2} K_M C_{\mathrm{src}} L^4 + \frac{\theta}{2K_\mu^{\mathrm{s}}} A^{\mathrm{red}} \right) e_{i+1}[\mathsf{g}].$$

Removing the scalar value doubles its array norm at most. The extraction forms add at most $K_\mu^{\mathrm{s}} - K_\mu$ times the pre-extraction norm, with the supremum on D_i taken at each form. To see this in the required order, a form satisfies

$$\sup_{D_i} \|P_{z;X,p}\| \le c_b^{-2} c_{\mathsf{S}} M^{-3} h_F^2 \bar{\lambda}_i^4 \|a_{X,p}^{[4]}(z)\|.$$

Apply the piecewise form estimate to the fourth coefficient of each original polymer. Its pieces use a rule independent of λ; $W_{i+1}(Z_p) \le W_{i+1}(X)$ since $Z_p \supset X$; and Cauchy's estimate bounds $\bar{\lambda}_i^4$ times that coefficient by the original polymer norm on D_i. This proves the stated form factor without interchanging a supremum and a sum. The total extraction factor is at most $\max\{2, 1 + K_\mu^{\mathrm{s}} - K_\mu\} \le 2K_\mu^{\mathrm{s}}$. With $\theta \le 1/2$ and the definition of A^{red}, the induction closes.

The fourth coefficient at a site. It is the fourth coefficient of $-\mathfrak{s}_i^{-1} \sum'_{X,p} a_{X,p}(y;\lambda)$ in the reduced system, by Lemma 14.20. The moment estimate and the bound just proved give

$$W_i(y) \sum_{X,p} \|a_{X,p}(y;\lambda)\|_{\mathrm{tr}} \le 3 C_{\mathrm{mo}} h_F^{-2} A^{\mathrm{red}} e_i[\mathsf{g}].$$

The restricted sum is holomorphic on D_{i-1}. Apply Cauchy's formula there and use $\mathfrak{s}_i^{-1} \le c_b^{-2}$ and operator norm at most trace norm. This proves the second assertion.

The constant part. Put $\hat{\mathsf{n}}_i = (\mathsf{bt}_i)^{-1}$, so $L^{i-l}\hat{\mathsf{n}}_l = \hat{\mathsf{n}}_i$, and set

$$\|\mathfrak{N}\|_{*,i} = \max_{l \le i} \sup_x \frac{|\mathfrak{N}_l^\infty[1](x)|}{\hat{\mathsf{n}}_l}.$$

The site supremum has only finitely many translation classes at each level. The constant-source identity of Lemma 14.15 gives

$$\mathfrak{N}_i^\infty[1](x) = \partial_c \sigma_i^{\mathrm{c}}(0) - \sum_{l \le i} L^{i-l} r_l(y_l(x)),$$
$$r_l(y) = \partial_c \nu_l^\infty - \mathfrak{n}_l^\infty[1](y).$$

Let $m_l(\lambda; y)$ be $-\mathfrak{s}_l^{-1}$ times the moment of $\widetilde{\mathbf{P}}^{\infty(l)}[1]$ at y. It is even and starts at order four, $m_l(1;y) = \partial_c \nu_l^\infty$, and its fourth coefficient is $\mathfrak{n}_l^\infty[1](y)$. Thus r_l is its tail of orders at least six at $\lambda = 1$. Cauchy's formula gives

$$|r_l(y)| \le \frac{8}{7} \bar{\lambda}_{l-1}^{-6} \sup_{D_{l-1}} |m_l(\cdot; y)|,$$

using $\bar{\lambda}_{l-1}^{-2} \le 1/8$. The constant chain is holomorphic on $|c| < 8\varrho$ and satisfies $|s_i + \sigma_i^{\mathrm{c}}(c)| \le 2\bar{B}_M/\mathsf{t}_i$. Hence $|\partial_c \sigma_i^{\mathrm{c}}(0)| \le \bar{B}_M/(4\varrho \mathsf{t}_i)$.

Separate the moment and its tail into their reduced and feedback parts. For the reduced part the preceding estimate gives

$$\sup_{D_{l-1}} |m_l^{\mathrm{red}}| \le K_{\mathfrak{n}} (c_{\mathsf{S}} h_F^2)^{-1} A^{\mathrm{red}} a_l^{-1}.$$

Since $a_l^{-1}/\hat{\mathsf{n}}_l = \mathsf{t}_l^2$, $\mathsf{t}_l \le 2x_l \le 2x_{l-1}$ and $\bar{\lambda}_{l-1}^{-2} = c_\lambda (\log x_{l-1})^{2\mathfrak{p}_0}/(2x_{l-1})$,

$$\frac{L^{i-l}|r_l^{\mathrm{red}}|}{\hat{\mathsf{n}}_i} \le \frac{4}{7} K_{\mathfrak{n}} (c_{\mathsf{S}} h_F^2)^{-1} A^{\mathrm{red}} c_\lambda^3 \frac{(\log x_{l-1})^{6\mathfrak{p}_0}}{x_{l-1}}.$$

Here $x_{l-1} \ge L^{i+1-l}x_i/2$ and $x_i \ge e^{18\mathfrak{p}_0}$. The function $(\log x)^{6\mathfrak{p}_0}/x$ decreases for $x \ge e^{6\mathfrak{p}_0}$. The geometric series obtained by substituting $L^{i+1-l}x_i/2$ is uniformly bounded. Let C_r be any bound for the sum of the displayed right sides; it depends only on the constants listed in the statement.

Run the feedback response on its own discs. At step $k-1$ its driving member is $\mathcal{V}^{\mathrm{s}}_{k-1}[\lambda^4\mathfrak{N}^{\infty}_{k-1}[1]]$. Relative to $e_k[1] = a_k^{-1}$, its derivative image is bounded by

$$\mathfrak{d}_k = \frac{3}{2}K_M C_{\mathrm{mn}} c'_\psi c_{\mathsf{S}} h_F^2 L\,\frac{25}{4}c_\lambda^{-2}(\log x_{k-1})^{-4\mathfrak{p}_0}\|\mathfrak{N}\|_{*,k-1}.$$

Indeed, $\hat{\mathfrak{n}}_{k-1}a_k = L/\mathfrak{t}^2_{k-1}$ and $\bar{\lambda}^4_{k-1} \le (25/4)\mathfrak{t}^2_{k-1}c_\lambda^{-2}(\log x_{k-1})^{-4\mathfrak{p}_0}$. The quantity $\mathfrak{d}_k$ is nondecreasing in k.

For the old feedback families the scalar and fermionic norms gain q_n on a smaller disc. Their trivial-pair quadratic norms and cell sums gain q_n^3, since their order is at least six. With the moved-piece convention, the former cost at most $2q_n^3$ and the latter at most $2C_{\mathrm{mo}}q_n^3$ times the old response norm. These are bounded by the terms used in the reduced induction. Thus the total old-family factor is at most $\theta/(2K^{\mathrm{s}}_\mu)$. No forms are extracted from a feedback family; removing the scalar value costs at most two. If R_k is its larger family or scalar-array norm divided by $e_k[1]$, then

$$R_k \le 2\mathfrak{d}_k + \theta\max_{k'<k}R_{k'}, \qquad R_k \le 4\mathfrak{d}_k.$$

The pre-extraction norm is at most $\mathfrak{d}_k + \frac{1}{2}\theta\max_{k'<k}R_{k'} \le 2\mathfrak{d}_k$. Its quadratic norm with attached moved pieces is therefore at most $4\mathfrak{d}_k e_k[1]$.

Apply the moment estimate and Cauchy's tail bound once more. Since $(8/7)\cdot 4\cdot(3/2)\cdot(25/4) = 300/7 \le 43$ and

$$\bar{\lambda}^{-6}_{l-1}\mathfrak{t}^2_l c_\lambda^{-2}(\log x_{l-1})^{-4\mathfrak{p}_0} \le \bar{\lambda}^{-2}_{l-1},$$

one obtains

$$\frac{L^{i-l}|r^{\mathrm{rsp}}_l|}{\hat{\mathfrak{n}}_i} \le 43LC_{\mathrm{mn}}K_{\mathfrak{n}}K_M c'_\psi\bar{\lambda}^{-2}_{l-1}\|\mathfrak{N}\|_{*,l-1}.$$

Use $\bar{\lambda}^{-2}_{l-1} \le q_{i+1-l}\bar{\lambda}_i^{-2}$, $\sum q_n \le S_\diamond$ and $K^{\mathrm{s}}_\mu \ge 1$. The sum is at most $C_{\mathrm{fb}}\bar{\lambda}_i^{-2}\|\mathfrak{N}\|_{*,i} \le \frac{1}{2}\|\mathfrak{N}\|_{*,i}$. Therefore

$$\|\mathfrak{N}\|_{*,i} \le \frac{\bar{B}_M\mathsf{b}}{4\varrho} + C_r + \frac{1}{2}\|\mathfrak{N}\|_{*,i},$$

which proves the constant-direction estimate.

The torus in the constant direction. At a nonwrapping level-l polymer X, subtract from $\widetilde{\mathbf{P}}^{\mathrm{red},(l)}[1](X)$ the infinite-lattice pre-extraction family in the corresponding local coordinates. Denote this difference by $\Delta^{(l)}(X)$. We prove inductively that its scalar part has joint order at least four and its fermionic part has joint order at least five. It is even, so its quadratic part at the trivial pair starts at order six.

The zero-source field-scale steps have the same local structure on these two polymers: the gauge-fixing operators, gauge vertices, scalar localization, normalizers, shift system and fermion objects are carried to each other by the local coordinate identification. The common coefficients are $\xi_i(\lambda)$, $\tau_i(\lambda)$ and $\lambda^2 s_i$. The extraction uses the same infinite-lattice numbers. Induction gives the same assertion for their old zero-source families. The direct constant source is also local and compatible with this identification. Thus $\Delta^{(1)} = 0$ on nonwrapping polymers.

At a later level, an old contribution filed at a nonwrapping polymer receives no term from a wrapping original polymer. Its difference from the infinite-lattice reading has precisely three parts:

(1) the preceding difference $\Delta^{(l')}(X')$ with its scalar value removed;
(2) the negative of the extraction forms of pieces that fit locally but whose original polymer has diameter at least $\ell_G/4$, so the torus rule omits them;
(3) the difference of the cell terms, $\bar{\Phi}(z_c)(m_c - m_c^{\infty})\Phi(z_c)$.

The first has the inductive orders; removing a constant has no effect on its variation. In the second, forms used by both rules agree: the induction gives equality of the fourth coefficients of their nonwrapping pieces. In the third,

$$m_c - m_c^{\infty} = \lambda^4 m_c^{[4]} + (m_c^{(6)} - m_c^{\infty}).$$

Apart from terms of order at least six, the last two parts are bilinears $\lambda^4 \bar{\Psi} K \Psi$ with K independent of λ and the fluctuation. Their variation with no W leg vanishes at $\lambda = 0$, hence has order at least five. Terms linear in W have zero Gaussian mean and require another member of positive order to contribute after expansion. Terms with two W legs have degree zero in the new generators and require an additional member with legs to contribute to their positive-degree part. Thus their derivative images have scalar order at least four and fermionic order at least five. This closes the induction for $\Delta^{(l)}$.

Consequently the fourth moments agree for pieces whose original and expanded polymers both fit in the torus. The difference $\mathfrak{n}_l[1](y) - \mathfrak{n}_l^{\infty}[1](\hat{y})$ is $-\mathfrak{s}_l^{-1}$ times the difference of two sums: the torus pieces retained by extraction whose expanded polymer wraps, and the infinite-lattice pieces whose expanded polymer does not fit or whose original polymer has diameter at least $\ell_G/4$. In either sum,

$$d_l(Z_p) \geq t'_G := \frac{\ell_G}{2c' M a_l} - 2.$$

The wrapping estimate is $d_l(Z_p) \geq \ell_G/(2Ma_l) - 2$; the large-diameter estimate is stronger than t'_G. Here $c' \geq 2$. The tail moment bound and Cauchy's formula for the fourth coefficient give

$$|\mathfrak{n}_l[1](y) - \mathfrak{n}_l^{\infty}[1](\hat{y})| \leq 2e^{\kappa} K_{\mathfrak{n}} (c_{\mathfrak{S}} h_F^2)^{-1} A^{\mathrm{red}} \bar{\lambda}_{l-1}^{-4} a_l^{-1} e^{-2c_{\mathrm{def}} \ell_G / a_l}.$$

For $t'_G < 0$ use the ordinary moment estimate; the displayed tail factor before its simplification then exceeds one.

Now

$$\bar{\lambda}_{l-1}^{-4} a_l^{-1} \leq 4c_{\lambda}^2 L^{-2} (\log x_{l-1})^{4\mathfrak{p}_0} \hat{\mathfrak{n}}_l, \qquad x_{l-1} \leq 2\mathfrak{b} L^{K-l+1}.$$

Sum with factors L^{i-l}. The identity $L^{i-l}\hat{\mathfrak{n}}_l = \hat{\mathfrak{n}}_i$ gives the asserted torus estimate with

$$C_{\mathrm{wr}}^{\mathfrak{N}} = 8e^{\kappa} K_{\mathfrak{n}} (c_{\mathfrak{S}} h_F^2)^{-1} A^{\mathrm{red}} c_{\lambda}^2 L^{-2} \times \sup_{\mathfrak{b} \in \bar{I}(g)} \sum_{u \geq 0} \big(\log(2\mathfrak{b} L^{u+1})\big)^{4\mathfrak{p}_0} e^{-2c_{\mathrm{def}} \ell_G L^u}.$$

This is finite. The translation periods of the infinite-lattice constant-direction response divide the torus periods, so its value is independent of the chosen representative $\hat{x}$.

A direction of the test pair. Linearity and $1 = \sum_p \delta_p$ give the exact identity, at every bare site x,

$$\mathfrak{N}_i[f](x) = f(x)\mathfrak{N}_i[1](x) + \sum_{1 \leq l \leq i} L^{i-l} \sum_p (f(x(p)) - f(x)) \mathfrak{n}_l[\delta_p](y_l(x)). \tag{14.xxxiv}$$

First suppose $a_i \leq \mathsf{d}$. Then $e_i[f] = \|f\|_{\infty} a_i^{-1}$ and $\mathfrak{l}_i = 1$. The constant-direction estimates bound the first term by $(A_c + C_{\mathrm{wr}}^{\mathfrak{N}})\|f\|_{\infty} \hat{\mathfrak{n}}_i$. Moreover $\bar{\lambda}_i^4 \hat{\mathfrak{n}}_i \leq (25/4) c_{\lambda}^{-2} a_i^{-1}$.

For the second term use $|f(x(p)) - f(x)| \leq \|f\|_{\infty} |x(p) - x|/(\vartheta \mathsf{d})$, with the torus distance. The direction δ_p is supported in the radius-a_K ball S_p about $x(p)$, and $e_l[\delta_p] = a_K^3 a_l^{-4}$. Since $\delta_l(y_l(x); S_p) \geq |x(p) - x|/(Ma_l) - 3$, the site estimate gives

$$\sum_p |x(p) - x| \, |\mathfrak{n}_l[\delta_p](y_l(x))| \leq C_{\mathrm{Lip}} M^4 K_{\mathfrak{n}} (c_{\mathfrak{S}} h_F^2)^{-1} A^{\mathrm{red}} \bar{\lambda}_{l-1}^{-4}, \qquad C_{\mathrm{Lip}} = 78\pi e \, \kappa_{\mathrm{s}}^{-4}.$$

For completeness, the barycentres form three translates of $a_K\mathbb{Z}^3$. Choose representatives that realize each torus distance and set $\kappa_l = \kappa_{\mathrm{s}}/(Ma_l) \le 1/(4a_K)$. On one translate their radial sum is at most

$$a_K^{-3}e^{1/4}\int_{\mathbb{R}^3}(|z|+a_K)e^{-\kappa_l|z|}\,dz \le 26\pi e^{1/4}a_K^{-3}\kappa_l^{-4}.$$

The three translates and $e^{3\kappa_{\mathrm{s}}} \le e^{3/4}$ give C_{Lip}.

Multiply eq. 14.xxxiv by $c_{\mathsf{S}}h_F^2\bar\lambda_i^4$ and use

$$\sum_{l\le i}L^{i-l}(\bar\lambda_i/\bar\lambda_{l-1})^4 \le \sum_{n\ge 1}L^{n-1}q_n^2 \le S_\diamond.$$

Since $\mathsf{d} \ge a_i$, this yields

$$c_{\mathsf{S}}h_F^2\bar\lambda_i^4|\mathfrak{N}_i[f](x)| \le B_f e_i[f],$$

where

$$B_f = \frac{25}{4}c_\lambda^{-2}c_{\mathsf{S}}h_F^2(A_c + C_{\mathrm{wr}}^{\mathfrak{N}}) + \frac{C_{\mathrm{Lip}}M^4K_{\mathfrak{n}}A^{\mathrm{red}}S_\diamond}{\vartheta}.$$

This estimate also has the required spatial decay. If $\Delta_x = \operatorname{dist}(x,S_f)/(Ma_i) \ge 1$, then $f(x) = 0$. Apply the site estimate directly to the direction f, using $e_l[f] = L^{i-l}e_i[f]$ and $\delta_l(y_l(x);S_f) \ge L^{i-l}\Delta_x - \sqrt{3}$. It gives

$$\begin{aligned}c_{\mathsf{S}}h_F^2\bar\lambda_i^4|\mathfrak{N}_i[f](x)| &\le 2K_{\mathfrak{n}}A^{\mathrm{red}}e_i[f]\sum_{u\ge 0}L^{2u}q_{u+1}^2e^{-\kappa_{\mathrm{s}}L^u\Delta_x}\\ &\le 2K_{\mathfrak{n}}A^{\mathrm{red}}\Sigma_{\mathrm{w}}e_i[f]e^{-\kappa_{\mathrm{s}}\Delta_x},\end{aligned}$$

where

$$\Sigma_{\mathrm{w}} = \sum_{u\ge 0}L^{2u}q_{u+1}^2e^{-\kappa_{\mathrm{s}}(L^u-1)} < \infty.$$

For $\Delta_x < 1$, use $e^{\kappa_{\mathrm{s}}} \le 2$. Thus, with $B_f' = 2(B_f + K_{\mathfrak{n}}A^{\mathrm{red}}\Sigma_{\mathrm{w}})$,

$$c_{\mathsf{S}}h_F^2\bar\lambda_i^4|\mathfrak{N}_i[f](x)| \le B_f'e_i[f]\,e^{-\kappa_{\mathrm{s}}\operatorname{dist}(x,S_f)/(Ma_i)} \qquad (a_i \le \mathsf{d}). \tag{14.xxxv}$$

On each level-i cube the exponential pays the source weight of its mean. This proves the assertion at these levels.

Now suppose $a_i > \mathsf{d}$. Let l_{d} be the last level with $a_{l_{\mathsf{d}}} \le \mathsf{d}$ and put $n_{\mathsf{d}} = i - l_{\mathsf{d}} \ge 1$. This level exists because $\mathsf{d} \ge a_K$. Then

$$e_i[f] = \|f\|_\infty \mathsf{d}^3a_i^{-4}, \qquad \mathfrak{N}_i[f] = L^{n_{\mathsf{d}}}\mathfrak{N}_{l_{\mathsf{d}}}[f] + \sum_{l_{\mathsf{d}}<l\le i}L^{i-l}\mathfrak{n}_l[f]\circ y_l.$$

Also $1 + n_{\mathsf{d}}\log L \le (1+\log L)\mathfrak{l}_i$ and $n_{\mathsf{d}} - 1 \le \mathfrak{l}_i - 1$.

For $l < i$ and x in a level-i cube $\square$,

$$e^{\kappa_{\mathrm{s}}\delta_i(\square;S_f)}e^{-\kappa_{\mathrm{s}}\operatorname{dist}(x,S_f)/(Ma_l)} \le e^{-\kappa_{\mathrm{s}}\operatorname{dist}(x,S_f)/(2Ma_l)}.$$

The mean of the right side over $\square$ is at most $C_{\mathcal{C}}L^{-3(i-l)}(1+\mathsf{d}/a_l)^3$. This follows by counting the M^3L^{3l} bare sites in each level-l cell and summing its distance weight, then dividing by M^3L^{3i}.

For the first term use eq. 14.xxxv at l_{d} and $|\lambda|^4 \le q_{n_{\mathsf{d}}}^2\bar\lambda_{l_{\mathsf{d}}}^4$ on D_i. If $l_{\mathsf{d}} = 0$, this term vanishes. Otherwise its weighted mean is at most

$$8C_{\mathcal{C}}L^{2n_{\mathsf{d}}}q_{n_{\mathsf{d}}}^2B_f'e_i[f] \le 32C_{\mathcal{C}}(1+\log L)^{4\mathfrak{p}_0}B_f'\mathfrak{l}_i^{4\mathfrak{p}_0}e_i[f].$$

We used $L^{2n}q_n^2 \le 4(1+n\log L)^{4\mathfrak{p}_0}$ and $(1+\mathsf{d}/a_{l_{\mathsf{d}}})^3 \le 8(\mathsf{d}/a_{l_{\mathsf{d}}})^3$.

For each $l_{\mathsf{d}} < l < i$, the site estimate, $e_l[f] = L^{4(i-l)}e_i[f]$, the preceding mean bound and $|\lambda|^4 \le q_{i+1-l}^2\bar\lambda_{l-1}^4$ give at most

$$16C_{\mathcal{C}}K_{\mathfrak{n}}A^{\mathrm{red}}L^{2(i-l)}q_{i+1-l}^2e_i[f] \le 64C_{\mathcal{C}}K_{\mathfrak{n}}A^{\mathrm{red}}L^{-2}(1+\log L)^{4\mathfrak{p}_0}\mathfrak{l}_i^{4\mathfrak{p}_0}e_i[f].$$

There are at most $\mathfrak{l}_i - 1$ such levels. The level $l = i$ contributes at most $2K_{\mathfrak{n}} A^{\mathrm{red}} q_1^2 e_i[f] \le K_{\mathfrak{n}} A^{\mathrm{red}} e_i[f]$. Adding these bounds proves the final assertion with

$$A_{\mathfrak{N}} = 64 C_{\mathcal{C}} (1 + \log L)^{4\mathfrak{p}_0} (B'_f + K_{\mathfrak{n}} A^{\mathrm{red}}).$$

It also bounds B'_f. The proof for h is identical. □

14.20. **First and mixed responses.**

Lemma 14.22 (First responses)**.** Put

$$A^{\mathrm{rsp}} = 6 K_\mu^{\mathrm{s}} K_M C_{\mathrm{mn}} c'_\psi A_{\mathfrak{N}} L^4, \qquad A_{\mathbf{P}} = A^{\mathrm{red}} + A^{\mathrm{rsp}}, \qquad A'_{\mathbf{P}} = C_{\mathrm{src}} + C_{\mathrm{mn}} A_{\mathfrak{N}} + A_{\mathbf{P}}.$$

For $1 \le i \le j$,

$$\|\mathbf{P}^{\mathrm{rsp},(i)}[f]\|_{i;D_{i-1};S_f} \le A^{\mathrm{rsp}} e_i[f] \mathfrak{l}_i^{4\mathfrak{p}_0+1},$$
$$\|\mathbf{P}^{(i)}[f]\|_{i;D_{i-1};S_f} \le A_{\mathbf{P}} e_i[f] \mathfrak{l}_i^{4\mathfrak{p}_0+1}.$$

The latter bound also holds for the pre-extraction family. These estimates hold in the scalar array norm and in the fermionic norm summing the individual summands, with each form at its own polymer. For $0 \le i < j$,

$$\epsilon_{S_f}(\delta_i[f]) \le A'_{\mathbf{P}} e_i[f] \mathfrak{l}_i^{4\mathfrak{p}_0+1} \quad \text{on } D_i.$$

The same estimates hold for h. The families $\mathbf{P}^{(l)}[f]$, $l \le i$, read together at level $i \le K$, have norm at most

$$(1 + C_{\mathrm{rd}} C_{\mathcal{C}} (1 + 2C_{\mathrm{mo}}) S_\diamond) A_{\mathbf{P}} e_i[f] \mathfrak{l}_i^{4\mathfrak{p}_0+1}$$

in $\|\cdot\|^{\mathrm{out}}_{i;D_{i-1};S_f}$. Their scalar parts have the same bound in the strong family norm on the full level-i cores. ■

Proof. The feedback recursion is driven by $\mathcal{V}_i^{\mathrm{s}}[\lambda^4 \mathfrak{N}_i[f]]$. It has no extraction forms. Its units are the differences of the full and reduced units: the kernel minus its moment at the original cell, and the moved pieces at the cells of their sites. Thus the old variations split term by term into reduced and feedback variations. At level one the feedback vanishes because $\mathfrak{N}_0 = 0$.

For the induction, let R_l be the larger of the feedback family norm and its scalar array norm, divided by $e_l[f] \mathfrak{l}_l^{4\mathfrak{p}_0+1}$. Lemmas 14.19 and 14.21 give

$$\epsilon_{S_f}(\mathcal{V}_i^{\mathrm{s}}[\lambda^4 \mathfrak{N}_i[f]]) \le C_{\mathrm{mn}} c'_\psi A_{\mathfrak{N}} L^4 e_{i+1}[f] \mathfrak{l}_{i+1}^{4\mathfrak{p}_0+1}.$$

For an old feedback family of lag $n = i + 1 - l$, Schwarz's lemma gives factors q_n in its scalar array and fermionic norms, $2q_n^3$ in its trivial-pair quadratic norm with attached pieces, and $2C_{\mathrm{mo}} q_n^3$ in its cell sum. The latter two use order six and the absence of forms, on both the torus and $\mathbb{Z}^3$. The lag estimate and spatial count therefore bound the sum of old variations by

$$\frac{\theta}{3 K_\mu^{\mathrm{s}} K_M} \max_{l \le i} R_l \, e_{i+1}[f] \mathfrak{l}_{i+1}^{4\mathfrak{p}_0+1}.$$

Indeed, $2q_n^3 \le 3q_n^2$ and $2C_{\mathrm{mo}} \le 3(1 + 2C_{\mathrm{mo}})$, so this is the same summable bracket as in the reduced induction. The derivative image has bound

$$\left(\tfrac{3}{2} K_M C_{\mathrm{mn}} c'_\psi A_{\mathfrak{N}} L^4 + \frac{\theta}{2 K_\mu^{\mathrm{s}}} \max_{l \le i} R_l \right) e_{i+1}[f] \mathfrak{l}_{i+1}^{4\mathfrak{p}_0+1}.$$

Removing the scalar value doubles the norm at most. It follows that $R_{i+1} \le A^{\mathrm{rsp}}/(2K_\mu^{\mathrm{s}}) + (\theta/K_\mu^{\mathrm{s}}) \max_{l \le i} R_l \le A^{\mathrm{rsp}}$. The pre-extraction bound is half this bound. Adding the reduced response proves the first two estimates.

For $\delta_i[f]$, the direct term costs $C_{\mathrm{src}} e_i[f]$. Its mass term costs at most $A^{\mathrm{rsp}} e_i[f] \mathfrak{l}_i^{4\mathfrak{p}_0+1}/(6 K_\mu^{\mathrm{s}} K_M L^4)$. The old full families have $\|F_2^{\mathrm{tr}}\| \le 3 A_{\mathbf{P}} e_l[f] \mathfrak{l}_l^{4\mathfrak{p}_0+1}$. Their cell sums are those of Lemma 14.20, including the torus defect. The reduced-family summation therefore bounds their total by $A_{\mathbf{P}} e_i[f] \mathfrak{l}_i^{4\mathfrak{p}_0+1}/6$,

using $e_{i+1} \le e_i$ and $K_\mu^{\mathrm{s}} K_M \ge 1$. The mass term is also at most one sixth of this size. These bounds imply the assertion for $A'_{\mathbf{P}}$.

For the readings at level i, the family with $l = i$ already has the required strong norm. For $l < i$, put $n = i - l$ and apply Lemma 14.18 on D_{i-1}. The scalar family vanishes at the trivial pair, so (R0) is its actual reading on the full core, including at $i = K$. The fermionic terms use the common-pair norm for long lags and the strong norm for short lags. Their combined bound is

$$C_{\mathrm{rd}} C_{\mathcal{C}} [2q_n + 3L^{(2-\alpha)n} q_n^2 + L^{2n}(6C_{\mathrm{mo}} q_n^3 + 3C_{\mathrm{def}} q_n^2 e^{-2Mc_{\mathrm{def}} L^n})] A_{\mathbf{P}} e_i[f] \mathfrak{l}_i^{4\mathfrak{p}_0+1}.$$

Its sum is bounded by the asserted factor $C_{\mathrm{rd}} C_{\mathcal{C}}(1 + 2C_{\mathrm{mo}}) S_\diamond$. The scalar contribution alone is bounded by $C_{\mathrm{rd}} C_{\mathcal{C}} q_n$ times the size, in the strong norm. This proves the final assertion without moving a pair supremum inside the long-lag fermionic sum. □

Lemma 14.23 (Mixed responses)**.** Set

$$C_{\mathfrak{l}} = \sup_{t \ge 1} t^{-7/2}(1 + \log t)^{8\mathfrak{p}_0+2}, \qquad A_{\mathbf{Q}} = 32 c_{\mathrm{cl}}^{-1} K_M (A'_{\mathbf{P}})^2 C_{\mathfrak{l}} L^{9/2}.$$

For $1 \le i \le j$,

$$\|\mathbf{Q}^{(i)}[f,h]\|_{i;D_{i-1};S_f,S_h} \le A_{\mathbf{Q}} e_i[f,h].$$

The same holds before extraction and in the scalar array norm. For $0 \le i < j$, on D_i,

$$\epsilon_{S_f,S_h}\left(\sum_{1\le l\le i} \mathcal{V}_i^{\mathrm{old}}[\mathbf{Q}^{(l)}[f,h]]\right) \le \frac{\theta}{3K_\mu^{\mathrm{s}} K_M} A_{\mathbf{Q}} e_{i+1}[f,h] \le A_{\mathbf{Q}} e_i[f,h].$$

The sum is zero at $i = 0$. The families with $l \le i$, read at level $i \le K$, have together norm at most

$$(1 + C_{\mathrm{rd}} C_{\mathcal{C}}(1 + 2C_{\mathrm{mo}}) S_\diamond) A_{\mathbf{Q}} e_i[f,h]$$

in $\|\cdot\|^{\mathrm{out}}_{i;D_{i-1};S_f,S_h}$. The scalar parts obey the same bound in the strong norm on the full cores. ■

Proof. The new mixed term is the second derivative of the step in the two first-response directions. Lemmas 14.16 and 14.22 bound it by

$$8 c_{\mathrm{cl}}^{-1} K_M (A'_{\mathbf{P}})^2 e_i[f] e_i[h] \mathfrak{l}_i^{8\mathfrak{p}_0+2}.$$

If $a_i \le \mathsf{d}$, the product of sizes is $e_i[f,h]$ and $\mathfrak{l}_i = 1$. If $a_i > \mathsf{d}$, set $t = a_i/\mathsf{d}$. Its ratio to $e_i[f,h]$ is $t^{-7/2}(1 + \log t)^{8\mathfrak{p}_0+2} \le C_{\mathfrak{l}}$. In either case the new term is at most $8 c_{\mathrm{cl}}^{-1} K_M (A'_{\mathbf{P}})^2 C_{\mathfrak{l}} L^{9/2} e_{i+1}[f,h]$.

For an old mixed family at lag $n = i + 1 - l$, its scalar and fermionic norms gain q_n on D_i. Its trivial-pair quadratic norm gains q_n^3, and its cell sum gains $C_{\mathrm{mo}} q_n^3$, since it has no forms and starts at order six there. The spatial count is now $\mathcal{N} e_l[f,h] \le C_{\mathcal{C}} L^{9n/2} e_{i+1}[f,h]$. The sum of its old variations is bounded by

$$C_{\mathrm{rd}} C_{\mathcal{C}} \sum_{n \ge 1} [(2s_\lambda + c'_\psi) L^{n/2} q_n + c'_\psi L^{(5/2-\alpha)n} q_n^3 + c'_\psi C_{\mathrm{mo}} L^{5n/2} q_n^3] A_{\mathbf{Q}} e_{i+1}[f,h].$$

Here $L^{(5/2-\alpha)n} q_n^3 = (L^{n/2} q_n) L^{(2-\alpha)n} q_n^2 \le L^{(2-\alpha)n} q_n^2$, because $L^{n/2} q_n \le 2^{10/9} L^{-7n/18} \le 1$. Thus this is at most $\theta A_{\mathbf{Q}} e_{i+1}/(3K_\mu^{\mathrm{s}} K_M)$. It proves the old-direction estimate as well.

Apply the step derivative estimate and remove the scalar value, at cost at most two. The induction closes because

$$\|\mathbf{Q}^{(i+1)}[f,h]\| \le 2\left(8 c_{\mathrm{cl}}^{-1} K_M (A'_{\mathbf{P}})^2 C_{\mathfrak{l}} L^{9/2} + \frac{\theta}{2K_\mu^{\mathrm{s}}} A_{\mathbf{Q}}\right) e_{i+1}[f,h] \le (\tfrac{1}{2} + \theta) A_{\mathbf{Q}} e_{i+1}[f,h].$$

The pre-extraction norm is at most half the middle expression, and the same argument uses scalar array norms throughout.

For the total reading, the term $l = i$ is already bounded. For $l < i$, the lag estimate and the same Schwarz factors give

$$C_{\mathrm{rd}} C_{\mathcal{C}} [2L^{n/2} q_n + L^{(5/2-\alpha)n} q_n^3 + C_{\mathrm{mo}} L^{5n/2} q_n^3] A_{\mathbf{Q}} e_i[f,h].$$

Its sum has the stated bound. The scalar part has only the first term with coefficient one, in the strong norm on the full core; the fermionic part uses the common-pair norm. The scalar estimate includes the top level. □

Lemma 14.24 (Sums of response families)**.** Let G be a level-i family and let S be a nonempty set in a closed ball of radius $r \ge a_K$. Then

$$\sum_{X'} \sup_{\lambda\in D} \|G(X')\| \le e^{\kappa/16} C_{\mathcal{C}} (1 + r/a_i)^3 \|G\|_{i;D;S}.$$

For the two supports of a test pair,

$$\sum_{X'} \sup_{\lambda\in D} \|G(X')\| \le e^{\kappa/8} \mathcal{C}'_i \|G\|_{i;D;S_f,S_h}.$$

The norms on the left include the individual pair suprema. For one pair P on a polymer Z, the analogous bound for $\sup_{\lambda\in D} \sum_{X'\subset Z} \|G(X'; P|_{X'}, \cdot; \lambda)\|_{h^{(i)}}$ holds with $\|\cdot\|^{\mathrm{out}}$ on the right.

For $1 \le i \le j$, on D_{i-1},

$$\left|\sum_{X'} \mathfrak{e}_i[f](X')\right| \le C_{\mathfrak{e},1} \|f\|_\infty a_i^{-1} (\mathsf{d}/a_i)^3 \mathfrak{l}_i^{4\mathfrak{p}_0+1},$$

$$\left|\sum_{X'} \mathfrak{e}_i[f,h](X')\right| \le C_{\mathfrak{e},2} \mathsf{n}_0(X; a_i) \begin{cases} (\mathsf{d}/a_i)^{5/2}, & a_i \ge \mathsf{d}, \\ (\mathsf{d}/a_i)^3 e^{-\kappa_{\mathrm{s}} \mathsf{d}/(2M a_i)}, & a_i < \mathsf{d}, \end{cases}$$

where $C_{\mathfrak{e},1} = 8e^{\kappa/16} C_{\mathcal{C}} A_{\mathbf{P}}$ and $C_{\mathfrak{e},2} = e^{\kappa/8} C_{\mathcal{C}} A_{\mathbf{Q}}$. On D_i the second sum is n_i^λ, so it has the same bounds there. ■

Proof. Overcount the polymer sum by $\sum_\square \sum_{X'\supset\square}$. For $X' \supset \square$, the diameter estimate gives

$$\delta_i(X'; S) \ge \delta_i(\square; S) - \tfrac{1}{2} c'(1 + d_i(X')).$$

Hence its inverse weight is bounded by $e^{\kappa/16} e^{\kappa d_i(X')/16} e^{-\kappa_{\mathrm{s}} \delta_i(\square;S)}$. The family norm pays the polymer factor and the remaining cube sum is bounded by $C_{\mathcal{C}}(1 + r/a_i)^3$. Two source weights give instead $e^{\kappa/8} e^{\kappa d_i(X')/8}$ and the cube sum $\mathcal{C}'_i$. For the common-pair assertion do this count at fixed λ, Z, P, using only the restrictions of P, then take the suprema.

A scalar value at the trivial pair is bounded by its family's norm. Apply the first part to the pre-extraction families in Lemmas 14.22 and 14.23. Use

$$e_i[f](1 + \mathsf{d}/a_i)^3 \le 8 \|f\|_\infty a_i^{-1} (\mathsf{d}/a_i)^3$$

and $e_i[f,h] = \mathsf{n}_0(X; a_i) \min\{1, (\mathsf{d}/a_i)^{5/2}\}$. The bounds on $\mathcal{C}'_i$ give the two mixed cases. The identity with n_i^λ is the total-family identity of Lemma 14.15 on D_i. □

14.21. **Responses joining different cells.** Use the board and sources of Definition 14.7, and let $n_{\mathfrak{J}} = \#\mathfrak{J} \le 16$. We work at $\lambda = 1$ throughout this subsection. The decay between different cells will absorb the growth of the source sizes under reading.

Define, in this order,

$$C_r^D = \frac{71}{64} \left(\frac{4r}{3}\right)^r, \qquad 1 \le r \le 16,$$

$$C_{\mathrm{v}} = \max\{1, 2s_\lambda + c'_\psi\} C_{\mathrm{rd}} (3 + C_{\mathrm{mo}}),$$

$$T_{\mathrm{x}} = \frac{2[39 \log L + \log(12 K_M C_{\mathrm{v}} M^3)]}{L - 1}.$$

Let $C_{\mathcal{C}}^{\mathrm{B}}$ bound the cube count

$$\sum_\square e^{-\kappa_{\mathrm{B}} \delta_l(\square;S)} \le C_{\mathcal{C}}^{\mathrm{B}} (1 + r_S/a_l)^3$$

for every S in a radius-r_S ball, $r_S \geq a_K$. Put

$$C^{\mathrm{x}}_{\sup} = \sup_{t\geq 0}(1+t)^3 e^{-c_1 t/2}, \qquad m_{\mathrm{x}} = \min\{m \geq 0 : c_1 \ell_G L^m/4 \geq T_{\mathrm{x}}\}.$$

For a one-cell block set $A'_{\{\mathsf{g}\}} = A'_{\mathbf{P}}$ and $A'_{\{f_c,h_c\}} = A_{\mathbf{Q}}$. For sets J joining cells define inductively $A'_{\mathsf{J}} = A_{\mathsf{J}}$, where

$$A_{\mathsf{J}} = 4K_M e^{c_E} C_{\mathfrak{l}}^8 L^{36} \sum_{\substack{\pi\in \mathrm{Part}(\mathsf{J})\\ \#\pi\geq 2}} C^D_{\#\pi} c_{\mathrm{cl}}^{1-\#\pi} \prod_{b\in\pi} A'_b,$$

$$A^{\mathrm{x}}_{n_{\mathfrak{J}}} = 4e^{\kappa/8} C^{\mathrm{B}}_{\mathcal{C}} C^{\mathrm{x}}_{\sup} \max_{\substack{\mathsf{J} \text{ joins cells}\\ \#\mathsf{J}\leq n_{\mathfrak{J}}}} A_{\mathsf{J}}.$$

Only the numbers of cells carrying one or two directions enter A_{J}. The constants through T_{x} are chosen after c_λ, ρ and before γ_{e}. The number m_{x} is chosen after g. The constants A_{J} are chosen after $A'_{\mathbf{P}}, A_{\mathbf{Q}}$ and depend on g, ϑ and the fixed constants. None depends on the cutoff, b, m, the distance scale or the positions of the sources, and none enters an earlier parameter choice.

Lemma 14.25. For every $\mathsf{J} \subset \mathfrak{J}$ joining cells:

(1) At every $l \leq K$,

$$\partial_{\mathsf{J}} \sum_X \mathfrak{E}_l(X; w, 0)\Big|_{w=0} = \sum_X \mathfrak{e}_l[\mathsf{J}](X), \qquad \partial_{\mathsf{J}} E_k(w,0)|_{w=0} = \sum_{l\leq k}\sum_X \mathfrak{e}_l[\mathsf{J}](X).$$

The derivative of the field-dependent total at level l, including the accumulated mass member, is the sum of the $\mathbf{R}^{(l')}[\mathsf{J}]$, $l' \leq l$, read at that level. This is an identity of functions at the backgrounds of real block fields passing the density test, with values in the level-l Grassmann algebra.

(2) For $1 \leq i \leq K - m_{\mathrm{x}}$,

$$\|\mathbf{R}^{(i)}[\mathsf{J}]\|_{i;\mathsf{J}} \leq A_{\mathsf{J}} \varsigma_i^{\mathrm{x}} \widehat{e}_i[\mathsf{J}].$$

The same bound holds before removing its scalar value, and $\epsilon_{\mathsf{J}}(\delta_{i-1}[\mathsf{J}]) \leq A_{\mathsf{J}} \varsigma_i^{\mathrm{x}} \widehat{e}_i[\mathsf{J}]$.

(3) For the same i,

$$\sum_{l\leq i}\sum_X |\mathfrak{e}_l[\mathsf{J}](X)| \leq A^{\mathrm{x}}_{n_{\mathfrak{J}}} e^{-c_{\mathrm{x}} D_{\mathrm{x}}/a_i} \widehat{e}_i[\mathsf{J}].$$

The same bound holds for the families $\mathbf{R}^{(l)}[\mathsf{J}]$, $l \leq i$, read at level i, evaluated at restrictions of one regular complex pair on any level-i polymer Z, and summed over their polymers contained in Z in the Grassmann norm at $h^{(i)}$. The whole torus is allowed as Z.

■

Proof. The source-neighbourhood construction in Proposition 14.14 at $\lambda = 1$ makes all steps through K holomorphic for $\|w\|_\sharp < \varepsilon_K(1)$. On the finite torus, $w = \sum_s z_s \mathsf{g}_s$ belongs to this neighbourhood for small z. Thus all the derivatives in the statement exist.

Derivatives of the totals. Induct on the level, simultaneously for all sets of directions. There is no activity or mass response at level zero. At each step the source enters through the weight members, linear in w, and through variation of the field-dependent total. The calibration, tests, backgrounds, vertices and covariance structure are independent of w. For every block b of a partition, the derivative of the potential and quadratic origin therefore has the total of $\delta_i[b]$. For a singleton this is the first-response identity; for a pair in one cell it is the mixed-response identity. For a block joining cells it is the level induction hypothesis. Higher derivatives of the direct weight vanish.

Including its Gaussian part, the sum of the step activities is the logarithm of the Gaussian integral of $e^{\mathcal{V}^\circ} \chi_i e^{\mathcal{V}}$. It depends on the potential and origin through their totals alone. For two systems of at most sixteen directions with equal totals, their expansions on a common parameter polydisc are

continuous logarithms of the same exponential. They agree at zero, so their difference, which lies in $2\pi i\mathbb{Z}$, is zero. Their mixed derivatives agree. The chain rule in the distinct source variables is the sum over partitions of J. It follows that

$$\partial_{\mathsf{J}} \sum_X \widetilde{E}^{(i+1)}(X;w,0)\Big|_{w=0} = \sum_X \widetilde{\mathbf{R}}^{(i+1)}[\mathsf{J}](X).$$

The mass extraction moves a local bilinear from the activity into the mass member and does not change their field-dependent total. Evaluation of the scalar part at the trivial configuration is linear, so its derivative is the sum of $\mathfrak{e}_{i+1}[\mathsf{J}]$. Subtracting this value and carrying the preceding total to level $i+1$ proves the first assertion.

A step derivative of order at most sixteen. For directions $\delta_1,\dots,\delta_r$, use the polydisc

$$|t_a| < \frac{3c_{\mathrm{cl}}}{4r\epsilon(\delta_a)}.$$

Zero-size directions are handled by their vanishing derivative or by a limiting radius. Its total perturbation size is less than $3c_{\mathrm{cl}}/4$. The potential size is less than $c_{\mathrm{cl}}/4 + 3c_{\mathrm{cl}}/4$ and the origin norm less than $(5/64+3/8)w_* < w_*/2$. Thus the expansion is admitted and holomorphic. The potential part, twice the tail and the Gaussian origin give the polymer bound

$$\left(\frac{1}{4}+\frac{1}{32}+\frac{5}{64}+\frac{3}{4}\right) c_{\mathrm{cl}} K'_M e^{-\kappa_+ d_{i+1}(X)} = \frac{71}{64} c_{\mathrm{cl}} K'_M e^{-\kappa_+ d_{i+1}(X)}.$$

The tail bound used here is $c_\lambda^{-1/2} \le c_{\mathrm{cl}}/64$. Cauchy's formula and locality give

$$\|D^r \widetilde{E}_i[\delta_1,\dots,\delta_r](X)\| \le C_r^D c_{\mathrm{cl}}^{1-r} K'_M \prod_a \epsilon(\delta_{a,X}) e^{-\kappa_+ d_{i+1}(X)}.$$

When J is the disjoint union of J_a, the local directions satisfy $\epsilon(\delta_{a,X}) \le (w_{i+1}^{\mathsf{J}_a}(X))^{-1}\epsilon_{\mathsf{J}_a}(\delta_a)$. The weights multiply to w_{i+1}^{J}. The decay excess $\kappa_+ - \kappa \ge 2c_E + 1$ permits the polymer sum, with $K_M = C_E K'_M$, and retains the scalar array rate $\kappa+1$. Consequently

$$\|D^r \widetilde{E}_i[\delta_1,\dots,\delta_r]\|_{i+1;\mathsf{J}} \le C_r^D c_{\mathrm{cl}}^{1-r} K_M \prod_a \epsilon_{\mathsf{J}_a}(\delta_a). \tag{14.xxxvi}$$

The cases $r=1,2$ give $71/48 \le 3/2$ and $71/9 \le 8$, consistently with the earlier estimates.

Reading an old family. Each member of $\widetilde{\mathbf{R}}^{(l)}$ is a Cauchy integral of step activities with holomorphic, jointly gauge-invariant inputs. It therefore has those properties on the full core. Its scalar part is local and pointed by equal shares. Its units are its polymers; no diagonal terms are moved. Thus Lemma 14.18 applies with $F_1 = G = 0$. Its cell sums are bounded by $C_{\mathrm{mo}} h_F^{-2}\|F\|_{l;\mathsf{J}}$, including the torus cell sums. The total source weight rate is $\#\mathsf{J}\,\kappa_{\mathrm{B}} \le 2\kappa_{\mathrm{s}}$. For $n = i+1-l \ge 1$, discard the decay factors $L^{-4n}, L^{-(2+\alpha)n}, L^{-2n} \le 1$ in the lag estimate and use the count $M^3 L^{3n}$. It gives

$$\epsilon_{\mathsf{J}}(\mathcal{V}_i^{\mathrm{old}}[\mathbf{R}^{(l)}[\mathsf{J}]]) \le C_{\mathrm{v}} M^3 L^{3n} \|\mathbf{R}^{(l)}[\mathsf{J}]\|_{l;\mathsf{J}}. \tag{14.xxxvii}$$

The reading itself has the same bound in $\|\cdot\|_{i+1;\mathsf{J}}^{\mathrm{pair}}$: the scalar value is zero at the trivial pair, so (R0) gives its reading even on the top core; the fermionic bracket costs at most $2 + C_{\mathrm{mo}}$ times the old norm.

The source sizes. The first-response size changes by at most L^{4n}, the mixed one by at most $L^{9n/2}$, and $\mathfrak{l}_l \le \mathfrak{l}_{i+1}$. There are at most eight cells. Hence

$$\widehat{e}_l[\mathsf{J}] \le L^{36n} \widehat{e}_{i+1}[\mathsf{J}].$$

For a partition π of J,

$$\prod_{\substack{b\in\pi \\ b \text{ in one cell}}} \widehat{e}_i[b] \prod_{\substack{b\in\pi \\ b \text{ joins cells}}} \widehat{e}_{i+1}[b] \le C_{\mathfrak{l}}^8 L^{36} \widehat{e}_{i+1}[\mathsf{J}]. \tag{14.xxxviii}$$

Indeed, a cell whose sources lie in one block costs at most $L^{9/2}$. If its two sources lie in different blocks, the product estimate in Lemma 14.23 costs at most $C_{\mathfrak{l}}L^{9/2}$ when both blocks lie in one cell, $C_{\mathfrak{l}}L^4$ when only one does, and $C_{\mathfrak{l}}$ when both join cells.

New terms joining cells. Induct on $\#\mathsf{J}$. In a partition with at least two blocks, every block joining cells is smaller than J. A one-cell block has its direction bounded by $A'_b\widehat{e}_i[b]$, by the first- and mixed-response estimates. A smaller block joining cells has bound $A'_b\varsigma^{\mathrm{x}}_{i+1}\widehat{e}_{i+1}[b]$. If such a block occurs, retain its factor $\varsigma^{\mathrm{x}}_{i+1}$ and discard the same factors from the other blocks. Equations eq. 14.xxxvi and eq. 14.xxxviii then give

$$C^D_{\#\pi}c_{\mathrm{cl}}^{1-\#\pi}K_M\prod_{b\in\pi}A'_b\,\varsigma^{\mathrm{x}}_{i+1}C^8_{\mathfrak{l}}L^{36}\widehat{e}_{i+1}[\mathsf{J}].$$

If every block lies within one cell, the inputs have rate κ_{s} at each support, whereas the output asks only for $\kappa_{\mathrm{B}}=\kappa_{\mathrm{s}}/8$. The unused weight is

$$\exp\left[-\frac{7}{8}\kappa_{\mathrm{s}}\sum_{s\in\mathsf{J}}\delta_{i+1}(X;S_s)\right].$$

Two of these supports lie in different cells. Their distances from X add to at least $D_{\mathrm{x}}-\operatorname{diam}X$. By the choices of η_{x} and c_1, this unused weight is at most

$$\varsigma^{\mathrm{x}}_{i+1}e^{c_E(1+d_{i+1}(X))}.$$

One c_E of the decay excess pays the polymer factor, another pays the polymer count, and the remaining unit keeps the scalar array rate. The preceding new-term bound therefore holds with the additional factor e^{c_E}. Sum over the partitions with at least two blocks. By the definition of A_{J} their total is at most $A_{\mathsf{J}}\varsigma^{\mathrm{x}}_{i+1}\widehat{e}_{i+1}[\mathsf{J}]/4$.

Induction over the levels. For $i<K-m_{\mathrm{x}}$ put

$$v_l=\frac{\|\mathbf{R}^{(l)}[\mathsf{J}]\|_{l;\mathsf{J}}}{\varsigma^{\mathrm{x}}_l\widehat{e}_l[\mathsf{J}]},\qquad t=\frac{c_1D_{\mathrm{x}}}{a_{i+1}}\ge T_{\mathrm{x}}.$$

The inequality for t uses $D_{\mathrm{x}}\ge\ell_G/4$ and $a_{i+1}\le L^{-m_{\mathrm{x}}}$. Since $\varsigma^{\mathrm{x}}_l=\varsigma^{\mathrm{x}}_{i+1}e^{-t(L^n-1)}$, the old-direction estimate is

$$\epsilon_{\mathsf{J}}(\delta_i[\mathsf{J}])\le C_{\mathrm{v}}M^3\sum_{n=1}^{i}L^{39n}e^{-t(L^n-1)}v_{i+1-n}\,\varsigma^{\mathrm{x}}_{i+1}\widehat{e}_{i+1}[\mathsf{J}].$$

Use $L^n-1\ge n(L-1)$ and $x=L^{39}e^{-t(L-1)}$. The definition of T_{x} implies $x\le L^{-39}(12K_MC_{\mathrm{v}}M^3)^{-2}$. Thus the sum of the kernel is at most $2x\le(6K_MC_{\mathrm{v}}M^3)^{-1}$. Apply eq. 14.xxxvi at $r=1$ to the old direction and remove the scalar value at cost two. The result is

$$v_{i+1}\le 2\left(\frac{3}{2}K_MC_{\mathrm{v}}M^3\sum_{n=1}^{i}L^{39n}e^{-t(L^n-1)}v_{i+1-n}+\frac{1}{4}A_{\mathsf{J}}\right)\le\frac{1}{2}\max_{l\le i}v_l+\frac{1}{2}A_{\mathsf{J}}.$$

The sum is empty at $i=0$, so induction gives $v_l\le A_{\mathsf{J}}$. Before removing the scalar value the norm is at most half this bound, and the old-direction bound is at most this size divided by $6K_M$. This proves the second assertion. The same finite recursive estimates, without the contraction restriction on i, show that all families needed for the total identities through K have finite size.

Summing the families. For $X\supset\square$, $w^{\mathsf{J}}_i(\square)\le e^{\kappa(1+d_i(X))/8}w^{\mathsf{J}}_i(X)$. The polymer decay pays the factor depending on $d_i(X)$. Keep any one support in the remaining cube count to obtain

$$\sum_X\sup\|G(X)\|\le e^{\kappa/8}C^{\mathrm{B}}_{\mathcal{C}}(1+\mathsf{d}/a_i)^3\|G\|_{i;\mathsf{J}}.$$

The same proof at fixed Z,P gives the common-pair version of this bound. Since $\mathsf{d}<D_{\mathrm{x}}$,

$$(1+\mathsf{d}/a_l)^3(\varsigma^{\mathrm{x}}_l)^{1/2}\le C^{\mathrm{x}}_{\mathrm{sup}}.$$

Apply this to the scalar values of the pre-extraction families. Their sum over levels is bounded by

$$e^{\kappa/8} C_{\mathcal{C}}^{\mathrm{B}} C_{\mathrm{sup}}^{\mathrm{x}} A_{\mathsf{J}} \sum_{l \le i} e^{-c_{\mathrm{x}} D_{\mathrm{x}}/a_l} \widehat{e}_l[\mathsf{J}].$$

The level-i term dominates this sum up to a factor two:

$$\sum_{m \ge 1} L^{36m} e^{-(t_i/2)(L^m-1)} \le 1, \qquad t_i = c_1 D_{\mathrm{x}}/a_i \ge T_{\mathrm{x}}.$$

Indeed, $L^{36} e^{-T_{\mathrm{x}}(L-1)/2} = L^{-3}(12 K_M C_{\mathrm{v}} M^3)^{-1} \le 1/12$.

For the read families use eq. 14.xxxvii without the variation, at lag $i-l$, and the same kernel estimate with exponent 39. Together with the level-i family their common-pair norm is at most $(1+(6K_M)^{-1}) A_{\mathsf{J}} \varsigma_i^{\mathrm{x}} \widehat{e}_i[\mathsf{J}]$, hence at most twice that size without the parenthesized factor. The fixed-pair polymer count just proved and the bound by $C_{\mathrm{sup}}^{\mathrm{x}}$ give the final assertion, with the stated $A_{n_{\mathfrak{J}}}^{\mathrm{x}}$. □

14.22. **Consequences of the complex field scale.** For the fixed test pair $X = (f, h)$ put

$$\mathfrak{f} = \|f\|_\infty \|h\|_\infty, \qquad \mathsf{n}_m(X; \mathsf{d}) = \mathsf{b}^{-m} \mathsf{d}^{m-2} \mathfrak{f}, \qquad \mathsf{l} = \max\{1, \log(2\mathsf{b}/\mathsf{d})\}. \tag{14.xxxix}$$

The cases $m = 2$ and $m = 3$ are the scales of the leading and next covariance coefficients.

Proposition 14.26 (One complex field scale)**.** Assume $g \le \gamma_{\mathrm{e}}$ and the parameter choices of Remark 14.9. For every datum on the least torus, the flow with no large-field region has the complex field-scale construction of Proposition 14.14. Every gauge fluctuation is multiplied by λ, the fermion fluctuations are unchanged, and the level-j action has weight

$$\lambda^{-2}(L^j \xi_j(\lambda) - \lambda^2 a_K^{-1} w), \qquad \xi_j(\lambda) = L^{-j}\beta + \lambda^2 (x_j - L^{-j}\beta).$$

The mass member is multiplied by λ^2. Each step removes the original gauge-action coefficient and λ^2 times its original mass increment, as well as its vacuum and source energies at λ. There are radii $\bar\lambda_j > 1$ and $0 < \rho \le 1/2$ with

$$\bar\lambda_j^{-2} \le c_\lambda (\log x_j)^{2\mathfrak{p}_0} \frac{a_j}{\mathsf{b}}, \qquad (\rho \bar\lambda_j)^2 \ge 2.$$

They depend on the datum only through the specified running coefficients.

(1) The zero-source energies and activities are even and holomorphic for $|\lambda| < \bar\lambda_j$. For each $1 \le \lambda_\circ < \bar\lambda_j$ they extend holomorphically in (λ, w) for $|\lambda| \le \lambda_\circ$ and $\|w\|_\sharp < \varepsilon_j(\lambda_\circ)$, with some positive radius. At $\lambda = 1$ the construction agrees with the original source flow, including at every source in its admitted polydisc. At $\lambda = 0$ the gauge field is its background and the fermion steps are Gaussian Berezin integrals.

(2) Let

$$\mathsf{n}_j^\lambda = \partial_{z_1}\partial_{z_2} \sum_{X'} \mathfrak{E}_j(X'; z_1 f + z_2 h, 0)\Big|_{z=0}, \qquad \mathsf{n}_j^\lambda = \sum_{m \ge 0} \mathsf{n}_{j,[2m]} \lambda^{2m}.$$

For $0 \le m \le 4$ and $0 \le p \le 3$,

$$\sum_{j=1}^{K} |\mathsf{n}_{j,[2m]}| \le C_m^{\mathrm{ii}} \mathsf{l}^{2m\mathfrak{p}_0} \mathsf{n}_m(X; \mathsf{d}),$$

$$\left| \sum_{j=1}^{K} \left(\mathsf{n}_j^1 - \sum_{m=0}^{p} \mathsf{n}_{j,[2m]} \right) \right| \le C_p^{\mathrm{ii}} \mathsf{l}^{2(p+1)\mathfrak{p}_0} \mathsf{n}_{p+1}(X; \mathsf{d}).$$

One may choose the same constant at index k for the coefficient and remainder estimates that use that index.

(3) On $|\lambda| < \bar{\lambda}_j$,

$$|\mathsf{n}_j^{\lambda}| \le C_{\mathrm{iii}}\mathsf{n}_0(X;a_j)\begin{cases}(\mathsf{d}/a_j)^{5/2}, & a_j \ge \mathsf{d},\\ (\mathsf{d}/a_j)^3 e^{-\kappa_{\mathrm{s}}\mathsf{d}/(2Ma_j)}, & a_j < \mathsf{d}.\end{cases}$$

The first source derivative is at most

$$C_{\mathrm{iii}}\|f\|_\infty a_j^{-1}(\mathsf{d}/a_j)^3 \mathfrak{l}_j^{4\mathfrak{p}_0+1}.$$

If $a_j \ge \mathsf{d}$, it is also at most $C_{\mathrm{iii}}\|f\|_\infty a_j^{-1}(\mathsf{d}/a_j)^{11/4}$. The same holds for h.

The first and mixed derivatives of the field-dependent total, including the mass member, are respectively

$$\sum_{l\le j}\sum_{X'} \mathbf{P}^{(l)}[f](X') - L^{-j}\mathsf{S}_1(\lambda^4\mathfrak{N}_j[f]\,\cdot), \qquad \sum_{l\le j}\sum_{X'} \mathbf{Q}^{(l)}[f,h](X'),$$

read at level j as total functions. The action-weight member is kept separately. On D_{j-1} their activity families have weighted read norms at most

$$C_{\mathrm{iii}}e_j[f]\mathfrak{l}_j^{4\mathfrak{p}_0+1}, \qquad C_{\mathrm{iii}}e_j[f,h],$$

respectively. Scalar terms use the strong norm on the full core. Short-lag fermionic terms use the strong norm; long-lag terms use the norm with a common pair for the sum. On D_j,

$$c_{\mathsf{S}}h_F^2|\lambda|^4\langle\mathfrak{N}_j[f]\rangle_{j;\{\lambda\};S_f} \le C_{\mathrm{iii}}e_j[f]\mathfrak{l}_j^{4\mathfrak{p}_0+1}.$$

All these response families are even and vanish at $\lambda = 0$. At the trivial configuration their first and mixed fermionic parts start at orders four and six, respectively.

The constants depend on g, ϑ and the fixed construction constants. ■

Proof. Proposition 14.14 proves the field-scale construction and its agreement with the source flow. Lemmas 14.15, 14.20, 14.21, 14.22, 14.23 and 14.24 give the total identities, orders, moment estimates, read norms and source-energy estimates in the third assertion. Their proofs preserve the distinction between the strong scalar norm and the common-pair fermionic norm at long lags. Increase C_{iii} by $\sup_{t\ge1} t^{-1/4}(1+\log t)^{4\mathfrak{p}_0+1}$ to obtain the first-derivative exponent $11/4$.

For an even holomorphic function H bounded by B on $|\lambda| < \bar{\lambda}_j$, Cauchy's formula gives

$$|H_{[2m]}| \le \mathsf{B}\bar{\lambda}_j^{-2m}, \qquad \left|H(1) - \sum_{m=0}^{p} H_{[2m]}\right| \le \frac{\mathsf{B}\bar{\lambda}_j^{-2p-2}}{1-\bar{\lambda}_j^{-2}}.$$

Here $\bar{\lambda}_j^{-2} \le \rho^2/2 \le 1/8$. For $a_j \ge \mathsf{d}$, the contribution to the coefficient of order $2m$ is at most

$$C_{\mathrm{iii}}c_\lambda^m(\log x_j)^{2m\mathfrak{p}_0}\mathsf{b}^{-m}\mathsf{d}^{5/2}a_j^{m-9/2}\mathfrak{f}.$$

In this range $\log x_j \le \mathsf{l}$. For $m \le 4$, the exponent $m - 9/2$ is negative, so the geometric sum begins at the scale $a_j \asymp \mathsf{d}$ and is bounded by a constant times $\mathsf{l}^{2m\mathfrak{p}_0}\mathsf{n}_m(X;\mathsf{d})$.

For $a_j < \mathsf{d}$, put $t = \mathsf{d}/a_j$. The bound is the same target size times a constant times

$$(1+\log t)^{2m\mathfrak{p}_0}t^{5-m}e^{-\kappa_{\mathrm{s}}t/(2M)}.$$

We used $\log x_j \le \mathsf{l}(1+\log(\mathsf{d}/a_j))$. The possible t are one number in $(1,L]$ times successive powers of L, so this sum is uniformly finite. For the remainder use the same calculation with $m = p+1 \le 4$ and the additional factor $(1-\bar{\lambda}_j^{-2})^{-1} \le 8/7$. Take at each index the larger of the coefficient and remainder constants. □

Remark 14.27 (The scalar specialization). The same field-scale argument applies to the pure gauge steps calibrated at the places t_i in eq. 14.xix. Omit the fermion vertex, fermion normalizer, mass member and all terms with fermion legs. The remaining terms satisfy the potential, origin and

scalar-family bounds of Proposition 14.14. The extraction moments vanish, and the first-response recursion is its reduced recursion.

Lemmas 14.16–14.24 then use only (R0) and the scalar variation in Lemma 14.18. For first derivatives the lag factor is $L^{4n}L^{-4n}q_n = q_n$. For mixed derivatives it is $L^{9n/2}L^{-4n}q_n = L^{n/2}q_n$. Both are summable, and no accumulated-mass logarithm occurs. Thus their proofs give the same first and mixed bounds with 1 replacing $\mathfrak{l}_i$; one may retain $A_{\mathbf{P}}, A'_{\mathbf{P}}, A_{\mathbf{Q}}$ as upper bounds for the constants. For $a_j \geq \mathsf{d}$, on $|\lambda| < \bar{\lambda}_j$, this gives

$$|\mathfrak{n}_j^{\lambda}| \leq C_{\mathfrak{e},2}\mathfrak{n}_0(X;a_j)(\mathsf{d}/a_j)^{5/2}, \qquad \left|\partial_z \sum_{X'} \mathfrak{E}_j(X';zf,0)\right|_{z=0}\Bigg| \leq C_{\mathfrak{e},1}\|f\|_\infty a_j^{-1}(\mathsf{d}/a_j)^3.$$

14.23. **The fermion polarization.** The quadratic part of the fermion determinant controls the leading change in the mean block action. We define it on the infinite lattice, including at bare mass zero. The estimate below separates its continuum singularity from a local quadratic term and a remainder. Throughout this subsection the lattice spacing is one and $0 \leq m_0 \leq 1$.

Lemma 14.28. Let $G = (D_+[1] + m_0)^{-1}$ denote the free kernel on $\mathbb{Z}^3$, for one flavour. If $U = 1$ off finitely many bonds, set

$$\psi^F(U) = -2\log\left|\det(1 + G(D_+[U] - D_+[1]))\right|.$$

The determinant is a finite determinant and is nonzero, also when $m_0 = 0$. Near $U = 1$, ψ^F is real analytic, has no linear term, and has quadratic part

$$\psi_2^F(A) = 2\,\mathrm{Re}\left\{\frac{1}{2}\,\mathrm{Tr}(G\mathbb{V}_1 G\mathbb{V}_1) - \mathrm{Tr}(G\mathbb{V}_2)\right\} = \frac{1}{2}\sum_{c=1}^{3}(A^c, \Pi A^c). \tag{14.xl}$$

Here $\mathbb{V}_j$ is the term of degree j in $D_+[e^{iA}] - D_+[1]$. The kernel Π is real, symmetric and translation invariant, and

$$|\Pi(b,b')| \leq C(1 + |b_- - b'_-|)^{-4}, \qquad \Pi\,\mathrm{d}\chi = 0 \tag{14.xli}$$

for every bounded complex function χ. The constant is uniform in m_0. ∎

Proof. The free estimates of [QCDM1, §6] give

$$\|\widehat{G}(p)\| \leq C|p|^{-1}, \qquad |G(v,w)| \leq C(1 + |x_v - x_w|)^{-2}.$$

The symbol is integrable, and its inverse Fourier transform is continuous in m_0, including at zero. Both inverse identities hold on vectors of finite support. Write $\mathbb{V} = D_+[U] - D_+[1]$. This operator is supported on the places over the endpoints of the finitely many altered bonds, so $\det(1 + G\mathbb{V}) = \det(1 + \mathbb{V}G)$ is finite dimensional. The other spin block gives its complex conjugate.

To check nonvanishing, suppose that $w + \mathbb{V}Gw = 0$, with $w \neq 0$ supported on these places. Then $v = Gw$ belongs to ℓ^2, since it decays as $(1 + |x|)^{-2}$, and $(D_+[U] + m_0)v = 0$. The real part of this equation gives

$$0 = \langle v, (\Delta[U] + m_0)v\rangle.$$

For the chamber operator this forces the component orthogonal to the chamber constants to vanish, and the remaining site field to be covariantly constant. Its norm is constant; membership in ℓ^2 therefore forces $v = 0$, a contradiction. On any fixed finite set of links the squared modulus of the determinant is thus a positive continuous function of the compact link variables. Its logarithm is smooth and is analytic near 1.

Expanding $\log\det(1 + G\mathbb{V})$ gives eq. 14.xl. Its linear term vanishes because the free operator is scalar in colour and $\mathrm{tr}\,\sigma_c = 0$. Constant colour rotations imply that the quadratic colour tensor is a multiple of $\delta_{cc'}$. Taking its real part gives the real symmetric kernel Π.

Gauge invariance can be checked without an infinite determinant. Put $A_1 = D_+[1] + m_0$, $B = D_+[U] + m_0$, and let $u - 1$ have finite support. The identity

$$1 + G(uBu^{-1} - A_1) = [1 + G(u-1)A_1]u^{-1}\, u[1 + G(B - A_1)]u^{-1}$$

is an identity between products of finite-rank perturbations of the identity. The first two factors have determinant one, since

$$\det[1 + G(u-1)A_1] = \det[1 + (u-1)A_1 G] = \det u = 1.$$

Thus $\psi^F(U^u) = \psi^F(U)$. Comparing quadratic terms under $u = e^{-i\chi}$ proves $\Pi \mathrm{d}\chi = 0$ for finitely supported χ.

The second-order vertex is diagonal in the bond index. The term with two first-order vertices contains two free propagators between the endpoints of the two bonds. Their decay gives eq. 14.xli. Translation invariance follows from that of the free operator. For bounded χ, replace it by its restriction to a ball of radius R and use

$$\sum_{|b'_- - b_-| \geq R-1} (1 + |b_- - b'_-|)^{-4} \longrightarrow 0.$$

The row sum is absolutely convergent, so the identity extends to bounded χ, and hence to plane waves. □

Define, with the value at $m = 0$ understood by continuity,

$$\Upsilon(k; m) = \frac{N_f}{2\pi}\left[\frac{2m}{|k|^2} + \frac{|k|^2 - 4m^2}{|k|^3}\arctan\frac{|k|}{2m}\right] = \frac{N_f}{2\pi}\int_{2m}^{\infty} \frac{1 + 4m^2/\mu^2}{\mu^2 + |k|^2}\, d\mu. \tag{14.xlii}$$

In particular, $\Upsilon(k; 0) = N_f/(4|k|)$.

Lemma 14.29. Place bond variables at their midpoints, so that

$$A_\mu(y) = e^{ik\cdot(y + e_\mu/2)}\alpha_\mu, \qquad (\Pi A)_\mu(y) = e^{ik\cdot(y+e_\mu/2)}(\widehat{\Pi}(k; m_0)\alpha)_\mu.$$

There are constants $c^{\mathrm{vp}} \in \mathbb{R}$ and C, depending only on the chamber operator, such that

$$\begin{aligned} &\left|\widehat{\Pi}_{\mu\nu}(k; m_0) - (|k|^2\delta_{\mu\nu} - k_\mu k_\nu)(c^{\mathrm{vp}} + N_f^{-1}\Upsilon(k; m_0))\right| \\ &\qquad \leq C|k|^2\left[(|k| + m_0)(1 + |\log|k||) + \frac{m_0}{|k| + m_0}\right] \end{aligned} \tag{14.xliii}$$

for $0 < |k| \leq \pi$ and $0 \leq m_0 \leq 1$. Away from $k = 0$, $\widehat{\Pi}$ is uniformly bounded and continuous in (k, m_0). ■

Proof. Write $\langle p\rangle = |p| + m_0$, $p_\pm = p \pm k/2$, and $\widehat{k}_\mu = 2\sin(k_\mu/2)$. We use $\int_p$ for normalized Lebesgue integration over either the momentum torus or $\mathbb{R}^3$, as specified. All constants in this proof depend only on the chamber operator.

Symmetry and the light component. The decay in eq. 14.xli gives an absolutely convergent Fourier series, uniformly in m_0. The same equation, applied to plane waves, gives

$$\widehat{\Pi}(k; m_0)\widehat{k} = 0. \tag{14.xliv}$$

The cubic isometries act on midpoint variables by their linear parts. Improper isometries exchange the two spin blocks, which enter the determinant symmetrically. Thus $\widehat{\Pi}(tk) = t\widehat{\Pi}(k)t^{\mathsf{T}}$ for $t \in W_3$; in particular $\widehat{\Pi}$ is even and real symmetric. Uniform boundedness settles the claimed estimate when $|k| \geq \delta/2$, for any fixed small $\delta > 0$.

Let Π_0 project onto the chamber constants and put $\Pi_1 = 1 - \Pi_0$. On the positive spin block set

$$H = \Pi_1(D_+ + m_0)\Pi_1, \quad X = \Pi_0 D_+ \Pi_1, \quad X' = \Pi_1 D_+ \Pi_0, \quad S = \Pi_0(D_+ + m_0)\Pi_0 - XH^{-1}X'.$$

The chamber gap gives $\operatorname{Re} H \geq \lambda_\rho + m_0$. The triangular factorization

$$D_+ + m_0 = (1 + XH^{-1})(S \oplus H)(1 + H^{-1}X')$$

splits the relative determinant into its S and H determinants. Indeed, at a finitely supported link perturbation all relative factors are finite-rank perturbations of the identity. The two triangular relative factors have determinant one. Both remaining determinants are gauge invariant, since Π_1 commutes with gauge transformations. Their quadratic kernels give $\Pi = \Pi^S + \Pi^{\mathrm{h}}$.

At $U = 1$, $\widehat{D}_+(p)$ is a trigonometric polynomial. Consequently $H(p)^{-1}$ is analytic on a complex strip of width $\min\{1, \lambda_\rho/(2C)\}$, with norm at most $2/\lambda_\rho$. Its mass derivative has norm at most $4/\lambda_\rho^2$. The heavy kernel and its required momentum and mass derivatives are therefore bounded. The free kernel of S^{-1}, being a compression of the full propagator, is bounded by $C(1 + |x - y|)^{-2}$. These bounds justify the determinant splitting also at $m_0 = 0$, by dominated convergence in its quadratic coefficients.

For $|p| \leq 4\delta$, the Schur complement has the expansion

$$\widehat{S}(p) = m_0 + i\sigma \cdot p + K_2(p) + K_3(p, m_0), \tag{14.xlv}$$

where K_2 is quadratic and independent of m_0, and

$$|K_3| \leq C|p|^2\langle p\rangle, quad|\partial_p^\alpha K_3| \leq C\langle p\rangle^{3-|\alpha|}\ (|\alpha| \leq 4), \quad |\partial_{m_0}\partial_p^\alpha K_3| \leq C\langle p\rangle^{2-|\alpha|}\ (|\alpha| \leq 3).$$

To see the mass dependence explicitly, write $H = H_\circ + m_0$ and let X_1, X_1' be the linear terms of X, X' at zero. Then

$$K_2 = \frac{r}{2}|p|^2 - X_1 H_\circ(0)^{-1} X_1'.$$

The remaining Schur terms are cubic in p, or contain $H^{-1} - H_\circ^{-1} = -m_0 H^{-1} H_\circ^{-1}$ between $X, X' = O(|p|)$. This proves all the displayed bounds by Taylor's formula.

The first vertex of $S[e^{iA}]$, at mean fermion momentum q and gauge momentum k, has the form

$$v_\mu(q, k) = i\sigma_\mu + v_\mu^{(1)}(q, k) + v_\mu^{(2)}(q, k; m_0).$$

Here $v^{(1)}$ is linear in (q, k) and independent of mass, and, for $|\alpha| \leq 3$,

$$|\partial^\alpha v^{(2)}| \leq C(|q| + |k| + m_0)^{2-|\alpha|}, \qquad |\partial_{m_0}\partial^\alpha v^{(2)}| \leq C(|q| + |k| + m_0)^{1-|\alpha|}.$$

In fact each mass-dependent first variation of the Schur term contains X or X'; hence $\partial_{m_0} v = O(|q| + |k|)$, and the bounds follow by Taylor expansion at $(q, k, m_0) = 0$. The second vertex $w_{\mu\nu}(q, k)$ and the derivatives used below are smooth and bounded. Formula eq. 14.xl now writes the light contribution as

$$\widehat{\Pi}^S(k) = \int_{[-\pi,\pi]^3} B(p, k)\, \frac{dp}{(2\pi)^3} + T(k).$$

The integrand B is the trace, with the normalization in that formula, of

$$\widehat{S}(p_-)^{-1} v_\mu(p, k) \widehat{S}(p_+)^{-1} v_\nu(p, -k).$$

The term T contains one propagator and the second vertex, with the sign of $-\operatorname{Tr}(G\mathbb{V}_2)$.

Parity and the expansion at zero momentum. On the two spin blocks an operator and its adjoint occur together. Their Hermitian parts multiply τ_0 and their anti-Hermitian parts multiply τ_3. Inversion shows that $m_0, K_2, v^{(1)}$ have the former parity and $i\sigma \cdot p, i\sigma_\mu$ the latter. A trace with an odd number of τ_3 factors vanishes.

Put $G^c(p) = (m_0 + i\sigma \cdot p)^{-1}$. For small δ, the Neumann expansion of eq. 14.xlv gives

$$\widehat{S}^{-1} = G^c + G^{(1)} + G^{(2)}, \qquad G^{(1)} = -G^c K_2 G^c,$$

$$G^{(2)} = G^c K_2 G^c (K_2 + K_3)\widehat{S}^{-1} - G^c K_3 \widehat{S}^{-1}.$$

For $|\alpha| \le 3$, these three terms satisfy bounds of degrees $-1-|\alpha|, -|\alpha|, 1-|\alpha|$, respectively, in $\langle p\rangle$. A mass derivative lowers each degree by one. The first two terms are homogeneous in (p, m_0) of degrees $-1, 0$.

Choose a radial smooth χ, equal to one on $|p| \le \delta$ and zero on $|p| \ge 2\delta$. On the support of $\chi(p_-)\chi(p_+)$, expand

$$B = B^c + B^{(1)} + B^{(2)}.$$

The term B^c uses G^c and the constant vertices; $B^{(1)}$ uses exactly one $G^{(1)}$ or $v^{(1)}$. They are homogeneous of degrees $-2, -1$ in (p, k, m_0). Every term of $B^{(2)}$ has total degree at least zero, with each of its propagator factors of degree at least -1. Parity gives $B^{(1)}(p, k; 0) = 0$. Every surviving term at positive mass contains $m_0/(p^2 + m_0^2)$, so $B^{(1)} = O(m_0|p|^{-2})$ at infinity, with one additional inverse power for each momentum derivative.

For differentiation of these integrals, separate the points $p_- = 0$ and $p_+ = 0$ by cutoffs $\zeta(p_-/|k|), \zeta(p_+/|k|)$, where $\zeta = 1$ on the ball of radius $1/4$ and zero outside radius $1/3$. On the first piece use p_- as variable, so derivatives fall on the other factor; do the converse on the second piece. The remaining piece stays away from both singularities. Derivatives of order j of these cutoffs cost $C_j|k|^{-j}$ on shells of volume $O(|k|^3)$. On those shells all momenta are comparable to $|k|$. In particular the cutoff terms from three derivatives of $B^{(2)}$ are bounded.

The continuum term. For one flavour the continuum kernel is

$$-2\int_p \mathrm{tr}_4[\gamma_\mu G_4^c(p)\gamma_\nu G_4^c(p+k)], \qquad G_4^c(p) = (m_0 + i\gamma\cdot p)^{-1}.$$

The factor two is the colour trace. The two spin blocks cancel the parity-odd trace. With $\gamma_\mu = \sigma_\mu \otimes \tau_3$, the continuum trace is

$$\mathrm{tr}_4[\gamma_\mu(-i\not{p} + m_0)\gamma_\nu(-i(\not{p}+\not{k}) + m_0)] = 4[-p_\mu(p+k)_\nu - p_\nu(p+k)_\mu + \delta_{\mu\nu}(p\cdot(p+k) + m_0^2)].$$

Feynman parametrization, $p = l - xk$, gives $\Delta = m_0^2 + x(1-x)|k|^2$. The rotational average of the numerator is

$$\delta_{\mu\nu}(l^2/3 + \Delta) + 2x(1-x)(k_\mu k_\nu - |k|^2\delta_{\mu\nu}).$$

The first term divided by $(l^2+\Delta)^2$ is $\partial_l \cdot [l/(3(l^2+\Delta))]$. To handle it by absolutely convergent integrals, subtract the same loop at mass $M > m_0$. Its boundary flux through $|l| = R$ is

$$\frac{4\pi}{3}R^3[(R^2 + \Delta_{m_0})^{-1} - (R^2 + \Delta_M)^{-1}] \longrightarrow 0.$$

The remaining integral uses $\int_l (l^2+\Delta)^{-2} = (8\pi\sqrt{\Delta})^{-1}$ and equals

$$(|k|^2\delta - kk^{\mathsf{T}})[\Upsilon(k; m_0) - \Upsilon(k; M)]/N_f.$$

This also proves eq. 14.xlii by elementary integration. Third momentum derivatives of the mass-M loop are $O(M^{-2})$, by homogeneity. Letting $M \to \infty$ shows that third momentum derivatives and $\partial_{m_0}\partial_k^2$ of the continuum loop agree with those of $(|k|^2\delta - kk^{\mathsf{T}})\Upsilon/N_f$. Consequently

$$R_c = \int_p \chi(p_-)\chi(p_+)B^c - (|k|^2\delta - kk^{\mathsf{T}})\Upsilon/N_f$$

is C^2 and satisfies $|\partial_k^3 R_c| + |\partial_{m_0}\partial_k^2 R_c| \le C$. Indeed, these derivatives are integrals of $[\chi(p_-)\chi(p_+) - 1]B^c$, away from both singularities.

The first correction and the remainder. Define the absolutely convergent integral on $\mathbb{R}^3$

$$F(k; m_0) = \int_p \left[\tfrac{1}{2}\{B^{(1)}(p,k) + B^{(1)}(p,-k)\} - B^{(1)}(p,0)\right].$$

It is even, homogeneous of degree two, and vanishes when either $k = 0$ or $m_0 = 0$. Its mass derivative is homogeneous of degree one and bounded for $|k| = 1$, $0 \le m_0 \le 1$. Thus $|F| \le Cm_0|k|$ when

$m_0 \le |k|$. For $|k| \le m_0$, use $F(k;m_0) = m_0^2 F(k/m_0;1)$; the function on the right is even and C^2, and vanishes at zero. Hence

$$|F(k;m_0)| \le C|k|^2 \frac{m_0}{|k|+m_0}. \tag{14.xlvi}$$

Writing $\Phi_1 = \int_p \chi(p_-)\chi(p_+)B^{(1)}$, the difference $R_1 = \Phi_1 - \Phi_1(0) - F$ is an integral away from the singularities and has derivatives through order three bounded by Cm_0.

For $R_2 = \int_p \chi(p_-)\chi(p_+)B^{(2)}$, consider a term of total degree $n \ge 0$, whose factor at p_- has degree $n_- \ge -1$. In $|p_-| \le |k|/4$, three derivatives give at most

$$C\langle p_-\rangle^{n_-}(|k|+m_0)^{n-n_--3}.$$

Its integral is bounded. The same holds near $p_+ = 0$ and on the remaining region $|p| \le |k|$. On $|k| \le |p| \le 2\delta$, the bound is $C\langle p\rangle^{n-3}$, whose integral is at most $C(1+|\log|k||)$. Including the cutoff terms gives

$$|\partial_k^3 R_2(k)| \le C(1+|\log|k||), \qquad |\partial_{m_0}\partial_k^2 R_2(0;m_0)| \le C(1+|\log m_0|).$$

In particular R_2 is C^2. The complementary integral $\Pi^{\mathrm{out}} = \int_p [1-\chi(p_-)\chi(p_+)]B$ is smooth with bounded derivatives. In T, momentum derivatives fall on the second vertex, and a mass derivative costs at most $C\langle p\rangle^{-2}$, which is integrable. The heavy term has already been bounded.

The quadratic local term. Collect these terms as

$$\widehat{\Pi} = (|k|^2\delta - kk^{\mathsf{T}})\Upsilon/N_f + F + Q, \quad Q = R_c + \Phi_1(0) + R_1 + R_2 + \Pi^{\mathrm{out}} + T + \widehat{\Pi}^{\mathrm{h}}.$$

The function Q is even and C^2, with

$$|\partial_k^3 Q(k;m_0)| \le C(1+|\log|k||), \qquad |\partial_k^2 Q(0;m_0) - \partial_k^2 Q(0;0)| \le Cm_0(1+|\log m_0|).$$

For the last bound integrate the mass derivative from zero. Its value at zero is justified as follows. At a fixed $k \neq 0$, dominated convergence gives continuity in mass of the second derivatives. The third-derivative estimate bounds the difference between those derivatives at k and at zero by $2C|k|(1+|\log|k||)$, uniformly in mass. This extends continuity to $k = 0$.

The Ward identity along every ray gives $Q(0;m_0) = 0$. Taylor's formula therefore yields

$$Q(k;m_0) = \tfrac{1}{2}\partial_k^2 Q(0;m_0)[k,k] + O(|k|^3(1+|\log|k||)).$$

At mass zero, $F = 0$ and $\Upsilon = N_f/(4|k|)$. Since $\widehat{k} - k = O(|k|^3)$, the Ward identity forces the homogeneous quadratic matrix $T_2(k) = \frac{1}{2}\partial_k^2 Q(0;0)[k,k]$ to satisfy $T_2(k)k = 0$. A cubic-invariant symmetric quadratic matrix has the form

$$\alpha|k|^2 I + \beta kk^{\mathsf{T}} + \gamma\,\mathrm{diag}(k_1^2, k_2^2, k_3^2).$$

Its product with k vanishes only when $\gamma = 0$, $\beta = -\alpha$. Thus $T_2(k) = c^{\mathrm{vp}}(|k|^2 I - kk^{\mathsf{T}})$. Combining this with the bounds for F and Q, and using

$$m_0(1+|\log m_0|) \le 2(|k|+m_0)(1+|\log|k||) \quad (|k|, m_0 \le 1),$$

proves eq. 14.xliii. □

14.24. **The determinant in the second covariance coefficient.** The finite-dimensional Laplace formula is Lem. D.7.

Lemma 14.30 (The constrained determinant)**.** On the bare torus with a flat block constraint $\mathcal{M}^n U = 1$, $n \ge 1$, put

$$\mathcal{K}_n[U;m_0] = D[U] + m_0 + \mathfrak{a}_n R_n(U)^* R_n(U), \qquad 0 \le m_0 \le 1.$$

For one flavour let $\mathcal{K}_{n,+}$ be its restriction to the e_+ block and define, near the trivial configuration,

$$\psi^{(\le n)}(U) = -2\,\mathrm{Re}\big[\log\det\mathcal{K}_{n,+}[U;m_0] - \log\det\mathcal{K}_{n,+}[1;m_0]\big].$$

This function is real analytic, vanishes at 1 and has zero linear part. If $G_n = \mathcal{K}_{n,+}[1;m_0]^{-1}$ and $\mathbb{V}_j^{(n)}$ is the degree-j part of $\mathcal{K}_{n,+}[e^{iA};m_0] - \mathcal{K}_{n,+}[1;m_0]$, then

$$\psi_2^{(\le n)}(A) = 2\,\mathrm{Re}\left\{\tfrac{1}{2}\,\mathrm{Tr}(G_n\mathbb{V}_1^{(n)}G_n\mathbb{V}_1^{(n)}) - \mathrm{Tr}(G_n\mathbb{V}_2^{(n)})\right\} = \tfrac{1}{2}\sum_{c=1}^{3}(A^c, \Pi^{(\le n)}A^c). \tag{14.xlvii}$$

The kernel is real and symmetric and annihilates exact forms.

In the setting of Lemma D.7, multiplying the density by $e^{-N_f\psi_{\mathcal{K}}}$, with $\psi_{\mathcal{K}} = \psi^F$ of Lemma 14.28 or $\psi^{(\le n)}$, leaves ζ_0 unchanged and changes ζ_2 by

$$-N_f\langle X_{1,0}; X_{2,0}; \psi_{\mathcal{K},2}\rangle.$$

There is no quadratic term in N_f and no contribution from $\psi_{\mathcal{K},3}$. ■

Proof. The averaging operators are real analytic near the trivial configuration. Their entries are finite sums of transports, with the normalization in R_n. At positive mass $\mathcal{K}_n$ is invertible. At zero mass its inverse at the trivial configuration is the antiperiodic image sum of the exponentially decaying constrained free kernel. It is therefore invertible there as well, and invertibility persists in a neighbourhood. The operator commutes with $e_\pm$; at real configurations its e_- block is the adjoint of its e_+ block. Thus $\det\mathcal{K}_n = |\det\mathcal{K}_{n,+}|^{2N_f}$. This proves analyticity of the stated logarithm.

The expansion of $\log\det(\mathcal{K}_{n,+}[1] + \mathbb{V})$ gives eq. 14.xlvii. At the trivial configuration the inverse acts as the identity on colour. Every term of the first variation has one colour matrix σ_c, including the first variations of the averaging transports. Its trace vanishes because $\mathrm{tr}\,\sigma_c = 0$. Constant colour rotations force the quadratic colour tensor to be a multiple of $\delta_{cc'}$.

Gauge covariance gives $R_n(U^u) = u|_{\mathbb{T}^{(n)}}R_n(U)\mathsf{u}^{-1}$ and $\mathcal{K}_n[U^u;m_0] = \mathsf{u}\mathcal{K}_n[U;m_0]\mathsf{u}^{-1}$. For $u = e^{-i\chi}$, the logarithmic field changes by $A \mapsto A + \mathrm{d}\chi + O(2)$. Comparison of quadratic terms, with no linear term present, proves $\psi_2^{(\le n)}(A+\mathrm{d}\chi) = \psi_2^{(\le n)}(A)$. Its polarized form gives $\Pi^{(\le n)}\mathrm{d}\chi = 0$.

Finally, in any smooth chart $A = A_{(1)} + A_{(2)} + \cdots$, the quadratic part of the logarithmic determinant is $\psi_{\mathcal{K},2}(A_{(1)})$: the potential contribution $\psi_{\mathcal{K},1}(A_{(2)})$ vanishes. In the Laplace formula W_1 is unchanged and W_2 gains $N_f\psi_{\mathcal{K},2}$. The latter occurs exactly once and linearly in ζ_2, proving the final assertion. □

Lemma 14.31 (The fermionic covariance coefficient)**.** In one colour let $C(p,q) = \langle F_pF_q\rangle_0$ and $\mathsf{Y}(p;b) = \langle F_pA_b\rangle_0$ be the infinite-lattice free curvature and curvature-potential covariances. Then

$$(\mathsf{Y}\Pi\mathsf{Y}^{\mathsf{T}})(p,q) = \sum_{b,b'}\mathsf{Y}(p;b)\Pi(b,b')\mathsf{Y}(q;b')$$

is absolutely convergent, and put

$$\mathsf{B}_1^F(p,q;m_0) = -3C(p,q)(\mathsf{Y}\Pi\mathsf{Y}^{\mathsf{T}})(p,q).$$

The second covariance coefficient of the Dirichlet box theory with its determinant relative to the infinite-lattice free operator has the iterated limit

$$\begin{aligned}\lim_{n\to\infty}\lim_{\beta\to\infty}\beta^3\left[\langle a_{p+o_n}; a_{q+o_n}\rangle_{n,\beta,m_0} - \beta^{-2}\mathsf{B}_0^{[n]}(p+o_n, q+o_n)\right]\\ = \mathsf{B}_1^{\mathrm{lat}}(p,q) + N_f\mathsf{B}_1^F(p,q;m_0).\end{aligned}$$

Here the box configurations equal 1 outside their free bonds, their density is proportional to $e^{-\beta S(U)}e^{-N_f\psi^F(U)}$, and o_n places the two fixed plaquettes in the interior.

For the flat block constraint at depth n on a torus of bare side N, write

$$A_{(1)} = \sum_{i<n} A_i,$$

$$C_N^{(n)}(p,q) = \langle (\mathrm{d}A_{(1)})_p (\mathrm{d}A_{(1)})_q \rangle_0 = (\Pi_0^{\mathrm{tp}} - \Pi_n^{\mathrm{tp}})(p,q),$$

$$\mathsf{Y}^{(n)}(p;b) = \langle (\mathrm{d}A_{(1)})_p A_{(1),b} \rangle_0.$$

The covariance is the projection onto Fl_n. The determinant changes the second Laplace coefficient by $N_f \mathsf{B}_N^{F,(n)}$, where

$$\mathsf{B}_N^{F,(n)}(p,q;m_0) = -3 C_N^{(n)}(p,q)(\mathsf{Y}^{(n)} \Pi^{(\le n)} \mathsf{Y}^{(n)\mathsf{T}})(p,q).$$

Each $\mathsf{Y}^{(n)}(p;\cdot)$ is a potential of $C_N^{(n)}(\cdot,p)$ whose sum over bonds in each direction is zero. In this formula it may be replaced by any potential with these properties. ∎

Proof. We first check that the determinant-weighted box law has the stated Laplace interpretation. By Lemma 14.28, its determinant ratio is a positive continuous function of finitely many compact links. It is bounded above and below by positive constants and is gauge invariant. The axial-gauge localization for the pure box measure therefore applies with each of $1, a_p, a_q, a_p a_q$ multiplied by this ratio. Outside a fixed small window it gives an $O(e^{-c\beta})$ error for fixed box and mass. Inside, the density is $e^{-\beta\phi - \Theta - N_f \psi^F}$, where

$$\phi(A) = S(e^{iA}) + \frac{1}{2\tau_{\mathrm{gf}}} \|\mathrm{d}^* A\|^2.$$

Its Hessian is $\mathrm{d}^*\mathrm{d} + \tau_{\mathrm{gf}}^{-1} \mathrm{d}\mathrm{d}^*$, positive definite on the box forms. The determinant logarithm is smooth there. Lemma D.7 and Lemma 14.30 thus apply, with the same pure-gauge leading coefficient.

The leading plaquette term is $a_p^{[2]} = \frac{1}{2}\sum_c (F_p^c)^2$. For one colour the connected Wick contractions in

$$\left\langle \tfrac{1}{2} F_p^2; \tfrac{1}{2} F_q^2; \tfrac{1}{2}(A, \Pi A) \right\rangle_0$$

join one F_p to one F_q in four ways, and join the other two to the two A factors in two ways. There are eight contractions, each contributing $C(p,q)(\mathsf{Y}\Pi\mathsf{Y}^{\mathsf{T}})(p,q)$. The prefactor is $1/8$, and three colours give the factor three. The same count on a finite box, with its free bonds, yields

$$\beta^3 \big[\langle a_p; a_q \rangle_{n,\beta,m_0} - \beta^{-2} \mathsf{B}_0^{[n]}(p,q) \big] = \mathsf{B}_1^{[n]}(p,q;\tau_{\mathrm{gf}}) \\ - 3 N_f C^{[n]}(p,q)(\mathsf{Y}^{[n]} \Pi \mathsf{Y}^{[n]\mathsf{T}})(p,q) + O(\beta^{-1}).$$

The left side is independent of the gauge-fixing parameter, so take $\tau_{\mathrm{gf}} = 1$.

The pure box kernels, after translating by o_n, converge to their infinite-lattice kernels, as does the pure coefficient $\mathsf{B}_1^{[n]}$ to $\mathsf{B}_1^{\mathrm{lat}}$. They obey uniformly $|\mathsf{Y}^{[n]}(p + o_n; b + o_n)| \le c_0 \langle x_p - b_- \rangle^{-2}$, where $\langle z \rangle = \max\{1, |z|\}$; set the term to zero if a bond is outside the free bonds of the box. Together with the polarization decay, this bounds the double-sum summand by

$$c_0^2 C_\Pi^{\mathbb{Z}} \langle x_p - b_- \rangle^{-2} \langle b_- - b'_- \rangle^{-4} \langle b'_- - x_q \rangle^{-2}.$$

This majorant is summable. Indeed, first convolve the last two factors: their sum is at most $C\langle b_- - x_q \rangle^{-2}(1 + \log\langle b_- - x_q \rangle)$, hence at most $C\langle b_- - x_q \rangle^{-3/2}$. Its product with the first factor is summable since $2 + 3/2 > 3$. Dominated convergence proves both absolute convergence of the infinite-lattice sum and the asserted outer limit. Gauge independence follows from the Ward identity $\Pi\,\mathrm{d} = 0$.

For the block constraint, the independent slices have curvature covariances $\mathcal{C}_{(i+1)}$ in each colour. Their sum is $\Pi_0^{\mathrm{tp}} - \Pi_n^{\mathrm{tp}}$, which is the projection onto Fl_n. The covariance applied to $\mathrm{d}^*\delta_p$ is $\mathsf{Y}^{(n)}(p;\cdot)$. It belongs to $T_n = \ker \mathbf{Q}_n^{\mathrm{lin}}$, and its coboundary is $C_N^{(n)}(\cdot,p)$.

Every $A \in T_n$ has zero sum in each bond direction. To see this, sum $\mathbf{Q}_n^{\mathrm{lin}} A = \mathbf{Q}_n^{\mathrm{s}} A - \mathrm{d}\bar{n}_n A = 0$ over the coarse bonds in that direction. The coboundary sums to zero, while the straight-path average sums to $L^{-2n}\sum_x A_\mu(x)$: each bare bond belongs to L^n of its straight paths. The same Wick calculation therefore gives the displayed constrained coefficient. Two potentials with the same curvature and these zero sums differ by a closed one-form with zero periods, hence an exact form. The constrained Ward identity annihilates their difference. □

Proposition 14.32 (The continuum polarization profile)**.** Use four-component Dirac matrices $\gamma_\mu = \sigma_\mu \otimes \tau_3$ and colour generators with $\mathrm{tr}(\sigma_c \sigma_d) = 2\delta_{cd}$.

(1) The gauge-invariant quadratic part of the continuum fermion determinant has transverse kernel $(k^2\delta_{\mu\nu} - k_\mu k_\nu)\Upsilon(k;m)$, up to a local transverse term $c(k^2\delta_{\mu\nu} - k_\mu k_\nu)$, where

$$\Upsilon(k;m) = \frac{N_f}{2\pi}\left[\frac{2m}{k^2} + \frac{k^2 - 4m^2}{|k|^3}\arctan\frac{|k|}{2m}\right] = \frac{N_f}{2\pi}\int_{2m}^{\infty} \frac{1 + 4m^2/\mu^2}{\mu^2 + k^2}\, d\mu.$$

It is positive for $k \neq 0$, $\Upsilon(k;0) = N_f/(4|k|)$, and $\Upsilon(0;m) = N_f/(3\pi m)$ for $m > 0$. For $m/|k| \to 0$,

$$\Upsilon(k;m) = \frac{N_f}{4|k|} - \frac{N_f m^2}{|k|^3} + \frac{16 N_f m^3}{3\pi|k|^4} + O(m^5|k|^{-6}).$$

(2) Set

$$\mathcal{F}(s) = \begin{cases} [1 - (1+s)e^{-2s}]/s, & s > 0, \\ 1, & s = 0. \end{cases}$$

In the formal continuum perturbative expansion of the action-density covariance, the term linear in N_f at order b^{-3}, away from coincident points, is

$$-\frac{3N_f}{8\pi^3}\mathsf{b}^{-3}|x|^{-5}\mathcal{F}(m|x|).$$

The function $\mathcal{F}$ decreases from one to zero, is positive, and satisfies

$$\mathcal{F}(s) = 1 - \tfrac{2}{3}s^2 + \tfrac{2}{3}s^3 + O(s^4), \qquad \mathcal{F}(s) = s^{-1} + O(e^{-2s}) \quad (s \to \infty).$$

At $m = 0$ this coefficient is $-(N_f/4)\mathfrak{k}|x|^{-5}$, where $\mathfrak{k} = 3/(2\pi^3)$. In the same formal expansion, the term quadratic in N_f at order b^{-4} and $m = 0$ is $3(\pi^2 + 8)N_f^2|x|^{-4}/(256\pi^4)$.

■

Proof. The determinant expansion gives

$$\Pi_{\mu\nu}(k) = -2N_f \int_p \mathrm{tr}_4[\gamma_\mu G_m(p)\gamma_\nu G_m(p+k)].$$

Traces with three gamma matrices vanish because $\mathrm{tr}\,\tau_3 = 0$, so there is no parity-odd term. The numerator trace is

$$4[-p_\mu(p+k)_\nu - p_\nu(p+k)_\mu + \delta_{\mu\nu}(p\cdot(p+k) + m^2)].$$

Introduce a Feynman parameter and put $p = l - xk$, $\Delta = m^2 + x(1-x)k^2$. After angular averaging, the expression inside the factor four is

$$\delta_{\mu\nu}(l^2/3 + \Delta) + 2x(1-x)(k_\mu k_\nu - k^2\delta_{\mu\nu}).$$

Divided by $(l^2 + \Delta)^2$, its first term is the divergence $\delta_{\mu\nu}\partial_l \cdot [l/(3(l^2+\Delta))]$. A gauge-invariant regularization removes this constant; on the lattice it is cancelled by the tadpole. Since $\int_l (l^2+\Delta)^{-2} =$

$(8\pi\sqrt{\Delta})^{-1}$, the remaining term gives $\Upsilon = (2N_f/\pi)\int_0^1 x(1-x)(m^2+x(1-x)k^2)^{-1/2}\,dx$. Elementary integration yields the first formula. For the second use

$$|k|^{-1}\arctan\frac{|k|}{2m} = \int_{2m}^{\infty}\frac{d\mu}{\mu^2+k^2}, \qquad \frac{k^2-4m^2}{k^2(\mu^2+k^2)} = -\frac{4m^2}{\mu^2k^2}+\frac{1+4m^2/\mu^2}{\mu^2+k^2}.$$

The integral of $4m^2/\mu^2$ is $2m$. The limits and expansion follow from these formulas, proving (a).

For (b), write the curvature covariance as $G = \mathsf{b}^{-1}G^0 - \mathsf{b}^{-2}G^1 + \mathsf{b}^{-3}G^2 + \cdots$, with $\hat{G}^j = (\delta - \hat{k}\hat{k})\Upsilon^j$. Away from zero, $G^0 = \mathbb{K}_3$. Wick's formula with three colours gives the coefficient $-3\operatorname{tr}(\mathbb{K}_3 G^1)$. The spectral representation reads

$$\frac{\Upsilon}{k^2} = \frac{N_f}{2\pi}\int_{2m}^{\infty}(1+4m^2/\mu^2)\mu^{-2}\left[\frac{1}{k^2}-\frac{1}{k^2+\mu^2}\right]d\mu.$$

The bracket transforms to $(1-e^{-\mu r})/(4\pi r)$, $r=|x|$. For $Y = e^{-\mu r}/(4\pi r)$,

$$-\Delta Y = -\mu^2 Y, \qquad \partial_i\partial_j Y = Y[\hat{x}_i\hat{x}_j(\mu^2+3\mu/r+3/r^2) - \delta_{ij}(\mu/r+1/r^2)]$$

away from zero. Use $(-\Delta\delta+\partial\partial)(4\pi r)^{-1} = \mathbb{K}_3$, $\operatorname{tr}\mathbb{K}_3 = 0$, and $\operatorname{tr}(\mathbb{K}_3\hat{x}\hat{x}^{\mathsf{T}}) = 2/(4\pi r^3)$. The contraction is

$$\operatorname{tr}\Big[\mathbb{K}_3(-\Delta\delta+\partial\partial)((4\pi r)^{-1}-Y)\Big] = \frac{6}{16\pi^2 r^6}\mathsf{h}(\mu r), \quad \mathsf{h}(t) = 1-e^{-t}(1+t+t^2/3).$$

Thus the coefficient is $-3N_f r^{-5}/(8\pi^3)$ times

$$\frac{3}{2}\int_{2s}^{\infty}(1+4s^2/t^2)t^{-2}\mathsf{h}(t)\,dt, \qquad s = mr.$$

Primitives for $\mathsf{h}(t)/t^2$ and $\mathsf{h}(t)/t^4$ are $-(1-e^{-t}(1+t/3))/t$ and $-(1-e^{-t}(1+t))/(3t^3)$. The integral before multiplication by $3/2$ is $2[1-(1+s)e^{-2s}]/(3s)$. This gives $\mathcal{F}$. Since $(s\mathcal{F}(s))'' = -4se^{-2s} \le 0$ and $s\mathcal{F}(s)\to 0$ at zero, $\mathcal{F}$ is decreasing. Positivity and the two expansions follow from its explicit expression. Its large-mass coefficient equals the first correction in $c_2(\mathsf{b}+N_f/(3\pi m))^{-2}r^{-6}$, since $2c_2/(3\pi) = 3/(8\pi^3)$.

At $m=0$, the transforms of $|k|^{-3}$ and $|k|^{-4}$, up to the irrelevant constant in the first, are $-(2\pi^2)^{-1}\log r$ and $-r/(8\pi)$. Hence

$$G^1 = \frac{N_f}{4\pi^2 r^2}\hat{x}\hat{x}^{\mathsf{T}}, \qquad G^2 = \frac{N_f^2}{128\pi r}(\delta+\hat{x}\hat{x}^{\mathsf{T}}).$$

The coefficient $\frac{3}{2}[2\operatorname{tr}(\mathbb{K}_3G^2)+\operatorname{tr}(G^1G^1)]$ is $3(\pi^2+8)N_f^2 r^{-4}/(256\pi^4)$.

These Gaussian expressions give the stated coefficients of the continuum expansion. Indeed, a connected graph with $n_x, n_0 \ge 2$ legs at the two observables, fermion loops with j_i legs, ghost loops with g_i legs, and gauge vertices with $v_i \ge 3$ legs has order b^{-e}, where

$$e = \tfrac{1}{2}(n_x+n_0+\sum_i j_i+\sum_i g_i)+\sum_i(v_i/2-1).$$

One-legged fermion and ghost loops vanish by their colour traces. With l fermion loops, $e \ge 2+l$, with equality only for two legs at each observable, no ghost or gauge vertex, and two legs at each fermion loop. The loops then lie in the two gauge lines joining the observables. For $l = 1, 2$ these are precisely the Gaussian insertions computed above. □

Proposition 14.33 (The fermionic coefficient in physical units)**.** Let $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$, $8a \le \mathsf{d}$, and $\mathsf{m} \ge 0$ with $\mathsf{md} \le 1$. The fermionic part of the second covariance coefficient on $a\mathbb{Z}^3$ satisfies

$$\begin{aligned} &a\mathsf{b}^{-3}\sum_{p,q} f(ax(p))h(ax(q))\,N_f\mathsf{B}_1^F(p,q;a\mathsf{m}) \\ &\quad = -\frac{3N_f}{8\pi^3}\mathsf{b}^{-3}\iint f(x)h(y)|x-y|^{-5}\mathcal{F}(\mathsf{m}|x-y|)\,dx\,dy + \epsilon_{\mathrm{vp}}, \end{aligned}$$

where p, q and their barycentres $x(p), x(q)$ are in the unit lattice, and $\mathcal{F}$ is the continuum polarization profile. More precisely,

$$\epsilon_{\mathrm{vp}} = -2c_2 c^{\mathrm{vp}} N_f a \mathsf{b}^{-3} \mathcal{K}_{\mathrm{free}}(f,h) + \epsilon'_{\mathrm{vp}},$$

and both errors obey

$$|\epsilon'_{\mathrm{vp}}|,\ |\epsilon_{\mathrm{vp}}| \le C_{\mathrm{vp}} \mathsf{b}^{-3}[a\log(\mathsf{d}/a) + a]\|f\|_\infty \|h\|_\infty.$$

The constant depends on the lattice fermion operator, its parameter r, N_f and ϑ. The local polarization constant therefore changes the coupling label by $N_f c^{\mathrm{vp}} a$, which tends to zero. ∎

Proof. Put $\mathsf{r} = \mathsf{d}/a \ge 8$ and $m_0 = a\mathsf{m} \le 1/8$. We use the notation $F, Q, R_2, B^{(1)}, B^{(2)}$ from the proof of Lemma 14.29. In Feynman gauge write

$$\mathsf{B}_1^F(p,q;m_0) = -3\Pi_\infty(p,q) X(p,q), \qquad X(p,q) = (\alpha_p, \Pi\alpha_q), \qquad \alpha_p = \mathrm{d}^* \Delta^{-1} \delta_p.$$

The expression is gauge independent by Lemma 14.31. Use the dual direction $\lambda(p)$ of a plaquette and put its coordinate at its barycentre. The orientation signs in the curvature covariance and X cancel in their product. For variables at bond midpoints and plaquette barycentres, set

$$\hat{k}_\mu = 2\sin(k_\mu/2), \qquad \widehat{\mathrm{d}}(k)\alpha = i\hat{k} \times \alpha, \qquad \varpi(k) = \frac{\hat{k}\hat{k}^{\mathsf{T}}}{\hat{k}^2}.$$

The matrix $\widehat{\mathrm{d}}$ is Hermitian, $\widehat{\mathrm{d}}\hat{k} = 0$ and $\widehat{\mathrm{d}}^{\,2} = \hat{k}^2 I - \hat{k}\hat{k}^{\mathsf{T}}$. Thus the symbols of Π_∞ and X are $I - \varpi$ and $\widehat{\mathrm{d}}\widehat{\Pi}\widehat{\mathrm{d}}/\hat{k}^4$. The corresponding kernel includes the phase $e^{ik\cdot(x(p)-x(q))}$; with the midpoint conventions its integrand is periodic. Write $\rho = |x(p) - x(q)|$ for $p \ne q$.

Four derivatives of the remainder. Choose a smooth radial χ_0 equal to one on $|k| \le \delta/8$ and zero on $|k| \ge \delta/4$. Set

$$R(k;m_0) = \widehat{\Pi}(k;m_0) - (\hat{k}^2 I - \hat{k}\hat{k}^{\mathsf{T}})(c^{\mathrm{vp}} + \chi_0(k) N_f^{-1} \Upsilon(k;m_0)),$$
$$\mathsf{w}(k) = (|k| + m_0)(1 + |\log|k||) + \frac{m_0}{|k| + m_0}.$$

We claim that for $|\alpha| \le 4$,

$$|\partial_k^\alpha R(k;m_0)| \le C|k|^{2-|\alpha|} \mathsf{w}(k) \quad (0 < |k| \le \delta/8), \tag{14.xlviii}$$

with uniformly bounded derivatives away from that ball.

First, the spectral integral for Υ gives

$$|\partial_k^\alpha \Upsilon(k;m)| \le C_\alpha N_f (|k| + m)^{-1-|\alpha|}.$$

Indeed, the spectral factor $1 + 4m^2/\mu^2$ is at most two for $\mu \ge 2m$ and $|\partial_k^\alpha (\mu^2 + k^2)^{-1}| \le C_\alpha (\mu^2 + k^2)^{-1-|\alpha|/2}$. Its integral over $[2m, \infty)$ is bounded by both $C_\alpha |k|^{-1-|\alpha|}$ and $C_\alpha m^{-1-|\alpha|}$.

Away from $k = 0$, split the loop integral into neighbourhoods of $p = \pm k/2$ and their complement. In each neighbourhood use the variable of the singular propagator. The k derivatives then fall on the other factors, smooth at a distance bounded below from their singularity. The one remaining singular factor is at most $C|p_\mp|^{-1}$ and is integrable. The free inverse derivative estimate, analytic vertices and analytic heavy term give uniform bounds through order four. This proves the away-from-zero assertion.

Near zero, the decomposition in the proof of Lemma 14.29 reads

$$R = \mathsf{D}(c^{\mathrm{vp}} + N_f^{-1}\Upsilon) + F + \widetilde{Q} + \tfrac{1}{2}[\partial_k^2 Q(0;m_0) - \partial_k^2 Q(0;0)][k,k],$$

where $\mathsf{D} = (k^2 I - kk^{\mathsf{T}}) - (\hat{k}^2 I - \hat{k}\hat{k}^{\mathsf{T}})$ and $\widetilde{Q} = Q - \tfrac{1}{2}\partial_k^2 Q(0;m_0)[k,k]$. The analytic matrix D vanishes to order four. The first term therefore has derivatives bounded by $C|k|^{3-|\alpha|}$.

For F, differentiate its symmetrized $B^{(1)}$ integral also in m_0. Its local singularities have degree at least -2. At infinity the differentiated integrand is $O(k^2|p|^{-4})$ at order zero, $O(|k||p|^{-4})$ at order one in k, and $O(|p|^{-2-|\alpha|})$ thereafter. The same partition around the two singular points shows that

$\partial_k^\alpha \partial_{m_0} F$ is homogeneous of degree $1-|\alpha|$ and uniformly bounded on $|k|=1$, $0 \le m_0 \le 1$. Since $F(k;0)=0$, integration in the mass gives $|\partial_k^\alpha F| \le C m_0 |k|^{1-|\alpha|}$ when $m_0 \le |k|$. When $|k| \le m_0$, use $F(k;m_0) = m_0^2 F(k/m_0;1)$. The last function is C^4 on the unit ball, vanishes at zero with its first derivative, and is even. Hence in this range $|\partial_k^\alpha F| \le C|k|^{2-|\alpha|}$. Together,

$$|\partial_k^\alpha F(k;m_0)| \le C|k|^{2-|\alpha|} \frac{m_0}{|k|+m_0}.$$

For $\widetilde{Q}$, extend the propagator and vertex estimates in that proof to four derivatives. The analytic remainders used there have four bounded derivatives. For a term of total degree $n \ge 0$ in $B^{(2)}$, the two regions $|p_\mp| \le |k|/4$ and the remaining part of $|p| \le |k|$ contribute at most $C|k|^3(|k|+m_0)^{n-4}$ to $\partial_k^4 R_2$. The region $|k| \le |p| \le 2\delta$ contributes $C\int (|p|+m_0)^{n-4}\, d^3p$. Derivatives falling on the cutoffs cost at most $C|k|^{-1}$. Thus $|\partial_k^4 R_2| \le C|k|^{-1}$. The other terms $R_c, R_1, \Pi^{\text{out}}, T$ and the heavy term have bounded fourth derivatives: their integrands are smooth on their supports and have integrable $O(|p|^{-6})$ tails after differentiation. The constant $\Phi_1(0)$ is independent of k. Together with the third-derivative bound already proved, this gives

$$|\partial_k^\alpha Q(k;m_0)| \le C|k|^{3-|\alpha|}(1+|\log|k||), \qquad |\alpha| = 3,4.$$

The function $\widetilde{Q}$ and its first two derivatives vanish at zero. Taylor's formula along the segment to k and $\int_0^1 (1+|\log(t|k|)|)\, dt \le 2(1+|\log|k||)$ give the same estimate for $\partial_k^\alpha \widetilde{Q}$ at every $|\alpha| \le 4$. Finally the coefficients of the last quadratic term are bounded by $Cm_0(1+|\log m_0|) \le 2C(|k|+m_0)(1+|\log|k||)$. These four estimates prove eq. 14.xlviii.

Position-space kernels. The matrix identity $\widehat{\mathrm{d}}(\hat{k}^2 I - \hat{k}\hat{k}^{\mathsf{T}})\widehat{\mathrm{d}} = \hat{k}^4(I-\varpi)$ gives

$$X = c^{\text{vp}}\Pi_\infty + N_f^{-1} X_\Upsilon + X_R.$$

The last two kernels have symbols $\chi_0 \Upsilon(I-\varpi)$ and $\widehat{\sigma}_R = \widehat{\mathrm{d}} R \widehat{\mathrm{d}}/\hat{k}^4$.

We use the following shell estimate. If $\widehat{\sigma}$ is supported near zero and $|\partial^\alpha \widehat{\sigma}(k)| \le A|k|^{-|\alpha|}\omega(|k|)$ for $|\alpha| \le 4$, decompose it into shells $|k| \asymp 2^{-j}$. Four integrations by parts on each shell bound its kernel by

$$CA \sum_{j \ge 0} 2^{-3j} \omega_j \min\{1, 2^{4j}|u|^{-4}\},$$

where ω_j is the supremum on the shell. For $|u| \ge 1$, this gives respectively

$$\begin{aligned}
&CA[|u|^{-4}(1+\log|u|)^2 + m_0|u|^{-3}(1+\log|u|)] && \text{if } \omega(t) = (t+m_0)(1+|\log t|),\\
&CAm_0|u|^{-2} && \text{if } \omega(t) = m_0/(t+m_0),\\
&CA|u|^{-4}(1+\log|u|) && \text{if } \omega(t) = t.
\end{aligned}$$

For the second estimate use $m_0/(t+m_0) \le \min\{1, m_0/t\}$.

The lattice curvature kernel satisfies

$$|\Pi_\infty(p,q) - \mathbb{K}_3(x(p)-x(q))_{\lambda(p)\lambda(q)}| \le C'_{\text{lat}}\rho^{-5}, \qquad \mathbb{K}_3(u) = \frac{3\hat{u}\hat{u}^{\mathsf{T}} - I}{4\pi|u|^3}.$$

In particular $|\Pi_\infty(p,q)| \le C\rho^{-3}$. Let $\mathcal{G}_1(u;m)$ be the continuum inverse Fourier transform of $\Upsilon(k;m)(I - kk^{\mathsf{T}}/k^2)$. The same shell argument over all integer shells, with three integrations by parts for the kernel and four for its gradient, gives

$$|\mathcal{G}_1(u;m)| \le CN_f|u|^{-2}, \qquad |\nabla \mathcal{G}_1(u;m)| \le CN_f|u|^{-3}.$$

These are uniform in the mass. Homogeneity gives $\mathcal{G}_1(u;m_0) = a^2 \mathcal{G}_1(au;\mathsf{m})$.

The difference between the symbol of X_Υ and this continuum symbol is

$$\chi_0 \Upsilon(\varpi^0 - \varpi) + (\chi_0 - 1)\Upsilon(I - \varpi^0), \qquad \varpi^0 = kk^{\mathsf{T}}/k^2.$$

Since $\varpi-\varpi^{0}$ has symbol order two, the first term has derivatives bounded by $CN_f|k|^{1-|\alpha|}$. Its kernel is at most $CN_f\rho^{-4}(1+\log\rho)$. The second vanishes near zero and has four integrable derivatives at infinity, so its kernel is at most $CN_f\rho^{-4}$. It follows, for $\rho\ge 1$, that

$$|X_{\Upsilon}(p,q)-\mathcal{G}_1(x(p)-x(q);m_0)_{\lambda(p)\lambda(q)}|\le CN_f\rho^{-3},\qquad |X_{\Upsilon}(p,q)|\le CN_f\rho^{-2}.$$

By eq. 14.xlviii, $|\partial^{\alpha}\widehat{\sigma}_R|\le C|k|^{-|\alpha|}\mathsf{w}(k)$ near zero. After removing a smaller ball its derivatives through order four are bounded; include the midpoint phase before integration by parts on the torus, so boundary terms cancel. The shell bounds give

$$\begin{aligned}|X_R(p,q)|&\le C[\rho^{-4}(1+\log\rho)^2+m_0\rho^{-3}(1+\log\rho)+m_0\rho^{-2}]\\&\le C[\rho^{-3}(1+\log\rho)+m_0\rho^{-2}].\end{aligned}$$

The plaquette sums. There are at most $81\mathsf{r}^3$ plaquettes in either source support. For a contributing pair, $\mathsf{r}\le\rho\le 10\mathsf{r}$. The preceding kernel estimates give, entry by entry in the two dual directions,

$$N_f\Pi_{\infty}X=N_fc^{\mathrm{vp}}\Pi_{\infty}^2+\mathbb{K}_3\mathcal{G}_1+\mathsf{E}+N_f\Pi_{\infty}X_R,$$

where $|\mathsf{E}|\le CN_f\rho^{-6}$ and $|N_f\Pi_{\infty}X_R|\le CN_f[\rho^{-6}(1+\log\rho)+m_0\rho^{-5}]$. Thus

$$\begin{aligned}&a\mathsf{b}^{-3}\sum_{p,q}f(ax(p))h(ax(q))N_f\mathsf{B}_1^{F}(p,q;m_0)\\&\quad=-3N_fc^{\mathrm{vp}}a\mathsf{b}^{-3}\Sigma_2-3a\mathsf{b}^{-3}\Sigma_F+\mathsf{E}',\end{aligned}$$

where Σ_2 is the sum with kernel $\Pi_{\infty}(p,q)^2$, Σ_F uses $(\mathbb{K}_3)_{\lambda(p)\lambda(q)}(\mathcal{G}_1)_{\lambda(p)\lambda(q)}$, and

$$|\mathsf{E}'|\le CN_f\mathsf{b}^{-3}[a+a\log(\mathsf{d}/a)+a\mathsf{m}\mathsf{d}]\mathfrak{f}.$$

The last term is at most a by hypothesis.

We spell out the Riemann-sum error. Suppose on $\mathsf{d}/2\le|u|\le 11\mathsf{d}$ that $|\kappa(u)|\le A\mathsf{d}^{-n}$ and $|\nabla\kappa(u)|\le A\mathsf{d}^{-n-1}$. For fixed dual directions, the product cubes of side a about the two barycentres tile $\mathbb{R}^6$. At most $(2\mathsf{r}+3)^6$ meet the source support. On each the function $f(x)h(y)\kappa(x-y)$ has Lipschitz constant at most $3A\vartheta^{-1}\mathsf{d}^{-n-1}\mathfrak{f}$. Summing its error from the centre gives

$$\begin{aligned}&\left|a^6\sum_{\lambda(p)=\lambda,\lambda(q)=\lambda'}f(ax(p))h(ax(q))\kappa(ax(p)-ax(q))-\iint f(x)h(y)\kappa(x-y)\,dx\,dy\right|\\&\qquad\le CA\vartheta^{-1}(a/\mathsf{d})\mathsf{d}^{6-n}\mathfrak{f}.\end{aligned}$$

For Σ_2, take $n=6$ and use $\mathbb{K}_3(u)=a^3\mathbb{K}_3(au)$ and $\operatorname{tr}\mathbb{K}_3(u)^2=3/(8\pi^2)|u|^{-6}$. The lattice-square error is $O(\rho^{-8})$. Consequently

$$\Sigma_2=\frac{3}{8\pi^2}\mathcal{K}_{\mathrm{free}}(f,h)+O(\vartheta^{-1}a/\mathsf{d}+\mathsf{r}^{-2})\mathfrak{f}.$$

Since $3\cdot 3/(8\pi^2)=2c_2$, this gives the stated term containing c^{vp}, with error at most

$$CN_f\vartheta^{-1}(a/\mathsf{d})a\mathsf{b}^{-3}\mathfrak{f}.$$

For Σ_F, use $n=5$, $(\mathbb{K}_3\mathcal{G}_1)(u;m_0)=a^5(\mathbb{K}_3\mathcal{G}_1)(au;\mathsf{m})$ and the continuum identity

$$-3\operatorname{tr}(\mathbb{K}_3\mathcal{G}_1)(u;\mathsf{m})=-\frac{3N_f}{8\pi^3}|u|^{-5}\mathcal{F}(\mathsf{m}|u|).$$

The Riemann-sum error is at most $CN_f\vartheta^{-1}a\mathsf{b}^{-3}\mathfrak{f}$. Adding these errors to E' gives the asserted bound for ϵ'_{vp}. The term containing c^{vp} is itself at most $Ca\mathsf{b}^{-3}\mathfrak{f}$ by the free-kernel test-pair bound, so the same form of estimate holds for ϵ_{vp}. □

14.25. **The leading covariance coefficient.** Fix a datum $(g,\ell_G,K,\mathsf{b},\mathsf{m})$ on the least torus, with $g\le\gamma_{\mathrm{e}}$, and $(f,h)\in\mathfrak{T}_{T_{\ell_G}}(\mathsf{d};\vartheta)$, where $\vartheta\in(0,1)$ and $8a_K\le\mathsf{d}\le 1$. Write $\mathfrak{f}=\|f\|_\infty\|h\|_\infty$. For $0\le i<K$, let B_i be the unscaled $\mathfrak{su}(2)$-valued fluctuation on the positively oriented non-pivot bonds $\mathfrak{X}_i^+$. Let μ_i be the centred Gaussian measure with covariance $\mathsf{t}_i^{-1}(\mathsf{A}_i^{\mathrm{cv}})^{-1}\otimes 1$, and let χ_i be the product of the bond tests at the calibration t_i. Put

$$\mu_i^\chi=\frac{\chi_i\mu_i}{\int\chi_i\,d\mu_i},\qquad \mu=\bigotimes_{i<K}\mu_i,\qquad \mu^\chi=\bigotimes_{i<K}\mu_i^\chi.$$

The curvature response of the ith chart is $\mathsf{T}_i^{\mathrm{cv}}=\mathsf{H}_i^{\mathrm{cv}}\mathsf{C}_{\mathbf{1},\mathbb{R}}$, acting separately on the three colour components. At a bare plaquette p, set

$$\Theta_p=\sum_{i<K}(\mathsf{T}_i^{\mathrm{cv}}B_i)(p),\qquad \mathcal{O}^\Theta(u)=\frac{1}{2a_K}\sum_p u(x(p))|\Theta_p|^2.$$

Proposition 14.34 (The coefficient of order four)**.** Let $\mathsf{n}_{j,[4]}$ be the coefficient of λ^4 in the mixed jet of Prop. 14.26. Then

(1)
$$\sum_{j\le K}\mathsf{n}_{j,[4]}=\mathrm{Cov}_{\mu^\chi}(\mathcal{O}^\Theta(f),\mathcal{O}^\Theta(h)).$$
This number depends only on the lattice, b, the test pair, and the thresholds $\delta(\mathsf{t}_i)=A_1\mathsf{t}_i^{-1/2}(\log\mathsf{t}_i)^{\mathfrak{p}_0}$. It reads the member, r, and N_f only through the previously chosen constants, and is independent of m_0,x_i,s_i,b_i.

(2)
$$\mathrm{Cov}_\mu(\mathcal{O}^\Theta(f),\mathcal{O}^\Theta(h))=\frac{3}{2}\mathsf{b}^{-2}\mathcal{K}[C^{\mathrm{fl}}_{K,0}](f,h).$$
The difference between this covariance and the tested covariance in (a) has absolute value at most $C_\chi\mathsf{b}^{-2}\mathsf{d}^3\mathfrak{f}$.

(3) With $C_{\mathrm{ord}}=3C_{\mathrm{tm}}+C_\chi$ and $X_g=L(g^{-2}+B_\sharp)\ge\mathsf{b}$,
$$\Big|\sum_{j\le K}\mathsf{n}_{j,[4]}-c_2\mathsf{b}^{-2}\mathcal{K}_{\mathrm{free}}(f,h)\Big|\le\mathsf{b}^{-2}\Big[\frac{3C_{\mathrm{tm}}}{2\vartheta}\frac{a_K}{\mathsf{d}}+C_{\mathrm{ord}}\mathsf{d}^3\Big]\mathfrak{f}$$
$$\le\Big[C_{\mathrm{ord}}X_g\mathsf{d}^2\,\mathsf{b}^{-3}\mathsf{d}+\frac{3C_{\mathrm{tm}}}{2\vartheta}\mathsf{b}^{-2}\frac{a_K}{\mathsf{d}}\Big]\mathfrak{f}.$$

(4) The assertions hold at every lattice translate of (f,h), with the same constants and the same $\mathcal{K}_{\mathrm{free}}(f,h)$, hence also after the translation average of Lem. 14.12. The conditions $\mathsf{d}\le\mathsf{d}_{\mathrm{e}}$ and $K>s(\mathsf{d})$ in Prop. 14.3 imply $\mathsf{d}<1$ and $8a_K\le\mathsf{d}$.

■

Here C_{tm} is the constant of the curvature-template comparison [QCDM1, Prop. A.2], enlarged to cover the constant $C^{\mathrm{fl}}_{\mathrm{tm}}$ in Lem. A.26. The remaining constants can be chosen explicitly:

$$C_\chi=\frac{841}{4}\Big(12C_\Theta K_{\chi,2}S_3^{\mathrm{lv}}+18K_{\chi,2}^2(S_3^{\mathrm{lv}})^2+9K_{\chi,4}S_6^{\mathrm{lv}}\Big),$$
$$S_q^{\mathrm{lv}}=\sum_{k\ge1}L^{qk}e^{-\frac12 c_{\mathrm{tl}}k^2(\log L)^2},\quad q=3,6,\qquad c_{\mathrm{tl}}=\tfrac12\gamma_{\mathrm{tail}}A_1^2,$$
$$K_{\chi,2}=\max\{(C_{\mathrm{tp}}^{\mathrm{row}})^2C_EK_{\mathrm{cp}},\sqrt{3}\,C_{\mathrm{sm}}C_{\mathrm{tp}}\},$$
$$K_{\chi,4}=\max\{(C_{\mathrm{tp}}^{\mathrm{row}})^4C_EK_{\mathrm{cp}}^{(4)},C_{\mathrm{sm}}(\sqrt{105}+6\sqrt{3}+9C_{\mathrm{sm}})C_{\mathrm{tp}}^2\},$$
$$C_\Theta=1+3C_{\mathrm{lat}}^\natural+C_{\mathrm{tp}}^{\mathrm{bg}},\qquad C_{\mathrm{sm}}=\tau_*^{1/2}(1+\tau_*^{1/2})e^{\tau_*},\qquad \tau_*=N_{\mathsf{H}}^3K_M.$$

The constants $C_{\mathrm{tp}}, C_{\mathrm{tp}}^{\mathrm{row}}, C_{\mathrm{tp}}^{\mathrm{bg}}$ are those of the slice estimates [YM3, Prop. 7.6]; $C_{\mathrm{lat}}^{\natural}$ is the absolute constant in the free-kernel estimate [QCDM1, eq. A.vi]. The test-tail rate is $\gamma_{\mathrm{tail}} = 2(LM)^{-3}$, $K_M = C_E K_M'$, and K_{cp} is the one-step tested-covariance constant. The fourth-order constant $K_{\mathrm{cp}}^{(4)}$ is specified below. Both series converge because their exponents are quadratic in k with negative leading coefficient. These constants are chosen after $A_1, K_M, K_{\mathrm{cp}}$. They impose no additional condition on an earlier constant and are independent of g, ϑ, P, K, the side, b, m, and the test pair.

Lemma 14.35 (One level under its test)**.** In the setting of Prop. 14.34, set

$$p_i = \exp[-\tfrac{1}{2}\gamma_{\mathrm{tail}}\mathfrak{b}(\mathsf{t}_i)^2].$$

Uniformly in the torus, the following hold for $0 \le i < K$.

(1) Both μ_i and μ_i^χ have zero odd moments. For non-pivot bonds b, b' and colours c, c',

$$\left|\mathsf{E}_{\mu_i^\chi} B_i^c(b) B_i^{c'}(b') - \mathsf{E}_{\mu_i} B_i^c(b) B_i^{c'}(b')\right| \le C_E K_{\mathrm{cp}} \mathsf{t}_i^{-1} p_i.$$

(2) For four bonds and colours, repetitions allowed,

$$\left|\kappa^{\mu_i^\chi}(B_i^{c_1}(b_1), \dots, B_i^{c_4}(b_4))\right| \le C_E K_{\mathrm{cp}}^{(4)} \mathsf{t}_i^{-2} p_i,$$

where

$$K_{\mathrm{cp}}^{(4)} = 16 K_P^{\Pi} \mathsf{C}_{\mathrm{ex}} ((L+2)M)^3 e^{4\lambda_1} M^6 w_*^{-4}.$$

The corresponding Gaussian cumulant is zero.

(3) If $N_{i+1}^{\square}$ is the number of cubes at level $i+1$ and $\tau_i^\chi = N_{i+1}^{\square} K_M p_i$, then

$$\int \chi_i \, d\mu_i \ge e^{-\tau_i^\chi}.$$

Every real polynomial P in B_i satisfies

$$|\mathsf{E}_{\mu_i^\chi} P - \mathsf{E}_{\mu_i} P| \le ((\tau_i^\chi)^{1/2} + \tau_i^\chi) e^{\tau_i^\chi} (\mathsf{E}_{\mu_i} P^2)^{1/2}.$$

■

Proof. At the trivial pair, $x_i = \mathsf{t}_i$, and zero quadratic potential, the one-step tested comparison has normalized measures μ_i^χ and μ_i. Its unscaled fluctuation is $B_i = \mathsf{t}_i^{-1/2} B$. The Gaussian and the bond tests are even; their odd moments therefore vanish. The one-step covariance comparison gives (a), after discarding its exponential decay factor.

For (b), introduce four distinct complex parameters, even if some bonds or colours coincide:

$$S(\sigma) = \sum_{r=1}^{4} \sigma_r B_i^{c_r}(b_r).$$

Place each summand on the cube containing its bond, or on the two-cube polymer containing its endpoints. These polymers have size zero. The one-step expansion admits these complex Borel functions as one quadratic origin: the name of that class does not require the function to be quadratic. Since

$$|B(b)| \le \tfrac{1}{2} M^{3/2} (1 + M^{-3} |B(b)|^2),$$

the norm of the rth summand is at most $\frac{1}{2}|\sigma_r| \mathsf{t}_i^{-1/2} M^{3/2}$. Thus, on the polydisc

$$|\sigma_r| < \rho_0, \qquad \rho_0 = \tfrac{1}{2} w_* \mathsf{t}_i^{1/2} M^{-3/2},$$

each summand has norm at most $w_*/4$. There are at most four summands on any polymer, so their family has norm at most w_*. The expansion applies with this single origin and with no potential origin.

The difference

$$\log \int \chi_i e^{S(\sigma)} d\mu_i - \log \int e^{S(\sigma)} d\mu_i$$

is the sum of the activities marked by a failed large-field test. Each is holomorphic on the polydisc and bounded by

$$K_P^{\Pi} p_* e^{-\kappa_+ d_{i+1}(X)}, \qquad p_* = \mathsf{C}_{\mathrm{ex}}((L+2)M)^3 e^{4\lambda_1} p_i.$$

This estimate is independent of the norm of the origin. The second logarithm is a quadratic polynomial in σ, so it has no coefficient of $\sigma_1\sigma_2\sigma_3\sigma_4$. That coefficient in the first logarithm is the fourth cumulant; its normalization contributes only a constant.

Locality forces a contributing polymer to contain all four bonds. Cauchy's estimate on radii $\rho < \rho_0$, followed by $\rho \uparrow \rho_0$, bounds its coefficient by

$$K_P^{\Pi} p_* \rho_0^{-4} e^{-\kappa_+ d_{i+1}(X)}, \qquad \rho_0^{-4} = 16 M^6 w_*^{-4} \mathsf{t}_i^{-2}.$$

Summing polymers containing the cube of b_1 costs C_E. This gives (b).

At zero potential, the scalar activities of the test logarithm have norm at most $K_M p_i$. Their sum is therefore bounded by $N_{i+1}^{\square} K_M p_i = \tau_i^{\chi}$. Consequently

$$\int \chi_i \, d\mu_i \geq e^{-\tau_i^{\chi}}, \qquad \int (1-\chi_i)\, d\mu_i \leq 1 - e^{-\tau_i^{\chi}} \leq \tau_i^{\chi}.$$

Finally,

$$\mathsf{E}_{\mu_i^{\chi}} P - \mathsf{E}_{\mu_i} P = \frac{\mathsf{E}_{\mu_i}[(\chi_i - 1)P] + \mathsf{E}_{\mu_i}[P] \int (1-\chi_i)\, d\mu_i}{\int \chi_i \, d\mu_i}.$$

Cauchy–Schwarz and $(1-\chi_i)^2 \leq 1 - \chi_i$ bound the first numerator term by $(\tau_i^{\chi})^{1/2} (\mathsf{E}_{\mu_i} P^2)^{1/2}$. The second is at most $\tau_i^{\chi} (\mathsf{E}_{\mu_i} P^2)^{1/2}$. The lower bound for the denominator proves (c). □

Proof of Prop. 14.34. The exact covariance at a real field scale. Let $0 < \lambda \leq 1$ and $w = z_1 f + z_2 h$ in the source neighbourhood of Prop. 14.14. Evaluate the level-K form at block field $\mathbf{1}$ and $\Psi_K = 0$. Its logarithm is the sum of its source energies and a source-independent number. Indeed, the gauge action vanishes at $\mathbf{1}$; the mass and fermion terms have fermion degree two; and the scalar value of each remaining activity has already been removed as its source energy. By Lem. 14.15, its mixed derivative is $\sum_{j \leq K} \mathsf{n}_j^{\lambda}$.

The exact chart identity in Prop. 14.14 expresses the same form as

$$\mathfrak{z}_\lambda \int_{\mathsf{Bx}} \rho^{[\lambda]}(B) \exp\left(a_K^{-1} \sum_p w(x(p)) a_p(U_\lambda(B)) \right) dB, \qquad \mathsf{Bx} = \prod_{i<K} \{\chi_i = 1\}.$$

Here $U_\lambda(B)$ is the composite chart over $\mathbf{1}$ with fluctuations λB_i, and $\mathfrak{z}_\lambda \neq 0$ is independent of w. The density is the product of the Wilson factor $e^{-\beta A(U_\lambda)/\lambda^2}$, the gauge fixings and Jacobians, the chart elimination factors z_i^{el}, and the fermion factor. All depend on B, λ but not on w. All density tests equal one on Bx: the final field $\mathbf{1}$ passes its test, and descending through a chart with a passing fluctuation preserves the density test. The fluctuation λB_i passes whenever B_i does.

The fermion tower is formed at the single bare configuration $U = U_\lambda(B)$. Its exact determinant identity gives

$$\mathcal{Z}_F(U) = \prod_{i<K} \theta^{-\dim H_{i+1}} \det \Gamma_i[U] = \frac{\det(D[U] + m_0)}{\det \mathbb{D}_K[U]}, \qquad \theta = b_F.$$

For real links it is nonzero because $\operatorname{Re} \Gamma_i[U] > 0$. The level-$K$ form has the logarithm supplied by the flow, so its integral at $w = 0$ is also nonzero. Define its normalized weight $\mu^{[\lambda]}$ on Bx and put

$$O^{[\lambda]}(u) = a_K^{-1} \sum_p u(x(p)) a_p(U_\lambda(B)).$$

Differentiating the logarithm gives the exact identity

$$\sum_{j \leq K} \mathsf{n}_j^{\lambda} = \operatorname{Cov}_{\mu^{[\lambda]}}(O^{[\lambda]}(f), O^{[\lambda]}(h)). \tag{14.xlix}$$

The derivative of the composite chart. Let $U_\lambda^{(i)}$ be the level-i background obtained from the fluctuations $\lambda B_{i'}$, $i \le i' < K$, over $\mathbf{1}$. Thus $U_\lambda^{(K)} = \mathbf{1}$ and $U_\lambda^{(0)} = U_\lambda(B)$. The chart expresses $U_\lambda^{(i)}$ as a gauge transform of $e^{i\mathcal{A}(\lambda B_i; U_\lambda^{(i+1)})} U_\lambda^{(i+1)}$. The shift $\mathcal{A}$ and the site gauge factor are holomorphic, and equal 0 and $\mathbf{1}$ at zero fluctuation. If $\mathfrak{y}_i$ is the first derivative at zero, the site gauge factor contributes an exact form. The first derivative of the shift has curvature $\mathsf{T}_i^{\mathrm{cv}} B_i$. Consequently

$$d\mathfrak{y}_i = d\mathfrak{y}_{i+1} + \mathsf{T}_i^{\mathrm{cv}} B_i, \qquad \mathfrak{y}_K = 0, \qquad d\mathfrak{y}_0 = \Theta.$$

At a plaquette the chart therefore has logarithm $\lambda\Theta_p + O(\lambda^2)$. Since $a_p = 1 - \cos|Y_p| = \frac{1}{2}|Y_p|^2 + O(|Y_p|^4)$,

$$\lambda^{-2} O^{[\lambda]}(u) \longrightarrow \mathcal{O}^{\Theta}(u).$$

This convergence is uniform on the closure of Bx. To see the required uniformity, keep the finite datum fixed. The zero fluctuation and trivial pair lie in the interiors of all chart domains. There are finitely many levels and the closure of Bx is compact. Hence all composed charts are holomorphic for $|\lambda| < 2\lambda_{\mathrm{h}}$ with one datum-dependent $\lambda_{\mathrm{h}} > 0$, and continuous on $|\lambda| \le \lambda_{\mathrm{h}}$ times that compact set. Cauchy's formula gives uniform Taylor remainders.

The limiting weight. The same argument applies to the gauge fixings, Jacobian factors, and fermion tower. The latter is holomorphic near $\mathbf{1}$, since its finitely many matrix inverses exist there. Its value $\mathcal{Z}_F(\mathbf{1})$ is nonzero. The elimination factor of the ith chart tends uniformly to $L^{9\#\Gamma_{i+1}^{\mathrm{b}}}$: at the trivial field every pivot coefficient is $L^{-3}\mathrm{id}$. Writing the gauge-fixing and Jacobian remainders as $\mathcal{V}_i^{\mathrm{G}}, \mathcal{V}_i^{\mathrm{J}}$, we have, uniformly on this compact set,

$$\frac{\beta}{\lambda^2} A(U_\lambda) \longrightarrow \frac{\beta}{2}|\Theta|^2, \quad \hat{\mathsf{G}}_i \longrightarrow \hat{\mathsf{G}}_{\mathbf{1},\mathbb{R}} \otimes 1, \quad \mathcal{V}_i^{\mathrm{G}}, \mathcal{V}_i^{\mathrm{J}} \longrightarrow 0.$$

The density limit is the product Gaussian μ. Here is the required orthogonality calculation. The range of $\mathsf{T}_i^{\mathrm{cv}}$ is the ith curvature slice $\mathsf{Bg}_i \cap \mathsf{Fl}_{i+1}$. The background condition follows from the curvature response. The fluctuation condition follows from

$$Q_{i+1}^{\mathrm{pl}} \mathsf{H}_i^{\mathrm{cv}} \mathsf{C}_{\mathbf{1},\mathbb{R}} B = \mathsf{W} d \mathsf{C}_{\mathbf{1},\mathbb{R}} B = d\mathbf{Q}^s \mathsf{C}_{\mathbf{1},\mathbb{R}} B = d(\mathbf{Q}^{\mathrm{lin}} + d\mathsf{k}_i) \mathsf{C}_{\mathbf{1},\mathbb{R}} B = 0.$$

The slices are orthogonal [YM3, Lem. 7.1]. With $\eta_i = L^{-i}$,

$$|\Theta|^2 = \sum_{i<K} \eta_i (B_i, (\mathsf{B}_i^{\mathrm{cv}} \otimes 1) B_i)_1.$$

Since $\beta\eta_i = \mathsf{t}_i$ and $\mathsf{A}_i^{\mathrm{cv}} = \mathsf{B}_i^{\mathrm{cv}} + \hat{\mathsf{G}}_{\mathbf{1},\mathbb{R}}$, the limiting Wilson and gauge-fixing exponents add to $\sum_{i<K} \frac{1}{2}\mathsf{t}_i (B_i, (\mathsf{A}_i^{\mathrm{cv}} \otimes 1) B_i)_1$.

More explicitly, put

$$R_\lambda = -\frac{\beta}{\lambda^2} A(U_\lambda) + \sum_{i<K} \Big(\mathcal{V}_i^{\mathrm{G}} + \mathcal{V}_i^{\mathrm{J}} + \tfrac{1}{2}\mathsf{t}_i (B_i, (\mathsf{A}_i^{\mathrm{cv}} \otimes 1 - \hat{\mathsf{G}}_i) B_i)_1 \Big)$$
$$+ \sum_{i<K} \log \frac{z_i^{\mathrm{el}}}{L^{9\#\Gamma_{i+1}^{\mathrm{b}}}} + \log \frac{\mathcal{Z}_F(U_\lambda)}{\mathcal{Z}_F(\mathbf{1})}.$$

The last logarithm is the branch vanishing at zero. For sufficiently small λ it is defined uniformly on the compact set. We have $\sup|R_\lambda| \to 0$ and

$$\mu^{[\lambda]} = \frac{e^{R_\lambda} \mu^\chi}{\int e^{R_\lambda}\, d\mu^\chi}.$$

Divide eq. 14.xlix by λ^4. The uniform observable and density limits identify the limit with the covariance in (a). The zero coefficients of lower order and holomorphy in Prop. 14.26 identify the same limit with $\sum_j \mathsf{n}_{j,[4]}$. Its stated parameter dependence follows directly from the resulting tested Gaussian formula.

Removing the Gaussian tests. The covariance of the ith curvature slice is

$$\mathsf{E}_\mu[(\mathsf{T}_i^{\mathrm{cv}}B_i)^c(p)(\mathsf{T}_i^{\mathrm{cv}}B_i)^{c'}(q)] = \delta_{cc'}\mathsf{t}_i^{-1}L^{-i}\mathcal{C}_{(i+1)}(p,q) = \delta_{cc'}\beta^{-1}\mathcal{C}_{(i+1)}(p,q).$$

The slices are independent and $\sum_{i<K}\mathcal{C}_{(i+1)} = C^{\mathrm{fl}}_{K,0}$ [YM3, eq. 7.ii]. Wick's formula, with three colours, now gives

$$\mathrm{Cov}_\mu(\mathcal{O}^\Theta(f),\mathcal{O}^\Theta(h)) = \frac{3}{2}a_K^{-2}\beta^{-2}\sum_{p,q} f(x(p))h(x(q))C^{\mathrm{fl}}_{K,0}(p,q)^2 = \frac{3}{2}\mathsf{b}^{-2}\mathcal{K}[C^{\mathrm{fl}}_{K,0}](f,h).$$

For the tested covariance, write $\Theta_i = \mathsf{T}_i^{\mathrm{cv}}B_i$ and

$$\delta S_i^{cc'}(p,q) = (\mathsf{E}_{\mu_i^\chi} - \mathsf{E}_{\mu_i})[\Theta_i^c(p)\Theta_i^{c'}(q)], \qquad \delta S = \sum_i \delta S_i.$$

Let κ_i denote the tested joint cumulant of $\Theta_i^c(p),\Theta_i^c(p),\Theta_i^{c'}(q),\Theta_i^{c'}(q)$. Independence between levels and the zero odd moments give

$$\begin{aligned}&\mathrm{Cov}_{\mu^\chi}(\mathcal{O}^\Theta(f),\mathcal{O}^\Theta(h)) - \mathrm{Cov}_\mu(\mathcal{O}^\Theta(f),\mathcal{O}^\Theta(h))\\ &= \frac{1}{4a_K^2}\sum_{p,q} f(x(p))h(x(q))\sum_{c,c'}\left[\sum_i \kappa_i + 4\delta_{cc'}\beta^{-1}C^{\mathrm{fl}}_{K,0}(p,q)\delta S^{cc'}(p,q) + 2(\delta S^{cc'}(p,q))^2\right].\end{aligned}$$

We estimate these terms separately according to the side at level i. If $i=0$, or if $\ell_G L^{K-i} \ge N_{\mathsf{H}}$, the row bound in [YM3, Prop. 7.6] is

$$\sum_b |\mathsf{T}_i^{\mathrm{cv}}(p,b)| \le C_{\mathrm{tp}}^{\mathrm{row}}L^{-2i}.$$

Multilinearity and Lem. 14.35(a),(b) imply

$$\begin{aligned}|\delta S_i^{cc'}(p,q)| &\le (C_{\mathrm{tp}}^{\mathrm{row}})^2 C_E K_{\mathrm{cp}}\beta^{-1}L^{-3i}p_i,\\ |\kappa_i| &\le (C_{\mathrm{tp}}^{\mathrm{row}})^4 C_E K_{\mathrm{cp}}^{(4)}\beta^{-2}L^{-6i}p_i.\end{aligned}$$

In the remaining case $i\ge 1$ and $N_{i+1}^{\square} < N_{\mathsf{H}}^3$. Thus $\tau_i^\chi \le \tau_* p_i$ and

$$\epsilon_i = ((\tau_i^\chi)^{1/2} + \tau_i^\chi)e^{\tau_i^\chi} \le C_{\mathrm{sm}}\sqrt{p_i}.$$

For $X = \Theta_i^c(p)$ and $Y = \Theta_i^{c'}(q)$, the Gaussian slice estimate gives variances at most $V_i = \beta^{-1}C_{\mathrm{tp}}L^{-3i}$. Wick's formula gives

$$\mathsf{E}_\mu X^4 \le 3V_i^2, \quad \mathsf{E}_\mu X^2Y^2 \le 3V_i^2, \quad \mathsf{E}_\mu X^4Y^4 \le 105V_i^4.$$

Part (c) of Lem. 14.35, first at XY and then at X^2, Y^2, yields

$$|\delta S_i| \le \sqrt{3}\,\epsilon_i V_i, \qquad \mathsf{E}_{\mu_i^\chi}X^2 \le (1+\sqrt{3}\,\epsilon_i)\mathsf{E}_{\mu_i}X^2.$$

Use the identity

$$\begin{aligned}\kappa_i = {}&(\mathsf{E}_{\mu_i^\chi} - \mathsf{E}_{\mu_i})X^2Y^2\\ &- (\mathsf{E}_{\mu_i^\chi}X^2\mathsf{E}_{\mu_i^\chi}Y^2 - \mathsf{E}_{\mu_i}X^2\mathsf{E}_{\mu_i}Y^2)\\ &- 2((\mathsf{E}_{\mu_i^\chi}XY)^2 - (\mathsf{E}_{\mu_i}XY)^2).\end{aligned}$$

The three lines are bounded respectively by $\sqrt{105}\epsilon_i V_i^2$, $\sqrt{3}\,\epsilon_i(2+\sqrt{3}\,\epsilon_i)V_i^2$, and $2\sqrt{3}\,\epsilon_i(2+\sqrt{3}\,\epsilon_i)V_i^2$. Both side cases therefore give the uniform bounds

$$|\delta S_i| \le K_{\chi,2}\beta^{-1}L^{-3i}\sqrt{p_i}, \qquad |\kappa_i| \le K_{\chi,4}\beta^{-2}L^{-6i}\sqrt{p_i}.$$

Put $k = K-i \ge 1$. Since $\mathsf{t}_i = \mathsf{b}L^k$, $\mathsf{b}\ge g^{-2}\ge 4$, and $\mathfrak{p}_0 \ge 1$,

$$\sqrt{p_i} \le e^{-\frac{1}{2}c_{\mathrm{tl}}k^2(\log L)^2}.$$

Hence

$$\sum_{i<K} L^{-3i}\sqrt{p_i} \le a_K^3 S_3^{\mathrm{lv}}, \qquad \sum_{i<K} L^{-6i}\sqrt{p_i} \le a_K^6 S_6^{\mathrm{lv}}.$$

On the supports of f, h, the physical separation is at least d. The free-kernel and background estimates, with the slice telescope, give

$$|C_{K,0}^{\mathrm{fl}}(p,q)| \le C_\Theta a_K^3 \mathsf{d}^{-3}.$$

Each support contains at most $3(2\mathsf{d}/a_K+1)^3 \le 29(\mathsf{d}/a_K)^3$ plaquette barycentres: there are three translates of a lattice of spacing a_K, and $8a_K \le \mathsf{d}$. Substitution into the covariance difference gives

$$\begin{aligned} |\operatorname{Cov}_{\mu^\chi} - \operatorname{Cov}_\mu| &\le \frac{841}{4} a_K^{-2} (\mathsf{d}/a_K)^6 \mathfrak{f}\, \beta^{-2} a_K^6 \\ &\quad \cdot [9K_{\chi,4} S_6^{\mathrm{lv}} + 12 C_\Theta K_{\chi,2} S_3^{\mathrm{lv}} \mathsf{d}^{-3} + 18 K_{\chi,2}^2 (S_3^{\mathrm{lv}})^2] \\ &\le C_\chi \mathsf{b}^{-2} \mathsf{d}^3 \mathfrak{f}. \end{aligned}$$

In the last step use $a_K\beta = \mathsf{b}$ and $\mathsf{d} \le 1$. This proves (b).

The continuum template. Apply Lem. A.26 to the fluctuation template with $n = K$ and $a_{K-n} = 1 \ge \mathsf{d}$. The hypotheses are $8a_K \le \mathsf{d}$, $20\mathsf{d} \le \ell_G$, and the prescribed test-pair condition, all of which hold on the least torus. It gives

$$\left| \mathcal{K}[C_{K,0}^{\mathrm{fl}}](f,h) - \frac{3}{8\pi^2} \mathcal{K}_{\mathrm{free}}(f,h) \right| \le C_{\mathrm{tm}} \left[\frac{a_K}{\vartheta \mathsf{d}} + (\mathsf{d}/\ell_G)^3 + \mathsf{d}^3 \right] \mathfrak{f} \le C_{\mathrm{tm}} \left[\frac{a_K}{\vartheta \mathsf{d}} + 2\mathsf{d}^3 \right] \mathfrak{f}.$$

Combine this with (a),(b) and $c_2 = \frac{3}{2}\frac{3}{8\pi^2}$. The second bound in (c) uses $\mathsf{b}^{-2}\mathsf{d}^3 \le X_g \mathsf{d}^2 \mathsf{b}^{-3} \mathsf{d}$.

All hypotheses and estimates are invariant under lattice translations. The continuum template is translation invariant as well, proving (d)'s first assertion. At the data of Prop. 14.3, $\mathsf{d} \le \mathsf{d}_{\mathrm{e}} \le \mathsf{d}_0(g,\vartheta) \le 1/200$. The definition of the stopping scale gives

$$\mathsf{d} = \check{\mathsf{d}}\, L^{K-s(\mathsf{d})} a_K, \qquad \check{\mathsf{d}} > 4L^2 M.$$

Since $K > s(\mathsf{d})$, this implies $\mathsf{d} > 4L^3 M a_K \ge 8a_K$. □

14.26. **The block constraint and the fermion loop.** All quantities in this subsection are in bare lattice units. We compare the operator under a flat block constraint with the infinite-lattice operator at the same bare mass. The first step is to compute the two derivatives of the composite fermion average.

Lemma 14.36 (The composite average and its current)**.** Let $n \ge 1$, on $\mathbb{Z}^3$ or on the bare torus at cutoff $K \ge n$. For a bare site x, let $y_0(x) = x$, and let $y_{m+1}(x)$ be the site whose level-$(m+1)$ block contains $y_m(x)$. For a bond b, let $m_x(b)$ be the least $m \ge 0$ such that both endpoints of b lie in $B^{m+1}(y_{m+1}(x))$. Set $m_x(b) = \infty$ if there is no such level, considering only levels at most K on a torus, and set $L^{-\infty} = 0$. Write $|x-b| = |x - b_-|_\infty$.

Let a be a real one-form and $U = e^{ia\sigma_3}$. For w at level $m+1$ and $z \in B(w)$, denote by $\bar{s}_m(w,z)$ and $\mathrm{Var}_m(w,z)$ the mean and variance of $(\mathbf{Q}_m^s a)(\Gamma)$ over $\Gamma \in \mathcal{G}(w,z)$. Put $\kappa_m(w) = L^{-3}\sum_{z\in B(w)} \bar{s}_m(w,z)$. Then the following hold.

(1) The matrix $R_n(U)$ is diagonal in the eigenbasis of σ_3. For $x \in B^n(y)$ its entries at eigenvalues 1 and -1 are $L^{-3n/2} \upsilon_x \iota^*$ and $L^{-3n/2}\bar{\upsilon}_x \iota^*$, respectively. The function υ_x is real analytic near $a = 0$, and, for $x, x' \in B^n(y)$,

$$\begin{aligned} \bar{\upsilon}_x \upsilon_{x'} = 1 &+ i(\tilde{\phi}(x') - \tilde{\phi}(x)) + \mathsf{p}(x) + \mathsf{p}(x') \\ &- \tfrac{1}{2}(\tilde{\phi}(x') - \tilde{\phi}(x))^2 + O(|a|^3), \end{aligned}$$

where

$$\tilde\phi(x) = \sum_{m<n} [\bar s_m(y_{m+1}(x), y_m(x)) - \kappa_m(y_{m+1}(x))] = \sum_b J_x(b) a(b),$$

$$\mathsf{p}(x) = -\frac{1}{2}\sum_{m<n}\left[\mathrm{Var}_m(y_{m+1}(x), y_m(x)) - L^{-3}\sum_{z\in B(y_{m+1}(x))}\mathrm{Var}_m(y_{m+1}(x), z)\right]$$
$$= \sum_{b,b'} \mathsf{p}_x(b,b') a(b) a(b').$$

The phase of υ_x is $\bar n_n(y) + \tilde\phi(x) + O(|a|^3)$, where $\bar n_n$ is linear in a and constant within each level-n block.

(2) With $C_J = 20L^2$,

$$|J_x(b)| \le 5L^{-2m_x(b)} \le C_J(1+|x-b|)^{-2},$$
$$|\mathsf{p}_x(b,b')| \le 9L^{-4\max\{m_x(b), m_x(b')\}}$$
$$\le C_J^2(1+|x-b|)^{-2}(1+|x-b'|)^{-2}.$$

These coefficients vanish unless both endpoints of each displayed bond lie in $B^n(y_n(x))$.

■

Proof. Constant gauge transformations $e^{i\theta\sigma_3}$ fix U, so gauge covariance shows that they fix $\mathcal{M}^m U$. Near the identity it therefore has the form $e^{ia_m\sigma_3}$. Conjugation by $i\sigma_1$ replaces a by $-a$, so a_m is odd. Analyticity and the derivative of the block average give $a_m = \mathbf{Q}_m^{\mathrm{lin}} a + O(|a|^3)$. At one step, $\mathbf{Q}^{\mathrm{lin}} = \mathbf{Q}^s - d\mathsf{k}$ and $\mathbf{Q}^{\mathrm{lin}} d\theta = d(\theta|)$. Since $\mathbf{Q}^s_{m+1} = \mathbf{Q}^s \mathbf{Q}^s_m$, induction yields

$$\mathbf{Q}^{\mathrm{lin}}_m = \mathbf{Q}^s_m - d\bar n_m, \qquad \bar n_0 = 0, \qquad \bar n_{m+1} = \mathsf{k}_m \mathbf{Q}^s_m + \bar n_m|_{m+1}.$$

Here $\mathsf{k}_m \mathbf{Q}^s_m a = \kappa_m$.

For one step at $V = e^{ia_m\sigma_3}$, let $\bar a$ and var be the path mean and variance of $a_m(\Gamma)$. The eigenvalue-1 entry of its transport average $S(V; w, z)$ is

$$e^{i\bar a}(1 - \tfrac{1}{2}\mathrm{var} + O(|a|^3)).$$

Its determinant is $1 - \mathrm{var} + O(|a|^4)$; the normalization has square $1 - L^{-3}\sum_{z\in B(w)} \mathrm{var}(w,z) + O(|a|^4)$. The other eigenvalue gives the complex conjugate. Along a path from w to z,

$$(\mathbf{Q}^{\mathrm{lin}}_m a)(\Gamma) = (\mathbf{Q}^s_m a)(\Gamma) - \bar n_m(z) + \bar n_m(w).$$

The last two terms are path independent. Thus $\mathrm{var} = \mathrm{Var}_m(w,z) + O(|a|^4)$ and $\bar a = \bar s_m(w,z) - \bar n_m(z) + \bar n_m(w) + O(|a|^3)$.

The composite average is the product of these normalized one-step entries:

$$R_n(U)(y,x) = L^{-3n/2} \prod_{m<n} q(\mathcal{M}^m U; y_{m+1}(x), y_m(x))\, \iota^*.$$

Its eigenvalue-1 entry consequently has the form $\upsilon_x = e^{i\phi_1(x)}(1 + \mathsf{p}(x)) + O(|a|^3)$, where

$$\phi_1(x) = \sum_{m<n} [\bar s_m(y_{m+1}, y_m) - \bar n_m(y_m) + \bar n_m(y_{m+1})].$$

Substitute $\bar n_m(y_{m+1}) = \bar n_{m+1}(y_{m+1}) - \kappa_m(y_{m+1})$. The sum telescopes to $\phi_1(x) = \bar n_n(y) + \tilde\phi(x)$. For x, x' in the same block the first term cancels. Expanding $e^{i(\phi_1(x') - \phi_1(x))}(1 + \mathsf{p}(x))(1 + \mathsf{p}(x'))$ proves (a).

For (b), a bare bond has coefficient at most $L^{-3m}L^m = L^{-2m}$ in the straight average of a level-m bond. It can occur only if the block containing its base point is incident to that level-m bond. A shortest contour inside $B(w)$ has at most two bonds incident to that site. The coefficients of a bare bond in $\bar s_m$, in κ_m, and in the difference of two contour sums are therefore at most $2L^{-2m}$, $2L^{-2m}$,

and $4L^{-2m}$. They vanish unless both endpoints lie in $B^{m+1}(w)$. The mth contribution to $J_x(b)$ is thus at most $4L^{-2m}$ and vanishes for $m < m_x(b)$. Since $L \geq 13$,

$$|J_x(b)| \leq \frac{4L^{-2m_x(b)}}{1-L^{-2}} \leq 5L^{-2m_x(b)}.$$

The variance is the mean, over pairs of paths, of half the square of their difference. Its quadratic kernel is at most $8L^{-4m}$. The mth contribution to $\mathsf{p}_x(b,b')$ obeys the same bound and vanishes below $\bar{m} = \max\{m_x(b), m_x(b')\}$. Hence

$$|\mathsf{p}_x(b,b')| \leq \frac{8L^{-4\bar{m}}}{1-L^{-4}} \leq 9L^{-4\bar{m}} \leq 9L^{-2m_x(b)}L^{-2m_x(b')}.$$

If $m_x(b) < \infty$, x and b lie in a cube of side $L^{m_x(b)+1}$. Thus $L^{-2m_x(b)} \leq 4L^2(1+|x-b|)^{-2}$. Finally $m_x(b) \leq n-1$ exactly when the endpoints of b lie in $B^n(y_n(x))$. This proves the support assertion and both spatial bounds. □

Proposition 14.37 (The fermion operator under a flat block constraint)**.** There is C_Π, depending only on the member, r, L, b_F, with the following properties for $n \geq 1$. Part (a) holds for $m_0 > 0$, and (b)–(d) for $0 \leq m_0 \leq L^{-n}$.

(1) On the history with no large-field region, at cutoff $K \geq n$ and every SU(2) configuration,

$$\prod_{i<n} \theta^{-\dim H_{i+1}} \det \Gamma_i[U] = \theta_n^{-\dim H_n} \det \mathcal{K}_n[U; m_0], \qquad \theta = b_F, \quad \theta_n = L^n \mathfrak{a}_n.$$

The prefactor is independent of U.

(2) On one colour and one flavour in the range of e_+, put $\mathcal{K} = \mathcal{K}_n[\mathbf{1}; m_0]$. Let $\mathbb{V}_j^{\mathrm{D}}$ be the term of order j in a in $D_+[e^{ia}]$. Write $1_{\mathrm{bl}}(x,x')$ for the indicator that x, x' belong to one level-n block, and set

$$\mathbb{V}_1^{\mathrm{R}}(x,x') = i\mathfrak{a}_n L^{-3n} 1_{\mathrm{bl}}(x,x')(\tilde{\phi}(x') - \tilde{\phi}(x))\iota^*,$$
$$\mathbb{V}_2^{\mathrm{R}}(x,x') = \mathfrak{a}_n L^{-3n} 1_{\mathrm{bl}}(x,x')[\mathsf{p}(x) + \mathsf{p}(x') - \tfrac{1}{2}(\tilde{\phi}(x') - \tilde{\phi}(x))^2]\iota^*, \qquad \mathbb{V}_j = \mathbb{V}_j^{\mathrm{D}} + \mathbb{V}_j^{\mathrm{R}}.$$

On a torus, $\Pi^{(\leq n)}$ is eight times the real part of the kernel of

$$\tfrac{1}{2}\operatorname{Tr}[\mathcal{K}^{-1}\mathbb{V}_1\mathcal{K}^{-1}\mathbb{V}_1] - \operatorname{Tr}[\mathcal{K}^{-1}\mathbb{V}_2].$$

This formula defines $\Pi^{(\leq n)}$ on $\mathbb{Z}^3$. The unconstrained Π is given by the same formula with $\mathcal{K}^{-1}$ replaced by G_1 and $\mathbb{V}_j$ by $\mathbb{V}_j^{\mathrm{D}}$.

(3) On $\mathbb{Z}^3$, with $|b-b'|$ the distance between bond base points,

$$|\Pi(b,b') - \Pi^{(\leq n)}(b,b')| \leq C_\Pi (L^n + |b-b'|)^{-2}(1+|b-b'|)^{-2},$$
$$|\Pi^{(\leq n)}(b,b')| \leq C_\Pi (1+|b-b'|)^{-4} e^{-\delta_f |b-b'|/(2L^n)},$$
$$|\Pi(b,b')| \leq C_\Pi (1+|b-b'|)^{-4}.$$

Both kernels annihilate $d\chi$ for bounded χ and annihilate constant one-forms. The kernel Π is translation invariant, and $\Pi^{(\leq n)}$ is invariant under $L^n\mathbb{Z}^3$. With $b = \langle x, x+e_\mu\rangle$ and $b' = \langle x+u, x+u+e_\nu\rangle$,

$$\Pi_{\mu\nu}(u) = \Pi_{\mu\nu}(e_\mu - e_\nu - u), \quad \sum_u \Pi_{\mu\nu}(u) = 0, \qquad |\hat{\Pi}_{\mu\nu}(k)| \leq C_\Pi |k|, \quad k \in [-\pi,\pi]^3.$$

(4) On a bare torus of side $N = \ell L^K \geq 16L^n$,

$$\Pi_N^{(\leq n)}(b,b') = \sum_{z \in N\mathbb{Z}^3} \Pi^{(\leq n)}(\tilde{b}, \tilde{b}' + z) + \mathcal{E}_N(b,b'), \qquad |\mathcal{E}_N(b,b')| \leq C_\Pi L^{4n} e^{-\delta_f N/(2L^n)}.$$

Here $\tilde{b}, \tilde{b}'$ are lifts to $\mathbb{Z}^3$.

■

Proof. The determinant identity. The empty history has $\mathcal{K}_n = D[U] + m_0 + \mathfrak{a}_n R_n^* R_n$. The exact Schur-complement identities of the fermion tower give $\Gamma_0 = \mathcal{K}_1$ and $\det \Gamma_j = \det c_j \det \mathcal{K}_{j+1} / \det \mathcal{K}_j$. They hold at every real configuration when $m_0 > 0$, where all the inverses exist. The determinant product thus telescopes to $\det \mathcal{K}_n \prod_{1 \le j < n} \det c_j$.

The operator $P = L q_{j+1}^+ q_{j+1}$ is idempotent, with rank $\dim H_{j+1}$, and

$$c_j = \theta_j (1 - P) + (\theta_j + \theta/L) P.$$

It follows that $\det c_j = \theta_j^{\dim H_j - \dim H_{j+1}} (\theta_j + \theta/L)^{\dim H_{j+1}}$ is field independent. For $n = 1$, the normalized product is $\theta^{-\dim H_1} = \theta_1^{-\dim H_1}$. The recursion $\theta_{n+1}^{-1} = \theta^{-1} + (L\theta_n)^{-1}$ implies $\theta_n + \theta/L = \theta\theta_n/\theta_{n+1}$. Multiplying the nth prefactor by $\theta^{-\dim H_{n+1}} \det c_n$ therefore gives $\theta_{n+1}^{-\dim H_{n+1}}$. This proves (a).

The quadratic determinant. At $A = a\sigma_3$, both $D_+[U]$ and $R_n(U)$ are diagonal in colour, and their two colour blocks are obtained by replacing a with $-a$. If $\ell(a)$ is the logarithm of the determinant of the eigenvalue-1 block, the two-colour logarithm is $\ell(a) + \ell(-a)$. The determinant convention in Lem. 14.30 gives

$$\tfrac{1}{2}(a, \Pi^{(\le n)} a) = -4 \operatorname{Re} \ell_2(a).$$

By Lem. 14.36, the first two terms of $\mathfrak{a}_n R_n^* R_n$ are precisely the displayed $\mathbb{V}_1^{\mathrm{R}}, \mathbb{V}_2^{\mathrm{R}}$. The logarithmic determinant expansion is

$$\ell_2 = \operatorname{Tr}[\mathcal{K}^{-1} \mathbb{V}_2] - \tfrac{1}{2} \operatorname{Tr}[\mathcal{K}^{-1} \mathbb{V}_1 \mathcal{K}^{-1} \mathbb{V}_1].$$

This proves (b), including its factor eight. Removing the block term gives the formula for Π.

Comparison of the propagators. Put $G = G_1$ on one colour block. The resolvent identity gives

$$\delta G = G - \mathcal{K}^{-1} = G\, \mathfrak{a}_n L^{-3n} \iota 1_{\mathrm{bl}} \iota^* \mathcal{K}^{-1}.$$

The elementary and constrained inverse estimates give

$$|G(v, v')| \le C_{\mathrm{el}} (1 + |x_v - x_{v'}|)^{-2}, \quad |\mathcal{K}^{-1}(v, v')| \le C_{\mathrm{fr}} (1 + |x_v - x_{v'}|)^{-2} e^{-\delta_f |x_v - x_{v'}|/L^n}.$$

They apply also at $m_0 = 0$ under the block constraint. For a level-n block B,

$$\sum_{z \in B} (1 + |u - z|)^{-2} \le C L^{3n} (L^n + \operatorname{dist}(u, B))^{-2}.$$

There are $O(1 + t^2)$ blocks at distances between tL^n and $(t+1)L^n$ from a point. Using $\mathfrak{a}_n \le b_F L^{-n}$ in the resolvent identity yields

$$\begin{aligned} |\delta G(v, v')| &\le C b_F L^{2n} \sum_B (L^n + \operatorname{dist}(x_v, B))^{-2} (L^n + \operatorname{dist}(B, x_{v'}))^{-2} e^{-\delta_f \operatorname{dist}(B, x_{v'})/L^n} \\ &\le C (L^n + |x_v - x_{v'}|)^{-2}. \end{aligned}$$

The quadratic form for $\Pi - \Pi^{(\le n)}$ is eight times the real part of $\Theta_1 + \cdots + \Theta_5$, where

$$\begin{aligned} \Theta_1 &= \tfrac{1}{2} \operatorname{Tr}[\delta G \mathbb{V}_1^{\mathrm{D}} G \mathbb{V}_1^{\mathrm{D}}] + \tfrac{1}{2} \operatorname{Tr}[\mathcal{K}^{-1} \mathbb{V}_1^{\mathrm{D}} \delta G \mathbb{V}_1^{\mathrm{D}}], \\ \Theta_2 &= -\operatorname{Tr}[\delta G \mathbb{V}_2^{\mathrm{D}}], \\ \Theta_3 &= -\operatorname{Tr}[\mathcal{K}^{-1} \mathbb{V}_1^{\mathrm{R}} \mathcal{K}^{-1} \mathbb{V}_1^{\mathrm{D}}], \\ \Theta_4 &= -\tfrac{1}{2} \operatorname{Tr}[\mathcal{K}^{-1} \mathbb{V}_1^{\mathrm{R}} \mathcal{K}^{-1} \mathbb{V}_1^{\mathrm{R}}], \\ \Theta_5 &= \operatorname{Tr}[\mathcal{K}^{-1} \mathbb{V}_2^{\mathrm{R}}]. \end{aligned}$$

The D_+ vertex at b acts only over its endpoints and has norm at most $c_{\mathrm{hop}} = 3(1 + r)$. An average vertex at b is supported on the block B_b containing it and carries the currents and quadratic kernels of Lem. 14.36. Writing $x = b - b'$, the first two terms satisfy

$$|\Theta_1(b, b')| \le C (L^n + |x|)^{-2} (1 + |x|)^{-2}, \qquad \Theta_2(b, b') = 0 \ (b \ne b'), \qquad |\Theta_2(b, b)| \le C L^{-2n}.$$

For the mixed term use

$$\sum_{u\in B_b}(1+|b'-u|)^{-2}(1+|u-b|)^{-2}\le C(1+|x|)^{-1},\qquad \sum_{u\in B}|\mathcal{K}^{-1}(u,v)|\le CL^n.$$

They give

$$|\Theta_3(b,b')|\le CL^{-3n}(1+|x|)^{-1}e^{-\delta_f(|x|-2L^n)_+/L^n}.$$

Each vertex in Θ_4 is a difference of currents at its two endpoints. When two currents are joined by a propagator, the corresponding term is bounded by

$$CL^{-8n}L^{4n}C_J^2\sum_{u,u'}(1+|u-b|)^{-2}(1+|u-u'|)^{-2}(1+|u'-b'|)^{-2}$$
$$\le CL^{-4n}\left(1+\log^+\frac{L^n}{1+|x|}\right).$$

The other terms are at most $CL^{-8n}(C_JL^n)^2(CL^n)^2$. All carry the factor $e^{-\delta_f\,\mathrm{dist}(B_b,B_{b'})/L^n}$. Finally Θ_5 vanishes unless $B_b=B_{b'}$. In that case $\sum_u|\mathsf{p}_u(b,b')|\le CC_J^2(1+|x|)^{-1}$, and the current products obey the same bound, so $|\Theta_5|\le CL^{-3n}(1+|x|)^{-1}$.

For $|x|\le L^n$, these estimates are bounded by $CL^{-2n}(1+|x|)^{-2}$. For the logarithm, use $s^2(1+\log(1/s))\le 1$ for $0<s\le 1$, with $s=(1+|x|)/L^n$; the short transition interval is absorbed into the same constant. For $|x|>L^n$, exponential decay and the boundedness of $t^4e^{-\delta_f t}$ give $C|x|^{-4}$. This proves the difference estimate. Applying the same estimates with constrained propagators throughout retains $e^{-\delta_f|x|/(2L^n)}$ and gives the bound for $\Pi^{(\le n)}$. The elementary inverse estimate gives the stated bound for Π.

Gauge identities and symmetries. For finitely supported χ, gauge covariance gives first derivative $-i[\chi,\mathcal{K}]$ at $d\chi$ and mixed derivative $-i[\chi,\mathbb{V}_1(a)]$ at $(a,d\chi)$. Substitution into the trace formula and cyclicity cancel the bilinear form at $(a,d\chi)$. The same argument applies with G^{-1}. The sums are justified by finite support and the kernel bounds just proved.

To extend to bounded χ, cut it off by $\eta_R(y)=\eta(y/R)$, equal to one for $|y|\le R$ and zero for $|y|\ge 2R$. At finitely supported a, the error is at most

$$CR^{-2}\sup_{R\le|y|\le 2R}|\chi(y)|+CR^{-1}\sup|d\chi|.$$

This tends to zero both for bounded χ and for $\chi(y)=\mathsf{c}\cdot y$. The latter proves annihilation of constant one-forms, hence zero sums in each direction.

The block operators, contour families, and their derivatives commute with translations in $L^n\mathbb{Z}^3$. Thus $J_{u+z}(b+z)=J_u(b)$ and the corresponding identity holds for p_u, proving the claimed translation covariance of $\Pi^{(\le n)}$. Without the block term there is full translation invariance. Point reflection exchanges e_+ and e_-, whose determinants are complex conjugates. The real fermion action is invariant under it. Keeping the orientations of the reflected bonds gives $\Pi_{\mu\nu}(u)=\Pi_{\mu\nu}(e_\mu-e_\nu-u)$. With $c=(e_\mu-e_\nu)/2$, the zero sum implies

$$e^{-ik\cdot c}\hat{\Pi}_{\mu\nu}(k)=\sum_u\Pi_{\mu\nu}(u)[\cos(k\cdot(u-c))-1].$$

Its absolute value is at most $\sum_u|\Pi_{\mu\nu}(u)|\min\{|k|^2|u-c|^2,2\}\le C|k|$. This finishes (c).

The torus comparison. The constrained inverse on antiperiodic sections is the signed image sum of its infinite-lattice kernel. The averages and their derivatives are block local and agree on the two lattices. In a trace with two propagators, the terms whose image displacements add to zero have sign one, since the signs form a character. They give the periodized infinite-lattice loop. In a trace with one propagator the vertex lies in one block; only its zero image does not wind, and, since $N\ge 16L^n$, only one periodized vertex contributes.

Every remaining term winds. Its propagators have total length $l \geq N - 4L^n \geq 3N/4$. Split their exponential factor as

$$e^{-\delta_f l/L^n} \leq e^{-\delta_f N/(2L^n)} e^{-\delta_f l/(3L^n)}.$$

The remaining image sums are bounded uniformly because $N \geq 16L^n$. At most four block point sums cost L^{12n}. Two average vertices contribute $\mathfrak{a}_n^2 L^{-6n} \leq b_F^2 L^{-8n}$; their currents and quadratic kernels are bounded by 5 and 9. Thus these terms cost at most CL^{4n}, and terms with fewer average vertices cost less. The bound in (d) follows. □

Lemma 14.38 (Curvature covariances under a flat block constraint)**.** Let $n \geq 1$ and $c_g = \delta_{\mathrm{tp}} L/4$, with the slice constants of [YM3, Prop. 7.6] chosen as functions of L. There is C_g, depending only on L, such that the following hold. Distances between plaquettes are distances between their barycentres.

(1) On a bare torus Λ_N at cutoff $K \geq n$, put

$$C_N^{(n)} = \Pi_0^{\mathrm{tp}} - \Pi_n^{\mathrm{tp}} = \sum_{j \leq n} \mathcal{C}_{(j)}, \qquad S_N^{(n)} = \Pi_n^{\mathrm{tp}}.$$

The first is the orthogonal projection onto Fl_n, and

$$|C_N^{(n)}(p,q)| \leq C_g (1 + |p-q|)^{-3} e^{-c_g |p-q|/L^n}, \qquad |S_N^{(n)}(p,q)| \leq C_{\mathrm{tp}} L^{-3n}.$$

(2) On tori of bare sides $N_s = L^s N$, the kernels $C_{N_s}^{(n)}$ and Π_0^{tp} converge pointwise to $C_\infty^{(n)}$ and $\Pi_{0,\infty}^{\mathrm{tp}}$. The latter is the orthogonal projection onto closed two-forms in $\ell^2(\mathbb{Z}^3)$ and has bound $C_g(1 + |p-q|)^{-3}$. The former obeys (a), is invariant under $L^n\mathbb{Z}^3$, and satisfies

$$C_N^{(n)}(p,q) = \sum_{z \in N\mathbb{Z}^3} C_\infty^{(n)}(\tilde{p}, \tilde{q} + z).$$

The kernel $S_\infty^{(n)} = \Pi_{0,\infty}^{\mathrm{tp}} - C_\infty^{(n)}$ satisfies $|S_\infty^{(n)}(p,q)| \leq C_g (L^n + |p-q|)^{-3}$. The columns of both kernels are closed.

(3) Let $\mathsf{g} = d^* \Delta^{-1}$ on two-forms of $\mathbb{Z}^3$, and set

$$\alpha_p = \mathsf{g}\delta_p, \qquad \beta_p = \mathsf{g} S_\infty^{(n)}(\cdot, p), \qquad \alpha_p^{(n)} = \alpha_p - \beta_p.$$

Then $d\alpha_p = \Pi_{0,\infty}^{\mathrm{tp}}(\cdot, p)$ and $d\alpha_p^{(n)} = C_\infty^{(n)}(\cdot, p)$. For $j = 0, 1, 2$, the jth differences of α_p obey $|\nabla^j \alpha_p(b)| \leq C_g (1 + |b-p|)^{-2-j}$, and

$$|\beta_p(b)| \leq C_g (L^n + |b-p|)^{-2} (1 + \log^+(|b-p|/L^n)).$$

(4) There is a potential α_p^{loc} with

$$d\alpha_p^{\mathrm{loc}} = C_\infty^{(n)}(\cdot, p), \qquad |\alpha_p^{\mathrm{loc}}(b)| \leq C_g L^{2n} e^{-c_g |b-p|/(8L^n)}.$$

It is covariant under simultaneous translation by $L^n\mathbb{Z}^3$, and $\alpha_p^{\mathrm{loc}} - \alpha_p^{(n)} = d\chi_p$ for a bounded function χ_p.

■

Proof. In bare units the slice estimate is

$$|\mathcal{C}_{(j)}(p,q)| \leq C_{\mathrm{tp}} L^{-3(j-1)} e^{-\delta_{\mathrm{tp}} r_\infty / L^{j-1}}, \qquad r_\infty \geq |p-q|/\sqrt{3}.$$

For $r_\infty \geq 1$, write $t_i = r_\infty / L^i$. For $i \leq n-1$,

$$L^{-3i} e^{-\delta_{\mathrm{tp}} t_i} \leq r_\infty^{-3} t_i^3 e^{-\delta_{\mathrm{tp}} t_i/2} e^{-\delta_{\mathrm{tp}} r_\infty/(2L^{n-1})}.$$

The sum of $t_i^3 e^{-\delta_{\mathrm{tp}} t_i/2}$ along the geometric sequence is bounded by a constant depending on L. For $r_\infty < 1$, the slice sum is at most $2C_{\mathrm{tp}}$. Summing the telescope proves the first bound in (a), since $\delta_{\mathrm{tp}} L/(2\sqrt{3}) \geq c_g$. The background bound in [YM3, Prop. 7.6] gives the second.

An exact form on $\Lambda_{N_{s+1}}$ which is N_s-periodic is exact on Λ_{N_s}: average any potential over the translates. The plaquette average commutes with these translations. Thus the N_s-periodic part of $\mathsf{Fl}_n(\Lambda_{N_{s+1}})$ is exactly $\mathsf{Fl}_n(\Lambda_{N_s})$. Its projection commutes with translations, and the two inner products on periodic forms differ only by L^3. Consequently

$$C^{(n)}_{N_s}(p,q) = \sum_{z \in N_s\mathbb{Z}^3/N_{s+1}\mathbb{Z}^3} C^{(n)}_{N_{s+1}}(\tilde{p}, \tilde{q}+z).$$

If $|\tilde{p}-\tilde{q}| \le N_s/3$, all nonzero images have distance at least $2N_s/3$. Part (a) bounds the successive difference by $C_g L^3 e^{-c_g N_s/(2L^n)}$, a summable sequence. This proves existence of $C^{(n)}_\infty$. The same identity and dominated convergence give its periodization formula and symmetries.

The Fourier formula for Π^{tp}_0 is a Riemann sum of a bounded symbol continuous away from zero. Its limit is the projection onto closed ℓ^2 two-forms. The symbol is smooth away from zero and its derivative of order j is bounded by $C|k|^{-j}$. The dyadic Fourier estimate therefore gives decay $(1+|p-q|)^{-3}$. The limit of (a) gives $|S^{(n)}_\infty| \le C_{\mathrm{tp}} L^{-3n}$; subtracting its two kernels also gives $|S^{(n)}_\infty| \le C(1+|p-q|)^{-3}$. Together these yield the stated bound. Closedness passes to the pointwise limit, since d has finite range.

The symbols of g and of its jth differences have degrees $j-1$ near zero. The same dyadic estimate gives decay $(1+|\cdot|)^{-2-j}$. Moreover $d\mathsf{g} = dd^*\Delta^{-1}$ has the symbol of the projection onto closed two-forms, giving the assertion about α_p. For a closed form F bounded by $C(L^n+|\cdot - p|)^{-3}$, absolute convergence allows $d\mathsf{g}F = F - d^*\Delta^{-1}dF = F$. Apply this to $S^{(n)}_\infty(\cdot, p)$.

For completeness, put $t = |b-p|$. When $t \le 2L^n$,

$$|\beta_p(b)| \le C \sum_{|u-p|\le 3L^n} (1+|b-u|)^{-2} L^{-3n} + C \sum_{|u-p|>3L^n} |u-p|^{-5} \le CL^{-2n}.$$

When $t > 2L^n$, points with $|u-p| \le t/2$ contribute at most $Ct^{-2}\sum_{|v|\le t/2}(L^n+|v|)^{-3} \le Ct^{-2}(1+\log(t/L^n))$. The remaining points contribute at most

$$Ct^{-3} \sum_{|u-b|\le 2t} (1+|b-u|)^{-2} + C \sum_{|u-b|>2t} |u-b|^{-5} \le Ct^{-2}.$$

This proves (c).

For (d), put $F = C^{(n)}_\infty(\cdot, p)$ and write $x = (x_\perp, x_3)$. Choose the axial potential $\alpha^{\mathrm{ax}}_3 = 0$ and $\alpha^{\mathrm{ax}}_i(x) = \pm\sum_{t\ge x_3} F_{i3}(x_\perp, t)$, $i = 1, 2$, with the sign fixed by the convention for d. Then $d\alpha^{\mathrm{ax}} = F$ on the $i3$ plaquettes. Closedness and decay give the same identity on the 12 plaquettes. As $x_3 \to -\infty$, its first two components tend to a closed one-form $\varphi_i(x_\perp)$ on $\mathbb{Z}^2$. Its line sums are independent of the transverse line by closedness, and tend to zero as the line goes to infinity. They therefore vanish. Hence

$$\chi(x_\perp) = \sum_{t<0} \varphi_1(x_\perp + te_1) = -\sum_{t\ge 0} \varphi_1(x_\perp + te_1), \qquad d_\perp \chi = \varphi.$$

If p_3 is the third coordinate of the base point of p, define

$$\alpha^{\mathrm{loc}}_p = \alpha^{\mathrm{ax}} - d(\chi(x_\perp) 1[x_3 < p_3]).$$

Its curvature remains F. Above the plane $x_3 = p_3$, its first two components are forward sums of F; below that plane they are the opposite backward sums. In each case all summands are at least as far from p as x. Its third component is zero except on bonds crossing that plane, where it equals χ, computed using the ray pointing away from $p_\perp$. The exponential bound in (b) therefore gives $C_g L^{2n} e^{-c_g|b-p|/(8L^n)}$. This construction commutes with the required translations. The difference $\alpha^{\mathrm{loc}}_p - \alpha^{(n)}_p$ is closed on $\mathbb{Z}^3$, so is $d\chi_p$. Define χ_p by integrating from a site beside p along coordinate paths. The bounds just proved and those in (c) bound these path sums uniformly in the endpoint. Thus χ_p is bounded. □

Lemma 14.39 (Removing the flat block constraint from the fermion loop)**.** There are $C_F < \infty$ and $c_F = \min\{c_g/48, \delta_f/12\}$, depending only on the member, r, L, b_F, with the following property. Let $n \geq 1$, $0 \leq m_0 \leq L^{-n}$, and $\mathsf{r} \geq 1$. Let Λ_N be a bare torus at cutoff $K \geq n$, with $N \geq \max\{16L^n, 30\mathsf{r}\}$. Let S_1, S_2 each lie in a closed ball of radius r, with their mutual distance between r and $6\mathsf{r}$. Suppose w_1, w_2 are real plaquette functions supported, through their barycentres, in S_1, S_2. Choose lifts $\tilde{w}_1, \tilde{w}_2$ supported in lifts of those sets at the same distance. Then

$$\left|\sum_{p,q} w_1(p)w_2(q)\mathsf{B}_N^{F,(n)}(p,q;m_0) - \sum_{p,q}\tilde{w}_1(p)\tilde{w}_2(q)\mathsf{B}_1^F(p,q;m_0)\right|$$
$$\leq C_F\left[\left(\frac{\mathsf{r}}{L^n}\right)^2(1+\log\mathsf{r}+n\log L) + \mathsf{r}^5 L^{14n}e^{-c_F N/L^n}\right]\mathsf{r}\,\|w_1\|_\infty\|w_2\|_\infty.$$

■

Proof. There are at most $(C\mathsf{r}^3)^2$ contributing pairs. For each pair put $\rho = |\tilde{p}-\tilde{q}|$; the support assumptions give $\mathsf{r} \leq \rho \leq 10\mathsf{r} \leq N/3$. In Feynman gauge on $\mathbb{Z}^3$, $\langle A_b A_{b'}\rangle_0 = \Delta^{-1}(b,b')$, so the curvature covariance is $\Pi^{\mathrm{tp}}_{0,\infty}$ and its potential is α_p. By Lem. 14.31,

$$\mathsf{B}_1^F(\tilde{p},\tilde{q}) = -3\Pi^{\mathrm{tp}}_{0,\infty}(\tilde{p},\tilde{q})X, \qquad X = (\alpha_{\tilde{p}}, \Pi\alpha_{\tilde{q}}).$$

On the torus use

$$\alpha_p^{\mathrm{per}} = \sum_{z\in N\mathbb{Z}^3}\alpha_{\tilde{p}}^{\mathrm{loc}}(\cdot+z) - \mathsf{c}_p,$$

where c_p is the constant one-form making its sum in each bond direction zero. It is a potential of $C_N^{(n)}(\cdot,p)$, so the same coefficient formula gives

$$\mathsf{B}_N^{F,(n)}(p,q) = -3C_N^{(n)}(p,q)(\alpha_p^{\mathrm{per}}, \Pi_N^{(\leq n)}\alpha_q^{\mathrm{per}}).$$

The infinite-lattice comparison. The Ward identity and reflection symmetry improve the decay of $\Pi\alpha_q$. For $b = \langle x, x+e_\mu\rangle$, put $g_\nu(u) = \alpha_{q,\nu}(x+u)$ and $c = (e_\mu - e_\nu)/2$. Part (c) of Prop. 14.37 yields

$$(\Pi\alpha_q)(b) = \frac{1}{2}\sum_{\nu,u}\Pi_{\mu\nu}(u)[g_\nu(u) + g_\nu(2c-u) - g_\nu(0) - g_\nu(2c)].$$

If $R = |x-q| \geq 8$ and $|u| \leq R/2$, the bracket is at most $C(1+|u|)^2R^{-4}$ by the second difference estimate of Lem. 14.38. For $|u| > R/2$, use $|\Pi_{\mu\nu}(u)| \leq C|u|^{-4}$ and $|g_\nu(u)| \leq C(1+|x+u-q|)^{-2}$. Summing the two regions gives

$$|(\Pi\alpha_q)(b)| \leq C(1+|b-q|)^{-3}, \qquad |X| \leq C\rho^{-2}(1+\log\rho).$$

Write the difference of the constrained and unconstrained loop pairings as

$$D := (\alpha_{\tilde{p}}^{(n)}, \Pi^{(\leq n)}\alpha_{\tilde{q}}^{(n)}) - X$$
$$= (\alpha_{\tilde{p}}^{(n)}, (\Pi^{(\leq n)} - \Pi)\alpha_{\tilde{q}}^{(n)}) - (\beta_{\tilde{p}}, \Pi\alpha_{\tilde{q}}) - (\Pi\alpha_{\tilde{p}}, \beta_{\tilde{q}}) + (\beta_{\tilde{p}}, \Pi\beta_{\tilde{q}}).$$

By Lem. 14.38,

$$|\alpha_p^{(n)}(b)| \leq C(1+|b-p|)^{-2}(1+\log^+(|b-p|/L^n)).$$

Convolving this estimate with the difference kernel of Prop. 14.37, splitting the sums at distance L^n, bounds the first term by $CL^{-2n}(1+\log^+(L^n/\rho))$. Up to that distance the two potential factors have power -2; beyond it the difference kernel has power -4. The second and third terms are bounded by

$$C\sum_b (L^n + |b-\tilde{p}|)^{-2}(1+\log^+(|b-\tilde{p}|/L^n))(1+|b-\tilde{q}|)^{-3} \leq CL^{-2n}(1+n\log L)$$

and the same estimate with the points exchanged.

For the last term, Plancherel and $|\hat{\Pi}(k)| \le C|k|$ combine with the $C|k|^{-1}$ symbol bound for g. It is at most

$$C \prod_{u\in\{\tilde p,\tilde q\}} \left(\sum_{v,v'} |S_\infty^{(n)}(v,u)|\, \kappa(v-v')\, |S_\infty^{(n)}(v',u)| \right)^{1/2}, \qquad \kappa(v) \le C(1+|v|)^{-2}.$$

The bound $|S_\infty^{(n)}(v,u)| \le C(L^n+|v-u|)^{-3}$ makes this at most CL^{-2n}. Thus $|D| \le CL^{-2n}(1+n\log L)$. As $C_\infty^{(n)} = \Pi_{0,\infty}^{\mathrm{tp}} - S_\infty^{(n)}$,

$$\begin{aligned} &|C_\infty^{(n)}(\tilde p,\tilde q)(X+D) - \Pi_{0,\infty}^{\mathrm{tp}}(\tilde p,\tilde q)X| \\ &\quad \le |\Pi_{0,\infty}^{\mathrm{tp}}|\,|D| + |S_\infty^{(n)}|(|X|+|D|) \\ &\quad \le C\rho^{-3}L^{-2n}(1+\log\rho + n\log L). \end{aligned}$$

For the last inequality use $|S_\infty^{(n)}| \le C(L^n+\rho)^{-3}$ and $\rho L^{2n} \le (L^n+\rho)^3$.

The torus comparison. All nonzero periodized images of the curvature kernel have distance at least $2N/3$, so

$$|C_N^{(n)}(p,q) - C_\infty^{(n)}(\tilde p,\tilde q)| \le Ce^{-c_g N/(2L^n)}.$$

By Prop. 14.37(d), the fermion loop kernel is the periodization of $\Pi^{(\le n)}$ plus $\mathcal{E}_N$. The periodized kernel annihilates constant one-forms. Absolute convergence, translation covariance, and the definition of c_p therefore give

$$\begin{aligned} (\alpha_p^{\mathrm{per}}, \Pi_N^{(\le n)}\alpha_q^{\mathrm{per}}) = &\sum_{z\in N\mathbb{Z}^3} (\alpha_{\tilde p}^{\mathrm{loc}}, \Pi^{(\le n)}\alpha_{\tilde q+z}^{\mathrm{loc}}) \\ &+ (\alpha_p^{\mathrm{per}}, \mathcal{E}_N\alpha_q^{\mathrm{per}}). \end{aligned}$$

The zero image is $X+D$. Indeed, $\alpha^{\mathrm{loc}} - \alpha^{(n)}$ is the coboundary of a bounded function, which the kernel annihilates by Prop. 14.37(c). All these pairings converge absolutely.

Set $\gamma_F = \min\{c_g/8, \delta_f/2\}$. The exponential bounds in Prop. 14.37 and Lem. 14.38 bound the term with image z by $CL^{7n}e^{-\gamma_F|\tilde p-\tilde q-z|/(2L^n)}$. The nonzero images therefore sum to at most $CL^{7n}e^{-\gamma_F N/(6L^n)}$. Also $\|\alpha_p^{\mathrm{per}}\|_1 \le 2\|\alpha_{\tilde p}^{\mathrm{loc}}\|_1 \le CL^{5n}$, so

$$|(\alpha_p^{\mathrm{per}}, \mathcal{E}_N\alpha_q^{\mathrm{per}})| \le CL^{14n}e^{-\delta_f N/(2L^n)}.$$

The periodized loop kernel is bounded, giving $|(\alpha_p^{\mathrm{per}}, \Pi_N^{(\le n)}\alpha_q^{\mathrm{per}})| \le CL^{14n}$. Combining the curvature and loop comparisons bounds their product difference by

$$CL^{14n}(e^{-c_g N/(2L^n)} + e^{-\gamma_F N/(6L^n)} + e^{-\delta_f N/(2L^n)}) \le 3CL^{14n}e^{-c_F N/L^n},$$

because $c_F = \gamma_F/6$.

Finally sum over at most $(C\mathsf{r}^3)^2$ pairs. The infinite-lattice difference costs $C\mathsf{r}^3L^{-2n}(1+\log\mathsf{r}+n\log L)$, and the image error costs $C\mathsf{r}^6L^{14n}e^{-c_F N/L^n}$. These are exactly the two terms in the claimed estimate. □

Lemma 14.40 (Removing the flat block constraint from the pure coefficient)**.** Let $0 < g \le \gamma_{\mathrm{e}}$, $P > 1$, and $\mathsf{d}_1 = \min\{(L\mathsf{m}_+)^{-2}, 1/64\}$. There is a nondecreasing function $\epsilon^{\mathrm{pure}} : (0,\mathsf{d}_1] \to [0,\infty)$, depending on g, P and the constants, and tending to zero at zero, with the following property. Let $0 < \mathsf{d} \le \mathsf{d}_1$, $K > s(\mathsf{d})$, and $L^K \le \mathsf{d}^{-P}$. On the least torus put $N = \ell_G L^K$, $\mathsf{r} = L^K\mathsf{d}$, and let n be the least level with $L^n a_K \ge \mathsf{d}^{1/2}$. Let $S_1, S_2, w_1, w_2, \tilde w_1, \tilde w_2$ have the support and lift properties in Lem. 14.39, and put $\eta = \|w_1\|_\infty\|w_2\|_\infty$. Then:

(1) $\mathsf{r} \ge 4L^3M$ and $2 \le n \le K$. The coefficient of β^{-3} in the Laplace expansion of the covariance of a_p, a_q on $\{\mathcal{M}^n U = \mathbf{1}\}$, with action

$$F_\kappa = A + \sum_{j<n} \kappa_j G_j(\mathcal{M}^j U), \qquad \kappa_j > 0,$$

is the chart-independent and strength-independent coefficient $\mathsf{g}_2[N,n](p,q)$ of Prop. D.10. Write $\mathsf{T}[N,n](w) = \sum_{p,q} w_1(p) w_2(q) \mathsf{g}_2[N,n](p,q)$.

(2)

$$\left| \mathsf{T}[N,n](w) - \sum_{p,q} \tilde{w}_1(p) \tilde{w}_2(q) \mathsf{B}_1^{\mathrm{lat}}(p,q) \right| \le \epsilon^{\mathrm{pure}}(\mathsf{d}) \mathsf{r} \eta.$$

■

Proof. The definition of the witness depth gives

$$\mathsf{r} \ge L^{s(\mathsf{d})+1} \mathsf{d} \ge L C_W(s(\mathsf{d})) \ge L \mathsf{c}_W \ge 4L^3 M.$$

Here $C_W(s) = \mathsf{c}_W (\log \bar{x}_s)^{2\mathfrak{p}_0}$, $\bar{x}_s \ge L$, and $\mathsf{c}_W \ge 4L^2 M$. Thus $a_K < \mathsf{d} < \sqrt{\mathsf{d}}$, whereas $L^K a_K = 1$. The least n exists and

$$L^n a_K \in [\sqrt{\mathsf{d}}, L\sqrt{\mathsf{d}}), \qquad L^n \ge \mathsf{r}\mathsf{d}^{-1/2} > L^2.$$

In particular $n \ge 3$.

Restricting the QCD constants to the pure parameters gives an admitted pure choice. In particular, the QCD floor on $B_\sharp$ dominates the pure floor, since its added fermion bound is nonnegative. The pure coupling ceiling included in γ_{e} ensures $\ell_G L \ge C_L M$ and $\ell_G \ge N_{\mathsf{H}}$. Hence the side condition for Prop. D.10 holds at every level $1 \le t \le K$, including $t = n$. That proposition proves (a).

The contributing plaquette barycentres have separation between r and $10\mathsf{r}$. Their base points have distance at most $10\mathsf{r} + 2 \le N/8$, since $80\mathsf{d} + 16 a_K \le 2 \le \ell_G$. Each support has at most $30\mathsf{r}^3$ plaquette barycentres, so there are at most $900\mathsf{r}^6$ pairs.

Uniform comparison bounds. Use $\check{C}_{\mathrm{X}}, \check{C}_{\mathrm{R}}, \check{C}'_{\mathrm{W}}, \check{C}_\star$ for the constants in Props. D.12, D.15, D.19 and D.20, respectively. Let $\check{\gamma}^{\mathrm{tr}}$ and $\check{\delta}_{\mathrm{tp}}$ be their Gaussian and curvature decay rates, and let ν_{W} be the side-ratio threshold in Prop. D.19. Set

$$\Upsilon_{\mathrm{X}}(x, \mathsf{N}) = \check{C}_{\mathrm{X}} \mathsf{N}^{36} \exp[-\tfrac{1}{8} \check{\gamma}^{\mathrm{tr}} \mathfrak{b}(x)^2], \qquad \mathfrak{b}(x) = A_1 (\log x)^{\mathfrak{p}_0}, \qquad \mathsf{b}_* = X_g - \bar{B} = \mathsf{b}_+.$$

The function decreases in x and increases in N. Choose the least $k_{\mathrm{X}} \ge 3$ and largest $\mathsf{d}_{\mathrm{X}} \in (0, L^{-5}]$ for which

$$\check{\gamma}^{\mathrm{tr}} A_1^2 (\tfrac{k}{78} \log L)^{2\mathfrak{p}_0} \ge \max\{80, 8\log(10240 \ell_G^3) + 24k \log L\}, \qquad k \ge k_{\mathrm{X}},$$

$$\check{\gamma}^{\mathrm{tr}} A_1^2 \left(\log \frac{3}{4\sqrt{\mathsf{d}}} \right)^{2\mathfrak{p}_0} \ge \max\{80, 8\log(10240 (\ell_G L)^3) + 576 P \log(1/\mathsf{d})\}, \qquad 0 < \mathsf{d} \le \mathsf{d}_{\mathrm{X}}.$$

They exist because $2\mathfrak{p}_0 \ge 2$. Put

$$\mathsf{d}_{\mathrm{p}} = \min\{\mathsf{d}_1, (\ell_G/(L\nu_{\mathrm{W}}))^2, \mathsf{d}_{\mathrm{X}}, L^{-k_{\mathrm{X}}}\}$$

and define

$$\begin{aligned} \varepsilon^{\mathrm{ch}}(\mathsf{d}) &= 900 \check{C}'_{\mathrm{W}} (\ell_G L)^{237} \mathsf{d}^{-5693P} e^{-\check{\delta}_{\mathrm{tp}} \ell_G \mathsf{d}^{-1/2}/48} \\ &\quad + \Upsilon_{\mathrm{X}}(\tfrac{3}{4}\mathsf{d}^{-1/2}, (\ell_G + \mathsf{b}_*) L \mathsf{d}^{-24P}) \\ &\quad + \check{C}_{\mathrm{R}} (\log(2\mathsf{b}_* L) + 24P \log(1/\mathsf{d}))^{6\mathfrak{p}_0} \mathsf{d}^{3/4}, \\ \varepsilon^{\mathrm{top}}(k) &= \Upsilon_{\mathrm{X}}(L^{k/78}, (\ell_G + \mathsf{b}_*) L^k) + 2700 \check{C}_\star \ell_G^{19} L^{20} k L^{-k/24}. \end{aligned}$$

For $0 < \mathsf{d} \le \mathsf{d}_{\mathrm{p}}$ take

$$\epsilon^{\mathrm{pure}}(\mathsf{d}) = \sup_{0<d' \le \mathsf{d}} \varepsilon^{\mathrm{ch}}(d') + \sup_{k > s(\mathsf{d})} \varepsilon^{\mathrm{top}}(k).$$

A second pure datum. Let

$$K' = \max\{K, \lceil 24 \log_L \mathsf{r} \rceil\}, \quad N' = \ell_G L^{K'}, \quad n_1 = K' - \lceil K'/78 \rceil.$$

Use the pure datum with cutoff K' and label b_*, whose running couplings are x'_j. This is admitted because b_* is in the pure label window. Put $\mathsf{N}' = (\ell_G + \mathsf{b}_*)L^{K'}$. Lift the sources to $\Lambda_{N'}$ as in Prop. D.19, writing them $\breve{w}_1, \breve{w}_2$; if $K' = K$, leave them unchanged. Their physical support radius there is $\mathsf{d}' = \mathsf{r}L^{-K'}$ and $8a_{K'} \le \mathsf{d}'$. The normalization $\mathsf{r}\eta$ is unchanged. Since $\mathsf{r} \le \mathsf{d}^{1-P}$,

$$\mathsf{r}^{24} \le L^{K'} \le \max\{\mathsf{d}^{-P}, L\mathsf{r}^{24}\} \le L\mathsf{d}^{-24P}.$$

Also $s(\mathsf{d}) \ge k_{\mathrm{X}}$, since $L^s\mathsf{d} < 1 \le C_W(s)$ for $s < k_{\mathrm{X}}$. Thus $K' \ge K \ge k_{\mathrm{X}} + 1 \ge 4$.

We need $2 \le n < n_1 \le K'$. If $K' = K$, write $\lambda_0 = \log_L(1/\mathsf{d}) \ge 5$. Then $K \ge 24(K - \lambda_0)$, so $K \le 24\lambda_0/23$. Consequently

$$\lceil K/78 \rceil < 24\lambda_0/1794 + 1 \le \lambda_0/2 - 1 < K - n.$$

The last inequality is $L^{n-K} < L\sqrt{\mathsf{d}}$. If $K' > K$, then $\mathsf{r} \le L^{K'/24}$ and $\mathsf{d} = \mathsf{r}L^{-K} \ge \mathsf{r}L^{1-K'}$. Hence

$$L^n < L\mathsf{r}\mathsf{d}^{-1/2} \le L^{1/2}\mathsf{r}^{1/2}L^{K'/2} \le L^{1/2+25K'/48}, \quad n_1 > 77K'/78 - 1 \ge \tfrac{1}{2} + 25K'/48.$$

This proves the required level ordering. We also have $L^n \ge \mathsf{r}$.

The five comparisons. First compare the two torus sizes at level n. If $K' > K$, Prop. D.19 applies because

$$\nu = N/L^n > \ell_G \mathsf{d}^{-1/2}/L \ge \nu_{\mathrm{W}}$$

and the pair base points lie within $N/8$. It gives

$$|\mathsf{T}[N, n](w) - \mathsf{T}[N', n](\breve{w})| \le 900\mathsf{r}^6 \check{C}'_{\mathrm{W}} N'^{237} e^{-\check{\delta}_{\mathrm{tp}} L\nu/48}\eta.$$

Divide by $\mathsf{r}\eta$ and use $\mathsf{r}^5 \le \mathsf{d}^{-5P}$ and $N' \le \ell_G L\mathsf{d}^{-24P}$. This is the first term of $\varepsilon^{\mathrm{ch}}(\mathsf{d})$. For $K' = K$ the difference is zero.

Second remove the Gaussian tests at level n in the pure progression. Its coefficient is $\mathsf{H}^{(n)}[D'] = \beta'^3 a_{K'}^2 \gamma_2^{\mathrm{r}}$, while $\mathsf{T}[N', n] = \beta'^3 a_{K'}^2 \gamma_2^{\circ}$, with $\beta' = L^{K'}\mathsf{b}_*$. By Prop. D.11,

$$x'_{n-1} \ge \tfrac{3}{4}\mathsf{b}_* L^{K'-n+1} > \tfrac{3}{4}\mathsf{d}^{-1/2}.$$

The two conditions of Prop. D.15 follow from $\mathsf{d} \le \mathsf{d}_{\mathrm{X}}$. That proposition and Prop. D.17 give

$$|\mathsf{H}^{(n)}[D'] - \mathsf{T}[N', n](\breve{w})| \le \Upsilon_{\mathrm{X}}(x'_{n-1}, \mathsf{N}')\eta.$$

It is bounded by the second term of $\varepsilon^{\mathrm{ch}}(\mathsf{d})\mathsf{r}\eta$.

Third, the pure tail estimate Prop. D.12 at $n \le n_1$ gives

$$|\mathsf{H}^{(n_1)}[D'] - \mathsf{H}^{(n)}[D']| \le \check{C}_{\mathrm{R}} \mathsf{l}'^{6\mathfrak{p}_0} (\mathsf{r}/L^n)^{3/2} \mathsf{r}\eta.$$

Here $\mathsf{r}/L^n \le \sqrt{\mathsf{d}}$ and

$$\mathsf{l}' = \max\{1, \log(2\mathsf{b}_*/\mathsf{d}')\} \le \log(2\mathsf{b}_* L) + 24P \log(1/\mathsf{d}).$$

This is the third term of $\varepsilon^{\mathrm{ch}}(\mathsf{d})\mathsf{r}\eta$.

Fourth remove the tests at level n_1. Now $x'_{n_1-1} \ge \frac{3}{4}\mathsf{b}_* L^{\lceil K'/78 \rceil + 1} \ge L^{K'/78}$, and the conditions of Prop. D.15 hold because $K' \ge k_{\mathrm{X}}$. The cost is at most $\Upsilon_{\mathrm{X}}(L^{K'/78}, (\ell_G + \mathsf{b}_*)L^{K'})\eta$.

Finally use Prop. D.20 on the lattice (ℓ', n_1), where $\ell' = N'/L^{n_1} = \ell_G L^{\lceil K'/78 \rceil}$. It has $\ell' \ge N_{\mathsf{H}}$ and $\ell' L \ge C_L M$. Summing its bound over the at most $900\mathsf{r}^6$ pairs costs

$$2700\check{C}_\star \mathsf{r}^6 \ell'^{19} n_1 L^{-n_1/2}\eta \le 2700\check{C}_\star \ell_G^{19} L^{20} K' L^{-K'/24}\mathsf{r}\eta.$$

Indeed, $\mathsf{r}^5 \le L^{5K'/24}$, $\ell' \le \ell_G L^{K'/78+1}$, and $n_1 > 77K'/78 - 1$, and the resulting exponent is $5/24 + 19/78 + 1/156 - 1/2 = -1/24$. The last two comparisons cost at most $\varepsilon^{\mathrm{top}}(K')\mathsf{r}\eta$. Since $K' > s(\mathsf{d})$, adding the five bounds proves (b) for $\mathsf{d} \le \mathsf{d}_{\mathrm{p}}$.

The modulus. The supremum formula is nondecreasing because $s(\mathsf{d})$ does not increase with d. Each term of $\varepsilon^{\mathrm{ch}}(\mathsf{d})$ tends to zero: its exponential terms dominate every displayed power of d^{-1}. Likewise $\varepsilon^{\mathrm{top}}(k) \to 0$, and $s(\mathsf{d}) \to \infty$. Thus $\epsilon^{\mathrm{pure}}(\mathsf{d}) \to 0$.

For $\mathsf{d}_{\mathrm{p}} < \mathsf{d} \le \mathsf{d}_1$, we have $K \le K_{\mathrm{p}} := \lfloor P \log_L(1/\mathsf{d}_{\mathrm{p}}) \rfloor$. Let G_{p} bound the finitely many $|\mathsf{g}_2[\ell_G L^k, t](p,q)|$ with $1 \le t \le k \le K_{\mathrm{p}}$ and the $|\mathsf{B}_1^{\mathrm{lat}}(p,q)|$ at base-point separation at most $10L^{K_{\mathrm{p}}} + 2$. The latter collection is finite modulo translation. The difference in (b) is at most $1800 G_{\mathrm{p}} \mathsf{r}^6 \eta$. Extend the modulus on this interval by

$$\epsilon^{\mathrm{pure}}(\mathsf{d}) = \epsilon^{\mathrm{pure}}(\mathsf{d}_{\mathrm{p}}) + 1800 G_{\mathrm{p}} L^{5K_{\mathrm{p}}}.$$

This preserves monotonicity and proves the remaining cases. All these choices precede the final small-distance ceiling. □

Lemma 14.41 (Removing tests from the coefficients at the split)**.** Assume the parameter choices of Prop. 14.26, including the floor (F1) and the admitted gauge-background conditions. Let $P > 1$ and $0 < \mathsf{d} \le \mathsf{d}_1 = \min\{(L\mathsf{m}_+)^{-2}, 1/64\}$. For a datum on the least torus with $K > s(\mathsf{d})$ and $L^K \le \mathsf{d}^{-P}$, let n be the least integer with $L^n a_K \ge \sqrt{\mathsf{d}}$. For $m = 1, 2, 3$, the coefficient $\sum_{j \le n} \mathsf{n}_{j,[2m]}$ differs from the coefficient at order λ^{2m} of the Gaussian Laplace expansion under the flat level-n constraint by at most

$$\epsilon_{\mathrm{t}}(\mathsf{d}) \mathsf{n}_m(X; \mathsf{d}), \qquad \epsilon_{\mathrm{t}}(\mathsf{d}) \longrightarrow 0.$$

The modulus depends on g, P and the constants, uniformly in the other parameters of the datum and the position of the test pair. ■

Proof. The exact finite-dimensional integral. At block field $\mathbf{1}$ and $\Psi_n = 0$, the logarithm of the level-n form has mixed source derivative $\sum_{j \le n} \mathsf{n}_j^{\lambda}$. The scalar values of all activities have been removed as source energies. The exact chart composition of Prop. 14.14 expresses the form as a source-independent nonzero number times the bare integral on $\{\mathcal{M}^n U = \mathbf{1}\}$. The gauge fluctuations enter as λB_i, $i < n$, with tests at t_i; the gauge fixings have coefficients t_i/λ^2, and the Wilson weight is $e^{-A(\beta/\lambda^2 - a_K^{-1} w, U)}$. The running action coefficients leave no counterterm, since

$$\lambda^{-2} L^i \xi_i(\lambda) + \sum_{l \le i} L^l b_l = \lambda^{-2} \beta.$$

By Prop. 14.37(a), the fermion factor at $\Psi_n = 0$ is

$$\prod_{i<n} \theta^{-\dim H_{i+1}} \det \Gamma_i[U] = \theta_n^{-\dim H_n} \det \mathcal{K}_n[U; m_0].$$

Thus the field-dependent determinant is exactly the constrained determinant in Lem. 14.30. At fixed finite lattice, the Taylor coefficients are unique. The tested coefficients are therefore cumulants in the product of the tested Gaussian measures of the levels $i < n$.

Let $\mu = \bigotimes_{i<n} \mu_i$ and $\chi = \prod_{i<n} \chi_i$. By Lem. 14.35,

$$\tau_n = \sum_{i<n} N_{i+1}^{\square} K_M e^{-\frac{1}{2}\gamma_{\mathrm{tail}} \mathfrak{b}(\mathsf{t}_i)^2} \le N^{\mathrm{b}} K_M e^{-\frac{1}{2}\gamma_{\mathrm{tail}} \mathfrak{b}(\mathsf{t}_n)^2}, \qquad \int \chi \, d\mu \ge e^{-\tau_n}.$$

Here N^{b} is the number of bare bonds. For a real polynomial $\mathfrak{P}$, the same Cauchy–Schwarz argument as in that lemma gives

$$\begin{aligned} |\mathsf{E}^{\chi}\mathfrak{P} - \mathsf{E}\mathfrak{P}| &\le (\sqrt{\tau_n} + \tau_n) e^{\tau_n} (\mathsf{E}\mathfrak{P}^2)^{1/2}, \\ |\mathsf{E}^{\chi}\mathfrak{P}| &\le e^{\tau_n} (\mathsf{E}\mathfrak{P}^2)^{1/2}. \end{aligned} \tag{14.l}$$

We next obtain polynomial bounds for the Taylor coefficients to which this is applied.

A chain at fixed radii. Start at the trivial triple $\mathscr{P}_n = (\mathbf{1}, 0, 0)$. For $1 \le i < n$, define $\mathscr{P}_i$ by substituting the step shift at fluctuation λB_i into the restriction of $\mathscr{P}_{i+1}$. All triples have the same

radii (α_0, α_1) and real factor $\mathbf{1}$. Here are the domain checks. The side condition is $m' < m_{\mathfrak{L}}(g)+K-i$, with $M = L^{m'}$; it follows from $M\check{R}_i \le L^{m_{\mathfrak{L}}(g)+K-i-1}$ and $\check{R}_i \ge 1$. The standing radii satisfy

$$\alpha_0 < \alpha_1, \quad \alpha_1^2 \le \mathsf{c}_\natural \alpha_0, \quad \alpha_0 + \alpha_1 \le a^\star \le \min\{a_\flat, a^\sharp\}.$$

Thus they are admitted one-level substitution radii over themselves. At $|\lambda| \le 1$ the field polydisc gives $\sup|\lambda B_i| < \beta_\lambda \le \beta_0/2$. The standing bound $c_h\beta_0 \le \epsilon_P \le \varepsilon_\flat/2$ places the pair and fluctuation in the shift domain, and

$$\frac{\mathsf{c}_{\mathrm{os}}\alpha_0}{c_h\beta_\lambda} = \frac{2a_{\mathrm{an}}}{c_h\beta_\lambda} \ge 6 > 1.$$

The substitution theorem [YM3, Prop. 4.51] therefore applies at amplitude one. Its conclusion uses the same radii as its hypothesis, so the iteration loses no radius. Heredity of the cores [YM3, Lem. 4.27] permits restrictions to every polymer and fine cube needed by the next step.

If $\mathscr{A}_{i+1}$ is the second member of $\mathscr{P}_{i+1}$ and $\mathbf{a}_i, \mathbf{K}_i$ are the shift members, the substituted triple has second member

$$(iL^{-i})^{-1}\log(e^{iL^{-i}\mathbf{a}_i}e^{iL^{-i-1}\mathscr{A}_{i+1}})$$

and current $L^{-3}J_{i+1} + \mathbf{K}_i$. Its pair has first member $e^{iL^{-i}\mathbf{a}_i}\mathscr{U}_{i+1}$. The fixed real factor and the holomorphic substitution make the chain holomorphic in λ. At level zero there is no shift system: the bare configuration is $U^{\mathrm{ch}} = e^{iB'}\mathscr{U}_1$, where B' is the completed field of λB_0. The completed-field bound $2C_{\mathrm{cf}}\beta_0 \le \min\{r_P^\circ, \alpha_1/2\}$ gives

$$U^{\mathrm{ch}}(b) = e^{iX(b;\lambda)}, \qquad |X(b;\lambda)| \le \alpha_1, \qquad X(b;0) = 0.$$

The logarithm is holomorphic for $|\lambda| < 1$.

The chain and the composite chart. The chain just constructed follows the fine fields of the steps; the composite chart follows their minimal backgrounds. Their integrals agree by the triangular change of variables used in Prop. D.17. The proof applies at the present calibrations t_i: its geometric hypotheses are the passing tests and regular pair cores, already checked, and the tests are radial in each fluctuation. To specify the change of variables, let $W_n = \mathbf{1}$ and $W_i = V_{W_{i+1}}[\lambda B_i]$ be the composite block fields. Descending from $\mathfrak{v}_n = \mathbf{1}$, choose real gauge transformations $\mathfrak{v}_i$ so that the pair of $\mathscr{P}_i$, at

$$\check{B}_i = \mathrm{Ad}(\mathfrak{v}_{i+1})B_i,$$

is the pair of the plain triple over W_i transformed by $\mathfrak{v}_i$. The covariance of the shift gives the induction: the fine field of a step lies in the minimal orbit over W_i, its current is $L^{-3}J_{i+1} + \mathbf{K}_i$, and a real gauge transformation carries that pair to the chosen minimal background. The transformation $\mathfrak{v}_i$ reads only heights at least i. At level zero covariance of the completed field gives the same conclusion for the bare configuration.

Every factor of the integrand is invariant under this simultaneous transformation: the gauge fixing and the J, G, A potentials are functions of the regular pair and fluctuation; the elimination factors depend on the averaged background; the action and observables are gauge invariant; and the fermion factor is the determinant of an operator conjugated by the gauge action. The plain triples, their real gauge transforms, and the substituted triples lie in the cores where these identities hold. At a fixed choice of the higher fluctuations, $B_i \mapsto \check{B}_i$ is a fixed rotation at each bond. It preserves the box and Lebesgue measure. Integrate from the lowest height upward, holding those rotations fixed at each integration, as in Prop. D.17. The two integrals agree for real $0 < \lambda \le 1$ and every admitted source.

Their Gaussian coefficients also agree. Apply the Laplace expansion to either coordinate system on a product of small balls $|\lambda B_i(b)| < r$. The frozen rotations preserve that product, the integrand, and the Euclidean norm. The positive quadratic phase from Prop. D.10 therefore remains positive in the chain coordinates. Uniqueness of the expansion identifies the Gaussian cumulants of the chain with the Laplace coefficients of the composite chart.

Uniform bounds for the gauge logarithm. On $|\lambda| \le 1$, the two observables are bounded by a power of N^{b} and β, times their test-function norms. The Wilson terms are polynomials in $e^{iX(b;\lambda)}$ and its inverse. They vanish to second order at zero, so $\lambda^{-2}a_p(U^{\mathrm{ch}})$ is holomorphic and bounded by the maximum principle. The Wilson logarithm is therefore at most $C\beta N^{\mathrm{b}}$.

At a step, write its scaled fluctuation as $B = \mathsf{t}_i^{1/2}B_i$. The logarithm of its gauge fixing is $-\frac{1}{2}(B, \hat{\mathsf{G}}(P)B)_1 + \sum_Y \mathbb{V}^{\mathrm{G},\lambda}(Y)$, and its Jacobian logarithm is $\sum_Y \mathbb{V}^{\mathrm{J},\lambda}(Y)$. The row bound for each fine-cube operator $\hat{\mathsf{G}}_\square$ gives

$$|(B, \hat{\mathsf{G}}_\square B)_1| \le c_{\mathsf{G}} n_M \mathsf{C}_{\mathcal{A}}^2 \mathfrak{b}(\mathsf{t}_i)^2,$$

where n_M is the number of bonds meeting one closed M-cube. Since

$$\mathfrak{b}(\mathsf{t}_i)^2 \le A_1^2(2\mathfrak{p}_0)^{2\mathfrak{p}_0}\mathsf{t}_i \le A_1^2(2\mathfrak{p}_0)^{2\mathfrak{p}_0}\beta,$$

the sum of these quadratic terms is at most $CL^{-3i}\beta N^{\mathrm{b}}$. The J and G vertices obey

$$|\mathbb{V}^{\natural,\lambda}(Y)| \le \tfrac{10}{9} K_P \delta(\mathsf{t}_i) Q_Y(B) e^{-(\kappa+1)d_i(Y)}.$$

There are at most $8(1 + d_i(Y))$ cubes in Y, and

$$\delta(\mathsf{t}_i)Q_Y(B) \le \delta(\mathsf{t}_i) + 8(1 + d_i(Y))M^{-3} n_M \mathsf{C}_{\mathcal{A}}^2 \delta(\mathsf{t}_i)\mathfrak{b}(\mathsf{t}_i)^2.$$

The last scalar factor is at most $A_1^3(6\mathfrak{p}_0)^{3\mathfrak{p}_0}$. The spare exponential absorbs $1 + d_i(Y)$; the polymer sum and $\sum_{i<n} L^{-3i} \le 2$ give a bound $C\beta N^{\mathrm{b}}$ for the gauge fixings and Jacobians, both at λ and at zero.

The elimination factor of a step is the product of $(\det \mathsf{m}_c)^{-1}$ over bonds of the level above. Its logarithm is $\mathsf{z}_c + 3\log L$, where $\mathsf{z}_c = -\operatorname{Log}\det(L\mathsf{m}_c)$ and $\mathsf{z}_c = 6\log L$ at the identity. The local pivot-log bound [YM3, eq. 4.dcclviii] and [YM3, eq. 4.dcclix] gives

$$|\mathsf{z}_c - 6\log L| \le C_{\mathsf{z}} a_{\mathrm{nf}}.$$

We check its applicability here. The standing constant condition supplies $a = L\max\{\alpha_0, \alpha_1\} \le a_{\mathrm{nf}}$. The frame is $(K, m_{\mathfrak{L}}(g), (\mathsf{t}_l))$, with $\mathsf{t}_l \ge \mathsf{b} > \gamma^{-2} > 1$, $\varepsilon(\mathsf{t}_l), \delta(\mathsf{t}_l) \le 1$, and $M\check{R}_l \le L^{m_{\mathfrak{L}}(g)+K-l-1}$. At $t = i + 1 \le n$ this is the empty history, so each level-t bond is in its regular set. The pair $(\mathscr{U}_{i+1}, 0)$ has decomposition $(\mathbf{1}, L^{-1}\mathscr{A}_{i+1})$ in the level-i units. Its curvature bound is $|\partial\mathscr{U}_{i+1} - \mathbf{1}| < \alpha_0 L^{-2(i+1)}$; its form and difference bounds are exactly those of the core after the scale factors L, L^2; its current is zero. Restrictions to the double block of c preserve these bounds. In the local gauge the logarithmic potential has norm less than $4a/L$. The averaged pivot matrix has $E = L^3\mathsf{m}_c - 1$ with $|E| \le 4C_{\mathrm{piv}}C_Q a/L \le 1/2$. Thus $|\operatorname{tr}\log(1 + E)| \le 6|E|$, and its imaginary part has modulus at most $\pi/2$; it is the principal determinant logarithm. One may take $C_{\mathsf{z}} = 24C_{\mathrm{piv}}C_Q/L$. The elimination map is consequently defined, $\det(L\mathsf{m}_c)$ avoids $(-\infty, 0]$, and z_c is holomorphic in $\mathscr{U}_{i+1}$ throughout this domain. Thus it is holomorphic in λ and is the logarithm of the actual elimination factor. There are at most N^{b} bonds over levels $1, \dots, n$, so their logarithmic change is at most $2C_{\mathsf{z}}a_{\mathrm{nf}}N^{\mathrm{b}}$.

The fermion logarithm and coefficient bounds. Schwarz's lemma applied to $X(b;\lambda)$ gives $|X(b;\lambda)| \le \alpha_1|\lambda|$. Thus, on $|\lambda| < L^{-n}$, $|X(b;\lambda)| < \alpha_1 L^{-n}$. The real background is $\mathbf{1}$ and is regular at every level. The admitted determinant expansion through depth n applies because $0 < m_0 = L^{-K}\mathsf{m} \le L^{-K}/2$. At each $j < n$ it gives

$$\log \delta Z_j^{\mathrm{F}}(U^{\mathrm{ch}}) - \log \delta Z_j^{\mathrm{F}}(\mathbf{1}) = \sum_{Y \in \mathcal{D}_{j+1}} \mathbf{N}^{\mathrm{F},(j+1)}(Y, U^{\mathrm{ch}}), \qquad |\mathbf{N}^{\mathrm{F},(j+1)}| \le 2C_N N_f e^{-(\kappa+1)d_{j+1}(Y)}.$$

These terms are holomorphic in X, hence in λ. Their sum is the logarithmic change of the entire fermion factor, and is bounded by $CN_f N^{\mathrm{b}}$. This also proves that the chain integral is holomorphic near zero.

At order λ^{2m} the mixed covariance coefficient is a finite sum, with coefficients depending on m, of products of at most $2m + 2$ Gaussian means. Each mean contains at most $2m + 2$ Taylor coefficients of the observables and the logarithm of the integrand. At order $k \le 2m$ those coefficients

are homogeneous fluctuation polynomials of degree k or $k+2$, the latter coming from the factor λ^{-2} in the action. Cauchy's estimate on $|\lambda| = L^{-n-1}$ costs $L^{(n+1)k} \le (L^3 N^{\mathrm{b}})^{k/3}$. The preceding bounds therefore control each coefficient by a power of $N^{\mathrm{b}}\beta$ times $(1+\varpi_B)^{2m+2}$, and the appropriate test-function norm, where

$$\varpi_B = \max_{i<n,b} \frac{\mathsf{t}_i^{1/2}|B_i(b)|}{\mathsf{C}_{\mathcal{A}}\mathfrak{b}(\mathsf{t}_i)}.$$

The rotations preserve these absolute values. The scaled Gaussian variances are bounded, so $\mathsf{E}\varpi_B^{2q} \le C_q N^{\mathrm{b}}$. Apply eq. 14.1 and Hölder to each factor in the cumulant. The tested and untested coefficients differ by at most

$$C_m^{\mathrm{t}}(\sqrt{\tau_n}+\tau_n)e^{C_m^{\mathrm{t}}\tau_n}(N^{\mathrm{b}}\beta)^{c_m^{\mathrm{t}}}\mathfrak{f}.$$

Finally $N^{\mathrm{b}} \le 3\ell_G^3\mathsf{d}^{-3P}$, $\beta \le \mathsf{b}\mathsf{d}^{-P}$, and $\mathsf{t}_n > \mathsf{b}\mathsf{d}^{-1/2}/L$. Since the label window bounds b above and below and $\mathfrak{p}_0 \ge 1$, τ_n tends to zero faster than every power of d. After division by $\mathsf{n}_m(X;\mathsf{d}) = \mathsf{b}^{-m}\mathsf{d}^{m-2}\mathfrak{f}$, the displayed bound is a modulus $\epsilon_{\mathrm{t}}(\mathsf{d}) \to 0$. □

Proposition 14.42 (Identification of the first three covariance coefficients)**.** Assume the constants of Prop. 14.26, including (F1) and the admitted gauge-background conditions. Let $g \le \gamma_{\mathrm{e}}$, $\vartheta \in (0,1)$, $P > 1$, and $\mathsf{d}_1 = \min\{(L\mathsf{m}_+)^{-2}, 1/64\}$. For every datum on the least torus with $0 < \mathsf{d} \le \mathsf{d}_1$, $K > s(\mathsf{d})$, and $L^K \le \mathsf{d}^{-P}$, the coefficient $\sum_j \mathsf{n}_{j,[2m]}$, $1 \le m \le 3$, agrees with the order-β^{-m} lattice covariance coefficient on $\mathbb{Z}^3$, including its factor β^{-m} and the physical observable normalization, up to

$$C_m\epsilon'_m(\mathsf{d})\mathsf{n}_m(X;\mathsf{d}), \qquad \epsilon'_m(\mathsf{d}) \longrightarrow 0.$$

The assertion holds also for the translation mean of the jets. Its constants and moduli depend only on g, ϑ, P and the chosen constants. ■

Proof. Split at the least n with $L^n a_K \ge \sqrt{\mathsf{d}}$. The bounds of Prop. 14.26(iii), followed by Cauchy's estimate for λ, give

$$\sum_{j>n} |\mathsf{n}_{j,[2m]}| \le C\mathsf{l}^{2m\mathfrak{p}_0}\mathsf{b}^{-m}\mathsf{d}^{5/2}(L^{n+1}a_K)^{m-9/2}\mathfrak{f} \le C\mathsf{l}_+^{6\mathfrak{p}_0}\mathsf{d}^{3/4}\mathsf{n}_m(X;\mathsf{d}),$$

where $\mathsf{l}_+ = \max\{1, \log(2\mathsf{b}_+/\mathsf{d})\}$. This is a convergent geometric sum for $m \le 3$. By Lem. 14.41, the levels at most n may be replaced by the Gaussian Laplace coefficient under the flat block constraint at cost $\epsilon_{\mathrm{t}}(\mathsf{d})\mathsf{n}_m$.

Put $\mathsf{r} = \mathsf{d}/a_K$ and $N = \ell_G/a_K$ in bare units. The supports lie in balls of radius r, with distance between r and $6\mathsf{r}$. By Lem. 14.40, $\mathsf{r} \ge 4L^3M$ and $2 \le n \le K$. The scale choice gives

$$\mathsf{r}/L^n \le \sqrt{\mathsf{d}}, \quad m_0 L^n < L\mathsf{m}_+\sqrt{\mathsf{d}} \le 1, \quad N/L^n > \ell_G\mathsf{d}^{-1/2}/L \ge 16.$$

Also $30\mathsf{r} \le N$ and

$$1 + \log\mathsf{r} + n\log L \le (1+\log L)(1+2P\log(1/\mathsf{d})).$$

These check all hypotheses for the comparisons of the preceding subsection.

For $m = 1$, both connected coefficients vanish: two quadratic observables require at least two gauge lines. For $m = 2$, the determinant contributes nothing at this order. Wick's formula gives the bare coefficient $\frac{3}{2}\sum_{p,q} f(x(p))h(x(q))C_N^{(n)}(p,q)^2$. On $\mathbb{Z}^3$ it gives the same expression with $C_N^{(n)}$ replaced by $\Pi_{0,\infty}^{\mathrm{tp}}$. By Lem. 14.38, their squares differ at a contributing pair by at most

$$C\left(L^{-3n}|p-q|^{-3} + e^{-c_g N/(2L^n)}\right).$$

There are at most $C\mathsf{r}^6$ such pairs. Multiplying by $\beta^{-2}a_K^{-2} = \mathsf{b}^{-2}$ therefore bounds the physical difference by

$$C\left[\mathsf{d}^{3/2} + \mathsf{d}^{-6P}e^{-c_g\ell_G\mathsf{d}^{-1/2}/(2L)}\right]\mathsf{n}_2(X;\mathsf{d}).$$

For $m = 3$, the determinant formula separates the coefficient into its pure part and

$$N_f \sum_{p,q} f(x(p))h(x(q))\mathsf{B}_N^{F,(n)}(p,q;m_0).$$

The infinite-lattice coefficient is the same sum with the kernels $\mathsf{B}_1^{\mathrm{lat}} + N_f\mathsf{B}_1^F$. We identify the pure part carefully. The composite chart over $W_n = \mathbf{1}$ is an adapted chart $U = \Psi(\mathbf{1}, (\lambda B_i)_{i<n})$ of $\{\mathcal{M}^n U = \mathbf{1}\}$. By the exact one-step measure identity, the product of its Jacobian and elimination factors is a constant times the Haar density $e^{-\theta(\mathbf{1},\cdot)}$ in this chart: compose the identities and use Fubini. The densities are continuous, so the identity holds throughout the chart. The gauge fixings contribute

$$\sum_{i<n} \frac{\mathsf{t}_i}{\lambda^2} G_i(\mathcal{M}^i U) = \frac{\beta}{\lambda^2} \sum_{i<n} \kappa_i G_i(\mathcal{M}^i U), \qquad \kappa_i = \mathsf{t}_i/\beta = L^{-i}.$$

Thus the pure integrand is a constant times $e^{-(\beta/\lambda^2)F_\kappa(U) - \theta(\mathbf{1},(\lambda B_i))}$, with exactly the phase and chart density of Prop. D.10. By Lem. 14.40(a), its order-three coefficient is $\mathsf{T}[N,n](w)$ for $w_1(p) = f(x(p))$, $w_2(q) = h(x(q))$. Part (b) of that lemma bounds its difference from the pure infinite-lattice coefficient by $\epsilon^{\mathrm{pure}}(\mathsf{d})\mathsf{r}\mathfrak{f}$. Since $\beta^{-3}a_K^{-2}\mathsf{r} = \mathsf{b}^{-3}\mathsf{d}$, its physical size is $\epsilon^{\mathrm{pure}}(\mathsf{d})\mathsf{n}_3$.

The fermion terms are compared at the same bare mass m_0 by Lem. 14.39. Using the checked scale bounds, its error in physical units is at most

$$CN_f \left[\mathsf{d}(1 + P\log(1/\mathsf{d})) + L^{14}\mathsf{d}^{-19P} e^{-c_F \ell_G \mathsf{d}^{-1/2}/L}\right] \mathsf{n}_3.$$

We can therefore take, after adjusting the constants C_m,

$$\begin{aligned}
\epsilon_1'(\mathsf{d}) &= \mathsf{l}_+^{6\mathfrak{p}_0}\mathsf{d}^{3/4} + \epsilon_{\mathrm{t}}(\mathsf{d}),\\
\epsilon_2'(\mathsf{d}) &= \epsilon_1'(\mathsf{d}) + \mathsf{d}^{3/2} + \mathsf{d}^{-6P} e^{-c_g \ell_G \mathsf{d}^{-1/2}/(2L)},\\
\epsilon_3'(\mathsf{d}) &= \epsilon_1'(\mathsf{d}) + \epsilon^{\mathrm{pure}}(\mathsf{d})\\
&\qquad + N_f[\mathsf{d}(1 + P\log(1/\mathsf{d})) + L^{14}\mathsf{d}^{-19P} e^{-c_F \ell_G \mathsf{d}^{-1/2}/L}].
\end{aligned}$$

Each tends to zero. The bounds hold at every position of the test pair. The translations in Lem. 14.12 are lattice translations, and the infinite-lattice coefficient is invariant under them. Averaging proves the final assertion. □

14.27. **Comparison between torus sizes.**

Proposition 14.43 (The local covariance is independent of the side to next order)**.** Assume the constants of Prop. 14.26, including (F1) and the admitted gauge-background conditions, and let $g \le \gamma_{\mathrm{e}}$, $\vartheta \in (0,1)$. Use $\ell_G, \mathsf{b}_+, \mathsf{Q}, \mathsf{d}_{\mathrm{R}}, m_*(\mathsf{d}), \omega_{\mathrm{r}}(\mathsf{d})$ from Lem. 14.12, and m_{x} from Lem. 14.25. There is $\mathsf{d}_{\mathrm{T}} \in (0, \mathsf{d}_{\mathrm{R}}]$ and a nondecreasing modulus $\epsilon_{\mathrm{T}}(\mathsf{d}) \to 0$ such that the following holds.

Fix $(g, K, \mathsf{b}, \mathsf{m})$, with $0 < \mathsf{d} \le \mathsf{d}_{\mathrm{T}}$ and $K > s(\mathsf{d})$. Let ν, ν' each be either a torus measure at these parameters and a side in $\mathfrak{L}(g)$, regarded as a law on periodic configurations of $a_K\mathbb{Z}^3$, or a torus-limit state in $\mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$. For every $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$, at any position, and $X = ((0,f),(0,h))$,

$$\begin{aligned}
|S_2^T[\nu](X) - S_2^T[\nu'](X)| &\le C^{\mathrm{T}}[\omega_{\mathrm{r}}(\mathsf{d})^{1/8} + \mathsf{d}]\,\mathsf{d}\,\mathfrak{f}\\
&\le \epsilon_{\mathrm{T}}(\mathsf{d})\mathsf{b}^{-3}\mathsf{d}\,\mathfrak{f}.
\end{aligned}$$

For torus measures the tuple is periodized. The constants and modulus depend on g, ϑ and the chosen constants, and are independent of the cutoff, labels, masses, sides, states, and test pair.

Every pair in $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ is the periodization of such a pair on $\mathbb{R}^3$. The values of $\mathcal{K}_{\mathrm{free}}, \mathcal{K}_5$ and $\mathfrak{f}$ agree on the two tori and on $\mathbb{R}^3$. ■

We choose d_{T} to be the largest number at most d_{R} such that $m_*(\mathsf{d}_{\mathrm{T}}) \geq m_{\mathrm{x}}$ and

$$2\mathsf{Q}^2\, 96^3 \frac{40M}{\varrho\vartheta}\mathsf{d}^{5/2}(1+\log(1/\mathsf{d}))^{4\mathfrak{p}_0+1} \leq 1 \qquad (0<\mathsf{d}\leq \mathsf{d}_{\mathrm{T}}).$$

The proof specifies C^{T}, after which we take

$$\epsilon_{\mathrm{T}}(\mathsf{d}) = \mathsf{b}_+^3 C^{\mathrm{T}} \sup_{0<d'\leq \mathsf{d}} [\omega_{\mathrm{T}}(d')^{1/8} + d'].$$

Proof. Lifts, placement, and packing. The pair is contained in a ball $\mathbb{B}(x_0, 5\mathsf{d})$. On a torus this ball has radius less than $\ell/4$, since $32\mathsf{d} \leq \ell$. It therefore lifts isometrically to $\mathbb{R}^3$. All template integrals use only distances at most $10\mathsf{d}$, so their values are unchanged by the lift or periodization.

The witness-depth inequality gives $a_K < \mathsf{d}/(4L^3M)$. Choose the board offset $o \in a_K\mathbb{Z}^3$ with $|x_0 - o|_\infty \leq a_K/2$. The pair then lies in $\mathbb{B}(o, 6\mathsf{d})$. Use the same offset and cell zero on every torus under comparison. Put

$$\delta = a_K\lceil 18\mathsf{d}/a_K\rceil \leq 19\mathsf{d}, \quad \mathcal{T}_{\mathrm{cell}} = \{t \in \delta\mathbb{Z}^3 : |t|_\infty \leq 1/(4\sqrt{3}) - 7\mathsf{d}\}, \quad N = \#\mathcal{T}_{\mathrm{cell}}.$$

The translated pair at t lies in $\mathbb{B}(o+t, 6\mathsf{d}) \subset \mathbb{B}_*$. The balls containing distinct translated pairs have mutual distance at least $18\mathsf{d} - 10\mathsf{d} \geq \mathsf{d}$. Together with the original separation, this makes all the f^t, h^t supports pairwise disjoint. Since $\mathsf{d} \leq \mathsf{d}_{\mathrm{R}} \leq 1/400$,

$$N \geq \left(2\frac{1/(4\sqrt{3}) - 7\mathsf{d}}{19\mathsf{d}} - 1\right)^3 \geq (96\mathsf{d})^{-3}.$$

Every plaquette with barycentre in $\mathbb{B}_*$ has its corners in cell zero: the cell half-side is $\ell_G/4 - a_K/2 > 1/4 + a_K$. Thus all the translated observables are bounded real functions of the links in the cell algebra.

Centre with the least-torus law. Write $\mu_G = \mu_{K,\ell_G,\beta,m_0}$, $\mu_f = \mu_G(O(f))$, $\mu_h = \mu_G(O(h))$, and $\sigma = S_2^T[\mu_G](X)$. Set

$$\mathsf{X} = \frac{1}{N}\sum_t [(O(f^t) - \mu_f)(O(h^t) - \mu_h) - \sigma], \qquad \mathsf{X}_f = \frac{1}{N}\sum_t (O(f^t) - \mu_f),$$

and define X_h in the same way. All are bounded real cell functions. Translation invariance gives, for every $\mu_\ell = \mu_{K,\ell,\beta,m_0}$,

$$S_2^T[\mu_\ell](X) - \sigma = \mu_\ell(\mathsf{X}) - \mu_\ell(\mathsf{X}_f)\mu_\ell(\mathsf{X}_h). \tag{14.li}$$

The chessboard inequality applies to the even cell algebra with antiperiodic fermions: the sides have an even number of sites, $\beta \geq 0$, $m_0 > 0$, and the partition function is positive. For the board with N_ℓ cells, write

$$\|Y\|_{\mathrm{ch},\ell} = \left(\frac{\Omega_\ell(\mathrm{Fold}\, Y)}{Z_\ell}\right)^{1/N_\ell}.$$

It gives $|\mu_\ell(Y)| \leq \|Y\|_{\mathrm{ch},\ell}$ for $Y = \mathsf{X}, \mathsf{X}_f, \mathsf{X}_h$.

The unnormalized two-volume folded inequality Prop. 14.11(C3) applies because the boards have the same odd cell side and the same cell zero, with $2L^{m-m_{\mathfrak{L}}(g)} \geq 2$ cells in each direction on $\ell = 2L^m$. As $N_\ell = 8(\ell/\ell_G)^3$, it yields

$$\|Y\|_{\mathrm{ch},\ell} \leq \|Y\|_{\mathrm{ch},\ell_G}\left(\frac{Z_{\ell_G}^{(\ell/\ell_G)^3}}{Z_\ell}\right)^{1/N_\ell}.$$

The zero-source partition-function estimates are

$$Z_{\ell_G} \leq e^{E_K(\ell_G) + (C+2)\ell_G^3}, \qquad Z_\ell \geq e^{E_K(\ell) - C_P(1+\log x_K)\ell^3}.$$

Here $E_K(\ell)=\ell^3\sum_{j\le K}e_jL^{3(K-j)}$ has side-independent coefficients and $x_K\le L(g^{-2}+B_\sharp)$. Thus the ratio is at most $e^{c_V\ell^3}$, with $c_V=C+2+C_P(1+\log[L(g^{-2}+B_\sharp)])$. Combining the estimates gives

$$|S_2^T[\mu_\ell](X)-\sigma|\le e^{c_V\ell_G^3/8}\|\mathsf{X}\|_{\mathrm{ch},\ell_G}+e^{c_V\ell_G^3/4}\|\mathsf{X}_f\|_{\mathrm{ch},\ell_G}\|\mathsf{X}_h\|_{\mathrm{ch},\ell_G}. \tag{14.lii}$$

The folded generating function. For the eight cells on T_{ℓ_G}, let θ_c and $\mathbb{B}_*^{(c)}$ be the reflections and balls of the board in Def. 14.7. Put $f_c^t=f^t\circ\theta_c$ and $h_c^t=h^t\circ\theta_c$. The cell copies preserve products and real coefficients and send a_p to $a_{\theta p}$. Thus $\mathrm{Fold}\,\mathsf{X}=\prod_c\mathsf{X}_c$, where $\mathsf{X}_c=N^{-1}\sum_t[(O(f_c^t)-\mu_f)(O(h_c^t)-\mu_h)-\sigma]$.

Introduce independent signs $\epsilon_{c,t}$, sixteen variables z_{c1},z_{c2}, and

$$F_c=\sum_t\epsilon_{c,t}f_c^t,\quad H_c=\sum_t\epsilon_{c,t}h_c^t,\quad w_c=z_{c1}F_c+z_{c2}H_c,\quad W=\sum_c w_c.$$

Let $\mathsf{p}(w)$ be the degree-one and degree-two terms of $\log(Z_K^{\mathrm{sub}}[w]/Z_K^{\mathrm{sub}}[0])$, and set $\mathsf{c}(w)=E_K(w)-E_K(0)+\mathsf{p}(w)$. Its derivatives through order two equal those of $\log(Z[w]/Z[0])$. Define

$$\mathsf{Z}^{\mathrm{c}}(z)=e^{-\sum_c\mathsf{c}(w_c)}\frac{Z[W]}{Z[0]}=\mu_G\left(\prod_c e^{O(w_c)-\mathsf{c}(w_c)}\right).$$

Let I be the sixteen variables and $\partial_I=\prod_c\partial_{z_{c1}}\partial_{z_{c2}}|_0$. Differentiating centres both observables in each cell and subtracts their covariance. The torus symmetries identify the means and covariance with μ_f,μ_h,σ. Averaging the signs retains only $t=t'$ within each cell. Therefore

$$N^{-8}\,\mathrm{mean}_\epsilon\,\partial_I\mathsf{Z}^{\mathrm{c}}(0)=\mu_G(\mathrm{Fold}\,\mathsf{X})=\|\mathsf{X}\|_{\mathrm{ch},\ell_G}^8. \tag{14.liii}$$

With all signs one and $z_{c2}=0$, the eight derivatives $\prod_c\partial_{z_{c1}}$ give $N^8\|\mathsf{X}_f\|_{\mathrm{ch},\ell_G}^8$. The analogous assertion holds for h.

Use the polydisc

$$\Delta=\{|z_{c1}|\le\|F_c\|_\circ^{-1},|z_{c2}|\le\|H_c\|_\circ^{-1}\},\qquad\|u\|_\circ=\|u\|_\sharp/\varrho.$$

A sum of functions on disjoint supports, each vanishing on the support boundary, has sharp norm equal to the largest of their sharp norms. For points in different supports, follow the segment between them to the two boundaries; the sum of the two travelled distances is at most the original distance. This proves the Lipschitz assertion as well as the supremum assertion. Consequently $\|F_c\|_\circ=\|f\|_\circ$, $\|H_c\|_\circ=\|h\|_\circ$. On Δ, $W,w_c,\zeta w_c$, $|\zeta|\le1$, all have sharp norm at most ϱ and lie in the interior of the admitted source domain.

The subtracted partition-function bound is $|Z_K^{\mathrm{sub}}[w]/Z_K^{\mathrm{sub}}[0]|\le e^{c_V\ell_G^3}\le\mathsf{Q}$. For $\psi(\zeta)=Z_K^{\mathrm{sub}}[\zeta w_c]/Z_K^{\mathrm{sub}}[0]$, Cauchy's estimate and $\psi(0)=1$ give

$$|\mathsf{p}(w_c)|=|\psi'(0)+\tfrac12(\psi''(0)-\psi'(0)^2)|\le3\mathsf{Q}^2.$$

Different cell balls have separation at least $\ell_G/4$. The separated-source energy estimate thus gives $|E_K(W)-\sum_c E_K(w_c)+7E_K(0)|\le C_{\mathrm{add}}$, where

$$C_{\mathrm{add}}=8\sup_{K,S}\mathscr{C}_K(S;\ell_G/4)<\infty.$$

Here S ranges over sets in a ball of radius $1/4$. The count is finite because its level-j cube count is at most $(4+1/(2Ma_j))^3$, and

$$C_{\mathscr{C}}\sum_{m\ge0}(4+L^m/(2M))^3e^{-\kappa_{\mathrm{E}}\ell_GL^m/4}<\infty.$$

It follows that

$$|\mathsf{Z}^{\mathrm{c}}|\le C_Z:=\mathsf{Q}e^{C_{\mathrm{add}}+24\mathsf{Q}^2}\quad\text{on }\Delta.$$

The eight-derivative identity and Cauchy's estimate give $\|\mathsf{X}_f\|_{\mathrm{ch},\ell_G} \le C_Z^{1/8} N^{-1} \|f\|_\circ$. Since $\|f\|_\sharp \le 40M\|f\|_\infty/(\vartheta \mathsf{d})$ and $N^{-1} \le (96\mathsf{d})^3$,

$$\|\mathsf{X}_f\|_{\mathrm{ch},\ell_G} \|\mathsf{X}_h\|_{\mathrm{ch},\ell_G} \le C_Z^{1/4} \left(\frac{96^3\, 40M}{\varrho\vartheta}\right)^2 \mathsf{d}^4 \mathfrak{f}. \tag{14.liv}$$

The remainder at the reduction scale. Set $k = K - m_*(\mathsf{d})$. By Lem. 14.12, $1 \le k < K$, $a_k \ge \mathsf{d}$, and $Z[W] = \mathsf{A}_k[W] + \mathsf{B}_k[W]$, with both terms holomorphic. Split Z^{c} accordingly. On Δ the remainder bound and the zero-source floor give

$$|e^{-E_K(W)} \mathsf{B}_k[W]| \le \omega_{\mathrm{r}}(\mathsf{d}) M_0, \qquad Z_K^{\mathrm{sub}}[0] \ge M_0/\mathsf{Q}.$$

Hence $|\mathsf{Z}_B^{\mathrm{c}}| \le \omega_{\mathrm{r}} C_Z$. Cauchy's estimate in sixteen variables gives

$$|N^{-8} \operatorname{mean}_\epsilon \partial_I \mathsf{Z}_B^{\mathrm{c}}(0)| \le \omega_{\mathrm{r}} C_Z u_0^8, \qquad u_0 = N^{-1}\|f\|_\circ \|h\|_\circ \le 96^3 (40M/\varrho)^2 \vartheta^{-2} \mathsf{d}\mathfrak{f}. \tag{14.lv}$$

The retained term and its partitions. Since $0 < \mathsf{A}_k[0] \le Z[0]$, near zero we may write

$$\mathsf{Z}_A^{\mathrm{c}} = \frac{\mathsf{A}_k[0]}{Z[0]} e^{\mathcal{L}}, \qquad \mathcal{L}(z) = \log \frac{\mathsf{A}_k[W]}{\mathsf{A}_k[0]} - \sum_c \mathsf{c}(w_c), \qquad \mathcal{L}(0) = 0.$$

After taking the sixteen derivatives, sign averaging keeps the same translation index for the two directions in each cell. Thus

$$N^{-8} \operatorname{mean}_\epsilon \partial_I \mathsf{Z}_A^{\mathrm{c}}(0) = N^{-8} \sum_{(t_c)_c} \frac{\mathsf{A}_k[0]}{Z[0]} \sum_{\pi \in \mathrm{Part}(I)} \prod_{b \in \pi} L_b(t).$$

Here $L_b(t)$ is the derivative $\partial_b \mathcal{L}(0)$ with the directions $f_c^{t_c}, h_c^{t_c}$. If b lies within one cell, the subtraction of $\mathsf{c}(w_c)$ gives

$$L_b(t) = -\,\partial_b \log(1 + \mathsf{B}_k/\mathsf{A}_k)[w_c]|_0\,.$$

It is one of $-\mathsf{r}_1(f_c^{t_c})$, $-\mathsf{r}_1(h_c^{t_c})$, $-\mathsf{r}_2(f_c^{t_c}, h_c^{t_c})$ from Lem. 14.12. If b joins two or more cells, the subtraction vanishes. Since $\mathsf{A}_k[W] = e^{E_k(W)} \mathsf{I}_k[W]$,

$$L_b(t) = \partial_b E_k(W)|_0 + \partial_b \log \langle e^{\mathsf{T}_k(W)} \rangle_k \Big|_0\,.$$

Derivatives joining different cells. Let J be the directions of such a block b. They belong to a collection of sixteen directions as in Def. 14.7. The support balls and scale assumptions have already been checked, and $m_*(\mathsf{d}) \ge m_{\mathrm{x}}$ gives $1 \le k \le K - m_{\mathrm{x}}$. Part (c) of Lem. 14.25, discarding its additional separation exponential, yields

$$|\partial_b E_k(W)|_0 \le A_{16}^{\mathrm{x}} \widehat{e}_k[\mathsf{J}].$$

For $\varphi(z) = \langle e^{\mathsf{T}_k(W)} \rangle_k$, we have $\varphi(0) = 1$. For a derivative set γ, write $\mathsf{T}_k[\gamma] = \partial_\gamma \mathsf{T}_k(W)|_0$. When γ joins cells, its direct linear source weight vanishes, and Lem. 14.25(a) identifies it with the sum of the response activities $\mathbf{R}^{(l)}[\mathsf{J}_\gamma]$, $l \le k$, read at the common background.

The norm estimate applies to that actual background. If V passes the level-k test and $\mathsf{U} \in \mathsf{Ev}_k(V)$, Lem. 14.8(a) supplies one regular pair on the torus whose restrictions are the pairs used to read every activity. Part (c) of that lemma bounds the fibre norm term by term by the corresponding Grassmann norm at those restrictions. The summed bound in Lem. 14.25(c) therefore gives $A_{16}^{\mathrm{x}} \widehat{e}_k[\mathsf{J}_\gamma]$, uniformly over V. For γ within one cell, Lem. 14.12 instead gives $C_{\mathrm{T}} \widehat{e}_k[\mathsf{J}_\gamma]$. Put $C_\star = \max\{1, C_{\mathrm{T}}, A_{16}^{\mathrm{x}}\}$.

The derivatives are even and commute. They pass under $\langle\cdot\rangle_k$, so

$$\partial_\gamma \varphi(0) = \left\langle \sum_{\rho \in \mathrm{Part}(\gamma)} \prod_{\eta \in \rho} \mathsf{T}_k[\eta] \right\rangle_k.$$

Apply the expectation bound of Lem. 14.12 to this one function, and use submultiplicativity of the fibre norm. Splitting the pair of a cell into two blocks costs at most C_{l}: at $a_k \geq \mathsf{d}$,

$$\widehat{e}_k[\{f_c\}]\widehat{e}_k[\{h_c\}] = e_k[f_c]e_k[h_c]\mathsf{l}_k^{8\mathfrak{p}_0+2} \leq C_{\mathsf{l}} e_k[f_c, h_c].$$

There are only eight cells. Hence

$$|\partial_\gamma \varphi(0)| \leq \#\operatorname{Part}(I)\, \mathsf{Q}_k C_\star^{\#\gamma} C_{\mathsf{l}}^8 \widehat{e}_k[\mathsf{J}_\gamma].$$

The moment-to-cumulant formula is

$$\partial_b \log \varphi(0) = \sum_{\pi' \in \operatorname{Part}(b)} (\#\pi' - 1)!(-1)^{\#\pi'-1} \prod_{\gamma \in \pi'} \partial_\gamma \varphi(0).$$

Using $\mathsf{Q}_k \geq 1$ and the combinatorial constant k_{16} gives

$$|L_b(t)| \leq C_{\mathrm{lk}} \mathsf{Q}_k^{\#b} \widehat{e}_k[\mathsf{J}], \qquad C_{\mathrm{lk}} = A_{16}^{\mathrm{x}} + \mathsf{k}_{16} \#\operatorname{Part}(I)^{16} C_\star^{16} C_{\mathsf{l}}^{136}.$$

This bound is uniform in the translation indices.

The four possibilities in one cell. Fix a partition π of the sixteen variables. Bound every block joining cells as above, and assign its factors $\mathsf{Q}_k^{\#b}\widehat{e}_k[\mathsf{J}]$ to the cells containing its directions. There are at most eight such blocks. The remaining factors depend on t_c alone, so their sums separate:

$$N^{-8} \sum_{(t_c)} \prod_{b \in \pi} |L_b(t)| \leq \max\{1, C_{\mathrm{lk}}\}^8 \prod_c v_c.$$

Set $u_A = \mathsf{Q}_k^2 e_k[f, h]$. There are four possible forms of v_c.

If both directions of c belong to blocks joining cells, then $v_c \leq C_{\mathsf{l}} u_A$. If they form one block within c, use complex numbers s_t of modulus one with $s_t \mathsf{r}_2(f_c^t, h_c^t) = |\mathsf{r}_2(f_c^t, h_c^t)|$. Sign averaging gives

$$\sum_t |\mathsf{r}_2(f_c^t, h_c^t)| = \operatorname{mean}_\epsilon \mathsf{r}_2 \left(\sum_t \epsilon_t f_c^t, \sum_t \epsilon_t s_t h_c^t \right).$$

The disjoint-support norm bound and Lem. 14.12 yield $v_c \leq 7\omega_{\mathrm{r}} \mathsf{Q}^3 u_0$.

If both directions are singleton blocks, choose phases separately for r_1. Linearity and the same norm bound give

$$\sum_t |\mathsf{r}_1(f_c^t)| \leq 2\omega_{\mathrm{r}} \mathsf{Q}^2 \|f\|_\circ, \qquad v_c \leq 4\omega_{\mathrm{r}}^2 \mathsf{Q}^4 u_0.$$

Finally, if one is a singleton and the other joins another cell, then

$$v_c \leq 2\omega_{\mathrm{r}} \mathsf{Q}^2 N^{-1} \|f\|_\circ \mathsf{Q}_k \widehat{e}_k[\{h\}],$$

or the same with f, h exchanged. Use $e_k[h] = \|h\|_\infty \mathsf{d}^3 a_k^{-4}$, $e_k[f, h] = \mathfrak{f} \mathsf{d}^{5/2} a_k^{-9/2}$, $\mathsf{l}_k \leq 1 + \log(1/\mathsf{d})$, $a_k \leq 1$, and $\mathsf{Q}_k \geq 1$. The ratio of this last bound to u_A is at most

$$2\mathsf{Q}^2\, 96^3 \frac{40M}{\varrho\vartheta} \mathsf{d}^{5/2} (1 + \log(1/\mathsf{d}))^{4\mathfrak{p}_0+1} \leq 1$$

by the choice of d_{T}. All four cases consequently satisfy

$$v_c \leq v := C_{\mathsf{l}} u_A + 11\omega_{\mathrm{r}} \mathsf{Q}^4 u_0.$$

Assembly and passage to states. Summing partitions and using $\mathsf{A}_k[0]/Z[0] \leq 1$ gives

$$|N^{-8} \operatorname{mean}_\epsilon \partial_I \mathsf{Z}_A^{\mathrm{c}}(0)| \leq \#\operatorname{Part}(I) \max\{1, C_{\mathrm{lk}}\}^8 v^8.$$

Combine this with eq. 14.liii and eq. 14.lv and $\omega_{\mathrm{r}} \leq \omega_{\mathrm{r}}^{1/8} \leq 1$:

$$\|\mathsf{X}\|_{\mathrm{ch}, \ell_G} \leq C_{\mathsf{X}} [\omega_{\mathrm{r}} (\mathsf{d})^{1/8} u_0 + u_A],$$

where

$$C_{\mathsf{X}} = C_Z^{1/8} + \#\operatorname{Part}(I)^{1/8} \max\{1, C_{\mathrm{lk}}\}(C_{\mathsf{l}} + 11\mathsf{Q}^4).$$

The definition of $m_*(\mathsf{d})$ gives

$$\mathsf{Q}_k \le \mathsf{d}^{-1/8}, \qquad a_k^{-3} \le \log(1/\mathsf{d}).$$

Therefore $u_A \le \mathfrak{f}\mathsf{d}^{9/4}(\log(1/\mathsf{d}))^{3/2} \le \mathsf{d}^2\mathfrak{f}$. For the last step, put $u = \log(1/\mathsf{d})$. The ceiling $\mathsf{d}_{\mathrm{R}} \le \mathsf{Q}^{-8}$ gives $u \ge 128$, and $u^{3/2}e^{-u/4}$ decreases on $[6,\infty)$ and is below one at 128.

Substitute these bounds and eq. 14.liv into eq. 14.lii. For every $\ell \in \mathfrak{L}(g)$,

$$|S_2^T[\mu_\ell](X) - \sigma| \le \tfrac{1}{2}C^{\mathrm{T}}[\omega_{\mathrm{r}}(\mathsf{d})^{1/8} + \mathsf{d}]\mathsf{d}\mathfrak{f},$$

where

$$C^{\mathrm{T}} = 2\Big[e^{c_V\ell_G^3/8}C_{\mathsf{X}}\left(1 + 96^3(40M/\varrho)^2\vartheta^{-2}\right) + e^{c_V\ell_G^3/4}C_Z^{1/4}\left(\frac{96^3\,40M}{\varrho\vartheta}\right)^2\Big].$$

Every torus-limit state is a weak limit of torus measures at these fixed parameters. The smeared lattice observables are bounded continuous functions of finitely many links, so their means and products pass to the limit. The same bound therefore holds for such a state in place of μ_ℓ. Joining two states through σ gives the factor two in the stated estimate. Finally $\mathsf{b} \le \mathsf{b}_+$ gives the estimate with ϵ_{T}.

The ceiling d_{T} is positive. Indeed $m_*(\mathsf{d}) \ge m_{\mathrm{x}}$ is equivalent to $8c_V^+(m_{\mathrm{x}})\ell_G^3L^{3m_{\mathrm{x}}} \le \log(1/\mathsf{d})$, and the other ceiling inequality holds for all sufficiently small d. The modulus is nondecreasing by construction. It tends to zero because $m_*(\mathsf{d}) \to \infty$ and the expression for ω_{r} has a factor L^{3m_*} multiplied by the exponential of a negative multiple of $(\log(\mathsf{b}_-L^{m_*}))^{2\mathfrak{p}_0}$, where $2\mathfrak{p}_0 > 1$. All constants were chosen after the cell-response and reduction constants and before the final distance ceiling. They do not depend on the cutoff exponent P or on either side. □

14.28. **Proof of the finite-cutoff expansion.**

Proof of Prop. 14.3. Fix g, ϑ, P. All constants used in this proof have already been chosen in Rem. 14.9. Put $d_\natural = \min\{d_1, d_{\mathrm{T}}\}$. The threshold in Prop. 14.43 satisfies

$$d_{\mathrm{T}} \le d_{\mathrm{R}} \le \min\{d_0(g,\vartheta), e^{-8c_V^+(1)\ell_G^3L^3}\},$$

so the reduction depth $m_*(d)$ is at least one. For $d \le d_\natural$ and $K > s(d)$ we have $a_K < d/8$ and $\mathsf{m}d \le 1/10$. Write $\tau = a_K/d$ and $\lg u = 1 + |\log u|$.

Averaging on the least torus. First suppose $\ell = \ell_G$. The two supports lie in a ball $B(x_0, 5d)$. Take

$$\delta = a_K\lceil 17d/a_K\rceil, \qquad n_\delta = \lfloor \ell_G/\delta \rfloor, \qquad \mathcal{T} = \delta\{0,\dots,n_\delta - 1\}^3.$$

Here $17d \le \delta \le 18d$. Two distinct translates differ in some coordinate by $k\delta$, $1 \le |k| < n_\delta$; their circular distance in that coordinate is at least $\min\{\delta, \ell_G - (n_\delta - 1)\delta\} \ge \delta$. Thus the balls $B(x_0 + t, 8d)$, $t \in \mathcal{T}$, are separated by at least d. Since $\ell_G \ge 2$ and $d \le 1/64$,

$$N_\delta := \#\mathcal{T} \ge \left(\frac{\ell_G}{18d} - 1\right)^3 \ge \left(\frac{\ell_G}{24d}\right)^3.$$

Apply Lem. 14.12(e) to these translates:

$$S_2^T((0,f),(0,h)) = \frac{1}{N_\delta}\sum_{t\in\mathcal{T}}\sum_{j\le K}\mathsf{n}_j^1(t) + R, \qquad |R| \le \epsilon_{\mathrm{R}}(d)\mathsf{b}^{-3}d\,\mathfrak{f}.$$

The mixed source jet $\mathsf{n}_j^\lambda(t)$ is the one in Prop. 14.26. At $\lambda = 1$ its source energy is the source energy of the original flow, so the finite sum of its mixed derivatives is the derivative appearing in the reduction.

The fourth and sixth coefficients. Each $\mathsf{n}_j^\lambda(t)$ is even and holomorphic in λ, and vanishes to order four at zero. At $p=3$, Prop. 14.26(ii) gives

$$\left|\sum_{j\le K}[\mathsf{n}_j^1(t)-\mathsf{n}_{j,[4]}(t)-\mathsf{n}_{j,[6]}(t)]\right| \le C_3^{\mathrm{ii}}\frac{d}{\mathsf{b}_-}\left(\max\left\{1,\log\frac{2\mathsf{b}_+}{d}\right\}\right)^{8\mathfrak{p}_0}\mathsf{b}^{-3}d\,\mathfrak{f}.$$

The bound is independent of t. By Prop. 14.34(c),(d), the average of the fourth coefficients differs from $c_2\mathsf{b}^{-2}\mathcal{K}_{\mathrm{free}}(f,h)$ by at most

$$\left[C_{\mathrm{ord}}X_g d^2\mathsf{b}^{-3}d+\frac{3C_{\mathrm{tm}}}{2\vartheta}\mathsf{b}^{-2}\tau\right]\mathfrak{f}.$$

The hypothesis $L^K\le d^{-P}$ is used in Prop. 14.42, with $m=3$, to identify the sixth coefficient. Its average differs by at most $C_3\epsilon_3'(d)\mathsf{b}^{-3}d\,\mathfrak{f}$ from

$$\mathsf{b}^{-3}\mathsf{P}_1^{\mathrm{lat}}(f,h;a_K)+a_K\mathsf{b}^{-3}N_f\sum_{p,q}f(x(p))h(x(q))\mathsf{B}_1^F(p,q;a_K\mathsf{m}).$$

The factor a_K comes from the physical normalization $O(f)=a_K^{-1}\sum_p f(x(p))a_p$ and $\beta^{-3}=a_K^3\mathsf{b}^{-3}$. The determinant part in this formula is precisely the quadratic fermion loop identified in Lem. 14.31.

By Props. D.46 and D.61, the pure contribution is

$$\frac{3}{2\pi^3}\mathsf{b}^{-3}\mathcal{K}_5(f,h)+R_{\mathrm{pure}},\qquad |R_{\mathrm{pure}}|\le C_{\mathrm{IV}}\vartheta^{-1}\tau\lg(1/\tau)\mathsf{b}^{-3}d\,\mathfrak{f}.$$

By Prop. 14.33, the fermion contribution is

$$-\frac{3N_f}{8\pi^3}\mathsf{b}^{-3}\iint f(x)h(y)|x-y|^{-5}\mathcal{F}(\mathsf{m}|x-y|)\,dx\,dy+R_{\mathrm{vp}},$$

where

$$|R_{\mathrm{vp}}|\le C_{\mathrm{vp}}\tau\lg(1/\tau)\mathsf{b}^{-3}d\,\mathfrak{f},\qquad \mathcal{F}(s)=\frac{1-(1+s)e^{-2s}}{s},\quad \mathcal{F}(0)=1.$$

All its hypotheses hold: $8a_K\le d$, $a_K\mathsf{m}\le 1$ and $\mathsf{m}d\le 1$.

The mass profile has

$$0\le 1-\mathcal{F}(s)\le\frac{2}{3}s^2,\qquad s\ge 0.$$

The first inequality is Prop. 14.32(b). For the second, the function $\varphi(s)=\frac{2}{3}s^3-s+1-(1+s)e^{-2s}$ satisfies $\varphi(0)=\varphi'(0)=0$ and $\varphi''(s)=4s(1-e^{-2s})\ge 0$. Using eq. 14.i at $q=3$, the error made by replacing $\mathcal{F}$ by one is at most

$$\frac{3N_f}{8\pi^3}\frac{2}{3}\mathsf{m}^2\mathcal{K}_3(f,h)\mathsf{b}^{-3}\le\frac{4N_f}{9\pi}(\mathsf{m}_+d)^2\mathsf{b}^{-3}d\,\mathfrak{f}.$$

The two coefficients now add to

$$\frac{3}{2\pi^3}-\frac{3N_f}{8\pi^3}=c_2\mathfrak{c}_1(N_f).$$

Combining the errors. Put $C_\diamond=C_{\mathrm{IV}}\vartheta^{-1}+C_{\mathrm{vp}}$ and

$$\Sigma_0(d)=\epsilon_{\mathrm{R}}(d)+C_3^{\mathrm{ii}}\frac{d}{\mathsf{b}_-}\left(\max\left\{1,\log\frac{2\mathsf{b}_+}{d}\right\}\right)^{8\mathfrak{p}_0}+C_{\mathrm{ord}}X_gd^2$$
$$+C_3\epsilon_3'(d)+\frac{4N_f}{9\pi}(\mathsf{m}_+d)^2.$$

The least-torus error is bounded by

$$\left[\Sigma_0(d)\mathsf{b}^{-3}d+C_\diamond\tau\lg(1/\tau)\mathsf{b}^{-3}d+\frac{3C_{\mathrm{tm}}}{2\vartheta}\mathsf{b}^{-2}\tau\right]\mathfrak{f}.$$

To remove the logarithm from the mesh error, split into two cases. If $\lg(1/\tau) \le \mathsf{b}/d$, its contribution is at most $C_\diamond \mathsf{b}^{-2}\tau\mathfrak{f}$. Otherwise $\tau < e^{1-\mathsf{b}/d}$. The function $u(1-\log u)$ increases on $(0,1]$, and xe^{1-x} decreases on $[1,\infty)$. Since $\mathsf{b} \ge \mathsf{b}_- > d$, this gives

$$\tau \lg(1/\tau) \le \frac{\mathsf{b}}{d} e^{1-\mathsf{b}/d} \le \eta_{\mathrm{lat}}(d) := \frac{\mathsf{b}_-}{d} e^{1-\mathsf{b}_-/d}.$$

Thus in both cases the error is at most

$$\left[(\Sigma_0(d) + C_\diamond \eta_{\mathrm{lat}}(d))\mathsf{b}^{-3} d + C_{\mathrm{lat}} \mathsf{b}^{-2} \frac{a_K}{d}\right] \mathfrak{f}, \qquad C_{\mathrm{lat}} = \max\left\{1, \frac{3C_{\mathrm{tm}}}{2\vartheta} + C_\diamond\right\}.$$

For another torus, lift the pair to $\mathbb{R}^3$ and periodize it on T_{ℓ_G}, as in Prop. 14.43. The supports lie in a ball of radius $5d < \ell_G/4$, and $32d \le \ell_G$, so the distances, integrals, suprema and Lipschitz bounds are preserved. A path from a support to its complement first meets its boundary, where the function vanishes; this also checks the Lipschitz bound across the period boundary. The same lemma bounds the change in the two-point function by $\epsilon_{\mathrm{T}}(d)\mathsf{b}^{-3} d\mathfrak{f}$. It gives exactly the same comparison for a torus-limit state on the infinite lattice.

Finally set

$$\Sigma(d) = \Sigma_0(d) + C_\diamond \eta_{\mathrm{lat}}(d) + \epsilon_{\mathrm{T}}(d).$$

Every term tends to zero. The number

$$d^* = \sup\{d' \le d_\natural : \Sigma(d) \le 1 \text{ for } 0 < d \le d'\}$$

is positive. Choose $d_{\mathrm{e}} = d^*/2$ and $\epsilon_1(d) = \sup_{0<d'\le d} \Sigma(d')$. These have all the asserted dependencies and prove eq. 14.ii. □

14.29. **Removing the cutoff on a fixed torus.**

Proof of Prop. 14.4. The finite-cutoff expansion is only available when L^K is bounded by a fixed power of d^{-1}. We choose a cutoff in that range and use the quantitative cutoff estimate to control the remaining tail.

The constants can be chosen jointly. Choose h_F after the constants on the left of eq. 14.viii, then choose the leg constants. The enlarged free-history constants precede A_0, and the energy constants precede the final choice of $\mathscr{C}$. The coupling ceilings are imposed last. Thus the floors and ceilings in Def. 11.1 do not invalidate Rem. 14.9.

At $n = 2$, let the rate and cutoff exponent from Thm. Q(iii) be

$$c^{\mathrm{fs}} = \frac{c_\#}{92}, \qquad P^{\mathrm{fs}} = \frac{920}{c_\#},$$

and write its constants as $K^{\mathrm{fs}}(g,\ell)$ and $C^{\mathrm{fs}}(g,\ell)$. Put $P_\star = P^{\mathrm{fs}} + 1$ and apply Prop. 14.3 with $P = P_\star + 1$.

Choose $d_W \in (0, 1/L]$ so that, for every $0 < d \le d_W$,

$$d^{1-P_\star} \ge L\mathsf{c}_W [\log(d^{-P_\star}(X_g + \bar{B}))]^{2\mathfrak{p}_0}. \tag{14.lvi}$$

Such a choice exists: with $v = \log(1/d)$, the two sides grow respectively as $e^{(P_\star-1)v}$ and a polynomial in v. Set

$$d_2 = \min\{d_{\mathrm{e}}, \ell/20, d_W, L^{-K^{\mathrm{fs}}(g,\ell)}\}.$$

For $d \le d_2$, choose

$$K(d) = \lceil P_\star \log_L(1/d) \rceil.$$

Then

$$d^{-P_\star} \le L^{K(d)} < Ld^{-P_\star} \le d^{-P},$$

because $d \le 1/L$. Also $K(d) > s(d)$. Indeed

$$L^{K(d)-1} d \ge L^{-1} d^{1-P_\star}, \qquad \bar{x}_{K(d)-1} < d^{-P_\star}(X_g + \bar{B}),$$

so eq. 14.lvi places $K(d)-1$ in the set defining $s(d)$. Finally,

$$K(d) \geq P^{\mathrm{fs}} \max\{0, \log_L(1/d)\} + K^{\mathrm{fs}}(g,\ell).$$

Thus both the finite-cutoff expansion and the cutoff tail estimate apply at $K(d)$.

By Cor. 11.6, the difference between the continuum two-point function and the one at this cutoff is at most

$$\frac{C^{\mathrm{fs}}(g,\ell)}{1-L^{-c^{\mathrm{fs}}}} d^{c^{\mathrm{fs}}P_\star} \|f\|_\sharp \|h\|_\sharp.$$

For the present class and $d \leq 1$,

$$\|f\|_\sharp \leq \frac{1+40M}{\vartheta d}\|f\|_\infty, \qquad \|h\|_\sharp \leq \frac{1+40M}{\vartheta d}\|h\|_\infty.$$

The mesh term in Prop. 14.3 is at most $C_{\mathrm{lat}}\mathsf{b}^{-2}d^{P_\star-1}\mathfrak{f}$. Since $\mathsf{b} \leq X_g$, their sum is bounded as in eq. 14.iii with

$$\epsilon_2(d) = \epsilon_1(d) + C_{\mathrm{lat}} X_g d^{P_\star-2} + \frac{(1+40M)^2 C^{\mathrm{fs}}(g,\ell) X_g^3}{\vartheta^2(1-L^{-c^{\mathrm{fs}}})} d^{c^{\mathrm{fs}}P_\star-3}.$$

Both exponents are positive: $P_\star - 2 > 0$ and $c^{\mathrm{fs}}P_\star - 3 = 7 + c_\#/92 > 0$. This proves (a), including the dependence and monotonicity of the error.

For (b), the bounds eq. 14.i give

$$\mathcal{K}_6 \geq 10^{-6}\vartheta^2\mathfrak{f}, \qquad \mathcal{K}_5 \geq 10^{-5}\vartheta^2 d\,\mathfrak{f}.$$

The error in the first quotient of eq. 14.iv is therefore at most

$$10^6\vartheta^{-2}\mathsf{b}^{-3}d\left[(4\pi/3)^2|\mathfrak{c}_1(N_f)| + \epsilon_2(d)/c_2\right],$$

and the error in the second is at most

$$\frac{10^5}{c_2\vartheta^2}\epsilon_2(d).$$

Both tend to zero, uniformly over the test pairs.

For (c), subtract the two expansions. Since distinct integer flavour numbers satisfy

$$c_2|\mathfrak{c}_1(N_f) - \mathfrak{c}_1(N_f')| \geq \frac{3}{8\pi^3},$$

the modulus of the difference is at least

$$\left[\frac{3\vartheta^2}{8\pi^3}10^{-5} - \epsilon_2(d) - \epsilon_2'(d)\right]\mathsf{b}^{-3}d\,\mathfrak{f}.$$

This is positive under the stated condition. □

14.30. **Changing the volume and identifying the coefficient.**

Proof of Thm. X. Comparison with the least torus. The comparison in Prop. 14.43 holds for every $K > s(d)$; it has no upper bound on L^K. At any real label $(\mathsf{b}', \mathsf{m}') \in \mathsf{W}(g)$ it bounds the difference between the two-point function on any allowed torus, or in any torus-limit lattice state, and the function on T_{ℓ_G}, by

$$C^{\mathrm{T}}[\omega_{\mathrm{r}}(d)^{1/8} + d]\,d\,\mathfrak{f} \leq \epsilon_{\mathrm{T}}(d)\mathsf{b}^{-3}d\,\mathfrak{f}.$$

In the last inequality b can be any label in $\bar{I}(g)$: the definition of ϵ_{T} uses the upper endpoint b_+. Lift and periodize the pair as in the proof of Prop. 14.3; the functions $\mathcal{K}_5, \mathcal{K}_6$ and $\mathfrak{f}$ are unchanged.

For a fixed torus, let $K \to \infty$ with the label held fixed. By Cor. 11.6, both lattice functions converge at these Lipschitz entries. Add the least-torus expansion from Prop. 14.4. This proves (a).

The moving-label convergence and the agreement with the C^1 continuum limits both follow from Cor. 11.6.

For (b), take a sequence defining S, either with sides tending to infinity or with torus-limit lattice states. After restricting the sequence, its labels converge to $(\mathsf{b},\mathsf{m})\in\mathsf{W}(g)$. On C^1 pairs the first two-point function converges to the defining value of S_2. On the least torus, Cor. 11.6 gives convergence along the entire sequence of labels. Passing to the limit in the comparison gives

$$|S_2((0,f),(0,h))-S_2^{(\ell_G,\mathsf{b},\mathsf{m})}((0,f^{\langle\ell_G\rangle}),(0,h^{\langle\ell_G\rangle}))|\le\epsilon_{\mathrm{T}}(d)\mathsf{b}^{-3}d\mathfrak{f}.$$

Adding the least-torus expansion proves eq. 14.v. The division estimates and the flavour gap in the proof of Prop. 14.4 now prove (c).

Comparison with the pure theory. Use the admitted pure-theory constants of (d)(i). The pure short-distance estimate and its extension through all sufficiently deep cutoffs are Prop. D.70(c). On the least pure torus it gives a threshold d'_{e} and an error $\bar\epsilon'(d)\to0$. The side comparison Prop. D.69 costs at most

$$C_2^{\mathrm{T}}(g_0)\vartheta^{-1}d^2(\mathsf{b}')^{-2}\mathfrak{f}$$

when $K\ge3$ and $a_K\le9d$. Put

$$\mathsf{b}_+^0=\bar L(g_0^{-2}+\bar B^0),\qquad d^0=\min\{d'_{\mathrm{e}},1/180,d_\star\},\qquad \epsilon^0(d)=\bar\epsilon'(d)+C_2^{\mathrm{T}}(g_0)\vartheta^{-1}\mathsf{b}_+^0d,$$

where $d_\star$ is the threshold in Prop. D.69. For each fixed $d\le d^0$, pass to the defining first-passage sequence of S^0. Its depths tend to infinity and its labels b_j tend to b, while $a_{K_j}/d\to0$. The pure expansion and the side comparison therefore give the displayed estimate for S^0. The coefficient gap from the matter theory is

$$c_2|\mathfrak{c}_1(N_f)-\mathfrak{c}_1(0)|=\frac{3N_f}{8\pi^3}\ge\frac{3}{8\pi^3}.$$

The same lower bound eq. 14.i proves the asserted inequality of the two-point functions.

The existence assertion at $\mathsf{b}\in I(g_0)$ is [YM3, Prop. 6.1]. For this label, the data $K,L^K\mathsf{b}$ are admitted at every cutoff; taking their sides to infinity and extracting a subsequence gives a pure growing-torus limit with the same label. The estimate just proved applies to each such limit. For the stated explicit choice of g_0, the lower bound on b gives the required coupling ceiling. It also gives $\mathsf{b}\ge6\bar B^0$, since the admitted pure constants have $(\gamma_W^{\mathrm{sd}})^{-2}\ge4\bar B^0$. Now $g_0^{-2}=\mathsf{b}-2\bar B^0$, so $\mathsf{b}>g_0^{-2}+\bar B^0$. For the upper end of $I(g_0)$, use $\bar L\ge13$:

$$\bar L(g_0^{-2}+B_\sharp^0)-\bar B^0-\mathsf{b}=(\bar L-1)\mathsf{b}-(2\bar L+1)\bar B^0+\bar LB_\sharp^0\ge0.$$

This verifies $\mathsf{b}\in I(g_0)$.

Dilations. The pair (f_λ,h_λ) belongs to $\mathfrak{T}_{\mathbb{R}^3}(\lambda d;\vartheta)$: the supports and separation scale by λ, the integrals by λ^3, the Lipschitz constants by λ^{-1}, and the suprema do not change. Changing variables gives

$$\mathcal{K}_q(f_\lambda,h_\lambda)=\lambda^{6-q}\mathcal{K}_q(f,h).$$

Thus eq. 14.v is exactly eq. 14.vi. Divide its remainder by λ to obtain the limit. Apply that formula at λ and $\lambda/2$. Monotonicity of ϵ_4 bounds the sum of the remainders by $\frac32\epsilon_4(\lambda d)\mathsf{b}^{-3}\lambda d\mathfrak{f}$. If $N_f\ne4$, then

$$\frac12c_2|\mathfrak{c}_1(N_f)|\mathcal{K}_5(f,h)\ge\frac{3\vartheta^2}{16\pi^3}10^{-5}d\mathfrak{f},$$

which proves the strict lower bound in (d)(ii). At $N_f=4$ the coefficient vanishes, leaving the stated remainder estimate. □

15. The condensate subtraction

We work under all the hypotheses of Thm. S. The two data A, B have cutoffs $K, K+1$, the same labels and the same physical torus. In the adjacent-cutoff construction use $n = 1$, hence

$$s = \lceil K/34 \rceil, \quad u = \lceil c_{\#} K/340 \rceil, \quad t_0^{\mathrm{A}} = K - s, \quad t_1^{\mathrm{A}} = K - u, \qquad t_i^{\mathrm{B}} = t_i^{\mathrm{A}} + 1.$$

We keep the original and restarted flows of Def. 11.2. At level j of datum $\bullet$ write

$$a_j^\bullet = L^{j-K^\bullet}, \qquad n_j^\bullet = \ell^3 (a_j^\bullet)^{-3}, \qquad \mathsf{t}_j = b/a_j^\bullet.$$

The constant source is $v_\eta = (0, \eta\Gamma_1^0)$, so that insertion of v_η replaces m by $m - \eta$. Write $[f]_{m_1}^{m_2} = f(m_2) - f(m_1)$ and $m_c = (m_- + m_+)/2$.

Throughout this proof $c_\lambda \geq 1$ is the field-scale constant in Rem. 14.9. The different constant $0 < c_\lambda \leq 1/8$ in the cutoff comparison enters only through its already fixed K_1, C_1 and through the differentiated-condensate estimate. The threshold A_1 in $\delta(\mathsf{t}) = A_1 \mathsf{t}^{-1/2} (\log \mathsf{t})^{\mathfrak{p}_0}$ is distinct from the free coefficient A_1 in the subtraction.

15.1. Contact energies and mass shifts.

Definition 15.1 (Contact coefficients and error bounds)**.** The following definitions are used when (k0) and $u \geq u_*$ hold in Def. 11.5, with $n = 1$.

(1) Let $\mathfrak{D}_j^\bullet = \{|\lambda| < \bar{\lambda}_j^\bullet\}$ be the original-flow discs of Prop. 14.14. For the restarted flow put

$$\hat{\xi}_i(\lambda) = \mathsf{t}_i + \lambda^2(\hat{x}_i - \mathsf{t}_i), \quad \hat{\tau}_i(\lambda) = \hat{\xi}_i(\lambda)/\mathsf{t}_i, \quad \hat{\lambda}_i^2 = \frac{4\hat{x}_i}{5c_\lambda (\log \hat{x}_i)^{2\mathfrak{p}_0}}, \quad \hat{\mathfrak{D}}_i^\bullet = \{|\lambda| < \hat{\lambda}_i\}.$$

Here $\hat{x}_{t_0^\bullet} = L^s b = \mathsf{t}_{t_0^\bullet}$.

(2) Let $e_l^\bullet(\lambda)$ be the infinite-lattice vacuum energy per site extracted at level l by the common-field-scale construction, and let $\hat{e}_l^\bullet(\lambda)$ be its restarted counterpart from Lem. 15.5. The original flow is formed on the least torus $\ell_G(g)$ and on $\mathbb{Z}^3$; the per-site numbers are extracted on $\mathbb{Z}^3$. They are independent of the side and the reference coupling at fixed K, b, m and construction constants, as proved below. Their values at $\lambda = 1$ are the unscaled energies. Set

$$\mathsf{v}^\bullet(\lambda; b, m) = \sum_{l=1}^{t_1^\bullet} (a_l^\bullet)^{-3} e_l^\bullet(\lambda) - \sum_{t_0^\bullet < l \leq t_1^\bullet} (a_l^\bullet)^{-3} \hat{e}_l^\bullet(\lambda).$$

For any of these even functions, a subscript $[2n]$ denotes the coefficient of λ^{2n} at zero. Define

$$\mathsf{q}_n(K; b; m_1, m_2) = b^n [\mathsf{v}_{[2n]}^{\mathrm{B}}(b, \cdot) - \mathsf{v}_{[2n]}^{\mathrm{A}}(b, \cdot)]_{m_1}^{m_2}, \qquad 0 \leq n \leq 3.$$

At zero source, the contact energy satisfies

$$c^\bullet(0) - \hat{c}^\bullet(0) = \ell^3 \mathsf{v}^\bullet(1; b, m).$$

This uses $E_{t_1^\bullet} = \sum_{l \leq t_1^\bullet} e_l n_l^\bullet$ and the finite-size assignment in [QCDM1, Lem. 9.15].

(3) Set $S_{\mathrm{en}} = 1 + 6600 E_0 + L^3$. Let $C_{\mathrm{fs}}^\natural$ be the wrapping constant in the proof of [QCDM1, Lem. 9.15]. Use the operator constant C_D from [QCDM1, Lemmas 6.3 and 6.5] and the mass-form constant $c_{\mathsf{S}} = M^3 \nu_P \nu C_H^2$ of Lem. 14.8.

(4) Use $K_1(g, \ell), C_1(g, \ell)$ and $c_{\mathrm{cut}} = c_{\#}/68$ from Def. 11.5 at $n = 1$, for the comparison with contact terms retained. Write the bound in Thm. R as

$$\sup_{W(g)} |\partial_m^2 (y_{K+1} - y_K)| \leq \varepsilon_\natural(K; g, \ell) := C_\natural(g, \ell)(1 + K)^{\mathfrak{p}_0} L^{-c_y K}, \qquad K \geq K_*(g, \ell).$$

Let K_{v} be the least integer $K \geq 2$ for which $r_J L^{\lceil c_{\#} K/340 \rceil} \geq 2$, and put

$$K_{\mathrm{o}}(g, \ell) = \max\{K_1(g, \ell), K_*(g, \ell), K_{\mathrm{v}}\}.$$

The errors used below are

$$\varepsilon_{\mathfrak{f}}(K;\ell) = C_{\mathrm{fs}}^{\natural} E_0 \sum_{d\ge u} (\ell L^d)^3 e^{-(\kappa+1-\log 9216)(\ell L^d/(2M)-2)},$$

$$\varepsilon_1(K) = \ell^{-3} N_f^{3/2} C_1(g,\ell) L^{-c_{\mathrm{cut}} K},$$

$$\varepsilon_{\mathrm{rem}}(K) = \frac{32 S_{\mathrm{en}}}{L-1}\left(\frac{5c_\lambda}{3}\right)^4 \frac{((K+2)\log L + \log(2b_+))^{8\mathfrak{p}_0}}{b_-^4} L^{1-u},$$

$$\varepsilon_{\mathrm{S}}(K) = (m_+ - m_-)\varepsilon_1(K) + 8\ell^{-3}\varepsilon_{\mathfrak{f}}(K;\ell) + \varepsilon_{\mathrm{rem}}(K).$$

Here $\kappa \ge \log 9216$. The error $\varepsilon_{\mathrm{box}}$ is defined in Lem. 15.7; it depends on g and the constants but not on the torus side. We include g, ℓ among the arguments of an error when comparing different windows or sides.

■

The exact rescaling identity supplies the recursion that will determine the divergent coefficients. It is an identity of the bare integral and does not require comparison of two renormalization flows.

Lemma 15.2 (Exact lattice rescaling)**.** For every $K \ge 1$, every side $\ell = 2L^j$, $j \ge 1$, and all complex labels,

$$Z_{K+1,\ell}(b', m') = Z_{K,L\ell}(Lb', m'/L).$$

For real labels with $m > 0$ both partition functions are positive, and

$$y_{K+1,\ell}(b, m) = L^2 y_{K,L\ell}(Lb, m/L).$$

The logarithmic-derivative identity also holds on every complex open set on which the partition function is nonzero. ■

Proof. Both integrals are over $(\mathbb{Z}/N\mathbb{Z})^3$ with $N = \ell L^{K+1}$ and the same antiperiodic spinor sections. In lattice units the action, operator and Haar measure are identical. Their only parameters are the action weight and bare mass, and

$$(b'/a_{K+1}, a_{K+1}m') = (Lb'/a_K, a_K m'/L).$$

This proves equality of the entire partition functions. For positive real mass the determinant is positive at every configuration by [QCDM1, Prop. 2.1], hence so is the integral. Differentiate its real logarithm and use $\ell^{-3}L^{-1} = L^2(L\ell)^{-3}$. Where the partition function has no zero, the same calculation applies to its holomorphic logarithmic derivative. □

The cutoff comparison expresses an increment of y_K through a derivative of the contact energy. A constant mass source is a change of the bare mass. The next lemma turns its integral over a mass interval into an ordinary difference of vacuum energies, retaining the finite-size correction.

Lemma 15.3 (Mass shifts in the restarted contact energy)**.** Assume (k0) and $u \ge u_*$ at $n = 1$. Fix a datum $\bullet$, $b \in I_g$, and $t_0^\bullet \le k \le t_1^\bullet$. At mass x let $\hat{c}_k^{(x)}$ be its restarted source energy, and let $\hat{\mathfrak{f}}_k^{(x)}$ be the sum of the scalar activities at the trivial pair over $t_0^\bullet < j \le k$. Write $E_k^{(x)}$ and $\mathfrak{f}_k^{(x)}$ for the corresponding original-flow quantities of Prop. 10.12. For $s = (w, J)$ set $s_\zeta = (w, J - \zeta 1)$.

(a) If $x, x' \in [m_-, m_+]$ and $\zeta = x' - x$, then

$$\hat{c}_k^{(x)}(s_\zeta) + \hat{\mathfrak{f}}_k^{(x)} - \hat{c}_k^{(x')}(s) - \hat{\mathfrak{f}}_k^{(x')}$$

is one fixed element of $2\pi i\mathbb{Z}$ on the set $s, s_\zeta \in \mathscr{O}_k$.

(b) Both $|\mathfrak{f}_k^{(x)}|$ and $|\hat{\mathfrak{f}}_k^{(x)}|$ are at most $\varepsilon_{\mathfrak{f}}(K;\ell)$.

(c) Suppose also $r_J L^u \geq 2$ and $m_- \leq m_1 < m_2 \leq m_+$. For the contact energy $\mathfrak{c}^{\bullet,(m)} = c^{\bullet,(m)} - \hat{c}^{\bullet,(m)}$ put $h(\eta) = \mathfrak{c}^{\bullet,(m_1)}[v_\eta]$. It is holomorphic for $|\eta| < r_J L^u$, and

$$\partial_\eta \mathfrak{c}^{\bullet,(m)}[v_\eta]\Big|_{\eta=0} = h'(m_1 - m).$$

Moreover

$$\operatorname{Re} \int_{m_1}^{m_2} \partial_\eta \mathfrak{c}^{\bullet,(m)}[v_\eta]\Big|_{\eta=0} \, \mathrm{d}m = -\left[\ell^3 \mathsf{v}^\bullet(1; b, \cdot) + \mathfrak{f}_{t_1^\bullet}^{(\cdot)} - \operatorname{Re} \hat{\mathfrak{f}}_{t_1^\bullet}^{(\cdot)}\right]_{m_1}^{m_2}.$$

■

Proof. The restarted start at (x, s_ζ) equals the start at (x', s). Its tower and background maps use the same fine mass matrix,

$$a_{K^\bullet} x - a_{K^\bullet}(\hat{J} - \zeta) = a_{K^\bullet} x' - a_{K^\bullet} \hat{J}.$$

The tests, weighted action and mass-member coefficient $\hat{\Sigma}_w$ depend on the gauge source and calibration, not on J or the mass. The step map is also the same: its kernel, chart integral and density test depend on b_F, the averages and calibrations, not on the mass or running coefficients. By the exact density identity in Prop. B.13, at each later level the two densities are therefore densities of the same measure.

Their scalar parts have equal integrals on every Borel set. As in Prop. 10.12, they are continuous on an open neighbourhood of the trivial block field on which the density test is one, so they agree at that field. There the action vanishes and the tower, mass member and terms with legs have degree at least two. The scalar activities contain no source-dependent constant. Their values are consequently

$$\exp[\hat{c}_k^{(x)}(s_\zeta) + \hat{\mathfrak{f}}_k^{(x)}], \qquad \exp[\hat{c}_k^{(x')}(s) + \hat{\mathfrak{f}}_k^{(x')}].$$

Their exponents differ by an element of $2\pi i\mathbb{Z}$. The difference is holomorphic, hence continuous, on the convex common source domain; it is constant there. This proves (a).

For (b), the original-flow estimate is the wrapping bound in [QCDM1, Lem. 9.15]. At level j its value is at most

$$C_{\mathrm{fs}}^{\natural} E_0 \ell_j^3 e^{-(\kappa+1-\log 9216)(\ell_j/(2M)-2)}, \qquad \ell_j = \ell L^{K^\bullet - j}.$$

The same proof applies to the restart. Its nonwrapping terms fit the infinite lattice; its pointed scalar totals are translation covariant, and the cell mean is removed as $\hat{e}_j$. Thus the remaining cell sum is zero. Its normalizers are the original ones, and its scalar arrays have bound E_0 by Prop. B.5. Only wrapping polymers and normalizer walks contribute. Their depths $K^\bullet - j \geq u$ are distinct, so summing the displayed bounds gives $\varepsilon_{\mathfrak{f}}$.

For (c), $a_{t_1^\bullet}^\bullet = L^{-u}$, so the constant source belongs to $\mathscr{O}_{t_1^\bullet}$ for $|\eta| < r_J L^u$; both source energies are holomorphic there. If $m \in [m_1, m_2]$ and $|z| \leq 1/2$, then $|z|, |z - (m - m_1)| \leq 1 < r_J L^u$. Apply Prop. 10.12 and (a) to these two sources. After subtracting the identities, their difference is constant modulo $2\pi i$. Differentiation at $z = 0$ gives the derivative formula in (c). Its integral is $h(0) - h(m_1 - m_2)$. At $z = 0, m = m_2$, the same identities give

$$\begin{aligned} h(m_1 - m_2) = c^{\bullet,(m_2)}(0) - \hat{c}^{\bullet,(m_2)}(0) + \mathfrak{f}_{t_1^\bullet}^{(m_2)} - \mathfrak{f}_{t_1^\bullet}^{(m_1)} \\ - \hat{\mathfrak{f}}_{t_1^\bullet}^{(m_2)} + \hat{\mathfrak{f}}_{t_1^\bullet}^{(m_1)} + 2\pi i k, \qquad k \in \mathbb{Z}. \end{aligned}$$

Use $c^\bullet(0) - \hat{c}^\bullet(0) = \ell^3 \mathsf{v}^\bullet(1)$ and take real parts. The original scalar correction and both zero-source extracted energies are real. This proves the final identity. □

Proposition 15.4 (The mass-integrated cutoff increment)**.** For $K \geq K_{\mathrm{o}}(g,\ell)$, $b \in I_g$ and $m_- \leq m_1 < m_2 \leq m_+$,

$$\left| \int_{m_1}^{m_2} (y_{K+1} - y_K)(b,m)\,\mathrm{d}m + [\mathsf{v}^{\mathrm{B}}(1;b,\cdot) - \mathsf{v}^{\mathrm{A}}(1;b,\cdot)]_{m_1}^{m_2} \right| \leq (m_2 - m_1)\varepsilon_1(K) + 8\ell^{-3}\varepsilon_{\mathfrak{f}}(K;\ell). \qquad \blacksquare$$

Proof. The threshold K_1 supplies the cutoff conditions at $n = 1$, and K_{v} gives $r_J L^u \geq 2$. Expand the flavour identity in its orthonormal Hermitian basis: $\Gamma_1^0 = \sum_j \mathrm{tr}(t_j)\Gamma_{0,j}$, where $\sum_j |\,\mathrm{tr}(t_j)| \leq N_f^{3/2}$. Use the adjacent-cutoff comparison of Thm. Q with contact terms retained, at the constant tuple $((\Gamma_{0,j}, 1))$. The test norm is one. Multilinearity and the constant-source identity give

$$\left| \ell^3(y_{K+1} - y_K) - \partial_\eta(\mathfrak{c}^{\mathrm{B}} - \mathfrak{c}^{\mathrm{A}})[v_\eta] \Big|_{\eta=0} \right| \leq N_f^{3/2} C_1(g,\ell) L^{-c_{\mathrm{cut}}K}.$$

This is the general contact-term estimate, with the B_1, C_1 majorant; no support-separation conclusion is used. Since y_K is real, one may take the real part of the derivative. Integrate over the mass interval and apply Lem. 15.3(c) for both data. The eight torus corrections at the two endpoints are bounded by its part (b). Division by ℓ^3 proves the result. □

15.2. **The field-scale expansion.** To identify the contact energies, we apply the common-scale construction to the restarted flow as well as to the original one. The restart has a nonzero initial mass member and begins above level zero. The proof below checks these two changes, including the scalar normalization needed when masses are subsequently compared.

Lemma 15.5 (A common field scale for the restarted flow)**.** Assume (k0) and $u \geq u_*$ at $n = 1$, and let $\bullet \in \{\mathrm{A}, \mathrm{B}\}$ and $(b,m) \in W(g)$. The restarted flow can be formed on $\hat{\mathfrak{D}}_i^\bullet$ at every step $t_0^\bullet \leq i < t_1^\bullet$, on $\mathbb{Z}^3$ and on every admitted torus with these data. Multiply every gauge fluctuation by λ and use the coefficients $\hat{\xi}_i(\lambda)$ of Def. 15.1. At the start, replace the action weight $L^s b$ by $\lambda^{-2}L^s b$, leave the tower form unscaled, and replace the mass member by

$$-L^{-t_0^\bullet}\mathsf{S}_1(\lambda^2 \hat{s}_{t_0^\bullet}\psi^{\mathrm{bg}}), \qquad \hat{s}_{t_0^\bullet} = F^{\mathrm{UV}}/(L^s b).$$

At step i, use the old restarted families at λ and remove the original coupling shift $\hat{b}_{i+1}$, the mass term $-L^{-(i+1)}\lambda^2\hat{\mu}_{i+1}\mathsf{S}_1$, and the new vacuum energy $\hat{e}_{i+1}(\lambda)$. All other terms remain in the activity. Then:

(a) The step is holomorphic in λ. Its potential satisfies $\mathfrak{V}_i(-\lambda,-B) = \mathfrak{V}_i(\lambda,B)$. The Gaussian data, integrated activities, pointed terms, vacuum energies and field-independent parts are even in λ. At $\lambda = 1$ they agree with the restarted zero-source flow of Prop. B.13, term by term. At $\lambda = 0$ the potential vanishes: only the calibration tests and Gaussian fermion integrals at the background remain.

(b) The induction bounds eq. 14.xxi hold for both restarted families, with $\eta = C_\eta c_\lambda^{-1/2}$ and $\hat{\mathfrak{D}}_{i-1}^\bullet$ in place of D_{i-1}. The full scalar activities have array bound E_0 at rate $\kappa + 1$, at every pointer.

(c) For real $0 < \lambda \leq 1$, the level-j form at a block field passing its density test is a nonzero scalar independent of field, source and mass times the composite chart integral over steps $t_0^\bullet \leq i < j$. Its fluctuations run over the calibration boxes $|B_i(b)| < \delta(\mathsf{t}_i)$ and enter each chart as λB_i. The integrand contains the elimination factors, chart Jacobians and gauge fixings at coefficients t_i/λ^2, together with

$$e^{-A(\beta^\bullet/\lambda^2, \mathsf{U}_{t_0^\bullet}^\bullet(W_{t_0^\bullet}))}, \qquad \beta^\bullet = L^{K^\bullet} b,$$

and the Berezin integral of the block kernels times

$$\exp\left[-\langle \bar{\Psi}, \mathbb{D}_{t_0^\bullet}^\bullet(\mathsf{U}_{t_0^\bullet}^\bullet(W_{t_0^\bullet}))\Psi\rangle - L^{-t_0^\bullet}\mathsf{S}_1(\lambda^2 \hat{s}_{t_0^\bullet}\psi^{\mathrm{bg}}) \right].$$

The integrated generators are those of levels $t_0^\bullet \le i < j$. The scalar in the original-flow composition of Prop. 14.14 is also independent of the mass.

(d) For $t_0^\bullet < l \le t_1^\bullet$,

$$|\hat e_l^\bullet(\lambda) - \hat e_l^\bullet(0)| \le S_{\mathrm{en}} \quad (\lambda \in \hat{\mathfrak{D}}_{l-1}^\bullet).$$

For the original flow the same bound holds for $e_l^\bullet$ on $\mathfrak{D}_{l-1}^\bullet$, $1 \le l \le t_1^\bullet$.

(e) These vacuum energies and their discs depend only on K, b, m, the level and the construction constants. They are unchanged when the reference coupling or torus side is changed while those data are fixed.

(f) On a nonwrapping polymer, the restarted activity at λ is the fitting part of its infinite-lattice counterpart, with the same pointers. The infinite-lattice family and its pointed totals are covariant under translations by $M\mathbb{Z}^3$ in level units. Moreover

$$\hat e_l(\lambda) + \frac{1}{2}\frac{\dim B_{l-1}}{n_l}\operatorname{Log}\hat\tau_{l-1}(\lambda)$$

is the cell mean of the scalar pointed totals of the new infinite-lattice activity, including its Gaussian part, at the trivial configuration before extraction. The same assertions hold for the original flow on its field-scale discs and on the least torus.

■

Proof. Fix $\bullet$ and abbreviate $t_i = t_i^\bullet$ throughout this proof; all unhatted and hatted operators refer to this same datum. We apply the induction of Prop. 14.14 from the restart level. We verify its numerical inputs, its exact composition, and its agreement between a torus and the infinite lattice.

The coefficient windows in Prop. B.13 give

$$|\hat x_i - \mathsf{t}_i| \le \bar B \le \mathsf{t}_i/4, \qquad |\hat s_i| \le 2\bar B_M/\mathsf{t}_i, \qquad \tfrac{3}{4}\mathsf{t}_i \le \hat x_i \le \tfrac{5}{4}\mathsf{t}_i.$$

Since $\mathsf{t}_i \ge L^u b \ge 13g^{-2}$, all the coefficient floors in Rem. 14.9 apply. In particular,

$$\hat x_i \ge e^{18\mathfrak{p}_0}, \qquad \mathsf{t}_i \ge 2\mathsf{t}_\flat, \qquad \hat x_i(\log \hat x_i)^{-2\mathfrak{p}_0} \ge c_\lambda/\rho^2,$$
$$C_w\phi(\hat x_i) \le \tfrac{1}{2}, \qquad e^{-\gamma_{\mathrm{tail}}\mathfrak{b}(\mathsf{t}_i)^2/2} \le c_\lambda^{-1/2}.$$

Thus $\hat\lambda_i^2 \ge (4/5)\rho^{-2} \ge 16/5$. The discs decrease with i, by monotonicity of $x(\log x)^{-2\mathfrak{p}_0}$ and of the running coefficients. Their upper radius is smaller than the one used in the original induction: $\hat\lambda_i^2 \le 4\hat x_i/(5c_\lambda) \le \mathsf{t}_i/c_\lambda$. Consequently

$$|\hat\xi_i(\lambda) - \hat x_i| \le (1+\hat\lambda_i^2)\bar B \le r_x\hat x_i, \qquad |\hat\tau_i(\lambda) - 1| \le \bar B/c_\lambda \le r_x/4,$$
$$\hat\lambda_i\delta(\mathsf{t}_i) \le A_1 2^{\mathfrak{p}_0+1}c_\lambda^{-1/2} \le \delta(\mathsf{t}_\flat).$$

The radius-ratio proof of eq. 14.xxiii uses only the same coefficient windows and $\log \hat x_i \ge 18\mathfrak{p}_0$. It therefore gives, for $n = k+1-l$,

$$(\hat\lambda_k/\hat\lambda_{l-1})^2 \le 2L^{-n}(1+n\log L)^{2\mathfrak{p}_0}, \qquad (\hat\lambda_k/\hat\lambda_{l-1})^2 \le 2^{10/9}L^{-8n/9} < \tfrac{1}{4}.$$

For the mass member and increment we obtain

$$|\lambda^2\hat s_i| \le 2\bar B_M/c_\lambda, \qquad |\lambda^2\hat\mu_{i+1}| \le 4L\bar B_M/c_\lambda \le 6L\bar B_M/c_\lambda.$$

For the second inequality use $\hat s_{i+1} = L\hat s_i + \hat\mu_{i+1}$ and $\mathsf{t}_{i+1} = \mathsf{t}_i/L$.

These are all the numerical inputs of the field-scale induction. The coefficient-polydisc assertion of Prop. B.5 constructs the structure, quadratic origin and Gaussian part at $\hat\xi_i(\lambda)$; at the first step $\hat\xi_{t_0} = \mathsf{t}_{t_0}$ and the quadratic origin is zero. The fixed-calibration Schwarz bounds apply because $\mathsf{t}_i \ge \mathsf{t}_\flat$ and the scaled box is contained in the fixed box.

The fermion tower and fluctuation operators are those of the original datum, on the same regular complex domain. The normalizer families $\mathbf{N}^{\mathrm{un},(l)}$, $\mathbf{N}^{\mathrm{F},(l)}$ and $\mathbf{c}_l$ are unchanged by the restart and do not depend on λ. The Gaussian family is evaluated at $|\hat\tau_{l-1} - 1| \le r_x/4$, and the new fluctuation

family has norm at most one. Their scalar arrays therefore obey the same E_0 bound, with the same K_{old}. Holomorphy, joint invariance and property (N) are preserved by expansion and extraction, so the old-activity estimate of [QCDM1, Lem. 7.9] applies.

For clarity, the moment calculation also survives the restart. Let $\widehat{Z}_l$ and Z_l be the quadratic zeroth moments before and after extraction, and let $\mathfrak{s}_l 1$ be the moment of $L^{-l}\mathsf{S}_1(\psi^{\mathrm{bg}})$. The restart extracts its own mass coefficient, hence $\widehat{Z}_l(1) = -\mathfrak{s}_l \hat{\mu}_l 1$. If z_j are its even Taylor coefficients, then

$$Z_l(\lambda) = \widehat{Z}_l(\lambda) - \lambda^2 \widehat{Z}_l(1) = \sum_{j\geq 2} z_j(\lambda^{2j} - \lambda^2).$$

The radius ratios give the same lag estimate $\varsigma_i(\lambda) \leq 3K_\mu C_\omega S_\omega \eta$. Initially there are no old families, so $\varsigma_{t_0} = 0$ and $\mathcal{V}^{\mathrm{old},\lambda} = 0$. The remaining base vertices are the gauge and fermion vertices and the scaled initial mass variation. The extracted local mass costs at most $6Lc_{\mathsf{S}} h_F^2 \bar{B}_M / c_\lambda$ per cube. The remaining gauge shift

$$\hat{b}^r_{i+1} = \hat{b}_{i+1} - e_{\mathsf{N},i+1} - b^F_{i+1}(\mathfrak{m}_i)$$

satisfies $|\hat{b}^r_{i+1}| \leq C_w \phi(\hat{x}_i) \leq 1/2$. The original closing estimate therefore proves (b) with the same C_η. Holomorphy and simultaneous-sign symmetry follow at each step; changing B to $-B$ in the symmetric Gaussian integral makes its output even. At $\lambda = 1$ the data and extraction rule are exactly the original restarted ones, and at zero every vertex vanishes. This proves (a).

We next prove the composition and the independence of its scalar factor from the mass. The tested chart identity, the four local summations and the Gaussian identity are unchanged; the fermion completion of squares is [QCDM1, Lem. 7.3], at the datum's fine mass. The running coefficients telescope:

$$L^i \hat{x}_i = L^{t_0}\hat{x}_{t_0} - \sum_{t_0 < l \leq i} L^l \hat{b}_l, \qquad \lambda^{-2} L^i \hat{\xi}_i(\lambda) + \sum_{t_0 < l \leq i} L^l \hat{b}_l = \lambda^{-2}\beta^\bullet.$$

At the start, the constant weighted action is $\mathsf{wab}(c, W) = L^{t_0} c\, A(\mathsf{U}_{t_0}(W))$. For $c = \lambda^{-2} L^s b$ this is the action in (c). The start's density test is one on the composite tested chart by [QCDM1, Lem. 7.1].

Every mass-dependent or running-coefficient factor is retained in the output form. Specifically, $\hat{\tau}_i^{-\dim B_i/2}$ enters the extracted vacuum energy and Gaussian family; the fermion normalizer enters $\mathbf{N}^{\mathrm{F},(i+1)}$ and its extracted per-site constant; the tower, mass member and old activities remain as their displayed forms at the new background; the joined integral is the exponential of its new activity; and the calibration Gaussian normalizer enters $\mathbf{N}^{\mathrm{un},(i+1)}$ and its extracted constant. The extracted gauge and mass terms enter the action and mass member. The scalar left outside is therefore a gauge-step number depending only on the lattice and calibration. It depends on no field, source, mass or running coefficient. This argument also applies at every original-flow step in Prop. 14.14. Compose the identities and use Fubini; Berezin integration is a finite operation on the generators. This proves (c).

The vacuum-energy bound requires only scalar counting. The fluctuation family before extraction has pointed norm at most

$$K^\bullet_M(C^\lambda_V + C^\lambda_F + C_\eta + 2)c_\lambda^{-1/2} \leq \tfrac{1}{4}.$$

The Gaussian family contributes at most $2048E_0 \sum_{d\geq 0} e^{-d} \leq 3300E_0$ to its pointed total, using $2048 \cdot 9216^d$ for the polymer count and $\kappa + 1 - \log 9216 \geq 1$. Also $\dim B_i / n_{i+1} \leq 9L^3$ and $|\operatorname{Log} \hat{\tau}_i| \leq 2|\hat{\tau}_i - 1| \leq r_x/2 \leq 2/9$. The normalizers and calibration numbers are independent of λ. Subtracting the value at zero gives

$$|\hat{e}_{i+1}(\lambda) - \hat{e}_{i+1}(0)| \leq 2(1/4 + 3300E_0) + L^3 \leq S_{\mathrm{en}}.$$

For the original flow the same calculation uses $|\tau_i - 1| \leq 5/64$, which also gives $|\operatorname{Log} \tau_i| \leq 2/9$. This proves (d).

For (e), perform the induction on $\mathbb{Z}^3$. Each step uses its earlier infinite-lattice families, the pair $(\mathsf{t}_i, \xi_i(\lambda))$, the fixed-calibration gauge vertices at $\lambda(\mathsf{t}_\flat/\mathsf{t}_i)^{1/2}B$, the fermion objects at the fine mass and field, the mass member $\lambda^2 s_i$, and the original extracted numbers. At $\lambda = 1$ these numbers are determined by the infinite-lattice step. Starting from $x_0 = L^{K^\bullet}b$, $s_0 = 0$, or from $\hat{x}_{t_0} = L^s b$, $\hat{s}_{t_0} = F^{\mathrm{UV}}/(L^s b)$, their recursions therefore involve no torus or reference coupling. The normalizer constants depend only on the calibration and fine mass. The radii depend only on the running coefficients. Finally, the additional term $-\frac{1}{2}(\dim B_i/n_{i+1})\operatorname{Log}\tau_i$ has a fixed number of fluctuation coordinates per site, determined by the same bond rule and L on every torus. The asserted independence follows.

For (f), induct simultaneously on torus fitting and translation covariance, starting with the empty family. The local structure, quadratic origin, gauge vertices, fermion objects and old scalar terms are the same under a nonwrapping chart on the torus and its lift. For the gauge vertices and old scalar arrays this is the chart covariance of [YM3, Prop. 4.40](a7) and [YM3, Prop. 4.52](d); the local fermion walks and background maps have the corresponding property by [QCDM1, §6]. For old terms with fermion legs and lag less than n_{lag}, the pair, fluctuation, old activity and fermion maps are carried together. At larger lags use the torus filing in [QCDM1, Lem. 7.9], with the nonzero cell moments of Lem. 14.13: the fitting lifted term retains its cell sum $M^3 Z_l(\lambda)$, and the discrepancy of torus cell sums is filed on wrapping polymers.

The new expansion at X depends only on potentials whose enlarged supports lie in X. Extraction removes infinite-lattice numbers, and the normalizer families are unchanged by λ. Thus the new pointed activity fits on every nonwrapping polymer. Translations by $M\mathbb{Z}^3$ carry all of these ingredients and the filing rule; by the induction the old arrays correspond to themselves. Expansion, pointing and extraction commute with these translations. This proves covariance of the new infinite-lattice family and its totals. The extraction rule takes their scalar cell mean and adds precisely $-\frac{1}{2}(\dim B_{l-1}/n_l)\operatorname{Log}\tau_{l-1}$, proving the last assertion. The original flow follows by the same induction from level zero. □

The growing field-scale discs make the Taylor remainder summable over the fine levels. In three dimensions the vacuum-volume weight costs a_l^{-3}; the first omitted term, of degree eight, supplies four powers of a_{l-1}.

Lemma 15.6 (Truncation at degree six)**.** Under (k0) and $u \geq u_*$ at $n = 1$, for $b \in I_g$ and $m_- \leq m_1 < m_2 \leq m_+$,

$$\left|[\mathsf{v}^{\mathrm{B}}(1;b,\cdot) - \mathsf{v}^{\mathrm{A}}(1;b,\cdot)]_{m_1}^{m_2} - \sum_{n=0}^{3} b^{-n}\mathsf{q}_n(K;b;m_1,m_2)\right| \leq \varepsilon_{\mathrm{rem}}(K).$$

■

Proof. For the original flow, $e_l(\lambda) - e_l(0)$ is even, holomorphic and bounded by S_{en} on the disc of radius $\bar{\lambda}_{l-1}$. Since $\bar{\lambda}_{l-1}^2 \geq 2\rho^{-2} \geq 8$, Cauchy's estimate and the geometric tail give

$$\left| e_l(1) - \sum_{n=0}^{3} e_{l,[2n]} \right| \leq 2S_{\mathrm{en}}\bar{\lambda}_{l-1}^{-8}.$$

The radius estimate gives

$$\bar{\lambda}_{l-1}^{-2} \leq c_\lambda(\log x_{l-1})^{2\mathfrak{p}_0} a_{l-1}/b, \quad \log x_{l-1} \leq (K+2)\log L + \log(2b_+).$$

For the restart use $\hat{\lambda}_{l-1}^2 \geq 16/5$ and $\sum_{n\geq 4}(5/16)^{n-4} \leq 2$. Its coefficient window gives

$$\hat{\lambda}_{l-1}^{-2} \leq \tfrac{5}{3} c_\lambda(\log \hat{x}_{l-1})^{2\mathfrak{p}_0} a_{l-1}/b,$$

with the same logarithmic bound. Now

$$a_{l-1}^4 a_l^{-3} = L^{-4} a_l, \qquad \sum_{l \le t_1^\bullet} a_l \le L^{1-u}/(L-1).$$

At each mass the weighted remainder of either flow is therefore at most

$$\frac{2S_{\mathrm{en}}}{L-1}\left(\frac{5}{3}c_\lambda\right)^4 ((K+2)\log L + \log(2b_+))^{8\mathfrak{p}_0} b_-^{-4} L^{-3-u}.$$

Sum over the two masses, two flows and two data. The definition of $\varepsilon_{\mathrm{rem}}$ dominates that sum. The coefficients of the retained polynomial are precisely $b^{-n}\mathsf{q}_n$, proving the claim. □

The expansion has reduced the problem to four coefficients. For the three nonconstant coefficients, the essential step is to remove the calibration tests. The untested Gaussian integrals have an exact homogeneity in b; the tests obscure that homogeneity but change these finitely many coefficients by a rapidly decreasing error.

Lemma 15.7 (Gaussian coefficients and the calibration tests)**.** Assume (k0) and $u \ge u_*$ at $n = 1$. Fix $\bullet \in \{\mathrm{A}, \mathrm{B}\}$, $n \in \{1, 2, 3\}$ and $m_- \le m_1 < m_2 \le m_+$. On the least torus put

$$j = t_1^\bullet, \quad N_K^{\mathrm{b}} = 3\ell_G^3 L^{3(K+1)}, \quad \beta_+ = L^{K+1} b_+, \quad \theta = b_F, \quad \gamma_{\mathrm{tail}} = 2(LM)^{-3}.$$

Let K_M be the cluster-expansion constant and set

$$\tau_K = N_K^{\mathrm{b}} K_M \exp[-\tfrac{1}{2}\gamma_{\mathrm{tail}} A_1^2 (u+1)^2 (\log L)^2],$$

$$\Pi_K = (1+\beta_+) N_K^{\mathrm{b}} (2L^{K+1})^6 \left(1 + \frac{32 \bar{B}_M C_D c_{\mathfrak{S}} \ell_G^3 L^{3s} (8\theta^2 C_D^2 + 1)^s}{L^s b_-}\right)^3.$$

There are constants $C^{\mathrm{v}}, c^{\mathrm{v}} \ge 1$, depending only on the construction constants, for which

$$\varepsilon_{\mathrm{box}}(K) = 8 b_+^3 \ell_G^{-3} \left[C^{\mathrm{v}} \Pi_K^{c^{\mathrm{v}}} (\tau_K^{1/2} + \tau_K) e^{c^{\mathrm{v}} \tau_K} + 2\varepsilon_{\mathfrak{f}}(K; \ell_G)\right]$$

has the following property. There are real numbers $\mathscr{L}_n^\bullet, \hat{\mathscr{L}}_n^\bullet$, independent of b, such that for every $b \in I_g$,

$$\left| b^n \sum_{l \le j} (a_l^\bullet)^{-3} [e_{l,[2n]}^\bullet]_{m_1}^{m_2} - \mathscr{L}_n^\bullet \right| \le \varepsilon_{\mathrm{box}}(K)/8,$$

$$\left| b^n \sum_{t_0^\bullet < l \le j} (a_l^\bullet)^{-3} [\hat{e}_{l,[2n]}^\bullet]_{m_1}^{m_2} - \hat{\mathscr{L}}_n^\bullet \right| \le \varepsilon_{\mathrm{box}}(K)/8.$$

These numbers may depend on the chamber operator, N_f, the construction constants, K, s, u, m_1, m_2 and ℓ_G. In particular,

$$|\mathsf{q}_n(K; b; m_1, m_2) - \mathsf{q}_n(K; b'; m_1, m_2)| \le \varepsilon_{\mathrm{box}}(K), \qquad b, b' \in I_g,$$

and $C_{\mathrm{box}}(g) = \sup_K L^K \varepsilon_{\mathrm{box}}(K) < \infty$, where the supremum is over the admitted cutoffs. ■

Proof. Evaluate the level-j form at the trivial block field and zero remaining fermion generators. The action, tower form, mass member and all terms with legs vanish there. The density test is one. Exact composition from Prop. 14.14 and Lem. 15.5 therefore gives

$$\exp\left[\sum_{l \le j} n_l^G e_l^\bullet(\lambda) + f_j(\lambda)\right] = \mathsf{n}_j I_j(\lambda; x), \qquad n_l^G = \ell_G^3 (a_l^\bullet)^{-3},$$

at mass $x \in [m_-, m_+]$. Here f_j sums the scalar activities at the trivial pair, and $\mathsf{n}_j \ne 0$ is independent of the mass. The fermion factor in I_j is $\theta_j^{-\dim H_j} \det \mathcal{K}_j[U; a_{K^\bullet} x]$. For the restart the same identity holds with hats and the sum restricted to $t_0^\bullet < l \le j$.

The wrapping estimate of Lem. 15.3 remains valid for $|\lambda| \le 1$. Indeed Lem. 15.5(b),(f) supplies the scalar array bound, fitting relation, translation covariance and cell-mean extraction used in that estimate. The normalizer families are independent of λ. Thus

$$|f_j(\lambda)|, |\hat{f}_j(\lambda)| \le \varepsilon_{\mathfrak{f}}(K; \ell_G).$$

The unit circle is inside both field-scale discs, so every Taylor coefficient has the same bound.

We now express the composite as a tested Gaussian mean. Use the chain of triples in the proof of Prop. 14.42, starting at $(\mathbf{1}, 0, 0)$ at level j. For the original flow descend to level one and the bare field; for the restart stop at level $t_0 = t_0^\bullet$. At real λ, the frozen rotations of that proof identify the composite integrand with the triple-chain integrand. They preserve the boxes and Lebesgue measure. The three restarted start terms also respect these rotations: the action is gauge invariant, the tower is covariant, and the mass member is a sum of $\bar\psi\psi$. Hence the two box integrals agree.

Write this integrand as its value at zero times $e^{R(\lambda,B)}$, with $R(0,B) = 0$. At zero its fluctuation measure is the product Gaussian $\mu = \prod_i \mu_i$ at the calibrations, with tests $\chi = \prod_i \chi_i$; the levels are $i < j$ for the original flow and $t_0 \le i < j$ for the restart. Consequently

$$I_j(\lambda; x)/I_j(0; x) = \mathsf{E}^\chi e^{R(\lambda,B)}, \qquad \mathsf{E}^\chi F = \frac{\int \chi F \, \mathrm{d}\mu}{\int \chi \, \mathrm{d}\mu},$$

and likewise with hats. We next justify the holomorphy and coefficient bounds used in this identity.

For the original chain take $r_j = L^{-j}/2$. The chart, gauge-fixing, Jacobian and elimination-factor bounds from the proof of Prop. 14.42 give $|R| \le C_R(1 + \beta^\bullet) N_K^{\mathrm{b}}$ on this disc. The action divided by λ^2 is bounded by the maximum principle, since its plaquette terms vanish to second order at zero. The fermion determinant logarithm is bounded by the regular-complex-field expansion in [QCDM1, Prop. 6.9].

For the restart the fine field at level t_0 is $\mathscr{U}_{t_0} = e^{iX}\mathbf{1}$ with $|X(b)| \le \alpha_1 L^{-t_0}$ on the field polydiscs. It is holomorphic in λ and $X(0) = 0$; Schwarz's lemma gives $|X(b)| \le \alpha_1 L^{-t_0}|\lambda|$. On $|\lambda| < L^{-(j-t_0)}$ it is therefore regular through depth j. The gauge factors obey the preceding bounds on this disc.

The restarted fermion factor needs an additional radius because its initial form includes the scaled mass member. Let M_λ be that member's kernel and form

$$\hat\Gamma_i = \hat{\mathbb{D}}_i + \theta q^* q, \qquad \hat{\mathbb{D}}_{i+1} = \theta - \theta^2 q \hat\Gamma_i^{-1} q^*.$$

At $\mathsf{M}_\lambda = 0$ these are the original tower operators at $\mathscr{U}_{t_0}$, and the regular complex bounds give

$$\operatorname{Re}\Gamma_i \ge (2C_D)^{-1}, \qquad \|\Gamma_i\| \le 2C_D + \theta.$$

The kernel of the unscaled mass form is $L^{-t_0}\bar{\mathsf{H}}\mathsf{H}$. The row bounds for each background map are $2C_H L^{-t_0}$. Summing over the $\nu_P L^{3t_0}\ell_G^3 L^{3s}$ bare places gives

$$\|L^{-t_0}\bar{\mathsf{H}}\mathsf{H}\| \le 4\nu_P C_H^2 \ell_G^3 L^{3s} \le c_{\mathsf{S}} \ell_G^3 L^{3s}.$$

The last inequality uses $M^3\nu \ge 4$. Since $|\hat{s}_{t_0}| \le 2\bar{B}_M/(L^s b)$,

$$\|\mathsf{M}_\lambda\| \le \frac{2|\lambda|^2 \bar{B}_M c_{\mathsf{S}} \ell_G^3 L^{3s}}{L^s b}.$$

If $d_i = \|\hat\Gamma_i - \Gamma_i\| \le (4C_D)^{-1}$, the resolvent identity yields $\|\hat\Gamma_i^{-1}\| \le 4C_D$ and $d_{i+1} \le 8\theta^2 C_D^2 d_i$. Starting with $d_{t_0} = \|\mathsf{M}_\lambda\|$, all these bounds therefore hold on $|\lambda| \le 2\hat{r}_j$, where

$$\hat{r}_j = \frac{1}{2}\min\left\{ L^{-(j-t_0)}, \left(\frac{L^s b_-}{8\bar{B}_M C_D c_{\mathsf{S}} \ell_G^3 L^{3s}(8\theta^2 C_D^2 + 1)^s} \right)^{1/2} \right\}.$$

The determinant product is holomorphic and nonzero there. Moreover

$$\sum_{t_0 \le i < j} \dim H_i \le 2\nu N_K^{\mathrm{b}}, \quad \|\hat\Gamma_i\| \le 2C_D + \theta + 1, \quad |\det \Gamma_i[\mathbf{1}]| \ge C_D^{-\dim H_i}.$$

The real part of its logarithmic change from zero is at most $2\nu N_K^{\mathrm{b}} \log((2C_D+\theta+1)C_D)$. Borel–Carathéodory bounds the modulus on the half disc by twice this number. Thus, for either chain, with $r_j = L^{-j}/2$ or $\hat{r}_j$ as appropriate,

$$|R(\lambda, B)| \le C_R(1+\beta^\bullet)N_K^{\mathrm{b}} \qquad (|\lambda| \le r_j).$$

This proves the asserted holomorphy uniformly on the boxes.

Let R_k be the Taylor coefficient of λ^k. Cauchy's estimate bounds it on the field polydiscs by $C_R(1+\beta^\bullet)N_K^{\mathrm{b}} r_j^{-k}$. It is a fluctuation polynomial of degree at most $k+2$; the additional degree comes from the action's factor λ^{-2}. Balancedness bounds each homogeneous part by the same number. Therefore, on arbitrary real fields,

$$|R_k(B)| \le (k+3)C_R(1+\beta^\bullet)N_K^{\mathrm{b}} r_j^{-k}(1+|B|_{\mathrm{sc}})^{k+2},$$

where

$$|B|_{\mathrm{sc}} = \max_{i,b} \frac{\mathsf{t}_i^{1/2}|B_i(b)|}{\mathsf{C}_{\mathcal{A}}\mathfrak{b}(\mathsf{t}_i)}.$$

The scaled Gaussian variances are uniformly bounded, so every fixed even moment of $|B|_{\mathrm{sc}}$ is at most a constant times N_K^{b}. Also

$$r_j^{-6} \le (2L^{K+1})^6 \left(1 + \frac{32\bar{B}_M C_D c_{\mathsf{S}} \ell_G^3 L^{3s}(8\theta^2 C_D^2+1)^s}{L^s b_-}\right)^3.$$

It follows that the squared Gaussian moments of products of at most six coefficients R_k, $k \le 6$, are bounded by a fixed power of Π_K.

The coefficient of order $2n \le 6$ in $\log \mathsf{E}^\chi e^R$ is a fixed polynomial in tested means of products of at most $2n$ such coefficients. Replacing every tested mean by its full Gaussian mean defines the corresponding Laplace coefficient. The probability of a failed test satisfies

$$\int (1-\chi)\,\mathrm{d}\mu \le \sum_i N_{i+1}^{\square} K_M e^{-\gamma_{\mathrm{tail}}\mathfrak{b}(\mathsf{t}_i)^2/2} \le \tau_K.$$

Here $\mathsf{t}_i \ge L^{u+1}b \ge L^{u+1}$ and $\mathfrak{p}_0 \ge 1$. Apply eq. 14.l and Hölder to each factor in the finite polynomial. The tested coefficient differs from the Laplace coefficient by at most

$$\tfrac{1}{2}C^{\mathrm{v}}\Pi_K^{c^{\mathrm{v}}}(\tau_K^{1/2}+\tau_K)e^{c^{\mathrm{v}}\tau_K}.$$

Taking the maximum over $n = 1, 2, 3$ fixes the two constants.

The full Gaussian coefficient has exact homogeneity. With $B_i' = (\beta^\bullet)^{1/2}B_i$ and $\epsilon = \lambda(\beta^\bullet)^{-1/2}$, the composite integrand depends on ϵ, B' and not on b. Indeed its arguments are λB_i, its action and gauge-fixing coefficients are $\beta^\bullet/\lambda^2$ and $L^{-i}\beta^\bullet/\lambda^2$, and the restarted mass coefficient is $\lambda^2 F^{\mathrm{UV}} L^{t_0}/\beta^\bullet$. In the new variables the Gaussian covariance is

$$\beta^\bullet \mathsf{t}_i^{-1}(\mathsf{A}_i^{\mathrm{cv}})^{-1} = L^i(\mathsf{A}_i^{\mathrm{cv}})^{-1}.$$

Thus the coefficient of λ^{2n} is $(\beta^\bullet)^{-n}$ times a number independent of b. Define $\mathscr{L}_n^\bullet$ to be $\ell_G^{-3}L^{-nK^\bullet}$ times the mass increment of the corresponding ϵ^{2n} coefficient, and define its hatted version in the same way. These are real: the real composite and its Gaussian Taylor coefficients have the conjugation symmetry of the chamber determinant.

Finally take the mass increment of the exact exponential identity at the start of the proof. Near $\lambda = 0$ the tested means lie in $|z-1| < 1$. The difference between the vacuum-energy sum plus f_j and $\mathrm{Log}\,\mathsf{E}^\chi e^R$ has constant mass increment, since its exponential is independent of λ after normalization at zero and its remaining prefactor is independent of the mass. Equivalently, on real λ this increment

lies in one coset of $2\pi i\mathbb{Z}$; holomorphy makes it constant. Its positive-order Taylor coefficients vanish. Hence

$$\ell_G^3 \sum_{l\le j} (a_l^\bullet)^{-3} [e^\bullet_{l,[2n]}]_{m_1}^{m_2} = [\log \mathsf{E}^\chi e^R]^{m_2}_{[2n];m_1} - [f_{j,[2n]}]_{m_1}^{m_2}.$$

Multiply by $b^n \ell_G^{-3}$ and use $b^n(\beta^\bullet)^{-n} = L^{-nK^\bullet}$ and $b^n \le b_+^3$. The test error at both masses and the two torus errors give the first estimate. The restarted identity gives the second. Four signed contributions form q_n, so comparison at two labels costs at most $8\varepsilon_{\mathrm{box}}/8$.

To verify the final assertion, Π_K grows at most as a fixed power of L^K, since $s \le K$. In contrast,

$$\tau_K^{1/2} \le (N_K^{\mathrm{b}} K_M)^{1/2} e^{-\gamma_{\mathrm{tail}} A_1^2 (c_\# K/340)^2 (\log L)^2/4}.$$

The wrapping error decreases faster than every power of L^{-K}, since its sum starts at $d \ge c_\# K/340$. These bounds prove $C_{\mathrm{box}}(g) < \infty$. □

The coefficient of degree zero is different: at zero field scale the original and restarted towers agree above their common start. Their difference is a free determinant through that start. This identifies the two largest divergences.

With $\nu = 8N_f$, $\nu_{\mathrm{c}} = 48\nu$ and the symbol of Lem. 11.21, put

$$I^{\mathrm{fr}}(\mu) = \int_{[-\pi,\pi]^3} \mathrm{tr}(\widehat{D}(p) + \mu)^{-1} \frac{\mathrm{d}^3 p}{(2\pi)^3},$$

$$I^{\mathrm{fr}\prime}(0) = -\int_{[-\pi,\pi]^3} \mathrm{tr}\, \widehat{D}(p)^{-2} \frac{\mathrm{d}^3 p}{(2\pi)^3}, \qquad y_K^{\mathrm{fr}}(m) = -L^{2K} I^{\mathrm{fr}}(L^{-K} m).$$

Choose the symbol threshold p_* and constants $C_{\mathrm{pr}}, C_{\mathrm{lt}}$ as in the proof of Lem. 11.21, and set

$$C_I = 2\nu_{\mathrm{c}} C_{\mathrm{pr}}^3 p_*^{-3} + \frac{2}{(2\pi)^3}\left(\frac{5}{2}\pi^2\nu + 12\pi\nu_{\mathrm{c}} C_{\mathrm{pr}}^2 C_{\mathrm{lt}} p_*\right).$$

The integrals and their mass regularity are justified in Lem. 15.8.

Lemma 15.8 (The free coefficient)**.** (a) I^{fr} is continuously differentiable on $[0, 1/2]$ and twice continuously differentiable on $(0, 1/2]$, with $|I^{\mathrm{fr}\prime\prime}| \le C_I$. In particular,

$$|I^{\mathrm{fr}}(\mu) - I^{\mathrm{fr}}(0) - \mu I^{\mathrm{fr}\prime}(0)| \le \tfrac{1}{2} C_I \mu^2,$$

and, for $0 < m \le 1/2$,

$$|y_K^{\mathrm{fr}}(m) + I^{\mathrm{fr}}(0) L^{2K} + I^{\mathrm{fr}\prime}(0) m L^K| \le \tfrac{1}{2} C_I m^2.$$

(b) Assume (k0), $u \ge u_*$ and $s \ge s_F$ at $n = 1$, and let $m_- \le m_1 < m_2 \le m_+$. Then $\mathsf{q}_0(K; b; m_1, m_2)$ is independent of b, and

$$\left|\mathsf{q}_0(K; b; m_1, m_2) + \int_{m_1}^{m_2} (y_{K+1}^{\mathrm{fr}} - y_K^{\mathrm{fr}})\, \mathrm{d}m\right| \le (m_2 - m_1) \frac{4\nu K_c}{c_b m_-^2} L^{5s-K}.$$

Here K_c is the block-comparison constant in Prop. A.41, and c_b is the accretivity constant used in Lem. 11.23.

(c) Consequently,

$$\left|\mathsf{q}_0(K; b; m_1, m_2) - I^{\mathrm{fr}}(0)(L^2 - 1)L^{2K}(m_2 - m_1) - \tfrac{1}{2} I^{\mathrm{fr}\prime}(0)(L-1)L^K(m_2^2 - m_1^2)\right|$$
$$\le (m_2 - m_1)\left(C_I m_+^2 + \frac{4\nu K_c}{c_b m_-^2} L^{5s-K}\right).$$

■

Proof. The symbol bound in Lem. 11.21 gives

$$|\operatorname{tr}(\widehat{D}(p)+\mu)^{-k}| \le \nu_{\mathrm{c}} C_{\mathrm{pr}}^{k} |p|^{-k}, \qquad k=1,2.$$

These majorants are integrable in three dimensions. Dominated convergence gives continuity at zero and $I^{\mathrm{fr}\prime}(\mu) = -\int \operatorname{tr}(\widehat{D}(p)+\mu)^{-2}\,\mathrm{d}^3p/(2\pi)^3$. For positive μ, differentiation once more gives $I^{\mathrm{fr}\prime\prime}(\mu) = 2\int \operatorname{tr}(\widehat{D}(p)+\mu)^{-3}\,\mathrm{d}^3p/(2\pi)^3$.

On $|p| \ge p_*$ the trace is at most $\nu_{\mathrm{c}} C_{\mathrm{pr}}^3 p_*^{-3}$. On $|p| \le p_*$ the light-block comparison in the proof of Lem. 11.21, at spacing one, bounds the difference of the lattice resolvent and the embedded continuum resolvent by C_{lt}; both resolvents have norm at most $C_{\mathrm{pr}}|p|^{-1}$. Hence their cubed traces differ by at most $3\nu_{\mathrm{c}} C_{\mathrm{pr}}^2 C_{\mathrm{lt}} |p|^{-2}$. Its integral over the small ball is at most $12\pi\nu_{\mathrm{c}} C_{\mathrm{pr}}^2 C_{\mathrm{lt}} p_*$. The continuum cubed trace is $\nu h^{\mathrm{c}}(p;\mu)$, and the change $p = \mu q$ gives

$$\begin{aligned}\left|\int_{|p|\le p_*} h^{\mathrm{c}}(p;\mu)\,\mathrm{d}^3p\right| &\le 4\pi \int_0^{\infty} \frac{q^2(1+3q^2)}{(1+q^2)^3}\,\mathrm{d}q \\ &= 4\pi(\pi/16 + 9\pi/16) = \tfrac{5}{2}\pi^2.\end{aligned}$$

Adding the three bounds gives C_I. Taylor's formula on $[\epsilon,\mu]$, followed by $\epsilon \downarrow 0$, proves (a).

At $\lambda = 0$ both flows have only the calibration gauge tests and Gaussian fermion integrals. The scalar energy at level l is its infinite-lattice fermion normalizer constant e_l^{F} plus gauge numbers independent of the mass and running coefficients. Above $t_0^{\bullet}$ both flows use the same tower, and their energies at zero agree. Therefore

$$[\mathsf{v}_{[0]}^{\bullet}]_{m_1}^{m_2} = \sum_{l \le t_0^{\bullet}} (a_l^{\bullet})^{-3} [e_l^{\mathrm{F}}]_{m_1}^{m_2}.$$

This proves independence of b.

We identify the sum by first putting the free tower on a torus of side $\ell' \in \mathfrak{L}(g)$ and then letting $\ell' \to \infty$. At level l,

$$\log \delta Z_{l-1}^{\mathrm{F}}(\mathbf{1}) = -\nu n_l' \log\theta + \mathsf{L}_l^{\mathrm{F}} = n_l' e_l^{\mathrm{F},0}(\ell'), \qquad n_l' = \ell'^3 (a_l^{\bullet})^{-3}.$$

The closed-walk expansion in [QCDM1, Prop. 6.9] bounds the difference from $n_l' e_l^{\mathrm{F}}$ by $C\nu n_l' e^{-c\ell' L^{K^{\bullet}-l}}$. For fixed K the sum of these differences, divided by ℓ'^3, tends to zero. The Schur determinant identity of Lem. 11.22 telescopes along finitely many mass segments of length less than $m_-/2$:

$$\begin{aligned}\sum_{l \le t_0^{\bullet}} [\mathsf{L}_l^{\mathrm{F}}]_{m_1}^{m_2} &= [\log\det(D^{\bullet}[1] + a_{K^{\bullet}} m)]_{m_1}^{m_2} \\ &\quad - [\log\det \mathbb{D}_{t_0^{\bullet}}^{\bullet}(1; a_{K^{\bullet}} m)]_{m_1}^{m_2}.\end{aligned}$$

For the fine determinant, with $a = a_{K^{\bullet}}$ and $N' = \ell'/a$, the normalized mass increment is the Riemann sum

$$a^{-3} N'^{-3} \sum_{p \in P_{N'}} \int_{m_1}^{m_2} a \operatorname{tr}(\widehat{D}(p) + am)^{-1}\,\mathrm{d}m.$$

Its integrand is continuous at these positive masses. It converges to $a^{-2}\int_{m_1}^{m_2} I^{\mathrm{fr}}(am)\,\mathrm{d}m = -\int_{m_1}^{m_2} y_{K^{\bullet}}^{\mathrm{fr}}\,\mathrm{d}m$.

It remains to compare the two block determinants at depth s. The proof of Lem. 11.23 uses the depth only through (k0), $s \ge s_F$, the spacing L^{-s} and $\xi = L^{s-K}$, so it applies at the present depth. Its logarithmic ratio φ^{bl} satisfies

$$|\varphi^{\mathrm{bl}}| \le \nu \ell'^3 L^{3s} K_c L^{s-K} / (c_b L^{-s} m/2) \qquad (|\zeta| < m/2).$$

Cauchy's first-derivative estimate on radii tending to $m/2$ gives

$$\ell'^{-3} |\partial_\zeta \varphi^{\mathrm{bl}}(0)| \le \frac{4\nu K_c}{c_b m_-^2} L^{5s-K}.$$

This is uniform in ℓ'. Integrate in m and take the infinite-volume limit of the free identity above. The resulting estimate is exactly (b). This auxiliary free-volume limit does not change the fixed torus of the interacting theorem. Part (a) bounds the remaining free increment by $C_I m^2$; integrating proves (c). □

We have now separated the integrated increment into four powers of $1/b$. The coefficient of the zeroth power is exactly independent of b; the other three are independent up to a rapidly decreasing error.

Proposition 15.9 (Expansion of the cutoff increment)**.** For $K \geq K_{\mathrm{o}}(g,\ell)$, $b \in I_g$ and $m_- \leq m_1 < m_2 \leq m_+$,

$$\left| \int_{m_1}^{m_2} (y_{K+1} - y_K)(b,m)\,\mathrm{d}m + \sum_{n=0}^{3} b^{-n} \mathsf{q}_n(K;b;m_1,m_2) \right| \leq \varepsilon_{\mathrm{S}}(K).$$

For $n = 1,2,3$, changing b to another label of I_g changes q_n by at most $\varepsilon_{\mathrm{box}}(K)$. For $n = 0$ it does not change it. ■

Proof. The threshold supplies (k0), $u \geq u_*$ and $s \geq s_F$. Combine the contact identity Prop. 15.4 with the Taylor remainder Lem. 15.6. Their errors sum to ε_{S}. The coefficient assertions are Lems. 15.7 and 15.8. □

15.3. **Rescaling the coefficients.** The mass estimate now supplies an affine approximation to each coefficient. Four distinct labels isolate the four powers of $1/b$. Exact rescaling then gives a scalar recursion for each affine coefficient. The strict condition $Lm_- < m_+$ is used here: both a mass interval and its rescaled interval must lie in the original window.

For $0 < \beta_1 < \beta_2$, take four equally spaced nodes $b_k = \beta_1 + (k-1)(\beta_2 - \beta_1)/3$. Let (ϖ_{nk}) invert the matrix (b_k^{-n}) $(1 \leq k \leq 4,\ 0 \leq n \leq 3)$, and put

$$\mathsf{N}_V[\beta_1,\beta_2] = \max_n \sum_k |\varpi_{nk}|.$$

The inverse exists because the nodes are distinct. With $g' = g/\sqrt{L}$, define

$$\begin{aligned} I_1(g) &= [b_-(g), Lg^{-2} + B_\sharp - \bar{B}/L] = I_g \cap L^{-1} I_{g'}, \\ b_*(g) &= Lg^{-2} + \bar{B} = b_-(g') \in I_g \cap I_{g'}, \\ \mathsf{N}_V(g) &= \mathsf{N}_V[b_-(g), b_+(g)], \qquad \mathsf{N}_{V,1}(g) = \mathsf{N}_V[I_1(g)], \qquad B_{\mathrm{lab}} = 4(1 + b_-^{-1})^3. \end{aligned}$$

The first interval has positive length $(L-1)g^{-2} - 2B_\sharp/(L-1)$: use $\bar{B} = LB_\sharp/(L-1)$ and $g^{-2} \geq L\bar{B}$. Also $2\bar{B} \leq LB_\sharp$ since $L \geq 13$, proving the asserted membership of b_*. Put

$$m_R(g) = \max\{m_{\mathfrak{L}}(g), m_{\mathfrak{L}}(g')\}, \qquad \ell_R(g) = 2L^{m_R(g)}.$$

Then $\ell_R \in \mathfrak{L}(g)$ and $L\ell_R \in \mathfrak{L}(g')$.

For these comparison sides set

$$K_R(g) = \max\{K_{\mathrm{o}}(g,\ell_R), K_{\mathrm{o}}(g', L\ell_R)\}.$$

Lemma 15.10 (Affine coefficients and their rescaling)**.** Put $h = (m_+ - m_-)/2$. For $0 \leq n \leq 3$ define

$$\begin{aligned} \sigma_n^-(K) &= h^{-1} \mathsf{q}_n(K; b_*(g); m_-, m_c), \qquad \sigma_n^+(K) = h^{-1} \mathsf{q}_n(K; b_*(g); m_c, m_+), \\ \mathsf{z}_n(K) &= h^{-1}(\sigma_n^-(K) - \sigma_n^+(K)), \\ \mathsf{u}_n(K) &= -\tfrac{1}{2}(\sigma_n^-(K) + \sigma_n^+(K)) - m_c \mathsf{z}_n(K). \end{aligned}$$

They depend on K, the construction constants, the mass window and g through $b_*(g)$, but not on ℓ. Set

$$E_{\mathrm{A}}(K) = \mathsf{N}_V(g)[(2h)^3\varepsilon_\natural(K) + \varepsilon_{\mathrm{S}}(K) + B_{\mathrm{lab}}\varepsilon_{\mathrm{box}}(K)],$$
$$\varepsilon_{\mathrm{A}}(K) = 7E_{\mathrm{A}}(K) + \varepsilon_{\mathrm{box}}(K),$$
$$\varepsilon'_{\mathrm{A}}(K) = \frac{3}{h}[\varepsilon_{\mathrm{S}}(K) + B_{\mathrm{lab}}\varepsilon_{\mathrm{A}}(K)] + 2h^2\varepsilon_\natural(K).$$

Then:

(a) For $K \ge K_{\mathrm{o}}(g,\ell)$ and $b \in I_g$,

$$\bigl|\mathsf{q}_n(K;b;m_1,m_2) + \mathsf{u}_n(K)(m_2-m_1) + \tfrac12\mathsf{z}_n(K)(m_2^2-m_1^2)\bigr| \le \varepsilon_{\mathrm{A}}(K),$$
$$\left|(y_{K+1}-y_K)(b,m) - \sum_{n=0}^{3} b^{-n}(\mathsf{u}_n(K) + m\mathsf{z}_n(K))\right| \le \varepsilon'_{\mathrm{A}}(K).$$

The first estimate holds for every interval in $[m_-,m_+]$; the second holds throughout $W(g)$.

(b) Let $h_1 = (m_+ - Lm_-)/2 > 0$, $g' = g/\sqrt{L}$, and $\ell_R = \ell_R(g)$. Define

$$E_{\mathrm{R}}(K) = \mathsf{N}_{V,1}(g)\bigl[\varepsilon_{\mathrm{S}}(K+1;g,\ell_R) + L^3\varepsilon_{\mathrm{S}}(K;g',L\ell_R) + B_{\mathrm{lab}}(\varepsilon_{\mathrm{box}}(K+1;g) + L^3\varepsilon_{\mathrm{box}}(K;g'))\bigr]$$
$$+ \varepsilon_{\mathrm{A}}(K+1;g,\ell_R) + L^3\varepsilon_{\mathrm{A}}(K;g,\ell_R), \qquad \varepsilon_{\mathrm{R}}(K) = 3E_{\mathrm{R}}(K)/h_1^2.$$

For $K \ge K_R(g)$,

$$|\mathsf{u}_n(K+1) - L^{2-n}\mathsf{u}_n(K)| \le \varepsilon_{\mathrm{R}}(K), \qquad |\mathsf{z}_n(K+1) - L^{1-n}\mathsf{z}_n(K)| \le \varepsilon_{\mathrm{R}}(K).$$

(c) There are limits

$$\mathsf{G}_0 = \lim_K L^{-2K}\mathsf{u}_0(K), \qquad \mathsf{H}_0 = \lim_K L^{-K}\mathsf{z}_0(K), \qquad \mathsf{G}_1 = \lim_K L^{-K}\mathsf{u}_1(K),$$
$$w_1 = \lim_K \mathsf{z}_1(K), \qquad w_2 = \lim_K \mathsf{u}_2(K).$$

Write $\bar\varepsilon_{\mathrm{R}}(K) = \sup_{j\ge K}\varepsilon_{\mathrm{R}}(j)$ and $\mathsf{M}_R = \max_n \max\{|\mathsf{u}_n(K_R)|, |\mathsf{z}_n(K_R)|\}$. For $K \ge K_R$,

$$|\mathsf{u}_0(K) - \mathsf{G}_0L^{2K}|, |\mathsf{z}_0(K) - \mathsf{H}_0L^K|, |\mathsf{u}_1(K) - \mathsf{G}_1L^K| \le \bar\varepsilon_{\mathrm{R}}(K),$$
$$|\mathsf{z}_1(K) - w_1|,\ |\mathsf{u}_2(K) - w_2| \le \sum_{j\ge K}\varepsilon_{\mathrm{R}}(j),$$
$$|\mathsf{u}_3(K)|,\ |\mathsf{z}_2(K)|,\ |\mathsf{z}_3(K)| \le L^{-(K-K_R)}\mathsf{M}_R + \sum_{K_R\le j<K} L^{-(K-1-j)}\varepsilon_{\mathrm{R}}(j).$$

(d) The free limits are

$$\mathsf{G}_0 = -(L^2-1)I^{\mathrm{fr}}(0), \qquad \mathsf{H}_0 = -(L-1)I^{\mathrm{fr}\prime}(0).$$

All five limits are independent of the reference coupling $g \in (0,\gamma_{\mathrm{e}}]$. ∎

Proof. For (a), let b_k be the four nodes of I_g and let ϖ_{nk} be their inverse Vandermonde weights. Put $\delta_K = y_{K+1} - y_K$. By Thm. R, $|\partial_m^2\delta_K| \le \varepsilon_\natural(K)$. At each node expand about m_c:

$$\delta_K(b_k,m) = \alpha_k + \beta_k(m-m_c) + r_k(m), \qquad |r_k(m)| \le \tfrac12 h^2\varepsilon_\natural(K).$$

Apply Prop. 15.9, replacing each coefficient's label by b_* at cost $\varepsilon_{\mathrm{box}}$. Since $\sum_{p=0}^3 b_k^{-p} \le B_{\mathrm{lab}}$ and $\sum_k \varpi_{nk}b_k^{-p} = \delta_{np}$, we obtain

$$\left|\mathsf{q}_n(K;b_*;m_1,m_2) + \int_{m_1}^{m_2}(\tilde\alpha_n + \tilde\beta_n(m-m_c))\,\mathrm{d}m\right| \le E_{\mathrm{A}}(K),$$

where $\tilde\alpha_n=\sum_k \varpi_{nk}\alpha_k$ and $\tilde\beta_n=\sum_k \varpi_{nk}\beta_k$. The integrated Taylor error is at most $h^3\varepsilon_\natural$, which is covered by $E_{\mathrm A}$. Apply this estimate to the two half intervals. It gives

$$\sigma_n^{\mp}=-\tilde\alpha_n\pm\tfrac12 h\tilde\beta_n+O(E_{\mathrm A}/h),$$

where each indicated error has modulus at most its bound. Thus

$$|\mathsf z_n-\tilde\beta_n|\le 2E_{\mathrm A}/h^2,\qquad |-(\sigma_n^-+\sigma_n^+)/2-\tilde\alpha_n|\le E_{\mathrm A}/h.$$

The two affine functions differ by at most $3E_{\mathrm A}/h$ on the window. Integration over its length $2h$, followed by the previous estimate, gives $7E_{\mathrm A}$. Changing the coefficient's label back to b costs $\varepsilon_{\mathrm{box}}$ and proves the first assertion.

For the pointwise assertion put $D(m)=\delta_K(b,m)-\sum_n b^{-n}(\mathsf u_n+m\mathsf z_n)$. Its integral over every subinterval is at most $E'=\varepsilon_{\mathrm S}+B_{\mathrm{lab}}\varepsilon_{\mathrm A}$, and $|D''|\le\varepsilon_\natural$. Its affine Taylor polynomial $\bar D$ at m_c differs from it by at most $h^2\varepsilon_\natural/2$. The two half-interval integrals of $\bar D$ are bounded by $E'+h^3\varepsilon_\natural/2$, so

$$|\bar D|\le\frac{3}{h}(E'+h^3\varepsilon_\natural/2)$$

throughout the window. Adding the Taylor error proves $|D|\le 3E'/h+2h^2\varepsilon_\natural=\varepsilon'_{\mathrm A}$.

For (b), take $b\in I_1(g)$ and $Lm_-\le m_1<m_2\le m_+$. Then $Lb\in I_{g'}$ and $m_i/L\in[m_-,m_+]$. Exact rescaling gives

$$\delta_{K+1,\ell_R}(b,m)=L^2\delta_{K,L\ell_R}(Lb,m/L),$$

so its mass integral acquires the factor L^3. Apply Prop. 15.9 on the two tori and change all coefficient labels to $b_*(g)$, which belongs to both windows. This gives

$$\left|\sum_{p=0}^{3}b^{-p}Y_p\right|\le\varepsilon_{\mathrm S}(K+1;g,\ell_R)+L^3\varepsilon_{\mathrm S}(K;g',L\ell_R)+B_{\mathrm{lab}}(\varepsilon_{\mathrm{box}}(K+1;g)+L^3\varepsilon_{\mathrm{box}}(K;g')),$$

where

$$Y_p=\mathsf q_p(K+1;b_*;m_1,m_2)-L^{3-p}\mathsf q_p(K;b_*;m_1/L,m_2/L).$$

These coefficients are independent of the variable b. Vandermonde inversion on $I_1(g)$ bounds each Y_n by $\mathsf N_{V,1}$ times the displayed right side. Part (a), used at coupling g for both occurrences of $\mathsf q_n$, then bounds the integral of the affine function

$$\Delta(m)=\mathsf u_n(K+1)-L^{2-n}\mathsf u_n(K)+m(\mathsf z_n(K+1)-L^{1-n}\mathsf z_n(K))$$

by $E_{\mathrm R}(K)$ on every subinterval of $[Lm_-,m_+]$. Here we used

$$L^{3-n}\int_{m_1/L}^{m_2/L}(\mathsf u+\mathsf z m')\,\mathrm dm'=\int_{m_1}^{m_2}(L^{2-n}\mathsf u+L^{1-n}\mathsf z m)\,\mathrm dm.$$

On the two halves of length h_1, these integral bounds give slope at most $2E_{\mathrm R}/h_1^2$ and midpoint value at most $E_{\mathrm R}/h_1$. The intercept is at most $E_{\mathrm R}/h_1+2m_+E_{\mathrm R}/h_1^2\le 3E_{\mathrm R}/h_1^2$, proving both recursions.

All errors in these recursions are summable. More precisely,

$$\varepsilon_{\mathrm R}(K),\ \varepsilon'_{\mathrm A}(K)\le C(g,\ell)(1+K)^{8\mathfrak p_0}L^{-c_yK}.$$

This follows from Thm. R, $u\ge c_\#K/340$, $c_{\mathrm{cut}}=c_\#/68>c_y$, $\varepsilon_{\mathrm{box}}\le C_{\mathrm{box}}L^{-K}$, and the faster-than-exponential wrapping bound. For the elementary recursion $x(K+1)=L^dx(K)+e_K$, $|e_K|\le\varepsilon_{\mathrm R}(K)$, if $d\ge1$ define

$$G=L^{-dK_R}x(K_R)+\sum_{j\ge K_R}L^{-d(j+1)}e_j.$$

Then $x(K)-GL^{dK}=-\sum_{j\ge K}L^{d(K-1-j)}e_j$, whose modulus is at most $\bar\varepsilon_{\mathrm R}(K)$. If $d=0$, the tail is at most $\sum_{j\ge K}\varepsilon_{\mathrm R}(j)$. If $d\le-1$, iterate forward:

$$x(K)=L^{d(K-K_R)}x(K_R)+\sum_{K_R\le j<K}L^{d(K-1-j)}e_j.$$

Use $d = 2 - n$ for u_n and $d = 1 - n$ for z_n. This proves (c).

To identify the free limits, apply Lem. 15.8(c) to the two half intervals. It gives

$$\sigma_0^{\mp} = (L^2 - 1)I^{\mathrm{fr}}(0)L^{2K} + \tfrac{1}{2}(L - 1)I^{\mathrm{fr}\prime}(0)L^K(m_c + m_{\mp}) + O(C_I m_+^2 + 4\nu K_c c_b^{-1} m_-^{-2} L^{5s-K}).$$

The error is bounded because $5s - K \le 5 - 29K/34$. Thus $\mathsf{z}_0 + (L - 1)I^{\mathrm{fr}\prime}(0)L^K$ and $\mathsf{u}_0 + (L^2 - 1)I^{\mathrm{fr}}(0)L^{2K}$ are bounded. For the latter use $(m_- + 2m_c + m_+)/4 = m_c$. Part (c) now gives the stated values of $\mathsf{G}_0, \mathsf{H}_0$.

Finally, connect two reference couplings $g_2 < g$ by a finite decreasing chain $g = g^{(0)} > \cdots > g^{(k)} = g_2$ with $(g^{(i+1)})^{-2} \le L(g^{(i)})^{-2}$. Successive label windows overlap at $(g^{(i+1)})^{-2} + \bar{B}$, since $2\bar{B} \le LB_\sharp$. The per-site coefficients at a common label coincide by Lem. 15.5(e). Apply Lem. 15.7 for $n = 1, 2, 3$ and Lem. 15.8(b) for $n = 0$ within each window. The two endpoint coefficients differ by at most $L^{-K} \sum_{i=0}^{k} C_{\mathrm{box}}(g^{(i)})$. The corresponding $\mathsf{u}_n, \mathsf{z}_n$ differ by at most $3h^{-2}$ times this bound. Their five limits are therefore equal, proving (d). □

15.4. **Analytic convergence and the subtraction.** The real convergence estimate alone does not give convergence of parameter derivatives. We use the same holomorphic parameter domain as for the differentiated condensate. The one-point function grows there at most as L^{2K}; a two-constants estimate transfers the small real increment to a fixed smaller complex neighbourhood.

Lemma 15.11 (Complex propagation and a coarse bound)**.** (a) Let $I = [x_1 - \delta, x_1 + \delta]$, $\delta > 0$, and $J(w) = x_1 + \frac{\delta}{2}(w + w^{-1})$. For $1 < r \le R'$ put $E_r = I \cup J(\{1 < |w| < r\})$. Then

$$\{z : \mathrm{dist}(z, I) < \delta(r - 1)^2/(2r)\} \subset E_r \subset \{z : \mathrm{dist}(z, I) \le \delta(r - r^{-1})/2\}.$$

If f is holomorphic near $\overline{E_{R'}}$, $|f| \le M$ there and $|f| \le \varepsilon \le M$ on I, then, for $0 < \vartheta \le 1$,

$$|f| \le M^{\vartheta}\varepsilon^{1-\vartheta} \quad \text{on } E_{R'^{\vartheta}}.$$

The assertion for $\varepsilon = 0$ follows by the identity theorem.

(b) At every cutoff y_K is holomorphic on $W^\flat$ and $|y_K| \le C_{\mathrm{cr}} L^{2K}$ there, where

$$C_{\mathrm{cr}} = 3\nu_{\mathrm{c}} C_{\mathrm{pr}} + 4\nu c_1 m_-^{-2} + \frac{2C_w^{\mathrm{s}}\phi(\frac{1}{2}L\gamma^{-2})}{LM^3 r_J(1 - L^{-2})} + \frac{2\log 2}{\ell^3 \rho_m}.$$

Here c_1 is the tower norm constant used in Lem. 11.22, and ρ_m is the original parameter radius in Prop. 9.3.

■

Proof. The image of $|w| = r$ is the ellipse with foci $x_1 \pm \delta$ and sum of focal distances $\delta(r + r^{-1})$. Its interior is E_r. For a point z on the ellipse and $y \in I$,

$$\delta(r + r^{-1}) = |z - x_1 + \delta| + |z - x_1 - \delta| \le 2|z - y| + 2\delta.$$

Thus the ellipse is at distance at least $\delta(r - 1)^2/(2r)$ from I, proving the first inclusion by convexity. For $w = \rho e^{i\phi}$, $1 < \rho < r$,

$$|J(w) - J(e^{i\phi})| \le \tfrac{\delta}{2}[(\rho - 1) + (1 - \rho^{-1})] \le \tfrac{\delta}{2}(r - r^{-1}),$$

and $J(e^{i\phi}) \in I$, proving the second. On the annulus $1 \le |w| \le R'$, the subharmonic function $\log|f(J(w))|$ is bounded above by

$$\log\varepsilon + (\log M - \log\varepsilon)\frac{\log|w|}{\log R'}.$$

The maximum principle proves the interpolation bound. On I the same conclusion follows from $\varepsilon \le M^{\vartheta}\varepsilon^{1-\vartheta}$.

For (b), use the proof of Lem. 11.25 with one mass derivative. Around a real centre (b_0, m_0) write $(b', m') = (b_0 + t, m_0 + \zeta)$. The zero-free representation of Prop. 9.3 gives

$$y_K(b', m') = -\ell^{-3}\partial_{\zeta'} E_K(-t1, -\zeta' 1)|_{\zeta} - \ell^{-3}\partial_{\zeta'} \log\psi(t, \zeta')|_{\zeta}, \qquad |\log\psi| \le \log 2.$$

Cauchy's estimate with radius $\rho_m/2$ bounds the last term by $2\ell^{-3}\log 2/\rho_m$. The energy decomposition of Lem. 11.22 has a fine determinant, a terminal determinant, scalar activity sums, and a Gaussian term independent of the mass.

For the fine determinant, the derivative is

$$-a_K^{-2}N^{-3}\sum_{p\in P_N}\operatorname{tr}(\widehat{D}(p)+a_K m')^{-1}.$$

The condition $|m'-m_0|\le m_0/(2C_{\mathrm{pr}})$ and the resolvent bound in Lem. 11.21 give inverse norm at most $2C_{\mathrm{pr}}/|p|$. On the cubes of side $2\pi/N$ centred at P_N, one has $|y|\le 2|p|$. Comparison with $\int_{|y|\le\sqrt{3}\pi}|y|^{-1}\,\mathrm{d}^3y = 6\pi^3$ therefore bounds the determinant contribution by $3\nu_{\mathrm{c}}C_{\mathrm{pr}}L^{2K}$. The terminal determinant costs $4\nu c_1 m_-^{-2}$ by the first-derivative estimate in Lem. 11.22.

At level j, the scalar sum is bounded by $C_w^{\mathrm{s}}\phi_{j-1}(\ell/(M\lambda_j))^3$. Its mass-source disc about ζ has radius $r_J/(2\lambda_{j-1})$. Cauchy's estimate bounds its normalized first derivative by $2C_w^{\mathrm{s}}(LM^3r_J)^{-1}\phi_{j-1}\lambda_j^{-2}$. Finally

$$\sum_{j\le K}\phi_{j-1}\lambda_j^{-2}\le\frac{\phi(\frac12 L\gamma^{-2})}{1-L^{-2}}L^{2K}.$$

Adding these bounds gives C_{cr}. □

Proof of Thm. S. First fix one admissible choice of the construction constants. Define the five limits by Lem. 15.10 and set

$$\mathsf{A}_0=-I^{\mathrm{fr}}(0),\qquad \mathsf{A}_1=-I^{\mathrm{fr}\prime}(0),\qquad \mathsf{A}_2=\mathsf{G}_1/(L-1).$$

We prove real convergence before comparing different choices of the constants.

The subtraction has increment

$$\mathsf{br}_{K+1}-\mathsf{br}_K=\mathsf{G}_0L^{2K}+(\mathsf{H}_0m+\mathsf{G}_1/b)L^K+w_1m/b+w_2/b^2.$$

By Lem. 15.10(a), the increment of y_K differs from $\sum_{n=0}^3 b^{-n}(\mathsf{u}_n+m\mathsf{z}_n)$ by at most $\varepsilon'_{\mathrm{A}}$. Part (c) of that lemma controls the eight coefficient remainders. Each is bounded by a supremum tail, a summed tail, or the stable convolution appearing there. Since $m\le 1/2$ and $\sum_{n=0}^3 b^{-n}\le B_{\mathrm{lab}}$, these estimates and

$$\sum_{j\ge K}(1+j)^{8\mathfrak{p}_0}L^{-c_yj}\le(1+K)^{8\mathfrak{p}_0}L^{-c_yK}S_q,$$
$$\sum_{j<K}L^{-(K-1-j)}(1+j)^{8\mathfrak{p}_0}L^{-c_yj}\le L(1+K)^{8\mathfrak{p}_0}L^{-c_yK}\sum_{i\ge0}L^{-(1-c_y)i},$$

give the claimed summable increment bound. Also $L^{-(K-K_R)}\le L^{K_R}L^{-c_yK}$. More precisely, take

$$K_\Xi(g,\ell)=\max\{K_{\mathrm{o}}(g,\ell),K_R(g)\},$$

and define C_Ξ as the finite supremum

$$\sup_{K\ge K_\Xi}(1+K)^{-q}L^{c_yK}\sup_{W(g)}|(y_{K+1}-\mathsf{br}_{K+1})-(y_K-\mathsf{br}_K)|.$$

The series of increments converges uniformly. Summing its tail proves (b), using $1+K+j\le(1+K)(1+j)$.

The coefficient formulas in (a) follow from the affine estimate in Lem. 15.10(a). For the first formula use $L^{-K}\mathsf{u}_1\to\mathsf{G}_1$ and $L^{-K}\mathsf{z}_1\to0$; for the second use $\mathsf{u}_2\to w_2$ and $\mathsf{z}_2\to0$. Subtracting the two adjacent mass-interval quotients eliminates u_1 and gives w_1. The errors ε_{A} tend to zero. Part (d) of that lemma makes the limits independent of the reference coupling and the chosen coefficient label.

They are real. The two free integrals are real because the two parity blocks at the trivial field are adjoints, so their traces are conjugate pairs. For the other coefficients use reality of δ_K on $W(g)$. Part (a) of Lem. 15.10 implies

$$\left|\sum_{n=0}^{3} b^{-n}(\operatorname{Im}\mathsf{u}_n + m\operatorname{Im}\mathsf{z}_n)\right| \le \varepsilon'_{\mathrm{A}}(K).$$

Vandermonde inversion bounds each affine expression by $\mathsf{N}_V(g)\varepsilon'_{\mathrm{A}}(K)$. Evaluation at $m_\pm$ bounds its slope by this quantity divided by h, and its intercept by this quantity times $1 + m_+/h$. Both tend to zero. The five limiting coefficients, with their stated powers of L^{-K}, are therefore real. This also proves that the real limit in (b) is real.

To prove independence of the auxiliary choices, take two choices with the same chamber operator, N_f and L; mark the second by primes. Since $Lm_- < m_+$, the intervals $L^{-j}(m_-, m_+)$, $j \in \mathbb{Z}$, overlap consecutively and cover $(0,\infty)$. After interchanging the choices if needed, choose $j \ge 0$ and $\mu_1 < \mu_2$ in both $L^{-j}[m_-, m_+]$ and $[m'_-, m'_+]$. For every sufficiently large b, there are admitted couplings with $b = b_-(g)$ and $L^j b = b'_-(g')$. Choose $\ell = 2L^k$ sufficiently large that $\ell \in \mathfrak{L}(g)$ and $L^j\ell \in \mathfrak{L}'(g')$. Iterating Lem. 15.2 gives

$$y_{K+j,\ell}(b, L^j\mu) = L^{2j} y_{K,L^j\ell}(L^j b, \mu).$$

The subtraction itself satisfies

$$L^{-2j}\mathsf{br}_{K+j}(b, L^j\mu) = \mathsf{br}_K(L^j b, \mu) + j\left(\frac{w_1\mu}{L^j b} + \frac{w_2}{(L^j b)^2}\right).$$

The extra term is independent of K. By (b), the one lattice function $y_{K,L^j\ell}$ minus either subtraction converges at $(L^j b, \mu_i)$, $i = 1, 2$. The free coefficients agree for the two choices, so the difference of their subtractions is

$$(\mathsf{A}_2 - \mathsf{A}'_2)L^K/\beta + [(w_1 - w'_1)\mu/\beta + (w_2 - w'_2)/\beta^2]K, \qquad \beta = L^j b.$$

Divide its convergent sequence first by L^K to obtain $\mathsf{A}_2 = \mathsf{A}'_2$, and then by K. The two distinct masses give $w_1 = w'_1$ and $w_2 = w'_2$. This completes (a).

For (c), Lem. 15.11(a) gives $W^\sharp \subset W^\flat$, since $r_b \le \rho_b^\flat/2$ and $r_m \le \rho_m^\flat/2$. On $W^\flat$, $\operatorname{Re} b' \ge b_-/2$ and $|m'| \le 1$. The first bound uses $\rho_b^\flat \le \varrho/4$ and $32\varrho \le \gamma^{-2} \le b_-$. Thus the subtraction is holomorphic and satisfies

$$|\mathsf{br}_K| \le C_{\mathsf{br}} L^{2K}, \qquad C_{\mathsf{br}} = L^2|\mathsf{A}_0| + 2L(|\mathsf{A}_1| + |\mathsf{A}_2|b_-^{-1}) + 8(|w_1| + |w_2|)(1 + b_-^{-2}).$$

We used $K \le L^K$. Put $\Delta_K = (y_{K+1} - \mathsf{br}_{K+1}) - (y_K - \mathsf{br}_K)$. By Lem. 15.11(b),

$$|\Delta_K| \le M_K := 2L^2(C_{\mathrm{cr}} + C_{\mathsf{br}})L^{2K} \quad \text{on } W^\flat,$$

whereas on $W(g)$ it is at most $\varepsilon_K = \min\{C_\Xi(1+K)^q L^{-c_y K}, M_K\}$.

Fix real $b \in I_g$ and apply Lem. 15.11(a) in the mass variable, with the ellipse determined by R'_m. Its closure is inside the original mass neighbourhood, and its smaller ellipse contains the r_m-neighbourhood of the mass interval. We obtain $|\Delta_K(b, m')| \le M_K^\vartheta \varepsilon_K^{1-\vartheta}$. For each such m', repeat in b with R'_b. This gives, on $W^\sharp$,

$$|\Delta_K| \le M_K^{1-\omega}\varepsilon_K^\omega, \qquad \omega = (1-\vartheta)^2.$$

Our choice makes $\omega = (4 + c_y)/(4 + 2c_y)$ and $2(1-\omega) - c_y\omega = -c_y/2$. Thus the stated complex increment bound holds with

$$C'_\Xi = [2L^2(C_{\mathrm{cr}} + C_{\mathsf{br}})]^{1-\omega} C_\Xi^\omega.$$

It is summable in K. Uniform convergence on the fixed open set gives a holomorphic limit. Cauchy's estimates on the bidiscs of radii $r_b/2, r_m/2$ about the real window give uniform convergence of every parameter derivative. Finally $\partial_m \mathsf{br}_K = \mathsf{A}_1 L^K + w_1 K/b$. □

Remark 15.12 (Order of the constants). No quantity introduced in this proof is a new floor or ceiling on a construction constant. The hypotheses are the existing source and cutoff-comparison choices, eq. 14.viii, the field-scale choices in Rem. 14.9, and the choice $Lm_- < m_+$ at the mass-window stage. The symbol constants and free integrals depend only on the chamber operator and N_f. The energy, wrapping and Gaussian-coefficient constants are formed from the already chosen construction constants. The exponents c_y, q, the sum S_q and the interpolation weights depend only on $L, c_\#, \mathfrak{p}_0$; K_{v} also uses the existing source radius r_J.

After choosing the mass window, g and ℓ, form the Vandermonde weights, comparison side, cutoff thresholds, error sequences and complex coarse bound. The five limits may first be formed at $g = \gamma_{\mathrm{e}}$; Lem. 15.10(d) and Thm. S(a) then identify them at every other coupling and every other admissible choice with the same L. Only after these limits are fixed are $C_{\mathsf{br}}, C_\Xi, C'_\Xi$ formed. The radii r_b, r_m are then fixed by the already available parameter neighbourhood. None of these choices depends on the cutoff or on a variable label within its window.

Appendix A. Comparing two meshes

We first compare the pure-gauge backgrounds and source cumulants on two bare lattices, then the fermion operators reconstructed at a common block scale. The operator comparison controls their inverses and the determinant at the start of the restarted flow.

A.1. **The pure-gauge comparison.** We first compare the pure-gauge constructions on consecutive bare lattices. The lower action bound comes from block averaging. For the upper bound, we refine a coarse configuration while preserving its straight bond products. Fibre coercivity then controls the discrepancy between the two classical fields. These estimates also control the change of links used below.

We use the real backgrounds, fibre charts, radii, and calibration of [YM3, eq. 6.xcvi]–[YM3, eq. 6.cii], the constants in [YM3, eq. 6.xcvii] and [YM3, eq. 6.xcix], and the two-cutoff setting of [YM3, eq. 6.civ] and [YM3, eq. 6.cv]. In particular, the whole-data hypotheses of [YM3, Prop. 4.58] and [YM3, Prop. 4.56] hold at the fixed radii. The notation $\mathsf{ab}_t(c, V)$ denotes the weighted action of the regular critical background over V; the unweighted action is $\mathsf{ab}_t(V)$. The norm $\|\cdot\|_2$ below is the counting norm on positively oriented bonds.

A.1.1. *Refinement.* For $W \in G$, put

$$\varphi(W) = \arccos\big(\tfrac{1}{2}\operatorname{tr} W\big), \qquad a(W) = 1 - \tfrac{1}{2}\operatorname{tr} W = \tfrac{1}{2}|W - \mathbf{1}|^2.$$

Thus $|W - \mathbf{1}| = 2\sin(\varphi(W)/2)$, and

$$\tfrac{1}{2}\varphi(W)^2\big(1 - \varphi(W)^2/12\big) \le a(W) \le \tfrac{1}{2}\varphi(W)^2. \tag{A.i}$$

The inequalities follow from $u^2/2 - u^4/24 \le 1 - \cos u \le u^2/2$ for $0 \le u \le \pi$. The lower bound follows by integrating $\sin u \ge u - u^3/6$, itself obtained by integrating $\cos u \ge 1 - u^2/2$.

We will also use the following logarithmic estimate. If $X_1, \dots, X_r \in \mathfrak{g}$ and $s = \sum_j |X_j| \le 1/4$, then $W = \prod_j e^{iX_j} \ne -\mathbf{1}$ and

$$Y = \tfrac{1}{i}\log W \in \mathfrak{g}, \qquad |Y| \le s, \qquad \Big|Y - \sum_j X_j\Big| \le s^2 e^s. \tag{A.ii}$$

Here and below the logarithm is intrinsic. To verify the first bound, note that $\varphi(W_1W_2) \le \varphi(W_1) + \varphi(W_2)$. When the sum on the right is less than π, the identity

$$W_1W_2 - \mathbf{1} = \tfrac{1}{2}(W_1 - \mathbf{1})(W_2 + \mathbf{1}) + \tfrac{1}{2}(W_1 + \mathbf{1})(W_2 - \mathbf{1})$$

and $|W + \mathbf{1}| = 2\cos(\varphi(W)/2)$ bound the left side by $2\sin((\varphi(W_1) + \varphi(W_2))/2)$. Monotonicity of sine on $[0, \pi/2]$ proves the assertion; when the sum is at least π, the assertion is immediate. Since $\varphi(e^{iX}) \le |X|$ and $|Y| = \varphi(W)$, iteration gives $|Y| \le s$. Expanding the product of exponentials in

its absolutely convergent series bounds the terms of total degree at least two by $e^s - 1 - s \le s^2 e^s/2$. Apply the same bound to e^{iY} and subtract the two first-order expansions to obtain eq. A.ii.

Let $\mathbb{Z}_{\underline{n}} = \prod_{\mu=1}^{3} \mathbb{Z}/n_\mu\mathbb{Z}$, where $n_\mu \ge 2$. A configuration is a family $U_\mu(x) \in G$. Write

$$P_{\alpha\beta}(x) = U_\alpha(x)U_\beta(x+e_\alpha)U_\alpha(x+e_\beta)^{-1}U_\beta(x)^{-1}, \qquad E_{\alpha\beta}(U) = \sum_x a(P_{\alpha\beta}(x)),$$

and set $E(U) = E_{12}(U) + E_{13}(U) + E_{23}(U)$. The reversal of the two directions inverts $P_{\alpha\beta}$ and does not change its angle or action. Let $\vartheta(U)$ be the largest plaquette angle. For a site x, let $c(x)$ be the largest angle of the six faces of its coordinate cube, namely the plaquettes $P_{\alpha\beta}(x)$ and $P_{\alpha\beta}(x+e_\gamma)$ with $\alpha < \beta$ and $\gamma \notin \{\alpha, \beta\}$.

Fix an integer $L \ge 2$ and a direction λ. Enlarge the λ side by L, and let ι_λ multiply that coordinate by L. Every site of the enlarged box has a unique expression $x^{(j)} = \iota_\lambda x + j e_\lambda$, $0 \le j < L$. If $\vartheta(U) < \pi$, define $\Theta_{\lambda\kappa}(x) = \frac{1}{i}\log P_{\lambda\kappa}(x)$ and set

$$\begin{aligned} &U'_\lambda(x^{(j)}) = \mathbf{1} && (j < L-1), \quad U'_\lambda(x^{(L-1)}) = U_\lambda(x), \\ &U'_\kappa(x^{(j)}) = e^{i(j/L)\Theta_{\lambda\kappa}(x)}U_\kappa(x) && (\kappa \ne \lambda). \end{aligned} \tag{A.iii}$$

We denote this refinement by $U' = \mathsf{Rfn}_\lambda U$.

Lemma A.1 (Refinement in one direction)**.** Suppose $\vartheta_0 = \vartheta(U) \le 10^{-4}$, and let μ, ν be the directions other than λ. The refinement preserves the straight coarse bond products and the transverse bonds at embedded sites:

$$\prod_{j=0}^{L-1} U'_\lambda(x^{(j)}) = U_\lambda(x), \qquad U'_\kappa(\iota_\lambda x) = U_\kappa(x) \quad (\kappa \ne \lambda).$$

Its plaquettes containing the refined direction satisfy $P'_{\lambda\kappa}(x^{(j)}) = e^{i\Theta_{\lambda\kappa}(x)/L}$. For the transverse plaquettes, put $Y_0 = \frac{1}{i}\log P_{\mu\nu}(x)$ and $Y_1 = \mathrm{Ad}(U_\lambda(x))\frac{1}{i}\log P_{\mu\nu}(x+e_\lambda)$. Then

$$\varphi(P'_{\mu\nu}(x^{(j)})) \le |(1-j/L)Y_0 + (j/L)Y_1| + 54c(x)^2. \tag{A.iv}$$

Consequently,

$$\begin{aligned} E_{\lambda\kappa}(U') &\le L^{-1}(1+\vartheta_0^2)E_{\lambda\kappa}(U), \\ E_{\mu\nu}(U') &\le L(1+\vartheta_0^2)E_{\mu\nu}(U) + 221L\vartheta_0 E(U), \\ \vartheta(U') &\le \vartheta_0 + 54\vartheta_0^2 \le 2\vartheta_0. \end{aligned} \tag{A.v}$$

■

Proof. The bond identities follow directly from eq. A.iii. For $j < L-1$, the two λ bonds of a plaquette equal the identity, so its two transverse factors give $e^{i\Theta_{\lambda\kappa}(x)/L}$. At $j = L-1$, the same product is $P_{\lambda\kappa}(x)e^{-i(L-1)\Theta_{\lambda\kappa}(x)/L}$, which has the same value.

For a transverse plaquette write

$$\begin{aligned} A_1 = \Theta_{\lambda\mu}(x), \quad A_2 = \Theta_{\lambda\nu}(x+e_\mu), \quad A_3 = \Theta_{\lambda\mu}(x+e_\nu), \quad A_4 = \Theta_{\lambda\nu}(x), \\ u_\mu = U_\mu(x), \quad u_\nu = U_\nu(x), \quad u'_\nu = U_\nu(x+e_\mu), \quad u'_\mu = U_\mu(x+e_\nu). \end{aligned}$$

For $0 \le t \le 1$ consider

$$W(t) = e^{itA_1}u_\mu e^{itA_2}u'_\nu u'^{-1}_\mu e^{-itA_3}u_\nu^{-1}e^{-itA_4}.$$

Then $W(j/L) = P'_{\mu\nu}(x^{(j)})$. Moving the unitary factors past the exponentials gives

$$W(t) = e^{itA_1}e^{it\,\mathrm{Ad}(u_\mu)A_2}e^{iY_0}e^{-it\,\mathrm{Ad}(u_\nu)A_3}e^{-itA_4}.$$

The five exponents and Y_1 have norms at most $c = c(x)$. By eq. A.ii,

$$\tfrac{1}{i}\log W(t) = Y_0 + t(A_1 + \mathrm{Ad}(u_\mu)A_2 - \mathrm{Ad}(u_\nu)A_3 - A_4) + e(t), \qquad |e(t)| \le 25e^{5c}c^2 \le 27c^2.$$

At $t = 1$, substitution of the four defining plaquette products cancels the three successive pairs of λ bonds and gives $W(1) = U_\lambda(x)P_{\mu\nu}(x + e_\lambda)U_\lambda(x)^{-1}$. Thus $\frac{1}{i}\log W(1) = Y_1$. Subtracting this identity from the preceding expression gives $\frac{1}{i}\log W(t) = (1-t)Y_0 + tY_1 + e(t) - te(1)$. This proves eq. A.iv and the angle bound in eq. A.v.

For an old plaquette angle $\varphi \le \vartheta_0$, eq. A.i implies

$$\tfrac{1}{2}\varphi^2 \le (1 + \vartheta_0^2)a. \tag{A.vi}$$

Indeed, $(1+y)(1-y/12) \ge 1$ for $0 \le y \le 1$. Each plaquette containing λ becomes L plaquettes of angle φ/L, which proves its energy estimate. For a transverse plaquette put $p = |(1-t)Y_0 + tY_1|$. Since $p \le c \le 10^{-4}$,

$$a(W(t)) \le \tfrac{1}{2}(p + 54c^2)^2 \le \tfrac{1}{2}((1-t)|Y_0|^2 + t|Y_1|^2) + 55c^3.$$

The last bound uses $54c^2p + 1458c^4 \le 55c^3$. Summing over $j = 0, \dots, L-1$ and then x, the quadratic term is $\frac{L}{2}\sum_x |Y_0(x)|^2$: the sums of $|Y_0|^2$ and $|Y_1|^2$ coincide by translation and unitary invariance. It is at most $L(1+\vartheta_0^2)E_{\mu\nu}(U)$ by eq. A.vi. Each plaquette belongs to two of the coordinate cubes, and hence

$$\sum_x c(x)^3 \le \vartheta_0 \sum_x \sum_{\text{faces of } x} \varphi(P)^2 \le 4(1+\vartheta_0^2)\vartheta_0 E(U).$$

The remaining contribution is at most $220(1+10^{-8})L\vartheta_0 E(U) \le 221 L\vartheta_0 E(U)$. □

Proposition A.2 (Prolongation)**.** Let $\vartheta_0 = \vartheta(U) \le \frac{1}{4}10^{-4}$ and define

$$\widetilde{U} = \mathsf{Prl}_L U = \mathsf{Rfn}_3\mathsf{Rfn}_2\mathsf{Rfn}_1 U$$

on $\mathbb{Z}_{L\underline{n}}$. Then

$$\prod_{j=0}^{L-1} \widetilde{U}_\mu(Lx + je_\mu) = U_\mu(x), \qquad LE(\widetilde{U}) \le (1 + 4000L^4\vartheta_0)E(U), \qquad \vartheta(\widetilde{U}) \le 8\vartheta_0. \tag{A.vii}$$

The bond product is ordered by increasing j. ■

Proof. After i refinements the largest angle is at most $2^i\vartheta_0$. Thus the input angle of each of the three steps is at most 10^{-4}, and Lem. A.1 applies. At the step refining direction μ, the straight product becomes the old μ bond. At the other two steps, the transverse bonds at embedded sites are preserved. Applying these identities in their order proves the first assertion.

Write $E^{(i)}_{\alpha\beta}$ and $E^{(i)}$ for the energies after i steps, and put $\epsilon = 16\vartheta_0^2 \le 10^{-8}$. The energy inequalities in eq. A.v give, at step i,

$$E^{(i)}_{i\kappa} \le \frac{1+\epsilon}{L}E^{(i-1)}_{i\kappa}, \qquad E^{(i)}_{\mu\nu} \le L(1+\epsilon)E^{(i-1)}_{\mu\nu} + 221\cdot 2^{i-1}L\vartheta_0 E^{(i-1)},$$

where $\mu, \nu \ne i$. In particular $E^{(i)} \le 2LE^{(i-1)}$. For the pairs 12, 13, and 23, the step with the error term is, respectively, the third, second, and first. The displayed recurrence therefore yields

$$\begin{aligned}
E^{(3)}_{12} &\le (1+\epsilon)^3 L^{-1}E^{(0)}_{12} + 3536L^3\vartheta_0 E^{(0)},\\
E^{(3)}_{13} &\le (1+\epsilon)^3 L^{-1}E^{(0)}_{13} + 884(1+\epsilon)L\vartheta_0 E^{(0)},\\
E^{(3)}_{23} &\le (1+\epsilon)^3 L^{-1}E^{(0)}_{23} + 221(1+\epsilon)^2 L^{-1}\vartheta_0 E^{(0)}.
\end{aligned}$$

After summing and multiplying by L, use $(1+\epsilon)^3 - 1 \le 4\epsilon \le \vartheta_0$, $884(1+\epsilon) \le 885$, $221(1+\epsilon)^2 \le 222$, and $885L^2 + 223 \le 464L^4$. These give the coefficient $4000L^4$ in eq. A.vii. □

A.1.2. *Weighted averaging.* For the comparison of smeared actions we need averaging on arbitrary sets of plaquettes. Suppose that a fine lattice of spacing h/L is averaged to a lattice of spacing h. For a coarse plaquette p', let $\mathcal{N}(p')$ be the fine plaquettes all of whose corners lie in the union of the four fine blocks centered at the corners of p'; put $\mathcal{N}(S) = \bigcup_{p'\in S} \mathcal{N}(p')$. The one-step bound [YM3, eq. 3.cxxviii] is

$$\sum_{p'\in S} a_{p'}(\mathcal{M}W) \le L(1 + C_{\mathrm{en}}\omega)^2 \sum_{p\in\mathcal{N}(S)} a_p(W), \qquad \omega = \max_{p\in\mathcal{N}(S)} |\partial W(p) - \mathbf{1}| \le \varpi'_L. \tag{A.viii}$$

This holds for every set S, not only all plaquettes. For clarity, the reason is that the positive linear curvature kernel of [YM3, Lem. 3.47]–[YM3, Lem. 3.48] has counting ℓ^2 norm at most $\sqrt{L}$ after restriction to $\mathcal{N}(S)$, while its nonlinear remainder at p' is at most $C_0^{\mathrm{SO}}\omega_{p'}^2$. A fine plaquette belongs to at most twelve coarse cones: its base lies in one coarse block, whose center must be a corner of each of the coarse plaquettes. Consequently

$$\sum_{p'\in S} \omega_{p'}^4 \le 12\omega^2 \sum_{p\in\mathcal{N}(S)} |\partial W(p) - \mathbf{1}|^2.$$

The triangle inequality gives the curvature bound with factor $\sqrt{L}(1 + C_{\mathrm{en}}\omega)$, where $C_{\mathrm{en}} = \sqrt{12}\, C_0^{\mathrm{SO}} L^{-1/2}$; squaring and using $2a_p = |\partial W(p) - \mathbf{1}|^2$ proves eq. A.viii.

For a plaquette p, let x_p denote its barycenter. If $p \in \mathcal{N}(p')$, choose a corner z of p in the fine block of a corner y of p'. In the torus distance,

$$d_\infty(x_p, x_{p'}) \le d_\infty(x_p, z) + d_\infty(z, y) + d_\infty(y, x_{p'}) < \frac{h}{2L} + h \le \frac{3}{2}h. \tag{A.ix}$$

This estimate uses only the defining union of blocks of the cone.

Lemma A.3 (Weighted one-step action)**.** Let W have curvature at most $\omega \le \varpi'_L$ on the fine lattice. Write η and η/L for the two dimensionless spacings, and set

$$A^\eta(c, V) = \eta^{-1} \sum_p c(x_p) a_p(V), \qquad q = (1 + C_{\mathrm{en}}\omega)^2.$$

For every nonnegative real Lipschitz function c on the torus,

$$A^\eta(c, \mathcal{M}W) \le q \left[A^{\eta/L}(c, W) + \tfrac{3}{2} h \,\mathrm{Lip}_\infty(c) A^{\eta/L}(1, W) \right]. \tag{A.x}$$

■

Proof. Put $v = \frac{3}{2} h \,\mathrm{Lip}_\infty(c)$ and $S_\lambda = \{p' : c(x_{p'}) > \lambda\}$. By eq. A.ix, $\mathcal{N}(S_\lambda) \subset \{p : c(x_p) + v > \lambda\}$. Apply eq. A.viii to S_λ and integrate over $\lambda \ge 0$. The elementary identity $c(x) = \int_0^\infty \mathbf{1}_{\{c(x)>\lambda\}}\, \mathrm{d}\lambda$ gives

$$\sum_{p'} c(x_{p'}) a_{p'}(\mathcal{M}W) \le Lq \sum_p (c(x_p) + v) a_p(W).$$

Multiplication by η^{-1} proves the assertion. □

A.1.3. *Coercivity and competitors.* Use the fibre chart $V_W[\mathcal{A}]$ and the calibration fixed above.

Proposition A.4 (Coercivity on the fibre)**.** Let $0 \le k < K$ and $\tau \ge \gamma^{-2}$, and suppose that W belongs to the actual global domain $\mathcal{W}_{k+1}(\tau; \alpha_0, \alpha_1)$ of [YM3, Prop. 4.58]. Put $g = \tau^{-1/2}$ and take a real nonpivot field B with $\sup|B| < \mathfrak{b}(\tau)$. For

$$F_W(B) = \mathsf{ab}_k(V_W[gB]), \qquad G_W(B) = \sum_y \sum_{x\in B(y)\setminus\{y\}} \left[1 - \tfrac{1}{2}\,\mathrm{Re\,tr}\, V_W[gB](y, x)\right],$$

one has

$$\tau(F_W(B) - F_W(0) + G_W(B)) \ge (\gamma_{\mathrm{r}}/4 - C_{\mathrm{cub}} g \sup|B|)\|B\|_2^2. \tag{A.xi}$$

Moreover $F_W(0) = L^{-1}\mathsf{ab}_{k+1}(W)$, $G_W(B) \ge 0$, and $G_W(B) = 0$ when the chart field is axial. ■

Proof. The global action and gauge sumrules, with their actual background identification, give

$$\tau(F_W(B) - F_W(0) + G_W(B)) = \tfrac{1}{2}(B, \mathsf{A}_W B)_1 - \sum_{\natural\in\{\mathrm{A},\mathrm{G}\}} \sum_{Y\in\mathcal{D}_k} \mathbb{V}^{\natural}(Y; P_W|_Y, B|_Y). \tag{A.xii}$$

Here P_W is the plain triple of the axial top background and $\mathsf{A}_W = \mathbf{C}_*(P_W)^{-1}$; see [YM3, eq. 4.cxciv], [YM3, eq. 4.cxcvi]–[YM3, eq. 4.cxcvii], [YM3, Prop. 4.56] and [YM3, Prop. 4.58]. All the triples on the right belong to their cores by that proposition. The reference covariance satisfies $(B, \mathbf{C}_{*\mathrm{r}}^{-1} B)_1 \geq \gamma_{\mathrm{r}} \|B\|_2^2$, and $\|\mathsf{A}_W - \mathbf{C}_{*\mathrm{r}}^{-1}\| \leq \gamma_{\mathrm{r}}/2$. The real part of the quadratic term is therefore at least $\gamma_{\mathrm{r}}\|B\|_2^2/4$.

It remains to use the third-order vanishing of the two potentials, which is stronger than their uniform smallness. For a domain Y, put $d = d_k(Y)$, $b_Y = \max_{b\in\mathfrak{X}^{\circ+}\langle Y\rangle} |B(b)|$, and $r = \mathsf{C}_{\mathcal{A}}\mathfrak{b}(\tau) \geq 3$. There are at most $24M^3(d+1)$ bonds in this domain. If $b_Y = 0$, its potential is zero. Otherwise $z \mapsto \mathbb{V}^{\natural}(Y; P_W|_Y, zB|_Y)$ is holomorphic for $|z| < r/b_Y$, vanishes to order three, and, by the actual potential bound [YM3, eq. 4.ccliv], has modulus at most

$$K_P \delta(\tau) e^{-\lambda_V d}(1 + 24(d+1)r^2) \leq 25 K_P \delta(\tau)(d+1) r^2 e^{-\lambda_V d}.$$

Apply the maximum principle to the function divided by z^3 on smaller concentric discs and let their radii tend to r/b_Y. This gives

$$|\mathbb{V}^{\natural}(Y; P_W|_Y, B|_Y)| \leq \frac{25 K_P}{\mathsf{C}_{\mathcal{A}}} g \sup|B| (d+1) e^{-\lambda_V d} \sum_{b\in\mathfrak{X}^{\circ+}\langle Y\rangle} |B(b)|^2.$$

For a fixed bond, the domains in this sum contain its initial point. Their weighted sum is at most $C_{\mathrm{E},1}$ by [YM3, Prop. 4.11], since $\lambda_V > \log 9216 + 2$. Summing over both potentials gives $C_{\mathrm{cub}} g \sup|B| \|B\|_2^2$, proving eq. A.xi.

The background over $V_k^{\mathrm{bf}}(W)$ is a representative of the background over W, by restriction of criticality and uniqueness as proved in [YM3, Prop. 4.58]. Changing the action spacing from η_k to η_{k+1} contributes the factor L, so $F_W(0) = L^{-1}\mathsf{ab}_{k+1}(W)$. Every term defining G_W is nonnegative and vanishes in axial gauge. □

Call a bare configuration U' a competitor of radius e at level t if $1 \leq t \leq K$ and

$$\varpi(\mathcal{M}^k U') \leq e L^{-2(t-k)} \qquad (0 \leq k \leq t). \tag{A.xiii}$$

Its excess is $\mathsf{exc}_t(U') = A^{\eta_t}(U') - \mathsf{ab}_t(\mathcal{M}^t U')$.

Proposition A.5 (Small competitors)**.** If $0 < e \leq e_{\mathrm{cp}}$ and U' satisfies eq. A.xiii, all backgrounds $U_k(V_k')$, $V_k' = \mathcal{M}^k U'$, exist. There are real nonpivot fields $\mathcal{A}_k$, $0 \leq k < t$, with

$$\sup|\mathcal{A}_k| \leq h_k := \tfrac{\pi}{2} C_{\mathrm{ax}}(1 + 2B_3) e L^{-2(t-k)} \leq h, \tag{A.xiv}$$

such that the axial representative of V_k' equals $V_{V_{k+1}'}[\mathcal{A}_k]$. One has

$$\mathsf{ab}_k(V_k') - L^{-1}\mathsf{ab}_{k+1}(V_{k+1}') \geq \tfrac{1}{8}\gamma_{\mathrm{r}} \|\mathcal{A}_k\|_2^2 \tag{A.xv}$$

and

$$\mathsf{exc}_t(U') \geq \tfrac{1}{8}\gamma_{\mathrm{r}} \sum_{k<t} L^{t-k} \|\mathcal{A}_k\|_2^2 \geq 0. \tag{A.xvi}$$

In particular $A^{\eta_t}(U_k(V_k'))$ is nonincreasing in k. ■

Proof. Since $e \leq \varepsilon_{\mathcal{W}}\alpha_0/2 < \bar{\varepsilon}$, the real-background theorem constructs $U_k(V_k')$ for $k \geq 1$. At $k = 0$ the identity background is regular: its curvature is at most e and its current at most $4e < \varepsilon_{\mathrm{v}}$. Fix $k < t$, put $W = V_{k+1}'$, and normalize V_k' to axial gauge by its contour means. This gauge equals the identity on the coarse sites and preserves the average W. [YM3, Prop. 4.58] puts W in the actual global domain at $\tau_\natural$ and constructs $V_0 = V_k^{\mathrm{bf}}(W)$. Its curvature estimate, applied with any radius strictly above $\varpi(W)$ and then taking the infimum, gives

$$\varpi(V_0) \leq 2B_3 L^{-2}\varpi(W) \leq 2B_3 e L^{-2(t-k)} \leq L^{-2}\varepsilon_{\mathrm{cp}}.$$

The two axial fields have the same average. Apply [YM3, eq. 6.ci]; their relative intrinsic logarithm is defined and has norm at most $h_k \le \delta(\tau_\natural)/2$. Its restriction to the nonpivot bonds is $\mathcal{A}_k$. The inverse property of the fibre chart identifies the axial configuration with $V_W[\mathcal{A}_k]$.

Apply Prop. A.4 with $B = \sqrt{\tau_\natural}\mathcal{A}_k$. The gauge term vanishes, and $C_{\mathrm{cub}} \sup|\mathcal{A}_k| \le \gamma_{\mathrm{r}}/8$ by [YM3, eq. 6.c]. Divide by $\tau_\natural$ and use gauge invariance of the background action to obtain eq. A.xv. Finally $A^{\eta_t}(U_k(V_k')) = L^{t-k}\mathsf{ab}_k(V_k')$. Multiply eq. A.xv by L^{t-k} and sum in k; the left sides telescope to the excess, proving eq. A.xvi. □

Lemma A.6 (Plaquette differences between successive backgrounds)**.** Under the hypotheses of Prop. A.5, for each $k < t$ one can choose representatives $\widehat{U} \in \mathcal{C}(\mathbf{B}_k, V_k')$ and $\check{U} \in \mathcal{C}(\mathbf{B}_{k+1}, V_{k+1}')$ and elements $u_p, R_p \in G$ such that

$$\partial\widehat{U}(p) = u_p R_p \partial\check{U}(p) u_p^{-1}, \qquad \eta_k^{-1}\sum_p |R_p - \mathbf{1}|^2 \le C_{\mathrm{nrg}}\|\mathcal{A}_k\|_2^2. \tag{A.xvii}$$

■

Proof. Put $V_0 = V_k^{\mathrm{bf}}(V_{k+1}')$ and use [YM3, eq. 6.cii] to write the axial version of V_k' as $e^{iB'}V_0$. Then

$$\sup|B'| \le 2C_{\mathrm{db}}h, \qquad \|B'\|_2^2 \le (1 + 24C_{\mathrm{db}}^2)\|\mathcal{A}_k\|_2^2.$$

For $k \ge 1$, the real response applies at datum radius $\varepsilon_{\mathrm{cp}}$: the base has curvature at most $L^{-2}\varepsilon_{\mathrm{cp}} < \varepsilon_{\mathrm{cp}}$, $C_{\mathfrak{P}}\mathfrak{M}B_3\varepsilon_{\mathrm{cp}} \le a_{\mathrm{R}}$, and $\sup|B'| \le \rho_{\mathrm{no}}/2$. The constant sequence has all pillars of level k and size one in level-k units. Thus [YM3, eq. 4.ccclxiii]–[YM3, eq. 4.ccclxiv] give representatives related, up to gauge, by $e^{i\eta_k H}$, where

$$|H(b)| \le B_5 m_{y(b)}, \qquad |\nabla_{\check{U}}H(x)| \le B_5 m_{y(x)}, \qquad m_y = \sum_{c>0} e^{-\delta_{\mathrm{lc}}d(y,c_-)}|B'(c)|.$$

In particular $\sup|H| \le B_5 S_{\mathrm{e}}\, 2C_{\mathrm{db}}h \le 1/132$. Let ϕ_p be the maximum of $|H|$ on the four bonds of p. The real plaquette formula of [YM3, Lem. 4.43], with $\eta_k \le 1/13$ and base curvature at most η_k^2, gives

$$|R_p - \mathbf{1}| \le \eta_k^2|D_{\check{U}}H(p)| + 11\eta_k^2\phi_p^2 + 6\eta_k\phi_p|\partial\check{U}(p) - \mathbf{1}| \le 3\eta_k^2 B_5 \max_{x \text{ corner of } p} m_{y(x)}.$$

Here the derivative and curl are covariant with respect to the unperturbed background $\check{U}$. Indeed $|D_{\check{U}}H(p)| \le 2B_5 \max m_y$, $\phi_p \le B_5 \max m_y$, $\phi_p \le 1/132$, and $2 + 11/132 + 6/13 < 3$. The product-expansion hypotheses hold since $4\eta_k\phi_p < 1/4$ and $\eta_k^2 < 1/4$. Every bare site is a corner of twelve plaquettes and each pillar contains η_k^{-3} sites. The row and column sum bounds on the response kernel therefore imply

$$\sum_p \max_{x \text{ corner of } p} m_{y(x)}^2 \le 12\eta_k^{-3}\sum_y m_y^2 \le 12\eta_k^{-3}S_{\mathrm{e}}^2\|B'\|_2^2.$$

After multiplying the squared plaquette bound by η_k^{-1}, this is eq. A.xvii. The unperturbed representative is also in the top background orbit, by the restriction identity used in Prop. A.4. Undoing the level-k axial gauge and the response gauge gives the stated representatives and the conjugations u_p.

At $k = 0$, take $\widehat{U} = U'$ and $\check{U} = V_0$. The four-factor telescoping identity gives $|R_p - \mathbf{1}| \le \sum_{b\in\partial p}|B'(b)|$. Each bond occurs in four plaquettes, so $\sum_p |R_p - \mathbf{1}|^2 \le 16\|B'\|_2^2$. Since $C_{\mathrm{nrg}} \ge 16(1 + 24C_{\mathrm{db}}^2)$, the same estimate holds. □

Proposition A.7 (Weighted excess)**.** For a competitor as in Prop. A.5 and any bounded complex plaquette weight c,

$$\left|A^{\eta_t}(c, U') - \mathsf{ab}_t(c, \mathcal{M}^t U')\right| \le 4\sup|c|\left[C_{\mathrm{nrg}}\gamma_{\mathrm{r}}^{-1}t\,\mathsf{exc}_t(U')A^{\eta_t}(U')\right]^{1/2}. \tag{A.xviii}$$

■

Proof. For two successive backgrounds use Lem. A.6 and set $s_k = \eta_k^{-1} \sum_p |R_p - \mathbf{1}|^2$. Unitary invariance and the difference of two squares give

$$|a_p(\widehat{U}) - a_p(\check{U})| \le \tfrac{1}{2}|R_p - \mathbf{1}|(|\partial\widehat{U}(p) - \mathbf{1}| + |\partial\check{U}(p) - \mathbf{1}|).$$

By Cauchy–Schwarz and the monotonicity in Prop. A.5, the difference of their weighted actions, both in level-t units, is at most $\sup|c|\,[2L^{t-k}s_k A^{\eta_t}(U')]^{1/2}$. Telescope these weighted actions in k and apply Cauchy–Schwarz once more. Equations eq. A.xvii and eq. A.xvi give

$$\sum_{k<t} L^{t-k} s_k \le C_{\mathrm{nrg}} \sum_{k<t} L^{t-k} \|\mathcal{A}_k\|_2^2 \le 8 C_{\mathrm{nrg}} \gamma_{\mathrm{r}}^{-1} \mathsf{exc}_t(U').$$

The resulting coefficient is $\sqrt{2 \cdot 8} = 4$. □

A.1.4. *The two classical fields.* Let $\mathrm{A} = (\ell, K)$ and $\mathrm{B} = (\ell, K+1)$, and compare level t of A to level $t+1$ of B, with $1 \le t \le K$. Their block lattices and averaging maps agree. For a common datum V with $\varpi(V) < \epsilon = 2c_\rho \varepsilon(x_t^{\mathrm{A}})$, where $x_t^{\mathrm{A}} \ge \Theta_\star$, write

$$U^{\mathrm{A}} = U_t^{\mathrm{A}}(V), \quad U^{\mathrm{B}} = U_{t+1}^{\mathrm{B}}(V), \quad a^{\mathrm{A}} = A^{\eta_t}(U^{\mathrm{A}}), \quad a^{\mathrm{B}} = A^{\eta_{t+1}}(U^{\mathrm{B}}).$$

Put $N_t = (2L^{m+K-t})^3$, $j_t = C_{\mathrm{en}} B_3 \epsilon L^{-2(t+1)}$, and $l_t = 4\pi^2 C_{\mathrm{pr}} B_3 \epsilon \eta_t$. The smallness conditions are exactly [YM3, eq. 6.civ]. We use the datum-response estimate [YM3, eq. 6.ciii].

Lemma A.8 (The two one-sided comparisons)**.** With these data, $U' = \mathcal{M}U^{\mathrm{B}}$ is a competitor at level t in A, of admitted radius $2B_3\epsilon$, and $\mathcal{M}^t U' = V$. Moreover

$$a^{\mathrm{A}} \le A^{\eta_t}(U') \le (1 + j_t)^2 a^{\mathrm{B}}. \tag{A.xix}$$

The reverse comparison is

$$a^{\mathrm{B}} \le (1 + 4000 L^4 \vartheta_0) a^{\mathrm{A}} + K_{\mathrm{up}} \epsilon^2 \eta_t N_t, \qquad \vartheta_0 = \vartheta(U^{\mathrm{A}}), \tag{A.xx}$$

where

$$K_{\mathrm{up}} = 24\pi^2 K_{\mathrm{B}}^{\mathrm{s}} C_{\mathrm{pr}} B_3^2 (1 + 4\pi^2 K_{\mathrm{B}}^{\mathrm{s}} C_{\mathrm{pr}}).$$

■

Proof. The curvature iteration of [YM3, Prop. 3.40] applies because $B_3\epsilon \le c_2^{\mathrm{SO}}$ and gives

$$\varpi(\mathcal{M}^{k+1} U^{\mathrm{B}}) \le 2B_3 \epsilon L^{-2(t-k)} \qquad (k < t).$$

At $k = t$ use $\mathcal{M}^{t+1} U^{\mathrm{B}} = V$. Since $2B_3\epsilon \le C_{\mathrm{cp}}\epsilon \le e_{\mathrm{cp}}$, Prop. A.5 proves the first inequality in eq. A.xix. The arbitrary-set averaging bound with all plaquettes proves the second one, since the input curvature is at most $B_3 \epsilon L^{-2(t+1)}$.

For the other direction, the angle of U^{A} is at most $\vartheta_0 \le \frac{\pi}{2} B_3 \epsilon \eta_t^2 \le 10^{-4}/4$. Set $\widetilde{U} = \mathsf{Prl}_L U^{\mathrm{A}}$. By Prop. A.2,

$$A^{\eta_{t+1}}(\widetilde{U}) \le (1 + 4000 L^4 \vartheta_0) a^{\mathrm{A}}, \qquad \varpi(\widetilde{U}) \le 4\pi \varpi(U^{\mathrm{A}}). \tag{A.xxi}$$

Put $U_1 = \mathcal{M}\widetilde{U}$ on the bare lattice of A. Normalize $\widetilde{U}$ to axial gauge by a gauge equal to one at the coarse sites. Its straight products remain $U^{\mathrm{A}}(c)$. Each of the $L-1$ internal bonds in such a product differs from one by at most $8A_L \varpi(\widetilde{U})$, and its crossing bond differs from $U_1(c)$ by at most $C_{\mathrm{AC}} \varpi(\widetilde{U})$. The product telescoping bound therefore gives

$$|U_1(c) - U^{\mathrm{A}}(c)| \le C_{\mathrm{pr}} \varpi(\widetilde{U}), \qquad \varpi(U_1) \le C'_{\mathrm{pr}} \varpi(U^{\mathrm{A}}). \tag{A.xxii}$$

The axial estimates apply because $\varpi(\widetilde{U}) \le 4\pi B_3 \epsilon \eta_t^2 < \omega_M$. The second bound uses four bond differences in a plaquette and eq. A.xxi.

At level zero in B, eq. A.xxi gives the competitor radius $4\pi L^2 B_3 \epsilon$. At levels $k \ge 1$, apply the curvature iteration to U_1 using $C'_{\mathrm{pr}} B_3 \epsilon \le c'_2 \le c_2^{\mathrm{SO}}$; this gives radius $2C'_{\mathrm{pr}} B_3 \epsilon$. Thus $\widetilde{U}$ is a competitor of radius $C_{\mathrm{cp}}\epsilon \le e_{\mathrm{cp}}$ at level $t+1$ of B.

The field

$$A(c) = \eta_t^{-1} \tfrac{1}{i} \log(U_1(c) U^{\mathrm{A}}(c)^{-1})$$

is real and has supremum at most $2\pi^2 C_{\mathrm{pr}} B_3 \epsilon \eta_t = l_t/2 < c_4$. The real centered averaging iteration gives

$$\mathcal{M}^t U_1 = (e^{i a_t} V)^u, \qquad \sup |a_t| \le 2 \sup |A| \le l_t,$$

where u is a real gauge on the top lattice. This is the factorization in the proof of [YM3, Prop. 4.37], also recorded in [YM3, eq. 4.cdxxx]; the reference curvature and field radii are exactly those in [YM3, eq. 6.civ]. Minimality for the competitor $\widetilde{U}$ and gauge invariance now imply

$$\mathsf{ab}^{\mathrm{B}}_{t+1}(e^{i a_t} V) \le A^{\eta_{t+1}}(\widetilde{U}).$$

By [YM3, eq. 6.civ], $l_t < \min\{r_{\mathrm{B}}, r^{\mathrm{s}}_{\mathrm{B}}/2\}$. The datum estimate [YM3, eq. 6.ciii] in B, with weight one, bounds the difference between this action and a^{B} by

$$6 K^{\mathrm{s}}_{\mathrm{B}} N_t l_t (B_3 \epsilon + K^{\mathrm{s}}_{\mathrm{B}} l_t) \le K_{\mathrm{up}} \epsilon^2 \eta_t N_t.$$

Combining this with eq. A.xxi proves eq. A.xx. □

Set

$$\begin{aligned} K_u &= K_{\mathrm{up}} + 10^4 L^4 B_3^3 + 5 C_{\mathrm{en}} B_3^3, \\ K_w &= 6 \left[2K_u + 5B_3^2 + 10 (C_{\mathrm{nrg}} \gamma_{\mathrm{r}}^{-1} K_u B_3^2)^{1/2} \right], \qquad \|c\|_{(t)} = \sup |c| + a_{K-t} \operatorname{Lip}_\infty(c). \end{aligned} \tag{A.xxiii}$$

Proposition A.9 (Classical action comparison)**.** For the data above, the two background actions satisfy

$$|a^{\mathrm{B}} - a^{\mathrm{A}}| \le K_u \epsilon^2 N_t L^{-t}, \tag{A.xxiv}$$

$$|A^{\eta_{t+1}}(c, U^{\mathrm{B}}) - A^{\eta_t}(c, U^{\mathrm{A}})| \le K_w \|c\|_{(t)} \epsilon^2 N_t \sqrt{t L^{-t}} \tag{A.xxv}$$

for every complex Lipschitz function c on T_ℓ. The constants K_u, K_w depend only on the fixed construction constants, not on K, m, b, t, V, or c. ■

Proof. Write $q = (1 + j_t)^2$. Since $j_t \le 2/5$, one has $1 \le q \le 2$, $q - 1 \le 3 j_t$, and

$$(q-1) a^{\mathrm{B}} \le 5 C_{\mathrm{en}} B_3^3 \epsilon^2 \eta_t N_t. \tag{A.xxvi}$$

Here we used [YM3, eq. 6.cv], $\epsilon \le 1$, and $L^{-2(t+1)} \le \eta_t$. The lower comparison eq. A.xix bounds $a^{\mathrm{A}} - a^{\mathrm{B}}$ by this quantity. In the upper comparison eq. A.xx, the prolongation error is at most

$$4000 L^4 \cdot \tfrac{\pi}{2} B_3 \epsilon \eta_t^2 \cdot \tfrac{3}{2} B_3^2 \epsilon^2 N_t \le 10^4 L^4 B_3^3 \epsilon^2 \eta_t N_t.$$

This proves eq. A.xxiv.

For the weighted bound first suppose $0 \le c \le c_{\max}$ and put $L_c = a_{K-t} \operatorname{Lip}_\infty(c)$, $v = \tfrac{3}{2} \eta_t L_c$. Apply Lem. A.3 to U^{B} with physical coarse spacing a_K. Its curvature satisfies the required bound, and its factor is at most q. Thus, for $X = A^{\eta_{t+1}}(c, U^{\mathrm{B}})$ and $Y = A^{\eta_t}(c, U')$,

$$Y \le q(X + v a^{\mathrm{B}}).$$

Apply the same bound to $c_{\max} - c$, and use $A^{\eta_t}(U') \ge a^{\mathrm{A}}$. Since $0 \le X \le c_{\max} a^{\mathrm{B}}$, the two inequalities give

$$-(q-1) c_{\max} a^{\mathrm{B}} - 2 v a^{\mathrm{B}} \le X - Y \le c_{\max}(a^{\mathrm{B}} - a^{\mathrm{A}}) + (q-1) c_{\max} a^{\mathrm{B}} + 2 v a^{\mathrm{B}}.$$

Equations eq. A.xxiv, eq. A.xxvi, and [YM3, eq. 6.cv] yield

$$|X - Y| \le (2 K_u c_{\max} + \tfrac{9}{2} B_3^2 L_c) \epsilon^2 \eta_t N_t. \tag{A.xxvii}$$

For the competitor U', the same inequalities give

$$\mathsf{exc}_t(U') \le 2 K_u \epsilon^2 \eta_t N_t, \qquad A^{\eta_t}(U') \le 3 B_3^2 \epsilon^2 N_t.$$

Apply Prop. A.7 to obtain

$$|Y - A^{\eta_t}(c, U^{\mathrm{A}})| \le 10 c_{\max} (C_{\mathrm{nrg}} \gamma_{\mathrm{r}}^{-1} K_u B_3^2)^{1/2} \epsilon^2 N_t \sqrt{t\eta_t}.$$

Since $\eta_t \le \sqrt{t\eta_t}$, summing this with eq. A.xxvii bounds the desired difference by $K_w(c_{\max} + L_c)\epsilon^2 N_t \sqrt{t\eta_t}/6$.

For a real signed weight, apply this result to $c + \sup|c|$ and subtract the constant weight using eq. A.xxiv. Since $K_u \le K_w/6$, the resulting bound is at most $K_w \|c\|_{(t)} \epsilon^2 N_t \sqrt{t\eta_t}/2$. Apply this separately to the real and imaginary parts of a complex weight and add their bounds. This proves eq. A.xxv. □

Proposition A.10. Use the two partner lattices above, with top levels t_* and $t_* + 1$, and assume $x_{t_*}^{\mathrm{A}} \ge \Theta_\star$. Let $1 \le k < t_*$ and $W \in \mathsf{Cut}_{k+1}^{\mathrm{A}} \cap \mathsf{Cut}_{k+2}^{\mathrm{B}}$. Then, on every common nonpivot bond,

$$\varphi\left(V_{k+1}^{\mathrm{bf,B}}(W)(b) V_k^{\mathrm{bf,A}}(W)(b)^{-1}\right) \le 8 c_\rho \varepsilon(x_k) \sqrt{K_u \gamma_{\mathrm{r}}^{-1} L^{-k} N_k}. \tag{A.xxviii}$$

■

Proof. Write $V^{\mathrm{A}}, V^{\mathrm{B}}$ for the two axial backgrounds and $\epsilon_j = 2c_\rho \varepsilon(x_j)$. Both average to W and, by [YM3, Lem. 6.33], have curvature less than $L^{-2}\epsilon_{k+1} \le L^{-3/2}\epsilon_k$. The axial comparison [YM3, eq. 6.ci] therefore bounds their relative intrinsic logarithm by $h' = \pi C_{\mathrm{ax}} L^{-3/2}\epsilon_k$. The first inequality in [YM3, eq. 6.civ], together with $C_{\mathrm{cp}} \ge 4\pi L^2$, puts h' below the chart and cubic-remainder thresholds in Prop. A.4. Thus V^{B} lies in the chart over W based at V^{A}. If A is its nonpivot logarithm, that proposition yields

$$\tfrac{1}{8}\gamma_{\mathrm{r}} \|A\|_2^2 \le \mathsf{ab}_k^{\mathrm{A}}(V^{\mathrm{B}}) - L^{-1}\mathsf{ab}_{k+1}^{\mathrm{A}}(W).$$

The restriction identity on lattice B gives $\mathsf{ab}_{k+1}^{\mathrm{B}}(V^{\mathrm{B}}) = L^{-1}\mathsf{ab}_{k+2}^{\mathrm{B}}(W)$. Apply eq. A.xxiv at level k to V^{B} and at level $k+1$ to W. These applications are valid since their curvatures are below $\epsilon_k, \epsilon_{k+1}$. The right side is at most

$$K_u\left(\epsilon_k^2 L^{-k} N_k + L^{-1}\epsilon_{k+1}^2 L^{-(k+1)} N_{k+1}\right) \le 2K_u \epsilon_k^2 L^{-k} N_k,$$

where $N_{k+1} = L^{-3}N_k$ and $\epsilon_{k+1}^2 \le L\epsilon_k^2$. Consequently $\|A\|_2 \le 4\epsilon_k \sqrt{K_u \gamma_{\mathrm{r}}^{-1} L^{-k} N_k}$. Each bond norm is bounded by this square-sum norm, and the intrinsic logarithm has norm equal to the angle. This is eq. A.xxviii; reversing a bond leaves that angle unchanged. □

A.2. **Sources and intermediate integration.** We use the direct source family, early systems, and recursion of [YM3, eq. 6.cx], [YM3, Lem. 6.35], and [YM3, eq. 6.cxvii]. The bounds and smallness conditions are [YM3, eq. 6.cxiv] and [YM3, eq. 6.cxv]. An early system at t_0 consists of pointed families $E^{(i)}[w]$, $1 \le i \le t_0$, on the fixed pair cores, with $\|E^{(i)}[w]\|_{i,\kappa+1} \le E_0^{\mathrm{s}}$. They are jointly holomorphic in w and every open pair slice; the member at X depends on w only through the plaquettes in X. The chain starts at $t_0 = 0$ with the empty array. A classical start at t_0 sets all its early families equal to zero; a hat denotes this flow.

Only the fourth potential changes with the source; the real backgrounds, Gaussian measures, gauge factors, and cuts stay fixed. All flows below stop at $t_* \le K - 1$. For every $t \le t_*$ the room condition gives

$$L^{m+K-t} \ge L^{m+1} \ge C_L M/2 \ge 4LM \ge 16M.$$

Thus the geometric hypothesis of the old-term estimate [YM3, Lem. 6.34] holds at every use below. Its constants are those of [YM3, eq. 6.cix].

Proposition A.11. For $1 \le t \le t_*$ and $V \in \mathsf{Cut}_t$, let $U = U_t(V)$ and let $\mathfrak{f} = ((D^{(i)}, 0)_{1 \le i \le t}, \emptyset)$ be a bare array at the fixed radii. Then its action is defined on the whole critical orbit, is constant there, and

$$|S[\mathfrak{f}](U)| \le C_{\mathrm{old}} N_t \sum_{i=1}^{t} L^{-(t-i)} \|D^{(i)}\|_{i,\kappa+1}. \tag{A.xxix}$$

■

Proof. Plain-core membership of the level-t background, its tower restrictions, and radius monotonicity put every level-i pair in the fixed core. The defining sum for the action is therefore

$$S[\mathfrak{f}](U) = \sum_{i=1}^{t} \sum_{z} \sum_{X \ni z} [D^{(i)}(X, U, z) - D^{(i)}(X, \mathbf{1}, z)].$$

Each summand is gauge invariant. The regularity statement of [YM3, Lem. 6.33] and the room assumption permit [YM3, Lem. 6.34] at every i. Sum its estimates. □

Proposition A.12. Every early system continues through level K by [YM3, eq. 6.cxvii]. Its new families have the same norm, holomorphy and source-locality properties as the early families. At each step the fourth family has norm at most $7w_\star/32 < w_\star/4$.

For $t_0 \le k \le t_*$ and $V \in \mathsf{Cut}_k$ put

$$\Phi_k^w(V) = -A^{\eta_k}(x_k^w, U_k(V)) + S[\mathfrak{f}_k[w]](U_k(V)), \qquad F_k^w = \mathbf{1}_{\mathsf{Cut}_k} e^{\Phi_k^w}.$$

These are bounded Borel functions. For $t_0 \le k < t_*$ set

$$c_{k+1}(w) = \log\{\mathsf{pf}_k(x_k)\mathsf{Z}^{(k)}[\mathbf{1}]\} + \sum_{X,z} E^{(k+1)}[w](X, \mathbf{1}, z), \tag{A.xxx}$$

where the logarithm is real. For every $W \in \mathsf{Cut}_{k+1}$,

$$\mathsf{I}[F_k^w](W) = e^{c_{k+1}(w) + \Phi_{k+1}^w(W)} \neq 0. \tag{A.xxxi}$$

Consequently, the finite complex measures

$$\nu_{t_0}^w = F_{t_0}^w \, \mathrm{d}V, \qquad \nu_{k+1}^w = \mathbf{1}_{\mathsf{Cut}_{k+1}} \mathcal{M}_*[\mathfrak{G}_k[x_k] \chi_k^{\mathrm{sf}} \nu_k^w]$$

satisfy

$$\nu_k^w = \exp\left(\sum_{t_0 < i \le k} c_i(w) \right) \mathbf{1}_{\mathsf{Cut}_k} e^{\Phi_k^w} \, \mathrm{d}V. \tag{A.xxxii}$$

The chain restarted with its own first t_0 families is the same chain. ■

Proof. Suppose the families through j have been constructed. Their weighted array size is at most $LE_0^{\mathrm{s}}/(L-1) \le N_\star^{\mathrm{s}}$. The actual old-potential rule of [YM3, Prop. 4.56] therefore has norm at most $K_\Lambda \mathsf{s}_j N_\star^{\mathrm{s}} \le w_\star/8$. The source contributes at most $3\mu_\star \le 3w_\star/32$. The three internal potentials have total norm at most $3K_P\delta(x_j) \le 3w_\star/8$. Thus [YM3, Lem. 5.21] applies on the full core. Its unperturbed bound, the tent mass, and the normalization bound of [YM3, Prop. 4.56] give respectively

$$\|\mathbf{F}[\mathbb{V}'[\mathfrak{f}_j]]\|_{j+1,\kappa+1} \le K_{\mathbf{M}} \left(\delta(x_j) + \mathsf{s}_j N_\star^{\mathrm{s}} + e^{-\gamma_2 \mathfrak{b}(x_j)^2/2} \right) \le E_0^{\mathrm{s}}/4, \qquad \|\mathbf{N}^{(j+1)}\|_{j+1,\kappa+1} \le E_0^{\mathrm{s}}/2.$$

Here the output rates exceed $\kappa + 1$ by $4\beta_{\mathrm{s}}\kappa \ge 1$ and the normalizer decay assumption. In [YM3, eq. 5.cvi], the coefficient is at most $162L^3K_\Pi < K^{\mathrm{d}}$ whenever $n_{\mathrm{int}} \le 3w_\star/8$, $n_{\mathrm{iv}} \le w_\star/4$ and $n_W \le w_\star/2$. The source perturbation therefore costs at most $3K^{\mathrm{d}}\mu_\star \le 3E_0^{\mathrm{s}}/16$. The new norm is at most $15E_0^{\mathrm{s}}/16$. The parameter statement of [YM3, Lem. 5.21], the linear old-potential rule and [YM3, Lem. 6.35] give holomorphy. If two sources agree on X, every earlier member used inside X agrees, as does the source family; the same local construction then gives the same new member. This proves the induction, including the first step, whose old array is empty.

For the density identity fix $W \in \mathsf{Cut}_{k+1}$. The cut theorem puts W in the actual global sumrule class. The old-potential sumrule and [YM3, eq. 6.cxiii] identify the fourth-family sum with the change along the fibre of

$$\mathcal{S}_k(U) = S[\mathfrak{f}_k[w]](U) - x_k A^{\eta_k}(v_k^w, U).$$

The three internal sumrules and the identity $\mathbf{C}_*(P_W) = \mathsf{A}_W^{-1}$ identify the chart integral with its truncated Gaussian integral. Its cluster sum is the exponential of the sum of the pointed terms by [YM3, eq. 5.civ]. The Gaussian normalization, localized by $\mathbf{N}^{(k+1)}$, contributes the first logarithm

in eq. A.xxx; this is the actual normalization identity of [YM3, Prop. 4.56], with the hypotheses verified in [YM3, Prop. 4.58]. Hence the chart value is

$$\exp\left\{\log(\mathsf{pf}_k \mathsf{Z}^{(k)}[\mathbf{1}]) + \sum_{X,z} E^{(k+1)}[w](X,U_+,z) - A^{\eta_k}(x_k^w,U_+) + S[\mathfrak{f}_k[w]](U_+)\right\}.$$

Subtract and add the trivial-pair values in this finite sum. The old members and the new centered member are precisely the action of $\mathfrak{f}_{k+1}[w]$; the weighted action rescales because $x_k^w/L = x_{k+1}^w$. This proves eq. A.xxxi, including its nonvanishing.

All these actions are defined on the critical orbit by the cut theorem, and are constant there by gauge invariance. Local analytic representatives of that orbit show that Φ_k^w is Borel on the cut. Its absolute value is bounded there by the cut action estimate and Prop. A.11. Extending its exponential by zero gives the asserted bounded Borel function. The physical image-density formula applies to it. Every chart over Cut_{k+1} lies in Cut_k, so the inserted cut is identically one in the chart. Induction with eq. A.xxxi gives eq. A.xxxii. Equality upon restarting follows directly from the deterministic recursion. □

For the decay of earlier interactions use $c_{\log} = \log L/\log\gamma^{-2}$ and

$$\Sigma_\phi = \sum_{a\ge 1} L^{-a/2}(1+ac_{\log})^{\mathfrak{p}_0}, \qquad C_{\mathrm{fg}} = \exp\{\tfrac{13}{72}(L-1)\Sigma_\phi\}.$$

The scale decay and contraction bounds are [YM3, eq. 6.cxviii] and [YM3, eq. 6.cxix].

Proposition A.13. For two early systems at t_0 and the same source, let

$$\Delta_i = \|E^{(i)}[w] - \widetilde{E}^{(i)}[w]\|_{i,\kappa+1}, \qquad n_j = \sum_{i\le j} L^{-(j-i)}\Delta_i.$$

For $j \ge t_0$,

$$n_j \le C_{\mathrm{fg}} L^{-(j-t_0)} n_{t_0} \le C_{\mathrm{fg}} L^{-(j-t_0)} \frac{L}{L-1} \max_{i\le t_0} \Delta_i. \tag{A.xxxiii}$$

In particular, the chain and classical start obey this bound with $\max \Delta_i \le E_0^{\mathrm{s}}$. ■

Proof. The normalizers and source families cancel in the difference of the two recursions. Linearity of the old-potential rule and its weighted-lag estimate give $\mathsf{n}^{\mathrm{f}}(\mathbb{V}'[\mathfrak{f}_j - \widetilde{\mathfrak{f}}_j]) \le K_\Lambda \mathsf{s}_j n_j$. Both fourth families have norm below $w_\star/4$, so their difference has norm below $w_\star/2$. Equation [YM3, eq. 5.cvi] therefore gives $\Delta_{j+1} \le K^{\mathrm{d}} K_\Lambda \mathsf{s}_j n_j$. Consequently

$$n_{j+1} \le (L^{-1} + K^{\mathrm{d}} K_\Lambda \mathsf{s}_j) n_j, \qquad n_j \le L^{-(j-t_0)} n_{t_0} \exp\left(L K^{\mathrm{d}} K_\Lambda \sum_{i=t_0}^{j-1} \mathsf{s}_i\right).$$

Equations [YM3, eq. 6.cxviii] and [YM3, eq. 6.cxix] bound the last exponential by C_{fg}. The geometric sum bounds n_{t_0}, proving the result. If $t_0 = 0$ the flows coincide. □

Define

$$\epsilon_{\mathrm{fg}}(t,t_0) = C_{\mathrm{old}} C_{\mathrm{fg}} \frac{L}{L-1} E_0^{\mathrm{s}} N_t L^{-(t-t_0)}, \qquad \mathsf{m}_t^S = C_{\mathrm{old}} \frac{L}{L-1} E_0^{\mathrm{s}} N_t. \tag{A.xxxiv}$$

Combining Prop. A.13, Prop. A.11, for $t_0 \le t \le t_*$ and $V \in \mathsf{Cut}_t$ we have

$$|\Phi_t^w(V) - \widehat{\Phi}_t^w(V)| \le \epsilon_{\mathrm{fg}}(t,t_0), \qquad |S[\mathfrak{f}_t[w]](U_t(V))| \le \mathsf{m}_t^S. \tag{A.xxxv}$$

The same action bound holds for every early system. The weighted Wilson actions cancel in the first difference; this estimate concerns the field-dependent exponents only.

A.2.1. *A uniform fluctuation tail.* Set

$$\kappa_{\mathrm{b}} = \frac{w_\star}{32M^3}e^{-\lambda_V}, \qquad C_{\mathrm{sh}} = \tfrac{1}{4}K^{\mathrm{d}}w_\star M^3 C_{\mathrm{E},1}, \quad C_{\mathrm{E},1} = 2048\sum_{n\ge 0}(n+2)e^{-n}. \tag{A.xxxvi}$$

These constants are independent of γ and of the cutoff, torus, level and source.

Proposition A.14. Let $t_0 \le k < t_*$, let b be a nonpivot bond at level k, and let $g = x_k^{-1/2}$. Let $\Sigma \in \mathbb{C}$ be a scalar such that $e^{\Sigma}F_k^w$ is real and nonnegative on Cut_k. Then, for $W \in \mathsf{Cut}_{k+1}$ and $a \ge 0$,

$$0 \le \mathsf{I}[e^{\Sigma}F_k^w \mathbf{1}\{|\mathcal{A}_k^\diamond(b)| \ge ga\}](W) \le e^{-\kappa_{\mathrm{b}}a^2 + C_{\mathrm{sh}}}\mathsf{I}[e^{\Sigma}F_k^w](W), \qquad \mathsf{I}[e^{\Sigma}F_k^w](W) > 0. \tag{A.xxxvii}$$

The indicator is set to zero where the intrinsic fluctuation is undefined; with this convention it is Borel. ∎

Proof. On the stable background domain the map $(V, W) \mapsto V(b)V_k^{\mathrm{bf}}(W)(b)^{-1}$ is continuous. The intrinsic logarithm is continuous away from $-\mathbf{1}$. Taking its norm superlevel set and composing with the Borel map $V \mapsto (V, \mathcal{M}V)$ proves the Borel assertion.

Let $\mathbb{Q}_b(Y; P, B) = \frac{1}{2}\operatorname{tr}(B(b)^2)$ if $Y = \square(b)$, and zero otherwise. This is holomorphic, gauge invariant and independent of P. Since $d_k(\square(b)) = 0$ and $|B(b)|^2 \le M^3 q_{\square(b)}(B)$,

$$\mathsf{n}_X^{\mathrm{f}}(\kappa_{\mathrm{b}}\mathbb{Q}_b) \le w_\star/32, \qquad \mathsf{n}_X^{\mathrm{f}}(\kappa_{\mathrm{b}}\mathbb{Q}_b) = 0 \quad \text{unless } \square(b)^\sharp \subset X.$$

Adding $\lambda\mathbb{Q}_b$, for $\lambda = 0, \kappa_{\mathrm{b}}$, leaves the fourth-family norm at most $w_\star/4$. The Gaussian sumrule used in Prop. A.12 now gives

$$\mathsf{I}[F_k^w e^{\lambda x_k|\mathcal{A}_k^\diamond(b)|^2}](W) = p_W \Xi_\lambda, \qquad \Xi_\lambda = \exp\sum_{X,z}(\mathbf{G}_1 + \mathbf{F}_\lambda)(X, P_W|_X, z). \tag{A.xxxviii}$$

The value assigned to the extra exponential off its domain is immaterial. Indeed every chart lies inside the cut, where $\mathcal{A}_k^\diamond(b) = gB(b)$, so the insertion is $e^{\lambda|B(b)|^2}$. In the Gaussian calculation

$$p_W = \mathsf{pf}_k(x_k)Z^{(k)}(W; \mathsf{A}_W)\exp[-x_k\mathbb{F}_W(0) + \mathcal{S}_k(U_{k+1}(W))] \ne 0$$

is independent of λ. The three internal sums are $\mathbb{P}^J, \mathbb{P}^G, \mathbb{P}^A[\mathsf{B}_W]$, the fourth sum is $\mathsf{s}_W[\mathcal{S}_k] + \lambda|B(b)|^2$, and the covariance is A_W^{-1}. At $\tau = 1$ the quadratic correction potential vanishes. These identities give eq. A.xxxviii directly from the physical Gaussian change of variables.

The local form of [YM3, eq. 5.cvi] gives

$$|(\mathbf{F}_{\kappa_{\mathrm{b}}} - \mathbf{F}_0)(X, P, z)| \le K^{\mathrm{d}}w_\star e^{-(\kappa+1)d_{k+1}(X)}/32,$$

and the difference vanishes unless $\square(b)^\sharp \subset X$. All contributing domains contain one fixed level-$(k+1)$ site z_b: choose it in the partition cube containing b_-. Each such domain has at most $8M^3(d_{k+1}(X)+1)$ base points. The rooted entropy bound therefore gives

$$|\log(\Xi_{\kappa_{\mathrm{b}}}/\Xi_0)| \le \frac{K^{\mathrm{d}}w_\star}{32}\sum_{X \ni z_b} 8M^3(d_{k+1}(X)+1)e^{-(\kappa+1)d_{k+1}(X)} \le C_{\mathrm{sh}},$$

where the logarithm denotes the displayed cluster-sum difference. Thus $|\Xi_{\kappa_{\mathrm{b}}}/\Xi_0| \le e^{C_{\mathrm{sh}}}$.

The chart measure is positive, and its fields lie in Cut_k. Apply $\mathbf{1}\{|B(b)| \ge a\} \le e^{-\kappa_{\mathrm{b}}a^2}e^{\kappa_{\mathrm{b}}|B(b)|^2}$ under that measure with the nonnegative factor $e^{\Sigma}F_k^w$. The denominator is nonzero by eq. A.xxxviii and nonnegative by the same chart formula, hence positive. The quotient of the tilted and untilted integrals is real and nonnegative, and its modulus is at most $e^{C_{\mathrm{sh}}}$. This proves eq. A.xxxvii. □

A.2.2. *Source radii and field-independent terms.* Let $f_1, \dots, f_n$ be nonzero real Lipschitz functions on T_ℓ and $w_z = \sum_i z_i f_i$. Write

$$D_R^\flat = \{z : \sum_i |z_i| \|f_i\|_{\flat;T_\ell} < R\}, \qquad r_i(t) = (8nM_t^{\mathrm{rd}} \|f_i\|_{\flat;T_\ell})^{-1},$$
$$M_t^{\mathrm{rd}} = \max\{\varrho^{-1}, \tfrac{3}{2}(e_t^{\mathrm{c}})^2 L^{K-t} N_t, \varrho^{-1} \mathsf{m}_t^S\}.$$

The polydisc $D_t^{\mathrm{rd}} = \{|z_i| < r_i(t)\}$ lies in $D^\flat_{\varrho/8}$. Recall that $0 < \|f_i\|_\flat \le \|f_i\|_\sharp$ and $\|w_z\|_\flat \le \sum_i |z_i| \|f_i\|_\flat$.

Lemma A.15. For any flow beginning at $t_0 < t$, $1 \le t \le t_*$, and $V \in \mathsf{Cut}_t$, the function $z \mapsto \Phi_t^{w_z}(V)$ is holomorphic on $D^\flat_{2\varrho}$, and

$$|\Phi_t^{w_z}(V) - \Phi_t^0(V)| \le 1/4 \qquad (z \in D_t^{\mathrm{rd}}). \tag{A.xxxix}$$

■

Proof. Fix a critical representative $U = U_t(V)$, independent of z. The finite sum $\Psi(w) = S[\mathfrak{f}_t[w]](U)$ is holomorphic and bounded by m_t^S on Src. The exponent difference is $L^{K-t} A^{\eta_t}(w_z, U) + \Psi(w_z) - \Psi(0)$. The cut action bound makes the first term at most $M_t^{\mathrm{rd}} \|w_z\|_\flat < 1/8$. Schwarz's lemma on the complex line through w_z, whose source radius is $2\varrho/\|w_z\|_\flat$, bounds the second term by $\mathsf{m}_t^S \|w_z\|_\flat / \varrho < 1/8$. Their sum proves the assertion. □

For $n \ge 2$ define the physical support spread

$$d_\infty(\underline{f}) = \min_{x_i \in \operatorname{supp} f_i} \max_{i,i'} |x_i - x_{i'}|_\infty.$$

Compactness shows that the minimum is attained and that it is positive exactly when the supports have no common point. For families with the norm and locality properties of the flows, put

$$c_*(z) = \sum_{j=j_0}^{j_1} \sum_{X,z'} E_*^{(j)}[w_z](X, \mathbf{1}, z'), \qquad 1 \le j_0 \le j_1 \le K.$$

Lemma A.16. The function c_* is holomorphic on $D^\flat_{2\varrho}$ and $|c_*| \le 2E_0^{\mathrm{s}} C_{\mathrm{E},0} N_{j_0}$ there. If $n \ge 2$ and $d_\infty L^{K-j_1} \ge 6M$, then

$$|\partial_1 \cdots \partial_n c_*(0)| \le \epsilon_{\mathrm{mx}}(j_1), \qquad \epsilon_{\mathrm{mx}}(j_1) = 2e^3 E_0^{\mathrm{s}} C_{\mathrm{E},0} N_{j_1} e^{-d_\infty L^{K-j_1}/M} \prod_i \frac{n\|f_i\|_\flat}{2\varrho}. \tag{A.xl}$$

■

Proof. Each member is bounded by $E_0^{\mathrm{s}} e^{-(\kappa+1)d_j(X)}$. Sum first over domains containing a fixed site and then over levels, using $N_j = L^{-3(j-j_0)} N_{j_0}$, to obtain the first bound. All sums are finite and holomorphic.

Apply Cauchy's formula on the polydisc with radii $2\varrho/(n\|f_i\|_\flat)$. A differentiated member vanishes unless, for every i, X contains a plaquette at whose barycentre f_i is nonzero: otherwise locality makes it independent of that coefficient. Choose one such plaquette for each i. Their corners are in X, are within $\eta_j/2$ of their barycentres, and have mutual sup distance at most $M(d_j(X) + 2)$ in level-j units. Hence a contributing domain satisfies

$$d_j(X) \ge d_\infty L^{K-j}/M - 3 =: D_j \ge 3.$$

The rooted tail and Cauchy's formula bound its level sum by

$$T_j = E_0^{\mathrm{s}} C_{\mathrm{E},0} N_j e^{3 - d_\infty L^{K-j}/M} \prod_i \frac{n\|f_i\|_\flat}{2\varrho}.$$

Since $T_{j-1}/T_j \le L^3 e^{-6(L-1)} \le 1/2$, the sum is at most $2T_{j_1}$, proving eq. A.xl. □

A.3. **Comparison of the positive densities.** We use the calibrated history and discarding estimates of [YM3, Prop. 6.37]. The comparison takes place on a common block lattice: the two backgrounds differ, but all integrations and gauge factors at equal depth agree.

Fix ℓ and a label b in the calibrated range. The lattices $\mathrm{A} = (\ell, K)$ and $\mathrm{B} = (\ell, K+1)$ use the lines $x_j^{\mathrm{A}} = L^{K-j}\mathsf{b}$ and $x_j^{\mathrm{B}} = L^{K+1-j}\mathsf{b}$. At equal depth d their configuration lattices, Haar measures, averaging maps, gauge factors and fluctuation thresholds coincide; their critical backgrounds need not coincide. Write

$$x_{[d]} = L^d\mathsf{b}, \qquad N_{[d]} = \ell^3 L^{3d}, \qquad y_{[d]} = (\log x_{[d]})^{2\mathfrak{p}_0}.$$

Choose integers $1 \le u < s < K$, put $t_0 = K - s$, $t_1 = K - u$, and require $x_{[u]} \ge \Theta_\star$. The partner of a level k of A is $k+1$ of B; a subscript (k) refers to this common depth.

On each lattice run the chain and a classical start at depth s to depth u. On the common final configuration space $\mathcal{X} = \mathcal{V}(\Lambda_{u,\ell})$ set

$$I^\bullet(z) = \int_{\mathsf{In}_{[u]}} e^{\Phi^{\bullet,w_z}}\,\mathrm{d}V, \quad \widehat{I}^\bullet(z) = \int_{\mathsf{In}_{[u]}} e^{\widehat{\Phi}^{\bullet,w_z}}\,\mathrm{d}V, \quad T^\bullet = e^{c^\bullet} I^\bullet, \quad \widehat{T}^\bullet = e^{\widehat{c}^\bullet}\widehat{I}^\bullet.$$

Here $c^\bullet$ sums all the chain increments through the final level, while $\widehat{c}^\bullet$ sums the increments strictly after the classical start. These identities are eq. A.xxxii. Define the contact function

$$\mathfrak{c}^\bullet(z) = \sum_{j=1}^{t_1^\bullet} \sum_{X,z'} E^{(j),\bullet}[w_z](X, \mathbf{1}, z') - \sum_{j=t_0^\bullet+1}^{t_1^\bullet} \sum_{X,z'} \widehat{E}^{(j),\bullet}[w_z](X, \mathbf{1}, z').$$

Then $c^\bullet - \widehat{c}^\bullet - \mathfrak{c}^\bullet$ is independent of z: it is the sum of the real Gaussian normalization logarithms up to the starting level.

Use the common radii $r_i = r_i(t_1)$ of Lem. A.15, and put

$$\epsilon_{\mathrm{fg}} = C_{\mathrm{old}} C_{\mathrm{fg}} \tfrac{L}{L-1} E_0 N_{[u]} L^{-(s-u)}, \tag{A.xli}$$

$$\epsilon_{\mathrm{cl}} = 4K_w c_\rho^2 A_0^2 (1 + 2\varrho/\mathsf{b}) y_{[K]} N_{[s]} \sqrt{K} L^{-(K-s)/2}, \tag{A.xlii}$$

$$\epsilon_{\mathrm{sh}} = 3(s-u) N_{[s]} \exp\{C_{\mathrm{sh}} - \kappa_{\mathrm{b}} A_1^2 y_{[u]}/4\}, \tag{A.xliii}$$

$$\epsilon_3 = 4\epsilon_{\mathrm{cl}} + 6\epsilon_{\mathrm{sh}}, \qquad M_c = 4E_0 C_{\mathrm{E},0} N_{[s]}. \tag{A.xliv}$$

In this section impose

$$\epsilon_{\mathrm{fg}} \le 1/8, \quad \epsilon_{\mathrm{cl}}, \epsilon_{\mathrm{sh}} \le 1/20, \quad 256 K_u c_\rho^2 A_0^2 N_{[s]} L^{-(K-s)} \le \gamma_{\mathrm{r}} A_1^2. \tag{A.xlv}$$

Proposition A.17. At every real point x of the polydisc $|z_i| < r_i$, $\widehat{T}^{\mathrm{A}}(x), \widehat{T}^{\mathrm{B}}(x) > 0$ and

$$\left| \widehat{T}^{\mathrm{B}}(x)/\widehat{T}^{\mathrm{A}}(x) - 1 \right| \le \epsilon_3 \le 1/2. \tag{A.xlvi}$$

■

Proof. On $\mathcal{Y} = \mathcal{V}(\Lambda_{s,\ell})$ write $W_k = \mathcal{M}^{k-t_0} W$, and let

$$k^\bullet(W) = \mathbf{1}_{\mathsf{In}_{[u]}}(W_{t_1}) \prod_{k=t_0}^{t_1-1} [\mathbf{1}_{\mathsf{Cut}^\bullet_{(k)}} \mathfrak{G}_{(k)} \chi^\bullet_{(k)}](W_k),$$

$$\widehat{F}^\bullet_z(W) = \mathbf{1}_{\mathsf{Cut}^\bullet_{(t_0)}}(W) \exp[-A^{\eta_{t_0^\bullet}}(L^s(\mathsf{b} - w_z), U^\bullet_{t_0^\bullet}(W))].$$

Iterating the image formula gives $\widehat{T}^\bullet(z) = \int k^\bullet \widehat{F}^\bullet_z\,\mathrm{d}W$. For real x the start is strictly positive on its cut. The positive chart kernel propagates nonnegativity, while the nonzero exponential density formula makes the final density positive on its cut. Since the common inner set is open and contains $\mathbf{1}$, it has positive Haar measure. Thus both final integrals are strictly positive.

We first bound the mass where the two weights differ. Suppose $k^\bullet(W) > 0$ and $k^\circ(W) = 0$. Starting from $W_{t_1} \in \mathsf{In}_{[u]}$, cut stability shows that there is a last index k with $\chi^\circ_{(k)}(W_k) = 0$, and its

next field lies in both next-level cuts. Otherwise all the other cuts would follow downwards from their fluctuation tests, and k° would be positive. By Prop. A.10 and eq. A.xlv, at each nonpivot bond the two background origins differ by an angle at most

$$8c_\rho\varepsilon(x_{(k)})(K_u\gamma_{\mathrm{r}}^{-1}L^{-k}N_k)^{1/2}\le\delta(x_{(k)})/2.$$

If all fluctuations of lattice $\bullet$ were smaller than $\delta(x_{(k)})/2$, the angular triangle inequality would make every fluctuation of lattice $\circ$ defined and smaller than $\delta(x_{(k)})<\pi$. Its test would equal one, a contradiction. Thus some bond has a fluctuation at least half the threshold.

Insert the indicator of this event at its step in the positive classical-start measure. By Prop. A.14 with $a=\mathfrak{b}(x_{(k)})/2$, the next measure is bounded by $\exp(C_{\mathrm{sh}}-\kappa_{\mathrm{b}}\mathfrak{b}(x_{(k)})^2/4)$ times the ordinary next measure. All later positive cut-and-image operations preserve this inequality. There are $s-u$ steps, at most $3N_{[s]}$ nonpivot bonds at each step, and $\mathfrak{b}(x_{(k)})^2\ge A_1^2y_{[u]}$. The union bound consequently gives both estimates

$$\int k^\bullet\mathbf{1}_{\{k^\circ=0\}}\widehat{F}_x^\bullet\,\mathrm{d}W\le\epsilon_{\mathrm{sh}}\widehat{T}^\bullet(x).\tag{A.xlvii}$$

This argument uses the image identity also for the marked measure, obtained by the same induction as the unmarked one; hence it bounds the final mass, without an assumption that the later gauge factors are at most one.

Where both weights are positive they equal the product of the common gauge factors. The two starts there satisfy $\widehat{F}_x^{\mathrm{B}}=\widehat{F}_x^{\mathrm{A}}e^{q_x}$. Apply Prop. A.9 at $t=t_0$ with $\varepsilon_{\mathrm{r}}=2c_\rho\varepsilon(x_{[s]})$ and $c_x=L^s(\mathsf{b}-w_x)$. Since $\|w_x\|_\flat<\varrho/8$, $\|c_x\|_{(t_0)}\le L^s(\mathsf{b}+2\varrho)$. The comparison estimate gives

$$|q_x|\le K_w\|c_x\|_{(t_0)}\varepsilon_{\mathrm{r}}^2N_{[s]}\sqrt{t_0}L^{-t_0/2}\le\epsilon_{\mathrm{cl}}.$$

Split the two integrals into their common positive set and their two exceptional sets. The latter have masses bounded by eq. A.xlvii; on the common set use $|e^{q_x}-1|\le2\epsilon_{\mathrm{cl}}$. First,

$$(1-\epsilon_{\mathrm{sh}})\widehat{T}^{\mathrm{B}}(x)\le e^{\epsilon_{\mathrm{cl}}}\widehat{T}^{\mathrm{A}}(x),\qquad\widehat{T}^{\mathrm{B}}(x)\le2\widehat{T}^{\mathrm{A}}(x).$$

Then the total variation of the two weighted densities is at most $(2\epsilon_{\mathrm{cl}}+3\epsilon_{\mathrm{sh}})\widehat{T}^{\mathrm{A}}(x)$. This implies the weaker bound eq. A.xlvi, with the stated constants. □

The real estimate does not by itself control derivatives in complex sources. The following elementary two-constants argument supplies the required passage.

Lemma A.18. If f is holomorphic on the polydisc $|z_i|<r_i$, bounded there by B, and bounded at its real points by $0<\epsilon\le B$, then, for $0<\sigma\le1$,

$$|\partial_1\cdots\partial_nf(0)|\le\epsilon^{1-\sigma}B^\sigma(4n/(\pi\sigma))^n\prod_i r_i^{-1}.\tag{A.xlviii}$$

■

Proof. First take a one-variable unit disc. In its upper half put $h(\zeta)=\frac{2}{\pi i}\operatorname{Log}\frac{1+\zeta}{1-\zeta}$. Its real part lies between zero and one, is zero on the diameter and tends to one on the circular arc. For $\nu>0$ apply the maximum principle to

$$f(\zeta)\epsilon^{-1}e^{-\log(B/\epsilon)h(\zeta)}((1-\zeta^2)/2)^\nu.$$

The boundary upper limits of its modulus are at most one; the last factor makes those at ±1 zero. Let $\nu\downarrow0$. For $|\zeta|\le\tau<1$,

$$\Re h(\zeta)=\frac{2}{\pi}\arctan\frac{2\Im\zeta}{1-|\zeta|^2}\le\delta_\tau:=\frac{4}{\pi}\arctan\tau.$$

Reflection gives the same lower-half bound, so $|f|\le\epsilon^{1-\delta_\tau}B^{\delta_\tau}$ on the smaller disc. Iterating in the n variables gives $|f|\le\epsilon^{(1-\delta_\tau)^n}B^{1-(1-\delta_\tau)^n}$ on $|z_i|\le\tau r_i$. Choose $\tau=\pi\sigma/(4n)$. Then $\delta_\tau\le\sigma/n$ and $(1-\delta_\tau)^n\ge1-\sigma$. Cauchy's formula on the smaller polydisc proves the result. The maximum

principle with boundary upper limits follows by taking a maximizing sequence in the bounded domain: a limit on the boundary obeys the assumed bound, whereas an interior maximum forces constancy. □

Proposition A.19. On the common source polydisc the two chain functions are zero-free. Moreover, for $0 < \sigma \le 1$,

$$\begin{aligned}&\left|\partial^n \log T^{\mathrm{B}}(0) - \partial^n \log T^{\mathrm{A}}(0) - \partial^n(\mathfrak{c}^{\mathrm{B}} - \mathfrak{c}^{\mathrm{A}})(0)\right| \\ &\quad \le \left[8\epsilon_{\mathrm{fg}} + 2^{n+1}(4n/(\pi\sigma))^n \epsilon_3^{1-\sigma}(1+2M_c)^\sigma\right] \prod_i r_i^{-1}.\end{aligned} \tag{A.xlix}$$

Here $\partial^n = \partial_1 \cdots \partial_n$. ■

Proof. The integrals are holomorphic by dominated differentiation: the quarter-disc bound supplies the integrable majorant $e^{1/4}|e^{\Phi^0}|$ for each of them. The positive classical-start final density at zero shows that $e^{\widehat{\Phi}^0}$ is a fixed nonzero complex scalar times a positive function. Thus $\widehat{I}(z)/\widehat{I}(0)$ is an average of $e^{\widehat{\Phi}^{w_z} - \widehat{\Phi}^0}$ with respect to a probability measure. The quarter-disc bound puts it in the convex region

$$\Re\zeta \ge e^{-1/4}\cos(1/4), \qquad |\zeta| \le e^{1/4}, \qquad |\Im\zeta| \le \tan(1/4)\Re\zeta.$$

In this region $|\operatorname{Log}\zeta| \le 1/2$. Also eq. A.xxxv implies

$$|I/\widehat{I} - 1| \le \frac{e^{1/2}}{\cos(1/4)}(e^{\epsilon_{\mathrm{fg}}} - 1) \le 2\epsilon_{\mathrm{fg}}.$$

Consequently the holomorphic functions $h^\bullet = \operatorname{Log}(I^\bullet/\widehat{I}^\bullet)$ and $l^\bullet = \operatorname{Log}(\widehat{I}^\bullet/\widehat{I}^\bullet(0))$ satisfy $|h^\bullet| \le 4\epsilon_{\mathrm{fg}}$, $|l^\bullet| \le 1/2$, and $\partial^n \log T^\bullet = \partial^n(c^\bullet + h^\bullet + l^\bullet)$.

The contact bound Lem. A.16 on the classical-start families gives $|\widehat{c}^\bullet(z) - \widehat{c}^\bullet(0)| \le M_c$. Define

$$H = (\widehat{c}^{\mathrm{B}} - \widehat{c}^{\mathrm{B}}(0)) - (\widehat{c}^{\mathrm{A}} - \widehat{c}^{\mathrm{A}}(0)) + l^{\mathrm{B}} - l^{\mathrm{A}}.$$

Then $H(0) = 0$, $|H| \le 1 + 2M_c$, and on the real polydisc

$$H(x) = \log(\widehat{T}^{\mathrm{B}}(x)/\widehat{T}^{\mathrm{A}}(x)) - \log(\widehat{T}^{\mathrm{B}}(0)/\widehat{T}^{\mathrm{A}}(0)).$$

Indeed their exponentials agree, and the continuous difference takes values in $2\pi i\mathbb{Z}$ and is zero at the origin. Equation eq. A.xlvi bounds this real value by $4\epsilon_3$. Apply Lem. A.18 with $\epsilon = 4\epsilon_3$ and $B = \max(2, 1 + 2M_c)$. The resulting constant satisfies $4^{1-\sigma}2^\sigma \le 2^{n+1}$. Finally Cauchy's formula bounds $\partial^n(h^{\mathrm{B}} - h^{\mathrm{A}})$ by $8\epsilon_{\mathrm{fg}}\prod_i r_i^{-1}$. The source-independent difference between $c^\bullet - \widehat{c}^\bullet$ and $\mathfrak{c}^\bullet$ has zero mixed derivative. Adding the estimates proves eq. A.xlix. □

A.4. **Summing the cutoff differences.** Write $S^T_{n;K}(\underline{f}) = S^T_{n;K,\ell,L^K\mathsf{b}}(\underline{f})$ and retain the calibrated setting of [YM3, §6.12]. For the comparison at cutoffs $K, K+1$, choose

$$u = \left\lceil \frac{K}{180(3n+4)} \right\rceil, \qquad s = \lceil K/9 \rceil, \qquad \mathsf{p}_n = 2\mathsf{p}_0(n+1) + \tfrac{1}{2}.$$

At this point the label is in a compact interval $J = [\mathsf{b}_-, \mathsf{b}_+]$ satisfying

$$\mathsf{b}_- \ge \gamma^{-2}, \qquad MR(\mathsf{b}_+) \le L^m, \qquad \ell = 2L^m, \quad m \ge 1. \tag{A.l}$$

All constants chosen before the cutoff, including the source radius ϱ, are fixed throughout J.

Proposition A.20. For an arbitrary nonzero Lipschitz tuple of order $n \geq 1$, assume the comparison hypotheses of Prop. A.17 and $R_{t_1} \leq 1/2$, where $t_1 = K - u$. Then

$$\begin{aligned} &\left| S^T_{n;K+1}(\underline{f}) - S^T_{n;K}(\underline{f}) - \partial^n(\mathfrak{c}^{\mathrm{B}} - \mathfrak{c}^{\mathrm{A}})(0) \right| \\ &\quad \leq \left[8\epsilon_{\mathrm{fg}} + q_n \epsilon_3^{79/80} (1 + 2M_c)^{1/80} \right] \prod_i r_i(t_1)^{-1} \\ &\qquad + 8\mathsf{op}_n R_{t_1} \prod_i (6\ell^3 L^{4(K+1)} \|f_i\|_\infty). \end{aligned} \tag{A.li}$$

Here $q_n = 2^{n+1}(320n/\pi)^n$. If also $n \geq 2$ and $d_\infty(\underline{f}) L^u \geq 6M$, then $|\partial^n \mathfrak{c}^\bullet(0)| \leq 2\epsilon_{\mathrm{mx}}(t_1)$ for each $\bullet$.

There are constants $C_n(J) > 0$, $K_0(\ell, n, J)$, and, for every $d > 0$, $K_1(\ell, n, J, d) \geq K_0$, such that all the preceding smallness hypotheses hold for $K \geq K_0$, uniformly in $\mathsf{b} \in J$. For $n \geq 2$, $K \geq K_1$, and all nonzero tuples with $d_\infty(\underline{f}) \geq d$,

$$|S^T_{n;K+1}(\underline{f}) - S^T_{n;K}(\underline{f})| \leq C_n(J) \ell^{3(n+1)} K^{\mathfrak{p}_n} L^{-K/10} \prod_i \|f_i\|_\sharp. \tag{A.lii}$$

Here $C_n(J)$ depends only on the fixed constants, n, and the endpoints of J. The thresholds have precisely the additional dependencies displayed, and K_1 may be chosen nonincreasing in d. ■

Proof. Apply [YM3, Prop. 6.37] at the common final depth u to both lattices. Their discarding factors are the same, since the final coupling, site count, and physical volume are the same. The two errors add to the last term of eq. A.li. Apply the holomorphic derivative estimate eq. A.xlix with $\sigma = 1/80$. Its contact term is exactly the one displayed here, since the omitted Gaussian normalization logarithms are source independent. This proves eq. A.li. Each $\mathfrak{c}^\bullet$ is a difference of two contact sums ending at depth u; Lem. A.16 bounds its derivative by $2\epsilon_{\mathrm{mx}}(t_1)$.

We give the uniform estimates, including the choice of thresholds. Put $y_{[j]} = (\log(L^j \mathsf{b}))^{2\mathfrak{p}_0}$, $N_{[j]} = \ell^3 L^{3j}$, and

$$\overline{C}_M = \max\left\{ \varrho^{-1}, \frac{3c_\rho^2 A_0^2}{2\mathsf{b}_-}, \varrho^{-1} C_{\mathrm{old}} \frac{L}{L-1} E_0 \right\}, \qquad \overline{y}_K = (\log(L^K \mathsf{b}_+))^{2\mathfrak{p}_0}.$$

The source radii of Lem. A.15 satisfy

$$r_i(t_1)^{-1} \leq 8n \overline{C}_M \overline{y}_K N_{[u]} \|f_i\|_\sharp. \tag{A.liii}$$

Indeed the three entries in $M^{\mathrm{rd}}_{t_1}$ are bounded by $\overline{C}_M y_{[K]} N_{[u]}$: for its classical entry use $(e^{\mathrm{c}}_{t_1})^2 L^u = c_\rho^2 A_0^2 y_{[u]}/\mathsf{b}$, and for the old-action entry use $m^S_{t_1} = C_{\mathrm{old}} L(L-1)^{-1} E_0 N_{[u]}$. Also $1 \leq y_{[u]} \leq y_{[K]} \leq \overline{y}_K$ and $\|f_i\|_\flat \leq \|f_i\|_\sharp$.

Replace $y_{[K]}$ by $\overline{y}_K$ and $1 + 2\varrho/\mathsf{b}$ by $1 + 2\varrho/\mathsf{b}_-$ in ϵ_{cl} to define $\bar{\epsilon}_{\mathrm{cl}}$. Define the other two majorants by

$$\bar{\epsilon}_{\mathrm{sh}} = 3(s-u) N_{[s]} \exp\left[C_{\mathrm{sh}} - \tfrac{1}{4}\kappa_b A_1^2 (\log(L^u \mathsf{b}_-))^{2\mathfrak{p}_0} \right],$$

$$\overline{R} = 2M^{-3} N_{[u]} \exp\left[-c' A_0^2 (\log(L^u \mathsf{b}_-))^{2\mathfrak{p}_0} + (C^\Sigma_{\mathrm{maj}} + 2 + C_{\mathrm{low}}(\mathsf{b}_+)) \ell^3 \right].$$

They dominate the corresponding errors throughout J; the last assertion uses that $C_{\mathrm{low}}(b)$ is increasing. Since $\log(L^u \mathsf{b}_-) \geq K \log L/[180(3n+4)]$ and $2\mathfrak{p}_0 \geq 2$, both majorants tend to zero after multiplication by any $K^a e^{a'K}$.

The ceiling bounds give

$$\begin{gathered} u(3n+4) - s \leq -19K/180 + 3n + 4, \\ 3s - \tfrac{79}{160}(K - s) + 3un \leq -K/10 + 7/2 + 3n, \\ 4u - s \leq -K/10 + 4, \qquad (7s - K)/2 \leq -K/9 + 7/2, \qquad 4s - K \leq -5K/9 + 4. \end{gathered} \tag{A.liv}$$

For example, in the second inequality the coefficient of s is $559/160$; replacing s by $K/9+1$ gives $-19K/180+559/160$, while $3un < K/180+3n$. The others follow directly from $K/[180(3n+4)] \le u < K/[180(3n+4)]+1$ and $K/9 \le s < K/9+1$. Thus $1 \le u < s < K$, $L^u \mathsf{b}_- \ge \Theta_\star$, $\epsilon_{\mathrm{fg}} \le 1/8$, and the axial comparison condition hold eventually. The latter's left side is a constant times L^{4s-K}. Also $\bar\epsilon_{\mathrm{cl}}$ is bounded by a constant times $K^{2\mathsf{p}_0+1/2}L^{-K/9}$, so that $\bar\epsilon_{\mathrm{cl}}, \bar\epsilon_{\mathrm{sh}} \le 1/20$ and $\overline{R} \le 1/2$ eventually.

Here is an explicit constant for the asserted rate. With $h = \log(L\mathsf{b}_+) > 1$, take

$$C_n(J) = 4(8n\overline{C}_M)^n h^{2\mathsf{p}_0(n+1)} L^{3n+4} \times \left[8C_{\mathrm{old}} C_{\mathrm{fg}} \frac{L}{L-1} E_0 + q_n \max\{1, 16K_w c_\rho^2 A_0^2 (1+2\varrho/\mathsf{b}_-)\}(1+8E_0 C_{\mathrm{E},0}) \right]. \tag{A.lv}$$

Let $P = (8n\overline{C}_M \overline{y}_K N_{[u]})^n$. Split the right side of eq. A.li, after removing $\prod_i \|f_i\|_\sharp$, into

$$8\epsilon_{\mathrm{fg}} P, \quad q_n(4\bar\epsilon_{\mathrm{cl}})^{79/80}(1+2M_c)^{1/80}P, \quad q_n(6\bar\epsilon_{\mathrm{sh}})^{79/80}(1+2M_c)^{1/80}P, \quad 8\mathsf{op}_n \overline{R}(6\ell^3 L^{4(K+1)})^n.$$

This is valid because $(a+b)^{79/80} \le a^{79/80} + b^{79/80}$. The first two terms together are at most $\frac{1}{4}C_n(J)\ell^{3(n+1)}K^{\mathsf{p}_n}L^{-K/10}$. For the first, use the first inequality of eq. A.liv and $\overline{y}_K \le h^{2\mathsf{p}_0}K^{2\mathsf{p}_0}$. For the second, use

$$(1+2M_c)^{1/80} \le (1+8E_0 C_{\mathrm{E},0})\ell^{3/80} L^{3s/80}.$$

This combines with the $79/80$ power of $\bar\epsilon_{\mathrm{cl}}$ to give exactly $\ell^3 L^{3s-79(K-s)/160}$; the second inequality of eq. A.liv gives the required exponential, and $\overline{y}_K^{n+1} K^{1/2} \le h^{2\mathsf{p}_0(n+1)} K^{\mathsf{p}_n}$ gives its polynomial.

Choose K_0 so that all preceding smallness conditions hold for every $K \ge K_0$, and the last two terms of the four-term display are at most a further quarter of the rate. Such a K_0 exists by the superexponential decay just proved. This bounds eq. A.li by half the rate, uniformly on J and in the tuple. For $n \ge 2$ choose $K_1 \ge K_0$ so that, for every $K \ge K_1$,

$$dL^u \ge 6M, \qquad 8e^3 E_0 C_{\mathrm{E},0} \left(\frac{n}{2\varrho}\right)^n N_{[u]} e^{-dL^u/M} \le \tfrac{1}{2} C_n(J)\ell^{3(n+1)} K^{\mathsf{p}_n} L^{-K/10}.$$

The left side decays doubly exponentially in K apart from $N_{[u]}$, so this choice exists. It bounds $4\epsilon_{\mathrm{mx}}(t_1)$ after the same norm product is removed. Taking the least eventual threshold makes K_1 nonincreasing in d. This proves the proposition. □

Corollary A.21. Under eq. A.l, fix $n \ge 2$ and $d > 0$. Put

$$C_n'(J) = C_n(J) \sum_{j \ge 0} (1+j)^{\mathsf{p}_n} L^{-j/10}.$$

For $K' \ge K \ge K_1(\ell, n, J, d)$ and every nonzero tuple with $d_\infty(\underline{f}) \ge d$,

$$|S^T_{n;K'}(\underline{f}) - S^T_{n;K}(\underline{f})| \le C_n'(J)\ell^{3(n+1)} K^{\mathsf{p}_n} L^{-K/10} \prod_i \|f_i\|_\sharp. \tag{A.lvi}$$

Thus the whole cutoff sequence has a limit, uniformly in the label on J and on every set of tuples satisfying $d_\infty(\underline{f}) \ge d$ and a fixed upper bound on $\prod_i \|f_i\|_\sharp$. The limit is continuous in the label. The same limit is obtained whenever $K_j \to \infty$ and $a_{K_j}\beta_j \in J$ tends to the specified label. ■□

Proof. For $k = K + j$ one has $k \le K(1+j)$, whence

$$\sum_{k \ge K} k^{\mathsf{p}_n} L^{-k/10} \le K^{\mathsf{p}_n} L^{-K/10} \sum_{j \ge 0} (1+j)^{\mathsf{p}_n} L^{-j/10}.$$

The last series converges by geometric domination. Telescoping eq. A.lii proves eq. A.lvi, and the uniform Cauchy criterion gives the limit. At fixed cutoff, the partition function and its source derivatives are entire functions of the bare coupling: the configuration space is compact and all plaquette actions are bounded. The real partition function is strictly positive. Hence every finite-cutoff cumulant is continuous in the positive real label, and its uniform limit on J is continuous there. Finally write $b_j = a_{K_j}\beta_j$. The difference from the asserted limit is bounded by the uniform

cutoff error at b_j, plus the difference of the two limiting values at b_j and its limit. Both tend to zero. □

A.5. **Positive labels and dilation.**

Lemma A.22. The constants can be chosen so that the calibrated setting used in Prop. A.20, Cor. A.21 holds on every label interval satisfying eq. A.l. The choices precede the torus, the cutoff, and the labels. ■

Proof. Use the actual canonical auxiliary data and the ordered choice of [YM3, Prop. 4.163]. Its real background, response, Gaussian, and normalizer hypotheses are the global hypotheses imposed above. The fixed radii are chosen with $\alpha_0 \le a_{\mathrm{H}}$, and the decay and normalizer floors include $4\beta_{\mathrm{s}}\kappa \ge 1$ and the rates used in the rooted sums. These choices precede γ. In particular [YM3, eq. 4.dcccxxxiv] gives $E_0 \ge 2E_{\mathsf{N}}$. Take $E_0^{\mathrm{s}} = E_0$ and $N_\star^{\mathrm{s}} = L(E_0 + \frac{3}{2}\alpha_0^2 B_\sharp)/(L-1)$.

Every coefficient in [YM3, eq. 6.cxv] is now fixed. Its three small quantities are constant multiples of

$$\gamma(\log\gamma^{-2})^{\mathfrak{p}_0}, \qquad \exp[-\gamma_2 A_1^2(\log\gamma^{-2})^{2\mathfrak{p}_0}/2],$$

and tend to zero as $\gamma \downarrow 0$. Thus the three inequalities hold after decreasing γ. Impose the direct-source half-radius requirement at the same stage; its left side is $C_{\mathcal{A}}\delta(\gamma^{-2})$. The response and classical calibration requirements are positive fixed-radius ceilings on ε and δ, so they hold uniformly above a finite threshold. The derived numbers $\tau_\natural, \Theta_\star$ are then finite; they are thresholds on the final intermediate coupling, which tends to infinity in the cutoff comparison.

Fix $\eth$ and the history constants before this last choice. The case-II and summability thresholds used in [YM3, Lem. 6.36] are finite and independent of γ. They can therefore be imposed on $\log\gamma^{-2}$ simultaneously. Also require $\gamma^{-2} > \bar{B}$ and $R(\gamma^{-2}) \ge C_L/(2L)$. If $MR(\mathsf{b}') \le L^m$ and $\mathsf{b}' \ge \gamma^{-2}$, monotonicity of R gives $2L^{m+1} \ge C_L M$, as required in [YM3, eq. 6.xcvi]. The line $x_j = L^{K-j}\mathsf{b}'$ has zero increments, is admissible, and has precisely the torus room required in [YM3, Thm. C]. The actual mass bounds and vacuum lower bound used in the discarding proof consequently apply to this same line and the same vacuum constant.

Finally $K_{\mathrm{Q}}^\vee, K_{\mathrm{L}}^\vee, w_\star, E_0$ and $K_{\mathbf{M}}^{\mathrm{f}}$ are positive and finite. Thus $\mu_\star, \sigma_{\mathrm{src}} > 0$, and the stated minimum defining ϱ is positive. Its two defining inequalities give [YM3, eq. 6.cxvi] for every $\mathsf{b}' \ge \gamma^{-2}$. At zero source the added family vanishes identically; the sourced recursion and the canonical real progression therefore coincide step by step. This verifies the interface used in the history comparison. The constants and this source radius involve no finite-lattice datum. □

For $s \in \mathbb{Z}$ and $m + s \ge 1$, let $f^{(s)}(x') = f(L^{-s}x')$ on $T_{L^s\ell}$. Dilation multiplies both the Euclidean and sup distances by L^s. Consequently

$$\|f^{(s)}\|_\infty = \|f\|_\infty, \qquad \operatorname{Lip} f^{(s)} = L^{-s}\operatorname{Lip} f, \qquad d_\infty(\underline{f}^{(s)}) = L^s d_\infty(\underline{f}). \tag{A.lvii}$$

Supports are carried onto their dilates. In particular the absence of a common support point is preserved, and $\|f^{(s)}\|_\sharp \le \|f\|_\sharp$ when $s \ge 0$.

Lemma A.23. For $K, m \ge 1$, $-m < s < K$, $\beta > 0$, and every Lipschitz tuple, one has

$$S^T_{n;K,\ell,\beta}(\underline{f}) = L^{sn} S^T_{n;K-s,L^s\ell,\beta}(\underline{f}^{(s)}), \qquad a_{K-s}\beta = L^s a_K \beta. \tag{A.lviii}$$

For every compact interval $J = [b_-, b_+] \subset (0,\infty)$, there is an integer $s \ge 0$ for which

$$L^s b_- \ge \gamma^{-2}, \qquad MR(L^s b_+) \le L^{m+s}. \tag{A.lix}$$

One may take any sufficiently large integer satisfying

$$L^s \ge \max\{\gamma^{-2}/b_-, b_+\}, \qquad L^{m+s/2} \ge LM(4r/e)^r. \tag{A.lx}$$

■

Proof. For $s \geq 0$, dilation is a bijection of the bonds and plaquettes and preserves their orientations, plaquette products, and product Haar measure. Corresponding source coefficients agree because $f^{(s)}(L^s x_p) = f(x_p)$ and $a_{K-s}^{-1} L^s = a_K^{-1}$. Thus

$$Z_{K,\ell,\beta}\left[\sum_i z_i f_i\right] = Z_{K-s,L^s\ell,\beta}\left[\sum_i L^s z_i f_i^{(s)}\right].$$

Differentiating local logarithms at zero gives the factor L^{sn}. If $s < 0$, apply the same nonnegative-dilation identity with exponent $-s$ to the data $(K-s, m+s, \beta, \underline{f}^{(s)})$; its dilated functions are the original f_i. This proves the identity in both directions. Equation eq. A.lvii follows by taking infima over lifts for the distances and then the minimum over the product of supports.

For eq. A.lix, put $x = L^s b_+$ and suppose eq. A.lx. Then $x \geq \gamma^{-2} > e$. The least-power definition of R gives $R(x) < L(\log x)^r$. Also $\log x \leq 2s \log L = 4y$, where $y = s \log L/2 > 0$. Since $y^r \leq (r/e)^r e^y$, we obtain

$$MR(x) < LM(4r/e)^r L^{s/2} \leq L^{m+s}.$$

The other inequality in eq. A.lix is already in eq. A.lx. Both conditions in eq. A.lx hold for all sufficiently large s, since their right sides are fixed. □

Proposition A.24. Fix $\ell = 2L^m$, $m \geq 1$. For every $\mathsf{b} > 0$, $n \geq 2$, and real Lipschitz functions $f_1, \dots, f_n$ with $\bigcap_i \operatorname{supp} f_i = \emptyset$, the limit

$$S_n^{(\ell,\mathsf{b})}(\underline{f}) = \lim_{K\to\infty} S^T_{n;K,\ell,L^K\mathsf{b}}(\underline{f}) \tag{A.lxi}$$

exists along all positive integer cutoffs. Convergence is locally uniform in $\mathsf{b} \in (0,\infty)$.

More precisely, fix $J = [b_-, b_+] \subset (0,\infty)$, $d > 0$, and an integer $s \geq 0$ satisfying eq. A.lix. Put

$$C_s' = C_n'([L^s b_-, L^s b_+]), \qquad k_s = K_1(L^s\ell, n, [L^s b_-, L^s b_+], L^s d).$$

For every nonzero tuple with $d_\infty(\underline{f}) \geq d$, every $\mathsf{b} \in J$, and $K \geq k_s + s$,

$$\left|S^T_{n;K,\ell,L^K\mathsf{b}}(\underline{f}) - S_n^{(\ell,\mathsf{b})}(\underline{f})\right| \leq L^{sn} C_s' (L^s\ell)^{3(n+1)} (K-s)^{\mathsf{p}_n} L^{-(K-s)/10} \prod_i \|f_i\|_\sharp. \tag{A.lxii}$$

For all $K \geq 1$, the finite-cutoff cumulant and its limit are bounded in modulus by

$$\mathsf{op}_n \prod_i (6\ell^3 L^{4(k_s+s)} \|f_i\|_\infty) + L^{sn} C_s' (L^s\ell)^{3(n+1)} k_s^{\mathsf{p}_n} L^{-k_s/10} \prod_i \|f_i\|_\sharp. \tag{A.lxiii}$$

The constants are independent of the tuple except for its lower spread bound d.

For every integer $s > -m$, the scaling law is

$$S_n^{(\ell,\mathsf{b})}(\underline{f}) = L^{sn} S_n^{(L^s\ell, L^s\mathsf{b})}(\underline{f}^{(s)}). \tag{A.lxiv}$$

If $K_j \to \infty$, $\beta_j > 0$, and $a_{K_j}\beta_j \to \mathsf{b} > 0$, then

$$S^T_{n;K_j,\ell,\beta_j}(\underline{f}) \longrightarrow S_n^{(\ell,\mathsf{b})}(\underline{f}). \tag{A.lxv}$$

No monotonicity of the sequence K_j is required. ■

Proof. If an entry is zero, multilinearity makes every cumulant zero. Otherwise compactness of the supports gives $d_\infty(\underline{f}) > 0$. For the dilated tuple, eq. A.lvii gives spread at least $L^s d$ and norms no larger than the original ones. The interval $[L^s b_-, L^s b_+]$ satisfies eq. A.l on $T_{L^s\ell}$. Apply Cor. A.21 there with cutoff $k = K - s$ and multiply its estimate by L^{sn} using eq. A.lviii. This is the Cauchy estimate with the right side of eq. A.lxii. It tends to zero uniformly in the label, so the whole sequence converges, and passing the second cutoff to infinity proves eq. A.lxii.

For $K \geq k_s + s$, compare to cutoff $k_s + s$ by the same Cauchy estimate. The finite-cutoff moment bound used in [YM3, Prop. 6.37] bounds that term by the first term of eq. A.lxiii. For $K < k_s + s$ the same finite bound is even smaller. This proves the uniform bound for all cutoffs and, by passage to the limit, for the limit itself. Taking $J = \{\mathsf{b}\}$ proves existence at every positive label. Taking any

compact interval proves local uniformity and hence continuity, since each finite-cutoff cumulant is continuous.

For any fixed integer $s > -m$, eq. A.lviii holds for all sufficiently large K. The index $K - s$ also runs through all sufficiently large integers, so passage to the limit gives eq. A.lxiv, including negative s. For the last assertion put $b_j = a_{K_j}\beta_j$. Eventually $b_j \in [\mathsf{b}/2, 2\mathsf{b}]$; subtract the limit at b_j and use the uniform cutoff estimate there. Continuity then replaces b_j by b. □

Corollary A.25. Let $0 < g \le \gamma$ and let $\ell = 2L^m$, $m \ge 1$. The cutoff limits for separated tuples converge uniformly for $\mathsf{b} \in [g^{-2} - \bar{B}, X_g + \bar{B}]$. In particular this holds on every side in $\mathfrak{L}(g)$. For $\mathsf{b} \in I(g)$ the pairs $(K, L^K\mathsf{b})$ are first-passage data at g for every K. More generally, a sequence (K_j, β_j) of first-passage data with $K_j \to \infty$ and convergent labels has the limit eq. A.lxv. ■□

Proof. The condition $\gamma^{-2} \ge 2B_\sharp$ gives $\bar{B} \le L[2(L-1)]^{-1}g^{-2} < g^{-2}$. The displayed label interval is consequently a compact subset of $(0, \infty)$, so Prop. A.24 applies. The assertion about $I(g)$ is [YM3, Cor. 5.8]. The same result bounds every first-passage label between $g^{-2} + B_\sharp - L^{-1}\bar{B}$ and $X_g + \bar{B}$; the lower endpoint is positive. Thus a convergent sequence of those labels has positive limit, as required in eq. A.lxv. □

A.6. **The coarsest fluctuation template.** The squared-kernel comparison for the free field also controls the fluctuation field after the coarsest background is removed. We record the extra error, since the fluctuation kernel is used in the fourth-order calculation.

Lemma A.26. Suppose $8a_K \le d \le 1$, $20d \le \ell$, and $(f, h) \in \mathfrak{T}_{T_\ell}(d; \vartheta)$. For $\mathcal{K}[k](f, h) = \sum_{p,q} f(x(p))h(x(q))k(p, q)^2$, there is a construction constant $C^{\mathrm{fl}}_{\mathrm{tm}}$ such that

$$\left|\mathcal{K}[C^{\mathrm{fl}}_{K,0}](f, h) - \frac{3}{8\pi^2}\mathcal{K}_{\mathrm{free}}(f, h)\right| \le C^{\mathrm{fl}}_{\mathrm{tm}}\left[\frac{a_K}{\vartheta d} + (d/\ell)^3 + d^3\right]\|f\|_\infty\|h\|_\infty.$$

■

Proof. The projection identity [YM3, eq. 7.ii] gives $C^{\mathrm{fl}}_{K,0} = C^{\mathrm{free}}_K - C^{\mathrm{bg}}_{K,0}$. The background estimate [YM3, eq. 7.xi], with $a_0 = 1$, gives

$$|C^{\mathrm{fl}}_{K,0}(p, q) - C^{\mathrm{free}}_K(p, q)| \le C^{\mathrm{bg}}_{\mathrm{tp}}a_K^3 = (C^{\mathrm{bg}}_{\mathrm{tp}}d^3)a_K^3 d^{-3}.$$

On the product of the test supports, the free-kernel bound [QCDM1, eq. A.vii] gives $|C^{\mathrm{free}}_K(p, q)| \le (1 + 3C_{\mathrm{lat}})a_K^3 d^{-3}$. Apply the squared-kernel inequality [QCDM1, eq. A.ix] with $A_1 = C^{\mathrm{bg}}_{\mathrm{tp}}d^3$ and $A_2 = 1 + 3C_{\mathrm{lat}}$. Since $d \le 1$, the difference of these two squared-kernel sums is at most

$$C_{\mathrm{fl}}d^3\|f\|_\infty\|h\|_\infty, \qquad C_{\mathrm{fl}} = 324C^{\mathrm{bg}}_{\mathrm{tp}}[2(1 + 3C_{\mathrm{lat}}) + C^{\mathrm{bg}}_{\mathrm{tp}}].$$

The free case of [QCDM1, Prop. A.2] supplies the other two errors. Take $C^{\mathrm{fl}}_{\mathrm{tm}} = \max\{C_{\mathrm{tm}}, C_{\mathrm{fl}}\}$. □

Lemma A.27 (Smooth reflection pairs)**.** For every $0 < \vartheta < 1$ and $d > 0$ there is $f \in C^\infty_c(\mathbb{R}^3)$ supported in $\mathbb{B}(2de_3, d)$ such that $(\theta f, f)$ is a test pair of scale d and shape ϑ, where $\theta(x_1, x_2, x_3) = (x_1, x_2, -x_3)$. ■

Proof. Put $x = 2de_3$ and

$$r_0 = \max\{\vartheta, (3\vartheta/\pi)^{1/3}\}, \qquad r = (1 + r_0)/2, \qquad \epsilon = \tfrac{1}{2}\min\{1 - r, r - \vartheta\}.$$

Then $r_0 < r < 1$, $0 < \epsilon < r$, $r + \epsilon < 1$, $r - \epsilon > \vartheta$, and $(\pi/3)r^3 > \vartheta$. Convolve $g_r(y) = \max\{0, 1 - |y - x|/(rd)\}$ with a nonnegative smooth mollifier of integral one supported in $\mathbb{B}(0, \epsilon d)$. The resulting function satisfies

$$0 \le f \le 1, \qquad f(x) \ge 1 - \epsilon/r > 0, \qquad \operatorname{Lip} f \le (rd)^{-1},$$
$$\int f(y)\,\mathrm{d}y = (\pi/3)r^3d^3, \qquad \operatorname{supp} f \subset \mathbb{B}(x, (r + \epsilon)d).$$

The integral follows by polar integration of g_r; convolution preserves it. Since $\|f\|_\infty \ge (r-\epsilon)/r > \vartheta/r$, these bounds imply $\operatorname{Lip} f \le \|f\|_\infty/(\vartheta d)$ and $\int f(y)\,\mathrm{d}y \ge \vartheta\|f\|_\infty d^3$. Reflection preserves the bounds. The supports lie in balls of radius d centered at $\pm x$, so their distance is at least $2d$; it is at most $4d$ because $f(x) > 0$. Their cross distances are at most $6d$, and both supports lie in $\mathbb{B}(0, 3d)$. These verify the remaining test-pair conditions. □

A.7. **Block operators in physical units.** We first record the estimates for the chamber operator that will be used to compare two meshes. Their constants do not require a lower bound on the mass. Fix $r = 1/2$, odd $L \ge 13$ and $b_F > 0$. In units in which the target block lattice has spacing one, let $\xi = L^{-j}$ be the bare spacing and ℓ' the side. The spinor norm is $\|f\|^2 = \xi^3 \sum_v |f(v)|^2$. Write $f = \iota\varphi + g$, where ι is the isometric inclusion of chamber-independent spinors and $g = \Pi_1 f$. Put

$$\mathcal{N}(f)^2 = \|\nabla^U\varphi\|^2 + \xi^{-2}\|g\|^2, \qquad (\nabla^U_e\varphi)(x) = \xi^{-1}(U(x, x+\xi e)\varphi(x+\xi e) - \varphi(x)),$$

with both orientations in the gradient norm.

Lemma A.28 (Energy estimates independent of the mesh)**.** Let $\alpha = \xi^{-2}\max_p |\partial U(p) - 1|$. There are constants $c_1', C_{\mathrm{en}} > 0$, depending only on the chamber operator, such that

$$\begin{aligned} \operatorname{Re}\langle f, D^\xi[U]f\rangle &\ge \tfrac{1}{2}\lambda_\rho\xi^{-1}\|g\|^2 + c_\Delta\xi\|\nabla^U\varphi\|^2, \\ \|D^\xi[U]f\|^2 &\ge \tfrac{1}{2}c_1'\mathcal{N}(f)^2 - C_{\mathrm{en}}(\alpha + \ell'^{-2})\|f\|^2. \end{aligned}$$

For $m \ge 0$, $\|(D^\xi[U] + m)f\| \ge \|D^\xi[U]f\|$ and $\operatorname{Re}(D^\xi[U] + m - \widehat{J}) \ge m - \sup|J|$. ■

Proof. The first inequality is the chamber coercivity in [QCDM1, §5], after inserting the powers of ξ in the norm and difference. The last two claims follow by expanding the square and using $\operatorname{Re} D^\xi \ge 0$.

For the second inequality at $U = 1$, the heavy block of $\widehat{D}(p)$ has real part at least λ_ρ, its light-heavy couplings are $O(|p|)$, and its light Schur complement is $i\gamma\cdot p + O(|p|^2)$. Thus its two block equations control $|p||\Pi_0 f| + |\Pi_1 f|$ near zero. Away from zero the symbol is invertible, and compactness gives a positive minimum. Equivalently,

$$c_1' = \inf_{p\ne0, f\ne0} \frac{|\widehat{D}(p)f|^2}{2\widehat{p}^2|\Pi_0 f|^2 + |\Pi_1 f|^2} > 0, \qquad \widehat{p}^2 = \sum_\mu (2 - 2\cos p_\mu).$$

Plancherel proves the free bound with coefficient c_1'.

Set $R = \min\{\alpha^{-1/2}, \ell'/4\}$. If $R < \xi$, the error in the statement absorbs $\mathcal{N}(f)^2 \le 24\xi^{-2}\|f\|^2$. Otherwise choose $\sum_i \chi_i^2 = 1$ subordinate to cubes of side $2R$, with bounded overlap and $\operatorname{Lip}\chi_i \le C/R$. The range-one property gives $\|[D^\xi, \chi_i]\| \le C/R$, while

$$\sum_i \chi_i[D^\xi, \chi_i] = -\tfrac{1}{2}\sum_i [[D^\xi, \chi_i], \chi_i]$$

has norm at most $C\xi/R^2$. Expanding the squared localized norms therefore gives

$$\|D^\xi f\|^2 \ge \sum_i \|D^\xi\chi_i f\|^2 - CR^{-2}\|f\|^2.$$

In a complete axial gauge on one cube, each link is a product of at most $6R/\xi$ plaquette holonomies, so it differs from one by at most $6R\xi\alpha$. The operator and gradient differ from their free versions by $CR\alpha$. Apply the free estimate to each $\chi_i f$ and use the gradient localization identity. Two applications of $(a-b)^2 \ge 2^{-1/2}a^2 - (1+\sqrt{2})b^2$ leave the coefficient $c_1'/2$ and error $C(R^{-2} + R^2\alpha^2) \le C(2\alpha + 16\ell'^{-2})$. The antiperiodic seam is absorbed in the local gauge; it changes no plaquette. This proves the claim. □

Lemma A.29 (Averages at the level of the spinor)**.** Let U have bare spacing $\xi_j = L^{-j}$ and satisfy $|\partial U(p) - 1| \le c_2^{\mathrm{SO}} \xi_j^2$ at every plaquette, so that all its successive averages are defined. Put

$$\alpha = \max_{0 \le n \le j} \xi_n^{-2} \max_p |\partial(\mathcal{M}^{j-n} U)(p) - 1|, \qquad \xi_n = L^{-n}.$$

Let $R(U)$ be the composite average onto the unit lattice, normalized by $RR^* = 1$. Then

$$\begin{aligned} R(U) &= \widetilde{R}(\mathcal{M}U) Q(U) \iota^*, \\ \|\nabla^{\mathcal{M}U}(Q(U)\varphi)\| &\le (1 + CL^2 \alpha \xi^2) \|\nabla^U \varphi\| + CL^2 \alpha \xi \|\varphi\|, \\ \|\nabla(R\iota\varphi)\| &\le C_A(\|\nabla^U \varphi\| + \alpha \|\varphi\|), \\ \|\varphi - (R\iota)^*(R\iota)\varphi\| &\le \tfrac{3}{4} \|\nabla^U \varphi\| + C'_A \alpha \|\varphi\|. \end{aligned}$$

The constants are independent of the number of steps. ■

Proof. The composite is formed with the averaged configuration at each successive level, so its first factor is $Q(U)\iota^*$ and its remaining factors form $\widetilde{R}(\mathcal{M}U)$. Their normalized adjoints are isometries, proving the first identity and $RR^* = 1$. Two contour transports in one block differ by at most $\frac{3}{2} L^2 \alpha \xi^2$: join them by the corresponding plaquette surface and telescope the product. Thus the normalized contour mean differs from any chosen contour by $O(L^2 \alpha \xi^2)$. The same comparison holds between a block link and a chosen joining path. Taking a covariant difference of the block average now gives the second inequality. At successive meshes $\xi_n = L^{-n}$ its errors contain ξ_n or ξ_n^2. Their sums and products are bounded, which proves the third inequality.

In a unit cube an axial gauge makes each bare link within $2\alpha\xi_j$ of one. The same gauge makes each averaged link within $C\alpha\xi_n$ of one, by [YM3, Prop. 3.40]. Summing the contour errors over the levels shows that $R\iota$ differs from the ordinary unit-cube mean by at most $C\alpha$. Also the transformed ordinary gradient differs from the covariant gradient by at most $C\alpha$. The cube Poincaré inequality has constant $\sqrt{3}/\pi < 3/4$. Apply it in this gauge and undo the gauge to prove the last estimate. □

Proposition A.30 (Block operators uniformly in the mass)**.** There are $\epsilon_S, \varrho_S, \varrho'_S, \delta_0 > 0$ and $C_{\mathrm{side}} \ge 1$, depending only on L, b_F and the chamber operator, with the following property. Suppose $\ell' \ge C_{\mathrm{side}}$, $|\partial U(p) - 1| \le \epsilon_S \xi^2$, $m \ge 0$, $a \in [a_-, a_+] = [b_F(1 - L^{-1}), b_F]$. Let J be a matrix field on the bare sites with values in the complex span of $\mathfrak{B}$, acting by the same matrix $\widehat{J}$ on every chamber at a site, and assume $\sup |J| \le m/2 + \varrho_S$. Then

$$\mathcal{S} = (D^\xi[U] + m - \widehat{J} + aR^*R)^{-1}, \qquad T = R\mathcal{S}R^*, \qquad D = a - a^2 T$$

exist, and $|T(y, y')| \le Ce^{-\delta_0 |y - y'|}$. If the target has a next block level, the bare curvature is at most $\epsilon_S \xi^2 / L^2$, and $\sup |J| \le m/2 + \min\{\varrho_S, \varrho'_S / L\}$, then $\Gamma = D + b_F q^* q$ has

$$\max \left\{ \sup_y \sum_{y'}, \sup_{y'} \sum_y \right\} e^{\delta_0 |y - y'|/(2L)} |\Gamma^{-1}(y, y')| \le C.$$

Here $q = L^{-1/2} Q$ at that level. ■

Proof. First take $\sup |J| \le m/2$. Put $G = (D^\xi + m - \widehat{J} + aR^*R) f$ and $P = (R\iota)^*(R\iota)$. The penalty lives on chamber-independent spinors. Pair the equation with f and take real parts to obtain

$$a\|P\varphi\|^2 \le \|f\| \|G\|, \qquad (m/2)\|f\|^2 \le \|f\| \|G\|.$$

In particular $\sup |J| \|f\| \le \|G\|$. The energy and averaging estimates give

$$\sqrt{c'_1/2}\, \mathcal{N}(f) \le 2\|G\| + a\|P\varphi\| + \sqrt{C_{\mathrm{en}}(\alpha + \ell'^{-2})}\, \|f\|,$$

and $\|f\| \le \|P\varphi\| + \frac{5}{4} \mathcal{N}(f) + C'_A \alpha \|f\|$. Choose $\epsilon_S \le c_2^{\mathrm{SO}}$ and the side floor so that

$$\tfrac{5}{4} \sqrt{2C_{\mathrm{en}}(\alpha + \ell'^{-2})/c'_1} + C'_A \alpha \le \tfrac{1}{2}.$$

The small-curvature averaging bound of [YM3, Prop. 3.40] gives $\alpha \le 2\epsilon_S$ at every level. The inequalities close and give

$$\|f\| \le C_{\mathcal{S}}\|G\|, \qquad \mathcal{N}(f) \le C_{\nabla}\|G\|,$$

with

$$C_{\mathcal{S}} = 4A^2 + 10\sqrt{2}(c_1')^{-1/2}, \qquad A = (1 + 5\sqrt{2}a_+/(4\sqrt{c_1'}))a_-^{-1/2}.$$

The finite-dimensional operator is therefore invertible.

Take $\varrho_S = (6C_{\mathcal{S}})^{-1}$. Split a larger source into a part of norm at most $m/2$ and a remainder of norm at most ϱ_S. Conjugating by e^h, with $|h(x) - h(x')| \le \delta_0|x - x'|$, changes the Dirac term by $C\delta_0$, because it has range one mesh. It changes the penalty by the same order, because it acts within unit cubes. For sufficiently small δ_0 the total change is at most $(2C_{\mathcal{S}})^{-1}$. The Neumann series bounds the conjugated inverse uniformly. Smearing by R, R^* then gives the exponential kernel bound for T.

For the final assertion set $b_F' = b_F/L$, $P_1 = Q^*Q$, $Y = (a + b_F'P_1)^{-1}$ and $a' = ab_F'/(a + b_F')$. The resolvent identity gives

$$\Gamma^{-1} = Y + a^2YT'Y, \qquad T' = R(D^{\xi} + m - \widehat{J} + a'(QR)^*(QR))^{-1}R^*.$$

Indeed $Y = a^{-1} - a^{-2}a'P_1$; multiplication by Γ verifies the identity. The parameter La' belongs to $[a_-, a_+]$. Dilate by L so the next block lattice has spacing one. The curvature is multiplied by L^2, the mass, source and penalty by L, and the side is divided by L. Choose C_{side} to include this extra factor. The stronger curvature and source hypotheses let the preceding inverse proof apply. Its decay is δ_0/L in the old units. The kernels of Y are supported in blocks of side L. Sum the identity with weight $e^{\delta_0|y-y'|/(2L)}$ to obtain the assertion. □

Remark A.31. The side floor is used even at zero curvature: the localization error contains ℓ'^{-2}. The mass does not enter the constants. Invertibility at small mass comes from the block penalty and Poincaré inequality.

A.8. **Comparing the fermion operators on two meshes.** At the restart level the two cutoffs have the same block lattice. Their quadratic forms nevertheless use different bare lattices and different classical backgrounds. We compare the forms in two steps: first change the background on one bare lattice, and then change the bare spacing. The mass term carried into the restart is obtained by differentiating the same quadratic form. A Cauchy estimate therefore gives its comparison without a separate expansion.

Definition A.32 (The common block lattice)**.** Use the positive-mass data of Def. 11.1, with chamber fermions, $r = 1/2$, $N_f \ge 1$, and $\nu = 8N_f$. In particular the constants are those of [QCDM1, Prop. 10.20], with its positive lower mass m_- and the source radii chosen there. The two data are

$$\mathrm{A} = (g, \ell, K, \mathsf{b}, \mathsf{m}), \qquad \mathrm{B} = (g, \ell, K+1, \mathsf{b}, \mathsf{m}).$$

Their classical comparison uses the hypotheses and constants of Prop. A.5, eq. A.xviii, and Prop. A.9. Thus the blocks and averaging maps agree at partner levels, although their classical backgrounds can differ. Choose $1 \le u < s < K$ and put

$$t_0 = K - s, \quad t_1 = K - u, \quad N_{[d]} = \ell^3 L^{3d}, \quad x_{[d]} = L^d\mathsf{b}, \quad \varepsilon_{\mathrm{r}} = 2c_\rho\varepsilon(L^s\mathsf{b}).$$

Level t_0 of A and level $t_0 + 1$ of B have the common lattice $\Lambda_{s,\ell}$. In units where its spacing is one, the bare spacings are $\xi^{\mathrm{A}} = \xi = L^{-t_0}$ and $\xi^{\mathrm{B}} = \xi/L$. The spinor inner product at spacing ξ' is $\xi'^3\sum f^*f'$. The common block space $\mathsf{H}_{[s]}$ has dimension $\nu N_{[s]}$. The mass and source in these units are

$$\mathsf{m}_s = L^{-s}\mathsf{m}, \qquad J_s = L^{-s}J, \qquad \mathrm{Lip}_s(J_s) = L^{-2s}\,\mathrm{Lip}(J).$$

Here J takes values in the complex span of the permitted bilinear matrices. This span contains the identity.

For either bare field U, let $R = R^\bullet(U)$ be the composite fermion average, normalized by $RR^* = 1$, and set

$$a^{\mathrm{A}} = \theta_{t_0}, \quad a^{\mathrm{B}} = \theta_{t_0+1}, \qquad a_- = b_F(1 - L^{-1}) \le a^\bullet \le a_+ = b_F.$$

Whenever the inverse exists, define

$$\begin{aligned} \mathcal{K}^\bullet[U; J'] &= D^{\xi^\bullet}[U] + \mathfrak{m}_s - \widehat{J}' + a^\bullet R^* R, \\ \mathcal{S}^\bullet &= (\mathcal{K}^\bullet)^{-1}, \qquad \mathsf{T}^\bullet = R\mathcal{S}^\bullet R^*, \qquad \mathsf{D}^\bullet = a^\bullet - (a^\bullet)^2 \mathsf{T}^\bullet, \\ \mathsf{M}^\bullet[U; J'][N] &= \partial_z \mathsf{D}^\bullet[U; J' - zN]\big|_{z=0} = (a^\bullet)^2 R\mathcal{S}^\bullet \widehat{N} \mathcal{S}^\bullet R^*. \end{aligned}$$

For $\sup|J'| \le \mathfrak{m}_s/2$ the inverse exists at every U, by accretivity. For $\sup|J'| \le \mathfrak{m}_s/2 + \varrho_S$ it exists under the curvature and side hypotheses of Prop. A.30.

For $\sup|w| < 8\varrho$ put

$$\hat{s}_w(x) = \frac{F^{\mathrm{UV}}}{L^s(\mathsf{b} - w(x))}.$$

Suppose $L^s\mathsf{b} \ge \Theta_\star$ and $\varpi(W) < \varepsilon_{\mathrm{r}}$. Choose $U^{\mathrm{A}} \in \mathcal{C}(\mathbf{B}^{\mathrm{A}}_{t_0}, W)$ and $U^{\mathrm{B}} \in \mathcal{C}(\mathbf{B}^{\mathrm{B}}_{t_0+1}, W)$, and define

$$\mathsf{D}^\bullet_{\mathrm{st}}(W) = \mathsf{D}^\bullet[U^\bullet; J_s] + \mathsf{M}^\bullet[U^\bullet; J_s][\hat{s}_w].$$

These are the two quadratic forms at the classical restart. ■

Corollary A.33 (The discarded mass slices)**.** Let $F_{<i_0}$, $F^{\mathrm{UV}}_{<i_0}$, F^{UV}, and B_5 be as in Prop. B.2. Then $0 \le F^{\mathrm{UV}} \le B_F$ and, for $t_0 \le j \le k$ and masses in $[0, 1]$,

$$\begin{aligned} 0 \le F^{\mathrm{UV}} - F^{\mathrm{UV}}_{<t_0} &\le B_5 L^{-t_0}, \\ |F_{<t_0}(k; m) - F^{\mathrm{UV}}| &\le B_5(L^{-t_0} + 3L^{-(k-t_0)}), \\ |F_{<t_0}(k; m) - F_{<t_0}(j; m')| &\le 6B_5 L^{-(j-t_0)}, \\ |F_{<t_0}(k; m) - F_{<t_0}(k; m'')| &\le B_5|m - m''|L^{-(k-t_0)}, \\ |\sigma_i(k; m) - \sigma_i(j; m')| &\le 8LB_5 L^{-(j-i)} \quad (i < j). \end{aligned}$$

At level t_1 the second bound is $3B_5(L^{-(K-s)} + L^{-(s-u)})$.

Here is the corresponding estimate with a complex constant source. Use the cutoff and levels of the datum under consideration, put $t_i = L^{K-i}\mathsf{b}$, $t'_k = L^{K-k}(\mathsf{b} - c)$, and assume $|x_i - t_i| \le \bar{B}$, $|c| < 8\varrho$, $i < i_0 \le k$, and $n = k - i_0$. For $x'_i = x_i - L^{K-i}c$,

$$\begin{aligned} |\mathsf{b} - c| &\ge \tfrac{3}{4}\mathsf{b}, \qquad |x'_i| \ge \tfrac{1}{4}t_i, \\ \left| G_{<i_0,k}(x') - \frac{F_{<i_0}(k; \mathfrak{m}_k)}{t'_k} \right| &\le \frac{16\bar{B}F(k; \mathfrak{m}_k)}{3t_k^2} L^{-(n+1)}, \\ |t'_k G_{<i_0,k}(x') - F^{\mathrm{UV}}| &\le B_5(L^{-i_0} + 3L^{-n}) + \frac{4\bar{B}F(k; \mathfrak{m}_k)}{t_k} L^{-(n+1)}. \end{aligned}$$

At $c = 0$ the constants $16/3$ and 4 can both be replaced by 2. Finally,

$$\begin{aligned} \sup|\hat{s}_w| &\le \frac{4F^{\mathrm{UV}}}{3L^s\mathsf{b}} \le \frac{c_b}{18}\mathfrak{m}_s, \\ \mathrm{Lip}_s(\hat{s}_w) &\le \frac{16F^{\mathrm{UV}}}{9L^{2s}\mathsf{b}^2}\,\mathrm{Lip}(w), \\ \sup|\hat{s}_w - \hat{s}_{w'}| &\le \frac{16F^{\mathrm{UV}}}{9L^s\mathsf{b}^2}\sup|w - w'|. \end{aligned}$$

■□

Proof. The partial-sum and slice bounds follow from Prop. B.2 by the triangle inequality. Since $32\varrho \le \gamma^{-2} < \mathsf{b}$ and $\mathsf{b} \ge 2\bar{B}$, the two denominator bounds follow. Writing $x_i' = L^{k-i}t_k' + (x_i - t_i)$ gives

$$\left|\frac{L^{k-i}}{x_i'} - \frac{1}{t_k'}\right| \le \frac{16\bar{B}}{3t_k^2}L^{-(k-i)}.$$

Multiply by the nonnegative σ_i and sum over $i < i_0$. The estimate with t_k' uses $|t_k'| \le 5t_k/4$; equivalently, estimate $|t_k'L^{k-i}/x_i' - 1|$ directly by $4\bar{B}L^{-(k-i)}/t_k$. At $c = 0$ use $x_i \ge t_i/2$. The estimates for $\hat{s}_w$ are the same denominator calculation. Its relative mass bound uses the original admission condition $\mathsf{bm} \ge 8C_{\mathcal{S},1}\bar{B}_{\mathsf{M}}/c_b$, $\bar{B}_{\mathsf{M}} = 3B_F + 1$, and $C_{\mathcal{S},1} \ge 1$. □

Lemma A.34 (Starting the source chain at a later level)**.** The source-chain estimate of [QCDM1, Lem. 10.4] holds from any levels $0 \le t_* \le t^* \le K$ for which its one-step maps satisfy the same three hypotheses. More precisely, suppose

$$0 < \phi_k \le 2L^{-1/2}\phi_{k+1}, \quad \kappa_\sigma \le \tfrac{1}{4}, \quad \kappa_\sigma M \le \tfrac{1}{4}\kappa L,$$

and

$$B_\sigma \ge 4B_0 + 3C_{\mathrm{var}} + 8C_{\mathrm{tor}}C_{\mathcal{L}}, \qquad B_\sigma \ge 2C_{\mathcal{L}}B_\nu.$$

Let the constant-source reference fields be holomorphic for $|c| < 8\varrho$ and satisfy $|\sigma_i^c(c)| \le B_0'\phi_i|c|/\varrho$, where $B_0' \le B_0 + C_{\mathrm{var}}/4$. Start with $\sigma_i(x) = \sigma_i^c(w(x))$ for $i \le t_*$ and then use

$$\sigma_{k+1}(x) = L\sigma_k(x) + \nu_{k+1}(c_k)(y_{k+1}(x)) - \mu_{k+1}.$$

Here x is a bare site and $y_{k+1}(x)$ is its level-$(k+1)$ block centre. The chain is holomorphic on its source domain, remains inside the coordinate polydiscs, and obeys

$$\|\sigma_k - \sigma_k^c(w)\| \le C_{\mathrm{var}}\phi_k\Xi_k + 2C_{\mathrm{tor}}C_{\mathcal{L}}\phi_k e^{-c_t\ell/(2\lambda_k)}\|w\|_\infty/\varrho, \qquad \|\sigma_k\| \le B_\sigma\phi_k.$$

The norms, local source variation Ξ_k, scale λ_k, and constant c_t are those of that source-chain estimate. ■

Proof. We give the changes to its induction. The initial error is zero at every level $i \le t_*$. Cauchy's estimate for the reference function gives, at a local comparison point x_0,

$$|\sigma_i^c(w(x)) - \sigma_i^c(w(x_0))| \le 2B_0'\phi_i \operatorname{Lip}(w)a_K|x - x_0|/\varrho.$$

Thus the coordinate difference used in the local one-step estimate has the same bound as before, including the coordinates preceding t_*. In the induction replace the torus error by

$$T_k = 2C_{\mathrm{tor}}\sum_{j=t_*}^{k} L^{k-j}\phi_j e^{-c_t\ell/(2\lambda_j)}\|w\|_\infty/\varrho.$$

Its recurrence is the required one, and the geometric scale estimate in the source-chain proof gives $T_k \le 2C_{\mathrm{tor}}C_{\mathcal{L}}\phi_k e^{-c_t\ell/(2\lambda_k)}\|w\|_\infty/\varrho$. The ratio entering the local variation sum is at most $8/(3\sqrt{L})$. The admission of the new coordinate follows from

$$4B_0' + 2C_{\mathrm{var}} + 8C_{\mathrm{tor}}C_{\mathcal{L}} \le B_\sigma.$$

These bounds close the same induction and keep each composition in the holomorphy domain of its next map.

For the restart, use zero reference fields before t_0 and

$$\sigma_{t_0}^c(c) = \frac{F^{\mathrm{UV}}c}{L^s\mathsf{b}(\mathsf{b} - c)}.$$

Its modulus is at most $(F^{\mathrm{UV}}/24)\phi(L^s\mathsf{b})|c|/\varrho$. The restarted one-step hypotheses and the required bound on B_0' are supplied by Prop. B.10; hence the lemma applies from t_0 to t_1 when $u \ge u_*$. No map at an earlier level is needed in this application. □

A.9. **Changing the classical background.** We use the classical comparison in Prop. A.5 in a form that also records the change of individual links. Its excess bound pays for the square of this change at every scale. This will allow the fermion resolvent to be compared in operator norm.

Proposition A.35 (The change of the links)**.** Let U' be a competitor of admitted radius e at level t in Prop. A.5. Thus $V'_k = \mathcal{M}^k U'$ satisfies $\varpi(V'_k) \le eL^{-2(t-k)}$. Let $\mathcal{A}_k$ be its nonpivot fields, v_k its axial gauges, and

$$h_k = \tfrac{\pi}{2} C_{\mathrm{ax}}(1+2B_3)eL^{-2(t-k)}, \qquad c_{\mathrm{db}} = (1+24C_{\mathrm{db}}^2)^{1/2}, \qquad \mathcal{E}_t = 8\gamma_{\mathrm{r}}^{-1}\mathsf{exc}_t(U').$$

The constants and the allowed radius are those of that proposition. There are backgrounds $\check{U}_k, \hat{U}_k$, real bond fields $B'_k, \mathcal{H}_k$, and gauges u_k for $0 \le k < t$ such that

$$(V'_k)^{v_k} = e^{iB'_k} V_k^{\mathrm{bf}}(V'_{k+1}), \qquad \sup|B'_k| \le 2C_{\mathrm{db}}h_k, \qquad \|B'_k\| \le c_{\mathrm{db}}\|\mathcal{A}_k\|.$$

For $k \ge 1$ put

$$m_y^k = \sum_{c'} e^{-\delta_0^{\mathrm{e}} d_{\mathbf{B}}(y, c'_-)} |B'_k(c')|,$$

where c' runs over positive level-k bonds; put $m_y^0 = \sup|B'_0|$. Then

$$\sum_y (m_y^k)^2 \le \mathsf{S}_{\mathrm{e}}^2 \|B'_k\|^2 \quad (k \ge 1), \qquad \sup_y m_y^k \le \min\{\mathsf{S}_{\mathrm{e}}^{1/2}\|B'_k\|, \mathsf{S}_{\mathrm{e}} \sup|B'_k|\}.$$

The backgrounds belong to

$$\check{U}_k \in \mathcal{C}(\mathbf{B}_{k+1}, V'_{k+1}), \qquad \hat{U}_k \in \mathcal{C}(\mathbf{B}_k, V'_k).$$

Extend v_k to the bare lattice by the identity away from the level-k sites and denote this extension by $\tilde{v}_k$. Then

$$\hat{U}_k^{\tilde{v}_k} = (e^{i\eta_k \mathcal{H}_k} \check{U}_k)^{u_k}, \qquad |\mathcal{H}_k(b)| \le B_5^{\mathrm{e}} m^k_{y(b_-)}, \qquad \sup|\mathcal{H}_k| \le 1/132,$$

and $|u_k(y) - 1| \le 4\kappa_6 B_5^{\mathrm{e}} m_y^k$ at level-k sites. At $k = 0$ these choices are $\hat{U}_0 = U'$, $\check{U}_0 = V_0^{\mathrm{bf}}(V'_1)$, $\mathcal{H}_0 = B'_0$, and $u_0 = 1$.

Set $\tilde{U}_k = e^{i\eta_k \mathcal{H}_k}\check{U}_k$. For $k < j \le t$,

$$\mathcal{M}^j \tilde{U}_k = (V'_j)^{u_k^{-1}}, \qquad \varpi(\check{U}_k), \varpi(\tilde{U}_k) \le B_3 e \eta_t^2.$$

Moreover, $\check{U}_k$ and $\hat{U}_{k+1}$ belong to the same residual gauge orbit, with gauges equal to one on the level-$(k+1)$ lattice. The last $\check{U}_{t-1}$ belongs to the prescribed minimal orbit over V'_t.

In units of level t, write $a_k(b) = \eta_t^{-1}\eta_k|\mathcal{H}_k(b)|$ and $\|a_k\|_2^2 = \eta_t^3 \sum_b a_k(b)^2$. Then

$$\sum_{k<t} L^{t-k}\|\mathcal{A}_k\|^2 \le \mathcal{E}_t,$$

$$\|a_k\|_2 \le \sqrt{3} B_5^{\mathrm{e}} \mathsf{S}_{\mathrm{e}} c_{\mathrm{db}} L^{-(t-k)/2} \|\mathcal{A}_k\|,$$

$$\sum_{k<t} \|a_k\|_2 \le \frac{\sqrt{3} B_5^{\mathrm{e}} \mathsf{S}_{\mathrm{e}} c_{\mathrm{db}}}{\sqrt{L^2-1}} \sqrt{\mathcal{E}_t},$$

$$\sum_{k<t} \sup_y m_y^k \le \frac{\mathsf{S}_{\mathrm{e}}^{1/2} c_{\mathrm{db}}}{\sqrt{L-1}} \sqrt{\mathcal{E}_t},$$

$$\sup_b a_k(b) \le B_5^{\mathrm{e}} \min\left\{ \mathsf{S}_{\mathrm{e}}^{1/2} c_{\mathrm{db}} L^{(t-k)/2} \sqrt{\mathcal{E}_t}, \pi \mathsf{S}_{\mathrm{e}} C_{\mathrm{db}} C_{\mathrm{ax}} (1+2B_3) e L^{-(t-k)} \right\}.$$

For the two cutoffs of Def. A.32, $U' = \mathcal{M}U^{\mathrm{B}}$ is such a competitor, with $e = 2B_3\varepsilon_{\mathrm{r}}$, and

$$\mathcal{E}_t \le 8\gamma_{\mathrm{r}}^{-1} \mathsf{K}_u \varepsilon_{\mathrm{r}}^2 N_t L^{-t}.$$

■

Proof. The construction in Lem. A.6, and its proof gives the fields B'_k, $\check{U}_k$, $\hat{U}_k$, and the displayed response identity. Its pivot estimate is eq. (2136), so the factor in the ℓ^2 bound is exactly c_{db}. The response bound uses the exponentially weighted sum in [YM3, eq. 4.ccclxiii]. Its row and column sums are at most S_{e}. The Schur test gives the sum of squares; the two supremum bounds follow, respectively, by Cauchy–Schwarz and the row sum. The same response construction bounds u_k at the top sites. At $k = 0$ the field is B'_0 itself, and the admitted radius gives $\sup |B'_0| \le 1/132$.

The average is gauge covariant. Since $v_k = 1$ on level $k+1$, applying $\mathcal{M}^j$ to the response identity gives $\mathcal{M}^j \tilde{U}_k = (V'_j)^{u_k^{-1}}$ for $j > k$. The background regularity estimate at level $k+1$ gives

$$\varpi(\check{U}_k) \le B_3 \varpi(V'_{k+1}) \eta_{k+1}^2 \le B_3 e \eta_t^2.$$

The same argument at level k, followed by gauge invariance of plaquette size, gives the estimate for $\tilde{U}_k$. These uses of the regularity estimate are valid at the admitted radius; its strict input bound can be decreased to the actual plaquette size. The uniqueness of the residual minimal orbit, used in the proof of the cited lemma, gives the orbit identifications. Its residual gauges are the identity at every later block site.

The competitor estimate in Prop. A.5 gives the first sum in the statement. For $k \ge 1$ there are three positive bare bonds per bare site and η_k^{-3} sites per pillar. Hence

$$\sum_b |\mathcal{H}_k(b)|^2 \le 3 \eta_k^{-3} (B_5^{\mathrm{e}} \mathsf{S}_{\mathrm{e}})^2 \|B'_k\|^2.$$

Multiplication by $\eta_t \eta_k^2$ proves the bound for $\|a_k\|_2$. At $k = 0$ use $\|a_0\|_2^2 = \eta_t \|B'_0\|^2$ directly. The two summed bounds now follow from

$$\sum_{k<t} L^{-(t-k)/2} \|\mathcal{A}_k\| \le (L^2 - 1)^{-1/2} \sqrt{\mathcal{E}_t}, \qquad \sum_{k<t} \|\mathcal{A}_k\| \le (L-1)^{-1/2} \sqrt{\mathcal{E}_t}.$$

For the supremum, use $a_k \le L^{t-k} B_5^{\mathrm{e}} \sup m^k$ and the two bounds for $\sup m^k$.

Finally Lem. A.8 makes $\mathcal{M} U^{\mathrm{B}}$ an admitted competitor of radius $2 B_3 \varepsilon_{\mathrm{r}}$. Writing $q = (1 + j_t)^2$ in its action comparison gives

$$\mathsf{exc}_t(U') \le (q-1) A^{\mathrm{B}} + (A^{\mathrm{B}} - A^{\mathrm{A}}).$$

The proof of Prop. A.9 bounds these two terms by $5 C_{\mathrm{en}} B_3^3 \varepsilon_{\mathrm{r}}^2 \eta_t N_t$ and $(10^4 L^4 B_3^3 + \mathsf{K}_{\mathrm{up}}) \varepsilon_{\mathrm{r}}^2 \eta_t N_t$. Their sum is $\mathsf{K}_u \varepsilon_{\mathrm{r}}^2 \eta_t N_t$. □

Lemma A.36 (The lattice Sobolev estimate)**.** On a cubic torus of spacing $\xi \le 1$ and side $\ell' \ge 1$,

$$\|h\|_6 \le \|h\|_2 + \tfrac{2}{3} \sum_{\mu=1}^{3} \|\nabla_\mu h\|_2, \qquad \|h\|_4 \le \|h\|_2 + \tfrac{1}{2} \sum_{\mu=1}^{3} \|\nabla_\mu h\|_2.$$

For chamber spinors, the energy seminorm of Lem. A.28 satisfies

$$\|f\|_4 \le \|f\|_2 + 2\mathcal{N}(f).$$

If $U_1(b) = e^{iX(b)} U_2(b)$ and $a(b) = \xi^{-1} |X(b)|$, with $\|a\|_2^2 = \xi^3 \sum_{b>0} a(b)^2$, then

$$|\langle \psi, (D^\xi[U_1] - D^\xi[U_2]) \varphi \rangle| \le c_D \|a\|_2 \|\psi\|_4 \|\varphi\|_4, \qquad c_D = 12(1+r) = 18.$$

■

Proof. On a periodic lattice line a nonnegative function satisfies $v \le \ell'^{-1} \int v + \frac{1}{2} \int |\nabla v|$. Apply this along each of the three coordinate lines. Cauchy–Schwarz in the three transverse pairs, followed by the arithmetic-geometric mean inequality, gives

$$\|v\|_{3/2} \le \|v\|_1 + \tfrac{1}{6} \sum_{\mu} \|\nabla_\mu v\|_1.$$

For completeness, if g_μ is the resulting bound independent of coordinate μ, then $v^{3/2} \le (g_1 g_2 g_3)^{1/2}$ and the repeated Cauchy–Schwarz inequality bounds its integral by $\prod_\mu (\int g_\mu)^{1/2}$. This is the finite lattice version of the inequality just used; the powers of the mesh cancel.

Set $v = |h|^4$. The difference of fourth powers is bounded by $2(a^3 + b^3)|a - b|$, so

$$\|\nabla_\mu |h|^4\|_1 \le 4\|h\|_6^3 \|\nabla_\mu h\|_2, \qquad \|h\|_4^4 \le \|h\|_2 \|h\|_6^3.$$

Division by $\|h\|_6^3$ proves the L^6 bound; the zero case is immediate. Interpolation with L^2 and the concavity of the power $3/4$ gives the L^4 bound.

For a covariant difference, unitarity gives $|\nabla|\varphi|| \le |\nabla^U \varphi|$. Since the gradient norm includes both orientations, $\sum_{\mu>0} \|\nabla_\mu^U \varphi\|_2 \le \sqrt{3/2}\|\nabla^U \varphi\|_2$. Write $f = \iota\varphi + g$. The chamber normalization gives $\|\iota\varphi\|_4 = 48^{-1/4}\|\varphi\|_4$, whereas $\|g\|_4 \le \xi^{-3/4}\|g\|_2 \le \mathcal{N}(f)$. Combining the two estimates proves the claimed chamber bound.

For the last assertion, each positive cubic bond occurs in sixteen chamber hopping terms. Each term has coefficient at most $3(1 + r)$, and $|e^{iX} - 1| \le |X|$. Cauchy–Schwarz in the chamber labels, then Hölder with exponents $2, 4, 4$ on the bonds, gives the stated constant $12(1 + r)$, including both hopping directions. □

Lemma A.37 (Solutions on the full source disc)**.** Under the side and curvature hypotheses of Prop. A.30, let $\sup |J| \le m/2 + \varrho_S$. Define

$$C'_{\mathcal{S}} = \tfrac{6}{5} C_{\mathcal{S}}, \qquad C'_\nabla = \tfrac{6}{5} C_\nabla, \qquad C_4 = C'_{\mathcal{S}} + 2C'_\nabla.$$

Solutions $f = \mathcal{S}G$, and solutions for the adjoint, obey

$$\|f\|_2 \le C'_{\mathcal{S}} \|G\|_2, \qquad \mathcal{N}(f) \le C'_\nabla \|G\|_2, \qquad \|f\|_4 \le C_4 \|G\|_2.$$

Consequently $\|\mathsf{T}\| \le C'_{\mathcal{S}}$ and $\|\mathsf{D}\| \le a_+ + a_+^2 C'_{\mathcal{S}}$. ■

Proof. Split $J = J_a + J_b$ pointwise by radial projection onto the ball $|J_a| \le m/2$. Then $|J_b| \le \varrho_S = 1/(6C_{\mathcal{S}})$. The equation $\mathcal{K}[J_a] f = G + \widehat{J}_b f$ and the estimates before enlargement of the source disc give

$$\|f\|_2 \le C_{\mathcal{S}} \|G\|_2 + \tfrac{1}{6}\|f\|_2, \qquad \mathcal{N}(f) \le C_\nabla (\|G\|_2 + \varrho_S \|f\|_2).$$

These give the factors $6/5$. The L^4 bound is Lem. A.36. The adjoint is conjugate by γ_4 to the same operator at $J^\sharp = \gamma_4 J^\dagger \gamma_4$; both the source norm and $\mathcal{N}$ are unchanged. □

Proposition A.38 (Changing the background in the block operator)**.** Use the competitor of Prop. A.35. Assume the side in units of level t is at least C_{side}, the source has norm at most $m/2 + \varrho_S$, and

$$B_3 e \le \min\{\epsilon_S, a_Q\}, \qquad \pi L^{-1} B_5^{\mathrm{e}} \mathsf{S}_{\mathrm{e}} C_{\mathrm{db}} C_{\mathrm{ax}} (1 + 2B_3) e \le \sigma_Q.$$

Here a_Q, σ_Q, C_Q are the curvature radius, perturbation radius, and derivative constant for the holomorphic averages in [QCDM1, Lem. 6.4]. Put

$$C_{\mathrm{bg}} = (2C'_{\mathcal{S}} + 2a_+ {C'_{\mathcal{S}}}^2)(4\kappa_6 + C_Q L) B_5^{\mathrm{e}} \frac{\mathsf{S}_{\mathrm{e}}^{1/2} c_{\mathrm{db}}}{\sqrt{L - 1}}$$
$$+ c_D C_4^2 \frac{\sqrt{3} B_5^{\mathrm{e}} \mathsf{S}_{\mathrm{e}} c_{\mathrm{db}}}{\sqrt{L^2 - 1}}.$$

For U'' in the minimal orbit over $\mathcal{M}^t U'$, at the same a and source,

$$\|\mathsf{T}[U'] - \mathsf{T}[U'']\| \le C_{\mathrm{bg}} \sqrt{\mathcal{E}_t}, \qquad \|\mathsf{D}[U'] - \mathsf{D}[U'']\| \le a^2 C_{\mathrm{bg}} \sqrt{\mathcal{E}_t}.$$

The kernel difference is also bounded by the sum of the two exponentially decreasing kernel bounds from Prop. A.30. For the two cutoffs, these hypotheses become

$$\begin{aligned} 2B_3^2 \varepsilon_{\mathrm{r}} &\le \min\{\epsilon_S, a_Q\}, \\ 2\pi L^{-1} B_5^{\mathrm{e}} \mathsf{S}_{\mathrm{e}} C_{\mathrm{db}} C_{\mathrm{ax}} (1 + 2B_3) B_3 \varepsilon_{\mathrm{r}} &\le \sigma_Q. \end{aligned} \tag{A.lxvi}$$

With $\mathsf{K}_b = a_+^2 C_{\mathrm{bg}}\sqrt{8\mathsf{K}_u/\gamma_{\mathrm{r}}}$, the resulting bound for $\mathsf{D}^{\mathrm{A}}[\mathcal{M}U^{\mathrm{B}}] - \mathsf{D}^{\mathrm{A}}[U^{\mathrm{A}}]$ is $\mathsf{K}_b\varepsilon_{\mathrm{r}}\sqrt{N_t L^{-t}}$. ■

Proof. Residual gauges equal one on the target lattice, so T is constant on each residual orbit. The chain in Prop. A.35 therefore telescopes as

$$\mathsf{T}[U'] - \mathsf{T}[U''] = \sum_{k<t}(\mathsf{T}[\hat{U}_k] - \mathsf{T}[\check{U}_k]).$$

Fix k. All averages above level $k+1$ are common after taking account of the gauge u_k. Write R_{top} for their composite, $w = u_k|_{\mathbb{T}^{(k+1)}}$, and

$$R' = R_{\mathrm{top}} w R_{k+1}(\tilde{U}_k), \qquad R = R_{\mathrm{top}} R_{k+1}(\check{U}_k).$$

Gauge covariance gives $\mathsf{T}[\hat{U}_k] = R'\mathcal{S}'R'^*$, where $\mathcal{S}'$ has bare field $\tilde{U}_k$ and penalty aR'^*R'. Both averages have norm one. The real averages coincide with their holomorphic extensions on this regular domain. The derivative bound for that extension and the estimate on u_k give

$$\|R' - R\| \le \rho_k, \qquad \rho_k = (4\kappa_6 + C_Q L)B_5^{\mathrm{e}} \sup_y m_y^k.$$

Indeed the field perturbation at level $k+1$ has norm $L\sup|\mathcal{H}_k|$; the second radius condition keeps it in the permitted disc. The first radius condition gives both the operator estimates and the regularity required for the averages.

The resolvent identity is

$$\begin{aligned} R'\mathcal{S}'R'^* - R\mathcal{S}R^* &= (R'-R)\mathcal{S}'R'^* + R\mathcal{S}'(R'-R)^* \\ &\quad - R\mathcal{S}'\{D^\xi[\tilde{U}_k] - D^\xi[\check{U}_k] + a(R'^*R' - R^*R)\}\mathcal{S}R^*. \end{aligned}$$

By Lems. A.36 and A.37, its norm is at most

$$(2C'_{\mathcal{S}} + 2a_+ C'^2_{\mathcal{S}})\rho_k + c_D C_4^2 \|a_k\|_2.$$

The two sums in Prop. A.35 give C_{bg}. The last assertion uses its excess estimate and $\mathsf{D} = a - a^2\mathsf{T}$. □

Corollary A.39 (A bound using only the radius)**.** Under the same hypotheses one also has

$$\|\mathsf{T}[U'] - \mathsf{T}[U'']\| \le C_{\mathrm{bg}}^{\infty} e,$$

where

$$\begin{aligned} C_{\mathrm{bg}}^{\infty} &= \pi B_5^{\mathrm{e}} \mathsf{S}_{\mathrm{e}} C_{\mathrm{db}} C_{\mathrm{ax}}(1+2B_3) \\ &\quad \times \left[\frac{(2C'_{\mathcal{S}} + 2a_+C'^2_{\mathcal{S}})(4\kappa_6 + C_Q L)}{L^2-1} + \frac{3(1+r)C'^2_{\mathcal{S}}}{L-1}\right]. \end{aligned}$$

Thus the two-cutoff background error is at most

$$\varepsilon_{\mathrm{r}} \min\{\mathsf{K}_b\sqrt{N_t L^{-t}}, \mathsf{K}_b^{\infty}\}, \qquad \mathsf{K}_b^{\infty} = 2B_3 a_+^2 C_{\mathrm{bg}}^{\infty}.$$

It is also bounded by $\max\{\mathsf{K}_b, \mathsf{K}_b^{\infty}\}\varepsilon_{\mathrm{r}}(N_t L^{-t})^{1/3}$. ■□

Proof. In the same resolvent identity use $\sup m^k \le \pi \mathsf{S}_{\mathrm{e}} C_{\mathrm{db}} C_{\mathrm{ax}}(1+2B_3) e L^{-2(t-k)}$ and the elementary hopping bound

$$\|D^\xi[\tilde{U}_k] - D^\xi[\check{U}_k]\| \le 3(1+r)\sup a_k.$$

The two geometric sums are $(L^2-1)^{-1}$ and $(L-1)^{-1}$. Finally $\min\{x^{1/2}, 1\} \le x^{1/3}$ for $x \ge 0$. □

A.10. **Changing the bare spacing.**

Lemma A.40 (Transport between the two meshes)**.** Let $|\partial U^{\mathrm{B}}(p)-1|\le\alpha(\xi/L)^2$, where $0\le\alpha\le 10^{-4}/(2\pi)$, and put $V=\mathcal{M}U^{\mathrm{B}}$. Let Q be the normalized first fermion average between the two bare lattices:

$$(Q\varphi)(y)=L^{-3}\sum_{x\in B(y)} q(U^{\mathrm{B}};y,x)\varphi(x),\qquad (Q^*\psi)(x)=q(U^{\mathrm{B}};y,x)^*\psi(y).$$

Then $QQ^*=1$. Write $T_e^{\mathrm{A}},T_e^{\mathrm{B}}$ for covariant translations by $\xi e,(\xi/L)e$ at V,U^{B}, and $\nabla_e^{\mathrm{A}}=\xi^{-1}(T_e^{\mathrm{A}}-1)$, $\nabla_e^{\mathrm{B}}=L\xi^{-1}(T_e^{\mathrm{B}}-1)$. With $C_1=234$ and $C_{\mathrm{tr}}=9C_1$,

$$\begin{aligned}
&\|T_e^{\mathrm{A}}Q-Q(T_e^{\mathrm{B}})^L\|\le C_{\mathrm{tr}}\alpha\xi^2,\\
&\|((T_e^{\mathrm{B}})^n-1)Q^*\psi\|\le 2\sqrt{n/L}\,\xi\|\nabla_e^{\mathrm{A}}\psi\|+C_{\mathrm{tr}}\alpha\xi^2\|\psi\|,\qquad 1\le n\le L,\\
&\|\nabla_e^{\mathrm{A}}Q\varphi\|\le\|\nabla_e^{\mathrm{B}}\varphi\|+C_{\mathrm{tr}}\alpha\xi\|\varphi\|.
\end{aligned}$$

■

Proof. The weighted inner products give the stated adjoint, and the normalization of the contour means gives $L^{-3}\sum_x q(y,x)q(y,x)^*=1$. Fix two neighboring blocks. In their complete axial gauge every link is a product of at most $2L-2$ conjugated plaquette variables, so its distance from one is at most $2\alpha\xi^2/L$. Each contour defining a fermion average has length at most $3(L-1)/2$. Its transport and its mean S are within $3\alpha\xi^2$ of one. The normalization satisfies $n(y)\ge(1-3\alpha\xi^2)^2$; hence $|q(y,x)-1|\le 12\alpha\xi^2$.

The same calculation for the gauge average uses the contour-mean formula in [YM3, eq. 3.xvii]. Its contour families have diameter at most $6\alpha\xi^2$, and their group means lie within $24\alpha\xi^2$ of their members. Thus those means are within $27\alpha\xi^2$ of one. The straight transport is within $2\alpha\xi^2$ of one. Each argument of the logarithm in the block-average formula is consequently within $58\alpha\xi^2$ of one, and its logarithm has norm at most $116\alpha\xi^2$. It follows that $|V(y,y+\xi e)-1|\le 234\alpha\xi^2$. All logarithms and contour means used here are in their defining small-field neighborhoods, since $\alpha\le 10^{-4}/(2\pi)$.

The products just compared transform covariantly. Thus any two such products with the same endpoints differ by at most $6C_1\alpha\xi^2$ in every gauge. The kernel of the first operator in the statement is L^{-3} times just such a difference:

$$V(y,y+\xi e)q(y+\xi e,x+\xi e)-q(y,x)U^{\mathrm{B}}(x\longrightarrow x+\xi e).$$

Cauchy–Schwarz within each block proves its norm bound.

For the second estimate, among the L^3 sites of a block, exactly nL^2 cross its boundary after n translations in direction e. At the other sites the difference is a transport error of size $6C_1\alpha\xi^2|\psi(y)|$. At the crossing sites it is the sum of a transport error and $q(y,x)^*\xi\nabla_e^{\mathrm{A}}\psi(y)$. Here $|q|\le 2$. The transport errors have combined norm at most $6\sqrt{2}C_1\alpha\xi^2\|\psi\|$, while the other term has norm at most $2\sqrt{n/L}\,\xi\|\nabla_e^{\mathrm{A}}\psi\|$. This proves the estimate with C_{tr}. Finally use

$$\nabla_e^{\mathrm{A}}Q=\xi^{-1}(T_e^{\mathrm{A}}Q-Q(T_e^{\mathrm{B}})^L)+\xi^{-1}Q\sum_{j<L}(T_e^{\mathrm{B}})^j(T_e^{\mathrm{B}}-1).$$

□

Proposition A.41 (The block operators at two spacings)**.** Assume $\ell L^s\ge C_{\mathrm{side}}$, $|\partial U^{\mathrm{B}}(p)-1|\le\alpha(\xi/L)^2$, and

$$0\le\alpha\le\min\{10^{-4}/(2\pi),\epsilon_S/2\},\qquad \sup|J'|\le\mathfrak{m}_s/2+\varrho_F,\qquad \varrho_F=\min\{\varrho_S,\lambda_\rho/2\}.$$

The same Lipschitz function J' is sampled on both bare lattices. Then

$$\|\mathsf{D}^{\mathrm{B}}[U^{\mathrm{B}};J']-\mathsf{D}^{\mathrm{A}}[\mathcal{M}U^{\mathrm{B}};J']\|\le\mathsf{K}_c\xi(1+\mathrm{Lip}_s(J')),$$

where

$$\mathsf{K}_c = b_F + 2b_F^2 C'_{\mathcal{S}} + b_F^2 \mathsf{K}_T,$$
$$\mathsf{K}_T = \left[(1+r)(1+3C_{\mathrm{tr}}) + \frac{2c_\flat^2}{\lambda_\rho}(2+\sqrt{6}C_{\mathrm{tr}})\right](C'_{\mathcal{S}} + C'_\nabla)^2 + (\sqrt{3}/2 + b_F){C'_{\mathcal{S}}}^2,$$
$$c_\flat^2 = 3(1+r)^2.$$

These constants require no lower bound on $\mathfrak{m}_s \geq 0$. ∎

Proof. Set $V = \mathcal{M}U^{\mathrm{B}}$. The curvature averaging estimate gives $|\partial V - 1| \leq 2\alpha\xi^2$, so Lem. A.37 applies on both lattices.

First eliminate the chamber complement at spacing ξ'. For $T_0 = D^{\xi'}[U] + \mathfrak{m}_s - \widehat{J}'$, write

$$E = \Pi_1 T_0 \Pi_1, \quad C = \Pi_1 T_0 \iota, \quad B = \iota^* T_0 \Pi_1.$$

The chamber coercivity and the source radius give $\operatorname{Re} E \geq \lambda_\rho/(2\xi')$. The off-diagonal bounds are $\|C\varphi\| \leq c_\flat \|\nabla^U \varphi\|$ and $|\langle \varphi', Bg\rangle| \leq c_\flat \|\nabla^U \varphi'\| \|g\|$. Consequently the site operator and heavy correction are

$$k = D^{\mathrm{W},\xi'}[U] + \mathfrak{m}_s - J' - H + aP_R, \qquad H = BE^{-1}C, \qquad P_R = (R\iota)^*(R\iota),$$

and

$$|\langle \varphi', H\varphi\rangle| \leq \frac{2c_\flat^2}{\lambda_\rho}\xi' \|\nabla^U \varphi'\| \|\nabla^U \varphi\|.$$

Since $R = R\iota\iota^*$, putting $R^{\mathrm{s}} = R\iota$ gives $\mathsf{T} = R^{\mathrm{s}} k^{-1} R^{\mathrm{s}*}$. Both $k^{-1}G$ and $(k^{-1})^*G$ have norm at most $C'_{\mathcal{S}}\|G\|$ and gradient norm at most $C'_\nabla\|G\|$, by Lem. A.37.

The composite averages satisfy $R^{\mathrm{s}}_{\mathrm{B}} = R^{\mathrm{s}}_{\mathrm{A}} Q$. Using $QQ^* = 1$ gives the exact identity

$$\mathsf{T}^{\mathrm{B}} - \mathsf{T}^{\mathrm{A}} = R^{\mathrm{s}}_{\mathrm{A}} k_{\mathrm{A}}^{-1}(k_{\mathrm{A}} Q - Q k_{\mathrm{B}}) k_{\mathrm{B}}^{-1} R^{\mathrm{s}*}_{\mathrm{B}}.$$

Decompose the middle factor as $E_{\mathrm{W}} + E_{\mathrm{M}} + E_{\mathrm{P}} - E_H$, corresponding to the Wilson term, source, penalty, and heavy correction. For unit block spinors it suffices to estimate its pairing with $\varphi = k_{\mathrm{B}}^{-1} R^{\mathrm{s}*}_{\mathrm{B}} \chi$ and $\varphi' = (k_{\mathrm{A}}^{-1})^* R^{\mathrm{s}*}_{\mathrm{A}} \chi'$.

For the Wilson term, with $T = T_e^{\mathrm{B}}$ use

$$(1 - T_e^{\mathrm{A}})Q - LQ(1-T) = QT^L - T_e^{\mathrm{A}} Q + Q\sum_{j=1}^{L-1}(T^j - 1)(1-T).$$

Move $T^j - 1$ to the first factor in the pairing and apply Lem. A.40 in direction $-e$. Sum $\sqrt{j/L} \leq L$, then sum the six directions. This yields

$$\begin{aligned}|\langle \varphi', E_{\mathrm{W}}\varphi\rangle| &\leq \tfrac{1}{2}(1+r)\xi\big[2\|\nabla^{\mathrm{A}}\varphi'\|\|\nabla^{\mathrm{B}}\varphi\| + \sqrt{6}C_{\mathrm{tr}}\|\varphi'\|\|\nabla^{\mathrm{B}}\varphi\| \\ &\qquad + 6C_{\mathrm{tr}}\|\varphi'\|\|\varphi\|\big] \\ &\leq (1+r)(1+3C_{\mathrm{tr}})\xi(\|\varphi'\| + \|\nabla^{\mathrm{A}}\varphi'\|)(\|\varphi\| + \|\nabla^{\mathrm{B}}\varphi\|).\end{aligned}$$

The source acts on spin and flavour, whereas q acts on colour. The distance from a block site to its center is at most $\sqrt{3}\xi/2$, and $L^{-3}\sum_x |q(y,x)|^2 = 1$. Thus blockwise Cauchy–Schwarz gives

$$\|E_{\mathrm{M}}\| \leq \tfrac{\sqrt{3}}{2}\xi \operatorname{Lip}_s(J').$$

For the penalty,

$$E_{\mathrm{P}} = (a^{\mathrm{A}} - a^{\mathrm{B}}) R^{\mathrm{s}*}_{\mathrm{A}} R^{\mathrm{s}}_{\mathrm{B}}, \qquad |a^{\mathrm{A}} - a^{\mathrm{B}}| \leq b_F \xi.$$

The heavy correction at spacing ξ is bounded using the third transport estimate. At spacing ξ/L use the second estimate with $n = 1$, which gives

$$\|\nabla^{\mathrm{B}} Q^* \varphi'\| \leq 2\sqrt{L}\|\nabla^{\mathrm{A}}\varphi'\| + \sqrt{6}C_{\mathrm{tr}} L\alpha\xi\|\varphi'\|.$$

Since $1 + 2L^{-1/2} \le 2$, their sum satisfies

$$|\langle \varphi', E_H \varphi \rangle| \le \frac{2c_\flat^2}{\lambda_\rho}(2 + \sqrt{6}C_{\mathrm{tr}})\xi(\|\varphi'\| + \|\nabla^{\mathrm{A}}\varphi'\|)(\|\varphi\| + \|\nabla^{\mathrm{B}}\varphi\|).$$

Together these bounds give $\|\mathsf{T}^{\mathrm{B}} - \mathsf{T}^{\mathrm{A}}\| \le \mathsf{K}_T \xi(1 + \mathrm{Lip}_s(J'))$. Finally expand

$$\mathsf{D}^{\mathrm{B}} - \mathsf{D}^{\mathrm{A}} = (a^{\mathrm{B}} - a^{\mathrm{A}}) - ((a^{\mathrm{B}})^2 - (a^{\mathrm{A}})^2)\mathsf{T}^{\mathrm{B}} - (a^{\mathrm{A}})^2(\mathsf{T}^{\mathrm{B}} - \mathsf{T}^{\mathrm{A}}).$$

Use $|(a^{\mathrm{B}})^2 - (a^{\mathrm{A}})^2| \le 2b_F^2\xi$ and $\|\mathsf{T}^{\mathrm{B}}\| \le C'_{\mathcal{S}}$. □

A.11. **The classical starts.**

Proposition A.42 (Comparison of the starting quadratic forms)**.** There is $\Theta_F \ge \Theta_\star$ such that eq. A.lxvi holds whenever $L^s\mathsf{b} \ge \Theta_F$. Assume this inequality and $\ell L^s \ge C_{\mathrm{side}}$ in Def. A.32. Take $\varpi(W) < \varepsilon_{\mathrm{r}}$, $\sup|J_s| \le \mathfrak{m}_s/2$, and $\sup|w| < 8\varrho$. Set

$$\begin{aligned}
\delta_D &= \mathsf{K}_c\xi(1 + \mathrm{Lip}_s(J_s)) + \mathsf{K}_b\varepsilon_{\mathrm{r}}\sqrt{N_{[s]}\xi},\\
\delta_F &= (1 + \sup|\hat{s}_w|/\varrho_F)\delta_D + \mathsf{K}_c\xi\,\mathrm{Lip}_s(\hat{s}_w),\\
\mathfrak{f}_s &= (1 - c_b/9)\frac{a_-\mathfrak{m}_s}{2a_- + \mathfrak{m}_s}, \qquad \epsilon_F = \nu N_{[s]}\delta_F/\mathfrak{f}_s.
\end{aligned}$$

Then the following assertions hold.

(i) Each $\mathsf{D}^\bullet_{\mathrm{st}}$ is independent of the representative of its minimal orbit, and

$$\mathrm{Re}\,\mathsf{D}^\bullet_{\mathrm{st}} \ge \mathfrak{f}_s \ge \tfrac{4}{9}c_b\mathfrak{m}_s \ge \tfrac{1}{4}c_b\mathfrak{m}_s.$$

If $\mathfrak{m}_s \le 2a_-$, then also $\mathfrak{f}_s \ge 2\mathfrak{m}_s/9$.

(ii) The two ordinary block operators differ by at most δ_D. For any Lipschitz matrix field N with values in the complex span of $\mathfrak{B}$,

$$\|\mathsf{M}^{\mathrm{B}}[N] - \mathsf{M}^{\mathrm{A}}[N]\| \le \frac{\sup|N|}{\varrho_F}\delta_D + \mathsf{K}_c\xi\,\mathrm{Lip}_s(N).$$

Thus $\|\mathsf{D}^{\mathrm{B}}_{\mathrm{st}} - \mathsf{D}^{\mathrm{A}}_{\mathrm{st}}\| \le \delta_F$. In all these bounds the background term can be replaced by

$$\varepsilon_{\mathrm{r}}\min\{\mathsf{K}_b\sqrt{N_{[s]}\xi}, \mathsf{K}_b^\infty\}.$$

(iii) If $C, \delta > 0$ are the kernel constants from Prop. A.30, the kernel difference of the starts at $x \ne y$ is bounded by

$$\min\{\delta_F, 2a_+^2 C(1 + \sup|\hat{s}_w|/\varrho_F)e^{-\delta d(x,y)}\}.$$

Its exponentially weighted row and column sums at rate $\delta/4$ are at most

$$\delta_F + C_\delta\sqrt{2a_+^2 C(1 + \sup|\hat{s}_w|/\varrho_F)\delta_F}, \qquad C_\delta = \sum_{z\in\mathbb{Z}^3} e^{-\delta|z|/4}.$$

The norm $|z|$ is the same one used in the kernel decay of Prop. A.30.

(iv) For $P_\tau = (1-\tau)\mathsf{D}^{\mathrm{A}}_{\mathrm{st}} + \tau\mathsf{D}^{\mathrm{B}}_{\mathrm{st}}$,

$$\varphi_F = \int_0^1 \mathrm{tr}[P_\tau^{-1}(\mathsf{D}^{\mathrm{B}}_{\mathrm{st}} - \mathsf{D}^{\mathrm{A}}_{\mathrm{st}})]\,d\tau$$

obeys $|\varphi_F| \le \epsilon_F$ and $\det\mathsf{D}^{\mathrm{B}}_{\mathrm{st}}/\det\mathsf{D}^{\mathrm{A}}_{\mathrm{st}} = e^{\varphi_F}$.

(v) If w is real and $\gamma_4 J^\dagger \gamma_4 = J$, both determinants are positive.

(vi) For two sources in the stated ranges set

$$\Delta_{\mathrm{src}} = (1 + c_b C'_{\mathcal{S}} \mathfrak{m}_s/9) \sup |J'_s - J_s| + \frac{16 F^{\mathrm{UV}}}{9 L^s \mathsf{b}^2} \sup |w' - w|.$$

On either lattice the starts differ by at most $b_F^2 {C'_{\mathcal{S}}}^2 \Delta_{\mathrm{src}}$. Their determinant logarithms, continued along the straight source segment, differ by at most $\nu N_{[s]} b_F^2 {C'_{\mathcal{S}}}^2 \Delta_{\mathrm{src}}/\mathfrak{f}_s$.

(vii) Put $c_w = L^s(\mathsf{b} - w)$ and

$$\widehat{F}^{\bullet} = e^{-A^{\eta_{t_0^{\bullet}}}(c_w, U^{\bullet})} \det \mathsf{D}^{\bullet}_{\mathrm{st}}.$$

Then $\widehat{F}^{\mathrm{B}} = \widehat{F}^{\mathrm{A}} e^{\zeta_F}$, where

$$|\zeta_F| \le \mathsf{K}_w \|c_w\|_{(t_0)} \varepsilon_{\mathrm{r}}^2 N_{[s]} t_0^{1/2} L^{-t_0/2} + \epsilon_F.$$

The norm and K_w are those of the weighted classical action comparison in Prop. A.9.

(viii) The starts are continuous in W on $\{\varpi < \varepsilon_{\mathrm{r}}\}$, Borel on the field cut, and holomorphic in the finitely many source values. Uniformly there,

$$\|\mathsf{D}^{\bullet}_{\mathrm{st}}\| \le a_+ + a_+^2 C'_{\mathcal{S}} + a_+^2 {C'_{\mathcal{S}}}^2 c_b \mathfrak{m}_s/18.$$

■

Proof. Since $\varepsilon(x)$ decreases to zero, the two positive bounds in eq. A.lxvi hold for all sufficiently large $x = L^s \mathsf{b}$. Choose Θ_F accordingly. The classical backgrounds then satisfy the hypotheses of Props. A.38 and A.41; their bare curvature coefficient is $B_3 \varepsilon_{\mathrm{r}}$. The extra small numerical bound in the transport lemma is already part of the classical comparison radius, eq. A.xviii.

For accretivity put $\mu = \mathfrak{m}_s/2$ and $\varphi = a \mathcal{S} R^* \chi$. The defining equation gives

$$\langle \chi, \mathsf{D}\chi \rangle = a \|\chi - R\varphi\|^2 + \langle \varphi, (D^{\xi} + \mathfrak{m}_s - \widehat{J}_s)\varphi \rangle.$$

Its real part is at least $a\|\chi - R\varphi\|^2 + \mu\|\varphi\|^2$. Since $\|R\| = 1$, minimizing this expression gives

$$\mathrm{Re}\langle \chi, \mathsf{D}\chi \rangle \ge \frac{a\mu}{a + \mu} \|\chi\|^2, \qquad \mu \|\varphi\|^2 \le \mathrm{Re}\langle \chi, \mathsf{D}\chi \rangle.$$

The same calculation for the adjoint gives this last bound for $\varphi' = a \mathcal{S}^* R^* \chi$. Hence

$$|\langle \chi, \mathsf{M}[\hat{s}_w]\chi \rangle| \le \frac{\sup |\hat{s}_w|}{\mu} \mathrm{Re}\langle \chi, \mathsf{D}\chi \rangle \le \frac{c_b}{9} \mathrm{Re}\langle \chi, \mathsf{D}\chi \rangle.$$

This proves the displayed floor. Its numerical consequences follow from the admitted c_b and $\mathfrak{m}_s \le 1$; if $\mathfrak{m}_s \le 2a_-$, use $a_-/(2a_- + \mathfrak{m}_s) \ge 1/4$ and $1 - c_b/9 \ge 8/9$. The residual gauges equal one on the block sites, giving the independence of the minimal-orbit representative.

For (ii), insert $\mathsf{D}^{\mathrm{A}}[\mathcal{M} U^{\mathrm{B}}; J_s]$ between the two operators and apply Props. A.38 and A.41. To differentiate, let $N \ne 0$ and use the disc $|z| < \varrho_F / \sup |N|$ in $J_s - zN$. This entire disc lies in the extended source domain. The previous comparison there is bounded by

$$\delta_D + \mathsf{K}_c \xi |z| \, \mathrm{Lip}_s(N).$$

Cauchy's estimate on smaller circles, followed by a limit to the displayed radius, proves the derivative estimate. For $N = 0$ it is immediate. Set $N = \hat{s}_w$ and add the ordinary comparison. The alternative background bound is Cor. A.39.

Each kernel also has the exponential bound of Prop. A.30. The same Cauchy argument gives its derivative bound, proving (iii). Off the diagonal use $\min\{\delta_F, A e^{-\delta d}\} \le \sqrt{A \delta_F} e^{-\delta d/2}$ and sum after multiplying by $e^{\delta d/4}$; the diagonal contributes δ_F.

Every P_τ has real part at least $\mathfrak{f}_s$, so it is invertible and $\|P_\tau^{-1}\| \le \mathfrak{f}_s^{-1}$. Jacobi's determinant formula proves (iv), including the bound with dimension $\nu N_{[s]}$.

For real w, at $J = 0$ the two parity blocks of the starting form are adjoints. Its determinant is therefore $|\det D_+|^{2N_f} > 0$. If $J^\sharp = J$, the same relation with γ_4 makes the determinant real along the path τJ. The accretive floor prevents a zero, so its sign remains positive.

For (vi), differentiate along the source segment. The resolvent derivative is $\dot{\mathcal{S}} = \mathcal{S}\widehat{\dot{J}_s}\mathcal{S}$. The derivative of $\mathsf{M}[\hat{s}_w]$ has three terms: the derivative of $\hat{s}_w$, and a resolvent derivative on each side. Thus Cor. A.33 and Lem. A.37 give

$$\|\dot{\mathsf{D}}_{\mathrm{st}}\| \le b_F^2 {C'_{\mathcal{S}}}^2 \left[(1 + 2C'_{\mathcal{S}} \sup|\hat{s}_w|) \sup|\dot{J}_s| + \sup|\dot{\hat{s}}_w|\right].$$

Integration gives Δ_{src}. Apply Jacobi's formula and the same floor along this segment to obtain its logarithmic form.

Part (vii) adds the logarithm in (iv) to the weighted action difference. The latter is exactly the bound of Prop. A.9, applicable to the same backgrounds, radius, and complex coefficient c_w.

The norm bound in (viii) follows directly from Lem. A.37. For continuity, fix W_0 in the stated open set. Nearby W have a principal logarithmic coordinate relative to W_0. The local real-background response used in Lem. A.6 supplies a continuous representative of the minimal orbit in this coordinate. Its plaquettes stay inside the curvature domain. The finite matrices D^ξ, R, and $\mathcal{K}^{-1}$ are continuous there. Independence of the representative then proves continuity of the displayed start as a function of W. The source denominators $\mathsf{b} - w$ do not vanish, and all inverses remain in their source domains, proving holomorphy. □

Remark A.43 (Size of the determinant error). The constants are chosen in the following order: the averaging constants, the mass-independent operator constants, the background comparison constants, and then Θ_F. All are formed after the original admission constants. None is a new restriction on their choice; $L^s\mathsf{b} \ge \Theta_F$ is a restriction on the restart depth.

For later summation it is useful to expand ϵ_F. Put $q_s = \mathfrak{m}_s/\mathfrak{f}_s$ and $y_{[s]} = (\log L^s\mathsf{b})^{2\mathfrak{p}_0}$. Then $q_s \le 9/(4c_b)$, and $q_s \le 9/2$ if $\mathfrak{m}_s \le 2a_-$. The estimate above gives

$$\begin{aligned}\epsilon_F \le \frac{q_s\nu}{\mathsf{m}}\left(1 + \frac{c_b\mathfrak{m}_s}{18\varrho_F}\right)\Big[&\mathsf{K}_c\ell^3 L^{5s-K}(1 + L^{-2s}\operatorname{Lip}(J)) \\ &+ 2c_\rho A_0 \mathsf{K}_b (y_{[s]}/\mathsf{b})^{1/2}\ell^{9/2}L^{5s-(K-s)/2}\Big] \\ &+ \frac{q_s\nu}{\mathsf{m}}\mathsf{K}_c\ell^3 L^{5s-K}\frac{16F^{\mathrm{UV}}}{9L^{2s}\mathsf{b}^2}\operatorname{Lip}(w).\end{aligned}$$

This identity keeps the dependence on the torus side visible.

Corollary A.44 (Uniform constants for the restart comparison)**.** Define

$$s_F = \min\{s' \ge 1 : L^{s'}\gamma^{-2} \ge \Theta_F,\ 2L^{s'} \ge C_{\mathrm{side}},\ 2a_-L^{s'} \ge \mathfrak{m}_+\},$$

and

$$\begin{aligned}K_F &= \max\left\{1, \left(1 + \frac{c_b\mathfrak{m}_+}{18\varrho_F}\right)\max\left\{\mathsf{K}_c\left(1 + \frac{\varrho_J}{40M}\right), \mathsf{K}_b, \mathsf{K}_b^\infty\right\} + \frac{\mathsf{K}_c B_F}{2880M}\right\}, \\ C_D &= \max\left\{1, b_F^2 {C'_{\mathcal{S}}}^2 \max\{1 + c_b C'_{\mathcal{S}}\mathfrak{m}_+/9, B_F/9\}\right\}.\end{aligned}$$

For the data and depth hypotheses of Prop. 11.7, assume $s \ge s_F$. Let $\|w\|_\sharp, \|w'\|_\sharp < \varrho$ and $\|J\|_\sharp, \|J'\|_\sharp < \varrho_J$, where $\|f\|_\sharp = \max\{\|f\|_\infty, 40M\operatorname{Lip}(f)\}$. The starting operator $\hat{\mathbb{D}}^\bullet$ of Def. 11.2 is exactly $\mathsf{D}^\bullet_{\mathrm{st}}$ above. It is continuous in W, Borel on the field cut, holomorphic in the sources, constant on the minimal orbit, and

$$\operatorname{Re}\hat{\mathbb{D}}^\bullet \ge \tfrac{2}{9}L^{-s}\mathsf{m}, \qquad \|\hat{\mathbb{D}}^\bullet\| \le b_F + b_F^2 C'_{\mathcal{S}} + b_F^2 {C'_{\mathcal{S}}}^2 c_b\mathfrak{m}_+/18.$$

Furthermore,

$$\|\hat{\mathbb{D}}^{\bullet}(w,J)-\hat{\mathbb{D}}^{\bullet}(w',J')\| \le C_D L^{-s}(\|J-J'\|_\infty+\|w-w'\|_\infty),$$
$$\|\hat{\mathbb{D}}^{\mathrm{B}}(w,J)-\hat{\mathbb{D}}^{\mathrm{A}}(w,J)\| \le K_F\left[\xi+\bar{\varepsilon}_{\mathrm{r}}\min\{(N_{[s]}\xi)^{1/2},1\}\right],$$

where $\bar{\varepsilon}_{\mathrm{r}}=2c_\rho\varepsilon(L^s\gamma^{-2})$. In particular the second bound is at most $K_F[\xi+\bar{\varepsilon}_{\mathrm{r}}(N_{[s]}\xi)^{1/3}]$. For real w and $J^\sharp=J$ its determinant is positive. These assertions prove Prop. 11.7. ■□

Proof. The definition of s_F, together with $\mathsf{b}>\gamma^{-2}$, $\ell\ge 2$, and $\mathsf{m}\le\mathsf{m}_+$, gives every depth, side, and mass condition in Prop. A.42. The field cut is contained in $\{\varpi<\varepsilon_{\mathrm{r}}\}$ by [YM3, Lem. 6.33]. Source admission gives $\varrho_J\le\mathsf{m}/4$, so $\sup|J_s|<\mathfrak{m}_s/4$. These statements hold along the segment between the two sources.

We verify the identification of the quadratic form, including its mass normalization. Put $k=t_0^{\bullet}$ and

$$n_z=m_0^{\bullet}-a_{K^{\bullet}}\widehat{J}+zL^{-k}\widehat{N}.$$

Then $\mathcal{K}^{\bullet}[J_s-zN]=L^k\mathcal{K}_k[n_z]$. If R_k is the average with plain counting inner products, the weighted normalization gives $R=L^{-3k/2}R_k$ and $R^*=L^{3k/2}R_k^*$. Hence

$$\mathsf{D}^{\bullet}[J_s-zN]=L^k\mathsf{D}^{(k)}[n_z].$$

The Schur tower exists near $z=0$ by accretivity of its initial form and then of each Schur complement. Differentiating one step gives

$$\partial_z\mathbb{D}_{j+1}=b_F^2q\Gamma_j^{-1}(\partial_z\mathbb{D}_j)\Gamma_j^{-1}q^*.$$

Iterating, starting from $\partial_z\mathbb{D}_0=L^{-k}\widehat{N}$, proves

$$\langle\bar{\Psi},\mathsf{M}^{\bullet}[N]\Psi\rangle=L^{-k}\sum_v\bar{\psi}^{\mathrm{bg}}(v)N(x_v)\psi^{\mathrm{bg}}(v).$$

For $N=\hat{s}_w$ this is exactly the mass member prescribed in Def. 11.2. This proves the identification on each lattice.

The first conclusions now follow from Prop. A.42. For source differences use $\sup|J_s-J'_s|\le L^{-s}\|J-J'\|_\infty$ and $16F^{\mathrm{UV}}/(9\mathsf{b}^2)\le B_F/9$, since $\mathsf{b}^2>\gamma^{-4}\ge 16$. For the two-cutoff bound use

$$\mathrm{Lip}_s(J_s)<\varrho_J/(40M),\qquad \mathrm{Lip}_s(\hat{s}_w)<B_F/(2880M),\qquad \varepsilon_{\mathrm{r}}\le\bar{\varepsilon}_{\mathrm{r}}.$$

The middle inequality follows from Cor. A.33, $\mathrm{Lip}(w)<\varrho/(40M)$, and $\varrho/\mathsf{b}^2<1/128$. Substitute these estimates into the alternative bound in Prop. A.42(ii). Finally $\min\{K_bx^{1/2},K_b^\infty\}\le\max\{K_b,K_b^\infty\}\min\{x^{1/2},1\}$ and $\min\{x^{1/2},1\}\le x^{1/3}$. □

Appendix B. Restarting the flow and comparing totals

The estimates in this appendix control the dependence on discarded scales. They also compare the scalar totals after the restart, including the errors from large fluctuations and from the final density test.

B.1. **The mass contributed by the discarded scales.** A restart at a fixed block scale must retain the mass generated before that scale. The relevant number is the ultraviolet part of the one-loop mass, rather than its full limiting value. We first identify this number and estimate the error in replacing the earlier slices by it.

We use the free operators and the three-square formula of [QCDM1, §8]. All Gaussian expectations below are per block, with the normalized trace over constant spinors and the three colour components included. Put $h=L^{-k}$ when the units are those of level k. In the operator formulas below $D[1]$ and V_A are in these units, while ∇_j is the unscaled lattice difference. Write $R=R_k(1)$, $P=R^*R$, and

$$\mathcal{K}(m)=D[1]+m+\theta_kP,\qquad \theta_k=\frac{b_F(1-L^{-1})}{1-L^{-k}}.$$

The linear variation of $D[e^{itA}]$ is V_A, and that of $R_k(e^{itA})^*c$ is κ_1. For an arbitrary representative A of its gauge class set

$$\zeta(m) = \mathcal{K}(m)^{-1}(V_A\chi_c - m(1-P)\kappa_1), \qquad \chi_c = R^*c.$$

In the three-square formula replace κ_1 by $(1-P)\kappa_1$; call the resulting form $\Phi_k^{\mathrm{nat}}(A;m)$. It agrees with Φ_k in the contour gauge used in [QCDM1, §8]. Its Gaussian mean is unchanged by a gauge transformation whose averaged gauge parameter vanishes. Thus it may be evaluated in the straight-average gauge. This replacement concerns the representative of the Gaussian field; it does not change the one-loop coefficients.

Lemma B.1 (The Gaussian for the earlier slices)**.** Let $k \ge i_0 \ge 1$ and $n = k - i_0$. The sum of the Gaussian curvature covariances for $i < i_0$ is the covariance of one Gaussian on $\ker Q^s_{i_0}$ with quadratic form

$$|dA|^2 + |(1-\widetilde{P}_{i_0})d^*A|^2.$$

Here $Q^s_{i_0}$ is the straight bond average and $\widetilde{P}_{i_0}$ is the corresponding projection for gauge parameters. Denote expectation for this Gaussian by $\mathbb{E}_{<i_0}$. In the units of level k its covariance satisfies

$$C_{<i_0} \le \Upsilon_0^{-1}\widehat{K}^{-1}, \qquad C_{<i_0} \le \frac{2}{\Upsilon_0}(\widehat{K} + L^{2n}/3)^{-1}, \qquad \Upsilon_0 = (2/\pi)^6,$$

where $\widehat{K}$ is the symbol of the scalar lattice Laplacian, acting on each component. Moreover,

$$\sum_{i<i_0} \sigma_i(k;m) = \mathbb{E}_{<i_0}\Phi_k^{\mathrm{nat}}(A;m), \qquad \mathbb{E}_{<i_0}|(1-P)\kappa_1|^2 \le C_\kappa L^{-n}.$$

The constants are independent of k and i_0. ■

Proof. First we prove the gap created by the constraint. Set $l = L^{i_0}$ and work temporarily in lattice units. For one component a of A, the average on a coarse bond issuing from y has the form $l\sum_z \rho_y(z)a(z)$, where $\rho_y \ge 0$, $\sum_z \rho_y(z) = 1$, $\rho_y \le l^{-3}$, and its support is the union $X(y)$ of the two adjacent blocks. This union has $2l^3$ sites. If $\bar{a}$ is the ordinary mean on $X(y)$, the constraint gives

$$|\bar{a}|^2 = \left|\sum_z \rho_y(z)(a(z)-\bar{a})\right|^2 \le l^{-3}\sum_{X(y)} |a-\bar{a}|^2.$$

Consequently $\sum_{X(y)}|a|^2 \le 3\sum_{X(y)}|a-\bar{a}|^2$. The Neumann Laplacian on a box of sides $2l, l, l$ has gap $4\sin^2(\pi/(4l)) \ge l^{-2}$. Each site belongs to two such boxes, and each edge to at most two. Summing the box estimate yields

$$\|A\|^2 \le 3L^{2i_0}\langle A, (dd^* + d^*d)A\rangle.$$

After changing units this is the gap $L^{2n}/3$.

For completeness, the Fourier constraint has the same uniformly bounded correction as the full constrained Gaussian. A level-k fibre $\xi \in [-\pi,\pi]^3 \setminus \{0\}$ splits into the L^{3n} classes

$$\xi' = \xi + 2\pi p, \qquad |p_\nu| \le (L^n-1)/2, \qquad q \equiv \xi' \pmod{2\pi L^n}.$$

In each class the covariance is

$$\widehat{K}^{-1/2}B'\widehat{K}^{-1/2}, \qquad B' = (1-\Pi') + \frac{|(1-\Pi')w'\rangle\langle(1-\Pi')w'|}{|\Pi' w'|^2}.$$

The three direction constraints give the orthonormal vectors spanning Π', and w' is the normalized site-average vector. The centered average symbols give $|\Pi' w'|^2 \ge (2/\pi)^6$, exactly as for the constrained covariance in [QCDM1, §8]. Hence $0 \le B' \le \Upsilon_0^{-1}$. On the constrained subspace the quadratic form is at least $\Upsilon_0\widehat{K}$. Combining this with the gap and inverting the forms gives the two asserted covariance bounds on the infinite lattice.

The orthogonal curvature projections telescope: the sum of the first i_0 slice covariances is the projection associated with $Q^s_{i_0}$. The Gaussian expectation of a quadratic form is linear in its covariance. Gauge invariance of Φ^{nat}_k therefore gives the identity for the partial sum.

Finally the variation of the composite fermion average telescopes as

$$(1-P)\kappa_1 = -i\sum_{j<i_0} n_A^{(j)}\chi_c.$$

There are no terms with $j \geq i_0$, since $Q^s_j A = 0$ there. Each contour defining $n_A^{(j)}$ has at most $3L$ bonds of level j. A straight-average bond at that level has variance at most $C_Q L^j$ in lattice units. Taking square roots before summing the levels gives the constant $6L\sqrt{C_Q}/(\sqrt{L}-1)$, and hence a mean square bounded by $C_\kappa L^{i_0}$. In level-k units this is $C_\kappa L^{-(k-i_0)}$, as required. □

Define the ultraviolet quadratic form by omitting the block part of the propagator:

$$\Phi^{\mathrm{UV}}_k(A) = \left\langle h^{-1}\sum_j \frac{\varrho_j}{2}|\nabla_j^{(1)}\chi_c - \nabla_j\zeta^{\mathrm{UV}}|^2 \right\rangle_c, \qquad \zeta^{\mathrm{UV}} = D[1]^{-1}V_A\chi_c.$$

Its slice means are σ_i^{UV}, and its mean for the unconstrained covariance $\widehat{K}^{-1}$ is F^{UV}. This number is independent of scale: the substitution $q = t/h$ gives

$$F^{\mathrm{UV}} = 3\Sigma(r), \qquad \Sigma(r) = \int_{[-\pi,\pi]^3}\frac{d^3t}{(2\pi)^3}\frac{1}{k_1(t)}\sum_j\frac{\varrho_j}{2}\|\widehat{\nabla}_j^{(1)}(t) - \widehat{\nabla}_j(t)\widehat{D}(t)^{-1}\widehat{V}(t)\|^2_{\mathrm{HS}}, \tag{B.i}$$

where $k_1(t) = \sum_\nu 4\sin^2(t_\nu/2)$. The normalization of the Hilbert–Schmidt norm is the normalized spin trace above. The identity $\operatorname{Re}\widehat{D}^{-1} \leq c_{\mathrm{re}}^{-1}$ bounds the integrand by $C/k_1(t)$; thus the integral is finite in dimension three.

Proposition B.2 (Partial one-loop sums)**.** There is $B_5 < \infty$, depending only on the chamber operator, r, L and b_F, with the following properties. Let $k \geq 1$, $0 \leq i_0 \leq k$, $n = k - i_0$, and $m, m' \in [0,1]$, in level-k units. Set

$$F_{<i_0}(k;m) = \sum_{i<i_0}\sigma_i(k;m), \qquad F^{\mathrm{UV}}_{<i_0} = \sum_{i<i_0}\sigma_i^{\mathrm{UV}}.$$

Empty sums are zero.

(a) The coefficients σ_i^{UV} are nonnegative and do not depend on $k > i$, the mass, or b_F. They satisfy

$$0 \leq F^{\mathrm{UV}} - F^{\mathrm{UV}}_{<i_0} \leq B_5 L^{-i_0}, \qquad 0 \leq \sigma_i^{\mathrm{UV}} \leq B_5 L^{-i}.$$

(b) The partial sum obeys

$$|F_{<i_0}(k;m) - F^{\mathrm{UV}}_{<i_0}| \leq B_5 L^{-n}(L^{-n} + L^{-k/2} + m),$$
$$|F_{<i_0}(k;m) - F^{\mathrm{UV}}| \leq B_5(L^{-i_0} + 3L^{-n}).$$

(c) Its mass dependence and its comparison at two levels satisfy

$$|F_{<i_0}(k;m) - F_{<i_0}(k;m')| \leq B_5|m-m'|L^{-n},$$
$$|F_{<i_0}(k;m) - F_{<i_0}(j;m')| \leq 6B_5 L^{-(j-i_0)} \quad (i_0 \leq j \leq k).$$

In particular, for $i < j \leq k$,

$$|\sigma_i(k;m) - \sigma_i^{\mathrm{UV}}| \leq 4LB_5 L^{-(k-i)},$$
$$0 \leq \sigma_i(k;m) \leq B_5(L^{-i} + 4L^{1-(k-i)}),$$
$$|\sigma_i(k;m) - \sigma_i(j;m')| \leq 8LB_5 L^{-(j-i)}.$$

(d) At every fixed depth n the limit

$$F^{[n]} = \lim_{k\to\infty} F_{<k-n}(k;0) = F^{\mathrm{UV}} + F^{\mathrm{GW,[n]}}$$

exists, with $0 \le F^{\mathrm{GW,[n]}} \le B_5 L^{-2n}$. Uniformly in i_0, n, m,

$$|F_{<i_0}(k;m) - F^{[n]}| \le B_5(L^{-i_0/4} + mL^{-n}).$$

Define $F_\infty := F^{[0]} = \lim_{k\to\infty} F(k;0)$. Then $F^{[n]} \downarrow F^{\mathrm{UV}}$, and $\bar\sigma_d = F^{[d-1]} - F^{[d]}$ is nonnegative with $\sum_{d\ge1} \bar\sigma_d = F_\infty - F^{\mathrm{UV}}$. The equal-depth increments satisfy

$$|F_{<i_0+1}(k+1;m) - F_{<i_0}(k;m)| \le 2B_5(L^{-i_0/4} + mL^{-n}),$$
$$|\sigma_{i+1}(k+1;m) - \sigma_i(k;m)| \le 4B_5(L^{-i/4} + mL^{1-(k-i)}).$$

(e) Let $1 \le i_0 \le k \le K$, let $x_i \ne 0$ for $i < i_0$, and let $t_i = L^{K-i}\mathfrak{b}$, $\tau \ne 0$, and

$$G_{<i_0,k}(x) = \sum_{i<i_0} \frac{L^{k-i}}{x_i} \sigma_i(k;\mathfrak{m}_k).$$

Then

$$\left| G_{<i_0,k}(x) - \frac{F_{<i_0}(k;\mathfrak{m}_k)}{\tau t_k} \right| \le \frac{F_{<i_0}(k;\mathfrak{m}_k)}{|\tau| t_k} \max_{i<i_0} \frac{|x_i - \tau t_i|}{|x_i|}.$$

If $\varpi > 0$, $|x_i - \tau t_i| \le \bar B$ and $|x_i| \ge \varpi t_i$, this gives

$$|\tau t_k G_{<i_0,k}(x) - F^{\mathrm{UV}}| \le B_5(L^{-i_0} + 3L^{-n}) + \frac{\bar B F(k;\mathfrak{m}_k) L^{-(n+1)}}{\varpi t_k}.$$

■

Proof. We give the estimates in level-k units. Constants denoted by C depend only on the parameters listed in the statement; their maximum will determine B_5.

The form Φ_k^{UV} is a nonnegative quadratic form in the curvature and is invariant under every addition $d\lambda$. Its density in lattice units does not depend on k. A slice-i law is periodic under translations of level $i+1$, so its mean per block is independent of the later level used to compute it. The curvature projections increase with i_0. Hence the partial sums increase and are bounded by F^{UV}. At $k = i_0$ the constrained covariance differs from $\widehat K^{-1}$ by at most four terms

$$\widehat K^{-1/2} |u\rangle\langle u| \widehat K^{-1/2}$$

per fibre, where $\|u\| = 1$ and each coefficient is bounded by Υ_0^{-1}. The ultraviolet form per unit of $|A_q|^2$ is bounded by Ch. Each correction therefore costs at most

$$Ch \sum_q \frac{|u_q|^2}{k_q} \le Ch\, C|\xi|^{-2}.$$

Its integral is finite. This proves (a), including the slice estimate by subtraction of successive partial sums.

We next record the estimates that use the gap of Lem. B.1. In the standard fibre notation let $g = D[1]^{-1}$, $X = Rg$, and $D_k = (\theta_k^{-1} + RgR^*)^{-1}$. Write v_q for the block-average symbol. The free symbol bounds give

$$\|g_q\| \le C|q|^{-1}, \qquad \|D_k\| \le \min(C|\xi|, \theta_k), \qquad \sum_q v_q^2 = 1.$$

The improved covariance implies

$$\mathbb{E}_{<i_0}|D_k X V_A \chi_c|^2 \le C\|D_k\|^2 \sum_q \frac{v_q^2}{|q|^2(k_q + L^{2n}/3)} \le CL^{-2n}\|D_k\|^2(|\xi|^{-2} + \pi^{-2}) \le C_1 L^{-2n}.$$

At zero mass,

$$\zeta_0 = \mathcal{K}(0)^{-1} V_A \chi_c = gV_A\chi_c - gR^* D_k X V_A \chi_c.$$

The mean square of its first term is bounded by

$$C\int_{\mathbb{R}^3} \frac{d^3q}{|q|^2(c|q|^2 + L^{2n}/3)} \le CL^{-n}.$$

That of the second is at most $CL^{-2n}\int_{[-\pi,\pi]^3}(|\xi|^{-2}+1)\,d^3\xi$. Together with the last estimate of Lem. B.1, this proves

$$\mathbb{E}_{<i_0}|\zeta_0|^2 \le CL^{-n}, \qquad \mathbb{E}_{<i_0}|f|^2 \le C_f L^{-n}, \qquad f = \zeta_0 + (1-P)\kappa_1. \tag{B.ii}$$

The free inverse is uniformly bounded, $\|\mathcal{K}(m)^{-1}\| \le C_{\mathcal{K}}$ for $m \ge 0$. Indeed $\mathcal{K}(0)$ is accretive and its inverse is bounded by the free symbol calculation in [QCDM1, §8]; multiplication by $(1+m\mathcal{K}(0)^{-1})^{-1}$ does not increase this bound. For $u = \mathcal{K}(m)^{-1}f$ we have

$$h^{-1}\sum_j \frac{\varrho_j}{2}|\nabla_j u|^2 + \theta_k|Ru|^2 + m|u|^2 = \mathrm{Re}\langle u, f\rangle.$$

We also need the stronger estimate for the first term alone:

$$h^{-1}\sum_j \frac{\varrho_j}{2}|\nabla_j u|^2 \le C_{\mathrm{en}} h|f|^2, \qquad C_{\mathrm{en}} = c_{\mathrm{re}}^{-1}(1 + b_F C_{\mathcal{K}})^2. \tag{B.iii}$$

To see this, write $u = (D+m)^{-1}(1 - \theta_k P\mathcal{K}(m)^{-1})f$. The second factor has norm at most $1 + b_F C_{\mathcal{K}}$, while $(D+m)^{-1} = D^{-1}(1+mD^{-1})^{-1}$ and the last factor is a contraction. The identity $\mathrm{Re}\, D^{-1} \le h c_{\mathrm{re}}^{-1}$ now gives eq. B.iii.

The resolvent identity is

$$\zeta(m) - \zeta_0 = -m\mathcal{K}(m)^{-1} f.$$

The first square of Φ_k^{nat} is consequently at most $C(L^{-2n} + m^2 L^{-n})$ in expectation, and the second is at most CmL^{-n}. In the third square, the correction to the ultraviolet vector has expected energy at most $C(hL^{-2n} + hm^2 L^{-n})$, by eq. B.iii. For a nonnegative quadratic form q,

$$|q(v+w) - q(v)| \le q(w) + 2\sqrt{q(v)q(w)}.$$

Applying this inequality and Cauchy–Schwarz to the Gaussian mean bounds the change in the third square by $C(h^{1/2}L^{-n} + mh^{1/2}L^{-n/2})$ together with smaller terms. Since $h^{1/2}L^{-n/2} = L^{-n}L^{-i_0/2} \le L^{-n}$, these bounds prove the first assertion in (b); its second assertion follows from (a).

For two masses use instead

$$\zeta(m) - \zeta(m') = -(m-m')\mathcal{K}(m)^{-1} f', \qquad f' = f - m'\mathcal{K}(m')^{-1} f.$$

Equation eq. B.ii bounds the mean square of f' by CL^{-n}. The energy identities, the displayed quadratic-form inequality, and the Lipschitz dependence of the coefficients $\theta_k + m$ and m give $C|m-m'|(L^{-n} + h^{1/2}L^{-n/2}) \le 2C|m-m'|L^{-n}$. This proves the first assertion of (c). Compare each of the two partial sums with $F^{\mathrm{UV}}_{<i_0}$ for the second. Taking differences at $i_0 = i$ and $i_0 = i+1$ then gives its three slice bounds.

It remains to prove the limit at fixed depth; this requires a quantitative comparison of the free symbols. The first square at zero mass has integrand

$$3\theta_k^{-1}\,\mathrm{Tr}(D_k y^* B_h' y D_k^*), \qquad y = \widehat{K}^{-1/2}\widehat{V}^* X^*.$$

Here B'_h is the direct sum of the brackets in the proof of Lem. B.1. At fixed n its classes ξ' are fixed. In one class the average symbol is

$$v'_q = \prod_\nu \frac{\sin(L^{-n}q_\nu/2)}{L^{i_0}\sin(hq_\nu/2)}.$$

The vectors defining Π' and w' are obtained from $v'_q/(\widehat{q}_\mu\sqrt{k_q})$ and $v'_q\widehat{q}_\mu/\sqrt{k_q}$ by normalization. Their limits replace $\widehat{q}$ by q, k_q by $|q|^2$, and the average symbol by the corresponding sinc product. Ratios of finite-mesh and limiting entries are uniformly bounded and differ from 1 by at most $C\min(1,h|q|)$; missing fine modes are assigned value zero.

Write $q = \xi' + 2\pi L^n l$ and $u(0) = 1$, $u(l) = (2|l|-1)^{-1}$ for $l \neq 0$. The explicit average symbol gives

$$|v'_q| \le \prod_\nu u(l_\nu), \qquad h|q| \le \sqrt{3}\pi L^{-i_0}(2|l|_\infty + 1).$$

The normalized constraint vectors have squares bounded by $C|v'_q|^2$. The inequalities $\min(1,s)^2 \le s^{1/2}$ and

$$\sum_{l\in\mathbb{Z}^3}(2|l|_\infty+1)^{1/2}\prod_\nu u(l_\nu)^2 < \infty$$

therefore bound their squared difference from the limiting vectors by $CL^{-i_0/2}$. The normalization factors have the same relative bound: their zero modes dominate their sums up to fixed constants. Since $|\Pi' w'|^2 \ge \Upsilon_0$, this proves

$$\|B'_h - B'_0\| \le CL^{-i_0/4}, \tag{B.iv}$$

uniformly in n and in the classes.

We spell out the other symbol errors as well. With $\widehat{\Pi}_0$ the light-space projection, the chamber symbols obey

$$\begin{aligned}
|\iota_0^* g_q \iota_0 - g_q^0| &\le C\min(1,h|q|)/|q|,\\
\|(1-\widehat{\Pi}_0)g_q\| &\le Ch,\\
|v_q(h) - v_q(0)| &\le C\min(1,h|q|)|v_q(0)|,\\
|\widehat{q} - q| \le C\min(1,h|q|)|q|, &\qquad |\theta_k - \theta_\infty| \le 2b_F L^{-k}.
\end{aligned}$$

The first two follow from the light/heavy block inversion of the free chamber symbol; the remaining formulas follow from the centered averages and lattice differences. The vertex is a trigonometric polynomial, so it differs from its zero-mesh value by $C\min(1,h|q|)$; its heavy components are multiplied by a row of g_q bounded by Ch. These are the same free symbol identities used to prove the uniform mass bound in [QCDM1, §8]. All sums defining y and D_k can now be compared term by term, including the modes present only in the limiting sum. Their majorant is

$$|y_q|^2 \le Cv_q^2|q|^{-2}k_q^{-1}.$$

Using $\min(1,h|q|) \le (h|q|)^{1/4}$ and Cauchy–Schwarz, their change in the trace is at most

$$Ch^{1/4}\|D_k\|^2\sum_q(1+|q|^{1/2})|y_q|^2 \le Ch^{1/4}(|\xi|^{-2}+1).$$

For the zero mode this follows from $|y_0|^2 \le C|\xi|^{-4}$ and $\|D_k\| \le C|\xi|$; the other modes have $|q| \ge \pi$ and sum to a bounded function. The same majorant bounds $\|D_k\|^2\|y\|^2$. Combining it with eq. B.iv proves convergence of the first square with error $CL^{-i_0/4}$. Its limit is nonnegative and at most C_1L^{-2n} by the earlier gap estimate.

The second square vanishes at zero mass. The third differs from $F^{\mathrm{UV}}_{<i_0}$ by $CL^{-k/2-n}$. Part (a) now identifies the limit as $F^{\mathrm{UV}} + F^{\mathrm{GW},[\mathrm{n}]}$ and proves the asserted rate at zero mass. Part (c) adds CmL^{-n}. For each k the partial sums decrease with n, so the limits do too. The bound $F^{\mathrm{GW},[\mathrm{n}]} \le CL^{-2n}$ identifies their decreasing limit as F^{UV} and proves the assertions concerning $\bar{\sigma}_d$.

Applying the rate to two successive cutoffs at equal depth, then subtracting two adjacent partial sums, proves the last two estimates in (d). The case $i_0 = 0$ follows by increasing B_5 to cover the bounded empty-sum cases.

Finally,

$$\frac{L^{k-i}}{x_i} - \frac{1}{\tau t_k} = \frac{\tau t_i - x_i}{\tau x_i t_k}.$$

Multiply by the nonnegative $\sigma_i(k;\mathfrak{m}_k)$ and sum. For $i < i_0$, $t_i \ge L^{n+1}t_k$, so the additional hypotheses in (e) bound the resulting error by $\bar{B}F(k;\mathfrak{m}_k)L^{-(n+1)}/(\varpi|\tau|t_k^2)$. Part (b) proves the final estimate. A single maximum of the constants in these arguments gives B_5. □

B.2. **Restarting the flow.** We compare the original flow with a flow that starts later and carries only a prescribed mass correction from the discarded scales. The background field and the fermion block operator at the starting level are retained. All activities and scalar normalizers generated before that level are discarded. The construction below checks that this modified initial density satisfies the hypotheses needed for each subsequent step.

Definition B.3 (Restart and mass mismatch)**.** Fix the datum $(g, \ell, K, \mathsf{b}, \mathsf{m})$ of the construction, and keep its calibrations and fine masses fixed:

$$\mathsf{t}_k = L^{K-k}\mathsf{b}, \qquad \mathfrak{m}_k = L^{k-K}\mathsf{m}.$$

Choose integers $1 \le u < s < K$, and set

$$t_0 = K - s, \quad t_1 = K - u, \quad \Lambda_k = L^{-(k-t_0)/2}, \quad \varepsilon_s = (L^s\mathsf{b})^{-1/2}(1 + \log(L^s\mathsf{b}))^{9\mathfrak{p}_0}.$$

For $k \ge t_0$ we have $\phi(\mathsf{t}_k)\Lambda_k \le \varepsilon_s$ and $\phi_\sharp(\mathsf{t}_k)\Lambda_k \le \varepsilon_s$.

A label is $c \in \mathbb{C}$ with $|c| < 8\varrho$. The ordinary flow at this label has bare coupling $L^K(\mathsf{b} - c)$ and coefficients $x_k(c), s_k(c)$. The calibrations remain t_k. In particular,

$$|x_i(c) - x_i| \le \tfrac{2}{3}r_x x_i, \qquad |x_i(c) - (x_i - L^{K-i}c)| \le 4C_w\phi(x_i).$$

For $|\zeta| \le 2B_F + 1$, define the restarted density at level t_0 by

$$\hat{\tau}^{\mathrm{F}}_{t_0}(dV) = \mathsf{Stay}(V)\exp\big[-A(L^{t_0}\hat{x}_{t_0}, \mathsf{U}) - \langle\bar{\Psi}, \mathbb{D}_{t_0}(\mathsf{U})\Psi\rangle - L^{-t_0}\hat{s}_{t_0}\mathsf{S}_1(\psi^{\mathrm{bg}})\big]\mathscr{M}^{\mathrm{fr}}_{h_\emptyset}(dV),$$

$$\hat{x}_{t_0} = L^s(\mathsf{b} - c), \qquad \hat{s}_{t_0} = \frac{\zeta}{L^s(\mathsf{b} - c)}, \qquad \hat{E}_{t_0} = 0.$$

The density test $\mathsf{Stay} = \chi_{t_0}(T_\ell)$, background $\mathsf{U} = \mathsf{U}_{t_0}(V)$, fermion tower $\mathbb{D}_{t_0}$, and reference measure are those of the original datum. There is no activity of any degree at a level $j \le t_0$, and no scalar normalizer $\mathbf{N}^{\mathrm{un},(j)}$, $\mathbf{N}^{\mathrm{F},(j)}$, $\mathbf{G}^{(j)}$, or $\mathbf{c}_j$ at these levels.

At later levels apply the usual small-field step and extract its own local parts:

$$\hat{x}_{k+1} = L^{-1}\hat{x}_k - \hat{b}_{k+1}, \qquad \hat{s}_{k+1} = L\hat{s}_k + \hat{\mu}_{k+1}, \qquad \hat{E}_{k+1} = \hat{E}_k + \hat{e}_{k+1}n_{k+1}.$$

The same local formulas define the activities on $\mathbb{Z}^3$ and on every admitted torus. The extracted numbers are read from the infinite-lattice family and are used on all these tori. Let $\hat{x}^\circ_k$ be the real restarted sequence with label zero and start $\mathrm{Re}\,\zeta$. Set

$$\hat{y}_i = 0 \quad (i < t_0), \qquad \hat{y}_i = 1/\hat{x}_i \quad (i \ge t_0), \qquad \zeta' = \frac{\mathsf{b}\zeta}{\mathsf{b} - c}, \qquad \pi = \zeta' - \mathsf{t}_{t_1}G_{<t_0,t_1}(x(c)).$$

The one-loop expressions in $\hat{y}$ mean their linear extensions from the reciprocal couplings. Define $\hat{\rho}_j = \hat{\mu}_j - \mu_j^{1\mathrm{L}}(\hat{y})$.

For corresponding quantities in the ordinary and restarted flows put $\delta X = X - \hat{X}$. Earlier restarted activities are zero, so $\delta E^{(j)} = E^{(j)}$ for $j \le t_0$. Their complete scalar parts satisfy

$$\sup_{X,z} e^{(\kappa+1)d_j(X)}\sup|E_0^{(j)}(X;\cdot;z)| \le E_0.$$

For $j > t_0$ the normalizers depending only on the calibration and fine mass cancel, and $\delta E_0^{(j)} = \delta F_0^{(j)} + \delta \mathbf{G}^{(j)}$. At the trivial pair on $\mathbb{Z}^3$ use the second-order decompositions $E_{\geq 2}^{(j)} = T^{(j)} + R^{(j)}$ and their restarted counterparts; both restarted terms are zero for $j \leq t_0$. ∎

We first isolate the linear operation used to compare activities. The RG step itself is nonlinear. Reading previously formed activities into its potential is linear, provided their cells, paths and hulls have been fixed by the same rules.

Proposition B.4 (Linear reading of old activities)**.** Let $0 \leq k < K$. Use the matrix-mass ball, pair cores and formal-leg norms of [QCDM1, §§5,7,10]. The pair radius is $a = \alpha_0$ at zero sources and $a_k^{\mathrm{rad}} \leq a \leq \alpha_0$ at sources. Let $G^{(j)} = G_0^{(j)} + G_{\geq 2}^{(j)}$, $1 \leq j \leq k$, satisfy the following conditions.

The scalar terms are local, holomorphic, strictly gauge invariant, and have finite array norm

$$\|G_0^{(j)}\|_{j,\kappa} = \sup_{X,z} e^{\kappa d_j(X)} \sup |G_0^{(j)}(X;\cdot;z)|.$$

Each term of $G_{\geq 2}^{(j)}$ is even, has degree at least two, is holomorphic on the whole pair core, and is jointly gauge invariant. Its cell units have the first-moment bound used in the old-activity lemma, with bound $\varepsilon_f(G^{(j)}) \geq \|G_{\geq 2}^{(j)}\|_{j;a}$ for the sum of their norms. The two auxiliary families F_1 and $G^\flat$ used to form moved diagonal terms also have norms at most $\varepsilon_f(G^{(j)})$. Any moved diagonal terms are formed from the same fixed row and column sums as in the source construction, and obey its cell-unit condition at lags at least n_{lag}.

At zero sources it suffices that each term is invariant under the common flavour phase $V_1 = V_2 = e^{i\alpha}1$. Then the cell units are the polymers, no diagonal terms are moved, and $\varepsilon_f(G^{(j)})$ may be its own activity norm. On a torus use a pair $(G^{(j),\mathrm{tor}}, G^{(j),\infty})$: on polymers which do not wrap, the first is the part of the second that fits in the torus, with the same cells and pointers. The second is covariant under the torus periods, with the fermionic seam sign. The norm bounds both families and m_c denotes the infinite-lattice cell sum.

At sources use the triple consisting of the torus family at the source, its value at zero source, and the corresponding zero-source family on $\mathbb{Z}^3$. The last two form a pair as above. File the terms of $m_c - m_c(0)$ at the hulls of their source-dependent summands, and use their weighted norm $\|\cdot\|_\natural$. The number ε_f bounds the sums of the norms of all families in this triple.

Let $\mathcal{V}_k[G]$ be the sum of the old scalar localization and the old terms with legs in the step potential. Then:

(a) The maps $G \mapsto \mathcal{V}_k[G](Y)$ are complex linear. They commute with absolutely convergent sums, Taylor coefficients and Schwarz quotients. Each localized term is holomorphic, jointly gauge invariant, and vanishes at $(B, \bar{W}, W) = 0$. It depends on the fields in its polymer alone. It depends on the source there alone as well, except for a cell term filed whole, which also reads the source through its cell sum. Filing its summands at their hulls removes this exception.

(b) Put

$$\omega_n = C_\omega L^{n/4}, \qquad c_\psi^{(a)} = c_\psi + \mathsf{C}_{\mathcal{A}} \frac{\alpha_0}{a} \frac{c_h \delta(\mathsf{t}_k)}{a_{\mathrm{an}}} + C_G \mathsf{C}_{\mathcal{A}} \delta(\mathsf{t}_k).$$

Then $c_\psi^{(\alpha_0)} \leq c'_\psi$ and $c_\psi^{(a)} \leq 9c'_\psi/8$ on the source radii, and

$$\|\mathcal{V}_{\geq 2}^{\mathrm{old}}[G]\| \leq c_\psi^{(a)} \sum_{j \leq k} \left[\omega_{k+1-j} \varepsilon_f(G^{(j)}) + 1_{\{k-j \geq n_{\mathrm{lag}}\}} C_\omega L^{k-j} h_F^2 \sup_c \|m_c(G^{(j)})\| \right],$$

$$|\mathcal{V}_0^{\mathrm{old}}[G](Y)| \leq K_\Lambda \frac{\mathsf{C}_{\mathcal{A}} \delta(\mathsf{t}_k)}{a} \sum_{j \leq k} L^{-(k-j)} \|G_0^{(j)}\|_{j,\kappa} e^{-(1-2\delta_{\mathrm{w}})\kappa d_k(Y)}.$$

At sources replace the cell term inside the first sum by

$$L^{k-j}h_F^2\left[C_\omega \sup_c \|m_c^\infty(0)\| + C_{\mathrm{hl}} \sup_c \|m_c - m_c(0)\|_\natural\right].$$

The zero-source torus discrepancy $m_c(0) - m_c^\infty(0)$ is included in the first term of the bound. Thus all cell terms vanish from the bound when the infinite-lattice quadratic cell sums vanish and there is no source discrepancy.

(c) Suppose a segment of potentials $\mathcal{V}^\vartheta = \mathcal{V} + \vartheta\delta\mathcal{V}$ satisfies $\|\mathcal{V}_0^\vartheta\| \le c_{\mathrm{cl}}/2$ and $\|\mathcal{V}_\psi^\vartheta\| \le c_{\mathrm{cl}}/20$ for $0 \le \vartheta \le 1$. Before extraction of local parts,

$$\|\delta\widetilde{E}_{\ge 2}(X)\| \le K'_M\left[\frac{50}{9}\|\delta\mathcal{V}_\psi\| + 40\frac{\sup_\vartheta \|\mathcal{V}_\psi^\vartheta\|}{c_{\mathrm{cl}}}\|\delta\mathcal{V}_0\|\right]e^{-\kappa_+ d_{k+1}(X)},$$

$$|\delta\widetilde{E}_0(X)| \le \frac{50}{9}K'_M(\|\delta\mathcal{V}_0\| + \|\delta\mathcal{V}_\psi\|)e^{-\kappa_+ d_{k+1}(X)}.$$

In the activity norm replace K'_M by $K_M = C_E K'_M$. The same bounds hold for the first variation and, pointer by pointer, for the pointed terms, except for pointers on bonds of the scalar system. Those terms do not change if $\delta\mathcal{V}_0 = 0$. If $\delta\mathcal{V}_0$ changes only the fourth family of that system, and its admission bounds obey $\mathsf{n}' \le w_*/2$ and $\mathsf{n}^{\mathrm{int}} + \mathsf{n}' \le w_*$ at both endpoints, the scalar pointed bound holds with $K'_M + K^{\mathrm{v}}_{\mathbf{M}}$ in place of K'_M, and $K^\bullet_M = K_M + C_E K^{\mathrm{v}}_{\mathbf{M}}$ in the activity norm.

(d) At the trivial pair on $\mathbb{Z}^3$, let P_j be totals represented by even local families $F^{(j)}$ with no scalar part, covariant under cube translations, and with quadratic cell sums zero. The coefficient linear in the old-family parameter has total

$$\left(\mathcal{J}_k\Big[\sum_{j\le k} P_j\Big] - \sum_{j\le k} P_j \circ (\mathsf{H}_k, \bar{\mathsf{H}}_k)\right)_{\ge 2},$$

and a representative of norm at most $K_M c_\psi \sum_{j\le k} C_7 L^{(k+1-j)/4}\|F^{(j)}\|_j$. It depends linearly on the totals P_j.

■

Proof. Fix the polymer, the cell at which it is filed, and a local or walk term in each of its legs. The reading construction first transports the legs to the cell centre along fixed paths. It then subtracts the value at the trivial pair, splits each leg into its value at the centre plus its difference from that value, and forms row sums, column sums and the cell sum of the quadratic kernel. Each operation is linear in the kernel. The trees and paths depend on the polymer and walks; the maps used along them depend on the pair, but none depends on the family being read. Moving diagonal terms uses fixed linear combinations of the same kernel sums. Finally localization uses the walk expansion and finite differences in the shift parameters, again linearly. Scalar localization is linear in its array for the same reason. This proves linearity at each filed term.

At zero source the common flavour phase eliminates the $\Psi\Psi$ and $\bar\Psi\bar\Psi$ quadratic terms. The remaining kernel has the form $\bar\Psi\, g_X \Psi$. Exponential decay of the background maps bounds its weighted trace norm by $Ch_F^{-2}\|G^{(j)}(X)\|$. The first moments about the cell centre are paid by these exponential weights and by the polymer diameter. This proves the cell-unit bound with the family's own norm. At sources the same calculation applies to each summand of the units and moved pieces, and the norms add. Phase invariance and the core properties are closed linear conditions; they therefore pass to sums, Taylor coefficients and Schwarz quotients. Absolute convergence permits the reading maps to commute with these operations.

On a torus the fitting relation and the filing of wrapping terms are also linear. At sources split the cell sum into

$$m_c = m_c^\infty(0) + (m_c(0) - m_c^\infty(0)) + (m_c - m_c(0)).$$

The first term is filed as on $\mathbb{Z}^3$, the second piece by piece on wrapping polymers, and the third at the source hulls. This gives the pair and triple versions without changing the argument. Localization preserves the stated holomorphy and invariance. A term vanishes at zero fluctuation because it contains either a fluctuation leg, a walk vertex that vanishes there, or a difference of two evaluations which agree there. These observations also prove the assertions about local dependence.

For the bounds, at lags below n_{lag} read the family without expanding its cell sums. At larger lags the old-activity estimate gives $\omega_{k+1-j}\varepsilon_f$ and $C_\omega L^{k-j}h_F^2\|m_c\|$. The field substitution contributes c_ψ for terms containing a fluctuation leg, the shift contributes $\mathsf{C}_{\mathcal{A}}(\alpha_0/a)c_h\delta/a_{\mathrm{an}}$, and the walk vertices contribute $C_G\mathsf{C}_{\mathcal{A}}\delta$. A cell sum is formed at the trivial pair and depends on the fluctuation only through its two legs, so it incurs no additional shift derivative. The wrapping discrepancy is exponentially small in the torus diameter and is included in $C_\omega\varepsilon_f$. For source-dependent cells the hull count costs C_{hl} times their weighted norm. The scalar bracket localization costs $K_\Lambda\mathsf{C}_{\mathcal{A}}\delta/a$ and $L^{-(k-j)}$ per old level. This proves (b), with exactly the radii and constants used in the old-activity and source estimates of [QCDM1, §§7,10].

For (c), the polymer expansion is holomorphic on $\|\mathcal{V}\| < c_{\mathrm{cl}}$ and is bounded there, including the test tail, by

$$(5/2)K'_M c_{\mathrm{cl}} e^{-\kappa_+ d(X)}.$$

Every point of the segment has norm at most $11c_{\mathrm{cl}}/20$. Cauchy's estimate on the remaining radius $9c_{\mathrm{cl}}/20$ gives the coefficient $50/9$. The part with legs vanishes when $\mathcal{V}_\psi = 0$. Schwarz's lemma in a leg ball of radius $c_{\mathrm{cl}}/4$, followed by Cauchy's estimate in a scalar ball of the same radius, bounds its scalar derivative by

$$40K'_M \frac{\|\mathcal{V}_\psi\|}{c_{\mathrm{cl}}} e^{-\kappa_+ d(X)}.$$

Change the leg part and then the scalar part and integrate these derivative bounds. Summing polymers costs C_E. The pointed expansion gives the same estimates term by term. For scalar bond pointers divide the segment into two halves. Convexity preserves the admission bounds, and the change of the fourth family on each half is at most $w_*/2$. The scalar-system Lipschitz estimate gives the additional $K^{\mathrm{v}}_{\mathbf{M}}$, proving the exception stated in (c).

At the trivial pair the coefficient linear in the old families is the first variation of the expansion at zero potential. Its Berezin integral is the difference between integrating the old total and evaluating that total at the background fermions. This is the displayed expression in (d). The zero cell sums remove its relevant quadratic term; the old-family bound gives its stated norm. Representatives within any positive error of the total norm may be used, after which that error tends to zero. □

Proposition B.5 (Existence of the restarted flow)**.** For every label c and every $|\zeta| \le 2B_F + 1$ the restarted small-field flow is defined through level K, simultaneously on $\mathbb{Z}^3$ and on all admitted tori. It has the same form and closed activity bounds as the original construction, with constants C_w, C_2, C_T, C_3. These bounds hold for both torus families and at every pointer. The density at each level is bounded and Borel. The extracted numbers do not depend on the side; on polymers which do not wrap, the torus family is the fitting part of the infinite-lattice family, with the same pointers.

Its explicit and remainder terms satisfy

$$\hat{T}^{(j)} = T^{(j)}(\hat{y}), \qquad \|\hat{R}^{(j)}\|_j \le C_3\phi(\mathsf{t}_{j-1})^3, \qquad |\hat{\rho}_j| \le C'_\rho\phi_\sharp(\mathsf{t}_{j-1})^3,$$

and

$$\mathsf{t}_k\hat{s}_k = \zeta' + \sum_{t_0\le i<k}\frac{\mathsf{t}_i}{\hat{x}_i}\sigma_i(k;\mathfrak{m}_k) + \sum_{t_0<j\le k}\mathsf{t}_j\hat{\rho}_j, \qquad \left|\sum_{t_0<j\le k}\mathsf{t}_j\hat{\rho}_j\right| \le \tfrac{1}{2}. \tag{B.v}$$

The coefficient windows are

$$|\hat{x}_k(0) - \mathsf{t}_k| \le (1 - L^{-(k-t_0)})\bar{B},$$
$$|\hat{x}_k(c) - \hat{x}^\circ_k| \le \tfrac{5}{12}r_x\hat{x}^\circ_k, \qquad |\hat{s}_k(c)| \le 2\bar{B}_M/\mathsf{t}_k.$$

If $\zeta \in [0, B_F]$ is real and $c = 0$, then

$$\zeta - \tfrac{1}{2} \le \mathsf{t}_k \hat{s}_k \le \zeta + 2B_F + \tfrac{1}{2}, \qquad |\hat{s}_k| \le \bar{B}_M / \mathsf{t}_k.$$

Every subsequent step obeys the original one-step estimates on the full coefficient polydiscs about $\hat{x}_k^\circ$. The outputs are holomorphic in c and are real when c and ζ are real. ■

Proof. We run the induction simultaneously in the label, the start and the torus side. This is necessary because the extracted numbers come from the infinite lattice and the symmetry argument compares all torus sizes.

At level t_0 all activity bounds and fitting relations are empty. The density test restricts V to an open set of small plaquette configurations. Locally on this set the minimizing background has a continuous representative. Two representatives differ by a residual gauge transformation equal to the identity at the level- t_0 sites. The action, the tower form and the background mass form take the same values on this orbit: the action is gauge invariant, and the other two forms are gauge covariant with their boundary arguments fixed. Their values are therefore continuous functions of V on the tested set. The regular-background inverse estimates give $\operatorname{Re} \Gamma_{t_0-1} \ge C_D^{-1}$, bounded background maps and a bounded tower $\mathbb{D}_{t_0} = \theta - \theta^2 q \Gamma_{t_0-1}^{-1} q^*$. The Grassmann exponential has finitely many coefficients on a torus, each bounded because

$$|e^{-L^K(\mathsf{b}-c)A(1,\mathsf{U})}| \le 1 \quad \text{and} \quad \operatorname{Re}(\mathsf{b} - c) > 0.$$

Multiplication by the density-test indicator thus gives a bounded Borel density. Flavour transformations, charge conjugation, complex conjugation and lattice isometries carry these three forms and the background orbit jointly with their data. This proves the covariance required at the initial level, including the tested configuration pairs used by the step.

We make explicit the parts of the original induction that use its initial level. They remain valid with t_0 as the initial level:

(i) Covariance on the tested pairs follows from the covariance of the initial density and from covariance of the integration kernels. It extends to complex coefficient polydiscs by holomorphy.

(ii) The linear symmetry spaces for the activity families are preserved by the step. Their initial elements are zero, on $\mathbb{Z}^3$ and on every torus; the norm bounds are uniform in the side.

(iii) Partial charge conjugation and the conjugation operation $\sharp$ preserve the activity totals and the real local coefficients. Their initial hypotheses are empty, and the initial action and mass coefficients are real for real c, ζ. This gives reality at each later level.

(iv) Nonwrapping torus terms agree with the terms that fit from $\mathbb{Z}^3$. This holds at the empty initial family and is preserved term by term by localization and the polymer expansion.

(v) On the full polydisc $|x_i' - \hat{x}_i^\circ| \le r_x \hat{x}_i^\circ$, $|s_i'| \le A\phi(\hat{x}_i^\circ)$, the invariant activity balls are preserved. The one-step potential stays below $c_{\mathrm{cl}}/2$ at the real fine mass; this argument uses only the earlier families in their balls and the coefficient windows, not the original bare action.

(vi) There is a common open complex neighbourhood of the fine mass on which these flows are jointly holomorphic and the potential is below $3c_{\mathrm{cl}}/4$. At the initial level a change of size ϱ_* in level-t_0 units has size at most ϱ_*/L at level $t_0 - 1$. It lies in the actual mass ball of the regular background inverse estimate, where $\operatorname{Re} \Gamma_{t_0-1} \ge (2C_D)^{-1}$ and the background kernel constants may be doubled. In particular $\|\mathbb{D}_{t_0}\| \le \theta + 2\theta^2 C_D$ there at real configurations. The local estimates are jointly holomorphic in configuration and fine mass on this ball. At later levels uniform Lipschitz bounds on the region with potential below $3c_{\mathrm{cl}}/4$, applied about the real-mass region below $c_{\mathrm{cl}}/2$, give a neighbourhood independent of the side. No inverse estimate is continued beyond its mass domain.

(vii) Translation covariance on a torus holds polymer by polymer and pointer by pointer, including wrapping terms. It is empty at level t_0, and the fixed shift, path and localization rules

commute with translations by $M\mathbb{Z}^3$ in the units of each level. Here $t_0 \geq 1$ ensures that every restarted step uses these rules.

These seven inductions justify application of the original local step to the changed initial density. The step gives a bounded Borel image density and the stated local form. They also justify using its full complex coefficient domains, rather than merely the real trajectory.

The closed second-order and cubic-remainder induction is unchanged once its initial mass lies in the sharp window. At the initial level,

$$|\hat{s}_{t_0}(c)| \leq \frac{64}{63}\frac{2B_F+1}{\mathsf{t}_{t_0}} \leq \frac{2\bar{B}_M}{\mathsf{t}_{t_0}},$$

since $|c|/\mathsf{b} < 1/64$ and $\bar{B}_M = 3B_F + 1$. The four-variable expansion therefore retains the mass parameter at weight two, with inverse radius

$$r_4^{-1} \leq \frac{2c'_\psi c_{\mathsf{S}} h_F^2 \bar{B}_M}{c_\diamond \mathsf{t}_k} \leq C_r \phi(\mathsf{t}_k)^2.$$

Its square has weight four and is included in the cubic remainder.

To identify the explicit part, observe that the second-order creation at level j is linear in $y_{j-1} = 1/x_{j-1}$; Gaussian integration, reading at the next level and mass extraction are all linear. Thus $T^{(j)}$, $\mu_j^{(2)}$ and the one-loop sum rule are linear identities in y. They hold on the reciprocal image of a coupling box with nonempty interior, so they hold for every complex y. With $\hat{y}_i = 0$ for $i < t_0$ their recursion is exactly the restarted recursion. This proves $\hat{T}^{(j)} = T^{(j)}(\hat{y})$.

The bound for these explicit terms can also be checked directly. Their quadratic zeroth moments vanish after mass extraction and remain zero at later levels. The new second-order creation is bounded by $4C_{\mathrm{sl}}C_V^2 h_F^4/\mathsf{t}_k$, since $|\hat{x}_k(c)| \geq 11\mathsf{t}_k/24$. By Prop. B.4(d), the contribution from earlier explicit terms is at most

$$K_M c_\psi C_7 C_T \frac{h_F^4}{\mathsf{t}_k} \sum_{t_0<j\leq k} L^{-3(k+1-j)/4} \leq K_M c_\psi C_7 S_2^\sharp C_T \frac{h_F^4}{\mathsf{t}_k}.$$

Mass extraction costs K_μ. The choices $C_T = 8K_\mu C_{\mathrm{sl}} C_V^2$ and $K_\mu K_M c_\psi C_7 S_2^\sharp \leq 1/4$ leave a total at most $3C_T h_F^4/(4\mathsf{t}_k)$. Thus a representative satisfying the closed bound exists. The same weighted expansion then gives $\hat{R}^{(j)}$ and $\hat{\rho}_j$ with the asserted bounds.

The mass recursion telescopes:

$$\hat{s}_k = L^{k-t_0}\hat{s}_{t_0} + \sum_{t_0<j\leq k} L^{k-j}(G_j(\hat{y}) - LG_{j-1}(\hat{y}) + \hat{\rho}_j).$$

Since $G_{t_0}(\hat{y}) = 0$ and $\mathsf{t}_k L^{k-i} = \mathsf{t}_i$, this is eq. B.v. The summable cubic bound gives $\mathsf{t}_k C'_\rho S_3^\sharp \phi_\sharp(\mathsf{t}_k)^3 \leq 1/2$ for its remainder sum.

It remains to verify the windows used in this induction. At label zero, $|\hat{b}_j| \leq B_\sharp$ gives $|\hat{x}_k - \mathsf{t}_k| \leq (1 - L^{-(k-t_0)})\bar{B}$. At a complex label compare with the real restart:

$$\hat{x}_k(c) - \hat{x}_k^\circ = -L^{K-k}c - \sum_{t_0<j\leq k} L^{-(k-j)}(\hat{b}_j(c) - \hat{b}_j^\circ).$$

The calibration and fermion normalizers cancel from the last difference. Its remainder is bounded by $2C_w\phi(\gamma^{-2})$, so the existing ceilings $8\varrho \leq r_x\mathsf{b}/8$ and $26C_w\phi(\gamma^{-2})\gamma^2 \leq r_x$ give

$$|\hat{x}_k(c) - \hat{x}_k^\circ| \leq (1/4 + 1/6) r_x \hat{x}_k^\circ.$$

In particular $|\hat{x}_k(c)| \geq 11\mathsf{t}_k/24$. Nonnegativity of the slice coefficients and $\sum_i \sigma_i \leq B_F$ in eq. B.v give the claimed real mass interval. In general they give

$$\mathsf{t}_k|\hat{s}_k(c)| \leq \tfrac{64}{63}(2B_F+1) + \tfrac{24}{11}B_F + \tfrac{1}{2} \leq 6B_F + 2 = 2\bar{B}_M.$$

These estimates close the windows at the next level. They put the entire flow inside the coefficient polydiscs with $A = 5B_s$. Composition of the holomorphic step maps proves holomorphy in c; the third induction above proves reality. □

B.3. **Loss of memory of the earlier activities.** The earlier mass is relevant and must be retained in the restart. The other earlier terms are small after sufficiently many further steps. We prove this by following the mass, coupling, scalar activities and activities with legs together. The cubic remainder must be compared separately: estimating each of its two values would lose the decay with the distance from the restart.

Lemma B.6 (Comparison of the cubic remainders)**.** For $t_0 \le k < K$ let $\widetilde{R}^{(k+1)}$ and $\widehat{\widetilde{R}}^{(k+1)}$ be the remainder totals before mass extraction, at the trivial pair on $\mathbb{Z}^3$. Put

$$q_k^{\mathrm{old}} = \frac{\sum_{j\le k} L^{(k+1-j)/4}\|\delta E_{\ge 2}^{(j)}\|_j}{\sum_{j\le k} L^{(k+1-j)/4} C_2 \phi(\mathsf{t}_{j-1})^2}, \qquad q_k^0 = \frac{K_\Lambda \mathsf{C}_{\mathcal{A}}}{\alpha_0 K_0^\sharp} \sum_{j\le k} L^{-(k-j)} \|\delta E_0^{(j)}\|_{j,\kappa}.$$

Then

$$\|\widetilde{R}^{(k+1)} - \widehat{\widetilde{R}}^{(k+1)}\|_{k+1} \le 8C_\chi \phi(\mathsf{t}_k)^3 \frac{|\delta x_k|}{r_x \mathsf{t}_k} + K_M c_\psi \sum_{j\le k} C_7 L^{(k+1-j)/4} \|\delta R^{(j)}\|_j$$
$$+ C_{\mathrm{rem}} \phi(\mathsf{t}_k)^3 \left(q_k^{\mathrm{old}} + q_k^0 + \frac{\mathsf{t}_k |\delta s_k|}{\bar{B}_M} + \frac{|\delta x_k|}{r_x \mathsf{t}_k} \right).$$

One may take

$$C_0^{\mathrm{rem}} = \frac{27}{4} K_M c_{\mathrm{cl}} C_r^3 \sum_{w\ge 3} (w+1)^2 (3/4)^{w-3},$$
$$C_1^{\mathrm{rem}} = \frac{9}{2} K_M c_{\mathrm{cl}} C_r^2 \sum_{N\ge 1} (N+1)^3 (3/4)^{N-1}, \qquad C_{\mathrm{rem}} = 4(C_0^{\mathrm{rem}} + C_1^{\mathrm{rem}} C_2).$$

The difference has the linear symmetries required for mass extraction, and

$$|\delta\rho_{k+1}| \le c_b^{-2} C_{am}^{\psi} h_F^{-2} \|\widetilde{R}^{(k+1)} - \widehat{\widetilde{R}}^{(k+1)}\|, \qquad \|\delta R^{(k+1)}\| \le K_\mu \|\widetilde{R}^{(k+1)} - \widehat{\widetilde{R}}^{(k+1)}\|.$$

■

Proof. Use the four deformation variables of the second-order expansion, with weights $(1,2,1,2)$. Before extraction its remainder is the sum of three terms:

$$\widetilde{R}^{(k+1)} = \left(\sum_X a_{(2,0,0,0)}(X) - \mathscr{A}_{k+1} \right)_{\ge 2}$$
$$+ \left(\mathcal{J}_k \Big[\sum_{j\le k} R^{(j)} \Big] - \sum_{j\le k} R^{(j)} \circ (\mathsf{H}_k, \bar{\mathsf{H}}_k) \right)_{\ge 2} + \sum_{|v|_w \ge 3} \left(\sum_X a_v(X) \right)_{\ge 2}.$$

The first term depends only on the coupling of the current step. It is holomorphic on the coupling disc and bounded on its inner 2/3 disc by $C_\chi \phi(\mathsf{t}_k)^3$, after the Gaussian test tail is included. The segment joining the two couplings stays within $5r_x x^\circ/12$ of real centres in the coupling window. It has a disc of radius at least $r_x \mathsf{t}_k/8$ inside those inner discs. Cauchy's estimate gives the first term in the asserted bound. For the second, $\delta R^{(j)}$ has zero quadratic cell sum, so Prop. B.4(d) gives the displayed old-remainder term.

For the third term compare the potentials on a common polydisc. Write their differences as $\Delta\mathcal{V}^{\mathrm{old}}_{\geq 2}$, $\Delta\mathcal{V}^{\mathrm{old}}_{0}$, and $\Delta\mathcal{V}^{s}$. By Prop. B.4,

$$\|\Delta\mathcal{V}^{\mathrm{old}}_{\geq 2}\| \leq c'_\psi \sum_{j\leq k} C_7 L^{(k+1-j)/4} \|\delta E^{(j)}_{\geq 2}\|_j,$$
$$\|\Delta\mathcal{V}^{\mathrm{old}}_{0}\| \leq K_\Lambda \frac{\mathsf{C}_{\mathcal{A}}\delta(\mathsf{t}_k)}{\alpha_0} \sum_{j\leq k} L^{-(k-j)} \|\delta E^{(j)}_{0}\|_{j,\kappa},$$
$$\|\Delta\mathcal{V}^{s}\| \leq c'_\psi c_{\mathsf{S}} h_F^2 |\delta s_k|.$$

The first and third bounds also hold with the fluctuation replaced by $z_1 B$ for $|z_1| \leq r_1$. Indeed the physical fluctuation is then at most $r_1 \mathsf{C}_{\mathcal{A}}\delta \leq \min(c_\diamond/(C_V h_F^2), \beta_0/2)$. The shift is at most $a_{\mathrm{an}}/4$ and each walk link costs at most $1/4$. The shift and walk contributions remain linear in this fluctuation size and their sum is included in c'_ψ.

Choose the ordinary radii r_1, r_3 and

$$r_2 = \frac{c_\diamond}{c'_\psi \sum_{j\leq k} C_7 L^{(k+1-j)/4} C_2 \phi(\mathsf{t}_{j-1})^2}, \qquad r_4 = \frac{c_\diamond \mathsf{t}_k}{2c'_\psi c_{\mathsf{S}} h_F^2 \bar{B}_M}.$$

Interpolate independently in the three potential differences, with radii R_2, R_3, R_4 equal to half the corresponding reference bound divided by the norm of the difference. A zero difference has infinite radius. The preceding estimates give

$$R_2^{-1} \leq 2q_k^{\mathrm{old}}, \qquad R_3^{-1} \leq 2q_k^0, \qquad R_4^{-1} \leq \mathsf{t}_k |\delta s_k|/\bar{B}_M.$$

On $|z_1| < r_1$, $|z_i| < 2r_i/3$ for $i = 2, 3, 4$, and $|\vartheta_i| < R_i$, each interpolated bracket is at most $3/2$ times its reference bound. The polymer expansion is therefore holomorphic and bounded by $2K_M c_{\mathrm{cl}}$. Its coefficient of z^v is bounded by

$$M_v = 2K_M c_{\mathrm{cl}} (3/2)^{v_2+v_3+v_4} r^{-v}.$$

The same bound holds when any interpolation parameter is set to either endpoint 0 or 1, since each endpoint is a valid flow. Change the parameters one at a time. If $R_i > 1$, Schwarz's lemma costs $2M_v/R_i$; if $R_i \leq 1$, the bound at the two endpoints costs $2M_v \leq 2M_v/R_i$. Changing the coupling last costs $4M_v|\delta x_k|/(r_x \mathsf{t}_k)$ by Cauchy's estimate.

Put $\varepsilon = \phi(\mathsf{t}_k)$. The radii satisfy $r_1^{-1}, r_3^{-1} \leq C_r\varepsilon$, $r_2^{-1} \leq C_r C_2 \varepsilon^2$ and $r_4^{-1} \leq C_r \varepsilon^2$. If $v_2 = 0$, group coefficients by their weight $w \geq 3$; there are at most $(w+1)^2$ and their sum is bounded by $C_0^{\mathrm{rem}}\varepsilon^3$. If $v_2 \geq 1$, remove one factor r_2^{-1} and put $N = v_1 + v_2 - 1 + v_3 + v_4 \geq 1$. There are at most $(N+1)^3$ terms. The ceilings $C_r\varepsilon \leq 1/2$ and $C_r^2 C_2 \varepsilon \leq 1$ bound their sum by $C_1^{\mathrm{rem}} C_2 \varepsilon^3$. Combining these coefficient sums with the three interpolation estimates proves the stated C_{rem} bound.

Both remainder totals have the same linear symmetries. The mass functional is linear, with $\delta\rho_{k+1} = -\eta_{k+1}^{-2} \mathsf{cf}_m(\widetilde{R}^{(k+1)} - \widehat{\widetilde{R}}^{(k+1)})$. The mass functional and subtraction bounds of the construction give the last two estimates. □

The constants in the next result are fixed after the construction constants, independently of K, s, u, and of the mass window chosen after them. We record their choices because the proof uses a floor on u, rather than a further bound depending on K.

Definition B.7 (Constants for the comparison)**.** Set

$$S_{\mathrm{fl}} = \sum_{m\geq 0} L^{-m}(1 + m\log L)^{\mathfrak{p}_0}, \qquad S^\sharp_{\mathrm{fl}} = \sum_{m\geq 0} L^{-m/2}(1 + m\log L)^{3\mathfrak{p}_0},$$
$$\Theta_L = \sup_{v\geq 0} (1+v) L^{-v/2} (1 + v\log L)^{9\mathfrak{p}_0}, \qquad \bar{\phi}_v = L^{-v/2}(1 + v\log L)^{\mathfrak{p}_0},$$
$$\bar{\phi}^\sharp_v = L^{-v/2}(1 + v\log L)^{3\mathfrak{p}_0}, \qquad \bar{\phi}^{(8)}_v = L^{-v}(1 + v\log L)^{8\mathfrak{p}_0}.$$

For $L^{v} \geq e^{18\mathfrak{p}_0}$ these functions decrease to zero, and bound the corresponding scale functions at every $\mathsf{t} \geq L^{v}$. In the following displays the constants are defined in their order:

$$
\begin{aligned}
C_{\mathbf{G}} &= 9K_{\mathbf{G}}(2+\sigma_\star/(4r_x)),\\
C_{\mathrm{nm}} &= 5r_x^{-1}\big[2^{\mathfrak{p}_0+2}(C_w+5\theta_s h_F^2 B_s)+2C_2+200\,4^{\mathfrak{p}_0}c_s h_F^2 B_s^2\big],\\
C_\psi^0 &= C_V h_F^2 \mathsf{C}_{\mathcal{A}} A_1 + 2c'_\psi c_{\mathsf{S}} h_F^2 \bar{B}_M + c'_\psi C_\omega S_\omega C_2,\\
C_{\mathrm{ld}} &= 40K_\mu K_M C_\psi^0/c_{\mathrm{cl}},\\
C_{\mathrm{dr}} &= \frac{L}{L-1}K_\Lambda \mathsf{C}_{\mathcal{A}} A_1 \alpha_0^{-1}\max\{\sqrt{2}(2+C_{am}r_\alpha^{-2})K_M^{\bullet}, C_{\mathrm{ld}}\},\\
C_{\mathrm{fc}} &= C_{\mathrm{nm}}+C_{\mathrm{dr}}, \qquad C_x^\delta = 1+C_{am}r_\alpha^{-2}, \qquad \beta_\star = \theta_E C_\omega S_\omega \leq 1/(4\sqrt{2}),\\
B_\star &= (7B_5+6(\bar{B}+4C_w)B_F)L^{1/2},\\
C_I &= 2+B_\star+E_0L^{1/2}+C_{\mathbf{G}}, \qquad A_{\mathrm{LM}} = \max\{C_2, C_w, 2C_{\mathrm{fc}}C_I\},\\
C_x &= C_x^\delta A_{\mathrm{LM}} S_{\mathrm{fl}},\\
C_{\mathrm{arr}} &= \frac{K_\Lambda \mathsf{C}_{\mathcal{A}}}{\alpha_0 K_0^\sharp}\left[\frac{13}{12}(E_0L^{1/2}+C_{\mathbf{G}})+LS_{\mathrm{fl}}A_{\mathrm{LM}}\right],\\
C_{\mathrm{fc}}^R &= \frac{8C_\chi+C_{\mathrm{rem}}}{r_x}+C_{\mathrm{rem}}\left[\frac{A_{\mathrm{LM}}}{C_2}L^{3/4}S_\omega + C_{\mathrm{arr}}+1+B_\star\right],\\
A_R &= \max\{C_3, 2K_\mu C_{\mathrm{fc}}^R\}, \qquad C_\rho^\delta = c_b^{-2}C_{am}^\psi h_F^{-2}L^{-1}.
\end{aligned}
$$

Choose

$$
\begin{aligned}
C_{\mathrm{LM}} = \max\{&A_{\mathrm{LM}}+A_R+LC_\rho^\delta A_R, C_x, 10B_F(\bar{B}+4C_w),\\
&2(5B_FC_x+C'_\rho S_3^\sharp+C_\rho^\delta A_R), 2(5B_FC_x+C_\rho^\delta A_R S_{\mathrm{fl}}^\sharp)\},\\
C_\Pi &= 2(5B_F+2)+C_{\mathrm{LM}}(1+\Theta_L/(2C'_\rho S_3^\sharp)),\\
B_\delta &= \bar{B}+4C_w+C_x(1+C_\Pi), \qquad \eta_u = (\bar{B}+4C_w)L^{-u}+LS_{\mathrm{fl}}C_x\bar{\phi}_u^3.
\end{aligned}
$$

For the zero-source comparison take u_* so that $L^{u_*} \geq e^{18\mathfrak{p}_0}$ and, for $u \geq u_*$,

$$
\begin{aligned}
(5B_FC_x+C_\rho^\delta A_R S_{\mathrm{fl}}^\sharp)\bar{\phi}_u^\sharp \leq \tfrac{1}{2}, &\qquad C_{\mathrm{LM}}\bar{\phi}_u^\sharp \leq 1,\\
16B_\delta \leq r_x L^u, &\qquad \bar{\phi}_u^\sharp \leq 1, \quad \eta_u \leq 1,\\
C_\psi^0\bar{\phi}_u \leq c_{\mathrm{cl}}/20, &\qquad C_{\mathrm{ld}}K_\Lambda \mathsf{C}_{\mathcal{A}} A_1 L\bar{\phi}_u \leq \alpha_0\theta_E C_\omega.
\end{aligned}
$$

Such a finite floor exists because all the left sides that involve $\bar{\phi}_u$ tend to zero. The source comparison below imposes further floors on the same u_*. ■

Proposition B.8 (Loss of memory at a constant label)**.** Suppose $u \geq u_*$. With the notation of Def. B.3, set

$$
\Pi_k = 2|\pi| + C_{\mathrm{LM}}[(k-t_0+1)\varepsilon_s + L^{-s}/\mathsf{b}], \qquad B'_k = 7B_5L^{-(k-t_0)} + 6(\bar{B}+4C_w)B_FL^{-(s+1)}/\mathsf{b}.
$$

For $t_0 \leq k \leq k' \leq K$,

$$
|\mathsf{t}_{k'}G_{<t_0,k'}(x(c)) - \mathsf{t}_kG_{<t_0,k}(x(c))| \leq B'_k,
$$

and

$$
\left|\pi - \frac{\mathsf{b}}{\mathsf{b}-c}(\zeta - F_{<t_0}(t_1;\mathfrak{m}_{t_1}))\right| \leq 3(\bar{B}+4C_w)B_FL^{-(s+1)}/\mathsf{b}.
$$

At $c=0$ one may replace $7B_5$ by $6B_5$ and the coefficients $6(\bar{B}+4C_w)B_F$, $3(\bar{B}+4C_w)B_F$ by $4\bar{B}B_F$, $2\bar{B}B_F$, respectively. In particular,

$$
|\pi(F^{\mathrm{UV}};0)| \leq B_5(L^{-t_0}+3L^{-(s-u)}) + 2\bar{B}B_FL^{-(s+1)}/\mathsf{b}.
$$

For $t_0 < k \le t_1$, both torus families obey the following bounds, including each pointed term:

$$\begin{aligned} |\delta x_k| &\le (\bar B + 4C_w)L^{-(k-t_0)} + C_{\mathrm{LM}}[\phi(\mathsf{t}_{k-1})\Lambda_k + \phi(\mathsf{t}_{k-1})^2\Pi_{k-1}], \\ \|\delta F_0^{(k)}\|_k &\le C_{\mathrm{LM}}\phi(\mathsf{t}_{k-1})[\Lambda_k + \phi(\mathsf{t}_{k-1})\Pi_{k-1}], \\ \|\delta E_{\ge 2}^{(k)}\|_k, \|\delta T^{(k)}\|_k &\le C_{\mathrm{LM}}\phi(\mathsf{t}_{k-1})^2[\Lambda_k + \phi(\mathsf{t}_{k-1})\Pi_{k-1}], \\ \|\delta R^{(k)}\|_k, |\delta\rho_k| &\le C_{\mathrm{LM}}\phi(\mathsf{t}_{k-1})^3[\Lambda_k + \Pi_{k-1}]. \end{aligned}$$

The coupling difference is at most $r_x\hat x_k^\circ/8$. The scalar activity bound holds polymer by polymer at rate $\kappa+1$. The explicit Gaussian term satisfies

$$e^{\kappa_+ d_k(X)}|\delta\mathbf{G}^{(k)}(X;\cdot;z)| \le C_{\mathbf{G}}|\delta x_{k-1}|/\mathsf{t}_{k-1}.$$

The scalar and leg bounds remain valid after summing the norms over all pointers, with the additional factor $8M^3$.

The mass difference has the more precise form

$$\begin{aligned} \mathsf{t}_k\delta s_k &= -\pi + [\mathsf{t}_k G_{<t_0,k}(x(c)) - \mathsf{t}_{t_1}G_{<t_0,t_1}(x(c))] + r_k, \\ |r_k| &\le C_{\mathrm{LM}}[(k-t_0+1)\varepsilon_s + L^{-s}/\mathsf{b} + \phi_\sharp(\mathsf{t}_k)|\pi|]. \end{aligned}$$

In particular $\mathsf{t}_k|\delta s_k| \le \Pi_k + B'_k$, and the bracket vanishes at $k = t_1$.

Finally the activities discarded at the restart, read at a level $t_0 \le k \le K$, satisfy

$$\sum_{j\le t_0} C_7 L^{(k-j)/4}\|E_{\ge 2}^{(j)}\|_j \le C_7C_2S_\omega\phi(\mathsf{t}_k)^2\Lambda_{k+1}.$$

For $t_0 \le k < K$, their scalar localization on $Y \in \mathcal{D}_k$ is at most

$$K_\Lambda \frac{\mathsf{C}_{\mathcal{A}}\delta(\mathsf{t}_k)}{\alpha_0}\frac{L}{L-1}E_0L^{-(k-t_0)}e^{-(1-2\delta_{\mathrm{w}})\kappa d_k(Y)}.$$

■

Proof. By Prop. B.2(e), with $\tau = 1 - c/\mathsf{b}$,

$$\left|\mathsf{t}_k G_{<t_0,k}(x(c)) - \frac{\mathsf{b}}{\mathsf{b}-c}F_{<t_0}(k;\mathfrak{m}_k)\right| \le 3(\bar B + 4C_w)B_F L^{-(s+1)}/\mathsf{b}.$$

Here $|x_i(c) - \tau\mathsf{t}_i| \le \bar B + 4C_w$, $|x_i(c)| \ge 11\mathsf{t}_i/24$ and $|\tau| \ge 63/64$. The partial-sum comparison in Prop. B.2(c) proves the first bound and the estimate for π. At label zero the stronger real coupling window gives the stated smaller constants. Taking $\zeta = F^{\mathrm{UV}}$ and applying Prop. B.2(b) gives its last estimate. Notice that $0 \le F^{\mathrm{UV}} \le F_\infty \le B_F$, so this is an allowed restart.

Subtract the mass identities of the two flows. The exact identity is

$$\begin{aligned} \mathsf{t}_k\delta s_k = [\mathsf{t}_k G_{<t_0,k}(x(c)) - \zeta'] &+ \sum_{t_0\le i<k}\sigma_i(k;\mathfrak{m}_k)\mathsf{t}_i\left(\frac{1}{x_i(c)} - \frac{1}{\hat x_i(c)}\right) \\ &+ \sum_{j\le t_0}\mathsf{t}_j\rho_j(c) + \sum_{t_0<j\le k}\mathsf{t}_j\delta\rho_j. \end{aligned} \tag{B.vi}$$

The second term is at most $5B_F\max_{t_0\le i<k}|\delta x_i|/\mathsf{t}_i$. The third is at most $C'_\rho S_3^\sharp\varepsilon_s$. These facts use slice positivity, the coupling windows, and

$$\phi(\mathsf{t}_{k-n}) \le L^{-n/2}(1+n\log L)^{\mathfrak{p}_0}\phi(\mathsf{t}_k), \quad \phi_\sharp(\mathsf{t}_{k-n}) \le L^{-n/2}(1+n\log L)^{3\mathfrak{p}_0}\phi_\sharp(\mathsf{t}_k).$$

We now derive the one-step inequalities. Set

$$X_j = |\delta x_j|/\mathsf{t}_j, \quad P_j = \mathsf{t}_j|\delta s_j|, \quad d_j^\psi = \frac{\|\delta E_{\ge 2}^{(j)}\|_j}{\phi(\mathsf{t}_{j-1})^2}, \quad d_j^0 = \frac{\|\delta F_0^{(j)}\|_j}{\phi(\mathsf{t}_{j-1})}, \quad d_j^R = \frac{\|\delta R^{(j)}\|_j}{\phi(\mathsf{t}_{j-1})^3}.$$

For $j \le t_0$ use $d_j^{\psi} \le C_2$, $d_j^R \le C_3$ and the full scalar array bound E_0; do not replace that bound by a bound for the fluctuation family alone.

First change the current coupling and mass while holding the restarted old families fixed. On the coefficient polydiscs the outputs have bounds $2C_w\phi + 2\theta_s h_F^2|s'|$ in degree zero and $2C_2\phi^2 + 2c_s h_F^2|s'|^2$ with legs. The coupling segment is at distance at least $11 r_x \mathtt{t}_k/48$ from its boundary; the mass segment is at distance at least $5B_s\phi(\hat{x}_k^{\circ})/2$. The definition of C_{nm} therefore bounds the relative changes of both activity degrees and of the coupling extraction by $C_{\mathrm{nm}}(X_k + \phi(\mathtt{t}_k)P_k)$. The explicit term $\mathbf{G}$ depends only on the coupling, so Cauchy's estimate on the same segment gives $C_{\mathbf{G}}X_k$ at rate κ_+, pointer by pointer.

Next change the old families, keeping the current numbers fixed. By Prop. B.4 the potential is affine on this segment and its norms are bounded by those at the endpoints. Its scalar part is at most $c_{\mathrm{cl}}/8$, and its leg part is at most $C_{\psi}^0\phi(\mathtt{t}_k) \le c_{\mathrm{cl}}/20$. The derivative estimates in Prop. B.4(c), after local extraction, give the following four kernels, with $n = k + 1 - j$:

$$K_n^{\psi\psi} = K_n^{\psi 0} = \theta_E C_\omega L^{-3n/4}(1 + n\log L)^{2\mathfrak{p}_0},$$
$$K_n^{0\psi} = \theta_E C_\omega L^{-n}, \qquad K_n^{00} = \theta_E C_\omega L^{-3n/2}(1 + n\log L)^{\mathfrak{p}_0}.$$

Each satisfies $\sum_{n\ge 1} K_n L^{n/2} \le \beta_\star$. For the scalar-to-leg kernel the derivative carries the additional factor $\phi(\mathtt{t}_k)$, because a scalar potential alone creates no legs. The last floor in Def. B.7 absorbs its constant. The ordinary scalar contraction uses the construction ceiling $\sqrt{2}L(2 + C_{am}r_\alpha^{-2})K_M^{\bullet}K_\Lambda \mathsf{C}_{\mathcal{A}}\delta(\gamma^{-2}) \le \alpha_0\theta_E C_\omega$. The old scalar arrays and $\delta\mathbf{G}$ contribute the forcing

$$H_k = C_{\mathrm{fc}}\left[X_k + \phi(\mathtt{t}_k)P_k + E_0 L^{-(k-t_0)} + C_{\mathbf{G}} \max_{t_0 \le i < k} X_i\right].$$

We have proved

$$d_{k+1}^{\psi} \le D_{k+1}^{\psi} := \sum_{j\le k} K_n^{\psi\psi} d_j^{\psi} + \sum_{t_0 < j \le k} K_n^{0\psi} d_j^0 + H_k,$$
$$d_{k+1}^{0} \le D_{k+1}^{0} := \sum_{j\le k} K_n^{\psi 0} d_j^{\psi} + \sum_{t_0 < j \le k} K_n^{00} d_j^0 + H_k,$$
$$|\delta x_{k+1}| \le L^{-1}|\delta x_k| + C_x^{\delta}\phi(\mathtt{t}_k) D_{k+1}^0.$$

All estimates include the local extraction factors. They hold for both torus families, and for each pointer, since the potential and its first variation have these properties.

For the remainder, Lem. B.6 gives

$$d_{k+1}^R \le D_{k+1}^R := \sum_{j\le k} K_n^{RR} d_j^R + K_\mu\left[\frac{8C_\chi + C_{\mathrm{rem}}}{r_x} X_k + C_{\mathrm{rem}}(q_k^{\mathrm{old}} + q_k^0 + P_k/\bar{B}_M)\right],$$
$$K_n^{RR} = K_\mu K_M c_\psi C_7 L^{-5n/4}(1 + n\log L)^{3\mathfrak{p}_0},$$
$$\mathtt{t}_{k+1}|\delta\rho_{k+1}| \le C_\rho^{\delta}\phi_\sharp(\mathtt{t}_k) D_{k+1}^R.$$

The weighted sum of K_n^{RR} is at most $1/4$ by the second-order contraction floor of the construction.

We close these inequalities by induction with the bounds

$$D_j^{\psi}, D_j^0 \le A_{\mathrm{LM}}[\Lambda_j + \phi(\mathtt{t}_{j-1})\Pi_{j-1}],$$
$$D_j^R \le A_R(\Lambda_j + \Pi_{j-1}),$$
$$|\delta x_j| \le (\bar{B} + 4C_w)L^{-(j-t_0)} + C_x[\phi(\mathtt{t}_{j-1})\Lambda_j + \phi(\mathtt{t}_{j-1})^2\Pi_{j-1}],$$
$$P_j \le \Pi_j + B_j'.$$

For $j \le t_0$ take $\Pi_{j-1} = 0$ and use $\Lambda_j \ge 1$. At t_0 the coupling bound follows from its ordinary window, and the mass bound follows from eq. B.vi with the later sums empty.

Suppose the induction holds through $k < t_1$. The bound $\Pi_j \le C_\Pi$ and $16B_\delta \le r_x L^u$ ensure $|\delta x_k| \le r_x \hat{x}_k^\circ/8$, justifying the Cauchy segments used above. The scale inequalities imply, for $i \le k$,

$$X_i \le \eta_u[\Lambda_{k+1} + \phi(\mathfrak{t}_k)\Pi_k], \qquad \phi(\mathfrak{t}_k)P_k \le \phi(\mathfrak{t}_k)\Pi_k + B_\star \Lambda_{k+1},$$

and $E_0 L^{-(k-t_0)} \le E_0 L^{1/2}\Lambda_{k+1}$. Each of the four kernel sums is at most $\beta_\star A_{\mathrm{LM}}[\Lambda_{k+1} + \phi(\mathfrak{t}_k)\Pi_k]$. Thus the two activity inequalities close because $2\beta_\star \le 1/2$ and $A_{\mathrm{LM}} \ge 2C_{\mathrm{fc}}C_I$.

For the remainder, the same scale sums give

$$q_k^{\mathrm{old}} \le (A_{\mathrm{LM}}/C_2)L^{3/4}S_\omega[\Lambda_{k+1} + \phi(\mathfrak{t}_k)\Pi_k],$$
$$q_k^0 \le C_{\mathrm{arr}}[\Lambda_{k+1} + \phi(\mathfrak{t}_k)\Pi_k].$$

For the first inequality, bound the numerator by the induction hypothesis and keep just its last term in the denominator, which is at least $C_2 L^{-3/4}\phi(\mathfrak{t}_k)^2$. For the second, split the scalar sum at t_0: the earlier part costs $(L/(L-1))E_0 L^{-(k-t_0)}$; the later fluctuation part costs $LS_{\mathfrak{fl}}A_{\mathrm{LM}}$ times the indicated rate; the explicit part costs $(L/(L-1))C_{\mathbf{G}} \max_i X_i$. As $L/(L-1) \le 13/12$, this is precisely C_{arr}. The remainder inequality now closes with coefficient $A_R/4 + K_\mu C_{\mathrm{fc}}^R \le A_R$.

Iterating the coupling recursion and using the newly proved scalar bound gives a geometric convolution bounded by $S_{\mathfrak{fl}}$. This proves its induction bound because $C_x = C_x^\delta A_{\mathrm{LM}} S_{\mathfrak{fl}}$.

For the mass, return to eq. B.vi. The later remainder sum is bounded by

$$C_\rho^\delta A_R \left[L^{-1/2}(k+1-t_0)\varepsilon_s + S_{\mathfrak{fl}}^\sharp \phi_\sharp(\mathfrak{t}_k)\Pi_k\right].$$

The reciprocal-coupling term is at most

$$5B_F(\bar{B} + 4C_w)L^{-s}/\mathfrak{b} + 5B_F C_x[L^{-1/2}\varepsilon_s + \phi(\mathfrak{t}_k)^4\Pi_k].$$

The terms containing Π_k sum to at most $\Pi_k/2$ by the first depth condition. The defining lower bounds on C_{LM} absorb the remaining ε_s and $L^{-s}/\mathfrak{b}$ terms, yielding $P_{k+1} \le \Pi_{k+1} + B'_{k+1}$. This closes the simultaneous induction.

The asserted activity estimates follow, since $C_{\mathrm{LM}} \ge A_{\mathrm{LM}}, C_x, A_R$. Use $\delta T = \delta E_{\ge 2} - \delta R$ for the explicit part and the remainder mass functional for $\delta\rho$; the additional terms in the definition of C_{LM} cover both. Retaining the signed first bracket of eq. B.vi, instead of its absolute bound, gives the stated formula and estimate for r_k.

The polymer estimates before extraction have rate κ_+. There are at most $8M^3 e^{d_k(X)}$ pointers in X, and $\kappa_+ - \kappa - 1 \ge 2c_E$ leaves the same summable polymer factor after their count. Summing pointers therefore costs $8M^3$. For local extraction the constant term is put at each pointer's cube; the action uses nonnegative pointer weights with sum one; and the mass bound already sums its cells and places. The resulting factors are

$$\frac{8M^3 + 1 + C_{am}r_\alpha^{-2}}{2 + C_{am}r_\alpha^{-2}} \le 8M^3, \qquad \frac{8M^3 + K_\mu - 1}{K_\mu} \le 8M^3.$$

This proves the summed-pointer assertion without replacing it by a bound for the norm of the signed sum.

Finally sum the closed bounds over $j \le t_0$. With $n = k + 1 - j$, retain $L^{-n/2} \le \Lambda_{k+1}$ and sum the remaining $L^{-n/4}(1 + n\log L)^{2\mathfrak{p}_0}$ into S_ω. For the scalar terms apply Prop. B.4(b) and sum $\sum_{j \le t_0} L^{-(k-j)} \le (L/(L-1))L^{-(k-t_0)}$. These are the two final estimates. □

B.4. **Comparison in the presence of sources.** The preceding comparison treats the constant part of a source. A varying source also changes the local mass increments and their cell sums. We first compare the two flows at the same source coordinates, then compare their actual source chains. The second comparison is needed because equal coordinates do not describe equal physical sources in the two flows.

Definition B.9 (Source coordinates for the restart)**.** Let ζ be real, $|\zeta| \le 2B_F + 1$, and use the real restarted sequence $\hat{x}_i, \hat{s}_i$. Write $\hat{\phi}_i = \phi(\hat{x}_i)$ and

$$\hat{\tau}_i(w; a) = 40Ma\hat{x}_i^{1/2}\lambda_i \operatorname{Lip}(w)/\mathsf{b}.$$

Hats on the source domains mean that ϕ_i, τ_i are replaced by $\hat{\phi}_i, \hat{\tau}_i$, and that the mass coordinates run only over $t_0 \le i \le k$. Thus a restarted coordinate is $\hat{\mathfrak{c}} = (w, \mathsf{n}, (\sigma_i)_{t_0 \le i \le k})$. If $\mathfrak{c}$ is an ordinary coordinate, write $\mathfrak{c}^\wedge$ for its restriction to these indices.

For $a_k^{\mathrm{rad}} \le a \le \alpha_0$ let $\mathfrak{D}_k(a)$ be the intermediate domain

$$\|w\|_\infty < 6\varrho, \quad C_{\mathrm{lin}}\tau_k(w; a) < 3c_{\mathrm{cl}}/32, \quad L^{k+1}\|\mathsf{n}\| < 3r_M/2, \quad \|\sigma_i\| < 3B_\sigma\phi_i/2.$$

In $\mathfrak{D}_k$ replace the first two requirements by $\|w\|_{\sharp,k} < 6\varrho$.

At level t_0 the source restart is

$$\hat{\tau}^{\mathrm{F}}_{t_0}[\hat{\mathfrak{c}}](dV) = \mathsf{Stay}(V)\exp\big[-A(L^{t_0}\hat{x}_{t_0} - a_K^{-1}w, \mathsf{U}) - \langle\bar{\Psi}, \mathbb{D}_{t_0}(\mathsf{U})\Psi\rangle \\ - L^{-t_0}\mathsf{S}_1((\hat{s}_{t_0} + \sigma_{t_0})\psi^{\mathrm{bg}})\big]\mathscr{M}^{\mathrm{fr}}_{h\emptyset}(dV).$$

The tower and background fermions have fine mass matrix $m_0 + \hat{\mathsf{n}}$, where $\mathsf{n} = -a_K J$. There are no earlier activities, normalizers, or source energies. Later format data use the same source step as the construction, with the real numbers of the restarted zero-source flow. At level i its coupling weight is $L^i\hat{x}_i - a_K^{-1}w$, its mass member is $\hat{s}_i + \sigma_i$, and its old activities are exactly those generated at levels $t_0 < j \le i$.

For a physical source $s = (w, J)$ define

$$\hat{\sigma}_{t_0}(s)(x) = \frac{\zeta}{L^s(\mathsf{b} - w(x))} - \frac{\zeta}{L^s\mathsf{b}},$$
$$\hat{\sigma}_{k+1}(s)(x) = L\hat{\sigma}_k(s)(x) + \hat{\nu}_{k+1}(\hat{\mathfrak{c}}_k(s))(y_{k+1}(x)) - \hat{\mu}_{k+1},$$
$$\hat{\mathfrak{c}}_k(s) = (w, -a_K J, (\hat{\sigma}_i(s))_{t_0 \le i \le k}).$$

The occurrence of L^s in the initial value uses the integer restart depth; the argument s denotes the source. At a constant label set $\hat{\sigma}^{\mathrm{c}}_k(c) = \hat{s}_k(c) - \hat{s}_k$ and $\hat{\mathfrak{c}}^{\mathrm{c}}_k(c) = (c, 0, (\hat{\sigma}^{\mathrm{c}}_i(c)))$. The source energies are

$$\hat{e}_j(X; s) = \hat{\mathfrak{E}}_j(X; \hat{\mathfrak{c}}_{j-1}(s)) - \hat{\mathfrak{E}}_j(X; 0), \qquad \hat{E}_k(s) = \hat{E}_k + \sum_{t_0 < j \le k}\sum_X \hat{e}_j(X; s).$$

Here $\hat{\mathfrak{E}}_j$ includes the value at the trivial pair of the scalar new activity before extraction and the localized normalizers and Gaussian term of that step.

At equal coordinates put $\delta X(\mathfrak{c}) = X(\mathfrak{c}) - \hat{X}(\mathfrak{c}^\wedge)$; along the chains put $\delta^{\mathrm{ch}}X(s) = X(\mathfrak{c}_k(s)) - \hat{X}(\hat{\mathfrak{c}}_k(s))$. Also set $d\nu_{k+1} = \nu_{k+1} - \hat{\nu}_{k+1} \circ \wedge$. Earlier restarted families are zero in these definitions.

Let c_{lm} be the largest number in $(0, 1/12]$ such that

$$\sum_{n \ge 1} L^{-n(1/4 - c_{\mathrm{lm}})}(1 + n)^{\mathfrak{p}_0} \le \tfrac{3}{2}S_{1/4}, \qquad L^{c_{\mathrm{lm}}}\left(\tfrac{8}{3}L^{-1/2} + \tfrac{27}{112}L^{-1}\right) \le \tfrac{9}{10}.$$

Such a number exists since both inequalities are strict at zero. Put

$$\Lambda^{\mathrm{s}}_k = L^{-c_{\mathrm{lm}}(k - t_0)}, \qquad \Pi^\flat_k = \Pi_k(\zeta; 0), \qquad \Pi^*_k = \sup_{|c| < 8\varrho}\Pi_k(\zeta; c),$$
$$m_k = \Lambda^{\mathrm{s}}_k + \phi(\mathfrak{t}_{k-1})\Pi^\flat_{k-1}, \qquad m^*_k = \Lambda^{\mathrm{s}}_k + \phi(\mathfrak{t}_{k-1})\Pi^*_{k-1}.$$

Take both Π sequences to be zero before t_0. Finally $\mathrm{tor}_k = e^{-c_{\mathrm{t}}\ell/(2\lambda_k)}$ on the torus and $\mathrm{tor}_k = 0$ on $\mathbb{Z}^3$. ∎

Proposition B.10 (Loss of memory with sources)**.** After the additional, fixed depth conditions specified below, the following hold for $u \ge u_*$. The norms of families with legs sum the norms of

their filed summands. The statements hold on every admitted torus; on $\mathbb{Z}^3$ the assertions about format data and equal coordinates concern constant coordinate fields.

(S1) For $t_0 \le k < t_1$ the restarted step exists on $2\hat{\mathscr{D}}_k(a)$, simultaneously for $a_k^{\mathrm{rad}} \le a \le \alpha_0$, and has the original source-step bounds with hats:

$$\hat{\epsilon}_{k+1}^{\mathrm{s},\psi}(a) \le 2C_w^{\mathrm{s}}\hat{\phi}_k, \qquad \hat{\epsilon}_{k+1}^{\mathrm{s},0}(a) \le 2C_w^{\mathrm{s}}\hat{\phi}_k + C'_{\mathrm{lin}}\hat{\tau}_k(w;a).$$

At $a = \alpha_0$ its full scalar arrays have bound E_0^{s}. The fluctuation scalar families have their stated bounds at rate $\kappa+1$ pointer by pointer. At $a = \alpha_0$, the norm of the leg families summed over pointers is at most $8M^3$ times the displayed size, before and after extraction. Its increments obey the original local holomorphy, Lipschitz and torus-defect estimates with B_ν and C_{tor}.

The restarted constant chain is holomorphic, real at real labels, and satisfies

$$|\hat{\sigma}_k^{\mathrm{c}}(c)| \le \tfrac{1}{4}C_{\mathrm{var}}\hat{\phi}_k|c|/\varrho.$$

For $t_0 \le k \le t_1$ its physical source chain is holomorphic on $\mathscr{O}_k$, belongs to $\hat{\mathscr{D}}_k$, and obeys

$$\|\hat{\sigma}_k(s) - \hat{\sigma}_k^{\mathrm{c}}(w(\cdot))\| \le C_{\mathrm{var}}\hat{\phi}_k\Xi_k(s) + 2C_{\mathrm{tor}}C_{\mathcal{L}}\hat{\phi}_k\mathrm{tor}_k\|w\|_\infty/\varrho.$$

It has the original source-locality tail estimates. The energies $\hat{e}_j$, $t_0 < j \le t_1$, have the original local holomorphy, conjugation and weighted bounds. At constant coordinates on $\mathbb{Z}^3$, $\hat{\nu}_{k+1}(\hat{\mathfrak{c}}_k^{\mathrm{c}}(c)) = \hat{\mu}_{k+1}(c)1$.

(S2) For $t_0 < k \le t_1$ and $\mathfrak{c} \in \mathfrak{D}_{k-1}(a)$, before and after extraction,

$$\|\delta E_{\ge 2}^{(k)}(\mathfrak{c})\|_{k;a} \le A_\psi^\sharp m_k\, 2C_w^{\mathrm{s}}\phi_{k-1},$$
$$\|\delta F_0^{(k)}(\mathfrak{c})\|_{k;a} \le m_k\big(A_0^{\mathrm{s}}\, 2C_w^{\mathrm{s}}\phi_{k-1} + 4C'_{\mathrm{lin}}\tau_{k-1}(w;a)\big).$$

(S3) For $t_0 \le k < t_1$, the function $d\nu_{k+1}$ is holomorphic on $\mathfrak{D}_k$. For $\mathfrak{c} \in \mathscr{D}_k$, $\mathfrak{c}' \in \mathfrak{D}_k$, and every level-$(k+1)$ site y,

$$|d\nu_{k+1}(\mathfrak{c})(y) - d\nu_{k+1}(\mathfrak{c}')(y)| \le 2A_\psi^\sharp m_{k+1}B_\nu\phi_{k+1}\|\mathfrak{c} - \mathfrak{c}'\|_{k,y},$$
$$|d\nu_{k+1}(\mathfrak{c})(y) - (\mu_{k+1} - \hat{\mu}_{k+1})| \le A_\psi^\sharp m_{k+1}B_\nu\phi_{k+1} \quad (\mathfrak{c} \in \mathfrak{D}_k).$$

(S4) For $t_0 \le k \le t_1$, along a physical source chain with $s \in \mathscr{O}_{t_1}$,

$$\mathfrak{t}_k|\delta^{\mathrm{ch}}(s_k + \sigma_k)(x)| \le \mathfrak{t}_k|s_k(w(x)) - \hat{s}_k(w(x))| + A_D\mathfrak{t}_k\phi_k\Xi_k(s)m_k^* + C_\Delta\mathfrak{t}_k\phi_k\mathrm{tor}_k\|w\|_\infty/\varrho.$$

The first term is at most $\Pi_k(\zeta; w(x)) + B'_k$. Moreover, with $N(s) = \mathrm{Lip}(w)/\varrho + L\|J\|_\infty/r_M$,

$$\mathfrak{t}_k\phi_k\Xi_k(s) \le c_{\mathrm{th}}\mathfrak{b}^{1/2}L^{-(K-k)/2}(1 + \log\mathfrak{t}_k)^{\mathfrak{p}_0}\pi_k N(s).$$

(S5) For $s \in \mathscr{O}_{t_1}$ and $t_0 < k \le t_1$, at $a = a_{k-1}^{\mathrm{rad}}$, and also at $a = \alpha_0$ when $\|w\|_\sharp < 4\varrho$,

$$\|\delta^{\mathrm{ch}}E_{\ge 2}^{(k)}(s)\|_{k;a} \le A_{\mathrm{ch}}(m_k^* + \mathrm{tor}_{k-1})2C_w^{\mathrm{s}}\phi_{k-1},$$
$$\|\delta^{\mathrm{ch}}F_0^{(k)}(s)\|_{k;a} \le A_{\mathrm{ch}}(m_k^* + \mathrm{tor}_{k-1})(2C_w^{\mathrm{s}}\phi_{k-1} + C'_{\mathrm{lin}}\tau_{k-1}(w;a)),$$
$$\|\varpi_k(s) - \hat{\varpi}_k(s)\| \le C_\varpi^\delta\phi_k(\Xi_k(s)m_k^* + \mathrm{tor}_k).$$

(S6) At the fixed pair radius, the scalar bounds in (S2) and (S5) hold polymer by polymer and pointer by pointer at rate $\kappa+1$, for $\|w\|_\sharp < 6\varrho$ and $\|w\|_\sharp < 4\varrho$, respectively. The leg bounds hold with factor $8M^3$ in the norm summed over pointers, before and after extraction. At coordinates in $2\mathscr{D}_{k-1}$ and along the chains,

$$e^{\kappa_+ d_k(X)}|\mathbf{G}^{(k)}(X;\cdot;z) - \hat{\mathbf{G}}^{(k)}(X;\cdot;z)| \le C_{\mathbf{G}}^{\mathrm{s}}|\delta x_{k-1}|/\mathfrak{t}_{k-1}.$$

■

Proof. We give the source induction, the comparison of the two format data, and finally the comparison of their chains. The constants and extra depth floors are specified following the proof.

At the initial level there are no activities or pointed terms, and the format datum is independent of the pair radius. The source step can therefore start simultaneously at all radii. Its zero- source data satisfy the bounds of Prop. B.5. In particular, for $t_0 \le i \le j \le t_1$,

$$|\hat{x}_j - L^{-(j-i)}\hat{x}_i| \le \bar{B}, \quad (1-L^{-1})\mathsf{t}_i \le \hat{x}_i \le (1+L^{-1})\mathsf{t}_i, \quad \hat{x}_i \ge \gamma^{-2}.$$

As $\mathsf{t}_i\hat{\phi}_i \ge 2$, its mass members and extracted increments satisfy

$$|\hat{s}_i| \le \bar{B}_M\hat{\phi}_i, \qquad |\hat{\mu}_i| \le 4\bar{B}_M/\mathsf{t}_i \le 3\bar{B}_M\hat{\phi}_i.$$

These are the numerical inputs to the source-step estimates; their proofs now run from t_0 with the hatted scale functions.

The scalar array bound needs the complete zero-source terms, not only their fluctuation parts. At coordinate zero the restart is exactly the flow of Prop. B.5. Its new scalar activity before extraction, its value at the trivial pair and its action coefficient have the original bounds with $\hat{s}_i$ and $\hat{\phi}_i$. The normalizer families generated after t_0 are the ordinary normalizers at the same calibration and fine mass; there are no earlier terms to estimate. The Gaussian terms at the weight obey the same bounds because

$$|\hat{v}_i| = L^{K-i}\|w\|_\infty/\hat{x}_i \le 8\varrho/(\mathsf{b}-\bar{B}) \le r_v/2, \qquad |\hat{v}_i| \le 16\varrho\gamma^2.$$

The scalar constant created by pointing the Gaussian determinant has bound

$$(K_{\mathbf{G}}\bar{B}\gamma^2 + C_w\phi(\gamma^{-2}))e^{-(\kappa+1)d_i(X)}.$$

Its nontrivial polymer part is the Gaussian term at $\hat{x}_{i-1}/\mathsf{t}_{i-1}$, and at a single cube the additional term is $-3(L^3-1)\log(\hat{x}_{i-1}/\mathsf{t}_{i-1})/2$. The same two unused shares of E_0 cover these terms. The scalar admission bounds and the source-radius floor therefore remain valid, giving E_0^{s} and the pointed scalar bounds.

There is one change in the symmetry proof along a source chain. The mass recursion gives

$$L^{-k}(\hat{s}_k + \hat{\sigma}_k(s)) = L^{-t_0}\hat{\Sigma}_w^\zeta + \sum_{t_0<i\le k} L^{-i}\hat{\nu}_i(\cdot;\hat{\mathfrak{c}}_{i-1}(s)), \qquad \hat{\Sigma}_w^\zeta = \frac{\zeta}{L^s(\mathsf{b}-w)}.$$

After reading on shell, cancellation with the extracted local parts leaves precisely the starting mass form with multiplier $\hat{\Sigma}_w^\zeta$. This scalar-valued field is invariant under flavour transformations and charge conjugation, and complex conjugation carries it to $\hat{\Sigma}_{\bar{w}}^\zeta$. Thus it has the same joint source symmetries as the other initial terms. At each step the transformed logarithm and the logarithm of the transformed integral differ by a scalar in $2\pi i\mathbb{Z}$. It is continuous on the convex source domain and zero at source zero, by Prop. B.5; hence it vanishes throughout. This proves the symmetry and conjugation assertions used for the energies.

At a constant source c the restarted source density at t_0 is exactly the density with label c. Their later splittings therefore agree. In particular the infinite-lattice increment is $\hat{\mu}_{k+1}(c)1$. Subtracting their mass identities gives

$$\mathsf{t}_k\hat{\sigma}_k^{\mathrm{c}}(c) = \zeta\left(\frac{\mathsf{b}}{\mathsf{b}-c}-1\right) + \sum_{t_0\le i<k}\sigma_i(k;\mathfrak{m}_k)\mathsf{t}_i\left(\frac{1}{\hat{x}_i(c)}-\frac{1}{\hat{x}_i}\right) + \sum_{t_0<j\le k}\mathsf{t}_j(\hat{\rho}_j(c)-\hat{\rho}_j(0)).$$

Schwarz's lemma on $|c| < 8\varrho$ bounds the coupling-extraction and cubic-remainder differences. Slice positivity and the coupling windows then give

$$|\hat{\sigma}_k^{\mathrm{c}}(c)| \le \left[\frac{9}{80}(B_F+1)c_{\mathrm{th}}\bar{\phi}_u + \frac{1}{2}C'_\rho S_3^\sharp\bar{\phi}_u^{(8)}\right]\hat{\phi}_k|c|/\varrho.$$

The extra depth floor makes the bracket at most $C_{\mathrm{var}}/4$. The torus increment differs from this constant infinite-lattice increment only through wrapping terms; the same torus count as in the source construction applies, now starting from an empty family at t_0.

The initial chain defect $\hat{\sigma}_{t_0}(s)(x) - \hat{\sigma}^{\mathrm{c}}_{t_0}(w(x))$ is zero. Apply the original local-increment recursion with this initial value and the bound $C_{\mathrm{var}}/4$ just proved. Its constant requirements are

$$B_\sigma \geq 3C_{\mathrm{var}} \geq 8B_\nu, \quad B_\sigma \geq 2C_{\mathcal{L}}B_\nu, \quad B_\sigma \geq 3C_{\mathrm{var}} + 8C_{\mathrm{tor}}C_{\mathcal{L}},$$

all among the original floors. The chain stays in $\hat{\mathscr{D}}_k$ and satisfies (S1). The initial value reads $w(x)$ alone, so the source-locality tail proof also starts with zero defect off the support of a source difference. The energy proof uses only these chain bounds, the local step and its normalizers, giving its stated conclusions for the restarted energies.

We next compare format data at equal coordinates. The depth floor and Prop. B.8 give $|\hat{x}_i/x_i - 1| \leq 1/64$. Hence

$$\tfrac{8}{9}\phi_i \leq \hat{\phi}_i \leq \tfrac{9}{8}\phi_i, \qquad \tfrac{8}{9}\tau_i \leq \hat{\tau}_i \leq \tfrac{9}{8}\tau_i, \qquad \mathfrak{D}_k(a)^\wedge \subset 2\hat{\mathscr{D}}_k(a).$$

Also $\mathscr{D}_k^\wedge \subset (9/8)\hat{\mathscr{D}}_k$. Both potentials have scalar norm at most $c_{\mathrm{cl}}/2$ and leg norm at most $(9/8)C^{\mathrm{s}}_\psi\phi_k \leq c_{\mathrm{cl}}/20$. Convexity preserves these bounds on the interpolation segment.

First vary the current coupling on $|x' - \hat{x}_k| \leq r_x\hat{x}_k/2$, holding the other format data fixed. The scalar origin depends on x' through $\tau = x'/\mathsf{t}_k$; its norm remains below $w_*/2$ because $|\tau - 1| \leq 3\sigma_\star/4$. The weighted part depends on $x'v' = -L^{K-k}w$ and is therefore fixed. The same is true of the linear source term; the ordinary gauge vertices are linear in x' and remain within their original bounds. The expansion is thus holomorphic on this whole disc. The segment from $\hat{x}_k$ to x_k has distance at least $3r_x\mathsf{t}_k/16$ from its boundary. Cauchy's estimate gives a change at most $14K_Mc_{\mathrm{cl}}X_k/r_x$, before extraction, and $C^{\mathrm{s}}_{\mathbf{G}}X_k$ for the explicit Gaussian term.

Next vary the families. For $n = k + 1 - j$, the four derivatives from (old leg, old scalar) to (new leg, new scalar), after extraction, are bounded respectively by

$$\frac{9}{2}\theta^{\mathrm{s}}_E(1 + c_{\mathrm{cs}})\omega_n, \quad \frac{15}{4}K^{\mathrm{s}}_\mu\frac{C^{\mathrm{s}}_\psi}{c_{\mathrm{cl}}}\phi_kL^{-(n-1)}, \quad C^{\mathrm{s}}_{\mathrm{dl}}\omega_n, \quad \frac{25}{27}L^{-(n-1)}.$$

These are Prop. B.4(b),(c), the hull count for the cell sums, and the source-radius floor

$$2K^\bullet_MK^{\mathrm{s}}_\Lambda\mathsf{s}_k(a) \leq 1/6.$$

The bounds apply to the maximum of the sums of the norms at the given coordinates, at zero on the torus, and at zero on $\mathbb{Z}^3$. We keep these three sizes together in the induction.

The remaining differences are forcing terms. There are five: the current coupling; the current mass member; earlier mass members and the coefficients μ_i of moved cell terms; the explicit scalar families; and the subtraction at zero source in the scalar extraction. The first two are bounded using

$$X_k \leq C^{\mathrm{s}}_x\phi_km_{k+1}, \qquad |\delta s_k| \leq \sqrt{2}(1 + S_{B'})\phi_km_{k+1}.$$

For the earlier mass terms, the discarded levels have $|\mu_i| \leq 3\bar{B}_M\phi_i$, while the later levels have $|\delta\mu_i| \leq (L+1)\sqrt{2}(1 + S_{B'})\phi_im_{k+1}$. Their ordinary reading is summed with ω_{k+1-i}. The additional cell terms involving $A^{\mathsf{n}}_i - A_i$ are summed separately: their lag factor is L^{k-i} and their mass variation contributes $L^i\|\mathsf{n}\|/r_M$, so their product is at most $3/(2L)$ on $\mathfrak{D}_k(a)$. These two sums give $C_{\mathrm{mm}}\phi_km_{k+1}$.

The discarded explicit scalar families have array bound E^{s}_0, and the later Gaussian differences have bound $C^{\mathrm{s}}_{\mathbf{G}}X_{j-1}$. Their localization on the fixed radius has size at most $C^\natural\phi_km_{k+1}$. The zero-source scalar subtraction has size at most $(2A_{\mathrm{LM}} + C_{\mathrm{LM}})\phi(\mathsf{t}_k)m_{k+1}$ by Prop. B.8. Wrapping cell discrepancies are included in the differences of the zero-source families. Altogether these terms cost $C^{\mathrm{fc}}_\psi m_{k+1}2C^{\mathrm{s}}_w\phi_k$ for the legs and $C^{\mathrm{fc}}_0m_{k+1}2C^{\mathrm{s}}_w\phi_k$ for the scalar part.

Induct with the slightly stronger bounds

$$\|\delta E^{(j)}_{\geq 2}\| \leq m_j(A_\psi\, 2C^{\mathrm{s}}_w\phi_{j-1} + A'_\psi\phi_{j-1}C'_{\mathrm{lin}}\tau_{j-1}),$$
$$\|\delta F^{(j)}_0\| \leq m_j(A^{\mathrm{s}}_0\, 2C^{\mathrm{s}}_w\phi_{j-1} + A_\tau C'_{\mathrm{lin}}\tau_{j-1}).$$

Before t_0 these follow from $m_j \geq 1$ and the ordinary source sizes. At the next step use $m_j \leq L^{c_{\mathrm{lm}}n}m_{k+1}$ and

$$\omega_n\phi_{j-1} \leq C_{\mathrm{pc}}L^{-n/4}(1+n)^{\mathfrak{p}_0}\phi_k, \quad \phi_{j-1} \leq (2L^{-1/2})^n\phi_k, \quad \tau_{j-1} \leq \sqrt{2}L^{-n/2}\tau_k.$$

The four parts of the induction close with the following coefficients: the leg ϕ part receives at most $(27/32)A_\psi$ from itself and $A_\psi/32$ from the scalar part; its forcing is at most $A_\psi/8$. The leg $\phi\tau$ part receives at most $(7/20+1/2)A'_\psi$. The scalar ϕ part receives at most

$$\tfrac{7}{10}A^{\mathrm{s}}_0 + \tfrac{3}{2}C^{\mathrm{s}}_{\mathrm{dl}}C_{\mathrm{pc}}S_{1/4}A_\psi + C^{\mathrm{fc}}_0 \leq A^{\mathrm{s}}_0,$$

and its τ part at most $3A_\tau/4$. Here the numerical estimates are $2L^{c_{\mathrm{lm}}-1/2} \leq 7/10$, $2L^{c_{\mathrm{lm}}-3/2} \leq 1/18$ and $\sqrt{2}L^{-1/2} \leq 2/5$; the small scalar-to-leg terms are covered by the additional depth conditions. Since $C'_{\mathrm{lin}}\tau_k < 3K_Mc_{\mathrm{cl}}/8$ on $\mathfrak{D}_k(a)$, this proves (S2) with $A^\sharp_\psi$.

Mass extraction is a fixed linear rule on the localized kernels. At equal n the pieces, walks and filing cells agree in the two flows. Applying that rule to their difference and using (S2) gives its ordinary increment bound multiplied by $A^\sharp_\psi m_{k+1}$. The line from $\mathfrak{c} \in \mathscr{D}_k$ towards $\mathfrak{c}' \in \mathfrak{D}_k$ stays in $\mathfrak{D}_k$ for the normalized complex parameter of modulus below $1/2$. Schwarz's lemma costs the factor two in (S3). Comparing with zero gives its second bound.

To compare the chains fix a bare site x_0, put $c = w(x_0)$, and set

$$\Delta_k(x_0) = [\sigma_k(s) - \hat\sigma_k(s)](x_0) - [\sigma^{\mathrm{c}}_k(c) - \hat\sigma^{\mathrm{c}}_k(c)],$$
$$g_i(c') = \sigma^{\mathrm{c}}_i(c') - \hat\sigma^{\mathrm{c}}_i(c').$$

We prove

$$|\Delta_k(x_0)| \leq A_D\phi_k\Xi_k(s)m^*_k + T^\Delta_k, \qquad T^\Delta_k = \frac{112}{31}C'_\Delta\sum_{j\leq k}L^{k-j}\phi_j\mathrm{tor}_j\|w\|_\infty/\varrho.$$

The ordinary source-chain estimate gives the initial value because the restarted chain agrees pointwise with its constant chain at t_0. It is bounded by the displayed right side since $A_D \geq C_{\mathrm{var}}$ and $2C_{\mathrm{tor}} \leq C'_\Delta$.

Let $A = \mathfrak{c}_k(s)^\wedge$ and $A' = \mathfrak{c}^{\mathrm{c}}_k(c)^\wedge$. Let δ_g have entries $g_i(w(x))$, let δ' have entries $g_i(c)$, and let δ_Δ have entries $\Delta_i(x)$, in their mass slots. All other entries are zero. The difference of increment differences in the recursion for Δ_{k+1} is

$$\begin{aligned} &d\nu(\mathfrak{c}_k(s)) - d\nu(\mathfrak{c}^{\mathrm{c}}_k(c)) \\ &+ \{[\hat\nu(A) - \hat\nu(A-\delta_g)] - [\hat\nu(A') - \hat\nu(A'-\delta_g)]\} \\ &+ [\hat\nu(A'-\delta') - \hat\nu(A'-\delta_g)] + [\hat\nu(A-\delta_g) - \hat\nu(A-\delta_g-\delta_\Delta)]. \end{aligned}$$

The first line is controlled by (S3). The middle braces are a mixed difference and must not be bounded as two unrelated first differences. The constant-label estimate and Cauchy's estimate in c' give

$$\|\delta_g\|_\infty \leq 4(1+S_{B'})m^*_{k+1}/B_\sigma,$$
$$\|\delta_g - \delta'\|_{k,y} \leq C_{\mathrm{tl}}(1+S_{B'})\Xi_{k+1}(s)m^*_{k+1}/B_\sigma,$$
$$\|\delta_\Delta\|_{k,y} \leq \tfrac{9}{8}\max_{t_0\leq i\leq k}|\Delta_i|/(B_\sigma\phi_i).$$

For the smallness needed below, (U7) gives the sharper bound

$$\|\delta_g\|_\infty \le \frac{4}{B_\sigma}(C_\Pi \bar{\phi}_u + S_{B'} L^{-u/4}) \le \frac{1}{16}.$$

The initial coordinate distance is at most $C_c \Xi_{k+1}(s)$ apart from the torus term. All the shifted coordinates above lie in $(19/16)\hat{\mathscr{D}}_k$, while the actual restarted chains lie in $\hat{\mathscr{D}}_k$.

For a localized increment piece bounded by M_0, let T_1 be its normalized size in direction $A - A'$ and $T_2 \le 1/16$ that in direction δ_g. If $T_1 \le 1/16$, the bidisc of radii $7/(16T_1)$ and $7/(16T_2)$ about A' stays in $2\hat{\mathscr{D}}_k$; the mixed Cauchy estimate is at most $2^{11} M_0 T_1 T_2$. If $T_1 > 1/16$, the two separate Schwarz estimates in the second direction give the same upper bound. Summing pieces therefore bounds the braces by

$$\tfrac{9}{7} C^{\mathrm{mix}} B_\nu \phi_{k+1} \|\mathfrak{c}_k(s) - \mathfrak{c}_k^{\mathrm{c}}(c)\|_{k,y} \|\delta_g\|_\infty, \qquad C^{\mathrm{mix}} = 2^{10}.$$

The last two brackets are bounded by $(9/7) B_\nu \phi_{k+1}$ times their coordinate distances. This gives the recursion

$$|\Delta_{k+1}(x_0)| \le \left[L^{c_{\mathrm{lm}}} \left(\tfrac{8}{3} L^{-1/2} + \tfrac{27}{112} L^{-1} \right) A_D + C_D^{\mathrm{fc}} \right] \phi_{k+1} \Xi_{k+1}(s) m_{k+1}^* + T_{k+1}^\Delta.$$

It closes because the bracket is at most A_D. The torus recursion closes with factor $112/31$: its feedback is at most $81/112$, by $B_\sigma \ge 2 C_{\mathcal{L}} B_\nu$. The torus scale sum gives $T_k^\Delta \le C_\Delta \phi_k \mathrm{tor}_k \|w\|_\infty / \varrho$. This proves (S4), including its last estimate by $\Xi_k(s) \le \lambda_k \pi_k N(s)$.

For (S5), split a family difference along the chains into its equal-coordinate difference and the change of the restarted family between the two chain coordinates. Their normalized mass distance is at most

$$T_k(s) \le \frac{4(1 + S_{B'}) + (9/4) A_D}{B_\sigma} m_k^* + \frac{9 C_\Delta}{2 B_\sigma} \mathrm{tor}_{k-1}.$$

The first coordinate lies in $(9/8)\hat{\mathscr{D}}_{k-1}$, the second in $\hat{\mathscr{D}}_{k-1}$. Schwarz's lemma on the line between them costs at most $(16/7) T_k(s)$ times the source size. It applies also to the sum of norms of summands: first apply it to a finite sum tested against arbitrary norm-one linear functionals, then take the supremum and the absolutely convergent limit. This proves the family bounds with A_{ch}.

For the zeroth moment the same kernel $A_k(y,z)$ occurs in both flows at the trivial pair. Thus

$$\delta^{\mathrm{ch}} \varpi_k(y) = \sum_z A_k(y,z) [\delta^{\mathrm{ch}} \nu_k(z) - \delta^{\mathrm{ch}} \nu_k(y)] + \delta^{\mathrm{ch}} r_k(y).$$

Freeze the constant label at $w(y)$. The four-bracket estimate above bounds the spatial variation of the increments by $C \phi_k \Xi_k m_k^*$ times $1 + |y - z|_k$, plus the torus term. The weighted kernel sum is at most C_A^{m}. The change of the kernel with n is at most $C_A^{\mathrm{n}} L^k \|\mathsf{n}\| / r_M$ in row-sum norm; it acts on the same increment difference. The remaining constant-label increment is bounded by $2\sqrt{2}(2L^{1/2} + L S_{B'}) \phi_k m_k^*$, using $\delta \mu_k(c) = \delta s_k(c) - L \delta s_{k-1}(c)$. Finally the two torus remainders cost $3 C_r^{\mathrm{t}} \phi_k \mathrm{tor}_k$. The explicit constant C_ϖ^δ below is the sum of these bounds.

The scalar comparison was made polymer by polymer at rate κ_+, apart from the zero-source subtraction, which has rate $\kappa + 1$ by Prop. B.8. The same induction and the chain Schwarz estimate therefore prove the scalar part of (S6). For legs, count at most $8M^3 e^{d_k(X)}$ pointers before extraction. The extra decay pays their count. Each source mass piece is pointed wholly at its own cell, and the local subtraction adds at most $K_\mu^{\mathrm{s}} - 1$ times the largest pre-extraction size at the three coordinates used in the induction. Hence $8M^3 + K_\mu^{\mathrm{s}} - 1 \le 8M^3 K_\mu^{\mathrm{s}}$ gives the same factor after extraction. Apply Schwarz's lemma in this summed norm for the chain comparison. The same argument for one flow proves the summed-pointer assertion in (S1). The earlier coupling Cauchy estimate proves the last Gaussian bound. □

Constants for the source comparison. Continue the choices in Def. B.7 in the following order. Set

$$S_{B'} = \bigl(7B_5 + 6(\bar{B} + 4C_w)B_F\bigr) \sup_{v \ge 0} L^{-v/4}(1 + v\log L)^{\mathfrak{p}_0}.$$

Then $\phi(\mathsf{t}_i)B'_i \le S_{B'}L^{-s/4}$ for $i \ge t_0$. Define

$$\begin{aligned}
C^{\mathrm{s}}_{\psi} &= C^{\mathrm{s}}_V h_F^2 A_1 + \tfrac{9}{8}c'_{\psi}c_{\mathsf{S}}h_F^2(\bar{B}_M + 2B_{\sigma}) + \tfrac{9}{8}c'_{\psi}C^{\mathrm{cs}}_0 h_F^2 \\
&\quad + \tfrac{9}{4}(1 + c_{\mathrm{cs}})c'_{\psi}C_{\mathrm{pc}}S_{1/4}C^{\mathrm{s}}_w, \\
C^{\mathrm{s}}_{\mathrm{dl}} &= \tfrac{25}{2}K^{\bullet}_M c'_{\psi}(1 + c_{\mathrm{cs}}), \qquad C^{\mathrm{s}}_{\mathbf{G}} = \tfrac{32}{3}(C^{v}_{\mathbf{G}} + 2r_x^{-1}), \\
C^{\mathrm{s}}_x &= \sqrt{2}[\bar{B} + 4C_w + C_x(LS_{\mathrm{fl}} + 1)], \\
C^{\natural} &= \tfrac{13\sqrt{2}}{12}K^{\mathrm{s}}_{\Lambda}\mathsf{C}_{\mathcal{A}}A_1\alpha_0^{-1}(C^{\mathrm{s}}_{\mathbf{G}} + L^{1/12}E^{\mathrm{s}}_0), \\
C_{\mathrm{mm}} &= \tfrac{9}{8}(1 + c_{\mathrm{cs}})c'_{\psi}c_{\mathsf{S}}h_F^2 L^{1/2}C_{\mathrm{pc}}S_{1/4}[6\bar{B}_M + 3B_{\sigma} + (L+2)\sqrt{2}(1 + S_{B'})] \\
&\quad + \tfrac{9}{8}c'_{\psi}C^{\mathrm{cs}}_0 h_F^2[L^{1/12} + (L+1)\sqrt{2}(1 + S_{B'})/(3\bar{B}_M)], \\
C^{\mathrm{fc}}_{\psi} &= \frac{K^{\mathrm{s}}_{\mu}K^{\bullet}_M}{C^{\mathrm{s}}_w}\left[\frac{7c_{\mathrm{cl}}C^{\mathrm{s}}_x}{r_x} + \frac{25\sqrt{2}}{8}c'_{\psi}c_{\mathsf{S}}h_F^2(1 + S_{B'}) + 3C_{\mathrm{mm}} + \frac{45}{2}\frac{C^{\mathrm{s}}_{\psi}}{c_{\mathrm{cl}}}C^{\natural}\right], \\
C^{\mathrm{fc}}_0 &= \frac{1}{C^{\mathrm{s}}_w}\left[K^{\bullet}_M\left(\frac{14c_{\mathrm{cl}}C^{\mathrm{s}}_x}{r_x} + \frac{25\sqrt{2}}{4}c'_{\psi}c_{\mathsf{S}}h_F^2(1 + S_{B'}) + 6C_{\mathrm{mm}} + 6C^{\natural}\right) + \frac{2A_{\mathrm{LM}} + C_{\mathrm{LM}}}{\sqrt{2}}\right].
\end{aligned}$$

The term C^{cs}_0 includes the separately summed cell contribution:

$$C^{\mathrm{cs}}_0 = \frac{6C_{\mathrm{hl}}M^3C^{\mathrm{w}}_{\mathrm{bg}}\bar{B}_M}{L(1 - 2L^{-1/2})}.$$

Next put

$$\begin{aligned}
A_{\psi} &= \max\{1, 8C^{\mathrm{fc}}_{\psi}, C^{\mathrm{fc}}_0\}, & A_{\tau} &= 4, \\
A'_{\psi} &= 16K^{\mathrm{s}}_{\mu}C^{\mathrm{s}}_{\psi}/c_{\mathrm{cl}}, & C^{\mathrm{s}}_A &= 5C^{\mathrm{s}}_{\mathrm{dl}}C_{\mathrm{pc}}S_{1/4} + 4, \\
A^{\mathrm{s}}_0 &= \max\{1, 5C^{\mathrm{s}}_{\mathrm{dl}}C_{\mathrm{pc}}S_{1/4}A_{\psi} + 4C^{\mathrm{fc}}_0\}, & A^{\sharp}_{\psi} &= A_{\psi} + \tfrac{3}{16}A'_{\psi}K_M c_{\mathrm{cl}}/C^{\mathrm{s}}_w.
\end{aligned}$$

These choices give $A^{\mathrm{s}}_0 \le C^{\mathrm{s}}_A A_{\psi}$ and all the closing inequalities in the four-part induction.

For the chain comparison set $C_c = C_{\mathrm{tl}} + 41M$, $C^{\mathrm{mix}} = 2^{10}$ and

$$\begin{aligned}
C^{\mathrm{fc}}_D &= 2A^{\sharp}_{\psi}B_{\nu}C_c + \tfrac{9}{7}\frac{B_{\nu}}{B_{\sigma}}(1 + S_{B'})(4C^{\mathrm{mix}}C_c + C_{\mathrm{tl}}), \\
A_D &= \max\{C_{\mathrm{var}}, 10C^{\mathrm{fc}}_D\}, \\
C'_{\Delta} &= 4C_{\mathrm{tor}} + (1 + C_{\Pi})\frac{B_{\nu}}{B_{\sigma}}C_{\mathrm{tor}}C_{\mathcal{L}}[4A^{\sharp}_{\psi} + \tfrac{72}{7}C^{\mathrm{mix}}(1 + S_{B'})/B_{\sigma}], \\
C_{\Delta} &= 4C_{\mathcal{L}}C'_{\Delta}, \\
A_{\mathrm{ch}} &= \max\{A^{\sharp}_{\psi}, A^{\mathrm{s}}_0, 4\} + \frac{18}{7B_{\sigma}}[4(1 + S_{B'}) + \tfrac{9}{4}A_D + \tfrac{9}{2}C_{\Delta}].
\end{aligned}$$

For the zeroth-moment comparison take

$$\begin{aligned}
C^{\mathrm{m}}_A &= \nu_P A_{\kappa}^2 C_H^2 \sup_{d \ge 0}(1 + c_{\mathcal{D}}M(1 + d))e^{-\kappa d}, \\
S_{\alpha} &= e^{3\alpha_D}\left(\frac{1 + e^{-\alpha_D}}{1 - e^{-\alpha_D}}\right)^3, \\
C^{\mathrm{n}}_A &= 2\nu_P S_{\alpha}^2 C_{\mathrm{bg}}(C_H + C_{\mathrm{bg}}), \qquad C^{\mathrm{t}}_r = C + C_A(3\bar{B}_M + B_{\nu}).
\end{aligned}$$

Here C is the original bound for the remainder r_k in the zeroth-moment formula, and C_A its kernel constant. The first constant bounds $\sum_z \|A_k(y,z)\|(1+|y-z|_k)$: its walk pieces have weighted sum $\nu_P A_\kappa^2 C_H^2$, and their polymers contain both endpoints. The second bounds the two sums of the exponential background decay. Indeed a bare site is within three path units of the origin of its unit cube, and $\sum_{v\in\mathbb{Z}^3} e^{-\alpha_D|v|_1} = ((1+e^{-\alpha_D})/(1-e^{-\alpha_D}))^3$.

To verify the third bound, write the mass dependence of each kernel product as the change of the first leg times the second, plus the first unchanged leg times the change of the second. The background-map difference has kernel bound $C_{\mathrm{bg}}(L^k\|\mathsf{n}\|/r_M)L^{-k}e^{-\alpha_D d}$. Summing one leg over bare places and the other over block sites, with the factor L^{-k} in the mass form, gives

$$\sup_y \sum_z \|(A_k^{\mathsf{n}} - A_k)(y,z)\| \le C_A^{\mathrm{n}} L^k \|\mathsf{n}\|/r_M.$$

This argument uses $L^k\|\mathsf{n}\|/r_M < 2$ and the two background decay sums above. On a torus the sums are bounded by the corresponding sums over all translates on $\mathbb{Z}^3$.

With these constants one may take

$$\begin{aligned} C_\varpi^\delta = {} & C_A^{\mathrm{m}}(2C_D^{\mathrm{fc}} + A_D + 8C'_\Delta + 2C_\Delta + 20C_{\mathrm{tor}}) \\ & + C_A^{\mathrm{n}}(2C_D^{\mathrm{fc}} + A_D + 6L^{1/2} + 6LS_{B'} + 8C'_\Delta + 2C_\Delta + 18C_{\mathrm{tor}}) + 3C_r^{\mathrm{t}}. \end{aligned}$$

Enlarge the depth floor u_* in Def. B.7 so that the following four further conditions hold for every $u \ge u_*$:

$$\begin{aligned} \tfrac{9}{8} c_{\mathrm{th}} C_\psi^{\mathrm{s}} \bar\phi_u &\le c_{\mathrm{cl}}/20, \\ \tfrac{9}{80}(B_F+1)c_{\mathrm{th}}\bar\phi_u + \tfrac{1}{2}C'_\rho S_3^\sharp \bar\phi_u^{(8)} &\le C_{\mathrm{var}}/4, \\ 3K_\mu^{\mathrm{s}}(C_\psi^{\mathrm{s}}/c_{\mathrm{cl}})c_{\mathrm{th}}C_A^{\mathrm{s}}\bar\phi_u &\le 1/32, \\ 10C_{\mathrm{dl}}^{\mathrm{s}} C_{\mathrm{pc}} S_{1/4} K_\mu^{\mathrm{s}}(C_\psi^{\mathrm{s}}/c_{\mathrm{cl}})c_{\mathrm{th}}\bar\phi_u &\le 1, \\ 4(C_\Pi \bar\phi_u + S_{B'}L^{-u/4}) &\le B_\sigma/16. \end{aligned}$$

The first line admits the potential interpolation; the second bounds the restarted constant chain; the next two close the scalar-to-leg estimates; and the last keeps the translated chain coordinates inside $(19/16)\hat{\mathscr{D}}_k$. All constants on their left have already been fixed, so this enlargement is finite and independent of K, the side, and the source.

B.5. **Differences along the source chains.** We record the estimates in the norms used to add differences to an RG flow. For a family with legs, its norm is the sum of the norms of its filed summands. The pointed norm sums these further over the pointers of each polymer. For scalar families use

$$|G_0|_j = \sup_{X,z} e^{(\kappa+1)d_j(X)} \sup |G_0(X;\cdot;z)|.$$

All pair radii in this subsection are the fixed radii α_0, α_1.

Corollary B.11 (Differences along the source chains)**.** Let $u \ge u_*$, $\zeta \in \mathbb{R}$, $|\zeta| \le 2B_F+1$, and $s = (w,J) \in \mathscr{O}_{t_1}$ with $\|w\|_\sharp < 4\varrho$. Use the restart of Def. B.9. Let

$$\delta^{\mathrm{ch}} E^{(j)}(s) = E^{(j)}(\cdot;\mathfrak{c}_{j-1}(s)) - \hat E^{(j)}(\cdot;\hat{\mathfrak{c}}_{j-1}(s)),$$

where the second term is zero for $j \le t_0$. Let $r_j^\psi(s)$ be the largest of the following four quantities: the norm of $\delta^{\mathrm{ch}} E_{\ge 2}^{(j)}(s)$; the norms of the zero-source difference on the torus and on $\mathbb{Z}^3$; and $c_{\mathsf{S}} h_F^2 |\mu_j - \hat\mu_j|$, with $\hat\mu_j = 0$ for $j \le t_0$.

The constants in Def. B.7 may be supplemented by

$$A_r = \max\left\{2C_w^{\mathrm{s}}A_{\mathrm{ch}}, 4\sqrt{2}L^{1/2}(2L^{1/2}+LS_{B'})c_{\mathsf{S}}h_F^2\right\},$$
$$A_r^{\mathrm{ch}} = \max\{2C_w^{\mathrm{s}}, 6L^{1/2}c_{\mathsf{S}}h_F^2\bar{B}_M\}.$$

Then the following hold.

(a) For $t_0 < j \le t_1$,

$$\|\delta^{\mathrm{ch}}E_{\ge 2}^{(j)}(s)\|_j \le 2C_w^{\mathrm{s}}A_{\mathrm{ch}}(m_j^* + \mathrm{tor}_{j-1})\phi_{j-1},$$
$$\|\delta^{\mathrm{ch}}E_{\ge 2}^{(j)}(s)\|_j^{\mathrm{pt}} \le 16M^3C_w^{\mathrm{s}}A_{\mathrm{ch}}(m_j^* + \mathrm{tor}_{j-1})\phi_{j-1},$$
$$r_j^{\psi}(s) \le A_r(m_j^* + \mathrm{tor}_{j-1})\phi_{j-1}.$$

(b) At these levels the calibration and fine-mass normalizers are identical in the two flows. Consequently

$$\delta^{\mathrm{ch}}E_0^{(j)}(s) = \delta^{\mathrm{ch}}F_0^{(j)}(s) + \mathbf{G}^{(j)} - \hat{\mathbf{G}}^{(j)}$$

and

$$|\delta^{\mathrm{ch}}E_0^{(j)}(s)|_j \le A_{\mathrm{ch}}(m_j^* + \mathrm{tor}_{j-1})(2C_w^{\mathrm{s}} + 8\alpha_0\varrho C'_{\mathrm{lin}})\phi_{j-1} + C_{\mathbf{G}}^{\mathrm{s}}C_x^{\mathrm{s}}m_j\phi_{j-1}.$$

Subtracting the value at the trivial pair multiplies this bound by at most two.

(c) For $j \le t_0$ the difference is the ordinary activity and

$$\|E_{\ge 2}^{(j)}\|_j \le 2C_w^{\mathrm{s}}\phi_{j-1}, \qquad \|E_{\ge 2}^{(j)}\|_j^{\mathrm{pt}} \le 16M^3C_w^{\mathrm{s}}\phi_{j-1}, \qquad r_j^{\psi}(s) \le A_r^{\mathrm{ch}}\phi_{j-1}.$$

Moreover $|E_0^{(j)}|_j \le E_0^{\mathrm{s}}$; subtracting its value at the trivial pair gives the bound $2E_0^{\mathrm{s}}$.

(d) For $t_0 < j \le t_1$, the rates satisfy

$$m_j \le m_j^* = L^{-c_{\mathrm{lm}}(j-t_0)} + \phi(\mathsf{t}_{j-1})\Pi_{j-1}^*, \qquad c_{\mathrm{lm}} \ge c_{\#},$$

where $\phi(\mathsf{t}_{j-1}) \le \bar{\phi}_u$ and

$$\Pi_{j-1}^* = 2\sup_{|c|<8\varrho}|\pi(\zeta;c)| + C_{\mathrm{LM}}[(j-t_0)\varepsilon_s + L^{-s}/\mathsf{b}].$$

For the ultraviolet start,

$$\sup_{|c|<8\varrho}|\pi(F^{\mathrm{UV}};c)| \le \frac{64}{63}B_5(L^{-t_0} + 3L^{-(s-u)}) + 3(\bar{B}+4C_w)B_FL^{-(s+1)}/\mathsf{b}.$$

For $j \le k \le t_1$, $m_j^* \le L^{c_{\mathrm{lm}}(k-j)}m_k^*$. On every admitted torus and for $j \le t_1$, $\mathrm{tor}_j \le e^{-c_{\mathrm{t}}L^{u+1}}$; on $\mathbb{Z}^3$ it vanishes.

■□

Proof. Part (a)'s first estimate is Prop. B.10(S5) at $a = \alpha_0$; the second is (S6). The difference is formed summand by summand, with the same cells and pointers, because the filing and extraction rules are linear. The two zero-source norms are at most $2C_w^{\mathrm{s}}A_\psi^\sharp m_j\phi_{j-1}$ by (S2), and $A_\psi^\sharp \le A_{\mathrm{ch}}$.

For the remaining entry of r_j^ψ, use

$$\mu_j - \hat{\mu}_j = \delta s_j - L\delta s_{j-1}.$$

The mass bound of Prop. B.8, including its initial-level bound, gives $\mathsf{t}_i|\delta s_i| \le \Pi_i^\flat + B'_i$. Since $\mathsf{t}_{j-1} = L\mathsf{t}_j$, $\Pi_i^\flat$ grows and B'_i decreases,

$$|\mu_j - \hat{\mu}_j| \le 2(\Pi_j^\flat + B'_{j-1})/\mathsf{t}_j.$$

Now $1/\mathsf{t}_j \le \phi(\mathsf{t}_j)^2$,

$$\phi(\mathsf{t}_j)\Pi^\flat_j \le 2L^{1/2}\phi(\mathsf{t}_{j-1})\Pi^\flat_{j-1}, \qquad \phi(\mathsf{t}_j)B'_{j-1} \le L^{1/2}S_{B'}L^{-s/4} \le LS_{B'}\Lambda^{\mathrm{s}}_j.$$

Here $\Pi^\flat_j \le 2\Pi^\flat_{j-1}$ follows from $\Pi^\flat_{j-1} \ge C_{\mathrm{LM}}\varepsilon_s$, while $j - t_0 \le s$ and $c_{\mathrm{lm}} \le 1/12$ give the last inequality. Finally $\phi(\mathsf{t}_j) \le \sqrt{2}\phi_j$ and $\phi_j \le 2L^{1/2}\phi_{j-1}$. Combining these inequalities gives

$$c_{\mathsf{S}}h_F^2|\mu_j - \hat{\mu}_j| \le 4\sqrt{2}L^{1/2}(2L^{1/2} + LS_{B'})c_{\mathsf{S}}h_F^2 m_j\phi_{j-1},$$

as required.

For (b), the scalar part is $F_0^{(j)} + \mathbf{N}^{\mathrm{un},(j)} + \mathbf{N}^{\mathrm{F},(j)} - \mathbf{c}_j + \mathbf{G}^{(j)}$. The calibration and fine mass determine $\mathbf{N}^{\mathrm{un},(j)}$ and $\mathbf{c}_j$; $\mathbf{N}^{\mathrm{F},(j)}$ depends on the fine mass matrix $m_0 - a_K\hat{J}$, which the two chains share. These three terms cancel. Apply (S5) and (S6), using $\tau_{j-1}(w;\alpha_0) \le 8\alpha_0\varrho\phi_{j-1}$, and then the estimates

$$|\mathbf{G}^{(j)} - \hat{\mathbf{G}}^{(j)}|_j \le C^{\mathrm{s}}_{\mathbf{G}}X_{j-1}, \qquad X_{j-1} \le C^{\mathrm{s}}_x\phi_{j-1}m_j$$

from the proof of Prop. B.10. Evaluation at the trivial pair is bounded by the same supremum.

For (c), the restarted family is absent. The ordinary source step bounds its summand norms by $2C^{\mathrm{s}}_w\phi_{j-1}$. The pointed count multiplies this by $8M^3$. The same bound holds at zero sources on both lattices, while

$$c_{\mathsf{S}}h_F^2|\mu_j| \le 3c_{\mathsf{S}}h_F^2\bar{B}_M\phi_j^2 \le 6L^{1/2}c_{\mathsf{S}}h_F^2\bar{B}_M\phi_{j-1}.$$

We used $|\mu_j| \le 3\bar{B}_M/x_j$, $x_j^{-1} \le \phi_j^2$, and $\phi_j \le 1$. The scalar bound is the scalar-source estimate of [QCDM1, §11], applied at $\mathfrak{c}_{j-1}(s) \in \mathscr{D}_{j-1}(\alpha_0)$.

For (d), $\Pi^\flat \le \Pi^*$ and the choice of c_{lm} gives $c_{\mathrm{lm}} \ge c_\#$. Also $\mathsf{t}_{j-1} \ge L^u$, so $\phi(\mathsf{t}_{j-1}) \le \bar{\phi}_u$. To estimate the initial mismatch, use the formula in Prop. B.8(A), $|\mathsf{b}/(\mathsf{b}-c)| \le 64/63$, and

$$|F^{\mathrm{UV}} - F_{<t_0}(t_1;\mathfrak{m}_{t_1})| \le B_5(L^{-t_0} + 3L^{-(s-u)}),$$

which is Prop. B.2(b) with $i_0 = t_0$, $k = t_1$. The geometric part of m_j^* changes by $L^{c_{\mathrm{lm}}(k-j)}$ between levels j and k; its floor increases with the level, since ϕ decreases on $[L^u,\infty)$ and Π^* increases. This proves the comparison of rates. Finally $\ell/\lambda_j = \ell L^{K-j} \ge \ell L^u \ge 2L^{u+1}$, since an admitted side is at least $2L$. □

B.6. **The restart used for two cutoffs.** The abstract restart now gives the concrete measure to which the two cutoff constructions will be compared. Its initial mass retains the ultraviolet part of the mass renormalization.

Lemma B.12 (The comparison restart)**.** Use the data of Def. 11.1, the condition (k0), and $u \ge u_*$. For either datum $\bullet \in \{\mathrm{A},\mathrm{B}\}$ write $t_0 = t_0^\bullet$, $t_1 = t_1^\bullet$ and set $\zeta = F^{\mathrm{UV}} \in [0, B_F]$. Then:

(a) At a source $s = (w, J)$ with $\|w\|_\infty < 8\varrho$,

$$\hat{x}_{t_0} = L^s\mathsf{b}, \qquad \hat{s}_{t_0} + \hat{\sigma}_{t_0}(s) = \frac{F^{\mathrm{UV}}}{L^s(\mathsf{b} - w)} =: \hat{\Sigma}_w.$$

Put $\mathsf{c}_w = L^s(\mathsf{b} - w)$, and let $W \in \mathcal{Y} = \mathcal{V}_{t_0}$. Define the weighted background action by

$$\mathsf{wab}(\mathsf{c}, W) = A^{\eta_{t_0}}(\mathsf{c}, \mathsf{U}_{t_0}(W)).$$

The level-t_0 term of the restart is

$$\chi_{t_0}(T_\ell)(W)\exp\big[-\mathsf{wab}(\mathsf{c}_w, W) - \langle\bar{\Psi}, \mathbb{D}_{t_0}(\mathsf{U}_{t_0}(W); m_0 - a_K\hat{J})\Psi\rangle - L^{-t_0}\mathsf{S}_1(\hat{\Sigma}_w\psi^{\mathrm{bg}})\big]\,dW.$$

Here dW is product Haar probability measure. No activities, numbers or normalizers of levels $\le t_0$ occur. This is the initial density $\hat{d}_{t_0}(s)$ of Def. 11.2, restricted by the density test $\chi_{t_0}(T_\ell)$.

For $s \in \mathscr{O}_{t_0}$ its density is a bounded Borel function on $\mathcal{Y}$ with values in $\mathrm{Gr}^{\mathrm{ev}}_{t_0}$. It is zero off the open set $\mathcal{V}^{\chi}_{t_0}$ of configurations passing the test and continuous on that set. At each such W, all three terms of the exponent have one value on the minimal orbit $\mathcal{C}(\mathbf{B}_{t_0}, W)$.

(b) The subsequent steps along $h_\emptyset$ use the calibrations, tests, backgrounds, reference measures and tower of the original datum. Their real numbers $(\hat{x}_i, \hat{s}_i, \hat{e}_i, \hat{b}_i, \hat{\mu}_i)$ are read from the new zero-source activity on $\mathbb{Z}^3$ at each step. The chain is the hatted chain of Def. B.9; the increments use $\hat{\mu}_{k+1} = \hat{\nu}_{k+1}(0)$. The extracted number is

$$\hat{E}_k(s) = \sum_{t_0 < j \le k} \left(\hat{e}_j n_j + \sum_{X \in \mathcal{D}_j} \hat{e}_j(X; s) \right).$$

Thus $\hat{E}_k(s) = \hat{c}_k(s)$ in Prop. B.13, and at $k = t_1$ it is $\hat{c}^{\bullet}(s)$ in Def. 11.2.

(c) At zero sources the same local start and steps define the restart on the torus and on $\mathbb{Z}^3$ of spacing a_K. Its extracted numbers do not depend on ℓ. On every polymer which does not wrap, the torus activity is the part of the infinite-lattice activity which fits, with the same pointers under the local chart. Both families satisfy the symmetry, locality, holomorphy and norm assertions of Prop. B.5 and Prop. B.10(S1), in their respective norms.

■

Proof. At $c = 0$ the definitions give $\hat{s}_{t_0} = F^{\mathrm{UV}}/(L^s \mathsf{b})$ and $\hat{\sigma}_{t_0}(s) = \hat{\Sigma}_w - \hat{s}_{t_0}$. Also

$$L^{t_0} \hat{x}_{t_0} - a_K^{-1} w = L^K(\mathsf{b} - w) = L^{t_0} \mathsf{c}_w.$$

The weighted background action is linear in its weight, read at the barycentres of the bare plaquettes, and $A^{\eta_{t_0}} = L^{t_0} A$. Hence the action in Def. B.9 is $\mathsf{wab}(\mathsf{c}_w, W)$. The history $h_\emptyset$ has no stored fluctuations; its free coordinates are precisely the level-t_0 configurations. Its free measure is therefore product Haar probability measure. The definition of the source restart now gives the displayed density and absence of earlier members.

We check its measurability and bounds, since later integral identities use this actual density. The tested-field lemma and the background construction used in Prop. B.5 give, near every $W_0 \in \mathcal{V}^{\chi}_{t_0}$, a continuous representative

$$\Upsilon(W) = U^{\circ}_{\mathbf{B}_{t_0}}(W) \in \mathcal{C}(\mathbf{B}_{t_0}, W).$$

The latter set is one orbit of the residual gauge group, whose elements are the identity at the level-t_0 sites. The weighted action is constant on this orbit, also for complex weights. Both the fine mass matrix $m_0 - a_K \hat{J}$ and the scalar field $\hat{\Sigma}_w$ commute with gauge transformations. Covariance of the tower and background maps therefore makes their two quadratic forms constant on the same orbit: the action on the external level-t_0 generators is trivial. All three forms can consequently be evaluated at $\Upsilon(W)$.

The regularity and source bounds needed for these forms hold on their stated domains. Indeed every minimizing background over a tested W satisfies

$$|\partial U(p) - 1| < \alpha_0 L^{-2t_0} \le \epsilon_* L^{-2(t_0+1)},$$

because $\alpha_0 \le \epsilon_* L^{-2}$. Thus it satisfies $(\mathrm{Reg}_{t_0})[\epsilon_*]$ on $h_\emptyset$. For $s \in \mathscr{O}_{t_0}$ and $\mathsf{n} = -a_K J$,

$$L^{t_0+1} \|\mathsf{n}\| < r_M, \qquad \sup_x \|L^{t_0-K} J(x)\| < r_M/L \le \varrho_* \le \varrho_\sharp.$$

The matrix-source tower estimate in [QCDM1, §7] now applies at $j = t_0$, $k = t_0 + 1 \le K$. Its matrix-source span condition is part of the definition of J. It gives $\|\mathbb{D}_{t_0}\| \le 3\mathsf{c}_{\mathrm{tw}}$ and holomorphy of its entries. The corresponding background-map estimate applies as well: regularity and the source bound at $t_0 - 1$ follow from the displayed ones, with an additional factor L^{-1}. It gives the uniform kernel bounds and holomorphy of those maps. Their compositions with Υ are thus continuous and bounded on the tested set.

Moreover $\|w\|_\infty < 4\varrho$ there and $64\varrho \le r_x\gamma^{-2} < \mathsf{b}$, whence

$$\operatorname{Re}\mathsf{c}_w > 0, \qquad |e^{-\mathsf{wab}(\mathsf{c}_w,W)}| \le 1, \qquad |\hat\Sigma_w| \le \frac{B_F}{L^s(\mathsf{b}-4\varrho)}.$$

The second inequality uses the nonnegative plaquette action. On the finite torus the Grassmann exponential is a polynomial in finitely many generators. Its coefficients are bounded continuous functions on $\mathcal{V}_{t_0}^{\chi}$. Extending them by zero gives the claimed bounded Borel density.

For (b), Prop. B.5 gives the restart's real extracted numbers at $c = 0$. Extraction reads the new infinite-lattice activity and depends on neither the torus nor the coupling weight. The source step uses these numbers; its local mass increment at the zero coordinate is $\hat\mu_{k+1}$. The source-dependent extracted scalar is

$$\hat e_j(X;s) = \hat{\mathfrak{E}}_j(X;\hat{\mathfrak{c}}_{j-1}(s)) - \hat{\mathfrak{E}}_j(X;0).$$

Summing it and the zero-source terms $\hat e_j n_j$, with $\hat E_{t_0} = 0$, gives precisely the number in the statement.

For (c), the step acts on the pair of infinite-lattice and torus families, preserving agreement on polymers which do not wrap and carrying pointers by the same chart. Both starts use the same local formulas and have no activity. Induction on the level proves the assertion, including the independence of the extracted numbers from ℓ. At zero sources the hatted chain is zero, since its initial value is zero and $\hat\nu_{k+1}(0) = \hat\mu_{k+1}$. The source restart at this coordinate is therefore the zero-source restart on each lattice, with identical source energies (all zero). The bounds of Prop. B.10(S1) apply on both lattices, including the scalar and pointed bounds. □

Proposition B.13 (The restarted flow)**.** Let the two-cutoff data satisfy (k0) and $u \ge u_*$. For either datum, let $t_0 \le k \le t_1$ and $s = (w, J) \in \mathscr{O}_k$. The restart with initial mass $F^{\mathrm{UV}}/[L^s(\mathsf{b}-w)]$ exists up to level k. Its term along $h_\emptyset$ is the exact image of the tested start in Lem. B.12 under the steps $t_0+1,\dots,k$. It has a bounded Borel density and the usual RG form, with

$$\text{coupling weight } L^k\hat x_k - a_K^{-1}w, \qquad \text{tower mass } m_0 - a_K\hat J,$$
$$\text{mass term } \; -L^{-k}\mathsf{S}_1((\hat s_k + \hat\sigma_k(s))\psi^{\mathrm{bg}}),$$
$$\text{activities } \hat E^{(j)}(X;\cdot;s), \quad t_0 < j \le k, \qquad \hat c_k(s) = \sum_{t_0<j\le k}\left(\hat e_j n_j + \sum_{X\in\mathcal{D}_j}\hat e_j(X;s)\right).$$

The activities have the locality, covariance, holomorphy and source extraction properties of the original construction. Their scalar parts contain no source-dependent constant. The zeroth moment at a level-j site y is

$$\hat\varpi_j(s)(y) = \sum_z A_j(y,z)[\hat\nu_j(z;\hat{\mathfrak{c}}_{j-1}(s)) - \hat\nu_j(y;\hat{\mathfrak{c}}_{j-1}(s))] + \hat r_j(y),$$

and $\|\hat\varpi_j(s)\| \le C_\varpi\hat\phi_j(\Xi_j(s) + \mathrm{tor}_j)$. All extracted numbers are real, and

$$\hat x_i > 1, \qquad |\hat x_i - \mathsf{t}_i| \le \bar B, \qquad |\hat s_i| \le 2\bar B_M/\mathsf{t}_i.$$

At $t_0 < j \le k$ the activities and their pointed terms obey

$$\|\hat E^{(j)}_{\ge 2}\|_{j;a_k^{\mathrm{rad}}} \le 2C_w^{\mathrm{s}}\hat\phi_{j-1},$$
$$\|\hat F^{(j)}_0\|_{j;a_k^{\mathrm{rad}}} \le 2C_w^{\mathrm{s}}\hat\phi_{j-1} + C'_{\mathrm{lin}}\hat\tau_{j-1}(w;a_k^{\mathrm{rad}}).$$

For $\|w\|_\sharp < 4\varrho$ both are at most $2(C_w^{\mathrm{s}} + 4\alpha_0\varrho C'_{\mathrm{lin}})\hat\phi_{j-1}$ at the fixed pair radii. At these sources the full scalar parts have bound E_0^{s}, pointer by pointer at rate $\kappa+1$. The chain belongs to $\hat{\mathscr{D}}_k$ and $\|\hat\sigma_k(s)\| < B_\sigma\hat\phi_k$.

The activities, chain and source energies are holomorphic on $\mathscr{O}_k$. At zero source they agree with the zero-source restart. The latter is defined on both lattices, with side-independent extracted numbers and the torus agreement of Lem. B.12(c). ■

Proof. By Lem. B.12 this is the source restart of Def. B.9 with $\zeta = F^{\mathrm{UV}} \in [0, B_F]$. We may therefore apply Prop. B.10. For $t_0 \le j < k$, $s \in \mathscr{O}_k \subset \mathscr{O}_j$, and its restarted coordinates belong to $\hat{\mathscr{D}}_j \subset 2\hat{\mathscr{D}}_j(a_j^{\mathrm{rad}})$. Thus (S1) forms each step with the original source-step constants. The step extracts the local mass $-L^{-(j+1)}\mathsf{S}_1(\hat{\nu}_{j+1}\,\cdot)$, the coupling shift $\hat{b}_{j+1}a_p$, and the scalar $\hat{e}_{j+1}n_{j+1}+\sum_X \hat{e}_{j+1}(X;s)$. The identities

$$\hat{s}_{j+1} + \hat{\sigma}_{j+1}(s) = L(\hat{s}_j + \hat{\sigma}_j(s)) + \hat{\nu}_{j+1}(\hat{\mathfrak{c}}_j(s)), \qquad \hat{x}_{j+1} = L^{-1}\hat{x}_j - \hat{b}_{j+1}$$

give the displayed form by induction.

This induction is an identity of densities. If $\hat{g}_j(s)$ is the density at level j, the next one is

$$\chi_{j+1}(T_\ell)\,\mathsf{I}[\mathrm{Ber}_j(\mathrm{Ker}_j\,\hat{g}_j(s))].$$

The tested chart identity and its image-density assertion preserve bounded Borel densities. Apply that assertion coefficient by coefficient; Berezin integration is a finite sum. The start has this property by Lem. B.12(a), proving the exact-image assertion at every level.

At each step the new scalar's source-dependent value at the trivial pair is extracted. The only remaining constant is the zero-source torus correction

$$\hat{\mathfrak{f}}_k = \sum_{t_0<j\le k} \left(\sum_{X\in\mathcal{D}_j} \hat{\mathfrak{E}}_j(X;0) - \hat{e}_j n_j \right),$$

which depends on neither fields nor sources; it is zero on $\mathbb{Z}^3$. Source-dependent action coefficients and first moments stay in the activities. The usual piecewise mass extraction consequently gives the displayed zeroth-moment formula.

Its bound uses only estimates already proved for the restart. The increments have the local bounds with B_ν, the constant-chain comparison and the torus defect with C_{tor}, all by (S1). The physical chain obeys the same bound with C_{var}. For the constant chain,

$$|\hat{\sigma}_i^{\mathrm{c}}(c)| \le \tfrac{1}{4}C_{\mathrm{var}}\hat{\phi}_i|c|/\varrho \le \tfrac{1}{4}B_\sigma\hat{\phi}_i|c|/\varrho,$$

which supplies the Cauchy estimate in the zeroth-moment argument of [QCDM1, §11]. Also $|\hat{\mu}_j| \le 3\bar{B}_M\hat{\phi}_j$ and the zero-source torus remainder is at most $C_{\mathrm{mt}}\hat{\phi}_j e^{-2c_{\mathrm{t}}\ell/\lambda_j}$. The omitted polymers are bounded by the (S1) family estimates. These are precisely the inputs to that argument, so it gives $C_\varpi\hat{\phi}_j(\Xi_j + \mathrm{tor}_j)$ with the original constant.

For the norms at level k, note that $a_{j-1}^{\mathrm{rad}} \le a_k^{\mathrm{rad}} \le \alpha_0$ and

$$C_{\mathrm{lin}}\hat{\tau}_{j-1}(w; a_k^{\mathrm{rad}}) \le \tfrac{9}{8}\sqrt{2}L^{-1/2}C_{\mathrm{lin}}\tau_k(w; a_k^{\mathrm{rad}}) < c_{\mathrm{cl}}/16.$$

Thus $\hat{\mathfrak{c}}_{j-1}(s)$ lies in $2\hat{\mathscr{D}}_{j-1}(a_k^{\mathrm{rad}})$, and (S1) gives both displayed bounds. If $\|w\|_\sharp < 4\varrho$, the coordinate also lies in $2\hat{\mathscr{D}}_{j-1}(\alpha_0)$ and

$$\hat{\tau}_{j-1}(w;\alpha_0) < 8\alpha_0\varrho\hat{x}_{j-1}^{-1/2} \le 8\alpha_0\varrho\hat{\phi}_{j-1}.$$

Here $40M\,\mathrm{Lip}\,w < 4\varrho$ and $\hat{x}_{j-1}\lambda_{j-1}/\mathsf{b} \le 2$. This proves the fixed-radius assertions, including the scalar bound of (S1).

Reality and the windows for $\hat{x}_i, \hat{s}_i$ follow from Prop. B.5(c),(d) at $c = 0$ and the real start $F^{\mathrm{UV}} \in [0, B_F]$. In particular

$$\hat{x}_i \ge \mathsf{t}_i - \bar{B} \ge (1 - L^{-1})\gamma^{-2} > 1,$$

using $\mathsf{t}_i \ge \mathsf{b} > \gamma^{-2} \ge L\bar{B}$ and $\gamma^{-2} \ge 4$. Finally, the source activities and energies are holomorphic in the coordinates, and the chain is holomorphic on $\mathscr{O}_k$, so their compositions are holomorphic

there. The chain vanishes at $s = 0$ and its source energies vanish there. Lem. B.12(c) supplies the two-lattice assertions. □

B.7. **Sizes of the differences.** We will add the difference between the ordinary and restarted flows as an earlier activity. Besides decay with the distance from the restart, this requires the precise extraction rule, the pointed scalar normalization, and estimates summed with the weights used by later steps. We verify these in turn.

Definition B.14 (The comparison at a base level)**.** Use the two-cutoff data with (k0), $u \geq u_*$, and

$$e^{-c_{\mathrm{t}} L^u} \leq L^{-c_{\#}(s-u)}.$$

Fix either datum, write its cutoff as K, and put $t_0 = K - s$, $t_1 = K - u$. A base level satisfies $t_0 < t_b \leq t_1$. The hatted flow is the restart of Prop. B.13, with $\hat{x}_{t_0} = L^s \mathfrak{b}$ and $\zeta = F^{\mathrm{UV}}$. Its earlier members are zero. Use the source set

$$S_b = \{s = (w, J) \in \mathscr{O}_{t_1} : \|w\|_\sharp < 4\varrho\}.$$

At level j, let $F_j(s)$ and G_j denote the parts with legs of the new torus activity *before extraction*, at $\mathfrak{c}_{j-1}(s)$ and at zero coordinates. Let $\tilde{G}_j^\infty$ be the corresponding zero-source family on $\mathbb{Z}^3$, and write

$$\mu_j = \mu_j[\tilde{G}_j^\infty]$$

for the linear mass extraction functional. Adding the cube members $l_\square$ at this number gives G_j^∞. Hats denote the same objects for the restart, and all five hatted objects are zero for $j \leq t_0$. Set

$$D_j(s) = \max\{\|F_j(s) - \hat{F}_j(s)\|_j, \|G_j - \hat{G}_j\|_j, \|\tilde{G}_j^\infty - \hat{\tilde{G}}_j^\infty\|_j\}.$$

These are the fixed-radius norms on each family's lattice. Let $\Delta^{(j)}$ be the difference of the activities along the two source chains, with the scalar value at the trivial pair subtracted pointer by pointer, as in Def. 11.3. Use its sizes r_j^ψ, r_j^0 from Def. C.2.

In the estimates of Prop. B.8 and Def. B.9 set $\Pi_k(c) = \Pi_k(F^{\mathrm{UV}}; c)$ and define

$$X_k = |x_k - \hat{x}_k|/\mathsf{t}_k, \qquad f_k = \phi(\mathsf{t}_k)\Pi_k^* + \mathrm{tor}_k \quad (t_0 \leq k \leq t_1), \qquad f_k = 0 \quad (k < t_0).$$

Thus for $t_0 < j \leq t_1$, $m_j^* + \mathrm{tor}_{j-1} = L^{-c_{\mathrm{lm}}(j-t_0)} + f_{j-1}$. ■

Definition B.15 (Constants for the difference bounds)**.** Form the following constants after γ, in the displayed order. They impose no new condition on previously fixed constants and depend on

neither a datum nor a base level:

$$
\begin{aligned}
K_r &= K_\mu^{\mathrm{s}} + 2(K_\mu - 1),\\
C_\pi &= \tfrac{64}{63}(4B_5 + 4\bar{B}B_F) + 3(\bar{B} + 4C_w)B_F,\\
\Theta_\gamma &= \gamma \sup_{v\ge 0}(v+1)L^{-v/4}(1 + v\log L + \log\gamma^{-2})^{9\mathfrak{p}_0},\\
C_\Pi^\circ &= 2C_\pi + C_{\mathrm{LM}}(\Theta_\gamma + 1), \qquad & C_{\mathrm{flo}} &= \tfrac12 C_\Pi^\circ + 1,\\
C_{\mathrm{dist}} &= \frac{3(1+S_{B'}) + \frac94 A_D + \frac92 C_\Delta}{B_\sigma}, \qquad & A'_{\mathrm{ch}} &= A_\psi^\sharp + \tfrac92 C_{\mathrm{dist}},\\
C_\Lambda^\psi &= 2C_{\mathrm{pc}}S_{1/4}[\tfrac32 K_r + K_r A'_{\mathrm{ch}}(\tfrac32 + C_{\mathrm{flo}})],\\
C_\Lambda^0 &= \tfrac{13}{6} + \frac{1}{E_0^{\mathrm{s}}}\Big\{\tfrac98 A_{\mathrm{ch}}(2C_w^{\mathrm{s}} + 8\alpha_0\varrho C'_{\mathrm{lin}})(\tfrac98 + \tfrac{13}{12}C_{\mathrm{flo}})\\
&\qquad + 2C_{\mathbf{G}}^{\mathrm{s}}[\tfrac98(L(\bar{B}+4C_w) + L^{1/2}C_{\mathrm{LM}}) + \tfrac{13}{12}C_{\mathrm{LM}}C_{\mathrm{flo}}]\Big\},\\
C_\Lambda^\sigma &= \frac{\frac98(C_{\mathrm{flo}} + S_{B'}) + 2A_D(1+C_{\mathrm{flo}}) + 2C_\Delta}{B_\sigma},\\
C_\Lambda^w &= \frac{\bar{B} + 4C_w + C_{\mathrm{LM}}(1+C_{\mathrm{flo}})}{\varrho},\\
C_\Lambda &= \max\{1, C_\Lambda^\psi, C_\Lambda^0, C_\Lambda^\sigma, C_\Lambda^w\}, \qquad & p_\Lambda &= 0.
\end{aligned}
$$

Here $K_\mu^{\mathrm{s}} = K_\mu + \eta_{\min}^{-2} c_{\mathsf{S}} M^{-3} C_{\mathrm{pw}}$ is the source extraction factor. The supremum defining Θ_γ is finite by exponential decay in v. ∎

Lemma B.16 (Depths and domains)**.** In the setting of Def. B.14:

(a) $t_0 \ge s$, $t_b - t_0 \le s - u$, and $k - t_0 + 1 \le s$ for $t_0 \le k \le t_1$.

(b) $0 < c_\# < 1/12$ and $c_\# \le c_{\mathrm{lm}}$.

(c) $\ell \ge 2L$ and, for $k \le t_1$, $\mathrm{tor}_k \le e^{-c_{\mathrm{t}}L^u} \le L^{-c_\#(s-u)}$.

(d) $12\mathsf{t}_i/13 \le x_i \le 14\mathsf{t}_i/13$ for $i \le K$, and the same inequalities hold for $\hat{x}_i$ where it is defined. Also $\phi(\mathsf{t}_i) \le 1/2$, $\mathsf{t}_i^{-1} \le \phi(\mathsf{t}_i)^2$, and $\phi(\mathsf{t}_i)$ increases with i. Any two of $\phi(\mathsf{t}_i), \phi_i, \hat{\phi}_i$ have ratio at most $9/8$, and

$$1/(\mathsf{t}_i\hat{\phi}_i) \le (14/13)^{1/2}\phi(\mathsf{t}_i).$$

(e) S_b is open and $\mathcal{P} \subset S_b \subset \mathscr{O}_{t_1} \subset \mathscr{O}_k$ for $k \le t_1$. On S_b,

$$\Xi_k(s) < 2, \quad \tau_k(w;\alpha_0) \le 8\alpha_0\varrho\phi_k, \quad \|\sigma_k(s)\| \le B_\sigma\phi_k.$$

The ordinary chain lies in $\mathfrak{D}_k(\alpha_0) \cap 2\mathscr{D}_k(\alpha_0)$; the restarted chain lies in $\hat{\mathscr{D}}_k \cap 2\hat{\mathscr{D}}_k(\alpha_0)$ for $t_0 \le k < t_1$.

(f) For $t_0 \le i \le j \le t_1$,

$$m_i^* \le L^{c_{\mathrm{lm}}(j-i)} m_j^*, \qquad \Xi_i(s)m_i^* \le \Xi_j(s)m_j^* \quad (s \in \mathscr{O}_j).$$

The quantities $\Pi_k(c), \Pi_k^*, \phi(\mathsf{t}_k)\Pi_k^*, \mathrm{tor}_k, f_k$ increase with k.

∎

Proof. For the smaller cutoff, $s = \lceil K/[2(6n+11)]\rceil$, with $1/[2(6n+11)] \le 1/34$. Condition (k0) gives $s > u \ge 1$, hence $s \ge 2$, $K \ge 35$, and

$$K - s \ge 33K/34 - 1 \ge K/34 + 1 \ge s.$$

For the other datum t_0 is larger by one. The remaining depth inequalities follow from $t_1 - t_0 = s - u$.

By the choice of $c_{\#}$, $L^{c_{\#}} \le 11/10$. Since $L \ge 13$, $c_{\#} < 1/12$. Its weighted-series condition is the first condition defining c_{lm}. The other condition holds because

$$\tfrac{8}{3}L^{-1/2} + \tfrac{27}{112}L^{-1} < 0.76 \quad (L \ge 13), \qquad (11/10)0.76 < 9/10.$$

Both expressions increase with the exponent, and c_{lm} was chosen maximal. Therefore $c_{\#} \le c_{\mathrm{lm}}$. An admitted side is $2L^m$ with $m \ge 1$; for $k \le t_1$, $\lambda_k \le L^{-u}$. These give (c).

The coupling windows, $\mathtt{t}_i \ge \mathtt{b} > \gamma^{-2} \ge L\bar{B}$, and $L \ge 13$ give (d)'s first bounds. The fixed coupling ceilings make ϕ decreasing on $[\gamma^{-2}, \infty)$, with value at most $1/2$ there. For $x, y \in [12\mathtt{t}/13, 14\mathtt{t}/13]$,

$$\frac{\phi(x)}{\phi(y)} \le (7/6)^{1/2}\left(1 + \frac{\log(7/6)}{18\mathfrak{p}_0}\right)^{\mathfrak{p}_0} \le (7/6)^{1/2+1/18} < 9/8.$$

The last bound in (d) follows directly from $\hat{x}_i \le 14\mathtt{t}_i/13$ and $(1+\log \hat{x}_i)^{-\mathfrak{p}_0} \le 1$.

All defining inequalities of S_b are strict and continuous. On $\mathcal{P}$, $\|w\|_\sharp < 2\varrho$ and $\mathcal{P} \subset \mathscr{O}_K$. The source-domain inclusions thus give the first assertions of (e). The ordinary chain obeys

$$C_{\mathrm{lin}}\tau_k(w;\alpha_0) < c_{\mathrm{cl}}/16, \qquad L^{k+1}\|\mathsf{n}\| < r_M, \qquad \|\sigma_i(s)\| \le B_\sigma \phi_i,$$

which are inside both specified coordinate domains. The restarted domains follow from Prop. B.10(S1) and Prop. B.13.

Finally the floor in m_i^* increases with i, whereas its geometric term changes by exactly $L^{c_{\mathrm{lm}}(j-i)}$. Since the source domains give $L^{j-i}\Xi_i \le (4/3)^{j-i}\Xi_j$,

$$\Xi_i m_i^* \le ((4/3)L^{c_{\mathrm{lm}}-1})^{j-i}\Xi_j m_j^* \le \Xi_j m_j^*.$$

The last inequality uses $c_{\mathrm{lm}} \le 1/12$ and $L \ge 13$. The other monotonicity statements follow from the formulas for Π, $\phi(\mathtt{t}_k)$, and λ_k. □

Lemma B.17 (The mismatch floor)**.** For $t_0 \le k \le t_1$ and $|c| < 8\varrho$,

$$\Pi_k(c) \le C_\Pi^\circ L^{-c_{\#}(s-u)}, \qquad f_k \le C_{\mathrm{flo}}L^{-c_{\#}(s-u)} \le C_{\mathrm{flo}}L^{-c_{\#}(t_b-t_0)}.$$

For $t_0 \le k \le K$, $\phi(\mathtt{t}_k)B'_k \le S_{B'}L^{-s/4}$, where

$$L^{-s/4} \le L^{-c_{\mathrm{lm}}(j-t_0)} \quad (t_0 \le j \le t_1), \qquad L^{-s/4} \le L^{-c_{\#}(s-u)}.$$

For $t_0 < k \le t_1$,

$$|x_k - \hat{x}_k| \le (\bar{B} + 4C_w)L^{-(k-t_0)} + C_{\mathrm{LM}}(L^{-(k-t_0)/2} + f_{k-1}).$$

The same right-hand side bounds X_k also at $k = t_0$. For $t_0 \le k \le t_1$ and $|c| < 4\varrho$,

$$\mathtt{t}_k|s_k(c) - \hat{s}_k(c)| \le \Pi_k^* + B'_k.$$

■

Proof. The mismatch formula of Prop. B.8(A) at zero label gives

$$|\pi(F^{\mathrm{UV}};0) - (F^{\mathrm{UV}} - F_{<t_0}(t_1;\mathfrak{m}_{t_1}))| \le 2\bar{B}B_F L^{-(s+1)}/\mathtt{b},$$
$$|\pi(F^{\mathrm{UV}};0)| \le B_5(L^{-t_0} + 3L^{-(s-u)}) + 2\bar{B}B_F L^{-(s+1)}/\mathtt{b}.$$

It follows that the partial-sum mismatch is at most the first term on the latter right-hand side plus $4\bar{B}B_F L^{-(s+1)}/\mathtt{b}$. At any label, $|\mathtt{b}/(\mathtt{b}-c)| \le 64/63$, $L^{-t_0} \le L^{-(s-u)}$, and $L^{-(s+1)}/\mathtt{b} \le L^{-(s-u)}$. The same mismatch formula therefore yields $|\pi(F^{\mathrm{UV}};c)| \le C_\pi L^{-(s-u)}$.

The function $x^{-1/2}(1+\log x)^{9\mathfrak{p}_0}$ decreases for $x \ge e^{18\mathfrak{p}_0}$. Thus

$$\varepsilon_s \le \gamma L^{-s/2}(1 + s\log L + \log\gamma^{-2})^{9\mathfrak{p}_0}, \qquad (k - t_0 + 1)\varepsilon_s \le \Theta_\gamma L^{-s/4}.$$

Insert these bounds in the definition of Π_k. Since $c_{\#} \le 1/8$, all its decays are bounded by $L^{-c_{\#}(s-u)}$. Taking the supremum in c, multiplying by $\phi(\mathtt{t}_k) \le 1/2$ and adding the torus bound of Lem. B.16 proves the assertions about Π_k and f_k.

For the bound on B'_k, write $m = k - t_0 \le s$, so $\mathsf{t}_k = L^{s-m}\mathsf{b}$. Then

$$\begin{aligned}\phi(\mathsf{t}_k)L^{-m} &\le \tfrac{1}{2}L^{-(s+m)/2}(1+(s-m)\log L)^{\mathfrak{p}_0}\\ &\le \tfrac{1}{2}L^{-s/4}L^{-(s-m)/4}(1+(s-m)\log L)^{\mathfrak{p}_0},\\ \phi(\mathsf{t}_k)L^{-(s+1)}/\mathsf{b} &\le \tfrac{1}{2}L^{-s}.\end{aligned}$$

The definition of B'_k and $S_{B'}$ gives the claim. The two comparisons of decay rates use $c_{\mathrm{lm}}(j-t_0) \le s/12$ and $c_{\#}(s-u) \le s/8$.

The coupling bound is Prop. B.8(B1), with $\phi(\mathsf{t}_{k-1}) \le 1/2$ and $\phi(\mathsf{t}_{k-1})^2\Pi^{\flat}_{k-1} \le f_{k-1}$. Since $\mathsf{t}_k \ge 1$ it also bounds X_k. At $k = t_0$, $X_{t_0} \le \bar{B}$ directly from the windows. For the mass bound apply Prop. B.10(S4) to the constant source $(c, 0) \in \mathscr{O}_{t_1}$: its first term is exactly the displayed constant-label mass difference. □

B.8. **Extraction and the form of the differences.**

Lemma B.18 (Boundedness of the extraction rule)**.** Let $1 \le j \le K$. Suppose that the data (F, G, ν) obey the free-family rule (f2) of Def. C.2, at the fine mass matrix of a source in $\mathscr{O}_{j-1}$. Let $\tilde{G}^\infty$ be an even infinite-lattice family of degrees at least two and finite norm. Assume (f3) for the family G^∞ obtained by adding its cube members $l_\square$ at $\nu = \mu_j[\tilde{G}^\infty]$. If $\Delta_{\ge 2}$ is formed from these data by (f2), then

$$c_{\mathsf{S}}h_F^2|\nu| \le (K_\mu - 1)\|\tilde{G}^\infty\|_j, \qquad r^\psi(\Delta) \le K_r \max\{\|F\|_j, \|G\|_j, \|\tilde{G}^\infty\|_j\}.$$

■

Proof. The functional $\mu_j[E]$ is $-\eta_j^{-2}$ times the normalized trace of the cell average of the row sums of the degree-two kernel of E at the trivial configuration. Condition (f3) gives translation covariance of $\tilde{G}^\infty$ and invariance under the common flavour phase. The latter excludes the $\Psi\Psi$ and $\bar{\Psi}\bar{\Psi}$ parts, since their phases are $e^{2i\alpha}$ and $e^{-2i\alpha}$. The row-sum estimate used in the mass extraction of [QCDM1, §9] therefore applies: these two properties are its only symmetry inputs. It bounds the normalized trace by $C^\psi_{am}(8N_f h_F^2)^{-1}\|E\|_j$. Since $\eta_j^2 \ge c_b^2$ and $K_\mu = 1 + c_{\mathsf{S}}C^\psi_{am}c_b^{-2}/(8N_f)$, this proves the first inequality.

The extraction rule is

$$\Delta_{\ge 2}(Z) = F(Z) + \sum_{(X,p):Z_p=Z}\sum_z P_{z;X,p} + 1_{\{Z=\square\}}\left(l_\square + \sum_{z\in\square} P^\nu_z\right),$$

where

$$P_{z;X,p} = -\mathfrak{s}_j^{-1}a_{X,p}(z)L^{-j}\mathsf{S}_1(1_z\psi^{\mathrm{bg}}), \qquad a_{X,p} = \operatorname{ord}_0 k_{X,p}(F) - \operatorname{ord}_0 k_{X,p}(G).$$

A form supported on the block of one site z has norm

$$\|L^{-j}\mathsf{S}_1(N1_z\psi^{\mathrm{bg}})\| \le c_{\mathsf{S}}M^{-3}h_F^2\|N\|.$$

This is the cube estimate with $\nu_P L^{3j}$ places in place of $\nu_P M^3 L^{3j}$; the same background bound holds at the admitted fine mass matrix.

For clarity, the piece count also applies to the present data, without assuming that they are an RG output. For any even element $E(X)$ its degree-two coefficients satisfy

$$\sum_{n,n',x,x'} h^{(j)}(n)h^{(j)}(n')\|g_X^{nn'}(x,x')\|_{\mathrm{tr}} \le \|E(X)\|.$$

Each homogeneous part has norm at most that of $E(X)$. The two localized legs use the same walks for F and G, with the original A_κ, C_H bounds: at a source in $\mathscr{O}_{j-1}$, $L^j\|\mathsf{n}\| = L\lambda_{j-1}\|J\|_\infty < r_M$.

The walk counts C_{wk} and $3C_{\mathrm{wk}}$ therefore give

$$\sum_{p}\sum_{y}\|a_{X,p}(y)\|e^{\kappa(d_j(Z_p)-d_j(X))} \le CA_\kappa^2 h_F^{-2}(\|F(X)\|+\|G(X)\|),$$
$$\sum_{Z_p\supset\square}\sum_{z}\|a_{X,p}(z)\|e^{\kappa d_j(Z_p)} \le C_{\mathrm{pw}} h_F^{-2}\max\{\|F\|_j,\|G\|_j\}.$$

The factor two is included in C_{pw}. Combining this with $\mathfrak{s}_j^{-1}\le\eta_{\min}^{-2}$ bounds the piece forms by $(K_\mu^{\mathrm{s}}-K_\mu)\max\{\|F\|_j,\|G\|_j\}$.

The cube member has $\|l_\square\|\le c_{\mathsf{S}}h_F^2|\nu|$. Each P_z^ν is the difference of two one-block forms, at n and zero; summing over the cube gives $\sum_{z\in\square}\|P_z^\nu\|\le 2c_{\mathsf{S}}h_F^2|\nu|$. A single cube has weight one. Thus the sum of the summand norms of $\Delta_{\ge 2}$ is at most

$$\|F\|_j+(K_\mu^{\mathrm{s}}-K_\mu)\max\{\|F\|_j,\|G\|_j\}+3c_{\mathsf{S}}h_F^2|\nu|.$$

The first inequality of the lemma makes this at most K_r times the stated maximum, since $1+K_\mu^{\mathrm{s}}-K_\mu+3(K_\mu-1)=K_r$. The other three entries in r^ψ are bounded respectively by K_μ, K_μ, and $K_\mu-1$ times the same maximum. All are at most K_r times it. □

Lemma B.19 (The differences are admissible free families)**.** For $s\in S_b$ and $j\le t_b$:

(a) Every pointed term $\Delta^{(j)}(X;\cdot;s;z)$ is even, jointly gauge invariant, local in X, Borel in the pair, and holomorphic on the regular complex pairs of radii α_0,α_1. It is holomorphic in $s\in S_b$. It is invariant under $\mathrm{U}(N_f)\times\mathrm{U}(N_f)$ acting jointly on the fields and sources:

$$V=e_+\otimes V_1+e_-\otimes V_2,\qquad (w,J,\Psi_n^{\mathrm{bg}},\bar\Psi_n^{\mathrm{bg}})\longmapsto(w,VJV^{-1},V\Psi_n^{\mathrm{bg}},\bar\Psi_n^{\mathrm{bg}}V^{-1}).$$

In particular the common phase fixes each source and leaves the pointed term invariant. Its scalar part vanishes at the trivial pair and obeys the pointed sum rule.

(b) It is a free family in the sense of Def. C.2. Its part with legs is formed by (f2) from

$$(F_j(s)-\hat F_j(s),G_j-\hat G_j,\mu_j-\hat\mu_j),$$

with (f3) for $G_j^\infty-\hat G_j^\infty$. Moreover

$$\mu_j-\hat\mu_j=\mu_j[\tilde G_j^\infty-\hat{\tilde G}_j^\infty].$$

For $j\le t_0$ these are simply the ordinary flow's data.

(c) The scalar removed from the differences obeys $d_b(s)=d_b(0)$.

■

Proof. Both ordinary and restarted activities are evaluated inside their fixed-radius coordinate domains by Lem. B.16(e). The source step and Prop. B.10(S1) give the stated properties for the unpointed activities. We check the pointed terms, as their scalar normalization is needed below.

For the terms pointed at a fermion place p, the insertion of an element $F=\sum_n F_n$ is

$$F_p=\sum_{n\ge1}n^{-1}\mathsf{N}_pF_n.$$

It preserves degree, vanishes unless p carries a leg, and commutes with every linear change of the fibre at each place. In particular it commutes with gauge transformations and with the joint flavour operation on the coordinates. The pointed output is the integral over $t\in[0,1]$ of the ϑ derivative at zero of the expansion with leg potential $t\mathcal{V}_\psi+\vartheta(\mathcal{V}_\psi)_p$. For complex $|t|<9/8$, $|\vartheta|<1$ the potential bound is

$$\varepsilon_{\mathrm{V}}+\varepsilon_{\natural}\le 49c_{\mathrm{cl}}/64<c_{\mathrm{cl}}.$$

Thus the expansion is holomorphic in these variables, the coordinates and the pair. Gauge covariance follows from the expansion's gauge identity. For flavour, each Gaussian and Berezin integral in the series is unchanged under the contragredient unitary pair V,V^{-1}, which preserves the fibre norms

and maps each cube to itself. It also conjugates the tower and the potential, term by term, at the transformed coordinates. The two absolutely convergent series therefore correspond term by term. Differentiation and integration preserve both identities. The output depends only on the potential pieces and localized covariances within X; if $p \notin X$ its ϑ derivative is zero. This proves locality and both invariances for the place-pointed terms.

The remaining pointed terms have the same properties for explicit reasons. The bond terms and gauge normalizers are local, separately gauge invariant scalar terms of the step. Their earlier scalar inputs are flavour invariant by induction. A diagonal block $N_p(X, \mathsf{U})$ of a closed fermion loop is a sum of traces of walk products within X, at the fine mass matrix. Gauge transformations conjugate every walk and average, and commute with that matrix. At the diagonal place p their trace is unchanged. The subtracted trivial-field term is independent of the configuration. Flavour conjugates these operators by V, so also preserves their traces, numbers per site and wrapping shares.

The extracted cube forms and $P_{z;X,p}$ are forms $\bar{\psi}^{\mathrm{bg}} N \psi^{\mathrm{bg}}$ with colour-scalar matrices. Under flavour their matrices become VNV^{-1}, because the degree-two kernels and background maps have that covariance. They are consequently jointly invariant. The pointed action members are

$$-L^{j} b_j \upsilon_z(\mathsf{p}) a_{\mathsf{p}}(\mathsf{U}), \qquad a_{\mathsf{p}}(\mathsf{U}) = 1 - \tfrac{1}{2} \operatorname{tr} \partial \mathsf{U}(\mathsf{p}).$$

Their polymer contains the four bonds of p; $\upsilon_z(\mathsf{p})$ is one quarter the number of its corners in the block of z and sums to one over z. These scalar terms are local, holomorphic, gauge invariant, source independent and zero at the trivial pair. The same argument uses $\hat{b}_j$ in the restart.

Every listed term is holomorphic in the source coordinates: the step output and its pointed variations by the admitted expansion, and the loop blocks, piece coefficients and cube forms by holomorphy of the tower and background maps on the fine-mass domain. Composing with the two holomorphic chains proves holomorphy on S_b. The pair dependence is holomorphic, hence continuous and Borel.

The source set is invariant under $J \mapsto VJV^{-1}$: the values remain in the specified matrix span and their operator norm is unchanged. The chain transforms in the same way, by induction on its increments; the restarted initial mass is scalar at each site. Thus joint coordinate invariance becomes joint source invariance on the chain. For the common phase $V = e^{i\alpha} 1$, J itself is fixed. Taking differences and subtracting the scalar values at the trivial pair preserves all these linear identities and the pointed sum rule. This proves (a) and the scalar part of (b).

For the part with legs, both flows use the rule (f2). The ordinary input is $(F_j(s), G_j, \mu_j)$ and the restarted input is its hatted counterpart. All these families are invariant under the common phase. At zero coordinates the forms $P_{z;X,p}$ and P^{ν}_{z} vanish, since $a_{X,p} = 0$ and $\mathsf{n} = 0$. Thus the infinite-lattice family after extraction is precisely $\tilde{G}^{\infty}_j$ plus its cube terms $l_{\square}$. It is translation covariant summand by summand, and its torus part which does not wrap is the part which fits.

Its cell sums vanish as matrices. Indeed the zero-source family lies in the linear symmetry space preserved by extraction; its cell sum is $M^3 \mathsf{cf}_m 1$, and extracting $\mu_j[\tilde{G}^{\infty}_j]$ makes $\mathsf{cf}_m = 0$. The same conclusion holds for the restart by the verified symmetry-space induction in Prop. B.5, with its own mass functional and extracted number. Thus both satisfy (f3), including matrix cancellation rather than only cancellation of a trace.

Their pieces, walks, trees, paths, cells and hulls are the same: these choices are independent of the activity and the pair, and the two flows share the tower and background maps at the fine mass matrix $m_0 - a_K \hat{J}$ and at zero source. The extraction rule and the functional μ_j are linear with these choices fixed. Taking differences therefore gives exactly the data in (b). All conditions of (f3), including the matrix cell cancellation, are closed linear conditions and pass to the difference. For $j \le t_0$ the hatted family is zero. The resulting data and family are holomorphic in s by the preceding coordinate argument.

Finally the scalar values at the trivial pair at level j sum to $\sum_X \mathfrak{E}_j(X;0) - e_j n_j$ for the ordinary flow, and to its hatted counterpart for the restart. Neither depends on the source. The finite sum defining $d_b(s)$ is therefore $d_b(0)$, proving (c). □

Lemma B.20 (Differences before extraction)**.** For $t_0 < j \le t_1$ and $s \in S_b$,

$$\|G_j - \hat{G}_j\|_j, \|\tilde{G}_j^{\infty} - \hat{\tilde{G}}_j^{\infty}\|_j \le 2C_w^{\mathrm{s}} A_\psi^\sharp m_j \phi_{j-1},$$
$$\|F_j(s) - \hat{F}_j(s)\|_j \le 2C_w^{\mathrm{s}} A'_{\mathrm{ch}}(m_j^* + \mathrm{tor}_{j-1})\phi_{j-1}.$$

In particular,

$$D_j(s) \le 2C_w^{\mathrm{s}} A'_{\mathrm{ch}}(L^{-c_{\mathrm{lm}}(j-t_0)} + f_{j-1})\phi_{j-1}.$$

■

Proof. At zero coordinates, (S2) of Prop. B.10 applies on both lattices and before extraction, giving the first two estimates. At the physical chains, put $\mathfrak{c} = \mathfrak{c}_{j-1}(s)$ and $\hat{\mathfrak{c}} = \hat{\mathfrak{c}}_{j-1}(s)$, and split

$$F_j(s) - \hat{F}_j(s) = [\tilde{E}_{\ge 2}^{(j)}(\mathfrak{c}) - \hat{\tilde{E}}_{\ge 2}^{(j)}(\mathfrak{c}^\wedge)] + [\hat{\tilde{E}}_{\ge 2}^{(j)}(\mathfrak{c}^\wedge) - \hat{\tilde{E}}_{\ge 2}^{(j)}(\hat{\mathfrak{c}})].$$

The first bracket is the equal-coordinate difference. Since $\mathfrak{c} \in \mathfrak{D}_{j-1}(\alpha_0)$, (S2) bounds it by $2C_w^{\mathrm{s}} A_\psi^\sharp m_j \phi_{j-1}$.

The second bracket changes only the mass coordinates. Set

$$T = \max_{t_0 \le i < j} \frac{\|\sigma_i(s) - \hat{\sigma}_i(s)\|}{B_\sigma \hat{\phi}_i}.$$

At a bare site x, write this difference as the change of the total mass $s_i + \sigma_i(s)(x)$ minus $s_i - \hat{s}_i$. By (S4) and Lem. B.17, its magnitude is at most

$$2(\Pi_i^* + B_i')/\mathsf{t}_i + A_D \phi_i \Xi_i(s) m_i^* + C_\Delta \phi_i \mathrm{tor}_i \|w\|_\infty/\varrho.$$

Divide by $B_\sigma \hat{\phi}_i$. Use Lems. B.16 and B.17, in particular

$$\phi(\mathsf{t}_i)\Pi_i^* \le m_j^*, \quad \phi(\mathsf{t}_i)B_i' \le S_{B'} m_j^*, \quad \Xi_i(s) m_i^* \le 2m_j^*,$$
$$\phi_i/\hat{\phi}_i \le 9/8, \quad 2(14/13)^{1/2} < 3, \quad \|w\|_\infty < 4\varrho.$$

This gives

$$T \le \frac{3(1 + S_{B'}) + \frac{9}{4}A_D}{B_\sigma} m_j^* + \frac{9C_\Delta}{2B_\sigma}\mathrm{tor}_{j-1} \le C_{\mathrm{dist}}(m_j^* + \mathrm{tor}_{j-1}).$$

If $T = 0$ the second bracket vanishes. For $T > 0$, the function

$$f(z) = \hat{\tilde{E}}_{\ge 2}^{(j)}(\hat{\mathfrak{c}} + z(\mathfrak{c}^\wedge - \hat{\mathfrak{c}}))$$

is holomorphic for $|z| < 1/T$. Its mass coordinates have norm less than $2B_\sigma\hat{\phi}_i$ there, while w and n are unchanged. The domain is thus contained in $2\hat{\mathscr{D}}_{j-1}(\alpha_0)$, and (S1) gives $\|f(z)\|_j \le 2C_w^{\mathrm{s}}\hat{\phi}_{j-1}$. For $T \le 1/2$, Cauchy's formula on $1 < r < 1/T$ gives

$$\|f(1) - f(0)\|_j \le \frac{2C_w^{\mathrm{s}}\hat{\phi}_{j-1}}{r-1} \longrightarrow \frac{T}{1-T}\, 2C_w^{\mathrm{s}}\hat{\phi}_{j-1} \le 2T\, 2C_w^{\mathrm{s}}\hat{\phi}_{j-1}.$$

If $T > 1/2$, both endpoints still belong to that coordinate domain, because $\|\sigma_i(s)\| \le (9/8)B_\sigma\hat{\phi}_i$. Their separate bounds give at most $4T\, 2C_w^{\mathrm{s}}\hat{\phi}_{j-1}$. In both cases $\hat{\phi}_{j-1} \le (9/8)\phi_{j-1}$ bounds the second bracket by $(9/2)T\, 2C_w^{\mathrm{s}}\phi_{j-1}$. Adding the first bracket gives the result with $A'_{\mathrm{ch}} = A_\psi^\sharp + (9/2)C_{\mathrm{dist}}$. □

B.9. **Summing the differences over levels.**

Proposition B.21 (The sizes at each level)**.** Let $t_0 < t_b \le t_1$ and $s = (w, J) \in \mathcal{P}$. For $j \le t_0$,

$$r_j^{\psi}(s) \le 2K_r C_w^{\mathrm{s}} \phi_{j-1}, \qquad r_j^{0}(s) \le 2E_0^{\mathrm{s}}.$$

For $t_0 < j \le t_b$,

$$\begin{aligned} c_{\mathsf{S}} h_F^2 |\mu_j - \hat{\mu}_j| &\le (K_\mu - 1) A_\psi^\sharp m_j \, 2C_w^{\mathrm{s}} \phi_{j-1}, \\ r_j^{\psi}(s) &\le K_r A'_{\mathrm{ch}} (L^{-c_{\mathrm{lm}}(j-t_0)} + f_{j-1}) \, 2C_w^{\mathrm{s}} \phi_{j-1}, \\ r_j^{0}(s) &\le \tfrac{9}{8} A_{\mathrm{ch}} (2C_w^{\mathrm{s}} + 8\alpha_0 \varrho C'_{\mathrm{lin}}) (L^{-c_{\mathrm{lm}}(j-t_0)} + f_{j-1}) + 2C_{\mathbf{G}}^{\mathrm{s}} X_{j-1}. \end{aligned}$$

The remaining mass field and coupling weight at the base obey

$$\|\delta_{t_b}(s)\| \le C_\Lambda^\sigma B_\sigma L^{-c_\#(t_b - t_0)} \phi_{t_b}, \qquad |\hat{w}_b| \le C_\Lambda^w \varrho L^{-c_\#(t_b - t_0)}.$$

■

Proof. For $j \le t_0$ the restarted family is zero. Each of $F_j(s), G_j, \tilde{G}_j^\infty$ is a summand of an ordinary activity after extraction. The ordinary source estimates bound the sums of their norms by $(2C_w^{\mathrm{s}} - C_w)\phi_{j-1} \le 2C_w^{\mathrm{s}} \phi_{j-1}$. Apply Lem. B.18 to these data. The scalar norm is bounded by E_0^{s} before subtracting the trivial value and by $2E_0^{\mathrm{s}}$ after that subtraction.

For $j > t_0$, Lem. B.19 identifies the data as differences and Lem. B.18 gives

$$c_{\mathsf{S}} h_F^2 |\mu_j - \hat{\mu}_j| \le (K_\mu - 1) \|\tilde{G}_j^\infty - \hat{\tilde{G}}_j^\infty\|_j, \qquad r_j^{\psi} \le K_r D_j.$$

Lem. B.20 proves the two corresponding estimates.

For the scalar terms, the gauge normalizers and $\mathbf{c}_j$ depend on the common calibration and are their values at zero source. The fermion loop normalizers use the common tower at $m_0 - a_K \hat{J}$; their constant wrapping shares are those of zero source. Both flows use the same pointing rule. These terms therefore agree pointer by pointer, as in Cor. B.11(b). There are also numbers $d(X, z)$ attached to the ratio x_{j-1}/t_{j-1}, and hatted numbers attached to $\hat{x}_{j-1}/\mathsf{t}_{j-1}$. Their difference is independent of the fields and source, so disappears when the trivial value is subtracted. Only $\delta^{\mathrm{ch}} F_0^{(j)}$ and $\mathbf{G}^{(j)} - \hat{\mathbf{G}}^{(j)}$ remain. Before the subtraction their pointed scalar bounds are

$$\begin{gathered} A_{\mathrm{ch}} (m_j^* + \mathrm{tor}_{j-1})(2C_w^{\mathrm{s}} + 8\alpha_0 \varrho C'_{\mathrm{lin}}) \phi_{j-1}, \\ C_{\mathbf{G}}^{\mathrm{s}} X_{j-1}. \end{gathered}$$

The latter holds at rate $\kappa_+ \ge \kappa + 1$. Subtracting the trivial value doubles these bounds. Since $2\phi_{j-1} \le (9/4)\phi(\mathsf{t}_{j-1}) \le 9/8$, this is the estimate for r_j^0.

At the base, (S4) gives at each bare site

$$|\delta_{t_b}(s)(x)| \le \frac{\Pi_{t_b}^* + B'_{t_b}}{\mathsf{t}_{t_b}} + A_D \phi_{t_b} \Xi_{t_b}(s) m_{t_b}^* + C_\Delta \phi_{t_b} \mathrm{tor}_{t_b} \frac{\|w\|_\infty}{\varrho}.$$

The first term is at most $\frac{9}{8}(C_{\mathrm{flo}} + S_{B'}) L^{-c_\#(s-u)} \phi_{t_b}$ by Lems. B.16 and B.17. On $\mathcal{P}$, $\Xi_{t_b} < 2$ and $\|w\|_\infty < 2\varrho$. Use

$$m_{t_b}^* \le L^{-c_\#(t_b - t_0)} + C_{\mathrm{flo}} L^{-c_\#(s-u)}, \qquad \mathrm{tor}_{t_b} \le L^{-c_\#(s-u)}.$$

The three terms add to the claimed bound with C_Λ^σ. Finally $\hat{w}_b = L^{-(K-t_b)}(x_{t_b} - \hat{x}_{t_b})$. The coupling estimate of Lem. B.17, the floor bound, and $L^{-(K-t_b)} \le 1$ give

$$|\hat{w}_b| \le [\bar{B} + 4C_w + C_{\mathrm{LM}}(1 + C_{\mathrm{flo}})] L^{-c_\#(t_b - t_0)},$$

which is the stated result. □

Proposition B.22 (The lag sums)**.** Let $\Lambda_b^{-1} = C_\Lambda L^{-c_\#(t_b-t_0)}$. For $s \in \mathcal{P}$ and $t_b \le k \le K$,

$$\sum_{j\le t_b} \omega_{k+1-j} r_j^\psi(s) \le C_\Lambda^\psi C_w^{\mathrm{s}} L^{-c_\#(t_b-t_0)}\phi_k \le \Lambda_b^{-1} C_w^{\mathrm{s}}\phi_k, \tag{L1}$$

$$\sum_{j\le t_b} L^{-(k-j)} r_j^0(s) \le \sum_{j\le t_b} L^{-(t_b-j)} r_j^0(s) \le C_\Lambda^0 E_0^{\mathrm{s}} L^{-c_\#(t_b-t_0)} \le \Lambda_b^{-1} E_0^{\mathrm{s}}, \tag{L2}$$

$$\|\delta_{t_b}(s)\| \le \Lambda_b^{-1} B_\sigma \phi_{t_b}, \qquad |\hat{w}_b| \le \Lambda_b^{-1}\varrho. \tag{L3}$$

■

Proof. For $n = k+1-j$ the power-counting inequality is

$$\omega_n \phi_{j-1} \le C_{\mathrm{pc}} L^{-n/4}(1+n)^{\mathfrak{p}_0}\phi_k.$$

The definition of $c_\#$ gives, for every $N \ge 1$,

$$\sum_{n\ge N} L^{-n/4}(1+n)^{\mathfrak{p}_0} \le L^{-c_\# N} \sum_{n\ge 1} L^{-n(1/4-c_\#)}(1+n)^{\mathfrak{p}_0} \le \tfrac{3}{2} S_{1/4} L^{-c_\# N}.$$

For $j \le t_0$, $n \ge k+1-t_0 > t_b - t_0$, so these levels contribute at most

$$2K_r C_w^{\mathrm{s}} C_{\mathrm{pc}} \tfrac{3}{2} S_{1/4} L^{-c_\#(t_b-t_0)}\phi_k.$$

For $t_0 < j \le t_b$, the geometric term satisfies

$$L^{-c_{\mathrm{lm}}(j-t_0)} \le L^{-c_\#(k+1-t_0)} L^{c_\# n}.$$

Its convolution is therefore at most $\frac{3}{2} S_{1/4} C_{\mathrm{pc}} L^{-c_\#(t_b-t_0)}\phi_k$ before the factor $2K_r A'_{\mathrm{ch}} C_w^{\mathrm{s}}$. The floor is at most $C_{\mathrm{flo}} L^{-c_\#(t_b-t_0)}$, and its convolution uses the unmodified sum $S_{1/4}$. Adding these three contributions gives exactly C_Λ^ψ and (L1).

For (L2), the first inequality follows from $k \ge t_b$. The levels $j \le t_0$ contribute at most

$$2E_0^{\mathrm{s}} \frac{L}{L-1} L^{-(t_b-t_0)} \le \tfrac{13}{6} E_0^{\mathrm{s}} L^{-c_\#(t_b-t_0)}.$$

For the later levels, Lem. B.17 gives

$$X_{j-1} \le [L(\bar{B} + 4C_w) + L^{1/2} C_{\mathrm{LM}}] L^{-c_\#(j-t_0)} + C_{\mathrm{LM}} C_{\mathrm{flo}} L^{-c_\#(t_b-t_0)}.$$

This includes $j-1 = t_0$. The geometric and constant parts are summed using

$$\sum_{t_0<j\le t_b} L^{-(t_b-j)} L^{-c_\#(j-t_0)} \le \tfrac{9}{8} L^{-c_\#(t_b-t_0)},$$

$$\sum_{t_0<j\le t_b} L^{-(t_b-j)} \le \tfrac{13}{12}.$$

Indeed $L^{1-c_\#} \ge 13^{11/12} \ge 9$. Substitution into the scalar bound of Prop. B.21 gives the braces in C_Λ^0. Adding the earlier levels proves (L2). Part (L3) is the last assertion of Prop. B.21, since the four constants are bounded by C_Λ. □

Proposition B.23 (Loss of memory as an admissible perturbation)**.** In the two-cutoff setting, assume (k0), $u \ge u_*$ and $e^{-c_{\mathrm{t}} L^u} \le L^{-c_\#(s-u)}$. At every base level $t_0 < t_b \le t_1$, the differences $\Delta^{(j)}$, $j \le t_b$, satisfy the format conditions of Def. 11.3, with the free-family data of Def. B.14. Their removed scalar $d_b(s)$ is constant on $\mathcal{P}$, and their lag sums hold with

$$\Lambda_b^{-1} = C_\Lambda L^{-c_\#(t_b-t_0)}.$$

Thus one may take $p_\Lambda = 0$. At the base t_1,

$$\Lambda_{t_1} = C_\Lambda^{-1} L^{c_\#(s-u)} = \Lambda,$$

with Λ as in Def. 11.5. ■

Proof. Lem. B.19 gives the pointed format and the free-family data on the open set $S_b \supset \mathcal{P}$, as well as $d_b(s) = d_b(0)$. The joint flavour map preserves the interior of $\mathcal{P}$: it preserves its defining norms, is linear and continuous, and has the continuous inverse with V^{-1}. This verifies the precise source-domain clause in the format. Prop. B.22 gives the three lag sums with the sizes of these actual data. Finally $t_1 - t_0 = s - u$ in either datum and $p_\Lambda = 0$, so the base value is exactly the displayed Λ. □

B.10. **Continuing the restart to the last level.** The zero-source restart can be continued beyond t_1 to level K. We need this continuation for two reasons: its last-level integral gives a positive lower bound for the normalizing total, and its histories bound the error from leaving the small-field region after the restart.

Definition B.24 (Histories above the restart)**.** Use the model and constants of Def. 11.1, including the two conditions on the gauge tests in the construction. Let $0 < g \le \gamma$, $\ell \in \mathfrak{L}(g)$ and $(\mathsf{b}, \mathsf{m}) \in \mathsf{W}(g)$; assume (k0). For either cutoff write $1 \le u < s < K$, $t_0 = K - s$, $t_1 = K - u$, $\mathsf{t}_i = L^{K-i}\mathsf{b}$, $\beta = L^K \mathsf{b}$, and $m_0 = L^{-K}\mathsf{m}$. All gauge data and the fermion tower are those of this datum. Put $\mathcal{Y} = \mathcal{V}_{t_0}$, with product Haar probability measure dW, and $W_k = \mathcal{M}^{k-t_0} W$. Write $\mathcal{V}_k^\chi = \{\chi_k(T_\ell) = 1\}$.

The start is

$$\hat\tau_{t_0}(dW) = \chi_{t_0}(T_\ell)(W) \exp[-\beta A(\mathsf{U}) - \langle \bar\Psi, \mathbb{D}_{t_0}(\mathsf{U})\Psi\rangle - L^{-t_0}\hat s_{t_0} \mathsf{S}_1(\psi^{\mathrm{bg}})]\, dW,$$

where $\mathsf{U} = \mathsf{U}_{t_0}(W)$, $\hat s_{t_0} = F^{\mathrm{UV}}/\mathsf{t}_{t_0}$ and $\psi^{\mathrm{bg}} = \mathcal{H}_{h_\emptyset}\Psi$. It has $\hat x_{t_0} = \mathsf{t}_{t_0}$, $\hat E_{t_0} = 0$, and no activity, normalizer or number of a level $\le t_0$. Let $\hat{\mathbb{D}}(W)$ be the quadratic form in its Grassmann exponent, and put

$$\hat F(W) = e^{-\beta A(\mathsf{U})} \det \hat{\mathbb{D}}(W), \qquad \hat\nu = \chi_{t_0}(T_\ell)\hat F\, dW.$$

Let $\hat\tau_k$ be the restarted small-field term at level k, with numbers $\hat x_i, \hat s_i, \hat e_i, \hat b_i, \hat\mu_i$ and $\hat E_k = \sum_{t_0 < j \le k} \hat e_j n_j$. If $\mathbf{1}_j$ is the indicator that all tests pass in the step from level j and $\mathfrak{G}_j \ge 0$ its gauge-fixing factor, set

$$\hat{\mathsf{k}}^{(k)}(W) = \prod_{t_0 \le j < k} (\mathbf{1}_j \mathfrak{G}_j)(W_j).$$

The histories above the restart are $\mathcal{H}_k^{(t_0)} = \{h \in \mathcal{H}_k : h|_{t_0} = h_\emptyset^{(t_0)}\}$. Their terms are the images of $\hat\tau_{t_0}$ under their subsequent weights, partial kernels and partial Berezin integrals. Denote their densities against $\mathscr{M}_h^{\mathrm{fr}}$ by $\hat{\mathsf{g}}_h$, and their fully Berezin-integrated masses by $\hat t(h)$. ■

Lemma B.25 (The start and the field cut)**.** Every $W \in \mathcal{V}_{t_0}^\chi$ has

$$\varpi(W) < 2\varepsilon(\mathsf{t}_{t_0}) \le \varepsilon_{\mathcal{W}}\alpha_0 \le \bar\varepsilon.$$

Its background exists, and $\mathsf{wab}(L^s\mathsf{b}, W) = \beta A(\mathsf{U}_{t_0}(W))$. Consequently the zero-source start of Def. 11.2 is $\hat\tau_{t_0}$. If $L^u\mathsf{b} \ge \Theta_\star$, the two-lattice gauge comparison is admitted and

$$\mathcal{V}_{t_0}^\chi \subset \mathsf{In}_{t_0} \subset \mathsf{Cut}_{(t_0)}.$$

The same action identity holds on the field cut. Moreover

$$\mathbf{1}_{t_0}\chi_{t_0}(T_\ell) = \mathbf{1}_{t_0}, \qquad \mathbf{1}_{t_0} 1_{\mathsf{Cut}_{(t_0)}} = \mathbf{1}_{t_0} \quad (L^u\mathsf{b} \ge \Theta_\star).$$

These identities also hold with the weight $\hat{\mathsf{k}}^{(t_1)} = \hat{\mathsf{k}}^\bullet$ in place of $\mathbf{1}_{t_0}$. ■

Proof. The tested-field estimate, applied at level $t_0 - 1$, gives the displayed plaquette bound. The lattice is large at that level: (k0) has $K^{\mathrm{A}} \ge 3$, and the two lattices of the gauge comparison are large at levels $K^{\mathrm{A}} - 1$ and K^{A} respectively. Their calibrated frames therefore admit the minimizing background at t_0. The weighted action is $A^{\eta_{t_0}}(\mathsf{c}, U) = L^{t_0}\sum_p \mathsf{c}(p) a_p(U)$, constant on the minimal orbit. At $\mathsf{c} = L^s\mathsf{b}$ it is $\beta A(U)$. This also identifies the tested start, including its fine mass, mass member, tower and Haar measure.

When $L^u\mathsf{b} \geq \Theta_\star$, the gauge comparison with top level t_1 gives its inner set $\mathsf{In}_{t_0} = \{\varpi < 2\varepsilon(\mathsf{t}_{t_0})\}$ and its inclusion in the field cut. This proves the assertion about the cut and the action there. The first small-field step contains the tests

$$\chi_{t_0+1}(T_\ell)(\mathcal{M}W), \mathsf{q}^0_{t_0}(W), \mathsf{q}_{t_0+1}(W), \chi^{\mathrm{sf}}_{\mathsf{t}_{t_0}}(W),$$

where $\mathsf{q}^0_{t_0} = 1$ since $t_0 \geq 1$. The density-test descent property of these calibrated tests gives $\chi_{t_0}(T_\ell)(W) = 1$ whenever this product is one. The thresholds satisfy the descent conditions because the construction imposes, with $\varepsilon_\gamma = A_0\gamma(\log\gamma^{-2})^{\mathfrak{p}_0}$,

$$\varepsilon_\gamma \leq \min\{\bar{\varepsilon}/2, \varepsilon_\flat\}, \qquad C_{\mathrm{reg}}\varepsilon_\gamma \leq \min\{\varepsilon_{\mathrm{v}}, L^{-2}c_2^{\mathrm{SO}}, L^{-2}\bar{\varepsilon}/2\}.$$

The remaining strip-comparison and density-descent bounds are precisely [YM3, eq. 4.ccclxxv], [YM3, eq. 4.ccclxxviii], [YM3, eq. 4.cccxcii], [YM3, eq. 4.cccxcvii], [YM3, Prop. 4.68], and [YM3, Prop. 4.74]; their uniform admission is checked in the proof of [YM3, Lem. 5.58]. Since $\varepsilon(\mathsf{t}_j) \leq \varepsilon_\gamma$ and the calibrated geometry is admitted at every step, these bounds apply to the present frames. This proves the first indicator identity; the inclusion just proved gives the second. The product weight contains this first-step indicator, so has the same identities. □

Proposition B.26 (The zero-source continuation)**.** The restart is defined to level K, on the given torus, every admitted torus with the same other data, and $\mathbb{Z}^3$. Its activities satisfy the original locality, symmetry, holomorphy and scalar-array conditions, and the closed bounds

$$\|\hat{F}_0^{(j)}\|_j \leq C_w\phi(\mathsf{t}_{j-1}), \qquad \|\hat{E}^{(j)}_{\geq 2}\|_j \leq C_2\phi(\mathsf{t}_{j-1})^2, \qquad \|\hat{R}^{(j)}\|_j \leq C_3\phi(\mathsf{t}_{j-1})^3.$$

The first two hold also for each pointed term, on both lattices. The explicit part on $\mathbb{Z}^3$ is $\hat{T}^{(j)} = T^{(j)}(\hat{y})$, with $\hat{y}_i = 0$ for $i < t_0$ and $\hat{y}_i = 1/\hat{x}_i$ later. The extracted numbers are real, read from that infinite-lattice family, and independent of the side and weight. They obey

$$\hat{x}_i = L^{-1}\hat{x}_{i-1} - \hat{b}_i, \qquad \hat{s}_i = L\hat{s}_{i-1} + \hat{\mu}_i, \qquad \hat{E}_i = \hat{E}_{i-1} + \hat{e}_i n_i,$$

and

$$|\hat{x}_k - \mathsf{t}_k| \leq (1 - L^{-(k-t_0)})\bar{B}, \qquad F^{\mathrm{UV}} - \tfrac{1}{2} \leq \mathsf{t}_k\hat{s}_k \leq F^{\mathrm{UV}} + 2B_F + \tfrac{1}{2},$$
$$|\hat{s}_k| \leq \bar{B}_M/\mathsf{t}_k, \qquad \hat{x}_k \geq g^{-2} \geq \gamma^{-2}, \qquad \hat{x}_k \geq \tfrac{7}{8}\mathsf{t}_k, \qquad \hat{x}_K \leq X_g.$$

Every step is the admitted zero-source RG step, including its invariant symmetry space. The density of $\hat{\tau}_k$ is bounded Borel, zero off $\mathcal{V}_k^\chi$, and is the RG expression on that set, with no member of a level $\leq t_0$. If $u \geq u_*$, this continuation agrees through t_1 with the zero-source flow of Prop. B.13. ■

Proof. Apply Prop. B.5 at $c = 0$ and $\zeta = F^{\mathrm{UV}} \in [0, B_F]$. That proposition forms the zero-source restart to K and verifies the original-step applicability from the new base. It gives the displayed norms, decomposition, recursions and the first two windows. The zero-source symmetry space gives matrix cell cancellation on $\mathbb{Z}^3$, so the old-activity estimates hold there as on a torus. For a scalar-array bound on a given infinite-lattice polymer, choose a sufficiently large admitted torus into which it fits. The local chart preserves the scalar values, pointers, norms and polymer sizes, and the torus estimate uses the same constant E_0. Thus it gives the infinite-lattice scalar bound.

The term has a bounded Borel density at the base by Lem. B.12; each tested step preserves that property and supplies its stated expression on $\mathcal{V}_k^\chi$. Its numbers are the linear extraction functionals on the infinite-lattice activity, which here coincide with the original extraction rules because its totals have the required symmetries.

The first coupling window and $\mathsf{b} \geq g^{-2} + \bar{B}$ imply $\hat{x}_k \geq g^{-2}$. The ceiling $4\bar{B}\gamma^2 \leq \sigma_\star \leq 1/2$ gives $\bar{B} \leq \mathsf{t}_k/8$, hence the 7/8 bound. At the last level $\mathsf{b} \leq X_g - \bar{B}$, proving $\hat{x}_K \leq X_g$. Finally the source restart's chain is zero at zero source: its initial value is zero and $\hat{\nu}_{k+1}(0) = \hat{\mu}_{k+1}$. Its source energies then vanish and its steps are the same zero-source steps with the same numbers. Induction identifies the two flows through t_1. □

Lemma B.27 (Admission above the empty earlier family)**.** Set $E^{(j)} = 0$ for $j \le t_0$ and $E^{(j)} = \hat{E}^{(j)}$ later. With the restarted numbers, these form a flow with free earlier activities as in Prop. 11.14, at base t_0, with $R^{(j)} = D^{(j)} = 0$ for $j \le t_0$. All earlier-family error parameters are zero. For $j > t_0$, on both lattices,

$$\sup_{\square} \sum_{X \supset \square} e^{\kappa d_j(X)} \sup_{\mathsf{U}} \sum_{z \in X} \|\hat{E}^{(j)}_{\ge 2}(X; \mathsf{U}, \cdot; z)\|_{h^{(j)}} \le C_2 \phi(\mathsf{t}_{j-1})^2,$$
$$\|\hat{E}^{(j)}_{\ge 2}\|^{\mathrm{pt}}_j \le C_2^{\mathrm{pt}} \phi(\mathsf{t}_{j-1})^2, \qquad C_2^{\mathrm{pt}} = 8M^3 C_2.$$

The exact terms in Prop. 11.14 are the restarted terms here, with scalar $\hat{E}_k$ and action coefficient $\beta_h = \beta - \sum_{t_0 < j \le k} L^j \hat{b}_j \varphi_j$. Its action bounds also hold with β itself. ■

Proof. The zero family satisfies all earlier-family conditions, and both lag sums are zero. For the later families, Prop. B.26 gives the required symmetries, localization, holomorphy, matrix cell cancellation and torus agreement. The scalar parts contain no action multiple and, on $\mathbb{Z}^3$, no constant; their torus constants are field independent. These are the first free-flow conditions.

We verify the norm summed over pointers. At a fixed pair the shares $n^{-1} \sum_{\mathsf{z}_j(p)=z} \mathsf{N}_p$ of a monomial with $n \ge 2$ legs lie in $[0,1]$ and sum to one over z. Taking the infimum over representations in the fibre norm gives

$$\sum_z \Big\| \sum_{\mathsf{z}_j(p)=z} F_p \Big\| \le \|F\|.$$

The expansion majorant is nonnegative and multilinear in the vertex norms, and the pointed term is the average of the terms with one leg vertex replaced by its insertion. Thus these sums pass under the gauge majorant integral, Berezin contractions and Cauchy integrals of the expansion. They obey the same coefficient bounds as the whole expansion. The vanishing coefficients in the deformation vanish pointer by pointer. The remaining mass coefficient is at most $K_\mu^{-1} c_s h_F^2 |\hat{s}_{j-1}|^2$. Mass extraction adds forms on disjoint site blocks; their norms sum to at most $c_{\mathsf{S}} h_F^2 |\hat{\mu}_j|$. Extraction therefore multiplies the bounds by at most K_μ, giving

$$\tfrac{1}{2} C_2 \phi(\mathsf{t}_{j-1})^2 + c_s h_F^2 |\hat{s}_{j-1}|^2 + \theta_E \sum_{t_0 < i < j} \omega_{j-i} \|\hat{E}^{(i)}_{\ge 2}\|_i.$$

The closed bounds and mass window make this at most $(1/2 + 1/4 + 1/(4\sqrt{2})) C_2 \phi(\mathsf{t}_{j-1})^2$, which proves the first displayed estimate. For the separate supremum at each pointer, apply the pointed-output estimate of [QCDM1] at the actual step. Its step-admission hypothesis is Prop. B.26; the mass is the original datum's fine mass, $|\hat{s}_{j-1}| \le \bar{B}_M / \mathsf{t}_{j-1}$, and all earlier leg activities have the closed C_2 bound. The estimate is therefore C_2^{pt}.

The remaining number bounds also hold: $|\hat{b}_i| \le B_\sharp$,

$$|\hat{\mu}_i| \le 2\bar{B}_M / \mathsf{t}_i \le 3\bar{B}_M / \hat{x}_i, \qquad |\hat{e}_i| \le C'_e (1 + \log \hat{x}_{i-1}).$$

The latter follows by comparing the extracted scalar with the free gauge and block-fermion vacuum numbers, using $|\hat{x}_{i-1} - \mathsf{t}_{i-1}| \le \bar{B}$. Thus all the free-flow hypotheses hold, including the pointed and scalar bounds and the fixed (Γ^{F}) potential ceilings. Its base density is exactly $\hat{\tau}_{t_0}$, so its terms and extracted scalar are those here. Finally $L^{t_0} \hat{x}_{t_0} = \beta$, giving the stated action coefficients and the bound with β. □

Proposition B.28 (History bounds after the restart)**.** Let $C_5^{\mathrm{F}}, C_E^{\mathrm{F}}, C_\flat^{\mathrm{F}}$ be the zero-source history constants of the construction, and set $\mathsf{C}^\dagger_{\mathrm{maj}} = 48 L^3 C_E^{\mathrm{F}} + C'_\star$, $\theta_{\mathrm{dis}} = (c_\star - c_\Theta)/2$. The restarted histories have the exact history form and bounds of Prop. 11.14, with the original constants, $\hat{E}_k$ in place of E_k and $\hat{f}_n = 0$ for $n \le t_0$. Their vacuum numbers obey

$$|\hat{f}_n(\mathcal{S})| \le C_\flat^{\mathrm{F}} (1 + \log \mathsf{t}_{n-1}) \#_n(Z_n(\mathcal{S})^\sim),$$

and

$$e^{-\hat{E}_k+\sum_n \operatorname{Re}\hat{f}_n(\mathcal{S})}\int \|\hat{\mathsf{g}}_h\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}} \le \exp[\mathsf{C}_{\mathrm{maj}}^{\dagger}(n_k+\Sigma_Z(h))] \cdot \prod_{t=1}^{k} e^{-c_\star p_0(g_t)^2 \mathsf{n}_t^{\mathrm{cr}}(h)}.$$

For every $t_0 < t \le K$,

$$\sum_{h\in\mathcal{H}_K^{(t_0)}:Z_t(h)\neq\emptyset}\int \|\hat{\mathsf{g}}_h\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}} \le e^{-\theta_{\mathrm{dis}}p_0(g_t)^2} e^{\hat{E}_K+(C+2)\ell^3},$$

$$\sum_{h\in\mathcal{H}_K^{(t_0)}}\int \|\hat{\mathsf{g}}_h\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}} \le e^{\hat{E}_K+(C+2)\ell^3}.$$

Here $Z_t(h)\neq\emptyset$ means that a large-field region has occurred by level t; such a history has a created cell at some level $t_0 < t' \le t$. ■

Proof. Apply Prop. 11.14 using Lem. B.27. All earlier-family errors vanish, so the constants retain their zero-source values. The initial coupling correction is also zero. This gives the form, vacuum bound and integral majorant.

Extend the normalized integral majorant by zero to histories outside $\mathcal{H}_K^{(t_0)}$, and call it $f(h)$. Put $\mathsf{e}(\mathcal{S}) = -\sum_n \operatorname{Re}\hat{f}_n(\mathcal{S})$ when $Z_{t_0}(\mathcal{S})=\emptyset$, and zero otherwise. Thus fe^{e} is the history integral multiplied by $e^{-\hat{E}_K}$ on the retained histories. The equivalence $Z_{t_0}(h)=\emptyset \iff h|_{t_0} = h_\emptyset^{(t_0)}$ ensures this extension respects the history labels. Define

$$f'(h) = f(h)\exp\left[\theta_{\mathrm{dis}}\sum_{t'=1}^{K} p_0(g_{t'})^2 \mathsf{n}_{t'}^{\mathrm{cr}}(h)\right].$$

Its decay rate is $c = c_\star - \theta_{\mathrm{dis}} = (c_\star + c_\Theta)/2$. The calibrated history-summation theorem applies at this rate, with deposit c, weight share $c_\Theta/c < 1$ and the already imposed thresholds $\log\gamma^{-2} \ge \mathsf{Y}_{\mathrm{f1}}$ or Y_{f2} according to $c_a = 0$ or $c_a > 0$. The required floors are those used for the original history sum. Its majorant and label constants are at least the ones above (the source constants and $C'_{\mathfrak{e}}$ only enlarge them). Consequently

$$\sum_h f'(h) e^{\mathsf{e}(\mathcal{S}(h))} e^{\Theta'(h)} \le \Pi_K(C,c'') \le e^{(C+2)\ell^3}, \qquad c'' = \tfrac{1}{2}(1-\eth)(c_\star - c_\Theta), \quad \Theta' \ge 0.$$

This is the same calibrated history sum proved in [QCDM1, §14]; its inputs have just been checked for the restarted families. Since $f \le f'$, it gives the second estimate. If $Z_t(h)\neq\emptyset$, some cell was created at $t_0 < t' \le t$ and $p_0(g_{t'})^2 \ge p_0(g_t)^2$. Thus $f'(h) \ge e^{\theta_{\mathrm{dis}}p_0(g_t)^2} f(h)$ on this set. The same sum gives the first estimate. Nesting of the large-field regions proves the last interpretation. □

B.11. **Positivity and a lower bound for the restarted total.**

Lemma B.29 (The Gaussian at the start)**.** For a bare configuration U, put $\mathfrak{a} = \mathfrak{a}_{t_0}$, $R = R_{t_0}(U)$, $\mathcal{K} = \mathcal{K}_{t_0}[U;m_0]$, and $\mathbb{D}_{t_0} = L^{t_0}\mathsf{D}^{(t_0)}[U;m_0]$. Define

$$\hat{\mathbb{M}} = \hat{s}_{t_0}\mathfrak{a}^2 R\mathcal{K}^{-2}R^*, \qquad \hat{\mathbb{D}} = \mathbb{D}_{t_0} + \hat{\mathbb{M}}, \qquad \epsilon_{\mathrm{st}} = \frac{F^{\mathrm{UV}}}{\mathsf{bm}}.$$

The mass form in the start is $L^{-t_0}\hat{s}_{t_0}\mathsf{S}_1(\psi^{\mathrm{bg}}) = \langle\bar{\Psi},\hat{\mathbb{M}}\Psi\rangle$ and

$$|\langle\chi,\hat{\mathbb{M}}\chi\rangle| \le \epsilon_{\mathrm{st}}\operatorname{Re}\langle\chi,\mathbb{D}_{t_0}\chi\rangle, \qquad \epsilon_{\mathrm{st}} \le c_b/24.$$

Consequently

$$\operatorname{Re}\hat{\mathbb{D}} \ge (1-\epsilon_{\mathrm{st}})\frac{\theta_{t_0}\mathfrak{m}_{t_0}}{\theta_{t_0}+\mathfrak{m}_{t_0}} > 0.$$

Starting with $\hat{\mathbb{D}}_{t_0} = \hat{\mathbb{D}}$ and $N_{t_0} = 1$, define, for $t_0 \le j < K$,

$$\hat{\Gamma}_j = \hat{\mathbb{D}}_j + \theta q^* q, \qquad \hat{\mathbb{D}}_{j+1} = \theta - \theta^2 q \hat{\Gamma}_j^{-1} q^*,$$
$$N_{j+1} = N_j \theta^{-\nu n_{j+1}} \det \hat{\Gamma}_j, \qquad \theta = b_F, \quad q = q_{j+1}(W_j), \quad W_j = \mathcal{M}^j U.$$

All these operators are defined and accretive, and

$$\hat{\mathbb{D}}_j^{-1} = \theta^{-1} \sum_{0 \le i < j - t_0} L^{-i} + Q_j \hat{\mathbb{D}}^{-1} Q_j^*, \qquad Q_j = q_j \cdots q_{t_0+1}, \quad Q_j Q_j^* = L^{-(j-t_0)}.$$

At the last level,

$$\operatorname{Re} \hat{\mathbb{D}}_K \ge (1 - \epsilon_{\mathrm{st}}) c_b \mathsf{m} \ge \tfrac{23}{24} c_b \mathsf{m} \ge \tfrac{1}{2} c_b \mathsf{m}_-.$$

Every singular value has the same lower bound.

The one-step Gaussian identity is

$$\mathrm{Ber}_j \left(\mathrm{Ker}_j(W_j) N_j e^{-\langle \bar{\Psi}_j, \hat{\mathbb{D}}_j \Psi_j \rangle} \right) = N_{j+1} e^{-\langle \bar{\Psi}_{j+1}, \hat{\mathbb{D}}_{j+1} \Psi_{j+1} \rangle},$$

and $N_j \det \hat{\mathbb{D}}_j = \det \hat{\mathbb{D}}$. The two chiral blocks remain adjoints, so

$$\det \hat{\mathbb{D}}_j = |\det \hat{\mathbb{D}}_{j,+}|^{2N_f} > 0, \qquad \det \hat{\Gamma}_j = |\det \hat{\Gamma}_{j,+}|^{2N_f} > 0, \qquad N_j > 0.$$

In particular $\det \hat{\mathbb{D}} \ge N_K (\frac{1}{2} c_b \mathsf{m}_-)^{\nu \ell^3}$. At a tested background this is the Gaussian of the start; it is constant on its minimal orbit. The measure $\hat{\nu}$ of Def. B.24 is finite and positive, with bounded Borel, gauge-invariant density, strictly positive on $\mathcal{V}_{t_0}^{\chi}$. ∎

Proof. Differentiate the original zero-source tower with respect to m_0, keeping U fixed. Since $\partial_{m_0} \mathbb{D}_0 = 1$ and

$$\partial_{m_0} \mathbb{D}_{j+1} = \theta^2 q \Gamma_j^{-1} (\partial_{m_0} \mathbb{D}_j) \Gamma_j^{-1} q^*,$$

iteration through the background maps gives $\langle \bar{\Psi}, \partial_{m_0} \mathbb{D}_{t_0} \Psi \rangle = \mathsf{S}_1(\psi^{\mathrm{bg}})$. The closed formula $\mathbb{D}_{t_0} = L^{t_0}(\mathfrak{a} - \mathfrak{a}^2 R \mathcal{K}^{-1} R^*)$ then proves the mass identity.

For its relative bound set $w = \mathcal{K}^{-1} R^* \chi$, $\varphi = \mathfrak{a} w$, and $v = \chi - \mathfrak{a} R w$. Then $R^* v = (D[U] + m_0) w$, and

$$\langle \chi, \mathsf{D}^{(t_0)} \chi \rangle = \mathfrak{a} |v|^2 + \mathfrak{a}^2 \langle w, (D[U] + m_0) w \rangle.$$

Accretivity of $D[U]$ gives $\operatorname{Re} \langle \chi, \mathbb{D}_{t_0} \chi \rangle \ge \mathfrak{m}_{t_0} |\varphi|^2$. Apply the same calculation with $\mathcal{K}^*$ and $\varphi' = \mathfrak{a} (\mathcal{K}^*)^{-1} R^* \chi$; the adjoint quadratic form has the same real part. Since $\langle \chi, \hat{\mathbb{M}} \chi \rangle = \hat{s}_{t_0} \langle \varphi', \varphi \rangle$, the desired bound follows by Cauchy–Schwarz. Here $\hat{s}_{t_0} / \mathfrak{m}_{t_0} = F^{\mathrm{UV}} / (\mathsf{bm})$. The original positive-mass coupling ceiling gives $\mathsf{bm} \ge 8 C_{\mathsf{S}_1} \bar{B}_M / c_b$, and $F^{\mathrm{UV}} \le B_F < \bar{B}_M / 3$, with $C_{\mathsf{S}_1} \ge 1$. Thus $\epsilon_{\mathrm{st}} \le c_b / 24$. The original block-Dirac floor $\operatorname{Re} \mathbb{D}_{t_0} \ge \theta_{t_0} \mathfrak{m}_{t_0} / (\theta_{t_0} + \mathfrak{m}_{t_0})$ proves the floor for $\hat{\mathbb{D}}$.

Let this floor be $c > 0$ and write $P_j = \theta^{-1} \sum_{0 \le i < j - t_0} L^{-i}$. If $\operatorname{Re} \hat{\mathbb{D}}_j > 0$, then $\hat{\Gamma}_j$ is invertible and the matrix inversion identity gives

$$\hat{\mathbb{D}}_{j+1}^{-1} = \theta^{-1} + q \hat{\mathbb{D}}_j^{-1} q^*.$$

Since $q q^* = L^{-1}$, this proves the stated inverse formula by induction. It also preserves a quantitative floor. Set $g = \hat{\mathbb{D}}_{j+1}^{-1} f$ and $y = \hat{\mathbb{D}}^{-1} Q_{j+1}^* f$. Then

$$\operatorname{Re} \langle g, \hat{\mathbb{D}}_{j+1} g \rangle \ge P_{j+1} |f|^2 + c |y|^2, \qquad |g|^2 \le (P_{j+1} + L^{-(j+1-t_0)}/c)(P_{j+1} |f|^2 + c |y|^2).$$

Hence $\operatorname{Re} \hat{\mathbb{D}}_{j+1} \ge (P_{j+1} + L^{-(j+1-t_0)}/c)^{-1}$. At K, put $a_- = b_F (1 - L^{-1})$. Since $\theta_{t_0} \ge a_-$,

$$P_K + L^{-s}/c \le (1 - \epsilon_{\mathrm{st}})^{-1} (a_-^{-1} + \mathsf{m}^{-1}) \le [(1 - \epsilon_{\mathrm{st}}) c_b \mathsf{m}]^{-1}.$$

The last inequality uses $c_b = a_- / (a_- + 1/2)$ and $\mathsf{m} \le 1/2$. This proves the last-level floor, and its singular-value version follows from $|\hat{\mathbb{D}}_K g| \, |g| \ge \operatorname{Re} \langle g, \hat{\mathbb{D}}_K g \rangle$.

Completing the square in the kernel with $\Psi_j = \hat{\Gamma}_j^{-1} \theta q^* \Psi_{j+1} + \Xi$ and the corresponding barred substitution gives the Gaussian identity. Berezin translation invariance and the kernel factor $\theta^{-\nu n_{j+1}}$

give exactly N_{j+1}. Integrating the next-level generators uses the kernel normalization one and gives $N_j \det \hat{\mathbb{D}}_j = N_{j+1} \det \hat{\mathbb{D}}_{j+1}$.

The projections $e_\pm$ commute with the averages, the isometry to the chamber fields and all real scalar coefficients. The two blocks of $D[U]$ are adjoints, and R^*R, q^*q are self-adjoint. Thus the same adjoint relation holds for $\mathcal{K}$, $\mathbb{D}_{t_0}$ and $\hat{\mathbb{M}}$, and is preserved by the recursion. Invertibility makes the displayed determinants strictly positive, and hence $N_j > 0$. The last determinant bound follows in dimension $\nu n_K = \nu \ell^3$.

Two backgrounds in the minimal orbit over W differ by a gauge transformation equal to one at level t_0. Covariance of D and R therefore leaves $\hat{\mathbb{D}}$ unchanged. A gauge transformation of W lifts to the bare blocks and conjugates this operator, leaving its determinant and Wilson action unchanged. Finally $\hat{\nu}$ is the Berezin integral of the bounded Borel start density from Prop. B.26; the positivity just proved gives all remaining assertions. □

Lemma B.30 (Positive masses and consistency)**.** With the operators of Lem. B.29, evaluated at $\mathsf{U}_{t_0}(W)$, the small-field term is

$$\hat{\tau}_k = (\mathcal{M}^{k-t_0})_* \left[\hat{\mathsf{k}}^{(k)} \chi_{t_0}(T_\ell) e^{-\beta A(\mathsf{U})} N_k e^{-\langle \bar{\Psi}_k, \hat{\mathbb{D}}_k \Psi_k \rangle} \, dW \right].$$

The integrand is zero off $\mathcal{V}^{\chi}_{t_0}$. Its Berezin integral and its degree-zero part are finite positive measures; the former is $(\mathcal{M}^{k-t_0})_*(\hat{\mathsf{k}}^{(k)}\hat{\nu})$. At K the Berezin integral is at least $(\frac{1}{2} c_b \mathsf{m}_-)^{\nu\ell^3}$ times the degree-zero part, as measures.

For each $h \in \mathcal{H}_k^{(t_0)}$ there is a bounded Borel weight $\mathfrak{w}_{h/t_0} \geq 0$ on $\mathcal{Y}$ such that, pointwise on bare configurations,

$$\mathfrak{w}_h(U) = \mathfrak{w}_{h^{(t_0)}_\emptyset}(U) \mathfrak{w}_{h/t_0}(\mathcal{M}^{t_0} U), \qquad \hat{t}(h) = \int_{\mathcal{Y}} \mathfrak{w}_{h/t_0} \, d\hat{\nu} \geq 0.$$

In particular $\mathfrak{w}_{h^{(k)}_\emptyset/t_0} = \hat{\mathsf{k}}^{(k)}$. There is a bounded Borel nonnegative gauge-invariant bare density $\hat{\varsigma}$ such that every fully Berezin-integrated restarted term is the ordinary gauge history term at that density. Consequently, for $t_0 \leq k < k' \leq K$ and $h \in \mathcal{H}_k^{(t_0)}$,

$$\sum_{h'|_k = h} \hat{t}(h') = \hat{t}(h), \qquad \hat{t}(h^{(k)}_\emptyset) \geq \hat{t}(h^{(K)}_\emptyset).$$

■

Proof. The first identity follows by induction from the Gaussian identity of Lem. B.29. The maps, weights and kernel entries are Borel. The operator recursion uses finitely many rational operations with nonvanishing determinants, so its entries are Borel as well. Multiplying a pushforward by a function of the output multiplies its original measure by that function composed with the map. This observation commutes the next step through the displayed image. The identity $N_k \det \hat{\mathbb{D}}_k = \det \hat{\mathbb{D}}$ then gives the Berezin integral. Positivity of N_k and the determinants gives both positive measures; the final determinant floor gives their stated comparison.

For a history above t_0, all earlier regions are the torus. Thus its later determining sets contain no bonds from levels below t_0. The test property (T5) makes each later outcome indicator a function only of W_j, $j \geq t_0$; it is Borel by (T2). The gauge-fixing factors read the same block fields and are bounded Borel. Their product defines $\mathfrak{w}_{h/t_0}$ at every W and gives the pointwise factorization.

We construct a bare density to justify consistency for these new terms. Let $\varsigma_0 = e^{-\beta A} \det(D + m_0)$, and let $\rho \, dW$ be the ordinary small-field image $(\mathcal{M}^{t_0})_*(\mathfrak{w}_{h^{(t_0)}_\emptyset} \varsigma_0 \, dU)$. Its density is bounded Borel and nonnegative almost everywhere. It is invariant under block gauge transformations as a measure: their constant-on-block lifts preserve the original density, Haar measure, tests and gauge-fixing

factors. Haar averaging gives

$$\bar\rho(W)=\int \rho(W^v)\,dv,\qquad \bar\rho=\rho\quad\text{almost everywhere,}$$

with $\bar\rho$ bounded Borel and pointwise invariant.

The original intermediate-level lower bound supplies $\rho\ge c_1\mathsf{sf}_{t_0}$ almost everywhere, where

$$c_1=\exp\left[E_{t_0}-\left(C_5^{\mathrm F}+\nu\log(\theta_{t_0}^{-1}+\mathfrak{m}_{t_0}^{-1})\right)n_{t_0}\right]>0.$$

On the tested set, $0\le a_p\le 2$ gives $\mathsf{sf}_{t_0}\ge c_2:=e^{-2L^{t_0}x_{t_0}N_{\mathrm P}}>0$. Thus $\bar\rho\ge c_1c_2\chi_{t_0}(T_\ell)$ almost everywhere. Define

$$r(W)=\frac{\chi_{t_0}(T_\ell)(W)\hat F(W)}{\bar\rho(W)}\quad\text{if }\chi_{t_0}(T_\ell)=1,\bar\rho\ge c_1c_2,\qquad r=0\quad\text{otherwise.}$$

This is bounded Borel, nonnegative and pointwise gauge invariant, with $r\rho=\chi_{t_0}\hat F$ almost everywhere. Therefore

$$\hat\varsigma(U)=\varsigma_0(U)r(\mathcal{M}^{t_0}U)$$

is a bounded Borel nonnegative density invariant under every bare gauge transformation. The factorization gives, for every retained history,

$$\int \mathfrak{w}_h\hat\varsigma\,dU=\int \mathfrak{w}_{h/t_0}r\rho\,dW=\int \mathfrak{w}_{h/t_0}\,d\hat\nu.$$

The same identity with any bounded Borel test at $\mathcal{M}^{t_0}U$ identifies the image measures.

It remains to identify the terms on each history's coordinates, since total masses alone do not prove their consistency. Write ξ_h for those Borel partial coordinates. The domain contains the support of $\mathfrak{w}_h$. At a successor history,

$$\xi_{(h,\mathsf o)}=\Sigma_{(h,\mathsf o)}\circ\xi_h,\qquad \mathfrak{w}_{(h,\mathsf o)}=\mathfrak{w}_h(\mathsf f_{(h,\mathsf o)}\circ\xi_h),$$

on the respective domains, with bounded Borel $\mathsf f_{(h,\mathsf o)}\ge 0$. These are the coordinate and weight identities of Lem. C.26. The partial fermion kernel integrates to one in its new generators, and later kernels do not change stored generators. Hence full Berezin integration of the step sends a scalar measure ν_h to $(\Sigma_{(h,\mathsf o)})_*(\mathsf f_{(h,\mathsf o)}\nu_h)$. At the base, ν_h is the image of $\hat\nu$ under the one-to-one Borel coordinate chart. Induction and composition of partial pushforwards therefore give

$$\nu_h=(\xi_h)_*(\mathfrak{w}_h\hat\varsigma\,dU).$$

At a successor the factor $1_{\operatorname{dom}\xi_h}$ can be omitted because the new weight vanishes off that domain. This establishes the asserted identification as positive measures and their masses.

The calibrated gauge history consistency theorem now applies to the bounded Borel gauge-invariant density $\hat\varsigma$. Its test assumptions (T1)–(T4),(T6) are those of the construction, and its threshold condition follows from $\varepsilon(\mathsf t_j)\le\varepsilon(\gamma^{-2})\le\varepsilon_\flat$. Its pushforward keeps total mass, giving $\sum_{h'|_k=h}\hat t(h')=\hat t(h)$. Every extension remains above t_0. For the empty history the sum includes the final empty history, and all masses are nonnegative. This proves the last inequality. □

Proposition B.31 (A lower bound for the restarted total)**.** Put

$$\hat{\mathsf{sf}}_K(V)=\chi_K(T_\ell)(V)e^{-L^K\hat x_K A(\mathsf U_K(V))}$$

on $\mathcal{V}_K^\chi$, and zero elsewhere. This is a Borel function with values in $[0,1]$. As measures,

$$\operatorname{Ber}\hat\tau_K\ge(\tfrac12 c_b\mathsf m_-)^{\nu\ell^3}e^{\hat E_K-C_5^{\mathrm F}\ell^3}\hat{\mathsf{sf}}_K\,dV.$$

For $t_0 \le k \le K$,

$$\begin{aligned}\hat{t}(h_\emptyset^{(k)}) &\ge \hat{t}(h_\emptyset^{(K)}) \\ &\ge e^{\hat{E}_K - C_P^{(K)}(1+\log \hat{x}_K)\ell^3} \ge e^{\hat{E}_K - C_P(1+\log X_g)\ell^3},\end{aligned}$$

where $C_P^{(K)}$ and C_P are the original last-level and partition lower-bound constants. ■

Proof. The degree-zero part of the RG expression for $\hat{\tau}_K$ has density

$$d(V) = e^{\hat{E}_K} \hat{\mathsf{sf}}_K(V) e^{\mathring{\mathsf{rg}}_K(V)}.$$

For the empty final history all regions are the torus, all profiles are one, and the boundary sum is empty. The regular scalar sum includes the trivial values of the activities. The history estimate of Prop. B.28 therefore gives $|\mathring{\mathsf{rg}}_K| \le C_5^{\mathrm{F}} \ell^3$ on the support of that history's weight.

We pass this bound to the density as a measure. Its degree-zero measure μ is positive and is bounded above by $(\frac{1}{2} c_b \mathsf{m}_-)^{-\nu\ell^3} \operatorname{Ber} \hat{\tau}_K$ by Lem. B.30. The latter is the block image of $\mathfrak{w}_{h_\emptyset^{(K)}} \hat{\varsigma}\, dU$. Thus μ vanishes on any Borel set containing no block image of a point where that weight times the density is positive. The Borel set

$$B = \{V \in \mathcal{V}_K^\chi : |d(V)| < e^{\hat{E}_K - C_5^{\mathrm{F}} \ell^3} \hat{\mathsf{sf}}_K(V)\}$$

contains no such image, by the regular-sum bound. Hence $\mu(B) = 0$. But $d \ge 0$ almost everywhere because μ is positive, and $d \ne 0$ everywhere on the tested set. Therefore B is Haar-null. It follows that $\mu \ge e^{\hat{E}_K - C_5^{\mathrm{F}} \ell^3} \hat{\mathsf{sf}}_K\, dV$. Combining this with the determinant comparison proves the first inequality.

The tested set is open and its background action is continuous. Indeed the fixed ceilings give

$$\varepsilon(\mathsf{t}_K) \le \varepsilon_\flat, \qquad 2\varepsilon(\mathsf{t}_K) \le \varepsilon_{\mathcal{W}} \alpha_0 \le \bar{\varepsilon},$$

the conditions for the tested-field background construction. This proves the Borel assertion. For its integral use the final block ball of radius $r_b = \hat{x}_K^{-1/2}$. The bounds $\hat{x}_K \ge \gamma^{-2}$, $\hat{x}_K \ge 3\mathsf{t}_K/4$ and the fixed ceilings $\gamma \le \bar{\varepsilon}/4$, $A_0(\log \gamma^{-2})^{\mathfrak{p}_0} \ge 8 B_3^{\mathrm{cr}}$ give

$$0 < r_b \le \tfrac{1}{4} \min\{\bar{\varepsilon}, A_0 (\log \mathsf{t}_K)^{\mathfrak{p}_0} \mathsf{t}_K^{-1/2} / B_3^{\mathrm{cr}}\}.$$

Thus the original final-block ball estimate applies: all density tests pass there, $L^K \hat{x}_K A(\mathsf{U}_K(V)) \le 24 (B_3^{\mathrm{cr}})^2 \ell^3$, and its Haar volume is at least $(\hat{x}_K^{-3/2}/7)^{3\ell^3}$. Consequently

$$\int \hat{\mathsf{sf}}_K\, dV \ge \exp[-(\tfrac{9}{2} \log \hat{x}_K + 3 \log 7 + 24 (B_3^{\mathrm{cr}})^2) \ell^3].$$

Taking total masses in the measure inequality proves the lower bound with

$$C_P^{(K)} = \max\{\tfrac{9}{2}, C_5^{\mathrm{F}} + 3 \log 7 + 24 (B_3^{\mathrm{cr}})^2 + 1 + \nu \log(2/(c_b \mathsf{m}_-))\}.$$

Finally $C_P \ge C_P^{(K)}$, $\hat{x}_K \le X_g$, and Lem. B.30 supplies the comparison with earlier empty histories. □

The restarted total at the comparison depth. Under $u \ge u_*$, $s \ge s_F$ and $L^u \gamma^{-2} \ge \Theta_\star$, these results give Prop. 11.15. Indeed Prop. B.26 continues the zero-source restart to K, and Prop. B.28 bounds its histories. On the initial field cut,

$$\hat{F}_0^\bullet = 1_{\mathsf{Cut}_{(t_0)}} e^{-\beta A(\mathsf{U}_{t_0})} \det \hat{\mathbb{D}} \ge 0.$$

It equals $\hat{F}$ on the tested set. Any history above t_1 contains the first small-field step, whose weight enforces both the density test and this cut by Lem. B.25. Therefore its mass is

$$\hat{t}(h) = \int \mathfrak{w}_{h/t_0} \hat{F}_0^\bullet\, dW \ge 0.$$

The masses at every later level sum to $\hat{T}^{\bullet}(0) = \hat{t}(h_{\emptyset}^{(t_1)})$ by Lem. B.30, and Prop. B.31 bounds this total below by $e^{\hat{E}_K - C_P(1+\log X_g)\ell^3}$.

For $t_0 < t \le t_1$, at any intermediate level the masses of histories having a large-field region by level t are at most their sum at K: each history is the sum of its extensions, the regions are nested, and all masses are nonnegative. Full Berezin integration is bounded by the fibre norm since $h_F \ge 1$. Hence Prop. B.28 and the lower bound give the relative estimate

$$e^{-\theta_{\mathrm{dis}} p_0(g_t)^2} e^{(C+2+C_P(1+\log X_g))\ell^3} \hat{T}^{\bullet}(0) = e^{-\theta_{\mathrm{dis}} p_0(g_t)^2} \mathsf{M}_{g,\ell} \hat{T}^{\bullet}(0).$$

B.12. **Source energies and separated supports.** We next estimate the scalar energies removed between the restart and the comparison level. Their dependence on a source is local up to an exponentially decreasing tail. This gives a small mixed derivative when two source supports are separated.

Definition B.32 (Energies after the restart)**.** Use the data of Def. 11.1, assume (k0) and $u \ge u_*$, and choose either cutoff. Write $t_0 = K - s$ and $t_1 = K - u$. Distances below are in the units of the bare lattice; a physical distance d is $\mathsf{d}L^K$ in these units. The support of a source includes its sites and its plaquette barycentres. For $t_0 < j \le t_1$, define

$$\hat{e}_j(X; s) = \hat{\mathfrak{E}}_j(X; \hat{\mathfrak{c}}_{j-1}(s)) - \hat{\mathfrak{E}}_j(X; 0).$$

Here $\hat{\mathfrak{E}}_j$ is the scalar energy of the restart of Def. B.9, with $\zeta = F^{\mathrm{UV}}$, at a free coordinate of level $j-1$. The domains and chains are those of Def. B.9 and Prop. B.10. ■

Lemma B.33 (Identification at source coordinates)**.** The flow of Def. 11.2(2) is the restart of Def. B.9 with $\zeta = F^{\mathrm{UV}}$ and label zero. The identification includes its activities, increments, source chains and scalar energies. At the initial level,

$$\hat{s}_{t_0} + \hat{\sigma}_{t_0}(s) = \hat{\Sigma}_w, \qquad \hat{\Sigma}_w(x) = \frac{F^{\mathrm{UV}}}{L^s(\mathsf{b} - w(x))}.$$

The identification holds on the source window $\|w\|_\infty < 8\varrho$ and $L\lambda_{t_0}\|J\|_\infty < 2r_M$. ■

Proof. This is the identification in Lem. B.12, on the larger source window stated there. Indeed, the action is read at the same weight, the tower at the same fine mass matrix, and the mass member at $\hat{\Sigma}_w$. Neither start contains earlier activities or normalizing factors. The density test and product Haar measure are identical. The inequalities $0 \le F^{\mathrm{UV}} \le B_F$, $|\mathsf{b} - w| \ge \mathsf{b}/2$ and $L^{t_0+1}\|\mathsf{n}\|_\infty < 2r_M$ put these expressions in the domains used in that proof. Equality on each minimal orbit gives equality of the bounded Borel densities, including at fields for which the density test vanishes. Induction under the same one-step map proves all the claimed identifications. □

Proposition B.34 (Locality of restarted source energies)**.** Let $t_0 < j \le t_1$ and $X \in \mathcal{D}_j$.

(a) The step from level $j-1$ is defined on $2\hat{\mathscr{D}}_{j-1}(a_{j-1}^{\mathrm{rad}})$, which contains $2\hat{\mathscr{D}}_{j-1}$. The function $\hat{\mathfrak{E}}_j(X; \cdot)$ is holomorphic there. It reads w and σ_i on X; it reads n on X and along the walks of the background maps. Those walks use cubes of side M_w in their level units and carry the factor $e^{-1/2}$ per step. On $2\hat{\mathscr{D}}_{j-1}$,

$$|\hat{\mathfrak{E}}_j(X; \hat{\mathfrak{c}}) - \hat{\mathfrak{E}}_j(X; 0)| \le C_{\mathfrak{E}} e^{-\kappa d_j(X)}.$$

(b) The map $s \mapsto \hat{\mathfrak{c}}_k(s)$ is holomorphic from $\mathscr{O}_k$ into $\hat{\mathscr{D}}_k$, and $\hat{\mathfrak{c}}_k(0) = 0$. For $t_0 \le i \le k \le t_1$,

$$|\hat{\sigma}_i(s)(x) - \hat{\sigma}_i(s')(x)| \le C_{\mathfrak{t}} \hat{\phi}_i \exp\left[-\frac{\kappa_\sigma \operatorname{dist}(x, \operatorname{supp}(s - s'))}{L^i}\right].$$

At $i = t_0$ the two values agree whenever $x \notin \operatorname{supp}(s - s')$.

(c) The function $\hat{e}_j(X;\cdot)$ is holomorphic on $\mathscr{O}_{j-1}$, vanishes at zero, and satisfies

$$|\hat{e}_j(X;s)| \le C_{\mathfrak{E}} e^{-\kappa d_j(X)}.$$

For $s, s' \in \mathscr{O}_{j-1}$,

$$|\hat{e}_j(X;s) - \hat{e}_j(X;s')| \le C_{\mathfrak{E}} e^{-\kappa d_j(X)} \exp\left[-\frac{\kappa_\sigma \operatorname{dist}(X, \operatorname{supp}(s-s'))}{L^j}\right].$$

All constants are the constants of the original source induction. ∎

Proof. By Prop. B.10(S1), the step at every free coordinate is the restarted step. By Lem. B.33, its restriction to a source chain is the actual restarted flow. Thus $\hat{e}_j(X;s)$ is the composition of a holomorphic function on the coordinate domain of level $j-1$ with the chain on $\mathscr{O}_{j-1}$. We verify the estimates and the two places in the symmetry proof that depend on the initial level.

Summing the mass recursion from t_0 gives

$$L^{-k}\big(\hat{s}_k + \hat{\sigma}_k(s)\big) = L^{-t_0}\hat{\Sigma}_w + \sum_{t_0 < i \le k} L^{-i}\hat{\nu}_i(\cdot;\hat{\mathfrak{c}}_{i-1}(s)).$$

Consequently, when the potential is read on shell, every local part extracted at a level $i > t_0$ cancels its copy in the mass term. The term left from the initial level is

$$-L^{-t_0}\mathsf{S}_1(\hat{\Sigma}_w\,\cdot\,) = -L^{-t_0}\sum_v \hat{\Sigma}_w(x_v)\bar{\psi}^{\mathrm{bg}}(v)\psi^{\mathrm{bg}}(v).$$

Its coefficient is scalar and reads only $w(x_v)$. Charge conjugation therefore preserves it exactly as it preserves the bare mass term. Antilinear conjugation replaces its coefficient by its conjugate; since F^{UV} and b are real, this is the coefficient at $s^\star = (\bar{w}, \gamma_4 J^\dagger \gamma_4)$.

The bases of the symmetry inductions are now at t_0. There is no activity at that level. The exponent consists of the action at its source weight, the tower at its fine mass matrix, the Gaussian form at its critical point, and the displayed mass term. The first three transform jointly with their source entries, as in the original step; the last transforms as just checked. At a tested covariance pair $V, \mathsf{t}V$, the lattice isometry t maps the minimal orbit above V onto that above $\mathsf{t}V$. The two backgrounds are therefore related by a residual gauge transformation. At constant matrix entries the entries commute with that transformation; the tower and mass matrices transform by

$$(\mathsf{n}, \sigma_{t_0}) \longmapsto (S_{\mathsf{t}}\mathsf{n}S_{\mathsf{t}}^{-1}, S_{\mathsf{t}}\sigma_{t_0}S_{\mathsf{t}}^{-1}).$$

The constant-weight action is invariant. The initial densities consequently have the required covariance on tested pairs. The one-step map reads only such pairs, up to gauge transformations, so the same induction proves the symmetry statements at all subsequent levels. These are precisely the covariance and conjugation properties needed by the local extraction.

We next check its quantitative domain. The restarted couplings satisfy

$$\hat{x}_k \ge \gamma^{-2}, \qquad \tfrac{1}{2}\hat{x}_k \le \mathsf{t}_k \le 2\hat{x}_k, \qquad \hat{x}_k \le 2L^{K-k}\mathsf{b}.$$

The original ceilings

$$4C_w^{\mathrm{s}}\phi(\gamma^{-2}/2) \le K_M c_{\mathrm{cl}}, \qquad 64 c_{\mathrm{th}} c_\pi C_{\mathrm{lin}} C_{\mathrm{rad}} A_1 \varrho (1+\log(2\gamma^{-2}))^{\mathfrak{p}_0}\gamma^2 \le c_{\mathrm{cl}}$$

give $2C_w^{\mathrm{s}}\hat{\phi}_k \le K_M c_{\mathrm{cl}}/2$ and $C_{\mathrm{lin}}\hat{\tau}_k(w; a_k^{\mathrm{rad}}) \le c_{\mathrm{cl}}/8$ on $\|w\|_{\sharp,k} < 8\varrho$. Hence $\hat{\mathscr{D}}_k \subset \hat{\mathscr{D}}_k(a_k^{\mathrm{rad}})$, also after doubling, and $C'_{\mathrm{lin}}\hat{\tau}_k \le K_M c_{\mathrm{cl}}/2$. The recursion and the ceiling $\gamma^{-2} \ge \max\{L\bar{B}, (2L)^{3\mathfrak{p}_0}\}$ give

$$\hat{\phi}_{i-1} \le 2L^{-1/2}\hat{\phi}_i.$$

Indeed, the square-root ratio contributes at most $\sqrt{2}L^{-1/2}$ and the logarithmic ratio at most $e^{1/3}$, with $\sqrt{2}e^{1/3} < 2$. Also $|\hat{x}_k - L^{-(k-i)}\hat{x}_i| \le \bar{B}$ and $\hat{x}_i \le 2L^{k-i}\hat{x}_k$. These are the coupling comparisons used in the original source estimates, with the same constants.

For completeness, the tail induction starts with an exactly local function: $\hat{\sigma}_{t_0}(s)$ depends on $w(x)$ alone. If $a_i(t)$ denotes the supremum of the difference of two $\hat{\sigma}_i$ whose source difference is at

distance at least tL^i, then $a_{t_0}(t) = 0$ for $t > 0$ and $a_{t_0}(0) \le 2B_\sigma \hat\phi_{t_0} \le C_{\mathrm t}\hat\phi_{t_0}$. The recursion for the local extraction closes with the same three inequalities as before:

$$Lq_\phi e^{-\kappa_\sigma(L-1)t_\flat} \le \tfrac14, \qquad (C_{\mathrm{loc}} + 2e^{2\kappa_\sigma})B_\nu \le C_{\mathrm t}/4, \qquad e^{2\kappa_\sigma}B_\nu/B_\sigma \le \tfrac12.$$

This proves (b), together with the holomorphic chain construction of Prop. B.10.

The scalar energy has three parts: the degree-zero part of the new activity after Gaussian integration, the localized normalizers, and the Gaussian contribution at the weight. For the first part the bound used in the original source induction is

$$2\|D\| + \|D_0\| + M^3|\widetilde e| + \|\mathcal L_0\| \le 2C_w^{\mathrm s}\hat\phi_k + C'_{\mathrm{lin}}\hat\tau_k.$$

It is at most $K_M c_{\mathrm{cl}}$ on the doubled domain. The gauge normalizer has no source-coordinate dependence. The localized fermion normalizer is given by the same walk expansion as before; it is holomorphic for $L^j\|\mathsf n\|_\infty < 2r_M$ and its difference is bounded by $2C_N^{\mathsf n}N_f r_M e^{-(\kappa+1)d_j(X)}$. For the Gaussian part, each pointed term is bounded by

$$C_{\mathbf G}^v(\bar B + 16\varrho)\gamma^2 e^{-\kappa_+ d_j(X)} \le e^{-\kappa_+ d_j(X)}.$$

There are at most $8M^3e^{d_j(X)}$ pointers, so its difference is at most $16M^3e^{-\kappa d_j(X)}$. The original definition of $C_{\mathfrak e}$ bounds these three contributions. This proves (a).

Finally let $S = \operatorname{supp}(s - s')$. If X meets S, the same bound applies to the difference. Otherwise w and $\mathsf n$ agree on X, and so do the initial mass entries. Regard all later σ_i as one collection, with the normalized supremum norm. Apply Schwarz's lemma along the single coordinate line joining its two values in the doubled domain. This bounds the total change by (b), with

$$e^{-\kappa_\sigma \operatorname{dist}(X,S)/L^i} \le e^{-\kappa_\sigma \operatorname{dist}(X,S)/L^j}.$$

A change of $\mathsf n$ outside X requires a background walk to reach S. Split its factor $e^{-1/2}$ per step equally between summing the walk and paying for this distance. The latter gives $e^{-\operatorname{dist}(X,S)/(12\sqrt3 M_w L^j)}$, which is at most the required tail because $\kappa_\sigma \le 1/(24M_w)$. The existing factor $2\max\{1, C_{\mathrm t}/B_\sigma\}$ in $C_{\mathfrak e}$ bounds the two source values and the chain change. This proves (c). □

Lemma B.35 (Mixed derivatives of a local family)**.** Let $1 \le j \le K$, $n \ge 2$, and let $\mathbf s_1, \dots, \mathbf s_n$ have pairwise disjoint supports. Suppose the first two supports are at physical distance at least $\mathsf d > 0$ and $s(z) = \sum_i z_i \mathbf s_i \in \mathcal P$ for $|z_i| \le \varrho_i$. For $X \in \mathcal D_j$, let $f(X; s(z))$ be holomorphic near this closed polydisc and suppose that

$$|f(X;s(z)) - f(X;s(z'))| \le C' e^{-\kappa d_j(X)} e^{-\kappa_\sigma \operatorname{dist}(X,\operatorname{supp}(s(z)-s(z')))/L^j}.$$

Then

$$\sum_{X\in\mathcal D_j} |\partial^{[n]} f(X;s(z))| \le 2^{n-1}\prod_i \varrho_i^{-1}\, C' C_{\mathscr C}^\natural\, \#_j^+(\operatorname{supp}\mathbf s_1) e^{-\kappa_{\mathrm E}\mathsf d/\lambda_j},$$

where the derivative is evaluated at $z = 0$,

$$\kappa_{\mathrm E} = \min\{\kappa_\sigma/8, \kappa/(4c_{\mathcal D}M)\}, \quad C_{\mathscr C}^\natural = e^{\kappa/2} C_{\mathrm{cnt}} \sum_{t\ge0}(2t+3)^3 e^{-\kappa_\sigma M t/2}.$$

In particular the hypotheses hold for a family holomorphic on $\mathscr O_{j-1}$ with the displayed source tail there. ■

Proof. The last assertion includes holomorphy near the closed polydisc. Indeed, on $\mathcal P$ we have $\|w\|_\sharp < 2\varrho$ and $\|J\|_\infty \le 2\varrho_J \le r_J/2$. Linearity in z shows that on $|z_i| < 3\varrho_i/2$ these bounds become 3ϱ and $3r_J/4$. Since $\|w\|_{\sharp,j-1} \le \|w\|_\sharp$ and $\lambda_{j-1} \le 1$, the larger polydisc is in $\mathscr O_{j-1}$.

Write $F_X(z) = f(X; s(z))$ and let $\mathsf P_i$ set $z_i = 0$. Inclusion and exclusion gives $\partial^{[n]}F_X = \partial^{[n]}\prod_i(1-\mathsf P_i)F_X$. Cauchy's estimate therefore gives

$$|\partial^{[n]}F_X| \le 2^{n-1}\prod_i \varrho_i^{-1} \min_i \sup|(1-\mathsf P_i)F_X|.$$

For $S_i = \operatorname{supp} \mathbf{s}_i$ and $a_i = \operatorname{dist}(X, S_i)$, the tail assumption, using $i = 1, 2$, bounds this by

$$2^{n-1} \prod_i \varrho_i^{-1} C' e^{-\kappa d_j(X)} e^{-\kappa_\sigma \max\{a_1,a_2\}/L^j}.$$

In bare lattice units, $\mathsf{d} L^K \le a_1 + \operatorname{diam} X + a_2$. Also $\operatorname{diam} X \le c_{\mathcal{D}} M L^j (1 + d_j(X))$. If $\operatorname{diam} X \ge \mathsf{d} L^K/2$, half the polymer decay gives $e^{\kappa/2} e^{-\kappa_{\mathrm{E}} \mathsf{d}/\lambda_j}$. Otherwise $\max(a_1, a_2) > \mathsf{d} L^K/4$ and half the source tail gives that factor. In both cases,

$$e^{-\kappa d_j(X)} e^{-\kappa_\sigma \max(a_1,a_2)/L^j} \le e^{\kappa/2} e^{-\kappa_{\mathrm{E}} \mathsf{d}/\lambda_j} e^{-\kappa d_j(X)/2} e^{-\kappa_\sigma a_1/(2L^j)}.$$

Assign X to a level-j cube in X nearest S_1. The polymer sum at any assigned cube is at most C_{cnt}. There are at most $(2t+3)^3 \#_j^+(S_1)$ cubes at distance between tML^j and $(t+1)ML^j$ from S_1. Summing the last display proves the stated bound. □

Proposition B.36 (Contact terms after the restart)**.** Under the hypotheses on sources and radii in Lem. B.35, for $t_0 < j \le t_1$,

$$\sum_{X \in \mathcal{D}_j} |\partial^{[n]} \hat{e}_j(X; s(z))| \le 2^{n-1} \prod_i \varrho_i^{-1} C_{\mathfrak{E}} C_{\mathscr{C}}^\natural \#_j^+(\operatorname{supp} \mathbf{s}_1) e^{-\kappa_{\mathrm{E}} \mathsf{d}/\lambda_j}.$$

Choose $C_{\mathscr{C}} \ge C_{\mathfrak{E}} C_{\mathscr{C}}^\natural$. Then the scalar removed between t_0 and t_1 satisfies

$$|\partial^{[n]} \hat{c}^\bullet(s(z))| \le 2^{n-1} \prod_i \varrho_i^{-1} (\mathscr{C}_{t_1} - \mathscr{C}_{t_0})(\operatorname{supp} \mathbf{s}_1; \mathsf{d}) \le 2^{n-1} \prod_i \varrho_i^{-1} \mathscr{C}_{t_1}(\operatorname{supp} \mathbf{s}_1; \mathsf{d}).$$

■

Proof. Apply Lem. B.35 to the functions in Prop. B.34(c). The identity

$$\hat{c}^\bullet(s) = \sum_{t_0 < j \le t_1} \left(\hat{e}_j n_j + \sum_{X \in \mathcal{D}_j} \hat{e}_j(X; s) \right)$$

is a finite sum of holomorphic functions on the torus. The numbers $\hat{e}_j n_j$ have zero mixed derivative. Sum the first bound and use

$$\mathscr{C}_k(S; \mathsf{d}) = C_{\mathscr{C}} \sum_{j \le k} \#_j^+(S) e^{-\kappa_{\mathrm{E}} \mathsf{d}/\lambda_j}.$$

□

Choice of the contact constant. The new constant $C_{\mathscr{C}}^\natural$ is fixed after κ_σ, from κ, κ_σ, M and C_{cnt}. The floor $C_{\mathscr{C}} \ge C_{\mathfrak{E}} C_{\mathscr{C}}^\natural$ is imposed at the original choice of $C_{\mathscr{C}} = 27 C_{\mathscr{C}}^0$, after h_F and $C_{\mathfrak{E}}$ and before the mass interval. It creates no circular dependence: no preceding constant depends on $C_{\mathscr{C}}$ and no ceiling bounds it above. The original proof of the separated-source estimate is the proof of Lem. B.35 applied to the unrestarted energies, so it remains valid with this larger constant. The other uses are upper bounds by nonnegative multiples of $\mathscr{C}$. In the overlap estimate the apparent denominator is paired with $C'_{\mathscr{C}}$: $C'_{\mathscr{C}}/C_{\mathscr{C}}$ bounds the same geometric sum independently of $C_{\mathscr{C}}$. Thus $C'_{\mathscr{C}}$ and all other dependent constants are formed afterwards at the enlarged value.

The preceding propositions prove the source-energy and contact-term assertions of Prop. 11.12, including holomorphy on $\mathscr{O}_{j-1}$ and the difference of counts $\mathscr{C}_{t_1} - \mathscr{C}_{t_0}$.

B.13. **Comparing two integration weights.** The weights in the two restart integrals can have different supports. The following lemma separates the error on their common support from the mass lost at its boundary.

Lemma B.37. Let ν be a finite measure on Y, and let $k^{\mathrm{A}}, k^{\mathrm{B}}$ be bounded nonnegative measurable functions. Let $D \subset \mathbb{C}^n$ contain zero. For $\bullet = \mathrm{A}, \mathrm{B}$, suppose that $F_z^\bullet$ is bounded and measurable for every $z \in D$. Assume that $0 \le \epsilon, \delta \le 1/8$ and that the following hold.

(i) On $\{k^\bullet > 0\}$, $F_0^\bullet \geq 0$ and $F_z^\bullet = F_0^\bullet e^{\psi_z^\bullet}$, where $|\psi_z^\bullet| \leq 1/4$.
(ii) On the common support, $k^{\mathrm{A}} = k^{\mathrm{B}}$ and $F_z^{\mathrm{B}} = F_z^{\mathrm{A}} e^{\xi_z}$, where $|\xi_z| \leq \epsilon$.
(iii) For either choice of $\bullet$, with $\circ$ the other,

$$\int k^\bullet \mathbf{1}_{\{k^\circ=0\}} |F_z^\bullet| \, d\nu \leq \delta \int k^\bullet |F_z^\bullet| \, d\nu.$$

Put $T^\bullet(z) = \int k^\bullet F_z^\bullet \, d\nu$, and assume $T^{\mathrm{A}}(0) > 0$. Then

$$\begin{aligned} \int |k^{\mathrm{A}} F_z^{\mathrm{A}} - k^{\mathrm{B}} F_z^{\mathrm{B}}| \, d\nu &\leq e^{1/4}(2\epsilon + 3\delta) T^{\mathrm{A}}(0), \\ |T^{\mathrm{A}}(z)| &\geq e^{-1/4} \cos(1/4) T^{\mathrm{A}}(0), \\ \left| \frac{T^{\mathrm{B}}(z)}{T^{\mathrm{A}}(z)} - 1 \right| &\leq 4\epsilon + 6\delta. \end{aligned} \tag{B.vii}$$

At $z = 0$, the first bound holds with the factor $e^{1/4}$ replaced by one. ■

Proof. Write $q_z = e^{1/4}$ for $z \neq 0$, and $q_0 = 1$. Positivity at zero and (i) give $\int k^\bullet |F_z^\bullet| \, d\nu \leq q_z T^\bullet(0)$. Split the difference of the two integrands over the common support and the two sets on which just one weight is positive. Conditions (ii) and (iii) yield

$$\int |k^{\mathrm{A}} F_z^{\mathrm{A}} - k^{\mathrm{B}} F_z^{\mathrm{B}}| \, d\nu \leq q_z \big[(e^\epsilon - 1 + \delta) T^{\mathrm{A}}(0) + \delta T^{\mathrm{B}}(0) \big]. \tag{B.viii}$$

At zero, the same decomposition and positivity give

$$T^{\mathrm{B}}(0) \leq e^\epsilon T^{\mathrm{A}}(0) + \delta T^{\mathrm{B}}(0).$$

Thus $T^{\mathrm{B}}(0) \leq e^{1/8}(1 - 1/8)^{-1} T^{\mathrm{A}}(0) \leq 2T^{\mathrm{A}}(0)$. Inserting this and $e^\epsilon - 1 \leq 2\epsilon$ into eq. B.viii proves the first bound, including its improvement at zero.

On $\{k^{\mathrm{A}} > 0\}$, condition (i) also gives $\operatorname{Re} F_z^{\mathrm{A}} \geq e^{-1/4} \cos(1/4) F_0^{\mathrm{A}}$. Integration proves the second bound. Divide the first by this lower bound and use $e^{1/2}/\cos(1/4) < 2$ to obtain the third. □

B.14. **Transferring fluctuation tails between weights.** The comparison uses the weight of the history with no large-field region, whereas the change of variables uses the small-field weight. The following observation transfers tail bounds between them. Its error is the total mass of discarded histories.

Proposition B.38 (Comparison of the two weights)**.** Use the data of Def. B.32 and assume $L^u \gamma^{-2} \geq \Theta_\star$. Use the notation of Def. B.39, and put

$$V_k = \mathcal{M}^k U, \qquad N_{\mathrm{bd}} = \sum_{k=t_0}^{t_1 - 1} \# \mathfrak{X}_k^+.$$

Let $\mathfrak{w}^{\mathrm{sf}}$ be the weight of the small-field chain up to level t_1, at the calibration $x_j = L^{K-j}\mathfrak{b}$, and set

$$\mathfrak{w}^\chi(U) = \chi_{t_1}(T_\ell)(V_{t_1}) \mathfrak{w}^{\mathrm{sf}}(U).$$

(a) The two weights are nonnegative bounded Borel functions,

$$\mathfrak{w}_\emptyset \leq \mathfrak{w}^\chi \leq \mathfrak{w}^{\mathrm{sf}},$$

and all three agree wherever $\mathfrak{w}_\emptyset \neq 0$. For every nonnegative gauge-invariant bounded Borel F,

$$\int F \mathfrak{w}_\emptyset \varsigma_0 \, dU \leq \int F \mathfrak{w}^\chi \varsigma_0 \, dU \leq \int F \mathfrak{w}^{\mathrm{sf}} \varsigma_0 \, dU \leq \int F \varsigma_0 \, dU.$$

In particular, $T(0) \leq \int \mathfrak{w}^\chi \varsigma_0 \leq \int \mathfrak{w}^{\mathrm{sf}} \varsigma_0 \leq Z$.

(b) Let $\mathfrak{w}'$ be any Borel weight satisfying $\mathfrak{w}' \geq \mathfrak{w}_\emptyset$ and $\int \mathfrak{w}' \varsigma_0 \, dU \leq Z$. For every Borel F with values in $[0,1]$,

$$\int F \mathfrak{w}' \varsigma_0 \, dU \leq \int F \mathfrak{w}_\emptyset \varsigma_0 \, dU + Z - T(0) \leq \int F \mathfrak{w}_\emptyset \varsigma_0 \, dU + \mathsf{R}^{\mathrm{F}}_{t_1} Z.$$

This applies to both weights in (a).

(c) For the uniform mark with parameter $r > 0$,

$$\sum_{k=t_0}^{t_1-1} \sum_{b \in \mathfrak{X}_k^+} \int \mathfrak{w}_\emptyset \bar{\chi}_{k,b}(V_k) \varsigma_0 \, dU \leq \frac{T_0(r) - T(0)}{r}, \qquad \int \mathfrak{w}_\emptyset \mathbf{1}_{\{N \geq 1\}} \varsigma_0 \, dU \leq \frac{T_0(r) - T(0)}{r}.$$

With $\mathfrak{w}'$ as in (b), the corresponding bounds are

$$\sum_{k,b} \int \mathfrak{w}' \bar{\chi}_{k,b}(V_k) \varsigma_0 \, dU \leq \frac{T_0(r) - T(0)}{r} + N_{\mathrm{bd}} \mathsf{R}^{\mathrm{F}}_{t_1} Z,$$

$$\int \mathfrak{w}' \mathbf{1}_{\{N \geq 1\}} \varsigma_0 \, dU \leq \frac{T_0(r) - T(0)}{r} + \mathsf{R}^{\mathrm{F}}_{t_1} Z.$$

Moreover $N_{\mathrm{bd}} \leq 4 N_{[s]}$.

Here r is the mark parameter, independently of the Wilson parameter fixed in the model. ■

Proof. The depth assumption gives the small-field and one-weight settings on both lattices: apply the gauge comparison admission at cutoff K^{A} with $t_* = K^{\mathrm{A}} - u$. Since $1 \leq u < s < K^{\mathrm{A}}$ and $\mathsf{b} \geq \gamma^{-2}$, we have $1 \leq t_* \leq K^{\mathrm{A}} - 1$ and $L^u \mathsf{b} \geq \Theta_\star$. The resulting top levels are t_1^{A} and $t_1^{\mathrm{A}} + 1 = t_1^{\mathrm{B}}$. The calibrations are the original $x_j = L^{K-j} \mathsf{b} = \mathsf{t}_j$ and $x_0 = \beta$. Thus the bare gauge density in the weight comparison is $e^{-\beta A}$, and its history weight is exactly $\mathfrak{w}_\emptyset$.

The gauge weight comparison gives a bounded nonnegative Borel $\mathfrak{w}^{\mathrm{sf}}$ with $\mathfrak{w}_\emptyset \leq \mathfrak{w}^{\mathrm{sf}}$ and equality on $\{\mathfrak{w}_\emptyset \neq 0\}$. The density test is Borel and $\mathcal{M}^{t_1}$ is Borel. The last step of this history factors as

$$\mathfrak{w}_\emptyset(U) = \mathfrak{w}_{h_\emptyset^{(t_1-1)}}(U) \chi_{t_1}(T_\ell)(V_{t_1}) \mathsf{q}^0_{t_1-1}(V_{t_1-1}) \mathsf{q}_{t_1}(V_{t_1-1}) \mathfrak{G}_{t_1-1}[x_{t_1-1}](V_{t_1-1}) \chi^{\mathrm{sf}}_{x_{t_1-1}}(V_{t_1-1}).$$

It follows that $\chi_{t_1}(T_\ell) = 1$ wherever $\mathfrak{w}_\emptyset \neq 0$; this proves the pointwise statements in (a).

To apply the gauge integral inequality, replace its test function by $\widetilde{F}(U) = F(U) \det(D[U] + m_0)$. This is nonnegative, bounded and Borel. Boundedness follows from the uniform finite-dimensional bound on $D[U]$, and the determinant is polynomial in the link entries. Gauge covariance $D[U^\theta] = \theta D[U] \theta^{-1}$ makes $\widetilde{F}$ gauge invariant. The gauge inequality therefore gives the final inequality of (a), because $e^{-\beta A} \det(D + m_0) = \varsigma_0$. The other inequalities follow pointwise.

For (b), positivity gives

$$\int F(\mathfrak{w}' - \mathfrak{w}_\emptyset) \varsigma_0 \, dU \leq \int (\mathfrak{w}' - \mathfrak{w}_\emptyset) \varsigma_0 \, dU \leq Z - T(0).$$

The last inequality in (b) is Prop. B.43(a) at $t = t_1$. Gauge invariance of F is not needed in this step.

Finally the uniform mark is $\mathsf{Mk}_r = (1+r)^N \geq 1 + rN$, with integer $N \geq 0$. Integrate this inequality against the positive measure $\mathfrak{w}_\emptyset \varsigma_0 \, dU$. Since $N = \sum_{k,b} \bar{\chi}_{k,b}(V_k)$ and $\mathbf{1}_{\{N \geq 1\}} \leq N$, it proves the first two bounds in (c). Each $\bar{\chi}_{k,b} \circ \mathcal{M}^k$ is a Borel indicator; consequently N and $\mathbf{1}_{\{N \geq 1\}}$ are Borel. Applying (b) to these indicators proves the bounds for $\mathfrak{w}'$. There are at most $3 n_k$ positively oriented bonds at level k, so

$$N_{\mathrm{bd}} \leq 3\ell^3 \sum_{k=t_0}^{t_1-1} L^{3(K-k)} \leq 3\ell^3 L^{3s} \sum_{i \geq 0} L^{-3i} \leq 4\ell^3 L^{3s} = 4 N_{[s]}.$$

□

B.15. **Comparing the logarithms of scalar totals.** The difference estimates permit interpolation between the original flow and its restart on a large complex disc. An upper bound on that disc, together with a positive lower bound at the original flow, controls the difference of their logarithms. The radius of the disc will supply the small factor.

Definition B.39 (Scalar totals and fluctuation marks)**.** Fix a datum $(g,\ell,K,\mathsf{b},\mathsf{m})$ and $1\le u<s<K$, and put

$$t_0=K-s,\quad t_1=K-u,\quad N_{[d]}=\ell^3L^{3d},\quad \mathsf{y}_{[d]}=(\log(L^d\mathsf{b}))^{2\mathfrak{p}_0}.$$

The calibration is $\mathsf{t}_k=L^{K-k}\mathsf{b}$, $n_k=\ell^3L^{3(K-k)}$, and the creation deposit is $p_0(g_k)^2=A_0^2(\log\mathsf{t}_k)^{2\mathfrak{p}_0}$. Let $h_\emptyset^{(k)}$ be the history without large-field regions and $\mathfrak{w}_\emptyset=\mathfrak{w}_{h_\emptyset^{(t_1)}}$.

For a tuple $X=((\alpha_i,f_i))_{i\le n}$, $n\ge1$, with real Lipschitz f_i and no support restriction, let $s(z)=(w_z,J_z)=\sum_i z_i\mathsf{s}_i$ be its source. Define

$$\varsigma_z(U)=e^{-A(\beta-a_K^{-1}w_z,U)}\det(D[U]+m_0-a_K\hat J_z),$$
$$Z(z)=\int\varsigma_z\,dU,\qquad T(z)=\int\mathfrak{w}_\emptyset\varsigma_z\,dU,\qquad c(z)=E_{t_1}(s(z)).$$

The last expression is used where $s(z)\in\mathscr{O}_{t_1}$. Write $Z=Z(0)$ and $\varsigma_0=e^{-\beta A}\det(D+m_0)\ge0$. The crude source sizes are

$$\mathsf{s}_i=\begin{cases}6\ell^3L^{4K}\|f_i\|_\infty, & \alpha_i=0,\\ \nu\,\#\mathsf{R}\,a_Km_0^{-1}\|f_i\|_\infty, & \alpha_i\in\mathfrak{B},\end{cases}\qquad \nu=8N_f.$$

When the one-weight gauge setting holds up to t_1, let $\mathfrak{X}_k^+$ be the positively oriented non-pivot bonds at level k, and set

$$\bar\chi_{k,b}=\mathbf{1}_{\{|\mathcal{A}_k^\diamond(b)|\ge\delta(\mathsf{t}_k)/2\}},\qquad t_0\le k<t_1.$$

These are Borel indicators. The setting holds for both cutoffs under (k0), (k1). Put

$$\mathsf{Mk}_\lambda(U)=\prod_{k=t_0}^{t_1-1}\prod_{b\in\mathfrak{X}_k^+}(1+\lambda_k\bar\chi_{k,b}(\mathcal{M}^kU)),$$
$$N(U)=\sum_{k=t_0}^{t_1-1}\sum_{b\in\mathfrak{X}_k^+}\bar\chi_{k,b}(\mathcal{M}^kU),$$
$$T_0(\lambda)=\int\mathfrak{w}_\emptyset\mathsf{Mk}_\lambda\varsigma_0\,dU.$$

For a scalar λ all level parameters are equal, so that $\mathsf{Mk}_\lambda=(1+\lambda)^N$ and $T_0(0)=T(0)$.

For $k<K$ write $a_k=\mathfrak{b}(\mathsf{t}_k)/2$; this a_k is the mark threshold, while $a_K=L^{-K}$ remains the lattice spacing. With $w_*=c_{\mathrm{cl}}$, set

$$\kappa_{\mathrm{mk}}=\frac{w_*}{32M^3},\qquad \mathsf{r}_k=\kappa_{\mathrm{mk}}a_k^2-\pi,\qquad \epsilon_k^{\mathrm{mk}}(\lambda)=\frac{w_*}{32}\frac{|\lambda|}{e^{\mathsf{r}_k}-1},$$
$$\Lambda_{\mathrm{mk}}=\exp\left[\frac{\kappa_{\mathrm{mk}}A_1^2}{4}\mathsf{y}_{[u]}\right],\quad \theta_{\mathrm{dis}}=\tfrac12(c_\star-c_\Theta)>0.$$

The expression for ϵ_k^{mk} requires $\mathsf{r}_k>0$. For $t_0\le k<t_1$, $e^{\mathsf{r}_k}\ge e^{-\pi}\Lambda_{\mathrm{mk}}$. Finally,

$$\mathsf{Q}_\ell=e^{[C+2+C_P(1+\log x_K)]\ell^3}\le\mathsf{M}_{g,\ell}:=e^{[C+2+C_P(1+\log X_g)]\ell^3},\qquad \mathsf{R}_t^{\mathrm{F}}=\mathsf{Q}_\ell e^{-\theta_{\mathrm{dis}}p_0(g_t)^2}.$$

■

Lemma B.40 (Two values of a holomorphic family)**.** Let $n \ge 1$, $R_i > 0$, $\mathsf{Q} \ge 1$ and $\rho \ge 1 + 8\mathsf{Q}$. Put

$$\theta = \frac{1}{4(\mathsf{Q}+1)}, \quad \mathbb{D}_\rho = \{|\tau - 1| < \rho\}, \quad \mathbb{P}(R) = \{|z_i| < R_i\}.$$

Suppose H is holomorphic on $\mathbb{D}_\rho \times \mathbb{P}(R)$, $H(1,0) > 0$, and $|H| \le \mathsf{Q} H(1,0)$ there. Then on $|\tau - 1| \le 1$, $|z_i| \le \theta R_i$,

$$\left| \frac{H(\tau, z)}{H(1,0)} - 1 \right| \le \tfrac{3}{8}.$$

In particular H has no zero there, and

$$\left| \frac{H(1,z)}{H(0,z)} - 1 \right| \le \frac{8\mathsf{Q}}{5(\rho - 1)} \le \tfrac{1}{5}.$$

For continuous logarithms on this polydisc,

$$\left| \partial^{[n]} [\log H(1,\cdot) - \log H(0,\cdot)](0) \right| \le \frac{(4(\mathsf{Q}+1))^{n+1}}{2(\rho-1)} \prod_i R_i^{-1}.$$

If $H(0,0)$ is real, then $H(0,0) \ge 5H(1,0)/8$. Moreover, if G is holomorphic on the same domain and $|G(\tau,z) - a - b\tau| \le B$, then

$$|\partial^{[n]}[G(1,\cdot) - G(0,\cdot)](0)| \le B(\rho-1)^{-1} \prod_i R_i^{-1}.$$

■

Proof. Normalize $h = H/H(1,0)$. Cauchy's estimate in τ on discs of radius approaching $\rho - 1$ gives $|\partial_\tau h| \le \mathsf{Q}/(\rho-1)$ for $|\tau - 1| \le 1$. Thus $|h(\tau,z) - h(1,z)| \le 1/8$. For $|z_i| \le \theta R_i$, Schwarz's lemma applied to $h(1, \zeta z/\theta) - 1$ gives $|h(1,z) - 1| \le (\mathsf{Q}+1)\theta = 1/4$. This proves the first estimate. The difference $h(1,z) - h(0,z)$ is at most $\mathsf{Q}/(\rho-1)$ and $|h(0,z)| \ge 5/8$, proving the quotient estimate. The principal logarithm of that quotient is bounded by $2\mathsf{Q}/(\rho-1)$, since $|\operatorname{Log}(1+w)| \le 5|w|/4$ for $|w| \le 1/5$. Cauchy's estimate on the source polydisc gives the claimed mixed derivative. Changing continuous branches adds a constant and does not affect it. The real lower bound follows from the first estimate. For the last assertion apply the same τ estimate to $G - a - b\tau$, integrate from zero to one, and apply Cauchy's estimate in z; $n \ge 1$ removes the constant b. □

Proposition B.41 (Readout at depth u)**.** Use Def. B.39. Suppose $s(z) \in \mathcal{P}$ on $\mathbb{P}(R)$, $\rho \ge 1 + 8\mathsf{Q}_\ell$, and $\hat{c}$ is holomorphic there. Let $I, E_>$ be holomorphic on $\mathbb{D}_\rho \times \mathbb{P}(R)$ and suppose

$$I(1,z) = e^{-c(z)} T(z), \qquad E_>(1,0) = E_K - E_{t_1},$$
$$|I(\tau,z)| \le e^{\operatorname{Re} E_>(\tau,z) + (C+2)\ell^3}, \qquad |E_>(\tau,z) - a - b\tau| \le B.$$

Set $\hat{T}(z) = e^{\hat{c}(z)} I(0,z)$ and $\theta_\ell = 1/[4(\mathsf{Q}_\ell + 1)]$. Then

$$I(1,0) e^{-E_>(1,0)} \ge e^{-C_P(1+\log x_K)\ell^3} > 0.$$

For $|z_i| \le \theta_\ell R_i$, T and $\hat{T}$ are nonzero and

$$\left| \frac{T(z) e^{-c(z) - E_>(1,z)}}{\hat{T}(z) e^{-\hat{c}(z) - E_>(0,z)}} - 1 \right| \le \frac{8\mathsf{Q}_\ell}{5(\rho-1)}.$$

Their continuous logarithms satisfy

$$\left| \partial^{[n]} [\log T - \log \hat{T} - (c - \hat{c})](0) \right| \le \left[\tfrac{1}{2}(4(\mathsf{Q}_\ell + 1))^{n+1} + B \right] (\rho-1)^{-1} \prod_i R_i^{-1}.$$

Also

$$|\hat{T}(0)| \ge \tfrac{5}{8} e^{\operatorname{Re}(\hat{c}(0) + E_>(0,0)) - C_P(1+\log x_K)\ell^3}.$$

If $\hat{c}(0)+E_{>}(0,0)$ and $\hat{T}(0)$ are real, the last estimate holds without absolute values. One may replace Q_ℓ by $\mathsf{M}_{g,\ell}$ throughout the radii and bounds. ■

Proof. At every level the threshold is at most $\varepsilon_\flat$, because $\mathtt{t}_j \ge \gamma^{-2}$ and the initial ceiling gives $\varepsilon(\gamma^{-2}) \le \varepsilon_\flat$. Thus history consistency applies between t_1 and K. The determinant density is nonnegative and gauge invariant, and all history weights are nonnegative. Hence

$$T(0) = \sum_{h|_{t_1}=h_\emptyset^{(t_1)}} t^{\mathrm{F}}(h) \ge t^{\mathrm{F}}(h_\emptyset^{(K)}) \ge e^{E_K - C_P(1+\log x_K)\ell^3}.$$

The last inequality is the positive last-level estimate of the original construction. Since $c(0)=E_{t_1}$, it proves the first assertion. Now $H = Ie^{-E_>}$ satisfies the hypotheses of Lem. B.40 with $\mathsf{Q}=\mathsf{Q}_\ell$. Its quotient gives the asserted quotient; the difference of logarithms of I is the sum of that for H and the difference of $E_>$. Apply both derivative estimates of the lemma. Finally $\hat{T}(0) = e^{\hat{c}(0)+E_>(0,0)}H(0,0)$ gives the last estimate and its real version. □

Proposition B.42 (The interpolated scalar family)**.** Assume (k0)–(k2) for either cutoff and the source tuple of Def. 11.1(5). At base $t_b = t_1$, use the interpolated measure $\rho_{s,\lambda}^{(t_1)}$ of Def. 11.3 and the continued energies of Prop. 11.13, with $\Lambda_b = \Lambda$. Define, for $\lambda = 1-\tau$,

$$\begin{aligned} I(\tau,z) &= e^{-E_{t_1}(s(z))}\operatorname{Ber}\rho^{(t_1)}_{s(z),1-\tau}(\mathcal{V}_{t_1}),\\ E_>(\tau,z) &= E_K(s(z),1-\tau) - E_{t_1}(s(z)),\\ \rho &= r_\lambda, \qquad a = E_K - E_{t_1} + d_b, \qquad b = -d_b,\\ B &= C'_E N_{[u]}, \qquad C'_E = \frac{C_{\mathfrak{E}} C_{\mathrm{cnt}}}{M^3(L^3-1)}. \end{aligned}$$

Here d_b is independent of s, and all generators are integrated in the Berezin integral. These functions satisfy Prop. B.41 on $\mathbb{D}_\rho \times \mathbb{P}(2R)$. Its function $\hat{T}$ is the actual restarted total $\hat{T}^\bullet$, with $\hat{c}(z) = \hat{c}^\bullet(s(z))$. ■

Proof. The conditions in (k1) give $u \ge u_*$, $e^{-c_{\mathrm{t}}L^u} \le L^{-c_\#(s-u)}$, and $s \ge s_F$. Thus Props. B.13 and B.23 apply. The lag sums at t_1 hold with Λ. By (k2), $r_\lambda \ge 16\mathsf{M}_{g,\ell} \ge 16$ and $\Lambda = 2(3L^{1/2})^u r_\lambda/c_\lambda \ge 32$. The perturbations therefore satisfy Prop. 11.13, whose parameter radius is $\rho_b = \frac{1}{2}(3L^{1/2})^{-u}c_\lambda\Lambda = r_\lambda$.

On $\mathbb{P}(2R)$ the source bounds are $\|w_z\|_\sharp < (1-c_\lambda)\varrho$ and $\|J_z\|_\infty < (1-c_\lambda)\varrho_J$. Consequently $(s(z),1-\tau) \in \mathcal{D}_b$ for $|\tau-1| \le r_\lambda$, and lies in its interior for strict inequality. More explicitly, the open set

$$\|w\|_\sharp < (2-c_\lambda)\varrho, \qquad \|J\|_\infty < 2\varrho_J, \qquad |\lambda| < \rho_b$$

lies in $\mathcal{D}_b$ and contains these points. The shift in the source above the base also stays in the permitted region, because (L3) gives

$$\|\lambda\hat{w}_b\|_\sharp \le r_\lambda\Lambda^{-1}\varrho = \tfrac{1}{2}(3L^{1/2})^{-u}c_\lambda\varrho < c_\lambda\varrho.$$

We spell out holomorphy of the integral because its density need only be Borel in the field. The exact history identity gives

$$e^{E_{t_1}(s(z))}I(\tau,z) = \sum_{h\in\mathcal{H}_K^{(t_1)}} \int \operatorname{Ber} g_h[s(z),1-\tau]\, d\mathscr{M}_h^{\mathrm{fr}}.$$

There are finitely many histories. Replace each density by the pointwise holomorphic Borel representative $\mathfrak{g}_h$ of Prop. C.21. The two representatives agree almost everywhere, coefficient by coefficient, since they represent the same measure; their integrals and integrated fibre norms

therefore agree. On the open domain just described, each coefficient of g_h is holomorphic in (s,λ) at every field point. Its integral in absolute value is bounded by

$$\exp[\operatorname{Re} E_K(s(z),1-\tau)+(C+2)\ell^3],$$

using $|\operatorname{Ber} F|\le \|F\|_{h_F}$ and Prop. 11.13(B3). This is locally bounded in (τ,z) by (B4). On a closed polydisc about any parameter point, Cauchy's formula on the distinguished boundary and Tonelli's theorem give an integrable majorant on the polydisc with half its radii. The integrands are jointly Borel, since they are Borel in the field and continuous in the parameters. Integration consequently preserves holomorphy. This proves that I is holomorphic.

The energy identity is

$$E_>(\tau,z)=(1-\tau)d_b+\sum_{t_1<j\le K}\left(e_jn_j+\sum_{X\in\mathcal{D}_j}e_j(X;s(z),1-\tau)\right).$$

It proves holomorphy of $E_>$. The source energies of the restart prove holomorphy of $\hat c$ by Prop. B.34. At $\lambda=0$ the measure is the original term, giving $I(1,z)=e^{-c(z)}T(z)$. At $\lambda=1$ it is $e^{E_{t_1}(s)-\hat c^\bullet(s)}$ times the restarted term, giving $I(0,z)=e^{-\hat c(z)}\hat T^\bullet(z)$.

The history bound gives $|I|\le e^{\operatorname{Re}E_>+(C+2)\ell^3}$. At $(\tau,z)=(1,0)$ the source energies vanish, so $E_>(1,0)=E_K-E_{t_1}$. Finally,

$$\begin{aligned}|E_>-a-b\tau|&\le C_{\mathfrak{E}}\sum_{t_1<j\le K}\sum_{X\in\mathcal{D}_j}e^{-\kappa d_j(X)}\\&\le C_{\mathfrak{E}}C_{\mathrm{cnt}}M^{-3}\sum_{t_1<j\le K}n_j\\&=C_{\mathfrak{E}}C_{\mathrm{cnt}}M^{-3}\ell^3\frac{L^{3u}-1}{L^3-1}\le C'_EN_{[u]}.\end{aligned}$$

Here each polymer is assigned to a cube it contains; there are at most n_j/M^3 cubes. Since $\rho=r_\lambda\ge 16\mathsf{M}_{g,\ell}\ge 1+8\mathsf{Q}_\ell$, all hypotheses of Prop. B.41 are verified. □

B.16. **Discarding histories and summing contact terms.**

Proposition B.43 (Discarding early large-field regions)**.** For $1\le t\le K$, put $T_t(z)=\int \mathfrak{w}_{h_\emptyset^{(t)}\varsigma_z}\,dU$. Then

$$Z-T_t(0)=\sum_{h\in\mathcal{H}_K:\,Z_t(h)\neq\varnothing}t^{\mathrm{F}}(h),\qquad 0\le Z-T_t(0)\le \mathsf{R}_t^{\mathrm{F}}Z.$$

For every source tuple of Def. B.39, with no restriction on its supports, set $\sigma(z)=\sum_i|z_i|\mathsf{s}_i$. Then

$$|Z(z)-T_t(z)|\le e^{\sigma(z)}(Z-T_t(0))\le e^{\sigma(z)}\mathsf{R}_t^{\mathrm{F}}Z.$$

If $\mathsf{R}_t^{\mathrm{F}}\le 1/8$, then T_t has no zero on $\{4n\mathsf{s}_i|z_i|\le 1 \text{ for all } i\}$, and

$$\left|S^T_{n;K,\ell,\beta,m_0}(X)-\partial^{[n]}\log T_t(0)\right|\le 3\mathsf{R}_t^{\mathrm{F}}\prod_i(4n\mathsf{s}_i).$$

■

Proof. History consistency gives $T_t(0)=\sum_{h|_t=h_\emptyset^{(t)}}t^{\mathrm{F}}(h)$. At $t=K$ this is the single term $t^{\mathrm{F}}(h_\emptyset^{(K)})$. The remaining histories are exactly those with $Z_t(h)\neq\varnothing$. Positivity of their scalar measures proves the identity and its lower bound.

We retain part of the creation deposit before summing. For $\mathcal{S}=\mathcal{S}(h)$, the original construction gives the pair

$$f(h)=e^{-E_K+\sum_n\operatorname{Re}f_n(\mathcal{S})}\int\|\mathsf{g}_h\|\,d\mathscr{M}_h^{\mathrm{fr}},\qquad \mathsf{e}(\mathcal{S})=-\sum_n\operatorname{Re}f_n(\mathcal{S})$$

at the rate $c_\star$ and the fixed majorant and label constants. Put $c = c_\star - \theta_{\mathrm{dis}} = (c_\star + c_\Theta)/2$ and

$$f'(h) = f(h) \exp\left[\theta_{\mathrm{dis}} \sum_{t'=1}^{K} p_0(g_{t'})^2 \mathsf{n}_{t'}^{\mathrm{cr}}(h)\right].$$

Each created cell now has rate c instead of $c_\star$, so (f', e) is an admissible pair at rate c. The enlarged majorant and label constants used in the source construction weaken its inequalities and remain admissible.

Use the history mark with deposit dials $(\mathsf{w}', c'_{\mathrm{Dep}}) = (c_\Theta/c, c)$. Then $\mathsf{w}' < 1$, including when $c_\Theta = 0$. The thresholds at rate c are among the original ceilings on γ; in the calibration frame, $y_K = \log \mathsf{b} \geq \log \gamma^{-2}$ meets them. The history sum gives

$$\sum_h f'(h) e^{\mathsf{e}(\mathcal{S}(h))} e^{\Theta'(h)} \leq \Pi_K(C, c'') \leq e^{(C+2)\ell^3}, \qquad c'' = \tfrac{1}{2}(1 - \eth)(c_\star - c_\Theta) > 0.$$

The constant C contains the enlarged majorant and, when $c_a > 0$, the additional $(6L^3 + 1)\log 2$ required by that sum. If $Z_t(h) \neq \varnothing$, at least one cell was created at a level $t' \leq t$. Since $p_0(g_{t'})^2 \geq p_0(g_t)^2$ and $\Theta' \geq 0$,

$$\sum_{Z_t(h) \neq \varnothing} e^{-E_K} \int \|\mathsf{g}_h\| \, d\mathscr{M}_h^{\mathrm{fr}} \leq e^{-\theta_{\mathrm{dis}} p_0(g_t)^2 + (C+2)\ell^3}.$$

Use $t^{\mathrm{F}}(h) \leq \int \|\mathsf{g}_h\| \, d\mathscr{M}_h^{\mathrm{fr}}$ and $Z \geq e^{E_K - C_P(1 + \log x_K)\ell^3}$ to obtain the first upper bound.

The same exact history identity holds with the complex density ς_z: it is bounded, Borel and gauge invariant. Since $m_0 > 0$, $\varsigma_z = \varsigma_0 G_z$, where

$$G_z(U) = e^{\mathcal{O}(w_z)(U)} \det(1 - \mathsf{X}_z(U)), \qquad \mathsf{X}_z = a_K (D[U] + m_0)^{-1} \hat{J}_z.$$

The action term is at most $\sum_{\alpha_i = 0} |z_i| \mathsf{s}_i$ in modulus. Accretivity gives $\|(D + m_0)^{-1}\| \leq m_0^{-1}$; the trace norm of X_z is consequently at most $\sum_{\alpha_i \neq 0} |z_i| \mathsf{s}_i$. Its eigenvalues have total absolute value at most this trace norm. Expanding the product $\det(1 - \mathsf{X}_z)$ therefore yields

$$|G_z| \leq e^{\sigma(z)}, \qquad |G_z - 1| \leq e^{\sigma(z)} - 1.$$

Integration against the nonnegative history weights proves the source bound.

On the indicated closed set $\sigma(z) \leq 1/4$, so $|Z(z)| \geq (2 - e^{1/4}) Z$. Thus

$$q(z) = \frac{Z(z) - T_t(z)}{Z(z)} \quad \text{satisfies} \quad |q| \leq 1.8 \mathsf{R}_t^{\mathrm{F}} \leq 0.225.$$

It follows that $T_t = Z(1 - q)$ is nonzero and $|\operatorname{Log}(1 - q)| \leq 3\mathsf{R}_t^{\mathrm{F}}$. Cauchy's estimate on the polydisc $|z_i| \leq (4n\mathsf{s}_i)^{-1}$ gives the mixed derivative bound. If some $\mathsf{s}_i = 0$, then $f_i = 0$ and both mixed derivatives and the right side vanish, so no radius need be assigned to that variable. □

Proposition B.44 (Summing the contact energies)**.** Let $n \geq 2$ and use the separated sources and radii of Lem. B.35. Suppose

$$\kappa_{\mathrm{E}} \mathsf{d} L^u (L - 1) \geq 3 \log L + \log 2. \tag{Sep}$$

Then

$$|\partial^{[n]} E_{t_1}(s(z))| \leq 2^n C_{\mathscr{C}} M^{-3} N_{[u]} e^{-\kappa_{\mathrm{E}} \mathsf{d} L^u} \prod_i \varrho_i^{-1}.$$

More generally let $\mathcal{J} \subset \{1, \dots, t_1\}$ and let $f_j(X; s(z))$ be holomorphic near the closed source polydisc, with source tails as in Lem. B.35 at constant $C' C_{\mathfrak{E}}$. Then, at each level,

$$\sum_X |\partial^{[n]} f_j(X; s(z))| \leq 2^{n-1} \prod_i \varrho_i^{-1} C' C_{\mathscr{C}} \#_j^+(\operatorname{supp} \mathsf{s}_1) e^{-\kappa_{\mathrm{E}} \mathsf{d}/\lambda_j}.$$

Its sum $\hat{c} = \sum_{j\in\mathcal{J}} \sum_X f_j$ satisfies

$$|\partial^{[n]}\hat{c}| \le 2^{n-1} C' \mathscr{C}_{t_1}(\operatorname{supp} \mathbf{s}_1; \mathsf{d}) \prod_i \varrho_i^{-1} \le 2^n C' C_{\mathscr{C}} M^{-3} N_{[u]} e^{-\kappa_{\mathrm{E}} \mathsf{d} L^u} \prod_i \varrho_i^{-1}.$$

In particular the difference $c - \hat{c}$ between the original and restarted scalar energies has this last bound with C' replaced by 2. ■

Proof. The level estimate is Lem. B.35, using $C_{\mathfrak{E}} C_{\mathscr{C}}^{\natural} \le C_{\mathscr{C}}$. It applies to the original source energies and, by Prop. B.34, to the restarted energies, with $C' = 1$. For every support S, $\#_j^+(S) \le n_j/M^3$. Write $j = t_1 - m$. Then $n_j = N_{[u]} L^{3m}$, $\mathsf{d}/\lambda_j = \mathsf{d} L^{u+m}$, and $L^m - 1 \ge m(L-1)$. Therefore

$$\mathscr{C}_{t_1}(S; \mathsf{d}) \le C_{\mathscr{C}} M^{-3} N_{[u]} e^{-\kappa_{\mathrm{E}} \mathsf{d} L^u} \sum_{m\ge 0} \left(L^3 e^{-\kappa_{\mathrm{E}} \mathsf{d} L^u (L-1)} \right)^m \le 2 C_{\mathscr{C}} M^{-3} N_{[u]} e^{-\kappa_{\mathrm{E}} \mathsf{d} L^u}.$$

This proves all assertions. The usual choice $\varrho_i = \varrho_{\alpha_i}/(n\|f_i\|_{\alpha_i})$ is admissible: it gives $\|w_z\|_\sharp \le \varrho$ and $\|J_z\|_\infty \le \varrho_J$. □

B.17. **Adding a fluctuation mark to one step.** The mark is a bounded Borel function of the real fluctuation. We first estimate it on a large torus of complex parameters. The individual terms of the expansion are polynomials in those parameters, so the maximum principle extends the estimate across the entire polydisc, including the value zero.

Lemma B.45 (The additional origin in the expansion)**.** Consider an admitted step from k to $k+1$, at radius a, with potential size $\varepsilon_{\mathrm{V}} + \varepsilon_\natural \le c_{\mathrm{cl}}$, and assume $\mathsf{r}_k > 0$. For $b \in \mathfrak{X}_k^+$ put

$$\chi_b(B) = \mathbf{1}_{\{\mathsf{t}_k |B(b)|^2 \ge a_k^2\}}, \qquad \mathcal{V}_\zeta^{\mathrm{mk}}(Y; B) = \sum_{\square(b) = Y} (\operatorname{Log} \zeta_b) \chi_b(B).$$

Here $\square(b)$ is the union of the fine cubes containing an endpoint of b. It is a polymer of size zero, and both endpoints lie in its interior bond set. Initially $\zeta_b \ne 0$ and Log is the principal logarithm. Adjoin $\mathcal{V}_\zeta^{\mathrm{mk}}$ to the three origins of the step, and write its new expansion as $\widetilde{E}_{\varpi,\zeta}(X)$. The original expansion is $\widetilde{E}_\varpi(X)$; put $\Delta_{\varpi,\zeta}^{\mathrm{mk}} = \widetilde{E}_{\varpi,\zeta} - \widetilde{E}_\varpi$. Let

$$\mathbb{T}_k = \{|\zeta_b| = e^{\mathsf{r}_k}\}, \qquad \mathbb{P}_k = \{|\zeta_b| \le e^{\mathsf{r}_k}\}.$$

Then:

(a) $\|\mathcal{V}_\zeta^{\mathrm{mk}}\|_{\star,\lambda_V} \le M^3 a_k^{-2} \max_b |\operatorname{Log} \zeta_b|$. On $\mathbb{T}_k$ it is at most $w_*/32$, and

$$\log \mathsf{E}_k \left[e^{\mathcal{V}_\varpi} \prod_b \zeta_b^{\chi_b} \right] = \sum_X \widetilde{E}_{\varpi,\zeta}(X), \qquad \zeta^\chi = 1 + (\zeta - 1)\chi.$$

(b) Each polymer term at a Mayer index is a polynomial of degree at most one in each ζ_b. A refined cluster term with m polymers has degree at most m in each variable. Their coefficients are holomorphic in ϖ and in the pair of radius a. The refined series converges absolutely and uniformly on $\mathbb{P}_k$, defining both expansions there. They are holomorphic in the interior and continuous on the closed polydisc, jointly with the other parameters. The identity in (a) holds throughout $\mathbb{P}_k$, with a nonzero integral.

(c) On $\mathbb{P}_k$,

$$\|\Delta_{\varpi,\zeta}^{\mathrm{mk}}(X)\| \le \tfrac{1}{32} w_* K_\Pi e^{-\kappa_+ d_{k+1}(X)}.$$

If $\zeta_b = 1 + \lambda_b$ and $\|\lambda\|_\infty \le e^{\mathsf{r}_k} - 1$, then

$$\|\Delta_{\varpi,\zeta}^{\mathrm{mk}}(X)\| \le K_\Pi \epsilon_k^{\mathrm{mk}}(\|\lambda\|_\infty) e^{-\kappa_+ d_{k+1}(X)}.$$

(d) The difference is even and local in the pair on X, and reads only ζ_b with $\square(b)^\sharp \subset X$. It is jointly gauge invariant if the original potential is so member by member. The construction commutes with each linear symmetry carrying cubes to cubes and satisfying $\zeta_{(\mathsf{g}b)^+} = \zeta_b$, and with $B \mapsto -B$. At real ζ the mark is real.

(e) If $\|\lambda\|_\infty \le (e^{\mathsf{r}_k} - 1)/2$, define

$$\Delta^{\mathrm{mk}}(X; b) = \int_0^1 \lambda_b \, \partial_{\lambda_b} \widetilde{E}_{\varpi, \mathbf{1}+t\lambda}(X) \, dt.$$

These terms sum to $\Delta^{\mathrm{mk}}_{\varpi,\mathbf{1}+\lambda}(X)$ and vanish unless $\square(b)^\sharp \subset X$. For any complex weights $|\mathsf{a}_b| \le 1$,

$$\left\| \sum_b \mathsf{a}_b \Delta^{\mathrm{mk}}(X; b) \right\| \le 2K_\Pi \epsilon_k^{\mathrm{mk}}(\|\lambda\|_\infty) e^{-\kappa_+ d_{k+1}(X)}.$$

They retain locality, evenness, holomorphy and joint gauge invariance; a symmetry carries the pointer b to $(\mathsf{g}b)^+$.

■

Proof. On $\{\chi_b = 1\}$, $1 \le \mathsf{t}_k |B(b)|^2 / a_k^2$. Thus

$$|\mathcal{V}^{\mathrm{mk}}_\zeta(Y; B)| \le \frac{M^3}{a_k^2} \max_b |\operatorname{Log} \zeta_b| \left(1 + M^{-3} \sum_{b' \in \mathfrak{X}^{\circ +}\langle Y \rangle} \mathsf{t}_k |B(b')|^2 \right).$$

Since $d_k(Y) = 0$, this is the required norm at every rate. On $\mathbb{T}_k$, $|\operatorname{Log} \zeta_b| \le \mathsf{r}_k + \pi = \kappa_{\mathrm{mk}} a_k^2$ and $M^3 \kappa_{\mathrm{mk}} = w_*/32$. The expansion admits four origins with these sizes. It requires only Borel dependence on the real fluctuation, so the indicator causes no change in its hypotheses. Exponentiating and multiplying over Y gives exactly $\prod_b \zeta_b^{\chi_b}$, proving (a).

A Mayer vertex of the mark contributes

$$\prod_{\square(b)=Y} (1 + (\zeta_b - 1)\chi_b) - 1 = \sum_{\varnothing \ne S \subset \{b : \square(b) = Y\}} \prod_{b \in S} (\zeta_b - 1)\chi_b.$$

In a single polymer term each vertex occurs once. The decoupled measures and tests do not read ζ. The term is therefore of degree at most one in each ζ_b, with holomorphic coefficients furnished by the original parameter-dependent integral. Clusters may repeat or overlap polymers, so their degree can be as large as their number of polymers. On $\mathbb{T}_k$ each polymer term is bounded by its usual positive majorant, formed with mark size $w_*/32$. The maximum principle for a polynomial extends this same bound to $\mathbb{P}_k$. The convergent majorant series then proves absolute uniform convergence, holomorphy and continuity there. On a finite torus the entire integral in (a) is a polynomial in the finitely many ζ_b. It agrees with the exponential of the convergent series on the distinguished boundary; Cauchy's formula extends the equality to the polydisc. The exponential never vanishes, including at $\zeta_b = 0$, where the factor is $1 - \chi_b$. This proves (b).

Subtracting the unmarked expansion leaves precisely the clusters meeting a marked vertex. The marked majorant has size

$$K_E \eta^{\mathfrak{m}}, \qquad \eta^{\mathfrak{m}} \le (L+2)^3 S_{\mathrm{s}} e^{4\lambda_1} w_0^{-1} \frac{w_*}{32}, \qquad K_E (L+2)^3 S_{\mathrm{s}} e^{4\lambda_1} / w_0 = K_\Pi.$$

This proves the first bound of (c), first on the boundary and then term by term inside. For $\lambda \ne 0$ apply Schwarz's lemma to $\xi \mapsto \Delta^{\mathrm{mk}}_{\varpi, \mathbf{1} + \xi\lambda/\|\lambda\|_\infty}(X)$ on $|\xi| < e^{\mathsf{r}_k} - 1$. It vanishes at zero, and its boundary bound is the first bound of (c). The case $\lambda = 0$ is immediate.

Locality follows from the regions of the marked vertices. The mark reads $|B(b)|$ and no pair. Gauge transformations act by orthogonal adjoint maps on $B(b)$, so they preserve it. Likewise a linear symmetry acts by $B((\mathsf{g}b)^+) = O_b B(b)$, with O_b orthogonal, and carries $\square(b)$ to $\square((\mathsf{g}b)^+)$. The expansion intertwines these transformations on $\mathbb{T}_k$, term by term. The resulting identities are

polynomial identities in finitely many ζ_b, and hence hold everywhere: their coefficients are Fourier coefficients of their restriction to $\mathbb{T}_k$. Uniform convergence carries the identities to the sums and their derivatives. This proves (d).

For $t \in [0,1]$, $|\xi| \le 1$ and $|\mathsf{a}_b| \le 1$, the family $\mathbf{1} + t\lambda + \xi(\mathsf{a}_b\lambda_b)_b$ lies in $\mathbb{P}_k$. By (c), its difference from the unmarked expansion is at most $2K_\Pi \epsilon_k^{\mathrm{mk}}(\|\lambda\|_\infty) e^{-\kappa_+ d_{k+1}(X)}$. Cauchy's estimate in ξ and integration in t give the weighted bound in (e). The fundamental theorem of calculus gives the sum rule. Locality and the symmetries commute with these derivatives and integrals, proving the remaining claims. □

Proposition B.46 (One marked step)**.** Let $h \ge h_F$, $\gamma \le \gamma(h, 5B_s)$, $\mathsf{r}_k > 0$, and $|\lambda| \le (e^{\mathsf{r}_k} - 1)/2$. Take a point of the open step domain at weight h and the fine mass of the datum. Assume its potential has size at most c_{cl} and its earlier families are covariant under $M\mathbb{Z}^3$. Insert the uniform factor $\prod_b (1 + \lambda\chi_b)$. The resulting step is defined on both the torus and the infinite lattice. Its new activity before local extraction is $\widetilde{E}^{(k+1)} + \Delta_\lambda^{\mathrm{mk}}$; all terms outside the expansion are unchanged.

Point the difference at level-$k+1$ sites by

$$\Delta^{\mathrm{mk}}(X; z) = \sum_b h_z(x_b) \Delta^{\mathrm{mk}}(X; b),$$

where $h_z(x_b)$ is the mean of the tent at the two endpoints of b. Then the pointed terms vanish unless $z \in X$, sum to $\Delta_\lambda^{\mathrm{mk}}$, and

$$\|\Delta_\lambda^{\mathrm{mk}}\|_{k+1} \le C_E K_\Pi \epsilon_k^{\mathrm{mk}}(\lambda), \qquad \sup_z \|\Delta^{\mathrm{mk}}(\cdot; z)\|_{k+1} \le 2C_E K_\Pi \epsilon_k^{\mathrm{mk}}(\lambda), \qquad C_E = 4096.$$

The changes in the three extracted numbers satisfy

$$|\delta e| \le M^{-3} C_E K_\Pi \epsilon_k^{\mathrm{mk}}(\lambda),$$
$$|\delta b| \le C_{am} r_\alpha^{-2} C_E K_\Pi \epsilon_k^{\mathrm{mk}}(\lambda),$$
$$|\delta \mu| \le c_b^{-2} C_{am}^{\psi} (\nu h^2)^{-1} C_E K_\Pi \epsilon_k^{\mathrm{mk}}(\lambda).$$

After extraction the changes in both new families, including their pointed norms, are at most $K_{\mathrm{D}} \epsilon_k^{\mathrm{mk}}(\lambda)$, where

$$K_{\mathrm{D}} = 2C_E K_\Pi \max\{2 + C_{am} r_\alpha^{-2}, K_\mu\}.$$

All objects are holomorphic in the parameters. They retain locality, parity, gauge and flavour invariance, and translation covariance. On the infinite lattice the extracted families have zero scalar mean per cell, zero action coefficient, and zero normalized trace of the cell mean of their degree-two zeroth moment.

For real λ, if the input numbers and the totals of the input activities are real on tested background pairs of a torus, the new total before extraction is real there. If the same input numbers and infinite lattice families occur on arbitrarily large admissible tori, with real data and uniform bounds, the extracted numbers are real as well. Extraction then preserves reality on each such torus. ■

Proof. Apply Lem. B.45 at the uniform $\zeta = \mathbf{1} + \lambda$. The mark enters only the joined fluctuation integral. It does not enter the operators, normalizers, calibration prefactor, power of the relative coupling or $\mathbf{G}_\tau$ outside that integral. The tents satisfy $0 \le h_z \le 1$ and $\sum_z h_z(x_b) = 1$. If $h_z(x_b) \ne 0$, an endpoint y of b has $|y - z|_\infty < 1$ in level-$k+1$ units. In level-k units this is less than $L \le M$; thus $z \in \square(b)^\sim \subset \square(b)^\sharp$. The support and sum rule follow, and the weighted bound in Lem. B.45(e) gives the pointed estimate. Summing over polymers uses

$$\sup_{\square} \sum_{X \supset \square} e^{-(\kappa_+ - \kappa) d_{k+1}(X)} \le C_E,$$

the same count on either lattice.

On a polymer fitting into an admitted torus, all four origins agree with their infinite lattice versions under the chart. Their terms agree on $\mathbb{T}_k$, and hence as polynomials in the marked bond

variables. They therefore agree throughout the polydisc, including after differentiating and pointing. This defines the infinite lattice activity independently of the fitting torus. The same local bounds and polymer count apply there.

The extracted numbers are the three continuous linear functionals of the infinite lattice activity. The scalar value per site costs M^{-3}, because each cube has M^3 sites. The action functional has norm at most $C_{am} r_\alpha^{-2}$. The mass functional is $-(\nu\eta_{k+1}^2)^{-1}$ times the trace of the cell mean of the zeroth moment; $|\eta_{k+1}| \geq c_b$, and ν^{-1} times the absolute trace of the cell mean is bounded by $C_{am}^{\psi}(\nu h^2)^{-1}$ times the activity norm. This proves the three estimates. Local extraction is linear and multiplies the pointed norm by at most $2 + C_{am} r_\alpha^{-2}$ in degree zero and K_μ in positive degree, proving the post-extraction bound. Each extraction functional equals one on its own local term and zero on the other two: the action vanishes at the trivial field, a constant has zero second derivative, and $L^{-(k+1)}\mathsf{S}_1(\psi^{\mathrm{bg}})$ has zeroth moment $\eta_{k+1}^2\,1$. This proves the three normalizations.

We give the reality argument because boundedness alone does not supply it. Write $\dagger = \alpha_{(i,-i)}\sharp$. Flavour invariance of the earlier terms makes their totals $\sharp$-invariant on the tested pairs. The potential at a real pair and real fluctuation is therefore fixed by $\sharp$. For $0 \leq q \leq 1$, multiply this potential by q and replace the mark by $\mathbf{1} + q\lambda$. These parameters stay in the admitted domains. Let $\mathcal{L}(q)$ be the total expansion and $\Xi(q)$ its exponential, the marked Gaussian integral. Conjugation commutes with both integrations and the exponential of even elements, so $e^{\mathcal{L}(q)^\sharp} = e^{\mathcal{L}(q)}$. Their difference is a scalar in $2\pi i\mathbb{Z}$. It varies continuously with q and is zero at $q = 0$, where the pure Gaussian logarithm is real. It is consequently zero at $q = 1$. The terms outside the expansion are real scalars. Flavour invariance converts $\sharp$-invariance into $\dagger$-invariance, and gauge transformations preserve it.

It remains to check the numbers read on the infinite lattice. On a torus of side N in level-$k+1$ units, the terms on nonwrapping polymers agree with their lifts. Their pointing agrees too: the torus tents are periodizations, and at most one lift of a pointer contributes to a fitting polymer, contained in a cuboid of side less than $N/2$. The remaining polymers have $d_{k+1} \geq N/(2M) - 2$. Their bounds, including pointed bounds, are uniform in N: the expansion and mark decay at κ_+; the gauge normalizer and closed loops decay at $\kappa + 1$; and

$$|\mathbf{G}_\tau(X, P, z)| \leq K_{\mathbf{G}} \sigma_\star e^{-(\kappa+1) d_{k+1}(X)}.$$

The field-independent per-site terms occupy size-zero polymers and do not contribute to these tails when $N > 4M$. Thus the coefficients at fixed fermion places, pointed totals, and the scalar second-derivative kernels converge to the infinite lattice ones. For the last assertion the scalar terms are strictly gauge invariant and have the same local bounds, so the usual linear-response kernel limit applies.

The scalar mean per site on each torus is real. Its limit is the cell mean on the infinite lattice, by $M\mathbb{Z}^3$ covariance and uniform tail bounds; thus $e_{k+1} + \delta e$ is real. The second-derivative kernels are real on every torus near the trivial field and hence in the limit. Their second moments use real weights, so $b_{k+1} + \delta b$ is real. For the degree-two kernel, conjugation gives

$$\mathsf{k}(y, y') = \gamma_4 \mathsf{k}(y', y)^\dagger \gamma_4.$$

Its row sums over a cell converge absolutely. Translation covariance and the decomposition $y' = y'' + v$, $y'' \in c$, $v \in M\mathbb{Z}^3$, give

$$\sum_{y \in c} \sum_{y'} \operatorname{tr} \mathsf{k}(y', y) = \sum_{y'' \in c} \sum_{y \in c} \sum_{v} \operatorname{tr} \mathsf{k}(y'', y - v) = \sum_{y'' \in c} \sum_{y'} \operatorname{tr} \mathsf{k}(y'', y').$$

The first and last expressions are conjugates. Their value is real, and η_{k+1} is real; therefore $\mu_{k+1} + \delta\mu$ is real. Subtracting the resulting real local terms preserves reality on every tested torus. All other symmetries and holomorphy follow from Lem. B.45 and linear extraction. □

B.18. **Iterating marked steps.** On the chart of an admitted step, the Borel indicator in Def. B.39 is the indicator χ_b of Lem. B.45. Indeed, a block field passing the step tests passes the density test, so its plaquette size is less than $2\varepsilon(\mathfrak{t}_{k+1})$. It lies in the inner ball and hence in the field cut. The inverse fluctuation coordinate there satisfies $\mathcal{A}_k^{\diamond}(b) = B(b)$. Since $\mathfrak{b}(\mathfrak{t})\mathfrak{t}^{-1/2} = \delta(\mathfrak{t})$, this gives

$$\bar{\chi}_{k,b}(V_W(B)) = \mathbf{1}_{\{\mathfrak{t}_k |B(b)|^2 \geq a_k^2\}}.$$

Thus the marked expansion represents the actual Borel mark under the history map.

Proposition B.47 (The marked flow)**.** Use Def. B.39 at $h = h_F$, $A = 5B_s$, and put

$$C_{\mathrm{Lip}} = \max\{2 + 2\sqrt{2}\theta_s h^2 A/C_w,\ 2 + 4c_s h^2 A^2/C_2,\ 2 + 8(C_w + \theta_s h^2 A)/r_x,\ L + 6(C_\mu + c_s A^2)/A\},$$
$$C_{\mathrm{mk}} = \max\{K_{\mathrm{D}}/(2C_w),\ K_{\mathrm{D}}/(2C_2),\ 2C_{am} r_\alpha^{-2} C_E K_\Pi / r_x,\ 3c_b^{-2} C_{am}^{\psi} C_E K_\Pi/(\nu h^2 A)\},$$
$$C_{e,\mathrm{Lip}} = 3L^3 r_x + 2M^{-3} K_M (c_{\mathrm{cl}} + 1),$$
$$\bar{\theta}_{\mathrm{mk}}(r) = C_{\mathrm{Lip}}^s (s-u) C_{\mathrm{mk}} \frac{w_*}{16} e^\pi L^s \mathfrak{b}\, \frac{r}{\Lambda_{\mathrm{mk}}}.$$

Suppose $r > 0$ satisfies

$$e^{-\pi}\Lambda_{\mathrm{mk}} \geq 2, \qquad 2r \leq e^{-\pi}\Lambda_{\mathrm{mk}} - 1, \qquad \bar{\theta}_{\mathrm{mk}}(r) \leq \tfrac{1}{2}. \tag{Mk1–2}$$

For $|\lambda| \leq r$, start from the original flow through t_0, insert the uniform mark in the steps $t_0 \leq k < t_1$, and continue without marks from t_1 to K. At each step extract its own three numbers and update

$$x_{k+1}(\lambda) = L^{-1} x_k(\lambda) - b_{k+1}(\lambda), \qquad s_{k+1}(\lambda) = L s_k(\lambda) + \mu_{k+1}(\lambda).$$

This constructs the flow through K in the open step domain. The families lie in the spaces $\mathfrak{F}_j$, including all linear symmetries and the full zero degree-two moment. At a marked level the old part uses its original pointing and Δ^{mk} uses the tents of Prop. B.46. All numbers, energies and families are holomorphic for $|\lambda| < r$ and continuous on the closed disc. For real λ the numbers and energies are real; conjugation carries the families at λ to those at $\bar{\lambda}$.

With $\bar{\theta} = \bar{\theta}_{\mathrm{mk}}(|\lambda|)$, for all $i, j \leq K$,

$$|x_i(\lambda) - x_i| \leq \tfrac{1}{2} r_x x_i \bar{\theta}, \qquad |s_i(\lambda) - s_i| \leq \tfrac{1}{3} A\phi(x_i)\bar{\theta},$$
$$\|F_0^{(j)}(\lambda) - F_0^{(j)}\|_j \leq 2C_w \phi(\mathfrak{t}_{j-1})\bar{\theta}, \qquad \|E_{\geq 2}^{(j)}(\lambda) - E_{\geq 2}^{(j)}\|_j \leq 2C_2 \phi(\mathfrak{t}_{j-1})^2 \bar{\theta}.$$

The bounds include pointed terms and both lattice families at each level. The history with no large-field region, for the density $\mathsf{Mk}_\lambda^{(<k)} e^{-\beta A - S_F}\, dU$, has the usual level-$k$ formula with these numbers and families, as an identity of measures. At t_1 its Berezin integral is $T_0(\lambda)$. Finally

$$|E_K(\lambda) - E_K| \leq N_{[s]} s \left[C_{e,\mathrm{Lip}} \bar{\theta} + M^{-3} C_E K_\Pi \frac{w_*}{16} e^\pi \frac{|\lambda|}{\Lambda_{\mathrm{mk}}} \right].$$

In particular this difference is at most one if the last right side at $|\lambda| = r$ is at most one; call this additional condition (Mk3). ∎

Proof. Let P_k^0 be the original point at level k. For a point P with primed coordinates, define

$$\theta_k(P) = \max_{i,j \leq k} \max \left\{ \frac{|x_i' - x_i|}{r_x x_i/2}, \frac{|s_i' - s_i|}{A\phi(x_i)/3}, \frac{\|F_0'^{(j)} - F_0^{(j)}\|_j}{2C_w \phi(\mathfrak{t}_{j-1})}, \frac{\|E_{\geq 2}'^{(j)} - E_{\geq 2}^{(j)}\|_j}{2C_2 \phi(\mathfrak{t}_{j-1})^2} \right\}.$$

On a torus include both families in this maximum. Let Ω_k consist of points with $\theta_k < 1$ whose families lie in $\mathfrak{F}_j$, with the prescribed remaining scalar terms at their coefficients. Since $|s_i| \leq \bar{B}_M/x_i \leq \bar{B}_M \phi(x_i)$ and $3\bar{B}_M/2 \leq A$, this set lies inside the open step domain: $|x_i' - x_i| < r_x x_i$ and $|s_i'| < A\phi(x_i)$. The potential there has size less than $c_{\mathrm{cl}}/2$.

The coupling windows give $\phi(x_k) \le \sqrt{2}\phi(\mathsf{t}_k)$, $\phi(\mathsf{t}_k) \le 1/2$, and $\phi(x_k) < 1$. The output bounds on Ω_k are

$$\begin{aligned} \|F_0^{(k+1)}\| &\le (2C_w + 2\sqrt{2}\theta_s h^2 A)\phi(\mathsf{t}_k), \\ \|E_{\ge 2}^{(k+1)}\| &\le (2C_2 + 4c_s h^2 A^2)\phi(\mathsf{t}_k)^2, \\ |b_{k+1} - e_{\mathsf{N},k+1} - b_{k+1}^F| &\le 2(C_w + \theta_s h^2 A)\phi(x_k), \\ |\mu_{k+1}| &\le (C_\mu + c_s A^2)\phi(x_k). \end{aligned}$$

The fixed numbers in the third line do not read P. The variable scalar contribution per site, $\widetilde{e}_{k+1}$, is bounded by $M^{-3}K_M(c_{\mathrm{cl}} + 1)$. The full scalar number is

$$e_{k+1} = e_{k+1}^{\mathrm{YM},0} + e_{k+1}^{\mathrm{F},0} - 3(L^3 - 1)\,\mathrm{Log}(x_k'/\mathsf{t}_k) + \widetilde{e}_{k+1}.$$

For $P \in \Omega_k$ with $\theta_k(P) = \theta$, the line $P_k^0 + \xi(P - P_k^0)/\theta$, $|\xi| < 1$, lies in Ω_k, because its family constraints are linear. Schwarz's lemma bounds the change of an output bounded by S on that line by $2S\theta$. For family norms this is applied to coefficients and bounded linear combinations before taking their suprema.

At a marked level, Prop. B.46 adds a perturbation with

$$\epsilon_k^{\mathrm{mk}}(\lambda) \le \frac{w_*}{16} e^{\pi} \frac{|\lambda|}{\Lambda_{\mathrm{mk}}}.$$

At unmarked levels put $\epsilon_k = 0$. The four scaled output distances are bounded respectively by

$$\begin{aligned} &(2 + 2\sqrt{2}\theta_s h^2 A/C_w)\theta_k + K_{\mathrm{D}}\epsilon_k/(2C_w\phi(\mathsf{t}_k)), \\ &(2 + 4c_s h^2 A^2/C_2)\theta_k + K_{\mathrm{D}}\epsilon_k/(2C_2\phi(\mathsf{t}_k)^2), \\ &(2 + 8(C_w + \theta_s h^2 A)/r_x)\theta_k + 2C_{am} r_\alpha^{-2} C_E K_\Pi \epsilon_k/(r_x x_{k+1}), \\ &(L + 6(C_\mu + c_s A^2)/A)\theta_k + 3c_b^{-2} C_{am}^{\psi} C_E K_\Pi \epsilon_k/(\nu h^2 A \phi(\mathsf{t}_k)). \end{aligned}$$

For the coefficient line we used $L^{-1}x_k \le 2x_{k+1}$ and $x_{k+1} \ge 1 \ge \phi(x_k)$. For the mass line we used $x_{k+1} \le \min\{x_k, \mathsf{t}_k\}$ and monotonicity of ϕ. Both follow from the original coupling windows and their ceilings on γ. Since $\phi(\mathsf{t})^{-1} \le \phi(\mathsf{t})^{-2} \le \mathsf{t}$ and $\mathsf{t}_k \le L^s \mathsf{b}$,

$$\theta_{k+1} \le C_{\mathrm{Lip}}\theta_k + C_{\mathrm{mk}}\mathsf{t}_k\epsilon_k.$$

Starting at $\theta_{t_0} = 0$, this yields

$$\theta_{k+1} \le \sum_{j=t_0}^{\min\{k,t_1-1\}} C_{\mathrm{Lip}}^{k-j} C_{\mathrm{mk}} \mathsf{t}_j \epsilon_j \le \bar{\theta}_{\mathrm{mk}}(|\lambda|) \le \tfrac{1}{2}.$$

It also gives holomorphy and continuity at each step.

We must verify the constraints defining $\mathfrak{F}_{k+1}$ before applying the next step. First run this verification at real λ, simultaneously on the infinite lattice and all admissible tori obtained by changing the side. The one-weight setting is available on those tori: its only side requirement is $L\ell' \ge C_L M$, which follows from their admissibility; its remaining conditions involve the fixed cutoff, label and $\mathsf{t}_{t_1} = L^u \mathsf{b}$. All estimates just proved are uniform in the side. Thus the hypotheses for the reality conclusion of Prop. B.46 hold simultaneously.

The internal symmetries act by maps α_S that fix scalar coefficient functions. Inductively they fix the potential and commute with both Gaussian integrations. For any marked bond parameters in the polydisc, let $\mathcal{L}_\zeta$ be the total expansion and $\Xi_\zeta = e^{\mathcal{L}_\zeta}$. Then $e^{\alpha_S \mathcal{L}_\zeta} = e^{\mathcal{L}_\zeta}$. Their difference is a scalar in $2\pi i\mathbb{Z}$, but α_S fixes scalar parts; hence the difference is zero. This proves invariance of the totals. For the pointed totals of the original step, the insertion formula

$$\int_0^1 \mathsf{E}_k[\dot{\mathcal{V}}_z e^{\mathfrak{V}_t}]\mathsf{E}_k[e^{\mathfrak{V}_t}]^{-1}\,dt$$

has invariant numerator and denominator. For the additional pointed terms, differentiate $\alpha_S \mathcal{L}_\zeta = \mathcal{L}_\zeta$ in ζ_b and use the bond pointing and tents. Thus the pointed totals are invariant as well.

The real logarithm argument in Prop. B.46 works for every real family ζ in the polydisc: use the path $\mathbf{1} + q(\zeta - \mathbf{1})$ while scaling the potential by $q \in [0,1]$. Consequently real derivatives in ζ_b are $\sharp$-invariant, and so are the new pointed totals. The numbers extracted on the infinite lattice are real by the torus limit argument there. At a tested covariance pair, the density at $\mathtt{g}V$ is the image of that at V. The chart substitution $B = \mathsf{O}B_1$ preserves the uniform mark, since it permutes the non-pivot bonds and preserves $|B(b)|$. Thus covariance of the new density follows. The scalar ambiguity between its two logarithms is zero, because both scalar parts are real at the tested pairs. This proves covariance on shell.

On nonwrapping polymers all terms and pointers agree with their lifts. The torus tents are periodizations, with at most one contributing lift for a fitting polymer. Uniform decay controls the remaining terms. Passing to increasing admissible tori therefore gives the same linear symmetries for the infinite lattice totals and pointed totals at the required pairs. These symmetries make the degree-two zeroth moment a scalar matrix, constant from cell to cell. The mass extraction has already removed its normalized trace, so the whole zeroth moment is zero. Likewise the action coefficient is the invariant coefficient and the scalar extraction removes the trivial-field value. This verifies all family-space constraints at real λ.

Every remaining constraint is a closed complex-linear condition on the families. Its value is holomorphic in λ and vanishes on the real interval, so it vanishes on the disc and by continuity on its boundary. The same continuation proves the complex-linear symmetry identities. For conjugation use that $\lambda \mapsto (\Upsilon_{\bar\lambda})^\dagger$ is holomorphic whenever Υ_λ is; the equality on real λ gives $\Upsilon_\lambda^\dagger = \Upsilon_{\bar\lambda}$. This also applies to the infinite lattice coefficients, pointed totals, cell means and second-derivative kernels, which are continuous linear functions in the family norms. The prescribed scalar remainder does not obstruct this assertion: only $\mathbf{G}^{(k+1)}$ depends on the coefficient; its pointed terms vanish at the trivial field, and its total vanishes near that field on every torus by the sum rule. Holomorphy extends the sum rule to complex λ, and the uniform kernel limit makes its infinite lattice second derivative zero. The other remainder terms are independent of λ.

We finish by checking the identity of measures. Suppose the level-k density has the stated form $d_k[P_k(\lambda)]$ on the density-test set, and is zero elsewhere. It is bounded and Borel. Multiply by $m_k = \prod_b (1 + \lambda \bar\chi_{k,b})$ for $k < t_1$ and by $m_k = 1$ afterwards. The exact image map is linear in this density, and applies coefficient by coefficient to $\mathrm{Ber}_k(\mathrm{Ker}_k\, m_k d_k)$. On its chart the preceding identity for $\bar\chi_{k,b}$ gives the marked Gaussian integral. The completed square, background substitutions and localization sum rules are unchanged. They are algebraic identities at complex as well as real coefficients. The complex action parameter satisfies $|x_k(\lambda)/\mathtt{t}_k - 1| < \sigma_\star$, so its real part is positive and the Gaussian change of variables applies. The marked expansion and subsequent extraction give exactly $d_{k+1}[P_{k+1}(\lambda)]$. This proves the measure identity by induction. At t_1, integration over all fermion generators gives $\det(D[U] + m_0)$ at each bare field and the image preserves total mass. The total is therefore $T_0(\lambda)$.

There are s possible changed scalar increments above t_0, each with $n_{k+1} \le N_{[s]}$. The logarithmic coupling term changes by at most $3L^3 r_x \theta_k$, because $|\operatorname{Log}(x_k(\lambda)/x_k)| \le r_x \theta_k$. The expansion value changes by at most $2M^{-3} K_M (c_{\mathrm{cl}} + 1)\theta_k$. At marked levels add $M^{-3} C_E K_\Pi \epsilon_k^{\mathrm{mk}}(\lambda)$. Summing gives the final energy bound. □

Remark B.48 (Admissible initial flows). The proof also applies to a flow started at t_0. What it requires is its bounded Borel density there; the same open step domains and bounds around its reference points; the coupling windows and $|s_i| \le \bar B_M / x_i$; the family-space and translation conditions on its earlier activities; and the following real-data properties on all admissible tori, with common numbers and infinite lattice families and uniform bounds. The initial numbers, energies and fine masses must be real; the earlier activity totals must be invariant under the internal symmetries

and $\sharp$ at their tested pairs; and the initial density must be covariant at tested covariance pairs. The proof uses the bare density only to identify its final scalar integral with T_0. For another start, replace T_0 by the Berezin integral of its marked term at t_1.

Proposition B.49 (A mark bounds fluctuation tails)**.** For every $r > 0$, the two integrals of fluctuation indicators in Prop. B.38(c) are bounded by $(T_0(r) - T(0))/r$. Under (Mk1)–(Mk3), put

$$\mathfrak{u}(r) = T_0(r) e^{-E_K(r) - (C+2)\ell^3}.$$

Then $T_0(r) \le e\,\mathfrak{u}(r)\mathsf{Q}_\ell T(0)$, so each tail is at most $e\,\mathfrak{u}(r)\mathsf{Q}_\ell r^{-1} T(0)$. The changes under a weight $\mathfrak{w}'$ are those in Prop. B.38(b),(c). ■

Proof. The first assertion is Bernoulli's inequality as proved in Prop. B.38. At real $r > 0$ the marked density is nonnegative, $E_K(r)$ is real, and $E_K(r) \le E_K + 1$. Combine its defining identity with $T(0) \ge e^{E_K - C_P(1+\log x_K)\ell^3}$. □

Lemma B.50 (Support of the difference of weights)**.** In the two-lattice gauge setting, assume $1 \le u < s < K$, $L^u \mathsf{b} \ge \Theta_\star$ and

$$256 \mathsf{K}_u c_\rho^2 A_0^2 N_{[s]} L^{-(K-s)} \le \gamma_{\mathrm{r}} A_1^2. \tag{k4}$$

For $(\bullet, \circ) = (\mathrm{A}, \mathrm{B})$ or (B, A) and any Borel $0 \le \varphi \le 1$ on the common base configuration space,

$$\mathbf{1}_{\{\mathsf{k}^\circ(W)=0\}} \varphi(W) \mathsf{k}^\bullet(W) \le \sum_{k=t_0}^{t_1-1} \sum_{b \in \mathfrak{X}^+_{(k)}} \varphi(W) \mathsf{k}^\bullet(W) \bar{\chi}^\bullet_{(k),b}(W_k).$$

■

Proof. The two weights consist solely of the gauge tests, field cuts, inner balls, gauge-fixing factors and fluctuation coordinates at the calibrations. The gauge comparison says that failure of the other weight, when one is nonzero, forces at least one moderate fluctuation. Its applicability at the present depths uses only the following bounds. The inequalities $1 \le t_0 < t_1 \le K - 1$ give the permitted levels; $L^u \mathsf{b} \ge \Theta_\star$ gives the two-weight and one-weight settings and all calibrations needed by the background comparison; and for $k \ge t_0$, $N_k \le N_{[s]}$ and $L^{-k} \le L^{-(K-s)}$, so (k4) gives its error bound at every such level. The particular fractional choices of the depths are not used by that argument. The resulting pointwise inequality, multiplied by the nonnegative φ, is the claim. □

Constants and large-cutoff conditions. The constants of the marked step and flow are formed after the constants in their displayed formulas. In particular K_{D} follows K_μ, and $C_{\mathrm{Lip}}, C_{\mathrm{mk}}, C_{e,\mathrm{Lip}}$ follow $h_F, A, C_w, \theta_s, C_2, c_s, C_\mu, r_x$ and precede γ. The C_{Lip} here is the constant of the four scaled output distances, independently of the Lipschitz constant for free averaging kernels. These definitions impose no new ceiling on an earlier constant.

For later use, if $u \ge \varphi_1 K$ with $\varphi_1 > 0$, $s \le K$, $\mathsf{b} \le \bar{\mathsf{b}}$, and $r = \Lambda_{\mathrm{mk}}^{1/2}$, then $\mathsf{R}^{\mathrm{F}}_{t_1} \le 1/8$ and (Mk1)–(Mk3) hold for sufficiently large K. Indeed, $\mathsf{y}_{[u]} \ge (\varphi_1 K \log L)^{2\mathfrak{p}_0}$, where $2\mathfrak{p}_0 \ge 2$. The logarithms of the left sides in (Mk2), (Mk3) and of $\mathsf{R}^{\mathrm{F}}_{t_1}$ are bounded, up to a fixed $\log \mathsf{Q}_\ell$, by

$$K(\log C_{\mathrm{Lip}} + 4 \log L) + 2 \log K + 3 \log \ell + \log(\bar{\mathsf{b}} C'') - \min\{\kappa_{\mathrm{mk}} A_1^2/8, \theta_{\mathrm{dis}} A_0^2\} (\varphi_1 K \log L)^{2\mathfrak{p}_0}.$$

This tends to $-\infty$, and (Mk1) follows from $\Lambda_{\mathrm{mk}} \to \infty$. Also $L^u \mathsf{b} \ge \Theta_\star$ for large K. For the adjacent cutoff $K + 1$ one can use $\varphi_1 = \phi_1(n)/2$ and $\bar{\mathsf{b}} = X_g$, so one threshold works for both lattices.
The two comparison estimates. At $t = t_1^\bullet$, Prop. B.43 gives

$$0 \le Z^\bullet - T^\bullet(0) \le R^F Z^\bullet,$$

because both calibrations at that level are $L^u \mathsf{b} \ge L^u \gamma^{-2}$ and $\mathsf{Q}_\ell \le \mathsf{M}_{g,\ell}$. This is Prop. 11.8.

Under (k0)–(k2), Prop. B.42 applies Prop. B.41 at radii $2R_i$ and gives

$$\left|\partial_z^n[\log T^\bullet - \log \hat{T}^\bullet - \mathfrak{c}^\bullet](0)\right| \le \left[\tfrac{1}{2}(4(\mathsf{Q}_\ell+1))^{n+1} + C'_E N_{[u]}\right](r_\lambda - 1)^{-1}\prod_i (2R_i)^{-1}.$$

The original total is entire in the sources. The interpolation gives holomorphy of $\hat{T}^\bullet$ and $\mathfrak{c}^\bullet$ on $\mathbb{Q} = \mathbb{P}(2R)$. The lower bound in Prop. B.41 gives $T^\bullet(0) > 0$, while Prop. 11.15 gives $\hat{T}^\bullet(0) \ge e^{\hat{E}_{K^\bullet} - C_P(1+\log X_g)\ell^3} > 0$. Since $\mathsf{Q}_\ell + 1 \le 2\mathsf{M}_{g,\ell}$ and $(r_\lambda - 1)^{-1} \le 2/r_\lambda$, the displayed bound is at most

$$\left[2(4\mathsf{M}_{g,\ell})^{n+1} + C'_E N_{[u]}\right] r_\lambda^{-1}\prod_i R_i^{-1} \le \epsilon_{\mathrm{ro}}\prod_i R_i^{-1}.$$

This proves Prop. 11.9. For $n \ge 2$, Prop. B.44, with $C' = 1$ for each flow, also gives under (Sep)

$$|\partial_z^n \mathfrak{c}^\bullet| \le 2^{n+1} C_{\mathscr{C}} M^{-3} N_{[u]} e^{-\kappa_{\mathrm{E}} \mathrm{d} L^u}\prod_i \varrho_i^{-1} \le 2\epsilon_{\mathrm{mx}}.$$

B.19. **Marks in the restarted flow.** We apply the marked-step construction to the zero-source restart. The required margins are verified below; in particular the restart is an admissible centre for the open-domain estimates.

Definition B.51 (Parameters for the marked restart)**.** Use the two-cutoff data under (k0), (k1). All cutoff thresholds in this subsection refer to K^{A}, from which s, u are chosen. Within either lattice write $K = K^\bullet$, $t_0 = K - s$, $t_1 = K - u$. Let $\hat{P}_k$ be the zero-source restart of Props. B.5 and B.26, starting with

$$\hat{x}_{t_0} = L^s \mathsf{b}, \qquad \hat{s}_{t_0} = F^{\mathrm{UV}}/(L^s\mathsf{b}), \qquad \hat{E}_{t_0} = 0,$$

and no earlier activity. Use $h = h_F$, $A = 5B_s$ and the constants of Prop. B.47. The mark at a level $t_0 \le k < t_1$ is

$$\mathsf{m}_{k,\lambda}(V) = \prod_{b\in\mathfrak{X}_k^+}(1 + \lambda\bar{\chi}_{k,b}(V)).$$

It is bounded and Borel in V and polynomial in λ. Its chart expression is the mark of Lem. B.45.

Hereafter Λ_{mk} uses γ^{-2}:

$$\Lambda_{\mathrm{mk}} = \exp[\kappa_{\mathrm{t}} A_1^2 (\log(L^u\gamma^{-2}))^{2\mathfrak{p}_0}], \qquad \kappa_{\mathrm{t}} = w_*/(128M^3).$$

It is no larger than its value with b in place of γ^{-2}. Set

$$K_{\mathrm{D}}^{\mathrm{arr}} = K_\Pi(2 + (M^{-3} + C_{am} r_\alpha^{-2})C_E),$$
$$C_{\mathrm{win}} = \max\{32, A, 2K_{\mathrm{D}}^{\mathrm{arr}}/C_{\mathrm{mk}}\}, \quad q_2 = \tfrac{3}{4} + \tfrac{1}{4\sqrt{2}}, \quad \sigma_* = \tfrac{23}{11}B_F + \tfrac{1}{2},$$
$$\bar{\theta}_{\mathrm{rs}}(r) = C_{\mathrm{Lip}}^s (s-u) C_{\mathrm{mk}} \frac{w_*}{16} e^\pi L^s \mathsf{b}\, r/\Lambda_{\mathrm{mk}},$$
$$r_{\mathrm{rs}} = 2\Lambda_{\mathrm{mk}}^{1/2}, \qquad \theta_{\mathrm{rs},K^{\mathrm{A}}} = 2C_{\mathrm{Lip}}^s s C_{\mathrm{mk}} \frac{w_*}{16} e^\pi L^s X_g \Lambda_{\mathrm{mk}}^{-1/2}.$$

Thus $\bar{\theta}_{\mathrm{rs}}(r_{\mathrm{rs}}) \le \theta_{\mathrm{rs},K^{\mathrm{A}}}$ for both lattices. For a radius $r > 0$ consider

(Mr1) $e^{-\pi}\Lambda_{\mathrm{mk}} \ge 2$, $\quad 2r \le e^{-\pi}\Lambda_{\mathrm{mk}} - 1$,
(Mr2) $\bar{\theta}_{\mathrm{rs}}(r) \le \frac{1}{2}$,
(Mr3) $N_{[s]} s\left[C_{e,\mathrm{Lip}}\bar{\theta}_{\mathrm{rs}}(r) + M^{-3}C_E K_\Pi \frac{w_*}{16} e^\pi r/\Lambda_{\mathrm{mk}}\right] \le 1$,
(Mr4) $C_{\mathrm{win}} L^s \mathsf{b}\, \bar{\theta}_{\mathrm{rs}}(r) \le 1$.

Condition (Mr4) implies (Mr2) and $\bar{\theta}_{\mathrm{rs}}(r) \le 1/32$. Under (Mr1),

$$\epsilon_k^{\mathrm{mk}}(\lambda) \le \frac{w_*}{16} e^\pi \frac{|\lambda|}{\Lambda_{\mathrm{mk}}} \le \frac{\bar{\theta}_{\mathrm{rs}}(|\lambda|)}{C_{\mathrm{mk}} L^s \mathsf{b}}.$$

■

Definition B.52 (Smaller mark conditions for the scalar total)**.** Put $c_{\mathrm{room}} = 3/4 + 1/(4\sqrt{2})$, $C_{\mathrm{arr}} = K_{\mathrm{D}}^{\mathrm{arr}}$ and

$$\epsilon_\flat = \frac{w_*}{16} e^{\pi} \Lambda_{\mathrm{mk}}^{-1/2}, \qquad \bar{\theta}_\flat = C_{\mathrm{Lip}}^{s}(s-u) C_{\mathrm{mk}} L^s X_g \epsilon_\flat .$$

The conditions for the scalar total are

$$\begin{aligned}
&(\mathrm{m1}) \quad \Lambda_{\mathrm{mk}}^{1/2} \geq 3e^{\pi}, \\
&(\mathrm{m2}) \quad \bar{\theta}_\flat \leq 1/32, \quad r_x L^s X_g \bar{\theta}_\flat \leq \bar{B}, \quad A L^s X_g \bar{\theta}_\flat \leq 3\bar{B}_M, \\
&(\mathrm{m3}) \quad N_{[s]} s [C_{e,\mathrm{Lip}} \bar{\theta}_\flat + M^{-3} C_E K_\Pi \epsilon_\flat] \leq 1, \\
&(\mathrm{m4}) \quad C_{\mathrm{arr}} \epsilon_\flat \leq 1.
\end{aligned}$$

Let $K_\flat(g, \ell, n)$ be the least integer from which on (k0), (k1) and (m1)–(m4) hold at every K^{A}. It exists: $\log \Lambda_{\mathrm{mk}}$ is bounded below by a positive multiple of $(K^{\mathrm{A}})^{2\mathfrak{p}_0}$, while the logarithms of all other factors are $O(K^{\mathrm{A}} + \log K^{\mathrm{A}})$. This threshold is chosen after γ and depends only on the constants, g, ℓ, n. ■

Lemma B.53 (The marked expansion after a restart)**.** The conclusions of Lem. B.45 hold at every admitted restarted step, with the same constants, on the torus and on the infinite lattice. The infinite lattice construction intertwines $M\mathbb{Z}^3$ translations, and on fitting polymers the torus terms and bond-pointed terms agree with their infinite lattice lifts. ■

Proof. The fourth origin reads the same calibration, threshold and real fluctuation as in Lem. B.45; its bound uses no activity before the current step. Thus that proof applies. For the additional chart assertion, a chart preserves orientations, non-pivot bonds and $\square(b)$ and carries the fluctuation by $B(b) \mapsto B(ib)$ with equal values. It therefore carries the fourth origin together with the other three on every polymer whose enlargement is contained in the given polymer. The expansion intertwines these charts on the distinguished boundary, and polynomial continuation extends this to the full polydisc and its derivatives. For a translation by $M\mathbb{Z}^3$ in level-$k+1$ units, place the polymer and its translate in one admitted torus and apply the same argument there. This proves independence of the fitting torus, translation covariance and the pointed assertions. □

Proposition B.54 (A marked restarted step)**.** At a point of the restarted open step domain with potential size at most c_{cl}, translation covariant earlier families, and $|\lambda| \leq (e^{\mathfrak{r}_k} - 1)/2$, all bounds of Prop. B.46 hold. In addition the scalar change after extraction has the array bound

$$\sup_{X,z} e^{(\kappa+1)d_{k+1}(X)} \sup |\Delta F_0(X; \cdot; z)| \leq K_{\mathrm{D}}^{\mathrm{arr}} \epsilon_k^{\mathrm{mk}}(\lambda).$$

The construction is holomorphic on $|\lambda| < e^{\mathfrak{r}_k} - 1$. If the earlier totals are invariant under all α_S, the new totals and pointed totals are invariant as well. For real data on all admitted tori, with potential size at most $3c_{\mathrm{cl}}/4$, the reality and infinite lattice conclusions of Prop. B.46 hold for totals and positive-degree pointed totals. ■

Proof. Apply Lem. B.53 and the extraction proof of Prop. B.46. It uses only the current open-domain point and the linear extraction functionals, available at the restart by Prop. B.5. For the array bound, the degree-zero pointed difference before extraction is bounded by $2K_\Pi \epsilon_k^{\mathrm{mk}} e^{-(\kappa+1)d_{k+1}}$. The scalar subtraction has size $M^{-3} C_E K_\Pi \epsilon_k^{\mathrm{mk}}$ on size-zero polymers. The action subtraction has size at most $C_{am} r_\alpha^{-2} C_E K_\Pi \epsilon_k^{\mathrm{mk}}$, also on size-zero polymers, since the pointed weight member is at most $6M^3\alpha_0^2 \leq 1$. Their sum is the stated constant. The bond derivative formula is holomorphic on the larger disc: for $t \in [0,1]$, $\mathbf{1} + t\lambda$ stays inside the polydisc. The invariance and reality proofs in Props. B.46 and B.47 use exactly the asserted earlier-total symmetries. For the real logarithm path, the potential-size bound $3c_{\mathrm{cl}}/4$ permits complex scaling by $|q| < 4/3$, containing the real path $[0,1]$. Differentiating at real bond parameters proves reality of the pointed totals. Uniform local decay and the fitting-chart identity then give the same infinite lattice limits and real extraction numbers as in Prop. B.46. □

Lemma B.55 (Margins around the zero-source restart)**.** The restart is defined through K, with real numbers and the symmetries, family spaces and bounded Borel densities of Prop. B.5. For $t_0 < j \le K$ and $t_0 \le i \le K$,

$$|\hat{b}_j| \le B_\sharp - \tfrac{1}{2}, \qquad |\hat{x}_i - \mathsf{t}_i| \le (1 - L^{-(i-t_0)})\bar{B}, \qquad \tfrac{11}{12}\mathsf{t}_i \le \hat{x}_i \le \tfrac{13}{12}\mathsf{t}_i,$$
$$-\tfrac{1}{2} \le \mathsf{t}_i \hat{s}_i \le \sigma_*, \qquad \hat{x}_i|\hat{s}_i| \le \bar{B}_M, \qquad |\hat{s}_i| \le \tfrac{2}{3} A\phi(\hat{x}_i),$$
$$\|\hat{F}_0^{(j)}\|_j \le q_2 C_w \phi(\mathsf{t}_{j-1}), \qquad \|\hat{E}_{\ge 2}^{(j)}\|_j \le q_2 C_2 \phi(\mathsf{t}_{j-1})^2.$$

The family estimates include each pointer and both lattice families. In particular $\hat{x}_i \ge g^{-2} \ge \gamma^{-2}$.

At level k, let $\mathscr{D}_k$ consist of the admitted families with

$$|x_i' - \hat{x}_i| \le r_x \hat{x}_i, \qquad |s_i'| \le A\phi(\hat{x}_i),$$
$$\|F_0'^{(j)} - \hat{F}_0^{(j)}\|_j \le 2C_w \phi(\mathsf{t}_{j-1}), \qquad \|E_{\ge 2}'^{(j)} - \hat{E}_{\ge 2}^{(j)}\|_j \le 2C_2 \phi(\mathsf{t}_{j-1})^2,$$

and the prescribed remaining scalar terms. The step is holomorphic on this domain and its potential has size less than $c_{\mathrm{cl}}/2$. Its outputs satisfy

$$\|F_0^{(k+1)}\| \le 2C_w\phi(\mathsf{t}_k) + 2\theta_s h^2 |s_k'|,$$
$$\|E_{\ge 2}^{(k+1)}\| \le 2C_2\phi(\mathsf{t}_k)^2 + 2c_s h^2 |s_k'|^2,$$
$$|b_{k+1} - e_{\mathsf{N},k+1} - b_{k+1}^F| \le 2C_w\phi(\hat{x}_k) + 2\theta_s h^2 |s_k'|,$$
$$|\mu_{k+1}| \le C_\mu \phi(\hat{x}_k) + c_s |s_k'|^2.$$

The scalar expansion value is bounded by $M^{-3} K_M (c_{\mathrm{cl}} + 1)$ per site. The full scalar activity after extraction has array bound $E_0 - 1/2$ at rate $\kappa + 1$. ∎

Proof. Since $0 \le F^{\mathrm{UV}} \le B_F$, Prop. B.5 constructs the flow with its real-data and symmetry properties. The internal symmetry induction also starts at t_0: there is no earlier activity, the action is scalar, the doubled tower operator has the required symplectic symmetry, and the initial mass term is a constant multiple of the invariant $\mathsf{S}_1(\psi^{\mathrm{bg}})$.

The step bound gives

$$|\hat{b}_j| \le (1 + L^{-1})(c_{2,\mathrm{loop}} + N_f c_2^F) + C_w \phi(\hat{x}_{j-1}).$$

The last term is at most $1/2$ by its original small-coupling ceiling; the floor on $B_\sharp$ therefore gives $B_\sharp - 1/2$. Summing the coefficient recursion from $\hat{x}_{t_0} = \mathsf{t}_{t_0}$ gives the stated geometric bound. Since $\mathsf{t}_i \ge g^{-2} + \bar{B}$ and $\gamma^{-2} \ge 12\bar{B}$, it implies the lower bound and the $11/12$, $13/12$ comparison. This is an induction: the bound at the previous level supplies the bound for the next increment.

The mass formula of Prop. B.5 is

$$\mathsf{t}_i \hat{s}_i = F^{\mathrm{UV}} + \sum_{t_0 \le i' < i} \frac{\mathsf{t}_{i'}}{\hat{x}_{i'}} \sigma_{i'}(i; \mathfrak{m}_i) + \sum_{t_0 < j \le i} \mathsf{t}_j \hat{\rho}_j, \qquad \left| \sum_{t_0 < j \le i} \mathsf{t}_j \hat{\rho}_j \right| \le \tfrac{1}{2}.$$

The slices are nonnegative and sum to at most B_F, and $\mathsf{t}_{i'}/\hat{x}_{i'} \le 12/11$. This gives the interval with upper endpoint σ_*. Consequently $\hat{x}_i|\hat{s}_i| \le (13/12)\sigma_* \le 3B_F + 1 = \bar{B}_M$. Use $\phi(x) \ge x^{-1}$ and $3\bar{B}_M/2 \le A$ for the last mass bound. In the activity recursion the direct term costs $1/2$ of the closed bound, the mass contribution at most $1/4$, and the lag sum at most $\theta_E C_\omega S_\omega \le 1/(4\sqrt{2})$. The two mass ceilings used here are $8\theta_s h^2 \bar{B}_M \gamma \le C_w$ and $16 c_s h^2 \bar{B}_M^2 \gamma^2 \le C_2$. Their sum gives q_2.

The open-domain extension in Prop. B.5 is exactly $\mathscr{D}_k$. Its family balls have sizes at most three times the closed bounds. Each of the four potential contributions is less than $c_{\mathrm{cl}}/8$ there. The displayed output estimates keep the mass contribution separate, as in that extension. For the scalar arrays, $|x_k' - \mathsf{t}_k| \le (2r_x + \bar{B}\gamma^2)\mathsf{t}_k$ and $|s_k'| \le \sqrt{2} A\phi(\mathsf{t}_k)$. The array estimate therefore gives

$$E_{\mathbf{N}} + C_{\mathsf{n}} + C(C_N + M^3)N_f + C(1 + L^{-1})(c_{2,\mathrm{loop}} + N_f c_2^F) + CK_{\mathbf{G}}(\bar{B}\gamma^2 + 2r_x) + 1.$$

The original floor on E_0 contains these fixed terms, the additional action-extraction term, $CK_{\mathbf{G}}$ and 2. With $CK_{\mathbf{G}}\bar{B}\gamma \le 1$, $2r_x \le 1$ and $\gamma \le 1/2$, the displayed bound is at most $E_0 - 1 + \gamma \le E_0 - 1/2$. □

Proposition B.56 (The marked restart)**.** Under (Mr1), (Mr2), start at the same $\hat{d}^{\bullet}_{t_0}(0)$, insert $\mathsf{m}_{k,\lambda}$ for $t_0 \le k < t_1$, and continue without marks through K. For $|\lambda| \le r$ this defines a flow whose points lie in $\mathscr{D}_k$, whose densities are bounded Borel functions and whose families belong to $\mathfrak{F}_j$. Its exact measure formula contains only the levels $j > t_0$. It agrees up to t_1 with the marked restart of Def. 11.4. The numbers and families are holomorphic for $|\lambda| < r$, continuous on the boundary, real at real λ, and at zero equal to the unmarked restart. All symmetry and pointing conclusions of Prop. B.47 hold.

The four distance estimates of Prop. B.47 hold with hatted reference values and $\bar{\theta} = \bar{\theta}_{\mathrm{rs}}(|\lambda|)$. At every $k \le K$,

$$|\hat{E}_k(\lambda) - \hat{E}_k| \le N_{[s]} s \left[C_{e,\mathrm{Lip}} \bar{\theta} + M^{-3} C_E K_\Pi \frac{w_*}{16} e^{\pi} \frac{|\lambda|}{\Lambda_{\mathrm{mk}}} \right],$$

which is at most one under (Mr3). Under (Mr4) one has in addition

$$|\hat{b}_j(\lambda)| < B_\sharp, \qquad |\hat{x}_i(\lambda) - \mathsf{t}_i| \le (1 - L^{-(i-t_0)})\bar{B},$$

$$\operatorname{Re} \hat{x}_i(\lambda) \ge g^{-2} > 1, \qquad |\hat{x}_K(\lambda)| \le X_g, \qquad |\hat{s}_i(\lambda)| \le \frac{12\bar{B}_M}{13\mathsf{t}_i} \le \frac{\bar{B}_M}{|\hat{x}_i(\lambda)|},$$

$$\|\hat{F}^{(j)}_0(\lambda)\|_j \le C_w \phi(\mathsf{t}_{j-1}), \qquad \|\hat{E}^{(j)}_{\ge 2}(\lambda)\|_j \le C_2 \phi(\mathsf{t}_{j-1})^2.$$

The scalar array bound is E_0. At a later level $k' \ge j$, the fermion terms assigned to a cube have total norm at most $C_\omega L^{(k'-j)/4} C_2 \phi(\mathsf{t}_{j-1})^2$. ■

Proof. Use the four scaled distances of Prop. B.47, with hatted centres and indices starting at t_0. The initial distance is zero. By Lem. B.55, its open unit ball lies in $\mathscr{D}_k$; its proof supplies all four output bounds and the mass margin used in the marked-flow argument. In particular $L^{-1}\hat{x}_k \le 2\hat{x}_{k+1}$, $\phi(\hat{x}_k) \le \phi(\hat{x}_{k+1})$, and $\phi(\mathsf{t}_k) \le \phi(\hat{x}_{k+1})$. Thus the same four normalized inequalities give

$$\theta_{k+1} \le C_{\mathrm{Lip}} \theta_k + C_{\mathrm{mk}} \mathsf{t}_k \epsilon_k, \qquad \theta_k \le \bar{\theta}_{\mathrm{rs}}(|\lambda|) \le \tfrac{1}{2}.$$

Here $\epsilon_k = \epsilon^{\mathrm{mk}}_k(\lambda)$ below t_1 and zero afterwards. The construction is simultaneous on all admitted tori, with common infinite lattice families: the start and all constants are independent of the side, and (k0), (k1) admit the one-weight setting on each of them.

We check the remaining restart hypotheses of Rem. B.48. The initial density is bounded and Borel by Lem. B.12; its action, tower and scalar mass term are real and invariant under the internal symmetries. There are no earlier activities. Its density transforms at tested covariance pairs by the minimal-orbit argument of Prop. B.5. The open-domain and size hypotheses are precisely Lem. B.55. The marked step is Prop. B.54, with potential size below $c_{\mathrm{cl}}/2 < 3c_{\mathrm{cl}}/4$. Accordingly the symmetry proof in Prop. B.47 applies in its stated order: internal symmetries of totals and pointed totals, real logarithm and covariance on shell, uniform torus limits, scalar degree-two moment, and its removal by mass extraction. Analytic continuation gives the closed linear conditions at complex λ. This verifies membership in $\mathfrak{F}_j$ before the next step.

The density induction also applies: the same completed square and background substitutions rewrite the complex exponent; the coefficient remains within $(2r_x + \bar{B}\gamma^2)\mathsf{t}_k \le 3\sigma_\star \mathsf{t}_k/4$ of the calibration, so the Gaussian chart formula has positive real relative coefficient. Multiplication by the bounded Borel mark precedes the linear image map; its chart expression is the marked Gaussian integral. Extraction rearranges the exponent and gives the next bounded Borel density, zero off its test set. This proves the exact measure identity and its identification with Def. 11.4. The scalar-energy proof in Prop. B.47 uses at most s increments of size at most $N_{[s]}$ and gives the displayed estimate.

For the additional conclusions let (Mr4) hold, so $\bar\theta \le 1/32$. Put $D_i = \hat x_i(\lambda) - \hat x_i$. Then

$$|D_i| \le \tfrac{13}{24} r_x L^s \mathsf{b}\, \bar\theta \le \tfrac{13}{24}\cdot\tfrac{1}{8}\cdot\tfrac{1}{32} < \tfrac{1}{400}.$$

Hence $|\hat b_j(\lambda) - \hat b_j| = |L^{-1}D_{j-1} - D_j| < 1/200$. The unmarked increment has margin $1/2$, proving $|\hat b_j(\lambda)| < B_\sharp$. Summing its recursion proves the coefficient window. For the mass, $\phi(\hat x_i) \le 1/2$ and $A\bar\theta \le \mathsf{t}_i^{-1}$ give

$$|\hat s_i(\lambda)| \le (\sigma_* + 1/6)\mathsf{t}_i^{-1} = (\tfrac{23}{11}B_F + \tfrac{2}{3})\mathsf{t}_i^{-1} \le \tfrac{12}{13}\bar B_M \mathsf{t}_i^{-1}.$$

Together with $|\hat x_i(\lambda)| \le 13\mathsf{t}_i/12$, this proves the mass window. The family bounds follow from $q_2 + 2\bar\theta \le q_2 + 1/16 < 1$.

The local family conditions include property (N) and vanishing matrix cell sums on the infinite lattice. On a torus their wrapping remainder is included in C_ω. The earlier-family readout lemma therefore gives the claimed $L^{(k'-j)/4}$ bound. For the scalar arrays, the unmarked step at a point of $\mathscr{D}_k$ has size at most $E_0 - 1/2$. The marked change is at most

$$K_{\mathrm{D}}^{\mathrm{arr}}\epsilon_k \le \frac{C_{\mathrm{win}}\bar\theta}{2L^s\mathsf{b}} \le \frac{1}{2(L^s\mathsf{b})^2} \le \tfrac{1}{2}.$$

This proves the array bound E_0. □

Lemma B.57 (A uniform radius for the marked restart)**.** Let $K_{\mathrm{rs}}(g,\ell,n)$ be the least integer from which on, at every K^{A},

$$e^{-\pi}\Lambda_{\mathrm{mk}} \ge 4\Lambda_{\mathrm{mk}}^{1/2} + 1,$$
$$C_{\mathrm{win}} L^s X_g \theta_{\mathrm{rs},K^{\mathrm{A}}} \le 1,$$
$$N_{[s]}s\left[C_{e,\mathrm{Lip}}\theta_{\mathrm{rs},K^{\mathrm{A}}} + 2M^{-3}C_E K_\Pi \frac{w_*}{16} e^{\pi}\Lambda_{\mathrm{mk}}^{-1/2}\right] \le 1.$$

It exists and depends only on the constants, g,ℓ,n. At such cutoffs satisfying (k0), $r_{\mathrm{rs}} = 2\Lambda_{\mathrm{mk}}^{1/2}$ satisfies (Mr1)–(Mr4), for both lattices and every allowed label. ■

Proof. Substitute $r = r_{\mathrm{rs}}$ and use $\mathsf{b} \le X_g$ and $\bar\theta_{\mathrm{rs}}(r_{\mathrm{rs}}) \le \theta_{\mathrm{rs},K^{\mathrm{A}}}$. The first condition gives (Mr1), the second (Mr4) and hence (Mr2), and the third (Mr3). For existence, $\log\Lambda_{\mathrm{mk}} \ge c(K^{\mathrm{A}})^2$ for large K^{A}, where $c = \kappa_{\mathrm{t}} A_1^2(\phi_1(n)\log L)^2 > 0$. Also

$$\theta_{\mathrm{rs},K^{\mathrm{A}}} \le 2C_{\mathrm{mk}}\frac{w_*}{16}e^{\pi}X_g K^{\mathrm{A}}(LC_{\mathrm{Lip}})^{K^{\mathrm{A}}}\Lambda_{\mathrm{mk}}^{-1/2}.$$

The last two left sides are therefore bounded by $ae^{bK^{\mathrm{A}}}\Lambda_{\mathrm{mk}}^{-1/2}$, with $b = 2 + 4\log L + \log C_{\mathrm{Lip}}$. They tend to zero, and the first condition holds once $\Lambda_{\mathrm{mk}}^{1/2} \ge 5e^{\pi}$. □

B.20. **Continuing the marked restart to the last level.** The estimate of the marked total uses all histories above t_1. We first continue the marked small-field flow to K, retaining the bounds needed for the history expansion.

Lemma B.58 (Room in the restarted bounds)**.** Under (k0), (k1), the zero-source restart satisfies

$$|\hat s_i| \le \bar B_M/\mathsf{t}_i \le \sqrt{2}\gamma\bar B_M\phi(\hat x_i) \le \tfrac{2}{3}A\phi(\hat x_i), \qquad t_0 \le i \le K.$$

For $t_0 < j \le K$,

$$\|\hat F_0^{(j)}\|_j \le c_{\mathrm{room}}C_w\phi(\mathsf{t}_{j-1}), \qquad \|\hat E_{\ge 2}^{(j)}\|_j \le c_{\mathrm{room}}C_2\phi(\mathsf{t}_{j-1})^2.$$

These bounds hold for the families, their pointed terms, and both torus families at each level. The second also holds for the sum over any set of pointers of each polymer. ■

Proof. The start has $\zeta = F^{\mathrm{UV}} \in [0, B_F]$: Prop. B.2 gives $0 \le F^{\mathrm{UV}} \le \lim_k F(k;0) \le B_F$. Thus Prop. B.5 gives $|\hat{s}_i| \le \bar{B}_M/\mathsf{t}_i$ and $|\hat{x}_i - \mathsf{t}_i| \le \bar{B}$. Since $\mathsf{t}_i \ge \gamma^{-2} \ge 12\bar{B}$,

$$\tfrac{1}{2}\mathsf{t}_i \le \hat{x}_i \le 2\mathsf{t}_i, \qquad \mathsf{t}_i^{-1} \le \gamma \mathsf{t}_i^{-1/2} \le \sqrt{2}\gamma\phi(\hat{x}_i).$$

The first assertion follows from $\sqrt{2}\gamma \le 1$ and $3\bar{B}_M/2 \le A$. Also $\mathsf{t}_i^{-1} \le \gamma\phi(\mathsf{t}_i)$.

There are no activities at levels at most t_0. Use the activity recursions with the inductive bounds stated in the lemma. The mass contributions obey

$$\theta_s h_F^2 |\hat{s}_j| \le \tfrac{1}{8} C_w \phi(\mathsf{t}_j), \qquad c_s h_F^2 |\hat{s}_j|^2 \le \tfrac{1}{16} C_2 \phi(\mathsf{t}_j)^2,$$

by the chosen ceilings on γ. For $m = j + 1 - i$ the contributions of level i gain

$$L^{-m/2}(1 + m\log L)^{\mathfrak{p}_0}, \qquad L^{-m}(1 + m\log L)^{2\mathfrak{p}_0},$$

respectively. Since $2\sqrt{2}\theta_E C_\omega S_\omega \le 1/2$, the lag sums are bounded by $1/(4\sqrt{2})$ times the corresponding closed size. Therefore

$$\epsilon_{j+1} \le \left(\tfrac{1}{2} + \tfrac{1}{8} + \tfrac{1}{4\sqrt{2}}\right) C_w \phi(\mathsf{t}_j),$$
$$\epsilon^{\psi}_{j+1} \le \left(\tfrac{1}{2} + \tfrac{1}{16} + \tfrac{1}{4\sqrt{2}}\right) C_2 \phi(\mathsf{t}_j)^2.$$

Both coefficients are at most c_{room}. This starts at $j = t_0$, where the lag sums are empty.

For a set S of output pointers, replace the insertion at one pointer by the sum of all insertions sent to S. It multiplies each monomial of the fermion potential by a number in $[0,1]$, and hence has norm at most that of the potential. It preserves parity in the fluctuation, factorization and colour invariance. On a one-cube domain its extracted mass term is a part of the entire mass term, so extraction has the same factor K_μ. The majorant is monotone in the vertex norms polymer by polymer. Thus the same recursion bounds these sums, even when S varies with the polymer. This proves the last assertion. □

Lemma B.59 (The mark on each polymer)**.** At a translation covariant point of the restarted open step domain, with potential size at most c_{cl}, let $\mathsf{r}_k > 0$ and $|\lambda| \le (e^{\mathsf{r}_k} - 1)/2$. The difference before extraction between the marked and unmarked new activities satisfies

$$\|\Delta^{\mathrm{mk}}_\lambda(X)\|_{h^{(k+1)}} \le K_\Pi \epsilon^{\mathrm{mk}}_k(\lambda) e^{-\kappa_+ d_{k+1}(X)}, \qquad \epsilon^{\mathrm{mk}}_k(\lambda) = \frac{w_*}{32}\frac{|\lambda|}{e^{\mathsf{r}_k} - 1}.$$

The assertion holds at every regular complex pair at the fixed radii, on the torus and on $\mathbb{Z}^3$. ■

Proof. For completeness, the estimate before pointing uses only one application of Schwarz's lemma. On the distinguished boundary $|\zeta_b| = e^{\mathsf{r}_k}$, the fourth origin is

$$\mathcal{V}^{\mathrm{mk}}_\zeta(Y; B) = \sum_{b:\, \square(b) = Y} (\mathrm{Log}\,\zeta_b) \mathbf{1}_{\{\mathsf{t}_k |B(b)|^2 \ge a_k^2\}}.$$

Because $\mathbf{1}_{\{\mathsf{t}_k|B(b)|^2 \ge a_k^2\}} \le \mathsf{t}_k |B(b)|^2/a_k^2$ and $\mathsf{r}_k + \pi = \kappa_{\mathrm{mk}} a_k^2$, its input norm is at most $M^3\kappa_{\mathrm{mk}} = w_*/32$. The refined terms meeting this origin, counted with multiplicity, have total norm at most $K_\Pi(w_*/32)e^{-\kappa_+ d_{k+1}(X)}$. The terms avoiding it are exactly those of the unmarked expansion.

As proved in Lems. B.45 and B.53, the Mayer terms are polynomials in the ζ_b. Their majorants extend to the closed polydisc $|\zeta_b| \le e^{\mathsf{r}_k}$ by the maximum principle, applied after a norm-one linear functional. Uniform convergence then extends the refined sum and the displayed bound to that polydisc. The Gaussian part subtracted from the whole integral is independent of ζ. Consequently $\Delta^{\mathrm{mk}}_\lambda(X)$ is holomorphic for $|\lambda| < e^{\mathsf{r}_k} - 1$, vanishes at zero, and is bounded by that majorant. Schwarz's lemma gives the assertion. Fitting charts give the same expansion on $\mathbb{Z}^3$. □

Proposition B.60 (The marked restart through the last level)**.** Assume (k0), (k1), and $K^{\mathrm{A}} \geq K_\flat(g,\ell,n)$. For either lattice, start at t_0 with the zero-source restart, insert the uniform mark for $t_0 \leq k < t_1$, and use unmarked steps for $t_1 \leq k < K$. At each step extract its own numbers and use

$$x_{k+1}(\lambda) = L^{-1}x_k(\lambda) - b_{k+1}(\lambda), \qquad s_{k+1}(\lambda) = Ls_k(\lambda) + \mu_{k+1}(\lambda).$$

Point $\Delta_\lambda^{\mathrm{mk}}$ by the tents and the remaining activity by the ordinary step rule. Put

$$E_k(\lambda) = \sum_{t_0<j\leq k} e_j(\lambda)n_j, \qquad \bar\theta(\lambda) = \bar\theta_\flat|\lambda|\Lambda_{\mathrm{mk}}^{-1/2}, \quad \epsilon(\lambda) = \epsilon_\flat|\lambda|\Lambda_{\mathrm{mk}}^{-1/2}.$$

For $|\lambda| \leq \Lambda_{\mathrm{mk}}^{1/2}$ this construction has the following properties.

(1) Each step lies in the open domain about the zero-source restart at the fixed fine mass. The families lie in $\mathfrak{F}_j$; fitting polymers on every admitted torus are the lifts of the same families on $\mathbb{Z}^3$. The image of the start along the history with no large-field region has the level-k form with these numbers and activities, as an identity of measures. There are no activities at levels at most t_0. Through t_1 this image is the marked restart of Def. 11.4.

(2) The numbers, energies and families are holomorphic in the open disc and continuous on its closure. The numbers and energies are real for real λ, with the corresponding conjugation identities for the families. At zero they equal the unmarked restart.

(3) For $t_0 \leq i \leq K$, $t_0 < j \leq K$,

$$|x_i(\lambda) - \hat x_i| \leq \tfrac{1}{2}r_x\hat x_i\bar\theta(\lambda), \qquad |s_i(\lambda) - \hat s_i| \leq \tfrac{1}{3}A\phi(\hat x_i)\bar\theta(\lambda),$$
$$\|F_0^{(j)}(\lambda) - \hat F_0^{(j)}\|_j \leq 2C_w\phi(\mathtt{t}_{j-1})\bar\theta(\lambda),$$
$$\|E_{\geq 2}^{(j)}(\lambda) - \hat E_{\geq 2}^{(j)}\|_j \leq 2C_2\phi(\mathtt{t}_{j-1})^2\bar\theta(\lambda).$$

These include the pointed bounds and both torus families.

(4) The activity sizes are at most $C_w\phi(\mathtt{t}_{j-1})$ and $C_2\phi(\mathtt{t}_{j-1})^2$, respectively. The parts with fermion legs have the later-level reading bound with zero cell sum on $\mathbb{Z}^3$. The scalar arrays have bound E_0 at rate $\kappa+1$, pointer by pointer; at $j > t_1$ they have bound $E_0 - 1$. Moreover

$$|x_i(\lambda) - \mathtt{t}_i| \leq \bar B, \qquad |s_i(\lambda)| \leq 2\bar B_M/\mathtt{t}_i.$$

At real λ the real coefficient windows hold.

(5)
$$|E_K(\lambda) - \hat E_K| \leq N_{[s]}s\big[C_{e,\mathrm{Lip}}\bar\theta(\lambda) + M^{-3}C_EK_\Pi\epsilon(\lambda)\big] \leq 1.$$

(6) For real λ, $|b_j(\lambda)| < B_\sharp$. For $t_1 < j \leq K$,

$$|\mu_j(\lambda)| \leq 3\bar B_M/x_j(\lambda), \qquad |e_j(\lambda)| \leq C_e'(1 + \log x_{j-1}(\lambda)).$$

At these levels the sum over any set of pointers of each polymer has fermion norm at most $C_2\phi(\mathtt{t}_{j-1})^2$, and

$$\|E_{\geq 2}^{(j)}(\lambda)\|_j^{\mathrm{pt}} \leq 8M^3C_2\phi(\mathtt{t}_{j-1})^2.$$

The last norm sums the norms at the individual pointers, each with its own supremum over pairs.

■

Proof. We give the changes needed to run the marked-flow proof from t_0 and to obtain the later history bounds. First, for $t_0 \leq k < t_1$,

$$e^{\mathtt{r}_k} \geq e^{-\pi}\Lambda_{\mathrm{mk}} \geq 3\Lambda_{\mathrm{mk}}^{1/2} \geq 2\Lambda_{\mathrm{mk}}^{1/2} + 1.$$

Thus the mark is admitted and

$$\epsilon_k^{\mathrm{mk}}(\lambda) \leq \frac{w_*}{16}e^{\pi}\frac{|\lambda|}{\Lambda_{\mathrm{mk}}} = \epsilon(\lambda).$$

Define $\theta_k(P)$ by the four scaled distances in the proof of Prop. B.47, centred now at $\hat{P}_k$, with indices $i \geq t_0$, $j > t_0$. Let Ω_k consist of points with $\theta_k < 1$ whose families belong to $\mathfrak{F}_j$ and whose remaining scalar terms are the prescribed functions of the coefficients. By Lem. B.58, $|\hat{s}_i| \leq 2A\phi(\hat{x}_i)/3$. Hence Ω_k lies inside the restarted open domain, with potential size below c_{cl}. On it the step outputs satisfy

$$\begin{aligned} \|F_0^{(k+1)}\|_{k+1} &\leq (2C_w + 2\sqrt{2}\theta_s h_F^2 A)\phi(\mathsf{t}_k), \\ \|E_{\geq 2}^{(k+1)}\|_{k+1} &\leq (2C_2 + 4c_s h_F^2 A^2)\phi(\mathsf{t}_k)^2, \\ |b_{k+1} - e_{\mathsf{N},k+1} - b_{k+1}^F| &\leq 2(C_w + \theta_s h_F^2 A)\phi(\hat{x}_k), \\ |\mu_{k+1}| &\leq (C_\mu + c_s A^2)\phi(\hat{x}_k). \end{aligned}$$

The subtracted terms in the third line do not read the point. The variable part of the vacuum number is

$$-3(L^3 - 1)\,\mathrm{Log}(x_k'/\mathsf{t}_k) + \widetilde{e}_{k+1}, \qquad |\widetilde{e}_{k+1}| \leq M^{-3} K_M (c_{\mathrm{cl}} + 1).$$

On the complex line from $\hat{P}_k$ to P, Schwarz's lemma bounds the difference of an output bounded by S by $2S\theta_k(P)$. The mark adds the extraction differences of Prop. B.54 and Lem. B.59. Using $L^{-1}\hat{x}_k \leq 2\hat{x}_{k+1}$, $\phi(\hat{x}_k) \leq \phi(\hat{x}_{k+1})$ and $\phi(\mathsf{t}_k)^{-2} \leq \mathsf{t}_k$, the four normalized bounds therefore give

$$\theta_{k+1} \leq C_{\mathrm{Lip}}\theta_k + C_{\mathrm{mk}}\mathsf{t}_k\epsilon_k, \qquad \epsilon_k = \begin{cases} \epsilon_k^{\mathrm{mk}}(\lambda), & k < t_1, \\ 0, & k \geq t_1. \end{cases}$$

Since $\theta_{t_0} = 0$,

$$\theta_{k+1} \leq \sum_{j=t_0}^{\min(k,t_1-1)} C_{\mathrm{Lip}}^{k-j} C_{\mathrm{mk}} \mathsf{t}_j \epsilon_j \leq C_{\mathrm{Lip}}^{s}(s-u) C_{\mathrm{mk}} L^s X_g \epsilon(\lambda) = \bar{\theta}(\lambda) \leq 1/32.$$

The induction must also retain the conditions defining $\mathfrak{F}_j$. The proof of Prop. B.47 applies with the following verifications. The initial density has no activities and intertwines the isometries, flavour action and conjugations. At each marked step, Lem. B.53 and Prop. B.54 give the same four-origin expansion and tent pointing; after t_1 use its value at the trivial mark. The scalar mark is fixed by all α_S. The pointed totals are real linear combinations of the derivatives in the bond parameters of the total logarithm, so they inherit these symmetries. For real λ, fix the logarithm along

$$\rho \longmapsto (\rho\mathcal{V},\, \mathbf{1} + \rho(\zeta - \mathbf{1})), \qquad 0 \leq \rho \leq 1.$$

If the potential size is $\sigma < c_{\mathrm{cl}}$, the first coordinate has the required holomorphic neighbourhood $|\rho|\sigma < c_{\mathrm{cl}}$; the second stays in the interior of the mark polydisc. The integral is fixed by $\sharp$. The difference between its logarithm and its image under $\sharp$ is a continuous scalar in $2\pi i\mathbb{Z}$, zero at $\rho = 0$, hence zero throughout. This includes negative $1 + \lambda$. Real difference quotients give the same conclusion for the pointed derivatives.

The uniform mark is preserved by the change of fluctuation variables for each lattice isometry: non-pivot bonds are permuted and their norms are unchanged. This proves covariance on shell. The bounds above are uniform in the torus side; fitting-chart identities and the exponential tails therefore give the infinite lattice symmetries and real extraction numbers, exactly as in Props. B.46 and B.47. The degree-two cell moment is a scalar matrix; the extracted mass cancels it, while the action and vacuum extractions have the required invariant forms. Thus the families satisfy all the defining linear conditions for real λ. Those conditions extend to complex λ by the identity theorem and to the boundary by continuity. This closes the induction in Ω_k and proves the first three assertions apart from the measure identity.

For that identity the argument in Prop. B.47 requires the level-k form, families in $\mathfrak{F}_j$, the prescribed scalar terms, an admitted Gaussian coefficient, and the expansion identity at the actual

mark. The first conditions have just been checked, and

$$|x_k(\lambda)/\mathsf{t}_k - 1| < r_x + \bar{B}\gamma^2 \le 2r_x \le \sigma_\star/2.$$

The last is Lem. B.53 at a marked step and the ordinary expansion otherwise. The start is a bounded Borel density of the asserted form. The chart identity for the integrand, followed by the partial Berezin integral, therefore proves the measure identity inductively. Through t_1 its tests and Borel marks are precisely those of Def. 11.4.

The sizes follow from Lem. B.58 and $c_{\mathrm{room}} + 2\bar{\theta}_\flat \le 1$. Membership in $\mathfrak{F}_j$ gives zero cell sums and the later-level reading estimate; the wrapping cell remainder is included in C_ω. For scalar arrays, the ordinary step on these balls has bound

$$E_{\mathbf{N}} + C_{\mathsf{n}} + C(C_N + M^3)N_f + C(1 + L^{-1})(c_{2,\mathrm{loop}} + N_f c_2^F) + CK_{\mathbf{G}}(\bar{B}\gamma^2 + 2r_x) + 1 \le E_0 - 1.$$

Here $\bar{B}\gamma^2 + 2r_x < 1$, and the last term bounds the fluctuation family by the stipulated ceiling. At a marked step the additional array bound is $C_{\mathrm{arr}}\epsilon(\lambda) \le 1$; after t_1 there is no such term. The coefficient window follows from

$$|x_i(\lambda) - \mathsf{t}_i| \le r_x \mathsf{t}_i \bar{\theta}_\flat + (1 - L^{-(i-t_0)})\bar{B} \le \bar{B},$$

since $\mathsf{t}_i L^{i-t_0} \le L^s X_g$ and (m2). The same condition gives $|s_i(\lambda)| \le \bar{B}_M/\mathsf{t}_i + A\bar{\theta}_\flat/3 \le 2\bar{B}_M/\mathsf{t}_i$.

The vacuum number changes by at most $C_{e,\mathrm{Lip}}\theta_k + M^{-3}C_E K_\Pi \epsilon_k$: use $|\operatorname{Log}(x_k(\lambda)/\hat{x}_k)| \le r_x\theta_k$ and the Schwarz bound on $\widetilde{e}_{k+1}$. There are s terms and $n_{k+1} \le N_{[s]}$. Condition (m3) gives the energy assertion.

It remains to verify the bounds for later histories. At real λ the ordinary increment differs from $\hat{b}_{k+1}$ by at most $2\sqrt{2}\bar{\theta}(\lambda)$. The mark adds at most $r_x C_{\mathrm{mk}}\epsilon(\lambda)/2 \le r_x\bar{\theta}(\lambda)/2$. Thus $|b_{k+1}(\lambda) - \hat{b}_{k+1}| \le 3\bar{\theta}_\flat \le 3/32$; Lem. B.55 gives $|\hat{b}_{k+1}| \le B_\sharp - 1/2$. More precisely, (m2) gives

$$|s_i(\lambda)| \le (1 + L^{-(i-t_0)})\bar{B}_M/\mathsf{t}_i.$$

For $j > t_1$, $j - t_0 \ge 2$, and hence

$$|\mu_j(\lambda)| \le \big(2 + (1+L)L^{-(j-t_0)}\big)\bar{B}_M/\mathsf{t}_j \le \tfrac{11}{4}\tfrac{13}{12}\bar{B}_M/x_j(\lambda) < 3\bar{B}_M/x_j(\lambda).$$

After t_1 there is no mark, so $|e_j(\lambda) - \hat{e}_j| \le C_{e,\mathrm{Lip}}\bar{\theta}_\flat \le 1$. The unmarked restart has

$$|\hat{e}_j| \le 3(L^3 - 1)\log \mathsf{t}_{j-1} + L^3 C_{\mathrm{co}} + C_{\mathsf{n}} + C_F\nu + C_e \sup_{x \ge 1} \phi(x).$$

Use $\log \mathsf{t}_{j-1} \le \log x_{j-1}(\lambda) + \log 2$ and $3(L^3 - 1)(1 - \log 2) \ge 1$ to obtain the stated C_e' bound.

For a sum over any set of output pointers in an unmarked step, the insertion argument of Lem. B.58 gives the same ball bound $(2C_2 + 4c_s h_F^2 A^2)\phi(\mathsf{t}_k)^2$. The Schwarz estimate bounds its difference from the corresponding unmarked-restart sum by $2C_2\phi(\mathsf{t}_k)^2 C_{\mathrm{Lip}}\theta_k \le 2C_2\phi(\mathsf{t}_k)^2\bar{\theta}(\lambda)$. The room estimate proves the required size. Finally apply the pointed-sum lemma the pointed-sum lemma of QCD3_M1, Lemma 8.8, checking its hypotheses: the step is unmarked with its ordinary pointing in the restarted open domain; its coefficient is real; $|s_k(\lambda)| \le 2\bar{B}_M/\mathsf{t}_k$; all earlier fermion activities have size $C_2\phi(\mathsf{t}_{i-1})^2$, and those before t_0 vanish. The step uses these families through their values on polymers, so their earlier marks impose no further condition. The lemma gives $C_2^{\mathrm{pt}} = 8M^3C_2$ with the separate suprema at each pointer, as claimed. □

B.21. **Summing the histories after the mark.**

Lemma B.61 (The marked flow as a flow with free past)**.** Under the hypotheses of Prop. B.60, let $0 \le \lambda \le \Lambda_{\mathrm{mk}}^{1/2}$. For $t_0 < j \le t_1$, point the fermion part of $E^{(j)}(\lambda)$ at the centre of the cell where its polymer is filed, in the same shares if it is filed at several cells. Keep its scalar pointing. Keep all pointing at levels $j > t_1$, and set $E^{(j)} = 0$ for $j \le t_0$. These families, the marked-flow numbers at and above t_1, and $E_{t_1}(\lambda)$ form a flow with free past at t_1 in the sense of Prop. 11.14, with $R^{(j)} = E^{(j)}$ and $D^{(j)} = 0$ for $j \le t_1$.

Its initial density is the level-t_1 term of the marked restart. Write $g_h[\lambda]$ for the density, against $\mathscr{M}_h^{\mathrm{fr}}$, of its subsequent history $h \in \mathcal{H}_k^{(t_1)}$. The history bounds of Prop. 11.14 hold with the original constants $C_5^F, C_E^F, C_\flat^F, c_\star, \mathsf{C}_{\mathrm{maj}}^\dagger$, with $E_k(\lambda)$ and with scalar history terms $f_j = 0$ for $j \le t_1$. The action coefficient is

$$\beta_h = \beta - \sum_{t_0<j\le k} L^j b_j(\lambda)\varphi_j, \qquad \beta = L^K \mathsf{b}.$$

Both versions of the bound, with $-\beta A$ and with $-A(\beta_h,\cdot)$, hold. In particular,

$$\sum_{h\in\mathcal{H}_K^{(t_1)}} \int \|g_h[\lambda]\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}} \le \exp[E_K(\lambda) + (C+2)\ell^3].$$

■

Proof. We verify each hypothesis of the free-past construction. The constants satisfy its original ceilings and (Γ^F), which belongs to the assumptions on the two-cutoff data. Condition (k0) gives $1 \le t_0 < t_1 < K$.

For (F1), Prop. B.60 gives local, even, Borel pointed terms, holomorphic on the regular complex pairs at fixed radii, joint gauge invariance, $\mathrm{U}(N_f)\times\mathrm{U}(N_f)$ invariance, translation covariance, the pointed sum rule, zero cell sums on $\mathbb{Z}^3$, and the scalar normalization. Filing the older fermion terms at their cell centres preserves all these properties: each new pointed term is a nonnegative share of the same polymer value, the shares add to one, the cell sums do not depend on pointing, and filing commutes with translations and charts. The zero earlier families satisfy these linear conditions.

For (F2), after t_1 the step is the ordinary small-field step with its original pointing. Repointing the older terms has not changed its input: the fermion potential and the filing read their values on polymers. The extraction rules therefore give exactly the same numbers and later families. Their sizes, arrays, windows, sums over subsets of pointers and $\|\cdot\|^{\mathrm{pt}}$ bounds, together with $|b_j| < B_\sharp$ and $|e_j| \le C_e'(1+\log x_{j-1})$, are all provided by Prop. B.60.

For (F3), take $R^{(j)} = E^{(j)}$, $D^{(j)} = 0$. On each older polymer, the norms of the repointed fermion terms add to the norm of its value, since they are nonnegative shares. Thus their pointed-sum size is $C_2\phi(\mathsf{t}_{j-1})^2$. The fluctuation families retain size $C_w\phi(\mathsf{t}_{j-1})$, and the later-level reading and scalar array bounds are unchanged. The lag differences vanish. The initial density reads the same polymer values and numbers as before repointing, so is exactly the marked term at t_1. This proves the asserted applicability of Prop. 11.14.

Its effective coefficient initially reads $L^{t_1}x_{t_1}(\lambda)$. The recurrence and $x_{t_0}(\lambda) = L^s\mathsf{b}$ give

$$\sum_{t_0<j\le t_1} L^j b_j(\lambda) = L^{t_1}(\mathsf{t}_{t_1} - x_{t_1}(\lambda)).$$

Every history under consideration has $\varphi_j = 1$ for $j \le t_1$. This converts the free-past action coefficient to β_h above. The corresponding bound on the absolute action correction follows because every plaquette action is nonnegative:

$$\left|\sum_{t_0<j\le k} L^j b_j(\lambda) A(\varphi_j,\mathsf{U})\right| \le L^{t_1}|x_{t_1}(\lambda) - \mathsf{t}_{t_1}|A(\mathsf{U}) + \sum_{t_1<j\le k} |L^j b_j(\lambda) A(\varphi_j,\mathsf{U})|.$$

For the final summation extend to all histories at level K by putting

$$f(h) = e^{-E_K(\lambda)+\sum_j \mathrm{Re}\, f_j(\mathcal{S}(h))} \int \|g_h[\lambda]\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}}$$

on $\mathcal{H}_K^{(t_1)}$, and zero elsewhere. On labels occurring there put $\mathsf{e}(\mathcal{S}) = -\sum_j \mathrm{Re}\, f_j(\mathcal{S})$, and put it zero on other labels. The history and label estimates make (f,e) an admitted pair for the history summation theorem at

$$c = (c_\star + c_\Theta)/2, \qquad \mathsf{C}_{\mathrm{maj}}^{\mathrm{th}} \ge \mathsf{C}_{\mathrm{maj}}^\dagger, \qquad C_\flat^{\mathrm{th}} \ge C_\flat^F.$$

Decreasing the rate and enlarging these constants weakens the required majorants. The prescribed floors on $\mathfrak{p}_0$ and the corresponding ceilings $\log \gamma^{-2} \geq \mathsf{Y}_{\mathrm{f1}}$ or Y_{f2} are exactly those of that summation theorem. Its nonnegative mark $\Theta^{(K)}$ gives

$$e^{-E_K(\lambda)} \sum_{h \in \mathcal{H}_K^{(t_1)}} \int \|g_h[\lambda]\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}} = \sum_{h \in \mathcal{H}_K} f(h) e^{\mathsf{e}(\mathcal{S}(h))}$$
$$\leq \sum_{h \in \mathcal{H}_K} f(h) e^{\mathsf{e}(\mathcal{S}(h)) + \Theta^{(K)}(h)} \leq \Pi_K(C, c'') \leq e^{(C+2)n_K},$$

where $c'' = (1-\eth)(c_\star - c_\Theta)/2 > 0$ and $n_K = \ell^3$. For $c_a > 0$, C includes $\lambda_L = (6L^3+1)\log 2$; for $c_a = 0$ this addition is unnecessary. These are the same constants and threshold choices used in the ordinary history sum. □

Lemma B.62 (Recovering the marked total)**.** With the hypotheses and notation of Lem. B.61,

$$Z_\lambda^\bullet = \sum_{h \in \mathcal{H}_K^{(t_1)}} \mathrm{Ber} \int g_h[\lambda]\, d\mathscr{M}_h^{\mathrm{fr}} \leq \sum_{h \in \mathcal{H}_K^{(t_1)}} \int \|g_h[\lambda]\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}}.$$

■

Proof. The issue is that the coordinate maps are partial. We show that they lose no marked mass, using the already established conservation at $\lambda = 0$.

On $\mathcal{Y} = \mathcal{V}(\Lambda_{s,\ell})$ write $W_j = \mathcal{M}^{j-t_0} W$. For $h \in \mathcal{H}_k^{(t_1)}$ let φ_h be $\hat{\mathsf{k}}^\bullet$ times the tests and gauge-fixing factors of the steps of h above t_1, read at these W_j. This is the nonnegative bounded Borel weight $\mathfrak{w}_{h/t_0}$ of Lem. B.30. Let

$$\mathsf{m}_\lambda(W) = \prod_{t_0 \leq k < t_1} \mathsf{m}_{k,\lambda}^\bullet(W_k).$$

For the present λ it lies in $[1, (1+\lambda)^{N_{\mathrm{mk}}}]$. The total Berezin mass of the term of h is

$$\int_{\mathcal{Y}} \varphi_h \mathsf{m}_\lambda \hat{F}_0^\bullet\, dW. \tag{*}$$

To check this identity, first push the restarted start through the marked steps to t_1. Multiplying a pushforward by a Borel function of its image is the same as multiplying its preimage measure by the pullback. Thus its Berezin integral is $(\mathcal{M}^{t_1-t_0})_*(\hat{\mathsf{k}}^\bullet \mathsf{m}_\lambda \hat{\nu})$. Use the bounded bare density $\hat{\varsigma}$ of Lem. B.30 and replace it by

$$\varsigma_\lambda = \hat{\varsigma}\,(\mathsf{m}_\lambda \circ \mathcal{M}^{t_0}).$$

The identity of that lemma, $(\mathcal{M}^{t_0})_*(\mathfrak{w}_h \hat{\varsigma}\, dU) = \mathfrak{w}_{h/t_0} \hat{\nu}$, then holds with the added mark. The same induction under the partial history maps, from the one-to-one Borel coordinates at t_1, identifies the subsequent term with the coordinate image at the density ς_λ. The no-loss property of these coordinates gives its mass $\int \mathfrak{w}_h \varsigma_\lambda\, dU$, which is (*). This argument uses only bounded Borel multiplication and the identities for partial Berezin integrals proved in Lem. B.30.

Consider now one step $t > t_1$ from a history h', and a plaquette outcome p. Write $P'' = P_t''(\mathsf{p})$, $\mathsf{N} = B(P'') \setminus P''$. Let $\mathbf{1}_t^{\mathrm{P}}(\mathsf{p})$ be its plaquette test and $\mathbf{1}_{h',\mathsf{p}}^{\mathrm{D}}$ the indicator that its averaged contours are defined. The product form of the history tests gives

$$\sum_{\mathsf{o}:\, \mathsf{p}(\mathsf{o}) = \mathsf{p}} \varphi_{(h',\mathsf{o})} = \varphi_{h'} \mathbf{1}_t^{\mathrm{P}}(\mathsf{p}) \mathfrak{G}_{t-1}(W_{t-1}; P''), \qquad \sum_{\mathsf{p}} \mathbf{1}_t^{\mathrm{P}}(\mathsf{p}) = 1.$$

For $v \in G^{\mathsf{N}}$ extend v to the bare lattice constantly on level-$(t-1)$ blocks, and write s_j for its restriction at level j. Then $s_{t-1} = v$, $s_t = 1$, and at earlier levels the values are constant on the corresponding blocks. The substitution $W \mapsto W^{s_{t_0}}$ preserves Haar measure, W_t, the earlier history tests and gauge factors, and the plaquette test. It also preserves $\hat{F}_0^\bullet$: minimal orbits transform

covariantly, the weighted action and regularity cut are unchanged, and the effective fermion operator is conjugated, preserving its determinant.

It preserves the mark as well. For $k < t_1$, the averaged background transforms by s_k, since s_k is constant on level-$(k+1)$ blocks. The intrinsic logarithm therefore carries the fluctuation to $\mathrm{Ad}(s_k(b_-))\mathcal{A}_k^\circ(b)$. Its norm, the threshold indicator and every factor of m_λ are unchanged. Average the substitution over G^{N}. The normalized gauge-fixing identity is

$$\int \mathfrak{G}_{t-1}(W_{t-1}^v; P'')\, dv = \mathbf{1}^{\mathrm{D}}_{h',\mathsf{p}}.$$

Fubini then gives

$$\int \varphi_{h'} \mathbf{1}_t^{\mathrm{P}}(\mathsf{p}) \mathfrak{G}_{t-1} \mathsf{m}_\lambda \hat{F}_0^\bullet \, dW = \int \varphi_{h'} \mathbf{1}_t^{\mathrm{P}}(\mathsf{p}) \mathbf{1}^{\mathrm{D}}_{h',\mathsf{p}} \mathsf{m}_\lambda \hat{F}_0^\bullet \, dW.$$

Consequently the loss in this step is

$$\int \varphi_{h'} \mathsf{m}_\lambda \hat{F}_0^\bullet \, dW - \sum_{\mathsf{o}} \int \varphi_{(h',\mathsf{o})} \mathsf{m}_\lambda \hat{F}_0^\bullet \, dW = \sum_{\mathsf{p}} \int g_{h',\mathsf{p}} \mathsf{m}_\lambda \hat{F}_0^\bullet \, dW,$$

where $g_{h',\mathsf{p}} = \varphi_{h'} \mathbf{1}_t^{\mathrm{P}}(\mathsf{p})(1 - \mathbf{1}^{\mathrm{D}}_{h',\mathsf{p}}) \geq 0$. Summing over steps and histories telescopes to

$$Z_\lambda^\bullet - \sum_{h \in \mathcal{H}_K^{(t_1)}} \int \varphi_h \mathsf{m}_\lambda \hat{F}_0^\bullet \, dW = \sum_{t=t_1+1}^{K} \sum_{h' \in \mathcal{H}_{t-1}^{(t_1)}} \sum_{\mathsf{p}} \int g_{h',\mathsf{p}} \mathsf{m}_\lambda \hat{F}_0^\bullet \, dW.$$

At zero the left side vanishes by the unmarked history identity in Prop. 11.15. Every term on the right is nonnegative since $\hat{F}_0^\bullet \geq 0$. Thus each $g_{h',\mathsf{p}} \hat{F}_0^\bullet$ vanishes almost everywhere, and remains zero after multiplication by the bounded mark. This proves the equality for every present λ. Finally $|\operatorname{Ber} F| \leq \|F\|_{h_F}$ because $h_F \geq 1$, proving the bound. □

Proposition B.63 (Bound for the marked total)**.** Under (k0), (k1), and $K^{\mathrm{A}} \geq K_\flat(g,\ell,n)$, the marked restart exists on $|\lambda| \leq \Lambda_{\mathrm{mk}}^{1/2}$ with the form and holomorphy of Prop. B.60. Moreover

$$Z^\bullet_{\Lambda_{\mathrm{mk}}^{1/2}} \leq e\, \mathsf{M}_{g,\ell} \hat{T}^\bullet(0), \qquad \partial_\lambda Z_\lambda^\bullet|_{\lambda=0} \leq e\, \mathsf{M}_{g,\ell} \Lambda_{\mathrm{mk}}^{-1/2} \hat{T}^\bullet(0).$$

The first bound holds also for every $0 \leq \lambda \leq \Lambda_{\mathrm{mk}}^{1/2}$. ■

Proof. The preceding two lemmas and Prop. B.60 give

$$Z^\bullet_{\Lambda_{\mathrm{mk}}^{1/2}} \leq \exp[\hat{E}_K + 1 + (C+2)\ell^3].$$

The positive unmarked-total estimate of Prop. 11.15 applies under (k0), (k1): $u \geq u_*$, $s \geq s_F$ and $L^u \gamma^{-2} \geq \Theta_\star$. It gives $\hat{T}^\bullet(0) \geq \exp[\hat{E}_K - C_P(1 + \log X_g)\ell^3]$. The quotient is therefore at most

$$e \exp[(C + 2 + C_P(1 + \log X_g))\ell^3] = e\, \mathsf{M}_{g,\ell}.$$

The total is a polynomial in λ with nonnegative coefficients, by its Borel mark representation and $\hat{F}_0^\bullet \geq 0$. It is increasing for $\lambda \geq 0$, and its linear coefficient is at most $\Lambda_{\mathrm{mk}}^{-1/2} Z^\bullet_{\Lambda_{\mathrm{mk}}^{1/2}}$. □

Corollary B.64 (The marked restart input)**.** Let $K_{\mathrm{mk}}(g,\ell,n) = \max\{K_{\mathrm{rs}}(g,\ell,n), K_\flat(g,\ell,n)\}$. Under (k0)–(k2) and $K^{\mathrm{A}} \geq K_{\mathrm{mk}}$, the marked restart through t_1 has the numbers, activities, closed bounds and level-k measure form of Prop. B.56 for $|\lambda| \leq 2\Lambda_{\mathrm{mk}}^{1/2}$. Its normalizers and scalar remainders use the actual pair $(\mathsf{t}_{j-1}, x_{j-1}(\lambda))$. It is holomorphic in the open disc, real at real λ, and agrees with the unmarked restart at zero. Furthermore

$$Z^\bullet_{\Lambda_{\mathrm{mk}}^{1/2}} \leq e\, \mathsf{M}_{g,\ell} \hat{T}^\bullet(0) \leq 2e\, \mathsf{M}_{g,\ell} \hat{T}^\bullet(0).$$

Thus it supplies the marked input of Prop. 11.11. ■□

Proof. The threshold lemma Lem. B.57 verifies (Mr1)–(Mr4) at $r = 2\Lambda_{\mathrm{mk}}^{1/2}$; apply Prop. B.56. The smaller-disc scalar bound is Prop. B.63, proved independently of this corollary. □

B.22. **The top density test.** The geometric weight $\mathsf{k}^\bullet$ need not include the density test at depth u. The restarted history weight $\hat{\mathsf{k}}^\bullet$ does. We estimate the difference by histories with a large-field region at that level. A regularity estimate descending from depth u ensures that the geometric weight also imposes the density test at the starting depth s.

Definition B.65 (Weights and starting test)**.** Fix one of the two lattices and omit $\bullet$. Assume (k0) and $L^u\gamma^{-2} \geq \Theta_\star$. Write $K = K^\bullet$, $t_0 = K - s$, $t_1 = K - u$, $\mathsf{t}_j = L^{K-j}\mathsf{b}$, and $W_j = \mathcal{M}^{j-t_0}W$ for $W \in \mathcal{Y}$. In particular $t_0 \geq 2$. Let In_j be the inner ball and Cut_j the background cut of radius $c_\rho\varepsilon(\mathsf{t}_j)$. With the fluctuation test χ_j^{sf} and gauge factor $\mathfrak{G}_j$,

$$\mathsf{k}(W) = \mathbf{1}_{\mathsf{In}_{t_1}}(W_{t_1}) \prod_{t_0 \leq j < t_1} [\mathbf{1}_{\mathsf{Cut}_j} \mathfrak{G}_j \chi_j^{\mathrm{sf}}](W_j).$$

Write $\chi_j(T_\ell)$ for the density test and $\mathcal{V}_j^\chi = \{\chi_j(T_\ell) = 1\}$. The frame assumptions give $\varpi(V) < 2\varepsilon(\mathsf{t}_j) \leq \varepsilon_{\mathcal{W}}\alpha_0$ on this set; hence $\mathcal{V}_j^\chi \subset \mathsf{In}_j \subset \mathsf{Cut}_j$ when $1 \leq j \leq t_1$.

The restart includes $\chi_{t_0}(T_\ell)$ in its starting density. Its Berezin integral is $\hat{F}_z^\chi(W)\,dW$, where

$$\hat{F}_z^\chi = \chi_{t_0}(T_\ell)\hat{F}_z.$$

Where the effective operator is formed, this is

$$\chi_{t_0}(T_\ell)(W) e^{-\mathsf{wab}(\mathsf{c}_{w_z}, W)} \det \hat{\mathbb{D}}(W; s(z)).$$

Set $\hat{F}_z$ and $\hat{F}_z^\chi$ to zero where the operator or its mass kernel is not formed. Thus the pointwise identities below make sense for every z; holomorphy and integral identities are asserted only on their stated source domains. Whenever $\hat{T}(z)$ is used assume $u \geq u_*$ and $s(z) \in \mathscr{O}_{t_1}$; whenever $T^{\mathrm{sf}}(z)$ is used assume also $s \geq s_F$ and the source bounds of Prop. 11.7.

For $h \in \mathcal{H}_k^{(t_0)}$ let $\hat{\mathfrak{w}}_h$ be the product of its outcome indicators and gauge factors for the steps above t_0, read on $\mathcal{Y}$. Set $\hat{\mathfrak{w}}_{h_\emptyset^{(t_0)}} = 1$. These are bounded nonnegative Borel functions. The tests above t_0 read no field below t_0: the earlier determining sets are empty because their regions are the full torus. For a bounded Borel function $\varsigma \geq 0$ put

$$\mathsf{m}_\varsigma(h) = \int_{\mathcal{Y}} \hat{\mathfrak{w}}_h \varsigma \, dW, \qquad \mathsf{m}(h) = \mathsf{m}_{\hat{F}_0^\chi}(h).$$

Use the floor on A_0

$$(1 - L^{-3/2})A_0 > 2B_3 C_H A_1. \tag{D}$$

Here C_H is the constant in the background descent estimate, namely $4C_{\mathrm{db}} \max\{2, K_{\mathrm{B}}^{\mathrm{s}}(\rho_{\mathrm{s}})\}$. Put

$$\sigma_D = \frac{1}{2B_3} - \frac{C_H A_1}{(1 - L^{-3/2})A_0} > 0$$

and let n_0 be the least nonnegative integer with $2B_3 L^{-3n_0/2} \leq \sigma_D$. Let $K_{\mathrm{tt}}(n)$ be the least integer from which on $s - u \geq n_0$. It exists and satisfies

$$K_{\mathrm{tt}}(n) \leq \max\left\{2, \left\lceil \frac{2(6n + 11)(n_0 + 1)}{1 - c_\#/10} \right\rceil \right\}.$$

Indeed $s - u \geq (1 - c_\#/10)\phi_2(n)K^{\mathrm{A}} - 1$. ■

Lemma B.66 (The two weights)**.** On $\mathcal{Y}$,

$$\hat{\mathsf{k}} = \prod_{t_0 \leq j < t_1} \chi_{j+1}(T_\ell)(W_{j+1})\, \mathsf{q}_{j+1}(W_j)\, \mathfrak{G}_j(W_j) \chi_j^{\mathrm{sf}}(W_j) = \hat{\mathfrak{w}}_{h_\emptyset^{(t_1)}}.$$

Both weights are bounded nonnegative Borel functions, and

$$\hat{\mathsf{k}} = \chi_{t_1}(T_\ell)(W_{t_1})\,\mathsf{k}, \qquad 0 \le \hat{\mathsf{k}} \le \mathsf{k}.$$

If $\hat{\mathsf{k}}(W) \neq 0$, all W_j, $t_0 \le j \le t_1$, pass their density tests and both weights equal $\prod_j \mathfrak{G}_j(W_j)$. Consequently

$$\hat{\mathsf{k}}\hat{F}_z = \hat{\mathsf{k}}\hat{F}^\chi_z, \qquad \hat{T}(z) = \int_{\mathcal{Y}} \hat{\mathsf{k}}\hat{F}^\chi_z\,dW = \int_{\mathcal{Y}} \hat{\mathsf{k}}\hat{F}_z\,dW$$

on the asserted source domain. ■

Proof. The small-field history weight of one step has exactly the factors displayed, together with $\mathsf{q}^0_j = 1$ for $j \ge 1$. This identity holds for arbitrary consecutive block fields: the tests and gauge factors are evaluated from these fields alone, as in the coordinate definitions of [YM3, §4.10.1] and [YM3, Prop. 4.75]. The tests are Borel indicators, the gauge factors are bounded Borel functions, the cuts are open, and the block map is Borel.

The density-test implication follows from [YM3, Prop. 4.74] and [YM3, Prop. 4.75]. Indeed a passed level-$(j+1)$ density test gives the regular next datum. The small-field test compares W with its background on each test cube; the small-bond argument in the proof of Prop. 53.14 makes every Q test equal to one. The pair then belongs to $\mathsf{PQ}_{j+1}(T_\ell)$, and Prop. 53.13 gives all level-j density tests. Thus if $\chi_{j+1}(T_\ell)(\mathcal{M}W) = 1$ and $\chi^{\mathrm{sf}}_j(W) = 1$, then $\chi_j(T_\ell)(W) = \mathsf{q}_{j+1}(W) = 1$. If $\hat{\mathsf{k}} > 0$, its factors first give every density test above t_0 and then this implication gives the one at t_0. All inner-ball and field-cut indicators in k are therefore one, proving equality on this support. Conversely, if $\mathsf{k} > 0$ and the density test at t_1 passes, descend successively from $t_1 - 1$ to t_0 using the same implication. All the remaining factors of $\hat{\mathsf{k}}$ then equal one. This proves the weight identity everywhere. It also gives $\hat{\mathsf{k}}\chi_{t_0}(T_\ell) = \hat{\mathsf{k}}$. Finally the total of the restarted image is its weight integrated against the Berezin integral of its start, $\hat{F}^\chi_z\,dW$, by Lem. 11.16. □

Lemma B.67 (The starting density)**.** For $s \ge s_F$, $\hat{F}_0$ and $\hat{F}^\chi_0$ are bounded nonnegative Borel functions, $\hat{F}^\chi_0 \le \hat{F}_0$, and they are invariant under all gauge transformations of the level-t_0 lattice. They vanish off Cut_{t_0} and $\mathcal{V}^\chi_{t_0}$, respectively. ■

Proof. On the open field cut the weighted action is continuous, real, and has modulus at most $\frac{3}{2}(c_\rho \varepsilon(\mathfrak{t}_{t_0}))^2 N_{[s]} L^s \mathfrak{b}$. The determinant is bounded, Borel and positive by Prop. 11.7. Multiplication by the Borel indicators proves the size and support claims.

A gauge transformation u preserves the density test and field cut. Extend it to the bare lattice by the identity away from level-t_0 sites. This maps the minimal orbit over W onto the orbit over W^u and preserves its weighted action. The Dirac operator, block averages and fermion background transform by conjugation with the restrictions of this extension. The mass function $F^{\mathrm{UV}}/(L^s\mathfrak{b})$ acts on no colour index. Therefore $\hat{\mathbb{D}}(W^u;0) = u\hat{\mathbb{D}}(W;0)u^{-1}$, so the determinant is unchanged. □

Lemma B.68 (Averaging the gauge factors)**.** If $\varsigma \ge 0$ is bounded, Borel and gauge invariant on $\mathcal{Y}$, then

$$\int \mathsf{k}\varsigma\,dW \le \int \prod_{t_0 \le j < t_1} \mathfrak{G}_j(W_j)\varsigma(W)\,dW \le \int \varsigma\,dW.$$

■

Proof. The first inequality discards indicators. Put $J_j = \int \prod_{t_0 \le i < j} \mathfrak{G}_i(W_i)\varsigma(W)\,dW$. For the full level-$(j+1)$ site set P'' let $\mathsf{N} = B(P'') \setminus P''$. Lift $v \in G^{\mathsf{N}}$ to level t_0 constantly on level-j blocks. The lift changes W_j to W^v_j, fixes W_{j+1}, and preserves all earlier gauge factors, ς and Haar measure. Averaging over v therefore replaces $\mathfrak{G}_j(W_j)$ in J_{j+1} by the indicator that its averaged contours exist, by [YM3, eq. 3.xxxv]; the block-constant lift and its invariance are those used in [YM3, eq. 4.xviii]. This indicator is at most one, so $J_{j+1} \le J_j$. Iterate. □

B.23. **The starting mass and the history masses.** We first record why the history coordinates can be read on $\mathcal{Y}$ without losing mass. This also supplies the lifting fact used in the subsequent gauge average.

Lemma B.69 (Mass of a restarted history)**.** Assume $u \geq u_*$ and $s \geq s_F$. For $h \in \mathcal{H}_k^{(t_0)}$, the Berezin integral of its restarted term is the image of $\hat{\mathfrak{w}}_h \hat{F}_0^{\chi}\, dW$ under the free coordinates of h read on $\mathcal{Y}$. Its total mass is $\mathsf{m}(h)$, and

$$0 \leq \mathsf{m}(h) = \int \operatorname{Ber} \hat{g}_h \, d\mathscr{M}_h^{\mathrm{fr}} \leq \int \|\hat{g}_h\|_{h_F} \, d\mathscr{M}_h^{\mathrm{fr}}.$$

■

Proof. First consider any $W \in \mathcal{V}_{t_0}^{\chi}$. It admits a bare lift whose small-field history through t_0 has positive weight. To construct it, set $V_{t_0} = W$ and $V_j = V_j^{\mathrm{bf}}(V_{j+1})$ successively for $j = t_0 - 1, \dots, 0$. If V_{j+1} passes its density test, its background is defined in the axial gauge, $\mathcal{M}(V_j) = V_{j+1}$, and the averaged background is stably defined by [YM3, Prop. 4.58] and [YM3, Lem. 6.33]. Its fluctuation is zero, so $\chi_j^{\mathrm{sf}}(V_j) = 1$. The density-test descent implication gives $\chi_j(T_\ell)(V_j) = \mathsf{q}_{j+1}(V_j) = \mathsf{q}_j^0(V_j) = 1$. All averaged contour variables used by the gauge factor equal the identity, where its factors are positive. The bounds needed here hold at every descended level since $\mathsf{t}_j \geq L^u \gamma^{-2} \geq \Theta_\star$. Thus $U = V_0$ has $\mathcal{M}^{t_0} U = W$ and $\mathfrak{w}_{h_\emptyset^{(t_0)}}(U) > 0$.

The weight factorization

$$\mathfrak{w}_h(U) = \mathfrak{w}_{h_\emptyset^{(t_0)}}(U)\, \hat{\mathfrak{w}}_h(\mathcal{M}^{t_0} U) \tag{$*$}$$

follows directly from its product over steps: the factors through t_0 are those of the empty history, and later tests read only levels at least t_0.

Let ξ_h denote the full history coordinates. The determining sets below t_0 and the external fluctuation bonds there are empty. At $h_0 = h_\emptyset^{(t_0)}$ the free coordinate is therefore the entire level-t_0 block field, and

$$\xi_{h_0} = (\Pi_{h_0}^{\mathrm{le}} \times \mathrm{id}) \circ \mathcal{M}^{t_0}.$$

For each later step let $\Sigma_{h|_t}^{\mathrm{co}\prime}$ be its Borel partial coordinate map. The coordinate composition property, including its domains, is

$$\xi_{h|_t} = \Sigma_{h|_t}^{\mathrm{co}\prime} \circ \xi_{h|_{t-1}},$$
$$\operatorname{dom} \xi_{h|_t} = \{U \in \operatorname{dom} \xi_{h|_{t-1}} : \xi_{h|_{t-1}}(U) \in \operatorname{dom} \Sigma_{h|_t}^{\mathrm{co}\prime}\}.$$

It applies to the canonical step data at the calibration frame: these data have the coordinate properties at every step and history, by [YM3, Prop. 4.75] and [YM3, eq. 4.dccxxiii], including equality of the new variables on their precise domains. Define on $\mathcal{Y}$

$$\hat{\xi}_{h_0} = \Pi_{h_0}^{\mathrm{le}} \times \mathrm{id}, \qquad \hat{\xi}_{h|_t} = \Sigma_{h|_t}^{\mathrm{co}\prime} \circ \hat{\xi}_{h|_{t-1}}$$

on the inverse image of the domain of the step map. These are Borel partial maps. Induction gives

$$\xi_h = \hat{\xi}_h \circ \mathcal{M}^{t_0}, \qquad \operatorname{dom} \xi_h = (\mathcal{M}^{t_0})^{-1}(\operatorname{dom} \hat{\xi}_h).$$

Composing with $\mathrm{res}_h^{\mathrm{le}} \times \mathrm{id}$ defines the free map $\hat{\mathsf{v}}_h$ on the same domain, with $\mathsf{v}_h = \hat{\mathsf{v}}_h \circ \mathcal{M}^{t_0}$. Completion and restriction have the same domain by the explicit recursive formulas [YM3, eq. 4.cccxlix], [YM3, eq. 4.cccl], and [YM3, eq. 4.cccli]: completion retains the free coordinates and fills each tied bond with its prescribed average.

The coordinate factor of a step carries $\hat{\mathfrak{w}}_{h|_{t-1}}$ to $\hat{\mathfrak{w}}_{h|_t}$ on $\mathcal{V}_{t_0}^{\chi} \cap \operatorname{dom} \hat{\xi}_{h|_{t-1}}$. Indeed lift W by the construction above, apply the coordinate factor identity on the bare lattice, and cancel the positive factor $\mathfrak{w}_{h_0}(U)$ in (*). The partial fermion kernel has Berezin integral one over the next-level generators; stored generators occur in no later kernel. Consequently induction over the steps, using

these coordinate maps and weight factors, identifies the Berezin integral of the restarted term with $(\hat{\mathsf{v}}_h)_*(\hat{\mathfrak{w}}_h \hat{F}_0^{\chi}\, dW)$.

This pushforward loses no mass. If $W \in \mathcal{V}_{t_0}^{\chi}$ and $\hat{\mathfrak{w}}_h(W) \neq 0$, choose the positive-weight bare lift U above. Then (*) gives $\mathfrak{w}_h(U) \neq 0$. The support of a history weight is in the domain of its free coordinates, as proved in [YM3, Prop. 4.75] and the setting of [YM3, Prop. 4.153]; thus $U \in \operatorname{dom} \mathsf{v}_h$ and $W \in \operatorname{dom} \hat{\mathsf{v}}_h$. The original measure is carried by this set. Its total mass is therefore $\mathsf{m}(h)$. Nonnegativity follows from Lem. B.67; the norm bound follows from $|\operatorname{Ber} F| \leq \|F\|_{h_F}$. □

Lemma B.70 (Summation above the starting level)**.** Let $\varsigma \geq 0$ be bounded, Borel, gauge invariant, and zero off $\mathcal{V}_{t_0}^{\chi}$. For $h' \in \mathcal{H}_{k'}^{(t_0)}$ and $t_0 \leq k' \leq k \leq K$,

$$\sum_{\substack{h \in \mathcal{H}_k^{(t_0)} \\ h|_{k'} = h'}} \mathsf{m}_\varsigma(h) = \mathsf{m}_\varsigma(h').$$

In particular

$$\sum_{h \in \mathcal{H}_K^{(t_0)}} \mathsf{m}_\varsigma(h) = \int_{\mathcal{Y}} \varsigma\, dW,$$

$$\sum_{\substack{h \in \mathcal{H}_K^{(t_0)} \\ h|_{t_1} = h_\emptyset^{(t_1)}}} \mathsf{m}_\varsigma(h) = \int_{\mathcal{Y}} \hat{\mathsf{k}}_\varsigma\, dW.$$

For these histories, $h|_{t_1} \neq h_\emptyset^{(t_1)}$ if and only if $Z_{t_1}(h) \neq \varnothing$. The bare weight factorization (*) in the proof of Lem. B.69 holds for every U and every history above t_0. ■

Proof. It suffices to prove the one-step identity. Let $t_0 < q \leq K$ and $h' \in \mathcal{H}_{q-1}^{(t_0)}$. All its extensions remain above t_0. For a plaquette outcome p, put $P'' = P_q''(\mathsf{p})$ and $\mathsf{N} = B(P'') \setminus P''$. The partition of the other outcomes gives

$$\sum_{\mathsf{o}\colon \mathsf{p}(\mathsf{o}) = \mathsf{p}} \hat{\mathfrak{w}}_{(h',\mathsf{o})}(W) = \hat{\mathfrak{w}}_{h'}(W) \mathbf{1}_q^{\mathrm{P}}(\mathsf{p})(W_q) \mathfrak{G}_{q-1}(W_{q-1}; P'').$$

Lift $v \in G^{\mathsf{N}}$ constantly on level-$(q-1)$ blocks to level t_0. The substitution preserves W_q, the earlier tests and gauge factors, the plaquette test, ς, and Haar measure. Its average replaces the last factor by $\mathbf{1}_{\{W_{q-1} \in \mathcal{D}(P'')\}}$. Hence the sum of the masses for this plaquette outcome is

$$\int \hat{\mathfrak{w}}_{h'}\, \mathbf{1}_q^{\mathrm{P}}(\mathsf{p})\, \mathbf{1}_{\{W_{q-1} \in \mathcal{D}(P'')\}} \varsigma\, dW.$$

The domain indicator is one on the support of $\hat{\mathfrak{w}}_{h'}\varsigma$. Here is the domain check. The smallness condition on the frames gives $\varepsilon(\mathsf{t}_{q-1}) \leq \varepsilon(\gamma^{-2}) \leq \varepsilon_\flat$. The contour-domain implication at $q \geq 3$ says that the plaquette test of the preceding outcome implies the present domain condition; see [YM3, §4.1.2], specifically the proof of [YM3, eq. 4.xviii]. The proof places each double block in the preceding passed region, uses its curvature bound [YM3, eq. 4.xvii], and applies the small-curvature contour estimate [YM3, Lem. 3.3]. It is an implication for bare fields. For a point in the support under consideration, $W \in \mathcal{V}_{t_0}^{\chi}$; lift it to U as in Lem. B.69. If $q \geq t_0 + 2$, the preceding outcome indicator is a factor of $\hat{\mathfrak{w}}_{h'}(W)$, so its plaquette test is one. If $q = t_0 + 1$, the preceding history is $h_\emptyset^{(t_0)}$; its plaquette test is exactly $\chi_{t_0}(T_\ell)(W) = 1$, since its input region and outcome are empty and its test cubes cover the torus. In both cases $q - 1 \geq 2$, so that plaquette test reads only $\mathcal{M}^{q-1}U = W_{q-1}$. The contour-domain implication therefore gives $W_{q-1} \in \mathcal{D}(P'')$. Discarding this now redundant indicator and summing the plaquette tests gives $\sum_{\mathsf{o}} \mathsf{m}_\varsigma(h', \mathsf{o}) = \mathsf{m}_\varsigma(h')$.

Iteration proves the first display. At $k' = t_0$ its weight is one; at $k' = t_1$ the empty history has weight $\hat{\mathsf{k}}$. Finally a history at t_1 has full small-field region if and only if it is the empty history. Its regions are unchanged by restriction from level K, which proves the assertion about Z_{t_1}. □

B.24. **Descent of the background regularity.**

Lemma B.71 (The starting test on the support of the weight)**.** If $\mathsf{k}(W) > 0$, the background over W_j exists for $t_0 \le j \le t_1$. On its minimal orbit define

$$r_j = \frac{\mathsf{m}_j(U)}{\varepsilon(\mathsf{t}_j)}, \qquad \mathsf{m}_j(U) = L^{2j} \max_p |\partial U(p) - 1|.$$

The numerator is constant on the orbit. Then

$$r_j < 2B_3 L^{-3(t_1-j)/2} + \frac{C_H A_1}{(1 - L^{-3/2})A_0}.$$

If $s - u \ge n_0$, W passes the density test at t_0. Consequently, everywhere on $\mathcal{Y}$,

$$\mathsf{k}\chi_{t_0}(T_\ell) = \mathsf{k}, \qquad \mathsf{k}\hat{F}_z = \mathsf{k}\hat{F}_z^{\chi}.$$

■

Proof. At the top, $W_{t_1} \in \mathsf{In}_{t_1}$, so $\varpi(W_{t_1}) < 2\varepsilon(\mathsf{t}_{t_1})$. The background estimate [YM3, Prop. 4.58] gives $r_{t_1} < 2B_3$. At a lower level, the weight imposes the field cut for W_{j+1} and $\chi_j^{\mathrm{sf}}(W_j) = 1$, with $\mathcal{M}(W_j) = W_{j+1}$. The inverse chart [YM3, Prop. 4.55] gives $W_j = V_{W_{j+1}}(B)$ for $B \in \mathsf{Rg}_{\mathsf{t}_j}$. The estimate in the proof of [YM3, Lem. 6.33] is

$$\mathsf{m}_j(U) \le L^{-2}\mathsf{m}_{j+1}(U'') + C_H\delta(\mathsf{t}_j).$$

Its hypotheses hold because the next field is in the cut and $\mathsf{t}_j \ge L^u\gamma^{-2} \ge \Theta_\star$; the same chart result puts W_j in its cut. Use $\delta(\mathsf{t}_j) = (A_1/A_0)\varepsilon(\mathsf{t}_j)$ and $L^{-2}\varepsilon(\mathsf{t}_{j+1}) \le L^{-3/2}\varepsilon(\mathsf{t}_j)$ to obtain

$$r_j \le L^{-3/2} r_{j+1} + C_H A_1/A_0.$$

Its fixed point is $C_H A_1/((1 - L^{-3/2})A_0)$. Iteration gives the stated bound.

If $t_1 - t_0 = s - u \ge n_0$, then $r_{t_0} < 1/(2B_3)$ by the definition of σ_D, n_0. Thus $W \in \mathcal{C}^{\flat}_{t_0}(\varepsilon(\mathsf{t}_{t_0})/(2B_3))$. This implies the density test in the calibration frame. To check the parameters of that implication, the radius is in $(0, 1]$. The history ceilings $\varepsilon_\gamma \le \min\{\bar{\varepsilon}/2, \varepsilon_\flat\}$, verified in the proof of [YM3, Lem. 5.58], and $B_3 \ge 1$ give

$$\varepsilon(\mathsf{t}_{t_0}) \le \varepsilon_\gamma \le B_3 \min\{\bar{\varepsilon}, 2c_2^{\mathrm{SO}}\}, \qquad c_2^{\mathrm{SO}} = \varepsilon_\flat.$$

The one-lattice background setting includes the admitted-constants condition required by the density-test implication, with its two specified auxiliary constants $a_\mathfrak{h}, \varepsilon_\mathfrak{h}$. Here is the implication explicitly. The global background has scaled curvature less than $\varepsilon(\mathsf{t}_{t_0})/(2B_3)$. Curvature averaging gives $\varpi(W) < \varepsilon(\mathsf{t}_{t_0})/B_3$. For every density cube, apply the local background bound [YM3, Prop. 4.2] at this radius. Its defining curvature is strictly less than $\varepsilon(\mathsf{t}_{t_0})L^{-2t_0}$, so its density test equals one. Taking their product gives $\chi_{t_0}(T_\ell)(W) = 1$. The pointwise identities follow on the support of k and are trivial elsewhere. □

Proposition B.72 (Removing the top test)**.** Assume (k0), $L^u\gamma^{-2} \ge \Theta_\star$, $u \ge u_*$ and $s \ge s_F$. Then

$$0 \le \int_{\mathcal{Y}} |\mathsf{k} - \hat{\mathsf{k}}|\hat{F}_0^{\chi}\, dW \le e^{-\theta_{\mathrm{dis}}A_0^2\mathsf{y}_{[u]}}\mathsf{M}_{g,\ell}\hat{T}(0)$$
$$\le R^F\hat{T}(0).$$

Moreover $0 \le T^{\mathrm{sf}}(0) - \hat{T}(0) = \int |\mathsf{k} - \hat{\mathsf{k}}| \hat{F}_0 \, dW$. If $s - u \ge n_0$, in particular if $K^{\mathrm{A}} \ge K_{\mathrm{tt}}(n)$, then

$$T^{\mathrm{sf}}(z) = \int_{\mathcal{Y}} \mathsf{k} \hat{F}_z^{\chi} \, dW,$$

$$\int_{\mathcal{Y}} |\mathsf{k} - \hat{\mathsf{k}}| \hat{F}_0 \, dW = \int_{\mathcal{Y}} |\mathsf{k} - \hat{\mathsf{k}}| \hat{F}_0^{\chi} \, dW \le R^F \hat{T}(0) \le \hat{R}^F \hat{T}(0).$$

The source assertion holds wherever $T^{\mathrm{sf}}(z)$ is defined. ■

Proof. By Lems. B.66 and B.67, $|\mathsf{k} - \hat{\mathsf{k}}| = \mathsf{k} - \hat{\mathsf{k}}$ and $\hat{T}(0) = \int \hat{\mathsf{k}} \hat{F}_0^{\chi} \, dW$. The gauge average Lem. B.68, followed by Lems. B.69 and B.70, gives

$$\begin{aligned} \int |\mathsf{k} - \hat{\mathsf{k}}| \hat{F}_0^{\chi} \, dW &\le \int \hat{F}_0^{\chi} \, dW - \hat{T}(0) \\ &= \sum_{\substack{h \in \mathcal{H}_K^{(t_0)} \\ Z_{t_1}(h) \neq \varnothing}} \mathsf{m}(h) \\ &\le \sum_{\substack{h \in \mathcal{H}_K^{(t_0)} \\ Z_{t_1}(h) \neq \varnothing}} \int \|\hat{g}_h\|_{h_F} \, d\mathscr{M}_h^{\mathrm{fr}}. \end{aligned}$$

The discarded-history estimate and positive total bound of Prop. 11.15 bound this last sum by

$$e^{-\theta_{\mathrm{dis}} p_0(g_{t_1})^2} e^{\hat{E}_K + (C+2)\ell^3} \le e^{-\theta_{\mathrm{dis}} A_0^2 \mathsf{y}_{[u]}} \mathsf{M}_{g,\ell} \hat{T}(0),$$

where $p_0(g_{t_1})^2 = A_0^2 (\log(L^u \mathsf{b}))^{2\mathfrak{p}_0} = A_0^2 \mathsf{y}_{[u]}$. Since $\mathsf{b} \ge \gamma^{-2}$, this is bounded by $R^F \hat{T}(0)$ with the stated definition of R^F. The difference $T^{\mathrm{sf}}(0) - \hat{T}(0)$ is the integral of the difference of their weights against $\hat{F}_0 \ge 0$. When $s - u \ge n_0$, Lem. B.71 permits replacement by $\hat{F}_z^{\chi}$ for k, and Lem. B.66 does so for $\hat{\mathsf{k}}$. This proves the remaining assertions. □

Choice of the descent floor. The constant A_0 is free upward after A_1 in the choice of constants. The numbers L, B_3, C_H in (D) are already fixed there: B_3 depends on L, and $C_H = 4C_{\mathrm{db}} \max\{2, K_{\mathrm{B}}^{\mathrm{s}}(\rho_{\mathrm{s}})\}$ is the constant fixed in Prop. A.5 and eq. A.x. It depends on the earlier choices $L, \mathfrak{M}, R_1$ through ρ_{s}. Thus (D) is a legitimate additional floor. Indeed, $L \ge 13$ gives $L^{-3/2} < 1/2$, so it holds for every $A_0 \ge 4B_3 C_H A_1$. Intersecting this ray with the finitely many other floors at A_0 leaves an unbounded ray, and the subsequent choices retain their admissible order. This argument uses the actual constant in the descent estimate; it does not identify it with a supply-table constant having the same name. The quantities σ_D, n_0 are chosen after A_0, and $K_{\mathrm{tt}}(n)$ after the fixed constants and n. No earlier constant reads these quantities.

Under (k0)–(k2) and $K^{\mathrm{A}} \ge K_{\mathrm{tt}}(n)$, all hypotheses above hold by (k1). The last display of Prop. B.72 is therefore the top-test input Prop. 11.10, for either lattice.

Appendix C. Histories with prescribed earlier terms

A restart retains functions of earlier block fields. We construct the history expansion with those functions prescribed, first at zero source and then with sources and an interpolation parameter. The coordinate integrals below give a common measure space for the comparison.

C.1. **Source induction with prescribed earlier activities.** The cutoff comparison changes activities that were formed before its base level. Those activities no longer need to be outputs of the same flow. They do need the cancellation and locality properties used when an old activity is read at a later scale. We first specify these properties, then prove that two weighted sums of their sizes suffice for the source induction. The parameter disc shrinks at successive steps to keep the additional mass field inside its coordinate domain.

Definition C.1 (Histories through the base)**.** Fix an admitted positive-mass datum and $1 \le t_1 \le K$. The constants satisfy the additional floor and ceiling in appendix C.2. Write

$$\mathcal{H}_k[t_1] = \{h \in \mathcal{H}_k : h|_{t_1} = h_\emptyset\}, \qquad t_1 \le k \le K.$$

For such histories, every region before the base is the whole torus, every early pair is regular, and there is no earlier rim, edge term, or boundary term. The layers $\mathcal{L}_n$ with $n < t_1$ are empty. If $k > t_1$, the first layer is $\mathcal{L}_{t_1} = T_\ell \setminus \Omega_{t_1+1}$; if $k = t_1$ it is the whole torus. The history and the ordinary source chain agree at every level through t_1. ■

Definition C.2 (Free families)**.** A free family of level j is an even local family $\Delta = (\Delta(X))_{X \in \mathcal{D}_j}$ on the fixed pair domains, with the following form. Its scalar part is pointed:

$$\Delta_0(X) = \sum_{z \in X \cap \mathbb{T}^{(j)}} \Delta_0(X; \cdot; z), \qquad r^0(\Delta) = \sup_{X,z} e^{(\kappa+1)d_j(X)} \|\Delta_0(X; \cdot; z)\|_X.$$

Each pointed member is a local term, holomorphic as a function of the pair, strictly gauge invariant, and zero at the trivial pair.

The part with fermion generators is specified by data (F, G, ν). Here F, G are even families of degrees at least two with formal background fields at levels at most j, satisfying the core holomorphy and joint invariance hypotheses of [QCDM1, Lem. 7.9]. They are invariant under $\Psi \mapsto e^{i\alpha}\Psi$, $\bar\Psi \mapsto e^{-i\alpha}\bar\Psi$; thus their quadratic parts contain one barred and one unbarred field. The fields in F are read at $m_0 - a_K \widehat{J}$, and those in G at zero source. Use the pieces and paths of [QCDM1, Prop. 10.1 and its proof] and set

$$a_{X,p} = \mathrm{ord}_0[k_{X,p}(F)] - \mathrm{ord}_0[k_{X,p}(G)].$$

These matrix moments act trivially on colour, by invariance under constant colour rotations at the trivial configuration. Define

$$\begin{aligned} P_{z;X,p} &= -\mathfrak{s}_j^{-1} L^{-j} \mathsf{S}_1(a_{X,p}(z) 1_z \psi^{\mathrm{bg}}), \\ l_\square &= L^{-j} \nu \, \mathsf{S}_1(1_\square \psi^{\mathrm{bg}})|_{\mathsf{n}=0}, \\ P_z^\nu &= L^{-j} \nu \, [\mathsf{S}_1(1_z \psi^{\mathrm{bg}}) - \mathsf{S}_1(1_z \psi^{\mathrm{bg}})|_{\mathsf{n}=0}], \\ \Delta_{\ge 2}(Z) &= F(Z) + \sum_{(X,p): Z_p = Z} \sum_z P_{z;X,p} + \mathbf{1}_{\{Z=\square\}}(l_\square + \sum_{z \in \square} P_z^\nu). \end{aligned}$$

On a torus the piece sum uses only polymers of diameter less than one quarter of the side, as in the cited extraction rule. The moments for the omitted pieces are still defined and enter the cell estimates below. File $F(Z)$ at its polymer's cell, $P_{z;X,p}$ and P_z^ν at the cell containing z, and $l_\square$ at its cube.

Finally, G together with $l_\square$ must be the restriction, on polymers that do not wrap around the torus, of a family G^∞ on $\mathbb{Z}^3$. This family has the same holomorphy and symmetry properties and is covariant, summand by summand, under $M\mathbb{Z}^3$ translations. With the same lifted paths and moved diagonal terms, its cell sum vanishes as a matrix in every cell.

Define $r^\psi(\Delta)$ as the maximum of four quantities: the sum of the norms of the displayed summands of $\Delta_{\ge 2}$; the corresponding sum for G with $l_\square$; that for G^∞; and $c_{\mathsf{S}} h_F^2 |\nu|$. All three family norms are the weighted norms for formal backgrounds used in [QCDM1, §7, in particular Lem. 7.9]. Holomorphic dependence on a source means holomorphic dependence of these data, coefficient by coefficient. The construction is linear in (F, G, ν); both sizes are subadditive and absolutely homogeneous. ■

The ordinary activity at a source has precisely this form: take its parts before extraction at the actual coordinates and at zero coordinates for F, G, and its zero-source mass coefficient for ν. The zero-source cell sum vanishes after extraction. Consequently differences of two such activities are

free families after their scalar values at the trivial pair have been subtracted. Those values are added to the scalar normalization.

Definition C.3 (Perturbation at the base)**.** Let S be an open source set. For $1 \le i \le p$ prescribe a Lipschitz gauge weight w_i^{Δ}, a colour-scalar matrix field $\sigma_i^{\Delta}(s)$, free families $\Delta_i^{(j)}(s)$ for $j \le t_1$, and a number $e_i^{\Delta}(s)$. The last three data are holomorphic in $s \in S$. Write $|\lambda| = \max_i |\lambda^i|$, $\mathrm{P}(R) = \{\lambda \in \mathbb{C}^p : |\lambda| < R\}$, and

$$\Delta_i = a_K^{-1} A(w_i^{\Delta}, \mathsf{U}) - L^{-t_1} \mathsf{S}_1(\sigma_i^{\Delta} \psi^{\mathrm{bg}}) + \sum_{j \le t_1} \sum_X \Delta_i^{(j)}(X) + e_i^{\Delta}.$$

The fields and formal backgrounds in this formula are read in the coordinates of the empty history at level t_1. Multiply its ordinary sourced density by $e^{\lambda \cdot \Delta}$ and call the resulting measure $\rho_{s,\lambda}$. Its gauge weight, early activities, mass field, and scalar normalization are

$$\begin{aligned} w_\lambda &= w + \lambda \cdot w^{\Delta}, & s_\lambda &= (w_\lambda, J), \\ E^{(j)}(s,\lambda) &= E^{(j)}(s) + \lambda \cdot \Delta^{(j)}(s), & j &\le t_1, \\ \sigma_{t_1}(s,\lambda) &= \sigma_{t_1}(s) + \lambda \cdot \sigma^{\Delta}(s), \\ E_{t_1}(s,\lambda) &= E_{t_1}(s) + \lambda \cdot e^{\Delta}(s). \end{aligned}$$

The fermion tower retains the fine mass matrix $m_0 - a_K \widehat{J}$.

Put

$$r_j^{\psi} = \sum_i r^{\psi}(\Delta_i^{(j)}), \quad r_j^0 = \sum_i r^0(\Delta_i^{(j)}), \quad \|\sigma^{\Delta}\| = \sum_i \sup_x \|\sigma_i^{\Delta}(x)\|, \quad \|w^{\Delta}\|_\sharp = \sum_i \|w_i^{\Delta}\|_\sharp.$$

For $k \ge t_1$ define

$$\Sigma_k^{\psi} = \sum_{j \le t_1} \omega_{k+1-j} r_j^{\psi}, \qquad \omega_l = C_\omega L^{l/4}, \qquad \Sigma_k^0 = \sum_{j \le t_1} L^{-(k-j)} r_j^0.$$

At radius R we require

$$\begin{aligned} R\Sigma_k^{\psi} &\le c_{\psi}^{\Delta} \phi_{k-1} \quad (t_1 \le k \le K), & R\Sigma_{t_1}^0 &\le c_0^{\Delta}, \\ \|w\|_\sharp + R\|w^{\Delta}\|_\sharp &\le 2\varrho, & R\|\sigma^{\Delta}\| &\le \tfrac{1}{2} B_\sigma \phi_{t_1}. \end{aligned} \tag{C.i}$$

The constants $c_{\psi}^{\Delta}, c_0^{\Delta}$ are given in appendix C.2. Set

$$Q_\sigma = 2\sqrt{L} + \tfrac{1}{4}, \qquad R_k = R Q_\sigma^{-(k-t_1)}.$$

The weight condition ensures $s_\lambda \in \mathcal{P}$ for $s \in \mathcal{P}$ and $\lambda \in \mathrm{P}(R)$. ■

Lemma C.4 (Room in the ordinary source estimates)**.** At the fixed pair radius α_0, the ordinary activities, their pointed terms, and their zero-source reference families obey

$$\epsilon_j^{\mathrm{s},\psi} \le (2C_w^{\mathrm{s}} - C_w)\phi_{j-1}.$$

The ordinary source chain obeys

$$\|\sigma_k(s)\| \le (2B_0 + C_{\mathrm{var}} + 4C_{\mathrm{tor}} C_{\mathcal{L}})\phi_k \le \tfrac{1}{2}(B_\sigma - C_{\mathrm{var}})\phi_k.$$

■

Proof. The closing estimate in [QCDM1, Prop. 10.1, proof] is

$$C_w^{\psi}\phi_k + 8\theta_E^{\mathrm{s}}(1 + c_{\mathrm{cs}}) C_w^{\mathrm{s}} C_{\mathrm{pc}} S_{1/4} \phi_k + 4\sqrt{L} C_{\mathrm{pc}} S_{1/4} \theta_E^{\mathrm{s}} c_{\mathsf{S}} h_F^2 (\bar{B}_M + 2B_\sigma)\phi_k.$$

By the definition of C_w^{s} the first and third terms sum to at most $(C_w^{\mathrm{s}} - C_w)\phi_k$. The middle term is at most $C_w^{\mathrm{s}}\phi_k$ by $4\theta_E^{\mathrm{s}} C_{\mathrm{pc}} S_{1/4}(1 + c_{\mathrm{cs}}) \le 1/2$. This includes the reference and the pointed terms; at the first step the old-activity sums are empty.

For the chain apply [QCDM1, Lem. 10.4]. On $\mathcal{P}$, $\|w\|_\infty < 2\varrho$ and the local source variation satisfies $\Xi_k < 1/(20M) + 1/2 < 1$. The claimed bound follows from $|\sigma_k^c(c)| \le B_0\phi_k|c|/\varrho$ and $B_\sigma \ge 4B_0 + 3C_{\mathrm{var}} + 8C_{\mathrm{tor}}C_{\mathcal{L}}$. □

C.2. **The constants and the count over scales.** We record the constants before applying the history induction. The additional restriction on the regular sum is a lower bound on two majorant constants, chosen before the later small-coupling ceilings. It does not impose a condition depending on the cutoff.

First put

$$c_{\mathrm{cs}}^{\Delta} = \frac{2C_{\mathrm{hl}}M^3C_{\mathrm{bg}}^{\mathrm{w}}}{Lc_{\mathsf{S}}\omega_1}, \qquad c_{\psi}^{\Delta} = C_w\omega_1\frac{c_{\mathrm{cs}}}{c_{\mathrm{cs}} + c_{\mathrm{cs}}^{\Delta}},$$

and

$$c_0^{\Delta} = \min\left\{\frac{5}{12}K_M^{\bullet}c_{\mathrm{cl}}, \frac{\alpha_0C_w^{\mathrm{s}}}{16\sqrt{2}K_M^{\bullet}K_\Lambda^{\mathrm{s}}\mathsf{C}_{\mathcal{A}}A_1}, (\sqrt{2}-1)E_0^{\mathrm{s}}, M^{-3}\mathsf{B}\right\}.$$

The first two choices preserve the scalar room in Prop. C.15; the other two will bound the regular scalar sum and its frame. Define also

$$c_\lambda = \min\left\{\tfrac{1}{8}, c_0^{\Delta}/E_0^{\mathrm{s}}, c_\psi^{\Delta}/(2\sqrt{L}C_w^{\mathrm{s}})\right\}.$$

The units used below are the quadratic kernels with their diagonal moments removed, the moved diagonal pieces at their new cells, and the mass forms in Def. C.2. Their terms without a difference belong to the cell sum and are counted separately. Set $\alpha = 3/4$, $c_T = 64L^2$, $C_t = 9$, and

$$\begin{aligned} N_{\mathrm{d}} &= 2(C(\alpha) + c_T\epsilon_*),\\ N_{\mathrm{u}} &= 32C_tMA_\kappa N_{\mathrm{d}}[(2A_\kappa C_H)^2 + \nu C_H^2(2C_{\mathrm{pw}} + C_{(N)})],\\ c_l^{\mathrm{u}} &= \begin{cases} N_{\mathrm{u}}L^{9/2}L^{l/4}, & l > n_{\mathrm{lag}},\\ N_{\mathrm{u}}L^{3l}, & 0 \le l \le n_{\mathrm{lag}}. \end{cases} \end{aligned}$$

All constants on the right are the local background, piece, and path constants in [QCDM1, Lem. 7.9 and Prop. 10.1]. They are fixed before h_F. In particular $N_{\mathrm{u}} \ge 1$.

For a family of size ε_{f} having the format in Def. C.2, its units, apart from their cell sums, contribute at most $c_{n-j}^{\mathrm{u}}\varepsilon_{\mathrm{f}}$ to a level-n cube. For $n - j > n_{\mathrm{lag}}$ this statement counts the pieces contained in Ω_{n-1}; the other pieces have an exponential polymer tail and will enter the summable kernels below. For $n - j \le n_{\mathrm{lag}}$ it counts the whole units. These statements also hold along a history: at short lags the needed size is the sum of the norms of the kernels and forms actually retained by its regular pointers.

To prove these assertions, fix a level-j cell c with center z_c. At the trivial configuration write each localized leg as

$$\Upsilon_\omega(x)\Phi(z_c) + \varrho_\omega(x), \qquad \varrho_\omega(x) = \sum_y H_\omega(1)(x,y)[\Phi^{\mathsf{u}}(y) - \Phi(z_c)].$$

The term with no ϱ is precisely its cell moment. The remaining three terms of a quadratic form are bounded by four times a whole leg times a leg with a difference: the second difference in the product of two differences costs at most two whole legs.

Put $l = n - j > n_{\mathrm{lag}}$. A whole leg within a piece contained in Ω_{n-1} has relative size at most $2C_Hq_{\sup}^{|\omega|}L^{1-l}$. Transport the difference to z_c along the piece's path. A neighboring difference of level j is bounded by

$$A_\kappa N_{\mathrm{d}}L^{2(1+\alpha)}L^{-(1+\alpha)l}.$$

Indeed the background Hölder estimate contributes $2A_\kappa C(\alpha)(L^j/\mathsf{s}_k)^{1+\alpha}$, with $\mathsf{s}_k \ge L^{n-2}$ on Ω_{n-1}. Replacing straight transport by the block field contributes $2c_T\epsilon_*L^{-2(n-1-j)}$ times a leg of size L^{1-l}. Since $L^{3-3l} \le L^{2(1+\alpha)}L^{-(1+\alpha)l}$, both contributions have the displayed bound.

If $D'_p = M(1 + d_j(X)) + 9M_w \sum_i(|\omega_i| + 1)$ is the piece diameter, its path length satisfies

$$1 + C_t D'_p \le 8C_t M e^{\kappa d_j(X)} e^{|\omega_1|+|\omega_2|}.$$

Here $M \ge 3M_w$, $1 + d \le e^{\kappa d}$, and $1 + |\omega| \le e^{|\omega|}$. The polymer norm pays the first exponential, and the two localized walk sums pay the others with A_κ each. The ordinary kernel pieces and forms in the cell therefore contribute at most

$$32C_t M A_\kappa N_{\mathrm{d}} (2A_\kappa C_H)^2 L^{1-l} L^{2(1+\alpha)} L^{-(1+\alpha)l} \varepsilon_{\mathrm{f}}.$$

A moved piece has norm at most $\nu \|a_{X,p}(y)\| C_H^2 h_F^2$. There are two assignments to count: its negative copy in its old unit and its positive copy at the cell of y. The polymerwise piece bound pays the path length for the first; the first weighted piece estimate pays the second, whose path lies within its cell. Each contributes $C_{\mathrm{pw}} h_F^{-2} \varepsilon_{\mathrm{f}}$ before the leg factors. The same difference calculation thus supplies $\nu C_H^2 \, 2C_{\mathrm{pw}}$ in place of $(2A_\kappa C_H)^2$. The additional $\nu C_H^2 C_{(N)}$ in N_{u} also covers the alternative count by the old-activity moment bound.

There are L^{3l} cells in the level-n cube. The resulting scale factor is exactly

$$L^{3l} L^{1-l} L^{2(1+\alpha)} L^{-(1+\alpha)l} = L^{9/2} L^{l/4}.$$

This proves the long-lag count. At short lags no differences are formed. Each leg has relative size at most one. The kernels and forms cost ε_{f} per cell, and the two copies of moved pieces cost at most $2\nu C_H^2 C_{\mathrm{pw}} \varepsilon_{\mathrm{f}}$. Multiplication by L^{3l} proves the short-lag count.

For histories with a rim at level j, a piece contained in Ω_{n-1} at a long lag lies in $\Lambda_j^{\sim -1}$: the separation is at least $7ML^{n-1}$ and $n_{\mathrm{lag}} \ge 1$. All its pointers and paths therefore agree with the unrestricted reading; the same cancellation applies. At short lags the retained kernels and forms have the pointed size bound of the activity. The retained moved pieces are subsets of the same nonnegative absolute sums. This proves the asserted count along histories.

For the other contributions define $K_{\mathrm{fin}} = \sum_{l \ge 0} \mathsf{k}_l$, where

$$\begin{aligned}
\mathsf{k}_l &= N_{\mathrm{rd}} L^{-l} + 2C_\omega C_{\mathrm{wr}} L^l e^{-\kappa(L^{l+1}-2)} \\
&\quad + \mathbf{1}_{\{l > n_{\mathrm{lag}}\}} N_{\mathrm{wh}} L^{3l+2} e^{-\kappa(8L^l-2)} + r_l, \\
r_l &= \begin{cases} [C_\omega((1+L^2)C_\omega + 2C_{\mathrm{pw}})L^l + N_{\mathrm{wh}} L^{3l+2}] e^{-\kappa(7L^l-2)}, & l > 0, \\ C_\omega((1+L^2)C_\omega + 2C_{\mathrm{pw}}) + N_{\mathrm{wh}} L^2, & l = 0. \end{cases}
\end{aligned}$$

Here N_{rd} is the constant for the higher-degree and field-dependent quadratic terms in the proof of [QCDM1, Lem. 10.8]; N_{wh} is the larger constant for whole pieces outside Ω_{n-1} and whole rim terms in that proof. They depend only on the already fixed local geometry, background and piece constants, and are chosen before the size of the regular sum. The four kernels have finite sums. The only terms involving a moved coefficient in N_{wh} have size $\nu \|a_{X,p}(y)\| C_H^2 h_F^2 L^2$; the weighted piece bound cancels h_F^2, so there is no dependence on a later choice of h_F in these constants.

Set

$$\begin{aligned}
S_{\mathrm{lag}} &= \frac{L^{3(n_{\mathrm{lag}}+1)} - 1}{L^3 - 1}, \\
c_{\mathrm{u}}^{\mathrm{leg}} &= 2C_w^{\mathrm{s}} \left[\frac{L^{9/2} C_{\mathrm{pc}} S_{1/4}}{C_\omega L^{1/4}} + S_{\mathrm{lag}} \right], \\
c_{\mathrm{o}}^{\mathrm{leg}} &= 2C_w^{\mathrm{s}} K_{\mathrm{fin}} + C_0^{\mathrm{cs}} h_F^2 + 2c_{\mathrm{cs}} C_{\mathrm{pc}} S_{1/4} C_w^{\mathrm{s}}.
\end{aligned}$$

Choose the two constants in the sourced regular-sum bound so that

$$C_{\mathrm{leg}}^{\square,\mathrm{s}} \ge c_{\mathrm{o}}^{\mathrm{leg}} + N_{\mathrm{u}} c_{\mathrm{u}}^{\mathrm{leg}}, \qquad C_{\mathrm{leg}}^{\mathrm{s}} \ge \phi(\tfrac{1}{2} e^{18\mathfrak{p}_0}) C_{\mathrm{leg}}^{\square,\mathrm{s}}. \tag{C.ii}$$

The count just proved shows that these choices also satisfy the ordinary regular-sum estimate. They are made after C_w^{s} and before the stored-layer constants, majorants, and mass floor. All constants depending on them are then fixed at these values.

For the additional tail estimate put

$$\theta_{\mathrm{tl}} = \tfrac{1}{4}(c_\star - c_\Theta), \qquad c_{\mathrm{tl}} = \tfrac{1}{4}(c_\star + 3c_\Theta), \qquad \mathsf{w}_{\mathrm{tl}} = c_\Theta/c_{\mathrm{tl}}, \qquad c''_{\mathrm{tl}} = (1-\eth)\theta_{\mathrm{tl}}.$$

Choose γ small enough that the history-summation threshold of [YM3, Prop. 4.160](f) holds at rate c_{tl}, majorant C, vacuum-boundary constant $C_\flat^F + C'_{\mathfrak{E}}$, and dials $(\mathsf{w}_{\mathrm{tl}}, c_a, C_{\mathrm{vol}}, c_{\mathrm{tl}})$. Explicitly this is $\log\gamma^{-2} \geq \mathsf{Y}_{\mathrm{f1}}$ if $c_a = 0$ and $\log\gamma^{-2} \geq \mathsf{Y}_{\mathrm{f2}}$ if $c_a > 0$, with those thresholds formed at the displayed arguments and the corresponding admitted floor on $\mathfrak{p}_0$. Both remaining rates, $\eth\theta_{\mathrm{tl}}$ and $(1-\eth)\theta_{\mathrm{tl}}$, are positive. Thus the threshold is finite and depends only on constants fixed before γ. This is compatible with all the earlier ceilings, each of which holds on an interval $(0, \gamma']$.

The constants $c_{\mathrm{cs}}^\Delta, c_\psi^\Delta, c_0^\Delta, Q_\sigma, c_\lambda$ are read after this choice. They place conditions on the perturbation through eq. C.i, and do not enter the earlier choices. This verifies the order of constants in Def. C.1.

C.3. **Perturbations on the base coordinates.** The estimates for prescribed activities must apply to measures, including complex sources and complex interpolation parameters. We first define these measures and prove the identities used in the induction. The quantitative restrictions on the interpolation parameter enter only when we estimate their densities.

Lemma C.5 (Coordinates at the base)**.** Let $h_\emptyset = h_\emptyset^{(t_1)}$. Its free coordinates and its completed coordinates both identify with $\mathcal{V}_{t_1}$. Their reference measure is Haar probability measure, and their coordinate map on $\mathcal{V}_0$ is $\xi = \mathcal{M}^{t_1}$. For a gauge transformation u at level t_1, write $\tilde{u}(x) = u(y)$ on each block $B^{t_1}(y)$. Then

$$\xi(U^v) = \xi(U)^{v|_{t_1}}, \qquad \mathsf{Ev}_{h_\emptyset}(V^u) = \{\mathsf{U}^{\tilde{u}} : \mathsf{U} \in \mathsf{Ev}_{h_\emptyset}(V)\}.$$

Here $\mathsf{Ev}_{h_\emptyset}(V)$ consists of the adapted elements of the minimal orbit $U_{t_1}(V)$, and is empty when that orbit does not exist. Haar measure on $\mathcal{V}_{t_1}$ is invariant under $V \mapsto V^u$.

If v is a gauge transformation at a level $n \geq t_1$, its bare lift $\tilde{v}$ is also the lift of $\tilde{v}|_{t_1}$. The map $U \mapsto U^{\tilde{v}}$ preserves Haar measure, the weight $\mathfrak{w}_{h_\emptyset}$, and the support of that weight. ■

Proof. Along $h_\emptyset$, the regions through t_1 are $\Omega_j = \Lambda_j = T_\ell$. All proper bonds, lettered datum bonds, and external bonds below t_1 are therefore absent. At t_1, the proper and lettered datum bonds are all the bonds, and there are no tied bonds. Thus restriction and completion are identities; no fluctuation coordinate remains; and both coordinate spaces and measures have the stated form. Gauge covariance of $\mathcal{M}$ proves the first identity.

The determining set of this constant sequence consists of all bonds at level t_1. Its constraint is $\mathcal{M}^{t_1}U = V$, and its residual gauge group consists of transformations equal to one at the level-t_1 sites. A gauge transformation with restriction u maps this constraint to the constraint at V^u. It preserves the action, constrained criticality, and regularity, and conjugates the residual gauge group to itself. It therefore maps the minimal orbit at V onto that at V^u, with existence on either side equivalent to existence on the other.

For the lift $\tilde{u}$, the adaptation conditions are also preserved. Indeed, the endpoints of each adaptation contour (y, a), $a \in B(y)$, have the same value of $\tilde{u}$. Its contour variable equals one before the transformation if and only if it equals one after it. Applying this observation also to u^{-1} proves the identity for Ev.

The lift from a level $n \geq t_1$ is constant on every $B^{t_1}(y)$. Each test through t_1 is invariant under this transformation, as is each gauge-fixing factor. Their product is $\mathfrak{w}_{h_\emptyset}$. Left and right invariance of Haar measure proves the measure assertion; invariance of the weight and invertibility of the transformation prove the assertion about its support. □

Definition C.6 (Holomorphic perturbations)**.** Let $\mathcal{O} \subset \mathbb{C}^{n_s}$ be open and

$$s(z) = s_0 + \sum_{i=1}^{n_s} z_i s_i = (w, J) \in \mathcal{P}.$$

For $n_s \geq 1$, require $s(z) \in \mathcal{P}^\circ$. For $n_s = 0$, allow any one source in $\mathcal{P}$. Fix $r_\lambda > 0$. A perturbation consists of $w^\Delta(s)$, a complex Lipschitz function on T_ℓ; a field $\delta\sigma(s)$ in the colour-scalar matrix slot $\mathbb{M}$; a scalar $\mathfrak{n}(s)$; and pointed arrays $\Delta^{(j)}(s)$, $1 \leq j \leq t_1$. All these data are holomorphic in z, pointwise for the first two and coefficient by coefficient for the arrays.

Each array has the local domains and joint gauge invariance of Def. C.2. Its scalar part vanishes at the trivial pair, and its part with generators is formed from data $(F_j(s), G_j(s), \nu_j(s))$ by the extraction rule in that definition, including its phase invariance, common paths, infinite-lattice family with zero matrix cell sums, and rules for assigning terms to cells. The coefficients are Borel on the fixed cores $\mathcal{U}_j^\circ(X; \alpha_0, \alpha_1)$. Write

$$\rho_j(s) = \|\Delta_0^{(j)}(s)\|_{j,\kappa+1}, \qquad r_j^\psi(s) = r^\psi(\Delta^{(j)}(s)).$$

For every $z \in \mathcal{O}$ and $|\lambda| < r_\lambda$, require

$$\begin{aligned} |\lambda| \sum_{j \leq t_1} \omega_{k+1-j} r_j^\psi(s) &\leq c_\psi^\Delta \phi_{k-1} && (t_1 \leq k \leq K), \\ |\lambda| \sum_{j \leq t_1} L^{-(t_1-j)} \rho_j(s) &\leq c_0^\Delta, \\ (w + \lambda w^\Delta(s), J) &\in \mathcal{P}, \qquad |\lambda| \|\delta\sigma(s)\| \leq \tfrac{1}{2} B_\sigma \phi_{t_1}. \end{aligned} \tag{C.iii}$$

The constants are those of appendix C.2; in particular $c_0^\Delta \leq M^{-3}\mathsf{B}$ and $c_0^\Delta \leq (\sqrt{2}-1)E_0^{\mathrm{s}}$. No geometrical constant denoted by c_0 is being changed.

At an evaluation configuration of the base coordinates, set

$$\Delta(s) = a_K^{-1} A(w^\Delta(s), \mathsf{U}) - L^{-t_1} \mathsf{S}_1(\delta\sigma(s)\psi^{\mathrm{bg}}) + \sum_{j \leq t_1} \sum_X \sum_{z \in X^{(j)}} \Delta^{(j)}(X; \mathsf{U}; z; s) + \mathfrak{n}(s). \tag{C.iv}$$

The formal backgrounds are evaluated at the fine mass matrix $m_0 - a_K \widehat{J}$. Adding $\lambda\Delta(s)$ to the ordinary exponent replaces its gauge weight by $\beta_{t_1} - a_K^{-1}(w + \lambda w^\Delta)$, its mass coefficient by $L^{-t_1}(s_{t_1} + \sigma_{t_1}^{h_\emptyset}(s) + \lambda\delta\sigma)$, and its earlier activities by $E^{(j)}(s) + \lambda\Delta^{(j)}(s)$. It adds $\lambda\mathfrak{n}(s)$ to the scalar normalization. ∎

The conditions give bounds on each array, uniform on $\mathcal{O}$:

$$\rho_j(s) \leq \frac{c_0^\Delta L^{t_1-j}}{r_\lambda}, \qquad r_j^\psi(s) \leq \frac{c_\psi^\Delta \phi_{t_1-1}}{r_\lambda \omega_{t_1+1-j}}.$$

Indeed, let $|\lambda| \uparrow r_\lambda$ in eq. C.iii. Each coefficient is bounded by its fibre norm times a constant depending only on its finite Grassmann algebra. These bounds permit integration in the parameters on every compact subset of $\mathcal{O} \times \mathbb{C}$.

C.4. **The perturbed measures.** Write

$$\mathsf{S}_\emptyset = \{V : \mathsf{Stay}_{h_\emptyset} \mathbf{1}_{h_\emptyset}^{\mathrm{T}} \chi_{t_1}(T_\ell)(V) \neq 0\}.$$

The exponential representation at zero source permits a Borel Haar-null set N_0 such that every point of $\mathsf{S}_\emptyset \setminus \mathsf{N}_0$ lies in $\xi(\{\mathfrak{w}_{h_\emptyset} \neq 0\})$, all its evaluation configurations lie in the required domains, and $\Delta(s)$ takes one value there. We justify these assertions below. Define

$$g_{s,\lambda}(V) = \begin{cases} e^{\lambda\Delta(s)(V)}, & V \in \mathsf{S}_\emptyset \setminus \mathsf{N}_0, \\ 1, & \text{otherwise}, \end{cases}$$

and let $G_{s,\lambda} = g_{s,\lambda} \circ \xi$ on the support of $\mathfrak{w}_{h_\emptyset}$, with value one elsewhere.

For $h \in \mathcal{H}_k[t_1]$, let $T^{\mathrm{F}}_{h;(t_1,k]}[U]$ denote the composition of the normalized fermion kernels and partial Berezin integrals after level t_1. Thus $T^{\mathrm{F}}_h = T^{\mathrm{F}}_{h;(t_1,k]} \circ T^{\mathrm{F}}_{h_\emptyset}$. Set

$$\begin{aligned} F_s(U) &= T^{\mathrm{F}}_{h_\emptyset}[U]\big(e^{-A(\beta - a_K^{-1} w, U) - S_F(U) + \Phi(J)}\big), \\ \varsigma^{s,\lambda}_h(U) &= T^{\mathrm{F}}_{h;(t_1,k]}[U]\big(G_{s,\lambda}(U) F_s(U)\big). \end{aligned} \tag{C.v}$$

Define $\check{\tau}^{\mathrm{F}}_h[s,\lambda]$ by pushing $\mathfrak{w}_h \varsigma^{s,\lambda}_h \, dU$ to the free coordinates, coefficient by coefficient. Its image on the completed coordinates is $\hat{\tau}^{\mathrm{F}}_h[s,\lambda]$.

Lemma C.7. For the data of Def. C.6, these definitions have the following properties.

(i) The exponent $\Delta(s)$ is Borel on $\mathsf{S}_\emptyset \setminus \mathsf{N}_0$, coefficient by coefficient, holomorphic in z, and bounded there uniformly for z in compact subsets of $\mathcal{O}$. The coefficients of $\varsigma^{s,\lambda}_h$ are bounded Borel functions of U, holomorphic in (z,λ), and bounded uniformly in U on compact parameter sets. Both measures are holomorphic in these parameters when evaluated on a Borel set. At $\lambda = 0$, they are the ordinary sourced measures.

(ii) At the base,

$$\begin{aligned} \hat{\tau}^{\mathrm{F}}_{h_\emptyset}[s,\lambda] = \mathsf{Stay}_{h_\emptyset} \mathbf{1}^{\mathrm{T}}_{h_\emptyset} \chi_{t_1}(T_\ell) e^{\mathbf{\Phi}^w_{h_\emptyset}} \\ \cdot \exp\big[A^{\mathrm{F}}_{t_1,h_\emptyset}(s) + \lambda \Delta(s)\big] \mathscr{M}_{h_\emptyset}. \end{aligned} \tag{C.vi}$$

This equality holds coefficient by coefficient off one Borel null set $\mathsf{N}_s \cup \mathsf{N}_0$, and the exponent takes one value at the evaluation configurations of every point of $\mathsf{S}_\emptyset$ outside that set.

(iii) Whenever a later term has an exponential representation with one null set, the density, support, and admission conclusions of [QCDM1, Lem. 10.7] apply to that representation.

(iv) Define the scalar function

$$\varsigma^{s,\lambda}(U) = \mathrm{Ber}\big(G_{s,\lambda}(U) F_s(U)\big).$$

Then, for every $h \in \mathcal{H}_k[t_1]$,

$$\mathrm{Ber}\, \check{\tau}^{\mathrm{F}}_h[s,\lambda] = \check{\tau}_h[\varsigma^{s,\lambda}], \qquad \mathrm{Ber}\, \check{\tau}^{\mathrm{F}}_h[s,\lambda](\mathsf{X}^{\mathrm{fr}}_h) = \int \mathfrak{w}_h \varsigma^{s,\lambda} \, dU.$$

The finite sum of these masses, denoted by $Z_k(z,\lambda)$, is holomorphic on $\mathcal{O} \times \mathbb{C}$. At zero interpolation parameter,

$$\varsigma^{s,0}(U) = e^{-A(\beta - a_K^{-1} w, U)} \det(D[U] + m_0 - a_K \widehat{J}).$$

■

Proof. We first choose N_0. The ordinary source induction and its density lemma [QCDM1, Lem. 10.7 and Prop. 10.13] give the base density at zero source off one null set. On $\mathsf{S}_\emptyset$ it has the form ce^G, with $c \neq 0$, so its scalar coefficient ce^{G_0} is nonzero. The same lemma gives a Borel set $S' \subset \xi(\{\mathfrak{w}_{h_\emptyset} \neq 0\})$ outside which the density has integral norm zero. The two coordinate measures agree by Lem. C.5. Hence $\mathsf{S}_\emptyset \setminus S'$ is Haar-null. Adjoin this set and a Borel null set containing the exceptional set for the zero-source evaluation domains. The resulting N_0 has all the required support properties.

At a remaining point, all earlier pairs are regular, since $\Lambda_j = T_\ell$ for $j \leq t_1$. The arrays are therefore evaluated on their fixed cores. The sourced fermion tower also exists, since

$$L^{t_1} \|a_K J\|_\infty \leq 2\varrho_J \leq r_J/2 \leq r_M.$$

Two evaluation configurations differ by a gauge transformation equal to one at the retained places. The action is gauge invariant plaquette by plaquette. The matrix $\delta\sigma$ commutes with colour, so its mass term is jointly gauge invariant; the arrays have the same property. Thus $\Delta(s)$ is independent of the evaluation configuration.

The Borel evaluation construction in the ordinary density induction applies to these terms: the action is continuous in the configuration, the mass coefficients are fixed functions of the sites, and the array coefficients are Borel on the same cores. At a fixed coordinate point, the background maps are holomorphic in $-a_K J$; every coefficient of the exponent is a finite sum of products of holomorphic functions of z. The individual array bounds following Def. C.6, the uniform background bounds, and finiteness of the lattice give uniform coefficient bounds on compact parameter sets. The plaquette terms are bounded on the compact configuration space, and only finitely many values of w^{Δ}, $\delta\sigma$, and $\mathfrak{n}$ occur.

For an even Grassmann element H, $e^H = e^{H_0}\sum_{j\ge 0}(H-H_0)^j/j!$, with a finite sum. This proves the asserted properties of g and G. The partial kernels have bounded Borel coefficients, and the bare sourced density has continuous coefficients on the compact configuration space. Products and Berezin integration are finite operations on coefficients, which proves the bounds for $\varsigma_h^{s,\lambda}$. Integrating over the preimage of a Borel set preserves holomorphy by these compact-parameter bounds, Fubini, and Morera's theorem. At $\lambda = 0$, $G = 1$, giving the ordinary sourced term.

At the base the partial map in eq. C.v is the identity. For bounded Borel g, pushforward commutes with multiplication by $g \circ \xi$, coefficient by coefficient. Consequently $\hat{\tau}^{\mathrm{F}}_{h\emptyset}[s,\lambda] = g_{s,\lambda}\hat{\tau}^{\mathrm{F}}_{h\emptyset}[s]$. Insert the ordinary exponential representation. Its density vanishes off $\mathsf{S}_\emptyset$, and the even Grassmann algebra is commutative. This gives eq. C.vi, after enlarging N_s by a Borel null set for the ordinary one-value assertion. Multiplication by bounded g preserves integrability.

For (iii), the proof of the cited density lemma uses of a coefficient density f only its boundedness. Indeed, if ρ' is the strictly positive zero-source bare density, then on a null set E for its positive history measure,

$$|\check{\tau}_h[f]|(E) \le \frac{\sup|f|}{\min\rho'}\check{\tau}_h[\rho'](E) = 0.$$

The coordinate and history hypotheses are unchanged. This proves the same support and admission conclusions for each bounded coefficient of the perturbed term.

Finally integrate the generators of the highest level first. Each normalized partial kernel integrates to one; its shifted generators do not change that integral. Iterating through the later levels gives $\mathrm{Ber}\,\varsigma_h^{s,\lambda} = \varsigma^{s,\lambda}$, independently of h. Berezin integration commutes with the coefficientwise pushforward. This proves (iv), including holomorphy of the finite sum. At $\lambda = 0$, the same normalization reduces the full Berezin integral to the stated determinant. □

The equality $Z_k = Z_{t_1}$ for every complex λ requires an additional point: the exceptional set used to define g need not itself be gauge invariant. We prove the required almost-everywhere invariance next.

C.5. **Gauge invariance and the sum over histories.**

Lemma C.8. Use the notation of Lem. C.7. For a bare gauge transformation v, let α_v denote its action on the Grassmann algebra. Then $F_s(U^v) = \alpha_v F_s(U)$ at every configuration. If $V, V^u \in \mathsf{S}_\emptyset \setminus \mathsf{N}_0$, then

$$\Delta(s)(V^u) = \alpha_u\Delta(s)(V), \qquad g_{s,\lambda}(V^u) = \alpha_u g_{s,\lambda}(V).$$

The set

$$\{U : \mathfrak{w}_{h_\emptyset}(U) \ne 0,\ \xi(U) \notin \mathsf{S}_\emptyset \setminus \mathsf{N}_0\}$$

is Haar-null. Consequently, for every gauge transformation v at a level $n \ge t_1$,

$$\mathfrak{w}_{h_\emptyset}(U)\big[\varsigma^{s,\lambda}(U^{\tilde{v}}) - \varsigma^{s,\lambda}(U)\big] = 0 \quad \text{for almost every } U. \tag{C.vii}$$

This assertion holds for every $(z,\lambda) \in \mathcal{O}\times\mathbb{C}$. ■

Proof. The bare Dirac operator transforms by conjugation, and $\widehat{J}$ commutes with colour. The gauge action is invariant. Each partial fermion kernel is jointly gauge invariant, block by block, because the block average is covariant. Berezin integration commutes with the gauge action: its barred and

unbarred changes of variables are contragredient. Applying these facts through the base proves the assertion for F_s.

By Lem. C.5, an evaluation configuration U at V gives $\mathsf{U}^{\tilde{u}}$ at V^u. The background maps satisfy

$$\mathsf{H}_{i,t_1}[U^v] = v|_i\, \mathsf{H}_{i,t_1}[U](v|_{t_1})^{-1}, \qquad \bar{\mathsf{H}}_{t_1,i}[U^v] = v|_{t_1}\, \bar{\mathsf{H}}_{t_1,i}[U](v|_i)^{-1}.$$

These identities follow along the tower from covariance of the Dirac operator and the block averages, with the same fine mass matrix. If sb_{U} substitutes the backgrounds at U, they imply

$$\mathsf{sb}_{\mathsf{U}^{\tilde{u}}} \circ \alpha_{\tilde{u}} = \alpha_u \circ \mathsf{sb}_{\mathsf{U}}.$$

Apply this identity to the jointly invariant arrays and to the mass term, whose matrix commutes with colour. The action and scalar normalization are invariant separately. This proves the assertion for Δ, and hence for its exponential.

Let $E = \mathcal{V}_{t_1} \setminus (\mathsf{S}_\emptyset \setminus \mathsf{N}_0)$. The base density at zero source vanishes off $\mathsf{S}_\emptyset$, and $E \cap \mathsf{S}_\emptyset$ is Haar-null. Every coefficient of its measure therefore vanishes on E. Taking the full Berezin integral gives

$$\int_{\xi^{-1}(E)} \mathfrak{w}_{h_\emptyset}(U)\rho'(U)\, dU = 0, \qquad \rho'(U) = e^{-\beta A(U)} \det(D[U] + m_0).$$

At positive bare mass, ρ' has a strictly positive lower bound on the compact configuration space. Since $\mathfrak{w}_{h_\emptyset} \geq 0$, the bad preimage in the support of that weight is null.

Now fix v. The map $U \mapsto U^{\tilde{v}}$ preserves Haar measure and the support of the base weight. Thus almost every U in this support and its transform both have coordinates outside the bad set. At these points the first two parts of the proof give

$$G_{s,\lambda}(U^{\tilde{v}})F_s(U^{\tilde{v}}) = \alpha_{\tilde{v}|_{t_1}}\big(G_{s,\lambda}(U)F_s(U)\big).$$

The full Berezin integral is unchanged by this action. Off the support the weight is zero. This proves eq. C.vii. □

Lemma C.9 (One step under almost-everywhere invariance)**.** Let $t_1 < k \leq K$, $h' \in \mathcal{H}_{k-1}[t_1]$, and let f be a bounded complex Borel function on $\mathcal{V}_0$. Suppose that

$$\mathfrak{w}_{h_\emptyset}(U)[f(U^{\tilde{v}}) - f(U)] = 0 \quad \text{for almost every } U$$

for every gauge transformation v at level $k-1$. Then the gauge defect vanishes and

$$\sum_{h:\, h|_{k-1} = h'} \tau_h[f] = \mathbf{T}\tau_{h'}[f], \qquad \mathbf{T} = \mathcal{M}_*.$$

If the hypothesis holds at each level above t_1, then

$$\sum_{h \in \mathcal{H}_k[t_1]} \tau_h[f] = \mathbf{T}^{k-t_1}\tau_{h_\emptyset}[f], \qquad \sum_{h \in \mathcal{H}_k[t_1]} \int \mathfrak{w}_h f\, dU = \int \mathfrak{w}_{h_\emptyset} f\, dU.$$

■

Proof. We check the use of invariance in the history decomposition of [YM3, Prop. 4.1], and its proof. The calibrated thresholds obey $\varepsilon_j \leq c_2^{\mathrm{SO}}$. Indeed, $\varepsilon(t) = A_0 t^{-1/2}(\log t)^{\mathfrak{p}_0}$ is decreasing for $\log t \geq 2\mathfrak{p}_0$, and the coupling ceilings give $\log \gamma^{-2} \geq 2\mathfrak{p}_0$ and $\varepsilon(\gamma^{-2}) \leq c_2^{\mathrm{SO}}$. All calibrations satisfy $t_j \geq \gamma^{-2}$. At these thresholds the gauge defect is zero for every bounded Borel density; it does not require invariance.

Fix a plaquette outcome p in the step. The factor

$$e_{\mathsf{p}}(U) = \mathfrak{w}_{h'}(U)\mathbf{1}_k^{\mathrm{P}}(\mathsf{p})(\Theta_k(U))$$

is invariant under the gauge variables integrated in that outcome. The earlier weight factors as $\mathfrak{w}_{h'} = \mathfrak{w}_{h_\emptyset} w'$ with bounded Borel w'. Hence, for each fixed such gauge transformation v,

$$e_{\mathsf{p}}(U)[f(U^{\tilde{v}}) - f(U)] = 0 \quad \text{for almost every } U.$$

In the gauge-averaging proof of the decomposition, make the Haar-preserving substitution $U \mapsto U^{\tilde{v}}$ for this fixed v. The displayed equality replaces the pointwise invariance of the density. The resulting integrand is bounded and jointly Borel in (U, v). Integrate the equality in v and apply Fubini; no common exceptional set for all v is needed. The normalized gauge-fixing integral and the partition by the remaining test outcomes now give exactly the one-step identity. Summing over all parents and iterating gives the measure identity. Pushforward preserves total mass, which gives its final assertion. □

Proposition C.10 (Preservation of the total integral)**.** For the data of Def. C.6, every $(z, \lambda) \in \mathcal{O} \times \mathbb{C}$, and every $t_1 \le k \le K$,

$$\sum_{h \in \mathcal{H}_k[t_1]} \tau_h[\varsigma^{s,\lambda}] = \mathbf{T}^{k-t_1} \tau_{h_\emptyset}[\varsigma^{s,\lambda}].$$

In particular,

$$\begin{aligned} Z_k(z,\lambda) &= \int \mathfrak{w}_{h_\emptyset} \varsigma^{s,\lambda}\, dU \\ &= \mathrm{Ber} \int_{\mathcal{V}_{t_1} \setminus (\mathsf{N}_s \cup \mathsf{N}_0)} \mathsf{Stay}_{h_\emptyset} \mathbf{1}^{\mathrm{T}}_{h_\emptyset} \chi_{t_1}(T_\ell) e^{\mathbf{\Phi}^w_{h_\emptyset}} \\ &\qquad \cdot e^{A^{\mathrm{F}}_{t_1,h_\emptyset}(s) + \lambda \Delta(s)}\, d\mathscr{M}_{h_\emptyset} = Z_{t_1}(z,\lambda). \end{aligned} \tag{C.viii}$$

■

Proof. The scalar density is bounded and Borel by Lem. C.7, and satisfies the hypothesis of Lem. C.9 at every later level by Lem. C.8. Apply the former lemma and then the base exponential representation eq. C.vi. Berezin integration commutes with this integrable coefficientwise representation. □

C.6. **Densities holomorphic in both parameters.** Holomorphy of a measure on every Borel set does not yet select densities holomorphic at every coordinate point. The positive zero-source measure provides a common dominating measure and allows such a selection.

Proposition C.11. For each $h \in \mathcal{H}_k[t_1]$, there is a Borel function $g_h(z, \lambda, v)$ on $\mathcal{O} \times \mathbb{C} \times \mathsf{X}^{\mathrm{fr}}_h$, with values in the even Grassmann algebra of h, such that every coefficient is holomorphic in (z, λ) at every v, and

$$\check{\tau}^{\mathrm{F}}_h[s(z), \lambda] = g_h(z, \lambda, \cdot)\, \mathscr{M}^{\mathrm{fr}}_h.$$

Moreover, $\int \|g_h\|_{h_F}\, d\mathscr{M}^{\mathrm{fr}}_h$ is bounded on compact parameter sets. Whenever the term has an exponential representation with one null set, g_h agrees almost everywhere with its density from [QCDM1, Lem. 10.7]. For all parameters,

$$Z_k(z,\lambda) = \sum_{h \in \mathcal{H}_k[t_1]} \mathrm{Ber} \int g_h(z,\lambda,v)\, d\mathscr{M}^{\mathrm{fr}}_h(v) = Z_{t_1}(z,\lambda).$$

■

Proof. Let $\rho' = e^{-\beta A} \det(D + m_0) \ge c' > 0$, as in Lem. C.8. The finite measure $\nu_h = \check{\tau}_h[\rho']$ is positive. The ordinary zero-source density and full Berezin integration give

$$\nu_h = f^0_h\, \mathscr{M}^{\mathrm{fr}}_h, \qquad f^0_h = \mathrm{Ber}\, g^0_h \ge 0.$$

We choose a Borel version of f^0_h, nonnegative everywhere, changing it on a null set if necessary. For a coefficient I, put $\mu_{z,\lambda} = \check{\tau}_h[\varsigma^{s,\lambda}_{h,I}]$. The definition by pushforward and nonnegativity of the history weight imply, for every Borel E,

$$|\mu_{z,\lambda}(E)| \le \frac{\sup_U |\varsigma^{s,\lambda}_{h,I}(U)|}{c'} \nu_h(E).$$

The factor is bounded on compact parameter sets, and $\mu_{z,\lambda}(E)$ is holomorphic by Lem. C.7.

We give the selection argument. Suppose that ν is a finite positive Borel measure on a separable metrizable space X, that $D \subset \mathbb{C}^m$ is open, and that complex measures μ_ζ satisfy the two properties just established: $\mu_\zeta(E)$ is holomorphic for every Borel E, and $|\mu_\zeta(E)| \le C_Q \nu(E)$ for $\zeta \in Q$, for each compact $Q \subset D$. Fix a closed polydisc $\bar P = \{|\zeta_i - c_i| \le r_i\} \subset D$. For a multi-index α, Cauchy's formula defines the complex measure

$$a_\alpha(E) = \frac{1}{(2\pi i)^m} \oint_{\partial_0 P} \mu_\zeta(E) \prod_i (\zeta_i - c_i)^{-\alpha_i - 1}\, d\zeta.$$

Dominated convergence gives countable additivity, and $|a_\alpha| \le C_{\bar P} r^{-\alpha} \nu$. Radon–Nikodym therefore gives Borel coefficients d_α, with $a_\alpha = d_\alpha \nu$ and $|d_\alpha| \le C_{\bar P} r^{-\alpha}$ everywhere after null-set changes. There are only countably many coefficients. The series

$$g_P(\zeta, x) = \sum_\alpha d_\alpha(x)(\zeta - c)^\alpha$$

converges absolutely and uniformly in x on every smaller concentric closed polydisc. It is jointly Borel, holomorphic in ζ at every x, and bounded there. Integration term by term gives $\int_E g_P(\zeta, x)\, d\nu(x) = \mu_\zeta(E)$.

Cover D by countably many polydiscs of half the radii of polydiscs P_i whose closures lie in D. On an overlap $P_i \cap P_j$, the densities g_{P_i} and g_{P_j} agree ν-almost everywhere at each point of a countable dense subset. Outside a single Borel null set for this pair they agree everywhere on the overlap, by continuity in ζ. Take the union of these null sets over all pairs. Off this union the functions glue to $\hat g$; set $\hat g = 0$ on the union. This defines a jointly Borel density, holomorphic at every x. Every compact subset of D is covered by finitely many of the smaller polydiscs, so $\hat g$ is uniformly bounded there.

Apply this construction with $D = \mathcal{O} \times \mathbb{C}$, $X = \mathsf{X}_h^{\mathrm{fr}}$, and $\nu = \nu_h$, separately to each of the finitely many coefficients. Set $(g_h)_I = \hat g_I f_h^0$. The integral bounds follow from finiteness of ν_h and $\|F\|_{h_F} \le \sum_I h_F^{|I|} |F_I|$. Uniqueness of densities gives agreement with any exponential representation. Finally sum the integrals and apply Prop. C.10. □

C.7. **The later steps depend only on the base measure.** The restart comparison identifies two measures at the base. To propagate that equality, we must check both the later integrands and the domains of their coordinate maps.

Lemma C.12. Let $h \in \mathcal{H}_k[t_1]$ and $\xi = \mathcal{M}^{t_1}$. For $V \in \mathcal{V}_{t_1}$, form $\Theta_t^\circ(V)$ by setting its level-j member equal to $\mathcal{M}^{j-t_1} V$ for $j \ge t_1$, and to one for $j < t_1$. Then the following hold.

(i) The history weight factors as

$$\mathfrak{w}_h = \mathfrak{w}_{h_\emptyset}(\mathfrak{w}_h^\circ \circ \xi), \qquad \mathfrak{w}_h^\circ(V) = \prod_{t_1 < t \le k} \mathbf{1}_t(\mathsf{o}_t)(\Theta_t^\circ(V)) \mathfrak{G}_{t-1}(\mathcal{M}^{t-1-t_1} V; P_t''(h)).$$

The function $\mathfrak{w}_h^\circ$ is bounded and Borel.

(ii) The later fermion integration is $T^{\mathrm{F}}_{h;(t_1,k]}[U] = T_h^\circ[\xi(U)]$, where $T_h^\circ[V]$ is the composition of the same partial kernels with $\mathcal{M}^{n-t_1} V$ in place of $\mathcal{M}^n U$. Its matrix coefficients $T_h^\circ[V]_{II'}$ are bounded Borel functions of V.

(iii) The free-coordinate map factors, with equality of domains, as

$$\operatorname{dom} \mathsf{v}_h = \xi^{-1}(\operatorname{dom} \mathsf{v}_h^\circ), \qquad \mathsf{v}_h = \mathsf{v}_h^\circ \circ \xi, \tag{C.ix}$$

where the Borel partial map v_h° is

$$V \longmapsto \left(\left((\mathcal{M}^{l-t_1} V)|_{\Gamma_l^{\mathrm{le}}(h)}\right)_{t_1 \le l \le k}, \left(g_l^{-1} \mathcal{A}_l'^{\diamond}(b)(\Theta_{l+1}^\circ(V))\right)_{t_1 \le l < k,\ b \in \mathfrak{X}_l^{\mathrm{E}+}(h)} \right).$$

Here $\Gamma_l^{\mathrm{le}}(h)$ is the set of lettered datum bonds, $\mathfrak{X}_l^{\mathrm{E}+}(h)$ consists of the positively oriented external bonds, and $\mathcal{A}_l'^{\diamond}$ is the new fluctuation coordinate, scaled by g_l.

(iv) For a finite Grassmann-valued measure ρ on $\mathcal{V}_{t_1}$, define

$$(\mathsf{St}_h\rho)_I(E) = \sum_{I'} \int_{(\mathsf{v}_h^\circ)^{-1}(E)} \mathfrak{w}_h^\circ(V) T_h^\circ[V]_{II'}\, d\rho_{I'}(V). \tag{C.x}$$

If $\rho = \xi_*\mu$, this is the image under v_h of $(\mathfrak{w}_h^\circ \circ \xi) T_h^\circ[\xi(\cdot)]\mu$. It is therefore the same for every μ with this pushforward. In particular,

$$\mathsf{St}_h \hat{\tau}^{\mathrm{F}}_{h_\emptyset}[s,\lambda] = \check{\tau}^{\mathrm{F}}_h[s,\lambda] \quad ((z,\lambda) \in \mathcal{O} \times \mathbb{C}).$$

■

Proof. The determining bonds at levels $j < t_1$ are absent: $\Omega_{j+1} = T_\ell$. A later test depends only on the fields at its two top levels and on the restrictions to the determining bonds. The families $\Theta_t(U)$ and $\Theta_t^\circ(\xi(U))$ agree on this data. Thus they give the same test outcome. The gauge-fixing factors after t_1 already depend only on $\mathcal{M}^{t-1}U = \mathcal{M}^{t-1-t_1}\xi(U)$. This proves (i); all its factors are bounded and Borel. The same identity for the averaged configuration proves (ii), since each partial kernel has bounded Borel coefficients and each partial Berezin integral is a finite operation on coefficients.

For (iii), the coordinate factors below t_1 are one-point spaces: both their datum bonds and their external bonds are empty. The retained group coordinates at the other levels plainly factor through ξ. We verify the domains of the fluctuation coordinates. The completed collared sequence through the base is constant, so its datum bonds below t_1 are also absent. Hence the completed datum of a later step is the same for $\Theta_{l+1}(U)$ and $\Theta^\circ_{l+1}(\xi(U))$.

On a strip bond in either shift zone, the new fluctuation is defined by

$$\mathcal{A}_l'^{\diamond}(b) = \frac{1}{i}\log W_l'^{\diamond}(b), \qquad W_l'^{\diamond}(b) = (\Psi_b^\diamond)^{-1}\big(V_l(b) V_l^{\mathrm{bg,loc}\diamond}(b)^{-1}\big).$$

Its domain requires the shift to exist, the resulting bond map $\Psi_b^\diamond$ to be a bijection, and its inverse value to differ from -1. The shift, the bond map, and the local background, including their domains, depend only on the completed datum. Outside these zones, the coordinate is

$$\frac{1}{i}\log\big(V_l(b) V_l^{\mathrm{bg}\diamond}(b)^{-1}\big),$$

whose domain requires the stable background to exist and the logarithm's argument to differ from -1. Again the background and its domain are determined by the completed datum. The two families have this same datum and the same $V_l(b)$. Each fluctuation coordinate is therefore defined for both or for neither, with the same value. This proves eq. C.ix.

Every component is a Borel partial map, being a composition of the original fluctuation map with the Borel map Θ°_{l+1}. Their finite intersection is a Borel domain. Their product is Borel because the coordinate spaces have countable bases.

For (iv), apply the pushforward identity to the bounded Borel function

$$\mathbf{1}_{(\mathsf{v}_h^\circ)^{-1}(E)} \mathfrak{w}_h^\circ T_h^\circ[\cdot]_{II'}.$$

Its preimage under ξ is the corresponding integrand on $\mathsf{v}_h^{-1}(E)$, by (iii). Sum over I'. For the final assertion take $\mu = \mathfrak{w}_{h_\emptyset} G_{s,\lambda} F_s\, dU$ and use (i), (ii), and the definitions of the terms. □

C.8. **Reading the early families.**

Lemma C.13 (A free family in a later step)**.** Let Δ be a free family of level $j \le k < K$, read at a source of $\mathcal{P}$ on the fixed pair domains. Its summands satisfy the old-activity hypotheses of [QCDM1, Lem. 7.9] with size $r^\psi(\Delta)$, including the moved-diagonal condition and its torus version with reference G^∞. Its moved pieces satisfy both weighted piece bounds of [QCDM1, Prop. 10.1, proof], with that size. At lags below n_{lag} it can be counted whole, with weight ω_{k+1-j}.

More explicitly, its cell sum satisfies

$$m_c[\Delta] - m_c[G] = \sum_{z\in c}\Big[\sum_y (A_j^{\mathsf{n}} - A_j)(y,z)\nu + \sum_{X,p}\Big(1 - \mathfrak{s}_j^{-1}\sum_y A_j^{\mathsf{n}}(y,z)\Big)a_{X,p}(z) + \sum_{X,p}{}' a_{X,p}(z)\Big].$$

Here $m_c[G]$ includes $l_\square$, and the last sum consists of pieces omitted by the torus rule. The cell sum $m_c[G]$ vanishes on $\mathbb{Z}^3$; on a torus its remainder is assigned to polymers that wrap around it. At a level-k cube, the displayed difference has total norm at most

$$(c_{\mathrm{cs}} + c_{\mathrm{cs}}^{\Delta})\omega_{k+1-j} r^{\psi}(\Delta).$$

This includes lag zero and remains true when the same members are counted at a cube of the final level K.

The scalar part enters the array of old scalar terms with Wilson coefficient zero. Its bracket localization at a polymer Y is bounded by

$$K_\Lambda \mathsf{s}_k(\alpha_0) L^{-(k-j)} r^0(\Delta) e^{-(1-2\delta_{\mathrm{w}})\kappa d_k(Y)},$$

and in the step's polymer norm by $K_\Lambda^{\mathrm{s}}\mathsf{s}_k(\alpha_0)L^{-(k-j)}r^0(\Delta)$. ■

Proof. The old-activity argument uses the row-sum bound on a quadratic kernel, decay of the localized backgrounds and their source variation, and the piecewise identities

$$u_X + \delta_{yy'} a_X(y) = k_X, \qquad \sum_y \mathrm{ord}_0[k_{X,p}](y) = m_p(k_{X,p}).$$

These identities are linear in (F, G, ν). The definition supplies their common paths, the reference cell cancellation, the phase invariance excluding two unbarred or two barred fields, and the bound $c_{\mathsf{S}} h_F^2 |\nu| \le r^{\psi}(\Delta)$. Thus the proof of [QCDM1, Lem. 7.9], with the source modifications in [QCDM1, Prop. 10.1], applies term by term. In particular, the two weighted piece bounds use only the local background estimates and the sums of the norms of F, G.

To check the cell estimate, add the units, moved pieces, and mass forms. The result is exactly the displayed identity. The term with ν has weighted norm at most $M^3 C_{\mathrm{bg}}^{\mathrm{w}} L^j \|\mathsf{n}\| |\nu| / r_M$. Moving its hull to level k costs at most $C_{\mathrm{hl}} L^{k-j} h_F^2$. Since $L^{k+1}\|\mathsf{n}\| < 2r_M$, its contribution is at most

$$\frac{2 C_{\mathrm{hl}} M^3 C_{\mathrm{bg}}^{\mathrm{w}}}{L c_{\mathsf{S}}} r^{\psi}(\Delta) = c_{\mathrm{cs}}^{\Delta}\omega_1 r^{\psi}(\Delta).$$

For the moved pieces the same computation bounds the contribution by

$$C_{\mathrm{hl}} C_{\mathrm{pw}} \Big[\eta_{\min}^{-2}\Big(\tfrac{2}{L} C_{\mathrm{bg}}^{\mathrm{w}} + C_A C_{\mathrm{dx}}\Big) + \max\{1, C'_{\mathrm{dx}}\}\Big] r^{\psi}(\Delta) = c_{\mathrm{cs}} C_\omega r^{\psi}(\Delta).$$

Here $C_{\mathrm{dx}}, C'_{\mathrm{dx}}$ bound $L^{k-j} e^{-4c_t \ell/\lambda_j}$ and $L^{k-j} e^{-c_t \ell/\lambda_j}$, respectively. Their bounds hold through $k = K$, since $\ell/\lambda_j \ge 2L^{k+1-j}$. At lag zero use the second weighted piece estimate; the omitted pieces are bounded with factor one instead of a tail. The same constant works because $\max\{1, C'_{\mathrm{dx}}\} \ge 1$. This proves the asserted bound at every lag.

Finally the scalar localization is linear in its array. Apply [YM3, Prop. 4.52] to the array with the one entry Δ_0 at level j. Its system ratios are one and its scale factor is at most $\mathsf{s}_k(\alpha_0)$. The coefficient of the Wilson action is zero, so the stated estimate follows directly. □

Lemma C.14 (The four size bounds)**.** Suppose the first two conditions in eq. C.i hold, $|\lambda| < R$, and $t_1 \le n \le K$. Let the later sizes satisfy $\epsilon_j \le (2C_w^{\mathrm{s}} - C_w)\phi_{j-1}$ for $t_1 < j \le n$, and put

$$\epsilon_j[\lambda] = \begin{cases} \epsilon_j^{\mathrm{s},\psi} + |\lambda| r_j^{\psi}, & j \le t_1, \\ \epsilon_j, & j > t_1. \end{cases} \qquad \Phi_n = \sum_{j\le n} \omega_{n+1-j}\phi_{j-1}.$$

Then

$$\sum_{j\le n}\omega_{n+1-j}\epsilon_j[\lambda]\le 2C_w^{\mathrm{s}}\Phi_n\le 2C_w^{\mathrm{s}}C_{\mathrm{pc}}S_{1/4}\phi_n,$$
$$c_{\mathrm{cs}}\sum_{j\le n}\omega_{n+1-j}\epsilon_j[\lambda]+c_{\mathrm{cs}}^{\Delta}|\lambda|\Sigma_n^{\psi}\le 2c_{\mathrm{cs}}C_w^{\mathrm{s}}\Phi_n,$$
$$\epsilon_j[\lambda]\le 2C_w^{\mathrm{s}}\phi_{n-1}\quad(j\le t_1),$$
$$|\lambda|\Sigma_n^0\le c_0^{\Delta}.$$

The early size bounds include the pointed terms, the reference, and the infinite-lattice reference family. ■

Proof. Let A be the weighted sum at $\lambda=0$ and $X=|\lambda|\Sigma_n^{\psi}$. By Lem. C.4, $A\le(2C_w^{\mathrm{s}}-C_w)\Phi_n$. Since $\Phi_n\ge\omega_1\phi_{n-1}$ and $c_{\psi}^{\Delta}=C_w\omega_1c_{\mathrm{cs}}/(c_{\mathrm{cs}}+c_{\mathrm{cs}}^{\Delta})$,

$$X\le C_w\frac{c_{\mathrm{cs}}}{c_{\mathrm{cs}}+c_{\mathrm{cs}}^{\Delta}}\Phi_n.$$

This proves the first two bounds, including the cell contribution. For an individual early level use $|\lambda|r_j^{\psi}\le X/\omega_1\le C_w\phi_{n-1}$ and $\phi_{j-1}\le\phi_{n-1}$. The scalar bound follows from $\Sigma_n^0=L^{-(n-t_1)}\Sigma_{t_1}^0$. □

Proposition C.15 (The step with early activities)**.** Assume the first three conditions in eq. C.i at radius R. For $t_1\le k<K$ let

$$\mathscr{D}_k'=\{(\sigma_i)_{t_1\le i\le k}:\|\sigma_i\|<B_\sigma\phi_i\}.$$

The small-field steps from the base are defined on $2\mathscr{D}_k'\times\mathrm{P}(R)$ and have the same local form, fixed pair domains, reference at zero sources, and pointed outputs as [QCDM1, Prop. 10.1]. They are jointly holomorphic in the coordinates, parameter, and source wherever the latter varies in an open set satisfying the lag conditions. They depend only on coordinates in their polymer, are jointly gauge invariant, and preserve the scalar phase symmetry.

Put $\tau_k^{\lambda}=\max\{\tau_k(w;\alpha_0),\tau_k(w_\lambda;\alpha_0)\}$. The new activities, including their pointed terms, obey

$$\|\mathcal{V}_0\|\le c_{\mathrm{cl}}/2,\qquad n^{\mathrm{int}}+n'\le w_*/2,\qquad \|\mathcal{V}_\psi\|\le b_k^{\mathrm{blk,s}}\le c_{\mathrm{cl}}/8,$$
$$\epsilon_{k+1}^{\mathrm{s},\psi}\le(2C_w^{\mathrm{s}}-C_w)\phi_k,\qquad \epsilon_{k+1}^{\mathrm{s},0}\le\tfrac{15}{8}C_w^{\mathrm{s}}\phi_k+\tfrac{3}{5}C_{\mathrm{lin}}'\tau_k^{\lambda}.$$

Their pointed scalar activities have the ordinary bound E_0^{s} at rate $\kappa+1$. The mass increment is defined by the same moments relative to the zero-source reference:

$$\nu_{k+1}(y;\mathfrak{c},\lambda)=\mu_{k+1}-\mathfrak{s}_{k+1}^{-1}\sum_{X,p}a_{X,p}(y).$$

It takes values in the colour-scalar matrix slot $\mathbb{M}$ and satisfies

$$|\nu_{k+1}-\mu_{k+1}|\le\tfrac12B_\nu\phi_{k+1},\qquad|\nu_{k+1}(y;\mathfrak{c},\lambda)-\nu_{k+1}(y;\mathfrak{c}',\lambda)|\le B_\nu\phi_{k+1}\|\mathfrak{c}-\mathfrak{c}'\|_{k,y}$$

for $\mathfrak{c}\in\mathscr{D}_k'$ and $\mathfrak{c}'\in2\mathscr{D}_k'$. The first bound and the matrix slot remain valid when any subset of the pieces is used. The source energies satisfy

$$|e_{k+1}(X;\mathfrak{c},\lambda)|\le C_{\mathfrak{E}}e^{-\kappa d_{k+1}(X)}.$$

At $\lambda=0$ and the ordinary source coordinates, every output is the corresponding output of the ordinary flow. ■

Proof. Use the one-step proof of [QCDM1, Prop. 10.1]. We verify the properties and numerical estimates changed by the early families. Their pointed scalar terms are local holomorphic functions on the same cores, while Lem. C.13 supplies the precise old-activity hypotheses for their fermion terms. They are affine in λ and independent of the later mass coordinates. Hence the Gaussian

integral, localization, and cluster series remain holomorphic, with locally uniform convergence once the following bounds are established.

The scalar potential has the ordinary gauge and explicit terms of size at most $c_{\mathrm{cl}}/4$, the linear term of size at most $c_{\mathrm{cl}}/8$, and the old scalar array. The latter has the old bound $13c_{\mathrm{cl}}/144$, plus

$$K_\Lambda^{\mathrm{s}}\mathsf{s}_k(\alpha_0)|\lambda|\Sigma_k^0 \le \frac{c_0^\Delta}{12K_M^\bullet} \le \frac{5}{144}c_{\mathrm{cl}}.$$

Their sum is $c_{\mathrm{cl}}/2$ and also proves the Gaussian admission $n^{\mathrm{int}} + n' \le w_*/2$.

For the fermion terms the new cell contribution is

$$\Sigma_k^{\mathrm{cs}}[\lambda] \le C_0^{\mathrm{cs}}h_F^2\phi_k + c_{\mathrm{cs}}\sum_{i\le k}\omega_{k+1-i}\epsilon_i[\lambda] + c_{\mathrm{cs}}^\Delta|\lambda|\Sigma_k^\psi.$$

Thus the part of the one-step bound depending on old activities is controlled by

$$C_0^{\mathrm{cs}}h_F^2\phi_k + (1+c_{\mathrm{cs}})\sum_{i\le k}\omega_{k+1-i}\epsilon_i[\lambda] + c_{\mathrm{cs}}^\Delta|\lambda|\Sigma_k^\psi \le [C_0^{\mathrm{cs}}h_F^2 + 2(1+c_{\mathrm{cs}})C_w^{\mathrm{s}}C_{\mathrm{pc}}S_{1/4}]\phi_k.$$

This is the ordinary bound. The new fermion size therefore obeys

$$\begin{aligned}\epsilon_{k+1}^{\mathrm{s},\psi} \le{}& C_w^\psi\phi_k + 4\theta_E^{\mathrm{s}}\sum_{i\le k}\omega_{k+1-i}[(1+c_{\mathrm{cs}})\epsilon_i[\lambda] + c_{\mathsf{S}}h_F^2(\bar{B}_M + 2B_\sigma)\phi_i]\\ &+ 4\theta_E^{\mathrm{s}}c_{\mathrm{cs}}^\Delta|\lambda|\Sigma_k^\psi.\end{aligned}$$

Apply Lem. C.14 and the calculation in Lem. C.4. This gives the claimed fermion size and $b_k^{\mathrm{blk,s}}$, also for the reference and pointed terms.

In the new scalar fluctuation family the additional variation has size at most

$$2K_M^\bullet K_\Lambda^{\mathrm{s}}\sqrt{2}\,\mathsf{C}_{\mathcal{A}}A_1\alpha_0^{-1}\phi_k c_0^\Delta \le \tfrac{1}{8}C_w^{\mathrm{s}}\phi_k.$$

The other five terms of the ordinary proof retain their estimates. They give, respectively, a combined $C_w^{\mathrm{s}}\phi_k$, the old-fermion contribution $C_w^{\mathrm{s}}\phi_k/2$, and the ordinary scalar-array contribution $C_w^{\mathrm{s}}\phi_k/4$. The linear terms contribute $(1/2+1/10)C'_{\mathrm{lin}}\tau_k^\lambda$. Adding these terms gives $15/8$ and $3/5$ as stated. The same calculation with $K_M^\bullet$ proves the pointed bounds. The ordinary small-coupling ceiling then bounds each scalar pointer by E_0^{s}.

Every operation commutes with the scalar phase symmetry: the gauge vertices contain no generators; the fermion Gaussian, its normalized Berezin integral, the background maps, and all mass forms preserve the contragredient pair. Pointing counts generators and also commutes with this symmetry. The new quadratic terms therefore still have one barred and one unbarred field. At the trivial configuration, constant colour rotations fix the pair and commute with the fine mass and background walks. Each piece's moment commutes with their irreducible action on colour, so it lies in $\mathbb{M}$. This proves the assertion even for a subset of pieces.

The sum of absolute moment bounds gives $|\nu_{k+1} - \mu_{k+1}| \le c_{\mathrm{inc}}h_F^{-2}2C_w^{\mathrm{s}}\phi_k \le B_\nu\phi_{k+1}/2$. For locality, a piece bounded by M_0 on $2\mathscr{D}'_k$ reads only coordinates in its polymer X. Put $T = \max_{i,x\in X}\|\sigma_i'(x) - \sigma_i(x)\|/(B_\sigma\phi_i)$. Schwarz's lemma on the line from $\mathfrak{c}$ in the direction of this restricted difference gives $2M_0T$ if $T \le 1$; the bound $2M_0$ suffices otherwise. Summing the pieces with their localized walks is exactly the ordinary locality count and yields the stated B_ν.

Finally the scalar value at the trivial pair is bounded by $K_M c_{\mathrm{cl}}$, the Gaussian source term by one per pointer, and the fermion normalizer by its ordinary source bound. Their difference from the fixed zero-source reference gives $C_{\mathfrak{C}}e^{-\kappa d(X)}$. All properties required at the next level have now been proved. Induction starts at $k = t_1$, where there are no later activities. At $\lambda = 0$ the input agrees with the ordinary flow, so the same construction gives identical outputs. □

C.9. **Chains and histories at the parameter.**

Lemma C.16 (The shrinking parameter window)**.** Assume all four conditions in eq. C.i. Starting with the mass field of Def. C.3, define

$$\sigma_{k+1}(s,\lambda)(x) = L\sigma_k(s,\lambda)(x) + \nu_{k+1}(y_{k+1}(x);\mathfrak{c}_k(s,\lambda),\lambda) - \mu_{k+1}.$$

For $t_1 \le k \le K$ this chain is holomorphic on $\mathrm{P}(R_k)$ and satisfies

$$\|\sigma_k(s,\lambda) - \sigma_k(s)\| \le \tfrac{1}{2}B_\sigma\phi_k, \qquad \|\sigma_k(s,\lambda)\| \le (B_\sigma - C_{\mathrm{var}}/2)\phi_k.$$

It equals the ordinary source chain at $\lambda = 0$ and is jointly holomorphic in the source on every open set on which the lag conditions hold at the same radius. ■

Proof. Set $D_k(\lambda) = \sigma_k(s,\lambda) - \sigma_k(s)$. The initial bound is the mass condition. If it holds through k, then Lem. C.4 puts every preceding coordinate in $\mathscr{D}'_k$ on $\mathrm{P}(R_k)$. The difference of the two increments is holomorphic there, zero at zero, and bounded by $B_\nu\phi_{k+1}$, by Prop. C.15. Schwarz's lemma on each line in the polydisc bounds a holomorphic norm-valued function that vanishes at zero by its supremum times $|\lambda|/R_k$. On the smaller disc this gives

$$\|D_{k+1}\| \le Q_\sigma^{-1}(LB_\sigma\phi_k/2 + B_\nu\phi_{k+1}) \le \tfrac{1}{2}B_\sigma\phi_{k+1},$$

because $L\phi_k \le 2\sqrt{L}\phi_{k+1}$, $B_\nu \le B_\sigma/8$, and $Q_\sigma = 2\sqrt{L} + 1/4$. Composition proves holomorphy and closes the induction. □

Lemma C.17 (The chain along a history)**.** For $h \in \mathcal{H}_k[t_1]$ and $|\lambda| < R_k$, start its chain at $\sigma_{t_1}(s,\lambda)$. At each later step use only the pieces whose polymer Z_p lies in Λ_j to form ν_j^h, and set

$$\sigma_j^h = L\sigma_{j-1}^h + \nu_j^h - \mu_j \quad \text{on } \Omega_j.$$

Off Ω_j use the ordinary chain at the parameter. Let $S_j = \bigcup_{t_1<i\le j}\tilde{Z}_i(h)$. With the ordinary constants $R'_\flat, C'_{\mathfrak{E}}$,

$$|\sigma_j^h - \sigma_j|(x) \le 8B_\nu\phi_j e^{-\kappa_\sigma \operatorname{dist}(x,S_j)/(2L^j)} \qquad (\operatorname{dist}(x,S_j) \ge R'_\flat L^j),$$
$$\sup_{\Omega_j}|\sigma_j^h - \sigma_j| \le 16B_\nu\phi_j, \qquad \sup_{\Omega_j}|\sigma_j^h| \le 2B_\sigma\phi_j,$$
$$\sum_{X\subset\Lambda_n}|e_n(X;\mathfrak{c}_{n-1},\lambda) - e_n(X;\mathfrak{c}_{n-1}^h,\lambda)| \le C'_{\mathfrak{E}}\#_n(\tilde{Z}_n).$$

The coordinates of h are jointly holomorphic and belong to $2\mathscr{D}'_j$. ■

Proof. The two chains agree at the base. The localized coordinate distance $a_j(t)$ in the ordinary history-chain proof therefore starts at zero and satisfies

$$a_j(t) \le La_{j-1}(t) + B_\nu\phi_j\left[C_{\mathrm{loc}}e^{-\kappa_\sigma t/L^j} + e^{\kappa_\sigma\sqrt{3}}\sup_{v\ge 0}e^{-\kappa_\sigma v/L^j}\max_{t_1\le i<j}\frac{a_i(t-v)}{B_\sigma\phi_i}\right].$$

This follows from the local bound in Prop. C.15 and the rule discarding the pieces outside Λ_j. The ordinary induction uses only $Le^{-\kappa_\sigma R'_\flat} \le 1/4$, $(C_{\mathrm{loc}} + 3e^{2\kappa_\sigma})e^{-\kappa_\sigma R'_\flat/4} \le 2$, and $B_\nu \le B_\sigma/6$. All remain unchanged. It gives the first two bounds, and $16B_\nu \le B_\sigma$ keeps the history coordinates in $2\mathscr{D}'_j$. The source-energy estimate follows by the same exponentially weighted polymer count, using $|e_n(X)| \le C_{\mathfrak{E}}e^{-\kappa d_n(X)}$ and the first bound. These are exactly the inputs of the history-chain proof in [QCDM1, §10, following Lem. 10.4]. □

Lemma C.18 (Regular sums and history frames)**.** Under these hypotheses the regular activities, including all early families, have total norm at most $C^{\mathrm{s}}_{\mathrm{leg}}\Sigma_{\mathcal{L}}(h)$. Their contribution to a cube of a nonempty layer $\mathcal{L}_n$ is at most $C^{\square,\mathrm{s}}_{\mathrm{leg}}\phi_n$. The mass member

$$\mathsf{M}_h(s,\lambda) = -\mathsf{S}_1((s_h+\sigma_h(s,\lambda))\psi^{\mathrm{bg}}), \qquad \sigma_h = L^{-t_1}\sigma_{t_1} + \sum_{t_1<j\le k} L^{-j}(\nu_j^h - \mu_j)\varphi_j$$

has the same ordinary mass bounds, namely the per-cube bound $w_0\nu C_H^2 h_F^2 L(\bar{B}_M + 2B_\sigma)\phi_{n-1}$ and its ordinary sum over layers. The scalar regular sum, together with the absolute Wilson extraction terms, is at most $C_5^{\mathrm{s}}\Sigma_{\mathcal{L}}(h)$.

Suppose the history representation through level k has these bounds and the ordinary bounds on all its later boundary terms and edge terms. Then its next frame has the same admission bounds as the ordinary source frame. Its regular scalar class has size at most $\varepsilon_\flat$. The first seven fermion classes have total size at most $b_k^{\mathrm{tot,s}}$, and the full class obeys

$$\mathsf{sz}^\natural_{k+1} \le \frac{13}{64}\max\left\{\sup_{j\le k}\mathsf{sz}^\natural_j, \sup_{j\le k}\bar{v}_j^{\mathrm{s}}\right\} + b_k^{\mathrm{tot,s}}.$$

For $j \le t_1$ the sizes here are those of the original empty history at the source s; for $j > t_1$ they are those of the new steps. Thus the recursion closes at

$$\mathsf{sz}^\natural_{k+1} \le \tfrac{64}{51} b^{\mathrm{tot,s}}(\gamma) \le \tfrac{16}{17}\varepsilon_\flat, \qquad \bar{v}^{\mathrm{s}}_{k+1} \le b^{\mathrm{blk,s}}(\gamma) \le c_{\mathrm{cl}}/8.$$

All frame members are Borel, jointly gauge invariant, analytic on their tiers, and functions of the pair there. ∎

Proof. Every nonempty layer has $n \ge t_1$, so the four estimates in Lem. C.14 apply at its level. For the terms counted with a summable lag kernel, their individual size is at most $2C_w^{\mathrm{s}}\phi_{n-1}$. Their sum per cube is therefore at most $2C_w^{\mathrm{s}}K_{\mathrm{fin}}\phi_{n-1}$. The cell terms add at most

$$[C_0^{\mathrm{cs}}h_F^2 + 2c_{\mathrm{cs}}C_{\mathrm{pc}}S_{1/4}C_w^{\mathrm{s}}]\phi_n,$$

by the second estimate of Lem. C.14, including lag zero. For the units at lags $n-j > n_{\mathrm{lag}}$, use the count proved in appendix C.2 and the whole positive lag sum:

$$\sum_{n-j>n_{\mathrm{lag}}} c^{\mathrm{u}}_{n-j}\epsilon_j[\lambda] \le \frac{N_{\mathrm{u}}L^{9/2}}{C_\omega L^{1/4}} 2C_w^{\mathrm{s}}C_{\mathrm{pc}}S_{1/4}\phi_n.$$

At the remaining lags use the individual current-scale bound; each lag occurs once, giving $2N_{\mathrm{u}}C_w^{\mathrm{s}}S_{\mathrm{lag}}\phi_{n-1}$. These terms sum to at most $(c_{\mathrm{o}}^{\mathrm{leg}} + N_{\mathrm{u}}c_{\mathrm{u}}^{\mathrm{leg}})\phi_n$. The floor eq. C.ii proves both claims, since $\phi_n \le \phi(\frac{1}{2}e^{18\mathfrak{p}_0})$. The early families have no rim; the later rim estimates are included in K_{fin} and the history version of the unit count.

For $n > t_1$ the coefficient of the mass member on $\mathcal{L}_n$ is the ordinary convex combination of the coefficients at levels $n-1, n$. The bounds of Lem. C.17 give $L^{-(n-1)}(\bar{B}_M + 2B_\sigma)\phi_{n-1}$. On the first layer the coefficient is simply $L^{-t_1}(s_{t_1} + \sigma_{t_1}(s,\lambda))$. Use $L^{-1}\phi_{t_1} \le \phi_{t_1-1}$ to obtain the same bound. The ordinary background-field count now proves the mass assertions.

For the scalar sum, the contribution of a level-j pointed bound g_j is, by [YM3, Prop. 4.147](a.3), at most

$$C_a^{\mathrm{sc}} g_j \sum_{n=j}^{k} L^{-(n-j)}\#_n(\mathcal{L}_n).$$

Here $g_j \le E_0^{\mathrm{s}} + |\lambda| r_j^0$ at an early level and $g_j \le E_0^{\mathrm{s}}$ later. Sum in j, use $\sum_{j\le n} L^{-(n-j)} \le 13/12$ and $|\lambda|\Sigma_n^0 \le c_0^\Delta \le M^{-3}\mathsf{B}$. The Wilson extraction terms cost at most $13B_\sharp\Sigma_{\mathcal{L}}(h)/4$, since $|b_j| \le B_\sharp$ and the admitted curvature factor is at most one. Thus the total is bounded by $C_m^{\mathrm{sc}}(E_0^{\mathrm{s}} + B_\sharp + M^{-3}\mathsf{B})\Sigma_{\mathcal{L}}(h) = C_5^{\mathrm{s}}\Sigma_{\mathcal{L}}(h)$.

We next check the changes in the frame. The scalar list enters its regular class through

$$\mathsf{size}^{E} \le C_{\mathrm{sz}} C_{\mathrm{ert}} (E_0^{\mathrm{s}} + |\lambda| \Sigma_k^0 + \alpha_0^2 B_\sharp).$$

This follows by summing the coefficient of an old scalar entry,

$$C_{\mathrm{ert}} L^{-(k-j)} \sum_{i=0}^{k-j} L^i \varpi_i \le C_{\mathrm{ert}} C_h' L^{-(k-j)},$$

in [YM3, Prop. 4.122]. Since $c_0^{\Delta} \le (\sqrt{2}-1) E_0^{\mathrm{s}}$, the ordinary estimate on the doubled source box bounds this class by $\varepsilon_\flat/2$. The cubic and higher gauge-weight terms cost the other half.

The first two fermion classes use the regular sums just proved and the stored-layer constant. For the third and fourth classes, the potential bound is that of Prop. C.15. The terms with stored generators carry $\Theta_k'^{\,\mathrm{s}}$ times

$$C_0^{\mathrm{cs}} h_F^2 \phi_k + (1 + c_{\mathrm{cs}}) \sum_{i \le k} \omega_{k+1-i} \epsilon_i[\lambda] + c_{\mathrm{cs}}^{\Delta} |\lambda| \Sigma_k^{\psi}.$$

The first two bounds of Lem. C.14 give exactly the ordinary stored-layer estimate for this expression. All other terms in these classes have a summable lag kernel or a short-lag count; the other two bounds of that lemma preserve those estimates. The fifth and sixth classes use the same lag sum and the per-cube regular and mass bounds on the stored layers. In particular the newly stored layer $\Omega_k \setminus \Omega_{k+1}$ is contained in the old top layer Ω_k, where those bounds have just been proved. The seventh class contains the mass and rim forms and uses $|\nu_j^h - \mu_j| \le B_\nu \phi_j/2$. Thus these seven classes total at most $b_k^{\mathrm{tot,s}}$. The remaining three classes read earlier boundary terms and insertions. Their ordinary estimate is $13/64$ times the largest earlier size. Before the base there is no boundary term and the insertion sizes are those of the original empty history. After the base the inductive bounds apply. This gives the stated recursion; $13/51 + 1 = 64/51$ closes it.

Finally, the new early members are holomorphic functions of the pair and jointly invariant on their cores; their mass forms have coefficients independent of the gauge configuration. Composing with the Borel evaluation maps gives Borel frame members. The first seven classes are analytic in the usual field polydisc. The last three are formed from the later edge and boundary terms using the same leveling functions and shifts as the ordinary frame; these functions depend on neither source nor parameter. The ordinary proof that their sum at each domain depends only on the pair therefore applies, including the split contributions of each boundary term through its coefficients and its legs. Their tier analyticity follows from the same substituted triples and locally uniform series. This verifies every frame property used by the next history step, rather than only its numerical size. □

C.10. **The history expansion from the base.**

Definition C.19 (Terms and vacuum energy)**.** For $h \in \mathcal{H}_k[t_1]$ split the fermion step operator at the base: $T_h^{\mathrm{F}} = T_{h,>t_1}^{\mathrm{F}} \circ T_{h\emptyset}^{\mathrm{F}}$. Use the Borel representative $G_{s,\lambda}$ constructed in Lem. C.7: when $\xi(U) \in \mathsf{S}_\emptyset \setminus \mathsf{N}_0$ it is the pullback of $\exp(\sum_i \lambda_i \Delta_i(s))$, and it is one elsewhere. The construction and proof there apply to this finite sum coefficient by coefficient. Define the perturbed operator by inserting $G_{s,\lambda}$ between the two factors:

$$T_h^{\mathrm{F},\lambda}[U] = T_{h,>t_1}^{\mathrm{F}}[U] \left(G_{s,\lambda}(U) T_{h\emptyset}^{\mathrm{F}}[U](e^{-A(\beta - a_K^{-1} w, U) - S_F(U) + \Phi(J)}) \right).$$

The history term $\hat{\tau}_h^{\mathrm{F}}[s,\lambda]$ is the image of $\mathfrak{w}_h(U) T_h^{\mathrm{F},\lambda}[U]\, dU$ under its completed coordinates. Its image on free coordinates is denoted $\check{\tau}_h^{\mathrm{F}}[s,\lambda]$. At the base this is $\rho_{s,\lambda}$; at $\lambda = 0$ it is the ordinary sourced term.

The coefficients of this operator, set to zero away from the support of the base history, are bounded Borel functions of U. Indeed, each free-family coefficient is continuous on its pair domain,

and the pair and background evaluation maps are Borel off the fixed exceptional set, where the representative is used. The finite fermion kernels are bounded Borel functions. The Grassmann exponential is its scalar exponential times a finite polynomial in its nilpotent part. On the support its coefficients are bounded by the finite norms of the data.

Define the common vacuum energy by

$$E_k(s,\lambda) = E_{t_1}(s) + \lambda \cdot e^{\Delta}(s) + (E_k - E_{t_1}) + \sum_{t_1 < n \le k} \sum_{X} e_n(X; \mathfrak{c}_{n-1}(s,\lambda), \lambda).$$

The inductive representation $\mathsf{I}^{\mathrm{F}}(k)[s,\lambda]$ is the ordinary sourced representation of [QCDM1, Def. 10.11], with the following explicit changes: the action has weight $\beta_k - a_K^{-1} w_\lambda$; the early regular families are $E^{(j)}(s,\lambda)$; the mass member is $\mathsf{M}_h(s,\lambda)$ of Lem. C.18; and all formed factors, edge terms, and boundary terms begin strictly above t_1. Their kept quadratic halves use the same weight w_λ. The label-dependent vacuum constant is

$$-\mathcal{E}^{\mathrm{F}}_{k,\mathcal{S}}(s,\lambda) = E_k(s,\lambda) - \sum_{t_1 < n \le k} f_n(\mathcal{S}) + \sum_{t_1 < n \le k} g_n(\mathcal{S}; s, \lambda),$$

where f_n are the ordinary zero-source label constants and $|g_n| \le C'_{\mathfrak{E}} \#_n(\tilde{Z}_n)$. All operator, stored-layer, regular-sum, frame, and exponent bounds are the ordinary bounds with the constants of Lem. C.18. The real action retains at least the fraction $1 - \mathsf{a}$ of its zero-source weight. The representation is an equality of measures coefficient by coefficient, and its exponent is independent of the evaluation representative outside the fixed null set of the ordinary coordinate construction. ■

Proposition C.20 (Induction from the base)**.** Under the four lag conditions, $\mathsf{I}^{\mathrm{F}}(k)[s,\lambda]$ holds for every $t_1 \le k \le K$ and $\lambda \in \mathrm{P}(R_k)$. The numbers $g_n(\mathcal{S}; s, \lambda)$ are jointly holomorphic on the corresponding open source sets and parameter discs. ■

Proof. At the base, adding $\lambda \cdot \Delta$ changes exactly the four quantities listed in Def. C.3. The weight condition and the ordinary ceiling on ϱ preserve the fraction $1 - \mathsf{a}$ of the action. The mass bound is Lem. C.16, and the regular and exponent bounds are Lems. C.14 and C.18. There are no pieces, stored lower layers, or boundary terms.

For the step, use the ordinary sourced history proof in [QCDM1, Prop. 10.13]. The gauge coordinates and reference measures are unchanged. The partial fermion kernel, at the fixed fine mass matrix of s, multiplies the old exponent. Its change to the new background has the same operator bounds. The frame is admitted by Lem. C.18; in its interior the expansion is exactly Prop. C.15 at the history coordinates, which belong to $2\mathscr{D}'_k$ by Lem. C.17.

Extract the local parts using only pieces within Λ_{k+1}. The increment enters the mass through its profile; the omitted profile part enters the edge terms. The zero-source gauge coefficient shifts the action, the zero-source energy and the interior source energies enter the vacuum constant. The explicit weight $a_K^{-1} A(w_\lambda, \mathsf{U})$ is carried plaquette by plaquette through this same identity.

The measure identity is the sourced change-of-coordinates theorem [YM3, Prop. 5.51], applied separately to each bounded Borel coefficient described in Def. C.19. Its source action is the one Wilson term at weight w_λ. The frame and the old representation supply its other hypotheses, just as in the cited sourced history proof. The new density is Borel because all evaluation maps are Borel and every new coefficient is a continuous function on its pair domain. Joint gauge invariance of the early and later families, and plaquettewise invariance of the action, give independence of evaluation representatives.

The numerical bounds now close as follows. The mass and regular sums are Lems. C.17 and C.18; new activity and pointed sizes are Prop. C.15; boundary and edge sizes use the frame recursion in Lem. C.18. The source-dependent label error consists of source energies outside Λ_{k+1}, boundary normalizers, and the difference between ordinary and history coordinates for the interior source energies. The first is bounded by the polymer tail, the third by Lem. C.17, and the boundary normalizers have the ordinary bound. Their sum is at most $C'_{\mathfrak{E}} \#_{k+1}(\tilde{Z}_{k+1})$.

We also check holomorphy of the boundary normalizers, since they are needed in the vacuum energy. They depend on the gauge source only through the Gaussian weight, and on the fermion source only through its fine mass matrix. They do not depend on the added activities or mass coordinate. If $\mathsf{bn}(w, J)$ denotes their value in the ordinary step, then

$$\mathsf{bn}(w, J) = \mathsf{bn}(w, 0) + \mathsf{bn}(0, J) - \mathsf{bn}(0, 0).$$

The ordinary history identity makes bn the difference of its holomorphic label energy and two holomorphic source-energy sums. Thus it is holomorphic on the interior of $\mathcal{P}$. At the new step its value is $\mathsf{bn}(w_\lambda, J)$. The weight condition puts $(w_\lambda, 0)$ in that interior, and on every open source set under consideration $(0, J)$ is also in the interior. The displayed splitting proves joint holomorphy. All other source-energy terms are holomorphic by composition. This proves the new representation and completes the induction. □

Proposition C.21 (Histories with perturbed earlier activities)**.** Suppose the four lag conditions hold at s and radius R. For every $t_1 \le k \le K$ and $\lambda \in \mathrm{P}(R_k)$:

(B1) There are Borel densities $\mathsf{g}_h[s, \lambda]$ on the free coordinates, with a common parameter-independent exceptional set, such that

$$\mathrm{Ber} \int \rho_{s,\lambda} = \sum_{h \in \mathcal{H}_k[t_1]} \mathrm{Ber} \int \mathsf{g}_h[s, \lambda]\, d\mathscr{M}_h^{\mathrm{fr}}.$$

The sum is absolutely convergent. Every density coefficient is holomorphic at every free-coordinate point on each product $S' \times \mathrm{P}(R_k)$, where S' is an open source set on which the lag conditions hold at R.

(B2) The representation $\mathsf{I}^{\mathrm{F}}(k)[s, \lambda]$ holds with the ordinary constants. In particular its later frames have size at most $64 b^{\mathrm{tot,s}}(\gamma)/51$ and the history coordinates remain in $2\mathscr{D}'_k$.

(B3) For the ordinary mark $\Theta^{(k)}$ and majorant C,

$$\sum_{h \in \mathcal{H}_k[t_1]} e^{\Theta^{(k)}(h)} e^{-\operatorname{Re} E_k(s,\lambda)} \int \|\mathsf{g}_h[s, \lambda]\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}} \le \Pi_k(C, c'') \le e^{(C+2)n_k},$$

where $c'' = (1 - \eth)(c_\star - c_\Theta)/2$. The modulus of the total mass is consequently at most $e^{\operatorname{Re} E_k(s,\lambda) + (C+2)n_k}$.

(B3′) For $t_1 < t'' \le k$, the same sum restricted to $Z_{t''}(h) \neq \varnothing$ is at most

$$e^{-\theta_{\mathrm{tl}} p_0(g_{t''})^2} \Pi_k(C, c''_{\mathrm{tl}}) \le e^{-\theta_{\mathrm{tl}} p_0(g_{t''})^2 + (C+2)n_k}.$$

For $t'' \le t_1$ there are no such histories.

(B4) The vacuum energy is the one in Def. C.19. Its source-energy terms obey $|e_n(X; s, \lambda)| \le C_{\mathfrak{E}} e^{-\kappa d_n(X)}$. The vacuum energy, these terms, and the label errors are jointly holomorphic on $S' \times \mathrm{P}(R_k)$, and at $\lambda = 0$ they equal their ordinary sourced values. The total mass $\mathrm{Ber} \int \rho_{s,\lambda}$ is entire in λ for each fixed s.

■

Proof. Integrate the generators above the base in descending order. Normalization of each partial block kernel removes it, leaving

$$\theta_s^\lambda(U) = \mathrm{Ber}\left(G_{s,\lambda}(U) T^{\mathrm{F}}_{h_\emptyset}[U](e^{-A(\beta - a_K^{-1} w, U) - S_F(U) + \Phi(J)}) \right).$$

This bounded Borel function is the same for every history through the base. Its invariance under lifts from levels $n \ge t_1$ is the weighted almost-everywhere invariance of Lem. C.8, proved using the positive base density to transport the exceptional set. That proof applies to $\exp(\sum_i \lambda_i \Delta_i)$ because each Δ_i is jointly invariant on its domain. It does not assert pointwise invariance of the arbitrarily extended representative. Lem. C.9 consequently gives the scalar mass-preserving history identity for θ_s^λ. The history partition and [YM3, Prop. 4.1] then give (B1).

We explain the common exceptional set in this identity. The tests, backgrounds, coordinate maps, and reference measures have no parameter dependence. Let N_h^0 be the Borel null set outside which every member of the ordinary exponent is evaluated in its domain. On its complement, the early families are evaluated in the same fixed pair cores, and the new activities are given on the larger radius-α_0 cores. The boundary outputs and formed factors use the same filing spaces. Hence the new exponent is defined there at every parameter and has one value by joint invariance. Set it to zero on N_h^0. The free-coordinate construction takes its Borel pullback and sets it to zero on the preimage of N_h^0. This is one null set for all sources and parameters. The coefficientwise measure identity in Prop. C.20 shows that this function is a density of the history term.

At each point outside that null set, every coefficient is holomorphic. The action is affine in w_λ. The kept and rejected quadratic halves are linear in that weight: their domains, localization operators, and shifts are fixed. The store depends holomorphically on J through the fine mass matrix. The mass and history coordinates are holomorphic by Lems. C.16 and C.17. The new regular families are holomorphic by Prop. C.15; the early ones are affine in λ. Boundary and edge terms are locally uniform limits of products of these frame members and the localized covariances. Their majorants are independent of the parameter. The vacuum and boundary numbers are holomorphic by Prop. C.20. Substitution of the finitely many background legs preserves holomorphy. Finally an even exponential is its scalar exponential times the finite exponential of its nilpotent part. On the exceptional set the chosen function is identically zero. This proves pointwise holomorphy everywhere.

The ordinary one-history majorant, applied to the representation just proved, is

$$e^{-\operatorname{Re} E_k(s,\lambda)+\sum_n \operatorname{Re} f_n(\mathcal{S})} \int \|\mathsf{g}_h[s,\lambda]\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}}$$
$$\le e^{C'_{\mathfrak{E}} \sum_n \#_n(\tilde{Z}_n)} e^{\mathsf{C}_{\mathrm{maj}}(n_k+\sum_{j\le k}\#_j(Z_j))} \prod_t e^{-(1-\mathsf{a})c_\star p_0(g_t)^2 \mathsf{n}_t^{\mathrm{cr}}(h)}.$$

All its inputs are the bounds in Lem. C.18 and Prop. C.20. Set the corresponding nonnegative function of histories to zero outside $\mathcal{H}_k[t_1]$. It satisfies the ordinary summation hypothesis with creation rate $c_{\mathsf{a}} = (c_\star + c_\Theta)/2$ and boundary constant $C_{\mathfrak{b}}^F + C'_{\mathfrak{E}}$. Apply [YM3, Prop. 4.160](f) with the ordinary dials. This proves (B3), integrability, and absolute convergence in (B1), using $|\operatorname{Ber} F| \le \|F\|_{h_F}$ and $h_F \ge 1$.

For the tail, multiply this function by $\exp[\theta_{\mathrm{tl}} \sum_t p_0(g_t)^2 \mathsf{n}_t^{\mathrm{cr}}(h)]$. Its remaining creation rate is $c_{\mathrm{tl}} = c_{\mathsf{a}} - \theta_{\mathrm{tl}}$. If $Z_{t''} \neq \varnothing$, some creation occurred by level t'', and monotonicity gives $\sum_t p_0(g_t)^2 \mathsf{n}_t^{\mathrm{cr}}(h) \ge p_0(g_{t''})^2$. The same mark is represented by the new dials because $\mathsf{w}_{\mathrm{tl}} c_{\mathrm{tl}} = c_\Theta$. Its deposit rate equals the available creation rate, as required by the summation theorem. The extra ceiling in appendix C.2 therefore gives the sum at rate $(1-\eth)(c_{\mathrm{tl}} - c_\Theta) = c''_{\mathrm{tl}}$. Removing the multiplying factor proves (B3′).

The vacuum statement follows from its displayed finite sum and the already established holomorphy. At fixed source the original base density times $e^{\lambda\cdot\Delta}$ is entire in λ and bounded on compact parameter sets by the finite norms of the data on its support. Its integral is therefore entire. □

Remark C.22 (Differences of two flows). The format is closed under differences. If another activity has data (F'_j, G'_j, ν'_j) and reference G'^{∞}_j, its difference from the ordinary activity, after subtracting scalar values at the trivial pair, has data

$$(F'_j - F_j, G'_j - G_j, \nu'_j - \mu_j), \qquad G'^{\infty}_j - G^{\infty}_j.$$

The scalar values are collected in e^{Δ}. The four entries of r^{ψ} must all be estimated, including the reference and its infinite-lattice family. When the second flow has no activity at j, these data are $(-F_j, -G_j, -\mu_j)$ and $-G^{\infty}_j$. Several interpolating directions are allowed by summing their sizes before imposing eq. C.i.

Corollary C.23 (Application to the restarted flow)**.** Use the data, formats, and lag estimates of Def. 11.3 at a base $t_0 < t_b \le t_1$ and $\Lambda_b \ge 1$. Suppose the scalar difference $d_b(s)$ is a single number d_b on $\mathcal{P}$. Let $S_b \supset \mathcal{P}$ be the open set in its free-family hypothesis. Set

$$S = S_b \cap \mathscr{O}_{t_b}, \qquad R = c_\lambda \Lambda_b, \qquad \rho_b = \tfrac{1}{2}(3\sqrt{L})^{-(K-t_b)} R.$$

Apply the preceding construction with base t_b, one parameter, and data

$$w^\Delta = \hat{w}_b, \qquad \sigma^\Delta(s) = -\delta_{t_b}(s), \qquad \Delta_1^{(j)}(s) = -\Delta^{(j)}(s), \qquad e^\Delta = d_b.$$

Then its base density is exactly $\rho_{s,\lambda}^{(t_b)}$ of Def. 11.3. At every source with $\|w\|_\sharp \le (2 - c_\lambda)\varrho$, the four lag conditions hold at radius R, and $\rho_b \le R_K/2 < R_K$. Consequently the history representation and all four bounds of Prop. C.21 hold at every level $t_b \le k \le K$ throughout the parameter set $\mathcal{D}_b$ of Prop. 11.13.

The history terms are the same measures as the restarted terms defined from the block field in Def. 11.3. Their densities agree almost everywhere, coefficient by coefficient, with the entire-in-λ representatives of Lem. C.63 wherever both constructions are defined. For every Borel set E, the integral over E of each coefficient is holomorphic on the interior of $\mathcal{D}_b$. In particular Prop. 11.13 holds with the explicit c_λ of appendix C.2. ■□

Proof. The negative differences are free families by Rem. C.22, with the same sizes. Their scalar parts have been centered at the trivial pair, as required. The gauge difference $\hat{w}_b = L^{-(K-t_b)}(x_{t_b} - \hat{x}_{t_b})$ is a real constant, so $\|w^\Delta\|_\sharp = |\hat{w}_b|$.

We check holomorphy of the mass field on S. The ordinary chain is holomorphic on $\mathscr{O}_{t_b}$. The restarted chain is

$$\hat{\sigma}_{t_b}(s)(x) = L^{t_b - t_0}\hat{\sigma}_{t_0}(s)(x) + \sum_{t_0 < j \le t_b} L^{t_b - j}(\hat{\nu}_j - \hat{\mu}_j)(y_j(x)),$$

where

$$\hat{\sigma}_{t_0}(s)(x) = \frac{F^{\mathrm{UV}}}{L^{K-t_0}(\mathsf{b} - w(x))} - \frac{F^{\mathrm{UV}}}{L^{K-t_0}\mathsf{b}}.$$

The denominators do not vanish on $\mathscr{O}_{t_b}$, since $\|w\|_\infty < 4\varrho < \mathsf{b}$. Each later increment is the fixed linear moment extraction from $\hat{F}_j(s)$ and $\hat{G}_j$. The first is holomorphic because $\hat{F}_j = F_j - (F_j - \hat{F}_j)$ and the difference hypothesis provides the second term on S_b; the reference is constant. The background walk coefficients in these moments are holomorphic in $\mathsf{n} = -a_K J$ and have locally uniform absolute sums, with the admitted factor $e^{-1/2}$ per walk step. Thus the increments, restarted chain, and δ_{t_b} are holomorphic.

The difference-of-exponents identity in Def. 11.3 reads

$$a_K^{-1} A(\hat{w}_b, \mathsf{U}) + L^{-t_b}\mathsf{S}_1(\delta_{t_b}\psi^{\mathrm{bg}}) - \sum_{j \le t_b}\sum_X \Delta^{(j)}(X) + d_b = \hat{\Phi}_s^{(t_b)} - \Phi_s^{(t_b)}.$$

Multiplying the ordinary base density by its exponential with coefficient λ proves the claimed base identification.

The hypotheses on the difference sizes give

$$\begin{aligned} R\Sigma_k^\psi &\le c_\lambda C_w^{\mathrm{s}} \phi_k \le 2\sqrt{L} c_\lambda C_w^{\mathrm{s}} \phi_{k-1} \le c_\psi^\Delta \phi_{k-1}, \\ R\Sigma_{t_b}^0 &\le c_\lambda E_0^{\mathrm{s}} \le c_0^\Delta, \\ R\|\sigma^\Delta\| &\le c_\lambda B_\sigma \phi_{t_b} \le B_\sigma \phi_{t_b}/2, \\ \|w\|_\sharp + R|\hat{w}_b| &\le (2 - c_\lambda)\varrho + c_\lambda \varrho = 2\varrho. \end{aligned}$$

Also $Q_\sigma \le 3\sqrt{L}$, so $\rho_b \le R_K/2$. Every point of $\mathcal{D}_b$ therefore belongs to the domains of the all-level theorem. Its interior lies in

$$\{\|w\|_\sharp < (2 - c_\lambda)\varrho, \|J\|_\infty < 2\varrho_J\} \times \mathrm{P}(R_K).$$

To identify the history measures, put $\xi = \mathcal{M}^{t_b}$. The factorization in Lem. C.12 gives

$$\mathfrak{w}_h = \mathfrak{w}_{h_\emptyset}(\mathfrak{w}_h^\circ \circ \xi), \qquad T^{\mathrm{F}}_{h,>t_b}[U] = T^\circ_h[\xi(U)], \qquad \mathsf{v}_h = \mathsf{v}^\circ_h \circ \xi.$$

Apply this identity coefficientwise to the base measure before its pushforward under ξ. The image of the base density under the block-field steps is precisely the image of $\mathfrak{w}_h T_h^{\mathrm{F},\lambda}\, dU$ under v_h. This is the term defined in Def. C.19. Two integrable Borel densities of the same coefficient measure agree almost everywhere. For example their real difference has integral zero on its positive and negative sets, so vanishes almost everywhere; do the same for its imaginary part. Their integrals, Berezin integrals, and integrals of the fibre norm therefore agree.

For integrated holomorphy, the pointwise holomorphic density in Prop. C.21 obeys, for every coefficient I,

$$\int_E |\mathsf{g}_{h,I}[s,\lambda]|\, d\mathscr{M}_h^{\mathrm{fr}} \le c_h^{\mathrm{Gr}} e^{\mathrm{Re}\, E_k(s,\lambda)+(C+2)n_k}.$$

On a closed polydisc in any finite-dimensional affine parameter family inside $\mathcal{D}_b$, the right side is bounded. The coefficient is jointly Borel and pointwise holomorphic. Its Cauchy formula can thus be integrated over E by Fubini, giving the Cauchy formula for the integral. This proves holomorphy on each such family, and local boundedness gives the asserted source holomorphy.

Finally the ordinary bounds in Prop. 11.13 follow directly at each level. Because $\hat{w}_b$ is constant, $\tau_k(w_\lambda;\alpha_0) = \tau_k(w;\alpha_0)$. The early scalar excess obeys

$$|\lambda| \sum_{j\le t_b} L^{-(n-j)} r_j^0 \le \tfrac{1}{2} c_\lambda E_0^{\mathrm{s}} \le M^{-3}\mathsf{B},$$

and the corresponding fermion excess is at most $c_\psi^\Delta \phi_{n-1}/2$. Hence all regular and stored-layer bounds hold with the stated constants at every level. At $k = K$, $n_K = \ell^3$ and $\Theta^{(K)} \ge 0$ give the total mass bound. The vacuum identity is

$$E_k(s,\lambda) = E_{t_b}(s) + \lambda d_b + \sum_{t_b<j\le k} \left[e_j n_j + \sum_X e_j(X;s,\lambda) \right].$$

The tail bound at level $t > t_b$ is

$$\sum_{h: Z_t(h)\neq\varnothing} \int \|\mathsf{g}_h[s,\lambda]\|_{h_F}\, d\mathscr{M}_h^{\mathrm{fr}} \le e^{-\theta_{\mathrm{tl}} p_0(g_t)^2} e^{\mathrm{Re}\, E_K(s,\lambda)+(C+2)\ell^3}.$$

These are the remaining clauses of Prop. 11.13. □

C.11. **Large fields after a prescribed past.** We next allow the activities before a level t_1 to be prescribed. Their individual sizes need not satisfy the induction bounds. What the later steps require are two sums: the scalar terms are weighted by $L^{-(k-j)}$, and the terms with fermion legs by $L^{(k+1-j)/4}$. We prove the induction with these sums as hypotheses. All histories considered here have no large-field region through level t_1.

Fix the chamber model with $r = 1/2$, $N_f \ge 1$, and the constants of [QCDM1, Prop. 10.20]. The datum, calibrations $\mathsf{t}_i = L^{K-i}\mathsf{b}$, tests, tower and background maps remain fixed. The two geometric conditions of the history induction in [QCDM1, Prop. 9.14] are assumed. Coefficients of the action introduced below are real. The gauge step uses such a coefficient x'_k through $\tau = x'_k/\mathsf{t}_k$ and $|\tau - 1| \le \bar{B}/\mathsf{t}_k$; its Gaussian and normalization are those of [QCDM1, Lem. 9.4].

Definition C.24 (A prescribed past)**.** Let $1 \le t_1 < K$ and

$$\mathcal{H}_k^{[t_1]} = \{h \in \mathcal{H}_k : h|_{t_1} = h_\emptyset^{(t_1)}\}, \qquad t_1 \le k \le K.$$

A free family $F^{(j)}$ is a pointed family on the infinite lattice and on the torus with the following properties.

(F1) Its members are even polynomials in the formal background generators of levels $\le j$ on their polymer. They are holomorphic functions of the regular pair on the whole core at (α_0, α_1), jointly gauge invariant, invariant under $\mathrm{U}(N_f) \times \mathrm{U}(N_f)$, and Borel in the real data. They depend on no history or label.

(F2) On the infinite lattice the pointed terms commute with translations by $M\mathbb{Z}^3$ in level-j units. On a polymer that does not wrap around the torus, its torus family is the part of the infinite-lattice family that fits in the chart, with the pointers carried by that chart.

(F3) On the infinite lattice, at the trivial configuration, substitute $\Psi_n^{\mathrm{bg}} = \mathsf{H}_{n,j}\Phi$ and write the degree-two part of $\sum_{X \ni z} F^{(j)}(X; z)$ as $\sum_{y,y'} \bar{\Phi}(y) K_z(y, y') \Phi(y')$. Then $q_j(z) = \sum_{y,y'} K_z(y, y')$ commutes with colour, is periodic under $M\mathbb{Z}^3$, and $\sum_{z \in c} q_j(z) = 0$ for every cell c.

(F4) The scalar parts form a pointed scalar family on the same cores and are Borel there.

With $h^{(j)}(n) = C_H L^{-(j-n)} h_F$, define

$$\epsilon_j^{\psi}(F) = \max \sup_{\square} \sum_{X \supset \square} e^{\kappa d_j(X)} \sum_z \sup_P \|F_{\ge 2}^{(j)}(X; P; z)\|_{h^{(j)}},$$

$$\mathsf{g}_j(F) = \max \sup_{X,z} e^{(\kappa+1) d_j(X)} \sup_P |F_0^{(j)}(X; P; z)|.$$

Here the maximum is over the two lattices and P ranges over the whole open core. A free past $\mathcal{F}$ at t_1 consists of such families for $1 \le j \le t_1$, with finite sizes. Set

$$\Lambda_k^{\psi}(\mathcal{F}) = \sum_{j \le t_1} L^{(k+1-j)/4} \epsilon_j^{\psi}(F), \qquad \Lambda_k^{0}(\mathcal{F}) = \sum_{j \le t_1} L^{-(k-j)} \mathsf{g}_j(F).$$

Thus $\Lambda_k^{\psi} = L^{(k-t_1)/4} \Lambda_{t_1}^{\psi}$ and $\Lambda_k^{0} = L^{-(k-t_1)} \Lambda_{t_1}^{0}$.

An exponent with this past at $h \in \mathcal{H}_k^{[t_1]}$ has the history form of [QCDM1, Prop. 9.14], with

$$\beta_h' = L^{t_1} x_{t_1}' - \sum_{t_1 < j \le k} L^j b_j' \varphi_j, \qquad s_h' = L^{-t_1} \sigma + \sum_{t_1 < j \le k} L^{-j} \mu_j' \varphi_j,$$

where $x_i' > 1$, $|x_i' - \mathsf{t}_i| \le \bar{B}$, $b_j' = L^{-1} x_{j-1}' - x_j'$ is real with $|b_j'| \le B_\sharp$, and σ, μ_j' may be complex. Its regular sum is

$$\mathfrak{f}' + \sum_{j \le k} \sum_{(z,X) \in \mathsf{lx}_j^{\mathrm{rs}}} [F^{(j)}(X; \mathsf{U}, \bar{\Psi}^{\mathrm{bg}}, \Psi^{\mathrm{bg}}; z) - F_0^{(j)}(X; \mathbf{1}; z)].$$

For $j \le t_1$ use $\mathcal{F}$; for $j > t_1$ use free families. The stored quadratic form is the unchanged tower form. The boundary sum has levels $t_1 < j \le k$ only. Its generated terms are the four kinds of [QCDM1, Prop. 9.13], with their strip rows. Their bounds use the size $\mathsf{sz}_{j-1}^{\natural}$ of the frame that generated them. The edge terms are sums of pointed activities outside $\Lambda_j^{\sim -1}$, with their plaquette shares removed, together with the action and mass parts left by $1 - \varphi_j$. Their bounds use $\mathsf{g}_j(F)$, $|b_j'|$, $|\mu_j'|$ and the insertion size $\bar{v}_j$, as specified in Lem. C.34. Their remainders after removing the action parts obey those bounds on the whole core. A complex vacuum constant completes the exponent. ∎

The norm ϵ_j^{ψ} bounds every partial sum of pointed terms on a polymer, including sums with shares of modulus at most one. Taking the supremum before summing pointers is needed when different pointers are evaluated at different pairs. Conditions (F2)–(F3) give the zero cell moment used in [QCDM1, Lem. 7.9]: translation of each pointer to a fixed cell preserves its moment and its filing, and the resulting absolutely convergent sum is $\sum_{z \in c} q_j(z) = 0$. The ordinary activities are free families; their pointed sizes are finite by [QCDM1, Lem. 8.8].

Lemma C.25 (Geometry before the first large field)**.** For $h \in \mathcal{H}_k^{[t_1]}$, the regions of levels $j \le t_1$ are the torus and $\varphi_j = 1$ there. Every pair at those levels is regular; their edge and generated-boundary index sets are empty. Stored layers $\mathcal{L}_m$ with $m < t_1$, and pieces and created cells of steps $t \le t_1$,

are absent. Restriction to a level $\geq t_1$ and extension by one outcome preserve this class of histories. The extracted vacuum differences $f_n(\mathcal{S})$ vanish for $n \leq t_1$. Moreover

$$\beta'_h = L^k x'_k + \sum_{t_1 < j \leq k} L^j b'_j (1 - \varphi_j),$$
$$w'(\mathsf{p}) = \sum_{t_1 < j \leq k} L^{j-k} b'_j \mathbf{1}\{X(\mathsf{p}; j) \not\subset \Lambda_j\}, \qquad |w'| \leq \bar{B}.$$

■

Proof. The first j outcomes determine every region at level j. For $j \leq t_1$ these outcomes are those of the empty history. An edge or generated boundary requires a polymer or pointer outside its region, so none occurs there. Also $\mathcal{L}_m = \Omega_m \setminus \Omega_{m+1}$, and pieces and created cells lie in the new large-field region. The two histories defining f_n agree through level $n \leq t_1$. Finally telescope $L^j b'_j = L^{j-1} x'_{j-1} - L^j x'_j$ and use $\sum_{i \geq 0} L^{-i} = L/(L-1)$ and $\bar{B} = B_\sharp L/(L-1)$. □

Lemma C.26 (The measure with a prescribed past)**.** Let A_{t_1} be the ordinary exponent on the empty history and let $A_{t_1}^{[\mathcal{F}]}$ be the exponent of Def. C.24 with initial mass σ, initial coefficient x'_{t_1}, no boundary sum, $\mathfrak{f}' = 0$, and constant c. Use the fixed exceptional set N_0 and Borel representative construction of Lem. C.7. When $\xi_{t_1}(U) \in \mathsf{S}_\emptyset \setminus \mathsf{N}_0$, define

$$R_{\mathcal{F}}(U) = \exp[(A_{t_1}^{[\mathcal{F}]} - A_{t_1})(\xi_{t_1}(U))],$$

and give it the value one elsewhere. For $h \in \mathcal{H}_k^{[t_1]}$ put

$$\varsigma_h^{\mathcal{F}}(U) = e^{-\beta A(U)} T^{\mathrm{F}}_{h/t_1}[U](R_{\mathcal{F}}(U) T^{\mathrm{F}}_{h_\emptyset^{(t_1)}}[U] e^{-S_F(U)}).$$

Each coefficient is bounded and Borel. Let $\check{\tau}_h^{[\mathcal{F}]}$ be the history measure with these coefficients and $\hat{\tau}_h^{[\mathcal{F}]}$ its coordinate image. Then

$$\hat{\tau}^{[\mathcal{F}]}_{h_\emptyset^{(t_1)}} = \mathsf{Stay}\, \mathbf{1}^{\mathrm{T}} \chi_{t_1}(T_\ell) e^{\mathbf{\Phi}} e^{A_{t_1}^{[\mathcal{F}]}} \mathscr{M}.$$

Its exponent has one value at the evaluation configurations off one null set. For every coefficient I and Borel set E,

$$|\check{\tau}_h[\varsigma^{\mathcal{F}}_{h,I}]|(E) \leq \frac{\sup |\varsigma^{\mathcal{F}}_{h,I}|}{\min e^{-\beta A}} \check{\tau}_h(E),$$

where $\check{\tau}_h$ is the positive pure-gauge history term. The exact one-step pushforward identity holds with $T^{\mathrm{F}}_{\mathsf{o}}$ applied to the density.

If the past, σ and c depend holomorphically on a parameter, with sizes locally bounded, so do all coefficient functions, locally uniformly in U. These conclusions also hold at an admitted source (w, J), using the same $R_{\mathcal{F}}$: the base exponent is then $A_{t_1}[w, J] + A_{t_1}^{[\mathcal{F}]} - A_{t_1}$. ■

Proof. Outside the fixed exceptional set, every evaluation pair lies in the core at half the radii. A background leg of level $n \leq j \leq t_1$ has column sum at most $C_H L^{-(t_1 - n)}$, so its cost at weight h_F is no larger than $h^{(j)}(n)$. The torus regular sums and mass term are finite, and the action difference is a bounded continuous function on the compact link space. Thus $\sup \|A_{t_1}^{[\mathcal{F}]} - A_{t_1}\| < \infty$ and $\|R_{\mathcal{F}}\| \leq \exp(\sup \|A_{t_1}^{[\mathcal{F}]} - A_{t_1}\|)$. The coordinate maps and all members are Borel on that set, and the extension is Borel and bounded. Each partial Berezin integral is a finite sum of products of bounded Borel coefficients, which proves the first assertion.

At the base, multiplication by $R_{\mathcal{F}}$ is multiplication by a bounded function of the coordinates. Pushforward commutes with this operation. The even exponents commute, so their product is $e^{A_{t_1}^{[\mathcal{F}]}}$. Two evaluation configurations differ by a gauge transformation fixing the stored data; joint invariance makes the extra factor the same at both. The positivity and domination argument in

[QCDM1, Lem. 10.7 and its proof] applies coefficient by coefficient: $\min e^{-\beta A} > 0$, and the pure term supplies the positive reference measure. In particular every density obtained in coordinates is integrable in the fibre norm, since its weighted coefficient variations are finite.

For an extension (h, o), the new weight is the old weight times the step tests and gauge-fixing factor; the new coordinates are the step map of the old coordinates; and $T^{\mathrm{F}}_{(h,\mathsf{o})/t_1} = T^{\mathrm{F}}_{\mathsf{o}} T^{\mathrm{F}}_{h/t_1}$. These facts prove the step identity by pushforward and finite Berezin integration. For a holomorphic parameter the finite base sum and its exponential are holomorphic and locally bounded; subsequent kernels are linear. At a source the bare scalar density remains continuous and bounded on the compact link space and the Grassmann coefficients remain bounded. The sourced base identity follows from [QCDM1, Prop. 10.13], with the same multiplication by $R_{\mathcal{F}}$. For the ordinary past, $\sigma = s_{t_1}$ and $c = E_{t_1} + \mathfrak{f}_{t_1}$, this factor is one on the support and the original term is recovered. □

C.12. **The contribution of one earlier level.** Put $\delta_k = \delta(\mathsf{t}_k)$ and $s_a = c_h \delta_k / a_{\mathrm{an}}$. We use the constants in the proof of [QCDM1, Prop. 9.14] and set

$$\begin{aligned}
\Theta'_k &= 4Lc_\psi + \frac{2C_{\mathcal{A}}\delta_k}{\beta_0}[9L^4 A_\kappa C_H + (2 + 6L^5 A_\kappa) C_{\mathfrak{r}} \beta_0], \\
\Theta_k &= c_\psi + C_{\mathcal{A}} s_a + C_G C_{\mathcal{A}} \delta_k + \Theta'_k, \\
\Theta^{\mathrm{B}}_{\circ} &= 1 + 9L^4 A_\kappa C_H + (2 + 6L^5 A_\kappa) C_{\mathfrak{r}} \beta_0, \\
\Theta^{\mathrm{B}}_k &= \frac{2C_{\mathcal{A}}\delta_k}{\beta_0} \Theta^{\mathrm{B}}_{\circ}, \qquad \Theta^{\sharp}_k = \Theta_k + \Theta^{\mathrm{B}}_k, \\
\varkappa_n &= \omega_{n+1} + \frac{12}{11} C_{\mathrm{lay}} L^{n/4} + C_\omega (c_{\mathrm{rim}} + c^{\mathrm{pt}}_{\mathrm{rim}}) + c_{\mathrm{geo}} + 39(1 + 2\mathsf{n}_{\mathrm{wk}}), \\
C_{\mathrm{tab}} &= C_\omega (1 + c_{\mathrm{rim}} + c^{\mathrm{pt}}_{\mathrm{rim}}) + \frac{12}{11} C_{\mathrm{lay}} + c_{\mathrm{geo}} + 39(1 + 2\mathsf{n}_{\mathrm{wk}}).
\end{aligned}$$

Here $\omega_n = C_\omega L^{n/4}$, $c_{\mathrm{geo}} = 9\mathsf{n}^2_{\mathrm{wk}}$, and $\varkappa_n \le C_{\mathrm{tab}} L^{(n+1)/4}$.

We retain the following division of the fermion terms in a step. (B1)–(B2) are the changes with the gauge fluctuation, including its collar; (B3) is the current mass term; (B4) consists of regular activities filed in current cells; (B5)–(B6) are their stored-layer parts and the stored mass term; (B7) contains the mass parts of edges; and (B8)–(B10) contain the generated boundary terms and the remaining edges. This is the division used in the proof of [QCDM1, Prop. 9.14].

Proposition C.27 (The lag kernels)**.** Let $t_1 \le k < K$, $h \in \mathcal{H}^{[t_1]}_k$, and $h' = (h, \mathsf{o})$. Form the step from an exponent of Def. C.24. All bounds below are per level-k cube on the admitted tiers, in the summed norm of rate $\kappa - 1$ or the polymer norm of rate at least r_V, with the step's external and integrated weights. For $j \le k$ put $n = k - j$ and $\epsilon^{\psi}_j = \epsilon^{\psi}_j(F)$.

(i) The activity contributions to (B1), (B2), and (B4)–(B6) are complex linear in each $F^{(j)}_{\ge 2}$. The level-j contributions satisfy

$$\begin{aligned}
(\mathrm{B4})_j &\le \Theta_k (\omega_{n+1} + C_\omega c_{\mathrm{rim}}) \epsilon^{\psi}_j, \\
(\mathrm{B5} + \mathrm{B6})_j &\le [\tfrac{12}{11} C_{\mathrm{lay}} L^{n/4} \Theta'_k + (C_\omega c^{\mathrm{pt}}_{\mathrm{rim}} + c_{\mathrm{geo}} + 39(1 + 2\mathsf{n}_{\mathrm{wk}})) \Theta_k] \epsilon^{\psi}_j, \\
(\mathrm{B1})_j, (\mathrm{B2})_j &\le \Theta^{\mathrm{B}}_k \varkappa_n \epsilon^{\psi}_j.
\end{aligned}$$

Their sum is at most $2\Theta^{\sharp}_k C_{\mathrm{tab}} L^{(k+1-j)/4} \epsilon^{\psi}_j$. The levels $j \le t_1$ therefore contribute at most $2\Theta^{\sharp}_k C_{\mathrm{tab}} \Lambda^{\psi}_k(\mathcal{F})$.

(ii) On one lattice, or on an interior polymer of a history, the fermion potential and the insertion size $\bar v_{k+1}$ are bounded by

$$C_V h_F^2 C_{\mathcal A}\delta_k + (c_\psi + C_G C_{\mathcal A}\delta_k) c_{\mathsf S} h_F^2 |s'_k| \\ + (c_\psi + C_{\mathcal A} s_a + C_G C_{\mathcal A}\delta_k) \sum_{j\le k} \omega_{k+1-j} \|F^{(j)}_{\ge 2}\|_j,$$

where $s'_k = L^{k-t_1}\sigma + \sum_{t_1<j\le k} L^{k-j}\mu'_j$. On a torus take the larger of the two activity norms in this sum. Its early part is at most $C_\omega \Lambda_k^\psi$.

(iii) Put $\Sigma_k^{[t_1]} = \sup_{t_1\le j<k} \mathsf{sz}_j^\natural$ and $\bar v_*^{[t_1]} = \sup_{t_1<j\le k} \bar v_j$, with empty suprema zero. Terms (B8)–(B9) are bounded by

$$C_{\rm bd} K_M (1+\bar\eta)(c_\psi + 2C_{\mathcal A}\delta_k/\beta_0) \max\{\Sigma_k^{[t_1]}, \bar v_*^{[t_1]}\},$$

and (B10) by

$$C^{\rm bd}\vartheta_{0,k}[\tfrac{9}{8} K^{\rm lb}\Sigma_k^{[t_1]} + 108 M^3 K_M \bar v_*^{[t_1]}].$$

All (B7)–(B10) vanish at $k = t_1$.

(iv) Terms involving the mass coefficients alone obey

$$\text{(B3)} \le (c_\psi + C_G C_{\mathcal A}\delta_k) c_{\mathsf S} h_F^2 |s'_k|,$$
$$\text{(B6)}_{\rm mass} \le \frac{L}{L-2} C_{\rm lay} \Theta'_k w_0 \nu C_H^2 h_F^2 \max_{t_1\le m\le k} \max\{|s'_m|, L|s'_{m-1}|\},$$
$$\text{(B7)} \le C_{\rm bd}(c_\psi + 2C_{\mathcal A}\delta_k/\beta_0) M^3 w_0 \nu C_H^2 h_F^2 \max_{t_1<j\le k} |\mu'_j|.$$

The entry s'_{m-1} occurs only for $m > t_1$. The gauge variation of the current and stored mass terms has the same current and stored counts, with factor $\Theta_k^{\rm B}$; both integrated and collar variations have this bound.

■

Proof. The operations before the cluster expansion are substitution, filing, taking differences, and extracting specified legs. Each is linear in an activity. A split leg costs its largest ratio to the original weight. For a current leg this ratio is $c_\psi L^{-(k-j)}$; in a layer $\mathcal L_m$ it is $4Lc_\psi(2L^{-4})^{k-m}$. The gauge response adds the two ratios

$$\frac{2C_{\mathcal A}\delta_k}{\beta_0} 8L^4 A_\kappa C_H (2L^{-4})^{k-m}, \qquad \frac{2C_{\mathcal A}\delta_k}{\beta_0} 6L^5 A_\kappa C_{\mathfrak r}\beta_0 (2L^{-4})^{k-m}.$$

Cauchy's estimate for an old coefficient adds $(2C_{\mathcal A}\delta_k/\beta_0)L^{-(k+1-m)}L^{-4(k-m)}$ when its bond places lie below level m. It is valid on the shifted core by the domain calculation in [QCDM1, Prop. 9.14, proof]; (F1) supplies holomorphy and the norm on that whole core.

For (B4), the difference terms at lags at least $n_{\rm lag}$ and the direct count at smaller lags are exactly the old-activity estimate [QCDM1, Lem. 7.9]. Its hypotheses are (F1)–(F3) and the norm; its bound at level j is $\Theta_k \omega_{n+1}\epsilon_j^\psi$. The zero moment used there is the cell sum in (F3). On a torus the wrapping and nonfitting pieces are retained as in that lemma. At levels $j \le t_1$ every piece is regular, so the history leaves no additional zero moment. Pieces taken whole near a rim have the bounds

$$C_\omega c_{\rm geo}(L^{3n+2} + L^n) e^{-\kappa(1-L^{-1})(8L^{n+1}-2)}\epsilon_j^\psi,$$
$$C_\omega c_{\rm geo}(L^{n+2} + L^n) e^{-\kappa(1-L^{-1})(8L^{n}-2)}\epsilon_j^\psi,$$

and the degree-four remainder has $c_{\rm geo}L^{3n}e^{-\kappa(1-L^{-1})(8L^n-2)}\epsilon_j^\psi$. These use respectively L^{-3}, L^{-2}, L^{-2} of the defining rim sum. Their sum is below $C_\omega c_{\rm rim}\epsilon_j^\psi$, since $L^{-3} + 2L^{-2} < 1$. Multiplying by Θ_k proves (B4).

For stored layers, before inserting any closed size, the proof of [QCDM1, Prop. 9.14] gives the following counts. Whole pointed totals at short lags cost

$$4.3(1+2\mathsf{n}_{\mathrm{wk}})L^{3(m'-j)}(2/L)^{k-m'}\Theta'_k\epsilon_j^{\psi}.$$

Long-lag differences cost $5C_\omega c_{\mathrm{geo}}(2A_\kappa C_H)^2L^{12+3/4}L^{(m-j)/4}(2/L)^{k-m}\Theta'_k\epsilon_j^{\psi}$. The regular stored-layer count and the cell-moment count have the same $L^{(m-j)/4}$ factor. The latter uses $\|q_j(z)\|_{\mathrm{tr}}\le C'^2h_F^{-2}\epsilon_j^{\psi}$, the coefficient estimate [QCDM1, Lem. 7.8] applied pointer by pointer, and $\sum_{z\in c}q_j(z)=0$. Degree at least four on $X\subset\Omega_{m-1}\setminus\Omega_{m+2}$ costs $2(1+4\mathsf{n}_{\mathrm{wk}})L^8L^{-(m-j)}(2/L)^{k-m}\Theta'_k\epsilon_j^{\psi}$. These four groups are covered by the four shares defining C_{lay}. Summing layers gives

$$\sum_{m=j}^{k}L^{(m-j)/4}(2/L)^{k-m}\le\frac{L^{n/4}}{1-2L^{-5/4}}\le\frac{12}{11}L^{n/4}.$$

The remaining degree-four polymers span at least $8L^{m-j}-2$, and also $8L^{m'-j}-2$ if a split leg reaches $m'\ge m+2$. Their layer count is bounded by

$$\sum_{d\ge1}2(d+2)L^{3(d+1)}e^{-\kappa(1-L^{-1})(8L^d-2)}\le1,$$

under $L\ge13$ and $\kappa\ge\log9216$. They cost $c_{\mathrm{geo}}\Theta_k\epsilon_j^{\psi}$. Wrapping remainders and whole pointed terms have total cost at most

$$3C_\omega c_{\mathrm{geo}}(1+L^2)\left(1+\sum_{d\ge1}L^{3d}e^{-\kappa(1-L^{-1})(7L^d-2)}\right)\Theta_k\epsilon_j^{\psi}\le C_\omega c_{\mathrm{rim}}^{\mathrm{pt}}\Theta_k\epsilon_j^{\psi}.$$

Here $\sum_{e\ge0}(2L^{-4})^e\le1.0001$ pays repeated stored layers. Boundary shares add $39(1+2\mathsf{n}_{\mathrm{wk}})\Theta_k\epsilon_j^{\psi}$. All pointer counts use the sum of the separate pointed norms; partial sums with shares are bounded by the same norm. This proves the stated (B5)–(B6) estimate.

For (B1)–(B2), replace the old configuration by the new one. The difference of an activity is linear in it. Cauchy's estimate contributes $2|B|_Y/\beta_0$; the change of a stored leg contributes the two response ratios already displayed and $(2|B|_Y/\beta_0)(3C_{\mathfrak{r}}\beta_0/2)(2L^{-4})^{k-m}$. Current legs contribute $C_G|B|_Y$ and $c_h|B|_Y/a_{\mathrm{an}}$, each at most one quarter of $2|B|_Y/\beta_0$. These factors sum to at most $(2|B|_Y/\beta_0)\Theta_\circ^{\mathrm{B}}(2L^{-4})^{(k-m)_+}$. The same counts therefore give $\Theta_k^{\mathrm{B}}\varkappa_n\epsilon_j^{\psi}$ on the tiers, also for the collar. Adding the three groups proves (i).

The one-lattice estimate is the same substitution estimate before filing across a boundary. Its three terms are the vertex, the mass term and the old activities. The original argument uses only (F1)–(F3), the two-lattice compatibility, and the real admitted ratio τ; hence it applies with their actual finite norms. An insertion at specified places has norm at most the fermion potential, proving (ii).

Boundary terms occur only above t_1. Their old reentry estimate [QCDM1, Prop. 9.13, proof] therefore uses $\Sigma_k^{[t_1]}$ and $\bar v_*^{[t_1]}$ and gives (iii). In a stored layer,

$$L^m s'_h=(1-\varphi_m)Ls'_{m-1}+\varphi_m s'_m\quad(m>t_1),\qquad L^{t_1}s'_h=\sigma.$$

Thus its modulus is bounded by the maximum in (iv). The layer sum is $\sum_{m\le k}(2/L)^{k-m}\le L/(L-2)$. At a rim pointer the mass part retained in the activity and the part left by the profile combine to $\varphi_jL^{-j}\mu'_j\bar\psi^{\mathrm{bg}}\psi^{\mathrm{bg}}$. The cube count gives $M^3w_0\nu C_H^2h_F^2|\mu'_j|$. The current mass estimate and its gauge variation use the same legs and coefficients. This proves (iv). □

Proposition C.28 (The scalar contribution)**.** In the setting of Prop. C.27, form the scalar list of [QCDM1, Def. 9.5] with entries $F_0^{(j)}$, zero counterterms, the generated boundary entries of the exponent, and the canonical remaining entries. Use $\tau=x'_k/\mathsf{t}_k$ and w' of Lem. C.25. Put

$\mathsf{g}_j = \|F_0^{(j)}\|_{j,\kappa} \le \mathsf{g}_j(F)$. The scalar identities, measurability and covariance of [QCDM1, Lem. 9.6] hold for this list. Its regular potential is complex linear in the entries. With

$$C_\varpi = \sum_{q\ge 0} (L^5/2)^q v_q, \qquad v_q = \begin{cases} 1 & q \le 3, \\ e^a e^{-aL^{q-3}} & q \ge 4, \end{cases} \qquad a = 2\delta\kappa,$$

the wide and octet parts satisfy

$$|\mathbb{V}^{\mathrm{E,wd}}(Y) + \mathbb{V}^{\mathrm{E,oc}}(Y)| \le C_{\mathrm{V}} C_{\mathcal{A}} s_k^{\mathrm{fx}} C_{\mathrm{ert}} C_\varpi \frac{L}{L-2} \sum_{j\le k} L^{-(k-j)} (\mathsf{g}_j + \alpha_0^2 B_\sharp) e^{-r_V d_k(Y)}.$$

Here $s_k^{\mathrm{fx}} = \delta_k/\alpha_0$. A part made by specified scalar entries $G^{(j)}$, $j \in J$, with sizes g_j^G, has the same bound with sum $\sum_{j\in J} L^{-(k-j)} \mathsf{g}_j^G$ and no $\alpha_0^2 B_\sharp$ term. The edge-tent potential is zero. On one lattice the old scalar variation is bracket localization of the array with entries $F_0^{(j)}$ and zero counterterms, and

$$|\mathcal{V}_0^{\mathrm{old}}(Y)| \le K_\Lambda (1+\beta_s) \frac{\delta_k}{\alpha_0} \sum_{j\le k} L^{-(k-j)} \mathsf{g}_j e^{-(1-2\delta_{\mathrm{w}})\kappa d_k(Y)}.$$

In both bounds the early entries enter through $\Lambda_k^0(\mathcal{F})$. ∎

Proof. Condition (F4) supplies a pointed scalar family on each core. The generated entries are holomorphic on their slices, Borel, gauge invariant, and functions of the pair. Their old bound is $\mathsf{B} e^{-\kappa d}$. Edge entries at level $j > t_1$ have bound $8M^3 \mathsf{g}_j(F) e^{-\kappa d}$; there are none at earlier levels. Thus the scalar-list hypotheses used in [QCDM1, Lem. 9.6] hold with $\mathsf{B}' = \max\{\mathsf{B}, 8M^3 \max_{j>t_1} \mathsf{g}_j(F)\}$. The remaining hypotheses concern the step geometry and the fixed constants. The list identities therefore apply. In particular the action terms left on the rims combine with $-A(\beta_h', \mathsf{U})$ to give

$$-L^{t_1} x'_{t_1} A(\mathsf{U}) + \sum_{t_1 < j \le k} L^j b'_j \sum_{\mathsf{p}: X(\mathsf{p};j) \subset \Lambda_j} a_{\mathsf{p}}(\mathsf{U}) = -L^k x'_k A(\mathsf{U}) - L^k A(w'; \mathsf{U}).$$

This follows plaquette by plaquette from the pointed shares and the telescoping identity in Lem. C.25.

The regular-frame estimate [YM3, Prop. 4.122], and its proof uses

$$\mathsf{size}^{\mathrm{E}} = C_{\mathrm{ert}} \sum_{m\le k} \varpi_{k-m} \left(\mathsf{g}'_m + \sum_{j<m} L^{-(m-j)} \mathsf{g}'_j \right), \qquad \mathsf{g}'_j = \mathsf{g}_j + \alpha_0^2 B_\sharp.$$

Its weights are $\varpi_i = L^{3i} \sum_{p+q=i} u_p v_q$. The levelling floor $\delta_\star^{\mathrm{pc}} L \ge 10 \log L$ gives $u_p = 2^p e^{-\delta_\star^{\mathrm{pc}} L p/2} \le 2^p L^{-5p}$, and hence $\varpi_i \le C_\varpi (2L^{-2})^i$. Interchanging the finite sums now gives

$$\mathsf{size}^{\mathrm{E}} \le C_{\mathrm{ert}} C_\varpi \sum_{j\le k} L^{-(k-j)} \mathsf{g}'_j \sum_{m=j}^{k} (2/L)^{k-m} \le C_{\mathrm{ert}} C_\varpi \frac{L}{L-2} \sum_{j\le k} L^{-(k-j)} \mathsf{g}'_j.$$

The defining series for C_ϖ converges because v_q decays exponentially in L^q.

To isolate specified entries, apply this bound to tG for $t > 0$. Linearity multiplies the potential by t; the additive $\alpha_0^2 B_\sharp$ bound stays fixed. Divide by t and let $t \to \infty$. This removes that additive term. Zero counterterms make the edge-tent part vanish. Finally [YM3, Prop. 4.52] gives the bracket localization bound on the absolute system, with factor $(1+\beta_s)\delta_k/\alpha_0$ and array size $\sum_j L^{-(k-j)} \mathsf{g}_j$. Its linearity also gives the stated bound for the early part. □

Proposition C.29 (One step with a prescribed past)**.** Let $t_1 \le k < K$, $h \in \mathcal{H}_k^{[t_1]}$, and $h' = (h, \mathsf{o})$. Suppose its term has the measure identity

$$(\mathrm{I}_k) \qquad \hat{\tau}_h^{[\mathcal{F}]} = \mathsf{Stay}_h \mathbf{1}_h^{\mathrm{T}} \chi_k(\Omega_k) e^{\mathbf{\Phi}_h} e^{A_{k,h}^{[\mathcal{F}]}} \mathscr{M}_h,$$

with an exponent of Def. C.24, independent of the evaluation configuration off one null set. Assume also:

(Fr) Each of the four frame classes, the fermion class, the scalar classes rg, bd, and $\mathcal{V}^{\mathrm{q}}$, has size at most $\varepsilon_\flat$ on the tiers at rate r_V.

(Sm) The sum

$$\begin{aligned}\tfrac{7}{2}K_P\delta_k + K_\Lambda(1+\beta_s)\frac{\delta_k}{\alpha_0}\sum_{j\le k}L^{-(k-j)}\mathsf{g}_j(F)\\ + C_V h_F^2 C_{\mathcal{A}}\delta_k + (c_\psi + C_G C_{\mathcal{A}}\delta_k)c_{\mathsf{S}}h_F^2|s'_k|\\ + (c_\psi + C_{\mathcal{A}}s_a + C_G C_{\mathcal{A}}\delta_k)\sum_{j\le k}\omega_{k+1-j}\|F^{(j)}_{\ge 2}\|_j\end{aligned}$$

is at most $c_{\mathrm{cl}}/4$.

Then the step is defined. On an interior polymer its output equals the one-lattice output with the same families, coefficient x'_k, and mass coefficient $L^{-k}s'_k$. If $\widetilde{E}^{(k+1)}(X;z)$ is its pointed output, subtract local parts at arbitrary numbers e', b', μ':

$$\begin{aligned}F^{(k+1)}(X;z) = \widetilde{E}^{(k+1)}(X;z) - \mathbf{1}_{X=\{z\}}\left(e' - L^{-(k+1)}\mu'\sum_v \bar{\psi}^{\mathrm{bg}}(v)\psi^{\mathrm{bg}}(v)\right)\\ - L^{k+1}b'\sum_{\mathsf{p}:X(\mathsf{p})=X}\upsilon_z(\mathsf{p})a_{\mathsf{p}}(\mathsf{U}).\end{aligned}$$

It has (F1), (F2), and (F4), and finite sizes. The new coefficients are $x'_{k+1} = L^{-1}x'_k - b'$ and $s'_{k+1} = Ls'_k + \mu'$. The action and mass parts left by $1-\varphi_{k+1}$ are placed in the edge terms. The extracted numbers enter the regular constant and vacuum constant by the rule of [QCDM1, Lem. 9.15]. The resulting exponent $A^{[\mathcal{F}]}_{k+1,h'}$ satisfies (I_{k+1}), with one value at the evaluation configurations, and the domination of Lem. C.26 holds. If also (F3) holds for the new family and the new real coefficients are in their windows, the new exponent has the format of Def. C.24. ∎

Proof. Since $k \ge t_1 \ge 1$, the first small-field region is the whole torus. The old configuration at the chain point is the shifted configuration of the usual step. The gauge objects, tests, coordinates and Gaussian are unchanged except for their admitted ratio $\tau = x'_k/\mathsf{t}_k$. Every evaluation pair lies in its core at half the radii. Thus (F1) and the boundary assumptions define every old member at the places where the step evaluates it.

Complete the fermion square at the new background. The Schur identity concerns only the tower. Substitution into each old member and changing the background are linear in that member. Consequently the exact algebra of [QCDM1, Prop. 9.11] applies. The estimates for its activity and mass variations are those of Prop. C.27; the scalar variations and the action-profile identity are those of Prop. C.28. No assertion about the birth sizes of early families enters this algebra.

On Λ_{k+1} all profiles are one, so the two coefficients are $L^k x'_k$ and $L^{-k}s'_k$. By (Sm), the scalar and fermion potential sizes add to at most $c_{\mathrm{cl}}/4$. The test tail is at most c_{cl}. The expansion [QCDM1, Lem. 7.5] therefore applies on both lattices, with its bounds and first variations. Its inputs are holomorphic in the pair, jointly invariant and flavour invariant. Substitution, integration and pointing preserve these properties. Translation covariance and compatibility on polymers that fit follow term by term from the same properties of the inputs. The scalar bracket localization is defined on the whole core by [YM3, Prop. 4.52]; hence the scalar output is a pointed family there. Local extraction preserves these properties. The output has decay $e^{-\kappa_+ d}$, and the local parts have size-zero support. Since $\kappa_+ \ge \kappa + 1$, the polymer and pointer counts give finite sizes. These statements prove (F1), (F2), and (F4); they do not assert (F3) for arbitrary extraction numbers.

On an interior domain, the two frames have the same potential values, covariances and cluster operations. The interior agreement argument of [QCDM1, Lem. 9.10] therefore identifies their outputs and pointed terms. The extraction identity is linear in e', b', μ', so its signs give precisely

the new coefficients displayed above. Generated boundary terms and edge terms have the properties proved in Lem. C.34.

It remains to justify the measure identity for complex coefficients. The coefficient functions of Lem. C.26 are bounded Borel functions. Both the old term and the coefficientwise measure carried through the next step are dominated by a constant times a positive pure-gauge term. Their variations are therefore integrable. Each change of variables in the exact history identity can be applied to the four positive parts of each complex coefficient. Null sets are carried by the same coordinate maps: absolute continuity with respect to the pure term replaces any use of positivity of the fermion coefficient. The finite Berezin integral then commutes with these operations. This proves (I_{k+1}) as a measure identity, not only as an equality of formal integrands.

Two evaluation configurations differ by a gauge transformation fixing all stored generators. The new regular members and boundary terms are jointly invariant, and the tower and mass term are covariant. Thus the new exponent has one value off the coordinate null set. Its old boundary terms retain their levels and its new ones have level $k+1 > t_1$. With the extra moment and window assumptions it has all clauses of the required format. □

C.13. **Closing the bounds.**

Remark C.30 (Constants for the prescribed past). Let $b_k^{\mathrm{blk}}, b_k^{\mathrm{lay}}, b_k^{\mathrm{ms}}$ be the three bounds in the proof of [QCDM1, Prop. 9.14]. Define

$$
\begin{aligned}
m_\omega &= \min\{1, S_\omega\}, \qquad \lambda_0 = m_\omega/34,\\
\lambda_0' &= \min\left\{1, \frac{K_\Lambda(1+\beta_s)(L-2)\varepsilon_\flat}{LC_{\mathrm{V}}C_{\mathcal{A}}C_{\mathrm{ert}}C_\varpi c_{\mathrm{cl}}}\right\},\\
K_0^{\mathrm{F}} &= \tfrac{7}{2}K_P + \frac{K_\Lambda(1+\beta_s)}{\alpha_0}\left(\frac{L}{L-1}E_0 + 1 + E_0\right),\\
C_2^{\mathrm{pt}} &= 8M^3C_2, \qquad b_k^{\mathrm{pt}} = 2\Theta_k^\sharp C_{\mathrm{tab}}S_\omega C_2^{\mathrm{pt}}\phi(\mathfrak{t}_k)^2.
\end{aligned}
$$

The pointed constant is that of [QCDM1, Lem. 8.8]; in particular its use precedes the constants that count stored pointers. A function of γ, such as $b^{\mathrm{pt}}(\gamma)$ or $\Theta^\sharp(\gamma)$, means the corresponding expression evaluated at $\delta(\gamma^{-2})$, $\phi(\gamma^{-2})$, and γ^2. Impose the additional ceiling

$$
(\Gamma^{\mathrm{F}}) \qquad K_0^{\mathrm{F}}\delta(\gamma^{-2}) + b^{\mathrm{blk}}(\gamma) \le \tfrac{1}{4}c_{\mathrm{cl}}, \quad b^{\mathrm{ms}}(\gamma) \le \tfrac{3}{16}\varepsilon_\flat, \quad b^{\mathrm{lay}}(\gamma) + b^{\mathrm{pt}}(\gamma) \le \tfrac{1}{4}\varepsilon_\flat.
$$

Each left side tends to zero with γ. Its constants have already been fixed: $C_\varpi, \Theta_\circ^{\mathrm{B}}, \lambda_0, \lambda_0', K_0^{\mathrm{F}}$ precede h_F; C_{tab} and C_2^{pt} use constants through h_F; and γ is chosen last. The bound $2C_{\mathcal{A}}\delta(\gamma^{-2}) \le \beta_0$ gives $\Theta_k^{\mathrm{B}} \le \Theta_\circ^{\mathrm{B}}$.

For the variation of the outputs put

$$
\begin{aligned}
K_{\mathrm{num}} &= \max\{1, c_b^{-2}C_{am}^\psi(\nu h_F^2)^{-1}, 2 + C_{am}r_\alpha^{-2}\},\\
K_{\mathrm{lin}} &= 32K_M(1+K_\mu)(1+K_{\mathrm{num}}) + C_E K_{\mathbf{M}}^{\mathrm{v}} K_\Lambda^{-1},\\
C_\Sigma &= K_{\mathrm{lin}}\max\{2\Theta^\sharp(\gamma)C_{\mathrm{tab}}, (c_\psi + C_{\mathcal{A}}s_a(\gamma) + C_GC_{\mathcal{A}}\delta(\gamma^{-2}))C_\omega\},\\
C_\Sigma' &= K_{\mathrm{lin}}\max\left\{C_{\mathrm{V}}C_{\mathcal{A}}\frac{\delta(\gamma^{-2})}{\alpha_0}C_{\mathrm{ert}}C_\varpi\frac{L}{L-2}, K_\Lambda(1+\beta_s)\frac{\delta(\gamma^{-2})}{\alpha_0}\right\}.
\end{aligned}
$$

Here $K_{\mathbf{M}}^{\mathrm{v}}$ is the scalar-system variation constant of [QCDM1, Lem. 7.4]. The first two constants are fixed at h_F; the last two after γ. None is used to choose an earlier constant.

Definition C.31 (A flow above the prescribed past)**.** A flow above t_1 consists of families $E^{(j)}$ on both lattices, real x_i for $t_1 \le i \le K$, complex initial s_{t_1}, E_{t_1}, and numbers e_j, b_j, μ_j for $t_1 < j \le K$, with the unchanged tower, tests and backgrounds. Assume:

(G1) Every family has (F1)–(F4) and the pointed sum rule. Its scalar part has no extracted action multiple; its infinite-lattice pointed values at the trivial pair have zero mean over a cell; and on the torus those values are field-independent numbers.

(G2) For $j > t_1$, the family and numbers are those produced by the one-lattice step and its extraction rules. Thus $x_j = L^{-1}x_{j-1} - b_j$ and $s_j = Ls_{j-1} + \mu_j$. The windows are $x_i > 1$, $|x_i - \mathsf{t}_i| \le \bar{B}$, and $|s_i| \le 2\bar{B}_M/\mathsf{t}_i$. Also $|b_j| \le B_\sharp$ and $|e_j| \le C_e'(1 + \log x_{j-1})$. The scalar size is at most E_0. The usual activity norm, including each pointed term and every partial sum of pointed terms on one polymer, is at most $C_2\phi(\mathsf{t}_{j-1})^2$, on both lattices. The stronger pointed size satisfies

$$\epsilon_j^\psi(E) \le C_2^{\mathrm{pt}}\phi(\mathsf{t}_{j-1})^2.$$

(G3) For $j \le t_1$ write $E^{(j)} = R^{(j)} + D^{(j)}$, with both summands satisfying (G1), and $D_0^{(j)}$ zero at the trivial pair. Require

$$\epsilon_j^\psi(R) \le C_2\phi(\mathsf{t}_{j-1})^2, \qquad \mathsf{g}_j(R) \le E_0,$$

and, for every $t_1 \le k \le K$,

$$\Lambda_k^\psi(D) \le \lambda_0 C_2\phi(\mathsf{t}_k)^2, \qquad \Lambda_k^0(D) \le \lambda_0'(1 + E_0).$$

The flow's free past is $(E^{(j)})_{j\le t_1}$. Write $\mathfrak{f}_k$ for the torus sum of all scalar values at the trivial pair through level k, and set

$$E_k = E_{t_1} + \sum_{t_1<j\le k} e_j n_j, \quad \mathfrak{d}^0 = \max_{t_1\le k\le K} \Lambda_k^0(D), \quad \mathfrak{d}^\psi = \max_{t_1\le k\le K} \Lambda_k^\psi(D).$$

Its term is that of Lem. C.26 with $\sigma = s_{t_1}$ and $c = E_{t_1} + \mathfrak{f}_{t_1}$. The windows give $|\mu_j| \le 4\bar{B}_M/\mathsf{t}_j$. At a step satisfying (G1), the extracted mean quadratic moment is a scalar matrix; the mass rule is its negative scalar coefficient divided by η_j^2. ∎

Remark C.32 (Which assumptions are used). The whole-core holomorphy and invariance define the shifted members and their common value in coordinates. Translation covariance and (F3) give the cell cancellations; (F4) gives the scalar list. The ordinary norm in (G2) controls the one-lattice potential. Its pointed norm is used when stored pointers have different evaluation pairs. In (G3) the pointed norm of $R^{(j)}$ has the smaller constant C_2, so its stored contribution lies inside the old bound; the contribution above t_1 is instead paid by b_k^{pt}. The real coefficient window admits the Gaussian. The mass window gives the factor 4, rather than 3, in the edge mass bound. Finally the vanishing scalar value of $D^{(j)}$ keeps the wrapping constant unchanged. These conditions are separate hypotheses; one does not infer them merely from finite sizes.

Lemma C.33 (The frame closes)**.** Let a flow of Def. C.31 be given. Suppose its term at $h \in \mathcal{H}_k^{[t_1]}$, $k < K$, has the format and identity (I_k), with its actual families and coefficients, and its boundary terms have the properties of Lem. C.34. Suppose earlier frame sizes are at most $\varepsilon_\flat$ and $\bar{v}_j \le b_{j-1}^{\mathrm{blk}}$. Then (Sm) and (Fr) hold at every extension h'. More precisely,

(i) On one lattice, $\varepsilon_{\mathrm{V}} \le K_0^{\mathrm{F}}\delta_k$, $\varepsilon_\natural \le b_k^{\mathrm{blk}}$ and $\bar{v}_{k+1} \le b_k^{\mathrm{blk}}$. The potential part linear in D has size at most

$$s_D = (c_\psi + C_{\mathcal{A}}s_a + C_G C_{\mathcal{A}}\delta_k)C_\omega\Lambda_k^\psi(D) + K_\Lambda(1 + \beta_s)\frac{\delta_k}{\alpha_0}\Lambda_k^0(D) \le \tfrac{1}{4}c_{\mathrm{cl}}.$$

The potential with D removed also has size at most $c_{\mathrm{cl}}/4$.

(ii) With $b_k^{\mathrm{tot}} = b_k^{\mathrm{blk}} + b_k^{\mathrm{lay}} + b_k^{\mathrm{ms}}$, the fermion class satisfies

$$\mathsf{sz}_k^\natural \le \tfrac{13}{64}\max\{\Sigma_k^{[t_1]}, \bar{v}_*^{[t_1]}\} + b_k^{\mathrm{tot}} + \tfrac{1}{3}b_k^{\mathrm{ms}} + b_k^{\mathrm{pt}} + P_D,$$

$$P_D \le \min\left\{2\Theta_k^\sharp C_{\mathrm{tab}}\Lambda_k^\psi(D), \frac{\lambda_0}{m_\omega}(b_k^{\mathrm{blk}} + \tfrac{59}{11}b_k^{\mathrm{lay}})\right\} \le \tfrac{35}{748}\varepsilon_\flat.$$

In particular $\mathsf{sz}_k^{\natural} \le \varepsilon_\flat$.

(iii) The sizes of rg, bd and $\mathcal{V}^{\mathrm{q}}$ are at most $3\varepsilon_\flat/4$, $2\varepsilon_\flat/5$, and $\varepsilon_\flat$, respectively. The part of rg linear in D is at most $C_{\mathrm{V}} C_{\mathcal{A}} s_k^{\mathrm{fx}} C_{\mathrm{ert}} C_\varpi L \Lambda_k^0(D)/(L-2) \le \varepsilon_\flat/4$.

■

Proof. For the ordinary families in (G2) and R in (G3), put $n = k+1-j$. The scale fall gives

$$\phi(\mathsf{t}_{j-1})^2 \le L^{-n}(1+n\log L)^{2\mathsf{p}_0}\phi(\mathsf{t}_k)^2.$$

After multiplication by $L^{n/4}$ their sum is at most $L^{-1/2}S_\omega\phi(\mathsf{t}_k)^2$. The D terms contribute at most $\lambda_0\phi(\mathsf{t}_k)^2$, and $\lambda_0 \le (1-L^{-1/2})S_\omega$. Thus the old-activity sum fits in the third term of b_k^{blk}. The mass window gives its second term. For scalars,

$$\sum_{j\le k} L^{-(k-j)}\mathsf{g}_j(E) \le \frac{L}{L-1}E_0 + \Lambda_k^0(D) \le \frac{L}{L-1}E_0 + 1 + E_0.$$

Apply Props. C.27 and C.28 and (Γ^{F}). The same estimates with $D=0$ apply after its removal. Each of the two summands of s_D is covered by the corresponding bound just used. An insertion costs no more than the fermion potential. This proves (i).

For (ii), split each member into three parts: the one formed with R below t_1 and zero families above it, including its vertex, mass and boundary terms; the terms from $E^{(j)}$ above t_1; and the terms from D. The first part has the closed bounds of [QCDM1, Prop. 9.14, proof]. Its regular pointed sizes are at most $C_2\phi^2 \le C_2^{\mathrm{pt}}\phi^2$. Stored mass coefficients obey $L^m|s_h| \le \max\{|s_m|, L|s_{m-1}|\} \le 2\bar{B}_M/\mathsf{t}_m$. The boundary terms have their induction bounds. The only changed bound is $|\mu_j| \le 4\bar{B}_M/\mathsf{t}_j$: it replaces the edge contribution b_k^{ms} by $4b_k^{\mathrm{ms}}/3$. Thus the first part is at most the first three terms in the displayed recurrence.

For the second part use the pointed bound of (G2) in Prop. C.27(i). Its sum is at most $2\Theta_k^{\sharp}C_{\mathrm{tab}}S_\omega C_2^{\mathrm{pt}}\phi(\mathsf{t}_k)^2 = b_k^{\mathrm{pt}}$. For the third part the same proposition gives the first bound for P_D. To obtain the second, write $\Phi = C_2\phi(\mathsf{t}_k)^2$, $\Lambda = \Lambda_k^{\psi}(D)$ and $\theta_1 = c_\psi + C_{\mathcal{A}}s_a + C_G C_{\mathcal{A}}\delta_k$. Since $\Theta_k^{\mathrm{B}} \le 2\Theta_k'$, the separate kernels give

$$\begin{aligned} P_D &\le (\theta_1 + 5\Theta_k')C_\omega\Lambda + \tfrac{60}{11}C_{\mathrm{lay}}\Theta_k' L^{-1/4}\Lambda \\ &\quad + 5\Theta_k[C_\omega(c_{\mathrm{rim}} + c_{\mathrm{rim}}^{\mathrm{pt}}) + c_{\mathrm{geo}} + 39(1+2\mathsf{n}_{\mathrm{wk}})]L^{-1/4}\Lambda. \end{aligned}$$

Here we used $\sum_j L^{(k-j)/4}\epsilon_j^{\psi}(D) = L^{-1/4}\Lambda$ and $\sum_j \epsilon_j^{\psi}(D) \le L^{-1/4}\Lambda$. The first θ_1 term is at most (λ_0/S_ω) times the third term of b_k^{blk}. The remaining terms are paid by b_k^{lay}: its stored-layer and interface factors satisfy $C_{\mathrm{lay}}\Theta_k'\Phi \le b_k^{\mathrm{sl}}$ and $39(1+2\mathsf{n}_{\mathrm{wk}})\Theta_k\Phi \le b_k^{\mathrm{if}}$; also $c_{\mathrm{geo}} \le C_\omega c_{\mathrm{rim}}^{\mathrm{pt}}$. Their largest multiplier is

$$\lambda_0 \max\{5/S_\omega, 10L^{-1/4}, (60/11)L^{-1/4}\} \le (59/11)\lambda_0/m_\omega.$$

This uses $L \ge 13$. Consequently $P_D \le (1/34)(70/11)(1/4)\varepsilon_\flat = 35\varepsilon_\flat/748 < 3\varepsilon_\flat/64$. The new ceiling gives $b_k^{\mathrm{blk}} + (b_k^{\mathrm{lay}} + b_k^{\mathrm{pt}}) + 4b_k^{\mathrm{ms}}/3 \le 3\varepsilon_\flat/4$. Since the earlier maximum is at most $\varepsilon_\flat$, the recurrence closes with $13/64 + 3/4 + 3/64 = 1$.

Finally split the scalar list into its closed entries and the entries $D_0^{(j)}$. The first list satisfies the ordinary scalar bounds and contributes at most $\varepsilon_\flat/2$ to rg. Prop. C.28 and the definition of λ_0' bound the second by $K_0^{\mathrm{F}}\delta_k\varepsilon_\flat/c_{\mathrm{cl}} \le \varepsilon_\flat/4$. The boundary scalar class reads only later generated terms and edges, so its ordinary $2\varepsilon_\flat/5$ bound persists. The quadratic gauge class reads only the configuration, tests and real τ, and retains its bound. The four sizes give the frame amplitude condition of [QCDM1, Lem. 9.9]. □

Lemma C.34 (The new boundary terms)**.** Assume (I_k), (Fr), and (Sm) in Prop. C.29, and the stated boundary properties at earlier levels. The generated boundary term of level $k+1$ is the sum of the four kinds in [QCDM1, Prop. 9.13]: the cluster remainder, the gauge and fermion normalizers,

and the rejected forms, with their strip rows. It is holomorphic in the pair on its slices, jointly invariant, Borel, and depends only on the pair and stored data. It vanishes on the interior in the same cases as those four kinds. Its scalar and full bounds are those of that proposition; the part with fermion legs is bounded by

$$(1+\eta_{k+1}^{\mathrm{sr}})K^{\mathrm{lb}}\mathsf{sz}_k^{\natural}e^{-\kappa_+ d_{k+1}(X)}.$$

For an edge domain, the new edge term has norm at most

$$M^3(8\mathsf{g}_{k+1}(F)+108K_M\bar v_{k+1}+3\bar C_{\mathrm{reg}}^2|b'|+w_0\nu C_H^2h_F^2|\mu'|)e^{-\kappa d_{k+1}(X)}.$$

Its scalar bound omits the insertion and mass terms. Its part with fermion legs is bounded more precisely by

$$4M^3(d_{k+1}(X)+3)^3K_M\bar v_{k+1}e^{-\kappa_+ d_{k+1}(X)},$$

plus $M^3w_0\nu C_H^2h_F^2|\mu'|$ on one-cube domains. Here $\bar v_{k+1}\le c_{\mathrm{cl}}/4$ and $\sum_{X\in\mathsf{lx}_{k+1}^{\mathrm{ed}}}e^{-\kappa d_{k+1}(X)}\le C_{\mathsf{B}}\#_{k+1}(Z_{k+1}^{\sim})$.

After the action parts left on the rim are removed, the edge term depends on the configuration on $\mathrm{Bd}(X)$ alone and obeys its bounds on the whole core. Its scalar part there is the scalar edge family with bound $8M^3\mathsf{g}_{k+1}(F)e^{-\kappa d_{k+1}(X)}$. The removed action parts combine with the action profile to give $-L^{k+1}A(w';\mathsf{U})$, even for complex b'. When b' is real with $|b'|\le B_\sharp$, w' is real and $|w'|\le\bar B$. ■

Proof. The four kinds are built from the step's potential, background maps, decoupled covariance and normalizers. The latter three use only the tower and regions. The potential inputs have the required properties by (F1), the previous boundary hypotheses and Props. C.28 and C.29. For the stored form and mass term, gauge covariance of the tower and background maps gives

$$\langle\bar\Psi u^{-1},\mathbb{D}_h[U^u]u\Psi\rangle=\langle\bar\Psi,\mathbb{D}_h[U]\Psi\rangle.$$

The mass coefficient is scalar at every place, so the same statement holds for it. The scalar list gives the regular and boundary scalar potentials as functions of the pair; the quadratic gauge class has that property by [QCDM1, Lem. 9.4]. Boundary levelling uses only their slice bounds and their whole-core edge remainders, as in [QCDM1, Lem. 9.7].

The expansion under (Fr) preserves holomorphy, measurability and invariance. Its cluster bound is $\mathsf{b}_{\mathrm{cs}}^{\mathfrak{F}}e^{-\kappa''d}$, with $\mathsf{b}_{\mathrm{cs}}^{\mathfrak{F}}\le 3/16$, and its fermion part is at most $K^{\mathrm{lb}}\mathsf{sz}_k^{\natural}e^{-\kappa''d}$. The strip row multiplies these bounds by η_{k+1}^{sr} and uses rate κ_+. The normalizer amplitudes are $24(LM)^3E_{\mathrm{nf}}$ and C_NN_f; the rejected forms and quadratic correction have the original unit bounds. All their rates are at least κ. Thus their sum is at most $\mathsf{B}^{\mathrm{F}}e^{-\kappa d}\le\mathsf{B}e^{-\kappa d}$, where $\mathsf{B}^{\mathrm{F}}=24(LM)^3E_{\mathrm{nf}}+C_NN_f+3$. On an interior polymer the cluster remainder vanishes by Prop. C.29; the other vanishing assertions concern only the regions. This proves the generated-term claims.

An edge domain lies inside Λ_{k+1}, so its pointed terms are outputs of the one-lattice step. Their insertion part costs $4K_M\bar v_{k+1}e^{-\kappa_+ d}$ per pointer, and there are at most $M^3(d+3)^3$ pointers. Since $(d+3)^3e^{-(\kappa_+-\kappa)d}\le 27$, this gives the coefficient $108M^3K_M\bar v_{k+1}$. The scalar edge bound is the pointed scalar count of [QCDM1, Prop. 9.13, proof], applied with its actual size $\mathsf{g}_{k+1}(F)$.

The action remainder has size-zero support. On such a domain at most $3M^3L^{3(k+1)}$ fine plaquettes contribute, each at most $\bar C_{\mathrm{reg}}^2L^{-4(k+1)}$. Their coefficient is $L^{k+1}b'$ and their shares sum to one, giving $3M^3\bar C_{\mathrm{reg}}^2|b'|$. At a rim pointer the mass part in the pointed activity and the part left by the profile add to $\varphi_{k+1}L^{-(k+1)}\mu'\sum_v\bar\psi^{\mathrm{bg}}(v)\psi^{\mathrm{bg}}(v)$. There are $M^3L^{3(k+1)}$ fine sites per cube and each leg costs $C_HL^{-(k+1)}h_F$. Hence its bound is $M^3w_0\nu C_H^2h_F^2|\mu'|$.

The insertion and scalar bounds hold uniformly on the whole core: the one-lattice expansion is defined there, and $\mathsf{g}_{k+1}(F)$ is a supremum over that core. The mass bound is independent of the configuration. Only the removed action terms used regular evaluation configurations; subsequent

steps place them with the action profile, whose exact plaquette identity is Prop. C.28. The edge-domain count depends only on the label. These observations prove every stated domain and norm assertion. □

Lemma C.35 (Agreement for equal history excerpts)**.** Let two histories $h', h'_1 \in \mathcal{H}_{k+1}^{[t_1]}$ use the same free past, coefficients, mass numbers and later free families. Assume their steps through level $k+1$ satisfy Lem. C.34. Suppose the torus has the chart floor $(\mathrm{Tor}^{\mathrm{c}})_{k+1}$. If their excerpts, including label bonds, agree at a generated domain X, then X is generated for both histories and their boundary terms at X are the same function on the same space, including their strip rows. ■

Proof. Use the identity chart on the vicinity of X. The chart correspondence in [QCDM1, Prop. 9.13, proof] descends to every earlier level: the keys and records of a restricted history are the same earlier keys and records of the whole history. It also restricts to each subdomain $Y \subset X$, since its vicinities are nested and its label bonds are the corresponding subset. The chart floor descends to those levels. Thus the triples, graded spaces, stored data spaces and boxes on every such Y agree.

Induct on the level $m > t_1$ of a generated term in X. At the base there are no generated or edge terms. At a later step, the frame on a subdomain Y uses: the prescribed regular families, evaluated at pairs selected by the regions; the coefficients and profiles; the generated terms at earlier levels; the edge terms; and the tower, shifts and Gaussian objects. The regular families are the same functions. Regularity of a pair reads its polymer and the nearby region, and the background walks read only their recorded vicinities. The coefficients are the same; the profile $\varphi_j(y)$ reads Z_j within $|x-y|_\infty < 3\mathsf{s}_j/4$, a subset of its recorded vicinity. The action weight reads whether a plaquette corner lies in Z_j. Earlier generated terms agree by induction. An edge term reads its family, the coefficients, its rim pointers and its profile, so it also agrees. Finally the tower and its decoupled covariances are carried by the same chart.

Consequently the fermion class agrees, term by term. The two scalar lists have the same regular entries, earlier generated entries and canonical factors on Y; the scalar localization and bracket construction therefore give the same regular and boundary potentials. The weight potential agrees plaquette by plaquette. The quadratic class reads the pair, τ and whether its cube belongs to Λ_m, all fixed by the excerpt. The Gaussian and fermion integration families also agree. The local cluster operations thus produce the same output on every subdomain of Y.

The cluster kind is this output at the stored field minus its interior one-lattice output. The strip row is the same finite integer combination of output values at the same external fields. The normalizer kinds use the tower and regions; the rejected forms and quadratic correction use those data and the same τ. Each of the four kinds therefore agrees. This closes the induction and proves the claim at X. □

Lemma C.36 (Dependence on the early error)**.** Under the hypotheses of Lem. C.33, put

$$\mathfrak{d}_k(D) = C_\Sigma \Lambda_k^{\psi}(D) + C'_\Sigma \Lambda_k^{0}(D).$$

With the later families, coefficients and boundary terms fixed, every frame member and every one-lattice potential member is affine in D. Its linear part has norm at most $\mathfrak{d}_k(D)$ in the norm used by the step. The output activity, its pointed terms and the extracted numbers are holomorphic functions of the input members. Their differences from the outputs with D removed are at most $\mathfrak{d}_k(D)$, in the ordinary activity and scalar array norms and in modulus for the numbers.

At the step with D removed, extraction uses the same bounded linear functionals of the new infinite-lattice activity. It removes the trace of the mean quadratic moment. Without an additional symmetry assumption this asserts trace zero, not that the entire matrix moment vanishes. ■

Proof. The linearity and the four relevant bounds are

$$2\Theta_k^{\sharp} C_{\mathrm{tab}} \Lambda_k^{\psi}(D), \quad C_{\mathrm{V}} C_{\mathcal{A}} s_k^{\mathrm{fx}} C_{\mathrm{ert}} C_{\varpi} \frac{L}{L-2} \Lambda_k^{0}(D),$$

and the two summands of s_D in Lem. C.33. The other members read no early family. These estimates give the first assertion and, by the definition of the constants, $K_{\mathrm{lin}} s_D \le \mathfrak{d}_k(D)$.

Replace the early family by $R + \zeta D$ and keep all other data fixed. If $s_D = 0$ the outputs do not change. Otherwise set $\rho = c_{\mathrm{cl}}/(4 s_D) \ge 1$. The potential has size at most $c_{\mathrm{cl}}/2$ on $|\zeta| \le \rho$ and less than $3c_{\mathrm{cl}}/4$ on $|\zeta| < 2\rho$. The expansion [QCDM1, Lem. 7.5] is holomorphic on the latter disc. On the former disc its outputs are bounded by $K'_M(3c_{\mathrm{cl}}/2)e^{-\kappa_+ d}$, including the test tail of size at most c_{cl}.

At a potential of size at most $c_{\mathrm{cl}}/2$, perturb in an even direction f by $\lambda c_{\mathrm{cl}} f/(4\|f\|)$. On $|\lambda| = 1$ the potential has size at most $3c_{\mathrm{cl}}/4$. Cauchy's estimate and the test tail give the variation bound $7K'_M\|f\|e^{-\kappa_+ d}$. The insertion at a pointer costs at most the fermion potential, hence at most $c_{\mathrm{cl}}/2$, so its pointed output is bounded by $(7/2)K'_M c_{\mathrm{cl}} e^{-\kappa_+ d}$. It is holomorphic in ζ: the joint family consisting of the pointing path plus λ times its insertion has potential size below c_{cl} for $|\zeta| < 2\rho$ and $|\lambda| < 1/3$. Differentiate in λ and integrate the pointing path. Uniform Cauchy bounds justify that integral. Summing polymers through a cube gives the bound $4K_M c_{\mathrm{cl}}$ for the output and each pointed output in both norms under consideration.

The extraction functionals require no linear symmetry of the families. If m_ζ is the cell mean of the row sums of the quadratic kernel, they give

$$\mu(\zeta) = -(\nu\eta_{k+1}^2)^{-1}\operatorname{tr}\mathsf{m}_\zeta, \quad b(\zeta) = b^\circ + \mathsf{cf}^\sharp[\widetilde{E}_{\zeta,0}], \quad e(\zeta) = e^\circ + \widetilde{e}(\zeta).$$

The constants b°, e° use only the normalizers, the fixed coefficient and fine mass. The coefficient bound [QCDM1, Lem. 7.8] gives $\|\mathsf{m}_\zeta\|_{\mathrm{tr}} \le C^\psi_{am} h_F^{-2}\|\widetilde{E}_{\zeta,\ge 2}\|$ and $|\eta_{k+1}| \ge c_b$. Thus the mass extraction multiplies the activity bound by at most $K_\mu = 1 + c_{\mathsf{S}} C^\psi_{am} c_b^{-2}/\nu$. The action functional has bound $C_{am} r_\alpha^{-2}$. Translation covariance gives $|\widetilde{e}| \le M^{-3}\|\widetilde{E}_{\zeta,0}\|$: the cell sum of its pointed totals is the sum over polymers through the cell divided by their number of cubes. Scalar extraction consequently has factor at most $2 + C_{am} r_\alpha^{-2}$. All three functionals are bounded and linear.

The extracted outputs, pointed terms and numbers, after their constant contributions are removed, are therefore holomorphic and bounded by $M_{\mathrm{out}} = 4K_M c_{\mathrm{cl}}(1+K_\mu)(1+K_{\mathrm{num}})$. The remaining scalar bond terms need separate mention. Only the fourth scalar-system entry changes with D; it is the bracket localization of its scalar array. The two scalar potentials have size at most $c_{\mathrm{cl}}/4 = w_*/4$, and their difference has size at most $s_D \le w_*/4$. The variation estimate [QCDM1, Lem. 7.4] therefore gives a pointed bond difference at most $K^{\mathrm{v}}_{\mathbf{M}} K_\Lambda^{-1} s_D e^{-\kappa_+ d}$, and after summation at most $C_E K^{\mathrm{v}}_{\mathbf{M}} K_\Lambda^{-1} s_D$.

For a normed-space holomorphic function on $|\zeta| < 2\rho$ bounded by M_{out} on $|\zeta| \le \rho$, Schwarz's lemma applied to $g - g(0)$ and norm-one functionals gives $\|g(1) - g(0)\| \le 2M_{\mathrm{out}}/\rho$. This equals $32K_M(1+K_\mu)(1+K_{\mathrm{num}})s_D$. Add the scalar bond contribution to obtain $K_{\mathrm{lin}} s_D \le \mathfrak{d}_k(D)$. At $\zeta = 0$ the remaining mean moment is $\mathsf{m}_0 + \mu(0)\eta_{k+1}^2\mathbf{1}$, whose trace is zero by the displayed extraction formula. This proves the final qualification as well. □

Lemma C.37 (Summation over the layers)**.** Let a flow of Def. C.31 have the identities and step bounds of Lems. C.33 and C.34 through level k. In the constants of the upper history bound [QCDM1, Thm. D], replace

$$E_0 \text{ by } E_0 + \mathfrak{d}^0, \qquad C_{\mathrm{leg}} \text{ by } C_{\mathrm{leg}} + C^{\square}_{\mathrm{leg}}\mathfrak{d}^\psi.$$

Here $C^{\square}_{\mathrm{leg}}$ is the per-cube constant in its regular-leg proof. With these constants, the scalar and fermion sums obey the same layer bounds. The action shift to be included in the scalar bound is

$$L^{t_1}|x_{t_1} - \mathsf{t}_{t_1}|A(\mathsf{U}) + \sum_{t_1 < j \le k} |L^j b_j A(\varphi_j, \mathsf{U})|.$$

Moreover $|\mathfrak{f}_k| \le C_{\mathrm{fs}}$, $f_n(\mathcal{S}) = 0$ for $n \le t_1$, and

$$|f_n(\mathcal{S})| \le C^{\mathrm{F}}_\flat(1 + \log \mathsf{t}_{n-1})\#_n(Z_n^\sim) \quad (n > t_1).$$

The last constant can already be taken at the original E_0.

The pointwise and integrated bounds of that upper history theorem hold with $\operatorname{Re} E_k$ in place of E_k and with the present action profile. In particular, for the free-coordinate density g_h,

$$\|g_h\| \le e^{-\beta A(\mathsf{U})+C_E^{\mathrm{F}}\Sigma'(h)+\operatorname{Re} E_k-\operatorname{Re}\sum_n f_n} \prod_{(t,D)} |\mathsf{u}_{t,D}[\mathscr{Q}_{t,D}]|,$$

$$e^{-\operatorname{Re} E_k+\operatorname{Re}\sum_n f_n}\int \|g_h\|\, d\mathscr{M}_h^{\mathrm{fr}} \le e^{C^{\dagger}_{\mathrm{maj}}(n_k+\Sigma_Z(h))}\prod_t e^{-c_\star p_0(g_t)^2 n_t^{\mathrm{cr}}(h)}, \qquad C^{\dagger}_{\mathrm{maj}} = 48L^3C_E^{\mathrm{F}} + C'_\star.$$

The pointwise bound also holds with the actual action profile, or with $L^{t_1}x_{t_1}A$, in place of βA. These bounds hold for every x_{t_1} in its window. ■

Proof. Apply the scalar sum estimate [YM3, Prop. 4.147] to one level at a time, putting zero families at the other levels. This is permitted because its size parameter is free and (F4) supplies each family's hypotheses. On a layer $\mathcal{L}_n$, $n \ge t_1$, the sum of the resulting sizes is bounded by

$$\sum_{j\le n} L^{-(n-j)}\mathsf{g}_j(E) \le \frac{L}{L-1}E_0 + \Lambda_n^0(D) \le \frac{L}{L-1}(E_0+\mathfrak{d}^0).$$

There are no earlier layers. For the action profiles the same per-level plaquette count gives $L^jA(\varphi_j,\mathsf{U}) \le 3\sum_{n\ge j} L^{-(n-j)}(C_{\mathrm{reg}}\varepsilon_n)^2 \#_n(\mathcal{L}_n)$. At $j=t_1$ its profile is one. The coefficient at each layer is therefore at most

$$3\left[\bar{B}L^{-(n-t_1)} + B_\sharp \sum_{t_1<j\le n} L^{-(n-j)}\right] = 3\bar{B} \le \tfrac{13}{4}B_\sharp.$$

This is precisely the profile share of the old scalar bound, including the initial coefficient displacement.

The regular-leg proof of [QCDM1, Thm. D] bounds the level-j contribution per cube in $\mathcal{L}_n$ by $C^{\square}_{\mathrm{leg}}L^{(n-j)/4}$ times its pointed norm at a pointer. It uses the zero cell moment, covariance, the whole-core invariance and the coefficient estimate for $q_j(z)$, all supplied by (G1). The families $R^{(j)}$ and the later families have the usual size $C_2\phi(\mathsf{t}_{j-1})^2$ at each pointer. Their sum is the one included in C_{leg}, since

$$\sum_{j\le n} L^{(n-j)/4}\phi(\mathsf{t}_{j-1})^2 \le \phi(\mathsf{t}_{n-1})^2 \sum_{d\ge 0} L^{-3d/4}(1+d\log L)^{2\mathfrak{p}_0}.$$

The extra family contributes at most $C^{\square}_{\mathrm{leg}}L^{-1/4}\Lambda_n^{\psi}(D) \le C^{\square}_{\mathrm{leg}}\mathfrak{d}^{\psi}$ per cube. The mass term has the old bound because $L^n|s_h| \le \max\{|s_n|, L|s_{n-1}|\} \le L\bar{B}_M$. Summing cubes proves the claimed replacements.

The scalar values of $D^{(j)}$ vanish at the trivial pair. Those of $R^{(j)}$ and later families have zero infinite-lattice cell mean. Their torus discrepancy is the difference of wrapping and nonfitting polymers, with decay at rate $\kappa+1$ and size E_0. The wrapping estimate of [QCDM1, Lem. 9.15] therefore gives $|\mathfrak{f}_k| \le C_{\mathrm{fs}}$. For $n > t_1$, its bound on f_n uses only that step: the vacuum increment, $|e_n| \le C'_e(1+\log x_{n-1})$, the new scalar size E_0, the collar and normalizer, and $|\log(x_{n-1}/\mathsf{t}_{n-1})| \le 2\bar{B}/\mathsf{t}_{n-1}$. These are unchanged.

The boundary sum is bounded by Lem. C.34, $\mathsf{g}_j \le E_0$, $|b_j| \le B_\sharp$, $|\mu_j| \le 4\bar{B}_M/\mathsf{t}_j \le (1+L)\bar{B}_M$ and $\bar{v}_j \le c_{\mathrm{cl}}/4$. Its domain count is unchanged. Adding the stored-factor bound gives $C_E^{\mathrm{F}}\Sigma'(h)$ with the two replacements above. All its defining terms are affine with nonnegative coefficients in E_0 and C_{leg}.

For the measure estimate use the bounded Borel coefficients and domination in Lem. C.26. The coordinate-density construction of [QCDM1, Lem. 10.7] applies coefficient by coefficient. The even exponentials commute and $\|e^G\| \le e^{\|G\|}$. In the action write

$$-A(\beta_h,\mathsf{U}) = -\beta A(\mathsf{U}) + L^{t_1}(\mathsf{t}_{t_1}-x_{t_1})A(\mathsf{U}) + \sum_{t_1<j\le k} L^j b_j A(\varphi_j,\mathsf{U}).$$

The last terms are bounded by the profile sum already proved, with no sign restriction on the displacement. The same argument keeps either the actual profile or $L^{t_1}x_{t_1}$ as reference coefficient. Finally the pure-gauge piece estimate used in [QCDM1, Thm. D, proof] depends only on the history's canonical background data. It supplies $e^{-c_\star\sum_t p_0(g_t)^2 n_t^{\mathrm{cr}}}$ and the integrable piece majorants. The inequalities $\Sigma'(h)\le 48\Sigma_{\mathcal{L}}(h)$ and $\Sigma_{\mathcal{L}}(h)\le n_k+L^3\Sigma_Z(h)$ then give the integrated bound. No positivity of the perturbed fermion term has been used. □

Proposition C.38 (The induction above the base)**.** Assume the constants of [QCDM1, Thm. D] and (Γ^{F}) in Rem. C.30. Let a flow of Def. C.31 be given. For every $t_1\le k\le K$ and $h\in\mathcal{H}_k^{[t_1]}$, its term has identity (I_k) with its actual families, coefficient $\beta_h=L^{t_1}x_{t_1}-\sum_{t_1<j\le k}L^j b_j\varphi_j$, stored tower form, mass profile, and regular constant $\mathfrak{f}_k$. Its boundary terms have levels above t_1 only and satisfy Lem. C.34. Its vacuum constant is

$$-\mathcal{E}_{k,\mathcal{S}}=E_k-\sum_{n\le k}f_n(\mathcal{S}),\qquad f_n=0\quad(n\le t_1).$$

The exponent is defined and has one value at evaluation configurations off one null set. Every coefficient of the measure is dominated by a constant times the pure history term.

Every next step satisfies (Sm), (Fr), $\mathsf{sz}_k^{\natural}\le\varepsilon_\flat$ and $\bar v_{k+1}\le b_k^{\mathrm{blk}}$, as well as the variation estimate of Lem. C.36. The boundary terms agree for equal excerpts under the torus floor of Lem. C.35. All the bounds of Lem. C.37 hold.

If the families and numbers depend holomorphically on a parameter on a domain where (G1)–(G3) hold, the frame members, outputs, extracted numbers and bounded Borel coefficient functions are holomorphic there. The real x_i, and hence b_i, τ and the action weights, are constant on each connected component. ■

Proof. At $k=t_1$ there is one history. Its identity is Lem. C.26 with $c=E_{t_1}+\mathfrak{f}_{t_1}$ and $\sigma=s_{t_1}$. There is no boundary term, and all earlier frame hypotheses are empty. The exponent has precisely the claimed regular and vacuum constants.

Assume the identity at level k and the step bounds at earlier levels. Lem. C.33 gives (Sm), (Fr) and the new size bounds. The one-lattice step in Prop. C.29 is the step required by (G2), on the torus and on the infinite lattice with the same families and coefficients. Its extraction numbers are read from the infinite-lattice activity, as in the definition of the flow. Thus its new family is $E^{(k+1)}$, and its numbers are $e_{k+1},b_{k+1},\mu_{k+1}$. These have (F3) and the required windows by (G1)–(G2). Interior agreement identifies the same output at every history. Lem. C.34 supplies the new boundary terms, and Prop. C.29 gives the measure identity at level $k+1$. The extraction rule adds the new trivial values to $\mathfrak{f}_k$ and $e_{k+1}n_{k+1}$ to E_k; its history difference is f_{k+1}. This proves the asserted constants.

The induction gives the hypotheses of Lem. C.35 simultaneously for both histories being compared, since the flow's families and numbers are common to all histories. It gives those of Lem. C.36 at every step and Lem. C.37 at every level. The definedness, common value and domination follow from Lem. C.26 and Prop. C.29.

For the last assertion, a real-valued holomorphic coefficient is constant on each connected component. Thus the Gaussian, quadratic correction, tower and fermion integration family have no parameter dependence. The remaining frame members are formed by linear operations, background substitutions and earlier boundary terms. The expansion and pointing are holomorphic in their input potentials, and extraction is bounded and linear. The cluster part of a generated boundary term is an output difference; its strip row is a finite integer combination of output values. Its other kinds have no parameter dependence. Edges are formed from pointed outputs and the holomorphic mass numbers. Induction therefore proves their holomorphy. Finally Lem. C.26 gives the bounded Borel coefficient functions, locally uniformly; the early sizes are bounded by (G3). □

To apply this theorem to Prop. 11.14, take $t_1 = t_b$. The norms of Def. 11.3 are exactly the two sizes in Def. C.24, with the maximum over both lattices. The zero cell moments, joint invariance and translation covariance in the hypotheses of Prop. 11.14 give (F3): the zeroth moment commutes with colour by constant gauge invariance, is periodic by covariance, and its cell sum is the stipulated zero moment. The other structural hypotheses give (F1), (F2), and (F4), and the pointed sum rule and scalar normalization give (G1). The post-base step and pointed-size hypotheses are (G2); its split into $R + D$ and the lag sums are (G3). Its history measures are the measures in Def. C.31, since both start with the same density and apply the same subsequent weights and partial kernels.

The linear-entry assertion of Prop. 11.14 is Lem. C.36; its history induction and locality are Prop. C.38; its scalar, leg and integrated estimates are Lem. C.37. Per layer cube the additional leg contribution is $C^{\square}_{\mathrm{leg}} L^{-1/4} \Lambda^{\psi}_m(D)$. When $D = 0$ both replacements vanish, so the constants reduce to the original upper-history constants. The parameter assertion includes the necessary constancy of the real action coefficients.

C.14. **Scalar estimates with a prescribed past.** The scalar estimates retain the size of each early level. This is needed because an interpolation may have large individual early terms while its weighted sum at the base is small.

Definition C.39 (Levels and histories)**.** Fix $1 \le u < s < K$, and put $t_1 = K - u$, $t_0 = K - s$. We use $\mathcal{H}_k[t_1]$ from Def. C.1: its histories agree with $h^{(t_1)}_{\emptyset}$ through t_1. For numbers $a_j \ge 0$, $j \le t_1$, write

$$\Lambda^0_n[a] = \sum_{j \le t_1} L^{-(n-j)} a_j, \qquad n \ge t_1.$$

Thus $\Lambda^0_n[a] = L^{-(n-t_1)} \Lambda^0_{t_1}[a]$. ■

Lemma C.40 (The early regions)**.** For $h \in \mathcal{H}_k[t_1]$, the regions through t_1 are the torus. All early pairs are regular, the layers below t_1 are empty, and no boundary or edge term is generated through t_1. Boundary terms and the corresponding vacuum corrections therefore have levels strictly greater than t_1. ■

Proof. These are the geometric statements of Lem. C.25. They depend on the regions of the history and not on the activities inserted in its exponent. □

Proposition C.41 (Scalar sums by level)**.** Assume the geometric, regularity, and domain hypotheses for the scalar sums in [YM3, Prop. 4.147]. At level j, let e_j bound the pointed scalar terms at rate $\kappa + 1$; let $B_j e^{-\kappa d(X)}$ bound each generated boundary term. Let $C_{\mathrm{ly}} \ge 1$ be the regular-field constant. For a history evaluation these hypotheses are applied to the glued field, with $C_{\mathrm{ly}} = C_{\mathrm{reg}}$. Write

$$N_n = \#_n(\mathcal{L}_n), \qquad A_n = \sum_{j \le n} L^{-(n-j)} e_j.$$

For a level-j pointer z lying in $\mathcal{L}_n$,

$$\sum_{X \colon z \in X \subset \Lambda_j} |E_j(X, U, z) - E_j(X, 1, z)| \le e_j L^{-4(n-j)} \min\{C^{\mathrm{sc}}_a, C'^{\,\mathrm{sc}}_a (C_{\mathrm{ly}} \epsilon_n / \alpha_0)^2\}.$$

The regular sum of level j is consequently at most

$$C^{\mathrm{sc}}_a e_j \sum_{n \ge j} L^{-(n-j)} N_n,$$

with the corresponding curvature factor available term by term. The profile term satisfies

$$L^j A_{\varphi_j} \le 3 \sum_{n \ge j} L^{-(n-j)} (C_{\mathrm{ly}} \epsilon_n)^2 N_n.$$

The boundary and edge terms of level j have total norm at most

$$C'_f (B_j + 8M^3 e_j) M^{-3} \#_j(\widetilde{Z}_j) \le 27 C'_f (M^{-3} B_j + 8 e_j) \#_j(Z_j).$$

In particular, if $|b'_j| \le B_\sharp$, then

$$|\text{regular sum}| + |\text{boundary sum}| \le \sum_n \left[C_a^{\mathrm{sc}} A_n + \tfrac{13}{4} B_\sharp (C_{\mathrm{ly}}\epsilon_n)^2\right] N_n + 27 C'_f \sum_j (M^{-3} B_j + 8e_j)\#_j(Z_j).$$

The increments b'_j may be complex. ■

Proof. Apply the scalar estimate of [YM3, Prop. 4.147] to the array with its sole nonzero entry at level j. Zero entries satisfy the local-term conditions, so this gives the first estimate with e_j in place of its common bound. Counting the level-j pointers in a level-n cube costs $L^{3(n-j)}$; the factor $L^{-4(n-j)}$ therefore leaves $L^{-(n-j)}$. This proves the regular estimate. The profile estimate in that proof is geometric and does not use the increments. Multiplication by $|b'_j|$ thus permits complex increments. The boundary count uses the generated terms through B_j, and the edge terms through $8M^3 e_j$. At an evaluation configuration the same scalar proof applies to the glued field, with $C_{\mathrm{ly}} = C_{\mathrm{reg}}$; the bounds on generated terms are coefficient bounds there.

Finally interchange the finite sums over j, n. For the profiles use $3\sum_{j\le n} L^{-(n-j)} \le 3L/(L-1) \le 13/4$. The other two terms are exactly the displayed sums. No sign of an activity or increment enters these estimates. □

Corollary C.42 (The early scalar sum)**.** Suppose $\Lambda_j = \mathbb{T}_\ell$ for every $1 \le j \le t$. Then the layers below t are empty, and the contribution of levels $j \le t$ to the regular sum is at most

$$C_a^{\mathrm{sc}} A_t \sum_{n \ge t} L^{-(n-t)} N_n.$$

The curvature version of Prop. C.41 holds as well. There is no early boundary contribution. ■□

Proof. For $n \ge t$, $\sum_{j\le t} L^{-(n-j)} e_j = L^{-(n-t)} A_t$. The boundary sets of those levels are empty by Lem. C.40. □

C.15. **The steps at the interpolation parameter.**

Definition C.43 (The interpolated flow)**.** Let the data be as in Def. C.6, and put $w_\lambda = w + \lambda w^\Delta(s)$. For the step estimates below, take $z \in \mathcal{O}$, $s = s(z)$, and $|\lambda| < r_\lambda$. The early arrays are $E^{(j)}(s) + \lambda\Delta^{(j)}(s)$. Above t_1 use the small-field step with these earlier arrays, the fine mass $m_0 - a_K \widehat{J}$, and the zero-source reference. Extract the ordinary numbers e_j, b_j, μ_j; form the mass increment from the moments relative to that reference. The scalar source energy is

$$e_j(X; \mathfrak{c}) = \mathfrak{E}_j(X; \mathfrak{c}) - \mathfrak{E}_j(X; 0).$$

Start with $\sigma_{t_1}(s, \lambda) = \sigma_{t_1}(s) + \lambda\delta\sigma(s)$; below t_1 keep the source chain. Above it set

$$\sigma_{j+1}(s,\lambda)(x) = L\sigma_j(s,\lambda)(x) + \nu_{j+1}(y_{j+1}(x); \mathfrak{c}_j(s,\lambda)) - \mu_{j+1}.$$

The vacuum energy is

$$E_k(s,\lambda) = E_{t_1}(s) + \lambda\mathfrak{n}(s) + \sum_{t_1 < j \le k} \left[e_j n_j + \sum_X e_j(X; \mathfrak{c}_{j-1}(s,\lambda))\right].$$

For a history use the same start and only the pieces with polymer inside Λ_j in its increment ν_j^h. Its coordinates use σ_j^h on Ω_j and σ_j off Ω_j. All these rules are applied only while the indicated steps are defined.

We use three induction conditions.

(i) $(\mathrm{Int}_i)[s,\lambda;h]$ means that the step from i at the history coordinates belongs to the fixed pair domain $2\mathscr{D}_i(\alpha_0)$, has $\|\mathcal{V}_0\| \le c_{\mathrm{cl}}/2$ and $\|\mathcal{V}_\psi\| \le b^{\mathrm{blk,s}}(\gamma)$, and produces the local, Borel, holomorphic pointed activities of Prop. C.15. Their scalar pointer bound is E_0^{s} at rate $\kappa+1$. The scalar fluctuation norm uses $\tau_i^\lambda = \max\{\tau_i(w;\alpha_0), \tau_i(w_\lambda;\alpha_0)\}$.

(ii) $(\mathrm{Sh}_k)[s,\lambda;h']$ means that the first seven fermion classes of the step along h' have total size at most $b^{\mathrm{tot,s}}(\gamma)$, on every tier at rate r_V and every external-field box.

(iii) $\mathsf{I}^{\mathrm{F}}(k)[s,\lambda]$ is the representation and the eight bounds of Def. C.19, now for the measures of Lem. C.7. Its gauge weight is $\beta_k - a_K^{-1} w_\lambda$, its mass field is

$$s_h + \sigma_h(s,\lambda), \qquad \sigma_h = L^{-t_1}\sigma_{t_1}(s,\lambda) + \sum_{t_1<j\le k} L^{-j}(\nu_j^h - \mu_j)\varphi_j,$$

and its early arrays are those above. All generated boundary terms have levels greater than t_1. The frame bound is $64 b^{\mathrm{tot,s}}(\gamma)/51$. Its vacuum label is

$$-\mathcal{E}_h = E_k(s,\lambda) - \sum_{t_1<n\le k} f_n(h) + \sum_{t_1<n\le k} g_n(h;s,\lambda), \qquad |g_n| \le C'_{\mathfrak{E}} \#_n(\widetilde{Z}_n).$$

The other bounds are the ordinary sourced bounds, including the kept halves, the loss in the gauge action, the stored factors, and the coefficientwise norm of the exponent.

■

Corollary C.44 (The scalar regular sum)**.** Suppose $(\mathrm{Int}_i)[s,\lambda;h]$ holds for $t_1 \le i < k$. At every evaluation configuration of $h \in \mathcal{H}_k[t_1]$, the scalar regular sum and the absolute profile terms are together at most

$$\left[\tfrac{13}{12} C_a^{\mathrm{sc}} E_0^{\mathrm{s}} + C_a^{\mathrm{sc}} |\lambda| \Lambda_{t_1}^0[\rho(s)] + \tfrac{13}{4} B_\sharp\right] \Sigma_{\mathcal{L}}(h) \le C_5^{\mathrm{s}} \Sigma_{\mathcal{L}}(h).$$

■□

Proof. Apply Prop. C.41 and Cor. C.42 with $e_j = E_0^{\mathrm{s}} + |\lambda|\rho_j$ for $j \le t_1$ and $e_j = E_0^{\mathrm{s}}$ otherwise. The ordinary part sums to at most $13E_0^{\mathrm{s}}/12$. The extra part is at most $c_0^\Delta \le M^{-3}\mathsf{B}$. The admitted curvature factor is at most one. Since

$$C_m^{\mathrm{sc}} = \tfrac{13}{12} C_a^{\mathrm{sc}} + 432 C'_f + \tfrac{13}{4}, \qquad C_5^{\mathrm{s}} = C_m^{\mathrm{sc}}(E_0^{\mathrm{s}} + B_\sharp + M^{-3}\mathsf{B}),$$

the stated bound follows. □

Lemma C.45 (The scalar class)**.** Under the interior and boundary hypotheses of Def. C.43, form the scalar list from the early arrays, the later activities, and the boundary terms already generated. The counterterm at the early levels is zero. Then its class rg, including the gauge-weight terms of order at least three, has size at most $\varepsilon_\flat$. ■

Proof. The coefficient of a scalar entry from level j, in the regular-frame bound of [YM3, Prop. 4.122], is

$$C_{\mathrm{ert}} L^{-(k-j)} \sum_{i=0}^{k-j} L^i \varpi_i.$$

Writing $C'_h = \sum_{i\ge 0}(i+1)L^i\varpi_i$ and $C_{\mathrm{sz}} = LC'_h/(L-1)$, its total gives

$$\mathsf{size}^E \le C_{\mathrm{sz}} C_{\mathrm{ert}} \left[E_0^{\mathrm{s}} + \alpha_0^2 B_\sharp + (1-L^{-1})L^{-(k-t_1)}|\lambda|\Lambda_{t_1}^0[\rho(s)]\right].$$

Hence this is at most $C_{\mathrm{sz}} C_{\mathrm{ert}}(E_0^{\mathrm{s}} + c_0^\Delta + \alpha_0^2 B_\sharp)$. The regular-frame estimate proved in Lem. C.18 applies, because $E_0^{\mathrm{s}} + c_0^\Delta \le \sqrt{2} E_0^{\mathrm{s}}$. The doubled source ceiling leaves $\varepsilon_\flat/2$ for these terms. The terms of order at least three cost the remaining $\varepsilon_\flat/2$, since $(w_\lambda, J) \in \mathcal{P}$. Empty edge and counterterm lists are allowed; generated terms enter only above the base. Thus the scalar-list hypotheses used in that estimate are unchanged. □

Lemma C.46 (The fermion frame recursion)**.** Assume (Sh_k), the preceding interior conditions, and the boundary clause of $\mathsf{I}^{\mathrm{F}}(k)$. Let $\Sigma_k^{\natural}$ and v_k be the largest frame and insertion sizes of levels $t_1 < j \le k$, with value zero if there are no such levels. Then

$$\mathsf{sz}_{k+1}^{\natural} \le \tfrac{13}{64}\max\{\Sigma_k^{\natural}, v_k\} + b^{\mathrm{tot,s}}(\gamma) \le \tfrac{64}{51} b^{\mathrm{tot,s}}(\gamma) \le \tfrac{16}{17}\varepsilon_{\flat}.$$

■

Proof. The first seven classes are bounded by (Sh_k). The eighth and ninth read the previously generated terms with factor $(1/8)(9/8)$; the tenth has factor $1/16$. Their sum is $13/64$, as in Lem. C.18. There is no generated term below the base, and $v_k \le b^{\mathrm{blk,s}}(\gamma) \le b^{\mathrm{tot,s}}(\gamma)$. The inequality $13/51 + 1 = 64/51$ closes the recursion. □

Lemma C.47 (Agreement in the interior)**.** Suppose the conditions in Def. C.43 hold through the step from k to $t = k+1$. For $X \subset \Lambda_t$, the terms, activities, and cluster sums of the history step are those of the small-field step at its history coordinates. The cluster sum is

$$E^{\mathrm{lab}}(X) = \mathbf{G}^{v}(X) + \widetilde{E}(X).$$

The pointed outputs are the pointed activities of that step, with the bounds in (Int_k). In a frame without a centered origin, with at most four classes each of size at most $\epsilon_{\flat}$, the first variation of an even analytic insertion W on the tiers has bound

$$K^{\mathrm{lb}}\|W\|_{r_V} e^{-\kappa'' d_t(X)}.$$

In the small-field polymer norm the corresponding bound is $4K_M\|W\|$. ■

Proof. Inside Λ_t all profiles equal one, the decoupled mean vanishes, and there is no edge or boundary term. Both steps use the same gauge covariance, fermion covariance, fine mass, scalar list, and background maps. Their mass member is $-L^{-k}\mathsf{S}_1((s_k + \sigma_k^h)\psi^{\mathrm{bg}})$. Every early array is read at all its regular pointers.

At center zero the two centered terms satisfy $\mathbb{W}^{\mathrm{q}} + \mathbb{W}^{\mathrm{l}} = \mathcal{V}^{v} + \mathcal{V}^{\mathrm{lin}}$, and $\mathbb{W}^{\times} = 0$. Merge their origins in the multilinear cluster series. This gives the same terms as the small-field series. The Gaussian origin is $\mathcal{V}^{\mathrm{q}} + \mathcal{V}^{v}$, so its contribution is $\mathbf{G}^{v}$. The linear term remains in the scalar potential. Pointing and differentiation of a finite insertion commute with the locally uniformly convergent series. This proves the identities, including their normalization, by the argument of [QCDM1, Lem. 9.10 and Prop. 10.14].

For the norm estimate, the potential on the interpolation path has size at most $(1/2 + 1/8)c_{\mathrm{cl}}$. Add $\zeta c_{\mathrm{cl}} W/(4\|W\|)$. For $|\zeta| < 3/2$ it remains below c_{cl}; on $|\zeta| = 1$ it is at most $7c_{\mathrm{cl}}/8$. The test tail costs at most $c_{\mathrm{cl}}/16$. Cauchy's estimate therefore gives $(15/4)K'_M\|W\| \le 4K_M\|W\|$. Under the stated restrictions on the frame and insertion, the same insertion in the joined frame gives the stated K^{lb} bound and its exponential polymer factor. □

Proposition C.48 (The centered history frame)**.** Assume $\mathsf{I}^{\mathrm{F}}(k)[s,\lambda]$, $(\mathrm{Int}_i)[s,\lambda;h]$ for $t_1 \le i \le k$, and $(\mathrm{Sh}_k)[s,\lambda;h']$, where $h' = (h,\mathsf{o}) \in \mathcal{H}_{k+1}[t_1]$. The step has a centered frame, with the constants of [QCDM1, Lem. 10.12]. For $v' = -w_\lambda/\mathsf{b}$, its centered members satisfy

$$\|\mathbb{W}^{\mathrm{q}} + \mathbb{W}^{\mathrm{l}}\|^{\mathrm{rad}}_{\mathrm{I}\star,r_V} \le \mathsf{k}_{\odot}(\sigma_s + \varrho'_s g_k) \le \tfrac{1}{2}\mathsf{m}_{\odot}, \qquad \|\mathbb{W}^{\times}\|^{\mathrm{rad}}_{\mathrm{I}\star,r_V} \le C_{\mathrm{cr}}^{V}\mathsf{C}_{\mathcal{A}} A_1 \varrho\, \mathsf{t}_k^{-1}(\log \mathsf{t}_k)^{\mathfrak{p}_0} \le \tfrac{1}{4}c_M^{\mathrm{O}}.$$

The centered class has size at most c_M^{O}. The other four classes have sizes

$$\mathsf{sz}^{\mathrm{rg}} \le \varepsilon_{\flat}, \quad \mathsf{sz}^{\mathrm{bd}} \le \tfrac{1}{2}\varepsilon_{\flat}, \quad \mathsf{sz}^{\mathrm{q}} \le \varepsilon_{\flat}, \quad \mathsf{sz}^{\natural} \le \tfrac{16}{17}\varepsilon_{\flat}.$$

Their amplitude obeys

$$\varepsilon_{\mathsf{w}}^{\mathfrak{F}+} L^3 \le \frac{5}{32\mathsf{K}_{\mathrm{ap}}^{+}}, \qquad \mathsf{b}_{\mathrm{cs}}^{\mathfrak{F}+} \le \tfrac{9}{32}.$$

The cluster sums are Borel coefficientwise, analytic, local, jointly gauge invariant, and functions of the pair. For $X \not\subset \Lambda_{k+1}$, the full cluster sum and its Gaussian part each have bound $9e^{-\kappa'' d_{k+1}(X)}/32$. Their difference has bound $9e^{-\kappa'' d_{k+1}(X)}/16$, and its part with generators has bound

$$K^{\mathrm{lb}} \mathsf{sz}^{\natural}_{k+1} e^{-\kappa'' d_{k+1}(X)}.$$

For $X \subset \Lambda_{k+1}$, Lem. C.47 applies. ■

Proof. The centered terms read the source only through its weight. Here

$$\sup |v'| \le 2\varrho/\mathsf{b} \le \sigma_s, \qquad \mathsf{t}_k |v'(p) - v'(p')| \le \sqrt{3} M \operatorname{Lip}(w_\lambda) \le \varrho'_s$$

for the neighboring plaquettes in the centered-frame proof. These follow from $\|w_\lambda\|_\sharp \le 2\varrho$. The proof of [QCDM1, Lem. 10.12] therefore gives the centered estimates on exactly the same tiers.

The regular class is Lem. C.45. The boundary class uses the generated-term bound $\mathsf{B}_\natural$, the formed-factor bound $\mathsf{B}^\flat_{\mathrm{fr}} + 1$, and $8M^3 E^{\mathrm{s}}_0 \le \mathsf{B}_\natural$ for its edges. These are clauses of $\mathsf{I}^{\mathrm{F}}(k)$; no early edge occurs. The quadratic class is the source-independent class of $\mathcal{V}^{\mathrm{q}}$. The fermion class is Lem. C.46.

Analyticity on the tiers follows as in the last paragraph of the proof of Lem. C.18. In the first seven classes, the old arrays are holomorphic on the fixed cores and the mass coefficient is independent of the configuration. The last three classes use later boundary coefficients, their background legs, the fixed leveling functions and shifts, and the same substituted triples as the ordinary frame. These operations preserve the pair dependence and give locally uniform series. They also preserve the Borel and invariance properties. The source-dependent normalizer is the one at the fine mass $m_0 - a_K \widehat{J}$.

The four sizes sum to less than $6\varepsilon_\flat$. The choices of the ordinary frame constants give

$$6e\varepsilon_\flat L^3 = \frac{3}{32\mathsf{K}^{\max}_{\mathrm{ap}}} \le \frac{3}{32\mathsf{K}^{+}_{\mathrm{ap}}}, \qquad e(w_0^{\mathrm{O}})^{-1} c_M^{\mathrm{O}} L^3 = \frac{1}{16\mathsf{K}^{+}_{\mathrm{ap}}}.$$

Thus the amplitude is at most $5/(32\mathsf{K}^{+}_{\mathrm{ap}})$; adding the centered contribution $1/8$ gives $9/32$. The joined expansion in [QCDM1, Lem. 9.1] uses only these sizes and the ordinary fermion Gaussian properties. It proves the stated bounds, also for the Gaussian subframe and for the marked class with generators. Its termwise operations give all the stated regularity properties. The interior identity is Lem. C.47. □

Proposition C.49 (The step as an identity of measures)**.** Under the hypotheses of Prop. C.48, the term of h' has, coefficientwise off one null set, the representation

$$\widehat{\tau}^{\mathrm{F}}_{h'}[s,\lambda] = \mathsf{Stay}_{h'} \mathbf{1}^{\mathrm{T}}_{h'} \chi_{k+1}(\Omega_{k+1}) e^{\mathbf{\Phi}^{w}_{h'}} \exp[A^{\mathrm{F}}_{k+1,h'}(s,\lambda)] \mathscr{M}_{h'}.$$

The exponent is the one produced by the step in Def. C.43, with action weight $\beta_{k+1} - a_K^{-1} w_\lambda$, the updated mass field, the new pointed activities in the regular region, and the generated boundary terms elsewhere. Its density is Borel and integrable in the fibre norm, and the exponent takes one value on each evaluation set off the null set. The free-coordinate density properties of [QCDM1, Lem. 10.7] hold. ■

Proof. Apply the exact coordinate identity of [YM3, Prop. 5.51] coefficientwise, as in [QCDM1, Prop. 10.14]. Its geometric chain and change-of-variables conditions are independent of the parameters. The Gaussian normalizers use the fine mass of s. The weighted Wilson identity holds for every plaquette weight, here w_λ. All arrays are jointly gauge invariant; the added mass matrices commute with colour. Thus the source and the exponent take one value on evaluation sets. The expansion condition is Prop. C.48.

The terms themselves have the required transport identity by Lem. C.7: the next partial Berezin kernel acts on the preceding transported density, and the new scalar tests multiply its weight. The representation at level k supplies the remaining hypothesis of the coordinate identity. Inside Λ_{k+1}, extraction removes e_{k+1}, b_{k+1}, the source energies, and the restricted moment $-L^{-(k+1)} \mathsf{S}_1(\nu^{h'}_{k+1} \psi^{\mathrm{bg}})$.

The profiles return the mass increment and b_{k+1} to the mass field and action. Their residual terms are precisely the edge terms. This gives the displayed exponent.

For integrability first apply Tonelli to its fibre norm and then Fubini to each coefficient. Every coefficient measure in Lem. C.7 is dominated by the positive scalar reference measure. Its null sets therefore suffice for the complex parameters as well. The finite Grassmann exponential is invertible, so the algebraic identities pass to the exponential representation. All new coefficients are finite products or locally uniform limits of Borel coefficients; the source exponent is continuous in the gauge field. This proves Borel measurability. The one-value property just checked and the free-coordinate change of variables prove the last assertion. □

Proposition C.50 (A history step at the parameter)**.** If $\mathsf{I}^{\mathrm{F}}(k)[s,\lambda]$ holds and every prolongation h' satisfies the interior and share conditions of Prop. C.48, then its next term has the representation of Prop. C.49, the frame and output bounds of Prop. C.48, and the scalar regular-sum bound of Cor. C.44. Its measure is holomorphic in the source parameters and λ. The total integral satisfies $Z_{k+1}(z,\lambda') = Z_{t_1}(z,\lambda')$ for every $\lambda' \in \mathbb{C}$. ■

Proof. The first assertions have just been proved. The regular activities through level $k+1$ are produced by the same steps along h and h', since their regions through k agree. Thus Cor. C.44 applies at the next level. Holomorphy of the coefficient measures follows from Lem. C.7; preservation of the scalar integral follows from Prop. C.10. Neither of these measure assertions requires the small parameter disc used for the RG bounds. □

C.16. **Closing the induction.**

Remark C.51 (The scalar allowance). All constants used here were fixed in appendix C.2. In particular c_0^{Δ} satisfies four inequalities:

$$c_0^{\Delta} \le M^{-3}\mathsf{B}, \quad c_0^{\Delta} \le (\sqrt{2}-1)E_0^{\mathrm{s}}, \quad c_0^{\Delta} \le \tfrac{5}{12}K_M^{\bullet}c_{\mathrm{cl}}, \quad c_0^{\Delta} \le \frac{\alpha_0 C_w^{\mathrm{s}}}{16\sqrt{2}K_M^{\bullet}K_\Lambda^{\mathrm{s}}\mathsf{C}_{\mathcal{A}}A_1}.$$

The first bounds the scalar regular sum; the second admits its regular frame. The last two leave room for the extra scalar array in the one-step potential and its new activity. They impose no later restriction on previously chosen constants. The only additional lower bounds on the ordinary constants are the two regular-sum bounds in eq. C.ii.

Definition C.52 (A closed datum)**.** Let $R > 0$, and set

$$\mathrm{P}(R) = \{|\lambda| < R\}, \qquad R_k = RQ_\sigma^{-(k-t_1)}, \qquad Q_\sigma = 2\sqrt{L} + \tfrac{1}{4}.$$

A closed datum over $\mathcal{O}$ consists of the objects of Def. C.6, with the following requirements at every $s = s(z)$. The extraction data $F_j(s), G_j(s), \nu_j(s)$ and the infinite-lattice reference $G_j^{\infty}(s)$ of Def. C.2 are holomorphic in z, coefficientwise at every pair, with locally uniform bounds. Their quadratic terms have one barred and one unbarred field. The matrix cell sums of G_j^{∞}, including its mass forms, vanish. The same walks are used in F_j, G_j; on a torus the moved moments use only the permitted polymers of diameter less than a quarter of the side. The size $r_j^{\psi}(s)$ bounds the sums of summand norms at the point, the reference, and the infinite-lattice reference, and $c_{\mathsf{S}}h_F^2|\nu_j(s)|$, as in Def. C.2. Finally require

$$\begin{aligned} R\Sigma_k^{\psi}(s) &\le c_{\psi}^{\Delta}\phi_{k-1}, & \Sigma_k^{\psi}(s) &= \sum_{j\le t_1}\omega_{k+1-j}r_j^{\psi}(s) \quad (t_1 \le k \le K), \\ R\Lambda_{t_1}^0[\rho(s)] &\le c_0^{\Delta}, & R\|\delta\sigma(s)\| &\le \tfrac{1}{2}B_\sigma\phi_{t_1}, \\ (w+\lambda w^{\Delta}(s), J) &\in \mathcal{P} & (|\lambda| &< R). \end{aligned} \tag{C.xi}$$

These imply the conditions of Def. C.6 at every radius $r_\lambda \le R$. ■

Lemma C.53 (Sizes of the early arrays)**.** Let a closed datum be given. For $t_1 \le n \le K$, put

$$\epsilon_j[\lambda] = \begin{cases} \epsilon_j^{\mathrm{s},\psi}(\alpha_0) + |\lambda| r_j^{\psi}(s), & j \le t_1, \\ \epsilon_j, & t_1 < j \le n, \end{cases} \qquad \Phi_n = \sum_{j \le n} \omega_{n+1-j}\phi_{j-1},$$

where $\epsilon_j \le (2C_w^{\mathrm{s}} - C_w)\phi_{j-1}$. The early array $E^{(j)}(s) + \lambda\Delta^{(j)}(s)$ is formed by the extraction rule from

$$(\widetilde{E}_{\ge 2}^{(j)}(\mathfrak{c}_{j-1}(s)) + \lambda F_j(s), \widetilde{E}_{\ge 2}^{(j)}(0) + \lambda G_j(s), \mu_j + \lambda\nu_j(s)).$$

Its infinite-lattice reference has zero matrix cell sums. Its summand norms, including both reference families, are bounded by $\epsilon_j[\lambda]$. For $|\lambda| < R$,

$$\sum_{j \le n} \omega_{n+1-j}\epsilon_j[\lambda] \le 2C_w^{\mathrm{s}}\Phi_n \le 2C_w^{\mathrm{s}}C_{\mathrm{pc}}S_{1/4}\phi_n,$$

$$c_{\mathrm{cs}} \sum_{j \le n} \omega_{n+1-j}\epsilon_j[\lambda] + c_{\mathrm{cs}}^{\Delta}|\lambda|\Sigma_n^{\psi}(s) \le 2c_{\mathrm{cs}}C_w^{\mathrm{s}}\Phi_n,$$

$$\epsilon_j[\lambda] \le 2C_w^{\mathrm{s}}\phi_{n-1} \quad (j \le t_1),$$

$$|\lambda|\Lambda_n^0[\rho(s)] \le c_0^{\Delta}.$$

■

Proof. Units, moved moments, and mass forms are linear in their three extraction data. Their sums are therefore the extraction of the displayed three data, with cell sums $0 + \lambda 0$. Phase invariance preserves the form of their quadratic terms. Subadditivity proves the summand bounds. The remaining calculation is Lem. C.14, pointwise in z. Its extra weighted size is at most

$$|\lambda|\Sigma_n^{\psi}(s) \le (c_{\psi}^{\Delta}/\omega_1)\Phi_n = C_w \frac{c_{\mathrm{cs}}}{c_{\mathrm{cs}} + c_{\mathrm{cs}}^{\Delta}}\Phi_n.$$

This is exactly the allowance for both the old-activity sum and its additional cell contribution. The individual bound uses $\omega_{n+1-j} \ge \omega_1$; the scalar bound uses the decay of Λ_n^0. □

Proposition C.54 (The small-field step for a closed datum)**.** Let $t_1 \le k < K$. For $\mathfrak{c}' = (\sigma_i)_{t_1 \le i \le k} \in 2\mathscr{D}_k'$, set

$$\mathfrak{c}_\lambda[\mathfrak{c}'] = (w_\lambda, -a_K J, (\sigma_i(s))_{i<t_1}, \mathfrak{c}').$$

The step is defined at every $(z, \mathfrak{c}', \lambda) \in \mathcal{O} \times 2\mathscr{D}_k' \times \mathrm{P}(R)$. All conclusions of Prop. C.15 hold, including its pointed, reference, infinite-lattice, and source-energy bounds. The outputs are jointly holomorphic in these variables and the pair, with values in the stated local domains. In particular,

$$\epsilon_{k+1}^{\mathrm{s},\psi} \le (2C_w^{\mathrm{s}} - C_w)\phi_k, \qquad \epsilon_{k+1}^{\mathrm{s},0} \le \tfrac{15}{8}C_w^{\mathrm{s}}\phi_k + \tfrac{3}{5}C'_{\mathrm{lin}}\tau_k^{\lambda}.$$

The increment and any restriction of its pieces take values in $\mathbb{M}$, and satisfy

$$|\nu_{k+1} - \mu_{k+1}| \le \tfrac{1}{2}B_\nu\phi_{k+1}.$$

At fixed λ, the locality bound of Prop. C.15 uses only differences of the coordinates σ_i, $t_1 \le i \le k$. ■

Proof. The embedding $\mathfrak{c}_\lambda[\mathfrak{c}']$ belongs to $2\mathscr{D}_k(\alpha_0)$: $\|w_\lambda\|_\sharp \le 2\varrho < 8\varrho$, $L^{k+1}\|a_K J\| < 2r_M$, and the earlier source coordinates have norm less than $B_\sigma\phi_i$. Apply Prop. C.15, by induction on k, to the early arrays in Lem. C.53. We spell out why this application also gives the additional parameter assertions.

The early arrays are independent of $\mathfrak{c}'$, affine in λ, and holomorphic in z on the fixed cores. Their scalar pointers are local terms with zero Wilson coefficient. Their fermion summands satisfy Lem. C.13, including the torus rule. The preceding newly generated activities have the same properties by induction. Thus all hypotheses used for the small-field construction in Prop. C.15 hold jointly on the displayed open set. No agreement between constructions at smaller pair radii is needed: this step and its increment are formed at α_0.

The extra scalar localization is at most

$$K_\Lambda^{\mathrm{s}}\mathfrak{s}_k(\alpha_0)c_0^\Delta \le 5c_{\mathrm{cl}}/144.$$

Together with the ordinary terms this gives $(1/4+1/8+13/144+5/144)c_{\mathrm{cl}} = c_{\mathrm{cl}}/2$. Its contribution to the new scalar activity is at most $C_w^{\mathrm{s}}\phi_k/8$, by the last inequality in Rem. C.51. These are precisely the two scalar calculations in Prop. C.15. For the fermion terms, Lem. C.53 bounds the ordinary lag sum and the extra cell sum by their ordinary values. Consequently the full recurrence and its pointed version give the stated sizes, also at the reference.

All localizations, Gaussian integrations, and cluster series are holomorphic in their inputs, with the uniform bounds just proved on compact parameter sets. Their limits are therefore jointly holomorphic in $z, \mathfrak{c}', \lambda$ and the pair. Phase invariance gives one barred and one unbarred field in degree two. Constant colour rotations at the trivial pair show that every extracted moment lies in $\mathbb{M}$, including an arbitrary subset of pieces. The locality proof in Prop. C.15 varies only $\mathfrak{c}'$; w_λ, J, and the earlier coordinates are fixed. Its same Cauchy disc and absolute moment sum give the stated increment bounds. Finally the scalar value at the trivial pair, the Gaussian part at w_λ, and the fermion normalizer at $m_0 - a_K\widehat{J}$ have the same bounds as in that proof. This gives its source-energy estimate. □

Lemma C.55 (The parameter window)**.** For a closed datum the chain in Def. C.43 is defined and holomorphic on $\mathcal{O} \times \mathrm{P}(R_k)$ through level k, and

$$\|\sigma_k(s,\lambda) - \sigma_k(s)\| \le \tfrac{1}{2}B_\sigma\phi_k, \qquad \|\sigma_k(s,\lambda)\| \le (B_\sigma - C_{\mathrm{var}}/2)\phi_k < B_\sigma\phi_k.$$

At $\lambda = 0$ it is the source chain. These assertions include $k = K$. ■

Proof. The proof of Lem. C.16 applies with the jointly holomorphic increments of Prop. C.54. For completeness its two shrinking-disc terms are

$$Q_\sigma^{-1}LB_\sigma\phi_k/2, \qquad Q_\sigma^{-1}B_\nu\phi_{k+1}.$$

Their sum is at most $B_\sigma\phi_{k+1}/2$, since $L\phi_k \le 2\sqrt{L}\phi_{k+1}$, $B_\nu \le B_\sigma/8$, and $Q_\sigma = 2\sqrt{L} + 1/4$. Both differences vanish at $\lambda = 0$, which is the hypothesis needed for Schwarz's lemma on each successive disc. The initial bound is the mass condition in eq. C.xi. At K, the coordinate embedding is still admissible: $L^{K+1}\|a_K J\| = L\|J\|_\infty < Lr_J = r_M$. Composition gives the asserted joint holomorphy. □

Lemma C.56 (History coordinates and the interior)**.** Let $h \in \mathcal{H}_k[t_1]$ and $|\lambda| < R_k$. The chain along h has the estimates in Lem. C.17. In particular, for $S_j = \bigcup_{t_1<i\le j}\tilde{Z}_i(h)$ and sites x where step j is made along h,

$$|\sigma_j^h - \sigma_j|(x) \le 8B_\nu\phi_j e^{-\kappa_\sigma \operatorname{dist}(x,S_j)/(2L^j)} \quad \text{if } \operatorname{dist}(x,S_j) \ge R_\flat' L^j,$$

and on Ω_j,

$$|\sigma_j^h - \sigma_j| \le 16B_\nu\phi_j, \qquad |\sigma_j^h| \le 2B_\sigma\phi_j.$$

The history coordinates belong to $2\mathscr{D}_j'$ and are holomorphic on $\mathcal{O} \times \mathrm{P}(R_k)$. Moreover

$$\sum_{X\subset\Lambda_n} |e_n(X;\mathfrak{c}_{n-1}) - e_n(X;\mathfrak{c}_{n-1}^h)| \le C_{\mathfrak{E}}'\#_n(\tilde{Z}_n).$$

The conditions $(\mathrm{Int}_i)[s,\lambda;h]$ hold for $t_1 \le i \le k$, $i < K$. ■

Proof. The full and restricted chains agree through the base. The proof of Lem. C.17 therefore starts with zero coordinate difference. Its only analytic inputs are the absolute piece bound, the Lipschitz locality estimate, and the source-energy bound in Prop. C.54. Thus its exponentially weighted recurrence and its polymer sum apply unchanged. The bound $16B_\nu \le B_\sigma$, together with Lem. C.55, keeps every history coordinate in $2\mathscr{D}_j'$. Its entries are the history chain on a fixed set Ω_j

and the full chain elsewhere; both are holomorphic by induction. Evaluating Prop. C.54 at these coordinates gives every clause of (Int_i). □

Lemma C.57 (Regular fermion sums)**.** For $h \in \mathcal{H}_k[t_1]$, $|\lambda| < R_k$, the regular activities have total norm at most $C^{\mathrm{s}}_{\mathrm{leg}}\Sigma_{\mathcal{L}}(h)$. Their contribution to a cube of $\mathcal{L}_n$ is at most $C^{\square,\mathrm{s}}_{\mathrm{leg}}\phi_n$. The mass member has per-cube bound

$$w_0\nu C_H^2 h_F^2 L(\bar{B}_M + 2B_\sigma)\phi_{n-1}$$

and the corresponding sum over all layers. ■

Proof. Apply the regular-sum proof in Lem. C.18. We verify the inputs for all families, including those at a regular subset of pointers along a history. The early families have no rim. By Lem. C.53 they have the extraction data, vanishing matrix cell sums, and summand sizes needed for the unit count in appendix C.2. The later families have them by Prop. C.54. Their moved pieces satisfy, polymer by polymer,

$$\sum_p \sum_y \|a_{X,p}(y)\| e^{\kappa(d_j(Z_p)-d_j(X))} \le C A_\kappa^2 h_F^{-2}(\|F_j(X)\| + \|G_j(X)\|).$$

Indeed, the localized-leg count bounds the double row sums of each piece by the right side. These row sums bound its zeroth moment; taking the difference of the two data gives the formula. Thus the count requires no additional symmetry of an early family.

At a later rim, let $F^h(X)$ be the sum of pointed terms at the regular pointers of X. Before extraction its part with generators is the integral, over the pointing parameter, of the first variation of the cluster sum. The sum of insertions over any subset of places has norm at most the original even element's norm. Hence the pointed estimate in Prop. C.54 bounds $F^h(X)$ by the activity size. The extraction terms it retains are a subset of the same summands. The kernels and forms actually used along the history therefore satisfy the short-lag version of the unit count.

The terms with summable kernels cost at most $2C^{\mathrm{s}}_w K_{\mathrm{fin}}\phi_{n-1}$. The cell terms cost at most $(C_0^{\mathrm{cs}} h_F^2 + 2c_{\mathrm{cs}} C_{\mathrm{pc}} S_{1/4} C^{\mathrm{s}}_w)\phi_n$. At long lags the units cost at most

$$\frac{N_{\mathrm{u}} L^{9/2}}{C_\omega L^{1/4}}\, 2C^{\mathrm{s}}_w C_{\mathrm{pc}} S_{1/4}\phi_n;$$

at short lags they cost at most $2N_{\mathrm{u}} C^{\mathrm{s}}_w S_{\mathrm{lag}}\phi_{n-1}$. These are exactly the four sums proved in Lem. C.18, using the four estimates in Lem. C.53. The floor eq. C.ii bounds their sum by $C^{\square,\mathrm{s}}_{\mathrm{leg}}\phi_n$, and then by $C^{\mathrm{s}}_{\mathrm{leg}}$ after summing over the layers.

The mass coefficient is a convex combination of the two adjacent scale coefficients, except on the first layer where it is the base coefficient. The bounds in Lems. C.55 and C.56, and $L^{-1}\phi_{t_1} \le \phi_{t_1-1}$ on the first layer, give the same bound in either case. The ordinary background count proves the assertion for the mass. □

Proposition C.58 (The first seven fermion classes)**.** For a closed datum, $|\lambda| < R_k$, and $h' = (h, \mathsf{o}) \in \mathcal{H}_{k+1}[t_1]$, the first seven fermion classes together have size at most

$$b_k^{\mathrm{tot,s}} = b_k^{\mathrm{blk,s}} + b_k^{\mathrm{lay,s}} + b_k^{\mathrm{ms,s}} \le b^{\mathrm{tot,s}}(\gamma).$$

Thus $(\mathrm{Sh}_k)[s, \lambda; h']$ holds. ■

Proof. The class estimates in Lem. C.18 use the old families through their holomorphy on the fixed cores, the per-cube bounds, and three sums: the weighted lag sum, the cell sum, and a summable kernel times the individual current-scale size. These are supplied by Lems. C.53 and C.57. More explicitly, the first two classes use Cauchy's estimate on the ordinary link polydisc, with the regular sums of Lem. C.57; all shifted pairs remain in the same cores. The stored-layer count uses

$$\sum_{j\le m} C_\omega L^{(m-j)/4}\epsilon_j[\lambda] \le \sum_{j\le m} \omega_{m+1-j}\epsilon_j[\lambda] \le 2C^{\mathrm{s}}_w C_{\mathrm{pc}} S_{1/4}\phi_m.$$

Short lags also satisfy this bound since $1 \le \omega_{m+1-j}$. These estimates retain the ordinary $C^{\mathrm{s}}_{\mathrm{lay}}$.

The third and fourth classes use the potential bound of Prop. C.54, followed by the same lag, cell, and summable-kernel estimates. Apply the individual bound of Lem. C.53 at $n = k+1$ when its size is read at the current level k. The fifth and sixth classes use these estimates at each stored layer and its per-cube bound. All pointers of an early polymer are regular; for later polymers the subset-of-pointers bound was verified in Lem. C.57. Hence the division of classes by the cell of a polymer remains valid. The seventh class contains only later rim and mass forms; it uses $|\nu_j^h - \mu_j| \le B_\nu \phi_j/2$. Each of the three ordinary shares therefore retains its bound. □

Proposition C.59 (Induction for a closed datum)**.** Let a closed datum at radius R over $\mathcal{O}$ be given. For $t_1 \le k \le K$, $z \in \mathcal{O}$, and $|\lambda| < R_k$, every $h \in \mathcal{H}_k[t_1]$ satisfies the interior conditions of Def. C.43. For $k < K$, every next step satisfies its share condition. Moreover

$$\mathsf{I}^{\mathrm{F}}(k)[s(z), \lambda] \quad (t_1 \le k \le K,\ |\lambda| < R_k).$$

Thus all the conclusions of Cor. C.44, Lems. C.45 to C.47, and Props. C.48 to C.50 hold at every level on any common disc $|\lambda| < r_\lambda \le R_K$. ■

Proof. The interior and share assertions are Lem. C.56 and Prop. C.58. At the base, Lem. C.7 gives the exponential representation with exponent $A^{\mathrm{F}}_{t_1,h_\emptyset}(s) + \lambda\Delta(s)$. Its changed weight, mass, arrays, and vacuum constant are exactly those in Def. C.43. There is no piece, stored lower layer, or boundary term. The weight condition gives the ordinary loss in the action. The regular and mass bounds follow from Cor. C.44 and Lem. C.57; these also bound the exponent, with the ordinary background estimate. Hence $\mathsf{I}^{\mathrm{F}}(t_1)$ holds on $\mathrm{P}(R)$.

Suppose it holds at k, and restrict to $\mathrm{P}(R_{k+1}) \subset \mathrm{P}(R_k)$. By the first assertions, Prop. C.50 applies to every prolongation. We check the eight bounds in the resulting representation. The kept halves and formed factors have the ordinary bounds at w_λ, since $(w_\lambda, J) \in \mathcal{P}$. The action retains the ordinary fraction of its weight. The stored-factor estimate uses the same fine-mass operators and the mass coefficient bound of Lem. C.57. The updated mass and regular sums obey that lemma at $k+1$. The early arrays keep their domains; the later pointed activities have the properties and E^{s}_0 bound of Prop. C.54. The generated boundary terms, rejected halves, reblocked linear term, and sourced strip terms obey the ordinary boundary bound $\mathsf{B}_\natural$, by Props. C.48 and C.49. Their frame size is $64 b^{\mathrm{tot,s}}(\gamma)/51$; their edge scalar size is E^{s}_0; their insertion size is at most $b^{\mathrm{blk,s}}(\gamma)$.

For the vacuum label, the discrepancy from the common $E_{k+1}(s,\lambda)$ consists of three terms: source energies at polymers outside Λ_{k+1}, boundary normalizers, and differences between full-chain and history-chain source energies inside Λ_{k+1}. The first has the polymer-tail bound of Prop. C.54. For the second, the fermion normalizer is the ordinary one at the fine mass of s; its Gaussian part at w_λ has bound $9e^{-\kappa'' d(X)}/32$ outside the regular region and the ordinary pointer bound inside. The third is bounded by Lem. C.56. The ordinary three constants add to $C'_{\mathfrak{E}}$. Thus their sum defines g_{k+1} with

$$|g_{k+1}| \le C'_{\mathfrak{E}} \#_{k+1}(\widetilde{Z}_{k+1}).$$

Finally the exponent bound uses the regular fermion and mass bounds, the scalar regular bound, E^{s}_0 for edges, and the ordinary estimate on formal background legs. These verify all eight clauses. Induction proves the representation on each shrinking disc. The last assertion follows from $R_K \le R_k$. □

Proposition C.60 (Holomorphy along a history)**.** For $h \in \mathcal{H}_k[t_1]$, all its chains, coordinates, restricted increments, mass field, pointed activities, source energies, and label corrections g_n are jointly holomorphic on $\mathcal{W}_k = \mathcal{O} \times \mathrm{P}(R_k)$. The activity assertion includes joint holomorphy with the pair. The vacuum energy is holomorphic there, and

$$E_k(s,0) = E_k(s), \qquad g_n(h;s,0) = g_n(h;s).$$

Every coefficient of its measure has a jointly Borel density holomorphic in (z,λ) at every free-coordinate point on $\mathcal{O}\times\mathbb{C}$. On $\mathcal{W}_k$ this density agrees almost everywhere with the exponential representation. ■

Proof. The full chain is holomorphic by Lem. C.55. A restricted increment is the sum of the extracted pieces whose polymer is inside the fixed region Λ_j. Each piece is a product of holomorphic activity coefficients and localized background walks at the fine mass. The absolute moment bound of Prop. C.54 sums their norms uniformly on compact parameter sets. Thus restriction of the piece sum preserves holomorphy. The history recurrence and its coordinate assembly then prove holomorphy of the history chain, restricted increments, and mass field by composition. Evaluation of Prop. C.54 gives the activity and source-energy assertions; the vacuum energy is their finite sum.

The three parts of g_n were described in the proof of Prop. C.59. Its source-energy parts are holomorphic by the preceding paragraph. The fermion boundary normalizer is holomorphic in $-a_K J$ on $L^n\|a_K J\| < 2r_M$. Its Gaussian part inside the regular region is the pointed Gaussian path at weight w_λ, formed from polynomial insertions affine in that weight. Outside the regular region, it is the centered cluster series with origins $\mathcal{V}^{\mathrm{q}}$ and $\mathbb{W}^{\mathrm{q}} = -F^{v'}_{Y,2}$. The first is independent of the weight and the second is complex linear in its values on the polymer. The uniform bounds in Prop. C.48 give locally uniform convergence of both series on $\mathcal{W}_k$. Their terms are holomorphic, so their sums are holomorphic. At $\lambda = 0$, induction gives the same early arrays, reference, subsequent activities, increments, and coordinates as the ordinary source flow. This also identifies the three parts of every g_n, proving both identities at zero.

The density assertion, including the common Borel choice and pointwise holomorphy on the full parameter space, is Prop. C.11. That proposition uses the positive zero-source measure and a countable gluing of Cauchy series; it is independent of the shrinking RG window. Uniqueness of densities identifies it with the exponential representation on $\mathcal{W}_k$. □

Proposition C.61 (Summing the histories)**.** For the datum of Def. C.6 and any $(z,\lambda)\in\mathcal{O}\times\mathbb{C}$, its scalar bare density $\varsigma^{s,\lambda}$ has a bounded Borel representative $\varsigma^\sharp$ invariant under every bare gauge transformation constant on level-t_1 blocks, such that

$$\mathfrak{w}_h\varsigma^\sharp = \mathfrak{w}_h\varsigma^{s,\lambda} \quad dU\text{-almost everywhere} \qquad (h\in\mathcal{H}_k[t_1],\ t_1\le k\le K).$$

For every $h''\in\mathcal{H}_{n-1}[t_1]$, $n > t_1$,

$$\sum_{h|_{n-1}=h''} \tau_h[\varsigma^{s,\lambda}] = \mathbf{T}\,\tau_{h''}[\varsigma^{s,\lambda}].$$

Consequently

$$Z_k(z,\lambda) = Z_{t_1}(z,\lambda) \qquad (t_1\le k\le K,\ \lambda\in\mathbb{C}).$$

■

Proof. Let $\mathcal{G}_{t_1}$ be the compact group of these block-constant transformations, and let Υ_v be its action on the base Grassmann algebra. Write

$$\mu = \xi_*(\mathfrak{w}_{h_\emptyset}\,dU), \qquad N_1 = \mathsf{N}_0\cup(\mathsf{X}^{\mathrm{co}}_{h_\emptyset}\setminus\mathsf{S}_\emptyset).$$

By Lems. C.5 and C.8, μ is invariant under $\mathcal{G}_{t_1}$ and $\mu(N_1) = 0$. Where both arguments belong to $\mathsf{S}_\emptyset\setminus\mathsf{N}_0$,

$$\Delta(s)(v\cdot V) = \Upsilon_v\Delta(s)(V).$$

This identity uses the evaluation-set covariance and the joint gauge invariance of each array and background, and the fact that the mass matrix commutes with colour.

Let dv be Haar probability measure and set

$$N^\sharp = \left\{V : \int \mathbf{1}_{N_1}(v\cdot V)\,dv > 0\right\}.$$

Tonelli and invariance show that $N^\sharp$ is an invariant Borel μ-null set. Off it define

$$\Delta^\sharp(V) = \int \Upsilon_v^{-1}\Delta(s)(v\cdot V)\mathbf{1}_{\mathsf{S}_\emptyset\setminus\mathsf{N}_0}(v\cdot V)\,dv.$$

The integrand is bounded and Borel. Haar invariance gives $\Delta^\sharp(u\cdot V)=\Upsilon_u\Delta^\sharp(V)$; on $(\mathsf{S}_\emptyset\setminus\mathsf{N}_0)\setminus N^\sharp$ it equals $\Delta(s)(V)$. Put $G^\sharp(U)=\exp[\lambda\Delta^\sharp(\xi(U))]$ when $\mathfrak{w}_{h_\emptyset}(U)\neq 0$ and $\xi(U)\notin N^\sharp$, and put $G^\sharp=1$ elsewhere.

The unperturbed transported base density $F(U)$ satisfies $F(U^v)=\Upsilon_v F(U)$: the Dirac and block kernels are gauge covariant, J commutes with colour, and partial Berezin integration preserves this covariance. The base weight and its support are invariant. Hence $\varsigma^\sharp(U)=\mathrm{Ber}[G^\sharp(U)F(U)]$ is invariant everywhere. On the base support it agrees with $\varsigma^{s,\lambda}$ off a set of $\mathfrak{w}_{h_\emptyset}dU$-measure zero. Every later weight is this base weight times a bounded Borel product of tests and gauge-fixing factors. The claimed agreement follows for all h.

At a step $n>t_1$, the gauge transformations used to sum its prolongations are constant on level-$(n-1)$ blocks. Each such block is a union of level-t_1 blocks, so $\varsigma^\sharp$ has precisely the invariance required by the ordinary history identity. Its gauge defects vanish for bounded Borel densities by [YM3, Prop. 4.1]. Applying the identity to $\varsigma^\sharp$ and then using its weighted almost everywhere agreement gives the displayed measure identity. Iteration and preservation of total mass give the last formula. This is also the scalar identity of Prop. C.10. □

Lemma C.62 (The comparison datum)**.** Take the base data of Def. 11.3 at t_b, with its difference format and lag bounds (L1)–(L3) at a number $\Lambda_b\geq 1$. Suppose $d_b(s)=d_b$ is constant. Choose $0<c\leq 1$ such that

$$cE_0^{\mathrm{s}}\leq 2c_0^\Delta,\qquad 2\sqrt{L}cC_w^{\mathrm{s}}\leq c_\psi^\Delta.$$

On an affine source domain with $\|w\|_\sharp\leq(2-c)\varrho$, set

$$t_1=t_b,\quad w^\Delta=\widehat{w}_b,\quad \delta\sigma=-\delta_{t_b}(s),\quad \mathfrak{n}=d_b,\quad \text{early arrays } -\Delta^{(j)}(s).$$

These give a closed datum at $R=c\Lambda_b/2$. Its exponent is $\widehat{\Phi}_s^{(t_b)}-\Phi_s^{(t_b)}$, and

$$\tfrac12(3\sqrt{L})^{-(K-t_b)}c\Lambda_b<R_K.$$

■

Proof. The difference format in Def. 11.3 supplies local holomorphic pointed arrays with the common pair domains. Their extraction data are the negatives of the differences of the two flows' data, with their common paths. The infinite-lattice reference cell sums are $0-0$ as matrices, since each flow has had its mass coefficient extracted. Phase invariance and the summand norm bounds are preserved. The mass difference is a holomorphic field in $\mathbb{M}$; the weight and vacuum difference are constants. Thus all format and parameter requirements in Def. C.52 hold.

For the four quantitative conditions, the comparison lag bounds give, respectively,

$$\begin{aligned}
R\Sigma_k^\psi &\leq \tfrac12 cC_w^{\mathrm{s}}\phi_k\leq c\sqrt{L}C_w^{\mathrm{s}}\phi_{k-1}\leq\tfrac12 c_\psi^\Delta\phi_{k-1},\\
R\Lambda_{t_b}^0[\rho] &\leq \tfrac12 cE_0^{\mathrm{s}}\leq c_0^\Delta,\\
|\lambda\widehat{w}_b| &< \tfrac12 c\varrho,\\
R\|\delta_{t_b}\| &\leq \tfrac12 cB_\sigma\phi_{t_b}\leq\tfrac12 B_\sigma\phi_{t_b}.
\end{aligned}$$

The third estimate, and the zero Lipschitz constant of $\widehat{w}_b$, keep $w+\lambda\widehat{w}_b$ inside the source weight ball. Its fermion source is unchanged. Substituting the four data in eq. C.iv gives exactly the stated difference of exponents, with its signs. Finally $Q_\sigma<3\sqrt{L}$ and $K-t_b\geq 1$ give the strict comparison of radii. □

C.17. **Application to the restart interpolation.**

Lemma C.63. Use the cutoff-comparison data of Def. 11.1, with condition (k0) and $u \geq u_*$ from Prop. B.13. Fix one of the two cutoffs and a base $t_0 < t_b \leq t_1$, and suppress its superscript. Let the differences $\Delta_b^{(j)}$ have the format and lag bounds (L1)–(L3) of Def. 11.3 with $\Lambda_b > 0$, at every $s \in \mathcal{P}$. Assume that $d_b(s) = d_b$ is independent of s. Choose

$$0 < c_\lambda \leq \tfrac{1}{8}, \qquad c_\lambda E_0^{\mathrm{s}} \leq c_0^{\Delta}, \qquad 2\sqrt{L} c_\lambda C_w^{\mathrm{s}} \leq c_\psi^{\Delta}, \qquad \rho_b = \tfrac{1}{2}(3\sqrt{L})^{-(K-t_b)} c_\lambda \Lambda_b.$$

Let $s(z) = (w, J)$ be an affine family on $\mathcal{O}$ as in Def. C.6, with $\|w\|_\sharp \leq (2 - c_\lambda)\varrho$. Then the following data satisfy Def. C.6 at base t_b and radius $2\rho_b$:

$$w^{\Delta} = \hat{w}_b, \qquad \delta\sigma = -\delta_{t_b}, \qquad \mathfrak{n} = d_b, \qquad \Delta^{(j)} = -\Delta_b^{(j)} \quad (j \leq t_b). \tag{C.xii}$$

Their exponent is $\hat{\Phi}_s^{(t_b)} - \Phi_s^{(t_b)}$. For every complex λ, their base measure is the interpolation measure $\rho_{s,\lambda}^{(t_b)}$ of Def. 11.3. Its later terms are $\check{\tau}_h^{\mathrm{F}}[s, \lambda]$, with densities $g_h(z, \lambda, \cdot)$ from Prop. C.11, and

$$\operatorname{Ber} \rho_{s,\lambda}^{(t_b)}(\mathcal{V}_{t_b}) = \sum_{h \in \mathcal{H}_K[t_b]} \operatorname{Ber} \int g_h(z, \lambda, v)\, d\mathscr{M}_h^{\mathrm{fr}}(v).$$

These same data form a closed datum in the sense of Def. C.52, with

$$R = c_\lambda \Lambda_b, \qquad R_K = R Q_\sigma^{-(K-t_b)} > 2\rho_b, \qquad Q_\sigma = 2\sqrt{L} + \tfrac{1}{4}.$$

Consequently the integration, shift, and density induction conditions Int_i, Sh_k, and $\mathsf{I}^{\mathrm{F}}(k)$ of Def. C.43 hold at all levels from t_b through K, on $|\lambda| < 2\rho_b$, with the indicated conditions only where the next step exists. In particular, all the hypotheses required for the estimates in Cor. C.44, Lems. C.45 to C.47, and Props. C.48 to C.50 hold there. ∎

Proof. The depth assumptions imply $1 \leq K - t_b < K - t_0 < K$, so the base is in the permitted range. The constant function $\hat{w}_b = L^{-(K-t_b)}(x_{t_b} - \hat{x}_{t_b})$ is real. The matrix field

$$\delta_{t_b}(s) = s_{t_b} - \hat{s}_{t_b} + \sigma_{t_b}(s) - \hat{\sigma}_{t_b}(s)$$

lies in $\mathbb{M}$ and is holomorphic on a neighborhood of $\mathcal{P}$, by the original and restarted source flows. The scalar d_b is constant.

The asserted difference format gives locality, core holomorphy, Borel coefficients, joint gauge invariance, and vanishing of the scalar part at the trivial pair. The boundedness condition on the fixed core also holds. For the two original families it follows from their activity bounds at α_0; for the full scalar part one uses the bound $E_0^{\mathrm{s}} e^{-(\kappa+1)d_j(X)}$, which includes the normalizers, closed loops, Gaussian part, and constant family. Bounding only the scalar fluctuation family would not suffice here.

The fermionic parts are formed from

$$-(F_j(s) - \hat{F}_j(s),\ G_j - \hat{G}_j,\ \mu_j - \hat{\mu}_j),$$

with infinite-lattice family $-(G_j^{\infty} - \hat{G}_j^{\infty})$. The extraction rule is linear. Common paths, phase invariance, and the zero matrix cell sums are preserved by this difference and sign. Its sizes are exactly r_j^{ψ} and $\rho_j = r_j^0$ in the lag bounds.

Those bounds give, with $R = c_\lambda \Lambda_b$,

$$R \sum_{j \leq t_b} L^{-(t_b - j)} \rho_j \leq c_\lambda E_0^{\mathrm{s}} \leq c_0^{\Delta},$$

$$R \sum_{j \leq t_b} \omega_{k+1-j} r_j^{\psi} \leq c_\lambda C_w^{\mathrm{s}} \phi_k \leq 2\sqrt{L} c_\lambda C_w^{\mathrm{s}} \phi_{k-1} \leq c_\psi^{\Delta} \phi_{k-1},$$

$$R \|\delta_{t_b}\| \leq c_\lambda B_\sigma \phi_{t_b} \leq \tfrac{1}{2} B_\sigma \phi_{t_b}.$$

For $|\lambda| < R$, the gauge-weight increment has norm at most $c_\lambda \varrho$, strictly less if it is nonzero. The assumed margin for w therefore keeps $(w + \lambda\hat{w}_b, J)$ in $\mathcal{P}$. This proves all the required inequalities, already at radius $R > 2\rho_b$.

Substitution of eq. C.xii into eq. C.iv gives

$$a_K^{-1} A(\hat{w}_b, \mathsf{U}) + L^{-t_b} \mathsf{S}_1(\delta_{t_b}\psi^{\mathrm{bg}}) - \sum_{j \le t_b} \sum_X \Delta_b^{(j)}(X) + d_b = \hat{\Phi}_s^{(t_b)} - \Phi_s^{(t_b)}.$$

At zero interpolation parameter the base measure is the ordinary sourced term. Multiplication by the exponential of this difference, using Lem. C.7, gives the claimed equality at every complex λ. Lemma C.12 then identifies every later term from the base measure alone. Proposition C.11 supplies its holomorphic density, and Prop. C.10 gives the equality of total integrals.

For the closed datum, its additional compact-parameter bounds follow from holomorphy of the difference data at each core pair. The preceding inequalities are the closed inequalities at radius R. Since $Q_\sigma < 3\sqrt{L}$ and $K - t_b \ge 1$, $R_K > 2\rho_b$. The all-level induction Prop. C.59 therefore applies throughout the required disc and gives the remaining assertions. □

Appendix D. Pure-gauge coefficients

We identify the order-six coefficient of the pure-gauge source energy with a lattice Gaussian coefficient and compute its continuum limit. The comparison retains the nonlinear constraint, the Haar density, and the gauge-fixing determinant.

The proof has four reductions. The field-scale expansion and its tail bound are given in Props. D.11 and D.12. Prop. D.17 identifies the retained coefficient with the Gaussian coefficient under a flat block constraint; the running counterterms cancel in this identity. The comparison estimates Props. D.19 and D.20 and the box exhaustion then remove the constraint and the finite volume. Finally the lattice graphs are matched with their homogeneous kernels. Their local terms of degree -4 cancel, and the six contributions of degree -5 give the coefficient $8/(3\pi)$ in Prop. D.61.

D.1. **The Gaussian coefficient on the lattice.** We compute the pure-gauge coefficient in covariant gauge with gauge parameter one. The gauge fixing contributes two quadratic terms in addition to the Haar density. These terms must be kept when the coefficient is compared with the constrained integral.

Identify the Hermitian traceless matrices with $\mathbb{R}^3$ by the Pauli basis, so that $[X, Y] = 2iX \times Y$. For an oriented plaquette p, write its four boundary bonds as b_i and their signs as s_i, and put $A_i = A(b_i)$. Define homogeneous polynomials by

$$a_p(e^{itA}) = \sum_{j \ge 2} t^j \mathsf{a}_p^{(j)}(A), \qquad \frac{1}{i} \log \prod_{i=1}^4 e^{its_i A_i} = t\, dA(p) + t^2 \mathfrak{z}_1(p) + t^3 \mathfrak{z}_2(p) + O(t^4).$$

The logarithm is taken near the identity. Thus the formulas do not require a choice of logarithm for an arbitrary configuration.

Lemma D.1 (Wilson, Haar and gauge-fixing terms)**.** The terms needed through order four are

$$\begin{aligned} \mathfrak{z}_1(p) &= -\sum_{i<j} s_i s_j A_i \times A_j, \\ \mathsf{a}_p^{(2)} &= \tfrac{1}{2}|dA(p)|^2, \qquad \mathsf{a}_p^{(3)} = \langle dA(p), \mathfrak{z}_1(p)\rangle, \\ \mathsf{a}_p^{(4)} &= \tfrac{1}{2}|\mathfrak{z}_1(p)|^2 + \langle dA(p), \mathfrak{z}_2(p)\rangle - \tfrac{1}{24}|dA(p)|^4. \end{aligned}$$

The negative logarithm of the Haar density in exponential coordinates has quadratic term $|A|^2/3$ and no odd terms. For a site function λ, let $\bar{\lambda}(b) = (\lambda(b_-) + \lambda(b_+))/2$. The derivative of the

logarithmic gauge action is

$$\frac{d}{ds}A^{s\lambda}(b)\Big|_0 = -d\lambda(b) + 2A(b)\times\bar\lambda(b) - \tfrac{1}{3}A(b)\times(A(b)\times d\lambda(b)) + O(A^4)d\lambda(b).$$

Define the Faddeev–Popov operator by $\mathsf{M}(A)\lambda = \frac{d}{ds}d^*(A^{s\lambda})\Big|_{s=0}$. Its homogeneous terms are

$$\mathsf{M}^{(0)} = -d^*d, \qquad \mathsf{M}^{(1)}(A)\lambda = 2d^*(A\times\bar\lambda),$$
$$\mathsf{M}^{(2)}(A)\lambda = -\tfrac{1}{3}d^*(A\times(A\times d\lambda)), \qquad \mathsf{M}^{(3)} = 0.$$

■

Proof. The convergent product of the four exponential series gives

$$\mathsf{a}_p^{(j)} = -\tfrac{1}{2}\sum_{j_1+\cdots+j_4=j}\operatorname{tr}\prod_{i=1}^{4}\frac{(is_iA_i)^{j_i}}{j_i!}.$$

Comparing its linear and quadratic terms with the exponential of its logarithm gives $dA(p)$ and $i\mathfrak{z}_1 = -\frac{1}{2}\sum_{i<j}s_is_j[A_i,A_j]$. The logarithm is analytic, and rescaling all four arguments shows that its next coefficient $\mathfrak{z}_2$ is homogeneous of degree three. Expanding $1-\cos|t\,dA + t^2\mathfrak{z}_1 + t^3\mathfrak{z}_2 + O(t^4)|$ gives the three Wilson terms. The Haar density is a constant times $(\sin|A|/|A|)^2$; its negative logarithm is $|A|^2/3 + O(|A|^4)$.

For the gauge derivative put $\mathsf{a}_Y = -2Y\times\cdot$. Conjugation by e^{isY} is $e^{s\mathsf{a}_Y}$, and the derivative of the exponential is $\mathsf{f}(\mathsf{a}_Y)$, where $\mathsf{f}(z) = (e^z-1)/z$. Differentiating $e^{iA^{s\lambda}(b)} = e^{is\lambda(b_-)}e^{iA(b)}e^{-is\lambda(b_+)}$ therefore gives

$$\mathsf{f}(\mathsf{a}_A)^{-1}(\lambda(b_-) - e^{\mathsf{a}_A}\lambda(b_+)).$$

Use $1/\mathsf{f}(z) = 1 - z/2 + z^2/12 + O(z^4)$ and $\mathsf{a}_A^2 = 4A\times(A\times\cdot)$. There is no cubic term in this expansion. Applying d^* gives the operator formulas. □

Definition D.2 (The free kernels and the coefficient)**.** On $\mathbb{Z}^3$ write $\mathsf{k}_\mu = e^{ik_\mu}-1$ and $d^\circ k = (2\pi)^{-3}dk$ on $[-\pi,\pi]^3$. Set

$$G(x-y) = \int e^{ik\cdot(x-y)}|\mathsf{k}|^{-2}\,d^\circ k, \qquad C((x,\mu),(y,\nu)) = \delta_{\mu\nu}G(x-y).$$

On a torus of side N, replace the integral by $N^{-3}\sum_{k\in(2\pi/N)\mathbb{Z}^3/2\pi\mathbb{Z}^3,\ k\neq0}$. These are the inverses of d^*d on functions and of d^*d+dd^* on forms, off their constant modes. On the lattice their symbols are integrable and positive; hence they define real positive Gaussian covariances. Let $\langle\cdot\rangle_0$ denote the centered Gaussian family with covariance $\delta_{aa'}C(b,b')$ in the Pauli basis.

Define

$$E_p(b) = \sum_i s_iC(b_i(p),b), \qquad \Pi(p,q) = \sum_j s_jE_p(b_j(q)),$$
$$g(b) = G(b_+,b_+) + G(b_-,b_-) - 2G(b_+,b_-),$$
$$D(b;b') = \tfrac{1}{2}\sum_{x\in\{b'_-,b'_+\}}[G(x,b_+) - G(x,b_-)].$$

Thus E and Π are the covariances with one and two curvature entries. Put

$$S_3 = \sum_p \mathsf{a}_p^{(3)}, \qquad S_4 = \sum_p \mathsf{a}_p^{(4)},$$
$$V_2 = S_4 + \tfrac{1}{3}\sum_b|A(b)|^2 + \tfrac{2}{3}\sum_b g(b)|A(b)|^2 - 4\sum_{b,b'}D(b;b')D(b';b)\langle A(b),A(b')\rangle. \tag{D.i}$$

On the infinite lattice these are formal sums. Their occurrence in a cumulant means the sum of the corresponding local cumulants. Absolute convergence for the expressions below is proved in Lem. D.6.

The joint cumulant of Gaussian polynomials is defined by the finite moment–partition formula; no exponential moment of a polynomial is assumed. Define

$$\begin{aligned}\mathsf{B}_1(p,q) = {} & \langle \mathsf{a}_p^{(3)}; \mathsf{a}_q^{(3)} \rangle_0 + \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(4)} \rangle_0 + \langle \mathsf{a}_p^{(4)}; \mathsf{a}_q^{(2)} \rangle_0 \\ & - \langle \mathsf{a}_p^{(3)}; \mathsf{a}_q^{(2)}; S_3 \rangle_0 - \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(3)}; S_3 \rangle_0 \\ & - \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; V_2 \rangle_0 + \tfrac{1}{2} \langle \mathsf{a}_p^{(2)}; \mathsf{a}_q^{(2)}; S_3; S_3 \rangle_0 .\end{aligned} \tag{D.ii}$$

Use the superscript lat for the infinite lattice. Write $x(p)$ for the plaquette barycentre. For compactly supported f, h put

$$\mathsf{P}_1^{\mathrm{lat}}(f,h;a) = a \sum_{p,q} f(ax(p)) h(ax(q)) \mathsf{B}_1^{\mathrm{lat}}(p,q). \tag{D.iii}$$

The factor a is $a^{-2}\beta^{-3}\mathsf{b}^3$ when $\beta = \mathsf{b}/a$: each smeared action observable has the factor a^{-1}. ∎

The last two terms of V_2 follow from the gauge-fixing Jacobian. On a finite torus, with operators restricted to the mean-zero site functions, $(\mathsf{M}^{(0)})^{-1} = -G$ and

$$\begin{aligned} -\operatorname{tr}[(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(2)}] &= \tfrac{2}{3} \sum_b g(b) |A(b)|^2, \\ \tfrac{1}{2} \operatorname{tr}[((\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)})^2] &= -4 \sum_{b,b'} D(b;b') D(b';b) \langle A(b), A(b') \rangle . \end{aligned}$$

For the first identity use $\operatorname{tr}[X \times (Y \times \cdot)] = -2\langle X, Y \rangle$ and cycle Gd^* past a bondwise operator. For the second put $R = Gd^*(A \times \bar{\cdot})$. Its kernel is

$$R(x,y) = \tfrac{1}{2} \sum_b [G(x,b_+) - G(x,b_-)](\delta_{y,b_-} + \delta_{y,b_+}) A(b) \times \cdot .$$

Thus $\operatorname{tr} R = 0$ and $\operatorname{tr} R^2 = -2 \sum_{b,b'} D(b;b') D(b';b) \langle A(b), A(b') \rangle$, which proves the claim since $(\mathsf{M}^{(0)})^{-1}\mathsf{M}^{(1)} = -2R$. In particular the Jacobian has no linear term. On $\mathbb{Z}^3$ these are identities for the local coefficients of the formal sums defining V_2; we do not take a trace over the whole infinite lattice. Coordinate symmetry gives $g(b) = 1/3$ there.

D.2. **Kernel bounds and the comparison with a torus.** Write $\langle x \rangle = \max\{1, |x|\}$ and $\lg t = 1 + |\log t|$. A periodic symbol σ, smooth off zero, belongs to $\mathcal{S}^s$ if

$$|\partial^\alpha \sigma(k)| \le C_\alpha |k|^{s-|\alpha|} \qquad (0 < |k| \le \sqrt{3}\pi)$$

for every multi-index α. Its kernel is $K(x) = \int e^{ik\cdot x} \sigma(k)\, d^\circ k$ when $s > -3$. On a torus of side N use the nonzero-mode sum K_N. The symbols of C, E, Π, D above belong respectively to $\mathcal{S}^{-2}, \mathcal{S}^{-1}, \mathcal{S}^0, \mathcal{S}^{-1}$. This follows by multiplying $|\mathsf{k}|^{-2}$ by the displayed difference symbols; $|\mathsf{k}| \ge 2|k|/\pi$ controls every derivative.

Lemma D.3 (Lattice and torus kernels)**.** If $s > -3$ and $\sigma \in \mathcal{S}^s$, then

$$|\nabla^\alpha K(x)| \le C_\alpha \langle x \rangle^{-3-s-|\alpha|}, \qquad |K_N(x)| \le C \langle x \rangle_N^{-3-s}.$$

Here $|x|_N$ is the least Euclidean norm among the lifts of x and $\langle x \rangle_N = \max\{1, |x|_N\}$. The constants involve finitely many symbol bounds and are independent of N. For each fixed x, $K_N(x) \to K(x)$. Moreover,

$$|K_N(x) - K(x)| \le C N^{-3-s} \qquad (|x|_\infty \le 3N/4).$$

∎

Proof. A difference of order α multiplies the symbol by k^α and raises its order by $|\alpha|$, so first consider $\alpha = 0$. At $x = 0$ integrability gives the bound. At $|x| \ge 1$, choose a smooth cutoff χ equal to one on the unit ball and zero outside the ball of radius two. The integral of $\chi(|x|k)\sigma(k)$ is at most $C|x|^{-3-s}$. The complementary symbol is smooth and periodic, with derivative bounds $C_\alpha |k|^{s-|\alpha|}$

and support where $|k| \geq |x|^{-1}$. Choose j with $|x_j| \geq |x|/\sqrt{3}$ and integrate by parts $r > s+3$ times in k_j. The resulting bound is

$$C|x|^{-r} \int_{|k|\geq|x|^{-1}} |k|^{s-r}\, dk \leq C|x|^{-3-s}.$$

This proves the lattice estimate and its difference versions.

For the torus put $h = 2\pi/N$. Comparison with the cubes of side h about nonzero modes gives

$$N^{-3} \sum_{0<|k|\leq\rho} |k|^t \leq C\rho^{3+t} \quad (t > -3), \qquad N^{-3} \sum_{|k|\geq\rho} |k|^t \leq C\rho^{3+t} \quad (t < -3).$$

Indeed $|k'|$ on such a cube lies between $(1-\sqrt{3}/2)|k|$ and $(1+\sqrt{3}/2)|k|$. Use a lift with $|x| = |x|_N$, so $|x_j| \leq N/2$. The part $\chi(|x|k)\sigma$ has the required bound by the first sum. For the complementary part σ'', summation by parts gives

$$(1 - e^{ihx_j})^r \sum_k e^{ik\cdot x}\sigma''(k) = \sum_k e^{ik\cdot x}\delta_j^r \sigma''(k), \quad \delta_j f(k) = f(k) - f(k - he_j).$$

The left factor has modulus at least $(4|x_j|/N)^r$. For $|k| \geq 2rh$, the integral representation of δ_j^r bounds it by $Ch^r|k|^{s-r}$; it vanishes unless $|k| \geq 2/(3|x|)$. The bounded number of modes with $|k| < 2rh$ contribute at most $Ch^r|x|^{r-s-3}$ after normalization. The second sum gives the same bound for the other modes. Dividing by the left factor proves the torus estimate.

For pointwise convergence split the symbol at a fixed radius ϵ. The part near zero contributes at most $C\epsilon^{3+s}$ to both the sum and the integral. The remaining smooth part converges by Riemann summation. Let first $N \to \infty$ and then $\epsilon \downarrow 0$.

Finally the finite Fourier sums give the exact identity

$$\sum_{v\in\{0,1\}^3} K_{2N}(x + Nv) = K_N(x).$$

The sum over v selects the modes of the smaller torus; the zero mode is omitted on both sides. If $|x|_\infty \leq 3N/4$, each of the seven nonzero images has distance at least $N/4$ on the torus of side $2N$. Thus $|K_N(x) - K_{2N}(x)| \leq CN^{-3-s}$. Iterate over 2^jN and use pointwise convergence. The geometric series converges because $s > -3$. □

Lemma D.4 (Scalar convolutions)**.** For $a > D$, $D \in \{3, 6\}$ and $R \geq 1$,

$$\sum_{z\in\mathbb{Z}^D} (R + |z|)^{-a} \leq C_a R^{D-a}.$$

For $a, a' \geq 0$ and $a + a' > 3$,

$$\sum_{z\in\mathbb{Z}^3} \langle z\rangle^{-a}\langle x - z\rangle^{-a'} \leq C_{a,a'}\langle x\rangle^{-\min\{a,a',a+a'-3\}} \lg\langle x\rangle.$$

■

Proof. On a unit cube the factor $R + |z|$ changes by at most $1 + \sqrt{D}$. Comparison with the integral and rescaling by R prove the first assertion. For the convolution, $|x| < 2$ follows from the first assertion since the two factors are comparable. Otherwise split into $|z| \leq |x|/2$, $|x - z| \leq |x|/2$ and their complement. On the first part use $\langle x - z\rangle^{-a'} \leq C|x|^{-a'}$ and sum the first factor in radial shells. This gives $C|x|^{-a'} \max\{1, |x|^{3-a}\} \lg|x|$. The second part gives the symmetric bound. On the complement $|x - z|$ and $|z|$ are comparable and at least $|x|/2$, so the tail is bounded by $C|x|^{3-a-a'}$. These are the three asserted exponents. □

D.3. **Connected pairings and their sums.** For a homogeneous polynomial F of degree r, use its polarization

$$F[\![X_1,\dots,X_r]\!] = \partial_{t_1}\cdots\partial_{t_r}F(\textstyle\sum_j t_jX_j)\Big|_{t=0}.$$

Thus $F[\![A,\dots,A]\!] = r!F(A)$. An index $i=(b,a)$ specifies a bond and a colour; δ_i is its coordinate form. Write $C(i,j)=\delta_{aa'}C(b,b')$ and $E_p^a(b)=E_p(b)e_a$.

Lemma D.5 (Connected pairings)**.** Let F_v, $1\le v\le r$, have degrees m_v. Their joint cumulant is the sum over symmetric matrices $n=(n_{vv'})$ of nonnegative integers satisfying

$$2n_{vv}+\sum_{v'\neq v} n_{vv'} = m_v,$$

whose graph on the vertices $1,\dots,r$ is connected. For each matrix choose one pairing of the slots with these numbers of lines. Its contribution is

$$\frac{\displaystyle\sum_{\text{slot indices}}\prod_v F_v[\![\delta_{i(v,1)},\dots,\delta_{i(v,m_v)}]\!]\prod_{\{e,e'\}}C(i(e),i(e'))}{\displaystyle\prod_{v<v'}n_{vv'}!\prod_v 2^{n_{vv}}n_{vv}!}.$$

For the scalar cubic Wilson vertex, contraction of any two slots with a colour-diagonal kernel is zero. In particular a self-contraction at this vertex vanishes, as do two lines from one quadratic curvature vertex to that cubic vertex. ■

Proof. Wick's formula expresses a moment as a sum over all pairings. Each pairing splits uniquely into its connected components on the polynomial vertices. The moment–partition identity for cumulants therefore selects the connected pairings. Expand each polynomial as $1/m_v!$ times its polarized form. The number of pairings with fixed matrix n is

$$\frac{\prod_v m_v!}{\prod_{v<v'}n_{vv'}!\prod_v 2^{n_{vv}}n_{vv}!}:$$

choose the slots at each vertex, match the slots at different vertices, and pair those remaining at each vertex. The factors $m_v!$ cancel those in the polynomial expansions.

The polarization of $\mathsf{a}_P^{(3)}$ is the sum over the six permutations of

$$-\left\langle dX(P),\sum_{i<j}s_is_jY(b_i)\times Z(b_j)\right\rangle.$$

If two slots have the same colour, every summand contains one of $\langle e_a,e_a\times Z\rangle$, $\langle e_a,Z\times e_a\rangle$ or $\langle Z,e_a\times e_a\rangle$, all zero. Two lines from $\mathsf{a}_p^{(2)}=\frac12\sum_a(dA)^a(p)^2$ identify the two colours and give the same cancellation. Equivalently every monomial of $\mathsf{a}_P^{(3)}$ has three different colours, so these cancellations hold for each pairing before summing positions. □

For use in the coefficient calculation, define the colour-diagonal kernel Σ by

$$\delta_{aa'}\Sigma(b,b') = \langle\partial_{b,a}\partial_{b',a'}V_2\rangle_0 - \langle\partial_{b,a}S_3;\partial_{b',a'}S_3\rangle_0.$$

The expression is invariant under simultaneous rotations of the three colours, so its colour matrix is a scalar multiple of the identity. At fixed positions this is an identity of finite sums; the convergence assertion below also justifies the summed expression.

The seven cumulants in eq. D.ii give six classes:

$$\mathsf{T}_{\mathrm{I}} = -3\Pi(p,q)\sum_{b,b'} E_p(b)\Sigma(b,b')E_q(b'),$$

$$\mathsf{T}_1 = \tfrac{1}{2}\sum_{P,P',a,a',i,j} \mathsf{a}_P^{(3)}[\![E_p^a, E_q^{a'}, \delta_i]\!]C(i,j)\mathsf{a}_{P'}^{(3)}[\![E_p^a, E_q^{a'}, \delta_j]\!],$$

$$\mathsf{T}_4 = -\tfrac{1}{4}\sum_{P,a,a'} \mathsf{a}_P^{(4)}[\![E_p^a, E_p^a, E_q^{a'}, E_q^{a'}]\!],$$

$$\mathsf{T}_{4'} = \tfrac{1}{4}\sum_{a,i,j} C(i,j)\{\mathsf{a}_q^{(4)}[\![E_p^a, E_p^a, \delta_i, \delta_j]\!] + \mathsf{a}_p^{(4)}[\![E_q^a, E_q^a, \delta_i, \delta_j]\!]\},$$

$$\mathsf{T}_5 = -\tfrac{1}{2}\sum_{P,a,i,i',j,j'} C(i,j)C(i',j')\{\mathsf{a}_q^{(3)}[\![E_p^a, \delta_i, \delta_{i'}]\!]\mathsf{a}_P^{(3)}[\![E_p^a, \delta_j, \delta_{j'}]\!] + \mathsf{a}_p^{(3)}[\![E_q^a, \delta_i, \delta_{i'}]\!]\mathsf{a}_P^{(3)}[\![E_q^a, \delta_j, \delta_{j'}]\!]\},$$

$$\mathsf{T}_6 = \tfrac{1}{6}\sum_{i_1,i_2,i_3,j_1,j_2,j_3} \mathsf{a}_p^{(3)}[\![\delta_{i_1}, \delta_{i_2}, \delta_{i_3}]\!]\mathsf{a}_q^{(3)}[\![\delta_{j_1}, \delta_{j_2}, \delta_{j_3}]\!]\prod_{v=1}^{3} C(i_v, j_v).$$

Here P, P' range over plaquettes. In particular

$$\mathsf{B}_1(p,q) = \mathsf{T}_{\mathrm{I}} + \mathsf{T}_1 + \mathsf{T}_4 + \mathsf{T}_{4'} + \mathsf{T}_5 + \mathsf{T}_6.$$

We give the enumeration to fix the symmetry factors. The two cubic observables have three joining lines, giving $1/3!$. A quadratic and a quartic observable have two joining lines and one self-contraction at the quartic vertex, giving $1/4$. For degrees $(2,3,3)$ the joining multiplicities are $(1,1,2)$ and give $1/2$. For two quadratic observables and a quadratic term of V_2, the three joining multiplicities are one; these terms enter T_{I}. With the quartic Wilson vertex the possibilities are one line joining p, q, one from each to P, and one self-contraction at P; or two from each to P. Their factors are $1/2$ and $1/4$. Finally, with two cubic vertices P, P', the degree conditions give $n_{PP'} = 1 + n_{pq}$. The case $n_{pq} = 2$ is disconnected. If $n_{pq} = 1$, there are two assignments of the remaining observable legs, each with factor $1/2$; together with the coefficient $1/2$ in eq. D.ii they give the cubic part of T_{I}. If $n_{pq} = 0$, two legs from one observable to one cubic vertex vanish by Lem. D.5; the remaining pattern gives T_1 with factor $1/2$.

Lemma D.6 (Absolute sums and one weakened line)**.** Suppose the real kernels are colour diagonal and, with $c_0 \geq 1$, satisfy

$$|C(b,b')| \leq c_0\langle b_- - b'_-\rangle^{-1}, \qquad |E_p(b)| \leq c_0\langle x_p - b_-\rangle^{-2},$$
$$|D(b;b')| \leq c_0\langle b_- - b'_-\rangle^{-2}, \qquad |\Pi(p,q)| \leq c_0\langle x_p - x_q\rangle^{-3}, \qquad |g(b)| \leq c_0.$$

Write every Wilson vertex using the displayed curvature factors in Lem. D.1. The sum of the absolute values of the resulting connected pairings, over their positions, is at most Cc_0^5, uniformly in p, q. The same bound holds if the decay exponent of any one line is lowered by $1/2$. It also holds for torus sums with periodic distances, with a constant independent of the side. In particular the infinite-lattice coefficient and the six classes above are absolutely convergent. ∎

Proof. A plain leg is a component of A; a curvature leg is a component of dA. A line with j curvature ends has decay exponent $1 + j$. The standard writing of $\mathsf{a}^{(3)}$ has one curvature and two plain legs. The three parts of $\mathsf{a}^{(4)}$ have respectively four plain legs, one curvature and three plain legs, and four curvature legs. Each graph has at most five lines. The finite numbers of component monomials, colour sums and pairings can therefore be included in C, leaving only the position sums. Changes of a base point within one plaquette cost a fixed factor in each decay bound.

Put $u = x_q - x_p$, and let z, w be the positions of the summed vertices relative to x_p. For T_1 the majorant has exponents at least $(2,2,1,2,2)$ on the lines $z, w, w - z, u - z, u - w$. Discard the

central factor, which is at most one. The sum factors into two convolutions. Each has exponents at least $2, 2$, or $3/2, 2$ if one outer line was weakened. Their sums are bounded by Lem. D.4, since the sum of the two exponents is strictly greater than three. Weakening the central line leaves it bounded by one.

The cubic and ghost-loop terms of T_{I} are bounded by

$$\langle u\rangle^{-3}\sum_{z,w}\langle z\rangle^{-n_1}\langle w-z\rangle^{-n_3}\langle u-w\rangle^{-n_5},\quad n_1,n_5\in\{2,3\},\quad n_3\geq 2,\quad n_1+n_3+n_5=8.$$

The four possibilities for (n_3, n_5) are $(2,3),(3,2),(3,3),(4,2)$. Apply Lem. D.4 first to w. Its remaining power in $u-z$, added to n_1, exceeds three by at least $1/2$, even if one of the three exponents was lowered by $1/2$. Absorb the logarithm by a power smaller than this margin and apply the same lemma to z. This gives a uniform bound. If instead the fixed line from p to q was weakened, its factor is still at most one and the argument is unchanged.

The other terms of T_{I} have one position sum bounded by a convolution with exponents $2, 2$; weakening one line leaves their sum at least $7/2$. The class T_4 has exponents $4, 4$. The class T_5, after bounding its fixed p–q line by c_0, has exponents at least $2, 2$. These sums are bounded in the same way. There are no position sums in $\mathsf{T}_{4'}$ and T_6. Pairings excluded by colour vanish individually, by Lem. D.5; their absolute values therefore also vanish.

For a torus the same convolution proof uses the nearest images in the three regions of its radial split. The number of such images is bounded in dimension three. Extending each positive sum to the corresponding lattice sum proves the same bound; the finite number of lines makes the image factor independent of the side.

For the kernels themselves, Lem. D.3 supplies the hypotheses. To justify the derivative grouping in Σ, first sum the plaquettes containing one fixed bond. Differentiating $\mathsf{a}_P^{(3)}$ either leaves one curvature leg and one plain leg, or puts the derivative on the curvature leg. The first case has total line degree at least three between the two differentiated vertices. In the second case, the two opposite plaquettes in each direction give a first difference of a product of two line kernels; the difference bound in Lem. D.3 again gives total degree at least three. The resulting majorant is $C\langle z\rangle^{-2}\langle w-z\rangle^{-3}\langle u-w\rangle^{-2}$, which is summable by the same two convolutions. Thus the regrouping into Σ and the formula for T_{I} preserve absolutely convergent sums. □

D.4. **The finite-dimensional expansion.**

Lemma D.7 (Two coefficients of a covariance)**.** Let $\mathcal{W}\subset\mathbb{R}^m$ be bounded and Borel, and contain a ball about zero. Let ϕ,θ,X,Y be smooth real functions on a neighbourhood of $\overline{\mathcal{W}}$. Suppose that $\phi(0)=0$, $d\phi(0)=0$, its Hessian H is positive definite, and ϕ has a positive infimum on the part of $\mathcal{W}$ outside every sufficiently small ball about zero. Suppose also that X, Y and their gradients vanish at zero. Write subscripts for homogeneous Taylor polynomials and set

$$W_j=\phi_{j+2}+\theta_j,\qquad X_{1,j}=X_{j+2},\quad X_{2,j}=Y_{j+2}.$$

The covariance for the normalized density $e^{-\beta\phi-\theta}$ on $\mathcal{W}$ satisfies

$$\mathrm{Cov}_\beta(X,Y)=\beta^{-2}\gamma_0+\beta^{-3}\gamma_2+O(\beta^{-4}),\qquad \beta\geq 1,$$

where the cumulants below are Gaussian with covariance H^{-1}, and

$$\begin{aligned}\gamma_0&=\langle X_{1,0};X_{2,0}\rangle,\\ \gamma_2&=\langle X_{1,2};X_{2,0}\rangle+\langle X_{1,0};X_{2,2}\rangle+\langle X_{1,1};X_{2,1}\rangle\\ &\quad-\langle X_{1,1};X_{2,0};W_1\rangle-\langle X_{1,0};X_{2,1};W_1\rangle-\langle X_{1,0};X_{2,0};W_2\rangle\\ &\quad+\tfrac12\langle X_{1,0};X_{2,0};W_1;W_1\rangle.\end{aligned}$$

The error constant may depend on all the stated data. In particular the result applies on a sufficiently small fixed window under the local hypotheses alone. ■

Proof. Choose $\rho > 0$ so that $B(0,\rho) \subset \mathcal{W}$ and $\phi(y) \geq (y, Hy)/4$ on this ball. Outside it, the integrals of $1, X, Y, XY$ are $O(e^{-c\beta})$. Set $t = \beta^{-1/2}$ and change variables $y = tz$. The functions

$$v(t,z) = t^{-2}\phi(tz) - \tfrac{1}{2}(z, Hz) + \theta(tz) - \theta(0),$$
$$x(t,z) = t^{-2}X(tz), \qquad y(t,z) = t^{-2}Y(tz)$$

extend smoothly to $t = 0$. Taylor's formula with integral remainder shows that their first four t derivatives are bounded by fixed polynomials in $|z|$ whenever $|tz| \leq \rho$. Also

$$e^{-(z,Hz)/2 - v(t,z)} \leq Ce^{-(z,Hz)/4}.$$

For $f \in \{1, x, y, xy\}$, expand fe^{-v} through degree three in t. Its remainder, multiplied by $e^{-(z,Hz)/2}$, is bounded by $Ct^4\langle z\rangle^N e^{-(z,Hz)/4}$ for a fixed N. The same bound applies along the Taylor segment $0 \leq s \leq t$. Integration and the Gaussian tail outside $|z| < \rho/t$ therefore give

$$\int_{\mathcal{W}} Fe^{-\beta\phi-\theta} = Z_t t^{2\nu}\left[\sum_{j=0}^{3} t^j \langle (fe^{-v})_{[j]}\rangle + O(t^4)\right],$$

where $F = 1, X, Y, XY$, respectively, $\nu = 0, 1, 1, 2$, and $Z_t = t^m e^{-\theta(0)} \int e^{-(z,Hz)/2}\, dz$. The exponentially small outer integral is absorbed in the remainder. For $F = 1$ the leading term is one, so the denominator is at least $Z_t/2$ for small t. Division of the four expansions is therefore legitimate with an $O(t^4)$ error after the factors $t^{2\nu}$ have been removed.

To identify the coefficients, introduce formal variables u_1, u_2. The coefficient of u_1u_2 in

$$\log\langle e^{u_1x + u_2y - v}\rangle$$

is the normalized covariance of x, y. The moment-partition formula, grouped by the numbers of blocks of each size, gives

$$\log\langle e^Z\rangle = \sum_{r\geq 1} \frac{1}{r!}\langle Z; \ldots; Z\rangle.$$

Taking that coefficient yields the formal series

$$\sum_{r\geq 0} \frac{(-1)^r}{r!}\langle x; y; \underbrace{v; \ldots; v}_{r}\rangle.$$

Its constant and quadratic terms are exactly γ_0, γ_2. Each polynomial $X_{a,j}$ and W_j has the parity of j. At an odd total order, every term of the moment-partition formula has an odd Gaussian moment and vanishes. Multiplying by t^4 gives the claimed remainder $O(t^8)$. For the remaining compact interval of $\beta \geq 1$, enlarge the constant using boundedness of X, Y. □

D.5. **Gauge fixing in a finite box.** Let $\Lambda_n = \{0, \ldots, n\}^3$, $n \geq 2$, and $\Lambda_n^\circ = \{1, \ldots, n-1\}^3$. A bond is free if both endpoints lie in the box and at least one is interior. Write B_n for the positively oriented free bonds and P_n for all plaquettes in the box. The configurations $\mathcal{U}_n$ equal one on all other bonds. The gauge group $\mathcal{G}_n$ equals one on the boundary. Let Form_n be the real Lie-algebra-valued forms vanishing on the nonfree bonds, and let X_n be the site functions vanishing on the boundary. On these spaces d^* is the adjoint of d for the sums over bonds and interior sites. Put

$$T_n = \{\langle x - e_1, x\rangle : x \in \Lambda_n^\circ\}, \qquad S(U) = \sum_{p\in P_n} a_p(U).$$

A superscript T means that a configuration equals one, or a form vanishes, on T_n. In exponential coordinates write $\sigma_1(|A|^2) = (\sin|A|/|A|)^2$ for the Haar density with its constant factor removed.

Lemma D.8 (A slice in the box)**.** The map $(u, V) \mapsto V^u$ is a diffeomorphism $\mathcal{G}_n \times \mathcal{U}_n^T \to \mathcal{U}_n$ and carries the product of the normalized Haar measures to the product Haar measure on $\mathcal{U}_n$. The only zero of S in $\mathcal{U}_n^T$ is one. For every $\tau > 0$, the form

$$\tfrac{1}{2}\|dA\|^2 + \tfrac{1}{2\tau}\|d^*A\|^2$$

is positive definite on Form_n; d^*d is positive definite on X_n.

For suitable $0 < \rho_2, \rho_3 \le \rho_1 \le 1/20$, the map $\Xi(\lambda, A) = A^\lambda$ is a diffeomorphism on a neighbourhood of

$$\overline{\mathcal{D}}, \qquad \mathcal{D} = \{(\lambda, A) \in \mathsf{X}_n \times \mathsf{Form}_n^T : |\lambda|_\infty < \rho_1, |A|_\infty < \rho_3\}.$$

Its image $\mathcal{W} = \Xi(\mathcal{D})$ has compact closure in $\{|A'|_\infty < 1\}$, and the Faddeev–Popov determinant is nonzero there. For each $|A|_\infty \le \rho_3$, the map $\lambda \mapsto d^*(A^\lambda)$ is injective on $|\lambda|_\infty \le \rho_1$, and its open image contains $|c|_\infty < \rho_2$. For bounded Borel functions Φ, w, with $w \ge 0$,

$$\begin{aligned}&\int_{\mathcal{W}} \Phi(A')w(d^*A')|\det \mathsf{M}(A')| \prod_{b\in B_n} \sigma_1(|A'(b)|^2)\, dA' \\ &= \int_{|A|_\infty<\rho_3} \prod_{b\in B_n\setminus T_n} \sigma_1(|A(b)|^2)\, dA \int_{d^*(A^{\{|\lambda|_\infty<\rho_1\}})} w(c)\Phi(A^{\lambda(c)})\, dc.\end{aligned} \tag{D.iv}$$

Here $\lambda(c)$ is the unique solution in that domain. All Lebesgue measures use the Euclidean scalar products just specified. ∎

Proof. The condition $U^{u'} = 1$ on T_n is the recursion

$$u'(x) = u'(x - e_1)U(\langle x - e_1, x\rangle), \qquad x \in \Lambda_n^\circ,$$

starting with $u' = 1$ on $x_1 = 0$. It determines u' uniquely. Then $V = U^{u'}$ and $u = u'^{-1}$ are the required inverse coordinates. Products and inverses in the group make both maps smooth. At fixed u, invariance of Haar measure permits the change $V(b) \mapsto U(b)$ off T_n. On each line parallel to e_1, change $u(x)$ to $u(x - e_1)u(x)^{-1}$ in decreasing order of x_1. Each substitution preserves Haar measure and leaves the earlier substitutions fixed. This proves the measure identity.

If $S(V) = 0$, every plaquette holonomy is one. The direction-one bonds with initial coordinate $x_1 \le n - 2$ already equal one by the constraint or the boundary condition. The 1ν plaquettes, $\nu = 2, 3$, propagate the boundary value one along direction one to all direction-ν bonds. Finally the 12 plaquettes at $x_1 = n - 1$ propagate the value one from $x_2 = 0$ to the remaining direction-one bonds. Hence $V = 1$.

The additive version proves that $dA = 0$ implies $A = d\lambda$ for a unique $\lambda \in \mathsf{X}_n$: choose λ recursively along T_n to cancel A there and use the same plaquette argument. If also $d^*A = 0$, then $\|d\lambda\|^2 = 0$, so $A = 0$. The same recursion shows that $d\lambda = 0$ implies $\lambda = 0$. These facts prove both positivity assertions.

In exponential coordinates the preceding diffeomorphism is Ξ. Restrict first to $|\lambda|_\infty, |A|_\infty < 1/10$, where $|A^\lambda|_\infty < 1$. Set $c(\lambda, A) = d^*(A^\lambda)$. Differentiating the group action gives

$$\partial_\lambda c(\lambda, A) = \mathsf{M}(A^\lambda)\vartheta_\lambda, \qquad \det \vartheta_\lambda = \prod_{x\in\Lambda_n^\circ} \sigma_1(|\lambda(x)|^2) > 0.$$

At zero this derivative is $-d^*d$, which is invertible. Apply the inverse function theorem to $(\lambda, A) \mapsto (c(\lambda, A), A)$. Choose a closed product neighbourhood inside its domain of injectivity, and then smaller radii ρ_2, ρ_3 whose product lies in its image. Compactness gives the asserted common radii, the nonvanishing determinant and the closure property for $\mathcal{W}$.

The Haar identity in exponential coordinates contains the product of the site densities $\sigma_1(|\lambda(x)|^2)$ and the unconstrained bond densities. The constant Haar factors cancel because $\#T_n = \#\Lambda_n^\circ$. At

each fixed A, change λ to c. The displayed derivative identity gives

$$dc = |\det \mathsf{M}(A^\lambda)| \prod_x \sigma_1(|\lambda(x)|^2)\, d\lambda.$$

Fubini's theorem and this substitution give eq. D.iv. □

Proposition D.9 (The box coefficient)**.** For the Wilson probability measure on $\mathcal{U}_n$ with density proportional to $e^{-\beta S}$, and for $p, q \in P_n$,

$$\langle a_p; a_q \rangle_{n,\beta} = \beta^{-2} \mathsf{B}_0^{[n]}(p,q) + \beta^{-3} \mathsf{B}_1^{[n]}(p,q;\tau) + O(\beta^{-4}).$$

The coefficients are the Gaussian coefficients eq. D.ii on the box, with covariance $(d^*d + \tau^{-1} dd^*)^{-1}$; the leading coefficient is $\mathsf{B}_0^{[n]} = 3\Pi^{[n]}(p,q)^2/2$. The error constant may depend on the box, p, q and $\tau > 0$. The coefficient $\mathsf{B}_1^{[n]}$ is independent of τ. ■

Proof. For $F = 1, a_p, a_q, a_p a_q$, use the Haar identity of Lem. D.8 and gauge invariance to reduce the integral to $\mathcal{U}_n^T$. Away from the window $|A|_\infty < \rho_3$, the action has a positive minimum by compactness and the uniqueness of its zero. The omitted integral is therefore $O(e^{-c\beta})$. The denominator on the window is at least $c\beta^{-d_1/2}$, where $d_1 = \dim \mathsf{Form}_n^T$: integrate over a ball of radius $\beta^{-1/2}$ and use $S(e^{iA}) \le C|A|^2$. Thus the same error is exponential after normalization.

In eq. D.iv, take $w(c) = e^{-\beta \|c\|^2/(2\tau)}$ and $\Phi(A') = F(e^{iA'}) e^{-\beta S(e^{iA'})}$. Gauge invariance makes the latter constant on each gauge orbit. The inner integral of w equals

$$(2\pi\tau/\beta)^{d_0/2}(1 - \epsilon(A)), \qquad 0 \le \epsilon(A) \le C e^{-c\beta}, \qquad d_0 = \dim \mathsf{X}_n,$$

uniformly in A, because its domain contains the fixed ball $|c|_\infty < \rho_2$. Consequently the original normalized expectations, up to exponentially small errors, have density $e^{-\beta\phi_\tau - \theta}$ on $\mathcal{W}$, where

$$\phi_\tau(A) = S(e^{iA}) + \|d^*A\|^2/(2\tau),$$
$$\theta(A) = -\sum_{b \in B_n} \log \sigma_1(|A(b)|^2) - \log|\det \mathsf{M}(A)|.$$

These functions are smooth on a neighbourhood of the compact closure. The Hessian is positive by Lem. D.8. A zero of $\phi_\tau \circ \Xi$ would have $S(e^{iA}) = 0$ and $d^*(A^\lambda) = 0$. The first condition gives $A = 0$; injectivity of the gauge function gives $\lambda = 0$. Compactness therefore supplies the lower bounds away from zero required by Lem. D.7.

The chart expansions of Lem. D.1 give $\phi_{\tau,3} = S_3$ and $\phi_{\tau,4} = S_4$. Expand the determinant as $\log\det(1+Z) = \operatorname{tr} Z - \operatorname{tr} Z^2/2 + O(Z^3)$, with $Z = (\mathsf{M}^{(0)})^{-1}(\mathsf{M}(A) - \mathsf{M}^{(0)})$. The linear trace is zero. The quadratic Haar and determinant terms give precisely $V_2 - S_4$ in eq. D.i. Thus $W_1 = S_3$, $W_2 = V_2$, and Lem. D.7 yields the claimed coefficients. The leading curvature covariance is independent of τ. Subtract it, multiply by β^3 and let $\beta \to \infty$ to see that the next coefficient is independent of τ as well. □

D.6. **The pure-gauge coefficient on a constrained fibre.** We need a coefficient determined by the lattice and its constraint, independently of the intermediate running couplings. We first define it by a finite-dimensional integral.

For polynomials $X_{a,j}$, $a = 1, 2$, $j = 0, 1, 2$, and W_1, W_2, put

$$\begin{aligned}\gamma_2[\nu] = {} & \langle X_{1,2}; X_{2,0} \rangle_\nu + \langle X_{1,0}; X_{2,2} \rangle_\nu + \langle X_{1,1}; X_{2,1} \rangle_\nu \\ & - \langle X_{1,1}; X_{2,0}; W_1 \rangle_\nu - \langle X_{1,0}; X_{2,1}; W_1 \rangle_\nu \\ & - \langle X_{1,0}; X_{2,0}; W_2 \rangle_\nu + \tfrac{1}{2} \langle X_{1,0}; X_{2,0}; W_1^2 \rangle_\nu .\end{aligned} \tag{D.v}$$

These are joint cumulants. If ν is even, $X_{a,0}$ is even, and W_1 is odd, the last cumulant equals $\langle X_{1,0}; X_{2,0}; W_1; W_1 \rangle_\nu$. Indeed, expanding a cumulant with the product entry W_1^2 adds only $2\langle X_{1,0}; W_1 \rangle \langle X_{2,0}; W_1 \rangle$ and $2\langle W_1 \rangle \langle X_{1,0}; X_{2,0}; W_1 \rangle$, both zero by parity.

Let ϕ, θ, X, Y be real and smooth near zero, with $\phi(0) = 0$, $d\phi(0) = 0$, and positive Hessian H. Assume X, Y and their gradients vanish at zero. Subscripts on these functions denote homogeneous Taylor polynomials. Put

$$W_j = \phi_{j+2} + \theta_j, \qquad X_{1,j} = X_{j+2}, \qquad X_{2,j} = Y_{j+2}.$$

The covariance under the normalized density $e^{-\beta\phi-\theta}$ on a sufficiently small fixed window about zero has expansion

$$\beta^{-2}\gamma_0^{\mathrm{L}} + \beta^{-3}\gamma_2^{\mathrm{L}} + O(\beta^{-4}), \qquad \gamma_0^{\mathrm{L}} = \langle X_{1,0}; X_{2,0}\rangle_{H^{-1}}, \quad \gamma_2^{\mathrm{L}} = \gamma_2[\nu_{H^{-1}}]. \tag{D.vi}$$

Here $\nu_{H^{-1}}$ is the centered Gaussian measure with covariance H^{-1}. This is Lem. D.7; the last term has the form in eq. D.v by the parity identity above. We write $\gamma_{m'}^{\mathrm{L}}[\phi, \theta, X, Y]$ for these two coefficients, $m' = 0, 2$.

Proposition D.10 (Composition of the fibre charts)**.** Let $N = \ell L^K$, $\ell = 2L^m$, and $1 \le t \le K$, with $2L^{m+K-t+1} \ge C_L M$. Let Y be the space of arrays $y = (\mathcal{A}_j)_{j<t}$, where $\mathcal{A}_j$ is a real $\mathfrak{su}(2)$-valued one-form on the non-pivot bonds at level j. There are a neighborhood O_t of one in $\mathcal{V}_t$ and $\rho_2 > 0$ such that

$$W_t = W, \qquad W_j = V_{W_{j+1}}[\mathcal{A}_j], \qquad \Psi(W, y) = W_0$$

define a real analytic diffeomorphism from $\mathsf{O}_t \times \mathsf{Y}_{\rho_2}$ onto an open neighborhood U of the trivial bare configuration. Here $\mathsf{Y}_{\rho_2} = \{y : |\mathcal{A}_j| < \rho_2\}$, and $V_W[\mathcal{A}]$ is the one-step fibre chart. It satisfies $\mathcal{M}W_j = W_{j+1}$. For a real analytic θ and $C_H > 0$,

$$\int H(U)\, dU = C_H \int_{\mathsf{O}_t} \int_{\mathsf{Y}_{\rho_2}} e^{-\theta(W,y)} H(\Psi(W, y))\, dy\, dW \tag{D.vii}$$

whenever H is bounded Borel and supported in U.

For positive strengths κ_j, let

$$F_\kappa(U) = A(U) + \sum_{j<t} \kappa_j G_j(\mathcal{M}^j U), \qquad \phi_\kappa(y) = F_\kappa(\Psi(1, y)),$$

where G_j is the block gauge-fixing function. Then $\phi_\kappa \ge 0$, $\phi_\kappa(0) = 0$, $d\phi_\kappa(0) = 0$, and its Hessian is positive definite and block diagonal by level.

The coefficients

$$\mathsf{g}_{m'}[N, t](p, q) = \gamma_{m'}^{\mathrm{L}}[\phi_\kappa, \theta(1, \cdot), a_p \circ \Psi(1, \cdot), a_q \circ \Psi(1, \cdot)], \qquad m' = 0, 2, \tag{D.viii}$$

are independent of κ and of the adapted chart. They depend only on N, t, L, p, q, with plaquettes read in bare lattice units.

An adapted chart here is a smooth chart $\Psi'(w, y)$ near $(0, 0)$ with $\Psi'(0, 0) = 1$ and $\mathrm{Lg}\, \mathcal{M}^t \Psi'(w, y) = w$, whose Haar formula is $C' \varrho'(w) e^{-\theta'(w,y)}\, dw\, dy$, with $C' > 0$, positive continuous ϱ', continuous θ', and smooth $\theta'(0, \cdot)$. ∎

Proof. The size assumption makes each lattice through level t admissible for the one-step chart. Choose one calibration t_* so large that $\varepsilon(\mathsf{t}_*) \le 1$ and $\delta(\mathsf{t}_*) \le \delta_{\mathrm{im}}$. It serves at every level. Near the trivial block field, the averaged background $V_j^{\mathrm{bf}}(W)$ is analytic, $V_j^{\mathrm{bf}}(1) = 1$, and $\mathcal{M} V_j^{\mathrm{bf}}(W) = W$. The chart has the form $V_W[\mathcal{A}] = e^{iB'} V_j^{\mathrm{bf}}(W)$, with $B' = \mathcal{A}$ on the non-pivot bonds.

We first remove the gauge-fixing factor from the one-step integration formula of [YM3, Prop. 4.55]. On its chart image this factor is positive and bounded below, and the small-field indicator is one. Apply that formula to the bounded function $H/\mathfrak{G}_j[\mathsf{t}_*]$, and set $B = \mathsf{t}_*^{1/2}\mathcal{A}$. The powers of t_* and the gauge normalizers cancel. The remaining constant is $(2\pi^2)^{-\#\mathfrak{x}_j^+} L^{-3\#\Gamma_j}$, and the density is the

product of the elimination normalizer and the chart Jacobian. Iterating downward and applying Fubini gives eq. D.vii, with

$$C_H = \prod_{j<t} (2\pi^2)^{-\#\mathfrak{X}_j^+} L^{-3\#\Gamma_j}, \qquad 0 < e^{-\theta} \le \prod_{j<t} (27L^9)^{\#\Gamma_j}.$$

Here $\mathfrak{X}_j^+$ denotes the positively oriented non-pivot bonds at level j, and Γ_j the positive bonds at level $j+1$. Continuity of the charts lets us choose $\mathsf{O}_t \times \mathsf{Y}_{\rho_2}$ on which every step lies in this domain.

The map

$$U \longmapsto \Big(\mathcal{M}^t U, \big(\mathrm{Lg}[\mathcal{M}^j U(b) V_j^{\mathrm{bf}}(\mathcal{M}^{j+1}U)(b)^{-1}] \big)_{j<t,\ b \text{ non-pivot}} \Big)$$

is an analytic left inverse of Ψ. Its differential is injective. Indeed, a vector in its kernel has zero variation at level t. Descending one level, it has zero variation on the non-pivot bonds; the invertibility of the averaging derivative at each pivot forces its variation there also to vanish. Induction reaches the bare level. The dimensions agree, so the inverse function theorem makes Ψ a local analytic diffeomorphism. The left inverse gives global injectivity on the chosen domain. The Haar density pulled back by this diffeomorphism is positive and analytic. It agrees almost everywhere with the continuous density above, hence everywhere; this also proves analyticity of θ.

For the Hessian, let $E_j(W)$ be the value of the bare action on the minimal orbit over W, and set $E_0 = A$. Along the chain, $E_{j+1}(W_{j+1}) = E_j(V_{W_{j+1}}[0])$. Since $E_t(1) = 0$,

$$A(W_0) = \sum_{j<t} \big(E_j(V_{W_{j+1}}[\mathcal{A}_j]) - E_j(V_{W_{j+1}}[0]) \big).$$

Each summand and its derivative in $\mathcal{A}_j$ vanish at $\mathcal{A}_j = 0$, for all higher fields: the background is critical on the constrained fibre. Its Hessian at zero therefore has only the (j,j) block. The same is true of $G_j(W_j)$, which is nonnegative and zero at $\mathcal{A}_j = 0$.

The Hessian is positive definite. A vector in its kernel gives linearized fields a_j with $a_t = 0$, $da_0 = 0$, and zero linearized contour variables at every level. Averaging preserves closed forms, so each a_j is closed. On a simply connected block it then vanishes on internal bonds, because its contour variables vanish. It is constant on bonds crossing any one adjacent block face, by the plaquette equation. Its average at that coarse bond is this constant. Descending from $a_t = 0$ gives $a_j = 0$ at every level. The chart differential is invertible, so the original vector is zero.

To compare adapted charts, their transition map has the form $(w, y) \mapsto (w, s(w, y))$, with $s(0, 0) = 0$. Equality of the two Haar formulas yields

$$C_H \varrho_H(w) e^{-\theta(e^{iw}, y)} = C' \varrho'(w) e^{-\theta'(w, s(w,y))} |\det D_y s(w, y)|.$$

Both sides are continuous, so this equality holds also on the fibre $w = 0$, although that fibre has ambient measure zero. The change $y \mapsto s(0, y)$ therefore identifies the two conditional densities up to a constant. Their observable covariances and hence their Laplace coefficients coincide.

Finally, gauge transformations equal to one on the top lattice preserve this fibre, the action, and both plaquette observables. Integrate the normalized gauge-fixing factors over their orbits, starting with the finest sites. Each integrates to one at every strength, by [YM3, eq. 3.xxxv]. Restricting to a fixed small chart changes this identity only by exponentially small terms in β, because the gauge-fixing phase has a strict minimum in the gauge directions. Thus neither coefficient in eq. D.viii depends on κ. In bare units all the levels through t are $L^j\mathbb{Z}^3/N\mathbb{Z}^3$; the constraint, Haar measure, action, and observables on them depend only on N, t, L, p, q. □

D.7. **The pure-gauge flow at a complex field scale.** For the pure-gauge comparison we use the running coefficients as calibrations. This fixes both the integration boxes and the Gaussian measures at scale zero. We specify this choice before identifying any Taylor coefficient with a constrained integral.

Let $D = (K, \beta)$ be an admitted pure-gauge first-passage datum at g, with physical spacing $a = L^{-K}$ and label $b = a\beta$. Write $x_0 = \beta$ and $x_{j+1} = L^{-1}x_j - e_{j+1}$ for its running coefficients. Put

$$D_j = L^j a, \quad t_j = L^{-j}\beta = b/D_j, \quad \mathfrak{b}(x) = A_1(\log x)^{p_0}, \quad \delta(x) = \mathfrak{b}(x)x^{-1/2}.$$

The notation D_j denotes a length, while D denotes the datum. At level j the calibration and the coefficient of the original action are both x_j. For sources $w_1 = f\circ x, w_2 = h\circ x$, put $\eta = \|w_1\|_\infty\|w_2\|_\infty$. Suppose their supports lie in balls of radius d whose distance is at least d, with $8a \le d$. Set $r = d/a$ and $l = \max\{1, \log(2b/d)\}$.

Proposition D.11 (Pure-gauge parameters)**.** There is a threshold $\gamma_s > 0$, depending only on the construction constants, such that every admitted datum with $g \le \gamma_s$ and $K \ge 3$ has the following properties. Every step lattice is large enough for its chart and curvature response, and $\ell \ge N_{\mathsf{H}}$. Moreover

$$x_0 > \cdots > x_K > g^{-2}, \qquad x_{K-1} > Lg^{-2}, \qquad \tfrac{3}{4}t_j \le x_j \le 2t_j, \quad \tfrac{3}{4} \le t_j/x_j \le \tfrac{4}{3} \quad (j < K).$$

The weighted small-field progression is admitted with calibrations x_j and coefficients x_j. One can require in addition

$$\log x_{K-1} \ge \max\{2p_0, 1 + \log L\}, \qquad \delta(x_{K-1}) \le \delta_{\mathrm{ch}}, \qquad \delta(x_{K-1})^2 \le \tfrac{1}{8}\rho^2\delta(t_\flat)^2,$$

where the positive numbers $t_\flat, \rho, \delta_{\mathrm{ch}}$ are chosen from the construction constants as in the proof below. They are independent of the datum and the sources. ■

Proof. The coupling recursion gives $|x_j - t_j| \le \bar{B}$. The first-passage condition gives $x_{K-1} > Lg^{-2}$ and $x_K > g^{-2}$. Choose γ_s so small that $x_{K-1} \ge 12\bar{B}$. Then $|t_j/x_j - 1| \le 1/12$, which implies the displayed comparisons. The admitted sides satisfy $LMR(x_K) \le L^m$; the side conditions in the gauge construction give the admissibility of all preceding step lattices and $\ell \ge N_{\mathsf{H}}$. The admission lemma [YM3, Lem. 7.24] applies since $MR(x_K) \le L^m$. It supplies the place and progression hypotheses, including the weighted-array bound, for [YM3, Prop. 7.7]. The weighted progression uses its one-step coefficient discs and the diagonal choice of their coefficients. Its step and chain domains have fixed positive thresholds, including δ_{ch}. Once $t_\flat$ and ρ below have been chosen, all the remaining conditions follow by decreasing γ_s: $x_{K-1} > L\gamma_s^{-2}$, and $\delta(x)$ decreases to zero for $\log x \ge 2p_0$. This order of choices does not use any source norm or any later comparison depth. □

At a common complex scale λ, keep the counterterms e_j and the boxes $|B_j(b)| < \mathfrak{b}(x_j)$ fixed, put the field $\lambda x_j^{-1/2}B_j$ into every chart, and scale the nonlinear action potentials as specified in the proof below. Set

$$\xi_j(\lambda) = t_j + \lambda^2(x_j - t_j), \qquad \tau_j(\lambda) = \xi_j(\lambda)/x_j.$$

Thus $\xi_{j+1} = L^{-1}\xi_j - \lambda^2 e_{j+1}$. At $\lambda = 1$ this is the original progression. At $\lambda = 0$ its Gaussian precision on the non-pivot bonds is

$$\mathsf{M}_j = (t_j/x_j)\mathsf{B}_j^{\mathrm{cv}} + \widehat{\mathsf{G}}, \qquad \mathsf{M}_j \ge \tfrac{3}{4}\gamma^{\mathrm{tr}}.$$

In particular its Gaussian is conditioned on the stated box. The parameter $\tau_j(0)$ is t_j/x_j.

Let $E_\mu^{(j),\lambda}$ be the resulting local terms, including the source $\mu_1 w_1 + \mu_2 w_2$, and define

$$\mathsf{n}_j^\lambda = \sum_{X,z} \partial_{\mu_1}\partial_{\mu_2} E_\mu^{(j),\lambda}(X, 1, z)\Big|_{\mu=0}, \qquad \mathsf{h}_{2m}^{(n)} = \sum_{j\le n} \mathsf{n}_{j,[2m]}, \qquad \mathsf{H}^{(n)}[D] = \beta^3 a^2 \mathsf{h}_6^{(n)}.$$

The sum over z is over the level-j sites in X. The subscript $[2m]$ denotes the coefficient of λ^{2m} at zero.

Proposition D.12 (The coefficient tail)**.** There are constants $\rho \in (0, 1/2]$, c_0, C_J, C_n, C_R and radii

$$R_j = \frac{\delta(t_\flat)}{2\delta(x_j)}, \qquad R_j^{-2} \le c_0(\log x_j)^{2p_0} D_j/b, \qquad (\rho R_j)^2 \ge 2,$$

such that the progression is holomorphic in its common field scale on $|\lambda| < R_j$ through step $j+1$. Its integrated terms are even in λ. The source derivatives vanish on domains missing the corresponding source supports. With

$$\psi_i = \min\{1, (d/D_i)^3\}, \qquad \chi_i = \min\{1, (d/D_i)^{5/2}\},$$
$$\Pi_{i+1} = \max\{1, (d/D_{i+1})^3\} e^{-[d/(4MD_{i+1})-2]_+},$$

their norms at rate $\kappa^+ = (1+4\beta_s)\kappa$ obey

$$\|\dot{E}_a^{(i+1),\lambda}\|_{\kappa^+} \le C_J |\lambda|^2 R_i^{-2} D_i^{-1} \psi_i \|w_a\|_\infty,$$
$$\|\ddot{E}^{(i+1),\lambda}\|_{\kappa^+} \le C_J |\lambda|^2 R_i^{-2} D_i^{-2} \chi_i \eta, \qquad i \le j, \quad |\lambda| \le \rho R_j.$$

In particular $\mathsf{n}_{j+1,[0]} = \mathsf{n}_{j+1,[2]} = 0$ and

$$|\mathsf{n}_{j+1}^\lambda| \le C_n |\lambda|^2 R_j^{-2} D_j^{-2} \chi_j \Pi_{j+1} \eta.$$

For $m = 2, 3$ and $D_n \ge d$,

$$\sum_{j=n}^{K-1} |\mathsf{n}_{j+1,[2m]}| \le C_R b^{-m} l^{2mp_0} d^{m-2} (d/D_n)^{5/2-(m-2)} \eta.$$

Consequently, for $n \le n' \le K$ and $L^n \ge r$,

$$|\mathsf{H}^{(n')}[D] - \mathsf{H}^{(n)}[D]| \le C_R l^{6p_0} (r/L^n)^{3/2} r\eta. \tag{D.ix}$$

■

Proof. Choose the lower calibration $t_\flat \ge \gamma^{-2}$ so that $\log t_\flat \ge 2p_0$, $K_P \delta(t_\flat) \le w_*/8$, $\bar{B}\delta(t_\flat)^2 \le \sigma_* A_1^2$, and

$$S_\omega(E^\flat + \tfrac{3}{2}\alpha_0^2 B_\sharp) + 2 \le N^\flat(t_\flat) := \frac{w_* \alpha_0}{4K_\Lambda(1+\beta_s)\delta(t_\flat)}.$$

Here S_ω is the sum of the localization weights, and $E^\flat$ bounds the regular local terms and the terms created by one step. These constants are fixed by the one-step bounds; the two extra units provide room for its parameters and for Cauchy differentiation. Since δ decreases to zero, this choice exists. The last condition in Prop. D.11 implies $t_\flat < x_j$ and $(\rho R_j)^2 \ge 2$. The comparison $x_j \ge 3b/(4D_j)$ proves the stated bound on R_j^{-2}.

The action coefficient remains in its one-step disc:

$$|\xi_j(\lambda) - x_j| = |\lambda^2 - 1|\,|x_j - t_j| < 2R_j^2 \bar{B} \le \tfrac{1}{2}\sigma_* x_j \qquad (|\lambda| < R_j).$$

In the normalized fields, the gauge-fixing quadratic term is $-(1-\tau_j)(B_j, \widehat{\mathsf{G}} B_j)/2$. Combining it with the Gaussian of precision $\tau_j \mathsf{A}_j^{\mathrm{cv}}$ gives exactly $\tau_j \mathsf{B}_j^{\mathrm{cv}} + \widehat{\mathsf{G}}$. We give the scaling of the remaining potentials and check their domains. Put $t = x_j$ and $\mu = \lambda(t_\flat/t)^{1/2}$. For the Jacobian, gauge-fixing, action and localized potentials, respectively, the formulas are

$$V^{\mathrm{J},\lambda}(B) = V^{\mathrm{J}}[t_\flat](\mu B), \qquad V'^{,\lambda}(B) = V'[t_\flat](\mu B),$$
$$V^{\natural,\lambda}(B) = \mu^{-2} V^\natural[t_\flat](\mu B), \qquad \natural \in \{\mathrm{G}, \mathrm{A}\}.$$

The last two potentials vanish to order at least three in the fluctuation field. Their quotients therefore extend holomorphically across $\lambda = 0$ and vanish there. The first and fourth potentials vanish there as well. These formulas preserve the original potentials at $\lambda = 1$.

To check the bounds, write $r = \mathsf{C}_{\mathcal{A}}\mathfrak{b}(t)$, $r_\flat = \mathsf{C}_{\mathcal{A}}\mathfrak{b}(t_\flat)$, and $R_\flat = \delta(t_\flat)/\delta(t) = 2R_j$. Then

$$|\mu| r/r_\flat = |\lambda|/R_\flat.$$

Thus dilation by μ maps the fluctuation domain at t into that at $t_\flat$. Schwarz's lemma applied to the first and fourth potentials gives a factor $|\lambda|/R_\flat$. For the cubic-vanishing potentials, apply it after dividing by the first two powers of the dilation parameter. Their quadratic growth bound then gives

$$|V^{\natural,\lambda}(Y;B)| \le |\lambda| K_P \delta(t)(1+M^3 r_\flat^{-2}) e^{-\lambda_V d(Y)}(Q_Y(B)-1), \qquad \natural \in \{\mathrm{G},\mathrm{A}\}.$$

Here Q_Y is the quadratic fluctuation bound in the one-step expansion. The lower-calibration inequalities give $M^3 r_\flat^{-2} \le 2/9$. All three intrinsic potentials are consequently bounded by $|\lambda| k_{\mathrm{int}} e^{-\lambda_V d(Y)} Q_Y(B)$, and the fourth by $|\lambda| k_{\mathrm{iv}} e^{-\lambda_V d(Y)}$, where

$$k_{\mathrm{int}} = 2K_P\delta(t), \qquad k_{\mathrm{iv}} = K_\Lambda s_{\mathfrak{a}}(t) n_\flat.$$

The number $n_\flat$ is the norm of the localized array at the lower calibration, including the two units of room above; $s_{\mathfrak{a}}(t) = \delta(t)/a_{0,j+1}$. The chosen room yields

$$R_j k_{\mathrm{int}} = K_P\delta(t_\flat) \le w_*/8, \qquad R_j k_{\mathrm{iv}} = \tfrac{1}{2} K_\Lambda s_{\mathfrak{a}}(t_\flat) n_\flat \le w_*/8.$$

Locality and gauge covariance are preserved by dilation. These bounds verify the hypotheses of the scalar step.

We recall the step estimate with its parameter dependence. The scalar cluster expansion in [YM3, Prop. 4.54], on these domains, has nonconstant bond contribution bounded by

$$K_\Pi |\lambda| (32 k_{\mathrm{int}} + 9 k_{\mathrm{iv}}) e^{-\lambda_{\mathrm{out}} d(X)}, \quad R_j k_{\mathrm{int}} \le w_*/8, \quad R_j k_{\mathrm{iv}} \le w_*/4.$$

Thus its bound on $|\lambda| < R_j$ is at most $41 K_\Pi w_*/8$. The series converges normally in the action coefficient, field scale and bare-array parameters. Its change under $(\lambda, B) \mapsto (-\lambda, -B)$ is zero after Gaussian integration. Its value at zero is independent of the bare-array variation. Evenness, vanishing at zero and Cauchy's formula therefore bound its derivative in that variation by $c|\lambda|^2 R_j^{-2}$ on the smaller disc. The derivative is taken in the localized-array seminorm at $t_\flat$; the spare unit in the preceding room bound supplies its Cauchy disc. The pointed version differs by the bounded tent sum $3L^3$. These facts prove holomorphy, evenness and the uniform step estimate used in the induction.

Here are the localization counts that close its first and second source derivatives. A source weight at level i has size at most $36 M^3 \alpha_0^2 D_i^{-1} \psi_i \|w_a\|_\infty$. A member from level $i' = i - n$ whose domains meet that source is read with factor at most

$$c' \left[L^{3n} e^{-[2L^{n-1}-2]_+} + (n+2) L^{4-4n} \min\{C L^{3n}, C' N_{i'}\} \right].$$

This is the bracket-localization count of [YM3, Prop. 4.52]: the first term counts the distant domains; in the second, count only places over a cube meeting the source. Here $N_{i'} \le 77 \max\{1, (d/D_{i'})^3\}$. The current-level reading costs at most two. Furthermore

$$R_{i-n}^{-2} \le \tfrac{4}{3}(1+n)^{2p_0} L^{-n} R_i^{-2}.$$

For $D_{i'} \ge d$, one has $N_{i'} \le 77$, $\psi_{i'-1} \le L^{3(n+1)} \psi_i$ and $\chi_{i'-1} \le L^{5(n+1)/2} \chi_i$. After the radius and length factors are included, the first derivative has the summable factor $77 L^{7-n}$, and the second has $L^{7/2-n/2}$, up to fixed powers of $n+2$. For $D_{i'} < d$, a domain meeting both supports has size at least $[d/(4MD_{i'}) - 2]_+$. Passing from rate κ^+ to κ supplies $e^{-4\beta_s \kappa [d/(4MD_{i'})-2]_+}$, which sums the smaller-scale terms against every power and logarithm. The product of first derivatives is controlled by $\psi_i^2 \le \chi_i$.

These two finite sums are the constants in the two derivative recursions. Choose $\rho \le 1/2$ so that ρ^2 times each is at most its induction margin. This choice precedes γ_s. Induction proves the two stated derivative bounds. Each new domain depends only on the old members and source plaquettes within it, so it also proves the support assertion. Summing the second derivative over domains meeting both supports gives Π_{j+1} and the bound for n_{j+1}^λ.

At order zero the activity has no source dependence. At order two its mixed derivative is a covariance of two linear forms in the field at the trivial configuration. Each form is invariant under

constant gauge rotations. Averaging the adjoint representation over SU(2) gives zero, so both forms vanish. This proves the two missing low orders.

Finally apply Cauchy's formula on $|\lambda| = \rho R_j$. The coefficient of order $2m$ is at most

$$cb^{-m}(\log x_j)^{2mp_0} D_j^{m-2} \chi_j \Pi_{j+1} \eta.$$

If $D_j \geq d$, then $\Pi_{j+1} = 1$, $\log x_j \leq l$ and $D_j^{m-2}\chi_j = d^{m-2}(d/D_j)^{5/2-(m-2)}$. For $m = 2, 3$ the exponent is positive, so summing the geometric progression gives the tail bound. Multiply its $m = 3$ case by $\beta^3 a^2$ and use $d/D_n = r/L^n$ and $\beta^3 a^2 b^{-3} d = r$. This proves eq. D.ix. □

D.8. **Removing Gaussian restrictions.**

Lemma D.13. Let ν be a centered Gaussian measure on $\mathbb{R}^I$, with coordinate variances at most σ_i^2. Let Q be open, bounded, convex, and symmetric, containing $\prod_i (-c_i, c_i)$. Set $t_i = c_i/\sigma_i$ and $\epsilon = \sum_i e^{-t_i^2/2}$. Suppose that $F_1, \dots, F_r$, $r = 2, 3$, are complex polynomials of degrees at most k_j, with $|F_j| \leq S_j$ on Q, and put $k = \sum_j k_j$. If $t_i^2 \geq \max\{4, 2k\}$ and $2^k \epsilon \leq 1$, then their cumulants in ν and $\nu(\cdot \mid Q)$ differ by at most

$$q_r(1 + 2^k)\epsilon \prod_j S_j, \qquad q_2 = 5, \qquad q_3 = 37.$$

■

Proof. For $z \notin Q$, let $s = \max_i |z_i|/c_i \geq 1$. The segment from $-z/s$ to z/s lies in $\bar{Q}$. The polynomial $\tau \mapsto F_j(\tau z/s)$ is bounded by S_j on $[-1, 1]$, so the Chebyshev bound gives $|F_j(z)| \leq S_j(2s)^{k_j}$. On Q^c,

$$s^k \leq \sum_i (|z_i|/c_i)^k \mathbf{1}_{\{|z_i| \geq c_i\}}.$$

The one-dimensional Gaussian tail moment satisfies $\mathbb{E}[(|z_i|/c_i)^k; |z_i| \geq c_i] \leq e^{-t_i^2/2}$ under the assumed bound on t_i. Consequently

$$\nu(Q^c) \leq \epsilon, \qquad \int_{Q^c} \prod_j |F_j| \, d\nu \leq 2^k \epsilon \prod_j S_j.$$

For any subproduct F, write S for the corresponding product of bounds and m, m^{r} for its full and conditional moments. Then

$$m^{\mathrm{r}} - m = m^{\mathrm{r}} \nu(Q^c) - \int_{Q^c} F \, d\nu, \qquad |m^{\mathrm{r}} - m| \leq (1 + 2^k)\epsilon S, \quad |m^{\mathrm{r}}| \leq S, \quad |m| \leq 2S.$$

Expand the cumulant over partitions. For a partition with p blocks, telescoping its product of moments costs at most $p 2^{p-1}(1 + 2^k)\epsilon \prod_j S_j$. Multiplication by $(p-1)!$ and summation give $q_r = \sum_p \mathrm{S}(r, p) p! 2^{p-1}$, which is five for $r = 2$ and thirty-seven for $r = 3$. □

Lemma D.14 (Coefficients of a logarithm)**.** If G is holomorphic on the unit disc, $G(0) = 0$, and $\operatorname{Re} G \leq E$, then $|G_{[m]}| \leq 2E$ for every $m \geq 1$. In particular, if $F(0) = 1$, $F \neq 0$, and $|F| \leq e^E$ there, and

$$F = 1 + \lambda R_1 + \lambda^2 R_2 + O(\lambda^3),$$

then $W_1 = -R_1$ and $W_2 = R_1^2/2 - R_2$ satisfy $|W_1|, |W_2| \leq 2E$. ■

Proof. For $0 < r < 1$ and $m \geq 1$,

$$G_{[m]} r^m = -\frac{1}{\pi} \int_0^{2\pi} (E - \operatorname{Re} G(re^{i\theta})) e^{-im\theta} \, d\theta.$$

The nonnegative factor has mean E, since $\operatorname{Re} G(0) = 0$. Let $r \uparrow 1$. For the second assertion use the holomorphic logarithm $G = \log F = -\lambda W_1 - \lambda^2 W_2 + O(\lambda^3)$. □

D.9. **The coefficient before and after removing the tests.** Use the pure-gauge progression of Props. D.11 and D.12, calibrated at the running coefficients x_i. Let $N = \ell L^K$, $\mathsf{N} = N + \beta$, and write $a_K = L^{-K}$, $b = a_K\beta$. The admitted coupling window gives

$$x_0 > x_1 > \cdots > x_K > g^{-2}, \qquad \tfrac{3}{4} \le \tau_i := L^{-i}x_0/x_i \le \tfrac{4}{3}.$$

The side is large enough for every step and for the free curvature estimates. In particular $\ell L \ge C_L M$ and $\ell \ge N_H$. Take the coupling ceiling small enough that the background chains at $|\lambda| \le 1$ lie in their fixed cores.

Let $\mathsf{B}_i^{\mathrm{cv}}$ and $\hat{\mathsf{G}}$ be the free curvature and gauge-fixing matrices on the non-pivot bonds at height i. Both are nonnegative, and

$$\mathsf{A}_i^{\mathrm{cv}} = \mathsf{B}_i^{\mathrm{cv}} + \hat{\mathsf{G}} \ge \gamma^{\mathrm{tr}} > 0, \qquad \mathsf{M}_i = \tau_i \mathsf{B}_i^{\mathrm{cv}} + \hat{\mathsf{G}} \ge \tfrac{3}{4}\gamma^{\mathrm{tr}}.$$

These are the free-template operators of [YM3, Prop. 7.3] and [YM3, Prop. 7.6]. Let ν_i° be the centered Gaussian with precision $\mathsf{M}_i \otimes 1_{\mathfrak{g}}$, and let ν_i^{r} be its restriction, normalized to mass one, to $\{|B_i(b)| < \mathfrak{b}(x_i)\}$, where $\mathfrak{b}(x) = A_1(\log x)^{p_0}$. The norm here is the Euclidean norm on each copy of $\mathfrak{g}$; these windows are products of balls.

For real bounded sources $w_a = f_a \circ x$ on plaquettes, let n_j^{λ} be the mixed source derivative of the scalar activity sum of level j at the trivial configuration and common field scale λ. Write

$$\mathsf{h}_{2m}^{(n)} = \sum_{j \le n} \mathsf{n}_{j,[2m]}.$$

The sources have supports in balls of radius d at distance at least d, with $8a_K \le d$. No sign or Lipschitz restriction is needed in the next proposition.

Proposition D.15. Let $2 \le n \le K$, and let $\mathsf{Bx} = \prod_{i<n}\{|B_i(b)| < \mathfrak{b}(x_i)\}$. There are bounded functions O_a^{λ} and D^{λ} on this set, continuous for $|\lambda| \le 1$ and holomorphic for $|\lambda| < 1$, such that

$$\sum_{j \le n} \mathsf{n}_j^{\lambda} = \langle \mathsf{O}_1^{\lambda}; \mathsf{O}_2^{\lambda} \rangle_{\mathsf{D}^{\lambda}\, dB}, \qquad \int \mathsf{D}^{\lambda}\, dB = 1, \qquad \mathsf{D}^0\, dB = \bigotimes_{i<n} \nu_i^{\mathrm{r}}. \tag{D.x}$$

The observables have the form $\mathsf{O}_a^{\lambda}(B) = o_a(\lambda B)$, with o_a analytic near zero. Uniformly on the window,

$$\mathsf{O}_a^{\lambda} = \lambda^2(X_{a,0} + \lambda X_{a,1} + \lambda^2 X_{a,2}) + O(\lambda^5),$$
$$\mathsf{D}^{\lambda}/\mathsf{D}^0 = \exp[-\lambda W_1 - \lambda^2 W_2 + O(\lambda^3)].$$

Here $X_{a,m}$ is homogeneous of degree $m+2$; W_1 is odd of degree at most three; and W_2, up to an additive constant, is even of degree at most four. Thus

$$\mathsf{h}_4^{(n)} = \langle X_{1,0}; X_{2,0} \rangle_{\otimes_i \nu_i^{\mathrm{r}}}, \qquad \mathsf{h}_6^{(n)} = \gamma_2^{\mathrm{r}} := \gamma_2[\otimes_i \nu_i^{\mathrm{r}}].$$

Put $\gamma_2^{\circ} = \gamma_2[\otimes_i \nu_i^{\circ}]$. If

$$\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2 \ge 80, \qquad 10240 N^3 e^{-\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2/8} \le 1,$$

then

$$\beta^3 a_K^2 |\gamma_2^{\mathrm{r}} - \gamma_2^{\circ}| \le C_X \mathsf{N}^{36} e^{-\gamma^{\mathrm{tr}}\mathfrak{b}(x_{n-1})^2/8} \|w_1\|_{\infty} \|w_2\|_{\infty}. \tag{D.xi}$$

The constant C_X depends only on the fixed RG constants. ∎

Proof. Form the background chain downward from the trivial point $v_n^{\lambda} = 0$, inserting $\lambda x_i^{-1/2} B_i$ at step i. Its radius satisfies

$$r_n = 0, \qquad r_i = L^{-1} r_{i+1} + \tfrac{11}{10} c_{\mathfrak{h}} \delta(x_i) \le \tfrac{8}{5} c_{\mathfrak{h}} \delta(x_i), \qquad \|v_i^{\lambda}\| \le |\lambda| r_i.$$

The coupling ceiling keeps these radii inside the chart domains, for complex as well as real λ. Write $\tilde{U}_i^{\lambda} = \exp(iL^{-i}A_i^{\lambda})$ for its backgrounds. Define O_a^{λ} as the action insertion $a_K^{-1} \sum_p w_a(p) a_p(\tilde{U}_1^{\lambda})$ plus

the source term in the height-zero step. Define D^{λ} as the product of the normalized step densities along this chain.

Iterating the one-step integral identity gives

$$\exp D_n^{\lambda,\mu} = \int \exp(\mu_1 \mathsf{O}_1^{\lambda} + \mu_2 \mathsf{O}_2^{\lambda}) \mathsf{D}^{\lambda}\, dB,$$

where $D_n^{\lambda,\mu}$ is the sourced activity exponent at the trivial top. At the bottom step its value in the chart equals its value at $\tilde{U}_1^{\lambda}$: the chart triple is in the level-one core, and its restrictions and the configuration have the same pairs. Local terms depend only on those pairs. This verifies the bottom step as well as all later steps.

The identity first follows for real $0 < \lambda \le 1$. Both sides are holomorphic in μ and λ. Indeed, the provision gives holomorphic conditioned covariances and connecting operators; their conditioning identity gives a holomorphic whole covariance and inverse. The gauge forms and potentials are holomorphic on the same cores. Bounded integration over the windows preserves this property, and the step normalizers have no zeros by the scalar step in Prop. D.12, with the bounds in Lem. D.16. The identity therefore extends to $|\lambda| < 1$. At zero sources it gives normalization of D^{λ}; its mixed logarithmic derivative gives eq. D.x.

At $\lambda = 0$, the chain is trivial and the potentials vanish. The step's two quadratic forms combine to

$$\tau_i \mathsf{A}_i^{\mathrm{cv}} + (1 - \tau_i)\hat{\mathsf{G}} = \mathsf{M}_i.$$

At height zero, $\tau_0 = 1$ and the whole covariance is $(\mathsf{A}_0^{\mathrm{cv}})^{-1} \otimes 1$, so the same formula holds there. This proves the Gaussian product assertion, including the bottom field.

The action insertions start at quadratic order, and the chain reads (λ, B) through λB. Their Taylor coefficients have the stated homogeneous degrees. To control the degrees in the logarithm of the density, telescope the activity terms against the normalizers of adjacent steps. Their sum leaves a constant, the running counterterms at the backgrounds, the localized Gaussian terms, and the determinants of the whole covariances. All remaining terms have one of the forms

$$\lambda^{-2} F(\lambda, \lambda B), \quad F(\lambda, \lambda B), \qquad F(\lambda, \zeta) = O(|\zeta|^2) \text{ in the first case.}$$

The first class consists of the quadratic forms and the action and gauge-fixing potentials; the second contains the other potentials, determinants, localized Gaussian terms, and counterterms. Their functions F are holomorphic near zero. Thus the coefficient of λ^m has degree at most $m + 2$ in the first class and at most m in the second. The simultaneous substitution $(\lambda, B) \mapsto (-\lambda, -B)$ preserves the density: the moved coefficient is even in λ, and induction preserves this parity in the earlier activity families. Hence W_1 is odd and W_2 is even. Constants in W_2 disappear from the relevant cumulants. Expanding the covariance now gives exactly eq. D.v and the two asserted coefficients.

It remains to justify the quantitative removal of the windows. The bounds in the next lemma give $|X_{a,m}| \le C_S \mathsf{N}^8 \|w_a\|_\infty$ and $|W_1|, |W_2| \le 2 C_S \mathsf{N}^7$ on Bx. Every coordinate variance is at most $4/(3\gamma^{\mathrm{tr}})$, and the window contains the component cube of half-side $\mathfrak{b}(x_{n-1})/\sqrt{3}$. Thus the parameters of Lem. D.13 satisfy

$$t_i^2 \ge \tfrac{1}{4} \gamma^{\mathrm{tr}} \mathfrak{b}(x_{n-1})^2 \ge 20, \qquad \epsilon \le 10 N^3 e^{-\gamma^{\mathrm{tr}} \mathfrak{b}(x_{n-1})^2/8}.$$

Apply that lemma to the seven cumulants in eq. D.v. Their total polynomial degrees are at most ten. The largest product of their bounds has power N^{30}. Finally $\beta^3 a_K^2 \le \mathsf{N}^3$, and the coordinate count contributes at most $10\mathsf{N}^3$. This gives eq. D.xi. □

Lemma D.16 (Bounds on the Gaussian windows)**.** In the preceding setting, for $|\lambda| < 1$,

$$|\mathsf{O}_a^{\lambda}| \le C_S \mathsf{N}^8 \|w_a\|_\infty, \qquad \mathsf{D}^{\lambda} \ne 0, \qquad |\mathsf{D}^{\lambda}| \le e^{C_S \mathsf{N}^7} |\mathsf{D}^0|.$$

Consequently $|X_{a,m}| \le C_S \mathsf{N}^8 \|w_a\|_\infty$ and $|W_1|, |W_2| \le 2 C_S \mathsf{N}^7$. ■

Proof. There are at most $3N^3$ plaquettes and $a_K^{-1} \le N$. The backgrounds stay in a fixed complex neighborhood of one. These facts bound the first part of each observable. The source family at height zero has size at most $CL^K\|w_a\|_\infty$ and at most $3N^3$ terms. Its field weight on the window is at most $CN^3\mathfrak{b}(\beta)^2$. Since $L^K \le N$ and $\mathfrak{b}(\beta)^2 \le C\beta$, this gives the displayed N^8 bound.

For the density, the real-reference comparison of the whole covariance on a complex core gives

$$\|\mathbf{C}_*^{-1}\| \le c_{\mathrm{r}} + \gamma_{\mathrm{r}}/2, \qquad \operatorname{Re}\mathbf{C}_*^{-1} \ge \gamma_{\mathrm{r}}/2, \qquad \|\mathbf{C}_*\| \le 2/\gamma_{\mathrm{r}}.$$

The gauge-fixing operators also have uniformly bounded row sums. Thus each quadratic form on a window is bounded by $CN^3\mathfrak{b}(x_i)^2$. The potential bound has the form $Ce^{-\lambda_V d_i(Y)}Q_Y(B_i)$, where $\lambda_V > \kappa_{\mathcal{D}} + \log 2$ and $\lambda_V > \kappa_{\mathcal{D}} + 2$. Summing over polymers, first at a fixed cube or bond, therefore gives the same bound. Each step density is an exponential and has no zero.

Its normalizing integral is $\det(2\pi\tau_i(\lambda)^{-1}\mathbf{C}_*)^{1/2}$ times the exponential of a sum of local terms. The latter sum has modulus at most CN^3. The covariance bounds, $1/2 < \operatorname{Re}\tau_i(\lambda) \le |\tau_i(\lambda)| < 3/2$, and dimension at most $9N^3$ bound the modulus of the determinant factor between e^{-CN^3} and e^{CN^3}. Thus all normalizers are nonzero and have both bounds. The same estimates at zero bound the reciprocal of D^0. Multiplying over at most $K \le N$ heights, with $\mathfrak{b}(x_i)^2 \le C\beta$, proves the claimed N^7 bound. Cauchy's estimate gives the observable coefficients; Lem. D.14 applied to $\mathsf{D}^\lambda/\mathsf{D}^0$ gives the logarithmic coefficients. □

D.10. **Identification with the flat-top coefficient.** Set

$$\mathsf{T}[N,n](w_1,w_2) = \sum_{p,q} w_1(p)w_2(q)\mathsf{g}_2[N,n](p,q).$$

The next identity is exact. In particular, no error depending on the running counterterms enters it.

Proposition D.17. In the setting of Prop. D.15,

$$\beta^3 a_K^2 \gamma_2^\circ = \mathsf{T}[N,n](w_1,w_2).$$

The right side depends only on the bare lattice, the depth n, L, and the two lattice sources. The identity holds for every size of the integration windows allowed in Prop. D.15. ■

Proof. For real $0 < \lambda \le 1$, write $\zeta_i = \lambda x_i^{-1/2} B_i$. The background chain used in Prop. D.15 depends only on these fields. We compare it with the composed chart of Prop. D.10, keeping track of the gauge transformations between their backgrounds.

Start at the flat top $W_n = 1$. At a level $i+1$, choose a background $U_{+,i+1}$ in the minimal orbit over W_{i+1}, axial on its blocks, and a gauge transformation u_{i+1} such that the chain's background is $\tilde{U}_{i+1}^\lambda = U_{+,i+1}^{u_{i+1}}$. With this transformation fixed, put

$$\check{\zeta}_i = \mathrm{Ad}(u_{i+1})^{-1}\zeta_i, \qquad W_i = V_{W_{i+1}}[\check{\zeta}_i].$$

The background-lowering identity gives an element of the minimal orbit over W_i gauge equivalent to the chain's next background. Put this element into the axial gauge and absorb that change into u_i. The chain bounds keep $\varpi(W_i) < \varepsilon_{\mathcal{W}}\alpha_0$ and all three triples—the axial triple, its transform by u_{i+1}, and the chain triple—in their cores. The last two have the same pair. This construction works downward for every ζ_i in its window, with u_{i+1} depending only on higher fields. At height zero it gives the bare configuration $W_0 = V_{W_1}[\check{\zeta}_0]$.

The provision is gauge covariant and depends on a triple only through its pair. Therefore the whole covariance, the gauge-fixing forms, the potentials, the source terms, and the localized Gaussian terms can all be read at the axial triple with field $\check{\zeta}_i$. For the whole covariance this also follows directly from the conditioning identity $\mathbf{C}_* = \mathbf{C} + \mathbf{C}\boldsymbol{\Gamma}\boldsymbol{\Gamma}^{\mathsf{T}}\mathbf{C}$: the two operators on the right transform by the same orthogonal conjugation.

At the axial triple, the sum rules identify the whole covariance with A_W^{-1}, where $\mathsf{A}_W = \mathsf{B}_W + \hat{\mathsf{G}}_W$. The quadratic form of the step is

$$-\frac{\tau_i(\lambda)}{2}(B_i, \mathsf{B}_W B_i) - \frac{1}{2}(B_i, \hat{\mathsf{G}}_W B_i).$$

Its second term combines with the gauge-fixing potential to give $-\lambda^{-2} x_i G_i(W_i)$. Its first combines with the action potential to give

$$-\lambda^{-2}\xi_i(\lambda) L^i [A(\tilde{U}_i^\lambda) - A(\tilde{U}_{i+1}^\lambda)].$$

The Jacobian potential gives the chart Jacobian. The localized Gaussian family and its determinant give the elimination normalizer and a constant. The ordinary activity families telescope against the normalizing denominators. Their remaining counterterms cancel the changes in ξ_i, since

$$\lambda^{-2}\xi_j(\lambda) L^j + \sum_{i \le j} L^i e_i = \lambda^{-2} x_0.$$

Using $A(\tilde{U}_n^\lambda) = 0$, we obtain

$$\mathsf{D}^\lambda(B) = c_n(\lambda) \exp\left[-\lambda^{-2}\left(x_0 A(W_0) + \sum_{j<n} x_j G_j(W_j)\right) - \theta\right], \tag{D.xii}$$

where $c_n(\lambda)$ is independent of B. The observables become $a_K^{-1}\sum_p w_a(p) a_p(W_0)$. The function θ is the composed chart density, up to a constant.

This comparison does not require a measurable choice of all the gauge transformations as a function of all fields. At a fixed choice of the higher fields, $B_i \mapsto \mathrm{Ad}(u_{i+1})^{-1} B_i$ is one orthogonal map. It preserves Lebesgue measure and every product of centered Euclidean balls. Integrate first in the finest field. Equality of the integrands at corresponding fields gives equality of these integrals by that fixed orthogonal change of variables. Repeat upward. Fubini applies to the bounded Borel integrands. Thus the two descriptions give equal integrals on every sufficiently small product of such balls, with observables or their product inserted. The use of Euclidean balls here is essential to this change of variables.

For completeness, identify the Gaussian polynomials in this equality. Write $(qB)_i = x_i^{-1/2} B_i$. In the chain variables, let Φ^{ch}, Θ^{ch}, and o_a^{ch} denote respectively the phase, the logarithm of the chart density, and the observable in eq. D.xii. They are analytic near zero. The zero-scale Gaussian identifies their Hessian as

$$H^{\mathrm{ch}} = \bigoplus_{i<n} x_i(\mathsf{M}_i \otimes 1) > 0.$$

Thus q maps the full Gaussian product to the Gaussian with covariance $(H^{\mathrm{ch}})^{-1}$. The polynomials in Prop. D.15 are

$$\begin{aligned} X_{a,m}(B) &= (o_a^{\mathrm{ch}})_{m+2}(qB), \\ W_1(B) &= (\Phi_3^{\mathrm{ch}} + \Theta_1^{\mathrm{ch}})(qB), \\ W_2(B) &= (\Phi_4^{\mathrm{ch}} + \Theta_2^{\mathrm{ch}})(qB) + \text{constant}. \end{aligned}$$

Cumulants of order at least two do not see the constant. Hence γ_2° is the Laplace coefficient for these functions.

Choose the common windows so small that both phases are bounded below by a positive multiple of the squared coordinate norm. The preceding integral identity gives one covariance for every Laplace parameter $\beta' > 0$. In the chain chart its order-three coefficient is γ_2°. In the composed chart the phase is $\beta\phi_\kappa$, with $\kappa_j = x_j/\beta$, and the observables carry the factors a_K^{-1}. Comparing eq. D.vi for these two descriptions gives

$$\gamma_2^\circ = \beta^{-3} a_K^{-2} \sum_{p,q} w_1(p) w_2(q) \mathsf{g}_2[N, n](p, q).$$

The coefficients do not depend on the chosen windows, and Prop. D.10 removes the gauge strengths. The bare lattice remains Λ_N on replacing (ℓ, K) by $(\ell L^{K-n}, n)$, so the stated identification follows. □

D.11. **Locality of the constrained coefficient.**

Lemma D.18. The following three elementary facts will be used.

(i) Suppose $0 < \sigma \le 1/120$, and $F(y,z)$ is holomorphic for $\|y\| < \sigma$, $|z| < \sigma$, with values in $\mathfrak{g}^{\mathbb{C}}$, satisfying $|F| \le 1/40$, $F(0,0) = 0$, and $D_z F(0,0) = 1$. For $\|y\| < \sigma^3$, the equation $F(y,z) = 0$ has a unique solution with $|z| \le \sigma^2$. It is holomorphic, satisfies $|z(y)| \le \|y\|/(15\sigma)$, and is real at real y when F is real there.

(ii) If a lattice box embeds injectively in a bare torus, every closed path γ in it that lifts to a closed path in the box has integer Stokes coefficients supported on its plaquettes, with

$$a(\gamma) = \sum_p k_\gamma(p)(da)(p), \qquad |k_\gamma(p)| \le |\gamma|.$$

If each side of the box has at most $S-1$ sites, where the torus has side S, every closed path in the box has this lifting property.

(iii) The joint Gaussian cumulant of homogeneous polynomials $P_1, \dots, P_r$ is the sum of Wick pairings connected on the set of polynomial entries. If their total degree is n and the covariance entries have modulus at most c, its modulus is at most

$$(n-1)!! \prod_a \|P_a\|_1 \, c^{n/2},$$

and it is zero for odd n. If every connected pairing has an entry bounded by ϵ, replace $c^{n/2}$ by $c^{n/2-1}\epsilon$. For two covariances bounded by c, the difference of cumulants is at most

$$(n-1)!! \frac{n}{2} \prod_a \|P_a\|_1 \, c^{n/2-1} \sup |C - C'|.$$

Here $\|P\|_1$ is the sum of the moduli of its coefficients.

■

Proof. For (i), set $g(y,z) = z - F(y,z)$. Then $|g| \le 1/30$, $g(0,0) = 0$, and $D_z g(0,0) = 0$. Cauchy and Schwarz estimates give

$$|D_z g(y,z)| \le \frac{2}{15}\sigma^{-2} \max\{\|y\|, |z|\}, \qquad |g(y,0)| \le \frac{\|y\|}{30\sigma}$$

on the half-radius domain. For $\|y\| < \sigma^3$, the map $z \mapsto g(y,z)$ preserves the closed ball of radius σ^2 and contracts by $2/15$. Its iterates give the holomorphic solution and the bound stated; uniqueness gives the reality assertion.

For (ii), join a corner of the box to each site by paths running successively in the three coordinate directions. For a bond $e = (z, z + e_\mu)$, the path to z, followed by e and the reverse path from $z + e_\mu$, is the boundary of a strip of plaquettes in the box. Sum these strips along γ; the corner paths cancel and each plaquette occurs at most $|\gamma|$ times. Under the additional size condition, two box sites joined by a torus bond differ by an actual unit vector: their coordinate differences have modulus at most $S-2$. This proves the lifting assertion.

For (iii), partition each Wick pairing by the connected components of its graph on the entries. The moment–cumulant formula identifies the connected sum with the cumulant. Count at most $(n-1)!!$ pairings. The comparison bound follows by telescoping the product of its $n/2$ covariance entries. □

Proposition D.19 (Comparison of two bare tori)**.** Let $N = 2L^{\mathsf{m}}$, $N' = L^s N$, $s \geq 1$, and $1 \leq t < \mathsf{m}$. Write $\pi : \Lambda_{N'} \to \Lambda_N$ for reduction, and assume

$$\nu = N/L^t \geq \nu_W = \max\left\{192, \frac{48(3\log L + \log 2)}{\delta_{\mathrm{tp}} L(L-1)}, \frac{C_L M}{L}\right\}.$$

If the base points of p', q' are within $N/8$ in the sup metric, then, for $m' = 0, 2$,

$$|\mathsf{g}_{m'}[N,t](\pi p', \pi q') - \mathsf{g}_{m'}[N',t](p',q')| \leq C'_W N'^{237} e^{-\delta_{\mathrm{tp}} L\nu/48}. \tag{D.xiii}$$

For sources supported in balls of radius $r \geq 8$, with all their base points within $N/8$, let $\breve{w}_a$ be their lift on one ball of radius $N/4$, extended by zero. Then

$$|\mathsf{T}[N,t](w) - \mathsf{T}[N',t](\breve{w})| \leq 900 r^6 C'_W N'^{237} e^{-\delta_{\mathrm{tp}} L\nu/48} \|w_1\|_\infty \|w_2\|_\infty.$$

The constant C'_W depends only on L, C_{tp}. ∎

Proof. Choose every auxiliary gauge-fixing strength $\kappa_j = 1$, as permitted by Prop. D.10. We construct a chart in which its local Taylor polynomials agree on the two tori. For $S = N, N'$, let Q_t be the linearized block average and $Y_S = \ker \mathsf{Q}_t$. For a top bond c, let $\mathbf{1}_c$ be one on bare bonds crossing from its first top block to its second, minus one on their reverses, and zero elsewhere. A bare bond either lies inside one top block or crosses one such interface. Moreover, $\mathsf{Q}_t \mathbf{1}_c$ is one at c and zero at every other positive top bond.

The average at c depends only on its double block. Solve its constraint independently by writing

$$\hat{y} = y + \sum_{c>0} h_c(y) \mathbf{1}_c, \qquad \mathcal{M}^t(e^{i\hat{y}}) = 1, \qquad y \in Y_S.$$

Use Lem. D.18(i) with $\sigma_t = (75L)^{-t}/(60L)$ for the function $\operatorname{Lg} \mathcal{M}^t(e^{i(y+z\mathbf{1}_c)})(c)$. Its derivative in z at zero is one. To check its complex domain, one averaging step takes a configuration whose variables and inverses lie within $\epsilon \leq 1/(200L)$ of one to variables and inverses within $75L\epsilon$ of one. Indeed the contour means are within $20L\epsilon$, and the products entering the logarithm are within $48L\epsilon < 1$. The average is

$$\exp\left(L^{-3} \sum_{x \in B(c_-)} \log W_x\right) U[c],$$

where W_x is the contour-transport product divided by the straight transport $U[c]$. This formula is holomorphic. At real configurations with $\epsilon \leq 1/(1000L^2)$, it agrees with the given block average, since the contour means are unique and the averaging plaquette threshold holds. Starting with $\epsilon_0 = 4\sigma_t$, its iterates satisfy $\epsilon_k = (75L)^{k-t}/(15L)$. Thus $\epsilon_{t-1} = 1/(1125L^2)$, every step is admitted, and the final logarithm has modulus less than $1/40$. The implicit-function lemma applies.

Each h_c depends only on the double block of c, is real at real fields, and satisfies $|h_c(y)| \leq \|y\|_\infty/(15\sigma_t)$. Since $y \in \ker \mathsf{Q}_t$, $\hat{y} = y + O(y^2)$. The map

$$A \longmapsto \left(\operatorname{Lg} \mathcal{M}^t(e^{iA}), A - \sum_c (\mathsf{Q}_t A)(c) \mathbf{1}_c\right)$$

has invertible differential at zero, with inverse $(w, y) \mapsto y + \sum_c w(c)\mathbf{1}_c$. Its inverse followed by exponentiation is an adapted chart, so Prop. D.10 allows us to compute the coefficients in it.

The action, gauge-fixing terms, Haar density, and Jacobian in this chart are sums of local holomorphic terms. Each is bounded by one on $\|y\|_\infty < \sigma_t^3$, involves at most $12L^{3t}$ bonds, and has support diameter at most $2L^t$. The Jacobian is a product of the derivatives of the top constraints in the directions $\mathbf{1}_c$, since the constraints use distinct interfaces. Reduction by π preserves blocks, contours, and these formulas on sets of diameter at most $N/2$. Thus the local terms are identical there after renaming variables.

We next estimate the Gaussian covariance in this chart. By independence of the gauge strengths, choose $\kappa_j = 1$. For a bare site off the top lattice, let $j(x)$ be its highest lattice level, let $\hat{x}$ be its parent at level $j(x)+1$, and let $\mathrm{anc}(x)$ be its successive ancestors below the top. Put $\xi_A(x) = (\mathsf{Q}_{j(x)}A)(\hat{x}, x)$. The Hessian form is

$$\mathsf{H}_S(A, A') = (dA, dA') + \sum_x \langle \xi_A(x), \xi_{A'}(x)\rangle.$$

The gauge directions are $d\chi$, with $\chi = 0$ on the top sites, and $\xi_{d\chi}(x) = \chi(x) - \chi(\hat{x})$. This is a bijection, with inverse $\chi(x) = \sum_{z\in\mathrm{anc}(x)} \xi(z)$. Consequently

$$Y_S = Y_S^{\mathrm{ax}} \oplus d\mathcal{X}_S, \qquad \mathsf{H}_S(y, y) = \|dy^{\mathrm{ax}}\|^2 + \sum_x |\xi_y(x)|^2.$$

The map d identifies Y_S^{ax} with the curvatures whose order-t plaquette averages vanish. For the range assertion, use $\mathsf{Q}_t = \mathbf{Q}_t^s - d\vartheta_t$, where $\mathbf{Q}_t^s$ is the straight average: a curvature with zero plaquette average has a potential with zero straight average, and adding a gradient adjusts its top values and then its contour variables. For injectivity, a closed potential is a gradient plus a constant one-form. Summing its zero top average around each period kills the constant; the zero contour variables then make the gradient zero. In particular $\mathsf{H}_S > 0$ on Y_S.

The Gaussian variables $\xi(x)$ are independent with unit variance, and independent of the axial curvature, whose covariance is Π_t^{fl}. For an axial field, $y(b)$ is the mean over closed paths of length at most $4(4L)^t$ in the top block or double block of b. Apply the Stokes formula in Lem. D.18(ii):

$$y(b) = \sum_p \tau(b, p)(dy)(p), \qquad |\tau(b, p)| \le 4(4L)^t,$$

with τ supported in that box. There are at most $6L^{3t}$ plaquettes there. The covariance, diagonal in colour, is therefore

$$\begin{aligned} C_S &= C_S^{\mathrm{ax}} + C_S^{\mathrm{g}}, \\ C_S^{\mathrm{ax}}(b, b') &= \sum_{p,p'} \tau(b, p)\tau(b', p')\Pi_t^{\mathrm{fl}}(p, p'), \\ C_S^{\mathrm{g}}(b, b') &= \sum_x \sigma(b, x)\sigma(b', x), \qquad \sigma(b, x) = \mathbf{1}_{\mathrm{anc}(b_+)}(x) - \mathbf{1}_{\mathrm{anc}(b_-)}(x). \end{aligned}$$

In particular, $\sup|C_S| \le c_\tau^2 + 2t \le 2c_\tau^2$, where $c_\tau = 24L^{3t}(4L)^t$. The gauge part is zero unless endpoints of the two bonds lie in a common top block.

The free-curvature estimate [YM3, Prop. 7.6] gives

$$|\Pi_t^{\mathrm{fl}}(p, q)| \le 2L^{3-3t} C_{\mathrm{tp}} e^{-\delta_{\mathrm{tp}} L|x(p)-x(q)|_\infty / L^t}$$

when $\delta_{\mathrm{tp}} L(L-1)|x(p) - x(q)|_\infty \ge (3\log L + \log 2)L^t$. For bonds separated by $R \ge 8L^t$, this bounds their covariance by $2L^{3-3t} C_{\mathrm{tp}} c_\tau^2 e^{-\delta_{\mathrm{tp}} L(R-4L^t)/L^t}$, provided the same threshold holds for $R - 4L^t$. At $R = N/24$, the assumptions on ν imply these conditions and give

$$|C_S(b, b')| \le \epsilon_C := 2L^{3-3t} C_{\mathrm{tp}} c_\tau^2 e^{-\delta_{\mathrm{tp}} L\nu/48}. \tag{D.xiv}$$

There is also an exact periodization identity. With $\Gamma = N\mathbb{Z}^3/N'\mathbb{Z}^3$,

$$C_N(\pi\iota, \pi\iota') = \sum_{\gamma\in\Gamma} C_{N'}(\iota, \iota' + \gamma).$$

Indeed pullback commutes with d, Q_t, ξ and multiplies the Hessian form by $|\Gamma|$. The summed covariance on the right is the unique solution of the associated Gaussian variational equation with a periodized point source. Uniqueness makes it Γ-invariant; testing against pullbacks gives precisely the covariance equation on Λ_N.

Finally use the parity identity following eq. D.v to replace its product entry W_1^2 by the four-entry cumulant with two entries W_1. Expand the resulting cumulants into local Taylor polynomials. There

are at most $64N'^6$ tuples of places, with total degree at most ten. A place is near if its support lies within $N/4$ of the base of p'. The two observable places are near, and near tuples have the same polynomials on both tori. Their covariance difference is a sum of fewer than $|\Gamma|$ lines of length at least $N/2$, by periodization. For a tuple with a far place, every connected pairing has a line of length at least $N/24$. In fact a path from the observable place to the far place has at most three edges; the diameters of the intervening supports cost at most $7L^t$, and $N/4 - 7L^t > N/8$.

Use eq. D.xiv and the connected pairing estimates in Lem. D.18(iii). Cauchy's estimate bounds the coefficient sum of a degree-k local polynomial by $(108L^{3t}\sigma_t^{-3})^k$. Collecting these factors, the tuple count, $|\Gamma| \le N'^3$, and $L^t \le N'$ gives eq. D.xiii, with power 237 and a constant depending only on L, C_{tp}. A ball of radius $r \ge 8$ contains at most $30r^3$ plaquette barycentres. Sum over the two supports to obtain the last bound. □

D.12. **Comparison with the unconstrained Gaussian coefficient.**

Proposition D.20. Let $\ell = 2L^m \ge N_{\mathsf{H}}$, $\ell L \ge C_L M$ and $K \ge 1$. Put $N = \ell L^K$. If p, q are plaquettes of the bare torus and their chosen lattice lifts $\tilde{p}, \tilde{q}$ have base points at sup-distance at most $N/8$, then

$$\begin{aligned}|\mathsf{g}_2[N,K](p,q) - \mathsf{B}_1^{\mathrm{lat}}(\tilde{p},\tilde{q})| &\le C_*[\ell^{29/2}KL^{-K/2} + \ell^{18}L^{-2K} + \ell^{19}K^2L^{-5K}] \\ &\le 3C_*\ell^{19}KL^{-K/2}.\end{aligned}$$

If $\ell^3 \le L^K$, the bound is $3C_*\ell^{29/2}KL^{-K/2}$. The constant is independent of K, the side and the plaquettes. ■

Proof. By Prop. D.10, the coefficient can be computed in any normalized gauge on the fibre $\mathcal{M}^K U = 1$. We choose covariant gauge with parameter one. The linearized constrained covariance differs from the ordinary torus covariance by the curvature determined at level K. There is a further correction because the constraint is nonlinear. We separate these two comparisons.

Gauge fixing and the constraint. Fix a top site o. Use bare gauge transformations equal to one at o, and allow the top field to range over their pure gauges

$$w(\mu')(c) = \mathrm{Lg}(e^{i\mu'(c_-)}e^{-i\mu'(c_+)}), \qquad \mu'(o) = 0.$$

This gives one representative of each coset of the constant transformations. Let

$$\mathsf{M}_U\epsilon = \frac{d}{dt}d^*\,\mathrm{Lg}\,U^{\exp(it\epsilon)}\Big|_0 .$$

Its determinant, from functions vanishing at o to mean-zero functions, differs from the determinant on mean-zero functions by a three-dimensional factor. Indeed, changing a mean-zero ϵ_0 to $\epsilon_0 - \epsilon_0(o)$ and using $\mathsf{M}_U\kappa = i[\kappa, d^*a]$ for constant κ gives the factor $\det_3(1-\tau)$, where

$$\tau\kappa = (\mathsf{M}_U'^{-1}i[\kappa, d^*a])(o).$$

The first-order trace is zero because the adjoint action is traceless. Every second-order term has a factor d^*a. In its connected cumulant with the two quadratic observables, that factor must pair with $da(p)$ or $da(q)$. The covariance below satisfies $\langle da\, d^*a\rangle = 0$, so the factor changes neither coefficient under consideration. We may therefore use the mean-zero Faddeev–Popov terms in Def. D.2.

Write the block average in straight frames. For a one-step field ω, let

$$\varphi_\omega(z) = L^{-3}\sum_{x\in B(z)}\bar{\omega}(z,x), \qquad v_\omega(z) = e^{i\varphi_\omega(z)},$$

where $\bar{\omega}(z,x)$ is the linearized contour average. The step

$$m^s(\omega)(c) = \mathrm{Lg}[v_\omega(c_-)^{-1}\mathcal{M}(e^{i\omega})(c)v_\omega(c_+)]$$

has linear part Q^s. This follows from the linearized block formula: its difference from $Q^s\omega$ is $-d\varphi_\omega$. Gauge covariance shows that the K-fold composition $s^{(K)}$ differs from $\mathrm{Lg}\,\mathcal{M}^K(e^{ia})$ by a top gauge transformation $u(a)$. Change the top variable by

$$e^{i\mu'_s(z)} = u(a)(z)^{-1} e^{i\mu'(z)} u(a)(o).$$

At each fixed a this preserves the product Haar measure on the top sites other than o, and the Haar delta function on the top bonds is gauge invariant. The constraint is therefore exactly $s^{(K)}(a) = w(\mu'_s)$. Rename μ'_s as μ'. Its linearization is $Q^s_K a + d\mu' = 0$.

Put

$$\widehat{E} = \{(a,\mu') : Q^s_K a + d\mu' = 0,\ \mu'(o) = 0\}, \qquad (e, He') = (da, da') + (d^*a, d^*a').$$

The form is positive definite on $\widehat{E}$. If it vanishes, a is constant. Summing the constraint around each straight top cycle gives $a = 0$, and then $\mu' = 0$. Let $\widehat{R}\omega = (a_\omega, \mu'_\omega)$ be the minimum-energy solution of $Q^s_K a_\omega + d\mu'_\omega = \omega$, $\mu'_\omega(o) = 0$. It is H-orthogonal to $\widehat{E}$.

Let $\widehat{q}_2, \widehat{q}_3$ be the quadratic and cubic terms of $s^{(K)} - w$. For the centered Gaussian field e on $\widehat{E}$, set

$$\begin{aligned}
\zeta &= \widehat{R}\widehat{q}_2(e,e),\\
\zeta_3 &= \widehat{R}[\widehat{q}_3(e,e,e) - 2\widehat{q}_2(e,\zeta)],\\
N_1 &= 2\widehat{q}_2(e, \widehat{R}\,\cdot),\\
N_2 &= 3\widehat{q}_3(e,e,\widehat{R}\,\cdot) - 2\widehat{q}_2(\zeta, \widehat{R}\,\cdot).
\end{aligned}$$

Solving the constraint along $\operatorname{ran}\widehat{R}$ gives $e - \zeta - \zeta_3 + O(e^4)$ and the delta-function factor $\det(1 + N_1 + N_2 + O(e^3))^{-1}$. Orthogonality gives $(e, H\zeta) = (e, H\zeta_3) = 0$. Also $\operatorname{tr} N_1 = 0$: $\widehat{R}$ preserves a single colour, and a quadratic equivariant term has no component in the colour of its second argument when that argument has only one colour.

Write $X_i = \mathsf{a}_p^{(i+2)}$ and $Y_i = \mathsf{a}_q^{(i+2)}$; their derivatives below are evaluated at e. The observable terms become

$$X_1 - DX_0[\zeta], \qquad X_2 - DX_1[\zeta] - DX_0[\zeta_3] + X_0(\zeta),$$

and likewise for Y. The first interaction is still S_3. The second gains

$$-DS_3[\zeta] + \tfrac{1}{2}(\zeta, H\zeta) + \operatorname{tr} N_2 - \tfrac{1}{2}\operatorname{tr} N_1^2 + \tfrac{1}{3}\sum_y |\mu'(y)|^2 - \tfrac{1}{3}\sum_c |d\mu'(c)|^2.$$

The last two terms are the top-site Haar density and the inverse top-bond density of the delta function. Both signs follow from $-\log(\sin r/r)^2 = r^2/3 + O(r^4)$.

Substitution in eq. D.v gives the exact decomposition

$$\mathsf{g}_2[N,K] = \mathsf{B}_1^{\widehat{E}} + \sum_{i=1}^{11} \mathsf{t}_i,$$

where $\mathsf{B}_1^{\widehat{E}}$ is eq. D.ii with the Gaussian on $\widehat{E}$ and the ordinary mean-zero torus ghost kernels. The corrections are

$$\begin{aligned}
\mathsf{t}_1 &= -\langle DX_0[\zeta]; Y_1\rangle - \langle X_1; DY_0[\zeta]\rangle,\\
\mathsf{t}_2 &= \langle DX_0[\zeta]; DY_0[\zeta]\rangle,\\
\mathsf{t}_3 &= -\langle DX_1[\zeta]; Y_0\rangle - \langle X_0; DY_1[\zeta]\rangle,\\
\mathsf{t}_4 &= -\langle DX_0[\zeta_3]; Y_0\rangle - \langle X_0; DY_0[\zeta_3]\rangle,\\
\mathsf{t}_5 &= \langle X_0(\zeta); Y_0\rangle + \langle X_0; Y_0(\zeta)\rangle,\\
\mathsf{t}_6 &= \langle DX_0[\zeta]; Y_0; S_3\rangle + \langle X_0; DY_0[\zeta]; S_3\rangle,\\
\mathsf{t}_7 &= \langle X_0; Y_0; DS_3[\zeta]\rangle,\\
\mathsf{t}_8 &= -\tfrac{1}{2}\langle X_0; Y_0; (\zeta, H\zeta)\rangle,\\
\mathsf{t}_9 &= -\langle X_0; Y_0; \operatorname{tr} N_2\rangle,\\
\mathsf{t}_{10} &= \tfrac{1}{2}\langle X_0; Y_0; \operatorname{tr} N_1^2\rangle,\\
\mathsf{t}_{11} &= -\left\langle X_0; Y_0; \tfrac{1}{3}\sum_y |\mu'(y)|^2 - \tfrac{1}{3}\sum_c |d\mu'(c)|^2 \right\rangle.
\end{aligned}$$

The constrained covariance. Let C^{tor} be the inverse of $d^*d + dd^*$ off constant forms, and put $S_0 = C^{\mathrm{tor}}d^*$ on exact curvatures. Then $dS_0 = 1$ on that space. The a components of $\widehat{E}$ are exactly $a = S_0F + d\lambda$, with vanishing level-K plaquette averages of F. The Gaussian factors into independent F, λ with covariances $\Pi_0^{\mathrm{tp}} - \Pi_K^{\mathrm{tp}}$ and $(G^{\mathrm{tor}})^2$. Consequently

$$C_{\widehat{E}} = C^{\mathrm{tor}} - S_0\Pi_K^{\mathrm{tp}}S_0^{\mathsf{T}}, \quad dC_{\widehat{E}}d^* = \Pi_0^{\mathrm{tp}} - \Pi_K^{\mathrm{tp}}, \quad \langle da\, d^*a\rangle = 0.$$

Orthogonality of the response to the gradients first gives $d^*a_\omega = 0$. Orthogonality to S_0F and the constraint then give

$$da_\omega = H_K^{\mathrm{cv}}\omega, \qquad a_\omega = S_0H_K^{\mathrm{cv}}\omega + \text{constant form}.$$

The constant in direction ν is L^{-K} times the mean of ω on top bonds in that direction: sum the constraint over those bonds. The gradient term sums to zero, as does $Q_K^s S_0 H_K^{\mathrm{cv}}\omega$.

The free-template bounds in [YM3, Prop. 7.3] and [YM3, Prop. 7.6] give $|\Pi_K^{\mathrm{tp}}(p,q)| \le CL^{-3K}$ and curvature response row sums at most CL^{-2K}. By Lem. D.3, the kernels $C^{\mathrm{tor}}, dC^{\mathrm{tor}}$ and $dC^{\mathrm{tor}}d^*$ have powers $1, 2, 3$ uniformly in N. The row sum of S_0 is at most cN. It follows that the three covariance corrections are bounded respectively by $c\ell^3N^{-1}, c\ell^3N^{-2}, c\ell^3N^{-3}$. Thus, with $c_0 = c\ell^3$,

$$\begin{gathered}
|C_{\widehat{E}}(b,b')| \le c_0\langle b - b'\rangle^{-1}, \qquad |dC_{\widehat{E}}(p,b)| \le c_0\langle x_p - b\rangle^{-2},\\
|(\Pi_0^{\mathrm{tp}} - \Pi_K^{\mathrm{tp}})(p,q)| \le \min\{1, c_0\langle x_p - x_q\rangle^{-3}\},\\
\|a_\omega\|_\infty \le C\ell L^{-K}\|\omega\|_\infty, \qquad \|da_\omega\|_\infty \le CL^{-2K}\|\omega\|_\infty.
\end{gathered}$$

The bound by one uses that the curvature covariance is an orthogonal projection. The ghost kernels satisfy the same bounds as in Lem. D.6.

The Gaussian comparison. For $|z|_\infty \le 3N/4$, Lem. D.3 and the covariance corrections show that a line of $\mathsf{B}_1^{\widehat{E}}$ differs from its lattice counterpart by at most $CN^{-1/2}$ times its majorant with the exponent lowered by $1/2$. Expand both coefficients into the connected pairings of Lem. D.5. Tuples whose places all lie within $N/4$ of x_p correspond on the two lattices. Telescope their line products and apply Lem. D.6; their total difference is at most $CN^{-1/2}c_0^5$. Every remaining tuple has a line of length at least $N/16$ from p or q to a summed place. Extract its factor $(16/N)^{1/2}$ and use the same weakened-line bound on each lattice. Therefore

$$|\mathsf{B}_1^{\widehat{E}}(p,q) - \mathsf{B}_1^{\mathrm{lat}}(\tilde{p},\tilde{q})| \le C\ell^{29/2}L^{-K/2}.$$

The nonlinear vertices. The one-step straight average is the same holomorphic map at every level and bond. Its value at a top bond depends on at most $6L^3$ bonds of the double block. There is a fixed $\tilde{r} > 0$ on whose sup-norm ball its modulus is at most one. Cauchy's formula and polarization therefore bound its homogeneous terms by

$$|\tilde{c}_k(\omega_1, \dots, \omega_k)| \le \frac{k^k}{k!} \tilde{r}^{-k} \prod_i \sup_{P_c} |\omega_i|.$$

Writing $\phi_j = Q_j^s \phi$, composition gives

$$s_2^{(i)}(\phi, \psi) = \sum_{j<i} Q^s_{j+1\to i} \tilde{c}_2(\phi_j, \psi_j),$$
$$s_3^{(i)}(\phi, \psi, \chi) = \sum_{j<i} Q^s_{j+1\to i} \left[\tfrac{2}{3} \sum_{\text{cyc}} \tilde{c}_2(\phi_j, s_2^{(j)}(\psi, \chi)) + \tilde{c}_3(\phi_j, \psi_j, \chi_j) \right].$$

The required vertices are $\widehat{q}_k = s_k^{(K)} - w_k$. The averaging weights are nonnegative, at most L^{-2j}, have row sum L^j, and are supported within $3L^j/2$ of their output bond.

There are three kinds of entries in these vertices: a covariance line from an external curvature, a Gaussian field, and a response. Denote them in the following estimates by E, A, R, respectively. At level j their bounds are

$$|Q_j^s dC_{\widehat{E}}(p, \cdot)(c)| \le C c_0 L^j (L^j + |c_- - x_p|)^{-2},$$
$$\|Q_j^s a(c)\|_{L^2} \le C c_0^{1/2} L^{j/2},$$
$$|Q_j^s a_\omega(c)| \le C \ell L^{j-K} \|\omega\|_\infty.$$

All moments of the finitely many Gaussian factors are bounded by their second moments, with constants depending only on their degree. The corresponding top-site functions vanish at o and have differences $-Q_K^s a$ and $\omega - Q_K^s a_\omega$. Their values are bounded by summing along paths of at most $3\ell/2$ bonds, or 3ℓ for a response.

The outer average must be summed with the spatial profiles of the external lines. In particular, the product of two inverse-square profiles is summable; with one such profile its sum over a block of side L^i is $O(L^i)$. Applying the displayed entry and row bounds to each term of $s_2^{(K)}$, and summing over $j < K$, gives the following scale factors:

entries of $\widehat{q}_2$	(E,E)	(E,A)	(E,R)	(A,A)	(A,R)	(R,R)
factor	KL^{-2K}	$L^{-K/2}$	L^{-K}	KL^K	$L^{K/2}$	1.

For $\widehat{q}_3$, the factors needed below are KL^{-2K} at (E, E, R) and $KL^{3K/2}$ at (A, A, A). The products of the powers of c_0 from the entries and of the path factors multiply these scale factors.

For clarity, two path counts needed in the trace terms can be kept sharper. The quadratic term of w is, up to its orientation sign, $\mu'(c_-) \times d\mu'(c)$. It uses only one undifferentiated top potential. Thus

$$\|\widehat{q}_2(E, E)\|_\infty \le C \ell c_0^2 K L^{-2K},$$
$$\|\widehat{q}_2(R\omega, R\omega')\|_\infty \le C \ell^3 \|\omega\|_\infty \|\omega'\|_\infty,$$
$$\|\widehat{q}_3(E, E, R\omega)\|_\infty \le C \ell^4 c_0^2 K L^{-2K} \|\omega\|_\infty.$$

For the last bound, use the composition formula for s_3. In its two-vertex terms retain the spatial profiles of the inner quadratic vertex. The averaging entries and column counts give

$$\sum_c |s_2^{(j)}(E,E)(c)| \le C c_0^2 j L^{-2j},$$

$$|s_2^{(j)}(E,R\omega)(c)| \le C\ell c_0 L^{-K} \frac{L^{2j}}{(L^j + |c_- - x_p|)^2} \|\omega\|_\infty.$$

Multiplying by the remaining entry and the outer average gives $C\ell c_0^2 j L^{j-3K}\|\omega\|_\infty$. Its sum over j is at most $C\ell c_0^2 K L^{-2K}\|\omega\|_\infty$. The term with one cubic vertex has the same bound by the product of its two external profiles. In w_3 every monomial contains a difference of the top potential: this follows also by setting its two endpoint values equal in the BCH formula. The two external potentials have bounds $C\ell c_0 L^{-K}$ and differences $C c_0 L^{-K}$; the response potential and its difference have bounds $C\ell^2\|\omega\|_\infty$ and $C\ell\|\omega\|_\infty$. These products are bounded by the stated ℓ^4 factor.

One pairing below involves two plain covariance lines $C_z = C_{\widehat{E}}(z,\cdot)$ rather than curvature lines. Averaging the inverse-distance bound gives $|Q_j^s C_z(c)| \le C c_0 L^j (L^j + |c_- - z|)^{-1}$. In the quadratic composition, the sum of the product of these profiles with the outer weights has scale factor L^{j-K}. Its sum over $j < K$ is bounded. The quadratic top-gauge term again has only one path factor, so

$$\|\widehat{q}_2(C_z, C_z)\|_\infty \le C\ell c_0^2.$$

With one curvature line the same sum has factor $(K-j+1)L^{j-2K}$, whose sum is $O(L^{-K})$. The top-gauge term has the same single path factor. Therefore

$$\|\widehat{q}_2(E_z, C_z)\|_\infty \le C\ell c_0^2 L^{-K}.$$

The eleven corrections. Expand a connected cumulant into connected Gaussian pairings. An entry quadratic in the Gaussian is equivalently treated by two Gaussian integrations by parts, in its two curvature directions. A quadratic vertex with two entries of the same colour vanishes: both the straight average and w are odd on a single-colour field. In addition, contracting two legs of the scalar Wilson cubic S_3 with a colour-diagonal kernel gives zero. These cancellations remove the pairing patterns without the decay factors used below.

The resulting estimates are

term	upper bound, with a common constant C
$\lvert\mathsf{t}_1\rvert$	$\ell^8 K L^{-K}$
$\lvert\mathsf{t}_2\rvert + \lvert\mathsf{t}_5\rvert$	$\ell^{10} K^2 L^{-2K}$
$\lvert\mathsf{t}_3\rvert$	$\ell^{11} L^{-3K/2}$
$\lvert\mathsf{t}_4\rvert$	$\ell^{19/2} K L^{-K/2}$
$\lvert\mathsf{t}_6\rvert$	$\ell^{11} K L^{-K} + \ell^{25/2} L^{-5K/2}$
$\lvert\mathsf{t}_7\rvert$	$\ell^{13} L^{-K/2} + \ell^{15} K L^{-3K}$
$\lvert\mathsf{t}_8\rvert$	$\ell^{16} L^{-2K} + \ell^{19} K^2 L^{-5K}$
$\lvert\mathsf{t}_9\rvert$	$\ell^{13} K L^{-2K}$
$\lvert\mathsf{t}_{10}\rvert$	$\ell^{18} L^{-2K}$
$\lvert\mathsf{t}_{11}\rvert$	$\ell^{11} L^{-2K}$.

We explain the counts, including the terms with a sum over an interaction vertex. For t_1, the quadratic vertex and curvature response give $C\ell^2 c_0 K L^{-K}$. The remaining cubic observable has bounded moments of size $C c_0$; hence the bound is $C\ell^2 c_0^2 K L^{-K}$. Squaring the first bound gives $C\ell^4 c_0^2 K^2 L^{-2K}$ for $\mathsf{t}_2, \mathsf{t}_5$. For t_4, the two parts of ζ_3 have moments bounded by $C\ell^5 c_0^{3/2} K L^{3K/2}$ before applying the curvature response $C L^{-2K}$. These are the first, second and fourth rows.

In t_3, a surviving contraction puts one of the external curvature lines in the quadratic vertex. Its (E,A) rate and the plain response give $L^{-3K/2}$. The remaining entry and path factors are bounded by $C\ell^2 c_0^3$, yielding the third row.

For $\mathtt{t}_6$, sum the position z of S_3 after its three colours have been contracted. If the curl of S_3 pairs with the other quadratic observable and its two plain fields pair with $\widehat{q}_2$, use the plain-line estimate just proved. The remaining curvature sum is at most $Cc_0 \log N$ and the curvature response contributes L^{-2K}. Since $\log N \le C_L \ell K$, this is bounded by the first term of the sixth row. There is a second pattern with the same fixed p–q line: the leg of S_3 paired with q is a plain field, and its other two legs enter $\widehat{q}_2$ as (E_z, C_z). Its bound is

$$C\ell c_0^2 L^{-K} \cdot c_0 N \cdot L^{-2K} = C\ell^2 c_0^3 L^{-2K} \le C\ell^{11} K L^{-K}.$$

Here $\sum_z |E(q,z)| \le Cc_0 N$. The other surviving patterns put an external curvature line in $\widehat{q}_2$. Use its (E, A) rate and sum the remaining two external profiles. Whenever one is a curvature-curvature line, use

$$\min\{1, c_0\langle z\rangle^{-3}\} \le c_0^{1/2}\langle z\rangle^{-3/2}.$$

Together with an inverse-square line this is summable in dimension three. The entry and path counts are bounded by the two terms displayed for $\mathtt{t}_6$. In $\mathtt{t}_7$, put $R = L^K$, so $N = \ell R$. A fixed p–q line leaves one external direction in $\widehat{q}_2$. Its L^2 bound is $C\ell^2 c_0^{3/2} R^{-1/2}$; the remaining local Gaussian factor costs $c_0^{1/2}$, and the inverse-square position sum costs $Cc_0 N$. Using the larger, plain-response bound $C\ell/R$ gives

$$C\frac{\ell}{R}(\ell^2 c_0^{3/2} R^{-1/2}) c_0^{1/2} (c_0 N) = C\ell^4 c_0^3 R^{-1/2}.$$

Without a fixed p–q line both directions in $\widehat{q}_2$ are external, with bound $C\ell^2 c_0^2 K/R^2$. The product of the two remaining external lines has sum at most Cc_0^2. Thus this case costs $C(\ell/R)(\ell^2 c_0^2 K/R^2) c_0^2 = C\ell^3 c_0^4 K/R^3$. Substituting $c_0 = C\ell^3$ gives the seventh row.

For $\mathtt{t}_8$, the response energy satisfies

$$|(\widehat{R}\omega, H\widehat{R}\omega')| \le C\ell^3 L^{-K} \|\omega\|_\infty \|\omega'\|_\infty :$$

there are $O(N^3)$ plaquettes and each curvature response is $O(L^{-2K})$. A fixed p–q line leaves one external direction in each quadratic vertex. The bound is $C\ell^3 L^{-K}(\ell^2 c_0^{3/2} L^{-K/2})^2$. With no fixed line, each quadratic vertex has two external directions, giving instead $C\ell^3 L^{-K}(\ell^2 c_0^2 K L^{-2K})^2$. The other patterns vanish by the equal-colour cancellation. Substituting $c_0 = C\ell^3$ gives the two stated terms.

For $\mathtt{t}_9$, the two external directions in N_2 use the sharper bounds above. Both $\widehat{q}_3(E, E, R\,\cdot)$ and $\widehat{q}_2(\widehat{R}\widehat{q}_2(E,E), \widehat{R}\,\cdot)$ are bounded by $C\ell^4 c_0^2 K L^{-2K}$. Taking the trace costs $9\ell^3$ and gives $C\ell^7 c_0^2 K L^{-2K}$. For $\mathtt{t}_{10}$, the two entries $\widehat{q}_2(E, R\,\cdot)$ have bound $C\ell^3 c_0 L^{-K}$. Summing the two top indices costs at most $C\ell^6$, which gives $C\ell^{12} c_0^2 L^{-2K}$. Finally $\mathtt{t}_{11}$ has $O(\ell^3)$ top sites and two external potential entries of size $C\ell c_0 L^{-K}$, giving $C\ell^5 c_0^2 L^{-2K}$.

Every use of a response on polynomial entries is justified by its row-sum bound: a real matrix bounded in the sup norm has absolute row sums bounded by that norm. Thus no inverse on the top lattice has been bounded by an operator norm depending on its side. The constants use only the fixed one-step radius, the kernel and response constants, and the bounded polynomial degrees.

Sum the rows. Since $KL^{-K} \le 1$, their sum is at most

$$C[\ell^{13} K L^{-K/2} + \ell^{18} L^{-2K} + \ell^{19} K^2 L^{-5K}].$$

Adding the Gaussian comparison proves the first bound. The remaining conclusions follow from $\ell, K \ge 1$, and, for the sharper one, $\ell^3 \le L^K$. □

D.13. **Zero momentum.** Write $\mathbb{T} = [-\pi, \pi]^3/(2\pi\mathbb{Z})^3$ and $d^\circ k = (2\pi)^{-3} dk$.

In this subsection we compute the values at zero momentum of the self-energy and of the vertex function of a plaquette action. Both are expressed through Hessians of the action at constant abelian fields, whose symbols can be computed explicitly.

Definition D.21 (Hessians and zero-momentum coefficients)**.** Write $\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$ for the Pauli basis. We work on $\mathbb{Z}^3$ and fix a gauge parameter $\tau_{\mathrm{gf}} = 1$. For a positively oriented bond b and a colour index a we write $\partial_{b,a}$ for the derivative with respect to the variable $A^a(b)$. On a formal sum $\mathsf{F} = \sum_z \mathsf{F}_z$ of local terms (Def. D.2) the derivative acts term by term, and the terms on which it vanishes are omitted. The derivatives $\partial_{b,a} S_3$, $\partial_{b,a}\partial_{b',a'} S_4$ and $\partial_{b,a}\partial_{b',a'} V_2$ which occur below are then finite sums: a bond is contained in exactly four plaquettes; of a sum over bonds b'' only the term with $b'' = b$ survives; and of the sum over pairs of bonds in V_2 only the terms of the pairs (b, b') and (b', b) survive.

(1) Hessians. Let B be a 1-form on $\mathbb{Z}^3$ with $\sup_b |B(b)| < 1$, and let q be a plaquette. We define real symmetric matrices $\mathsf{N}(B)$ and $\mathsf{h}_q(B)$, indexed by the pairs (b, a) of a positively oriented bond b and a colour index a, by

$$\mathsf{N}(B)_{(b,a),(b',a')} := \partial_{b,a}\partial_{b',a'}\Big[\sum_p a_p(e^{iA}) + \frac{1}{2\tau_{\mathrm{gf}}}\sum_x |\mathrm{d}^* A(x)|^2\Big]_{A=B},$$

$$\mathsf{h}_q(B)_{(b,a),(b',a')} := \partial_{b,a}\partial_{b',a'}\, a_q(e^{iA})\big|_{A=B}.$$

The infinite sums over all plaquettes p and over all sites x are understood as follows: under $\partial_{b,a}\partial_{b',a'}$ only the finitely many terms which depend on $A(b)$ contribute, namely those of the plaquettes containing b and those of the two end points of b. Consequently an entry of either matrix vanishes unless b and b' lie in a common plaquette or have a common end point. For fixed (b, a) only finitely many entries $\mathsf{N}(B)_{(b,a),(b',a')}$ and $\mathsf{h}_q(B)_{(b,a),(b',a')}$ are different from 0, and both matrices act on bounded complex 1-forms A by $(\mathsf{N}(B)A)^a(b) := \sum_{(b',a')} \mathsf{N}(B)_{(b,a),(b',a')} A^{a'}(b')$, and likewise for $\mathsf{h}_q(B)$.

(2) Constant abelian fields. For $\gamma \in \mathbb{R}^3$ with $|\gamma| < 1$ let Y_γ be the constant 1-form $Y_\gamma((x,\mu)) := \gamma_\mu \mathsf{e}_3$. We put $\mathsf{e}_\pm := 2^{-1/2}(\mathsf{e}_1 \pm i\mathsf{e}_2)$ and $\mathsf{e}_0 := \mathsf{e}_3$. These three vectors form a Hermitian-orthonormal basis of $\mathfrak{g} \otimes \mathbb{C}$, and

$$[\mathsf{e}_3, \mathsf{e}_e] = 2e\, \mathsf{e}_e \qquad (e \in \{+1, -1, 0\});$$

we call e the *charge*, and we write $v := 2e\gamma \in \mathbb{R}^3$. For $k \in \mathbb{R}^3$ we define two 3×3 matrices $\hat{\mathsf{N}}_e(k;\gamma)$, $\hat{\mathsf{h}}_{q,e}(k;\gamma)$ and a number $\hat{\mathsf{M}}_e(k;\gamma)$ by

$$\mathsf{N}(Y_\gamma)(e^{ik\cdot x}\alpha_\mu \mathsf{e}_e) = e^{ik\cdot x}(\hat{\mathsf{N}}_e\alpha)_\mu\, \mathsf{e}_e,$$

$$(e^{ik\cdot x}\alpha' \mathsf{e}_e,\ \mathsf{h}_q(Y_\gamma)\, e^{ik\cdot x}\alpha\, \mathsf{e}_e) = \alpha'^* \hat{\mathsf{h}}_{q,e}\alpha,$$

$$\mathsf{M}(Y_\gamma)(e^{ik\cdot x}\mathsf{e}_e) = \hat{\mathsf{M}}_e\, e^{ik\cdot x}\mathsf{e}_e$$

for all $\alpha, \alpha' \in \mathbb{C}^3$. Here $e^{ik\cdot x}\alpha_\mu \mathsf{e}_e$ stands for the complex 1-form $(x,\mu) \mapsto e^{ik\cdot x}\alpha_\mu \mathsf{e}_e$ and $e^{ik\cdot x}\mathsf{e}_e$ for the complex site function $x \mapsto e^{ik\cdot x}\mathsf{e}_e$; the bracket $(\cdot,\cdot)$ is Hermitian, and the sum which defines it is finite, since $\mathsf{h}_q(Y_\gamma)$ has non-zero entries only at bonds of q.

The operator $\mathsf{M}(A)\lambda$ is d^* of the derivative at $s = 0$ of the logarithm divided by i of

$$e^{is\lambda(b_-)} e^{iA(b)} e^{-is\lambda(b_+)}.$$

It is extended complex linearly. That the left sides have these forms is shown in Lem. D.22, which also gives explicit formulas.

By these formulas, for fixed k, the three objects are real analytic functions of γ in $\{|\gamma| < 1\}$.

A second index j denotes the homogeneous polynomial of degree j in γ in the Taylor expansion at $\gamma = 0$, so that $\hat{\mathsf{N}}_e = \sum_j \hat{\mathsf{N}}_{e,j}$, $\hat{\mathsf{h}}_{q,e} = \sum_j \hat{\mathsf{h}}_{q,e,j}$ and $\hat{\mathsf{M}}_e = \sum_j \hat{\mathsf{M}}_{e,j}$.

For $k \notin 2\pi\mathbb{Z}^3$ we put $\hat{C} := \hat{\mathsf{N}}_{e,0}^{-1}$. By Lem. D.22 at $\gamma = 0$ we have $\hat{\mathsf{N}}_{e,0}(k) = |\mathsf{k}|^2 \mathbf{1} - (1 - \tau_{\mathrm{gf}}^{-1})\mathsf{k}\mathsf{k}^*$ with $\mathsf{k} = \mathsf{k}(k)$, so that $\hat{C}$ does not depend on e and is the matrix with the entries

$$\hat{C}_{\mu\nu}(k) = \frac{\delta_{\mu\nu}}{|\mathsf{k}|^2} + (\tau_{\mathrm{gf}} - 1)\frac{\mathsf{k}_\mu \overline{\mathsf{k}_\nu}}{|\mathsf{k}|^4}$$

which appears in the formula for C in Def. D.2; moreover $\hat{\mathsf{M}}_{e,0} = -|\mathsf{k}|^2$.

(3) Five numbers of the self-energy. For $k \in [-\pi,\pi]^3 \setminus \{0\}$ and $\gamma \in \mathbb{R}^3$ we put, with all functions taken at $(k;\gamma)$,

$$\begin{aligned}
\mathfrak{s}^{(1)} &:= -\tfrac{1}{4}\sum_e \operatorname{tr}\big[(\hat{C}\hat{\mathsf{N}}_{e,1})^2\big] && \text{(gluon loop)},\\
\mathfrak{s}^{(2)} &:= \tfrac{1}{2}\sum_e \operatorname{tr}\big[\hat{C}\hat{\mathsf{N}}_{e,2}\big] && \text{(gluon tadpole)},\\
\mathfrak{s}^{(3)} &:= \tfrac{1}{2}\sum_e \operatorname{Re}\big[(\hat{\mathsf{M}}_{e,1}/\hat{\mathsf{M}}_{e,0})^2\big] && \text{(ghost loop)},\\
\mathfrak{s}^{(4)} &:= -\sum_e \operatorname{Re}\big[\hat{\mathsf{M}}_{e,2}/\hat{\mathsf{M}}_{e,0}\big] && \text{(ghost tadpole)},
\end{aligned}$$

and $\mathfrak{s}^{(5)}(k;\gamma) := \frac{1}{3}|\gamma|^2$ (Haar term). For fixed k each $\mathfrak{s}^{(i)}(k;\cdot)$ is a homogeneous polynomial of degree 2 in γ, bounded by $C|\gamma|^2|k|^{-2}$ (for $i \le 4$ by Prop. D.23(i), for $i = 5$ because $|k|$ is bounded on the cube); hence it is integrable over $\mathbb{T}$, and its integral is a quadratic form in γ. For $i = 1,\dots,5$ we let $c^{(i)}_{\mu\nu}(\tau_{\mathrm{gf}})$, $1 \le \mu,\nu \le 3$, be the entries of the symmetric matrix with

$$\frac{1}{2}\sum_{\mu\nu} c^{(i)}_{\mu\nu}(\tau_{\mathrm{gf}})\,\gamma_\mu\gamma_\nu = \int_{\mathbb{T}} \mathfrak{s}^{(i)}(k;\gamma)\,d^\circ k \qquad (\gamma \in \mathbb{R}^3).$$

We also need the number

$$Z_0 := \int_{\mathbb{T}} \frac{d^\circ k}{|\mathsf{k}(k)|^2} = \int_{[-\pi,\pi]^3} \frac{dk}{(2\pi)^3}\Big(\sum_\mu 4\sin^2\frac{k_\mu}{2}\Big)^{-1}. \tag{D.xv}$$

(4) Four numbers of the vertex function of a_q. For a plaquette q, $k \in [-\pi,\pi]^3 \setminus \{0\}$ and $\gamma \in \mathbb{R}^3$ we put, again with all functions taken at $(k;\gamma)$,

$$\begin{aligned}
\mathfrak{v}^{(1)}_q &:= \tfrac{1}{2}\sum_e \operatorname{tr}\big[\hat{\mathsf{h}}_{q,e,2}\hat{C}\big], & \mathfrak{v}^{(2)}_q &:= -\tfrac{1}{2}\sum_e \operatorname{tr}\big[\hat{\mathsf{h}}_{q,e,1}\hat{C}\hat{\mathsf{N}}_{e,1}\hat{C}\big],\\
\mathfrak{v}^{(3)}_q &:= -\tfrac{1}{2}\sum_e \operatorname{tr}\big[\hat{\mathsf{h}}_{q,e,0}\hat{C}\hat{\mathsf{N}}_{e,2}\hat{C}\big], & \mathfrak{v}^{(4)}_q &:= \tfrac{1}{2}\sum_e \operatorname{tr}\big[\hat{\mathsf{h}}_{q,e,0}\hat{C}\hat{\mathsf{N}}_{e,1}\hat{C}\hat{\mathsf{N}}_{e,1}\hat{C}\big].
\end{aligned}$$

These are homogeneous polynomials of degree 2 in γ, bounded by $C|\gamma|^2|k|^{-2}$ (Prop. D.24). For $j = 1,\dots,4$ we let $\mathsf{g}^{(j)}_{q;\mu\nu}(\tau_{\mathrm{gf}})$ be the entries of the symmetric matrix with

$$\frac{1}{2}\sum_{\mu\nu} \mathsf{g}^{(j)}_{q;\mu\nu}(\tau_{\mathrm{gf}})\,\gamma_\mu\gamma_\nu = \int_{\mathbb{T}} \mathfrak{v}^{(j)}_q(k;\gamma)\,d^\circ k \qquad (\gamma \in \mathbb{R}^3).$$

(5) Kernels. Let $\langle\cdot\rangle_0$ refer to the Gaussian family of $\mathbb{Z}^3$ and τ_{gf} (Def. D.2), and write $\partial = \partial_{b,a}$, $\partial' = \partial_{b',a'}$. We define

$$\begin{aligned}
\Sigma(b,a;b',a') &:= \langle \partial\partial' V_2\rangle_0 - \langle \partial S_3;\partial' S_3\rangle_0,\\
\Gamma_q(b,a;b',a') &:= \langle \partial\partial' \mathsf{a}^{(4)}_q\rangle_0 - \langle \partial\mathsf{a}^{(3)}_q;\partial' S_3\rangle_0 - \langle \partial'\mathsf{a}^{(3)}_q;\partial S_3\rangle_0 - \langle \mathsf{a}^{(2)}_q;\partial\partial' S_4\rangle_0 + \langle \mathsf{a}^{(2)}_q;\partial S_3;\partial' S_3\rangle_0.
\end{aligned}$$

These are means and cumulants (Def. D.2) of polynomials in the Gaussian family, and both kernels are symmetric under the exchange $(b,a) \leftrightarrow (b',a')$.

We determine the colour structure of the two kernels. For $g \in G$ the map $\operatorname{Ad} g : X \mapsto gXg^{-1}$ is orthogonal on $(\mathfrak{g},\langle\cdot,\cdot\rangle)$; let $(\operatorname{Ad} g)_{aa'}$ be its matrix in the basis $\mathsf{e}_1,\mathsf{e}_2,\mathsf{e}_3$. The components $(gA(b)g^{-1})^a = \sum_{a'}(\operatorname{Ad} g)_{aa'}A^{a'}(b)$ of the $\mathfrak{g}$-valued family $(gA(b)g^{-1})_b$ are real linear combinations of the variables of the Gaussian family, so that this family is again centred Gaussian. By Def. D.2 and the orthogonality of $\operatorname{Ad} g$, its covariance is

$$\big\langle (gA(b)g^{-1})^a\,(gA(b')g^{-1})^c\big\rangle_0 = \sum_{a'}(\operatorname{Ad} g)_{aa'}(\operatorname{Ad} g)_{ca'}\,C(b,b') = \delta_{ac}\,C(b,b'),$$

which is the covariance of $(A(b))_b$; as a centred Gaussian family is determined in law by its covariance, $(gA(b)g^{-1})_b$ has the law of $(A(b))_b$. By Lem. D.1 the polynomials $\mathsf{a}_p^{(m)}$ are invariant under $A \mapsto gAg^{-1}$, and so are S_3, S_4 and, term by term, V_2, whose remaining terms involve A only through $|A(b)|^2$ and $\langle A(b), A(b')\rangle$. By the chain rule, each of the two kernels, regarded at fixed b, b' as a matrix K in (a, a'), satisfies $K = (\operatorname{Ad} g)^{\mathsf{T}} K \operatorname{Ad} g$, that is, it commutes with every $\operatorname{Ad} g$. For $g = e^{isY}$, differentiation at $s = 0$ shows that K commutes with $\mathsf{e}_c \times \cdot$ for $c = 1, 2, 3$. Such a matrix is a multiple of $\mathbf{1}$: it preserves $\ker(\mathsf{e}_c \times \cdot) = \mathbb{R}\mathsf{e}_c$, so it is diagonal in the basis $\mathsf{e}_1, \mathsf{e}_2, \mathsf{e}_3$, with entries m_1, m_2, m_3 say; applying it to $\mathsf{e}_1 \times \mathsf{e}_2 = \mathsf{e}_3$ and to $\mathsf{e}_3 \times \mathsf{e}_1 = \mathsf{e}_2$ gives $m_3 = m_2$ and $m_2 = m_1$. Hence

$$\Sigma(b, a; b', a') = \delta_{aa'}\, \Sigma(b, b'), \qquad \Gamma_q(b, a; b', a') = \delta_{aa'}\, \Gamma_q(b, b'),$$

and these identities define the kernels $\Sigma(b, b')$ and $\Gamma_q(b, b')$. Finally, the kernels are covariant under the translations of $\mathbb{Z}^3$. Indeed, $C((x, \mu), (y, \nu))$ and $G(x, y)$ depend on the sites x, y only through $x - y$ (Def. D.2), so that the law of the Gaussian family is invariant under translations; the coefficients of the polynomials $\mathsf{a}_p^{(m)}$ do not depend on p (Lem. D.1); and $g(b) = \frac{1}{3}$ as well as $D(b; b')$, which are built from G (Def. D.2), do not change when b and b' are translated together. Hence the formal sums S_3, S_4, V_2 are invariant under translations as well, and $\Sigma(b + y, b' + y) = \Sigma(b, b')$ and $\Gamma_{q+y}(b + y, b' + y) = \Gamma_q(b, b')$ for all $y \in \mathbb{Z}^3$. ∎

The symbols of Def. D.21(2) can be computed explicitly.

Lemma D.22 (The Hessian at a constant abelian field)**.** Let $\gamma \in \mathbb{R}^3$ with $|\gamma| < 1$, let $e \in \{\pm 1, 0\}$, $\upsilon = 2e\gamma$ and $k \in \mathbb{R}^3$. Let $j(\vartheta) := (e^{i\vartheta} - 1)/i\vartheta$ for real $\vartheta \neq 0$ and $j(0) := 1$, and put

$$J := \operatorname{diag}\big(j(\upsilon_1), j(\upsilon_2), j(\upsilon_3)\big), \qquad P(k) := |\mathsf{k}(k)|^2 \mathbf{1} - \mathsf{k}(k)\mathsf{k}(k)^*;$$

the matrix J is invertible, since $|\upsilon_\mu| < 2$ and $j(\vartheta) \neq 0$ for $|\vartheta| < 2\pi$. For a plaquette $q = p_{\mu\nu}(x)$ let $D_q(k)$ be the 1×3 matrix with $D_q(k)\alpha = \mathsf{k}_\mu(k)\alpha_\nu - \mathsf{k}_\nu(k)\alpha_\mu$, $\alpha \in \mathbb{C}^3$. Then the left sides of the three identities in Def. D.21(2) have the forms stated there, with

$$\begin{aligned} \hat{\mathsf{N}}_e(k; \gamma) &= J^* P(k + \upsilon) J + \tau_{\mathrm{gf}}^{-1} \mathsf{k}(k)\mathsf{k}(k)^*, \\ \hat{\mathsf{h}}_{q,e}(k; \gamma) &= J^* D_q(k + \upsilon)^* D_q(k + \upsilon) J, \\ \hat{\mathsf{M}}_e(k; \gamma) &= -\mathsf{k}(k)^* J^{-1} \mathsf{k}(k + \upsilon). \end{aligned}$$

Moreover the following hold.

(a) We have

$$\det \hat{\mathsf{N}}_e(k; \gamma) = \tau_{\mathrm{gf}}^{-1} \,|\det J|^2\, |\mathsf{k}(k + \upsilon)|^2\, |\hat{\mathsf{M}}_e(k; \gamma)|^2, \qquad \prod_e |\det J| = \prod_\mu \sigma_1(\gamma_\mu^2), \tag{D.xvi}$$

where $\sigma_1(t) = (\sin\sqrt{t}/\sqrt{t})^2$, continuously extended at zero, and in the second identity J is taken at $\upsilon = 2e\gamma$ for each $e \in \{+1, -1, 0\}$.

(b) If $\mathsf{k}(k + \upsilon) \neq 0$ and $\hat{\mathsf{M}}_e(k; \gamma) \neq 0$, then $\hat{\mathsf{N}}_e(k; \gamma)$ is invertible, and for $q = p_{\mu\nu}(x)$

$$\operatorname{tr}\big[\hat{\mathsf{h}}_{q,e}(k; \gamma) \hat{\mathsf{N}}_e(k; \gamma)^{-1}\big] = \varpi_q(k + \upsilon), \qquad \varpi_q(k) := \frac{|\mathsf{k}_\mu(k)|^2 + |\mathsf{k}_\nu(k)|^2}{|\mathsf{k}(k)|^2}. \tag{D.xvii}$$

(c) With $\mathsf{k} = \mathsf{k}(k)$,

$$\hat{\mathsf{M}}_{e,1} = -\sum_\mu \upsilon_\mu \sin k_\mu, \qquad \hat{\mathsf{M}}_{e,2} = \frac{1}{12} \sum_\mu \upsilon_\mu^2 |\mathsf{k}_\mu|^2, \qquad \operatorname{tr} \hat{\mathsf{N}}_{e,2} = \sum_\mu \upsilon_\mu^2 \Big[2 \cos k_\mu - \frac{1}{12}\big(|\mathsf{k}|^2 - |\mathsf{k}_\mu|^2\big)\Big].$$

∎

Proof. Second variation. Let a have finite support and $u_\mu = e^{i\gamma_\mu \mathsf{e}_3}$. For small real s the matrix $e^{i(Y_\gamma(b)+sa(b))}e^{-iY_\gamma(b)}$ is near the identity. Its local logarithm defines a smooth $\eta_s(b)$, with $\eta_0 = 0$. Termwise differentiation of the exponential series gives

$$\frac{d}{ds}e^{i(Y+sa)}e^{-iY}\Big|_{s=0} = i\int_0^1 e^{itY}ae^{-itY}\,dt = i\,\mathsf{dE}(Y)a.$$

Thus

$$\eta_s(b) = s\,\mathsf{dE}(Y_\gamma(b))\,a(b) + O(s^2).$$

Let $p = p_{\mu\nu}(x)$ have the bonds $b_1,\dots,b_4$, and put $\eta_i := \eta_s(b_i)$. Since $Y_\gamma(b_1) = Y_\gamma(b_3) = \gamma_\mu \mathsf{e}_3$ and $Y_\gamma(b_2) = Y_\gamma(b_4) = \gamma_\nu \mathsf{e}_3$, since u_μ and u_ν commute, and since $ue^{iX}u^{-1} = e^{iuXu^{-1}}$, we find

$$\partial\big(e^{i(Y_\gamma+sa)}\big)(p) = e^{i\eta_1}u_\mu\, e^{i\eta_2}u_\nu\, u_\mu^{-1}e^{-i\eta_3}\, u_\nu^{-1}e^{-i\eta_4} = e^{i\eta_1}\, e^{iu_\mu\eta_2u_\mu^{-1}}\, e^{-iu_\nu\eta_3u_\nu^{-1}}\, e^{-i\eta_4} = e^{iZ},$$

where $Z \in \mathfrak{g}$ is the image of this product under $\frac{1}{i}\log$; for small s the product is close to $\mathbf{1}$, and by the diffeomorphism property recalled above Z is a smooth function of s, with $Z = 0$ at $s = 0$. To first order

$$Z = \eta_1 + u_\mu\eta_2u_\mu^{-1} - u_\nu\eta_3u_\nu^{-1} - \eta_4 + O(s^2).$$

Since $a_p(U) = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$ and $\frac{1}{2}\operatorname{tr}e^{iZ} = \cos|Z|$, we have $a_p(e^{i(Y_\gamma+sa)}) = 1 - \cos|Z| = \frac{1}{2}|Z|^2 + O(|Z|^4)$. We put $(\mathsf{J}_\gamma a)(b) := \mathsf{dE}(Y_\gamma(b))a(b)$ and, for a 1-form a', $(D_\gamma a')(p) := a'(b_1) + u_\mu a'(b_2)u_\mu^{-1} - u_\nu a'(b_3)u_\nu^{-1} - a'(b_4)$; by the expansion of $\eta_s(b)$, $Z = s\,(D_\gamma \mathsf{J}_\gamma a)(p) + O(s^2)$, and therefore

$$a_p(e^{i(Y_\gamma+sa)}) = \tfrac{1}{2}s^2\big|(D_\gamma\mathsf{J}_\gamma a)(p)\big|^2 + O(s^3).$$

The constant form Y_γ satisfies $\mathrm{d}^*Y_\gamma = 0$, so that $\frac{1}{2\tau_{\mathrm{gf}}}\sum_x|\mathrm{d}^*(Y_\gamma+sa)(x)|^2 = \frac{s^2}{2\tau_{\mathrm{gf}}}\|\mathrm{d}^*a\|^2$. Taking second derivatives in s at $s = 0$ we obtain the quadratic forms of the two Hessians:

$$(a,\mathsf{h}_p(Y_\gamma)a) = \big|(D_\gamma\mathsf{J}_\gamma a)(p)\big|^2, \qquad (a,\mathsf{N}(Y_\gamma)a) = \sum_p\big|(D_\gamma\mathsf{J}_\gamma a)(p)\big|^2 + \tau_{\mathrm{gf}}^{-1}\|\mathrm{d}^*a\|^2,$$

the sum over p being finite. A real symmetric matrix of the kind considered in Def. D.21(1) is determined by its quadratic form on the forms of finite support; by polarisation, therefore,

$$\mathsf{h}_p(Y_\gamma) = (\delta_pD_\gamma\mathsf{J}_\gamma)^*(\delta_pD_\gamma\mathsf{J}_\gamma), \qquad \mathsf{N}(Y_\gamma) = (D_\gamma\mathsf{J}_\gamma)^*D_\gamma\mathsf{J}_\gamma + \tau_{\mathrm{gf}}^{-1}\mathrm{dd}^*,$$

where δ_p is the evaluation at p and the adjoints are taken with respect to the brackets $(\cdot,\cdot)$.

Step 2: symbols. We extend all operators complex linearly. From $[\mathsf{e}_3,\mathsf{e}_e] = 2e\,\mathsf{e}_e$ we get $u_\mu\mathsf{e}_eu_\mu^{-1} = e^{2ie\gamma_\mu}\mathsf{e}_e = e^{iv_\mu}\mathsf{e}_e$, and hence

$$\mathsf{dE}(\gamma_\mu\mathsf{e}_3)\,\mathsf{e}_e = \int_0^1 e^{isv_\mu}\,ds\;\mathsf{e}_e = j(v_\mu)\,\mathsf{e}_e.$$

So J_γ maps the form $e^{ik\cdot x}\alpha\,\mathsf{e}_e$ to $e^{ik\cdot x}J\alpha\,\mathsf{e}_e$, and for $\alpha' \in \mathbb{C}^3$

$$D_\gamma(e^{ik\cdot x}\alpha'\mathsf{e}_e)(p_{\mu\nu}(x)) = e^{ik\cdot x}\big[\alpha'_\mu + e^{i(v_\mu+k_\mu)}\alpha'_\nu - e^{i(v_\nu+k_\nu)}\alpha'_\mu - \alpha'_\nu\big]\mathsf{e}_e = e^{ik\cdot x}D_{p_{\mu\nu}(x)}(k+v)\,\alpha'\,\mathsf{e}_e.$$

Thus J_γ, D_γ and dd^* commute with the translations of $\mathbb{Z}^3$ and map a plane wave of momentum k and charge e to a plane wave of the same momentum and charge, the coefficient vector being multiplied by J, by the 1×3 matrices $D_{p_{\mu\nu}(x)}(k+v)$ $(\mu<\nu)$, and by $\mathsf{k}(k)\mathsf{k}(k)^*$, respectively; the last symbol follows from the plane-wave formulas of Def. D.2. The adjoint of such an operator acts on plane waves through the adjoint matrix. By Lagrange's identity,

$$\sum_{\mu<\nu}|v_\mu\alpha_\nu - v_\nu\alpha_\mu|^2 = |v|^2|\alpha|^2 - |v^*\alpha|^2 = \alpha^*\big(|v|^2\mathbf{1} - vv^*\big)\alpha \qquad (v,\alpha\in\mathbb{C}^3),$$

so that $\sum_{\mu<\nu}D_{p_{\mu\nu}(x)}(k')^*D_{p_{\mu\nu}(x)}(k') = P(k')$ for $k' \in \mathbb{R}^3$. Together with Step 1 this gives the formula for $\hat{\mathsf{N}}_e$; for $\hat{\mathsf{h}}_{q,e}$ only the plaquette q enters, and the factors $e^{ik\cdot x}$ have modulus 1.

For $\hat{\mathsf{M}}_e$, differentiate the defining gauge action in Def. D.21 and invert the differential $\mathsf{dE}(Y)$ of the exponential. At $A = Y_\gamma$, $\lambda = e^{ik\cdot x}\mathsf{e}_e$ and the bond (x,μ) it gives

$$\begin{aligned}&\frac{d}{ds}A^{s\lambda}((x,\mu))\Big|_{s=0}\\ &= \mathsf{dE}(\gamma_\mu \mathsf{e}_3)^{-1}\big(e^{ik\cdot x}\mathsf{e}_e - e^{iv_\mu}e^{ik\cdot(x+e_\mu)}\mathsf{e}_e\big)\\ &= j(v_\mu)^{-1}\big(1-e^{i(v_\mu+k_\mu)}\big)e^{ik\cdot x}\mathsf{e}_e\\ &= -j(v_\mu)^{-1}\mathsf{k}_\mu(k+v)\,e^{ik\cdot x}\mathsf{e}_e.\end{aligned}$$

Since $\mathsf{M}(A)\lambda$ is d^* of this derivative and d^* multiplies plane waves by $\mathsf{k}(k)^*$ (Def. D.2), we obtain $\hat{\mathsf{M}}_e = -\mathsf{k}(k)^* J^{-1}\mathsf{k}(k+v)$. Here we have used that $j(\vartheta)\neq 0$ for $|\vartheta|<2\pi$.

Step 3: proof of (a). Let $v := \mathsf{k}(k+v)$ and $\tilde{u} := (J^*)^{-1}\mathsf{k}(k)$. Then

$$\hat{\mathsf{N}}_e = J^*\big(P(k+v) + \tau_{\mathrm{gf}}^{-1}\tilde{u}\tilde{u}^*\big)J, \qquad v^*\tilde{u} = (J^{-1}v)^*\mathsf{k}(k) = -\overline{\hat{\mathsf{M}}_e},$$

and $P(k+v) = |v|^2\mathbf{1} - vv^*$. For $v,w\in\mathbb{C}^3$ we have $\det(|v|^2\mathbf{1} - vv^* + ww^*) = |v|^2|v^*w|^2$: for $v=0$ both sides vanish; for $v\neq 0$ we use an orthonormal basis whose first vector is $v/|v|$; in it the matrix is $|v|^2\,\mathrm{diag}(0,1,1) + ww^*$, the determinant is an affine function of the rank-one summand, and its value is $0 + |v|^4|w_1|^2$ with $w_1 = v^*w/|v|$. With $w = \tau_{\mathrm{gf}}^{-1/2}\tilde{u}$ this gives $\det\hat{\mathsf{N}}_e = |\det J|^2\,|v|^2\,\tau_{\mathrm{gf}}^{-1}|v^*\tilde{u}|^2$, the first identity in eq. D.xvi. For the second, $|j(\vartheta)|^2 = (2-2\cos\vartheta)/\vartheta^2 = \big(\sin\frac{\vartheta}{2}/\frac{\vartheta}{2}\big)^2$; since $\det J = \prod_\mu j(2e\gamma_\mu)$, the product over the three charges contributes, for each μ, the factor $|j(2\gamma_\mu)|^2 = (\sin\gamma_\mu/\gamma_\mu)^2 = \sigma_1(\gamma_\mu^2)$.

Step 4: proof of (b). By (a), $\hat{\mathsf{N}}_e$ is invertible. By the cyclicity of the trace,

$$\mathrm{tr}\big[\hat{\mathsf{h}}_{q,e}\hat{\mathsf{N}}_e^{-1}\big] = D_q\big(P + \tau_{\mathrm{gf}}^{-1}\tilde{u}\tilde{u}^*\big)^{-1}D_q^*,$$

with D_q and P taken at $k+v$. Now $D_q v = v_\mu v_\nu - v_\nu v_\mu = 0$, hence $v^*D_q^* = 0$ and $PD_q^* = |v|^2 D_q^*$; moreover $Pv = 0$, and $\tilde{u}^*v \neq 0$ because $v^*\tilde{u} = -\overline{\hat{\mathsf{M}}_e} \neq 0$. The vector

$$\mathsf{X} := |v|^{-2}D_q^* - \frac{\tilde{u}^*D_q^*}{|v|^2\,\tilde{u}^*v}\,v$$

satisfies $P\mathsf{X} = D_q^*$ and $\tilde{u}^*\mathsf{X} = 0$, hence $(P+\tau_{\mathrm{gf}}^{-1}\tilde{u}\tilde{u}^*)\mathsf{X} = D_q^*$. Therefore the trace equals $D_q\mathsf{X} = |v|^{-2}D_qD_q^* = |v|^{-2}(|v_\mu|^2+|v_\nu|^2) = \varpi_q(k+v)$.

Step 5: proof of (c). We expand in powers of ϑ and of v; the dots stand for terms of degree at least 3:

$$j(\vartheta) = 1 + \frac{i\vartheta}{2} - \frac{\vartheta^2}{6} + \dots, \qquad j(\vartheta)^{-1} = 1 - \frac{i\vartheta}{2} - \frac{\vartheta^2}{12} + \dots, \qquad |j(\vartheta)|^2 = 1 - \frac{\vartheta^2}{12} + \dots,$$

$$\begin{aligned}\mathsf{k}_\mu(k+v) &= \mathsf{k}_\mu + e^{ik_\mu}\big(iv_\mu - \tfrac{1}{2}v_\mu^2 + \dots\big),\\ |\mathsf{k}_\mu(k+v)|^2 &= 2 - 2\cos(k_\mu+v_\mu)\\ &= |\mathsf{k}_\mu|^2 + 2v_\mu\sin k_\mu + v_\mu^2\cos k_\mu + \dots.\end{aligned}$$

In $\hat{\mathsf{M}}_e = -\sum_\mu \overline{\mathsf{k}_\mu}\,j(v_\mu)^{-1}\mathsf{k}_\mu(k+v)$ the terms of degree 1 are

$$-i\sum_\mu v_\mu\overline{\mathsf{k}_\mu}\big[e^{ik_\mu} - \tfrac{1}{2}\mathsf{k}_\mu\big] = -i\sum_\mu v_\mu\overline{\mathsf{k}_\mu}\,\tfrac{1}{2}\big(e^{ik_\mu}+1\big) = -\sum_\mu v_\mu\sin k_\mu,$$

because $\overline{\mathsf{k}_\mu}(e^{ik_\mu}+1) = -2i\sin k_\mu$, and the terms of degree 2 are

$$-\sum_\mu v_\mu^2\,\overline{\mathsf{k}_\mu}\big[-\tfrac{1}{12}\mathsf{k}_\mu + \tfrac{1}{2}e^{ik_\mu} - \tfrac{1}{2}e^{ik_\mu}\big] = \frac{1}{12}\sum_\mu v_\mu^2|\mathsf{k}_\mu|^2.$$

The term $\tau_{\mathrm{gf}}^{-1}\mathsf{k}\mathsf{k}^*$ of $\hat{\mathsf{N}}_e$ does not depend on γ. Hence $\operatorname{tr}\hat{\mathsf{N}}_{e,2}$ is the homogeneous polynomial of degree 2 in the expansion of

$$\operatorname{tr}\big[J^*P(k+\upsilon)J\big] = \sum_\mu |j(\upsilon_\mu)|^2 \sum_{\nu\neq\mu} |\mathsf{k}_\nu(k+\upsilon)|^2,$$

namely

$$\sum_\mu \Big[-\frac{1}{12}\upsilon_\mu^2 \sum_{\nu\neq\mu}|\mathsf{k}_\nu|^2 + \sum_{\nu\neq\mu}\upsilon_\nu^2\cos k_\nu\Big] = \sum_\mu \upsilon_\mu^2\Big[2\cos k_\mu - \frac{1}{12}\big(|\mathsf{k}|^2 - |\mathsf{k}_\mu|^2\big)\Big].$$

This is the asserted formula for $\operatorname{tr}\hat{\mathsf{N}}_{e,2}$, and (c) is proved. □

With these formulas we evaluate the self-energy at zero momentum.

Proposition D.23 (The five contributions to the self-energy at zero momentum)**.** Let $\tau_{\mathrm{gf}} = 1$.

(i) For $i = 1,\dots,4$, all $k \in [-\pi,\pi]^3 \setminus \{0\}$ and all $\gamma \in \mathbb{R}^3$ we have $|\mathfrak{s}^{(i)}(k;\gamma)| \le C|\gamma|^2|k|^{-2}$, with $C = C(\tau_{\mathrm{gf}})$.

(ii) $\sum_{i=1}^5 c_{\mu\nu}^{(i)}(\tau_{\mathrm{gf}}) = 0$ for all μ,ν.

(iii) At $\tau_{\mathrm{gf}} = 1$,

$$(c_{\mu\nu}^{(1)},\dots,c_{\mu\nu}^{(5)})(1) = \delta_{\mu\nu}\cdot\Big(-20Z_0+\tfrac{8}{3},\quad 16Z_0-\tfrac{28}{9},\quad 4Z_0-\tfrac{2}{3},\quad \tfrac{4}{9},\quad \tfrac{2}{3}\Big). \tag{D.xviii}$$

■

Proof. Step 1: bounds. By the formulas of Lem. D.22, the entries of $J^{\pm1}$ and of $\mathsf{k}(k+\upsilon)$ are real analytic functions of (k,γ). On $[-\pi,\pi]^3$ we have $P(k) = O(|k|^2)$ and $\partial_k P(k) = O(|k|)$. The terms of degree 1 in γ of $J^*P(k+\upsilon)J$ are terms containing the factor $P(k)$ and the term $\sum_\mu \upsilon_\mu\partial_\mu P(k)$; hence $|\hat{\mathsf{N}}_{e,1}| \le C|\gamma||k|$, while $|\hat{\mathsf{N}}_{e,2}| \le C|\gamma|^2$. Every term of $\hat{\mathsf{M}}_e$ contains the factor $\mathsf{k}(k)^*$, so that $|\hat{\mathsf{M}}_{e,j}| \le C|\gamma|^j|k|$. Finally $|\hat{C}| \le C|k|^{-2}$ by the formula for $\hat{C}$ in Def. D.21(2), and $|\hat{\mathsf{M}}_{e,0}|^{-1} = |\mathsf{k}(k)|^{-2} \le C|k|^{-2}$, as $|\mathsf{k}(k)|^2 \ge \frac{4}{\pi^2}|k|^2$. Inserting these bounds into the definitions of $\mathfrak{s}^{(1)},\dots,\mathfrak{s}^{(4)}$ we obtain $|\hat{C}\hat{\mathsf{N}}_{e,1}|^2 \le C|\gamma|^2|k|^{-2}$, $|\hat{C}\hat{\mathsf{N}}_{e,2}| \le C|\gamma|^2|k|^{-2}$, $|\hat{\mathsf{M}}_{e,1}/\hat{\mathsf{M}}_{e,0}|^2 \le C|\gamma|^2|k|^{-2}$ and $|\hat{\mathsf{M}}_{e,2}/\hat{\mathsf{M}}_{e,0}| \le C|\gamma|^2|k|^{-1} \le C'|\gamma|^2|k|^{-2}$, which proves (i).

Step 2: identity of the integrands. Fix $k \in [-\pi,\pi]^3\setminus\{0\}$, and let γ be so close to 0 that $\mathsf{k}(k+\upsilon) \neq 0$ and $\hat{\mathsf{M}}_e(k;\gamma) \neq 0$ for all e. We divide the first identity in eq. D.xvi by its value at $\gamma = 0$, take one half of its logarithm, sum over e and use the second identity in eq. D.xvi:

$$\sum_e \Big\{\tfrac{1}{2}\log\det\big[\hat{C}\hat{\mathsf{N}}_e(k;\gamma)\big] - \log\big|\hat{\mathsf{M}}_e(k;\gamma)/\hat{\mathsf{M}}_{e,0}\big|\Big\} - \sum_\mu \log\sigma_1(\gamma_\mu^2) = \tfrac{1}{2}\sum_e \log\frac{|\mathsf{k}(k+2e\gamma)|^2}{|\mathsf{k}(k)|^2}.$$

Both sides are real analytic in γ near 0, and we compare their terms of degree 2. On the left, the expansion $\log\det(\mathbf{1}+\mathsf{A}) = \operatorname{tr}\mathsf{A} - \frac{1}{2}\operatorname{tr}\mathsf{A}^2 + \dots$ with $\mathsf{A} = \hat{C}(\hat{\mathsf{N}}_{e,1}+\hat{\mathsf{N}}_{e,2}+\dots)$ yields $\mathfrak{s}^{(2)}+\mathfrak{s}^{(1)}$; the expansion $\log|1+z| = \operatorname{Re}(z-\frac{1}{2}z^2+\dots)$ with $z = (\hat{\mathsf{M}}_{e,1}+\hat{\mathsf{M}}_{e,2}+\dots)/\hat{\mathsf{M}}_{e,0}$ yields, with the sign in front, $\mathfrak{s}^{(4)}+\mathfrak{s}^{(3)}$; and by Lem. D.1, applied to $Y = \gamma_\mu\mathsf{e}_3$, the term $-\sum_\mu\log\sigma_1(\gamma_\mu^2)$ yields $\frac{1}{3}|\gamma|^2 = \mathfrak{s}^{(5)}$. On the right, with $\mathsf{l}(k) := \log|\mathsf{k}(k)|^2$, the terms of degree 2 are $\frac{1}{2}\sum_{e=\pm1}\frac{1}{2}\sum_{\mu\nu}4\gamma_\mu\gamma_\nu\,\partial_\mu\partial_\nu\mathsf{l}(k)$. Hence

$$\sum_{i=1}^5 \mathfrak{s}^{(i)}(k;\gamma) = 2\sum_{\mu\nu}\gamma_\mu\gamma_\nu\,\partial_\mu\partial_\nu\mathsf{l}(k) \qquad (k\neq 0),$$

an identity between homogeneous polynomials of degree 2 in γ, valid for all $\gamma\in\mathbb{R}^3$.

Step 3: flux argument. The function l is smooth on $\mathbb{R}^3\setminus 2\pi\mathbb{Z}^3$ and periodic, and on the cube $|\partial\mathsf{l}| \le C|k|^{-1}$ and $|\partial\partial\mathsf{l}| \le C|k|^{-2}$. For $0<\varepsilon<1$ we apply Gauss' divergence theorem on the cube

$[-\pi,\pi]^3$ less the ball $\{|k|\le\varepsilon\}$; the contributions of opposite faces of the cube cancel by periodicity, and

$$\int_{[-\pi,\pi]^3\cap\{|k|>\varepsilon\}} \partial_\mu\partial_\nu \mathsf{I}\, dk = -\int_{|k|=\varepsilon} \frac{k_\mu}{\varepsilon}\,\partial_\nu \mathsf{I}\, dS = O(\varepsilon).$$

Letting $\varepsilon\to 0$ we get $\int_{\mathbb{T}} \partial_\mu\partial_\nu \mathsf{I}\, d^\circ k = 0$. By Step 2 and the definition of the numbers $c^{(i)}_{\mu\nu}$, $\frac12\sum_{\mu\nu}(\sum_i c^{(i)}_{\mu\nu})\gamma_\mu\gamma_\nu = 0$ for all γ; as the matrices are symmetric, (ii) follows.

Step 4: elementary integrals. We write $\mathsf{k} = \mathsf{k}(k)$. By the symmetry under permutations of the coordinates, $\int_{\mathbb{T}} |\mathsf{k}_\mu|^2|\mathsf{k}|^{-2} d^\circ k = \frac13$. As $\cos k_\mu = 1 - \frac12|\mathsf{k}_\mu|^2$, we get $\int_{\mathbb{T}} \cos k_\mu |\mathsf{k}|^{-2} d^\circ k = Z_0 - \frac16$. Since $\partial_\mu|\mathsf{k}|^2 = 2\sin k_\mu$,

$$\partial_\mu(\sin k_\mu |\mathsf{k}|^{-2}) = \cos k_\mu |\mathsf{k}|^{-2} - 2\sin^2 k_\mu |\mathsf{k}|^{-4};$$

the function $\sin k_\mu|\mathsf{k}|^{-2}$ is periodic and $O(|k|^{-1})$, so the flux argument of Step 3 shows that the integral of the left side over $\mathbb{T}$ vanishes, and $\int_{\mathbb{T}} \sin^2 k_\mu |\mathsf{k}|^{-4} d^\circ k = \frac12(Z_0 - \frac16)$. Finally $\int_{\mathbb{T}} \sin k_\mu \sin k_\nu |\mathsf{k}|^{-4} d^\circ k = 0$ for $\mu\neq\nu$, the integrand being odd under $k_\mu\to -k_\mu$.

Step 5: values. We use Lem. D.22(c) with $\upsilon = 2e\gamma$, and $\hat{\mathsf{M}}_{e,0} = -|\mathsf{k}|^2$. For $e=0$ we have $\upsilon = 0$, so that $\hat{\mathsf{N}}_{0,j} = 0$ and $\hat{\mathsf{M}}_{0,j} = 0$ for $j\ge 1$; the charge 0 contributes nothing, and for $e=\pm1$ we have $\upsilon_\mu^2 = 4\gamma_\mu^2$. We write δ for the matrix $(\delta_{\mu\nu})$.

- $\mathfrak{s}^{(5)} = \frac13|\gamma|^2$, whose integral is $\frac13|\gamma|^2$; hence $c^{(5)} = \frac23\delta$.
- $\mathfrak{s}^{(4)} = \sum_{e=\pm1}\frac{4}{12}\sum_\mu \gamma_\mu^2|\mathsf{k}_\mu|^2|\mathsf{k}|^{-2}$, whose integral is $\frac29|\gamma|^2$ by Step 4; hence $c^{(4)} = \frac49\delta$.
- $\mathfrak{s}^{(3)} = 4\big(\sum_\mu \gamma_\mu \sin k_\mu\big)^2|\mathsf{k}|^{-4}$, whose integral is $4|\gamma|^2\cdot\frac12(Z_0-\frac16)$; hence $c^{(3)} = (4Z_0 - \frac23)\delta$.
- At $\tau_{\mathrm{gf}} = 1$ we have $\hat C = |\mathsf{k}|^{-2}\mathbf{1}$ and

$$\mathfrak{s}^{(2)} = \tfrac12\sum_{e=\pm1}|\mathsf{k}|^{-2}\operatorname{tr}\hat{\mathsf{N}}_{e,2} = 4\sum_\mu \gamma_\mu^2\Big[2\cos k_\mu - \frac{1}{12}\big(|\mathsf{k}|^2 - |\mathsf{k}_\mu|^2\big)\Big]|\mathsf{k}|^{-2},$$

  whose integral is $4|\gamma|^2[2Z_0 - \frac13 - \frac{1}{12}\cdot\frac23] = |\gamma|^2(8Z_0 - \frac{14}{9})$; hence $c^{(2)} = (16Z_0 - \frac{28}{9})\delta$.
- By (ii), $c^{(1)} = -\sum_{i\ge2} c^{(i)} = -(20Z_0 - \frac{28}{9} - \frac23 + \frac49 + \frac23)\delta = (-20Z_0 + \frac83)\delta$.

This proves (iii). □

The same method applies to the vertex function of a plaquette action.

Proposition D.24 (The vertex function at zero momenta)**.** Let $\tau_{\mathrm{gf}} = 1$ and let q be a plaquette. Then $|\mathfrak{v}_q^{(j)}(k;\gamma)| \le C|\gamma|^2|k|^{-2}$ for $j = 1,\dots,4$, all $k\in[-\pi,\pi]^3\setminus\{0\}$ and all $\gamma\in\mathbb{R}^3$, where C depends on τ_{gf} only; and

$$\sum_{j=1}^4 \mathsf{g}^{(j)}_{q;\mu\nu}(\tau_{\mathrm{gf}}) = 0 \qquad \text{for all } \mu,\nu.$$

■

Proof. Step 1: bound. The 1×3 matrix $D_q(k)$ is smooth in k, satisfies $D_q(k) = O(|k|)$, and depends on q only through the two directions of q. In the expansion of $\hat{\mathsf{h}}_{q,e} = J^* D_q(k+\upsilon)^* D_q(k+\upsilon)J$ in powers of γ, each term of degree $j\le1$ retains at least $2-j$ factors $D_q(k)$ or $D_q(k)^*$; hence $|\hat{\mathsf{h}}_{q,e,j}| \le C|\gamma|^j|k|^{2-j}$ for $j=0,1,2$, with C independent of q. As in Step 1 of the proof of Prop. D.23, Lem. D.22 gives $|\hat{\mathsf{N}}_{e,1}| \le C|\gamma||k|$ and $|\hat{\mathsf{N}}_{e,2}| \le C|\gamma|^2$, and Def. D.2 gives $|\hat C| \le C|k|^{-2}$. Inserted into the definitions of $\mathfrak{v}_q^{(1)},\dots,\mathfrak{v}_q^{(4)}$, these bounds give the asserted bound.

Step 2: identity of the integrands. Fix $k\in[-\pi,\pi]^3\setminus\{0\}$. For small γ we write $\hat{\mathsf{N}}_e = \hat{\mathsf{N}}_{e,0}(\mathbf{1} + \hat C(\hat{\mathsf{N}}_e - \hat{\mathsf{N}}_{e,0}))$, and the Neumann series gives

$$\hat{\mathsf{N}}_e^{-1} = \hat C - \hat C\hat{\mathsf{N}}_{e,1}\hat C - \hat C\hat{\mathsf{N}}_{e,2}\hat C + \hat C\hat{\mathsf{N}}_{e,1}\hat C\hat{\mathsf{N}}_{e,1}\hat C + O(|\gamma|^3).$$

Hence the terms of degree 2 in γ of $\frac{1}{2}\sum_e \operatorname{tr}[\hat{\mathsf{h}}_{q,e}\hat{\mathsf{N}}_e^{-1}]$ are $\sum_{j=1}^4 \mathfrak{v}_q^{(j)}(k;\gamma)$. On the other hand, for small γ the hypotheses of Lem. D.22(b) hold, and by eq. D.xvii the same expression equals $\frac{1}{2}\sum_e \varpi_q(k+2e\gamma)$, whose terms of degree 2 are $2\sum_{\mu\nu}\gamma_\mu\gamma_\nu\partial_\mu\partial_\nu\varpi_q(k)$. Therefore

$$\sum_{j=1}^{4}\mathfrak{v}_q^{(j)}(k;\gamma) = 2\sum_{\mu\nu}\gamma_\mu\gamma_\nu\,\partial_\mu\partial_\nu\varpi_q(k) \qquad (k\neq 0,\ \gamma\in\mathbb{R}^3).$$

Step 3: flux argument. The function ϖ_q is smooth on $\mathbb{R}^3\setminus 2\pi\mathbb{Z}^3$ and periodic, and on the cube $|\partial\varpi_q|\le C|k|^{-1}$ and $|\partial\partial\varpi_q|\le C|k|^{-2}$, since it is a quotient of smooth functions vanishing to second order at 0 whose denominator is at least $\frac{4}{\pi^2}|k|^2$. As in Step 3 of the proof of Prop. D.23,

$$\int_{[-\pi,\pi]^3\cap\{|k|>\varepsilon\}}\partial_\mu\partial_\nu\varpi_q\,dk = -\int_{|k|=\varepsilon}\frac{k_\mu}{\varepsilon}\,\partial_\nu\varpi_q\,dS = O(\varepsilon),$$

hence $\int_{\mathbb{T}}\partial_\mu\partial_\nu\varpi_q\,d^\circ k=0$. By Step 2, $\frac{1}{2}\sum_{\mu\nu}(\sum_j \mathsf{g}^{(j)}_{q;\mu\nu})\gamma_\mu\gamma_\nu=0$ for all γ, and the symmetry of the matrices gives the claim. □

We next record the behaviour of these objects under point reflections.

Lemma D.25 (Point reflections)**.** Let $c\in\frac{1}{2}\mathbb{Z}^3$, and let $R_c: x\mapsto 2c-x$ on $\mathbb{Z}^3$ (note that $2c\in\mathbb{Z}^3$). We let R_c act on bonds by $R_c\langle x,x'\rangle := \langle R_cx, R_cx'\rangle$, on plaquettes through their corners (the corners of R_cp are the images of the corners of p), and on configurations U, 1-forms A and site functions λ by

$$(R_cU)(b) := U(R_cb), \qquad (R_cA)(b) := A(R_cb), \qquad (R_c\lambda)(x) := \lambda(R_cx);$$

thus $(R_cA)_\mu(x) = -A_\mu(2c-x-e_\mu)$. The following hold.

(a) For all plaquettes p, configurations U, 1-forms A on $\mathbb{Z}^3$ and $m\ge 0$: $a_p(R_cU)=a_{R_cp}(U)$ and $\mathsf{a}_p^{(m)}(R_cA)=\mathsf{a}_{R_cp}^{(m)}(A)$. Moreover $\mathrm{d}^*R_c=R_c\mathrm{d}^*$ on 1-forms, and $\mathrm{d}R_c=R_c\mathrm{d}$ on site functions.

(b) For every $\tau_{\mathrm{gf}}=1$ the family $\big((R_cA)^a(b)\big)_{b,a}$ has the law of the Gaussian family of $\mathbb{Z}^3$ and τ_{gf}. Moreover $D(R_cb;R_cb')=D(b;b')$ for all bonds b,b', where D is taken with oriented arguments, that is, it is given by the formula of Def. D.2 for arbitrary bonds; in particular $D(\bar b;b')=-D(b;b')$ and $D(b;\bar b')=D(b;b')$.

(c) For a positively oriented bond b let b^R be the positively oriented one of the bonds R_cb and $\overline{R_cb}$, that is, $(x,\mu)^R=(2c-x-e_\mu,\mu)$. Then for all positively oriented bonds b,b' and all plaquettes q

$$\Sigma(b^R,b'^R)=\Sigma(b,b'), \qquad \Gamma_{R_cq}(b^R,b'^R)=\Gamma_q(b,b').$$

■

Proof. Step 1: proof of (a). Let $p=p_{\mu\nu}(x)$. The loop $\partial p=(b_1,b_2,\bar b_3,\bar b_4)$ passes through x, $x+e_\mu$, $x+e_\mu+e_\nu$, $x+e_\nu$. The map R_c carries it to the loop through $y+e_\mu+e_\nu$, $y+e_\nu$, y, $y+e_\mu$, where $y:=2c-x-e_\mu-e_\nu$ is the base point of $R_cp=p_{\mu\nu}(y)$; this image loop is $\partial(R_cp)$, traversed in the same sense but started at the opposite corner. Since $\partial(R_cU)(p)$ is the product of the values of U along the image loop, it is conjugate in G to $\partial U(R_cp)$, and the two matrices have the same trace; hence $a_p(R_cU)=a_{R_cp}(U)$. As $R_c(e^{itA})=e^{itR_cA}$, comparison of the Taylor coefficients in Lem. D.1 gives $\mathsf{a}_p^{(m)}(R_cA)=\mathsf{a}_{R_cp}^{(m)}(A)$. Next, $(\mathrm{d}^*A)(x)=-\sum_{b:\,b_-=x}A(b)$, the sum over the six bonds starting at x, and R_c maps the bonds starting at x bijectively onto those starting at R_cx; therefore $\mathrm{d}^*R_cA=R_c\mathrm{d}^*A$. Finally $(\mathrm{d}R_c\lambda)(b)=\lambda(R_cb_+)-\lambda(R_cb_-)=(\mathrm{d}\lambda)(R_cb)$.

Step 2: proof of (b). The variables $(R_cA)_\mu(x)=-A_\mu(2c-x-e_\mu)$ form a centred Gaussian family, with the same colour structure as A. By Def. D.2 the covariance of $(R_cA)_\mu(x)$ and $(R_cA)_\nu(y)$, for equal colours, is

$$C\big((2c-x-e_\mu,\mu),(2c-y-e_\nu,\nu)\big)=\int_{\mathbb{T}}e^{-ik\cdot(x-y)}\,e^{-i(k_\mu-k_\nu)}\,\hat C_{\mu\nu}(k)\,d^\circ k.$$

Since $e^{-ik_\mu}\mathsf{k}_\mu(k) = -\mathsf{k}_\mu(-k)$, and correspondingly $e^{ik_\nu}\overline{\mathsf{k}_\nu(k)} = -\overline{\mathsf{k}_\nu(-k)}$, we have $e^{-i(k_\mu-k_\nu)}\hat{C}_{\mu\nu}(k) = \hat{C}_{\mu\nu}(-k)$, also for $\mu=\nu$. The substitution $k\to -k$ shows that the covariance equals $C((x,\mu),(y,\nu))$, and a centred Gaussian family is determined in law by its covariance. Moreover $G(R_cx,R_cy) = G(x,y)$, again by the substitution $k\to -k$ in the formula for G; with $(R_cb)_\pm = R_c(b_\pm)$ the formula for D in Def. D.2 gives $D(R_cb;R_cb') = D(b;b')$. The same formula shows that reversing the first argument changes the sign of D, while reversing the second does not change it.

Step 3: proof of (c). By (a) and (b), S_3 and V_2 are invariant under $A\to R_cA$, term by term: the term of a plaquette p is carried to that of R_cp, and the terms of a bond b and of a pair (b,b') in V_2 to those of b^R and of (b^R,b'^R); here $g(b)=\frac{1}{3}$ is constant on $\mathbb{Z}^3$, $|A(R_cb)| = |A(b^R)|$, and the signs coming from reversed arguments of D cancel in pairs. Likewise $\mathsf{a}_q^{(m)}(A) = \mathsf{a}_{R_cq}^{(m)}(R_cA)$. Since $(R_cA)(b^R) = A(\bar{b}) = -A(b)$, the chain rule gives

$$\partial_{b,a}(\mathsf{F}\circ R_c) = -(\partial_{b^R,a}\mathsf{F})\circ R_c$$

for every polynomial F. Each term of the kernels in Def. D.21(5) contains exactly two derivatives, so that the signs cancel, and by (b) the means and cumulants do not change under the substitution $A\to R_cA$. Hence $\Sigma(b,a;b',a') = \Sigma(b^R,a;b'^R,a')$ and $\Gamma_q(b,a;b',a') = \Gamma_{R_cq}(b^R,a;b'^R,a')$, which is (c). □

D.14. **Homogeneous parts of lattice kernels.** Fix a smooth function $\theta:[0,\infty)\to[0,1]$, equal to one on $[0,\frac{1}{2}]$ and zero on $[1,\infty)$. Fix a smooth radial function χ, equal to one on $\{|k|\le 1\}$ and zero on $\{|k|\ge 2\}$. Since $2<\pi$, it also defines a function on the Fourier torus. An elementary line of degree $n=s+3$, where $s\ge -2$ is an integer, is the Fourier kernel of a symbol $\sigma\in\mathcal{S}^s$ with

$$\sigma-\chi\sigma^h\in\mathcal{S}^{s+1},\qquad \sigma^h(tk)=t^s\sigma^h(k),\qquad \sigma^h(-k)=(-1)^s\sigma^h(k),$$

where σ^h is smooth off zero. Products and translations of line symbols preserve this definition whenever their resulting order is at least -2; their leading symbols multiply. In particular the symbols of G,C,E,Π,D have leading parts obtained by replacing k_μ by ik_μ and $|\mathsf{k}|^{-2}$ by $|k|^{-2}$. Indeed, $|\mathsf{k}|^2-|k|^2$ vanishes to fourth order at zero, and $|\mathsf{k}|^2\ge 4|k|^2/\pi^2$.

Lemma D.26 (Homogeneous kernels)**.** Let $s>-3$, and let σ^h be smooth off zero and homogeneous of degree s. There is a unique smooth function

$$K^h = \mathcal{F}_{\neq 0}^{-1}\sigma^h$$

on $\mathbb{R}^3\setminus\{0\}$, homogeneous of degree $-3-s$, such that for every smooth compactly supported $\rho=1$ near zero,

$$\int e^{ik\cdot x}\rho(k)\sigma^h(k)\,d^\circ k - K^h(x) = O(|x|^{-N})\quad(|x|\ge 1)$$

for every N. The construction preserves parity and rotations, and multiplication of the symbol by $(ik)^\gamma$ gives $\partial^\gamma K^h$. For an elementary line of degree n,

$$|\nabla^\gamma(K-K^h)(x)|\le C_\gamma|x|^{-n-1-|\gamma|}\quad(|x|\ge|\gamma|+1).$$

■

Proof. Choose $2N>s+3$, and define

$$K^h_\rho(x) = \int e^{ik\cdot x}\rho\sigma^h\,d^\circ k + |x|^{-2N}\int e^{ik\cdot x}(-\Delta_k)^N[(1-\rho)\sigma^h]\,d^\circ k.$$

The second integral converges absolutely. Integration by parts over balls shows that it does not change on replacing N by $N+1$; the boundary terms tend to zero since $2N>s+3$. The same argument shows independence of ρ: the difference of two cutoffs times σ^h is smooth and compactly supported away from zero. Taking N arbitrarily large proves the rapid remainder. Taking $2N>s+3+|\gamma|$ permits differentiation under the integrals, proving smoothness and the derivative

rule. Rescaling k proves homogeneity. The substitutions $k \mapsto -k$ and $k \mapsto Ok$ prove parity and rotation covariance. Uniqueness follows because a homogeneous difference of degree $-3-s$ cannot decay faster than every power on every ray unless it vanishes.

For the lattice line, split its symbol into $\chi\sigma^h$ and its remainder of order $s+1$. The kernel bound of Lem. D.3 gives $K - K^h = O(|x|^{-4-s})$. Apply the same argument to $\mathsf{k}^\gamma\sigma$, whose leading symbol is $(ik)^\gamma\sigma^h$. It gives the estimate with $\partial^\gamma K^h$ in place of $\nabla^\gamma K^h$. Taylor's formula replaces the former by the latter with the same error: on the finite difference stencil, the relevant derivatives of K^h are bounded by their homogeneity. This proves the final assertion. □

Definition D.27 (Lines with a leading part)**.** For an integer $n \geq 2$, write $K \in \mathfrak{K}(n)$ if it is the Fourier kernel of $\sigma \in L^1(\mathbb{T})$, and has leading data (σ^h, K^h) satisfying

$$\sigma^h(tk) = t^{n-3}\sigma^h(k), \qquad |\sigma - \chi\sigma^h|(k) \leq C|k|^{n-2}\lg|k|,$$

with σ^h continuous off zero, and

$$K^h(tx) = t^{-n}K^h(x), \qquad |\nabla^\gamma(K-K^h)(x)| \leq C_\gamma|x|^{-n-1-|\gamma|}\lg|x| \quad (|x| \geq |\gamma|+1),$$

with K^h smooth off zero. The constants include the displayed bounds, $\|\sigma\|_1$, and the sphere bounds of σ^h and the derivatives of K^h. Here $\lg t = 1 + |\log t|$. The leading data are unique: rescale a ray to zero for σ^h, and integer multiples of rational rays to infinity for K^h, then use continuity. Every elementary line of degree $n \geq 2$ belongs to $\mathfrak{K}(n)$, by Lem. D.26. ■

Lemma D.28 (Comparing a sum and an integral)**.** For $D \in \{3, 6\}$, suppose $F \in C^1(\mathbb{R}^D)$ and $\sum_{z\in\mathbb{Z}^D} \sup_{z+[0,1]^D}(|F| + |\nabla F|) < \infty$. Then

$$\left|\sum_z F(z) - \int F(x)\,dx\right| \leq \sqrt{D}\sum_z \sup_{z+[0,1]^D}|\nabla F|.$$

■

Proof. On each unit cube the mean value theorem bounds the difference between $F(z)$ and its integral by the displayed supremum times the cube diameter $\sqrt{D}$. The hypotheses permit summation. □

For $D = 3, 6$, $R \geq 1$, and $g : \mathbb{Z}^D \to \mathbb{C}$, put

$$\|g\|_{(R)} = \max_{|\gamma|\leq 10} R^{|\gamma|}\sup|\nabla^\gamma g|.$$

Let $\mathcal{H}_s(D)$ denote the nonnegative Borel functions homogeneous of degree $s > -D$ and integrable over the unit ball. If $s < 9 - D$, then

$$\int H(\kappa)\lg|\kappa|\langle\kappa\rangle^{-10}\,d\kappa < \infty \qquad (H \in \mathcal{H}_s(D)).$$

This follows by radial integration.

Lemma D.29 (A kernel against a slowly varying function)**.** Suppose g vanishes outside $\{|z| \leq R\}$, and write $\check{g}(\kappa) = \sum_z g(z)e^{-i\kappa\cdot z}$. Then

$$|\check{g}(\kappa)| \leq CR^D\langle R\kappa\rangle^{-10}\|g\|_{(R)}.$$

Let $\widehat{T} \in L^1(\mathbb{T}^{D/3})$, $0 < c_0 \leq 1$, $-D < s < 9 - D$, $H \in \mathcal{H}_s(D)$, and $\iota = 0$ or 1. If

$$|\widehat{T}(\kappa)| \leq H(\kappa)(\lg|\kappa|)^\iota \quad (0 < |\kappa| \leq c_0),$$

then

$$\left|\int \widehat{T}(\kappa)\check{g}(-\kappa)\,d^\circ\kappa\right| \leq CR^{-s}(\lg R)^\iota\|g\|_{(R)}.$$

Suppose further that $s < 8 - D$, that $\widehat{T}^h$ is homogeneous of degree s, and that

$$|\widehat{T}^h| \leq H, \qquad |\widehat{T} - \widehat{T}^h| \leq H'\lg|\kappa| \quad (0 < |\kappa| \leq c_0), \qquad H' \in \mathcal{H}_{s+1}(D).$$

If $G \in C_c^\infty(\{|x| \le 1\})$, $a \in \mathbb{R}$, and $\|g - R^{-a}G(\cdot/R)\|_{(R)} \le AR^{-a-1}\lg R$, then

$$\left|\int \widehat{T}\check{g} - R^{-a-s}\int_{\mathbb{R}^D} \widehat{T}^h(\kappa)\widehat{G}(-\kappa)\,d^\circ\kappa\right| \le C(A + \|G\|_{C^{10}})R^{-a-s-1}\lg R.$$

In the first integral $\check{g}$ is evaluated at $-\kappa$. The homogeneous integral is bounded by $C\|G\|_{C^{10}}$. Constants depend only on the displayed symbol bounds and c_0. ■

Proof. Finite differences give

$$(e^{i\kappa_j} - 1)^N \check{g}(\kappa) = \sum_z \nabla_j^N g(z) e^{-i\kappa\cdot z}.$$

For $N \le 10$ the right side has at most CR^D terms, each bounded by $R^{-N}\|g\|_{(R)}$. Choose $|\kappa_j| \ge |\kappa|/\sqrt{D}$, and use $|e^{i\kappa_j} - 1| \ge 2|\kappa_j|/\pi$. The estimates at $N = 0, 10$ prove the Fourier bound.

Outside $|\kappa| \le c_0$, that bound gives $CR^{D-10}c_0^{-10}\|\widehat{T}\|_1\|g\|_{(R)}$. Inside, substitute $\kappa = \kappa'/R$. Homogeneity and $\lg(|\kappa'|/R) \le \lg|\kappa'|\lg R$ give the second assertion by the integrability stated above.

For the last assertion set $g^h = R^{-a}G(\cdot/R)$. Near zero, $|\widehat{T}| \le H + H'/|\kappa|$, since $t\lg t \le 1$ for $0 < t \le 1$. This majorant belongs to $\mathcal{H}_s(D)$. The preceding estimate applied to $g - g^h$ therefore gives the error $CAR^{-a-s-1}\lg R$. Apply it again, now at order $s+1$, to $\widehat{T} - \chi(\cdot/c_0)\widehat{T}^h$ and g^h. This costs $CR^{-a-s-1}\lg R\|G\|_{C^{10}}$.

Poisson summation gives

$$\check{g}^h(\kappa) = R^{D-a}\sum_{m\in\mathbb{Z}^D} \widehat{G}(R(\kappa + 2\pi m)).$$

For $|\kappa| \le 2c_0 \le 2$, every nonzero m has $|\kappa + 2\pi m| \ge \pi|m|$. Since $|\widehat{G}(\xi)| \le C\langle\xi\rangle^{-10}\|G\|_{C^{10}}$, their total contribution is $O(R^{D-a-10}\|G\|_{C^{10}})$, within the claimed error. The zero term, after rescaling, is

$$R^{-a-s}\int \chi(\kappa/(c_0R))\widehat{T}^h(\kappa)\widehat{G}(-\kappa)\,d^\circ\kappa.$$

Removing χ costs at most $CR^{-a-s}R^{s+D-10}\|G\|_{C^{10}}$. The assumption $s < 8 - D$ makes this smaller than the stated error and also proves the absolute bound on the homogeneous integral. □

Lemma D.30 (One-loop functions)**.** Let $m = 2$ or 3, $D = 3(m-1)$, and let σ_i be elementary line symbols of integer orders $s_i \ge -2$. For signs ϵ_i and linear maps $\lambda_i : \mathbb{R}^D \to \mathbb{R}^3$, set $l_i = -\epsilon_i\lambda_i$. Assume the maps l_i are distinct and, for some $c_\lambda \ge 1$,

$$\max_i |l_i(\kappa)| \le c_\lambda|\kappa| \le c_\lambda^2 \max_{i<j} |l_i(\kappa) - l_j(\kappa)|.$$

Assume also that $(l,\kappa) \mapsto (\epsilon_i l + \lambda_i(\kappa))_{i=1}^m$ has an integer matrix of determinant ± 1. For $m = 3$ require $s_i + s_j \ge -3$ for $i \ne j$. Put

$$\omega = 3 + \sum_i s_i \ge -1, \qquad I(\kappa) = \int_{\mathbb{T}} \prod_i \sigma_i(\epsilon_i l + \lambda_i(\kappa))\,d^\circ l,$$

$$\Lambda(\kappa) = 1 + \sum_{i<j} \log_+ \frac{|\kappa|}{|l_i(\kappa) - l_j(\kappa)|}.$$

The integral exists almost everywhere and $\|I\|_1 \le \prod_i \|\sigma_i\|_1$. Near zero, away from the collision subspaces, there are a polynomial Q of degree at most ω, zero if $\omega = -1$, and a homogeneous Borel function I^h such that

$$I^h(t\kappa) = t^\omega I^h(\kappa), \qquad I^h(-\kappa) = (-1)^{\omega+1}I^h(\kappa), \qquad |I^h| \le C|\kappa|^\omega\Lambda(\kappa),$$

$$|I - Q - I^h|(\kappa) \le C|\kappa|^{\omega+1}\lg|\kappa|\Lambda(\kappa).$$

Set $I^h = 0$ on the collision subspaces and at zero. It depends multilinearly only on the leading symbols σ_i^h and the linear maps. It commutes with simultaneous rotations whenever the maps do, and is continuous off zero when $m = 2$. If $\omega \geq 1$, then

$$Q(0) = \int_{\mathbb{T}} \prod_i \sigma_i(\epsilon_i l)\, d^\circ l.$$

The pair (Q, I^h) is uniquely determined by these asymptotics and the indicated opposite parities at degree ω. Finally $|\kappa|^s \Lambda(\kappa) \in \mathcal{H}_s(D)$ for every $s > -D$. ∎

Proof. Write $\varepsilon = c_\lambda |\kappa|$. For exponents $t_i > -3$, with pair sums at least -3 when $m = 3$, the elementary collision estimate is

$$\int_{|l| \leq 2\varepsilon} \prod_i |l - l_i(\kappa)|^{t_i}\, dl \leq C \varepsilon^{3 + \sum t_i} \Lambda(\kappa). \tag{D.xix}$$

To prove it rescale to $\varepsilon = 1$. All centers lie in the unit ball and some pair is separated by c_λ^{-2}. For two centers the singularities are uniformly separated. For three, let d be the smallest pair distance. If d is bounded below, each singularity is uniformly isolated. Otherwise the third center stays a fixed distance from that pair. Near the third center use its integrable single power. Near the pair, the ball of radius $2d$ contributes $C d^{3+t_i+t_j} \leq C$; outside it the two distances are comparable, and radial integration gives at most $C(1 + \log(1/d))$. This proves the estimate. For each nonzero linear map $l_i - l_j$, one of its linear components bounds a multiple of $|v \cdot \kappa|$. The logarithm of the inverse of this component is integrable on the sphere. This proves the assertion about $\mathcal{H}_s(D)$.

For $|l| \geq 2\varepsilon$, $|\kappa'| \leq |\kappa|$, periodic distances of all $\epsilon_i l + \lambda_i(\kappa')$ are comparable to $|l|$. Hence, for symbols of orders t_i,

$$\left| \partial_\kappa^\alpha \prod_i \sigma_i(\epsilon_i l + \lambda_i(\kappa')) \right| \leq C_\alpha |l|^{\sum t_i - |\alpha|}. \tag{D.xx}$$

The same estimate holds for the homogeneous symbols on $\mathbb{R}^3$. The unimodular change of variables on the product torus proves the L^1 assertion by Tonelli. The collision estimate and the outer bound prove absolute convergence at small noncollision κ.

We first record a Taylor estimate for arbitrary symbols of integer orders $t_i \geq -2$, with $\omega' = 3 + \sum t_i \geq 0$. Split at 2ε. On the inner ball use eq. D.xix. On its complement expand through degrees less than ω'. The remainder is bounded by $C|\kappa|^{\omega'}|l|^{-3}$, so its integral costs $C|\kappa|^{\omega'} \lg|\kappa|$. Every Taylor coefficient of degree $j < \omega'$ is integrable at zero. Restoring its omitted inner ball costs $O(|\kappa|^{\omega'})$. Thus the integral is a polynomial of degree below ω' plus $O(|\kappa|^{\omega'} \lg|\kappa| \Lambda)$. When $\omega' \geq 1$, its constant is the integral at zero.

Now put $A^h(l, \kappa) = \prod_i \sigma_i^h(\epsilon_i l + \lambda_i(\kappa))$. For $|l| > 2\varepsilon$, expand

$$A^h = \sum_{j \leq \omega} A_j^h + A_{\mathrm{rem}}^h, \qquad |A_{\mathrm{rem}}^h| \leq C|\kappa|^{\omega+1}|l|^{-4}.$$

Here A_j^h is homogeneous of degree j in κ and $\omega - 3 - j$ in l. Empty sums are understood if $\omega = -1$. Define

$$I^h(\kappa) = \int_{|l| \leq 2\varepsilon} A^h\, d^\circ l + \int_{|l| > 2\varepsilon} A_{\mathrm{rem}}^h\, d^\circ l - \sum_{j < \omega} \int_{|l| \leq 2\varepsilon} A_j^h\, d^\circ l.$$

Each integral is absolutely convergent and bounded by $C|\kappa|^\omega \Lambda$. Rescaling, changing signs, and rotating l prove the homogeneity, parity, and covariance assertions. The formula is Borel and multilinear. For two centers, continuity follows by isolating each singularity with a fixed smooth partition, translating its center to zero, and using the integrable majorant $C|l|^{s_i}$. The outer remainder has the fixed integrable majorant $C|l|^{-4}$, and the subtracted terms are polynomials.

To compare with the cut off homogeneous integral, put $X(l, \kappa) = \prod_i \chi(\epsilon_i l + \lambda_i(\kappa))$ and $\chi_*(l) = \chi(l)^m$. For small κ, $X - \chi_*$ vanishes near zero and outside a fixed ball. Its integral against A^h

therefore has an ordinary Taylor expansion. For the integral with χ_*, substitute the preceding expansion outside 2ε. The term of degree $j = \omega$ integrates to zero: A^h_ω is odd in l, whereas the cutoff and domain are even. The lower terms supply a polynomial; the omitted outer remainder is $O(|\kappa|^{\omega+1})$. Consequently $\int X A^h = Q^{(0)} + I^h + O(|\kappa|^{\omega+1})$.

Finally expand every $\sigma_i = \chi\sigma_i^h + \sigma_i^\flat$. Each term containing a remainder has total order at least $\omega + 1$; the crude Taylor estimate applies to it. Keeping its polynomial through degree ω gives the stated error and Q. If $\omega \geq 1$, the constant terms add to the integral at zero.

For uniqueness compare two proposed expansions along $t\kappa$, $t \downarrow 0$, on an open cone avoiding collisions. Successive lower powers of t identify the polynomial through degree $\omega - 1$. At the final degree its polynomial term has parity $(-1)^\omega$, and the homogeneous term has the opposite parity. Comparing κ and $-\kappa$ separates them. A polynomial vanishing on an open set vanishes identically. This also proves independence of c_λ and that replacing $I(\kappa)$ by $I(-\kappa)$ replaces its leading data by the corresponding reflected data. □

Lemma D.31 (Differences of a leading term)**.** Suppose $n > 0$, and for every multi-index γ,

$$\nabla^\gamma f(x) = H_\gamma(x) + O(|x|^{-n-|\gamma|-1} \lg |x|)$$

for large lattice x, where H_γ is continuous off zero and homogeneous of degree $-n - |\gamma|$. Then H_0 is smooth, $\partial^\gamma H_0 = H_\gamma$, and

$$\nabla^\gamma (f - H_0)(x) = O(|x|^{-n-|\gamma|-1} \lg |x|).$$

■

Proof. Fix a segment $y + [0, t]e_\mu$ away from zero. Put $x_R = \lfloor Ry \rfloor$ and $N_R = \lfloor Rt \rfloor$. The telescoping identity

$$\nabla^\gamma f(x_R + N_R e_\mu) - \nabla^\gamma f(x_R) = \sum_{j<N_R} \nabla^{\gamma+e_\mu} f(x_R + j e_\mu)$$

multiplied by $R^{n+|\gamma|}$ has, as $R \to \infty$, the limit

$$H_\gamma(y + te_\mu) - H_\gamma(y) = \int_0^t H_{\gamma+e_\mu}(y + se_\mu)\, ds.$$

The summed remainder is $O(R^{-1} \lg R)$; the principal sum is a Riemann sum on a compact set. The identity also holds for negative small t. It proves the derivative relation and hence smoothness. Taylor's formula compares the continuous derivatives and finite differences of H_0 with error $O(|x|^{-n-|\gamma|-1})$, proving the last assertion. □

Lemma D.32 (Convolutions and bubbles)**.** Write $(X * Y)(u) = \sum_z X(z)Y(u - z)$.

(i) Suppose X, Y have integrable Fourier symbols with

$$|\widehat{X}(k)| \leq c|k|^{a_X} \lg |k|, \qquad |\nabla^\gamma X(x)| \leq c\langle x\rangle^{-n_X-|\gamma|} \quad (|\gamma| \leq 10),$$

and the analogous bounds for Y, where $-3 < a_X, a_Y < 3$, $n_X, n_Y \geq 0$, and $n_X + n_Y > 3$. Then the convolution converges absolutely and, for $r = |u| \geq 1$,

$$|(X * Y)(u)| \leq C(r^{-a_X-n_Y} + r^{-a_Y-n_X} + r^{3-n_X-n_Y}) \lg r.$$

(ii) If $X \in \mathfrak{K}(n_X)$, $Y \in \mathfrak{K}(n_Y)$, $n_X, n_Y \leq 4$, and $n_X + n_Y \geq 5$, then

$$X * Y \in \mathfrak{K}(n_X + n_Y - 3).$$

Its symbol is $\widehat{X}\widehat{Y}$, its leading symbol is $\widehat{X}^h\widehat{Y}^h$, and its leading kernel depends bilinearly and rotation covariantly only on the two sets of leading data.

(iii) Let K_a, K_b be elementary lines of degrees $n_a, n_b \in \{1,2,3\}$. The product $X = K_a K_b$ has symbol

$$I(k) = \int \sigma_a(k-l)\sigma_b(l)\, d^\circ l.$$

Let Q, I^h be its data in Lem. D.30, and let q have symbol

$$\widetilde{Q}(k) = Q(0) + \sum_\mu \partial_\mu Q(0) \sin k_\mu.$$

Then q is supported on $0, \pm e_\mu$, and, for $n = n_a + n_b$,

$$|\nabla^\gamma X(x)| \le C_\gamma \langle x\rangle^{-n-|\gamma|}, \qquad |I - \widetilde{Q}|(k) \le C|k|^{\min\{n-3,2\}}.$$

If $n \le 4$, then $X - q \in \mathfrak{K}(n)$, with leading data $(I^h, K_a^h K_b^h)$.

(iv) In $\mathfrak{K}(n)$, $n \le 6$, the leading kernel is determined linearly by the leading symbol. In particular a zero leading symbol gives a zero leading kernel.

■

Proof. For the first two assertions split the convolution into neighborhoods of its two endpoints and their complement. Choose a smooth radial θ, equal to one on $[0,1/2]$ and zero on $[1,\infty)$, and put

$$\theta_1(z) = \theta(4|z|/r), \qquad \theta_2(z) = \theta(4|u-z|/r).$$

Their supports are disjoint. The first endpoint sum is

$$T_1 = \int \widehat{X}(k)\check{g}_1(-k)\, d^\circ k, \qquad g_1(z) = \theta_1(z) Y(u-z).$$

The other has X, Y interchanged. The discrete Leibniz rule gives $\|g_1\|_{(r)} \le C r^{-n_Y}$. The ten additional differences cause only bounded shifts: on their stencils $\langle u - z\rangle \ge r/40$. Thus Lem. D.29 bounds the endpoint sums by the first two terms in (i). Away from both endpoints the product is bounded by $C(r+|z|)^{-n_X - n_Y}$. The scalar sum in Lem. D.4 gives its third term. It also proves absolute convergence and covers bounded r.

For (ii), $X, Y \in \ell^2$, so Plancherel identifies the symbol as the integrable product $\widehat{X}\widehat{Y}$. Multiplying their symbol expansions proves the required leading-symbol estimate. Put $n = n_X + n_Y - 3$. For large r, $\hat{u} = u/r$, define

$$G_1(z) = \theta(4|z|) Y^h(\hat{u} - z), \qquad G_2(z) = \theta(4|z|) X^h(\hat{u} - z).$$

Their C^{10} norms are bounded uniformly in $\hat{u}$. The defining estimates of $\mathfrak{K}$ give

$$\|g_1 - r^{-n_Y} G_1(\cdot/r)\|_{(r)} \le C r^{-n_Y - 1} \lg r,$$

and likewise for g_2. Apply the final assertion of Lem. D.29 at symbol orders $n_X - 3, n_Y - 3$, which lie in $[-1,1]$. In the middle sum replace both kernels by their leading parts; the error is

$$C \sum_z (r+|z|)^{-n_X - n_Y - 1} \lg(r+|z|) \le C r^{-n-1} \lg r.$$

The resulting smooth function vanishes near both singular points. The sum-to-integral estimate Lem. D.28 costs $O(r^{-n-1})$. Hence $X * Y = H_0 + O(r^{-n-1} \lg r)$, where

$$H_0(u) = r^{-n} \Big[\int \widehat{X}^h(k) \widehat{G}_1(-k)\, d^\circ k + \int \widehat{Y}^h(k) \widehat{G}_2(-k)\, d^\circ k + \int (1 - \theta(4|z|) - \theta(4|\hat{u} - z|)) X^h(z) Y^h(\hat{u} - z)\, dz \Big].$$

The endpoint integrals are continuous in $\hat{u}$ by the C^{10} bound in Lem. D.29. The middle integrand is dominated by an integrable power uniformly in $\hat{u}$. Thus H_0 is continuous and homogeneous of degree $-n$. The formula proves bilinearity and rotation covariance.

For derivatives, write $\nabla^\gamma(X*Y)=X*(\nabla^\gamma Y)$ and make the same split, taking $r\ge 200+8|\gamma|$. At the first endpoint replace the smooth factor by

$$\theta(4|z|)(\partial^\gamma Y^h)(\hat u-z).$$

At the second endpoint move all differences onto $\theta_2 X$ by summation by parts; its rescaled leading factor is

$$\partial_y^\gamma[\theta(4|\hat u-y|)X^h(y)]\Big|_{y=\hat u-z}.$$

In the middle region use $X^h\partial^\gamma Y^h$. The same estimates give a continuous homogeneous H_γ of degree $-n-|\gamma|$, with remainder $O(r^{-n-|\gamma|-1}\lg r)$. All differentiated kernels are evaluated away from zero on the indicated stencils. Lem. D.31 identifies these functions with the derivatives of H_0. Enlarging constants on the finitely many remaining lattice points proves the full $\mathfrak{K}(n)$ estimate.

For (iii), Fourier multiplication gives the displayed one-loop symbol. The maps in Lem. D.30 are $(l,k)\mapsto(k-l,l)$, with $c_\lambda=1$, $\Lambda=1$, and $\omega=n-3$. The discrete Leibniz rule gives the kernel bounds. Near zero use

$$I-\widetilde Q=(Q-\widetilde Q)+I^h+O(|k|^{n-2}\lg|k|).$$

Here $Q-\widetilde Q=O(|k|^2)$, and even $O(|k|^3)$ if $n\le 4$. Away from zero the two integrable singularities are separated, so I is bounded. For $n\le 4$, these are the symbol conditions of $\mathfrak{K}(n)$. The kernel conditions follow by expanding

$$K_aK_b-K_a^hK_b^h=(K_a-K_a^h)K_b+K_a^h(K_b-K_b^h)$$

and using Lem. D.26; the finitely supported q changes only finitely many points in any fixed difference stencil.

Finally suppose $\sigma^h=0$, and choose $G\in C_c^\infty(\{1/2<|x|<1\})$. The symbol bound and Lem. D.29 give

$$\sum_x K(x)G(x/R)=O(R^{2-n}\lg R).$$

The kernel bound and Lem. D.28 give the same sum as

$$R^{3-n}\int K^h(x)G(x)\,dx+O(R^{2-n}\lg R).$$

Divide by R^{3-n} and let $R\to\infty$. The integral vanishes for every such G, hence $K^h=0$ by continuity and homogeneity. Applying this to differences proves the last assertion. □

Proposition D.33 (The kite graph)**.** Let $K_1,\dots,K_5$ be elementary lines with symbols $\sigma_1,\dots,\sigma_5$ and degrees $n_1,\dots,n_5$, where

$$n_1,n_2,n_4,n_5\in\{2,3\},\quad n_3\in\{1,2,3\},\quad n_1+n_2+n_3\le 7,\quad n_3+n_4+n_5\le 7,$$

and put $\Delta:=\sum_{i=1}^5 n_i-6$. For $u\in\mathbb{Z}^3$ let $r:=|u|$ and

$$\mathsf{T}(u):=\sum_{z,w\in\mathbb{Z}^3}K_1(z)\,K_2(w)\,K_3(w-z)\,K_4(u-z)\,K_5(u-w).$$

(i) There is a constant C, depending on $n_1,\dots,n_5$ only, such that for every $u\in\mathbb{Z}^3$

$$\sum_{z,w\in\mathbb{Z}^3}\langle z\rangle^{-n_1}\langle w\rangle^{-n_2}\langle w-z\rangle^{-n_3}\langle u-z\rangle^{-n_4}\langle u-w\rangle^{-n_5}\le C.$$

In particular the series defining $\mathsf{T}(u)$ converges absolutely.

(ii) For $\kappa=(\kappa_1,\kappa_2)\in\mathbb{R}^3\times\mathbb{R}^3$ put

$$\hat{\mathsf{T}}_{\mathsf{p}}(\kappa_1,\kappa_2):=\int_{\mathbb{T}}\sigma_1(\kappa_1+l)\,\sigma_2(\kappa_2-l)\,\sigma_3(l)\,d^\circ l,$$

$$\hat{\mathsf{T}}_{\mathsf{q}}(\kappa_1,\kappa_2):=\int_{\mathbb{T}}\sigma_4(\kappa_1-l)\,\sigma_5(\kappa_2+l)\,\sigma_3(l)\,d^\circ l.$$

Both are functions I of the kind treated in Lem. D.30, with $\mathsf{m}=3$ and with data satisfying the hypotheses of that lemma; their numbers ω are

$$\omega_{\mathsf{p}} := n_1+n_2+n_3-6 \qquad \text{and} \qquad \omega_{\mathsf{q}} := n_3+n_4+n_5-6,$$

respectively, and both lie in $\{-1,0,1\}$. For every function g of finite support on $\mathbb{Z}^6$

$$\begin{aligned}\sum_{z,w} K_1(z)\,K_2(w)\,K_3(w-z)\,g(z,w) &= \int_{\mathbb{T}^2} \hat{\mathsf{T}}_{\mathsf{p}}(\kappa)\,\check{g}(-\kappa)\,d^\circ\kappa,\\ \sum_{z,w} K_4(z)\,K_5(w)\,K_3(z-w)\,g(z,w) &= \int_{\mathbb{T}^2} \hat{\mathsf{T}}_{\mathsf{q}}(\kappa)\,\check{g}(-\kappa)\,d^\circ\kappa.\end{aligned} \tag{D.xxi}$$

(iii) Let Q_{p} and Q_{q} be the polynomials of $\hat{\mathsf{T}}_{\mathsf{p}}$ and of $\hat{\mathsf{T}}_{\mathsf{q}}$ in the sense of Lem. D.30, let

$$\tilde{\mathsf{Q}}_\bullet(\kappa) := \mathsf{Q}_\bullet(0) + \sum_{j=1}^{6} \partial_j \mathsf{Q}_\bullet(0)\,\sin\kappa_j \qquad (\bullet\in\{\mathsf{p},\mathsf{q}\}),$$

and let $\mathsf{q}_{\mathsf{p}}, \mathsf{q}_{\mathsf{q}}$ be the kernels on $\mathbb{Z}^6$ with the symbols $\tilde{\mathsf{Q}}_{\mathsf{p}}, \tilde{\mathsf{Q}}_{\mathsf{q}}$, that is, $\mathsf{q}_\bullet(\zeta) = \int_{\mathbb{T}^2} e^{i\kappa\cdot\zeta}\,\tilde{\mathsf{Q}}_\bullet(\kappa)\,d^\circ\kappa$; they vanish off $\{0,\pm e_1,\dots,\pm e_6\}$. There are constants r_0 and C, depending on $\sigma_1,\dots,\sigma_5$ only, and a function $\mathsf{U}\in C^1(\mathbb{R}^3\setminus\{0\})$, homogeneous of degree $-\Delta$, such that for every u with $r\ge r_0$

$$\begin{aligned}\Big|\mathsf{T}(u) &- \sum_{z,w} \mathsf{q}_{\mathsf{p}}(z,w)\,K_4(u-z)\,K_5(u-w)\\ &- \sum_{z,w} \mathsf{q}_{\mathsf{q}}(z,w)\,K_1(u-z)\,K_2(u-w) - \mathsf{U}(u)\Big| \le C\,r^{-\Delta-1}\lg r.\end{aligned}$$

(iv) The function U depends only on the leading parts $\sigma_1^h,\dots,\sigma_5^h$, and it is linear in each of them. If every σ_i^h is replaced by $\sigma_i^h\circ O^{-1}$, with one and the same rotation O, then U is replaced by $\mathsf{U}\circ O^{-1}$.

■

Proof. Step 1: convergence. The symbol σ_i of K_i is a line symbol of degree $s_i := n_i-3 \ge -2 > -3$. Lem. D.3, applied with $s=s_i$ and $\gamma=0$, therefore gives $|K_i(y)|\le C\langle y\rangle^{-n_i}$ for all $y\in\mathbb{Z}^3$, and the absolute convergence of the series for $\mathsf{T}(u)$ follows from the bound in (i). To prove that bound we estimate $\langle w-z\rangle^{-n_3}\le 1$; what remains factorises into

$$\Big(\sum_z \langle z\rangle^{-n_1}\langle u-z\rangle^{-n_4}\Big)\Big(\sum_w \langle w\rangle^{-n_2}\langle u-w\rangle^{-n_5}\Big).$$

Lem. D.4 applies to the first factor with $(a,a',x)=(n_1,n_4,u)$, since $a,a'\ge 2$ and $a+a'\ge 4>3$; it bounds that factor by $C\langle u\rangle^{-\min\{n_1,n_4,n_1+n_4-3\}}\lg\langle u\rangle \le C\langle u\rangle^{-1}\lg\langle u\rangle \le C$, because $t^{-1}\lg t = t^{-1}(1+\log t)\le 1$ for $t\ge 1$. The second factor is bounded in the same way, with (n_2,n_5) in place of (n_1,n_4).

Step 2: the two one-loop functions. For $\hat{\mathsf{T}}_{\mathsf{p}}$ we take $\mathsf{m}=3$, $D=6$, the line symbols $\sigma_1,\sigma_2,\sigma_3$ of degrees s_1,s_2,s_3, the signs $(\epsilon_1,\epsilon_2,\epsilon_3)=(1,-1,1)$ and the linear maps $\lambda_1(\kappa)=\kappa_1$, $\lambda_2(\kappa)=\kappa_2$, $\lambda_3(\kappa)=0$. The integrand of $\hat{\mathsf{T}}_{\mathsf{p}}$ is then $\prod_i \sigma_i(\epsilon_i l+\lambda_i(\kappa))$, and $(l_1,l_2,l_3)(\kappa)=(-\kappa_1,\kappa_2,0)$. Up to signs, the pairwise differences of the $l_i(\kappa)$ are $\kappa_1+\kappa_2$, κ_1 and κ_2. Hence $\max_i|l_i(\kappa)|\le|\kappa|\le 2|\kappa|$ and $|\kappa|\le\sqrt{2}\max\{|\kappa_1|,|\kappa_2|\}\le 2\max_{i<j}|l_i(\kappa)-l_j(\kappa)|$, so that the condition on the l_i holds with $c_\lambda=2$; the three maps l_i are pairwise different. The map $(l,\kappa_1,\kappa_2)\mapsto(\kappa_1+l,\kappa_2-l,l)$ of $(\mathbb{R}^3)^3$ acts on the three components in $\mathbb{R}^3$ by the integer matrix

$$\begin{pmatrix} 1 & 1 & 0\\ -1 & 0 & 1\\ 1 & 0 & 0\end{pmatrix},$$

whose determinant is 1. Since $n_1, n_2 \geq 2$ and $n_3 \geq 1$, we have $s_1, s_2 \geq -1$ and $s_3 \geq -2$, hence $s_i + s_j \geq -3$ for $i \neq j$. Finally $\omega = 3 + s_1 + s_2 + s_3 = n_1 + n_2 + n_3 - 6 = \omega_{\mathsf{p}}$, and this number lies between $2 + 2 + 1 - 6 = -1$ and $7 - 6 = 1$. For $\hat{\mathsf{T}}_{\mathsf{q}}$ the line symbols are $\sigma_4, \sigma_5, \sigma_3$, the signs are $(-1, 1, 1)$ and the maps are $\kappa \mapsto \kappa_1$, $\kappa \mapsto \kappa_2$, $\kappa \mapsto 0$, so that $(l_1, l_2, l_3)(\kappa) = (\kappa_1, -\kappa_2, 0)$; the matrix of $(l, \kappa_1, \kappa_2) \mapsto (\kappa_1 - l, \kappa_2 + l, l)$ is $\left(\begin{smallmatrix} -1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{smallmatrix}\right)$, again of determinant 1, and the remaining conditions follow as before, now with $n_3 + n_4 + n_5 \leq 7$.

Step 3: proof of eq. D.xxi. We insert $K_1(z) = \int_{\mathbb{T}} e^{ik_1 \cdot z} \sigma_1(k_1)\, d^\circ k_1$, $K_2(w) = \int_{\mathbb{T}} e^{ik_2 \cdot w} \sigma_2(k_2)\, d^\circ k_2$ and $K_3(w - z) = \int_{\mathbb{T}} e^{il \cdot (w - z)} \sigma_3(l)\, d^\circ l$. The sum over (z, w) is finite and the three symbols are integrable, so Fubini's theorem gives

$$\sum_{z,w} K_1(z)\, K_2(w)\, K_3(w - z)\, g(z, w) = \iiint \sigma_1(k_1)\, \sigma_2(k_2)\, \sigma_3(l)\, \check{g}\big(-(k_1 - l), -(k_2 + l)\big)\, d^\circ k_1\, d^\circ k_2\, d^\circ l,$$

since $\sum_{z,w} g(z, w)\, e^{i(k_1 - l) \cdot z + i(k_2 + l) \cdot w} = \check{g}\big(-(k_1 - l), -(k_2 + l)\big)$. For fixed l we substitute $k_1 = \kappa_1 + l$ and $k_2 = \kappa_2 - l$, which are translations of $\mathbb{T}$, and we then carry out the integration over l first; this is the first identity. The function $y \mapsto K_3(-y)$ is the kernel with the symbol $l \mapsto \sigma_3(-l)$, because $K_3(-y) = \int e^{-il \cdot y} \sigma_3(l)\, d^\circ l = \int e^{il \cdot y} \sigma_3(-l)\, d^\circ l$. The derivation of the first identity used only the integrability of the three symbols. Applied to K_4, K_5 and $K_3(-\cdot)$ in place of K_1, K_2 and K_3, it represents $\sum_{z,w} K_4(z) K_5(w) K_3(z - w) g(z, w)$ by the function $\kappa \mapsto \int_{\mathbb{T}} \sigma_4(\kappa_1 + l)\, \sigma_5(\kappa_2 - l)\, \sigma_3(-l)\, d^\circ l$, which becomes $\hat{\mathsf{T}}_{\mathsf{q}}(\kappa)$ after the substitution $l \to -l$.

Step 4: short and long lines. Let $\varepsilon := \frac{1}{8}$ and, for $r > 0$, $\theta^r(y) := \theta(|y|/\varepsilon r)$ and $\bar{\theta}^r := 1 - \theta^r$, as functions on $\mathbb{R}^3$ and, by restriction, on $\mathbb{Z}^3$. By the choice of θ, $\theta^r(y) = 1$ for $|y| \leq \frac{1}{2}\varepsilon r$ and $\theta^r(y) = 0$ for $|y| \geq \varepsilon r$. For the arguments of the five lines we write

$$(y_1, y_2, y_3, y_4, y_5) := (z,\ w,\ w - z,\ u - z,\ u - w).$$

For $\mathsf{S} \subset \{1, \dots, 5\}$ let $\mathsf{T}_{\mathsf{S}}(u)$ be the series defining $\mathsf{T}(u)$, with the additional factor $\prod_{i \in \mathsf{S}} \theta^r(y_i)\ \prod_{i \notin \mathsf{S}} \bar{\theta}^r(y_i)$ in each term; the lines with an index in S are called short, the others long. These series converge absolutely by (i), the additional factors taking values in $[0, 1]$, and multiplying out $\prod_{i=1}^5 (\theta^r(y_i) + \bar{\theta}^r(y_i)) = 1$ we obtain $\mathsf{T} = \sum_{\mathsf{S}} \mathsf{T}_{\mathsf{S}}$.

The arguments satisfy

$$u = y_1 + y_4 = y_2 + y_5 = y_1 + y_3 + y_5 = y_2 - y_3 + y_4.$$

If S contains one of the sets $\{1, 4\}$, $\{2, 5\}$, $\{1, 3, 5\}$, $\{2, 3, 4\}$, then a non-vanishing term of $\mathsf{T}_{\mathsf{S}}(u)$ requires $|y_i| < \varepsilon r$ for the indices i of that set, hence $r = |u| < 3\varepsilon r < r$; thus $\mathsf{T}_{\mathsf{S}} = 0$. There are three possibilities for $\mathsf{S} \cap \{1, 4\}$, three for $\mathsf{S} \cap \{2, 5\}$ and two for $\mathsf{S} \cap \{3\}$; of the resulting $3 \cdot 3 \cdot 2 = 18$ sets, two, namely $\{1, 3, 5\}$ and $\{2, 3, 4\}$, contain one of the forbidden triples, and 16 sets remain.

The substitutions $(z, w) \to (w, z)$ and $(z, w) \to (u - z, u - w)$ are bijections of $\mathbb{Z}^6$; they carry $(y_1, \dots, y_5)$ into $(y_2, y_1, -y_3, y_5, y_4)$ and into $(y_4, y_5, -y_3, y_1, y_2)$, respectively. As θ^r is even, they show that T_{S}, computed for the lines $(K_1, K_2, K_3, K_4, K_5)$, equals $\mathsf{T}_{\mathsf{S}'}$ computed for $(K_2, K_1, K_3(-\cdot), K_5, K_4)$, where S' is the image of S under the permutation (12)(45), and that it equals $\mathsf{T}_{\mathsf{S}''}$ computed for $(K_4, K_5, K_3(-\cdot), K_1, K_2)$, where S'' is the image of S under (14)(25). The new families satisfy the hypotheses of the proposition: $K_3(-\cdot)$ is an elementary line of degree n_3, its symbol $\sigma_3(-k)$ being a line symbol with leading part $\sigma_3^h(-k)$ (the cut-off χ is even), and the conditions on the degrees are invariant under both permutations. Under the group generated by (12)(45) and (14)(25) the sixteen sets fall into seven classes: the set $\emptyset$ alone; the sets $\{1\}, \{2\}, \{4\}, \{5\}$; the set $\{3\}$ alone; the sets $\{1, 2\}, \{4, 5\}$; the sets $\{1, 5\}, \{2, 4\}$; the sets $\{1, 3\}, \{2, 3\}, \{3, 4\}, \{3, 5\}$; and the sets $\{1, 2, 3\}, \{3, 4, 5\}$. We choose the representatives

$$\emptyset, \quad \{1\}, \quad \{3\}, \quad \{1, 2\}, \quad \{1, 5\}, \quad \{1, 3\}, \quad \{1, 2, 3\}.$$

It therefore suffices to expand T_S for these seven sets, for an arbitrary family of lines obeying the hypotheses; for each of the remaining nine sets we use the expansion of its representative, computed for the transformed family.

Step 5: long factors. Let $L_i := \bar\theta^r K_i$ on $\mathbb{Z}^3$, let $K_i^h := \mathcal{F}_{\neq 0}^{-1}\sigma_i^h$, which by Lem. D.26 is smooth on $\mathbb{R}^3\setminus\{0\}$ and homogeneous of degree $-3-s_i=-n_i$, and let $L_i^h := \bar\theta^r K_i^h$, extended by 0 at the origin. Since $\bar\theta^r$ vanishes on $\{|y|\le \frac12\varepsilon r\}$, the function L_i^h is smooth on $\mathbb{R}^3$, and by homogeneity $L_i^h(y) = r^{-n_i}\,\bar\theta^1(y/r)\,K_i^h(y/r)$. We claim that there are r_0 and C such that for $r\ge r_0$, $i=1,\dots,5$ and $|\gamma|\le 10$

$$\begin{aligned} |\nabla^\gamma L_i(y)| + |\partial^\gamma L_i^h(y)| &\le C\,(r+|y|)^{-n_i-|\gamma|},\\ |\nabla^\gamma (L_i - L_i^h)(y)| &\le C\,(r+|y|)^{-n_i-1-|\gamma|};\end{aligned} \tag{D.xxii}$$

here $y\in\mathbb{Z}^3$, except in the term $\partial^\gamma L_i^h(y)$, where $y\in\mathbb{R}^3$. Indeed, Lem. D.3 gives $|\nabla^\gamma K_i(x)|\le C\langle x\rangle^{-n_i-|\gamma|}$, and Lem. D.26 gives $|\nabla^\gamma(K_i-K_i^h)(x)|\le C|x|^{-n_i-1-|\gamma|}$ for $|x|\ge|\gamma|+1$. The derivatives of order γ of K_i^h are homogeneous of degree $-n_i-|\gamma|$, hence bounded by $C|x|^{-n_i-|\gamma|}$; the derivatives of order $j\ge1$ of $\bar\theta^r$ are bounded by Cr^{-j} and vanish outside $\{\frac12\varepsilon r\le|y|\le\varepsilon r\}$. The discrete Leibniz rule $\nabla_j(fg) = (\nabla_j f)\,g(\cdot+e_j) + f\,\nabla_j g$ distributes ∇^γ over the two factors, the arguments being shifted by at most $|\gamma|\le10$. Only points y' with $|y'|\ge\frac12\varepsilon r$ contribute, and at such points, for $r\ge r_0$ with r_0 large enough, we have $|y'|\ge|\gamma|+1$ and $r+|y|\le C|y'|$ whenever $|y-y'|\le10$. These facts together give eq. D.xxii.

Step 6: the scheme for $\mathsf{S}\neq\emptyset$. For each of the six non-empty representatives we write

$$\mathsf{T}_\mathsf{S}(u) = \sum_{\zeta\in\mathbb{Z}^D} \mathsf{T}^\mathsf{S}(\zeta)\,g_\mathsf{S}(\zeta),$$

where the kernel T^S on $\mathbb{Z}^D$ collects the short lines, and we let s be the degree of the leading part $\hat{\mathsf{T}}^{\mathsf{S},h}$ of the symbol of T^S. The data are as follows.

- $\mathsf{S}=\{1\}$, $D=3$, $\zeta=z$: $\mathsf{T}^\mathsf{S}(z)=K_1(z)$, with symbol $\sigma_1(\kappa)$ and $s=n_1-3$; and $g_\mathsf{S}(z)=\theta^r(z)\,L_4(u-z)\sum_w L_2(w)\,L_3(w-z)\,L_5(u-w)$.
- $\mathsf{S}=\{3\}$, $D=3$, $\zeta=y$: $\mathsf{T}^\mathsf{S}(y)=K_3(y)$, with symbol $\sigma_3(\kappa)$ and $s=n_3-3$; and $g_\mathsf{S}(y)=\theta^r(y)\sum_z L_1(z)\,L_2(z+y)\,L_4(u-z)\,L_5(u-z-y)$.
- $\mathsf{S}=\{1,2\}$, $D=6$, $\zeta=(z,w)$: $\mathsf{T}^\mathsf{S}(z,w)=K_1(z)K_2(w)$, with symbol $\sigma_1(\kappa_1)\sigma_2(\kappa_2)$ and $s=n_1+n_2-6$; and $g_\mathsf{S}(z,w)=\theta^r(z)\,\theta^r(w)\,L_3(w-z)\,L_4(u-z)\,L_5(u-w)$.
- $\mathsf{S}=\{1,5\}$, $D=6$, $\zeta=(z,w')$: $\mathsf{T}^\mathsf{S}(z,w')=K_1(z)K_5(w')$, with symbol $\sigma_1(\kappa_1)\sigma_5(\kappa_2)$ and $s=n_1+n_5-6$; and $g_\mathsf{S}(z,w')=\theta^r(z)\,\theta^r(w')\,L_2(u-w')\,L_3(u-w'-z)\,L_4(u-z)$.
- $\mathsf{S}=\{1,3\}$, $D=6$, $\zeta=(z,y)$: $\mathsf{T}^\mathsf{S}(z,y)=K_1(z)K_3(y)$, with symbol $\sigma_1(\kappa_1)\sigma_3(\kappa_2)$ and $s=n_1+n_3-6$; and $g_\mathsf{S}(z,y)=\theta^r(z)\,\theta^r(y)\,L_2(z+y)\,L_4(u-z)\,L_5(u-z-y)$.
- $\mathsf{S}=\{1,2,3\}$, $D=6$, $\zeta=(z,w)$: $\mathsf{T}^\mathsf{S}(z,w)=K_1(z)K_2(w)K_3(w-z)$, with symbol $\hat{\mathsf{T}}_\mathsf{p}(\kappa)$ (by eq. D.xxi, applied to $g=\delta_\zeta$) and $s=\omega_\mathsf{p}$; and $g_\mathsf{S}(z,w)=\theta^r(z)\,\theta^r(w)\,\theta^r(w-z)\,L_4(u-z)\,L_5(u-w)$.

In the cases $\{1,5\}$ and $\{1,3\}$ we have substituted $w=u-w'$, resp. $w=z+y$, in the sum over w; in the cases $\{1\}$ and $\{3\}$ the inner sums converge absolutely by Step 1. Each g_S vanishes off $\{|\zeta|\le r\}$, since the short factors confine every component of ζ to $\{|\cdot|<\varepsilon r\}$ and $\sqrt{2}\,\varepsilon<1$.

Let g_S^h be the function on $\mathbb{R}^D$ obtained from g_S by replacing every L_i by L_i^h and, in the cases $\{1\}$ and $\{3\}$, the inner sum by the integral over $\mathbb{R}^3$. Since $\theta^r(y)=\theta^1(y/r)$ and $L_i^h(y)=r^{-n_i}(\bar\theta^1K_i^h)(y/r)$, the substitution of r times the variable in the inner integrals gives

$$g_\mathsf{S}^h(\zeta) = r^{-a}\,G_\mathsf{S}(\zeta/r), \qquad \text{where}\quad a := \begin{cases}\sum_{i\notin\mathsf{S}} n_i & \text{if } D=6,\\ \sum_{i\notin\mathsf{S}} n_i - 3 & \text{if } \mathsf{S}=\{1\},\{3\},\end{cases}$$

so that $a+s=\Delta$ in all six cases. Here $G_{\mathsf S}$ is the function $g^h_{\mathsf S}$ computed with $r=1$ and with $\hat u := u/r$ in place of u; it belongs to $C^\infty_c(\mathbb R^D)$ and has its support in $\{|\zeta|\le 1\}$. We claim

$$\|g_{\mathsf S} - g^h_{\mathsf S}\|_{(r)} \le C\, r^{-a-1} \qquad (r \ge r_0), \tag{D.xxiii}$$

where $g^h_{\mathsf S}$ is restricted to $\mathbb Z^D$ and $\|\cdot\|_{(r)}$ is the norm used in Lem. D.29.

In the cases $\{1,2\}$, $\{1,5\}$, $\{1,3\}$, $\{1,2,3\}$ the functions $g_{\mathsf S}$ and $g^h_{\mathsf S}$ are products of the same short factors with the long factors L_i, resp. L^h_i, $i\notin\mathsf S$, evaluated at linear functions of ζ. We write the difference as a sum of terms in each of which one factor L_i has been replaced by $L_i - L^h_i$, and we distribute the differences ∇^γ, $|\gamma|\le 10$, by the Leibniz rule. By eq. D.xxii a difference falling on a long factor gains a factor $C(r+|y_i|)^{-1} \le Cr^{-1}$, a difference falling on a short factor gains Cr^{-1}, and the factor $L_i - L^h_i$ gains $(r+|y_i|)^{-1}\le r^{-1}$ in comparison with L_i; moreover $\prod_{i\notin\mathsf S}(r+|y_i|)^{-n_i} \le r^{-a}$. This gives $r^{|\gamma|}|\nabla^\gamma(g_{\mathsf S} - g^h_{\mathsf S})| \le Cr^{-a-1}$.

In the case $\mathsf S=\{1\}$ let

$$\mathsf w(z) := \sum_w L_2(w)\, L_3(w-z)\, L_5(u-w),$$
$$\mathsf w^h(z) := \int_{\mathbb R^3} L^h_2(w)\, L^h_3(w-z)\, L^h_5(u-w)\, dw.$$

For $|z|\le r$ the differences in z fall on $L_3(w-z)$ only. By eq. D.xxii, by $r+|u-w| \ge \frac12(r+|w|)$ (second assertion of Lem. D.4, as $|u|=r$) and by the first assertion of Lem. D.4 with $D=3$, $R=r$ and exponent $n_2+n_5\ge 4>3$, we obtain

$$|\nabla^\gamma \mathsf w(z)| \le Cr^{-n_3-|\gamma|}\sum_w (r+|w|)^{-n_2-n_5} \le Cr^{3-n_2-n_3-n_5-|\gamma|}.$$

Each replacement of an L_i by L^h_i in $\mathsf w$ costs a factor Cr^{-1}, by the second inequality of eq. D.xxii. The replacement of the sum over w by the integral costs a factor Cr^{-1} as well: we apply Lem. D.28, with $D=3$, to $F(w) := L^h_2(w)\,(\nabla^\gamma_z L^h_3)(w-z)\,L^h_5(u-w)$, whose gradient is bounded by $Cr^{-n_3-|\gamma|-1}(r+|w|)^{-n_2-n_5}$ on each cube $w+[0,1]^3$. Since $g_{\{1\}}(z) = \theta^r(z) L_4(u-z)\mathsf w(z)$, since $|\nabla^\gamma L_4(u-\cdot)| \le Cr^{-n_4-|\gamma|}$ on $\{|z|\le r\}$ by eq. D.xxii, and since the differences of order γ of θ^r are bounded by $Cr^{-|\gamma|}$, the Leibniz rule gives eq. D.xxiii with $a = n_2+n_3+n_4+n_5-3$.

In the case $\mathsf S=\{3\}$ let $\mathsf w(y) := \sum_z L_1(z)\,L_2(z+y)\,L_4(u-z)\,L_5(u-z-y)$, and let $\mathsf w^h$ be the integral over $\mathbb R^3$ of the corresponding product of the L^h_i. For $|y|\le r$ the differences in y fall on L_2 and on L_5 (and, in $g_{\{3\}}$, on the factor $\theta^r(y)$), each difference gaining a factor Cr^{-1}. By Lem. D.4 we have $r+|z+y|\ge\frac12(r+|z|)$, $r+|u-z|\ge\frac12(r+|z|)$, and, with $R=2r\ge|u-y|$, also $r+|u-y-z|\ge\frac14(r+|z|)$; the first assertion of that lemma, with exponent $n_1+n_2+n_4+n_5\ge 8$, gives $\sum_z (r+|z|)^{-n_1-n_2-n_4-n_5} \le Cr^{3-n_1-n_2-n_4-n_5}$. Hence $|\nabla^\gamma \mathsf w(y)| \le Cr^{3-n_1-n_2-n_4-n_5-|\gamma|}$, and, as in the case $\{1\}$, the replacements $L_i\to L^h_i$ and the replacement of the sum by the integral (Lem. D.28) cost a factor Cr^{-1} each. This proves eq. D.xxiii in all six cases.

Step 7: symbols, and the pairing with slowly varying functions. We verify the hypotheses of Lem. D.29 for $\hat{\mathsf T}$, the symbol of $\mathsf T^{\mathsf S}$, and $\hat{\mathsf T}^h := \hat{\mathsf T}^{\mathsf S,h}$. In the cases $\{1\}$ and $\{3\}$ ($D=3$, $i=1$, resp. $i=3$, $s=s_i$) we take $c_0=1$. On $\{|\kappa|\le 1\}$ we have $\chi=1$, so $\sigma_i - \chi\sigma^h_i \in \mathcal S^{s+1}$ gives $|\sigma_i-\sigma^h_i|\le C|\kappa|^{s+1}$ there; the majorants are $\mathsf H := C|\kappa|^s \in \mathcal H_s(3)$ for σ^h_i and $\mathsf H' := C|\kappa|^{s+1}\in\mathcal H_{s+1}(3)$, which belong to these classes since $s>-3$; the required bound $|\sigma_i-\sigma^h_i|\le \mathsf H' \lg|\kappa|$ holds on $\{|\kappa|\le 1\}$, as $\lg|\kappa|\ge 1$. In the cases $\{1,2\}$, $\{1,5\}$, $\{1,3\}$ ($D=6$) the symbol is $\sigma_i(\kappa_1)\sigma_j(\kappa_2)$ with $(i,j)=(1,2),(1,5),(1,3)$, its leading part is $\sigma^h_i(\kappa_1)\sigma^h_j(\kappa_2)$, and $s=s_i+s_j$ with $s_i,s_j>-3$. Then $|\sigma^h_i(\kappa_1)\sigma^h_j(\kappa_2)|\le C|\kappa_1|^{s_i}|\kappa_2|^{s_j}$, a function in $\mathcal H_{s_i+s_j}(6)$, and on $\{|\kappa|\le 1\}$

$$|\sigma_i(\kappa_1)\sigma_j(\kappa_2) - \sigma^h_i(\kappa_1)\sigma^h_j(\kappa_2)| \le C\big(|\kappa_1|^{s_i+1}|\kappa_2|^{s_j} + |\kappa_1|^{s_i}|\kappa_2|^{s_j+1}\big),$$

a function in $\mathcal{H}_{s_i+s_j+1}(6)$; here we wrote the difference as $(\sigma_i-\sigma_i^h)\sigma_j+\sigma_i^h(\sigma_j-\sigma_j^h)$ and used $|\sigma_j(\kappa_2)|\le C|\kappa_2|^{s_j}$. In the case $\{1,2,3\}$ we use Lem. D.30, whose hypotheses hold by Step 2, with the number c_0 given there. By the asymptotic assertion of that lemma, $|\hat{\mathsf{T}}_{\mathsf{p}}-\mathsf{Q}_{\mathsf{p}}-\mathsf{I}_{\mathsf{p}}^h|\le C|\kappa|^{\omega_{\mathsf{p}}+1}\lg|\kappa|\,\Lambda(\kappa)$ for almost every κ with $|\kappa|\le c_0$, where $\mathsf{I}_{\mathsf{p}}^h$ is the homogeneous part of $\hat{\mathsf{T}}_{\mathsf{p}}$. The polynomial Q_{p} has degree at most $\omega_{\mathsf{p}}\le 1$, so $\mathsf{Q}_{\mathsf{p}}-\tilde{\mathsf{Q}}_{\mathsf{p}}=\sum_j\partial_j\mathsf{Q}_{\mathsf{p}}(0)(\kappa_j-\sin\kappa_j)=O(|\kappa|^3)$, and $|\kappa|^3\le|\kappa|^{\omega_{\mathsf{p}}+1}\lg|\kappa|\,\Lambda(\kappa)$ for $|\kappa|\le 1$. We therefore take $\hat{\mathsf{T}}:=\hat{\mathsf{T}}_{\mathsf{p}}-\tilde{\mathsf{Q}}_{\mathsf{p}}$, which is integrable on $\mathbb{T}^2$: Lem. D.30 bounds the L^1 norm of $\hat{\mathsf{T}}_{\mathsf{p}}$ by the product of the L^1 norms of $\sigma_1,\sigma_2,\sigma_3$, and $\tilde{\mathsf{Q}}_{\mathsf{p}}$ is a trigonometric polynomial. We further take $\hat{\mathsf{T}}^h:=\mathsf{I}_{\mathsf{p}}^h$ and the majorants $\mathsf{H}:=C|\kappa|^{\omega_{\mathsf{p}}}\Lambda$ (Lem. D.30) and $\mathsf{H}':=C|\kappa|^{\omega_{\mathsf{p}}+1}\Lambda$. They belong to $\mathcal{H}_{\omega_{\mathsf{p}}}(6)$ and to $\mathcal{H}_{\omega_{\mathsf{p}}+1}(6)$, respectively, since the same lemma asserts that $|\kappa|^s\Lambda(\kappa)$ belongs to $\mathcal{H}_s(6)$ for every real $s>-6$, and since $\omega_{\mathsf{p}}\ge -1$. In all cases $-D<s<8-D$, as Lem. D.29 requires: for $D=3$ we have $s\in\{-2,-1,0\}\subset(-3,2)$, and for $D=6$ we have $s\in\{-3,\dots,1\}\subset(-6,2)$.

Inserting the Fourier integrals of the short lines, we have $\mathsf{T}_{\mathsf{S}}(u)=\int_{\mathbb{T}^{D/3}}\hat{\mathsf{T}}^{\mathsf{S}}(\kappa)\,\check{g}_{\mathsf{S}}(-\kappa)\,d^\circ\kappa$, where $\hat{\mathsf{T}}^{\mathsf{S}}$ is the symbol of T^{S}: for the product kernels this follows from Fubini's theorem as in Step 3, and for $\{1,2,3\}$ it is eq. D.xxi. We apply Lem. D.29 with $R=r$, $g=g_{\mathsf{S}}$, which vanishes off $\{|\zeta|\le r\}$, $G=G_{\mathsf{S}}$, the number a of Step 6, and $A=C$, which is admissible by eq. D.xxiii because $\lg r\ge 1$. The norm $\|G_{\mathsf{S}}\|_{C^{10}}$ is bounded uniformly in $\hat{u}\in S^2$, by the smoothness in $(\zeta,\hat{u})$ established in Step 9 and the compactness of S^2. In the first five cases, $\{1\}$, $\{3\}$, $\{1,2\}$, $\{1,5\}$, $\{1,3\}$, we obtain

$$\mathsf{T}_{\mathsf{S}}(u)=\mathsf{U}_{\mathsf{S}}(u)+O(r^{-\Delta-1}\lg r),\qquad \mathsf{U}_{\mathsf{S}}(u):=r^{-\Delta}\int_{\mathbb{R}^D}\hat{\mathsf{T}}^{\mathsf{S},h}(\kappa)\,\hat{G}_{\mathsf{S}}(-\kappa)\,d^\circ\kappa,$$

because $a+s=\Delta$. In the case $\{1,2,3\}$, where $\hat{\mathsf{T}}^{\mathsf{S},h}:=\mathsf{I}_{\mathsf{p}}^h$, the lemma is applied to $\hat{\mathsf{T}}_{\mathsf{p}}-\tilde{\mathsf{Q}}_{\mathsf{p}}$, and the remaining term is $\int_{\mathbb{T}^2}\tilde{\mathsf{Q}}_{\mathsf{p}}(\kappa)\check{g}_{\mathsf{S}}(-\kappa)\,d^\circ\kappa=\sum_{z,w}\mathsf{q}_{\mathsf{p}}(z,w)\,g_{\mathsf{S}}(z,w)$; thus

$$\mathsf{T}_{\{1,2,3\}}(u)=\sum_{z,w}\mathsf{q}_{\mathsf{p}}(z,w)\,g_{\{1,2,3\}}(z,w)+\mathsf{U}_{\{1,2,3\}}(u)+O(r^{-\Delta-1}\lg r).$$

On the support of q_{p}, that is, for $(z,w)\in\{0,\pm e_1,\dots,\pm e_6\}$, we have $|z|,|w|,|w-z|\le 1\le\frac12\varepsilon r$ and $|u-z|,|u-w|\ge r-1\ge\varepsilon r$ as soon as $r\ge r_0\ge 16$; hence $g_{\{1,2,3\}}(z,w)=K_4(u-z)K_5(u-w)$ there, which produces the first local term in (iii). The image of $\{1,2,3\}$ under $(z,w)\to(u-z,u-w)$ is $\{3,4,5\}$, and the transformed family $(K_4,K_5,K_3(-\cdot),K_1,K_2)$ has, in place of $\hat{\mathsf{T}}_{\mathsf{p}}$, the function $\int_{\mathbb{T}}\sigma_4(\kappa_1+l)\sigma_5(\kappa_2-l)\sigma_3(-l)\,d^\circ l=\hat{\mathsf{T}}_{\mathsf{q}}(\kappa)$; its local kernel is therefore q_{q}, and $\mathsf{T}_{\{3,4,5\}}$ produces the second local term in (iii).

Step 8: the case $\mathsf{S}=\emptyset$. Here $\mathsf{T}_\emptyset(u)=\sum_{z,w}\prod_{i=1}^5 L_i(y_i)$. By eq. D.xxii with $\gamma=0$, by $(r+|y_3|)^{-n_3}\le r^{-n_3}$ and by Lem. D.4 (as in Step 6, with $r+|u-z|\ge\frac12(r+|z|)$ and $r+|u-w|\ge\frac12(r+|w|)$),

$$\sum_{z,w}\prod_{i=1}^5(r+|y_i|)^{-n_i}\le C\,r^{-n_3}\,r^{3-n_1-n_4}\,r^{3-n_2-n_5}=C\,r^{-\Delta}.$$

Replacing each L_i by L_i^h costs a factor Cr^{-1}, by the second inequality of eq. D.xxii; replacing the sum over $(z,w)\in\mathbb{Z}^6$ by the integral over $\mathbb{R}^6$ costs a factor Cr^{-1} by Lem. D.28 with $D=6$, applied to $F(z,w):=\prod_i L_i^h(y_i)$, whose gradient is bounded by $Cr^{-1}\prod_i(r+|y_i|)^{-n_i}$ on each cube of side 1. Hence $\mathsf{T}_\emptyset(u)=\mathsf{U}_\emptyset(u)+O(r^{-\Delta-1})$ with

$$\mathsf{U}_\emptyset(u):=\iint_{\mathbb{R}^3\times\mathbb{R}^3}\prod_{i=1}^5 L_i^h(y_i)\,dz\,dw,$$

which, after the substitution $(z,w)=r(z',w')$, is seen to be homogeneous of degree $6-\sum_i n_i=-\Delta$.

Step 9: the function U. Let U be the sum of the sixteen functions U_{S}, where for a set S which is not a representative, U_{S} is the function of its representative computed for the transformed family (Step 4).

By Steps 4, 7 and 8, the inequality in (iii) holds for $r \geq r_0$. Every U_S is homogeneous of degree $-\Delta$, and so is U. We let a vector $v \in \mathbb{R}^3$ take the place of $\hat{u}$ in the definition of G_S, and we write $G_\mathsf{S}(\zeta; v)$. In the factors $\bar{\theta}^1 K_i^h$ the argument of K_i^h has modulus at least $\frac{1}{2}\varepsilon$ wherever $\bar{\theta}^1$ does not vanish, so these factors are smooth on $\mathbb{R}^3$; the inner integrals in the cases $\{1\}$ and $\{3\}$ may be differentiated under the integral sign, with integrable majorants that are uniform for (ζ, v) in compact sets. Hence $(\zeta, v) \mapsto G_\mathsf{S}(\zeta; v)$ is smooth, with support in $\{|\zeta| \leq 1\}$ for every v, and $v \mapsto G_\mathsf{S}(\cdot\,; v)$ is of class C^1 with respect to the norm $\|\cdot\|_{C^{10}}$. The last bound of Lem. D.29, $|\int \hat{\mathsf{T}}^h \hat{G}(-\kappa)\, d^\circ\kappa| \leq C\|G\|_{C^{10}}$, which is linear in G, then shows that $\Phi_\mathsf{S}(v) := \int \hat{\mathsf{T}}^{\mathsf{S},h}(\kappa)\, \hat{G}_\mathsf{S}(-\kappa; v)\, d^\circ\kappa$ is of class C^1 in v. Since $\mathsf{U}_\mathsf{S}(u) = |u|^{-\Delta}\, \Phi_\mathsf{S}(u/|u|)$, the function U_S is of class C^1 on $\mathbb{R}^3 \setminus \{0\}$. For $\mathsf{U}_\emptyset$ we differentiate under the integral sign.

The functions U_S are built from the leading parts σ_i^h of the short lines, from the functions $K_i^h = \mathcal{F}_{\neq 0}^{-1}\sigma_i^h$ of the long lines, which depend linearly on σ_i^h (Lem. D.26), from the homogeneous parts I_p^h and I_q^h, which depend only on the leading parts and multilinearly on them (Lem. D.30), and from radial cut-offs; every line enters each U_S exactly once. Hence U depends only on $\sigma_1^h, \dots, \sigma_5^h$ and is linear in each of them. If every σ_i^h is replaced by $\sigma_i^h \circ O^{-1}$, then K_i^h is replaced by $K_i^h \circ O^{-1}$ (Lem. D.26), and I_p^h, I_q^h are replaced by their compositions with O^{-1} acting on both arguments, since the linear maps λ_i of Step 2 commute with this action (Lem. D.30). The substitutions $\zeta \to O\zeta$ in the continuum integrals defining G_S and $\mathsf{U}_\emptyset$, and $\kappa \to O\kappa$ in the integral defining U_S, together with the invariance of the radial cut-offs, show that U is replaced by $\mathsf{U} \circ O^{-1}$. This proves (iii) and (iv). □

Write $p_{[1]} = p_{23}$, $p_{[2]} = p_{13}$, $p_{[3]} = p_{12}$, and $\lambda(p_{[i]}) = i$. For $p = p_{[i]}(x_p)$, x_p is its base point and $x(p)$ its barycenter.

Lemma D.34 (The lines of the lattice)**.** Let $\tau_{\mathrm{gf}} = 1$; we work on $\mathbb{Z}^3$. For $\iota \in \{1, 2, 3\}$ write $p_{[\iota]} = p_{\alpha\beta}$ with $\alpha < \beta$, and put

$$\mathsf{c}_{\iota,\nu}(k) := \mathsf{k}_\alpha(k)\, \delta_{\nu\beta} - \mathsf{k}_\beta(k)\, \delta_{\nu\alpha} \qquad (\nu \in \{1, 2, 3\},\ k \in \mathbb{R}^3),$$

so that for $v \in \mathbb{C}^3$ the form $(x, \nu) \mapsto e^{ik\cdot x} v_\nu$ satisfies $\mathrm{d}(e^{ik\cdot x} v_\nu)(p_{[\iota]}(x)) = e^{ik\cdot x} \sum_\nu \mathsf{c}_{\iota,\nu}(k)\, v_\nu$. Let $\hat{C}_{\mu\nu}(k)$ be the bracket in the formula for C in Def. D.2,

$$\hat{C}_{\mu\nu}(k) = \frac{\delta_{\mu\nu}}{|\mathsf{k}|^2} + (\tau_{\mathrm{gf}} - 1)\, \frac{\mathsf{k}_\mu \overline{\mathsf{k}_\nu}}{|\mathsf{k}|^4}.$$

Then for every $x \in \mathbb{Z}^3$, all directions μ, ν and all ι, ι', with $p_{[\iota']} = p_{\alpha'\beta'}$, $\alpha' < \beta'$, the following functions of $y \in \mathbb{Z}^3$ are elementary lines, with the symbols, the degrees n and the leading parts of the symbols listed:

- $y \mapsto C((x, \mu), (x + y, \nu))$: symbol $\hat{C}_{\nu\mu}(k)$; $n = 1$; leading part $\delta_{\mu\nu}|k|^{-2} + (\tau_{\mathrm{gf}} - 1)\, k_\mu k_\nu |k|^{-4}$;
- $y \mapsto E_{p_{[\iota]}(x)}((x + y, \nu))$: symbol $\sum_\mu \overline{\mathsf{c}_{\iota,\mu}(k)}\, \hat{C}_{\nu\mu}(k) = \overline{\mathsf{c}_{\iota,\nu}(k)}\, |\mathsf{k}|^{-2}$; $n = 2$; leading part $-i(k_\alpha \delta_{\nu\beta} - k_\beta \delta_{\nu\alpha})|k|^{-2}$;
- $y \mapsto \Pi(p_{[\iota]}(x), p_{[\iota']}(x + y))$: symbol $\sum_\nu \overline{\mathsf{c}_{\iota,\nu}(k)}\, \mathsf{c}_{\iota',\nu}(k)\, |\mathsf{k}|^{-2}$; $n = 3$; leading part $\sum_\nu (k_\alpha \delta_{\nu\beta} - k_\beta \delta_{\nu\alpha})(k_{\alpha'} \delta_{\nu\beta'} - k_{\beta'} \delta_{\nu\alpha'})|k|^{-2}$;
- $y \mapsto D((x, \mu); (x + y, \nu))$: symbol $\frac{1}{2}(1 + e^{ik_\nu})\, \overline{\mathsf{k}_\mu(k)}\, |\mathsf{k}|^{-2}$; $n = 2$; leading part $-ik_\mu |k|^{-2}$.

If $K(y)$ is an elementary line with symbol $\sigma(k)$ and $y_0 \in \mathbb{Z}^3$, then $K(-y)$ and $K(y + y_0)$ are elementary lines of the same degree, with the symbols $\sigma(-k)$ and $e^{ik\cdot y_0}\sigma(k)$. Consequently, the contraction Def. D.2 of a leg at a bond or a plaquette with base point $x + \xi$ with a leg at a bond or a plaquette with base point $x + y + \xi'$ equals $\delta_{aa'}$ times an elementary line in y of degree $1 + \#\{\text{curl legs among the two}\}$. ■

Proof. The matrix $\hat C(k)$ is Hermitian, and $\hat C(-k) = \overline{\hat C(k)}$ because $\mathsf{k}_\mu(-k) = \overline{\mathsf{k}_\mu(k)}$. By Def. D.2 and the substitution $k \to -k$,

$$\begin{aligned} C((x,\mu),(x+y,\nu)) &= \int_{\mathbb{T}} e^{-ik\cdot y}\, \hat C_{\mu\nu}(k)\, d^\circ k \\ &= \int_{\mathbb{T}} e^{ik\cdot y}\, \overline{\hat C_{\mu\nu}(k)}\, d^\circ k = \int_{\mathbb{T}} e^{ik\cdot y}\, \hat C_{\nu\mu}(k)\, d^\circ k. \end{aligned}$$

The plaquette $p = p_{[\iota]}(x) = p_{\alpha\beta}(x)$ has the bonds $(x,\alpha),(x+e_\alpha,\beta),(x+e_\beta,\alpha),(x,\beta)$, with the signs $+,+,-,-$. By the formula just proved, $C((x+\xi,\mu'),(x+y,\nu)) = \int e^{ik\cdot(y-\xi)}\hat C_{\nu\mu'}(k)\, d^\circ k$, and therefore

$$\begin{aligned} E_p((x+y,\nu)) &= \sum_i s_i\, C(b_i(p),(x+y,\nu)) \\ &= \int_{\mathbb{T}} e^{ik\cdot y}\Big[\hat C_{\nu\alpha}(k)(1-e^{-ik_\beta}) + \hat C_{\nu\beta}(k)(e^{-ik_\alpha}-1)\Big]\, d^\circ k. \end{aligned}$$

As $\overline{\mathsf{k}_\mu(k)} = e^{-ik_\mu} - 1$, the bracket equals $\sum_\mu \overline{\mathsf{c}_{\iota,\mu}(k)}\, \hat C_{\nu\mu}(k)$. In this sum the part of $\hat C_{\nu\mu}$ containing $\mathsf{k}_\nu\overline{\mathsf{k}_\mu}$ drops out, since $\sum_\mu \overline{\mathsf{c}_{\iota,\mu}}\, \mathsf{k}_\nu\overline{\mathsf{k}_\mu} = \mathsf{k}_\nu\, \overline{\sum_\mu \mathsf{c}_{\iota,\mu}\mathsf{k}_\mu}$ and $\sum_\mu \mathsf{c}_{\iota,\mu}\mathsf{k}_\mu = \mathsf{k}_\alpha\mathsf{k}_\beta - \mathsf{k}_\beta\mathsf{k}_\alpha = 0$; what remains is $\overline{\mathsf{c}_{\iota,\nu}}\,|\mathsf{k}|^{-2}$. The kernel Π arises from E_p by the operation d in the second argument (Def. D.2). With $q = p_{[\iota']}(x+y) = p_{\alpha'\beta'}(x+y)$, the four bonds $(x+y,\alpha')$, $(x+y+e_{\alpha'},\beta')$, $(x+y+e_{\beta'},\alpha')$, $(x+y,\beta')$ of q contribute, with the signs $+,+,-,-$, the symbols $\overline{\mathsf{c}_{\iota,\alpha'}}|\mathsf{k}|^{-2}$, $e^{ik_{\alpha'}}\overline{\mathsf{c}_{\iota,\beta'}}|\mathsf{k}|^{-2}$, $e^{ik_{\beta'}}\overline{\mathsf{c}_{\iota,\alpha'}}|\mathsf{k}|^{-2}$, $\overline{\mathsf{c}_{\iota,\beta'}}|\mathsf{k}|^{-2}$, and their sum is

$$(\mathsf{k}_{\alpha'}\,\overline{\mathsf{c}_{\iota,\beta'}} - \mathsf{k}_{\beta'}\,\overline{\mathsf{c}_{\iota,\alpha'}})|\mathsf{k}|^{-2} = \sum_\nu \overline{\mathsf{c}_{\iota,\nu}}\,\mathsf{c}_{\iota',\nu}\,|\mathsf{k}|^{-2};$$

that is, d in the second argument multiplies the symbol by $\mathsf{c}_{\iota',\nu}(k)$ and sums over ν. Finally, with $G(x',x'') = \int e^{ik\cdot(x'-x'')}|\mathsf{k}|^{-2}\, d^\circ k$,

$$\begin{aligned} D((x,\mu);(x+y,\nu)) &= \tfrac12 \sum_{x'\in\{x+y,\,x+y+e_\nu\}} [G(x',x+e_\mu) - G(x',x)] \\ &= \int_{\mathbb{T}} e^{ik\cdot y}\, \tfrac12(1+e^{ik_\nu})(e^{-ik_\mu}-1)\,|\mathsf{k}|^{-2}\, d^\circ k. \end{aligned}$$

All four symbols are sums of products of the examples of Def. D.27, namely $e^{ik\cdot y}$ (degree 0, leading part 1), k_μ and $\overline{\mathsf{k}_\mu}$ (degree 1, leading parts ik_μ and $-ik_\mu$) and $|\mathsf{k}|^{-2}$ (degree -2, leading part $|k|^{-2}$), and no product of degree below -2 occurs: the term $\mathsf{k}_\nu\overline{\mathsf{k}_\mu}|\mathsf{k}|^{-4}$ is the product $(\mathsf{k}_\nu|\mathsf{k}|^{-2})(\overline{\mathsf{k}_\mu}|\mathsf{k}|^{-2})$ of two line symbols of degree -1. By the rules for sums and products in Def. D.27, the four symbols are line symbols of the degrees $-2,-1,0,-1$, so the kernels are elementary lines of the degrees $1,2,3,2$; their leading parts arise by replacing k_μ, $\overline{\mathsf{k}_\mu}$, $|\mathsf{k}|^{-2}$, e^{ik_ν} by ik_μ, $-ik_\mu$, $|k|^{-2}$, 1, which gives the leading parts listed.

If K has the symbol σ, then $K(-y) = \int e^{-ik\cdot y}\sigma(k)\, d^\circ k = \int e^{ik\cdot y}\sigma(-k)\, d^\circ k$ and $K(y+y_0) = \int e^{ik\cdot y}e^{ik\cdot y_0}\sigma(k)\, d^\circ k$. The function $\sigma(-k)$ is a line symbol of the same degree, with leading part $\sigma^h(-k)$, as χ is even; and $e^{ik\cdot y_0}\sigma(k)$ is the product of σ with a line symbol of degree 0 and leading part 1. For the consequence we recall from Def. D.2 that the contraction of two plain legs is $\delta_{aa'}C$, that of a curl leg with a plain leg is $\delta_{aa'}E$, and that of two curl legs is $\delta_{aa'}\Pi$. These kernels are invariant under simultaneous translations of their arguments, by their Fourier representations. Hence, by the list above and by the two rules just proved (which take care of the order of the legs and of the shifts ξ,ξ'), each contraction is an elementary line in y of degree 1, 2, 3, respectively. □

Lemma D.35 (The vertex insertion)**.** The kernel Γ_q of Def. D.21 satisfies

$$\frac32 \sum_{b,b'} \Gamma_q(b,b')E_p(b)E_p(b') = \mathsf{T}_{4'}^{(q)} + \mathsf{T}_5^{(q)} + \mathsf{T}_4 + \mathsf{T}_1. \tag{D.xxiv}$$

Here the superscript (q) denotes the first half of the corresponding expression in Lem. D.5: the quartic, respectively cubic, observable is at q. The identity holds separately for the four groups in the definition of Γ_q, grouping its two mixed derivatives. Exchanging p and q gives the second halves. ∎

Proof. Write $\partial = \partial_{b,a}$, $\partial' = \partial_{b',a'}$, $\delta = \delta_{(b,a)}$, and $\delta' = \delta_{(b',a')}$. The connected-pairing formula of Lem. D.5 gives

$$\langle \partial\partial' \mathsf{a}_q^{(4)} \rangle_0 = \frac{1}{2} \sum_{i,j} C(i,j) \mathsf{a}_q^{(4)} [\![\delta, \delta', \delta_i, \delta_j]\!],$$

$$\langle \partial \mathsf{a}_q^{(3)}; \partial' S_3 \rangle_0 = \frac{1}{2} \sum_{P} \sum_{i,i',j,j'} C(i,j) C(i',j') \mathsf{a}_q^{(3)} [\![\delta, \delta_i, \delta_{i'}]\!] \mathsf{a}_P^{(3)} [\![\delta', \delta_j, \delta_{j'}]\!],$$

$$\langle \mathsf{a}_q^{(2)}; \partial\partial' S_4 \rangle_0 = \frac{1}{2} \sum_{P,c} \mathsf{a}_P^{(4)} [\![\delta, \delta', E_q^c, E_q^c]\!],$$

$$\langle \mathsf{a}_q^{(2)}; \partial S_3; \partial' S_3 \rangle_0 = \sum_{P,P',c} \sum_{i,j} C(i,j) \mathsf{a}_P^{(3)} [\![\delta, E_q^c, \delta_i]\!] \mathsf{a}_{P'}^{(3)} [\![\delta', E_q^c, \delta_j]\!].$$

For the last line the three vertices are quadratic. The only connected pairing matrix has one line between each pair of vertices, with coefficient one. In each line set $a' = a$, multiply by $\frac{1}{2} E_p(b) E_p(b')$, and sum over b, b', a. The colour identity $\Gamma_q(b,a;b',a) = \Gamma_q(b,b')$ gives the left side of eq. D.xxiv. The first line gives $\mathsf{T}_{4'}^{(q)}$. The second line and its transpose have equal contractions; their negative sum gives $\mathsf{T}_5^{(q)}$. The negative of the third line gives T_4, and the fourth gives T_1, with exactly the coefficients of Lem. D.5. Absolute convergence is supplied by Lem. D.6. □

D.15. **Elementary terms, their expansions, the local parts.**

Lemma D.36 (Elementary terms and their degrees)**.** Let $\tau_{\mathrm{gf}} = 1$, let us work on $\mathbb{Z}^3$, let $p = p_{[\iota]}(x_p)$ and $q = p_{[\iota']}(x_q)$ be plaquettes, and put $u' := x_q - x_p$. We write every vertex occurring in the formulas of Lem. D.5 in the standard writing (Def. D.2, Lem. D.5), we carry out the sums over the colours, and we write every contraction as an elementary line by Lem. D.34. Then there are finite index sets $\mathfrak{G}_\bullet = \mathfrak{G}_\bullet(\iota,\iota')$, real numbers c_G and elementary lines $K_\bullet^G$, all independent of x_p and x_q, such that the following representations hold; every G is called an elementary term. We omit the index G on the lines, and $*$ denotes the convolution of Lem. D.32.

(1) $\mathsf{T}_1 = \sum_{G \in \mathfrak{G}_1} c_G \, \mathsf{T}^G(u')$, where every T^G is a kite as in Prop. D.33, with degrees satisfying the hypotheses of that proposition and with $\Delta = 5$. Here K_1 and K_2 are contractions of $(\mathrm{d}A)(p)$ with legs at P and at P', K_4 and K_5 are contractions of legs at P and at P' with $(\mathrm{d}A)(q)$, and K_3 is the contraction of a leg at P with a leg at P'.

(4) $\mathsf{T}_4 = \sum_{G \in \mathfrak{G}_4} c_G \, (X * Y)(u')$, where $X = K_1 K_2$ and $Y = K_3 K_4$ with $n_i \in \{2,3\}$, and $(n_X, n_Y) := (n_1 + n_2, n_3 + n_4) \in \{(4,4),(5,4),(4,5),(6,6)\}$, with $(n_X, n_Y) = (4,4)$ for the monomials of $\frac{1}{2}|\mathfrak{z}_1|^2$.

(4′) The first half of $\mathsf{T}_{4'}$ equals $\sum_{G \in \mathfrak{G}_{4'}} c_G \, K_1(u') \, K_2(u')$, where $n_i \in \{2,3\}$ and K_1, K_2 are contractions of $(\mathrm{d}A)(p)$ with legs at q.

(5) The first half of T_5 equals $\sum_{G \in \mathfrak{G}_5} c_G \, K_0(u') \, (X * Y)(u')$ with $Y = K_{\mathrm{a}} K_{\mathrm{b}}$. Here K_0 and X are contractions of $(\mathrm{d}A)(p)$ with a leg at q, resp. at P, and $n_0, n_X \in \{2,3\}$; $K_{\mathrm{a}}, K_{\mathrm{b}}$ are contractions of legs at P with legs at q, and $n_{\mathrm{a}}, n_{\mathrm{b}} \in \{1,2,3\}$; finally $n_0 + n_X + n_{\mathrm{a}} + n_{\mathrm{b}} = 8$, so that $n_Y := n_{\mathrm{a}} + n_{\mathrm{b}} \in \{2,3,4\}$.

(6) $\mathsf{T}_6 = \sum_{G \in \mathfrak{G}_6} c_G \, K_1(u') \, K_2(u') \, K_3(u')$, with $n_1 + n_2 + n_3 = 5$.

(I) $\mathsf{T}_{\mathrm{I}} = -3 \, \Pi(p,q) \sum_{\mu,\nu} (K_{\iota\mu}^{\rightarrow} * \Sigma_{\mu\nu} * K_{\nu\iota'}^{\leftarrow})(u')$, where $K_{\iota\mu}^{\rightarrow}(y) := E_{p_{[\iota]}(0)}((y,\mu))$ and $K_{\nu\iota'}^{\leftarrow}(y) := E_{p_{[\iota']}(y)}((0,\nu))$ are elementary lines of degree 2, and $\Sigma_{\mu\nu}(y) := \Sigma((x,\mu),(x+y,\nu))$, which

does not depend on x. With $\mathsf{s}_P(b) := s_i$ if $b = b_i(P)$ and $\mathsf{s}_P(b) := 0$ otherwise,

$$\partial_{b,a} S_3 = \mathsf{d}_{b,a} + \mathsf{u}_{b,a},$$
$$\mathsf{d}_{b,a} := \sum_{P \ni b} \mathsf{s}_P(b)\, \langle \mathsf{e}_a, \mathfrak{z}_1(P)\rangle, \qquad \mathsf{u}_{b,a} := \sum_{P \ni b} \langle \mathrm{d}A(P), \partial_{b,a}\mathfrak{z}_1(P)\rangle; \tag{D.xxv}$$

the monomials of $\mathsf{d}_{b,a}$ have two plain legs, those of $\mathsf{u}_{b,a}$ a curl leg and a plain leg. Moreover

$$\Sigma_{\mu\nu} = \mathsf{q}^0_{\mu\nu} + \sum_{G \in \mathfrak{G}_{\mathrm{I}}(\mu,\nu)} c_G\, \mathsf{f}_G(\nabla)\big[K_{\mathrm{a}} K_{\mathrm{b}}\big], \tag{D.xxvi}$$

where $\mathsf{q}^0_{\mu\nu}$ has finite support (it comes from the line between two slots of the quartic Wilson vertex, from the Haar term and from the ghost tadpole); $\mathsf{f}_G(k)$ is the product of two trigonometric polynomials, each a line symbol of degree 0 or 1, of total degree $d_G \in \{0,1,2\}$; $\mathsf{f}_G(\nabla)$ is the finite difference operator which multiplies symbols by f_G; $n_{\mathrm{a}}, n_{\mathrm{b}} \in \{1,2,3\}$; and $n_{\mathrm{a}} + n_{\mathrm{b}} + d_G = 4$. The terms G are the ghost loop $-8\, D(b;b')\, D(b';b)$, with $d_G = 0$, and the terms of $-\langle \partial S_3; \partial' S_3\rangle_0$ multiplied out by eq. D.xxv, with d_G the number of entries d.

The second halves of $\mathsf{T}_{4'}$ and T_5 have the same forms, with p and q exchanged and u' replaced by $-u'$. ∎

Proof. We use two counting rules: every monomial of $\mathsf{a}^{(3)}$ has exactly one curl leg and $\mathsf{a}^{(2)}$ has two (Def. D.2); and the degree of a line is $1 + \#$(its curl legs) (Lem. D.34). By the consequence stated in Lem. D.34, every contraction is an elementary line in the difference of the base points of the two vertices which it joins, its symbol depending only on the two legs; the shifts between the base points of the legs and those of the vertices are absorbed into the lines.

(1) A kite has five lines. There are four curl legs at p and q and one at each of P and P', hence $\sum_i n_i = 5 + 6 = 11$ and $\Delta = 5$. For $i \in \{1,2,4,5\}$ one end of the line i is a curl leg of p or of q, so that $n_i \geq 2$, and $n_i \leq 3$. Besides the two curl legs of p, the lines $1,2,3$ carry at most two further curl legs, one of P and one of P'; hence $n_1 + n_2 + n_3 \leq 2 + 2 + 1 + 2 = 7$, and the same count applies to the lines $3,4,5$. With the base points $x_p + z$ and $x_p + w$ of P and P', the sum over P, P' becomes the sum over z, w and over the dual directions of P, P', and the five lines become functions of z, w, $w - z$, $u' - z$, $u' - w$.

(4) Each of the four lines carries a curl leg at p or at q; it has degree 2, or degree 3 if its leg at P is a curl leg. The three kinds of monomials of $\mathsf{a}^{(4)}_P$ (Lem. D.5) give $(n_X, n_Y) = (4,4)$; $(5,4)$ or $(4,5)$; and $(6,6)$. With the base point $x_p + z$ of P, the two lines from p make up $X(z)$, the two lines to q make up $Y(u' - z)$, and the sum over z is $(X * Y)(u')$.

(4′) There are two lines from p to q, each with its curl leg at p; the line between two slots of the vertex at q is a number, independent of the positions, and it is absorbed into c_G.

(5) There are four lines, and there are $2 + 1 + 1$ curl legs, at p, q and P; hence $\sum n = 4 + 4 = 8$. With the base point $x_p + z$ of P, the line from p to P is $X(z)$, the two lines from P to q make up $Y(u' - z)$, and the line from p to q is $K_0(u')$.

(6) There are three lines, and there are $1 + 1$ curl legs, at p and q; hence $\sum n = 3 + 2 = 5$.

(I) In the formula for T_{I} we write $b = (x, \mu)$ and $b' = (x', \nu)$ with $x = x_p + z$, $x' = x_p + w$. By translation invariance $E_p(b) = K^{\rightarrow}_{\iota\mu}(z)$ and $E_q(b') = K^{\leftarrow}_{\nu\iota'}(u' - w)$, and by the translation covariance in Def. D.21(5) $\Sigma(b,b') = \Sigma_{\mu\nu}(w - z)$; this gives the formula for T_{I}. The functions $K^{\rightarrow}_{\iota\mu}$ and $K^{\leftarrow}_{\nu\iota'}$ are elementary lines of degree 2 by Lem. D.34, the second because $K^{\leftarrow}_{\nu\iota'}(y) = E_{p_{[\iota']}(0)}((-y,\nu))$. By Def. D.21(5) and Def. D.2, with $\partial = \partial_{b,a}$ and $\partial' = \partial_{b',a}$,

$$\Sigma(b,b') = \Big[\langle \partial\partial' S_4\rangle_0 + (\tfrac{2}{3} + \tfrac{4}{3} g(b))\delta_{bb'}\Big] \\ - 4\big[D(b;b')D(b';b) + D(b';b)D(b;b')\big] - \langle \partial S_3; \partial' S_3\rangle_0.$$

The bracket has finite range and is translation invariant; it gives $\mathsf{q}^0_{\mu\nu}$. The middle term is the ghost loop $-8\,D(b;b')\,D(b';b)$, a product of two elementary lines of degree 2 in y (Lem. D.34), with $\mathsf{f}_G = 1$. From Lem. D.1 and the Leibniz rule we have

$$\partial_{b,a}\langle \mathrm{d}A(P), \mathfrak{z}1(P)\rangle = \mathsf{s}_P(b)\,\langle \mathsf{e}_a, \mathfrak{z}1(P)\rangle + \langle \mathrm{d}A(P), \partial_{b,a}\mathfrak{z}1(P)\rangle,$$

which gives eq. D.xxv. For a function F on the plaquettes of one dual direction, the map $b \mapsto \sum_{P\ni b} \mathsf{s}_P(b)\,\mathsf{F}(P)$ is a difference operator whose symbol is $\pm\overline{\mathsf{k}_\alpha}$ up to a factor $e^{ik\cdot y_0}$, a line symbol of degree 1; this operator yields the factor of f_G which belongs to an entry d. An entry u we split into its monomials and treat each of them separately. By the consequence stated in Lem. D.34, the contraction of a leg with base point $x+\xi$ with a leg with base point $x+y+\xi'$ is an elementary line in y in which the shifts ξ and ξ' are already contained; hence the factor of f_G which belongs to an entry u is 1, a line symbol of degree 0 with leading part 1, just as for the ghost loop. By Lem. D.5, the cumulant of two quadratic monomials is a sum of two products of two lines. Hence every term of $-\langle \partial S_3; \partial' S_3\rangle_0$, multiplied out by eq. D.xxv, has the form $c_G\,\mathsf{f}_G(\nabla)[K_\mathrm{a}K_\mathrm{b}]$, with d_G the number of entries d and with $n_\mathrm{a}+n_\mathrm{b} = 2 + \#(\text{curl legs on the two lines})$. As d has no curl leg and u has one, $n_\mathrm{a}+n_\mathrm{b} = 2+(2-d_G)$. The second halves of $\mathsf{T}_{4'}$ and T_5 are treated in the same way, the roles of p and q being exchanged. □

Lemma D.37 (Expansions of the elementary terms, with errors)**.** Let the setting be that of Lem. D.36, let $r' := |u'|$ be large, and let $K^h := \mathcal{F}^{-1}_{\neq 0}\sigma^h$ for every line K with symbol σ. For the bubbles X and Y let q_X and q_Y be the kernels q of Lem. D.32(iii), and put $X' := X - \mathsf{q}_X$, $Y' := Y - \mathsf{q}_Y$. Then the elementary terms of Lem. D.36 have the following expansions.

(6) $K_1K_2K_3(u') = K_1^hK_2^hK_3^h(u') + O(r'^{-6})$.

(5) $K_0(u')\,(X{*}Y)(u') = K_0(u')\,(\mathsf{q}_Y{*}X)(u') + K_0^h(u')\,(X{*}Y')^h(u') + O(r'^{-6}\lg r')$, where $(X{*}Y')^h$ is the second component of the leading data of $X*Y' \in \mathfrak{K}(n_X+n_Y-3)$.

(4) $(X*Y)(u') = (\mathsf{q}_X*Y)(u') + (\mathsf{q}_Y*X)(u') + \mathsf{U}_G(u') + O(r'^{-6}\lg r')$, where $\mathsf{U}_G := (X'*Y')^h$ if $(n_X,n_Y) = (4,4)$ and $\mathsf{U}_G := 0$ otherwise.

(1) The expansion of Prop. D.33(iii) holds, with the error $O(r'^{-6}\lg r')$.

(∗) In all four cases the function of degree -5 in the expansion (that is, $K_1^hK_2^hK_3^h$, $K_0^h(X*Y')^h$, U_G, resp. the function U of Prop. D.33) is determined by the leading parts σ^h of the lines alone; it is multilinear in them, it belongs to $C^1(\mathbb{R}^3\setminus\{0\})$, and it is rotation covariant as in Prop. D.33(iv).

■

Proof. (6) By Lem. D.26, $K_i(u') = O(r'^{-n_i})$, $K_i^h(u') = O(r'^{-n_i})$ and $K_i(u') = K_i^h(u') + O(r'^{-n_i-1})$. As $n_1+n_2+n_3 = 5$, the product $K_1K_2K_3(u')$ differs from $K_1^hK_2^hK_3^h(u')$ by $O(r'^{-6})$, and $K_1^hK_2^hK_3^h$ is homogeneous of degree -5.

(5) The line X is an elementary line of degree $n_X \in \{2,3\}$, so $X \in \mathfrak{K}(n_X)$ by Lem. D.26. The bubble $Y = K_\mathrm{a}K_\mathrm{b}$ has $n_\mathrm{a}, n_\mathrm{b} \in \{1,2,3\}$ and $n_Y \le 4$, so $Y' \in \mathfrak{K}(n_Y)$ by Lem. D.32(iii). We write $X*Y = \mathsf{q}_Y*X + X*Y'$. Lem. D.32(ii) applies to X and Y', as $n_X, n_Y \le 4$ and $n_X+n_Y = 8-n_0 \ge 5$; it gives $X*Y' \in \mathfrak{K}(n_X+n_Y-3)$, and the spatial estimate in Def. D.27 with $\gamma = 0$ gives $X*Y' = (X*Y')^h + O(r'^{-(n_X+n_Y-3)-1}\lg r')$. Moreover $K_0 = K_0^h + O(r'^{-n_0-1})$ by Lem. D.26, and K_0, K_0^h, $X*Y'$, $(X*Y')^h$ are bounded by the corresponding powers of r'. As $n_0 + (n_X+n_Y-3) = 8-3 = 5$, the claim follows.

(4) Multiplying out $X*Y = (\mathsf{q}_X + X')*(\mathsf{q}_Y+Y')$ we obtain

$$X*Y = \mathsf{q}_X*Y + \mathsf{q}_Y*X - \mathsf{q}_X*\mathsf{q}_Y + X'*Y',$$

and $\mathsf{q}_X*\mathsf{q}_Y$ vanishes off $\{|u'|\le 2\}$, because q_X and q_Y vanish off $\{0,\pm e_j\}$. If $(n_X,n_Y) = (4,4)$, then $X',Y' \in \mathfrak{K}(4)$ by Lem. D.32(iii), and Lem. D.32(ii) gives $X'*Y' \in \mathfrak{K}(5)$ and $X'*Y' = (X'*Y')^h + O(r'^{-6}\lg r')$. Otherwise we apply Lem. D.32(i) to X' and Y', with $a_X = \min\{n_X-3,2\}$ and $a_Y = \min\{n_Y-3,2\}$. Its hypotheses hold by Lem. D.32(iii), which bounds the differences of a

bubble of degree n by $C_\gamma\langle x\rangle^{-n-|\gamma|}$ (the kernels X', Y' differ from X, Y at finitely many points only) and the symbol of X', resp. Y', by $|\mathsf{I}-\tilde{\mathsf{Q}}| \le C|k|^{\min\{n-3,2\}}$. The three exponents in the bound of Lem. D.32(i) are $-2-4$, $-1-5$, $3-9$ for $(n_X, n_Y) = (5,4)$, the same by symmetry for $(4,5)$, and at most -8 for $(6,6)$; so $X' * Y' = O(r'^{-6}\lg r')$ in these cases.

(1) This is Prop. D.33(iii) with $\Delta = 5$ (Lem. D.36(1)), so that $r^{-\Delta-1}\lg r = r'^{-6}\lg r'$.

$(*)$ In the case (6) this follows from Lem. D.26. In the cases (5) and (4), Lem. D.32(ii) shows that the second component of the leading data of a convolution depends only on the leading data of the two factors, bilinearly and rotation covariantly. By Lem. D.32(iii) the leading data of Y', and of X' in the case (4), are $(\mathsf{I}^h, K^h_{\mathrm{a}}K^h_{\mathrm{b}})$, where I^h depends only on $\sigma^h_{\mathrm{a}}, \sigma^h_{\mathrm{b}}$, bilinearly and rotation covariantly, by Lem. D.30; the leading data of the lines X and K_0 are (σ^h, K^h) with $K^h = \mathcal{F}^{-1}_{\neq 0}\sigma^h$, by Lem. D.26. The functions K^h and the second components of leading data are smooth off the origin. In the case (1) the assertions are Prop. D.33(iii),(iv). □

Lemma D.38 (A local insertion with vanishing moments)**.** (a) Let $\gamma_{\mu\nu}$, $1 \le \mu, \nu \le 3$, be functions of finite support on $\mathbb{Z}^3 \times \mathbb{Z}^3$ whose symbols

$$\hat\gamma_{\mu\nu}(\kappa_1, \kappa_2) := \sum_{y,y'} \gamma_{\mu\nu}(y, y')\, e^{-i(\kappa_1\cdot y + \kappa_2\cdot y')}$$

vanish at $\kappa = 0$ together with their gradients, and let K_1, K_2, K_3 be elementary lines of degree 2. Then

$$\Big|\sum_{\mu,\nu}\sum_{y,y'} \gamma_{\mu\nu}(y,y')\, K_\mu(u+y)\, K_\nu(u+y')\Big| \le C\,|u|^{-6} \qquad (u \in \mathbb{Z}^3 \setminus \{0\}),$$

where C depends on the $\gamma_{\mu\nu}$ and on the symbols of the K_μ.

(b) Let q be a function of finite support on $\mathbb{Z}^3$ with $\hat{\mathsf{q}}(k) := \sum_y \mathsf{q}(y)\, e^{-ik\cdot y} = O(|k|^2)$ at $k = 0$, and let K_1, K_2 be elementary lines of degree 2. Then $|(K_1 * \mathsf{q} * K_2)(u)| \le C\langle u\rangle^{-3}$ for all $u \in \mathbb{Z}^3$, where C depends on q and on the symbols of K_1, K_2. ■

Proof. (a) As the $\gamma_{\mu\nu}$ have finite support, only points y, y' in a fixed ball occur; let $|u|$ be large. By Lem. D.3 with $s = -1$, $K_\mu(u) = O(|u|^{-2})$, $\nabla K_\mu(u) = O(|u|^{-3})$, and the second differences of K_μ near u are $O(|u|^{-4})$. We write $K_\mu(u+y) - K_\mu(u)$ as a sum of first differences along a nearest-neighbour path from u to $u+y$ and compare each of them with $\nabla_j K_\mu(u)$; this gives

$$K_\mu(u+y) = K_\mu(u) + \sum_j y_j\, \nabla_j K_\mu(u) + O(|u|^{-4}).$$

We insert this expansion for $K_\mu(u+y)$ and for $K_\nu(u+y')$. The terms of the orders $|u|^{-4}$ and $|u|^{-5}$ then carry the coefficients

$$\sum_{y,y'} \gamma_{\mu\nu}(y,y') = \hat\gamma_{\mu\nu}(0), \qquad \sum_{y,y'} \gamma_{\mu\nu}(y,y')\, y_j, \qquad \sum_{y,y'} \gamma_{\mu\nu}(y,y')\, y'_j,$$

and the last two are i times the derivatives of $\hat\gamma_{\mu\nu}$ with respect to $\kappa_{1,j}$, resp. $\kappa_{2,j}$, at 0. All three vanish by hypothesis, and the remaining terms are $O(|u|^{-6})$. For $u \neq 0$ in a bounded set the left side is bounded and $|u|^{-6}$ is bounded below, so the bound holds for all $u \neq 0$ after enlarging C.

(b) The trigonometric polynomial $\hat{\mathsf{q}}$ is smooth and vanishes at 0 together with its gradient, hence $\hat{\mathsf{q}} \in \mathcal{S}^2$. The kernel $\mathsf{q} * K_2$, a finite combination of translates of K_2, has the symbol $\hat{\mathsf{q}}\sigma_2 \in \mathcal{S}^1$, where $\sigma_2 \in \mathcal{S}^{-1}$ is the symbol of K_2; by Lem. D.3 it is $O(\langle x\rangle^{-4})$ and lies in $\ell^2(\mathbb{Z}^3)$, as does $K_1 = O(\langle x\rangle^{-2})$. By Plancherel's theorem $K_1 * (\mathsf{q} * K_2)$ is the kernel with the symbol $\sigma_1\hat{\mathsf{q}}\sigma_2 \in \mathcal{S}^0$, and Lem. D.3 with $s = 0$ gives the bound $C\langle u\rangle^{-3}$. □

Lemma D.39 (The local parts vanish to the second order)**.** Let $\tau_{\mathrm{gf}} = 1$; we work on $\mathbb{Z}^3$.

(a) (Self-energy.) In eq. D.xxvi let q_G be the kernel q of the bubble $K_{\mathrm{a}}K_{\mathrm{b}}$ in the sense of Lem. D.32(iii), and put

$$\mathsf{q}^{\Sigma}_{\mu\nu} := \mathsf{q}^{0}_{\mu\nu} + \sum_G c_G\, \mathsf{f}_G(\nabla)\mathsf{q}_G, \qquad \mathsf{W}_{\mu\nu} := \sum_G c_G\, \mathsf{f}_G(\nabla)\big[K_{\mathrm{a}}K_{\mathrm{b}} - \mathsf{q}_G\big],$$

so that $\Sigma_{\mu\nu} = \mathsf{q}^{\Sigma}_{\mu\nu} + \mathsf{W}_{\mu\nu}$. Then $\mathsf{q}^{\Sigma}_{\mu\nu}$ has finite support. Each $\mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}} - \mathsf{q}_G]$ belongs to $\mathfrak{K}(4)$, and the first component of its leading data is $\mathsf{f}^h_G\, \mathsf{l}^h_G$, where f^h_G is the leading part of f_G and l^h_G is the homogeneous part of the symbol l_G of the bubble (Lem. D.32(iii), Lem. D.30). Hence $\mathsf{W}_{\mu\nu} \in \mathfrak{K}(4)$, and its leading data are functions of the leading parts σ^h and of the f^h_G occurring in eq. D.xxvi alone. Moreover the symbol of $\mathsf{q}^{\Sigma}_{\mu\nu}$ satisfies $\hat{\mathsf{q}}^{\Sigma}_{\mu\nu}(k) = O(|k|^2)$ at $k = 0$.

(b) (Vertex.) Let $q = p_{[\iota']}(x_q)$. Let $\gamma_q(b, b')$ be the symmetric kernel of finite range defined below; then, for every plaquette p with $|u'|$ large, $\frac{3}{2}\sum_{b,b'} \gamma_q(b,b')\, E_p(b)\, E_p(b')$ is the sum of the following terms of Lem. D.37: the first half of $\mathsf{T}_{4'}$; the terms $c_G\, K_0(u')\, (\mathsf{q}_Y * X)(u')$ of the first half of T_5; the terms $c_G\, (\mathsf{q}_Y * X)(u')$ of T_4; and the terms $c_G \sum_{z,w} \mathsf{q}_{\mathsf{q}}(z,w)\, K_1(u'-z)\, K_2(u'-w)$ of T_1. The kernel γ_q is defined as follows. Each of K_0, X, K_1, K_2, at the points where it is evaluated in these terms, is either a value $E_p(b)$ with b a bond near q, or a value $\Pi(p,P) = \sum_i s_i\, E_p(b_i(P))$ with P a plaquette near q. Substituting this term by term in the four sums we obtain a quadratic expression in the values $E_p(b)$, and $\frac{3}{2}\gamma_q$ is its coefficient kernel, symmetrised. This kernel does not depend on p. Indeed, by the rule for $\mathsf{a}^{(2)}_p$ in the proof of Lem. D.5, the plaquette p, with its base point and its dual direction, enters the four sums only through the forms E^a_p inserted into the two slots joined to p, that is, only through the values of K_0, X, K_1, K_2 just described. An elementary term G of these sums is given by the monomials at the vertices at q, P, P' and by the choice of their legs which are contracted with p, with q and with one another, so that the numbers c_G do not involve p; and the kernels q_Y, q_{q} and the line between two slots of the vertex at q are built from lines among q, P, P' alone. We define γ_q through this way of writing, since the numbers $\sum_{b,b'} \gamma_q(b,b') E_p(b) E_p(b')$ alone would not fix a kernel. Let $\gamma_{q;\mu\nu}(y,y') := \gamma_q((x_q+y,\mu),(x_q+y',\nu))$. Then the symbols $\hat{\gamma}_{q;\mu\nu}$, defined as in Lem. D.38(a), vanish at $\kappa = 0$ together with their gradients.

■

Proof. Step 1: a trace formula. Let $\mathsf{m}^1, \dots, \mathsf{m}^{\mathsf{r}}$, $\mathsf{r} \le 3$, be real symmetric matrices indexed by $\mathsf{i} = (x, \mu, a)$, that is, by a bond (x,μ) and a colour a, each with finitely many non-zero entries. For $y \in \mathbb{Z}^3$ let m_y be the translate of m by y, $(\mathsf{m}_y)_{(x,\mu,a),(x',\mu',a')} := \mathsf{m}_{(x-y,\mu,a),(x'-y,\mu',a')}$, and let $\hat{\mathsf{m}}(l', l)$ be the 9×9 matrix, indexed by (μ, a) and (μ', a'), with the entries

$$\sum_{x,x'} e^{-il'\cdot x}\, \mathsf{m}_{(x,\mu,a),(x',\mu',a')}\, e^{il\cdot x'}.$$

Then $\hat{\mathsf{m}}_y(l',l) = e^{-i(l'-l)\cdot y}\, \hat{\mathsf{m}}(l',l)$, and $\hat{\mathsf{m}}(l,l)$ is the symbol of the translation invariant matrix $\sum_y \mathsf{m}_y$, in the sense that $\sum_y \mathsf{m}_y$ maps $e^{il\cdot x}\alpha$ to $e^{il\cdot x}\, \hat{\mathsf{m}}(l,l)\, \alpha$. We insert $C(\mathsf{i},\mathsf{i}') = \delta_{aa'} \int_{\mathbb{T}} e^{il\cdot(x-x')}\, \hat{C}_{\mu\mu'}(l)\, d^\circ l$ and let $\hat{C}(l)$ act on the nine-dimensional space as $\hat{C}(l)$ in the directions and as the identity in the colours. For $\mathsf{r} = 3$ we obtain

$$\begin{aligned} &\mathrm{Tr}\big[\mathsf{m}^1 C\, \mathsf{m}^2_{y_2} C\, \mathsf{m}^3_{y_3} C\big] \\ &\quad = \iint e^{i(\kappa_2\cdot y_2 + \kappa_3\cdot y_3)} \Big\{ \int \mathrm{tr}\big[\hat{\mathsf{m}}^1(l_3,l)\, \hat{C}(l)\, \hat{\mathsf{m}}^2(l,l_2)\, \hat{C}(l_2) \\ &\qquad\qquad \hat{\mathsf{m}}^3(l_2,l_3)\, \hat{C}(l_3)\big]\, d^\circ l \Big\}\, d^\circ\kappa_2\, d^\circ\kappa_3, \end{aligned} \tag{D.xxvii}$$

with $l_2 := l + \kappa_2$, $l_3 := l + \kappa_2 + \kappa_3$, all integrals being over $\mathbb{T}$. Indeed, the sums over the indices are finite and the integrals over the variables l, l_2, l_3 of the three covariances converge absolutely; Fubini's

theorem allows us to carry out the sums first and then to pass to the variables $\kappa_2 = l_2 - l$, $\kappa_3 = l_3 - l_2$. For $\mathsf{r} = 2$ one omits m^3, the last factor $\hat{C}$ and κ_3; for $\mathsf{r} = 1$ one has $\mathrm{Tr}[\mathsf{m}^1 C] = \int \mathrm{tr}[\hat{\mathsf{m}}^1(l,l)\,\hat{C}(l)]\,d^\circ l$. For the quadratic forms $\mathsf{F}_j := \frac{1}{2}(A, \mathsf{m}^j A)$, Lem. D.5 gives

$$\langle \mathsf{F}_1 \rangle_0 = \tfrac{1}{2}\,\mathrm{Tr}[\mathsf{m}^1 C], \qquad \langle \mathsf{F}_1; \mathsf{F}_2 \rangle_0 = \tfrac{1}{2}\,\mathrm{Tr}[\mathsf{m}^1 C\, \mathsf{m}^2 C],$$
$$\langle \mathsf{F}_1; \mathsf{F}_2; \mathsf{F}_3 \rangle_0 = \mathrm{Tr}[\mathsf{m}^1 C\, \mathsf{m}^2 C\, \mathsf{m}^3 C].$$

The formula eq. D.xxvii also identifies integrands. For a leg ℓ of colour a we define $\mathsf{v}_\ell(k) \in \mathbb{C}^3$ by $\ell(e^{ik\cdot x}\alpha\,\mathsf{e}_a) = \mathsf{v}_\ell(k)\cdot\alpha$, where $e^{ik\cdot x}\alpha\,\mathsf{e}_a$ denotes the form $(x,\mu) \mapsto e^{ik\cdot x}\alpha_\mu \mathsf{e}_a$; thus $\mathsf{v}_\ell(k) = e^{ik\cdot\xi}e_\nu$ for the plain leg at the bond (ξ,ν), and $\mathsf{v}_\ell(k) = e^{ik\cdot\xi}(\mathsf{c}_{\iota,\nu}(k))_{\nu=1,2,3}$ for the curl leg at $p_{[\iota]}(\xi)$. The contraction of ℓ with the translate by y of a leg ℓ' of the same colour is $\int e^{-ik\cdot y}\,\mathsf{v}_\ell(k)^{\mathsf{T}}\,\hat{C}(k)\,\mathsf{v}_{\ell'}(-k)\,d^\circ k$, and the Hessian m of the product $\ell_1\ell_2$ of two legs has

$$\hat{\mathsf{m}}(l',l) = \mathsf{v}_{\ell_1}(-l')\,\mathsf{v}_{\ell_2}(l)^{\mathsf{T}} + \mathsf{v}_{\ell_2}(-l')\,\mathsf{v}_{\ell_1}(l)^{\mathsf{T}},$$

the colour indices being those of the legs. The matrices m^j used below are sums of Hessians of monomials in legs. Multiplying out, we find that for all (l,κ) the integrand in braces in eq. D.xxvii is a sum of products of the symbols of r lines evaluated at $\pm l, \pm l_2, \pm l_3$, and that these products are, term by term, the integrands of the one-loop functions (Lem. D.30, Lem. D.32(iii), Prop. D.33) of the elementary terms of Lem. D.36, up to substitutions $l \to \pm l + \lambda(\kappa)$ with λ linear. At $\kappa = 0$ the integrand in braces is the sum of the integrands of these one-loop functions at $\kappa = 0$, up to the substitution $l \to -l$ in single terms. We express this by saying that the two sides agree integrand by integrand.

Step 2: the Hessians at a constant abelian field. By Lem. D.1 with $t = 1$, $a_p(e^{iA}) = \sum_m \mathsf{a}_p^{(m)}(A)$. Let $\mathsf{N}_j(B)$ and $\mathsf{h}_q^{(j)}(B)$ be the parts of degree j in B of the Hessians $\mathsf{N}(B)$ and $\mathsf{h}_q(B)$ of Def. D.21(1). Since S_3 is cubic and S_4 quartic, $\mathsf{N}_1(B)$ is the Hessian of the quadratic polynomial $\sum_{b,a} B^a(b)\,\partial_{b,a}S_3$, and $\mathsf{N}_2(B)$ is that of $\frac{1}{2}\sum B^a(b)\,B^{a'}(b')\,\partial_{b,a}\partial_{b',a'}S_4$; the matrix $\mathsf{h}_q^{(j)}(B)$ is obtained in the same way from $\mathsf{a}_q^{(2+j)}$. Let H_b be the Hessian of $\partial_{b,3}S_3$, and let $\mathsf{H}_b^{\mathsf{u}}$ and $\mathsf{H}_b^{\mathsf{d}}$ be those of $\mathsf{u}_{b,3}$ and $\mathsf{d}_{b,3}$ in eq. D.xxv; these matrices are translation covariant, $\mathsf{H}_{b+y} = (\mathsf{H}_b)_y$. For the constant field Y_γ of Def. D.21(2), with $\mu(b)$ the direction of b, we have

$$\sum_b \gamma_{\mu(b)}\,\mathsf{d}_{b,3} = \sum_P \langle \mathrm{d}Y_\gamma(P), \mathfrak{z}1(P)\rangle = 0$$

(as a quadratic form, every entry of its Hessian being a finite sum), because $\sum_b \gamma_{\mu(b)}\,\mathsf{s}_P(b)\,\mathsf{e}_3 = \mathrm{d}Y_\gamma(P)$ and $\mathrm{d}Y_\gamma = 0$. Hence $\mathsf{N}_1(Y_\gamma) = \sum_b \gamma_{\mu(b)}\,\mathsf{H}_b^{\mathsf{u}} = \sum_\mu \gamma_\mu \sum_y (\mathsf{H}^{\mathsf{u}}_{(0,\mu)})_y$, and by Step 1 and Def. D.21(2) its symbol is

$$\sum_\mu \gamma_\mu\,\hat{\mathsf{H}}^{\mathsf{u}}_{(0,\mu)}(l,l) = \bigoplus_e \hat{\mathsf{N}}_{e,1}(l;\gamma)$$

in the basis e_e of the colours.

Step 3: the structure in (a). Let $K \in \mathfrak{K}(n)$ have the symbol σ and the leading data (σ^h, K^h), and let $y_0 \in \mathbb{Z}^3$. The kernels with the symbols $e^{ik\cdot y_0}\sigma$, $\mathsf{k}_\mu\sigma$ and $\overline{\mathsf{k}_\mu}\sigma$ lie in $\mathfrak{K}(n)$, $\mathfrak{K}(n+1)$ and $\mathfrak{K}(n+1)$, respectively, with the leading data (σ^h, K^h), $(ik_\mu\sigma^h, \partial_\mu K^h)$ and $(-ik_\mu\sigma^h, -\partial_\mu K^h)$. Indeed, the symbol $\mathsf{k}_\mu\sigma$ belongs to the kernel $\nabla_\mu K$. For the symbol estimate in Def. D.27 we write

$$\mathsf{k}_\mu\sigma - \chi\, ik_\mu\sigma^h = \mathsf{k}_\mu(\sigma - \chi\sigma^h) + (\mathsf{k}_\mu - ik_\mu)\,\chi\sigma^h = O(|k|^{n-1}\lg|k|),$$

and for the spatial estimate in Def. D.27, by Taylor's formula and the homogeneity of the derivatives of K^h,

$$\nabla^\gamma(\nabla_\mu K - \partial_\mu K^h) = \nabla^{\gamma+e_\mu}(K - K^h) + \nabla^\gamma(\nabla_\mu - \partial_\mu)K^h = O(|x|^{-n-2-|\gamma|}\lg|x|);$$

at the finitely many points with $|\gamma|+1 \le |x| < |\gamma|+2$ the left side is a finite number, which is absorbed into the constant. Next, $\overline{\mathsf{k}_\mu} = -e^{-ik_\mu}\mathsf{k}_\mu$. For the shift, $(e^{ik\cdot y_0} - 1)\,\chi\sigma^h = O(|k|^{n-2})$ and

$\nabla^\gamma[K^h(\cdot+y_0)-K^h]=O(|x|^{-n-1-|\gamma|})$. By Lem. D.32(iii), $K_\mathrm{a}K_\mathrm{b}-\mathsf{q}_G\in\mathfrak{K}(n_\mathrm{a}+n_\mathrm{b})$, with the leading data $(\mathsf{I}^h_G, K^h_\mathrm{a}K^h_\mathrm{b})$; that lemma applies, since $n_\mathrm{a},n_\mathrm{b}\in\{1,2,3\}$ and $n_\mathrm{a}+n_\mathrm{b}=4-d_G\le 4$. A trigonometric polynomial f is a finite sum of terms $c\,e^{ik\cdot y_0}$. As a line symbol of degree 0 it has the leading part $\mathsf{f}(0)$. As a line symbol of degree 1 it has $\mathsf{f}(0)=0$, its leading part is its linear part at 0, and f is a finite sum of terms $c\,e^{ik\cdot y_0}\mathsf{k}_\mu$: indeed f is a linear combination of the functions $e^{ik\cdot y}-1$, and each of these is a telescoping sum, along a nearest-neighbour path from 0 to y, of terms $\pm e^{ik\cdot y_0}\mathsf{k}_\mu$ (with $\overline{\mathsf{k}_\mu}=-e^{-ik_\mu}\mathsf{k}_\mu$ for the steps in the negative direction). Since $\mathfrak{K}(n)$ is a linear space and the leading data depend linearly on the kernel, multiplication of the symbol by f maps $\mathfrak{K}(n)$ into $\mathfrak{K}(n)$, resp. $\mathfrak{K}(n+1)$, and multiplies the first component of the leading data by f^h. Applying this once for each of the two factors of f_G, we obtain $\mathsf{f}_G(\nabla)[K_\mathrm{a}K_\mathrm{b}-\mathsf{q}_G]\in\mathfrak{K}(n_\mathrm{a}+n_\mathrm{b}+d_G)=\mathfrak{K}(4)$, with $\mathsf{f}^h_G\,\mathsf{I}^h_G$ as the first component of its leading data. Hence $\mathsf{W}_{\mu\nu}\in\mathfrak{K}(4)$. The first component of its leading data, $\sum_G c_G\,\mathsf{f}^h_G\,\mathsf{I}^h_G$, depends only on the leading parts σ^h (Lem. D.30) and on the f^h_G; by Lem. D.32(iv) the second component is determined by the first, linearly. Finally $\mathsf{q}^\Sigma_{\mu\nu}$ has finite support, since $\mathsf{q}^0_{\mu\nu}$ and the q_G have and $\mathsf{f}_G(\nabla)$ is a finite difference operator.

Step 4: the value of $\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ at 0. The symbol of q_G is $\tilde{\mathsf{Q}}_G$, whose value at 0 is $\mathsf{Q}_G(0)$, and $\mathsf{f}_G(0)=0$ if $d_G\ge 1$. Hence

$$\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)=\hat{\mathsf{q}}^0_{\mu\nu}(0)+\sum_{G:\,d_G=0}c_G\,\mathsf{f}_G(0)\,\mathsf{Q}_G(0).$$

For $d_G=0$ we have $n_\mathrm{a}+n_\mathrm{b}=4$, so that the bubble symbol has $\omega=1$, and Lem. D.30, with the data of the bubble given in the proof of Lem. D.32(iii), yields $\mathsf{Q}_G(0)=\int_{\mathbb{T}}\sigma_\mathrm{a}(-l)\,\sigma_\mathrm{b}(l)\,d^\circ l$. The terms with $d_G=0$ are the ghost loop and the terms of $-\langle\mathsf{u}_{b,3};\mathsf{u}_{b',3}\rangle_0=-\frac12\,\mathrm{Tr}[\mathsf{H}^\mathsf{u}_b C\,\mathsf{H}^\mathsf{u}_{b'}C]$ (Step 1). We compute $\frac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu$ times each contribution to $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)$.

- Gluon loop. By eq. D.xxvii, $-\frac12\,\mathrm{Tr}[\mathsf{H}^\mathsf{u}_{(0,\mu)}C\,\mathsf{H}^\mathsf{u}_{(y,\nu)}C]$ is the kernel in $y=b'_--b_-$ with the symbol $\kappa_2\mapsto-\frac12\int\mathrm{tr}[\hat{\mathsf{H}}^\mathsf{u}_{(0,\mu)}(l_2,l)\,\hat C(l)\,\hat{\mathsf{H}}^\mathsf{u}_{(0,\nu)}(l,l_2)\,\hat C(l_2)]\,d^\circ l$, and this symbol is $\sum c_G\,\mathsf{f}_G\,\mathsf{I}_G$, summed over the corresponding G, integrand by integrand. At zero momentum the integrand is $-\frac12\,\mathrm{tr}[\hat{\mathsf{H}}^\mathsf{u}_{(0,\mu)}(l,l)\,\hat C(l)\,\hat{\mathsf{H}}^\mathsf{u}_{(0,\nu)}(l,l)\,\hat C(l)]$, and by Step 2

$$\begin{aligned}&\tfrac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\Big(-\tfrac12\int\mathrm{tr}\,[\hat{\mathsf{H}}^\mathsf{u}_{(0,\mu)}(l,l)\,\hat C\,\hat{\mathsf{H}}^\mathsf{u}_{(0,\nu)}(l,l)\,\hat C]\,d^\circ l\Big)\\&\qquad=-\tfrac14\sum_e\int\mathrm{tr}\,[(\hat C\hat{\mathsf{N}}_{e,1})^2]\,d^\circ l=\int_{\mathbb{T}}\mathfrak{s}^{(1)}(l;\gamma)\,d^\circ l.\end{aligned}$$

- Gluon tadpole. Let $\mathsf{H}_{bb'}$ be the Hessian of $\partial_{b,3}\partial_{b',3}S_4$, and let $\mathsf{m}:=\frac12\sum_{\mu,\nu}\sum_y\gamma_\mu\gamma_\nu\,\mathsf{H}_{(0,\mu),(y,\nu)}$, a finite sum. By Step 2 and translation covariance $\mathsf{N}_2(Y_\gamma)=\sum_x\mathsf{m}_x$, hence $\hat{\mathsf{m}}(l,l)=\bigoplus_e\hat{\mathsf{N}}_{e,2}(l;\gamma)$, and Step 1 with $\mathsf{r}=1$ gives

$$\begin{aligned}\tfrac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\sum_y\langle\partial_{(0,\mu),3}\partial_{(y,\nu),3}S_4\rangle_0&=\tfrac12\,\mathrm{Tr}[\mathsf{m}\,C]\\&=\tfrac12\int\mathrm{tr}\,[\hat{\mathsf{m}}(l,l)\,\hat C(l)]\,d^\circ l=\int_{\mathbb{T}}\mathfrak{s}^{(2)}(l;\gamma)\,d^\circ l.\end{aligned}$$

- Haar term: $\frac12\cdot\frac23|\gamma|^2=\frac13|\gamma|^2=\int\mathfrak{s}^{(5)}$. Ghost tadpole: $\frac12\cdot\frac43\cdot\frac13|\gamma|^2=\frac29|\gamma|^2=\int\mathfrak{s}^{(4)}$, since by Lem. D.22(c) $\int\mathfrak{s}^{(4)}=\sum_{e=\pm1}\frac{4}{12}\sum_\mu\gamma_\mu^2\int|\mathsf{k}_\mu|^2|\mathsf{k}|^{-2}\,d^\circ k$, and $\int|\mathsf{k}_\mu|^2|\mathsf{k}|^{-2}\,d^\circ k=\frac13$.
- Ghost loop. Here $\mathfrak{s}^{(3)}=4\big(\sum_\mu\gamma_\mu\sin k_\mu\big)^2|\mathsf{k}|^{-4}$. By Lem. D.34 the symbols of $y\mapsto D((x,\mu);(x+y,\nu))$ and of $y\mapsto D((x+y,\nu);(x,\mu))$ are $\frac12(1+e^{ik_\nu})\,\overline{\mathsf{k}_\mu}\,|\mathsf{k}|^{-2}$ and its image under $\mu\leftrightarrow\nu$, $k\to-k$. Since $(1+e^{ik_\nu})\,\overline{\mathsf{k}_\nu}=-2i\sin k_\nu$, the ghost loop G, for which $c_G=-8$ and $\mathsf{f}_G=1$, has $\mathsf{Q}_G(0)=\int\frac14(-2i\sin l_\mu)(-2i\sin l_\nu)\,|\mathsf{k}(l)|^{-4}\,d^\circ l$, and $\frac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\,(-8)\,\mathsf{Q}_G(0)=\int\mathfrak{s}^{(3)}$.

Summing up, for small γ we find

$$\frac{1}{2}\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\,\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)=\sum_{i=1}^{5}\int\mathfrak{s}^{(i)}=\frac{1}{2}\sum_{\mu,\nu}(\sum_i c^{(i)}_{\mu\nu})\gamma_\mu\gamma_\nu$$

(Def. D.21(3)); hence $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)+\hat{\mathsf{q}}^\Sigma_{\nu\mu}(0)=2\sum_i c^{(i)}_{\mu\nu}$, and this number vanishes, because by Prop. D.23 the five matrices $c^{(i)}$ add up to zero, $\sum_{i=1}^5 c^{(i)}_{\mu\nu}=0$ for all μ,ν. It remains to show that $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)=\hat{\mathsf{q}}^\Sigma_{\nu\mu}(0)$. Let $\hat\Sigma_{\mu\nu}:=\hat{\mathsf{q}}^\Sigma_{\mu\nu}+\hat{\mathsf{W}}_{\mu\nu}$ be the symbol of $\Sigma_{\mu\nu}$. Since Σ is symmetric and translation covariant, $\Sigma_{\mu\nu}(y)=\Sigma_{\nu\mu}(-y)$, and the uniqueness of Fourier coefficients gives $\hat\Sigma_{\mu\nu}(k)=\hat\Sigma_{\nu\mu}(-k)$ for almost every k. As $\hat{\mathsf{q}}^\Sigma$ is continuous and $\hat{\mathsf{W}}(k)\to 0$ for $k\to 0$, by the symbol estimate in Def. D.27 for $\mathsf{W}\in\mathfrak{K}(4)$, letting $k\to 0$ gives the symmetry, and therefore $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)=0$.

Step 5: the gradient of $\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ at 0. The function $\hat\Sigma_{\mu\nu}$ is integrable on $\mathbb{T}$, with the Fourier coefficients $\Sigma_{\mu\nu}(y)$. Lem. D.25(c) with $c=0$, for which $(x,\mu)^R=(-x-e_\mu,\mu)$, gives

$$\Sigma_{\mu\nu}(y)=\Sigma((-e_\mu,\mu),(-y-e_\nu,\nu))=\Sigma_{\mu\nu}(-y+e_\mu-e_\nu),$$

hence $\hat\Sigma_{\mu\nu}(k)=e^{-ik\cdot(e_\mu-e_\nu)}\,\hat\Sigma_{\mu\nu}(-k)$ for almost every k, by the uniqueness of Fourier coefficients. With $\varphi(k):=\frac{1}{2}(k_\mu-k_\nu)$ the function $\tilde\Sigma(k):=e^{i\varphi(k)}\hat\Sigma_{\mu\nu}(k)$ is therefore even. By the symbol estimate in Def. D.27 for $\mathsf{W}_{\mu\nu}\in\mathfrak{K}(4)$, $\hat{\mathsf{W}}_{\mu\nu}=\hat{\mathsf{W}}^h+O(|k|^2\lg|k|)$ near 0, where $\hat{\mathsf{W}}^h=\sum_G c_G\,\mathsf{f}^h_G\,\mathsf{I}^h_G$ is homogeneous of degree 1 and even: by Lem. D.30 the parity of $\mathsf{f}^h_G\mathsf{I}^h_G$ is $(-1)^{d_G}(-1)^{\omega_G+1}$, where $\omega_G=n_\mathrm{a}+n_\mathrm{b}-3$, and $\omega_G+d_G=1$. Consequently the odd part of the smooth function $e^{i\varphi}\hat{\mathsf{q}}^\Sigma_{\mu\nu}=\tilde\Sigma-e^{i\varphi}\hat{\mathsf{W}}_{\mu\nu}$ equals the odd part of $-e^{i\varphi}\hat{\mathsf{W}}^h+O(|k|^2\lg|k|)$, which is $O(|k|^2\lg|k|)$, because $\hat{\mathsf{W}}^h$ is even and $(e^{i\varphi}-1)\hat{\mathsf{W}}^h=O(|k|^2)$. So the gradient of $e^{i\varphi}\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ at 0 vanishes; as $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(0)=0$ by Step 4, the gradient of $\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ at 0 vanishes as well. Since $\hat{\mathsf{q}}^\Sigma_{\mu\nu}$ is a trigonometric polynomial, this proves $\hat{\mathsf{q}}^\Sigma_{\mu\nu}(k)=O(|k|^2)$ and completes the proof of (a).

Step 6: set-up for (b). Let $\Gamma_{q;\mu\nu}(y,y'):=\Gamma_q((x_q+y,\mu),(x_q+y',\nu))$. In the rest of the proof c_G denotes $\frac{2}{3}$ times the number c_G of Lem. D.36; the reason is that the sums with the numbers c_G of that lemma equal $\frac{3}{2}\Gamma_q$, by eq. D.xxiv, resp. $\frac{3}{2}\gamma_q$, by the definition of γ_q in (b). By the first, second, third and fourth summand we mean the four groups of summands of Γ_q as in eq. D.xxiv. In each summand the derivatives $\partial_{b,a}$ and $\partial_{b',a'}$ of Def. D.21(5) act on two legs of the vertices at q, P, P'; we call these two legs the outer legs. In the terms on the right of eq. D.xxiv the outer legs are the legs which are contracted with the two curl legs $(\mathrm{d}A)(p)$, namely the legs at which the lines K_1, K_2 of Lem. D.36(4′), (4) and (1), resp. the lines K_0, X of Lem. D.36(5), end. An outer leg is either plain or a curl leg; in the second case we speak of an outer curl leg. The symbol of a kernel on $\mathbb{Z}^6$, such as $\Gamma_{q;\mu\nu}$, is defined as in Lem. D.38(a), so that κ_1 is the variable dual to y and κ_2 the variable dual to y'; accordingly we say that the outer leg on which $\partial_{b,a}$ acts, $b=(x_q+y,\mu)$, carries the momentum κ_1, and that the outer leg on which $\partial_{b',a'}$ acts, $b'=(x_q+y',\nu)$, carries the momentum κ_2. By eq. D.xxiv and Lem. D.36, $\Gamma_{q;\mu\nu}$ is a finite sum of kernels on $\mathbb{Z}^6$ with the symbols

$$c_G\,\mathsf{f}_{G,1}(\kappa_1)\,\mathsf{f}_{G,2}(\kappa_2)\,\mathsf{J}_G(\kappa),$$

symmetrised under $(\mu,\kappa_1)\leftrightarrow(\nu,\kappa_2)$. Here the $\mathsf{f}_{G,i}$ are trigonometric polynomials, line symbols of degree 0 if the corresponding outer leg is plain and of degree 1 if it is a curl leg. Further, $\mathsf{J}_G=1$ for the first summand; J_G is the bubble symbol $\mathsf{I}_G(\pm\kappa_2)$ of Y for the second summand, the outer leg at q carrying the momentum κ_1; $\mathsf{J}_G=\mathsf{I}_G(\pm(\kappa_1+\kappa_2))$ for the third summand, both outer legs sitting at P; and $\mathsf{J}_G=\hat{\mathsf{T}}_\mathsf{q}(\pm\kappa)$ for the fourth summand, the Fourier representation being eq. D.xxi. By the definition of γ_q in (b), the symbol $\hat\gamma_{q;\mu\nu}$ is the same sum with J_G replaced by $\tilde{\mathsf{Q}}_G$, where $\tilde{\mathsf{Q}}_G:=1$ for the first summand, which is entirely local (the line between two slots of the vertex at q is a number, a factor of c_G), and otherwise $\tilde{\mathsf{Q}}_G$ is the function $\tilde{\mathsf{Q}}$ of Lem. D.32(iii), resp. $\tilde{\mathsf{Q}}_\mathsf{q}$ of Prop. D.33, taken at the argument of J_G. Indeed, each of the local terms listed in (b) arises from the corresponding

full term of Lem. D.36 by replacing a single kernel by its local kernel, while the lines which end at the outer legs are left as they are. In the first half of T_5 the full term is $K_0(u')\sum_z Y(z)\,X(u'-z)$, and the local term has q_Y in place of Y, in accordance with $X*Y=\mathsf{q}_Y*X+X*Y'$; in T_4 the full term is $\sum_z Y(z)\,(K_1K_2)(u'-z)$, and again Y is replaced by q_Y. In T_1 the substitution $(z,w)\to(u'-z,u'-w)$ turns the full term into

$$\sum_{z,w}\big[K_4(z)\,K_5(w)\,K_3(z-w)\big]\,K_1(u'-z)\,K_2(u'-w),$$

where the bracket, a kernel in (z,w), has the symbol $\hat{\mathsf{T}}_{\mathsf{q}}$ by the second identity of eq. D.xxi; the local term has $\mathsf{q}_{\mathsf{q}}(z,w)$, with the symbol $\tilde{\mathsf{Q}}_{\mathsf{q}}$, in place of the bracket. The first half of $\mathsf{T}_{4'}$ is entirely local and is its own local term. In each case the symbol J_G of the replaced kernel turns into $\tilde{\mathsf{Q}}_G$, while the number c_G, the factors $\mathsf{f}_{G,1}$, $\mathsf{f}_{G,2}$ and the argument of J_G remain the same. Let $\hat\Gamma:=\hat\Gamma_{q;\mu\nu}$ be the sum of the symbols of $\Gamma_{q;\mu\nu}$; in $\hat\Gamma-\hat\gamma_{q;\mu\nu}$ the first summand cancels.

Step 7: counting degrees. Let $d_{G,i}\in\{0,1\}$ be the degree of $\mathsf{f}_{G,i}$ and $d_G:=d_{G,1}+d_{G,2}$, the number of outer curl legs; for G in the second, third and fourth summand let ω_G be the number ω of J_G. We use again that the degree of a line is $1+\#$(its curl legs) (Lem. D.34), and that $\mathsf{a}^{(2)}$ has two curl legs and every monomial of $\mathsf{a}^{(3)}$ one (Def. D.2). In the second summand the vertices at q and P have two curl legs, of which $2-d_G$ lie on the two bubble lines, so that $\omega_G=2+(2-d_G)-3=1-d_G$. In the fourth summand the vertices at q, P, P' have four curl legs, of which $4-d_G$ lie on the three lines of $\hat{\mathsf{T}}_{\mathsf{q}}$, so that $\omega_G=3+(4-d_G)-6=1-d_G$. In the third summand the bubble lines carry the two curl legs at q: for the monomials of $\frac12|\mathfrak{z}_1|^2$ we get $(d_G,\omega_G)=(0,1)$; for the monomials of $\mathsf{a}^{(4)}_P$ with one curl leg we get $(1,1)$ or $(0,2)$, according as this leg is outer or not; for those with four curl legs we get $(2,3)$. Thus $d_G+\omega_G=1$, except for these last three kinds, for which $d_G+\min\{\omega_G,2\}\ge2$ and $\omega_G\ge1$.

Step 8: expansion of $\hat\Gamma-\hat\gamma_{q;\mu\nu}$. We show that for almost every κ with $|\kappa|\le c$, c small,

$$\hat\Gamma-\hat\gamma_{q;\mu\nu}=\hat\Gamma^h+O(|\kappa|^2\lg|\kappa|\,\Lambda(\kappa)),$$
$$\hat\Gamma^h:=\sum_{G:\,d_G+\omega_G=1}c_G\,\mathsf{f}^h_{G,1}\,\mathsf{f}^h_{G,2}\,\mathsf{J}^h_G,\qquad|\hat\Gamma^h|\le C|\kappa|\,\Lambda(\kappa).$$

Here J^h_G is the homogeneous part of J_G (Lem. D.30), taken at the argument of J_G, and $\Lambda\ge1$ is the function Λ of Lem. D.30 for $\hat{\mathsf{T}}_{\mathsf{q}}$. The function $\hat\Gamma^h$ is homogeneous of degree 1 and even, each of its terms having the parity $(-1)^{d_G}(-1)^{\omega_G+1}=1$. Let κ' denote the argument of J_G; then $|\kappa'|\le\sqrt2|\kappa|$. For $d_G+\omega_G\ge2$, Lem. D.32(iii) gives

$$|\mathsf{f}_{G,1}\mathsf{f}_{G,2}\,(\mathsf{J}_G-\tilde{\mathsf{Q}}_G)|\le C|\kappa|^{d_G}\,|\kappa'|^{\min\{\omega_G,2\}}\le C|\kappa|^2.$$

For the other G we have $\omega_G\le1$, hence $\deg\mathsf{Q}_G\le1$ and $\mathsf{Q}_G-\tilde{\mathsf{Q}}_G=O(|\kappa'|^3)$, and Lem. D.30 gives

$$\mathsf{J}_G-\tilde{\mathsf{Q}}_G=\mathsf{J}^h_G+O(|\kappa'|^{\omega_G+1}\lg|\kappa'|\,\Lambda_G(\kappa'))+O(|\kappa'|^3),$$

where Λ_G is bounded for the bubbles and equals Λ for $\hat{\mathsf{T}}_{\mathsf{q}}$. Moreover $\mathsf{f}_{G,i}(\kappa_i)=\mathsf{f}^h_{G,i}(\kappa_i)+O(|\kappa_i|^{d_{G,i}+1})$, and the functions $t\lg t$ and $t^2\lg t$ are increasing on $(0,1]$, their derivatives being $-\log t$ and $t(1-2\log t)$. We go through the summands.

- Third summand, $\omega_G=1$, $d_G=0$: $|\mathsf{J}^h_G|\le C|\kappa'|\le C|\kappa|$, and the errors are $O(|\kappa|^2\lg|\kappa|)$.
- Fourth summand: $\kappa'=\pm\kappa$, $|\mathsf{J}^h_G|\le C|\kappa|^{\omega_G}\Lambda$ and $|\mathsf{f}_{G,1}\mathsf{f}_{G,2}|\le C|\kappa|^{d_G}$; the leading term is $O(|\kappa|\Lambda)$, and the errors are $O(|\kappa|^2\lg|\kappa|\,\Lambda)$.
- Second summand: $\kappa'=\pm\kappa_2$, and $\omega_G=-1$ occurs. We take the factor of the outer leg at P, which carries the momentum κ_2, together with J_G. As $d_{G,2}+\omega_G=1-d_{G,1}\ge0$, we have $|\mathsf{f}^h_{G,2}(\kappa_2)\,\mathsf{J}^h_G|\le C|\kappa_2|^{1-d_{G,1}}$ and $|\mathsf{f}^h_{G,1}\mathsf{f}^h_{G,2}\,\mathsf{J}^h_G|\le C|\kappa_1|^{d_{G,1}}|\kappa_2|^{1-d_{G,1}}\le C|\kappa|$. Replacing $\mathsf{f}_{G,i}$ by

its leading part costs a factor $|\kappa_i|$, and

$$|\mathsf{f}_{G,1}\mathsf{f}_{G,2}|\,|\kappa_2|^{\omega_G+1}\lg|\kappa_2| \le C|\kappa_1|^{d_{G,1}}\,|\kappa_2|^{2-d_{G,1}}\lg|\kappa_2| \le C|\kappa|^2\lg|\kappa|.$$

The symmetrised terms are treated in the same way, with the indices 1 and 2 exchanged.

Step 9: the value of $\hat\gamma_{q;\mu\nu}$ at 0. We have $\hat\gamma_{q;\mu\nu}(0) = \sum c_G\,\mathsf{f}_{G,1}(0)\,\mathsf{f}_{G,2}(0)\,\tilde{\mathsf{Q}}_G(0)$, the sum extending over the G with two plain outer legs, as an outer curl leg gives $\mathsf{f}_{G,i}(0)=0$. For such G in the first summand, $\tilde{\mathsf{Q}}_G = 1$, and the line between two slots of the vertex at q, contained in c_G, is the integral of its symbol (the case $\mathsf{r}=1$ of Step 1). For such G in the other summands, $d_G = 0$, hence $\omega_G \ge 1$ by Step 7, and $\tilde{\mathsf{Q}}_G(0) = \mathsf{Q}_G(0)$ is the integral of the integrand at $\kappa = 0$ (Lem. D.30). By Step 1 the four summands of $\Gamma_q(b,3;b',3)$ have the trace forms

$$\tfrac12\operatorname{Tr}[\mathsf{m}_{bb'}C],\qquad -\tfrac12\operatorname{Tr}[\mathsf{m}_b C\,\mathsf{H}_{b'}C] - (b\leftrightarrow b'),$$
$$-\tfrac12\operatorname{Tr}[\mathsf{h}_q^{(0)}C\,\mathsf{H}_{bb'}C],\qquad \operatorname{Tr}[\mathsf{h}_q^{(0)}C\,\mathsf{H}_bC\,\mathsf{H}_{b'}C],$$

where $\mathsf{m}_{bb'}$, m_b and $\mathsf{H}_{bb'}$ are the Hessians of $\partial\partial'\mathsf{a}_q^{(4)}$, $\partial\mathsf{a}_q^{(3)}$ and $\partial\partial' S_4$ ($\partial=\partial_{b,3}$, $\partial'=\partial_{b',3}$), H_b is the Hessian of $\partial_{b,3}S_3$ (Step 2), and $\mathsf{h}_q^{(0)}$ is the Hessian of $\mathsf{a}_q^{(2)}$. By Step 2,

$$\sum_{b,b'}\tfrac12\gamma_{\mu(b)}\gamma_{\mu(b')}\,\mathsf{m}_{bb'} = \mathsf{h}_q^{(2)}(Y_\gamma),\qquad \sum_b\gamma_{\mu(b)}\,\mathsf{m}_b = \mathsf{h}_q^{(1)}(Y_\gamma),$$
$$\sum_\mu\gamma_\mu\,\hat{\mathsf{H}}_{(0,\mu)}(l,l) = \bigoplus_e\hat{\mathsf{N}}_{e,1}(l;\gamma),$$
$$\sum_{\mu,\nu}\sum_y\tfrac12\gamma_\mu\gamma_\nu\,\hat{\mathsf{H}}_{(0,\mu),(y,\nu)}(l,l) = \bigoplus_e\hat{\mathsf{N}}_{e,2}(l;\gamma),$$

the sums in the first line being finite; and $\hat{\mathsf{h}}(l,l)$ is the matrix of the form $(e^{il\cdot x}\alpha'\mathsf{e}_e, \mathsf{h}\,e^{il\cdot x}\alpha\,\mathsf{e}_e)$, which for $\mathsf{h}=\mathsf{h}_q^{(j)}(Y_\gamma)$ is $\bigoplus_e\hat{\mathsf{h}}_{q,e,j}(l;\gamma)$ by Def. D.21(2). The parts of the Hessians which come from an outer curl leg drop out here, since $\mathrm{d}Y_\gamma = 0$; they belong to the G with $\mathsf{f}_{G,1}(0)\,\mathsf{f}_{G,2}(0)=0$. We use eq. D.xxvii at $\kappa=0$, integrand by integrand, and write $\hat{\mathsf{h}}_j := \bigoplus_e\hat{\mathsf{h}}_{q,e,j}$ and $\hat{\mathsf{N}}_j := \bigoplus_e\hat{\mathsf{N}}_{e,j}$. The four summands contribute to $\frac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\,\hat\gamma_{q;\mu\nu}(0)$ the amounts

$$\tfrac12\int\operatorname{tr}[\hat{\mathsf{h}}_2\hat C],\qquad \tfrac12\cdot2\cdot(-\tfrac12)\int\operatorname{tr}[\hat{\mathsf{h}}_1\hat C\hat{\mathsf{N}}_1\hat C],$$
$$-\tfrac12\int\operatorname{tr}[\hat{\mathsf{h}}_0\hat C\hat{\mathsf{N}}_2\hat C],\qquad \tfrac12\int\operatorname{tr}[\hat{\mathsf{h}}_0\hat C\hat{\mathsf{N}}_1\hat C\hat{\mathsf{N}}_1\hat C]$$

(integrals over $\mathbb{T}$ with respect to $d^\circ l$), and these are $\int\mathfrak{v}_q^{(j)}(l;\gamma)\,d^\circ l$, $j=1,\dots,4$, by Def. D.21(4). Hence, by Def. D.21(4) and Prop. D.24,

$$\tfrac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\,\hat\gamma_{q;\mu\nu}(0) = \tfrac12\sum_{\mu,\nu}\gamma_\mu\gamma_\nu\sum_{j=1}^4\mathsf{g}_{q;\mu\nu}^{(j)} = 0$$

for all small γ. Since γ_q is a symmetric kernel, $\hat\gamma_{q;\mu\nu}(0)$ is symmetric in μ,ν, and therefore $\hat\gamma_{q;\mu\nu}(0)=0$.

Step 10: the gradient of $\hat\gamma_{q;\mu\nu}$ at 0. Lem. D.25(c) with $c=x(q)$, for which $R_cq=q$, gives

$$\Gamma_{q;\mu\nu}(y,y') = \Gamma_{q;\mu\nu}(\mathsf{o}-e_\mu-y,\ \mathsf{o}-e_\nu-y'),\qquad \mathsf{o} := 2(x(q)-x_q),$$

and the uniqueness of Fourier coefficients gives $\hat\Gamma(\kappa) = e^{-2i\varphi(\kappa)}\,\hat\Gamma(-\kappa)$ for almost every κ, with the real linear form $\varphi(\kappa) := \frac12[\kappa_1\cdot(\mathsf{o}-e_\mu)+\kappa_2\cdot(\mathsf{o}-e_\nu)]$; hence $e^{i\varphi}\hat\Gamma$ is even almost everywhere. Since $\hat\Gamma^h$ is even, the odd part of $e^{i\varphi}\hat\Gamma^h$ is $O(|\kappa|^2\Lambda)$. By Step 8 the odd part of the smooth function $e^{i\varphi}\hat\gamma_{q;\mu\nu}$ is therefore $O(|\kappa|^2\lg|\kappa|\,\Lambda)$ almost everywhere near 0. The function Λ is bounded on an open cone. Indeed, for $\hat{\mathsf{T}}_{\mathsf{q}}$ we have $(l_1,l_2,l_3)(\kappa) = (\kappa_1,-\kappa_2,0)$ (Step 2 of the proof of Prop. D.33), so that the differences $l_i(\kappa)-l_j(\kappa)$, $i<j$, are $\kappa_1+\kappa_2$, κ_1 and $-\kappa_2$. Let

$$\mathsf{K} := \{\kappa\in\mathbb{R}^6:\ \min\{|\kappa_1|,|\kappa_2|,|\kappa_1+\kappa_2|\} > \tfrac12|\kappa|\}.$$

This set is open, it is invariant under $\kappa \to t\kappa$ for real $t \neq 0$, and it is not empty, since it contains (v, v) for every $v \in \mathbb{R}^3 \setminus \{0\}$: there $|\kappa| = \sqrt{2}\,|v|$, while $|\kappa_1| = |\kappa_2| = |v|$ and $|\kappa_1 + \kappa_2| = 2|v|$. On K each of the three terms $\log_+(|\kappa|/|l_i(\kappa) - l_j(\kappa)|)$ in the definition of Λ is at most $\log 2$, so that $\Lambda \leq 1 + 3\log 2$ there. On K the odd part is therefore $O(|\kappa|^2 \lg|\kappa|)$ almost everywhere near 0. The odd part of the smooth function $e^{i\varphi}\hat\gamma_{q;\mu\nu}$ differs from its linear part $\kappa \mapsto \kappa\cdot\nabla(e^{i\varphi}\hat\gamma_{q;\mu\nu})(0)$ by $O(|\kappa|^3)$. Hence $|\kappa\cdot\nabla(e^{i\varphi}\hat\gamma_{q;\mu\nu})(0)| \leq C|\kappa|^2\lg|\kappa|$ for almost every, and by continuity for every, $\kappa \in \mathsf{K}$ with $|\kappa| \leq c$. Replacing κ by $t\kappa$ and letting $t \downarrow 0$, we see that this linear function vanishes on K, a non-empty open set, and hence everywhere: the gradient of $e^{i\varphi}\hat\gamma_{q;\mu\nu}$ at 0 vanishes. As $\hat\gamma_{q;\mu\nu}(0) = 0$ by Step 9, the gradient of $\hat\gamma_{q;\mu\nu}$ at 0 vanishes as well. This proves (b). □

D.16. **The six terms, the continuum model, the kernel of degree -5.** In this subsection we introduce six continuum two-loop integrands and a model built from them, with a smooth ultraviolet cut-off on every line. We show that the model has the same parts of degree -5 as the lattice classes of Lem. D.36, that these parts are multiples of $|u|^{-5}$, and we deduce the large-distance asymptotics of $\mathsf{B}_1^{\mathrm{lat}}$ and its consequence for smeared plaquette observables.

Definition D.40 (The six two-loop integrands; the model)**.** *(1) Tensors and variables.* For $k_1, k_2, k_3 \in \mathbb{R}^3$ we put

$$v_2(k_1,k_2)_{\mu\nu} := -(k_1\cdot k_2)\,\delta_{\mu\nu} + k_{1\nu}k_{2\mu},$$
$$v_3(k_1,k_2,k_3)_{\alpha\beta\gamma} := \delta_{\alpha\gamma}(k_1-k_3)_\beta + \delta_{\alpha\beta}(k_2-k_1)_\gamma + \delta_{\beta\gamma}(k_3-k_2)_\alpha.$$

The tensor v_2 is even and the tensor v_3 is odd under the simultaneous change of sign $k_i \to -k_i$ of all arguments, and $|v_2(k_1,k_2)_{\mu\nu}| \leq 2|k_1||k_2|$ for all μ,ν. For $k, l, \mathsf{p} \in \mathbb{R}^3$ we write

$$\mathsf{x}_0 := \mathsf{p}^2, \quad \mathsf{x}_1 := k^2, \quad \mathsf{x}_1' := (k-\mathsf{p})^2, \quad \mathsf{x}_2 := l^2, \quad \mathsf{x}_2' := (l-\mathsf{p})^2, \quad \mathsf{x}_3 := (k-l)^2.$$

(2) The six integrands. With summation over repeated tensor indices we define

$$\begin{aligned}
F_1 &:= -\tfrac12\, v_2(k,\mathsf{p}-k)_{\mu\nu}\, v_2(-l,l-\mathsf{p})_{\rho\sigma} \\
&\qquad \times v_3(-k,l,k-l)_{\mu\rho\alpha}\, v_3(k-\mathsf{p},\mathsf{p}-l,l-k)_{\nu\sigma\alpha} \,/\, \mathsf{x}_1\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_2'\mathsf{x}_3, \\
F_2 &:= -\tfrac12\, v_2(k,\mathsf{p}-k)_{\mu\nu}\, v_2(-k,k-\mathsf{p})_{\rho\nu} \\
&\qquad \times v_3(-k,l,k-l)_{\mu\beta\gamma}\, v_3(k,-l,l-k)_{\rho\beta\gamma} \,/\, \mathsf{x}_1^2\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_3, \\
F_3 &:= -\,v_2(k,\mathsf{p}-k)_{\mu\nu}\, v_2(-k,k-\mathsf{p})_{\rho\nu}\, l_\rho(l-k)_\mu \,/\, \mathsf{x}_1^2\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_3, \\
F_4 &:= -\tfrac12\big[v_2(k,\mathsf{p}-k)_{\mu\mu} v_2(l,\mathsf{p}-l)_{\nu\nu} \\
&\qquad -\tfrac12 v_2(k,\mathsf{p}-k)_{\mu\nu}\big(v_2(l,\mathsf{p}-l)_{\mu\nu} + v_2(l,\mathsf{p}-l)_{\nu\mu}\big)\big] / \mathsf{x}_1\mathsf{x}_1'\mathsf{x}_2\mathsf{x}_2', \\
F_5 &:= v_3(\mathsf{p}-k,l,k-l)_{\alpha\beta\gamma}\, v_3(k,-l,l-k)_{\sigma\beta\gamma}\, v_2(k-\mathsf{p},-k)_{\alpha\sigma} \,/\, \mathsf{x}_1'\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3, \\
F_6 &:= -\tfrac16\, v_3(l,k-l,\mathsf{p}-k)_{\alpha\beta\gamma}\, v_3(-l,l-k,k-\mathsf{p})_{\alpha\beta\gamma} \,/\, \mathsf{x}_2\mathsf{x}_3\mathsf{x}_1'.
\end{aligned} \tag{D.xxviii}$$

We call F_1 the kite, F_2 the gluon loop on a line, F_3 the ghost loop on a line, F_4 the quartic vertex, F_5 the term of $\mathsf{a}^{(3)}$ with S_3, and F_6 the term of $\mathsf{a}^{(3)}$ with $\mathsf{a}^{(3)}$. Each F_Γ, $\Gamma \in \{1,\dots,6\}$, is a rational function of (k,l,p), homogeneous of degree -4, even under $(k,l,\mathsf{p}) \to -(k,l,\mathsf{p})$, and invariant under simultaneous rotations of k, l and p. The *lines of* Γ are the vectors whose squares occur in the denominator of F_Γ, counted with multiplicity: for $\Gamma = 1$ the vectors $k, \mathsf{p}-k, l, \mathsf{p}-l, k-l$; for $\Gamma = 2,3$ the vector k twice and the vectors $\mathsf{p}-k, l, k-l$; for $\Gamma = 4$ the vectors $k, \mathsf{p}-k, l, \mathsf{p}-l$; for $\Gamma = 5$ the vectors $k, \mathsf{p}-k, l, k-l$; for $\Gamma = 6$ the vectors $\mathsf{p}-k, l, k-l$.

(3) The model. Let $\rho_0 := \frac{\pi}{16}$, and let $\rho(\mathsf{q}) := \theta(|\mathsf{q}|/2\rho_0)$ for $\mathsf{q} \in \mathbb{R}^3$, with the function θ of Def. D.27; thus $\rho(\mathsf{q}) = 1$ for $|\mathsf{q}| \leq \rho_0$ and $\rho(\mathsf{q}) = 0$ for $|\mathsf{q}| \geq 2\rho_0$. Let $R_\Gamma(k,l,\mathsf{p})$ be the product of the values of ρ at the lines of Γ. We define

$$\Psi_\Gamma(\mathsf{p}) := \iint R_\Gamma F_\Gamma\, d^\circ k\, d^\circ l \quad (\mathsf{p} \in \mathbb{R}^3 \setminus \{0\}), \qquad \mathsf{T}_\Gamma^{\mathrm{mod}}(u) := 24 \int_{\mathbb{R}^3} e^{i\mathsf{p}\cdot u}\, \Psi_\Gamma(\mathsf{p})\, d^\circ\mathsf{p} \quad (u \in \mathbb{Z}^3),$$

the first integral extending over $\mathbb{R}^3 \times \mathbb{R}^3$.

These integrals converge absolutely. Indeed, every entry of $v_3(k_1, k_2, k_3)$ is bounded by $C \max_i |k_i|$; together with the bound on the entries of v_2 and with the denominators this gives, on every bounded set of points (k, l, p) and with a constant C depending on that set,

$$|F_1| \le \frac{C}{|k|\,|\mathsf{p}-k|\,|l|\,|\mathsf{p}-l|\,|k-l|^2}, \qquad |F_2|,\ |F_3| \le \frac{C(|k|+|l|)}{|k|^2|l|^2|k-l|^2},$$
$$|F_4| \le \frac{C}{|k|\,|\mathsf{p}-k|\,|l|\,|\mathsf{p}-l|}, \qquad |F_5| \le \frac{C}{|k|\,|\mathsf{p}-k|\,|l|^2|k-l|^2}, \qquad |F_6| \le \frac{C}{|\mathsf{p}-k|^2|l|^2|k-l|^2}. \tag{D.xxix}$$

We integrate these majorants first over l and then over k. For v, w in a bounded subset of $\mathbb{R}^3$, and with integrals over a bounded set of l, we have the elementary bounds

$$\int |l-v|^{-2}|l-w|^{-2}\,dl \le C|v-w|^{-1}, \qquad \int |l-v|^{-1}|l-w|^{-2}\,dl \le C \lg|v-w|,$$

with constants depending on these sets; moreover $|l|^{-1}|\mathsf{p}-l|^{-1} \le |\mathsf{p}|^{-1}(|l|^{-1}+|\mathsf{p}-l|^{-1})$, since $|\mathsf{p}| \le |l|+|\mathsf{p}-l|$, and $|k|+|l| \le 2|k|+|k-l|$. For $\Gamma = 1$ the majorant is at most $C|\mathsf{p}|^{-2}(|k|^{-1}+|\mathsf{p}-k|^{-1})(|l|^{-1}+|\mathsf{p}-l|^{-1})|k-l|^{-2}$; its integral over l is at most $C|\mathsf{p}|^{-2}(|k|^{-1}+|\mathsf{p}-k|^{-1})(\lg|k|+\lg|k-\mathsf{p}|)$, which is integrable over bounded sets of k, with a bound uniform for p in a bounded set. For $\Gamma = 4$ the same argument applies without the factor $|k-l|^{-2}$. For $\Gamma = 2, 3$ the majorant is at most $2C|k|^{-1}|l|^{-2}|k-l|^{-2} + C|k|^{-2}|l|^{-2}|k-l|^{-1}$, whose integral over l is at most $C|k|^{-2}\lg|k|$, a function integrable over bounded sets of k. For $\Gamma = 5$ the integral over l is at most $C|k|^{-2}|\mathsf{p}-k|^{-1}$, and the subsequent integral over k is at most $C\lg|\mathsf{p}|$. For $\Gamma = 6$ the integral over l is at most $C|k|^{-1}|\mathsf{p}-k|^{-2}$, and again the integral over k is at most $C\lg|\mathsf{p}|$. Hence, for p in a bounded set, the integral of each majorant over a bounded set of (k, l) is at most $C|\mathsf{p}|^{-2}$, since 1 and $\lg|\mathsf{p}|$ are bounded by $C|\mathsf{p}|^{-2}$ there.

Where $R_\Gamma \neq 0$, every line of Γ has modulus less than $2\rho_0$. Hence $|l| < 2\rho_0$ and $|k| < 4\rho_0$ there (by the lines l and k, and for $\Gamma = 6$ by the lines l and $k-l$), and $|\mathsf{p}| \le |\mathsf{p}-k| + |k| < 6\rho_0$, by the line $\mathsf{p}-k$, which occurs for every Γ. Consequently the integral defining $\Psi_\Gamma(\mathsf{p})$ converges absolutely, $|\Psi_\Gamma(\mathsf{p})| \le C|\mathsf{p}|^{-2}$, and Ψ_Γ is integrable, radial (by the rotation invariance of F_Γ and of R_Γ, and the substitution $(k, l) \to (Ok, Ol)$), and equal to 0 for $|\mathsf{p}| \ge 6\rho_0 = \frac{3\pi}{8}$; in particular $\mathsf{T}^{\mathrm{mod}}_\Gamma(u)$ is well defined. ∎

In order to compare the model with the elementary terms of Lem. D.36, we write these terms, too, as integrals over momenta.

Lemma D.41 (Momentum-space forms)**.** (a) Let $K_1, \dots, K_5$ be elementary lines as in Prop. D.33, with symbols $\sigma_1, \dots, \sigma_5$, and let T be their kite. Then, for every $u \in \mathbb{Z}^3$,

$$\mathsf{T}(u) = \int_{\mathbb{T}} e^{i\mathsf{p}\cdot u} \Big[\iint_{\mathbb{T}^2} \sigma_1(k)\,\sigma_2(\mathsf{p}-k)\,\sigma_3(k-l)\,\sigma_4(l)\,\sigma_5(\mathsf{p}-l)\,d^\circ k\,d^\circ l \Big] d^\circ \mathsf{p},$$

and the triple integral converges absolutely.

(b) If two kernels have symbols $\sigma_\mathrm{a}, \sigma_\mathrm{b} \in L^1(\mathbb{T})$, then their product is the kernel whose symbol is the convolution $k \mapsto \int_{\mathbb{T}} \sigma_\mathrm{a}(k-l)\,\sigma_\mathrm{b}(l)\,d^\circ l$ of the two symbols. If $X, Y \in \ell^2(\mathbb{Z}^3)$ are kernels with symbols $\hat{X}, \hat{Y} \in L^1(\mathbb{T})$, then $X * Y$ (Lem. D.32) is the kernel with the symbol $\hat{X}\hat{Y}$. ∎

Proof. We begin with (b). Let $K_\mathrm{a}, K_\mathrm{b}$ be the two kernels. The function $(k, l) \mapsto \sigma_\mathrm{a}(k)\sigma_\mathrm{b}(l)$ is integrable over $\mathbb{T}^2$; by Fubini's theorem and the substitution $k \to k-l$ on $\mathbb{T}$,

$$K_\mathrm{a}(x)\,K_\mathrm{b}(x) = \iint e^{i(k+l)\cdot x}\,\sigma_\mathrm{a}(k)\,\sigma_\mathrm{b}(l)\,d^\circ k\,d^\circ l = \int_{\mathbb{T}} e^{ik\cdot x} \Big[\int_{\mathbb{T}} \sigma_\mathrm{a}(k-l)\,\sigma_\mathrm{b}(l)\,d^\circ l \Big] d^\circ k,$$

and the bracket is an integrable function of k. For the second assertion we note that $X \in \ell^2(\mathbb{Z}^3)$ is, by Plancherel's theorem, the sequence of Fourier coefficients of a function in $L^2(\mathbb{T})$, which

coincides with $\hat X$ by the uniqueness of Fourier coefficients; thus $\hat X \in L^2(\mathbb{T})$, and likewise $\hat Y \in L^2(\mathbb{T})$. The kernel $z \mapsto Y(u-z)$ has the symbol $k \mapsto e^{-ik\cdot u}\hat Y(-k)$, and Plancherel's theorem in the form $\sum_z a(z)b(z) = \int_{\mathbb{T}} \hat a(-k)\,\hat b(k)\,d^\circ k$, for square summable kernels a, b with symbols $\hat a, \hat b$, gives $(X * Y)(u) = \int_{\mathbb{T}} e^{ik\cdot u}\hat X(k)\hat Y(k)\,d^\circ k$.

We turn to (a). *Step 1.* For $i \neq 3$ the degree n_i is at least 2, so that $|\sigma_i(\mathsf{q})| \le C|\mathsf{q}|^{-1}$ on $[-\pi,\pi]^3$ (Def. D.27); hence $\sigma_i \in L^2(\mathbb{T})$ for $i \neq 3$, while $\sigma_3 \in L^1(\mathbb{T})$. The functions

$$\mathsf{A}(v) := \int_{\mathbb{T}} \sigma_1(v+\mathsf{q})\,\sigma_4(\mathsf{q})\,e^{iu\cdot\mathsf{q}}\,d^\circ\mathsf{q}, \qquad \mathsf{B}(v) := \int_{\mathbb{T}} \sigma_2(v+\mathsf{q})\,\sigma_5(\mathsf{q})\,e^{iu\cdot\mathsf{q}}\,d^\circ\mathsf{q}$$

are bounded, by the Cauchy–Schwarz inequality, and continuous, since translations act continuously on $L^2(\mathbb{T})$. Consequently

$$\Phi(\eta_1,\eta_2) := \int_{\mathbb{T}} \sigma_3(\mathsf{q})\,\mathsf{A}(\eta_1+\mathsf{q})\,\mathsf{B}(\eta_2-\mathsf{q})\,d^\circ\mathsf{q}$$

is bounded and continuous on $\mathbb{T}^2$, by dominated convergence, and the integral over $(\mathsf{q}_3,\mathsf{q}_4,\mathsf{q}_5) \in \mathbb{T}^3$ which results from inserting the definitions of A and B (with q_3 in place of q and $\mathsf{q}_4, \mathsf{q}_5$ the variables of A, B) converges absolutely.

Step 2. Let $\phi \in C_c^\infty(\mathbb{R}^3)$ with support in $\{|x| \le 1\}$ and $\phi(0) = 1$, and let $\phi_R := \phi(\cdot/R)$. By Prop. D.33, the double sum defining $\mathsf{T}(u)$ converges absolutely; by dominated convergence, $\mathsf{T}(u) = \lim_{R\to\infty} \mathsf{T}_R(u)$, where $\mathsf{T}_R(u)$ denotes the same sum with the additional factor $\phi_R(z)\phi_R(w)$. For fixed R we insert $K_i(y) = \int_{\mathbb{T}} e^{i\mathsf{q}_i\cdot y}\sigma_i(\mathsf{q}_i)\,d^\circ\mathsf{q}_i$. The sums over z and w are then finite, and by Fubini's theorem they produce the factor

$$\check\phi_R(-(\mathsf{q}_1-\mathsf{q}_3-\mathsf{q}_4))\,\check\phi_R(-(\mathsf{q}_2+\mathsf{q}_3-\mathsf{q}_5))\,e^{iu\cdot(\mathsf{q}_4+\mathsf{q}_5)}.$$

We pass to the variables $\eta_1 := \mathsf{q}_1-\mathsf{q}_3-\mathsf{q}_4$, $\eta_2 := \mathsf{q}_2+\mathsf{q}_3-\mathsf{q}_5$, q_3, q_4, q_5; this is a linear change of variables with integer matrix of determinant ±1, hence a measure preserving bijection of $\mathbb{T}^5$, and it gives

$$\mathsf{T}_R(u) = \iint_{\mathbb{T}^2} \check\phi_R(-\eta_1)\,\check\phi_R(-\eta_2)\,\Phi(\eta_1,\eta_2)\,d^\circ\eta_1\,d^\circ\eta_2.$$

Here $\int_{\mathbb{T}} \check\phi_R(-\eta)\,d^\circ\eta = \phi_R(0) = 1$, and $|\check\phi_R(\eta)| \le CR^3\langle R\eta\rangle^{-10}$ by Lem. D.29, applied with $D = 3$ to the restriction of ϕ_R to $\mathbb{Z}^3$; this function vanishes off $\{|\zeta| \le R\}$, and its norm $\|\phi_R\|_{(R)}$ is at most $\|\phi\|_{C^{10}}$, a difference being an integral of a derivative. Thus the functions $\eta \mapsto \check\phi_R(-\eta)$ have integral 1, bounded L^1 norms, and integrals over $\{|\eta| \ge \delta\}$ tending to 0 as $R \to \infty$, for every $\delta > 0$. As Φ is bounded and continuous at $(0,0)$, $\mathsf{T}_R(u) \to \Phi(0,0)$.

Step 3. In the triple integral $\Phi(0,0)$ we change variables to $l = \mathsf{q}_4$, $k = \mathsf{q}_3 + \mathsf{q}_4$, $\mathsf{p} = \mathsf{q}_4 + \mathsf{q}_5$, again a measure preserving bijection of $\mathbb{T}^3$. Then $\mathsf{q}_3 = k - l$, $\mathsf{q}_5 = \mathsf{p} - l$ and $\mathsf{q}_5 - \mathsf{q}_3 = \mathsf{p} - k$, so that the integrand becomes $\sigma_1(k)\,\sigma_2(\mathsf{p}-k)\,\sigma_3(k-l)\,\sigma_4(l)\,\sigma_5(\mathsf{p}-l)\,e^{i\mathsf{p}\cdot u}$. The triple integral converges absolutely by Step 1, and Fubini's theorem yields the formula of (a). □

The next lemma computes the weights with which the leading parts of the elementary terms add up to the six integrands of Def. D.40.

Lemma D.42 (The weights of the six terms)**.** Let $\tau_{\mathrm{gf}} = 1$, and let $p = p_{[\iota]}(x_p)$, $q = p_{[\iota']}(x_q)$. To an elementary term G of Lem. D.36, with its lines named as there, we attach its *leading integrand* $\mathsf{F}_G(k,l,\mathsf{p})$, where σ^h denotes the leading part of the symbol of the line in question:

$$\begin{aligned}
&G \in \mathfrak{G}_1: && \sigma_1^h(k)\,\sigma_2^h(\mathsf{p}-k)\,\sigma_3^h(k-l)\,\sigma_4^h(l)\,\sigma_5^h(\mathsf{p}-l);\\
&G \in \mathfrak{G}_4,\ (n_X,n_Y) = (4,4): && \sigma_1^h(k)\,\sigma_2^h(\mathsf{p}-k)\,\sigma_3^h(l)\,\sigma_4^h(\mathsf{p}-l);\\
&G \in \mathfrak{G}_5: && \sigma_0^h(\mathsf{p}-k)\,\sigma_X^h(k)\,\sigma_{\mathrm{a}}^h(k-l)\,\sigma_{\mathrm{b}}^h(l);\\
&G \in \mathfrak{G}_6: && \sigma_1^h(\mathsf{p}-k)\,\sigma_2^h(k-l)\,\sigma_3^h(l);\\
&G \in \mathfrak{G}_{\mathrm{I}}(\mu,\nu): && \sigma_{\Pi}^h(\mathsf{p}-k)\,\sigma^h_{\rightarrow,\iota\mu}(k)\,\mathsf{f}_G^h(k)\,\sigma_{\mathrm{a}}^h(k-l)\,\sigma_{\mathrm{b}}^h(l)\,\sigma^h_{\leftarrow,\nu\iota'}(k);
\end{aligned}$$

here σ_Π, $\sigma_\rightarrow$ and $\sigma_\leftarrow$ are the symbols of $y \mapsto \Pi\big(p_{[\iota]}(0), p_{[\iota']}(y)\big)$, of $K^{\rightarrow}_{\iota\mu}$ and of $K^{\leftarrow}_{\nu\iota'}$. Let $\mathfrak{G}_2 = \mathfrak{G}_2(\iota,\iota')$, resp. $\mathfrak{G}_3 = \mathfrak{G}_3(\iota,\iota')$, be the set of the triples (μ,ν,G) such that $G \in \mathfrak{G}_{\mathrm{I}}(\mu,\nu)$ is a term of $-\langle \partial S_3; \partial' S_3\rangle_0$, resp. of the ghost loop, in eq. D.xxvi; such a triple carries the coefficient $-3\,c_G$, which takes the place of c_G in the sums below. Then, with sums over $\iota,\iota' \in \{1,2,3\}$ and over $G \in \mathfrak{G}_\Gamma(\iota,\iota')$, where for $\Gamma = 4$ only the terms with $(n_X, n_Y) = (4,4)$ are taken,

$$\sum_{\iota,\iota'}\sum_{G} c_G\,\mathsf{F}_G = 24\,F_\Gamma \quad (\Gamma = 1,2,3,4,6), \qquad \sum_{\iota,\iota'}\sum_{G\in\mathfrak{G}_5} c_G\,\mathsf{F}_G = 12\,F_5.$$

■

Proof. Step 1 (Feynman rules). Let $K(y) = \int e^{i\mathsf{q}\cdot y}\sigma(\mathsf{q})\,d^\circ\mathsf{q}$ be a line, where y is the base point of its second leg minus that of its first leg; we say that the line carries the momentum q from the first vertex to the second. The momentum entering a vertex through this line is then q at the second vertex and $-\mathsf{q}$ at the first. Leading parts of line symbols are additive and multiplicative (Def. D.27); hence the sum over the monomials of a vertex may be carried out on the leading parts. By Lem. D.34 at $\tau_{\mathrm{gf}} = 1$, every line contributes a factor $\delta_{\mu\nu}|\mathsf{q}|^{-2}$, where at a plain leg the index of the line is the direction of the bond of that leg; a curl leg at a plaquette $p_{\alpha\beta}$ contributes the factor $i(\mathsf{q}'_\alpha\delta_{\nu\beta} - \mathsf{q}'_\beta\delta_{\nu\alpha})$, where q' is the momentum entering the vertex; and shifts of base points contribute the factor 1, the leading part of $e^{ik\cdot y_0}$.

Step 2 (the vertex $\mathsf{a}^{(2)}$). Two curl legs at $p_{[\iota]}$, with entering momenta k_1, k_2 and line indices μ,ν, give after summation over ι, that is, over $\alpha < \beta$,

$$\begin{aligned} i^2 \sum_{\alpha<\beta} &(k_{1\alpha}\delta_{\mu\beta} - k_{1\beta}\delta_{\mu\alpha})(k_{2\alpha}\delta_{\nu\beta} - k_{2\beta}\delta_{\nu\alpha}) \\ &= -\big[(k_1\cdot k_2)\,\delta_{\mu\nu} - k_{1\nu}k_{2\mu}\big] = v_2(k_1,k_2)_{\mu\nu}. \end{aligned}$$

Step 3 (the vertex $\mathsf{a}^{(3)}$). By Lem. D.5, $\mathsf{a}^{(3)}_P[\![X,Y,Z]\!]$ is the sum of t_P over the six orders of X,Y,Z. Let $P = p_{\mu\nu}$. To leading order, $(Y(b_1),\dots,Y(b_4))$ is replaced by $(Y_\mu, Y_\nu, Y_\mu, Y_\nu)$; hence

$$\sum_{i<j} s_i s_j\, Y(b_i)\times Z(b_j) \to Y_\mu \times Z_\nu - Y_\nu \times Z_\mu - Y_\mu \times Z_\mu - Y_\nu \times Z_\nu,$$

and, by the antisymmetry of the cross product,

$$\mathsf{t}_P(X,Y,Z) + \mathsf{t}_P(X,Z,Y) \to -2\big\langle (\mathrm{d}X)_{\mu\nu},\, Y_\mu \times Z_\nu - Y_\nu \times Z_\mu\big\rangle.$$

Both factors are antisymmetric in $\mu\nu$. Summing over $\mu < \nu$ and inserting $(\mathrm{d}X)_{\mu\nu} \to i(k_{1\mu}X_\nu - k_{1\nu}X_\mu)$, we obtain

$$-2i\sum_{\mu,\nu} k_{1\mu}\langle X_\nu,\, Y_\mu \times Z_\nu - Y_\nu \times Z_\mu\rangle.$$

For legs of colours a,b,c, with indices α,β,γ and entering momenta k_1,k_2,k_3, this equals

$$-2i\,\epsilon_{abc}\big[\delta_{\alpha\gamma}k_{1\beta} - \delta_{\alpha\beta}k_{1\gamma}\big].$$

We add the terms in which the second, resp. the third, leg is the curl leg; they arise by cyclic permutation, under which ϵ_{abc} is invariant. The leading part of $\sum_\iota \mathsf{a}^{(3)}_{p_{[\iota]}}[\![\cdot,\cdot,\cdot]\!]$ is therefore

$$-2i\,\epsilon_{abc}\,v_3(k_1,k_2,k_3)_{\alpha\beta\gamma}.$$

Step 4 ($\Gamma = 1$). By Lem. D.41(a) the five lines $1,\dots,5$ of the kite carry k (from p to P), $\mathsf{p}-k$ (from p to P'), $k-l$ (from P to P'), l (from P to q) and $\mathsf{p}-l$ (from P' to q). The entering momenta are $(-k, k-\mathsf{p})$ at p; $(k,-l,l-k)$ at P, for the legs towards p, q, P'; $(\mathsf{p}-k, l-\mathsf{p}, k-l)$ at P'; and

$(l, \mathsf{p}-l)$ at q. At both cubic vertices the colours are (a, a', c), and $\sum_{a,a',c} \epsilon^2_{aa'c} = 6$. With the factor $\frac{1}{2}$ of T_1 in Lem. D.5 we obtain

$$\tfrac{1}{2}(-2i)^2 \cdot 6 \cdot v_2(-k, k-\mathsf{p})_{\mu\nu}\, v_2(l, \mathsf{p}-l)_{\rho\sigma} \\ \times v_3(k, -l, l-k)_{\mu\rho\alpha}\, v_3(\mathsf{p}-k, l-\mathsf{p}, k-l)_{\nu\sigma\alpha} \,/\, \mathsf{x}_1\mathsf{x}'_1\mathsf{x}_2\mathsf{x}'_2\mathsf{x}_3,$$

and $\frac{1}{2}(-4)\cdot 6 = 24\cdot(-\frac{1}{2})$. Since v_2 is even, and the product of the two factors v_3 is even under the change of sign of all their arguments, this is $24F_1$.

Step 5 ($\Gamma = 6$). The factor is $\frac{1}{6}(-2i)^2\cdot 6 = 24\cdot(-\frac{1}{6})$. The entering momenta are $-(\mathsf{p}-k)$, $-(k-l)$, $-l$ at p and the opposite ones at q. The tensor v_3 is invariant under the cyclic permutations of its arguments together with its indices, and it changes sign under a transposition, like ϵ_{abc}; this gives $24F_6$.

Step 6 ($\Gamma = 5$). The line K_0 carries $\mathsf{p}-k$ from p to q, the line X carries k from p to P, and K_{a}, K_{b} carry $k-l$ and l from P to q. The factor is $-\frac{1}{2}(-2i)^2\cdot 6 = 12$, with the factor $-\frac{1}{2}$ of T_5, and the tensors are

$$v_2(k-\mathsf{p}, -k)_{\alpha\sigma}\, v_3(\mathsf{p}-k, l, k-l)_{\alpha\beta\gamma}\, v_3(k, -l, l-k)_{\sigma\beta\gamma};$$

with the factors $|\mathsf{q}|^{-2}$ of the four lines this is $12F_5$.

Step 7 ($\Gamma = 2$). By Lem. D.5, applied to two quadratic polynomials,

$$-\langle \partial_{b,3} S_3; \partial_{b',3} S_3\rangle_0 = -\tfrac{1}{2}\sum \mathsf{a}^{(3)}_P[\![\delta_{b,3}, \delta_{\mathsf{i}}, \delta_{\mathsf{i}'}]\!]\, \mathsf{a}^{(3)}_{P'}[\![\delta_{b',3}, \delta_{\mathsf{j}}, \delta_{\mathsf{j}'}]\!]\, C(\mathsf{i},\mathsf{j})\, C(\mathsf{i}',\mathsf{j}'),$$

the sum extending over P, P' and over the indices $\mathsf{i}, \mathsf{i}', \mathsf{j}, \mathsf{j}'$. Here $\sum_{c,c'} \epsilon^2_{3cc'} = 2$, and together with the factor -3 of T_{I} the total factor is $-3\cdot(-\frac{1}{2})(-2i)^2\cdot 2 = -12 = 24\cdot(-\frac{1}{2})$. The line Π carries $\mathsf{p}-k$ from p to q, and the chain carries k. The curl legs give $v_2(-k, k-\mathsf{p})_{\mu\nu}$ at p and $v_2(k, \mathsf{p}-k)_{\rho\nu}$ at q, where ν is the index of the line Π, and the cubic vertices give $v_3(k, -l, l-k)_{\mu\beta\gamma}\, v_3(-k, l, k-l)_{\rho\beta\gamma}$. With the factors of the five lines, whose squares give the denominator $\mathsf{x}_1^2\mathsf{x}'_1\mathsf{x}_2\mathsf{x}_3$, and with the parities of v_2 and v_3, this is $24F_2$.

Step 8 ($\Gamma = 3$). The ghost loop enters Σ through the kernel $-8\, D(b;b')\, D(b';b)$. By Lem. D.34, the leading parts of the symbols of $y \mapsto D((x,\mu);(x+y,\rho))$ and of $y \mapsto D((x+y,\rho);(x,\mu))$ are $-i\mathsf{q}_\mu|\mathsf{q}|^{-2}$ and $+i\mathsf{q}_\rho|\mathsf{q}|^{-2}$. With $\mathsf{q} = k-l$, resp. $\mathsf{q} = l$, and with the factor -3 of T_{I} we obtain

$$-3\,(-8)\,(k-l)_\mu\, l_\rho \,/\, \mathsf{x}_3\mathsf{x}_2 = 24\cdot(-1)\, l_\rho (l-k)_\mu \,/\, \mathsf{x}_2\mathsf{x}_3;$$

together with the factors at p and q and on the lines Π, $K^{\rightarrow}$, $K^{\leftarrow}$, as in Step 7, this is $24F_3$.

Step 9 ($\Gamma = 4$). By Lem. D.36(4), the terms $G \in \mathfrak{G}_4$ with $(n_X, n_Y) = (4,4)$ are exactly those which arise from the monomials of $\frac{1}{2}|\mathfrak{z}_1|^2$ in the standard writing of $\mathsf{a}^{(4)}_P$ (Lem. D.5); here, by Lem. D.1 in Lem. D.1, with $A_i := A(b_i(P))$,

$$\mathsf{a}^{(4)}_P = \tfrac{1}{2}|\mathfrak{z}_1|^2 + \langle \mathrm{d}A(P), \mathfrak{z}_2\rangle - \tfrac{1}{24}|\mathrm{d}A(P)|^4, \qquad \mathfrak{z}_1 = -\sum_{i<j} s_i s_j\, A_i \times A_j.$$

Thus the vertex of the class $\Gamma = 4$ is $\frac{1}{2}|\mathfrak{z}_1|^2$. At $P = p_{\alpha\beta}$ the replacement of Step 3, $(A_1, \dots, A_4) \to (A_\alpha, A_\beta, A_\alpha, A_\beta)$, gives to leading order

$$\mathfrak{z}_1 \to -(A_\alpha \times A_\beta - A_\beta \times A_\alpha - A_\alpha \times A_\alpha - A_\beta \times A_\beta) = -2A_\alpha \times A_\beta;$$

since $\frac{1}{2}|2A_\alpha \times A_\beta|^2 = 2|A_\alpha \times A_\beta|^2$ and the terms with $\alpha = \beta$ vanish, the sum over ι, that is, over $\alpha < \beta$, gives $\sum_\iota \frac{1}{2}|\mathfrak{z}_1|^2 \to \mathsf{Q}(A) := \sum_{\alpha,\beta} |A_\alpha \times A_\beta|^2$, where $|Z|^2 := \langle Z, Z\rangle$ with the complex bilinear extension of $\langle\cdot,\cdot\rangle$. For $X = \xi\mathsf{e}_a$ and $Y = \eta\mathsf{e}_{a'}$ with $\xi, \eta \in \mathbb{C}^3$, that is, $X_\alpha = \xi_\alpha \mathsf{e}_a$ and $Y_\alpha = \eta_\alpha \mathsf{e}_{a'}$, we

have, with the bilinear dot product on $\mathbb{C}^3$,

$$\mathsf{Q}(sX+tY) = s^2t^2(1-\delta_{aa'})\sum_{\alpha,\beta}(\xi_\alpha\eta_\beta-\xi_\beta\eta_\alpha)^2,$$

$$\sum_{\alpha,\beta}(\xi_\alpha\eta_\beta-\xi_\beta\eta_\alpha)^2 = 2\big[(\xi\cdot\xi)(\eta\cdot\eta)-(\xi\cdot\eta)^2\big].$$

Polarising, with $X_i=\xi_i\mathsf{e}_a$ and $Y_j=\eta_j\mathsf{e}_{a'}$ $(i,j\in\{1,2\})$, we find

$$\mathsf{Q}[\![X_1,X_2,Y_1,Y_2]\!] = 4(1-\delta_{aa'})\cdot 2\big[(\xi_1\cdot\xi_2)(\eta_1\cdot\eta_2)-\tfrac12(\xi_1\cdot\eta_1)(\xi_2\cdot\eta_2)-\tfrac12(\xi_1\cdot\eta_2)(\xi_2\cdot\eta_1)\big].$$

Since $\sum_{a,a'}(1-\delta_{aa'})=6$, the factor is $-\frac14\cdot4\cdot2\cdot6=-12=24\cdot(-\frac12)$, with the factor $-\frac14$ of T_4. The pairs of legs contracted with p and with q give $\xi_{1\mu}\xi_{2\nu}\to v_2(k,\mathsf{p}-k)_{\mu\nu}/\mathsf{x}_1\mathsf{x}_1'$ and $\eta_{1\mu}\eta_{2\nu}\to v_2(l,\mathsf{p}-l)_{\mu\nu}/\mathsf{x}_2\mathsf{x}_2'$, and the result is $24F_4$. □

We now show that the model $\mathsf{T}_\Gamma^{\mathrm{mod}}$ of Def. D.40 has, up to a part of degree -4, the same part of degree -5 as the corresponding lattice terms. For $K\in\mathfrak{K}(n)$, write K^h for its spatial leading kernel in Def. D.27; it is smooth away from zero and homogeneous of degree $-n$.

Lemma D.43 (The model has the degree -5 parts of the lattice classes)**.** Let $\tau_{\mathrm{gf}}=1$. For $G\in\mathfrak{G}_1\cup\mathfrak{G}_4\cup\mathfrak{G}_5\cup\mathfrak{G}_6$ let U_G be the function of degree -5 attached to G in Lem. D.37. For $(\mu,\nu,G)\in\mathfrak{G}_2\cup\mathfrak{G}_3$ let

$$\mathsf{U}_G := \Pi^h\cdot\big(K^{\rightarrow}_{\iota\mu}*\mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}}-\mathsf{q}_G]*K^{\leftarrow}_{\nu\iota'}\big)^h, \qquad \Pi^h := \mathcal{F}^{-1}_{\neq0}\sigma^h_\Pi,$$

where the chain lies in $\mathfrak{K}(2)$ by Lem. D.39(a) and two applications of Lem. D.32(ii). Let

$$\mathsf{U}_\Gamma := \sum_{\iota,\iota'}\sum_{G\in\mathfrak{G}_\Gamma} c_G\,\mathsf{U}_G \quad (\Gamma\neq5),$$

$$\mathsf{U}_5(u) := \sum_{\iota,\iota'}\sum_{G\in\mathfrak{G}_5} c_G\big[\mathsf{U}_G(u)+\mathsf{U}_G(-u)\big],$$

with the coefficients $-3\,c_G$ of Lem. D.42 for $\Gamma=2,3$. Then for each $\Gamma=1,\dots,6$ there are a continuous function I_Γ on $\mathbb{R}^3\setminus\{0\}$, homogeneous of degree -4, and a pure constant C such that

$$\big|\mathsf{T}^{\mathrm{mod}}_\Gamma(u)-\mathsf{I}_\Gamma(u)-\mathsf{U}_\Gamma(u)\big| \le C|u|^{-6}\lg|u| \qquad (u\in\mathbb{Z}^3\setminus\{0\}).$$

■

Proof. Step 0 (model lines). For a line whose symbol has the leading part σ^h, of degree s, let σ^{mod} be the periodic continuation of $\rho\sigma^h$, with ρ of Def. D.40(3). As $\rho\in C^\infty_c(\mathbb{R}^3)$, $\operatorname{supp}\rho\subset\{|k|\le2\rho_0\}\subset\{|k|<\pi\}$ and $\rho=1$ near 0, the function σ^{mod} is a line symbol with the same leading part (Def. D.27), and its parity is exact: $\sigma^{\mathrm{mod}}(-k)=(-1)^s\sigma^{\mathrm{mod}}(k)$. Its kernel $K^{\mathrm{mod}}(x)=\int_{\mathbb{R}^3}e^{ik\cdot x}\rho\,\sigma^h\,d^\circ k$ satisfies $K^{\mathrm{mod}}=K^h+O(|x|^{-N})$ for every N, with $K^h=\mathcal{F}^{-1}_{\neq0}\sigma^h$, by Lem. D.26. For $(\mu,\nu,G)\in\mathfrak{G}_2\cup\mathfrak{G}_3$ the elementary term is, by Lem. D.36(I), the value at $u=u'$ of the function

$$u\mapsto K_\Pi(u)\,(K^{\rightarrow}_{\iota\mu}*\mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}}]*K^{\leftarrow}_{\nu\iota'})(u), \qquad K_\Pi(y):=\Pi(p_{[\iota]}(0),p_{[\iota']}(y)) \quad (y\in\mathbb{Z}^3);$$

here K_Π is the elementary line of degree 3 of Lem. D.34, with the symbol σ_Π of Lem. D.42, and $\Pi(p,q)=K_\Pi(u')$ by translation invariance. For an elementary term G let $G^{\mathrm{mod}}(u)$ be the same expression with every line K replaced by K^{mod}, in particular K_Π by K^{mod}_Π; for G of type (I) we moreover drop the operator $\mathsf{f}_G(\nabla)$ and replace $K^{\rightarrow}$ by the elementary line K'_1 whose symbol is the periodic continuation of $\rho\,\sigma^h_{\rightarrow}\mathsf{f}^h_G$; it has degree $2+d_G$.

Step 1 (the model in momentum space). We show that $\mathsf{T}^{\mathrm{mod}}_\Gamma=\sum c_G\,G^{\mathrm{mod}}$, the sum extending over ι,ι' and G as in Lem. D.42, where for $\Gamma=5$ the term $G^{\mathrm{mod}}(u)$ is replaced by $G^{\mathrm{mod}}(u)+G^{\mathrm{mod}}(-u)$. By Lem. D.41,

$$G^{\mathrm{mod}}(u)=\int_{\mathbb{T}}e^{i\mathsf{p}\cdot u}\Big[\iint_{\mathbb{T}^2}\prod\sigma^{\mathrm{mod}}\,d^\circ k\,d^\circ l\Big]d^\circ\mathsf{p},$$

where the product extends over the lines of G and the arguments of the symbols are those in F_G. For the kites this is assertion (a) of that lemma, the model lines having the degrees of the lattice lines. For the other types we apply assertion (b) to the products and convolutions of which G^{mod} is built, one at a time, and we verify its hypotheses. A model line of degree n is bounded by $C\langle x\rangle^{-n}$ (Lem. D.3), and its symbol is integrable. A kernel in $\ell^2(\mathbb{Z}^3)$ with an integrable symbol has its symbol in $L^2(\mathbb{T})$, as in the proof of Lem. D.41(b); hence the symbol of the convolution of two such kernels, a product of two functions in $L^2(\mathbb{T})$, is integrable, and so is the symbol of the product of two kernels with integrable symbols, a convolution of two integrable functions.

- For $G \in \mathfrak{G}_6$, G^{mod} is a product of three model lines, and the rule for products is used twice.
- For $G \in \mathfrak{G}_4$ with $(n_X, n_Y) = (4,4)$, $G^{\mathrm{mod}} = X * Y$, where X and Y are products of two model lines of degree 2. Their symbols are given by the rule for products, and $|X(x)|, |Y(x)| \le C\langle x\rangle^{-4}$, so that $X, Y \in \ell^2(\mathbb{Z}^3)$ and the rule for $X * Y$ applies.
- For $G \in \mathfrak{G}_5$, $G^{\mathrm{mod}}(u) = K_0(u)\,(X * Y)(u)$ with model lines K_0, X and a product Y of two model lines, where $n_X \in \{2,3\}$ and $n_Y \in \{2,3,4\}$. Thus $|X(x)| \le C\langle x\rangle^{-n_X}$ and $|Y(x)| \le C\langle x\rangle^{-n_Y}$ with exponents at least 2, so that $X, Y \in \ell^2(\mathbb{Z}^3)$; the kernel $X * Y$ has the symbol $\hat{X}\hat{Y}$, and the product with K_0 is treated by the rule for products.
- For G of type (I), $G^{\mathrm{mod}} = K_\Pi^{\mathrm{mod}} \cdot (K_1' * X * K^{\leftarrow,\mathrm{mod}})$ with $X := K_{\mathrm{a}}^{\mathrm{mod}} K_{\mathrm{b}}^{\mathrm{mod}}$; the degrees of K_1', X and $K^{\leftarrow,\mathrm{mod}}$ are $2 + d_G$, $4 - d_G$ and 2, all at least 2. Hence $K_1' * X$ has the symbol $\hat{K}_1'\hat{X}$, and Lem. D.4, with $(a, a') = (2 + d_G, 4 - d_G)$ and $a + a' - 3 = 3$, gives

$$|(K_1' * X)(x)| \le C\langle x\rangle^{-\min\{2+d_G,\, 4-d_G,\, 3\}} \lg\langle x\rangle;$$

  the exponent is at least 2, so that $K_1' * X \in \ell^2(\mathbb{Z}^3)$, the logarithm included. The rule for convolutions applies once more to $(K_1' * X) * K^{\leftarrow,\mathrm{mod}}$, and the product with K_Π^{mod} is treated by the rule for products.

For $k, l, \mathsf{p} \in [-\pi, \pi]^3$ the periodic continuations may be replaced by $\rho\sigma^h$. We use the following observation: if $v \in [-\pi,\pi]^3$, $|w| < 4\rho_0$ and $|v - w|_{\mathbb{T}} < 2\rho_0$, then $|v - w| < 2\rho_0$, because the distance of $v - w$ from any point $2\pi\mathsf{m}$ with $\mathsf{m} \in \mathbb{Z}^3 \setminus \{0\}$ is at least $\pi - 4\rho_0 > 2\rho_0$. Let $k, l, \mathsf{p} \in [-\pi,\pi]^3$, and assume first that every line q of Γ satisfies $|\mathsf{q}|_{\mathbb{T}} < 2\rho_0$; this is the case wherever all the factors σ^{mod} are non-zero. Then $|l| = |l|_{\mathbb{T}} < 2\rho_0$, since $l \in [-\pi,\pi]^3$. For $\Gamma \neq 6$ we have likewise $|k| < 2\rho_0$; then $|k - l| < 4\rho_0 < \pi$, so that $|k - l| = |k - l|_{\mathbb{T}} < 2\rho_0$ wherever $k - l$ is a line ($\Gamma = 1, 2, 3, 5$), and the observation with $v = \mathsf{p}$ and $w = k$, resp. $w = l$, gives $|\mathsf{p} - k| < 2\rho_0$, resp. $|\mathsf{p} - l| < 2\rho_0$, wherever these vectors are lines. For $\Gamma = 6$ the observation with $(v, w) = (k, l)$ gives $|k - l| < 2\rho_0$, hence $|k| < 4\rho_0$, and with $(v, w) = (\mathsf{p}, k)$ it gives $|\mathsf{p} - k| < 2\rho_0$. Thus every line satisfies $|\mathsf{q}| < 2\rho_0$, and $\sigma^{\mathrm{mod}}(\mathsf{q}) = \rho(\mathsf{q})\sigma^h(\mathsf{q})$. If, on the other hand, some line satisfies $|\mathsf{q}|_{\mathbb{T}} \ge 2\rho_0$, then $\sigma^{\mathrm{mod}}(\mathsf{q}) = 0$ and also $\rho(\mathsf{q}) = 0$, since $|\mathsf{q}| \ge |\mathsf{q}|_{\mathbb{T}}$. In either case the product of the factors σ^{mod} equals $R_\Gamma \mathsf{F}_G$ at (k, l, p), the lines of G being the lines of Γ (for type (I) the factor $\rho\,\sigma_\rightarrow^h \mathsf{f}_G^h$ of K_1' provides the factors $\sigma_\rightarrow^h(k)\,\mathsf{f}_G^h(k)$ of F_G). Since R_Γ vanishes unless $k, l, \mathsf{p} \in [-\pi,\pi]^3$ (Def. D.40(3)), we obtain

$$G^{\mathrm{mod}}(u) = \int_{\mathbb{R}^3} e^{i\mathsf{p}\cdot u} \Big[\iint R_\Gamma\, \mathsf{F}_G\, d^\circ k\, d^\circ l \Big] d^\circ \mathsf{p}.$$

By Lem. D.42, $\sum c_G \iint R_\Gamma \mathsf{F}_G\, d^\circ k\, d^\circ l = 24\Psi_\Gamma$ for $\Gamma \neq 5$, whence $\sum c_G G^{\mathrm{mod}} = \mathsf{T}_\Gamma^{\mathrm{mod}}$. For $\Gamma = 5$ the term $G^{\mathrm{mod}}(-u)$ carries the factor $e^{-i\mathsf{p}\cdot u}$, and $\sum c_G \iint R_5 \mathsf{F}_G\, d^\circ k\, d^\circ l = 12\Psi_5$ is even; hence $\sum c_G[G^{\mathrm{mod}}(u) + G^{\mathrm{mod}}(-u)] = 12\int (e^{i\mathsf{p}\cdot u} + e^{-i\mathsf{p}\cdot u})\Psi_5\, d^\circ \mathsf{p} = \mathsf{T}_5^{\mathrm{mod}}(u)$.

Step 2 (expansions). Lem. D.37 applies to G^{mod}, the model lines being elementary lines of the same degrees as the lattice lines; the functions of degree -5 which it provides are determined by the leading parts of the symbols of the lines alone (Lem. D.37), and therefore coincide with the U_G. For G of type (I) let $X := K_{\mathrm{a}}^{\mathrm{mod}} K_{\mathrm{b}}^{\mathrm{mod}}$, of degree $n_X = 4 - d_G$, and write $X = \mathsf{q}_X + X'$ with $X' \in \mathfrak{K}(n_X)$ (Lem. D.32(iii)). Since K_1' is a line of degree $2 + d_G \le 4$, two applications of Lem. D.32(ii) give $K_1' * X' * K^{\leftarrow,\mathrm{mod}} \in \mathfrak{K}(2)$, and the first component of its leading data is $\sigma_\rightarrow^h \mathsf{f}_G^h \mathsf{I}^h \sigma_\leftarrow^h$,

where the homogeneous part I^h of the bubble depends only on σ_{a}^h, σ_{b}^h (Lem. D.30). The lattice chain $K_{\iota\mu}^{\rightarrow} * \mathsf{f}_G(\nabla)[K_{\mathrm{a}}K_{\mathrm{b}} - \mathsf{q}_G] * K_{\nu\iota'}^{\leftarrow}$ has the same first component of its leading data, by Lem. D.39(a) and by Lem. D.32(ii), used for each of its two convolutions. Within $\mathfrak{K}(2)$ the leading kernel is determined by the first component of the leading data (Lem. D.32(iv)); hence the two chains have the same leading kernel. Now $G^{\mathrm{mod}} = K_{\Pi}^{\mathrm{mod}} \cdot (K_1' * \mathsf{q}_X * K^{\leftarrow,\mathrm{mod}}) + K_{\Pi}^{\mathrm{mod}} \cdot (K_1' * X' * K^{\leftarrow,\mathrm{mod}})$, and the first term is local. In the second term, $K_{\Pi}^{\mathrm{mod}} = \Pi^h + O(|u|^{-N})$ with $\Pi^h = O(|u|^{-3})$, and $K_1' * X' * K^{\leftarrow,\mathrm{mod}}$ differs from its leading kernel, which is $O(|u|^{-2})$, by $O(|u|^{-3}\lg|u|)$, by the spatial estimate in Def. D.27. We obtain in all cases

$$G^{\mathrm{mod}} = [\text{local terms}] + \mathsf{U}_G + O(|u|^{-6}\lg|u|).$$

Step 3 (local terms). In this step all lines are model lines, and we omit the superscript mod. Because of the exact parity of the model symbols, every one-loop function I built from model lines satisfies $\mathsf{I}(-\kappa) = (-1)^{\omega+1}\mathsf{I}(\kappa)$, as one sees by the substitution $l \to -l$; its homogeneous part I^h has the same parity (Lem. D.30). By Lem. D.30 the polynomial $\mathsf{Q}(\kappa) - (-1)^{\omega+1}\mathsf{Q}(-\kappa)$, of degree at most ω, is therefore $O(|\kappa|^{\omega+1}\lg|\kappa|\,\Lambda(\kappa))$, hence equal to 0. Consequently $\mathsf{Q} = 0$ if $\omega \le 0$, and $\mathsf{Q} = \mathsf{Q}(0)$ if $\omega = 1$; in both cases $\mathsf{q} = \mathsf{Q}(0)\,\delta_0$. The local terms are thus: $\mathsf{Q}_{\mathsf{p}}(0)K_4K_5(u) + \mathsf{Q}_{\mathsf{q}}(0)K_1K_2(u)$ for the kites, where $n_4 + n_5 = 4$ if $\omega_{\mathsf{p}} = 1$ and $n_1 + n_2 = 4$ if $\omega_{\mathsf{q}} = 1$; $\mathsf{Q}_Y(0)K_0X(u)$, with $n_0 + n_X = 4$, for $\mathfrak{G}_5$; $\mathsf{Q}_X(0)Y(u) + \mathsf{Q}_Y(0)X(u)$, with $n_X = n_Y = 4$, for $\mathfrak{G}_4$; and, for type (I) with $d_G = 0$, $\mathsf{Q}_X(0)K_\Pi(u)\,(K_1' * K^{\leftarrow})(u)$, where $K_1' * K^{\leftarrow}$ is, by Lem. D.41(b), the elementary line of degree 1 whose symbol is the periodic continuation of $\rho^2\sigma_{\rightarrow}^h\mathsf{f}_G^h\sigma_{\leftarrow}^h$ (a line symbol by Def. D.27, with ρ^2 in place of ρ), $\mathsf{f}_G^h = \mathsf{f}_G(0)$ being a number. By Step 0, and by Lem. D.26 with ρ^2 in place of ρ for the line $K_1' * K^{\leftarrow}$, each of these local terms is, up to $O(|u|^{-N})$, its factor $\mathsf{Q}_\bullet(0)$ times the product of the functions $K^h = \mathcal{F}_{\neq 0}^{-1}\sigma^h$ of its lines, a function homogeneous of degree -4 and smooth off 0. We let I_Γ be the sum of these functions, each taken with the coefficient with which its term G^{mod} enters $\mathsf{T}_\Gamma^{\mathrm{mod}}$ in Step 1. For $\Gamma = 5$, where $\mathsf{T}_5^{\mathrm{mod}}(u) = \sum c_G[G^{\mathrm{mod}}(u) + G^{\mathrm{mod}}(-u)]$, the expansion of Step 2 is used at u and at $-u$; accordingly, every such function h of a term $G \in \mathfrak{G}_5$ enters I_5 together with its reflection $u \mapsto \mathsf{h}(-u)$, just as U_5 contains $\mathsf{U}_G(u)$ and $\mathsf{U}_G(-u)$. Adding the expansions of Step 2 with these coefficients, we obtain the claimed bound for large $|u|$; for small $|u|$ all terms are bounded. All data being fixed, the constant C is pure. □

The functions I_Γ and U_Γ are in fact multiples of $|u|^{-4}$ and $|u|^{-5}$, since $\mathsf{T}_\Gamma^{\mathrm{mod}}$ is radial.

Lemma D.44 (Rotation invariance of the homogeneous parts; definition of $\kappa_\Gamma^{\mathrm{lat}}$)**.** Let $\mathsf{t}\colon [0,\infty) \to \mathbb{C}$, let I, U be continuous functions on $\mathbb{R}^3 \setminus \{0\}$, homogeneous of degrees -4 and -5 respectively, and assume that for some C

$$|\mathsf{t}(|u|) - \mathsf{I}(u) - \mathsf{U}(u)| \le C|u|^{-6}\lg|u| \qquad (u \in \mathbb{Z}^3 \setminus \{0\}).$$

Then $\mathsf{I}(u) = c_4|u|^{-4}$ and $\mathsf{U}(u) = \kappa|u|^{-5}$ with constants c_4, κ. In particular the functions of Lem. D.43 satisfy $\mathsf{I}_\Gamma(u) = c_{4,\Gamma}|u|^{-4}$ and

$$\mathsf{U}_\Gamma(u) = \kappa_\Gamma^{\mathrm{lat}}\,|u|^{-5} \qquad (\Gamma = 1, \dots, 6);$$

this defines the numbers $c_{4,\Gamma}$ and $\kappa_\Gamma^{\mathrm{lat}}$. ■

Proof. Let O be a rotation with rational entries, N a common denominator of these entries, and $\mathsf{m} \in \mathbb{Z}^3 \setminus \{0\}$. For integers $j \ge 1$ the points $j\mathsf{N}\mathsf{m}$ and $j\mathsf{N}O\mathsf{m}$ lie in $\mathbb{Z}^3$ and have the same norm jr_1, $r_1 := \mathsf{N}|\mathsf{m}|$. Let $\hat{\mathsf{m}} := \mathsf{m}/|\mathsf{m}|$. Subtracting the hypothesis at these two points, at which t takes the same value, we obtain by homogeneity

$$\left|(jr_1)^{-4}[\mathsf{I}(\hat{\mathsf{m}}) - \mathsf{I}(O\hat{\mathsf{m}})] + (jr_1)^{-5}[\mathsf{U}(\hat{\mathsf{m}}) - \mathsf{U}(O\hat{\mathsf{m}})]\right| \le 2C(jr_1)^{-6}\lg(jr_1).$$

Multiplying by $(jr_1)^4$ and letting $j \to \infty$ we obtain $\mathsf{I}(\hat{\mathsf{m}}) = \mathsf{I}(O\hat{\mathsf{m}})$; then, multiplying the remaining difference by $(jr_1)^5$, we obtain $\mathsf{U}(\hat{\mathsf{m}}) = \mathsf{U}(O\hat{\mathsf{m}})$. The directions $\mathsf{m}/|\mathsf{m}|$ are dense in S^2, since $\mathbb{Q}^3$ is

dense in $\mathbb{R}^3$. The rational rotations are dense in $SO(3)$: the Cayley map $\mathsf{A} \mapsto (\mathbf{1}-\mathsf{A})(\mathbf{1}+\mathsf{A})^{-1}$ is continuous from the real antisymmetric 3×3 matrices onto the rotations without eigenvalue -1, which are dense in $SO(3)$; the antisymmetric matrices with rational entries are dense among the real ones, and by the formula they are mapped to rotations with rational entries, since the inverse of an invertible matrix with rational entries has rational entries. By continuity, I and U are invariant under all rotations; as $SO(3)$ acts transitively on S^2, they are constant on S^2, and homogeneity gives the claim. For the last assertion we note that $\mathsf{T}_\Gamma^{\mathrm{mod}}$ is the restriction to $\mathbb{Z}^3$ of a radial function, since Ψ_Γ is radial (Def. D.40(3)); thus $\mathsf{T}_\Gamma^{\mathrm{mod}}(u) = \mathsf{t}(|u|)$ for a function t, and Lem. D.43 provides the hypothesis. □

Proposition D.45 (The kernel of degree -5: large-distance asymptotics of $\mathsf{B}_1^{\mathrm{lat}}$)**.** There are functions $\mathsf{U}_{ij} \in C^1(\mathbb{R}^3\setminus\{0\})$, $i,j\in\{1,2,3\}$, homogeneous of degree -5, and pure constants C, r_0 such that for all plaquettes p, q of $\mathbb{Z}^3$, with $u := x(q)-x(p)$ and $r := |u| \geq r_0$,

$$\begin{aligned} &|\mathsf{B}_1^{\mathrm{lat}}(p,q) - \mathsf{U}_{\lambda(p)\lambda(q)}(u)| \leq C\, r^{-6} \lg r, \\ &\sum_{i,j} \mathsf{U}_{ij}(u) = \kappa^{\mathrm{lat}} |u|^{-5}, \qquad \kappa^{\mathrm{lat}} := \sum_{\Gamma=1}^{6} \kappa_\Gamma^{\mathrm{lat}}, \end{aligned} \tag{D.xxx}$$

with the numbers $\kappa_\Gamma^{\mathrm{lat}}$ of Lem. D.44. Moreover $\mathsf{B}_1^{\mathrm{lat}}$ is bounded. ■

Proof. We work at $\tau_{\mathrm{gf}} = 1$; recall that $\mathsf{B}_1^{\mathrm{lat}}(p,q) = \mathsf{B}_1^{\mathrm{lat}}(p,q;1)$ by eq. D.iii. Boundedness is contained in Lem. D.6, which gives $\sup_{p,q}|\mathsf{B}_1^{\mathrm{lat}}(p,q)| < \infty$ for every $\tau_{\mathrm{gf}} = 1$. Let $p = p_{[\iota]}(x_p)$, $q = p_{[\iota']}(x_q)$ and $u' := x_q - x_p$; then $|u-u'| \leq \sqrt{2}$, and we take r large. By Lem. D.5 and Lem. D.36, $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ is the sum of T_{I} and of finitely many elementary terms, to which Lem. D.37 applies.

Step 1 (local terms at q). These are the first half of $\mathsf{T}_{4'}$ and the terms with q_Y in the first half of T_5, with q_Y in T_4 and with q_{q} in T_1, in the expansions of Lem. D.37. By Lem. D.39(b), and since $E_p((x_q+y,\mu)) = K^{\rightarrow}_{\iota\mu}(u'+y)$ by translation invariance, their sum equals

$$\tfrac{3}{2}\sum_{b,b'} \gamma_q(b,b')\, E_p(b)\, E_p(b') = \tfrac{3}{2}\sum_{\mu,\nu}\sum_{y,y'} \gamma_{q;\mu\nu}(y,y')\, K^{\rightarrow}_{\iota\mu}(u'+y)\, K^{\rightarrow}_{\iota\nu}(u'+y').$$

The kernels $\gamma_{q;\mu\nu}$ have finite support, and their symbols vanish at $\kappa = 0$ together with their gradients (Lem. D.39(b)); the $K^{\rightarrow}_{\iota\mu}$ are elementary lines of degree 2. Lem. D.38(a), applied with $\gamma_{\mu\nu} := \gamma_{q;\mu\nu}$ and $K_\mu := K^{\rightarrow}_{\iota\mu}$, shows that the sum is $O(|u'|^{-6}) = O(r^{-6})$. By the translation invariance in Lem. D.36, the kernels $\gamma_{q;\mu\nu}$ depend on q only through ι', so that only finitely many constants occur.

Step 2 (local terms at p). These are the second half of $\mathsf{T}_{4'}$ and the terms with q_Y in the second half of T_5, with q_X in T_4 and with q_{p} in T_1. We show that they are the local terms at the second plaquette of the ordered pair (q,p); by Step 1, applied to this pair, their sum is then $\frac{3}{2}\sum_{b,b'}\gamma_p(b,b')\,E_q(b)\,E_q(b') = O(r^{-6})$. For the second halves of $\mathsf{T}_{4'}$ and T_5 this holds because they are the first halves for (q,p) (Lem. D.36). The formulas for T_1 and T_4 in Lem. D.5 are symmetric under $p \leftrightarrow q$. An elementary term G of T_1 or T_4 is given by the monomials at P, P' and by the choice of the legs contracted with p, with q, and with one another; taken for the ordered pair (q,p), it becomes a term $\tilde{G}$ with $c_{\tilde{G}} = c_G$, and $G \mapsto \tilde{G}$ is a bijection $\mathfrak{G}_\bullet(\iota,\iota') \to \mathfrak{G}_\bullet(\iota',\iota)$. In $\tilde{G}$ the base points are counted from x_q, u' is replaced by $-u'$, and the contractions, which are symmetric in their two legs, are taken from the other end: for the kite

$$\tilde{K}_1 = K_4(-\cdot), \quad \tilde{K}_2 = K_5(-\cdot), \quad \tilde{K}_3 = K_3, \quad \tilde{K}_4 = K_1(-\cdot), \quad \tilde{K}_5 = K_2(-\cdot),$$

and for T_4 we have $\tilde{X} = Y(-\cdot)$, $\tilde{Y} = X(-\cdot)$. Since $K(-\cdot)$ has the symbol $\sigma(-k)$, the function $\hat{\mathsf{T}}_{\mathsf{q}}$ of $\tilde{G}$ is

$$\int_{\mathbb{T}} \sigma_1(-\kappa_1+l)\,\sigma_2(-\kappa_2-l)\,\sigma_3(l)\,d^\circ l = \hat{\mathsf{T}}_{\mathsf{p}}(-\kappa),$$

and the symbol of $\tilde{Y}$ is $\mathsf{I}_X(-k)$, after the substitution $l \to -l$. By the last assertion of Lem. D.30, whose hypothesis holds here for all κ, the corresponding polynomials are $\mathsf{Q}_\mathsf{p}(-\kappa)$ and $\mathsf{Q}_X(-k)$. The passage from a polynomial Q to $\tilde{\mathsf{Q}}(\kappa) = \mathsf{Q}(0) + \sum_j \partial_j \mathsf{Q}(0) \sin \kappa_j$ in Prop. D.33 and Lem. D.32(iii) commutes with the change of sign of the argument, the sine being odd. Hence the kernel q_q of $\tilde{G}$ is $(z,w) \mapsto \mathsf{q}_\mathsf{p}(-z,-w)$, and $\mathsf{q}_{\tilde{Y}} = \mathsf{q}_X(-\cdot)$, so that

$$\sum_{z,w} \mathsf{q}_\mathsf{p}(-z,-w)\, \tilde{K}_1(-u'-z)\, \tilde{K}_2(-u'-w) = \sum_{z,w} \mathsf{q}_\mathsf{p}(z,w)\, K_4(u'-z)\, K_5(u'-w),$$

$$(\mathsf{q}_{\tilde{Y}} * \tilde{X})(-u') = (\mathsf{q}_X * Y)(u').$$

These are the local terms at p in T_1 and in T_4.

Step 3 (the class T_I). By Lem. D.39(a),

$$K^{\rightarrow} * \Sigma_{\mu\nu} * K^{\leftarrow} = K^{\rightarrow} * \mathsf{q}^{\Sigma}_{\mu\nu} * K^{\leftarrow} + K^{\rightarrow} * \mathsf{W}_{\mu\nu} * K^{\leftarrow}.$$

The first term is $O(r^{-3})$ by Lem. D.38(b), since $\hat{\mathsf{q}}^{\Sigma}_{\mu\nu}(k) = O(|k|^2)$. The second lies in $\mathfrak{K}(2)$, by Lem. D.32(ii) applied to each of its two convolutions, and it equals its leading kernel, of degree -2, up to $O(r^{-3} \lg r)$; by the linearity of the leading data, this leading kernel is the sum over G, with the coefficients c_G, of the leading kernels of the chains in the definition of U_G in Lem. D.43. Moreover $\Pi(p,q) = \Pi^h(u') + O(r^{-4})$, with Π^h homogeneous of degree -3 (Lem. D.34, Lem. D.26). Therefore

$$\mathsf{T}_\mathrm{I} = \sum_{(\mu,\nu,G) \in \mathfrak{G}_2 \cup \mathfrak{G}_3} (-3\, c_G)\, \mathsf{U}_G(u') + O(r^{-6} \lg r).$$

Step 4 (assembly). Let

$$\mathsf{U}_{\iota\iota'}(u) := \sum_{\Gamma \neq 5} \sum_{G \in \mathfrak{G}_\Gamma(\iota,\iota')} c_G\, \mathsf{U}_G(u) + \sum_{G \in \mathfrak{G}_5(\iota,\iota')} c_G\, \mathsf{U}_G(u) + \sum_{G \in \mathfrak{G}_5(\iota',\iota)} c_G\, \mathsf{U}_G(-u),$$

with the coefficients $-3\, c_G$ for $\Gamma = 2, 3$; the last sum comes from the second half of T_5, which is the first half for (q,p) with u' replaced by $-u'$. The function $\mathsf{U}_{\iota\iota'}$ is homogeneous of degree -5 and of class C^1 off 0, by Lem. D.37 and since the leading kernels of the classes $\mathfrak{K}(n)$ are smooth off 0. By Steps 1–3 and Lem. D.37, $\mathsf{B}^{\mathrm{lat}}_1(p,q) = \mathsf{U}_{\iota\iota'}(u') + O(r^{-6} \lg r)$. The gradient of $\mathsf{U}_{\iota\iota'}$ is homogeneous of degree -6, and all points $u' + t(u-u')$, $0 \le t \le 1$, have modulus at least $r/2$; hence $|\mathsf{U}_{\iota\iota'}(u') - \mathsf{U}_{\iota\iota'}(u)| \le C r^{-6}$. With $\mathsf{U}_{ij} := \mathsf{U}_{\iota\iota'}$ for $(i,j) = (\iota,\iota') = (\lambda(p), \lambda(q))$ we obtain the bound in eq. D.xxx; as $\tau_{\mathrm{gf}} = 1$, the finite sets $\mathfrak{G}_\bullet(\iota,\iota')$, the coefficients c_G and the lines are fixed once and for all, the constants C, r_0 are pure. Finally, exchanging ι and ι' in the last sum and using Lemmas D.43 and D.44,

$$\sum_{\iota,\iota'} \mathsf{U}_{\iota\iota'} = \sum_{\Gamma=1}^{6} \mathsf{U}_\Gamma = \sum_{\Gamma=1}^{6} \kappa^{\mathrm{lat}}_\Gamma\, |u|^{-5}.$$

□

For the smeared observables of eq. D.iii the theorem gives the following rate of convergence on test pairs. Here $\mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ is the set of test pairs at the scale d with the shape parameter ϑ of Def. 14.1, taken on $\mathbb{E} = \mathbb{R}^3$.

Proposition D.46 (Rate of convergence of $\mathsf{P}^{\mathrm{lat}}_1$ to $\kappa^{\mathrm{lat}}\mathcal{K}_5$)**.** There is a pure constant C such that the following holds for all $\mathsf{d} > 0$, $\vartheta \in (0,1)$, $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ and $0 < a \le \mathsf{d}$. Recall

$$\mathcal{K}_5(f,h) = \iint f(x)\, h(y)\, |x-y|^{-5}\, dx\, dy,$$

the integrand being 0 where $f(x)h(y) = 0$. Then

$$|\mathsf{P}_1^{\mathrm{lat}}(f,h;a) - \kappa^{\mathrm{lat}}\,\mathcal{K}_5(f,h)| \le C\,\vartheta^{-1}\,a\,\lg(\mathsf{d}/a)\,\|f\|_\infty\|h\|_\infty. \tag{D.xxxi}$$

■

We recall three properties of a test pair $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$: the supports of f and h lie in closed balls $\mathbb{B}(x_f,\mathsf{d})$ and $\mathbb{B}(x_h,\mathsf{d})$ of radius d; $\mathsf{d} \le |v-w| \le 10\,\mathsf{d}$ for $v \in \operatorname{supp} f$ and $w \in \operatorname{supp} h$; and $\operatorname{Lip} f \le \|f\|_\infty/\vartheta\mathsf{d}$, $\operatorname{Lip} h \le \|h\|_\infty/\vartheta\mathsf{d}$.

Proof. Let $\varrho := \mathsf{d}/a \ge 1$, $\mathsf{A} := \|f\|_\infty\|h\|_\infty$ and $\varrho_0 := \max\{r_0, 8\}$, with r_0 of Prop. D.45. A ball of radius $\varrho+2$ contains at most $C\varrho^3$ points of $\frac{1}{2}\mathbb{Z}^3$. The barycentres $x(p)$ lie in $\frac{1}{2}\mathbb{Z}^3$, distinct plaquettes have distinct barycentres, and $f(ax(p)) \ne 0$ implies $x(p) \in \mathbb{B}(x_f/a, \varrho)$; likewise for h.

Step 1 ($\varrho \le \varrho_0$). Then the sum defining $\mathsf{P}_1^{\mathrm{lat}}(f,h;a)$ has at most C non-zero terms, and since $\mathsf{B}_1^{\mathrm{lat}}$ is bounded (Prop. D.45), $|\mathsf{P}_1^{\mathrm{lat}}(f,h;a)| \le Ca\mathsf{A}$. Moreover $|x-y| \ge \mathsf{d}$ where $f(x)h(y) \ne 0$, whence $|\mathcal{K}_5(f,h)| \le (\frac{4\pi}{3}\mathsf{d}^3)^2\mathsf{d}^{-5}\mathsf{A} \le C\varrho_0 a\mathsf{A}$. As $\vartheta^{-1} > 1$ and $\lg(\mathsf{d}/a) \ge 1$, eq. D.xxxi holds in this case.

Step 2 ($\varrho > \varrho_0$, first error). If $f(ax(p))h(ax(q)) \ne 0$, then $r = |x(q)-x(p)| \in [\varrho, 10\varrho]$ by the separation of the supports; in particular $r \ge r_0$. There are at most $C\varrho^6$ such pairs, and $r^{-6}\lg r \le \varrho^{-6}\lg(10\varrho)$ for these r. By eq. D.xxx, the replacement of $\mathsf{B}_1^{\mathrm{lat}}(p,q)$ by $\mathsf{U}_{\lambda(p)\lambda(q)}(x(q)-x(p))$ in $\mathsf{P}_1^{\mathrm{lat}}(f,h;a)$ costs at most

$$a \cdot C\varrho^6 \cdot C\varrho^{-6}\lg(10\varrho)\,\mathsf{A} \le Ca\lg\varrho\,\mathsf{A}.$$

Step 3 (second error). We write $p = p_{[i]}(x)$, $q = p_{[j]}(y)$, so that $x(p) = x + \mathsf{o}_i$ with $\mathsf{o}_i := x(p_{[i]}(0))$, $|\mathsf{o}_i| = 2^{-1/2}$; all terms below are set equal to 0 where the product of the values of f and h vanishes. After Step 2 we have to compare

$$\mathfrak{R}_1 := a\sum_{i,j}\sum_{x,y\in\mathbb{Z}^3} f(a(x+\mathsf{o}_i))\,h(a(y+\mathsf{o}_j))\,\mathsf{U}_{ij}(y-x+\mathsf{o}_j-\mathsf{o}_i)$$

with

$$\mathfrak{R}_2 := a\sum_{x,y} f(ax)\,h(ay)\sum_{i,j}\mathsf{U}_{ij}(y-x) = a\,\kappa^{\mathrm{lat}}\sum_{x,y}\frac{f(ax)\,h(ay)}{|y-x|^5},$$

the last equality by eq. D.xxx. We replace f, h and U_{ij} one after the other. In every non-zero term, the arguments of U_{ij} which occur, and the points between them, have modulus at least $\varrho - 2 \ge \varrho/2$, and x, y lie in balls of radius $\varrho+1$. By homogeneity, $|\mathsf{U}_{ij}| \le C\varrho^{-5}$ and $|\nabla\mathsf{U}_{ij}| \le C\varrho^{-6}$ at these points, and the Lipschitz bound gives $|f(a(x+\mathsf{o}_i)) - f(ax)| \le a\|f\|_\infty/\vartheta\mathsf{d}$, and likewise for h. Therefore

$$|\mathfrak{R}_1 - \mathfrak{R}_2| \le a\,C\varrho^6\Big[\frac{a}{\vartheta\mathsf{d}}\,\varrho^{-5} + \varrho^{-6}\Big]\mathsf{A} \le Ca\vartheta^{-1}\mathsf{A}.$$

Step 4 (third error). Let $\mathsf{F}(v,w) := f(v)h(w)|v-w|^{-5}$, equal to 0 where $f(v)h(w) = 0$. Then $\mathfrak{R}_2 = \kappa^{\mathrm{lat}}\sum_{x,y} a^6\mathsf{F}(ax,ay)$ and $\mathcal{K}_5(f,h) = \iint \mathsf{F}$. Let $(v,w) \in (ax,ay) + [0,a)^6$. Unless F vanishes at both points (v,w) and (ax,ay), we have $|v-w| \ge \mathsf{d} - 2\sqrt{3}a \ge \mathsf{d}/2$ at both points and at all points $(1-t)(ax,ay) + t(v,w)$, $0 \le t \le 1$ (recall $\varrho > 8$), and the Lipschitz bounds for f, h together with the bound $5|z|^{-6}$ for the gradient of $|z|^{-5}$ give

$$|\mathsf{F}(v,w) - \mathsf{F}(ax,ay)| \le \sqrt{3}a\Big[\frac{2}{\vartheta\mathsf{d}}\Big(\frac{\mathsf{d}}{2}\Big)^{-5} + 10\Big(\frac{\mathsf{d}}{2}\Big)^{-6}\Big]\mathsf{A}.$$

At most $C\varrho^6$ of the cubes $(ax,ay)+[0,a)^6$, of volume a^6, contain points where $\mathsf{F} \ne 0$. Hence

$$|\mathfrak{R}_2 - \kappa^{\mathrm{lat}}\mathcal{K}_5(f,h)| \le C\varrho^6 a^6 \cdot a\vartheta^{-1}\mathsf{d}^{-6}\mathsf{A} = Ca\vartheta^{-1}\mathsf{A}.$$

Adding the bounds of Steps 2–4 and using $\vartheta^{-1} > 1$, $\lg\varrho \ge 1$, we obtain eq. D.xxxi. □

Proposition D.47 (The boxes exhaust the lattice)**.** Let $\tau_{\mathrm{gf}} = 1$, let p, q be plaquettes of $\mathbb{Z}^3$ and $j \in \{0,1\}$, and write $\mathsf{B}_0^{\Omega}(p,q;\tau_{\mathrm{gf}}) := \mathsf{B}_0^{\Omega}(p,q)$. The numbers $\mathsf{B}_j^{[n]}(p+o_n, q+o_n;\tau_{\mathrm{gf}})$ are defined for all $n \geq 2$ with $p+o_n, q+o_n \in \mathsf{Pl}_n$, that is, for all but finitely many n, and

$$\lim_{n\to\infty} \mathsf{B}_j^{[n]}(p+o_n, q+o_n;\tau_{\mathrm{gf}}) = \mathsf{B}_j^{\mathrm{lat}}(p,q;\tau_{\mathrm{gf}}). \tag{D.xxxii}$$

■

Proof. We use the method of images. Fix n and write $\tau := \tau_{\mathrm{gf}}$.

Step 1 (reflections). We regard forms and functions on the torus $(\mathbb{Z}/2n)^3$ as $2ne_\nu$-periodic forms and functions on $\mathbb{Z}^3$, and we write $(\cdot,\cdot)_{\mathsf{t}}$ for the sums over one period. Let ϱ_ν be the reflection $z_\nu \to -z_\nu$, acting on bonds by $\varrho_\nu\langle z, z'\rangle = \langle \varrho_\nu z, \varrho_\nu z'\rangle$, and put $(\varrho_\nu^* \tilde A)(b) := \tilde A(\varrho_\nu b)$ and $\varrho_\nu^*\lambda := \lambda \circ \varrho_\nu$; these maps commute with d and d^*. Let F^{as} and X^{as} be the spaces of real periodic forms $\tilde A$ and functions λ with $\varrho_\nu^*\tilde A = -\tilde A$ and $\varrho_\nu^*\lambda = -\lambda$ for all ν; in components, $\tilde A_\mu(\varrho_\nu z) = -\tilde A_\mu(z)$ for $\nu \neq \mu$, and $\tilde A_\mu(\varrho_\mu z - e_\mu) = \tilde A_\mu(z)$. We claim that restriction to the box is a bijection of F^{as} onto $\mathsf{Form}_n^{\mathbb{R}}$ and of X^{as} onto $\mathsf{X}_n^{\mathbb{R}}$. If $z_\nu \in \{0, n\}$, then $\varrho_\nu z \equiv z$ modulo $2n$, hence $\tilde A_\mu(z) = 0$ for $\nu \neq \mu$, and $\lambda(z) = 0$; by Lem. D.8(a), $\tilde A$ vanishes on the non-free bonds of the box, and λ vanishes on $\partial\Lambda_n$. For $\varepsilon \in \{0,1\}^3$ let $R_\varepsilon := \prod_\nu \varrho_\nu^{\varepsilon_\nu}$, $|\varepsilon| := \sum_\nu \varepsilon_\nu$, and $R_\varepsilon(z,\mu) := (R_\varepsilon z - \varepsilon_\mu e_\mu, \mu)$, the positively oriented one among the image bond and its reverse. Every positively oriented bond of the torus has the form $R_\varepsilon(x,\mu)$ with $x_\mu \in \{0,\dots,n-1\}$ and $x_\nu \in \{0,\dots,n\}$ for $\nu \neq \mu$; here (x,μ) is unique, and ε is unique if (x,μ) is free. Hence every $A \in \mathsf{Form}_n^{\mathbb{R}}$ has the unique continuation $\tilde A(R_\varepsilon b) = (-1)^{|\varepsilon|+\varepsilon_\mu} A(b)$, $b = (x,\mu)$, and every $\lambda \in \mathsf{X}_n^{\mathbb{R}}$ the unique continuation $\tilde\lambda(R_\varepsilon x) = (-1)^{|\varepsilon|}\lambda(x)$.

Step 2 (operators). The eight boxes $R_\varepsilon[0,n]^3$ cover the torus, and a plaquette lying in two of them lies in a hyperplane $\{z_\nu = 0\}$ or $\{z_\nu = n\}$, on whose bonds $\tilde A = 0$. The functions $|\mathrm{d}\tilde A|$ and $|\mathrm{d}^*\tilde A|$ are invariant under the ϱ_ν, and $\mathrm{d}^*\tilde A \in \mathsf{X}^{\mathrm{as}}$ vanishes on these hyperplanes. Hence

$$(\tilde A, \tilde A')_{\mathsf{t}} = 8(A, A'), \qquad \|\mathrm{d}\tilde A\|_{\mathsf{t}}^2 = 8\|\mathrm{d}A\|^2, \qquad \|\mathrm{d}^*\tilde A\|_{\mathsf{t}}^2 = 8\|\mathrm{d}^*A\|^2.$$

The operator $\mathsf{L}_\tau := \mathrm{d}^*\mathrm{d} + \tau^{-1}\mathrm{d}\mathrm{d}^*$ on the torus commutes with the ϱ_ν^* and preserves F^{as}. By polarisation, $(\tilde A', \mathsf{L}_\tau \tilde A)_{\mathsf{t}} = 8(A', \mathsf{L}_\tau^{[n]} A)$ for all $A, A' \in \mathsf{Form}_n^{\mathbb{R}}$, where $\mathsf{L}_\tau^{[n]}$ is the operator on $\mathsf{Form}_n^{\mathbb{R}}$ of the quadratic form in Def. D.2; hence $\mathsf{L}_\tau\tilde A$ is the continuation of $\mathsf{L}_\tau^{[n]} A$. The forms in F^{as} have mean 0, and on periodic forms of mean 0 the inverse of L_τ is, by the finite Fourier inversion formula, the convolution with

$$C_{\mu\nu}^{(n)}(z) := (2n)^{-3} \sum_{k \in \mathbb{T}_n \setminus \{0\}} e^{ik\cdot z}\, \hat C_{\mu\nu}(k).$$

The continuation of $\delta_{b'}$, for a free bond $b' = (x',\mu')$, is $\sum_\varepsilon (-1)^{|\varepsilon|+\varepsilon_{\mu'}} \delta_{R_\varepsilon b'}$. For functions we have in the same way $(\tilde\lambda, \tilde\lambda')_{\mathsf{t}} = 8(\lambda,\lambda')$ and $\|\mathrm{d}\tilde\lambda\|_{\mathsf{t}}^2 = 8\|\mathrm{d}\lambda\|^2$, since $\mathrm{d}\tilde\lambda \in \mathsf{F}^{\mathrm{as}}$ is the continuation of $\mathrm{d}\lambda$; the elements of X^{as} have mean 0, and the continuation of $\delta_{x'}$, $x' \in \Lambda_n^\circ$, is $\sum_\varepsilon (-1)^{|\varepsilon|}\delta_{R_\varepsilon x'}$. By polarisation, as for forms, $\mathrm{d}^*\mathrm{d}\tilde\lambda$ is the continuation of $\mathrm{d}^*\mathrm{d}\lambda$, and on periodic functions of mean 0 the inverse of $\mathrm{d}^*\mathrm{d}$ is the convolution with $G^{(n)}(z) := (2n)^{-3}\sum_{k\in\mathbb{T}_n\setminus\{0\}} e^{ik\cdot z}|\mathsf{k}(k)|^{-2}$. We apply this to $B := (\mathsf{L}_\tau^{[n]})^{-1}\delta_{b'}$, whose value at a free bond b is $C_\tau^{[n]}(b,b')$; its continuation $\tilde B$ has mean 0 and satisfies $\mathsf{L}_\tau\tilde B = \tilde\delta_{b'}$, the continuation of $\delta_{b'}$, so that $\tilde B$ is the convolution of $C^{(n)}$ with $\tilde\delta_{b'}$. In the same way we treat $(\mathrm{d}^*\mathrm{d})^{-1}\delta_{x'}$ on $\mathsf{X}_n^{\mathbb{R}}$, whose value at $x \in \Lambda_n^\circ$ is $G^{[n]}(x,x')$. Consequently, for free bonds $(x,\mu),(x',\mu')$ and for $x, x' \in \Lambda_n^\circ$,

$$\begin{aligned} C_\tau^{[n]}((x,\mu),(x',\mu')) &= \sum_\varepsilon (-1)^{|\varepsilon|+\varepsilon_{\mu'}}\, C_{\mu\mu'}^{(n)}(x - R_\varepsilon x' + \varepsilon_{\mu'} e_{\mu'}), \\ G^{[n]}(x,x') &= \sum_\varepsilon (-1)^{|\varepsilon|}\, G^{(n)}(x - R_\varepsilon x'). \end{aligned} \tag{D.xxxiii}$$

In the first argument the right sides are values of elements of F^{as}, resp. X^{as}, and therefore vanish at non-free bonds, resp. on $\partial\Lambda_n$; in the second argument the same follows by symmetry. Since the box kernels vanish there by definition, eq. D.xxxiii holds for all bonds and sites of Λ_n.

Step 3 (uniform bounds). The kernels $E^{[n]}, \Pi^{[n]}, D^{[n]}, \mathsf{g}^{[n]}$ arise from eq. D.xxxiii by the differences and means of Def. D.2. A difference in the first argument acts on $C^{(n)}$ and $G^{(n)}$. In the second argument, R_ε maps the four bonds of q onto those of $R_\varepsilon q$, and $\sum_j s_j(-1)^{\varepsilon\mu_j}\delta_{R_\varepsilon b_j(q)} = \pm\sum_j s_j\delta_{b_j(R_\varepsilon q)}$, with μ_j the direction of $b_j(q)$, since d commutes with the ϱ^*_ν; the same holds for the end points of a bond. Hence each of these kernels is a sum of eight terms $\pm K^{(n)}(x - R_\varepsilon x' + y_0)$ with $|y_0| \le 3$, where x and x' are the base points of its two arguments, as in Lem. D.6 (the base point b_- of a bond b, the base point x_p of a plaquette p; for $\mathsf{g}^{[n]}(b)$ both are b_-), and where $K^{(n)}$ is built as in Lem. D.3 with the symbol of the corresponding line of Lem. D.34, an element of $\mathcal{S}^{n_\ell-3}$, n_ℓ being the degree of the line. For $x, x' \in \Lambda_n$ we have $|x - R_\varepsilon x'|_n \ge |x - x'|$: if $\varepsilon_\nu = 0$, then $|x_\nu - x'_\nu| \le n$, and if $\varepsilon_\nu = 1$, then $\min\{x_\nu + x'_\nu, 2n - x_\nu - x'_\nu\} \ge |x_\nu - x'_\nu|$. By Lem. D.3, the kernels of all boxes therefore satisfy the bounds Lem. D.6 with one constant $\mathsf{c}_0 = \mathsf{c}_0(\tau_{\mathrm{gf}})$.

Step 4 (pointwise limits). The arguments are translates by o_n of fixed bonds, sites and plaquettes. For $\varepsilon = 0$ the term $K^{(n)}(x - x' + y_0)$ has a fixed argument and converges to the corresponding kernel of $\mathbb{Z}^3$, by Lem. D.3 and Lem. D.34. If $\varepsilon_\nu = 1$, the ν-th coordinate of the argument is $\tilde{x}_\nu + \tilde{x}'_\nu + 2\lfloor n/2\rfloor + O(1)$, where $\tilde{x} = x - o_n$ and $\tilde{x}' = x' - o_n$ are fixed; its distance from $2n\mathbb{Z}$ is therefore at least $n - C$, and the term tends to 0 by Lem. D.3.

Step 5 (conclusion). For $j = 0$, $\mathsf{B}_0^{[n]} = \frac{3}{2}(\Pi^{[n]})^2$ by Def. D.2, which converges by Step 4 to $\frac{3}{2}(\Pi^{\mathrm{lat}})^2 = \mathsf{B}_0^{\mathrm{lat}}$. For $j = 1$, the formulas of Lem. D.5 in the box, which are finite sums, express $\mathsf{B}_1^{[n]}(p + o_n, q + o_n; \tau_{\mathrm{gf}})$ as a sum, over monomials, pairings and positions, of products of box kernels. We write the positions in the box as translates $P + o_n$, $P' + o_n$, $b + o_n, \dots$ of plaquettes P, P' and bonds $b, \dots$ of $\mathbb{Z}^3$, and we index the terms by the monomial, the pairing and the positions $P, P', b, \dots$ in $\mathbb{Z}^3$; this set of indices does not depend on n, and a term whose positions do not all lie in the box is set equal to 0. For fixed $P, P', b, \dots$ all positions lie in the box for every large n, and by Step 4 the term converges, as $n \to \infty$, to the term of the formulas of Lem. D.5 on $\mathbb{Z}^3$ with the same monomial, the same pairing and the positions $P, P', b, \dots$. By Step 3, the box kernels satisfy Lem. D.6 with one constant c_0; these bounds depend only on differences of base points, which the translation by o_n does not change, and the uniform form of Lem. D.6, valid for any kernels with these bounds, therefore dominates the moduli of the terms, uniformly in n, by a family of majorants over the same set of indices whose sum is at most $C\mathsf{c}_0^5$. By dominated convergence, $\mathsf{B}_1^{[n]}(p + o_n, q + o_n; \tau_{\mathrm{gf}})$ tends to the sum of the terms on $\mathbb{Z}^3$, which converges absolutely and equals $\mathsf{B}_1^{\mathrm{lat}}(p, q; \tau_{\mathrm{gf}})$ by Lem. D.6. This is eq. D.xxxii. □

D.17. **Evaluating the coefficient.** We extract the coefficient of $|p|^2 \log|p|$ from the six continuum integrals. In this calculation $(\mathcal{F}^{-1}\phi)(p) = \int e^{ip\cdot x}\phi(x)\, d^\circ x$. Write Γ_{E} for Euler's gamma function.

Lemma D.48 (Two Fourier constants)**.** Let $\phi \in \mathcal{S}(\mathbb{R}^3)$ vanish near 0.

(a) For $s \in \mathbb{C}$ with $\operatorname{Re} s > -3$,

$$\int_{\mathbb{R}^3} |p|^s\, (\mathcal{F}^{-1}\phi)(p)\, dp = c(s) \int_{\mathbb{R}^3} |x|^{-3-s}\, \phi(x)\, dx, \qquad c(s) := \frac{2^s\, \Gamma_{\mathrm{E}}(\frac{3+s}{2})}{\pi^{3/2}\, \Gamma_{\mathrm{E}}(-\frac{s}{2})}, \tag{D.xxxiv}$$

where $1/\Gamma_{\mathrm{E}}$ is entire. Moreover $c(s) = 0$ for $s = 0, 2, 4, \dots$, and $c(1) = -\pi^{-2}$.

(b) We have

$$\int_{\mathbb{R}^3} p^2 \log|p|\, (\mathcal{F}^{-1}\phi)(p)\, dp = c'(2) \int_{\mathbb{R}^3} |x|^{-5}\, \phi(x)\, dx, \qquad c'(2) = \frac{3}{2\pi}. \tag{D.xxxv}$$

■

Proof. Step 1. Let $\psi := \mathcal{F}^{-1}\phi \in \mathcal{S}(\mathbb{R}^3)$, and assume first $-3 < \operatorname{Re} s < 0$. The substitution $t \to t/p^2$ in Euler's integral gives $\int_0^\infty t^{-s/2-1}e^{-tp^2}\,dt = \Gamma_{\mathrm{E}}(-\frac{s}{2})\,|p|^s$ for $p \neq 0$. Since

$$\iint t^{-\operatorname{Re} s/2-1}e^{-tp^2}\,|\psi(p)|\,dt\,dp = \Gamma_{\mathrm{E}}(-\tfrac{1}{2}\operatorname{Re} s)\int |p|^{\operatorname{Re} s}|\psi(p)|\,dp < \infty,$$

Fubini's theorem yields

$$\int |p|^s\psi(p)\,dp = \Gamma_{\mathrm{E}}(-\tfrac{s}{2})^{-1}\int_0^\infty t^{-s/2-1}\Big[\int e^{-tp^2}\psi(p)\,dp\Big]dt.$$

By Fubini's theorem and the Gaussian integral $\int_{\mathbb{R}^3} e^{-tp^2+ip\cdot x}\,dp = (\pi/t)^{3/2}e^{-x^2/4t}$,

$$\int e^{-tp^2}\psi(p)\,dp = \int \phi(x)\Big[\int e^{-tp^2+ip\cdot x}\,d^\circ p\Big]dx = \int (4\pi t)^{-3/2}e^{-x^2/4t}\,\phi(x)\,dx.$$

For $x \neq 0$ the substitution $t \to x^2/4t$ gives

$$\int_0^\infty t^{-(s+3)/2-1}e^{-x^2/4t}\,dt = \Gamma_{\mathrm{E}}\big(\tfrac{s+3}{2}\big)\Big(\frac{x^2}{4}\Big)^{-(s+3)/2},$$

as $\operatorname{Re} s + 3 > 0$. The same identity with $\operatorname{Re} s$ in place of s,

$$\int_0^\infty t^{-(\operatorname{Re} s+3)/2-1}e^{-x^2/4t}\,dt = \Gamma_{\mathrm{E}}\big(\tfrac{\operatorname{Re} s+3}{2}\big)\Big(\frac{x^2}{4}\Big)^{-(\operatorname{Re} s+3)/2},$$

together with $\int |x|^{-\operatorname{Re} s-3}|\phi(x)|\,dx < \infty$, which holds as ϕ vanishes near 0 and lies in $\mathcal{S}(\mathbb{R}^3)$, shows that the modulus $(4\pi)^{-3/2}t^{-(\operatorname{Re} s+3)/2-1}e^{-x^2/4t}|\phi(x)|$ of the integrand $t^{-s/2-1}(4\pi t)^{-3/2}e^{-x^2/4t}\phi(x)$ is integrable over $(0,\infty)\times\mathbb{R}^3$. Hence Fubini's theorem applies once more, and

$$\int |p|^s\psi(p)\,dp = \Gamma_{\mathrm{E}}(-\tfrac{s}{2})^{-1}(4\pi)^{-3/2}\,2^{s+3}\,\Gamma_{\mathrm{E}}\big(\tfrac{s+3}{2}\big)\int |x|^{-s-3}\phi(x)\,dx,$$

with $(4\pi)^{-3/2}2^{s+3} = 2^s\pi^{-3/2}$. This is eq. D.xxxiv for $-3 < \operatorname{Re} s < 0$.

Step 2. The left side of eq. D.xxxiv is holomorphic in s on $\{\operatorname{Re} s > -3\}$, since $|p|^{\operatorname{Re} s}|\psi|$ provides integrable majorants locally uniformly in s; the function $s \mapsto \int |x|^{-3-s}\phi(x)\,dx$ is entire, since ϕ vanishes near 0 and lies in $\mathcal{S}(\mathbb{R}^3)$; and c is holomorphic on $\{\operatorname{Re} s > -3\}$, since Γ_{E} is holomorphic on the right half plane and $1/\Gamma_{\mathrm{E}}$ is entire. Both sides agree for $-3 < \operatorname{Re} s < 0$, hence on the connected set $\{\operatorname{Re} s > -3\}$ by the identity theorem. The entire functions $1/\Gamma_{\mathrm{E}}(z)$ and $z/\Gamma_{\mathrm{E}}(z+1)$ agree for $\operatorname{Re} z > 0$, hence for all z; therefore $1/\Gamma_{\mathrm{E}}$ vanishes at 0, and then successively at $-1,-2,\dots$, which gives $c(s) = 0$ for $s = 0,2,4,\dots$. Moreover $1/\Gamma_{\mathrm{E}}(-\frac{1}{2}) = -\frac{1}{2}/\Gamma_{\mathrm{E}}(\frac{1}{2}) = -\frac{1}{2\sqrt{\pi}}$, whence $c(1) = \frac{2\,\Gamma_{\mathrm{E}}(2)}{\pi^{3/2}}\cdot(-\frac{1}{2\sqrt{\pi}}) = -\pi^{-2}$.

Step 3. For (b) we differentiate eq. D.xxxiv with respect to s at $s = 2$. On the left we obtain $\int p^2\log|p|\,\psi(p)\,dp$, by differentiation under the integral; on the right

$$c'(2)\int |x|^{-5}\phi - c(2)\int |x|^{-5}\log|x|\,\phi,$$

where $c(2) = 0$. From $1/\Gamma_{\mathrm{E}}(z) = z(z+1)/\Gamma_{\mathrm{E}}(z+2)$ the derivative of $1/\Gamma_{\mathrm{E}}$ at $z = -1$ is $(2z+1)/\Gamma_{\mathrm{E}}(z+2)|_{z=-1} = -1$; with $z = -\frac{s}{2}$ this gives $\partial_s\big[1/\Gamma_{\mathrm{E}}(-\frac{s}{2})\big]_{s=2} = \frac{1}{2}$. As $1/\Gamma_{\mathrm{E}}(-1) = 0$, only this derivative contributes, and $c'(2) = 2^2\,\Gamma_{\mathrm{E}}(\frac{5}{2})\,\pi^{-3/2}\cdot\frac{1}{2}$ with $\Gamma_{\mathrm{E}}(\frac{5}{2}) = \frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi} = \frac{3}{4}\sqrt{\pi}$; thus $c'(2) = \frac{3}{2\pi}$. □

Lemma D.49 (Two Fourier series: Epstein-type sums near $\mathsf{p} = 0$)**.** For $n \in \{4,5\}$ and $\mathsf{p} \in \mathbb{R}^3$ let $\mathsf{s}_n(\mathsf{p}) := \sum_{u\in\mathbb{Z}^3\setminus\{0\}} |u|^{-n}e^{-i\mathsf{p}\cdot u}$. There are functions $\mathsf{e}_4, \mathsf{e}_5 \in C^\infty(\{|\mathsf{p}| < 1\})$, invariant under the 48 signed permutations of the coordinates, such that

$$\mathsf{s}_4(\mathsf{p}) = -\pi^2|\mathsf{p}| + \mathsf{e}_4(\mathsf{p}), \qquad \mathsf{s}_5(\mathsf{p}) = \tfrac{2\pi}{3}\,\mathsf{p}^2\log|\mathsf{p}| + \mathsf{e}_5(\mathsf{p}) \qquad (0 < |\mathsf{p}| < 1).$$

■

Proof. We treat the two cases $(\varpi, n, \mathsf{c}) = (|p|, 4, -\pi^{-2})$ and $(\varpi, n, \mathsf{c}) = (p^2 \log|p|, 5, \frac{3}{2\pi})$ together. The function ϖ is continuous, radial and smooth off 0, and its derivatives of order j are $O(|p|^{2-j} \lg|p|)$ at infinity. For an integer $\mathsf{N} \geq 3$ let

$$\mathsf{K}(x) := \int e^{ip\cdot x} \chi(p)\, \varpi(p)\, d^\circ p, \qquad E(x) := |x|^{-2\mathsf{N}} \int e^{ip\cdot x} (-\Delta_p)^{\mathsf{N}} [(1-\chi)\varpi](p)\, d^\circ p \quad (x \neq 0).$$

Step 1. The integrand of E vanishes for $|p| \leq 1$ and is $O(|p|^{2-2\mathsf{N}} \lg|p|)$, with $2\mathsf{N}-2 > 3$; hence the integral converges absolutely. The value of $E(x)$ does not depend on N: we write $|x|^2 e^{ip\cdot x} = -\Delta_p e^{ip\cdot x}$ and integrate by parts twice over balls (Gauss' theorem); the boundary terms on the sphere of radius r are $O\big((1+|x|)\mathsf{r}^2 \cdot \mathsf{r}^{2-2\mathsf{N}} \lg \mathsf{r}\big)$, which tends to 0 as $\mathsf{r} \to \infty$ because $\mathsf{N} \geq 3$; hence N and $\mathsf{N}+1$ give the same value. Consequently E is continuous off 0 and radial, and $|E(x)| \leq C_{\mathsf{N}} |x|^{-2\mathsf{N}}$ for every $\mathsf{N} \geq 3$.

Step 2. We show $\mathsf{K} + E = \mathsf{c}\,|x|^{-n}$ on $\mathbb{R}^3 \setminus \{0\}$. Let $\phi \in C_c^\infty(\mathbb{R}^3 \setminus \{0\})$. By Fubini's theorem, $\int \mathsf{K}\phi\, dx = \int \chi\varpi\, \mathcal{F}^{-1}\phi\, dp$. Let $\phi_{\mathsf{N}} := |x|^{-2\mathsf{N}}\phi \in C_c^\infty(\mathbb{R}^3 \setminus \{0\})$. By Fubini's theorem and $2\mathsf{N}$ integrations by parts, which are permitted as $\mathcal{F}^{-1}\phi_{\mathsf{N}} \in \mathcal{S}(\mathbb{R}^3)$, and by $(-\Delta_p)^{\mathsf{N}} \mathcal{F}^{-1}\phi_{\mathsf{N}} = \mathcal{F}^{-1}(|x|^{2\mathsf{N}}\phi_{\mathsf{N}}) = \mathcal{F}^{-1}\phi$,

$$\int E\phi\, dx = \int (-\Delta_p)^{\mathsf{N}} [(1-\chi)\varpi]\, \mathcal{F}^{-1}\phi_{\mathsf{N}}\, dp = \int (1-\chi)\varpi\, \mathcal{F}^{-1}\phi\, dp.$$

The sum is $\int \varpi\, \mathcal{F}^{-1}\phi\, dp = \mathsf{c} \int |x|^{-n}\phi\, dx$, by Lem. D.48(a) at $s = 1$ in the first case and by Lem. D.48(b) in the second. As $\mathsf{K} + E - \mathsf{c}|x|^{-n}$ is continuous on $\mathbb{R}^3 \setminus \{0\}$, it vanishes there.

Step 3. The periodic continuation of $\chi\varpi$, whose support lies in $\{|p| \leq 2\}$ and thus in the interior of $[-\pi, \pi]^3$, is continuous, and its Fourier coefficients are the numbers $\mathsf{K}(u)$, $u \in \mathbb{Z}^3$. By Steps 1 and 2, $\sum_u |\mathsf{K}(u)| < \infty$; hence the continuous function $\sum_u \mathsf{K}(u) e^{-i\mathsf{p}\cdot u}$ has the same Fourier coefficients and, both functions being continuous, coincides with the periodic continuation of $\chi\varpi$. For $|\mathsf{p}| < 1$ we have $\chi(\mathsf{p}) = 1$, and therefore

$$\varpi(\mathsf{p}) = \mathsf{K}(0) + \mathsf{c}\,\mathsf{s}_n(\mathsf{p}) - \sum_{u \neq 0} E(u)\, e^{-i\mathsf{p}\cdot u}.$$

We put $\mathsf{e}_n(\mathsf{p}) := \mathsf{c}^{-1}\big[\sum_{u\neq 0} E(u) e^{-i\mathsf{p}\cdot u} - \mathsf{K}(0)\big]$. This function is smooth, by the rapid decay of E, and invariant under the signed permutations of the coordinates, since E is radial. Finally $\mathsf{c}^{-1} = -\pi^2$, resp. $\mathsf{c}^{-1} = \frac{2\pi}{3}$, which gives the claim. □

Lemma D.50 (Coefficient of $\mathsf{p}^2 \log|\mathsf{p}|$ in Ψ_Γ; scaling of the cut-off)**.** Let $\Gamma \in \{1, \dots, 6\}$.

(a) There are numbers $\mathsf{a}_0, \mathsf{a}_1, \mathsf{a}_2$ and a function o on $\{|\mathsf{p}| < 1\}$ with $\mathsf{o}(\mathsf{p})/\mathsf{p}^2 \to 0$ as $\mathsf{p} \to 0$, depending on Γ only, such that for almost every p with $|\mathsf{p}| < 1$

$$\Psi_\Gamma(\mathsf{p}) = \mathsf{a}_0 + \mathsf{a}_1|\mathsf{p}| + \mathsf{a}_2\mathsf{p}^2 + \tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\, \mathsf{p}^2 \log|\mathsf{p}| + \mathsf{o}(\mathsf{p}).$$

(b) For real $\Lambda > 1$ let $R_\Gamma^\Lambda(k, l, \mathsf{p}) := R_\Gamma(k/\Lambda, l/\Lambda, \mathsf{p}/\Lambda)$. Then $\iint R_\Gamma^\Lambda F_\Gamma\, d^\circ k\, d^\circ l = \Lambda^2 \Psi_\Gamma(\mathsf{p}/\Lambda)$ for every $\mathsf{p} \neq 0$, and for almost every p with $|\mathsf{p}| < 1$

$$\iint (R_\Gamma^\Lambda - R_\Gamma) F_\Gamma\, d^\circ k\, d^\circ l = (\Lambda^2 - 1)\mathsf{a}_0 + (\Lambda - 1)\mathsf{a}_1|\mathsf{p}| - \tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}} \log\Lambda\, \mathsf{p}^2 + \Lambda^2 \mathsf{o}(\mathsf{p}/\Lambda) - \mathsf{o}(\mathsf{p}).$$

■

Proof. Step 1. By Lemmas D.43 and D.44,

$$\mathsf{T}_\Gamma^{\mathrm{mod}}(u) = c_{4,\Gamma}|u|^{-4} + \kappa_\Gamma^{\mathrm{lat}}|u|^{-5} + \mathsf{r}(u),$$
$$|\mathsf{r}(u)| \leq C|u|^{-6} \lg|u| \qquad (u \in \mathbb{Z}^3 \setminus \{0\});$$

we put $\mathsf{r}(0) := \mathsf{T}_\Gamma^{\mathrm{mod}}(0)$. Hence $\sum_u |\mathsf{T}_\Gamma^{\mathrm{mod}}(u)| < \infty$. The function Ψ_Γ is integrable and vanishes for $|\mathsf{p}| \geq \frac{3\pi}{8}$ (Def. D.40(3)), that is, outside a ball contained in $[-\pi,\pi]^3$. Therefore the periodic function which equals $24\Psi_\Gamma$ on $[-\pi,\pi]^3$ has the Fourier coefficients $24\int_{[-\pi,\pi]^3} e^{i\mathsf{p}\cdot u}\Psi_\Gamma(\mathsf{p})\, d^\circ\mathsf{p} = \mathsf{T}_\Gamma^{\mathrm{mod}}(u)$, which are also the Fourier coefficients of the continuous function $\sum_u \mathsf{T}_\Gamma^{\mathrm{mod}}(u)e^{-i\mathsf{p}\cdot u}$. By the uniqueness of Fourier coefficients the two periodic functions agree almost everywhere; as the first one coincides with $24\Psi_\Gamma$ on $[-\pi,\pi]^3$, we obtain, for almost every $\mathsf{p} \in [-\pi,\pi]^3$,

$$24\,\Psi_\Gamma(\mathsf{p}) = c_{4,\Gamma}\,\mathsf{s}_4(\mathsf{p}) + \kappa_\Gamma^{\mathrm{lat}}\,\mathsf{s}_5(\mathsf{p}) + \hat{\mathsf{r}}(\mathsf{p}),$$
$$\hat{\mathsf{r}}(\mathsf{p}) := \sum_u \mathsf{r}(u)\, e^{-i\mathsf{p}\cdot u}.$$

Step 2. Since $\sum_u |u|^2|\mathsf{r}(u)| < \infty$, the function $\hat{\mathsf{r}}$ is of class C^2. As $\mathsf{T}_\Gamma^{\mathrm{mod}}$ is the restriction to $\mathbb{Z}^3$ of a radial function, r and $\hat{\mathsf{r}}$ are invariant under the signed permutations of the coordinates. A function f of class C^2 near 0 with this invariance has vanishing gradient at 0 (invariance under $-\mathbf{1}$), a Hessian at 0 with vanishing off-diagonal entries (change of sign of one coordinate) and equal diagonal entries (permutations); hence $\mathsf{f}(\mathsf{p}) = \mathsf{f}(0) + \mathsf{a}\,\mathsf{p}^2 + o(\mathsf{p}^2)$. We apply this to $\hat{\mathsf{r}}$, e_4 and e_5 and insert Lem. D.49; this gives (a), the coefficient of $\mathsf{p}^2\log|\mathsf{p}|$ being $\frac{1}{24}\cdot\frac{2\pi}{3}\,\kappa_\Gamma^{\mathrm{lat}} = \frac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}$, and o being $\frac{1}{24}$ times the sum of the remainders $o(\mathsf{p}^2)$ of $c_{4,\Gamma}\mathsf{e}_4$, $\kappa_\Gamma^{\mathrm{lat}}\mathsf{e}_5$ and $\hat{\mathsf{r}}$.

Step 3. For (b) we substitute $k \to \Lambda k$, $l \to \Lambda l$ and use $F_\Gamma(\Lambda k, \Lambda l, \Lambda\mathsf{q}) = \Lambda^{-4}F_\Gamma(k,l,\mathsf{q})$ with $\mathsf{q} = \mathsf{p}/\Lambda$; this produces the factor $\Lambda^{6-4} = \Lambda^2$ and gives the first assertion. For the second we insert (a) at p/Λ and at p, both of modulus less than 1. The terms with a_2 cancel, the terms with a_0, a_1 give $(\Lambda^2-1)\mathsf{a}_0 + (\Lambda-1)\mathsf{a}_1|\mathsf{p}|$, and

$$\Lambda^2\,\tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\Lambda^{-2}\mathsf{p}^2(\log|\mathsf{p}| - \log\Lambda) - \tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\,\mathsf{p}^2\log|\mathsf{p}| = -\tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\log\Lambda\;\mathsf{p}^2.$$

□

Definition D.51 (Finite parts (analytic regularisation); cut-offs vanishing near the origin)**.** Let $\mathbb{D} := \{\mathsf{z} \in \mathbb{C} : \mathrm{Re}\,\mathsf{z} > -\frac{1}{2}\} \setminus \{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots\}$; this set is open and connected, and it contains 0.

(1) One variable. A *homogeneous factor of degree* $d \in \mathbb{Z}$ is a function $\mathsf{h} \in C^\infty(\mathbb{R}^3 \setminus \{0\})$ with $\mathsf{h}(tk) = t^d\mathsf{h}(k)$ for $t > 0$ and $\mathsf{h}(-k) = (-1)^d\mathsf{h}(k)$. Let $\mathfrak{P}$ be the set of functions on $\mathbb{R}^3\setminus\{0\}$ of the form $\mathsf{g} = \sum_\iota \mathsf{h}_\iota\mathsf{g}_\iota$, a finite sum, with homogeneous factors h_ι of degrees d_ι and with $\mathsf{g}_\iota \in C_c^\infty(\mathbb{R}^3)$. For such g the integral $\int |k|^{2\mathsf{z}}\mathsf{g}(k)\,dk$ converges absolutely and is holomorphic in z for $2\,\mathrm{Re}\,\mathsf{z} + \min_\iota d_\iota > -3$; it has a holomorphic continuation to $\mathbb{D}$ (Lem. D.52(a)), which is unique. The *finite part* $\mathrm{f.p.}\int \mathsf{g}\,dk$ is the value of this continuation at $\mathsf{z} = 0$; it depends on g only, and it is linear in g.

(2) Two variables. Let $c > 0$ and $\eta_c(k) := \theta(4|k|/c)$, so that $\eta_c(k) = 1$ for $|k| \leq \frac{c}{8}$ and $\eta_c(k) = 0$ for $|k| \geq \frac{c}{4}$. Let $\mathfrak{P}_2(c)$ be the set of Borel functions Φ on $\{(k,l) : k \neq 0,\ l \neq 0,\ k \neq l\}$ for which there is a constant C_Φ with the following properties.

- (Φ0) $\Phi(k,l) = 0$ if $|k| + |l| < c$ or $|k| + |l| > C_\Phi$.
- (Φ1) For $0 < |k| < \frac{c}{3}$, $\Phi(k,l) = \sum_\iota \mathsf{h}_\iota(k)\Phi_\iota(k,l)$, a finite sum with homogeneous factors h_ι and with functions $\Phi_\iota \in C^\infty(\{|k| < \frac{c}{3}\} \times \mathbb{R}^3)$ which vanish where $|k| + |l| < c$ or $|k| + |l| > C_\Phi$.
- (Φ2) For $0 < |l| < \frac{c}{3}$, $\Phi(k,l) = \sum_\iota \mathsf{h}'_\iota(l)\Phi'_\iota(k,l)$, with the same properties after the exchange of k and l.
- (Φ3) $|\Phi(k,l)| \leq C_\Phi|k-l|^{-2}$ for $|k|, |l| \geq \frac{c}{8}$.

For $\Phi \in \mathfrak{P}_2(c)$ the function

$$k \mapsto \eta_c(k)\int \Phi(k,l)\,dl = \sum_\iota \mathsf{h}_\iota(k)\,\eta_c(k)\int \Phi_\iota(k,l)\,dl$$

lies in $\mathfrak{P}$, since η_c vanishes for $|k| \geq \frac{c}{4}$ and the Φ_ι are smooth and vanish for $|l| > C_\Phi$; likewise $l \mapsto \eta_c(l) \int \Phi(k,l)\,dk$ lies in $\mathfrak{P}$. We define

$$\text{f.p.}\iint \Phi\,dk\,dl := \iint (1-\eta_c(k)-\eta_c(l))\Phi\,dk\,dl + \text{f.p.}\int \Big[\eta_c(k)\int \Phi(k,l)\,dl\Big]dk + \text{f.p.}\int \Big[\eta_c(l)\int \Phi(k,l)\,dk\Big]dl.$$

On the support of Φ, $|k| < \frac{c}{8}$ implies $|l| > \frac{7}{8}c$, hence $\eta_c(k) = 1$ and $\eta_c(l) = 0$; the same holds with k and l exchanged. Therefore the first integrand vanishes unless $|k|, |l| \geq \frac{c}{8}$, and the first integral converges absolutely, by (Φ0) and (Φ3): its integrand is bounded by $C_\Phi |k-l|^{-2}$, which is integrable over bounded sets of (k,l), and it vanishes for $|k|+|l| > C_\Phi$. This finite part is linear on $\mathfrak{P}_2(c)$ and symmetric under the exchange of k and l.

(3) Cut-offs. Let $\mathcal{Y}(c)$ be the set of functions $\Upsilon \in C^\infty(\mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3)$ for which there is a constant C_Υ with the following properties.

(Y1) $\Upsilon(k,l,\mathsf{p}) = 0$ if $|\mathsf{p}| < c$ and ($|k|+|l| < c$ or $|k|+|l| > C_\Upsilon$).
(Y2) $\Upsilon(-k,-l,-\mathsf{p}) = \Upsilon(k,l,\mathsf{p}) = \Upsilon(Ok, Ol, O\mathsf{p})$ for every rotation O.

Then $\mathcal{Y}(c') \subset \mathcal{Y}(c)$ for $c < c'$. For $\Lambda > 1$ and a function Υ of (k,l,p) or of (k,l), let Υ^Λ denote Υ evaluated at the arguments divided by Λ; if $\Upsilon \in \mathcal{Y}(c)$, then $\Upsilon^\Lambda \in \mathcal{Y}(\Lambda c) \subset \mathcal{Y}(c)$. For a function $G(k,l,\mathsf{p})$ which is of class C^2 in p near 0 we put

$$\overline{G_2}(k,l) := \tfrac{1}{6}\,\Delta_{\mathsf{p}} G(k,l,\mathsf{p})\big|_{\mathsf{p}=0};$$

this is the mean over the directions of p of the term of second order in the Taylor expansion of G in p, divided by p^2, since the mean of $\mathsf{p}_i\mathsf{p}_j$ over $\{|\mathsf{p}| = t\}$ is $\frac{1}{3}t^2\delta_{ij}$. Finally, for $\Upsilon \in \mathcal{Y}(c)$ and $0 < |\mathsf{p}| < c$ we put

$$\Xi_\Gamma[\Upsilon](\mathsf{p}) := \iint \Upsilon F_\Gamma\,d^\circ k\,d^\circ l;$$

this integral converges absolutely by eq. D.xxix and the estimates following it in Def. D.40(3), since by (Y1) the integrand vanishes for $|k|+|l| > C_\Upsilon$. ■

Lemma D.52 (Properties of finite parts)**.** (a) (Continuation with a parameter.) Let h be a homogeneous factor of degree d, and let $\mathsf{g}\colon \mathbb{R}^3 \times \mathbb{C} \to \mathbb{C}$ have the following properties: for every $\mathsf{z} \in \mathbb{C}$ the function $\mathsf{g}(\cdot;\mathsf{z})$ is smooth, and its support lies in a compact set which does not depend on z; for every multi-index α the derivative $\partial_k^\alpha \mathsf{g}$ is continuous on $\mathbb{R}^3 \times \mathbb{C}$ and, for each fixed k, an entire function of z. Then the function

$$\mathsf{z} \longmapsto \int |k|^{2\mathsf{z}}\,\mathsf{h}(k)\,\mathsf{g}(k;\mathsf{z})\,dk,$$

which is defined for $2\,\mathrm{Re}\,\mathsf{z} + d > -3$, has a holomorphic continuation to $\mathbb{D}$, and the value of this continuation at $\mathsf{z} = 0$ is $\text{f.p.}\int \mathsf{h}\,\mathsf{g}(\cdot;0)\,dk$.

(b) If $\mathsf{g} \in \mathfrak{P}$ is integrable, then $\text{f.p.}\int \mathsf{g}\,dk = \int \mathsf{g}\,dk$.

(c) Let h be a homogeneous factor of degree $d \leq -4$, and let $\eta(k) = \theta_1(|k|)$, where $\theta_1 \in C^\infty([0,\infty))$ has compact support and is equal to 1 near 0. Then

$$\text{f.p.}\int \eta\,\mathsf{h}\,dk = -\int (1-\eta)\,\mathsf{h}\,dk.$$

(d) (A soft line.) Let $\mu_0 > 0$ and $\eta(k) := \theta(|k|/2\mu_0)$. Let

$$S(k,\mathsf{p}) = \mathsf{P}(k,\mathsf{p})\,|k|^{-2\mathsf{a}}\,|k-\mathsf{p}|^{-2\mathsf{a}'},$$

where $\mathsf{a}, \mathsf{a}' \geq 0$ are integers and P is a polynomial, homogeneous of degree $2\mathsf{a}+2\mathsf{a}'-2$, and assume that $|S(k,\mathsf{p})| \leq C_S(|k|^{-2} + |k-\mathsf{p}|^{-2})$. Let $J \in C^\infty(\{|k| < 3\mu_0\} \times \{|\mathsf{p}| < \mu_0\})$ satisfy

$J(-k,-\mathsf{p}) = J(k,\mathsf{p})$, and for $k \neq 0$ and $i \in \{0,1,2\}$ put

$$[SJ]_i(k;\mathsf{p}) := \sum_{|\alpha|=i} \frac{\mathsf{p}^\alpha}{\alpha!}\, \partial_{\mathsf{p}}^\alpha (SJ)(k,0).$$

Then $\eta\,[SJ]_i(\cdot;\mathsf{p}) \in \mathfrak{P}$. Moreover, there are a Borel function h_S on $\mathbb{R}^3$, which depends on S only, is bounded on the unit sphere and satisfies $\mathsf{h}_S(t\mathsf{p}) = t\,\mathsf{h}_S(\mathsf{p})$ for $t>0$, and a constant $C = C(\mu_0, S, J)$, such that for $0 < |\mathsf{p}| \leq \frac{1}{2}\mu_0$

$$\Big| \int \eta SJ\, dk - \text{f. p.} \int \eta\,[SJ]_0\, dk - \text{f. p.} \int \eta\,[SJ]_2(k;\mathsf{p})\, dk - J(0,0)\,\mathsf{h}_S(\mathsf{p}) \Big| \leq C\,|\mathsf{p}|^3 \lg|\mathsf{p}|.$$

(e) (Two variables.) Let $\Phi \in \mathfrak{P}_2(c)$ and $(\epsilon,\epsilon') \in \{(1,1),(1,0),(0,1)\}$. If $\epsilon = 0$, assume that all degrees of the homogeneous factors in (Φ1) are ≥ -2; if $\epsilon' = 0$, assume that all degrees of the homogeneous factors in (Φ2) are ≥ -2. Then the integral

$$\iint |k|^{2\epsilon\mathsf{z}}\, |l|^{2\epsilon'\mathsf{z}}\, \Phi\, dk\, dl$$

converges absolutely for large $\operatorname{Re}\mathsf{z}$, and it has a holomorphic continuation to $\mathbb{D}$ whose value at $\mathsf{z}=0$ is f. p.$\iint \Phi\, dk\, dl$. If $0 < c' < c$, then $\mathfrak{P}_2(c) \subset \mathfrak{P}_2(c')$, and every $\Phi \in \mathfrak{P}_2(c)$ has the same finite part whether it is regarded as an element of $\mathfrak{P}_2(c)$ or of $\mathfrak{P}_2(c')$. If $\Phi \in \mathfrak{P}_2(c)$ is integrable, then f. p.$\iint \Phi\, dk\, dl = \iint \Phi\, dk\, dl$.

■

Proof. Proof of (a). Put $m := \max\{0, -d-3\}$; then $d+m+1 \geq -2$. For the proof of (a) let $\eta(k) := \theta(|k|)$, with the function θ of Def. D.27, so that $\eta = 1$ for $|k| \leq \frac{1}{2}$ and $\eta = 0$ for $|k| \geq 1$, and let

$$T\mathsf{g}(k;\mathsf{z}) := \sum_{|\alpha|\leq m} \frac{k^\alpha}{\alpha!}\, \partial_k^\alpha \mathsf{g}(0;\mathsf{z})$$

be the Taylor polynomial of degree m of $\mathsf{g}(\cdot;\mathsf{z})$ at $k=0$. For $2\operatorname{Re}\mathsf{z} + d > -3$ we decompose

$$\int |k|^{2\mathsf{z}} \mathsf{h}\,\mathsf{g}\, dk = \int |k|^{2\mathsf{z}} \mathsf{h}\,[\mathsf{g} - \eta\, T\mathsf{g}]\, dk + \sum_{|\alpha|\leq m} \frac{\partial_k^\alpha \mathsf{g}(0;\mathsf{z})}{\alpha!} \int |k|^{2\mathsf{z}}\, \eta(k)\, \mathsf{h}(k)\, k^\alpha\, dk.$$

The integrand of the first integral on the right vanishes outside a compact set which does not depend on z. For $|k| \leq \frac{1}{2}$ we have $\eta = 1$, and Taylor's formula, together with the continuity of the derivatives of order $m+1$ of g on $\mathbb{R}^3 \times \mathbb{C}$, bounds the modulus of the integrand by $C|k|^{2\operatorname{Re}\mathsf{z}+d+m+1}$, with a constant C that stays bounded as long as z ranges over a compact set. Since $d+m+1 \geq -2$, the exponent is larger than -3 as soon as $\operatorname{Re}\mathsf{z} > -\frac{1}{2}$. For each $k \neq 0$ the integrand is an entire function of z; differentiating under the integral sign, we see that the first integral is holomorphic on $\{\operatorname{Re}\mathsf{z} > -\frac{1}{2}\}$.

In the integrals of the sum we pass to polar coordinates and integrate by parts once. With

$$\mathsf{m}_\alpha := \int_{S^2} \mathsf{h}(\omega)\, \omega^\alpha\, d\omega$$

we obtain, for $2\operatorname{Re}\mathsf{z} + d + |\alpha| > -3$,

$$\int |k|^{2\mathsf{z}}\, \eta\, \mathsf{h}\, k^\alpha\, dk = \mathsf{m}_\alpha \int_0^\infty \theta(s)\, s^{2\mathsf{z}+d+|\alpha|+2}\, ds = -\frac{\mathsf{m}_\alpha}{2\mathsf{z}+d+|\alpha|+3} \int_0^\infty \theta'(s)\, s^{2\mathsf{z}+d+|\alpha|+3}\, ds.$$

As θ' is supported in $[\frac{1}{2},1]$, the last integral is an entire function of z. If $d+|\alpha|$ is odd, then $\omega \mapsto \mathsf{h}(\omega)\omega^\alpha$ is odd and $\mathsf{m}_\alpha = 0$. If $d+|\alpha|$ is even, the denominator vanishes only at $\mathsf{z} = -\frac{1}{2}(d+|\alpha|+3)$, which is half an odd integer and hence not a point of $\mathbb{D}$. Consequently the right side of the decomposition is holomorphic on $\mathbb{D}$; it is the asserted continuation, which is unique because $\mathbb{D}$ is connected.

The value of this continuation at $\mathsf{z}=0$ involves g only through $\mathsf{g}(\cdot;0)$: it contains the function $\mathsf{g}(\cdot;0) - \eta\, T\mathsf{g}(\cdot;0)$ and the numbers $\partial_k^\alpha \mathsf{g}(0;0)$. It is therefore equal to the value at 0 of the same

expression built with the function $\mathsf{g}(\cdot;0)$, which does not depend on z. By the argument just given, applied to this function, that expression continues $\mathsf{z}\mapsto\int|k|^{2\mathsf{z}}\mathsf{h}\,\mathsf{g}(\cdot;0)\,dk$ to $\mathbb{D}$, and by Def. D.51(1) its value at 0 is $\mathrm{f.p.}\int\mathsf{h}\,\mathsf{g}(\cdot;0)\,dk$.

Proof of (b). Let $\mathsf{g}\in\mathfrak{P}$ be integrable. Like every element of $\mathfrak{P}$, the function g vanishes outside a bounded set; hence for every $\mathsf{N}>0$ there is C_{N} with $|k|^{2\,\mathrm{Re}\,\mathsf{z}}|\mathsf{g}(k)|\le C_{\mathsf{N}}|\mathsf{g}(k)|$ whenever $0\le\mathrm{Re}\,\mathsf{z}\le\mathsf{N}$. By differentiation under the integral sign and by dominated convergence, $\int|k|^{2\mathsf{z}}\mathsf{g}\,dk$ is holomorphic on $\{\mathrm{Re}\,\mathsf{z}>0\}$ and continuous on $\{\mathrm{Re}\,\mathsf{z}\ge0\}$. The open set $\mathbb{D}\cap\{\mathrm{Re}\,\mathsf{z}>0\}$ is connected, and on it this function coincides with the continuation of Def. D.51(1), because the two functions agree for large $\mathrm{Re}\,\mathsf{z}$ (identity theorem). Letting $\mathsf{z}\downarrow0$ along the real axis, we obtain $\mathrm{f.p.}\int\mathsf{g}\,dk=\int\mathsf{g}\,dk$.

Proof of (c). Let $\mathsf{m}_0:=\int_{S^2}\mathsf{h}\,d\omega$. For $2\,\mathrm{Re}\,\mathsf{z}+d>-3$, polar coordinates and one integration by parts give

$$\int|k|^{2\mathsf{z}}\,\eta\,\mathsf{h}\,dk=\mathsf{m}_0\int_0^\infty\theta_1(s)\,s^{2\mathsf{z}+d+2}\,ds=-\frac{\mathsf{m}_0}{2\mathsf{z}+d+3}\int_0^\infty\theta_1'(s)\,s^{2\mathsf{z}+d+3}\,ds.$$

The support of θ_1' is a compact subset of $(0,\infty)$, so the last integral is an entire function of z. As in the proof of (a), either d is odd and $\mathsf{m}_0=0$, or the zero of the denominator is half an odd integer and lies outside $\mathbb{D}$. Hence the right side is the continuation of the left side to $\mathbb{D}$, and its value at $\mathsf{z}=0$ is

$$\mathrm{f.p.}\int\eta\,\mathsf{h}\,dk=-\frac{\mathsf{m}_0}{d+3}\int_0^\infty\theta_1'(s)\,s^{d+3}\,ds.$$

On the other hand, $1-\theta_1$ vanishes near 0 and $d+3<0$; an integration by parts without boundary terms therefore yields

$$\int(1-\eta)\,\mathsf{h}\,dk=\mathsf{m}_0\int_0^\infty(1-\theta_1(s))\,s^{d+2}\,ds=\frac{\mathsf{m}_0}{d+3}\int_0^\infty\theta_1'(s)\,s^{d+3}\,ds.$$

Comparing the last two displays, we obtain (c).

Proof of (d)*: preparation.* The function S is smooth on $\{k\neq0,\ k\neq\mathsf{p}\}$, homogeneous of degree -2 in (k,p) jointly, and even, since the polynomial P is homogeneous of even degree. The set $\{(k,\mathsf{q}):|k|=1,\ |\mathsf{q}|\le\frac12\}$ is compact, and S is smooth in a neighbourhood of it; together with homogeneity this gives, for every multi-index α,

$$|\partial_{\mathsf{p}}^\alpha S(k,\mathsf{q})|\le C_\alpha\,|k|^{-2-|\alpha|}\qquad(|\mathsf{q}|\le\tfrac12|k|).$$

For every multi-index β the function $\partial_{\mathsf{p}}^\beta S(\cdot,0)$ is a homogeneous factor of degree $-2-|\beta|$: it is smooth off 0, homogeneous of this degree, and of parity $(-1)^{|\beta|}$ because S is even. By the Leibniz rule, $\eta\,[SJ]_i(\cdot;\mathsf{p})$ is a finite sum of products of such factors with the functions $\eta\,\partial_{\mathsf{p}}^\gamma J(\cdot,0)$; these belong to $C_c^\infty(\mathbb{R}^3)$, since η vanishes for $|k|\ge2\mu_0$. Hence $\eta\,[SJ]_i(\cdot;\mathsf{p})\in\mathfrak{P}$.

Since J is even, the terms of odd degree in its Taylor expansion at $(0,0)$ vanish, and its Taylor polynomial of degree 3 is $J(0,0)+J_2$ with a quadratic form J_2 in (k,p). Thus $J=J(0,0)+J_2+J_4$, and Taylor's formula with integral remainder, applied to J and to its derivatives, gives

$$|\partial_{\mathsf{p}}^\alpha J_4(k,\mathsf{p})|\le C\,(|k|+|\mathsf{p}|)^{4-|\alpha|}\qquad(|\alpha|\le3)\quad\text{on }\{|k|\le2\mu_0,\ |\mathsf{p}|\le\tfrac12\mu_0\}.$$

Each of the four expressions inside the modulus in the asserted estimate, namely $\int\eta SJ\,dk$, $\mathrm{f.p.}\int\eta\,[SJ]_0\,dk$, $\mathrm{f.p.}\int\eta\,[SJ]_2(k;\mathsf{p})\,dk$ and $J(0,0)\,\mathsf{h}_S(\mathsf{p})$, depends linearly on J, because $J\mapsto[SJ]_i$ and the finite part are linear. We insert $J=J(0,0)+J_2+J_4$. The quadratic form J_2 and the function J_4 are even and vanish at $(0,0)$, so that for them the last of the four expressions is 0. It therefore suffices to consider, in place of SJ, the three functions

$$N:=S,\qquad N:=SJ_2,\qquad N:=SJ_4,$$

and to show that the difference $\int\eta N\,dk-\mathrm{f.p.}\int\eta\,[N]_0\,dk-\mathrm{f.p.}\int\eta\,[N]_2\,dk$ is equal to $\mathsf{h}_S(\mathsf{p})+O(|\mathsf{p}|^3)$ for $N=S$, the case which enters with the factor $J(0,0)$, and that it is $O(|\mathsf{p}|^3\lg|\mathsf{p}|)$ for $N=SJ_2$ and for $N=SJ_4$. For each of the three functions we write $N_i:=[N]_i$ $(i=0,1,2)$, and for $|k|\ge2|\mathsf{p}|$ we put $N_{\mathrm{r}}:=N-N_0-N_1-N_2$; moreover, $\mathsf{B}:=\{|k|\le2|\mathsf{p}|\}$. Since $|\mathsf{p}|\le\frac12\mu_0$, we have $\eta=1$ on B.

In all three cases $N(-k,-\mathsf{p}) = N(k,\mathsf{p})$, as S, J_2 and J_4 are even; differentiating with respect to p at $\mathsf{p}=0$, we see that N_1 is odd in k while N_0 and N_2 are even.

The case $N = S$. By the bound on the derivatives of S, $|N_i| \le C|\mathsf{p}|^i|k|^{-2-i}$. On the complement B^c of B, Taylor's formula in p, applied at the points $t\mathsf{p}$ with $0 \le t \le 1$, all of which satisfy $|t\mathsf{p}| \le \frac12|k|$, gives $|N_{\mathrm{r}}| \le C|\mathsf{p}|^3|k|^{-5}$. We write

$$\int \eta N\,dk = \int_{\mathsf{B}} N\,dk + \int_{\mathsf{B}^c} \eta\,(N_0+N_1+N_2+N_{\mathrm{r}})\,dk$$

and consider the four terms of the second integral one after the other. First, $\int_{\mathsf{B}^c}\eta N_0 = \int \eta N_0 - \int_{\mathsf{B}} N_0$. Second, $\int_{\mathsf{B}^c}\eta N_1 = 0$, because N_1 is odd and integrable over $\mathsf{B}^c \cap \operatorname{supp}\eta$, while η and B^c are invariant under $k \to -k$. Third, $N_2(\cdot;\mathsf{p})$ is a homogeneous factor of degree -4; assertion (c), applied with $\theta_1(s) := \theta(s/2\mu_0)$, and the vanishing of $1-\eta$ on B give

$$\int_{\mathsf{B}^c}\eta N_2 = \int_{\mathsf{B}^c} N_2 - \int (1-\eta)N_2 = \int_{\mathsf{B}^c} N_2 + \mathrm{f.\,p.}\int \eta N_2.$$

Fourth, $\int_{\mathsf{B}^c}\eta N_{\mathrm{r}} = \int_{\mathsf{B}^c} N_{\mathrm{r}} - \int(1-\eta)N_{\mathrm{r}}$, and

$$\Big|\int (1-\eta)N_{\mathrm{r}}\Big| \le C|\mathsf{p}|^3 \int_{|k|\ge\mu_0} |k|^{-5}\,dk \le C|\mathsf{p}|^3.$$

Collecting the terms, we arrive at

$$\int \eta S\,dk = \int \eta N_0\,dk + \mathrm{f.\,p.}\int \eta N_2\,dk + \mathsf{h}_S(\mathsf{p}) + O(|\mathsf{p}|^3),$$

$$\mathsf{h}_S(\mathsf{p}) := \int_{|k|\le 2|\mathsf{p}|} (S-N_0)\,dk + \int_{|k|>2|\mathsf{p}|} (N_2+N_{\mathrm{r}})\,dk.$$

Both integrals in the definition of h_S converge absolutely, because $|S|+|N_0| \le C(|k|^{-2}+|k-\mathsf{p}|^{-2})$, and $|N_2+N_{\mathrm{r}}| \le C|\mathsf{p}|^2|k|^{-4}$ on B^c. The substitution $k \to tk$, together with $N_i(tk;t\mathsf{p}) = t^{-2}N_i(k;\mathsf{p})$, yields $\mathsf{h}_S(t\mathsf{p}) = t\,\mathsf{h}_S(\mathsf{p})$ for $t>0$; the same bounds, taken at $|\mathsf{p}|=1$, show that h_S is bounded on the unit sphere $\{|\mathsf{p}|=1\}$, so that $|\mathsf{h}_S(\mathsf{p})| \le C|\mathsf{p}|$; and h_S is a Borel function by Fubini's theorem. By its definition h_S depends on S only. Finally, $\eta N_0 \in \mathfrak{P}$ is integrable, so that $\int \eta N_0\,dk$ is its finite part by (b).

The case $N = SJ_2$. Now N is homogeneous of degree 0 in (k,p), and as above $|\partial_{\mathsf{p}}^{\alpha} N(k,\mathsf{q})| \le C|k|^{-|\alpha|}$ for $|\mathsf{q}| \le \frac12|k|$. Hence $|N_i| \le C|\mathsf{p}|^i|k|^{-i}$, which is integrable near $k=0$ for $i \le 2$; further $|N_{\mathrm{r}}| \le C|\mathsf{p}|^3|k|^{-3}$ on B^c, and $|N| \le C|\mathsf{p}|^2(|k|^{-2}+|k-\mathsf{p}|^{-2})$ on B. Therefore

$$\int \eta N\,dk = \sum_{i\le 2}\int \eta N_i\,dk + \Big[\int_{\mathsf{B}}(N-N_0-N_1-N_2)\,dk + \int_{\mathsf{B}^c}\eta N_{\mathrm{r}}\,dk\Big],$$

and the bracket is bounded in modulus by

$$C|\mathsf{p}|^3 + C|\mathsf{p}|^3\int_{2|\mathsf{p}|\le|k|\le 2\mu_0} |k|^{-3}\,dk \le C|\mathsf{p}|^3\lg|\mathsf{p}|.$$

The term with $i=1$ vanishes because N_1 is odd, and the terms with $i=0$ and $i=2$ are finite parts by (b).

The case $N = SJ_4$. By the Leibniz rule and the bounds on the derivatives of S and of J_4,

$$|\partial_{\mathsf{p}}^{\alpha} N(k,\mathsf{q})| \le C|k|^{2-|\alpha|} \qquad (|\alpha|\le 3,\ |\mathsf{q}| \le \tfrac12|k|,\ |\mathsf{q}| \le \tfrac12\mu_0,\ |k| \le 2\mu_0).$$

Hence $|N_i| \le C|\mathsf{p}|^i|k|^{2-i}$, $|N_{\mathrm{r}}| \le C|\mathsf{p}|^3|k|^{-1}$ on $\mathsf{B}^c \cap \operatorname{supp}\eta$, and $|N| \le C|\mathsf{p}|^4(|k|^{-2}+|k-\mathsf{p}|^{-2})$ on B. Arguing as in the previous case, we find $\int \eta N\,dk = \int \eta N_0\,dk + \int \eta N_2\,dk + O(|\mathsf{p}|^3)$, and the two integrals on the right are finite parts by (b).

Adding the three cases, the first one multiplied by $J(0,0)$, and using $|\mathsf{p}|^3 \le |\mathsf{p}|^3\lg|\mathsf{p}|$, we obtain the estimate of (d), with a constant depending on μ_0, S and J.

Proof of (e). We split $1 = \big(1-\eta_c(k)-\eta_c(l)\big)+\eta_c(k)+\eta_c(l)$ and treat the three resulting integrals separately; each of them converges absolutely for large $\operatorname{Re}\mathsf{z}$, as the following discussion shows.

On the support of Φ, $|k|<\frac{c}{8}$ implies $|l|>\frac{7}{8}c$ by (Φ0), hence $\eta_c(k)=1$ and $\eta_c(l)=0$; the same holds with k and l exchanged. Therefore the first integrand, which contains the factor $1-\eta_c(k)-\eta_c(l)$, vanishes unless $|k|,|l|\geq\frac{c}{8}$, and by (Φ0) it vanishes unless $|k|+|l|\leq C_\Phi$. On this bounded set the factor $|k|^{2\epsilon\mathsf{z}}|l|^{2\epsilon'\mathsf{z}}$ is an entire function of z, bounded for z in compact sets, while $|\Phi|\leq C_\Phi|k-l|^{-2}$ by (Φ3). Hence the first integral is an entire function of z, whose value at $\mathsf{z}=0$ is $\iint\big(1-\eta_c(k)-\eta_c(l)\big)\Phi\,dk\,dl$.

In the second integral, $\eta_c(k)\neq 0$ forces $|k|\leq\frac{c}{4}<\frac{c}{3}$, so that ($\Phi$1) applies, and Fubini's theorem turns this integral into

$$\sum_\iota \int |k|^{2\epsilon\mathsf{z}}\,\mathsf{h}_\iota(k)\,\mathsf{g}_\iota(k;\mathsf{z})\,dk, \qquad \mathsf{g}_\iota(k;\mathsf{z}) := \eta_c(k)\int |l|^{2\epsilon'\mathsf{z}}\,\Phi_\iota(k,l)\,dl.$$

Since Φ_ι vanishes where $|k|+|l|<c$ or $|k|+|l|>C_\Phi$, the integrand defining g_ι vanishes unless $\frac{3}{4}c\leq|l|\leq C_\Phi$; consequently g_ι has the properties required in (a). If $\epsilon=1$, assertion (a) provides a holomorphic continuation to $\mathbb{D}$ whose value at $\mathsf{z}=0$ is $\mathrm{f.p.}\int \mathsf{h}_\iota\,\mathsf{g}_\iota(\cdot;0)\,dk$. If $\epsilon=0$, all degrees of the h_ι are ≥ -2; the k-integral then converges absolutely for every z and defines an entire function, whose value at 0 is again the finite part, by (b). Summing over ι, we obtain at $\mathsf{z}=0$ the number $\mathrm{f.p.}\int\big[\eta_c(k)\int\Phi(k,l)\,dl\big]dk$.

The hypotheses of (e) are symmetric under the exchange of k with l, of ϵ with ϵ' and of (Φ1) with (Φ2): the set $\{(1,1),(1,0),(0,1)\}$ is invariant under $(\epsilon,\epsilon')\mapsto(\epsilon',\epsilon)$, conditions ($\Phi$0) and ($\Phi$3) do not change when k and l are interchanged, and the assumptions on the degrees in (Φ1) and (Φ2) are interchanged. Under this exchange the third integral, which contains the factor $\eta_c(l)$, becomes the second one, and the argument just given applies to it. In detail: $\eta_c(l)\neq 0$ forces $|l|\leq\frac{c}{4}<\frac{c}{3}$, so that ($\Phi$2) applies, and Fubini's theorem turns the third integral into

$$\sum_\iota \int |l|^{2\epsilon'\mathsf{z}}\,\mathsf{h}'_\iota(l)\,\mathsf{g}'_\iota(l;\mathsf{z})\,dl, \qquad \mathsf{g}'_\iota(l;\mathsf{z}) := \eta_c(l)\int |k|^{2\epsilon\mathsf{z}}\,\Phi'_\iota(k,l)\,dk.$$

Since Φ'_ι vanishes where $|k|+|l|<c$ or $|k|+|l|>C_\Phi$, the integrand defining g'_ι vanishes unless $\frac{3}{4}c\leq|k|\leq C_\Phi$, and g'_ι has the properties required in (a). If $\epsilon'=1$, assertion (a) provides a holomorphic continuation to $\mathbb{D}$ whose value at $\mathsf{z}=0$ is $\mathrm{f.p.}\int \mathsf{h}'_\iota\,\mathsf{g}'_\iota(\cdot;0)\,dl$. If $\epsilon'=0$, all degrees of the h'_ι are ≥ -2; the l-integral then converges absolutely for every z and defines an entire function, whose value at 0 is the finite part, by (b). Summing over ι, we obtain at $\mathsf{z}=0$ the number $\mathrm{f.p.}\int\big[\eta_c(l)\int\Phi(k,l)\,dk\big]dl$. The sum of the three values at $\mathsf{z}=0$ is the right side of the definition of $\mathrm{f.p.}\iint\Phi\,dk\,dl$ in Def. D.51(2).

Change of the parameter c. Let $0<c'<c$ and $\Phi\in\mathfrak{P}_2(c)$. Conditions (Φ0), (Φ1) and (Φ2) at c' follow from the same conditions at c by restriction. For (Φ3) at c', let $|k|,|l|\geq\frac{c'}{8}$ and $\Phi(k,l)\neq 0$. If $|k|,|l|\geq\frac{c}{8}$, (Φ3) at c applies. Both $|k|$ and $|l|$ smaller than $\frac{c}{8}$ is impossible by (Φ0). If $\frac{c'}{8}\leq|k|<\frac{c}{8}$, we use $\Phi=\sum_\iota\mathsf{h}_\iota\Phi_\iota$ from (Φ1): the h_ι are bounded on $\{\frac{c'}{8}\leq|k|\leq\frac{c}{8}\}$, the Φ_ι are bounded there as well (they are continuous on $\{|k|\leq\frac{c}{8}\}\times\mathbb{R}^3$ and vanish for $|l|>C_\Phi$), and $|k-l|\leq|k|+|l|\leq C_\Phi$; hence $|\Phi|\leq C\leq C\,C_\Phi^2|k-l|^{-2}$. The case $\frac{c'}{8}\leq|l|<\frac{c}{8}$ is symmetric, with (Φ2). Thus $\mathfrak{P}_2(c)\subset\mathfrak{P}_2(c')$. The integral with $(\epsilon,\epsilon')=(1,1)$ does not involve c, so the value at $\mathsf{z}=0$ of its continuation, which is the finite part with respect to c and also the finite part with respect to c', is the same.

Finally, let Φ be integrable. By Fubini's theorem the functions $k\mapsto\eta_c(k)\int\Phi(k,l)\,dl$ and $l\mapsto\eta_c(l)\int\Phi(k,l)\,dk$ are integrable elements of $\mathfrak{P}$, so by (b) their finite parts are their integrals, and the definition of $\mathrm{f.p.}\iint\Phi$ reduces to $\iint\Phi\,dk\,dl$. □

Proposition D.53 (Expansion by regions of $\Xi_\Gamma[\Upsilon]$ at small p)**.** Let $\Gamma\in\{1,\dots,6\}$, $c>0$ and $\Upsilon\in\mathcal{Y}(c)$. Then $\overline{(\Upsilon F_\Gamma)_2}\in\mathfrak{P}_2(c)$, and there are numbers α, β and C, depending only on Γ, c and Υ,

such that

$$|\Xi_\Gamma[\Upsilon](\mathsf{p}) - \alpha - \beta\,|\mathsf{p}| - \mathfrak{f}_\Gamma[\Upsilon]\,\mathsf{p}^2| \le C\,|\mathsf{p}|^3 \lg|\mathsf{p}| \qquad (0 < |\mathsf{p}| \le \tfrac{c}{16}),$$

where

$$\mathfrak{f}_\Gamma[\Upsilon] := (2\pi)^{-6}\,\mathrm{f.\,p.}\iint \overline{(\Upsilon F_\Gamma)_2}\,dk\,dl.$$

The map $\Upsilon \mapsto \mathfrak{f}_\Gamma[\Upsilon]$ is linear on $\mathcal{Y}(c)$, and $\mathfrak{f}_\Gamma[\Upsilon^\Lambda] = \mathfrak{f}_\Gamma[\Upsilon]$ for every $\Lambda > 1$. ■

Proof. Step 1: two factorisations of F_Γ. We write the integrand F_Γ of eq. D.xxviii in two ways,

$$F_\Gamma = \sum S \cdot H = \sum S' \cdot H',$$

where the sums extend over the tensor indices carried by S, resp. by S', the factor S depends on (k, p) and S' on (l, p) as in the following table, and H, resp. H', denotes the product of all the other factors of F_Γ:

Γ	$S(k,\mathsf{p})$	$S'(l,\mathsf{p})$
1	$v_2(k,\mathsf{p}-k)_{\mu\nu}/\mathsf{x}_1\mathsf{x}'_1$	$v_2(-l,l-\mathsf{p})_{\rho\sigma}/\mathsf{x}_2\mathsf{x}'_2$
2, 3	$v_2(k,\mathsf{p}-k)_{\mu\nu}v_2(-k,k-\mathsf{p})_{\rho\nu}/\mathsf{x}_1^2\mathsf{x}'_1$	$1/\mathsf{x}_2$
4	$v_2(k,\mathsf{p}-k)_{\mu\nu}/\mathsf{x}_1\mathsf{x}'_1$	$v_2(l,\mathsf{p}-l)_{\mu\nu}/\mathsf{x}_2\mathsf{x}'_2$
5	$v_2(k-\mathsf{p},-k)_{\alpha\sigma}/\mathsf{x}_1\mathsf{x}'_1$	$1/\mathsf{x}_2$
6	$1/\mathsf{x}'_1$	$1/\mathsf{x}_2$

For $\Gamma = 4$ this means

$$H_{\mu\nu} = -\tfrac{1}{2}[\delta_{\mu\nu}v_2(l,\mathsf{p}-l)_{\lambda\lambda} - \tfrac{1}{2}v_2(l,\mathsf{p}-l)_{\mu\nu} - \tfrac{1}{2}v_2(l,\mathsf{p}-l)_{\nu\mu}]/\mathsf{x}_2\mathsf{x}'_2,$$

and H' has the same form, with $v_2(k,\mathsf{p}-k)$ and $\mathsf{x}_1\mathsf{x}'_1$ in place of $v_2(l,\mathsf{p}-l)$ and $\mathsf{x}_2\mathsf{x}'_2$. Every component of S has the form required in Lem. D.52(d): it is a homogeneous polynomial divided by $|k|^{2\mathsf{a}}|k-\mathsf{p}|^{2\mathsf{a}'}$, with $(\mathsf{a},\mathsf{a}') = (1,1)$ for $\Gamma \in \{1,4,5\}$, $(2,1)$ for $\Gamma \in \{2,3\}$ and $(0,1)$ for $\Gamma = 6$, and since $|v_2(k_1,k_2)| \le 2|k_1||k_2|$ its modulus is at most $2/|k||k-\mathsf{p}| \le |k|^{-2} + |k-\mathsf{p}|^{-2}$ for $\Gamma \in \{1,4,5\}$, at most $4|k|^{-2}$ for $\Gamma \in \{2,3\}$, and equal to $|k-\mathsf{p}|^{-2}$ for $\Gamma = 6$. The same holds for the components of S', as functions of l. The factor S contains all factors $\mathsf{x}_1, \mathsf{x}'_1$ of the denominator of F_Γ, and S' all factors $\mathsf{x}_2, \mathsf{x}'_2$. Hence H is a polynomial divided by powers of $\mathsf{x}_2, \mathsf{x}'_2, \mathsf{x}_3$, and H' a polynomial divided by powers of $\mathsf{x}_1, \mathsf{x}'_1, \mathsf{x}_3$; in both, x_3 occurs at most to the first power. Moreover, H and H' are even under the reflection of all their arguments: they are quotients of homogeneous polynomials of even degree, because besides factors v_2 each of them contains two factors v_3, or the factor $l_\rho(l-k)_\mu$, or neither.

Step 2: where the lines are long. Put $\mu_0 := \frac{c}{8}$ and $\eta := \eta_c = \theta(|\cdot|/2\mu_0)$. Let $|\mathsf{p}| < \mu_0$. By (Y1), $\Upsilon(k,l,\mathsf{p}) \neq 0$ implies $|k| + |l| \ge 8\mu_0$. If, in addition, $|k| < 3\mu_0$, then

$$|l| > 5\mu_0, \qquad |l-\mathsf{p}| > 4\mu_0, \qquad |k-l| > 2\mu_0;$$

these are the lengths whose squares $\mathsf{x}_2, \mathsf{x}'_2, \mathsf{x}_3$ occur in the denominator of H. Consequently the product ΥH, extended by 0 where $\Upsilon = 0$, is smooth on $\{|k| < 3\mu_0\} \times \mathbb{R}^3 \times \{|\mathsf{p}| < \mu_0\}$: this set is covered by the open set $\{|k| + |l| < 8\mu_0,\ |\mathsf{p}| < c\}$, on which Υ vanishes identically, and by the open set $\{|k| < 3\mu_0,\ |l| > 5\mu_0,\ |\mathsf{p}| < \mu_0\}$, on which H is smooth. Symmetrically, if $\Upsilon(k,l,\mathsf{p}) \neq 0$ and $|l| < 3\mu_0$, then $|k| \ge 8\mu_0 - |l|$, and

$$|k| > 5\mu_0, \qquad |k-\mathsf{p}| > 4\mu_0, \qquad |k-l| > 2\mu_0;$$

these are the lengths whose squares $\mathsf{x}_1, \mathsf{x}'_1, \mathsf{x}_3$ occur in the denominator of H'. The set $\mathbb{R}^3 \times \{|l| < 3\mu_0\} \times \{|\mathsf{p}| < \mu_0\}$ is covered by the open set $\{|k|+|l| < 8\mu_0,\ |\mathsf{p}| < c\}$, on which Υ vanishes identically, and by the open set $\{|k| > 5\mu_0,\ |l| < 3\mu_0,\ |\mathsf{p}| < \mu_0\}$, on which H' is smooth; hence $\Upsilon H'$, extended by 0 where $\Upsilon = 0$, is smooth on it. Finally, if $|k|, |l| \ge \mu_0$ and $|\mathsf{p}| \le \frac{1}{2}\mu_0$, then $\mathsf{x}_1, \mathsf{x}'_1, \mathsf{x}_2, \mathsf{x}'_2 \ge \frac{1}{4}\mu_0^2$, and therefore $F_\Gamma = F^\sharp/\mathsf{x}_3$ with a function $F^\sharp$ which is smooth on this set.

Step 3: $\Phi := \overline{(\Upsilon F_\Gamma)_2}$ *belongs to* $\mathfrak{P}_2(c)$. Condition (Φ0) follows from (Y1): where $|k|+|l|<c$ or $|k|+|l|>C_\Upsilon$, the function ΥF_Γ vanishes for all $|\mathsf{p}|<c$, and so does Φ. For (Φ1) let $0<|k|<\frac{c}{3}$; then $|k|<3\mu_0$, and

$$\Phi = \frac{1}{6}\sum_j \sum \Big[\partial_{\mathsf{p}_j}^2 S\cdot \Upsilon H + 2\,\partial_{\mathsf{p}_j} S\cdot \partial_{\mathsf{p}_j}(\Upsilon H) + S\cdot \partial_{\mathsf{p}_j}^2(\Upsilon H)\Big]_{\mathsf{p}=0}$$

by the Leibniz rule. Here $\partial_{\mathsf{p}}^\beta S(\cdot,0)$ is a homogeneous factor of degree $-2-|\beta|$, and $\partial_{\mathsf{p}}^\beta(\Upsilon H)(k,l,0)$ is smooth by Step 2. The latter function vanishes wherever $\Upsilon(\cdot,\cdot,\mathsf{p})=0$ for all small p, that is, by (Y1), where $|k|+|l|<c$ or $|k|+|l|>C_\Upsilon$; this is the support condition of (Φ1), with any $C_\Phi \geq C_\Upsilon$. Condition (Φ2) is obtained from the factorisation $F_\Gamma = \sum S'\cdot H'$ by the same argument with the roles of k and l exchanged: for $0<|l|<\frac{c}{3}$ the Leibniz rule writes Φ as the corresponding sum of products of the functions $\partial_{\mathsf{p}}^\beta S'(\cdot,0)$, which are homogeneous factors of degree $-2-|\beta|$ in l, with the functions $\partial_{\mathsf{p}}^\beta(\Upsilon H')(k,l,0)$, which are smooth on $\mathbb{R}^3\times\{|l|<3\mu_0\}$ by Step 2 and vanish where $|k|+|l|<c$ or $|k|+|l|>C_\Upsilon$. For (Φ3) let $|k|,|l|\geq\frac{c}{8}=\mu_0$; by Step 2,

$$\Phi = \tfrac{1}{6}\,\Delta_{\mathsf{p}}(\Upsilon F^\sharp)\big|_{\mathsf{p}=0}/\mathsf{x}_3,$$

and the numerator is bounded, being smooth with compact support in (k,l).

Step 4: three regions. For $0<|\mathsf{p}|\leq\frac{1}{2}\mu_0$ we split $\Xi_\Gamma[\Upsilon]=\Xi_{\mathrm{H}}+\Xi_{\mathrm{A}}+\Xi_{\mathrm{B}}$, inserting into the integral the factors $1-\eta(k)-\eta(l)$, $\eta(k)$ and $\eta(l)$, respectively. If $\Upsilon\neq 0$ and $|k|<\mu_0$, then $|l|>7\mu_0$, hence $\eta(k)=1$ and $\eta(l)=0$; thus on the support of Υ the first factor vanishes unless $|k|,|l|\geq\mu_0$, and the same holds with k and l exchanged. Each of the three terms is invariant under rotations of p, by (Y2), by the rotation invariance of F_Γ, and because η is radial.

Step 5: the hard region. The integrand of Ξ_{H} is $(1-\eta(k)-\eta(l))\,\Upsilon F^\sharp/\mathsf{x}_3$; it is supported in $\{\mu_0\leq|k|,|l|\leq C_\Upsilon\}$, and all its derivatives with respect to p are bounded by $C|k-l|^{-2}$. Hence Ξ_{H} is smooth on $\{|\mathsf{p}|<\frac{1}{2}\mu_0\}$; it is even, by (Y2) and the evenness of F_Γ, and rotation invariant. At $\mathsf{p}=0$ its gradient and its third derivatives therefore vanish, and its Hessian is a multiple of the identity, so that

$$\Xi_{\mathrm{H}}(\mathsf{p}) = \Xi_{\mathrm{H}}(0) + \tfrac{1}{6}\,\Delta\Xi_{\mathrm{H}}(0)\,\mathsf{p}^2 + O(|\mathsf{p}|^4), \qquad \tfrac{1}{6}\,\Delta\Xi_{\mathrm{H}}(0) = (2\pi)^{-6}\iint (1-\eta(k)-\eta(l))\,\Phi\,dk\,dl.$$

Step 6: k soft. Let $J(k,\mathsf{p}) := \int \Upsilon H\, d^\circ l$, one function for each value of the tensor indices of S. By Step 2, and since ΥH vanishes for $|l|>C_\Upsilon$, the function J is smooth on $\{|k|<3\mu_0\}\times\{|\mathsf{p}|<\mu_0\}$, and it is even, as the substitution $l\to -l$ shows. By Fubini's theorem $\Xi_{\mathrm{A}} = \sum \int \eta S J\, d^\circ k$. We apply Lem. D.52(d), with the present μ_0, to each component of S and the corresponding component of J; for $0<|\mathsf{p}|\leq\frac{1}{2}\mu_0$ this gives

$$(2\pi)^3\,\Xi_{\mathrm{A}}(\mathsf{p}) = \sum\Big\{\text{f. p.}\int \eta\,[SJ]_0\,dk + \sum_{i,j}\mathsf{p}_i\mathsf{p}_j\,\tfrac{1}{2}\,\text{f. p.}\int \eta\,\partial_{\mathsf{p}_i}\partial_{\mathsf{p}_j}(SJ)(k,0)\,dk + J(0,0)\,\mathsf{h}_S(\mathsf{p})\Big\} + O(|\mathsf{p}|^3\lg|\mathsf{p}|).$$

We take the mean of both sides over the sphere $|\mathsf{p}|=t$. The left side is constant there, by Step 4. The mean of h_S is t times its mean over the unit sphere, by homogeneity, and the mean of $\mathsf{p}_i\mathsf{p}_j$ is $\frac{1}{3}t^2\delta_{ij}$. For $k\neq 0$, differentiation under the integral sign gives $\sum\frac{1}{6}\Delta_{\mathsf{p}}(SJ)(k,0) = \int\Phi(k,l)\,d^\circ l$. Since the finite part is linear, we conclude that

$$\Xi_{\mathrm{A}}(\mathsf{p}) = \alpha_{\mathrm{A}} + \beta_{\mathrm{A}}|\mathsf{p}| + (2\pi)^{-6}\,\mathsf{p}^2\,\text{f. p.}\int\Big[\eta(k)\int\Phi(k,l)\,dl\Big]dk + O(|\mathsf{p}|^3\lg|\mathsf{p}|)$$

with numbers α_{A}, β_{A}.

Step 7: l soft, and the sum. For Ξ_{B} we exchange the roles of k and l in Step 6. We use the factorisation $F_\Gamma=\sum S'\cdot H'$ of Step 1 and put $J'(l,\mathsf{p}) := \int\Upsilon H'\,d^\circ k$, one function for each value of

the tensor indices of S'. By Step 2, and since $\Upsilon H'$ vanishes for $|k| > C_\Upsilon$, the function J' is smooth on $\{|l| < 3\mu_0\} \times \{|\mathsf{p}| < \mu_0\}$, and it is even, as the substitution $k \to -k$ shows; by Fubini's theorem $\Xi_{\mathrm{B}} = \sum \int \eta S' J' \, d^\circ l$. We apply Lem. D.52(d), with l in place of k, to each component of S' and the corresponding component of J', and we take the mean over the sphere $|\mathsf{p}| = t$ as in Step 6, now with $\sum \frac{1}{6}\Delta_{\mathsf{p}}(S'J')(l,0) = \int \Phi(k,l)\, d^\circ k$ for $l \neq 0$. This gives

$$\Xi_{\mathrm{B}}(\mathsf{p}) = \alpha_{\mathrm{B}} + \beta_{\mathrm{B}}|\mathsf{p}| + (2\pi)^{-6}\,\mathsf{p}^2\,\mathrm{f.p.}\int \Big[\eta(l)\int \Phi(k,l)\,dk\Big]dl + O(|\mathsf{p}|^3 \lg|\mathsf{p}|) \qquad (0 < |\mathsf{p}| \leq \tfrac{1}{2}\mu_0)$$

with numbers α_{B}, β_{B}. Since $\eta = \eta_c$ and $\Phi \in \mathfrak{P}_2(c)$ by Step 3, Def. D.51(2) shows that the sum of the three coefficients of p^2 obtained in Steps 5–7 is $\mathfrak{f}_\Gamma[\Upsilon]$. Adding the three expansions, and noting that $\frac{1}{2}\mu_0 = \frac{c}{16}$, we obtain the asserted estimate, with $\alpha := \Xi_{\mathrm{H}}(0) + \alpha_{\mathrm{A}} + \alpha_{\mathrm{B}}$ and $\beta := \beta_{\mathrm{A}} + \beta_{\mathrm{B}}$.

Step 8: linearity and scaling. The map $\Upsilon \mapsto \overline{(\Upsilon F_\Gamma)_2}$ and the finite part are linear; hence so is $\mathfrak{f}_\Gamma$. Let $\Lambda > 1$. The substitution $k \to \Lambda k$, $l \to \Lambda l$, together with the homogeneity of degree -4 of F_Γ, gives $\Xi_\Gamma[\Upsilon^\Lambda](\mathsf{p}) = \Lambda^2\,\Xi_\Gamma[\Upsilon](\mathsf{p}/\Lambda)$, so that by the estimate already proved

$$\Xi_\Gamma[\Upsilon^\Lambda](\mathsf{p}) = \Lambda^2\alpha + \Lambda\beta\,|\mathsf{p}| + \mathfrak{f}_\Gamma[\Upsilon]\,\mathsf{p}^2 + O(|\mathsf{p}|^3 \lg|\mathsf{p}|).$$

On the other hand $\Upsilon^\Lambda \in \mathcal{Y}(c)$, and the estimate applied to Υ^Λ gives $\Xi_\Gamma[\Upsilon^\Lambda](\mathsf{p}) = \alpha' + \beta'|\mathsf{p}| + \mathfrak{f}_\Gamma[\Upsilon^\Lambda]\,\mathsf{p}^2 + O(|\mathsf{p}|^3 \lg|\mathsf{p}|)$ with numbers α', β'. If $\mathsf{a} + \mathsf{b}t + \mathsf{c}t^2 = o(t^2)$ as $t \downarrow 0$, then $\mathsf{a} = \mathsf{b} = \mathsf{c} = 0$; comparing the two expansions, we obtain in particular $\mathfrak{f}_\Gamma[\Upsilon^\Lambda] = \mathfrak{f}_\Gamma[\Upsilon]$. □

Definition D.54 (The functional $\mathcal{R}$)**.** We use the variables $\mathsf{x}_0, \mathsf{x}_1, \mathsf{x}_1', \mathsf{x}_2, \mathsf{x}_2', \mathsf{x}_3$ of Def. D.40(1).

(1) The scalar products of k, l, p are linear in these variables:

$$k \cdot l = \tfrac{1}{2}(\mathsf{x}_1 + \mathsf{x}_2 - \mathsf{x}_3), \qquad k \cdot \mathsf{p} = \tfrac{1}{2}(\mathsf{x}_1 - \mathsf{x}_1' + \mathsf{x}_0), \qquad l \cdot \mathsf{p} = \tfrac{1}{2}(\mathsf{x}_2 - \mathsf{x}_2' + \mathsf{x}_0).$$

(2) For $n \in \mathbb{Z}$ let $\mathcal{L}_n$ be the real linear span of the functions $\mathsf{x}_1^{-\mathsf{a}}\mathsf{x}_2^{-\mathsf{b}}\mathsf{x}_3^{-\mathsf{c}}$, where $\mathsf{a}, \mathsf{b}, \mathsf{c} \in \mathbb{Z}$ and $\mathsf{a} + \mathsf{b} + \mathsf{c} = -n$, on the set $\{(k,l) : k \neq 0,\ l \neq 0,\ k \neq l\}$. These functions are linearly independent. Indeed, the image of the map $(k,l) \mapsto (\mathsf{x}_1, \mathsf{x}_2, \mathsf{x}_3)$ contains a non-empty open set: for $k = (t_1, t_2, 0)$ and $l = (0, t_3, 0)$ the map becomes

$$(t_1, t_2, t_3) \longmapsto (t_1^2 + t_2^2,\ t_3^2,\ t_1^2 + (t_2 - t_3)^2),$$

whose differential at $(1,0,1)$ has determinant 8, so that the inverse function theorem applies. A vanishing finite linear combination of the functions, multiplied by a sufficiently high power of $\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3$, becomes a polynomial in $(\mathsf{x}_1, \mathsf{x}_2, \mathsf{x}_3)$ which vanishes on a non-empty open set, and such a polynomial is zero. We denote by $r_{111} \colon \mathcal{L}_{-3} \to \mathbb{R}$ the linear map which assigns to an element of $\mathcal{L}_{-3}$ its coefficient of $(\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$.

(3) Let $\mathfrak{F}$ be the real linear span of the monomials

$$\mathsf{x}_0^n\,\mathsf{x}_1^{-a_1}\mathsf{x}_1'^{-a_2}\mathsf{x}_2^{-a_3}\mathsf{x}_2'^{-a_4}\mathsf{x}_3^{-a_5}, \qquad \text{integers } n \geq 0,\ a_1, \ldots, a_5, \quad \textstyle\sum_i a_i = 2 + n;$$

these monomials are homogeneous of degree -4. The numerators of the integrands in eq. D.xxviii are complete contractions of tensors built from k, l, p and Kronecker symbols; hence they are polynomials in the scalar products of k, l, p, and by (1) polynomials in the six variables. Since moreover x_3 occurs in the denominators of eq. D.xxviii at most to the first power, and not at all in F_4, each F_Γ belongs to $\mathfrak{F}$: it is a linear combination of monomials with $a_5 \leq 1$, and F_4 is a linear combination of monomials with $a_5 \leq 0$.

(4) For $F \in \mathfrak{F}$ we put

$$\mathcal{R}[F] := r_{111}(\overline{F_2}), \tag{D.xxxvi}$$

with $\overline{F_2}$ as in Def. D.51(3); by Lem. D.55 below, $\overline{F_2} \in \mathcal{L}_{-3}$.

■

Lemma D.55 (Values of $\mathcal{R}$ on monomials)**.** For every $F \in \mathfrak{F}$ the function $\overline{F_2}$ belongs to $\mathcal{L}_{-3}$; hence $\mathcal{R}$ is well defined on $\mathfrak{F}$, and it is linear. If F is a linear combination of monomials with $a_5 \le 1$, then $\overline{F_2}$ is a linear combination of functions $\mathsf{x}_1^{-\mathsf{a}}\mathsf{x}_2^{-\mathsf{b}}\mathsf{x}_3^{-\mathsf{c}}$ with $\mathsf{c} \le 1$. For integers $n \ge 0$ and $a_1, \dots, a_5$ with $\sum_i a_i = 2 + n$, put

$$(\alpha_1, \alpha_2, \alpha_3) := (a_1 + a_2,\ a_3 + a_4,\ a_5), \qquad \mathsf{q}(t) := \tfrac{1}{3}t(2t-1).$$

Then

$$\mathcal{R}\big[\mathsf{x}_0^n\, \mathsf{x}_1^{-a_1}\mathsf{x}_1'^{-a_2}\mathsf{x}_2^{-a_3}\mathsf{x}_2'^{-a_4}\mathsf{x}_3^{-a_5}\big] = \begin{cases} 1 & n = 1,\ (\alpha_1,\alpha_2,\alpha_3) = (1,1,1), \\ \mathsf{q}(a_2) + \frac{2}{3}a_2a_4 & n = 0,\ (\alpha_1,\alpha_2,\alpha_3) = (0,1,1), \\ \mathsf{q}(a_4) + \frac{2}{3}a_2a_4 & n = 0,\ (\alpha_1,\alpha_2,\alpha_3) = (1,0,1), \\ -\frac{2}{3}a_2a_4 & n = 0,\ (\alpha_1,\alpha_2,\alpha_3) = (0,0,2), \\ 0 & \text{otherwise.} \end{cases} \tag{D.xxxvii}$$

■

Proof. The map $F \mapsto \overline{F_2}$ is linear; it therefore suffices to consider a monomial F as on the left of eq. D.xxxvii. We write $x := k \cdot \mathsf{p}$ and $y := l \cdot \mathsf{p}$; then $\mathsf{x}_1' = \mathsf{x}_1 - 2x + \mathsf{x}_0$ and $\mathsf{x}_2' = \mathsf{x}_2 - 2y + \mathsf{x}_0$. For an integer t, the binomial series $(1+w)^{-t} = 1 - tw + \frac{1}{2}t(t+1)w^2 + O(w^3)$, taken at $w = (\mathsf{x}_0 - 2x)/\mathsf{x}_1$, gives for fixed (k,l) and small p

$$\mathsf{x}_1'^{-t} = \mathsf{x}_1^{-t}\Big[1 + 2t\,\frac{x}{\mathsf{x}_1} - t\,\frac{\mathsf{x}_0}{\mathsf{x}_1} + 2t(t+1)\,\frac{x^2}{\mathsf{x}_1^2} + O(|\mathsf{p}|^3)\Big],$$

and $\mathsf{x}_2'^{-t}$ has the same expansion, with y and x_2 in place of x and x_1. We use these expansions at $t = a_2$ and at $t = a_4$. On the terms of second order the operator $\frac{1}{6}\Delta_{\mathsf{p}}$ acts by

$$\tfrac{1}{6}\Delta_{\mathsf{p}}\mathsf{x}_0 = 1, \qquad \tfrac{1}{6}\Delta_{\mathsf{p}}x^2 = \tfrac{1}{3}\mathsf{x}_1, \qquad \tfrac{1}{6}\Delta_{\mathsf{p}}y^2 = \tfrac{1}{3}\mathsf{x}_2, \qquad \tfrac{1}{6}\Delta_{\mathsf{p}}(xy) = \tfrac{1}{3}\,k\cdot l = \tfrac{1}{6}(\mathsf{x}_1 + \mathsf{x}_2 - \mathsf{x}_3),$$

the last equality by Def. D.54(1).

If $n \ge 2$, then $F = O(|\mathsf{p}|^4)$ and $\overline{F_2} = 0$. If $n = 1$, the term of second order in p of F is $\mathsf{x}_0\,\mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}$, and $\overline{F_2} = \mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}$. If $n = 0$, the terms of second order are

$$\mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}\Big\{\Big[-a_2\frac{\mathsf{x}_0}{\mathsf{x}_1} + 2a_2(a_2+1)\frac{x^2}{\mathsf{x}_1^2}\Big] + \Big[-a_4\frac{\mathsf{x}_0}{\mathsf{x}_2} + 2a_4(a_4+1)\frac{y^2}{\mathsf{x}_2^2}\Big] + 4a_2a_4\,\frac{xy}{\mathsf{x}_1\mathsf{x}_2}\Big\},$$

and since $-t + \frac{2}{3}t(t+1) = \mathsf{q}(t)$, the rules above give

$$\overline{F_2} = \mathsf{x}_1^{-\alpha_1}\mathsf{x}_2^{-\alpha_2}\mathsf{x}_3^{-\alpha_3}\Big\{\mathsf{q}(a_2)\,\mathsf{x}_1^{-1} + \mathsf{q}(a_4)\,\mathsf{x}_2^{-1} + \tfrac{2}{3}a_2a_4\big[\mathsf{x}_2^{-1} + \mathsf{x}_1^{-1} - \mathsf{x}_3\mathsf{x}_1^{-1}\mathsf{x}_2^{-1}\big]\Big\}.$$

In each case $\overline{F_2} \in \mathcal{L}_{-3}$, because $\alpha_1 + \alpha_2 + \alpha_3 = 2 + n$; and the exponent c of x_3 in the terms obtained is a_5 or $a_5 - 1$, which proves the assertion on $\mathsf{c} \le 1$. Finally, the monomial $(\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$ occurs for $n = 1$ exactly when $(\alpha_1,\alpha_2,\alpha_3) = (1,1,1)$, with coefficient 1. For $n = 0$ it arises from the terms containing x_1^{-1} exactly when $(\alpha_1,\alpha_2,\alpha_3) = (0,1,1)$, from the terms containing x_2^{-1} exactly when $(\alpha_1,\alpha_2,\alpha_3) = (1,0,1)$, and from the last term exactly when $(\alpha_1,\alpha_2,\alpha_3) = (0,0,2)$. Collecting the coefficients, we obtain eq. D.xxxvii. □

Lemma D.56 (The residue)**.** Let $\mathsf{M} = \mathsf{x}_1^{-\mathsf{a}}\mathsf{x}_2^{-\mathsf{b}}\mathsf{x}_3^{-\mathsf{c}}$ with integers $\mathsf{a}, \mathsf{b}, \mathsf{c}$ such that $\mathsf{a} + \mathsf{b} + \mathsf{c} = 3$ and $\mathsf{c} \le 1$. Let $\phi \in C_c^\infty(\mathbb{R}^3 \times \mathbb{R}^3)$ be equal to 1 near 0, let $\Lambda > 1$, and let ϕ^Λ be as in Def. D.51(3). Then $(\phi^\Lambda - \phi)\mathsf{M} \in \mathfrak{P}_2(c)$ for all sufficiently small $c > 0$, and

$$\text{f. p.}\iint (\phi^\Lambda - \phi)\,\mathsf{M}\,\mathrm{d}k\,\mathrm{d}l = \begin{cases} 4\pi^4 \log\Lambda & (\mathsf{a},\mathsf{b},\mathsf{c}) = (1,1,1), \\ 0 & \text{otherwise;} \end{cases}$$
$$\oint_{S^5} \frac{\mathrm{d}\omega}{\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3} = 4\pi^4, \qquad \frac{4\pi^4}{(2\pi)^6} = \frac{1}{16\pi^2}. \tag{D.xxxviii}$$

where $d\omega$ denotes the surface measure of the sphere $S^5 \subset \mathbb{R}^3 \times \mathbb{R}^3$. ■

Proof. Step 0: the class. Let $\Upsilon_0 \in C_c^\infty(\mathbb{R}^6)$ vanish for $|k|+|l| < c$. We show that $\Upsilon_0 \mathsf{M} \in \mathfrak{P}_2(c)$. Condition ($\Phi$0) holds, since Υ_0 vanishes for $|k|+|l| < c$ and has compact support. If $|k| < \frac{c}{3}$ and $\Upsilon_0(k,l) \neq 0$, then $|l| \geq c - |k| > \frac{2}{3}c$ and $|k-l| \geq |l| - |k| > \frac{c}{3}$. Hence $\Upsilon_0 \mathsf{x}_2^{-\mathsf{b}} \mathsf{x}_3^{-\mathsf{c}}$, extended by 0 where Υ_0 vanishes, is smooth on $\{|k| < \frac{c}{3}\} \times \mathbb{R}^3$ (it vanishes on the open set $\{|k|+|l| < c\}$, and it is smooth on the open set $\{|k| < \frac{c}{3},\ |l| > \frac{2}{3}c\}$), and it vanishes where $|k|+|l| < c$ or where (k,l) lies outside the support of Υ_0. Thus (Φ1) holds with the single homogeneous factor $|k|^{-2\mathsf{a}}$, of degree $-2\mathsf{a}$. Symmetrically, if $|l| < \frac{c}{3}$ and $\Upsilon_0(k,l) \neq 0$, then $|k| \geq c - |l| > \frac{2}{3}c$ and $|k-l| \geq |k| - |l| > \frac{c}{3}$. Hence $\Upsilon_0 \mathsf{x}_1^{-\mathsf{a}} \mathsf{x}_3^{-\mathsf{c}}$, extended by 0 where Υ_0 vanishes, is smooth on $\mathbb{R}^3 \times \{|l| < \frac{c}{3}\}$ (it vanishes on the open set $\{|k|+|l| < c\}$, and it is smooth on the open set $\{|k| > \frac{2}{3}c,\ |l| < \frac{c}{3}\}$), and it vanishes where $|k|+|l| < c$ or where (k,l) lies outside the support of Υ_0. Thus (Φ2) holds with the single homogeneous factor $|l|^{-2\mathsf{b}}$, of degree $-2\mathsf{b}$. Condition (Φ3) holds because $\mathsf{c} \leq 1$ and Υ_0 has compact support. We apply this to $\Upsilon_0 = \phi^\Lambda - \phi$, which vanishes near 0, since ϕ and ϕ^Λ are both equal to 1 there.

Step 1: independence of the cut-off. Let Υ_0 be as in Step 0. For large $\operatorname{Re}\mathsf{z}$ the substitution $(k,l) \to (\Lambda k, \Lambda l)$ and the homogeneity of degree -6 of M give

$$\iint (|k||l|)^{2\mathsf{z}}\, \Upsilon_0^\Lambda\, \mathsf{M}\, dk\, dl = \Lambda^{4\mathsf{z}+6-6} \iint (|k||l|)^{2\mathsf{z}}\, \Upsilon_0\, \mathsf{M}\, dk\, dl.$$

Since Υ_0^Λ also vanishes for $|k|+|l| < c$, Lem. D.52(e), with $(\epsilon, \epsilon') = (1,1)$, continues both sides holomorphically to $\mathbb{D}$, and at $\mathsf{z} = 0$ we obtain $\operatorname{f.p.}\iint \Upsilon_0^\Lambda \mathsf{M} = \operatorname{f.p.}\iint \Upsilon_0 \mathsf{M}$. If $\tilde\phi$ is a second function with the properties of ϕ, we take $\Upsilon_0 := \phi - \tilde\phi$; then

$$(\phi^\Lambda - \phi) - (\tilde\phi^\Lambda - \tilde\phi) = \Upsilon_0^\Lambda - \Upsilon_0,$$

and the finite part of $(\phi^\Lambda - \phi)\mathsf{M}$ does not depend on the choice of ϕ. We may therefore choose ϕ conveniently in each case. In the following steps we use the function $\varphi(s) := \theta(s)$ of one variable $s \geq 0$; it is equal to 1 on $[0, \frac{1}{2}]$ and to 0 on $[1, \infty)$.

Step 2: the case $\mathsf{c} \leq 0$. We choose $\phi(k,l) := \varphi(|k|)\varphi(|l|)$. Since $\mathsf{x}_3^{-\mathsf{c}} = (k^2 - 2k \cdot l + l^2)^{|\mathsf{c}|}$ is a polynomial, M is a finite sum

$$\mathsf{M} = \sum c_{\alpha\beta}\, |k|^{-2\mathsf{a}_\alpha} k^\alpha \cdot |l|^{-2\mathsf{b}_\beta} l^\beta$$

with numbers $c_{\alpha\beta}$, multi-indices α, β and integers $\mathsf{a}_\alpha, \mathsf{b}_\beta$. For large $\operatorname{Re}\mathsf{z}$ put

$$\mathsf{Z}(\mathsf{z}) := \iint (|k||l|)^{2\mathsf{z}}\, \phi\, \mathsf{M}\, dk\, dl = \sum c_{\alpha\beta} \int |k|^{2\mathsf{z}-2\mathsf{a}_\alpha} k^\alpha \varphi(|k|)\, dk \int |l|^{2\mathsf{z}-2\mathsf{b}_\beta} l^\beta \varphi(|l|)\, dl.$$

As M is homogeneous of degree -6, we have $\iint (|k||l|)^{2\mathsf{z}} \phi^\Lambda \mathsf{M}\, dk\, dl = \Lambda^{4\mathsf{z}} \mathsf{Z}(\mathsf{z})$. The functions $|k|^{-2\mathsf{a}_\alpha} k^\alpha$ and $|l|^{-2\mathsf{b}_\beta} l^\beta$ are homogeneous factors, so by Lem. D.52(a) each factor in the expression for Z continues holomorphically to $\mathbb{D}$. Hence the continuation of $\iint (|k||l|)^{2\mathsf{z}} (\phi^\Lambda - \phi)\mathsf{M}\, dk\, dl$ is $(\Lambda^{4\mathsf{z}} - 1)\mathsf{Z}(\mathsf{z})$, which vanishes at $\mathsf{z} = 0$; by Lem. D.52(e) this value is the finite part.

Step 3: the case $\mathsf{c} = 1$, $(\mathsf{a}, \mathsf{b}) \neq (1,1)$. Now $\mathsf{a} + \mathsf{b} = 2$. The finite part is symmetric under $k \leftrightarrow l$ (Def. D.51(2)), and ϕ may be chosen freely; we may therefore assume $\mathsf{b} \leq 0$ and $\mathsf{a} \geq 2$. The degree of the homogeneous factor in (Φ2) is then $-2\mathsf{b} \geq 0$, so that Lem. D.52(e) applies with $(\epsilon, \epsilon') = (1,0)$. We choose $\phi(k,l) := \varphi(|k|)\varphi(|l-k|)$ and write $\mathsf{m} := l - k$. Then

$$\mathsf{M} = |k|^{-2\mathsf{a}}\, |k+\mathsf{m}|^{2|\mathsf{b}|}\, |\mathsf{m}|^{-2} = \sum c_{\alpha\beta}\, |k|^{-2\mathsf{a}} k^\alpha \cdot \mathsf{m}^\beta |\mathsf{m}|^{-2},$$

and for large $\operatorname{Re}\mathsf{z}$

$$\mathsf{Z}(\mathsf{z}) := \iint |k|^{2\mathsf{z}}\, \phi\, \mathsf{M}\, dk\, dl = \sum c_{\alpha\beta} \int |k|^{2\mathsf{z}-2\mathsf{a}} k^\alpha \varphi(|k|)\, dk \int \mathsf{m}^\beta |\mathsf{m}|^{-2} \varphi(|\mathsf{m}|)\, d\mathsf{m},$$

while $\iint |k|^{2\mathsf{z}}\phi^{\Lambda}\mathsf{M}\,dk\,dl = \Lambda^{2\mathsf{z}}\mathsf{Z}(\mathsf{z})$. The integrals over m converge absolutely, and the integrals over k continue to $\mathbb{D}$ by Lem. D.52(a). The continuation $(\Lambda^{2\mathsf{z}}-1)\mathsf{Z}(\mathsf{z})$ vanishes at $\mathsf{z}=0$.

Step 4: the case $(\mathsf{a},\mathsf{b},\mathsf{c}) = (1,1,1)$. We choose $\phi(k,l) := \varphi\big((k^2+l^2)^{1/2}\big)$. All degrees of the homogeneous factors in (Φ1) and (Φ2) are now -2; hence $(\phi^{\Lambda}-\phi)\mathsf{M}$ is integrable, and by Lem. D.52(e) its finite part is its integral. In polar coordinates of $\mathbb{R}^6$,

$$\iint (\phi^{\Lambda}-\phi)\,\mathsf{M}\,dk\,dl = \oint_{S^5} \mathsf{M}\,d\omega \cdot \int_0^{\infty} \big[\varphi(s/\Lambda)-\varphi(s)\big]\frac{ds}{s}.$$

The integrand on the right vanishes for $s \le \frac{1}{2}$, and for $0<\varepsilon\le\frac{1}{2}$

$$\int_{\varepsilon}^{\infty} \big[\varphi(s/\Lambda)-\varphi(s)\big]\frac{ds}{s} = \int_{\varepsilon/\Lambda}^{\varepsilon} \varphi(s)\,\frac{ds}{s} = \log\Lambda.$$

Together with the value of the integral over S^5, computed in Step 5, this gives $4\pi^4\log\Lambda$.

Step 5: the integral over the sphere. We parametrise S^5 by $k=\cos\psi\,\omega_1$, $l=\sin\psi\,\omega_2$ with $\psi\in(0,\frac{\pi}{2})$ and $\omega_1,\omega_2\in S^2$. In $\mathbb{R}^6$ we have $dk\,dl = \mathsf{r}_1^2 d\mathsf{r}_1\,d\omega_1\,\mathsf{r}_2^2 d\mathsf{r}_2\,d\omega_2$, with $\mathsf{r}_1 = s\cos\psi$, $\mathsf{r}_2 = s\sin\psi$ and $d\mathsf{r}_1\,d\mathsf{r}_2 = s\,ds\,d\psi$; hence

$$d\omega = \cos^2\psi\,\sin^2\psi\,d\psi\,d\omega_1\,d\omega_2.$$

On S^5 we have $\mathsf{x}_1\mathsf{x}_2 = \cos^2\psi\sin^2\psi$ and $\mathsf{x}_3 = 1-\sin 2\psi\;\omega_1\cdot\omega_2$. For Borel functions $g\ge 0$ on $[-1,1]$ we have $\iint_{S^2\times S^2} g(\omega_1\cdot\omega_2)\,d\omega_1\,d\omega_2 = 4\pi\cdot 2\pi\int_{-1}^{1} g(\tau)\,d\tau$, and for $0\le z<1$

$$\int_{-1}^{1}\frac{d\tau}{1-z\tau} = \sum_{j\ge0} z^j \int_{-1}^{1}\tau^j\,d\tau = 2\sum_{n\ge0}\frac{z^{2n}}{2n+1}.$$

Applying this with $z=\sin 2\psi$ for $\psi\neq\frac{\pi}{4}$, and interchanging sum and integral, which is allowed since all terms are non-negative, we find

$$\begin{aligned}\oint_{S^5}\frac{d\omega}{\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3} &= 8\pi^2\int_0^{\pi/2} 2\sum_{n\ge0}\frac{\sin^{2n}2\psi}{2n+1}\,d\psi\\ &= 16\pi^2\sum_{n\ge0}\frac{\mathsf{w}_n}{2n+1},\\ \mathsf{w}_n := \int_0^{\pi/2}\sin^{2n}2\psi\,d\psi &= \int_0^{\pi/2}\sin^{2n}\psi\,d\psi.\end{aligned}$$

An integration by parts gives $2n\,\mathsf{w}_n = (2n-1)\,\mathsf{w}_{n-1}$, and $\mathsf{w}_0=\frac{\pi}{2}$. Hence $\mathsf{w}_n = \frac{\pi}{2}\,\mathsf{b}_n$ with

$$\mathsf{b}_n := \prod_{j=1}^{n}\frac{2j-1}{2j},$$

the coefficients of the binomial series $(1-t^2)^{-1/2} = \sum_n \mathsf{b}_n t^{2n}$, $|t|<1$. Integrating this series over $[0,x]$ and letting $x\uparrow 1$ (monotone convergence), we obtain

$$\sum_n \frac{\mathsf{b}_n}{2n+1} = \int_0^1 (1-t^2)^{-1/2}\,dt = \frac{\pi}{2},$$

where we have substituted $t=\sin\psi$, $0\le\psi<\frac{\pi}{2}$, so that $(1-t^2)^{-1/2}dt = d\psi$. Therefore the integral over S^5 equals $16\pi^2\cdot\frac{\pi}{2}\cdot\frac{\pi}{2} = 4\pi^4$, and $4\pi^4/(2\pi)^6 = 1/16\pi^2$. □

Proposition D.57 (The bridge: $\kappa_\Gamma^{\mathrm{lat}}$ from $\mathcal{R}[F_\Gamma]$)**.** Let $\Gamma\in\{1,\dots,6\}$ and $\Lambda>1$; let $\mathfrak{f}_\Gamma$ be as in Prop. D.53, R_Γ^{Λ} as in Lem. D.50(b), and $\kappa_\Gamma^{\mathrm{lat}}$ as in Lem. D.44. Then

$$\mathfrak{f}_\Gamma\big[R_\Gamma^{\Lambda}-R_\Gamma\big] = d_{2,\Gamma}\log\Lambda,\quad d_{2,\Gamma} := \frac{\mathcal{R}[F_\Gamma]}{16\pi^2};\qquad \kappa_\Gamma^{\mathrm{lat}} = -\frac{3}{2\pi}\cdot 24\cdot d_{2,\Gamma} = -\frac{3}{32\pi^3}\cdot 24\cdot\mathcal{R}[F_\Gamma]. \tag{D.xxxix}$$

■

Proof. Step 1: cut-offs. The function R_Γ is smooth, even and rotation invariant. If $|k|+|l|<\frac12\rho_0$ and $|\mathsf{p}|<\frac12\rho_0$, then each of the vectors k, l, $k-l$, $\mathsf{p}-k$, $\mathsf{p}-l$, among which are the lines of Γ, has length less than ρ_0, and therefore $R_\Gamma=1=R_\Gamma^\Lambda$ there. If $R_\Gamma\neq0$, then $|l|<2\rho_0$ and $|k|<4\rho_0$: for $\Gamma\neq6$ because l and k are lines of Γ, and for $\Gamma=6$ because l and $k-l$ are. Put $\phi(k,l):=\theta((k^2+l^2)^{1/2})$; then $\phi=1$ for $|k|+|l|\le\frac12$. Consequently the functions

$$\Theta:=R_\Gamma^\Lambda-R_\Gamma,\qquad \Upsilon_0:=R_\Gamma-\phi,\qquad \phi^\Lambda-\phi,$$

regarded as functions of (k,l,p), belong to $\mathcal{Y}(c)$ with $c:=\frac12\rho_0$. Indeed, they are smooth and satisfy (Y2); for $|\mathsf{p}|<c$ and $|k|+|l|<c$ they vanish, because R_Γ, R_Γ^Λ, ϕ and ϕ^Λ are all equal to 1 there ($c<\frac12$); and they vanish when $|k|+|l|$ is large, because R_Γ, R_Γ^Λ, ϕ and ϕ^Λ do. We note that, by Lem. D.52(e), the finite part, and with it $\mathfrak{f}_\Gamma$, does not depend on the parameter c with respect to which an element of $\mathcal{Y}(c)$ is considered.

Step 2: $\mathfrak{f}_\Gamma[\Theta]=-\frac{\pi}{36}\kappa_\Gamma^{\mathrm{lat}}\log\Lambda$. The function $\Xi_\Gamma[\Theta](\mathsf{p})$ is the left side of the identity in Lem. D.50(b). We thus have two expansions of it: that of Lem. D.50(b), valid for almost every p with $|\mathsf{p}|<1$, and that of Prop. D.53, valid for every p with $0<|\mathsf{p}|\le\frac{c}{16}$. Since the remainder $\Lambda^2\mathsf{o}(\mathsf{p}/\Lambda)-\mathsf{o}(\mathsf{p})$ of the former is $o(\mathsf{p}^2)$, they give

$$(\Lambda^2-1)\,\mathsf{a}_0+(\Lambda-1)\,\mathsf{a}_1|\mathsf{p}|-\tfrac{\pi}{36}\,\kappa_\Gamma^{\mathrm{lat}}\log\Lambda\;\mathsf{p}^2+o(\mathsf{p}^2)=\alpha+\beta\,|\mathsf{p}|+\mathfrak{f}_\Gamma[\Theta]\,\mathsf{p}^2+o(\mathsf{p}^2).$$

Along a sequence $\mathsf{p}\to0$ of points at which both expansions hold, we compare first the constant terms, then the coefficients of $|\mathsf{p}|$, and then those of p^2.

Step 3: $\mathfrak{f}_\Gamma[\Theta]=\mathfrak{f}_\Gamma[\phi^\Lambda-\phi]$. We have $\Theta=(\phi^\Lambda-\phi)+(\Upsilon_0^\Lambda-\Upsilon_0)$. By the linearity of $\mathfrak{f}_\Gamma$ and the identity $\mathfrak{f}_\Gamma[\Upsilon_0^\Lambda]=\mathfrak{f}_\Gamma[\Upsilon_0]$ of Prop. D.53, the second difference contributes nothing.

Step 4: the value. Since ϕ does not depend on p, we have $\overline{((\phi^\Lambda-\phi)F_\Gamma)_2}=(\phi^\Lambda-\phi)\,\overline{(F_\Gamma)_2}$. By Def. D.54(3) and Lem. D.55, $\overline{(F_\Gamma)_2}=\sum_{\mathsf{M}}r_{\mathsf{M}}\mathsf{M}$ is a finite linear combination of monomials M as in Lem. D.56, and the coefficient of $\mathsf{M}=(\mathsf{x}_1\mathsf{x}_2\mathsf{x}_3)^{-1}$ is $r_{\mathsf{M}}=\mathcal{R}[F_\Gamma]$. The finite part is linear and independent of the parameter c (Lem. D.52(e)); hence eq. D.xxxviii yields

$$\mathfrak{f}_\Gamma[\phi^\Lambda-\phi]=(2\pi)^{-6}\cdot4\pi^4\log\Lambda\cdot\mathcal{R}[F_\Gamma]=d_{2,\Gamma}\log\Lambda.$$

Step 5: conclusion. By Steps 2–4, $\mathfrak{f}_\Gamma[R_\Gamma^\Lambda-R_\Gamma]=\mathfrak{f}_\Gamma[\Theta]=d_{2,\Gamma}\log\Lambda$ and $-\frac{\pi}{36}\kappa_\Gamma^{\mathrm{lat}}=d_{2,\Gamma}$. Hence

$$\kappa_\Gamma^{\mathrm{lat}}=-\frac{36}{\pi}\,d_{2,\Gamma}=-\frac{3}{2\pi}\cdot24\cdot d_{2,\Gamma},$$

and $\frac{3}{2\pi}\cdot\frac{1}{16\pi^2}=\frac{3}{32\pi^3}$ gives the last expression in eq. D.xxxix. □

Proposition D.58 (The six numbers $\mathcal{R}[F_\Gamma]$)**.** For the integrands $F_1,\dots,F_6$ of eq. D.xxviii,

$$\begin{aligned}
\mathcal{R}[F_1]&=-\tfrac43,\\
\mathcal{R}[F_2]&=-\tfrac{13}{12},\\
\mathcal{R}[F_3]&=-\tfrac{1}{12},\\
\mathcal{R}[F_4]&=0,\\
\mathcal{R}[F_5]&=+\tfrac{13}{6},\\
\mathcal{R}[F_6]&=-\tfrac13;
\end{aligned}\qquad\qquad(\text{D.xl})$$

$$\sum_{\Gamma=1}^{6}\mathcal{R}[F_\Gamma]=-\tfrac23.$$

■

Proof. Step 0: notation. Throughout this proof we use the abbreviations

$$A,\ A',\ B,\ B',\ C,\ P:=\mathsf{x}_1,\ \mathsf{x}_1',\ \mathsf{x}_2,\ \mathsf{x}_2',\ \mathsf{x}_3,\ \mathsf{x}_0,$$

and we put $k' := k - \mathsf{p}$, $l' := l - \mathsf{p}$ and $q := k - l = k' - l'$, so that $A' = k'^2$, $B' = l'^2$ and $C = q^2$. The relations $P = (k-k')^2 = (l-l')^2$, $C = (k-l)^2 = (k'-l')^2$, $k\cdot\mathsf{p} = A - k\cdot k'$ and $l\cdot\mathsf{p} = B - l\cdot l'$ express the remaining scalar products through these variables:

$$a := k\cdot k' = \tfrac{1}{2}(A + A' - P),$$
$$b := l\cdot l' = \tfrac{1}{2}(B + B' - P),$$
$$e := k\cdot l = \tfrac{1}{2}(A + B - C),$$
$$e' := k'\cdot l' = \tfrac{1}{2}(A' + B' - C),$$
$$f := k\cdot l' = e - k\cdot\mathsf{p} = \tfrac{1}{2}(A' + B - C - P),$$
$$f' := k'\cdot l = e - l\cdot\mathsf{p} = \tfrac{1}{2}(A + B' - C - P).$$

We shall use two symmetries. The exchange of primes, $(k,l) \leftrightarrow (k',l')$, interchanges $A \leftrightarrow A'$, $B \leftrightarrow B'$, $e \leftrightarrow e'$, $f \leftrightarrow f'$ and fixes a and b. The exchange $(k,k') \leftrightarrow (l,l')$ interchanges $(A,A') \leftrightarrow (B,B')$, $a \leftrightarrow b$, $f \leftrightarrow f'$ and fixes e and e'. Finally, $\mathsf{q}(1) = \frac{1}{3}$, $\mathsf{q}(0) = 0$ and $\mathsf{q}(-1) = 1$.

Step 1: two identities for v_3. Let

$$V := v_3(k_1,k_2,k_3)_{\alpha\beta\gamma} = \delta_{\alpha\gamma}\mathsf{a}_\beta + \delta_{\alpha\beta}\mathsf{b}_\gamma + \delta_{\beta\gamma}\mathsf{c}_\alpha, \qquad \mathsf{a} := k_1 - k_3, \quad \mathsf{b} := k_2 - k_1, \quad \mathsf{c} := k_3 - k_2.$$

(i) Since $\mathsf{a} + \mathsf{b} + \mathsf{c} = 0$,

$$V_{\alpha\beta\gamma}V_{\alpha\beta\gamma} = 3(\mathsf{a}^2+\mathsf{b}^2+\mathsf{c}^2)+2(\mathsf{a}\cdot\mathsf{b}+\mathsf{a}\cdot\mathsf{c}+\mathsf{b}\cdot\mathsf{c}) = 2(\mathsf{a}^2+\mathsf{b}^2+\mathsf{c}^2), \qquad \mathsf{a}^2+\mathsf{b}^2+\mathsf{c}^2 = 3\sum_i k_i^2 - \Big(\sum_i k_i\Big)^2.$$

(ii) If $k_1 + k_2 + k_3 = 0$, then

$$k_{1\alpha}V_{\alpha\beta\gamma} = (\mathsf{P}_2 - \mathsf{P}_3)_{\beta\gamma}, \qquad \mathsf{P}_i := k_i^2\,\delta_{\beta\gamma} - k_{i\beta}k_{i\gamma}.$$

Indeed, $k_{1\alpha}V_{\alpha\beta\gamma} = k_{1\gamma}(k_1-k_3)_\beta + k_{1\beta}(k_2-k_1)_\gamma + \delta_{\beta\gamma}\,k_1\cdot(k_3-k_2)$; here $k_{1\gamma}(k_1-k_3)_\beta + k_{1\beta}(k_2-k_1)_\gamma = k_{1\beta}k_{2\gamma} - k_{1\gamma}k_{3\beta}$, which equals $k_{3\beta}k_{3\gamma} - k_{2\beta}k_{2\gamma}$ because $k_1 = -k_2 - k_3$, and for the same reason $k_1\cdot(k_3 - k_2) = k_2^2 - k_3^2$.

Step 2: $\Gamma = 4$. By Def. D.54(3), F_4 is a linear combination of monomials with $a_5 \le 0$, whereas every non-zero case of eq. D.xxxvii has $\alpha_3 = a_5 \ge 1$. Hence $\mathcal{R}[F_4] = 0$.

Step 3: $\Gamma = 6$. Since v_3 is odd, $v_3(-l, l-k, k-\mathsf{p}) = -v_3(l, k-l, \mathsf{p}-k)$, and therefore

$$F_6 = +\tfrac{1}{6}\,V_{\alpha\beta\gamma}V_{\alpha\beta\gamma}/A'BC, \qquad V := v_3(l, q, -k'),$$

where the three arguments of v_3 add up to $l+q-k' = \mathsf{p}$. By Step 1(i), $V_{\alpha\beta\gamma}V_{\alpha\beta\gamma} = 2\big[3(B+C+A')-P\big]$, hence

$$F_6 = \frac{1}{A'C} + \frac{1}{A'B} + \frac{1}{BC} - \frac{1}{3}\,\frac{P}{A'BC}.$$

We evaluate $\mathcal{R}$ by eq. D.xxxvii. The monomial $1/A'C$ has $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (1,0,1)$ and $a_2 = 1$, $a_4 = 0$, so its value is $\mathsf{q}(0) = 0$; the monomial $1/A'B$ has $(\alpha_1,\alpha_2,\alpha_3) = (1,1,0)$ and value 0; the monomial $1/BC$ has $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (0,1,1)$ and $a_2 = 0$, so its value is $\mathsf{q}(0) = 0$; and $P/A'BC$ has $n = 1$ and $(\alpha_1,\alpha_2,\alpha_3) = (1,1,1)$, so its value is 1. Therefore $\mathcal{R}[F_6] = -\frac{1}{3}$.

Step 4: $\Gamma = 2, 3$, *a common factor.* Put

$$T_{\mu\nu} := v_2(k, \mathsf{p}-k)_{\mu\nu} = v_2(k, -k')_{\mu\nu} = a\,\delta_{\mu\nu} - k_\nu k'_\mu;$$

as v_2 is even, also $v_2(-k, k-\mathsf{p})_{\rho\nu} = T_{\rho\nu}$. Let

$$H_{\mu\rho} := T_{\mu\nu}T_{\rho\nu}/A' = \big[a^2\delta_{\mu\rho} - a\,(k_\rho k'_\mu + k_\mu k'_\rho) + A\,k'_\mu k'_\rho\big]/A',$$

and $V := v_3(-k, l, q)_{\mu\beta\gamma}$, so that $v_3(k, -l, -q) = -V$. In this notation

$$F_2 = \tfrac{1}{2}\,H_{\mu\rho}V_{\mu\beta\gamma}V_{\rho\beta\gamma}/A^2BC, \qquad F_3 = -H_{\mu\rho}\,l_\rho(l-k)_\mu/A^2BC,$$

and p enters these expressions through H only. We compute $\overline{(H_{\mu\rho})_2}$. We insert $k' = k - \mathsf{p}$, $a = A - x$ and $A' = A - 2x + P$, with $x := k\cdot\mathsf{p}$, into H and expand: the term of second order in p is a sum of

products of $\mathsf{p}_i\mathsf{p}_j$ with tensors built from δ and k, the coefficients being powers of A. Applying $\frac{1}{6}\Delta_{\mathsf{p}}$, we obtain

$$\overline{(H_{\mu\rho})_2} = \mathsf{a}_1\,\delta_{\mu\rho} + \mathsf{a}_2\,k_\mu k_\rho/A$$

with numbers $\mathsf{a}_1, \mathsf{a}_2$, since H is homogeneous of degree 2 in (k, p). Next, $k_\mu T_{\mu\nu} = a\,k_\nu - k_\nu(k\cdot k') = 0$ for all p; hence $k_\mu\overline{(H_{\mu\rho})_2} = 0$, which gives $\mathsf{a}_2 = -\mathsf{a}_1$. To determine a_1 we take the trace. From $T_{\mu\nu}T_{\mu\nu} = 3a^2 - 2a^2 + AA'$ we get $H_{\mu\mu} = a^2/A' + A$, and with $a = A - x$, $A' = A - 2x + P$,

$$\frac{a^2}{A'} = \Big(A - 2x + \frac{x^2}{A}\Big)\Big(1 + \frac{2x}{A} - \frac{P}{A} + \frac{4x^2}{A^2} + O(|\mathsf{p}|^3)\Big),$$

whose term of second order is $-P + 4x^2/A - 4x^2/A + x^2/A$; its image under $\frac{1}{6}\Delta_{\mathsf{p}}$ is $-1 + \frac{1}{3}$. Since the trace of $\overline{(H_{\mu\rho})_2}$ is $3\mathsf{a}_1 + \mathsf{a}_2 = 2\mathsf{a}_1$, we obtain $2\mathsf{a}_1 = -\frac{2}{3}$, that is,

$$\overline{(H_{\mu\rho})_2} = -\tfrac{1}{3}\,(\delta_{\mu\rho} - k_\mu k_\rho/A).$$

Step 5: $\Gamma = 2$. By Step 4,

$$\overline{(F_2)_2} = -\tfrac{1}{6}\,\mathsf{W}/A^2BC, \qquad \mathsf{W} := V_{\mu\beta\gamma}V_{\mu\beta\gamma} - |k_\mu V_{\mu\beta\gamma}|^2/A.$$

The function $A\mathsf{W}$ is a polynomial in A, B, C, homogeneous of degree 2, and $\overline{(F_2)_2} = -\frac{1}{6}A\mathsf{W}/A^3BC$. A monomial $A^iB^jC^m$ of $A\mathsf{W}$ thus contributes a multiple of $A^{i-3}B^{j-1}C^{m-1}$, and only A^2 yields $(ABC)^{-1}$; we may therefore set $B = C = 0$. By Step 1(i), applied to the arguments $-k, l, q$, whose sum is 0, $V_{\mu\beta\gamma}V_{\mu\beta\gamma} = 6(A + B + C)$, which becomes $6A$. By Step 1(ii), applied with $(k_1, k_2, k_3) = (-k, l, q)$, we have $|k_\mu V_{\mu\beta\gamma}|^2 = \operatorname{tr}(\mathsf{P}_l - \mathsf{P}_q)^2$ with $\mathsf{P}_l = B\delta - ll^{\mathsf{T}}$ and $\mathsf{P}_q = C\delta - qq^{\mathsf{T}}$. Here $\operatorname{tr}\mathsf{P}_l^2 = 2B^2$, $\operatorname{tr}\mathsf{P}_q^2 = 2C^2$, $\operatorname{tr}\mathsf{P}_l\mathsf{P}_q = BC + (l\cdot q)^2$ and $l\cdot q = e - B = \frac{1}{2}(A - B - C)$; thus $|k_\mu V_{\mu\beta\gamma}|^2 = 2B^2 + 2C^2 - 2BC - 2(l\cdot q)^2$ is a polynomial in A, B, C, which becomes $-2(\frac{1}{2}A)^2 = -\frac{1}{2}A^2$ at $B = C = 0$. Hence $A\mathsf{W}$ becomes $6A^2 + \frac{1}{2}A^2$, and

$$\mathcal{R}[F_2] = -\tfrac{1}{6}\cdot\tfrac{13}{2} = -\tfrac{13}{12}.$$

Step 6: $\Gamma = 3$. By Step 4,

$$\begin{aligned}\overline{(F_3)_2} &= +\tfrac{1}{3}\,\mathsf{W}'/A^2BC,\\ \mathsf{W}' &:= (\delta_{\mu\rho} - k_\mu k_\rho/A)\,l_\rho(l-k)_\mu\\ &= l\cdot(l-k) - \frac{(k\cdot l)\,(k\cdot(l-k))}{A}\\ &= B - e - \frac{e(e-A)}{A}.\end{aligned}$$

Again $A\mathsf{W}'$ is a polynomial in A, B, C, homogeneous of degree 2, and as in Step 5 only its coefficient of A^2 matters. At $B = C = 0$ we have $e = \frac{1}{2}A$, and $A\mathsf{W}'$ becomes $-\frac{1}{2}A^2 + \frac{1}{4}A^2$. Hence

$$\mathcal{R}[F_3] = \tfrac{1}{3}\cdot(-\tfrac{1}{4}) = -\tfrac{1}{12}.$$

Step 7: values for the denominator $AA'BC$. For a monomial $\mathsf{m} = P^nA^iA'^{i'}B^jB'^{j'}C^m$ of degree 2 we put $\mathsf{L}_5(\mathsf{m}) := \mathcal{R}[\mathsf{m}/AA'BC]$. In the notation of Lem. D.55, the monomial $\mathsf{m}/AA'BC$ has the exponents $(a_1, \dots, a_5) = (1-i, 1-i', 1-j, -j', 1-m)$, and $(\alpha_1, \alpha_2, \alpha_3) = (2-i-i',\ 1-j-j',\ 1-m)$. By eq. D.xxxvii, $\mathsf{L}_5(\mathsf{m}) \neq 0$ requires $m = 0$, and the values are as follows.

- $n = 1$ (then $i + i' = 1$, $j = j' = 0$): $\mathsf{L}_5(PA) = \mathsf{L}_5(PA') = 1$.
- $n = 0$, $(\alpha_1, \alpha_2, \alpha_3) = (0, 1, 1)$ (then $i + i' = 2$, $a_4 = 0$, value $\mathsf{q}(1 - i')$): $\mathsf{L}_5(A^2) = \frac{1}{3}$, $\mathsf{L}_5(AA') = 0$, $\mathsf{L}_5(A'^2) = 1$.
- $n = 0$, $(\alpha_1, \alpha_2, \alpha_3) = (1, 0, 1)$ (then $i + i' = 1$, $j + j' = 1$, value $\mathsf{q}(-j') - \frac{2}{3}(1 - i')j'$): $\mathsf{L}_5(AB) = \mathsf{L}_5(A'B) = 0$, $\mathsf{L}_5(AB') = 1 - \frac{2}{3} = \frac{1}{3}$, $\mathsf{L}_5(A'B') = 1$.

All other monomials of degree 2 have the value 0; in particular, every monomial containing B or C has the value 0.

Step 8: $\Gamma = 5$. Put

$$\tilde{T}_{\alpha\sigma} := v_2(k-\mathsf{p},-k)_{\alpha\sigma} = v_2(k',-k)_{\alpha\sigma} = a\,\delta_{\alpha\sigma} - k_\alpha k'_\sigma.$$

Then $k'_\alpha \tilde{T}_{\alpha\sigma} = 0 = \tilde{T}_{\alpha\sigma} k_\sigma$; we write $\tilde{T}(X,Y) := X_\alpha \tilde{T}_{\alpha\sigma} Y_\sigma = a\,X\cdot Y - (k\cdot X)(k'\cdot Y)$, and we note $\tilde{T}_{\alpha\alpha} = 2a$. Further,

$$\begin{aligned}
v_3(\mathsf{p}-k,l,k-l) = v_3(-k',l,q) &= \delta_{\alpha\gamma}u_{1\beta} + \delta_{\alpha\beta}v_{1\gamma} + \delta_{\beta\gamma}w_{1\alpha},\\
u_1 = -k'-q, \qquad & v_1 = l+k', \qquad w_1 = q-l,\\
v_3(k,-l,l-k) &= \delta_{\sigma\gamma}u_{2\beta} + \delta_{\sigma\beta}v_{2\gamma} + \delta_{\beta\gamma}w_{2\sigma},\\
u_2 = k+q, \qquad & v_2 = -(k+l), \quad w_2 = l-q.
\end{aligned}$$

Hence $F_5 = N_5/AA'BC$, where, after multiplying out the nine products and summing over β,γ,

$$\begin{aligned}
N_5 = {} & 2a\,(u_1\cdot u_2 + v_1\cdot v_2) + \tilde{T}(v_2,u_1) + \tilde{T}(u_1,w_2)\\
& + \tilde{T}(u_2,v_1) + \tilde{T}(v_1,w_2) + \tilde{T}(w_1,u_2)\\
& + \tilde{T}(w_1,v_2) + 3\,\tilde{T}(w_1,w_2).
\end{aligned}$$

The first argument of $\tilde{T}$ matters only modulo multiples of k', the second only modulo multiples of k. Using $q = k-l$ we find, in the first argument,

$$u_1 \equiv l-k, \qquad v_1 \equiv l, \qquad w_1 = k-2l, \qquad u_2 = 2k-l, \qquad v_2 = -(k+l),$$

and in the second argument

$$u_1 \equiv l-k', \qquad v_1 = l+k', \qquad u_2 \equiv -l, \qquad v_2 \equiv -l, \qquad w_2 \equiv 2l.$$

Therefore

$$\begin{aligned}
\tilde{T}(u_1,w_2) &= -2\tilde{T}(k,l) + 2\tilde{T}(l,l),\\
\tilde{T}(v_1,w_2) &= 2\tilde{T}(l,l),\\
\tilde{T}(w_1,u_2) = \tilde{T}(w_1,v_2) &= -\tilde{T}(k,l) + 2\tilde{T}(l,l),\\
3\,\tilde{T}(w_1,w_2) &= 6\tilde{T}(k,l) - 12\tilde{T}(l,l),\\
\tilde{T}(v_2,u_1) + \tilde{T}(u_2,v_1) = -\tilde{T}(k+l,l-k') + \tilde{T}(2k-l,l+k') &= \tilde{T}(k,l) - 2\tilde{T}(l,l) + 3\tilde{T}(k,k'),
\end{aligned}$$

and the sum of these seven terms is $3\tilde{T}(k,l) - 6\tilde{T}(l,l) + 3\tilde{T}(k,k')$. We regard N_5 as a polynomial in A, A', B, B', C, P; by the last assertion of Step 7 we may set $B = C = 0$ in it. Then $e = \frac{1}{2}A$, $f' = \frac{1}{2}(A + B' - P)$, $k\cdot q = A - e = \frac{1}{2}A$ and $k'\cdot q = a - f'$, and with the values of Step 7 we find:

$$\begin{aligned}
\tilde{T}(k,l) = ae - Af' &= \tfrac{1}{4}A\,(-A + A' - 2B' + P);\\
\mathsf{L}_5 &= \tfrac{1}{4}(-\tfrac{1}{3} + 0 - \tfrac{2}{3} + 1) = 0;\\
\tilde{T}(l,l) = aB - ef' &= -\tfrac{1}{4}A\,(A + B' - P);\\
\mathsf{L}_5 &= -\tfrac{1}{4}(\tfrac{1}{3} + \tfrac{1}{3} - 1) = \tfrac{1}{12};\\
\tilde{T}(k,k') = a^2 - AA' &= \tfrac{1}{4}(A + A' - P)^2 - AA';\\
\mathsf{L}_5 &= \tfrac{1}{4}(\tfrac{1}{3} + 1 - 2 - 2) = -\tfrac{2}{3};\\
u_1\cdot u_2 + v_1\cdot v_2 &= -(a + k'\cdot q + k\cdot q + C) - (e + B + a + f')\\
&= -(3a + A);\\
a\,(3a + A) &= \tfrac{3}{4}(A + A' - P)^2 + \tfrac{1}{2}A\,(A + A' - P);\\
\mathsf{L}_5 &= \tfrac{3}{4}(-\tfrac{8}{3}) + \tfrac{1}{2}(\tfrac{1}{3} - 1) = -\tfrac{7}{3}.
\end{aligned}$$

In the third line the four numbers in the bracket come from the monomials A^2, A'^2, $-2PA$, $-2PA'$ of $(A+A'-P)^2$, the others having the value 0; thus $\mathsf{L}_5((A+A'-P)^2) = \frac{1}{3}+1-2-2 = -\frac{8}{3}$, the value used in the last line. Consequently

$$\mathcal{R}[F_5] = -2(-\tfrac{7}{3}) + 3\cdot 0 - 6\cdot\tfrac{1}{12} + 3(-\tfrac{2}{3}) = \tfrac{14}{3} - \tfrac{5}{2} = \tfrac{13}{6}.$$

Step 9: values for the denominator $AA'BB'C$. For a monomial $\mathsf{m} = P^n A^i A'^{i'} B^j B'^{j'} C^m$ of degree 3 we put $\mathsf{L}(\mathsf{m}) := \mathcal{R}[\mathsf{m}/AA'BB'C]$. Now $(a_1,\dots,a_5) = (1-i, 1-i', 1-j, 1-j', 1-m)$ and $(\alpha_1,\alpha_2,\alpha_3) = (2-i-i',\, 2-j-j',\, 1-m)$, and again $\mathsf{L}(\mathsf{m}) \neq 0$ requires $m = 0$. Let $X \in \{A, A'\}$ and $Y \in \{B, B'\}$. The values are as follows.

- $n = 1$: $\mathsf{L}(PXY) = 1$.
- $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (0,1,1)$ (then $i+i' = 2$, $j+j' = 1$, value $\mathsf{q}(1-i') + \frac{2}{3}(1-i')(1-j')$): $\mathsf{L}(A^2B) = \frac{1}{3}+\frac{2}{3} = 1$, $\mathsf{L}(A^2B') = \frac{1}{3}$, $\mathsf{L}(A'^2B) = 1-\frac{2}{3} = \frac{1}{3}$, $\mathsf{L}(A'^2B') = 1$, $\mathsf{L}(AA'Y) = 0$.
- $n = 0$, $(\alpha_1,\alpha_2,\alpha_3) = (1,0,1)$ (then $i+i' = 1$, $j+j' = 2$, value $\mathsf{q}(1-j') + \frac{2}{3}(1-i')(1-j')$): $\mathsf{L}(AB^2) = 1$, $\mathsf{L}(A'B^2) = \frac{1}{3}$, $\mathsf{L}(AB'^2) = \frac{1}{3}$, $\mathsf{L}(A'B'^2) = 1$, $\mathsf{L}(XBB') = 0$.

All other monomials of degree 3 have the value 0: those containing C or P^2, and those of the forms PX_1X_2, PY_1Y_2, $X_1X_2X_3$, $Y_1Y_2Y_3$ with $X_i \in \{A, A'\}$, $Y_i \in \{B, B'\}$. These values are invariant under the exchange of primes and under $(A, A') \leftrightarrow (B, B')$.

Step 10: $\Gamma = 1$, reduction. As in Step 4, $T_{\mu\nu} = v_2(k, \mathsf{p}-k)_{\mu\nu} = a\,\delta_{\mu\nu} - k_\nu k'_\mu$, and $k_\mu T_{\mu\nu} = 0 = T_{\mu\nu}k'_\nu$. Similarly,

$$S_{\rho\sigma} := v_2(-l, l-\mathsf{p})_{\rho\sigma} = v_2(-l,l')_{\rho\sigma} = b\,\delta_{\rho\sigma} - l_\sigma l'_\rho, \qquad l_\rho S_{\rho\sigma} = 0 = S_{\rho\sigma}l'_\sigma.$$

The tensor $v_3(-k,l,q)_{\mu\rho\alpha} = \delta_{\mu\alpha}(l-2k)_\rho + \delta_{\mu\rho}(k+l)_\alpha + \delta_{\rho\alpha}(k-2l)_\mu$ is contracted with $T_{\mu\nu}S_{\rho\sigma}$, which annihilates the terms with l_ρ and with k_μ; in this contraction it may therefore be replaced by

$$\mathsf{W}_{\mu\rho\alpha} := -2k_\rho\,\delta_{\mu\alpha} + (k+l)_\alpha\,\delta_{\mu\rho} - 2l_\mu\,\delta_{\rho\alpha}.$$

Likewise, $v_3(k-\mathsf{p}, \mathsf{p}-l, l-k)_{\nu\sigma\alpha} = v_3(k', -l', l'-k')_{\nu\sigma\alpha} = \delta_{\nu\alpha}(2k'-l')_\sigma - \delta_{\nu\sigma}(k'+l')_\alpha + \delta_{\sigma\alpha}(2l'-k')_\nu$, contracted with $T_{\mu\nu}S_{\rho\sigma}$, may be replaced by $-\mathsf{W}'_{\nu\sigma\alpha}$, where W' denotes W with (k', l') in place of (k, l). Hence

$$F_1 = +\tfrac{1}{2}\,N'/AA'BB'C, \qquad N' := T_{\mu\nu}S_{\rho\sigma}\,\mathsf{W}_{\mu\rho\alpha}\mathsf{W}'_{\nu\sigma\alpha}.$$

We use the bilinear forms

$$T(X,Y) := X_\mu T_{\mu\nu} Y_\nu = a\,X\cdot Y - (k'\cdot X)(k\cdot Y), \qquad S(X,Y) := X_\rho S_{\rho\sigma} Y_\sigma = b\,X\cdot Y - (l'\cdot X)(l\cdot Y),$$

the vectors $\mathsf{s}_\sigma := k_\rho S_{\rho\sigma} = (bk - fl)_\sigma$ and $\mathsf{t}_\rho := S_{\rho\sigma}k'_\sigma = (bk' - f'l')_\rho$, and the identities $T_{\mu\mu} = 2a$, $S_{\rho\rho} = 2b$, $T_{\mu\nu}S_{\mu\nu} = 3ab - ab - ab + ee' = ab + ee'$. Let N'_{ij} be the contribution to N' of the i-th term of W and the j-th term of W'. Then

$$\begin{gathered}
N'_{11} = 4T_{\mu\mu}S(k,k') = 8a(ab - ff'),\\
N'_{33} = 4T(l,l')S_{\rho\rho} = 8b(ab - ff'),\\
N'_{22} = (k+l)\cdot(k'+l')\,(ab+ee'),\\
N'_{13} = 4T(\mathsf{s}, l'),\\
N'_{31} = 4T(l, \mathsf{t}),\\
N'_{12} = -2T(k'+l', \mathsf{s}),\\
N'_{21} = -2T(\mathsf{t}, k+l),\\
N'_{23} = -2\,T_{\mu\nu}l'_\nu\,S_{\mu\sigma}(k+l)_\sigma,\\
N'_{32} = -2\,l_\mu T_{\mu\nu}\,(k'+l')_\rho S_{\rho\nu}.
\end{gathered}$$

Under the exchange of primes, $T_{\mu\nu}$ goes over into $T_{\nu\mu}$, so that $T(X,Y)$ becomes the value of T at the images of Y and X, in this order, and s goes over into t; hence N'_{13} goes over into N'_{31}, and

N'_{12} into N'_{21}. Under the exchange $(k,k') \leftrightarrow (l,l')$, the tensors $T_{\mu\nu}$ and $S_{\mu\nu}$ are interchanged; hence $N'_{12} = -2(k'+l')_\mu T_{\mu\nu} k_\rho S_{\rho\nu}$ goes over into N'_{32}, $N'_{21} = -2S_{\mu\sigma} k'_\sigma T_{\mu\nu} (k+l)_\nu$ into N'_{23}, and N'_{11} into N'_{33}. By the invariance of the values in Step 9,

$$\mathsf{L}(N'_{33}) = \mathsf{L}(N'_{11}), \qquad \mathsf{L}(N'_{31}) = \mathsf{L}(N'_{13}), \qquad \mathsf{L}(N'_{21}) = \mathsf{L}(N'_{32}) = \mathsf{L}(N'_{23}) = \mathsf{L}(N'_{12}).$$

Step 11: $\Gamma = 1$, *values.* By Step 9 we may set $C = 0$. Then

$$\begin{aligned} 4(ab - ff') &= (A + A' - P)(B + B' - P) - (A' + B - P)(A + B' - P) \\ &= AB' + A'B - AA' - BB' =: D. \end{aligned}$$

- $N'_{11} = (A + A' - P)\, D$. Of its monomials only A^2B', A'^2B, $-PAB'$ and $-PA'B$ have non-zero values, and $\mathsf{L}(N'_{11}) = \frac{1}{3} + \frac{1}{3} - 1 - 1 = -\frac{4}{3}$.
- N'_{13}: since $\mathsf{s} \cdot l' = bf - fb = 0$ and $k' \cdot \mathsf{s} = ab - ff'$, we have $N'_{13} = -4f(ab - ff') = -\frac{1}{2}(A' + B - P)\, D$. In $(A' + B - P)\, D$ the monomials with non-zero values are A'^2B, $A'B^2$, $-PAB'$ and $-PA'B$; they give $\frac{1}{3} + \frac{1}{3} - 2 = -\frac{4}{3}$, and therefore $\mathsf{L}(N'_{13}) = \frac{2}{3}$.
- N'_{22}: here $(k+l)\cdot(k'+l') = a + f + f' + b = A + A' + B + B' - 2P$, and

$$\begin{aligned} 4(ab + ee') &= (A + A' - P)(B + B' - P) + (A + B)(A' + B') \\ &= E + AA' + BB' - P(A + A' + B + B') + P^2, \\ E &:= AB + 2AB' + 2A'B + A'B'. \end{aligned}$$

  We have $\mathsf{L}\big[(A+A'+B+B')\, E\big] = (1+\frac{2}{3})+(\frac{2}{3}+1)+(1+\frac{2}{3})+(\frac{2}{3}+1) = \frac{20}{3}$, the four contributions coming from $A(AB+2AB')$, $A'(2A'B+A'B')$, $B(AB+2A'B)$ and $B'(2AB'+A'B')$. Further, $\mathsf{L}\big[(A + A' + B + B')(AA' + BB')\big] = 0$, $\mathsf{L}\big[-P(A + A' + B + B')^2\big] = -2 \cdot 4 = -8$ and $\mathsf{L}[-2P\, E] = -2 \cdot 6 = -12$, while the remaining monomials, which contain P^2 or have one of the forms PX_1X_2, PY_1Y_2, have the value 0. Hence $\mathsf{L}(N'_{22}) = \frac{1}{4}(\frac{20}{3} - 20) = -\frac{10}{3}$.
- $N'_{12} = -2\big[a\,(k' + l') \cdot \mathsf{s} - \big(k' \cdot (k' + l')\big)(k \cdot \mathsf{s})\big]$. Since $(k' + l') \cdot \mathsf{s} = ab - ff'$, the first term is $a(ab - ff') = \frac{1}{8}(A + A' - P)\, D$, with the value $\frac{1}{8} \cdot (-\frac{4}{3}) = -\frac{1}{6}$. Next, $k' \cdot (k' + l') = A' + e' = \frac{1}{2}(3A' + B')$ and

$$k \cdot \mathsf{s} = bA - fe = \tfrac{1}{4}\big[2A(B+B'-P) - (A'+B-P)(A+B)\big] = \tfrac{1}{4}\big[AB + 2AB' - AA' - A'B - B^2 - PA + PB\big].$$

  In $(3A'+B')\big[AB+2AB'-AA'-A'B-B^2-PA+PB\big]$ the monomials with non-zero values are $-3A'^2B$, $-3A'B^2$, $+3PA'B$, $+2AB'^2$ and $-PAB'$; they give $-1-1+3+\frac{2}{3}-1 = \frac{2}{3}$. The second term, $\frac{1}{8}(3A' + B')\big[\cdots\big]$, therefore has the value $\frac{1}{8} \cdot \frac{2}{3} = \frac{1}{12}$, and $\mathsf{L}(N'_{12}) = -2(-\frac{1}{6} - \frac{1}{12}) = \frac{1}{2}$.
- Summing, with the identities at the end of Step 10,

$$\mathsf{L}(N') = 2(-\tfrac{4}{3}) + 2 \cdot \tfrac{2}{3} - \tfrac{10}{3} + 4 \cdot \tfrac{1}{2} = -\tfrac{8}{3}, \qquad \text{hence} \qquad \mathcal{R}[F_1] = \tfrac{1}{2}(-\tfrac{8}{3}) = -\tfrac{4}{3}.$$

Step 12: the sum. Adding the six values obtained in Steps 2, 3, 5, 6, 8 and 11, we find

$$\sum_{\Gamma=1}^{6} \mathcal{R}[F_\Gamma] = -\tfrac{4}{3} - \tfrac{13}{12} - \tfrac{1}{12} + 0 + \tfrac{13}{6} - \tfrac{1}{3} = \frac{-16 - 13 - 1 + 26 - 4}{12} = -\tfrac{2}{3},$$

which completes the proof of eq. D.xl. □

D.18. **The value: κ^{lat} and c_1.** Adding the six contributions of Prop. D.57 and Prop. D.58, we now obtain the value of the constant κ^{lat} of Prop. D.45, and with it the number c_1.

Proposition D.59 (The value of κ^{lat})**.** The number κ^{lat} of Prop. D.45 is

$$\kappa^{\text{lat}} = \frac{3}{2\pi^3}.$$

■

Proof. In eq. D.xxx, κ^{lat} is the sum of the numbers $\kappa_\Gamma^{\mathrm{lat}}$ of Lem. D.44. Expressing each of them by eq. D.xxxix and inserting the sum from eq. D.xl, we obtain

$$\kappa^{\mathrm{lat}} = \sum_{\Gamma=1}^{6} \kappa_\Gamma^{\mathrm{lat}} = -\frac{3}{32\pi^3} \cdot 24 \cdot \sum_{\Gamma=1}^{6} \mathcal{R}[F_\Gamma] = -\frac{3}{32\pi^3} \cdot 24 \cdot \Big(-\frac{2}{3}\Big) = \frac{48}{32\pi^3} = \frac{3}{2\pi^3},$$

as asserted. □

Definition D.60 (The number c_1)**.** We put

$$c_1 := \frac{\kappa^{\mathrm{lat}}}{\mathsf{c}_2},$$

where κ^{lat} is the number of Prop. D.45 and $\mathsf{c}_2 = 9/(16\pi^2)$ is the free Gaussian coefficient for $\dim \mathfrak{g} = 3$.

■

Proposition D.61 (The number c_1)**.** (a) $c_1 = \dfrac{8}{3\pi}$, and $\mathsf{c}_2 c_1 = \dfrac{3}{2\pi^3}$.

(b) There is a pure constant C with the following property. Let $\mathsf{d} > 0$, $\vartheta \in (0,1)$, $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(\mathsf{d};\vartheta)$ and $0 < a \le \mathsf{d}$, the class of test pairs and the functional $\mathcal{K}_5$ being those of Prop. D.46. Then

$$\Big| \frac{\mathsf{P}_1^{\mathrm{lat}}(f,h;a)}{\mathsf{c}_2\, \mathcal{K}_5(f,h)} - c_1 \Big| \le C\, \vartheta^{-3}\, \frac{a}{\mathsf{d}}\, \lg \frac{\mathsf{d}}{a}. \tag{D.xli}$$

Consequently, $\lim_{a\to 0} \mathsf{P}_1^{\mathrm{lat}}(f,h;a)/[\mathsf{c}_2 \mathcal{K}_5(f,h)] = c_1$ for every such pair.

■

Proof. (a) By Prop. D.59,

$$c_1 = \frac{3}{2\pi^3} \cdot \frac{16\pi^2}{9} = \frac{48}{18\pi} = \frac{8}{3\pi},$$

and $\mathsf{c}_2 c_1 = \kappa^{\mathrm{lat}} = \frac{3}{2\pi^3}$.

(b) By eq. 14.i with $q = 5$,

$$\mathcal{K}_5(f,h) \ge 10^{-5}\, \vartheta^2\, \mathsf{d}\, \|f\|_\infty \|h\|_\infty,$$

and the right side is positive, since neither f nor h vanishes identically. We divide eq. D.xxxi by $\mathsf{c}_2 \mathcal{K}_5(f,h)$ and use $\kappa^{\mathrm{lat}} = \mathsf{c}_2 c_1$. The left side of eq. D.xli is then at most

$$C\, \vartheta^{-1}\, a\, \lg(\mathsf{d}/a) \cdot 10^5\, \mathsf{c}_2^{-1}\, \vartheta^{-2}\, \mathsf{d}^{-1},$$

which is the asserted bound, with a pure constant. Since $\frac{a}{\mathsf{d}} \lg \frac{\mathsf{d}}{a} \to 0$ as $a \to 0$, the limit follows. □

D.19. **The pure-gauge law at short distances.** The pure comparison in Thm. X requires an expansion valid along every sufficiently deep cutoff. We first estimate the source coefficients. We then remove the upper cutoff restriction and compare the torus sides. The calculation above supplies the term of order six.

D.20. **Sources on the pure-gauge progression.** We use the pure-gauge constants and canonical data fixed for Prop. D.11. The constants are the restriction of the choice used in the stability theorem [YM3, Thm. D]; in particular the two constructions use the same running coefficients, auxiliary maps, and tests. Fix their source radius $\varrho > 0$ before choosing g. Decrease the coupling ceiling, if necessary, so that all first-passage data under consideration have the source and history hypotheses below. Throughout Lems. D.62 and D.63 and Prop. D.64 we work on the least torus, $\ell = \ell_g$. Thus constants allowed to depend on g may depend on this fixed side, but are uniform in the datum.

For a datum $D = (K, \beta)$ put

$$b_\pm = x_K \pm \bar{B}, \qquad v_j^w = -\frac{w}{D_j x_j} \quad (0 \le j < K), \qquad o_{\mathrm{src}}(x) = \frac{\mu_\star}{K_{\mathrm{L}}^\vee \sqrt{x}}, \qquad \sigma_{\mathrm{src}} = \frac{\mu_\star}{K_{\mathrm{Q}}^\vee}.$$

Here $D_j = L^{j-K}$, and $K_{\mathrm{Q}}^\vee, K_{\mathrm{L}}^\vee$ and $\mu_\star$ are the constants of the direct source bound [YM3, Lem. 6.35] and [YM3, eq. 6.cxiii]. The oscillation $\mathrm{sw}_j(v)$ is the maximum of $|v(p) - v(p')|$ for plaquettes whose barycentres are at sup-distance at most M in level-j units. Define

$$\|w\|_{\mathrm{lw}} = \max_{0 \le j < K} \max \left\{ \frac{|v_j^w|_\infty}{\sigma_{\mathrm{src}}}, \frac{\mathrm{sw}_j(v_j^w)}{o_{\mathrm{src}}(x_j)} \right\}, \qquad \mathcal{W} = \{w : \|w\|_{\mathrm{lw}} < 1\}. \tag{D.xlii}$$

This norm tests the source on every step of the given progression. The source energies are

$$\begin{aligned} e_j(X; w) &= \sum_{z \in X \cap \mathbb{T}^{(j)}} (E^{(j)}[w] - E^{(j)}[0])(X, 1, z), \\ \mathcal{E}_t(w) &= \sum_{j=1}^{t} \sum_{X} e_j(X; w), \qquad \mathcal{E} = \mathcal{E}_K. \end{aligned} \tag{D.xliii}$$

The family $E^{(j)}[w]$ uses the fixed counterterms $e_i = L^{-1} x_{i-1} - x_i$ and relative weights v_i^w. Thus the source changes the plaquette coefficient, not the running sequence used to calibrate the charts.

Lemma D.62. The progression is defined for every $w \in \mathcal{W}$. Its families are holomorphic in w, have norm at most E_0 at rate $\kappa + 1$, and the member on X depends on w only on the plaquettes of X. Moreover

$$\|(E^{(j)}[w] - E^{(j)}[0])(X, \cdot, z)\|_X \le 2E_0 \|w\|_{\mathrm{lw}} e^{-(\kappa+1) d_j(X)}, \tag{D.xliv}$$

$$|e_j(X; w)| \le 16 M^3 E_0 \|w\|_{\mathrm{lw}} e^{-\kappa d_j(X)}. \tag{D.xlv}$$

In particular the energies in eq. D.xliii are holomorphic and vanish at $w = 0$. ∎

Proof. All suprema in eq. D.xlii are maxima of finitely many absolute values of linear functions. The first entry is a norm, so $\mathcal{W}$ is an open convex neighbourhood of zero. The first-passage estimates give $b_- \le D_j x_j \le b_+$, with $b_- > 0$. The source bounds of the canonical progression apply on a sufficiently small ordinary source ball about zero: one may choose its radius $\varrho_\circ > 0$ so that

$$2\varrho_\circ \le \sigma_{\mathrm{src}} b_-, \qquad 2K_{\mathrm{L}}^\vee \varrho_\circ \gamma \le 40 \mu_\star / \sqrt{3}.$$

We now give the induction on the larger set $\mathcal{W}$; the recursion itself is unchanged.

Suppose the families through j have the stated bound. Their bare array satisfies $N_j(\mathfrak{f}_j[w]) \le N_1$, where the source-progression choice has

$$8 K_\Lambda s_\gamma N_1 \le w_\star, \qquad K_{\mathbf{M}} \{\delta(\gamma^{-2}) + s_\gamma N_1 + e^{-\gamma_2 \mathfrak{b}(\gamma^{-2})^2/2}\} \le E_0/4.$$

The ordinary Gaussian part has norm at most $E_0/2$. For $w \in \mathcal{W}$, the direct source estimate gives

$$\mathsf{n}^{\mathrm{f}}(\mathbb{W}^{\mathrm{src}}[v_j^w]) \le 2K_{\mathrm{Q}}^\vee |v_j^w|_\infty + K_{\mathrm{L}}^\vee \sqrt{x_j}\, \mathrm{sw}_j(v_j^w) < 3\mu_\star.$$

The weighted step therefore exists on every required pair core and has output norm bounded by

$$\tfrac{1}{2} E_0 + K_{\mathbf{M}} \{\delta(x_j) + s_j N_j(\mathfrak{f}_j[w]) + e^{-\gamma_2 \mathfrak{b}(x_j)^2/2}\} + 3K^{\mathrm{d}} \mu_\star \le (\tfrac{1}{2} + \tfrac{1}{4} + \tfrac{3}{16}) E_0 < E_0.$$

Here $K^{\mathrm{d}} \le K_{\mathbf{M}}^{\mathrm{f}}$ and $\mu_\star \le E_0/(16 K_{\mathbf{M}}^{\mathrm{f}})$. These are exactly the step and array estimates in the proof of Prop. A.12. Their use here requires only the displayed stepwise bounds on v_j^w. The nonzero counterterms are fixed entries of the array and are included in its ceiling N_1.

The step is jointly holomorphic in its input array, weight, and pair variables. The input array is holomorphic by induction, and $w \mapsto v_j^w$ is linear, proving holomorphy at level $j + 1$. If two sources agree on X, their earlier arrays agree on every $X' \subset X$ and their relative weights agree on X. Locality of the step proves locality at level $j + 1$. This starts with the empty array at $j = 0$.

The difference from $w=0$ is bounded by $2E_0e^{-(\kappa+1)d_j(X)}$ and vanishes at zero. Schwarz's lemma on the line through w proves eq. D.xliv. The sum over its base points is at most $8M^3e^{d_j(X)}$, giving eq. D.xlv. On the small ordinary source ball the two constructions agree, since they apply the same step to the same array at every level. □

Let $F_i=\sum_\iota h_{i,\iota}$ be real functions, where the nonzero sets of all the bumps $h_{i,\iota}$ are pairwise disjoint. Suppose

$$\|h_{i,\iota}\|_\infty\le A_i,\qquad \operatorname{Lip}h_{i,\iota}\le A_i/(\vartheta d),\qquad A_i>0,\quad \vartheta d\le 1.$$

For $z=(z_i)$ put $w_z=(\sum_i z_iF_i)\circ x$, and set

$$A_\star(d)=\min\left\{\sigma_{\mathrm{src}}b_-,\ \frac{\mu_\star}{K_{\mathrm{L}}^\vee}\frac{b_-}{\sqrt{b_+}}\left(\frac{\vartheta d}{4\sqrt{3}M}\right)^{1/2}\right\},\qquad u_i=A_i/A_\star(d). \tag{D.xlvi}$$

Then

$$\|w_z\|_{\mathrm{lw}}\le\max_i|z_i|u_i,\qquad |w|_\infty\le\sigma_{\mathrm{src}}b_+\|w\|_{\mathrm{lw}}. \tag{D.xlvii}$$

Indeed, write $A=\max_i|z_i|A_i$. At most one bump is nonzero at a point, so $|w_z|_\infty\le A$. Two plaquette barycentres at level-j sup-distance at most M have physical Euclidean distance at most $\sqrt{3}MD_j$. If the same bump is nonzero at both, use its Lipschitz bound; otherwise at most two bumps contribute. In both cases

$$|w_z(p)-w_z(p')|\le A\min\{2,2\sqrt{3}MD_j/(\vartheta d)\}.$$

Using $\min\{2,cD_j\}\le(2cD_j)^{1/2}$ and $b_-\le D_jx_j\le b_+$ gives precisely the second entry of eq. D.xlvi. The first entry controls the supremum norm. The last bound in eq. D.xlvii follows by taking $j=K-1$ in eq. D.xlii.

D.21. **Taylor coefficients after removing the source energy.** Let m_t be the positive measure carried by the history with no large-field region through level t, and let $\tau_t[\rho_0^w]$ denote its sourced version. Its support is the set $\mathcal{V}_t^X$ on which the level-t test is one. At a passing field W, write $U_t(W)$ for any representative of its minimal orbit and put

$$R_t(w,W)=\mathcal{O}(w)(U_t(W))+S[\mathfrak{f}_t[w]](U_t(W))-S[\mathfrak{f}_t[0]](U_t(W))\quad(t\ge1),\qquad R_0(w,U)=\mathcal{O}(w)(U).$$

The array action $S[\mathfrak{f}]$ subtracts the value at the trivial field, as in the sum rule.

Lemma D.63. For $w\in\mathcal{W}$, the function $R_t(w,\cdot)$ is bounded and Borel on $\mathcal{V}_t^X$, with one value on each minimal orbit, and

$$\tau_t[\rho_0^w]=e^{\mathcal{E}_t(w)}e^{R_t(w,\cdot)}m_t\qquad(0\le t\le K). \tag{D.xlviii}$$

Suppose w vanishes outside N balls of physical radius one. There are numbers $G_t^{[N]},H_t^{[N]}$ such that

$$|R_t(w,W)|\le\|w\|_{\mathrm{lw}}G_t^{[N]},\qquad|\mathcal{E}(w)-\mathcal{E}_t(w)|\le\|w\|_{\mathrm{lw}}H_t^{[N]}, \tag{D.xlix}$$

and, uniformly over the data at g,

$$G_K^{[N]}\le C(g)N,\qquad G_{t-1}^{[N]}+H_{t-1}^{[N]}\le NC(g)L^{4(K-t+1)}\quad(1\le t\le K). \tag{D.l}$$

The number $C(g)\ge1$ depends only on g and the fixed construction constants. ■

Proof. The admitted first-passage setting gives

$$d_*(t)=\min\left\{\frac{c_{\mathrm{A}}\alpha_0}{2MB_3\varepsilon_t},\frac{L^{m+K-t}-4}{2M}\right\}-1\ge3,\quad 8MB_3\varepsilon_t\le c_{\mathrm{A}}\alpha_0,\quad\varepsilon_t\le\varepsilon_K\le\varepsilon(g^{-2}).$$

In particular every passing background is in all the pair cores needed to evaluate the array. It is regular at radius $2B_3\varepsilon_t$, and

$$0\le a_p(U_t(W))\le2B_3^2\varepsilon_t^2L^{-4t}. \tag{D.li}$$

These are the background and array estimates used in [YM3, Prop. 7.9] and [YM3, eq. 7.xxv]. Gauge invariance makes their values constant on the minimal orbit.

On the small ordinary source neighbourhood in the proof of Lem. D.62, the exact step density gives eq. D.xlviii by induction. At level zero it is $\rho_0^w = e^{\mathcal{O}(w)}\rho_0$. At the next level, the preceding scalar $\mathcal{E}_t(w)$ and the new sum $\sum_X e_{t+1}(X;w)$ combine to $\mathcal{E}_{t+1}(w)$. The source-independent chart normalizer cancels, and the ratio of the resulting densities is $e^{\mathcal{E}_{t+1}(w)}e^{R_{t+1}(w,\cdot)}$. This proves the identity on that neighbourhood, including Borel measurability of $e^{R_t(w,\cdot)}$, since the unsourced density has no zero on $\mathcal{V}_t^X$.

We extend the identity to $\mathcal{W}$. For a passing field, the condition that an array action is defined depends on the field and its pair cores, not on the entries of the array. The zero-source chart identity at step $t-1$ verifies this condition through $t=K$. Thus both array actions defining $R_t(w,W)$ are defined for every $w\in\mathcal{W}$. Their counterterms are the same and cancel in their difference. If $\mathcal{Q}_j=\{(X,z): z\in X\cap\mathbb{T}^{(j)}\}$, that difference is the finite sum

$$\sum_{j=1}^{t}\sum_{(X,z)\in\mathcal{Q}_j}[E^w_{X,z}(U)-E^w_{X,z}(1)],\qquad E^w_{X,z}=(E^{(j)}[w]-E^{(j)}[0])(X,\cdot,z).$$

Every term has one value on the minimal orbit. It is holomorphic in w, and the uniform family bound gives the crude estimate

$$|R_t(w,W)|\le 2L^K\sum_p|w(p)|+4E_0\sum_{j=1}^{t}\#\mathcal{Q}_j.$$

For $w\neq 0$, let $\mathbb{D}_w=\{\zeta:|\zeta|\|w\|_{\mathrm{lw}}<1\}$. For each passing W, the function $\phi_W(\zeta)=R_t(\zeta w,W)$ is holomorphic on $\mathbb{D}_w$, vanishes at zero, and has a bound independent of W. Write $\phi_W(\zeta)=\sum_{r\ge1}c_r(W)\zeta^r$. Its coefficients are Borel. Indeed, assuming this for the coefficients below r, the functions

$$\exp\left(\phi_W(1/n)-\sum_{l<r}c_l(W)n^{-l}\right)$$

are Borel for large n, by the small-source identity. Subtract one and multiply by n^r. The resulting functions converge pointwise to $c_r(W)$; the error in replacing the exponential by its linear term tends to zero because its argument is $O(n^{-r})$. This proves the induction. The convergent series now shows that $R_t(\zeta w,\cdot)$ is Borel for every $\zeta\in\mathbb{D}_w$.

For a bounded Borel test F, both sides of eq. D.xlviii, integrated against F at source ζw, are holomorphic on $\mathbb{D}_w$. On the left this follows from the finite-lattice integral with factor $e^{\zeta\mathcal{O}(w)}$. On the right it follows from the preceding uniform bound and the finiteness of m_t. They agree near zero and hence on the whole disc. At $\zeta=1$, with F the indicator of any Borel set, this proves the asserted measure identity. The case $w=0$ is immediate.

To obtain the quantitative bounds, count the sampled plaquettes in the N source balls. Each ball contains at most $81L^{3K}$ barycentres, including all three orientations. By eq. D.xlvii, at level zero one may therefore take

$$G_0^{[N]}=162N\sigma_{\mathrm{src}}b_+L^{4K}.$$

For $1\le t\le K$, eq. D.li instead gives

$$|\mathcal{O}(w)(U_t(W))|\le 162NB_3^2\varepsilon_t^2\sigma_{\mathrm{src}}b_+L^{4(K-t)}\|w\|_{\mathrm{lw}}.$$

The array-action estimate at a regular background, [YM3, eq. 7.xxv], applied to eq. D.xliv, gives an additional

$$27NC_\Delta E_0\left(\frac{4B_3^2\varepsilon_t^2}{\alpha_0^2}+e^{-d_*(t)}\right)L^{3(K-t)}\|w\|_{\mathrm{lw}}.$$

One may take

$$C_\Delta = 32M^5 \max\left\{\frac{13}{12}C_{\mathrm{A}}C_{\mathrm{E},3}, 2C_{\mathrm{E},1}\sum_{r\ge 0} L^{3r}e^{-4(L^r-1)}\right\}.$$

In this application locality leaves only domains meeting the source balls. The number of level-t cubes they meet is at most $27NL^{3(K-t)}$; the anchored polymer estimate gives the displayed coefficient. The exponential term controls the domains too large for the local regular-field bound.

Similarly, summing eq. D.xlv over domains anchored at one of these balls gives

$$H_t^{[N]} = 3456NM^6E_0C_{\mathrm{E},1}\sum_{j=t+1}^{K} L^{3(K-j)}.$$

Use the preceding bounds as the definitions of $G_t^{[N]}$, with $G_0^{[N]}$ given separately. The geometric sum is at most $2L^{3(K-t)}$. Since $b_+ \le X_g + \bar{B}$, $\varepsilon_t \le \varepsilon(g^{-2})$, and $e^{-d_*(t)} \le e^{-3}$, these formulas prove eq. D.l. □

Proposition D.64. Let the bumps used in eq. D.xlvii vanish outside N unit balls, and let q be their number of source variables. On the polydisc $\mathbb{P} = \{z : |z_i|u_i < 1\}$, put

$$\Phi(z) = Z[w_z]e^{-\mathcal{E}(w_z)}/Z[0].$$

Then Φ is holomorphic, $\Phi(0) = 1$, and there is $Q^{[N,q]}(g) \ge 1$, independent of the datum, the bumps, and their positions, such that

$$\left|\prod_{i\in B}\partial_{z_i}\Phi(0)\right| \le Q^{[N,q]}(g)\prod_{i\in B}u_i \qquad (\varnothing \ne B \subset \{1,\dots,q\}). \tag{D.lii}$$

■

Proof. Partition histories by the first step at which a large-field region appears. The exact identity [YM3, eq. 7.xxxii] gives

$$Z[w] = \tau_K[\rho_0^w](\mathcal{V}_K) + \sum_{t=1}^{K}\mathrm{Cr}_t(w), \qquad \mathrm{Cr}_t(w) = \int g_t^{\mathrm{w}}\, d\tau_{t-1}[\rho_0^w]. \tag{D.liii}$$

Here $g_t^{\mathrm{w}} \ge 0$ is the sum of the step tests with a large-field output. Define

$$w_t = Z[0]^{-1}\int g_t^{\mathrm{w}}\, dm_{t-1} \ge 0.$$

Consistency of the history measure gives $\sum_{i\le t} w_i = \tilde{\nu}_K\{Z_t \ne \varnothing\}$ and $m_K(\mathcal{V}_K) \le Z[0]$.

After multiplication by $e^{-\mathcal{E}(w)}/Z[0]$, the first term of eq. D.liii has modulus at most $e^{\|w\|_{\mathrm{lw}}G_K^{[N]}}$. Its t-th remaining term has modulus at most

$$w_t \exp\{\|w\|_{\mathrm{lw}}(G_{t-1}^{[N]} + H_{t-1}^{[N]})\}.$$

Every term is holomorphic on $\mathcal{W}$. Set $A_t = NC(g)L^{4(K-t+1)} \ge 1$. The remaining term is bounded by ew_t on $\{|z_i|u_iA_t < 1\}$. Cauchy's formula, applied separately to each term, therefore gives

$$|\partial_B\Phi(0)| \le \left[e^{NC(g)} + e\{NC(g)L^4\}^q\sum_{t=1}^{K}L^{4q(K-t)}w_t\right]u^B. \tag{D.liv}$$

Shrinking the polydisc separately for each creation level is essential: a uniform bound on the whole of $\mathbb{P}$ would lose the useful dependence on t.

The history tail and partition lower bound eq. A.xxxix and [YM3, eq. 4.dccclii] imply

$$w_t \le 2M^{-3}\ell_g^3L^{3(K-t)}\exp\{-c'A_0^2(\log x_t)^{2p_0} + B(x_K)\ell_g^3\},$$

where $c' > 0$ is fixed and $B(x) = C^{\Sigma}_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + (9/2)\log x$. The hypotheses are the actual first-passage history hypotheses: the tests are the real canonical tests, the sequence has its fixed increments, and the first-passage frame is admitted through K. They hold at the same small-coupling ceiling used above. Since

$$x_t \geq L^{K-t}(x_K - \bar{B}) > L^{K-t}(g^{-2} - \bar{B}), \qquad x_K \leq X_g,$$

the sum in eq. D.liv is bounded by

$$2M^{-3}\ell_g^3 e^{B(X_g)\ell_g^3} \sum_{n\geq 0} L^{(4q+3)n} \exp\{-c' A_0^2[\log(g^{-2} - \bar{B}) + n\log L]^{2p_0}\}.$$

This series is finite, since $2p_0 \geq 2$. It depends only on g, q and the fixed constants. Taking its bound in eq. D.liv proves the proposition. Holomorphy and $\Phi(0) = 1$ follow from Lem. D.62 and $Z[0] > 0$. □

D.22. **The covariance on the least torus.** Write $b_+(g) = L(g^{-2} + \bar{B})$, and let

$$s(d) = \min\{s \geq 0 : L^s d \geq C_W(s)\},$$
$$C_W(s) = C_{W,0} A_0^2 \{\log[L^{s+1}(g^{-2} + \bar{B})]\}^{2p_0},$$
$$C_{W,0} = \max\{1, 4L^2 M/A_0^2\}.$$

This is the witness depth for the pure-gauge datum. For $d \leq L^{-1}$ and $K > s(d)$, its elementary bounds give

$$s(d) \geq 2, \quad K \geq 3, \quad a_K < d/13, \qquad d/a_K \geq L(\log(1/d))^2.$$

We use the pair class in Def. 14.1 and $\eta = \|f\|_\infty \|h\|_\infty$.

Proposition D.65. There are $\gamma_{\mathrm{C}} > 0$, and for each $0 < g \leq \gamma_{\mathrm{C}}$ numbers $d_{\mathrm{A}}(g) > 0$ and $C_{\mathrm{A}}(g) < \infty$, independent of the datum and test pair, with the following property. Let (K, β) be a first-passage datum at g, $\ell = \ell_g$, $s(d) \leq K$, and $d \leq d_{\mathrm{A}}(g)$. For $(f, h) \in \mathfrak{T}_{T_\ell}(d; \vartheta)$, set

$$\delta = a_K \lceil 18d/a_K \rceil, \qquad \mathsf{Pk} = \{v \in \delta\mathbb{Z}^3 : |v_\mu| \leq 0.575 - 9d,\ \mu = 1, 2, 3\}.$$

Then $\#\mathsf{Pk} \geq (30d)^{-3}$, and

$$\left| S^T_{2;K,\ell,\beta}(f, h) - \frac{1}{\#\mathsf{Pk}} \sum_{v\in\mathsf{Pk}} \mathsf{J}_2(\tau_v f, \tau_v h) \right| \leq C_{\mathrm{A}}(g)\vartheta^{-1} d^2 b^{-2} \eta. \tag{D.lv}$$

Here $\mathsf{J}_2 = \sum_{j\leq K} \mathsf{n}^1_j$ is the mixed source derivative of the field-scale progression in Prop. D.12. ■

Proof. Decrease d_{A} so that $K \geq 3$, $a_K \leq d$, and the following packing fits inside the least torus. Both supports lie in a ball $B(x_1, 9d)$. The balls $B(x_1 + v, 9d)$, $v \in \mathsf{Pk}$, have disjoint interiors and lie in $B(x_1, 1)$. Since $\delta \leq 19d$, counting points in the three coordinate intervals gives the stated lower bound after decreasing d_{A} once more. Every translation vector is in $a_K\mathbb{Z}^3$.

We first identify the mixed derivative of $\mathcal{E}$ with J_2. The two sourced recursions use the same calibrations, canonical provisions, and fixed counterterms. One places the source in the relative plaquette weight; the other in the weight member of its bare array. Near source zero both are defined on the common pair cores. Their exponentials agree by their chart integral identities, as follows.

At level one both are the integral of $\rho_0^{w_\mu}$ over the first averaging fibre. Suppose equality holds through level $t < K$ on the chain class of that level. At level $t + 1$, take a minimal representative above its block field. A fixed real gauge transformation carries this representative to the chain potential; the same transformation sends each fluctuation box to itself. For every point of the box the lowering formula puts the resulting potential in the preceding chain class. Thus the two

integrands agree there, and their chart integrals agree. All evaluations are independent of the chosen minimal representative by gauge invariance. The radii obey

$$r_K = 0, \qquad r_t = L^{-1} r_{t+1} + \tfrac{11}{10} c_{\mathfrak{h}} \delta(x_t) \le \tfrac{8}{5} c_{\mathfrak{h}} \delta(x_t).$$

These are the admitted radii of Prop. D.15; they satisfy the three hypotheses of the lowering map [YM3, Prop. 7.12]. Below any comparison depth the same bounds hold because $\delta(x_t)$ increases with t. The induction therefore reaches the flat top. There the weighted action and the subtracted array action vanish. Dividing by the identity at zero source gives equality of the exponentials of

$$\sum_{j=1}^{K} \sum_{X,z} \{E_\mu^{(j),1} - E_0^{(j),1}\}(X,1,z) \quad \text{and} \quad \mathcal{E}(w_\mu).$$

Both exponents are continuous and vanish at zero, so they agree on a connected neighbourhood of zero. Their mixed derivatives are therefore equal.

For signs $\epsilon_v \in \{-1,1\}$, put

$$F^\epsilon = \sum_v \epsilon_v \tau_v f, \qquad H^\epsilon = \sum_v \epsilon_v \tau_v h, \qquad w_z = (z_1 F^\epsilon + z_2 H^\epsilon) \circ x.$$

These bumps have disjoint nonzero sets. For a bound uniform in the datum use eq. D.xlvi with b_- replaced by $g^{-2} - \bar{B}$, and b_+ by $b_+(g)$; call the resulting amplitude $A_\star^\circ(d)$. Then $\|w_z\|_{\mathrm{lw}} \le \max\{|z_1|u_1, |z_2|u_2\}$, where $u_1 = \|f\|_\infty / A_\star^\circ$ and $u_2 = \|h\|_\infty / A_\star^\circ$. Prop. D.64, with one ball and two variables, gives

$$|\partial_1 \partial_2 \log\{Z[w_z] e^{-\mathcal{E}(w_z)} / Z[0]\}_{z=0}| \le (Q + Q^2) u_1 u_2 \le 2Q^2 u_1 u_2.$$

This follows directly from $\Phi(0) = 1$; no nonvanishing claim for Φ away from zero is required.

Average this derivative over all signs. Bilinearity eliminates the pairs of distinct translates and leaves the sum of $(S_2^T - \mathsf{J}_2)(\tau_v f, \tau_v h)$. Translation invariance makes its first term $\#\mathsf{Pk}\, S_2^T(f,h)$. Hence the error in eq. D.lv is at most $(30d)^3\, 2Q^2 A_\star^\circ(d)^{-2}\eta$. For sufficiently small d, the square-root entry of the amplitude is the smaller one. Its inverse square is a constant depending on g, times $(\vartheta d)^{-1}$. Since $b \in [g^{-2}, b_+(g)]$, this is the stated bound. If either sampled source vanishes, its observable and every translated source jet vanish, and the same conclusion holds without this calculation. □

D.23. **The fourth-order coefficient.**

Proposition D.66. Let $g \le \gamma_s$, $\ell \in \mathfrak{L}(g)$, $d \le L^{-1}$, $K > s(d)$, and $20d \le \ell$. For every pair $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$,

$$|\mathsf{h}_4 - c_2 b^{-2} \mathcal{K}_6|(f,h) \le \left[\bar{\epsilon}_4(g,d) b^{-3} d + C_{\mathrm{lat}} b^{-2} a_K / d\right] \eta, \qquad C_{\mathrm{lat}} = \tfrac{3}{2} C_{\mathrm{tm}} \vartheta^{-1}. \tag{D.lvi}$$

The same function $\bar{\epsilon}_4$ works at every side and datum at g, and $\bar{\epsilon}_4(g,d) \le C_4(g) d^{1/2}$. More explicitly, with $l' = \max\{1, \log(2b'/d)\}$, one may take

$$\bar{\epsilon}_4(g,d) = \sup_{g^{-2} < b' \le b_+(g)} \frac{b'}{d} \left\{ 3C_{\mathrm{tm}} d^3 + C_{\mathrm{Bx}} \left[d^{3/2} + e^{-c_\chi \{\log(3b'/(4d^{1/2}))\}^2} \right] + C_{\mathrm{Bt}} (l')^{4p_0} e^{-c_\flat L (\log(1/d))^2} \right\},$$

where $c_\flat = \min\{(8ML)^{-1}, \delta_{\mathrm{tp}}/(2\sqrt{3})\}$. The constants $C_{\mathrm{Bx}}, C_{\mathrm{Bt}}, c_\chi$ depend only on the fixed construction. ■

Proof. Fix $2 \le n \le K$ with $D_n \ge d$. Use the downward chain of Prop. D.15, starting at $A_n^\lambda = 0$, and let $\widetilde{U}_i^\lambda = e^{iL^{-i} A_i^\lambda}$. For $0 < \lambda \le 1$, the recursion gives

$$A_i^\lambda = A^{\mathrm{lw}}(v_{A_{i+1}^\lambda}^{[i+1]}, \lambda x_i^{-1/2} B_i), \qquad \|v_{A_i^\lambda}^{[i]}\| \le \lambda r_i, \qquad r_i \le \tfrac{8}{5} c_{\mathfrak{h}} \delta(x_i).$$

The same recursion is holomorphic for $|\lambda| \le 1$. The boxes are fixed, convex, and symmetric. The bounds of Prop. D.11 put every chart point in its pair core, including $r_1 \le r_\alpha/2$.

Let $D_i^{\lambda,\mu}$ be the observable $\mathcal{O}(w_\mu)$ plus the sum of the source differences of all families through level i. The quotient of the sourced and unsourced chart integrals, followed by the array sum rule, gives

$$e^{D_{i+1}^{\lambda,\mu}(\widetilde{U})} = \int e^{D_i^{\lambda,\mu}(\widetilde{U}^{\mathrm{lw}}(B))} d\nu_i^\lambda[v](B).$$

Here $\nu_i^\lambda[v]$ is the unsourced step integrand divided by its nonzero integral; it has mass one. This identity is valid on the entire chain class. In its derivation, the localized array difference sums to $D_i^{\lambda,\mu}(\widetilde{U}^{\mathrm{lw}}(B)) - D_i^{\lambda,\mu}(\widetilde{U})$, while the new family is evaluated at the same background. Thus its iteration uses the actual lowering maps.

Iterate down to level one and differentiate twice in μ. With

$$\mathbb{Q}_n^\lambda = \prod_{1\le i<n} \nu_i^\lambda[v^{[i+1]}_{A_{i+1}^\lambda}], \quad \mathcal{F}_a^\lambda = \mathcal{O}(w_a) + \mathsf{Sum}[\dot{E}_a^{(1),\lambda}], \quad F_1^\lambda = \mathsf{Sum}[\ddot{E}^{(1),\lambda}],$$

we obtain

$$\sum_{j=2}^{n} \mathsf{n}_j^\lambda = \langle \mathcal{F}_1^\lambda(\widetilde{U}_1^\lambda); \mathcal{F}_2^\lambda(\widetilde{U}_1^\lambda)\rangle_{\mathbb{Q}_n^\lambda} + \int \{F_1^\lambda(\widetilde{U}_1^\lambda) - F_1^\lambda(1)\}\, d\mathbb{Q}_n^\lambda. \tag{D.lvii}$$

We identify its fourth-order coefficient at fixed lattice. As $\lambda \to 0$, the potentials vanish uniformly on the boxes, the chart points tend uniformly to zero, and $\tau_i(\lambda) \to \tau_i = t_i/x_i$. The whole inverse covariance at the trivial pair and the extra gauge-fixing factor combine to

$$\mathsf{M}_i = \tau_i \mathsf{A}_i^{\mathrm{cv}} + (1-\tau_i)\widehat{\mathsf{G}} = \tau_i \mathsf{B}_i^{\mathrm{cv}} + \widehat{\mathsf{G}}.$$

Its gap is at least $\frac{3}{4}\gamma^{\mathrm{tr}}$. The normalizers converge to positive Gaussian normalizers. Hence the density converges uniformly, and $\mathbb{Q}_n^\lambda$ converges in total variation to $\bigotimes_{1\le i<n} \nu_i^{\mathrm{r}}$, the product of the Gaussians with inverse M_i conditioned on their boxes.

The curvature derivative of each lowering map is $\mathsf{sg}_i \mathsf{T}_i$, with $\mathsf{sg}_i \in \{-1, 1\}$. The plaquette expansion therefore gives, uniformly on the finite product of boxes,

$$a_p(\widetilde{U}_1^\lambda) = \tfrac{1}{2}\lambda^2 |\Theta_p|^2 + O(\lambda^3), \qquad \Theta_p = \sum_{1\le i<n} \mathsf{sg}_i x_i^{-1/2} (\mathsf{T}_i B_i)(p).$$

Thus $\lambda^{-2}\mathcal{O}(w_a) \to q_a = \frac{1}{2} a_K^{-1} \sum_p w_a(p) |\Theta_p|^2$.

The level-one families cause no additional covariance at this order. To check this explicitly, for a family f_1 on a pair core,

$$|f_1(X, P[v], z) - f_1(X, 1, z)| \le 2\|f_1\|_{1,\kappa'} e^{-\kappa' d_1(X)} \lambda^2 (r_1/r_\alpha)^2.$$

Indeed, $f_1(X, P[\zeta v], z)$ is holomorphic on $|\zeta| < r_\alpha/\|v\|$; its first derivative at zero vanishes by constant-gauge invariance and $\int_{\mathrm{SU}(2)} \mathrm{Ad}(u) Y\, du = 0$. Cauchy's estimate bounds the remaining series. The first source jets have norm $O(\lambda^2)$ by Prop. D.12, so their variation along the chain is $O(\lambda^4)$; their constant values cancel inside the covariance. For the mixed jet the same estimate and locality give

$$|F_1^\lambda(\widetilde{U}_1^\lambda) - F_1^\lambda(1)| \le c\lambda^4 R_0^{-2} a_K^{-2} (r_1/r_\alpha)^2 \Pi_1 \eta.$$

The coefficient exists by holomorphy. Together with the Cauchy bound on $\mathsf{n}_{1,[4]}$, it yields

$$\left| \mathsf{h}_4^{(n)} - \langle q_1; q_2\rangle_{\otimes \nu_i^{\mathrm{r}}} \right| \le C b^{-2} l^{4p_0} e^{-c_\flat d/a_K} \eta. \tag{D.lviii}$$

Here we used

$$R_0^{-2} \le c(\log(b/a_K))^{2p_0} a_K/b,$$
$$r_1^2 \le c(\log(b/a_K))^{2p_0} L a_K/b,$$
$$\Pi_1 \le \max\{1, (d/a_K)^3\} e^{2-d/(4ML a_K)}.$$

The polynomial factors are absorbed by half of this exponential. The signs sg_i may be removed, since every ν_i^{r} is invariant under $B_i \mapsto -B_i$.

We need a comparison with the unconditioned Gaussian uniform in the number of bonds. Its component cumulants of orders two and four differ from those under ν_i^{r} by at most

$$\epsilon_i = C_\chi e^{-c_\chi (\log x_i)^{2p_0}}.$$

Here is the relevant test-removal estimate. Introduce at most four distinct source variables in $\sum_\alpha z_\alpha B_\alpha$, allowing the components to repeat. Each insertion has quadratic-growth norm at most $\frac{1}{2}M^{3/2}|z_\alpha|$. On the polydisc $|z_\alpha| < w_\star/(4n_f)$, $n_f = M^{3/2}/2$, they make one admitted origin. Remove the bond tests from the Gaussian expansion. The difference of the connected activities on X is bounded by

$$K_P^{\Pi} C_{\mathrm{ex}}((L+2)M)^3 e^{4\lambda_1} e^{-\gamma_{\mathrm{tail}}\mathfrak{b}(x_i)^2/2} e^{-\kappa_+ d_{i+1}(X)}.$$

The proof is the Gaussian-part comparison in [YM3, Prop. 7.15], with the two potential origins zero. It applies to the present quadratic system: $3/4 \le \tau_i \le 4/3$, its precision has the stated uniform gap, and it is the admitted Gaussian system at scale zero in Prop. D.12. Cauchy's formula and the sum over polymers anchored at one insertion give the claimed ϵ_i. Odd moments vanish by symmetry.

In an orthonormal colour basis put

$$F_i^\alpha(p) = x_i^{-1/2}(\mathsf{T}_i B_i)^\alpha(p), \qquad c_i^{\alpha\beta}(p,q) = \langle F_i^\alpha(p) F_i^\beta(q)\rangle,$$
$$\kappa_i^{\alpha\beta}(p,q) = \kappa(F_i^\alpha(p), F_i^\alpha(p), F_i^\beta(q), F_i^\beta(q)).$$

Independence across levels and the moment–cumulant identity give

$$\langle q_1; q_2\rangle = \tfrac{1}{4}a_K^{-2}\sum_{p,q} w_1(p)w_2(q)\sum_{\alpha,\beta}\left\{2\left(\sum_i c_i^{\alpha\beta}(p,q)\right)^2 + \sum_i \kappa_i^{\alpha\beta}(p,q)\right\}.$$

The slice identity [YM3, Prop. 7.3] is

$$\mathsf{T}_i \mathsf{M}_i^{-1}\mathsf{T}_i^{\mathsf{T}} = L^{-i}\tau_i^{-1}\mathcal{C}_{(i+1)}.$$

Since $x_i^{-1}L^{-i}\tau_i^{-1} = \beta^{-1}$, the Gaussian covariance in the last formula is $\delta_{\alpha\beta}\beta^{-1}\mathcal{C}_{(i+1)}$. Its Gaussian value is therefore

$$\tfrac{3}{2}b^{-2}\sum_{p,q} w_1(p)w_2(q)\Sigma(p,q)^2, \qquad \Sigma = \sum_{1\le i<n}\mathcal{C}_{(i+1)} = C^{\mathrm{fl}}_{K,K-n} - \mathcal{C}_{(1)}.$$

Let C_{row} be the row-sum constant of [YM3, Prop. 7.6]: $\sup_p \sum_b |\mathsf{T}_i(p,b)| \le C_{\mathrm{row}}L^{-2i}$. Multilinearity gives

$$|c_i^{\alpha\beta} - \delta_{\alpha\beta}\beta^{-1}\mathcal{C}_{(i+1)}| \le \tfrac{4}{3}C_{\mathrm{row}}^2\beta^{-1}L^{-3i}\epsilon_i, \qquad |\kappa_i^{\alpha\beta}| \le \tfrac{16}{9}C_{\mathrm{row}}^4\beta^{-2}L^{-6i}\epsilon_i.$$

There are at most $35^2(d/a_K)^6$ contributing plaquette pairs. Put $u_i = d/D_i$, $E_k = \sum_{1\le i<n} u_i^k\epsilon_i$, and

$$S_\Sigma = \sup_{u>0}\sum_{s\in\mathbb{Z}}(uL^s)^3 e^{-\delta_{\mathrm{tp}}uL^s/\sqrt{3}} < \infty.$$

On those pairs the slice bound gives $(d/a_K)^3(|\Sigma| + |\mathcal{C}_{(1)}|) \le C_{\mathrm{tp}}S_\Sigma$. Expanding the square now bounds the test-removal error by

$$\frac{35^2}{4}b^{-2}\{16C_{\mathrm{tp}}S_\Sigma C_{\mathrm{row}}^2 E_3 + 32C_{\mathrm{row}}^4E_3^2 + 16C_{\mathrm{row}}^4E_6\}\eta.$$

For $D_i < d$, write $\log(3b/(4D_i)) = \Lambda_b + \log u_i$, where $\Lambda_b = \log(3b/(4d))$. Then $(\Lambda_b + t)^2 \ge \Lambda_b^2 + t^2$, and the $\log u_i$ are spaced by $\log L$. The sum is bounded using the finite number

$$S_\chi = \sup_{t_0\ge 0}\sum_{s\ge 0} e^{6(t_0+s\log L) - c_\chi(t_0+s\log L)^2}.$$

For $d \le D_i \le d^{1/2}$, use $\sum u_i^3 \le 2$ and $\epsilon_i \le C_\chi e^{-c_\chi\{\log(3b/(4d^{1/2}))\}^2}$. For $D_i > d^{1/2}$, use $\sum u_i^3 \le 2d^{3/2}$. Thus, for $k = 3, 6$,

$$E_k \le C_\chi(S_\chi + 2)\left[d^{3/2} + e^{-c_\chi\{\log(3b/(4d^{1/2}))\}^2}\right].$$

This proves the term with C_{Bx}. The omitted first slice has

$$(d/a_K)^3|\mathcal{C}_{(1)}(p,q)| \le C_{\mathrm{tp}}(d/a_K)^3 e^{-\delta_{\mathrm{tp}}d/(\sqrt{3}a_K)}.$$

Together with eq. D.lviii it gives the term with C_{Bt}.

Take $n = K$, so $D_n = 1$. The coarsest fluctuation-template estimate Lem. A.26, gives

$$\left|\sum_{p,q} w_1(p)w_2(q)(C^{\mathrm{fl}}_{K,0}(p,q))^2 - \frac{3}{8\pi^2}\mathcal{K}_6(f,h)\right| \le C_{\mathrm{tm}}\left[\frac{a_K}{\vartheta d} + (d/\ell)^3 + d^3\right]\eta.$$

The colour factor is $\frac{3}{2}\cdot 3/(8\pi^2) = c_2$. Use $d/a_K \ge L(\log(1/d))^2$, and write $b^{-2} = (b/d)b^{-3}d$; this gives the stated $\bar\epsilon_4$. Its Gaussian-logarithmic exponentials decrease faster than any power of d, uniformly for $g^{-2} < b' \le b_+(g)$. The slowest remaining term is $b'd^{1/2}$, proving $\bar\epsilon_4(g,d) \le C_4(g)d^{1/2}$. □

Lemma D.67. In the setting of Prop. D.12, write $\mathsf{J}_2 = \sum_{j=1}^K \mathsf{n}_j^1$ and $\mathsf{h}_{2m} = \mathsf{h}_{2m}^{(K)}$. There is a constant C_E, depending only on the construction constants, such that

$$|\mathsf{h}_{2m}| \le C_E b^{-m} d^{m-2} l^{2mp_0}\eta \quad (m = 2, 3), \qquad |\mathsf{J}_2 - \mathsf{h}_4 - \mathsf{h}_6| \le C_E b^{-4} d^2 l^{8p_0}\eta.$$

■

Proof. If an even holomorphic function H is bounded by B on $|\lambda| < R$, then

$$|H_{[2m]}| \le BR^{-2m}, \qquad \left|H(1) - \sum_{m=0}^{p} H_{[2m]}\right| \le \frac{BR^{-2p-2}}{1 - R^{-2}}.$$

Apply these inequalities to n_{j+1}^λ on the disc of radius ρR_j. Its squared radius is at least two. The bound in Prop. D.12, evaluated on that disc, therefore gives

$$c\, b^{-m}(\log x_j)^{2mp_0} D_j^{m-2}\chi_j\Pi_{j+1}\eta$$

for its coefficient of order $2m$, and the same expression with $m = 4$ for its remainder after order six. The coefficients of orders zero and two vanish.

For $D_j \ge d$, one has $\Pi_{j+1} = 1$, $\log x_j \le l$, and

$$D_j^{m-2}\chi_j = d^{m-2}(d/D_j)^{5/2-(m-2)}.$$

The exponents are $5/2, 3/2, 1/2$ for $m = 2, 3, 4$; each gives a convergent geometric sum. For $D_j < d$, write D_j/d as a power of L^{-1}, up to a factor in $[1, L]$. The factor Π_{j+1} decays exponentially in d/D_j, whereas the other factors grow by powers of d/D_j and $\log(d/D_j)$. Their sum is bounded by a constant times $d^{m-2}l^{2mp_0}$. Summing the two ranges proves both claims. □

D.24. **A cutoff sufficient for the pure-gauge limit.**

Proposition D.68. Fix $0 < g \le \gamma$ and $\ell \in \mathfrak{L}(g)$. There are $K_0, K_\flat \in \mathbb{N}$, $C_\diamond < \infty$, and $d_{\mathrm{K}} > 0$, depending only on g, ℓ and the fixed constants, such that

$$K_\sharp(d) = \max\{K_0, K_\flat, \lceil 3600\log_L(\sqrt{3}M/d)\rceil\}$$

has the following property. For $0 < d \le 1$, $20d \le \ell$, $b \in [\gamma^{-2}, L^2(g^{-2} + \bar B)]$, $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$, and $K' \ge K \ge K_\sharp(d)$,

$$|S^T_{2;K',\ell,L^{K'}b} - S^T_{2;K,\ell,L^K b}|(f,h) \le C_\diamond\vartheta^{-2}d^{178}\eta.$$

The same bound holds with the full cutoff limit $S_2^{(\ell,b)}$ in the first term. For $d \le d_{\mathrm{K}}$,

$$s(d) < K_\sharp(d), \qquad (\sqrt{3}M/d)^{3600} \le L^{K_\sharp(d)} \le d^{-3601}.$$

■

Proof. The displayed label interval is an admitted compact interval for the fixed-torus cutoff theorem Cor. A.21 and Prop. A.24. At order two that result gives, uniformly in this interval,

$$|S^T_{2;K'} - S^T_{2;K}|(f,h) \le C'\ell^9 K^{6p_0+1/2} L^{-K/10} \|f\|_\sharp \|h\|_\sharp, \qquad \|f\|_\sharp = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\}.$$

It applies above K_0 once, with $u = \lceil K/1800 \rceil$ and support sup-distance $d_\infty \ge d/\sqrt{3}$,

$$d_\infty L^u \ge 6M, \qquad c_9 \ell^3 L^{3u} e^{-d_\infty L^u/M} \le \tfrac{1}{2} C \ell^9 K^{6p_0+1/2} L^{-K/10}.$$

All constants here are those of that theorem on the indicated compact interval; in particular $C > 0$. Put $y = K/1800$ and $c_{10} = \max\{0, \log(2c_9/C)\}$. Choose $K_\flat = \lceil 1800 y_0 \rceil$ so that

$$L^{y/2} \ge c_{10} + 3\log L + 200 y \log L \quad (y \ge y_0).$$

The other defining bound for $K_\sharp$ gives $L^{y/2} \ge \sqrt{3}M/d$. Their product proves both required inequalities, after increasing $K_\flat$ to make the first right side at least six. Also

$$K^{6p_0+1/2} L^{-K/10} \le cL^{-K/20} \le c(d/(\sqrt{3}M))^{180}, \qquad \|f\|_\sharp \|h\|_\sharp \le (40M)^2 (\vartheta d)^{-2} \eta.$$

These give the claimed d^{178} bound. The fixed-torus theorem gives the full cutoff limit and permits $K' \to \infty$ in it.

Minimality of $s(d)$ gives

$$L^{s(d)} d < L C_{W,0} A_0^2 \{\log L^{s(d)} + \log b_+(g)\}^{2p_0}.$$

For small d this implies $L^{s(d)} \le d^{-2}$. Once the logarithmic term in $K_\sharp$ exceeds the two fixed thresholds,

$$(\sqrt{3}M/d)^{3600} \le L^{K_\sharp(d)} < L(\sqrt{3}M/d)^{3600} \le d^{-3601}.$$

Decreasing d_{K} proves both final assertions. □

D.25. **Comparison with the least pure-gauge torus.** We require the following comparison only for torus measures. Write $\widetilde{\mu}_{K,\ell,\beta}$ for their periodic extension to the infinite lattice.

Proposition D.69. Let $g \le \gamma_W^{\mathrm{sd}}$, where this ceiling ensures the source and history hypotheses of Lems. D.62 and D.63 and Prop. D.64. We include $(\gamma_W^{\mathrm{sd}})^{-2} \ge 4\bar{B}$ in this choice. Let (K,β) be a first-passage datum at g, $K \ge 3$, and $a_K \le 9d \le 1/20$. Suppose real Lipschitz functions $\tilde{f}, \tilde{h}$ have disjoint nonzero sets, each vanishes outside a ball of radius d, both vanish outside one ball of radius $9d$, and

$$\operatorname{Lip} \tilde{f} \le \|\tilde{f}\|_\infty/(\vartheta d), \qquad \operatorname{Lip} \tilde{h} \le \|\tilde{h}\|_\infty/(\vartheta d).$$

They need not be nonnegative or satisfy integral lower bounds. There are $Q_2^{\mathrm{T}}(g) \ge 1$ and $C_2^{\mathrm{T}}(g) < \infty$, independent of the cutoff, datum, side, pair, and position, such that for every $\ell' \in \mathfrak{L}(g)$,

$$|S_2^T[\widetilde{\mu}_{K,\ell',\beta}] - S_2^T[\widetilde{\mu}_{K,\ell_g,\beta}]|(\tilde{f},\tilde{h}) \le (30d)^3\, 2Q_2^{\mathrm{T}}(g)^2 A_\star(d)^{-2}\eta.$$

Here $A_\star$ is the amplitude in eq. D.xlvi. In particular, if also

$$d \le d_\star(g,\vartheta) = 4\sqrt{3}M\vartheta^{-1}(K_{\mathrm{L}}^\vee/K_{\mathrm{Q}}^\vee)^2 g^{-2},$$

then

$$|S_2^T[\widetilde{\mu}_{K,\ell',\beta}] - S_2^T[\widetilde{\mu}_{K,\ell_g,\beta}]|(\tilde{f},\tilde{h}) \le C_2^{\mathrm{T}}(g)\vartheta^{-1} d^2 b^{-2}\eta. \tag{D.lix}$$

These constants may depend on the fixed source radius and on the stability assignment. ■

Proof. Put $\mu' = \widetilde{\mu}_{K,\ell_g,\beta}$ and $l = \ell_g/2 = L^m$. Choose $o \in a_K\mathbb{Z}^3$ within coordinate distance $a_K/2$ of the centre x_1 of the ball of radius $9d$. The eight cells

$$Q_{\mathbf{k}} = o + l\mathbf{k} + [-l/2, l/2]^3, \qquad \mathbf{k} \in \{0,1\}^3,$$

form a board on the least torus. Reflection in each plane $x_\mu = o_\mu \pm l/2$ preserves the lattice. The unit ball about x_1 lies strictly inside Q_0. Its reflected copies have centres at sup-distance at least $l - a_K$.

For a function F of the bonds inside Q_0, the pure-gauge chessboard comparison [YM3, Cor. 5.70] gives

$$|\widetilde{\mu}_{K,\ell',\beta}(F)| \le Q_0 \|F\|_{\mathrm{ch}}, \qquad \|F\|_{\mathrm{ch}}^8 = \mu'\left(\prod_{\mathbf{k}} F^{[\mathbf{k}]}\right) \ge 0.$$

Its pressure factor is at most $\exp\{l^3(p_{\ell_g} - p_{\ell'})\}$. The zero-source upper and lower partition bounds of [YM3, Thm. D], including their vacuum-energy comparison, bound this difference by $C_{\mathrm{pr}}(g)$. Thus $Q_0 = e^{l^3 C_{\mathrm{pr}}(g)}$ works for every side. This is the sole use here of the source radius in the stability theorem.

For functions F_1, F_2 in the unit ball, write

$$X_a = \mathcal{O}(F_a), \quad P_a = X_a - \mu'(X_a), \quad P_{12} = P_1 P_2 - S_2^T[\mu'](F_1, F_2).$$

For $\mu = \widetilde{\mu}_{K,\ell',\beta}$,

$$(S_2^T[\mu] - S_2^T[\mu'])(F_1, F_2) = \mu(P_{12}) - \mu(P_1)\mu(P_2).$$

In the joint-cumulant expansion of the chessboard product of the eight P_{12}'s, singleton blocks cancel by centering, and a two-entry block contained in one cell cancels with its subtracted covariance. Every remaining block therefore meets at least two cells. The analogous assertion for a product of the P_a's follows from centering.

We bound these linking cumulants on the least torus. Assign one variable to each of the at most sixteen reflected observables, and let W_z be their summed source. By eq. D.xlvii, $\|W_z\|_{\mathrm{lw}} \le \max_i |z_i| u_i$, where the u_i's are the corresponding suprema divided by $A_\star(d)$. Decompose its logarithm as

$$\log \frac{Z[W_z]}{Z} = \log \Phi(z) + \left\{ \mathcal{E}(W_z) - \sum_{\mathbf{k}} \mathcal{E}(w^{[\mathbf{k}]}) \right\} + \sum_{\mathbf{k}} \mathcal{E}(w^{[\mathbf{k}]}).$$

All derivatives used here are at zero, where $\Phi(0) = 1$. A linking derivative kills the last sum. By Prop. D.64, with eight balls and sixteen variables, the squarefree derivatives of Φ are bounded by $Q_{\mathrm{D}} u^B$. The finite moment–cumulant formula bounds those of $\log \Phi$ by $\mathrm{cu}_{16} Q_{\mathrm{D}}^{16} u^B$.

Locality cancels from the bracket every domain meeting at most one reflected ball. Any remaining level-j domain contains two points at sup-distance at least $l - 3$. A nonwrapping domain fits in a cuboid of lengths at most $M D_j (d_j(X) + 2)$; a wrapping domain has $M(d_j(X) + 2) \ge l/D_j$. In either case

$$d_j(X) \ge (l - 3)/(M D_j) - 2.$$

Use eq. D.xlv and sum domains anchored at a source ball. Retaining half of the exponential for this size lower bound leaves a summable rooted-domain count, since $\kappa > 2(\kappa_{\mathcal{D}} + 2)$. The remaining sum over levels is bounded by a constant times

$$\sum_{r \ge 0} L^{3r} \exp\{-\tfrac{1}{2}\kappa[(l-3)L^r/M - 2]\} < \infty.$$

Thus the bracket is bounded on the unit source polydisc by $C_\Delta \max_i |z_i| u_i$. Cauchy's estimate bounds its linking derivatives by $C_\Delta u^B$. Enlarge $C_{\mathrm{LK}} = \mathrm{cu}_{16} Q_{\mathrm{D}}^{16} + C_\Delta$ to be at least one.

There are at most Bl_{16} relevant partitions. Every block has size at least two, so taking the eighth root costs at most one factor C_{LK} per pair of entries. The chessboard estimate and the preceding cancellation give

$$|S_2^T[\mu] - S_2^T[\mu']|(F_1, F_2) \le 2 Q_1^2 u_1 u_2, \qquad Q_1 = Q_0 \mathrm{Bl}_{16}^{1/8} C_{\mathrm{LK}}.$$

Apply this to the signed packed sums of $\tilde{f}, \tilde{h}$. The grid $\delta = a_K \lceil 18d/a_K \rceil \le 27d$ packs at least $(30d)^{-3}$ translates of their radius-$9d$ carrier inside the unit ball for $d \le 1/180$. Averaging the signs removes cross translates, and lattice translation invariance makes the surviving terms equal. This proves the first estimate with $Q_2^{\mathrm{T}} = Q_1$.

Finally $x_K > g^{-2}$, and $\sigma_{\mathrm{src}} = \mu_\star/K_{\mathrm{Q}}^\vee$. The square-root entry of $A_\star$ is the smaller one whenever $\vartheta d \le 4\sqrt{3}M(K_{\mathrm{L}}^\vee/K_{\mathrm{Q}}^\vee)^2 b_+$. The stated $d_\star$ ensures this. Hence

$$A_\star^{-2} = (K_{\mathrm{L}}^\vee/\mu_\star)^2 \frac{b_+}{b_-^2}\frac{4\sqrt{3}M}{\vartheta d}, \qquad \frac{b_+}{b_-^2} \le b^{-2}\frac{(X_g+\bar B)^3}{(g^{-2}-\bar B)^2}.$$

Inserting this into the first bound proves eq. D.lix. □

D.26. **The pure-gauge short-distance law.** The preceding estimates identify the first two terms of the covariance. The remaining issue is uniformity in the ultraviolet cutoff. We first compare the sixth-order coefficient with its infinite-lattice value at cutoffs of polynomial size in d^{-1}. The fixed-torus cutoff estimate then removes this restriction for a common interval of labels. Finally, comparison of two torus sizes covers the other labels of a first-passage datum.

Put

$$c_2 = \frac{9}{16\pi^2}, \qquad c_1 = \frac{8}{3\pi}, \qquad \varkappa = c_1c_2 = \frac{3}{2\pi^3}, \qquad L_b^+(f,h) = c_2 b^{-2}\{\mathcal{K}_6(f,h) + c_1 b^{-1}\mathcal{K}_5(f,h)\}.$$

All constants and auxiliary data used to define the pure-gauge progression are fixed once, before choosing g. In particular, when a pair (K,β) is admitted at two coupling thresholds, it defines the same progression and the same torus measure at either threshold. The interval

$$I(g) = (g^{-2}+\bar B, L(g^{-2}+B_\sharp) - \bar B]$$

has the useful property that every $(K, L^K b)$, $b \in I(g)$, is a first-passage datum at g, for every $K \ge 1$; see [YM3, Cor. 5.8].

Proposition D.70. There is a threshold $\gamma_{\mathrm{e}} > 0$, depending only on the construction constants, with the following properties. Fix $0 < g \le \gamma_{\mathrm{e}}$, $\vartheta \in (0,1)$, $P \ge 3601$, and $\ell = \ell_g$. Set $C_{\mathrm{lat}} = \frac{3}{2}C_{\mathrm{tm}}\vartheta^{-1}$, where C_{tm} is chosen to bound both the constant of [QCDM1, Prop. A.2] and $C_{\mathrm{tm}}^{\mathrm{fl}}$ in Lem. A.26. There are $d_{\mathrm{e}} > 0$, $c_{1/2} < \infty$, and a nondecreasing function $\bar\varepsilon : (0, d_{\mathrm{e}}] \to [0,\infty)$ tending to zero at zero such that, for $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$ and $d \le d_{\mathrm{e}}$, the following hold.

(a) For a first-passage datum (K,β) at g with $s(d) < K$ and $L^K \le d^{-P}$, set $b = a_K\beta$. Then

$$|S^T_{2;K,\ell,\beta} - L_b^+|(f,h) \le \{\bar\varepsilon(d) b^{-3} d + C_{\mathrm{lat}} b^{-2} a_K/d\}\eta. \tag{D.lx}$$

(b) For every $b \in I(g)$, eq. D.lx holds at $(K, L^K b)$ for every $K > s(d)$. Its full fixed-torus cutoff limit satisfies

$$|S_2^{(\ell,b)} - L_b^+|(f,h) \le \min\{\bar\varepsilon(d), c_{1/2}d^{1/2}\} b^{-3} d\,\eta.$$

(c) Suppose also that

$$g \le L^{-1/4}\min\{\gamma_{\mathrm{e}}, \gamma_W^{\mathrm{sd}}\},$$

where γ_W^{sd} is the threshold in Prop. D.69. There are $d_{\mathrm{e}}' > 0$ and $\bar\varepsilon'(d) \to 0$ such that eq. D.lx, with $\bar\varepsilon'$ in place of $\bar\varepsilon$, holds for every first-passage datum at g with $K > s(d)$ and every $d \le d_{\mathrm{e}}'$, without a restriction on L^K.

These constants and functions depend only on g, ϑ, P and the fixed construction data. In (c) this includes the source radius and the choice of constants used in the source progression. They are independent of the cutoff, label, pair, and position of its supports. ■

Proof. Take $\gamma_{\mathrm{e}} = \min\{\gamma_s, \gamma_{\mathrm{C}}\}$. All the smallness requirements on d below involve only the permitted dependencies. Their minimum will define d_{e}.

Reduction to the sixth-order coefficient. Let $D = (K,\beta)$ be as in (a), and put

$$r = d/a_K, \qquad l = \max\{1, \log(2b/d)\}, \qquad b_+ = L(g^{-2}+\bar B).$$

The witness-depth bounds give $K \geq 3$, $8a_K \leq d$, and $r \geq (\log(1/d))^2 \geq 8$, after restricting d. Apply Props. D.65 and D.66 and Lem. D.67 at every translate in Pk. Since L_b^+ is invariant under translations,

$$\begin{aligned} |S^T_{2;K,\ell,\beta} - L_b^+|(f,h) \leq \Big[(E_1+E_2+E_3)b^{-3}d + C_{\mathrm{lat}}b^{-2}\frac{a_K}{d}\Big]\,\eta \\ + b^{-3}a_K \max_{v\in\mathsf{Pk}} \Big|\mathsf{H}^{(K)}[D] - \varkappa\mathcal{K}_5/a_K\Big|\,(\tau_v f, \tau_v h), \end{aligned} \tag{D.lxi}$$

where

$$E_1 = C_{\mathrm{A}}(g)\vartheta^{-1}bd, \qquad E_2 = C_E b^{-1}dl^{8p_0}, \qquad E_3 = \bar{\epsilon}_4(g,d).$$

Here $\mathsf{h}_6 = b^{-3}a_K\mathsf{H}^{(K)}[D]$. It remains to bound the last absolute value by $r\eta$ times a function tending to zero. We work at any one translate and suppress it from the notation. If one sampled source vanishes, every source jet and the lattice coefficient $\mathsf{P}_1^{\mathrm{lat}}$ vanish. The last comparison below, Prop. D.46, still bounds their difference from $\varkappa\mathcal{K}_5(f,h)$; only that last comparison is needed in this case.

Depths for the comparison. Set

$$K' = \max\{K, \lceil 24\log_L r\rceil\}, \qquad N_0 = \ell L^K, \qquad N' = \ell L^{K'}, \qquad n_1 = K' - \lceil K'/78\rceil.$$

Then

$$r^{24} \leq L^{K'} \leq \max\{d^{-P}, Lr^{24}\} \leq Ld^{-24P}.$$

First suppose $K' > K$. Then $r^{24} > r/d$, so $r > d^{-1/23}$. The following auxiliary datum and estimates are used only in this case. Fix $b_* \in I(g)$, and set $D' = (K', L^{K'}b_*)$; this is a datum by [YM3, Cor. 5.8]. Choose n_0 so that

$$D_{n_0} \in [d^{1/2}, Ld^{1/2}), \qquad L^{n_0} \in [rd^{-1/2}, Lrd^{-1/2}).$$

For small d, $2 \leq n_0 \leq K$ and

$$L^{n_0} < Lr^{12.5}, \qquad L^{n_1} \geq L^{77K'/78-1} \geq r^{23.6}/L,$$

so $n_0 < n_1$. The bare plaquette base points lie within $12r+2$ of one another, and $12r+2 \leq N_0/8$. Their lifts to $\Lambda_{N'}$ are sources $\breve{w}_a$ with the same supremum norms and lattice support size. In the physical units of D' their support radius and separation are $d' = ra_{K'}$, with $8a_{K'} \leq d'$. Thus Props. D.11 and D.12 apply at D'. These propositions require neither the witness-depth condition nor Lipschitz estimates for the lifted sources. The unit $r\eta$ is unchanged.

Comparison of the coefficients. When $K' > K$, the sequence is

$$\begin{aligned} \mathsf{H}^{(K)}[D] &\longrightarrow \mathsf{H}^{(n_0)}[D] \longrightarrow \mathsf{T}[N_0,n_0](w) \longrightarrow \mathsf{T}[N',n_0](\breve{w}) \\ &\longrightarrow \mathsf{H}^{(n_0)}[D'] \longrightarrow \mathsf{H}^{(n_1)}[D'] \longrightarrow \mathsf{T}[N',n_1](\breve{w}) \\ &\longrightarrow a_K^{-1}\mathsf{P}_1^{\mathrm{lat}}(\tilde{f},\tilde{h};a_K) \longrightarrow \varkappa\mathcal{K}_5(f,h)/a_K. \end{aligned} \tag{D.lxii}$$

We justify each step, including its domain conditions. The first and fifth arrows use the coefficient tail eq. D.ix. The second, fourth, and sixth use Props. D.15 and D.17 and Lem. D.16. Specifically, $\mathsf{h}_6^{(n)} = \gamma_2^{\mathrm{r}}$, $\beta^3 a_K^2\gamma_2^\circ = \mathsf{T}[N,n](w)$, and their difference is at most

$$\mathcal{U}(x_{n-1},\mathsf{N})\eta, \qquad \mathcal{U}(x,\mathsf{N}) = C_X\mathsf{N}^{36}\exp[-\gamma^{\mathrm{tr}}A_1^2(\log x)^{2p_0}/8], \quad \mathsf{N} = N+\beta.$$

This removal of the Gaussian windows requires

$$\gamma^{\mathrm{tr}}A_1^2(\log x_{n-1})^{2p_0} \geq 80, \qquad 10240N^3 e^{-\gamma^{\mathrm{tr}}A_1^2(\log x_{n-1})^{2p_0}/8} \leq 1.$$

Their uniform validity is checked below. The two occurrences of $\mathsf{T}[N',n_0](\breve{w})$ represent the same number: this coefficient depends on no running coupling or counterterm.

For the third arrow use Prop. D.19 with

$$\nu = \ell/D_{n_0} > \ell d^{-1/2}/L \geq \nu_W.$$

The last condition follows from $d \le (\ell/(L\nu_W))^2$; the required separation from the boundary follows from $12r+2 \le N_0/8$. For the seventh arrow apply Prop. D.20 to the lattice with top side $\ell' = N'/L^{n_1} = \ell L^{\lceil K'/78\rceil}$. The size conditions $\ell' \ge N_{\mathsf{H}}$ and $\ell' L \ge C_L M$ follow from Prop. D.11. For small d, the plaquette base points also lie within $N'/8-1$ of one another. There are at most $900r^6$ pairs. The bound $3C_*\ell'^{19} n_1 L^{-n_1/2}$ in that proposition, and

$$\frac{19}{78} + \frac{1}{156} - \frac{1}{2} = -\frac{1}{4}, \qquad r^5 \le L^{5K'/24},$$

give the error E_{10} below. The remaining rescaling identity is $a_K \mathsf{T}^{\mathrm{lat}}(\tilde{w}) = \mathsf{P}_1^{\mathrm{lat}}(\tilde{f}, \tilde{h}; a_K)$.

Finally, Props. D.46, D.59 and D.61 give

$$|\mathsf{P}_1^{\mathrm{lat}}(\tilde{f}, \tilde{h}; a_K) - \varkappa \mathcal{K}_5(f,h)| \le C_T \vartheta^{-1} a_K \lg(r)\eta.$$

The lifted pair belongs to $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$. Here the computed continuum coefficient is $\varkappa = 3/(2\pi^3)$.

In units of $r\eta$, the successive errors are

$$\begin{aligned} E_4 &= C_R l^{6p_0} d^{3/4}, & E_5 &= \mathcal{U}(x_{n_0-1}, \mathsf{N}_0)/r, \\ E_6 &= 900 r^5 C'_W N'^{237} e^{-\delta_{\mathrm{tp}} \ell d^{-1/2}/48}, & E_7 &= \mathcal{U}(x'_{n_0-1}, \mathsf{N}')/r, \\ E_8 &= C_R (l')^{6p_0} d^{3/4}, & E_9 &= \mathcal{U}(x'_{n_1-1}, \mathsf{N}')/r, \\ E_{10} &= 2700 C_* \ell^{19} L^{20} K' L^{-K'/24}, & E_{11} &= C_T \vartheta^{-1} \lg(r)/r. \end{aligned} \tag{D.lxiii}$$

The notation x' refers to D'; $\mathsf{N}_0 = N_0 + \beta$ and $\mathsf{N}' = N' + L^{K'} b_*$. The parameter bounds give

$$\mathsf{N}_0 \le (\ell + b_+) d^{-P}, \qquad \mathsf{N}' \le (\ell + b_+) L r^{24},$$
$$x_{n_0-1} \ge \tfrac{3}{4} d^{-1/2}, \qquad x'_{n_0-1} \ge \tfrac{3}{4} r^{23} d^{1/2} \ge \tfrac{3}{4} \max\{r^{11.5}, d^{-1/2}\},$$
$$x'_{n_1-1} \ge L^{K'/78}, \qquad l' \le \log(2b_+) + K' \log L.$$

For E_4, E_8 we used $(r/L^{n_0})^{3/2} \le d^{3/4}$. When $K' = K$, set $D' = D$ and $\mathsf{N}' = \mathsf{N}_0 = N_0 + \beta$. Use only

$$\mathsf{H}^{(K)}[D] \longrightarrow \mathsf{H}^{(n_1)}[D] \longrightarrow \mathsf{T}[N_0, n_1](w) \longrightarrow a_K^{-1} \mathsf{P}_1^{\mathrm{lat}} \longrightarrow \varkappa \mathcal{K}_5 / a_K.$$

The first error, divided by $r\eta$, is at most $C_R l^{6p_0} L^{3(1-0.94K)/2}$, since $r/L^{n_1} \le L^{1-0.94K}$; the other errors are E_9, E_{10}, E_{11}, with $x_{n_1-1} \ge L^{K/78}$ and $\mathsf{N}' \le (\ell + b_+) L^K$.

Uniform control for (a). At each removal of Gaussian windows, write $x_\flat$ for the preceding lower bound on the relevant running coupling. The two required conditions hold if

$$\tfrac{1}{8} \gamma^{\mathrm{tr}} A_1^2 (\log x_\flat)^{2p_0} \ge 10 + \log(10240 \mathsf{N}^3).$$

Indeed, we may use

$$\log \mathsf{N} \le C_1 + C_2 \log x_\flat, \qquad C_1 = \log((\ell + b_+)L) + P,$$

with $C_2 = 2P, 3, 78$ at the first, second, and third such comparison. The first bound uses $\log(1/d) \le 2 \log x_\flat + 0.6$; the second uses $\log x_\flat \ge 11 \log r$, and the third uses $\log x_\flat \ge K' \log L/78$. When $K' = K$, only the third bound is needed, with $x_\flat = L^{K/78}$ and $\mathsf{N}' = N_0 + \beta$. Every $x_\flat$ used here tends uniformly to infinity as $d \to 0$. Since $2p_0 \ge 2$,

$$\mathcal{U}(x_\flat, \mathsf{N}) \le C_X \exp\Big[36 C_1 + 36 C_2 \log x_\flat - \tfrac{1}{8} \gamma^{\mathrm{tr}} A_1^2 (\log x_\flat)^{2p_0}\Big]$$

tends to zero faster than every negative power of $x_\flat$.

The remaining errors have uniform bounds

$$E_6 \le 900 C'_W (\ell L)^{237} d^{-5693P} e^{-\delta_{\mathrm{tp}} \ell d^{-1/2}/48},$$
$$E_4 + E_8 \le 2 C_R [\log(2b_+ L) + 24P \log(1/d)]^{6p_0} d^{3/4},$$
$$E_{11} \le C_T \vartheta^{-1} \sup_{\rho \ge (\log(1/d))^2} \lg(\rho)/\rho.$$

The error E_{10}, and the first error in the case $K' = K$, tend to zero since $K' \geq K > s(d) \to \infty$. The same is true of E_1, E_2, E_3, uniformly for $g^{-2} \leq b \leq b_+$. Taking the supremum of the sum of these bounds over all smaller positive d defines a nondecreasing function $\bar{\varepsilon}_{\mathrm{a}}(d)$ tending to zero. Inserting it into eq. D.lxi proves (a).

Arbitrary cutoffs for $b \in I(g)$. Every $(K, L^K b)$, $b \in I(g)$, is a datum at g. For $s(d) < K \leq K_\sharp(d)$, (a) applies because Prop. D.68 gives $L^K \leq d^{-3601} \leq d^{-P}$. For $K > K_\sharp(d)$, compare with $K_\sharp$ using that proposition and then apply (a). Since $a_{K_\sharp} \leq (d/(\sqrt{3}M))^{3600}$,

$$|S^T_{2;K,\ell,L^K b} - L^+_b|(f,h) \leq \left[\bar{\varepsilon}_{\mathrm{a}}(d) + C_\diamond \vartheta^{-2} b_+^3 d^{177} + C_{\mathrm{lat}} b_+ d^{3598}\right] b^{-3} d\,\eta.$$

The same estimate holds for the full cutoff limit. Add the nondecreasing envelope of the two last errors to $\bar{\varepsilon}_{\mathrm{a}}$ to obtain $\bar{\varepsilon}$.

For the stated rate in the limit, use just the cutoff $K_\sharp$, where $r \geq d^{-3599}$ and $K' > K$. The errors E_5, E_6, E_7, E_9 are smaller than every power of d; $E_{10} + E_{11} \leq C d^{3598} \log(1/d)$; $E_1 + E_2 \leq C d (\log(1/d))^{8p_0}$; and $E_4 + E_8 \leq C d^{3/4} (\log(1/d))^{6p_0}$. By Prop. D.66, $E_3 \leq C_4(g) d^{1/2}$. The cutoff comparison errors also decay faster than $d^{1/2}$. Enlarging a constant therefore gives the claimed $c_{1/2} d^{1/2}$ bound.

All first-passage labels. Assume the additional smallness condition in (c). Set

$$g_{\mathrm{a}} = L^{1/4} g, \qquad g_{\mathrm{b}} = L^{-1/4} g, \qquad x = g^{-2}.$$

Each of $g, g_{\mathrm{a}}, g_{\mathrm{b}}$ is at most $\min\{\gamma_{\mathrm{e}}, \gamma_W^{\mathrm{sd}}\}$. The choice of γ_W^{sd} gives $x \geq 4 L^{1/2} \bar{B} \geq 5\bar{B}$. At the fixed construction data the flow is a function of (K, β), independently of which of these thresholds admits the datum. The source progression and its torus measures in (a), (b), and Prop. D.69 consequently agree whenever both are defined.

The labels at g lie in $[x, L(x + \bar{B})]$, while

$$I(g) = (x + \bar{B}, L(x + B_\sharp) - \bar{B}].$$

The omitted lower interval is contained in $I(g_{\mathrm{a}})$, and the omitted upper interval is contained in $I(g_{\mathrm{b}})$. To check the endpoints, use $x \geq 5\bar{B}$ and $L \geq 13$:

$$L^{-1/2} x + \bar{B} < x, \qquad x + \bar{B} \leq L^{1/2} x + L B_\sharp - \bar{B},$$
$$L^{1/2} x + \bar{B} \leq L(x + B_\sharp) - \bar{B}, \quad L(x + \bar{B}) \leq L(L^{1/2} x + B_\sharp) - \bar{B}.$$

For each of the three thresholds put $C_w(g'') = \sup_{s \geq 0} C_W(s) L^{-s/2}$, with C_W evaluated at that threshold. Choose d'_{e} below the minimum, over the three thresholds, of

$$1/180, \quad L^{-2}, \quad d_{\mathrm{e}}(g''), \quad d_\star(g'', \vartheta), \quad (L C_w(g''))^{-2}.$$

The witness depth at any of them then satisfies $L^{s(d)} \leq d^{-3}$. Indeed, its minimality gives $L^{s-1} d < C_w L^{(s-1)/2}$.

Now let (K, β) be any datum at g with $K > s_g(d)$. If $L^K \leq d^{-P}$, use (a). If $b \in I(g)$, use (b). In the remaining case $b \in I(g_\bullet)$, with $g_\bullet = g_{\mathrm{a}}$ or g_{b}. The same pair $(K, L^K b)$ is a datum at $g_\bullet$, and $K > s_{g_\bullet}(d)$ because $L^K > d^{-P} \geq d^{-3}$. Part (b) gives the asserted law on $T_{\ell_{g_\bullet}}$ for the pair with the same lift. This lift preserves $\eta, \mathcal{K}_5, \mathcal{K}_6$.

Both ℓ_g and $\ell_{g_\bullet}$ are of the form $2L^m$. Let $g_\circ$ correspond to the smaller side. The larger side is admitted at $g_\circ$, the datum is admitted at both thresholds, and one torus is the least torus at $g_\circ$. Thus eq. D.lix gives the difference of their covariances as at most

$$C_2^T(g_\circ) \vartheta^{-1} d^2 b^{-2} \eta \leq C_2^T(g_\circ) \vartheta^{-1} b_+ d\, b^{-3} d\,\eta.$$

The lattice error $C_{\mathrm{lat}} b^{-2} a_K / d$ is unchanged. Hence (c) holds with

$$\bar{\varepsilon}'(d) = \max_{g'' \in \{g, g_{\mathrm{a}}, g_{\mathrm{b}}\}} \bar{\varepsilon}_{g''}(d) + \max_{g'' \in \{g, g_{\mathrm{a}}, g_{\mathrm{b}}\}} C_2^T(g'') \vartheta^{-1} b_+ d.$$

This completes the proof. □

Appendix E. Index of notation and conventions

The table lists the principal notation in the order in which it is used. A cited result includes the definitions immediately preceding it.

Notation	Meaning and location
$a = L^{-K}$, λ_j, β, m_0	The fine spacing, the level spacing $\lambda_j = L^{j-K}$, and the bare parameters $\beta = \mathsf{b}/a$, $m_0 = a\mathsf{m}$; section 2.
I_g, $\mathfrak{L}(g)$, T_ℓ	The admitted interval for b, the admitted torus sides, and $(\mathbb{R}/\ell\mathbb{Z})^3$; section 2.
$D[U]$, $D_\pm$, $e_\pm$, $\mathsf{G}_\pm$	The massless chamber operator, its two internal blocks, their projections $(1 \pm \tau_3)/2$, and the physical resolvents $a(D_\pm + m_0)^{-1}$; eq. 2.i.
$\mathfrak{B}$, $O_K(w)$, $\Phi_K(J)$	The reflection-real meson basis, the action insertion, and the site bilinear insertion; eq. 2.ii–eq. 2.iii. The label 0 denotes the action density.
$\|f\|_\alpha$, $\|f\|_\sharp$	$\|f\|_0 = \|f\|_\sharp = \max(\|f\|_\infty, 40M \operatorname{Lip} f)$ and $\|f\|_\Gamma = \|f\|_\infty$ for $\Gamma \in \mathfrak{B}$; section 2.
ν_K, $\mathcal{G}^{\mathrm{tor}}_{K,\beta,m_0}$	The finite-torus gauge law and the set of its infinite-lattice torus limits; section 3.
$\mathsf{L}(f)$, $\Xi^f_\nu(z)$	The loop $\operatorname{Tr}(\mathsf{G}_+\hat{f})$ and its determinant moment; eq. 3.i. The hat denotes multiplication by the test function.
D^{mix}, E^{mix}_K, h_K	Mixed extraction by inclusion–exclusion, the resulting energy in eq. 3.xii, and the holomorphic normalization in the circle bound eq. 3.iii.
F_s, $Z_K(s)$, $\nu_{K,s}$, $\mathfrak{l}_s$	The determinant at physical mass s, its partition function, its gauge law, and $\mathfrak{l}_s = \partial_s \log \lvert F_s\rvert^2$; eq. 3.xx–eq. 3.xxii.
R_b, R^{mix}, P_{col}	Deletion of species b, the mixed projection $(1-R_1)(1-R_2)$, and independent constant colour averaging in the two species at the trivial pair; Lem. 4.2.
t_j, x_j, ϕ_j, δ_j, $\mathfrak{l}_j$	The calibration coefficient $\lambda_j^{-1}\mathsf{b}$, the running coefficient, $\phi(x_j)$, and the gauge fluctuation radius; section 4. The additional scale weight is $\mathfrak{l}_j = \mathsf{t}_j^{-1/2}(\log \mathsf{t}_j)^{-p_0}$; Prop. 3.4.
$\mathfrak{c}$, $\mathscr{D}^\circ_k$, σ_i	The free coordinates $\mathfrak{c} = (0, \mathsf{n}, (\sigma_i)_{i\le k})$, their domain at zero gauge source, and the extracted mass-source corrections; eq. 4.i.
H_k, $\bar{H}_k$, Γ_k^{-1}, W	The unbarred background map, its row twin, the fermion fluctuation covariance and the fermion fluctuation; section 4. The gauge fluctuation is B.
ϵ_F, $\|m_c^{\mathrm{tor}}\|_{\mathrm{nat}}$	The weighted norm of the chosen local decomposition of F, and the norm of its grouped cell zeroth moments; Lem. 4.5.
$\mathscr{S}$, M^*, $\mathscr{S}_\mathbb{R}$	The enlarged bilinear label space, the involution $M^* = \tau_3 M^\dagger \tau_3$, and its real form; eq. 5.i. The insertion with label M is $\Phi[M](f)$.
$G_\mathbb{C}$, G, $V(S)$	$\mathrm{Sp}(2N_f,\mathbb{C})^2$, its compact form $\mathrm{USp}(2N_f)^2$, and the unitary action of $S \in G$; eq. 5.ii and Prop. 5.19.
S^{ext}, σ^{ext}, $\mathcal{H}^{\mathrm{ext}}$, H, P_1, P_2	The extended truncated functions, their centered functional, the reflection Hilbert space, and its time and spatial translation generators; Prop. 5.18.
n, Q, $\mathcal{H}^{\mathrm{ext}}_n$, $\mathcal{H}^{\mathrm{ext}}_\pi$	A species weight, the quark-number operator Q with eigenvalue $q = \sum_f n_f$, the corresponding weight space, and the π-isotypic space. Here $\lvert\pi\rvert$ is the largest ℓ^1 norm of a weight of π; Prop. 5.19 and Thm. H.
j^T, j^T_μ, $J^T(f)$	The chamber-link current, its sum across a site bond, and its smeared physical insertion, including its factor i; eq. 8.i.
$\mathcal{M}$, n, $\mathscr{O}^{\mathcal{M}}_k$, $\mathcal{P}^{\mathcal{M}}$, $\mathcal{O}^{\mathrm{lev},\mathcal{M}}_j$	The matrices acting on spin, internal and flavour indices and trivially on colour; the bare source $\mathsf{n} = -aJ$; and the chain, physical and level source domains; the definitions preceding Prop. 8.6.
$\mathcal{D}_j$, $d_j(Y)$, $E_K(s)$, $e_j(Y;s)$, $W_K[s]$	The polymers and their block-tree lengths, the extracted energy, its local increments, and the remaining factor in $Z_K[s] = e^{E_K(s)}W_K[s]$; eq. 9.i.
$\mathcal{P}$, ρ_b, ρ_m, $W_\mathbb{C}$	The physical source ball and the complex coupling–mass neighbourhood of the real parameter window; section 9 and eq. 9.v.

Notation	Meaning and location (continued)
$d_\cap(S_1,\dots,S_n)$	$\min_{x\in T_\ell}\max_i \operatorname{dist}(x,S_i)$. It is positive exactly when the closed supports have empty total intersection; the definition preceding Lem. 9.2.
I, U_r, $\mathcal{E}_K(z)$, $h_K(z)$	The mass interval, its complex r-neighbourhood, the common vacuum normalization, and $h_K(z)=Z_K(z)e^{-\mathcal{E}_K(z)}$; section 10 and Prop. 10.13.
I_ϵ, ρ_ℓ, w_ℓ, Ω_ϵ	The longer real mass interval, the radius used in its zero-free estimate, and its complex neighbourhood of width $w_\ell=\rho_\ell/[2(e+1)]$; eq. 10.vii and Prop. 10.16.
h, $\mathcal{Z}$, $\mathcal{S}_n(z;X)$	A subsequential limit of the normalized partition functions, its zero set, and the common meromorphic continuation of the separated cumulants; Thm. O.
s,u,t_0,t_1, ϕ_1,ϕ_2	The comparison depths for adjacent cutoffs and their fixed fractions of K; eq. 11.i.
y_K, br_K, $y_\infty^\flat$	The scalar condensate $-\ell^{-3}\partial_m \log Z_{K,\ell}$, its divergent subtraction and the limit after subtraction; eq. 11.xxv and Thm. S.
$\mathsf{A}_0,\mathsf{A}_1,\mathsf{A}_2,w_1,w_2$, q_n	The five subtraction coefficients and the differences of vacuum-energy coefficients used to determine them; Thm. S and Def. 15.1.
$\mathcal{M}$, $\mathcal{V}_{[d]}$, $\mu_d^{F,K}$, μ_d^F	The block map, the configuration space at physical depth d, the pushed-forward finite-cutoff law and its total-variation limit; eq. 12.i and Thm. U.
$\mathcal{V}_{\mathrm{thr}}$, μ_{thr}^F	The product of the block configuration spaces and the probability law supported on compatible block fields; Thm. U.
$K_d^\sharp$, $T_d^\sharp$, $m_d^\sharp$, π_d	The cutoff chosen for parameter recovery, the action statistic, its pure-gauge mean and the projection onto the mean's range; Def. 13.32.
$\hat b_d$, $\hat b$, Δ_d, $\hat N$	The coupling estimators, the rescaled mean shift and the flavour estimator; Def. 13.32.
$\mathfrak{T}_E(d;\vartheta)$, $\mathcal{K}_q$, $\mathfrak{f}$	The separated nonnegative test pairs, their kernel integrals, and $\mathfrak{f}=\|f\|_\infty\|h\|_\infty$; Defs. 14.1 and 14.2.
$s(d)$, c_2, $\mathfrak{c}_1(N_f)$	The witness depth and the coefficients $9/(16\pi^2)$ and $(8-2N_f)/(3\pi)$; Def. 14.2.
λ, $\xi_j(\lambda)$, $\bar\lambda_j$	The complex field scale, the interpolated gauge coefficient and the radii for its holomorphic flow; Prop. 14.26.
n_j^λ, $\mathsf{n}_{j,[2m]}$	The mixed source derivative of the level-j energy and its Taylor coefficients in the common field scale; Prop. 14.26.
$\mathsf{h}_{2m}^{(n)}$, D^λ	The sums of the pure-gauge coefficients through level n and the weight in their constrained covariance formula; Prop. D.15.

Conventions. The barred and unbarred variables are independent Grassmann generators. Haar measure has mass one, and the Berezin integral of $e^{-\bar\psi A\psi}$ is $\det A$. The traces defining L and $\mathfrak{l}_s$ include position, spin and colour but no flavour; when a source acts on flavour, the propagator is extended by the identity on that factor.

Operator formulas use lattice coordinates; supports and distances are physical unless stated otherwise. Links on T_ℓ are periodic and spinors are antiperiodic. The volume-first and growing-torus limits have the meanings given in section 2; each theorem specifies the limits it uses. The complex parameter results keep T_ℓ fixed.

The symbol M^* on $\mathscr{S}$ denotes the displayed reflection involution. Elsewhere $*$ on an operator denotes its Hilbert-space adjoint. The functions called h_K in the circle bound and in mass continuation have the separate definitions listed above. The former subtracts a one-species energy; the latter is a normalized partition function. Likewise $\mathfrak{l}_s$ in section 3.7 is a random loop at physical mass s, whereas $\mathfrak{l}_j$ in the scale estimates is the deterministic weight listed above. In section 8, $G_U=(D_+[U]+m_0)^{-1}=a^{-1}\mathsf{G}_+$ denotes the lattice propagator; G_+ includes the physical factor a.

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